

Collision Geometry Is Linear

Diophantine Certificates for Rational Signal Machines and Conservative Particle Dynamics

Event-sparse chambers, canonical quadratic and quartic certificates, two routes to complete chronology, a conservative universal frontend, sparse lattice certificates, conserved-mass thresholds, mass four in every dimension, unbounded-time certificates for exact three-mass targets, a reversible timing threshold, periodic five-signal returns, branching clocks and native-gap compilers

AI-assisted research report prepared for Vladimir Reshetnikov from twenty-nine manuscripts

(ProveIt’s incoming reports: batch 78, manuscripts 07 and 11; batch 79, 16 and 19;

batch 80, 10, 02, 05 and 03; batch 82, 19, 06, 17, 01, 21 and 02; batch 91, 01, 03, 08–20), 2 October 2026

Abstract

A rational signal machine computes by the collisions of finitely many kinds of particles moving at fixed rational speeds on a line. Its first two Parts print two independent manuscripts that prove one theorem by two routes: once a complete finite collision history is fixed, its realizations are exactly the natural zeros of a sum of squares of integer affine residuals, of degree two, with exactly one witness tuple for each realization. In both, typed endpoint conditions on maximal adjacency intervals exclude unrecorded earlier collisions, omitted simultaneous inputs and omitted simultaneous sites, and causal depth controls the denominators of the event coordinates.

Part I (source 07) eliminates every event coordinate. The realizing initial-gap vectors form a relatively open rational polyhedral cone cut out by at most $3(n - 1) + (3m + 1)E$ linear conditions, and the only witnesses are the slacks of the strict rows. The chamber geometry yields small integral realizations, an NP upper bound for bounded synthesis, a robustness–codimension criterion, a canonical first-hit quartic for a fixed event budget, exact quasi-polynomial seed counts, an exactly solved first-collision tournament and a four-speed accumulating clock with exponentially growing denominators. Part II (source 11) keeps batch times and event positions as natural witnesses on a fixed rational grid, adds halting semantics, proves that the witness matrix has full column rank (so the certificate is strictly convex in its witnesses), gives linear witness and residual ledgers and a full-layer reference construction, and bounds the canonical witnesses by $60^K \max(1, M_0)$ for the four-speed interface of Durand-Lose’s universal machine.

Parts III and IV (sources 12 and 13, from batch 79) certify conservative dynamics at a fixed horizon. Part III writes out a reversible number-conserving universal signal machine (Morita’s universal Turing machine through Durand-Lose’s stacks) with 18 live signals, whose event dynamics is an 18-dimensional integer-linear map with a canonical quadratic step certificate; Part IV compiles one-dimensional number-conserving lattice dynamics into a quartic sum of squares with a unique natural witness and proves decidability at total mass at most two.

Parts V and VI (sources 14–17, batch 80) find the least conserved mass of an undecidable observation. With many weight-one symbols, a fixed reversible automaton simulates a universal reversible two-counter machine at total mass three, also for exact pairs (Part V); with at most one, masses three and four stay decidable, so integer-state number-conserving automata need five particles (Part VI). Source 18 and Part VII (sources 19–21, batch 82) add timed quartics and mass-four decidability on every $\mathbb{Z}^d$. Part VIII (sources 22–23) removes Part V’s horizon for fixed sources, with infinite witness fibres.

Parts IX–XIII (sources 24–38, batch 91) add a reversible binary four-particle clock and conditional uniform charts at mass four (IX), and return to signal machines: periodic macros of at most four live signals have quadratic sign certificates, five give nonsemialgebraic valid sets with degree-twelve integer certificates (X), realize every invertible planar return (XI) and run exactly clocked counter programs (XII), whose native-gap certificates need one to three witnesses per horizon (XIII).

None of the thirteen Parts compiles universal unbounded halting into one fixed-arity polynomial or improves the repository’s universal operation bounds (75 and 87 operations when Parts I–VIII were written, 84 now); Parts I and II do not transcribe the published universal machine, and Part III transcribes a different one. Re-derived devices are printed as pointers or second routes. Part I’s convex-quadratic barrier is a special case of Part XI of the collection’s report *canonical-diophantine-certificates*. The report is AI-assisted and unrefereed; nothing in it is formalized in Lean or Rocq.

Status. Conventional proofs and finite exact-arithmetic checks; not kernel-checked. Section 1 compares the two routes, introduces Parts III–XIII, and states what is not claimed; Appendix E gives the provenance.

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1 Scope, the two routes, and conventions

[write] This section was written when the two manuscripts were merged. Part I and Part II print the two sources; text there without a [write] marker is the source’s own. Sections 1.7–1.11 were added on 2 October 2026, when two manuscripts of batch 79 became Parts III and IV; where they make a sentence of the earlier subsections stale, a dated note says so. Sections 1.12–1.16 were added later the same day, when four manuscripts of batch 80 became Parts V and VI, again with dated notes where earlier sentences became stale. Sections 1.17–1.21 were added on 3 October 2026, when four manuscripts of batch 82 became source 18 of Part VI and Part VII, likewise with dated notes. Sections 1.22–1.26 were added later that day, when two further manuscripts of batch 82 became Part VIII, again with dated notes. Sections 1.27–1.31 and 1.32–1.36 were added on 4 October 2026, when fifteen manuscripts of batch 91 became Parts IX–XIII, again with dated notes.

1.1 The two sources

This report is built from two AI-assisted research manuscripts that arrived in one delivery as batch 78 of ProveIt’s incoming reports (manuscripts 07 and 11; Appendix E gives the full provenance). They are called *source 07* and *source 11* after their batch-78 manuscript numbers, which are also the file prefixes of their shipped programs and data. Neither cites the other, and a text-overlap check at placement found almost no shared wording (at most 0.5 per cent of either text’s eight-word sequences occur in the other). They nevertheless prove the same theorem, by the same mechanism, in two different witness formats (Section 1.2).

- **Part I (source 07, the base), *Collision geometry is linear*.** Arbitrary rational speeds, of either sign. Every event time and position is eliminated as a rational linear form in the initial gaps, so a complete chronological skeleton becomes a homogeneous rational chamber (Theorem 5.3) and a convex quadratic with unique slack witnesses (Theorem 6.1). The chamber geometry then gives denominator and height bounds, small integral realizations, an NP upper bound, robustness and codimension, finite-horizon semilinearity, a canonical first-hit quartic, exact Ehrhart counts, two exact examples and a convex-quadratic barrier.
- **Part II (source 11), *Direct Diophantine certificates*.** Speeds and positions normalized to natural numbers by an external Galilean change of coordinates. Batch times and event positions stay in the certificate as natural witnesses on the grid $B = D^K$, and the certificate has prefix and halting versions (Theorem 18.3). Its additions are the halting semantics with terminal rays, full column rank of the witness matrix and hence strict convexity in the witnesses (Lemma 22.1), an explicit witness and residual ledger, a full-layer reference construction with exact counts (Section 23), the witness-height bound for Durand-Lose’s four-speed interface (Theorem 25.2) and a positive-witness translation.

Source 07 is the base because it assumes less (signed rational speeds, no normalization interface) and proves more about the chamber; source 11 is printed in full as the second route together with its additions. Each Part keeps its source’s definitions, letters and numbering of statements. ([write] Added 2 October 2026: two further Parts, from batch 79, follow Part II; Section 1.7 describes them. They do not prove the common theorem of this subsection and the next again.) ([write] Added 2 October 2026, batch 80: two more Parts, from four manuscripts of batch 80, follow Part IV, so that the report now prints eight sources in six Parts; Section 1.12 describes them.) ([write] Added 3 October 2026, batch 82: a third source joins Part VI and a seventh Part follows it, so that the report now prints twelve sources in seven Parts; Section 1.17 describes them.) ([write] Added 3

October 2026, batch 82, cluster M2: an eighth Part follows Part VII, so that the report now prints fourteen sources in eight Parts; Section 1.22 describes it.) ([write] Added 4 October 2026, batch 91: five more Parts follow Part VIII, so that the report now prints twenty-nine sources in thirteen Parts; Sections 1.27 and 1.32 describe them.) The common theorem is stated in both Parts, because its two statements have different inputs and witnesses; where a lemma of Part II coincides with one of Part I, a [write] note at the lemma says so, and source 11's proof is kept as a marked second route.

1.2 One theorem by two routes

The common theorem. Fix a rational signal machine, an initial label word and a finite chronological collision history (a *skeleton* in Part I, a *schema* in Part II). There is an effective sum of squares of integer affine residuals, in the input coordinates and natural witnesses, whose natural zeros are exactly the realizations of that history, with exactly one witness tuple for each realization; its degree is at most two. This is Theorems 5.3 and 6.1 of Part I and Theorem 18.3 of Part II (for prefix semantics).

The common mechanism. Both sources record each pair of adjacent signals once, over the maximal interval during which it stays adjacent, and impose a *typed* condition at each end of that interval: equality exactly when both segments are born (or die) at the same collision there, and strict positivity otherwise, including for a pair that survives the last listed batch. Affineness of the gap then excludes every interior meeting (Part I, Lemma 5.2; Part II, Lemma 21.1), and the strict conditions at a batch boundary make the declared blocks exactly the collision sites (the soundness part of the proof of Theorem 5.3; Lemma 21.2). The two sources then prove the same auxiliary statements: the lifetime count $(n - 1) + \sum_e (q_e + 1)$ ((5.2) and (18.2), where Part II writes $|\text{out}(e)|$ for q_e), causal-depth denominators dividing D^h (Theorem 7.1, Lemma 19.1), and semilinearity of the inputs accepted under a fixed event bound (Theorem 10.1, Proposition 26.1). Table 1 compares the two formats.

Table 1: The two routes compared.

	Part I (source 07)	Part II (source 11)
Speeds	arbitrary rationals, of either sign; time rescaled by $\tau = t/L$ to make them integers (Section 3)	natural numbers, after the external change of coordinates (19.1) (a Galilean shift and two scalings)
Input	the initial gap vector g , translation removed; real in the chamber theorem, positive integers in the certificate	the absolute initial positions $\mathbf{p}$, natural numbers in increasing order
Event times and positions	eliminated: rational linear forms in g , computed data (Section 4)	witnesses: natural T_i and X_e , scaled to the grid $B = D^K$ (18.1)
Witnesses	one slack $z_j = G_j g - 1$ per retained strict row; s in all	batch times T_i , time slacks d_i , event positions X_e , input-order slacks u_j , endpoint slacks $h^\pm$; $W \leq E + 2K + 2L + n_-$
Size	at most $3(n - 1) + (3m + 1)E$ raw conditions (5.3); a row may be dense in g	$R \leq I + K + 2L + n_- \leq (r + 2s + 3)E + 3n_-$ residuals, each of bounded width for a fixed machine (18.3)–(18.4)
Denominators	D^h at an event of causal depth h , for the coefficients of its forms	D^h for coordinates of depth h ; the compiler clears with D^K , a multiple of D^H

	Part I (source 07)	Part II (source 11)
Convexity	convex, Hessian $2A^T A \succeq 0$; unique slack by (6.3)	strictly convex in the witnesses for fixed input: full column rank over $\mathbb{R}$, even for unrealizable schemas (Lemma 22.1)
Semantics	prefixes: exactly the first B batches; nothing after them	prefixes, and halting realizations certified by terminal rays
Heights	row coefficients at most $2d(1+V)(DK)^h$ (Proposition 7.3); a realizing seed of size at most $d^{d/2}A^d$ (Theorem 8.1)	every canonical input and witness at most $60^K \max(1, M_0)$ for speeds in $\{0, 2, 3, 4\}$ (Theorem 25.2)
Shuttle	four-speed clock, speeds $0, 2, -1, -2$; denominator exactly $2 \cdot 3^k$ at event $2k$ for the seed $(1, 2)$; Zeno time $S = u + v$ (Section 12)	bounded shuttle, speeds $3, 2, 4, 0$; denominator exactly 3^{k-1} at the k th left event; Zeno time 2 (Proposition 25.3)
Missed simultaneity	two independent sites, word P, N, P, N , gaps $(2, 5, 2)$ (Section 11.3)	hidden collision at the final cut, word d, a, d, a at $(0, 4, 10, 14)$ (Section 24.2)
Only here	chamber geometry, small realizations, real = rational = integer realizability, NP bound, robustness and codimension, first-hit quartic, Ehrhart counts, the tournament, the convex-quadratic barrier	halting semantics, strict convexity, the full-layer construction and its exact counts, the four-speed height bound, positive witnesses

Normalizations. The two definitions of D agree. Part I multiplies all speeds by an integer L and leaves their signs; Part II subtracts a common speed c and multiplies by an integer m . A common shift does not change speed differences, so with $m = L$ both take D to be the least common multiple of the same nonzero integer differences. Part II needs nonnegative speeds and an origin because its inputs are natural absolute positions; Part I needs neither because its input is the gap vector. Part I’s integer seeds come from homogeneity (scaling a rational chamber point); Part II fixes the scale outside the witness tuple and warns that a scale witness or variable denominators would not preserve its canonical quadratic claim without a new argument (Section 19.1).

Two shuttles, one mechanism. Part I’s clock and Part II’s shuttle are both a particle reflected between two converging walls, each round trip shrinking the remaining time to the accumulation by a factor three, so that the reduced denominators grow exponentially in the number of events while their bit lengths grow linearly. They are different machines: Part I’s has a stationary left wall and a right wall of speed -1 , and its every finite prefix has the whole positive quadrant as chamber (robust chronology with exponentially growing precision); Part II’s lives entirely in $0 < t < 2$, $0 < x < 6$ and is stated for its natural-speed interface. Neither source gives the accumulation point a continuation. Source 07 credits the accumulation phenomenon to Becker et al. [6].

Two missed simultaneities, one failure mode. In both examples a proposed batch omits a second, spatially separate collision that happens at the same time, and in both the omitted pair is an unchanged adjacent pair whose gap reaches zero exactly at the last listed boundary. The strict upper endpoint of a surviving pair rejects it: in Part I through the chamber row that must be positive there, in Part II through the residual $G(T_1) - h^+ - 1 = -h^+ - 1$. Part II adds the observation that weaker conditions (two nonnegative endpoint gaps with positive sum) would accept the false schema.

Two meanings of “sparse”. In Part I “event-sparse” counts geometric conditions: the bound

(5.3) is linear in n and E , but a propagated intercept may depend on every initial gap, so the rows can be dense and storing them can take $O(dR_\sigma)$ coefficients (the paragraph after Theorem 5.4). In Part II the sparse certificate also has residuals of bounded width, because event coordinates are witnesses; its coefficients then contain the grid constant $B = D^K$. *Neither* size statement is a constant arithmetic-operation count for an unbounded computation, as both sources say.

1.3 Notions and letters

Each Part keeps its source's letters. Many letters carry different meanings in the two Parts; Table 2 lists every letter used in both, so that a statement of one Part is never read with the other Part's meaning. Inside a Part the meaning is always the one in that Part's column. No symbol was renamed and no normalization was changed: source 11's macro for the natural numbers is called `\NN` in this file and still prints $\mathbb{N}$, and Part I prints $\mathbb{N}_0$; both denote $\{0, 1, 2, \dots\}$. ([write] The letters of Parts III and IV are listed in Section 1.8, and those of Parts V and VI in Section 1.13.)

Table 2: Letters used in both Parts.

Letter	Part I (source 07)	Part II (source 11)
$\mathbb{N}_0, \mathbb{N}$	$\mathbb{N}_0 = \{0, 1, 2, \dots\}$	$\mathbb{N} = \{0, 1, 2, \dots\}$ (the same set)
A	coefficient bound (7.2); a matrix in proofs	witness coefficient matrix (Lemma 22.1); birth-position expression A_σ ; start coordinates $A_{i,j}$
B	number of batches; a nonsingular matrix in the proof of Theorem 8.1; a label of the tournament	the grid modulus $B = D^K$ (18.1), <i>not</i> a number of batches; a label of the shuttle
C	the chamber $\mathcal{C}$; the count $C(H)$; a label	matrix of input coefficients; end coordinates $C_{i,j}$
D	lcm of nonzero speed differences (7.1)	the same lcm (18.1) (Section 1.2)
d	$d = n - 1$	time slacks d_i ; a label
E	number of collision sites; $\mathbf{E}$ the equality matrix	number of events (a site of Part I)
G	$\mathbf{G}$ the strict-row matrix	the gap function $G_{\sigma\tau}(T)$ (20.5)
g	the initial gap vector	number of blocks g_i of a slab (Section 23); a gap in proofs
H, h	h causal depth; H the side of a counting box	h a depth bound, H the maximal depth; $h, h^\pm$ endpoint slacks
K	$K = 2 + 4V$ (Lemma 7.2); number of first-hit chambers	number of batches, <i>not</i> a number of chambers
L	integer clearing the speeds; a label of the clock	number of recorded lifetimes; a label of the shuttle
M	the label set	largest position M_j ; $\mathcal{M}$ the machine
m	number of distinct speeds	the speed multiplier of (19.1); number of generators of a linear set
n	number of initial signals	the same; $n_- = \max(n - 1, 0)$
p, q	p_e, q_e incoming and outgoing counts at site e	initial positions p_j ; the position scale q
R	raw row count R_σ ; a label	residual count; a label
r, s	numbers of retained equality and strict rows	largest incoming arity and largest fanout; residuals r_a
S	the accumulation time $u + v$ of the clock	a label; $\mathcal{S}$ the schema
T	the site budget	scaled batch times T_i

Letter	Part I (source 07)	Part II (source 11)
u, v	initial gaps of the examples; $v(a)$ a speed	input-order slacks u_j ; $u(\ell)$ an original speed, $v(\ell)$ a normalized speed
V	$\max(1, \max v(a))$	the maximal speed (remark after Theorem 25.2)
W	$W = 3u + 2v$ of the clock	number of witnesses
z, y	z the slack witnesses	$\mathbf{y}$ the witnesses; $\mathbf{z}$ a test vector

Three words need the same care.

- **Skeleton** (Part I) and **schema** (Part II) carry the same information: an ordered list of batches, each a set of disjoint contiguous blocks of the live list. A skeleton lists only the collision blocks (at least two members); a schema partitions the whole live list and marks singletons as persisting signals. A *site* of Part I is an *event* of Part II.
- **Realization.** Part I’s realization and Part II’s prefix realization are the same notion: exactly the listed batches as the first batches, nothing asserted afterwards. Part II’s *halting* realization additionally excludes every future collision. The “halting relation” of Part I’s Section 13 is the undecidable halting of a universal machine, a different thing.
- **Convex.** Part I’s certificate is convex (positive semidefinite Hessian) jointly in seed and slacks; Part II’s is *strictly* convex in the witnesses with the input held fixed, and Part II states that joint strict convexity is neither needed nor generally true.

1.4 Relation to the repository and to the literature

Neither source cites the related material below; this subsection supplies the cross-references (pins and dates in Appendix E). Reports of the collection are cited by label name, since their labels do not resolve in this file.

- **Report *canonical-diophantine-certificates*, Part XI** [11], “Order-free self-assembly and a positivity-restricted degree threshold”. Its Lemma `cdc:of:lem:affinezero`, Theorem `cdc:of:thm:cubicsemilinear` and Theorem `cdc:of:thm:classification` ($D_2^+(k) = D_3^+(k) = \text{SL}(k)$) prove, for polynomials of degree at most three that are nonnegative only on the real nonnegative orthant, that the full natural zero set is a finite union of natural solution sets of rational affine systems, hence semilinear and effectively so. Part I’s Theorem 13.1 and Corollary 13.2 are the globally nonnegative quadratic special case; they are printed with source 07’s direct proof, as a second route, and credited in Remark 13.3. No novelty is claimed for them. The same Part’s Theorem `cdc:of:thm:singlefoldsl` (a single-fold orthant-nonnegative quadratic for every semilinear set) bears on Part I’s first-hit quartic; see Remark 10.3. The bounded-search argument at the end of Part I’s Section 13 is the standard one of Proposition `cdc:bd:prop:nobound` there, for a different statement (site counts of accepting runs).
- **Report *liveness-beyond-halting*** [12]. Its Part V prints a question of its source 11, “Optimal degree under stronger geometric restrictions”, which asks for the minimal degree and dimension of a witness-free, binary-branching local polynomial that is globally nonnegative on all real inputs, and remarks that the globally nonnegative quadratic case “has a much more rigid

affine geometry”. Part I’s Theorem 13.1 (and, more generally, Part XI of [11]) states that geometry: such a quadratic has a rational affine zero set. This report does not settle that degree question.

- **The Hilbert-tenth-problem programme has already reviewed both manuscripts.** The review of the delivery of 2 October 2026 [13], with its detailed signal-machine part [14], replayed both suites from the original archives, read both articles in full and reports “PASS for both reports’ stated finite-schema, exact-domain theorems and generated examples”, with no mandatory correction. It also proves and checks two reductions that neither source claims: deleting source 11’s identically zero same-birth rows and its same-death rows (differences of retained flight rows), and erasing the initial-order slacks when $K > 0$ or the certificate is halting, which takes the worked annihilation from six residuals and five witnesses to four and four (a bijection of natural zero sets, not of nonnegative rational ones, with a rational counterexample); and deleting redundant strict rows, with computed slacks, from the five exported packets of source 07. Those reductions are the review’s, are not re-proved here, and do not change any theorem of either Part. This report’s intake is therefore not a first review; it adds the merge, the cross-references and the credit of Remark 13.3.
- **The programme’s audits of finite-history models.** Source 07 cites the reversible finite-history audit [2] for its distinction between cheap local relations and a complete computation certificate; that file is unchanged from source 07’s pin to the write. A sibling audit [15] notes that “finite particles can in principle store unbounded values in their separation” and that “a new construction would have to establish that interface”. Durand-Lose’s halting universal signal machine [3] uses a finite initial configuration and a halting event (Part II, Section 17), so signal machines are a candidate for such an interface, and both Parts compile its finite collision histories in principle; but neither transcribes its universal rule table or its input loader (Part I, Section 14; Part II, Question 1), so this report does not establish that interface. ([write] Added 2 October 2026: for signal machines Part III now establishes it, with a different universal machine and a prepared rational loader; see Section 1.9.)
- **The repository’s universal bounds.** The project README [1] still states the established bounds of 75 operations for a complete certificate and 87 for one universal polynomial. Neither Part improves or bears on them; both say so. ([write] Nor do Parts III–VI.) ([write] Added 3 October 2026: nor does Part VIII, whose horizon-free polynomials are for four fixed nonuniversal sources; Section 1.24.) ([write] Added 4 October 2026: nor do Parts IX–XIII. The universal ordinary-input construction of the programme now stands at 84 operations (commit 20aafb9a5); the figures 75 and 87 above are dated; Sections 1.29 and 1.34.)
- **Other reports of the category.** No result of *probabilistic-quantum-and-continuous-computation* or *group-theoretic-substrates* is re-proved here. Apart from the programme’s review described above, nothing in the repository treated signal machines before this report.

1.5 Status and what is not claimed

The report is AI-assisted and unrefereed, and **nothing in it is formalized in Lean or Rocq**. Both sources give conventional proofs together with finite exact-arithmetic checks that are regression evidence, not proofs. Every limitation stated by either source is kept in its Part; collected here, neither Part claims

- historical priority, or exhaustive novelty, for its constructions (Part I, Section 2; Part II,

Section 17); Part II adds that a general linearizability of signal-machine evolution is already in Durand-Lose’s linear real-register simulation [4];

- one fixed-arity polynomial for unbounded halting, a single-fold or finite-fold Diophantine representation of c.e. sets, or a fixed-universal arithmetic-operation bound: the skeleton or schema, the machine and the input length are compiler data, and the row and residual counts are not operation counts ([write] added 3 October 2026: still true of Parts I and II and of the report as a whole; Part VIII prints fixed-arity polynomials without a horizon for the first-target relation of four fixed nonuniversal sources of Part V’s three-mass automaton, whose witness fibres are infinite, and gives no universal source and no finite-fold representation; Section 1.25) ([write] added 4 October 2026: still true with Parts IX–XIII, whose certificates are for fixed machines, fixed rules or fixed horizons; Sections 1.30 and 1.35);
- that the published universal machine (13 meta-signals, 21 non-blank rules) or its input loader has been transcribed, executed or certified;
- any semantics at or after an accumulation point; the clock and the shuttle are tests of finite prefixes, and the clock is not a claim to have discovered accumulation ([write] added 4 October 2026: nor do Parts X–XIII give one; Part X certifies the infinite pre-accumulation validity of supplied periodic macros only);
- NP-hardness (Corollary 8.3 is an upper bound), an efficient enumeration of first-hit chambers or of their volumes, or an optimized exhaustive enumerator in the code;
- that the denominator-growth examples are lower bounds for arbitrary Diophantine representations: Part II’s is a lower bound for common-grid encodings;
- that integer scaling of a rational seed preserves a prescribed physical length or deadline;
- that the finite checks are exhaustive, or that the shared simulator of Part II’s two compilers is independently verified;
- any Lean or Rocq verification; neither source’s code is a verified compiler.

The report itself claims no novelty for Part I’s convex-quadratic barrier (Remark 13.3) and none for the reductions of the programme’s review. ([write] The non-claims of Parts III and IV are collected in Section 1.10, those of Parts V and VI in Section 1.15, those of source 18 and Part VII in Section 1.20, those of Part VIII in Section 1.25, those of Part IX in Section 1.30, and those of Parts X–XIII in Section 1.35.)

1.6 Research questions

Both agendas are printed in full: the nine research directions of Part I (Section 15) and the five questions of Part II (Section 27.1), each in its Part and with its source’s wording. Four of Part II’s questions coincide with, or overlap, directions of Part I; Table 3 pairs them, so that the two formulations are read together. None of them is answered by the other source, by the repository, or by the programme’s review; the one partial overlap is that Part II’s Question 3 begins with causal-depth clearing, which both sources already prove (a [write] note there re-scopes it). ([write] Added 2 October 2026: this sentence describes the report before batch 79. Part III now answers Part I’s direction 15.2, except for an ordinary-integer loader, and Parts III and IV answer Part II’s Question 2 in part; Section 1.11 and dated notes at the questions give the details.) ([write] Added 2 October 2026, batch 80: Parts V and VI answer two of Part IV’s questions and the remark after

its mass-two theorem; Section 1.16 gives the details.) ([write] Added 3 October 2026, batch 82, cluster M2: Part VIII answers the unbounded-time questions of sources 14 and 15 for fixed sources; Section 1.26.) ([write] Added 4 October 2026, batch 91: Part IX answers two of source 17’s questions conditionally (Section 1.31); Part X answers Part I’s direction 15.9 in part, and Parts XII and XIII touch Part I’s directions 15.6 and 15.4 and Part II’s Questions 2 and 4 (Section 1.36).)

Table 3: Concordance of the research questions.

Part I (source 07)	Part II (source 11)	Relation
15.2 An audited universal signal-machine frontend	Q1 Instantiate the universal table and input map	the same task; [write] answered by Part III for a different universal machine, except the ordinary-integer loader; the 13-label table of Q1 is still not transcribed
15.1 Formalizing the chronology theorem	Q5 Formalize geometric completeness	the same target (endpoint lemma and sweep invariant; typed endpoints and atomic replacement)
15.3 Sharper denominator invariants; 15.4 From few rows to small arithmetic circuits	Q3 Sharpen the denominator and cost ledger	overlapping: prime-valuation denominators, and a literal gate ledger; [write] 15.4 touched by Part XIII for bounded counter traces
15.5 Avoiding exhaustive first-hit enumeration	Q4 Compress histories without losing canonicity	overlapping: both separate bounded compression from fixed-arity unbounded compression; [write] Q4 touched by Part XIII at fixed horizons
15.6, 15.7, 15.8, 15.9	—	Part I only; [write] 15.9 answered in part by Part X, 15.6 touched by Part XII
—	Q2 Exploit conservative bounded populations carefully	Part II only; [write] answered in part, at a fixed horizon, by Parts III and IV, and further by Part XII
—	—	[write] Part IV’s five questions (Section 48); Part III states none
—	—	[write] Part IV’s fourth and fifth questions are answered, in substance and for the weighted three-mass automaton respectively, by Parts V and VI (Section 1.16); its first stays open
—	—	[write] Part V’s questions (Sections 63 and 73; source 14’s third is answered by source 15, both sources’ fourth in part by Part VI) and Part VI’s (Sections 81 and 92; source 16’s four-versus-five question is answered by source 17)

Part I (source 07)	Part II (source 11)	Relation
—	—	[write] Source 17's seventh question is answered for a fixed input by source 18; its second, fourth and sixth are touched by sources 18, 20 and 21, and its first, third and fifth stay open (Section 1.21)
—	—	[write] The questions of sources 18–21 (Sections 100, 117, 127 and 140); source 20's first is answered for binary rules by source 21, the others are open
—	—	[write] Source 14's Question 5 and source 15's Question 5 are answered for fixed eligible sources by Part VIII, and Part II's Question 2 is touched (Section 1.26); sources 22 and 23 state no questions
—	—	[write] Added 3 October 2026 (batch 82, reciprocal notes): outside this report, five-particle-binary-automata answers Part IV's first question in substance (its Part II, Report 16, for Report 15's guarded syntax) and source 14's Question 1 and source 15's Question 1 in part; with source 17, its Part I (Reports 14 and 15) answers both sources' fourth questions (Section 1.16)
—	—	[write] Added 4 October 2026 (batch 91): source 17's second question and the uniform half of its fourth are answered conditionally by source 25, its sixth is not answered (source 24 adds a reversible example of radius at most 90), and source 20's first is touched by source 25; source 24's second question is answered conditionally by source 25, and the other questions of sources 24 and 25 (Sections 167, 179 and 181) are open (Section 1.31)

Part I (source 07)	Part II (source 11)	Relation
—	—	[write] Added 4 October 2026 (batch 91): the questions of Parts X–XIII (Sections 238, 293.2, 315, 302, 317 and 355); several are answered by later sources of the same Parts, the rest are open (Section 1.36)

1.7 Parts III and IV: conservative frontends (batch 79)

[write] This subsection and the four that follow were written on 2 October 2026, when two manuscripts of batch 79 of ProveIt’s incoming reports were added to the report as Parts III and IV. They are called *source 12* and *source 13* after the file prefixes of their shipped programs and data, which continue this report’s local sequence after 07 and 11. Their batch-79 manuscript numbers are different: **source 12 is batch-79 manuscript 16, and source 13 is batch-79 manuscript 19.** Appendix E gives the full provenance. Neither manuscript cites Parts I and II, the other new manuscript, or any repository file.

- **Part III (source 12), *Conservative signal machines as Diophantine frontends*.** A fixed reversible, number-conserving universal signal machine $\mathcal{U}_{18}$, written out completely: 114 meta-signals, 445 explicit injective two-to-two collision rules, speeds $\{-12, -9, -4, 0, 4, 9, 12\}$ and exactly 18 live signals on every encoded input (Theorem 29.1). Its source is Morita’s 15-state, 6-symbol reversible universal Turing machine, compiled through Durand-Lose’s 2012 conservative rational stacks; the input of an explicitly halting cyclic-tag program is loaded into two rational distances. At event cuts the complete simultaneous-event dynamics of any fixed-population, equal-cardinality machine is a partial piecewise homogeneous integer-linear map with the pivot-scaled gap matrices $N_j = c_j I - c e_j^T$, $N_j^2 = c_j N_j$, plus one mode coordinate (Sections 34–35). A selector/copy construction then gives one explicit quadratic polynomial with a unique natural witness for each legal step, and for each fixed horizon (Theorem 36.1); for $\mathcal{U}_{18}$ the step polynomial has exactly 4,196,998 auxiliary variables over a closed family of 49,700 modes and 80,501 branches (Appendix H). A ten-signal binary version is proved to exist but not printed (Theorem 29.2).
- **Part IV (source 13), *Sparse Diophantine certificates for finite mass lattice dynamics*.** The discrete counterpart: one-dimensional partitioned number-conserving cellular automata with three lanes. A conserved mass M bounds the number of unit records but not the gaps between them, and a fixed horizon T of any supplied total conservative table compiles into one quartic sum of squares with exactly one natural zero, paying for collision detection, simultaneous redistribution, sorting, table lookup and empty space, with exact witness and residual ledgers (Theorem 38.1) and an option that is exact on the nonnegative real orthant. A source interface to Morita and Imai’s reversible conservative automaton corrects a printed sign, adds a reachable-legality hypothesis that an explicit false observer shows to be necessary, pays for a loader and a first-pulse observer, and works a fully specified, nonuniversal doubling example. Decidability results bound what any such frontend can do: under a bounded orbit diameter, and without it at total mass at most two (Theorem 44.1); at mass at most two the

relation is effectively semilinear (Theorem 45.1).

How they relate to Parts I and II. Neither new Part proves the common theorem of Section 1.2 again; both certify a step, or a fixed horizon, of a *deterministic* dynamics rather than the realizations of a prescribed collision history. Part III meets Parts I and II in one lemma: its complete event selection (Lemma 34.1) is their typed-endpoint argument for a single batch, and its uniform lift is their common-denominator device ([write] notes at both places). Part III is also the instantiated universal frontend that Part I’s research direction 15.2 asks for, though for a different universal machine than the one named there (Section 1.11). Part IV shares no lemma with Parts I–III; it answers, for lattice dynamics, the conservative-population question of Part II.

Two Morita sources, no shared data. Both new Parts draw their universal source from Morita, but from different constructions: Part III from Morita’s 2008 reversible universal Turing machine UTM(15,6), through Durand-Lose’s signal-machine stacks; Part IV from Morita and Imai’s 2001 number-conserving reversible partitioned automaton, built on Morita’s 1996 reversible two-counter machine. They share no table, program or data file. Part III prints its universal machine in full; Part IV deliberately leaves its safe universal source as an existence construction and prints only a nonuniversal example.

1.8 Notions and letters of Parts III and IV

[write] Parts III and IV keep their sources’ letters and numbering of statements, by section, like Parts I and II; no symbol was renamed and no normalization changed. Both sources’ macro for the natural numbers is written `\NN` in this file, as for source 11, and prints $\mathbb{N} = \{0, 1, 2, \dots\}$; Part I’s $\mathbb{N}_0$ is the same set. The table lists every letter of Parts III and IV that has another meaning in the other new Part or in Parts I–II (Table 2 gives the meanings in Parts I and II in full). Inside a Part the meaning is always that Part’s.

Letter	Part III (source 12)	Part IV (source 13)	Parts I–II
<i>A</i>	auxiliary-variable count $A = B(d + 1) + T + W$; loader numerators A_L, A_R ; the projection A_j	—	coefficient bound; witness matrix
<i>B</i>	number of branches of the step map, <i>not</i> batches	the control baseline $B = m + 16$ of the source interface	batches (I); grid modulus D^K (II)
<i>b</i>	base $\ell + 1$; selectors b_r ; the blank b	comparison flag <i>b</i> ; output masses b_i ; baselines b_i	—
<i>C</i>	the set C_ℓ of stack codes	centre lane <i>C</i> ; number <i>C</i> of congruence atoms	chamber (I); input matrix (II)
<i>c</i>	gadget speed <i>c</i> ; speed-difference vector <i>c</i> , $c_i = v_i - v_{i+1}$; pivot speed c_j	output masses c_i ; a configuration <i>c</i>	—
<i>D</i>	lcm of the nonzero speed differences, <i>the same D</i> as in Parts I–II	the diameter of the initial support	lcm of speed differences
<i>d</i>	dimension of a branch system; a move direction	comparison slack <i>d</i> ; a modulus	$n - 1$ (I); time slacks (II)
<i>E</i>	number of equality guards; escape tokens $E_{q,a}$	residual count E_Φ of the Presburger compiler	sites (I); events (II)
<i>F</i>	the partial step map on $\mathbb{N}^d$	the global cellular-automaton map	—

Letter	Part III (source 12)	Part IV (source 13)	Parts I–II
g	rational gap vector	the local table $g : \{0, \dots, K\}^3 \rightarrow \{0, \dots, K\}^3$; an initial gap	gap vector (I); blocks (II)
H	halt labels $H_{q,a}$	the witness-height bound (38.4)	box side (I); depth (II)
h	integer gap vector h_k , loader h_0	comparison flags h_i ; a time	depth (I); depth, end-point slacks (II)
K	the horizon (number of steps)	the alphabet cap, $q = K + 1$; witness count K_Φ	$2 + 4V$, chambers (I); batches (II)
L	a guard form $L(X)$; the move direction L	left lane L ; an affine atom L ; the term count $\mathcal{L}$	speed-clearing integer (I); lifetimes (II)
M	uniform-lift matrices M_j ; branch matrices $\widehat{M}_r$	the conserved mass	labels (I); largest position (II)
m	the padded word length of the loader	number of source states	distinct speeds (I); speed multiplier (II)
n	the population (live signals)	a counter value; counters n_0, n_1	initial signals
p	timestamp numerators p_k ; productions p_i	the period of a type cycle	in-counts (I); positions (II)
Q	the loader scale ℓb^m , or a clearing denominator	a finite cellular-automaton alphabet	—
q	mode code; Morita's states q_i	$q = K + 1$; source states q_i ; quotients $q^\pm$	out-counts (I); position scale (II)
R	the layout half-width $\ell + 2$, and the residual count $1 + 2d + E + T + W$	right lane R ; residual count; the right walker's position $R(t)$	raw rows (I); residuals (II)
r	branch index; the length of a run in (33.1)	ranks r_i ; the radius r ; a table entry	retained rows (I); arity (II)
S	the span $S = \mathbf{1}^\top h$; the source alphabet	number of allowed rows of a restricted list	accumulation time (I)
s	guard slacks $s_{r,L}$; Morita's target state	table selectors $s_{i,\tau}$; comparison slacks	strict rows (I); fanout (II)
T	number of strict guards, <i>not</i> a time	the horizon (number of lattice steps)	site budget (I); batch times (II)
u	mode coordinate $u = qS$	incoming masses u_i	gaps (I); slacks (II)
V	a bound on speed differences	witness count V	speed bound (both)
W	number of weak guards	—	$3u + 2v$ (I); witnesses (II)
w	a written symbol; the input word	incoming masses w_i ; the weight function	—
z	packet witnesses $\mathbf{z}$	channel index of a record	slacks (I); test vector (II)

Several words need the same care.

- **Horizon.** Part III's horizon is K steps of the event map, Part IV's is T lattice steps; Part II's K is a number of collision batches of a prescribed schema. In Parts III and IV the horizon is a compiler parameter that fixes the number of variables; it is never a free unknown of one polynomial.
- **Batch.** Part III's “signal batches” (Section 37) are the collision batches of Parts I–II, counted along a run. Its *mode* is an ordered label word, and a *germ* is a post-event state whose zero gaps separate.

- **Source.** “Source 12” and “source 13” name manuscripts; inside Parts III and IV “the source” is the simulated machine (Morita’s Turing machine, Morita’s two-counter machine).
- **Conservative.** Part III’s rules conserve the number of signals and a positive energy; Part IV’s local tables conserve the total lane mass. Both are reversible in the cases printed, but neither certificate theorem needs reversibility.
- **Sparse.** Part III’s “sparse scaled projections” have at most $2(n - 2)$ nonzero entries; Part IV’s “sparse” certificate keeps M records however large the gaps between them. Neither is Part I’s event-sparse row count, and none of them is a fixed-arity statement.
- **Loader.** Part III’s loader writes a prepared tape into two rational distances at a cost linear in the tape length (Section 33); Part IV’s uniform loader leaves two counter positions as free natural parameters and pays $4 + 3M(M - 1)$ witnesses and residuals (42.2). Neither converts an ordinary binary integer input.

1.9 Relation of Parts III and IV to the repository

[write] Neither new source cites the material below; these cross-references are added. Results of other reports are cited by label name.

- **Re-derived results, printed as pointers or second routes.** Part III’s complete event selection (Lemma 34.1) is the one-batch case of Part I’s Theorem 5.3, and its uniform lift is the denominator device of Theorem 7.1 and Part II’s grid. Part IV’s comparison gadget is Lemma `cdc:mem:lem:compare` and its sorting comparator the two-field record comparator (`cdc:mem:prop:comparator-cost`) of Part V of *canonical-diophantine-certificates* [11]; Part IV’s horizon-free quartic (Corollary 46.1) is a degree-four second route to the single-fold degree-two representation of semilinear sets of that report’s Part XI (`cdc:of:thm:singlefoldsl`); and the finite-search step of Proposition 43.2 is `cdc:bd:prop:nobound`. Each place carries a [write] note; no novelty is claimed for any of them.
- **The degree-two orthant format.** Part III’s step polynomial is literally of the form $\sum A_i^2 + \sum B_j C_j$ of the spine of *quadratic-orthant-certificates* (`qoc:eq:spine`), nonnegative on the real orthant; its existence also follows from `cdc:of:thm:singlefoldsl`, since a step relation with finitely many linear branches is semilinear; source 12’s contribution is the explicit construction and its ledger. Its selector/copy construction is Balas’s disjunctive formulation [26, 27, 28], credited at the place. (Added 2 October 2026, batch 79: that report’s Parts V and VI use the same selector device with the selector added to the right factor of each product, which makes their nonnegative real fibres natural; the note after Theorem 36.1 compares the two gates and shows that the real fibre of Part III’s packet is natural at every natural input $X \neq 0$, and at $X = 0$ as well under the theorem’s disjoint homogeneous branch hypothesis; the weak gates alone do not force one-hot selection at zero.) Part IV’s certificates are sums of squares of quadratic residuals, of degree four; its orthant-exact option restores exactness on the nonnegative real orthant with norm-one residuals.
- **The finite-input interface asked for by a programme audit.** The block-automaton audit [15] notes that finite particles could store unbounded values in their separations and that a new construction would have to establish that interface. For signal machines, Part III establishes it: a fixed machine with 18 live signals, finite support, stack values stored in two

distances, and observable halt and null outcomes with an escape of eight crossings. It does so with a prepared rational loader, not from an ordinary integer, and not for billiard-ball or block automata, which remain as the audit leaves them. (This supersedes, for signal machines, the last sentence of the corresponding item of Section 1.4.)

- **Other reports.** Part IV’s mass-two theorem is related only loosely to Part I’s rigidity theorem (both end in semilinearity, for different reasons). Part IV bears on the question “Local certificates on adaptive support geometry” of Part XVI of *canonical-diophantine-certificates* without answering it (note at Part IV’s questions). The fixed-horizon form of Part III is the same kind of statement as the trace theorem `pqc:pm:thm:trace` of *probabilistic-quantum-and-continuous-computation*, for different dynamics.
- **The programme’s reviews.** The Hilbert-tenth-problem programme reviewed the earlier edition of source 12 (batch-79 manuscript 07) and wrote the patch that the corrected edition printed here applies [29], and then reviewed the corrected edition itself, with a natural-domain projection of its packet [31]. It reviewed source 13 completely and wrote a constructor patch [30], which was not applied here; the corrected code edition of source 13 (batch 80) contains an equivalent repair of its own and is now shipped, and the programme’s correction audit [36] confirms it. It later reduced two of source 13’s ledgers [32, 33]. Its review index [34] lists all of these as complete, and its placement audit [35] authenticates every file of both sources placed by `a7ae02511` against the delivered archives (five of source 13’s have since been replaced by the corrected edition’s bytes). Sections 37 and 47 give the details. No review changes a theorem of this report, and the reductions are the programme’s, not re-proved here.
- **The repository’s universal bounds.** The 75- and 87-operation bounds stated in the project README [1] are untouched by Parts III and IV, which say so themselves (no fixed-arity polynomial for unbounded histories). The programme’s review of source 12’s earlier edition records the same conclusion.
- **Formal status.** Nothing in Parts III and IV is formalized in Lean or Rocq; the Hilbert-tenth-problem Lean development has no signal-machine, cellular-automaton or Presburger-compiler module that states any of their results. One classical ingredient that Part IV imports is formalized elsewhere in the repository: Cooper’s quantifier elimination for Presburger arithmetic, used in Section 45, is the Lean development `Logic/PresburgerArithmetic` (for instance `PresburgerArithmetic.Formula.holds_iff_quantifierEliminate` and the decision procedure `PresburgerArithmetic.Formula.presburgerArithmetic_decidable`). That formalizes the imported step only, over the integers, and none of Part IV’s own statements.

1.10 Status and what Parts III and IV do not claim

[write] Both new sources give conventional proofs and finite exact-arithmetic checks, which are regression evidence, not proofs; both are AI-assisted and unrefereed. Every limitation they state is kept in its Part. Collected here, they do not claim

- priority, or an exhaustive literature search, for the precise synthesis (Part III), the sparse constructions or the mass-two theorem (Part IV); a new universal computational mechanism or new classical universality; the stack, selector/copy and circuit-to-sum-of-squares devices as new;
- a fixed-arity polynomial for an unknown, unbounded horizon or for arbitrary halting histories;

the horizon fixes the number of variables in both Parts, and “there exists some finite horizon” is an unbounded union that needs a separate compression theorem;

- for Part III: that the ten-signal binary interpreter is printed or instantiated; optimality of 18 or ten live signals, of the loader or of the dimension; injectivity of the integer event map; that every path of the 49,700-mode graph is realizable from one input (it is a mode-only overapproximation), or that the graph bounds the length of a universal run; a flat monomial expansion (the trillion-term count is derived, not emitted); a minimal gate count; any semantics for malformed rational configurations or at accumulation points; that the finite replays prove universality, which is imported from Morita’s theorem; that the polynomial is convex or a sum of squares over all reals;
- for Part IV: a printed universal source table or its state count; faithful interpretation of arbitrary counter pairs; intrinsic universality, an efficient simulation, exact-target undecidability, or global halting (the pulse is an anchored observation); a minimum universal mass, or that three particles suffice; any statement over unrestricted reals; that the displayed counts are lower bounds; an arbitrary-rule quantifier eliminator; and anything about arbitrary c.e. sets from the decidable mass-two subclass ([write] added 2 October 2026, batch 80: these remain true of Part IV; Parts V and VI now prove exact-target undecidability and the sufficiency of three units for a weighted automaton with many weight-one symbols, and decidability at masses three and four with a single one; Section 1.16);
- a proof-assistant verification; neither source’s code is a verified compiler.

1.11 Research questions answered or touched by Parts III and IV

[write] Dated notes at the questions themselves record the following; Table 3 carries the same information in its relation column.

- **Part I’s direction 15.2** (an audited universal signal-machine frontend) is *answered by Part III, except for the ordinary-integer loader*: Part III instantiates a published finite-input universal construction with its exact labels, speeds, all collision rules, forbidden (not transparent) undefined encounters, input encoding and marked acceptance, audits simultaneous and triple collisions, and proves the correspondence of its loader with the simulated machine. Two differences from the direction as worded remain. The machine is Morita’s UTM(15,6) through Durand-Lose’s 2012 stacks, not the 13-type construction of [3] named as a candidate; and the input is a prepared cyclic-tag tape loaded as two rational distances, so the equivalence with “an ordinary integer program input” would still need a digit-extraction and base-conversion loader, whose cost Part III leaves explicit (Section 33).
- **Part II’s Question 1** (transcribe the 13-label, 21-rule machine) stays open as stated: Part III transcribes a different universal machine.
- **Part II’s Question 2** (exploit conservative bounded populations) is *answered in part, at a fixed horizon*, by both new Parts. Part III instantiates every finite transition branch of a reversible conservative universal signal machine, pays for strict guards and for its loader, and excludes accumulation on encoded runs; Part IV does the same for lattice dynamics, where conserved mass bounds the records but not the gaps. Neither encodes unbounded histories in a fixed arity, which the question also mentions. ([write] Added 3 October 2026: Part VIII does, for fixed sources of Part V’s three-mass automaton, not for signal machines; Section 1.26.)

- **Part II’s Questions 3 and 4** are not answered; Part III’s literal ledger is a variable-and-residual count, not the gate ledger asked for.
- **Part IV’s own five questions** (Section 48) are open; Part III states no separate questions. ([write] Added 2 October 2026, batch 80: the fourth and fifth are now answered, in substance and for the weighted three-mass automaton respectively, by Parts V and VI; Section 1.16.)

1.12 Parts V and VI: conserved-mass thresholds (batch 80)

[write] This subsection and the four that follow were written on 2 October 2026, when four manuscripts of batch 80 of ProveIt’s incoming reports were added to the report as Parts V and VI. They are called *sources 14–17* after the file prefixes of their shipped programs and data, which continue this report’s local sequence after 13. Their batch-80 manuscript numbers are different: **source 14 is batch-80 manuscript 10, source 15 is manuscript 02, source 16 is manuscript 05 and source 17 is manuscript 03**. A fifth archive of the batch, manuscript 06, is the superseded original of manuscript 05 and is not printed. Appendix E gives the full provenance. None of the four manuscripts names a repository commit; source 16 cites Part IV (source 13) by its title, for its post-elimination compiler, and sources 15 and 17 cite sources 14 and 16 as their “companion” reports. All four are dated 3 October 2026.

Source	Batch 80	Archive and title	Printed as
14	manuscript 10	<code>Three_Mass_Reversible_Computatio n.zip</code> ; <i>Three mass units in a reversible weighted cellular automaton</i>	Part V, Sections 49–63
15	manuscript 02	<code>Exact_Targets_Three_Mass_Units.zip</code> ; <i>Exact targets with three conserved mass units</i> , a declared extension of source 14	Part V, Sections 64–73
16	manuscript 05	<code>Single_Unit_Three_Mass_Decidability (1).zip</code> ; <i>Three mass units with a single unit symbol</i> , the attribution revision of manuscript 06	Part VI, Sections 74–82
17	manuscript 03	<code>Four_Mass_Decidability_Package.zip</code> ; <i>Four mass units and exact reachability</i> , which builds on source 16	Part VI, Sections 83–92
—	manuscript 06	<code>Single_Unit_Three_Mass_Decidability. zip</code> ; the original of manuscript 05	not printed (superseded)

- **Part V (sources 14 and 15), *Three conserved mass units suffice***. The model is a power-set alphabet 2^T of Boolean channel types, the weight of a symbol being its cardinality, so that many distinct symbols have weight one. Source 14 compiles every separated reversible two-counter machine into a globally reversible, mass-conserving cellular automaton of forward and inverse radius at most four whose canonical configurations have total mass exactly three and gap $12 \cdot 2^a 3^b$, and in which a clean one-cell symbol $\text{Ready}(q_f)$ occurs exactly when the source halts (Theorem 49.1); radius one follows by four-site spatial blocking at the same clock, or by a four-phase dilation at four times the time. With Part IV’s mass-two theorem this makes three the least mass with fixed-rule undecidable pattern occurrence in that model (Theorem 49.2). Source 14 also gives exact type and collision-row counts, exact microtimes, and a quadratic natural-number certificate at a fixed source horizon with a unique witness (Theorem 58.1). Source 15 wraps a fresh-entry source so that it runs forward, then its

logical inverse, and then relabels its state: for one fixed reversible cellular automaton, exact reachability $\exists t A^t(x_N) = y_N$ between two computable configurations of mass three and the same support is undecidable (Theorems 64.1 and 67.1); it adds an exact first-target clock and a compact certificate whose natural zero fibres are in affine bijection with those of the full wrapped certificate (Theorem 70.2).

- **Part VI (sources 16 and 17), *A single unit symbol*.** With at most one weight-one symbol, as in an ordinary integer-state number-conserving automaton, source 16 proves that the timed evolution relation at total mass at most three is effectively Presburger, uniformly in all positions and the time for each fixed rule (Theorem 74.1); exact reachability and pattern occurrence are decidable, and horizon-free canonical quartics follow. Source 17 proves that exact reachability and pattern occurrence remain decidable at mass four (Theorem 83.1), but that the timed relation is no longer Presburger there: a binary radius-six shuttle hits an anchored pattern exactly at the times $k^2 + (2d - 11)k$ (Proposition 89.2), with a one-witness quartic for that timing. For integer-state number-conserving automata the numerical particle lower bound is therefore five (Corollaries 79.1 and 83.2). ([write] Added 3 October 2026: batch 82 adds a third source to Part VI and a Part VII, which extends the decidability at mass four to every dimension; Section 1.17.)

The thresholds proved in this report, by model (“undecidable” means fixed-rule undecidable for a computable family of initial configurations):

Model	Mass at most 2	Mass 3	Mass 4	Mass 5
Weighted alphabet, several weight-one symbols	decidable (Part IV, Theorem 44.1)	undecidable pattern occurrence and exact-pair reachability, fixed reversible rule (Part V)	not treated	not treated
At most one weight-one symbol (includes integer-state NCCA)	decidable (Part IV)	decidable, timed relation Presburger (Part VI, source 16)	decidable, timed relation not Presburger (Part VI, source 17); [write] the same for a globally reversible binary rule (Part IX)	universal at large radius by the Alhazov–Imai abstract [24]; reversibility not asserted, full paper not checked. [write] Universal for binary and for globally reversible binary rules by Reports 14 and 15 (Part I of five-particle-binary-automata), so five is exact (added 3 October 2026)
[write] The same, on $\mathbb{Z}^d$ for every finite d (added 3 October 2026)	decidable (Part VII, source 19)	decidable, fixed-input timed Presburger (source 19, Lemma 104.1)	decidable (Theorem 101.1); timed relation not Presburger (source 17’s shuttle embeds linewise)	source 19 infers universality by embedding the one-dimensional binary five-particle construction of batch-82 Reports 16 and 26 linewise (Corollary 101.2); Report 16 is Part II of five-particle-binary-automata, Report 26 is printed there as Part VI (commit d2043bbd0), and this report does not establish it

How they relate to Parts I–IV. Both new Parts work in Part IV’s weighted model (Section 43: finite alphabet, finite radius, positive integer weights, a unique zero-weight vacuum, conservation on finite configurations), which allows several symbols of weight one. Part IV’s Theorem 44.1 is the lower half of Part V’s threshold, and source 14 proves it again (a pointer replaces that proof, Section 59). Part VI restricts the same model to at most one weight-one symbol, so its mass-two statements are

special cases of Part IV’s; source 16 reruns Part IV’s semilinearity architecture one mass higher, with a new no-split lemma, and reuses Part IV’s post-elimination compiler by name. The two new Parts do not contradict each other: Part V keeps its finite control in weight-one controller types and changes its gap multiplicatively, neither of which a single stateless unit symbol permits, as sources 14 and 16 both say. Parts V and VI share no lemma with Parts I–III, except that the certificate gate of sources 14 and 15 is the selector device of Part III and of *quadratic-orthant-certificates* (Section 1.14).

Morita and Imai again. Part IV and Part V both rest on Morita and Imai’s 2001 paper [19], in different ways: Part IV interfaces with their number-conserving reversible partitioned automaton itself (and corrects a sign in its rule (8.2)), while Part V uses only their reversible two-counter normal form (Definitions 3.1–3.3, Proposition 3.4 and, for source 15, Lemma 3.5). No table, program or data file is shared, except that source 14 ships a byte copy of one source-13 receipt, which is not shipped twice.

1.13 Notions and letters of Parts V and VI

[write] Parts V and VI keep their sources’ letters and their numbering of statements, by section; equations are numbered by section (sources 16 and 17 tagged theirs (1), (2), $\dots$). The four sources’ macro for the natural numbers is written `\NN` in this file; it prints $\mathbb{N} = \{0, 1, 2, \dots\}$, as each source defines it. No other symbol was renamed and no normalization changed. Inside Part V, source 15 restates source 14’s interface in source 14’s letters; inside Part VI, source 17 restates source 16’s stationary frame in source 16’s letters. The table lists the letters of Parts V and VI that carry another meaning in the other new Part or in Part IV (the table of Section 1.8 gives Part IV’s meanings in full). Inside a Part the meaning is always that Part’s.

Letter	Part V (sources 14, 15)	Part VI (sources 16, 17)	Part IV
R	the stationary right-marker channel type R ; in source 15 also a source program R , with R_{cl} its wrapped program	the radius R of F	right lane; residual count
r	residue $N \bmod 6$; clock phase in $\mathbb{Z}_{144}$; the radius in the mass-two summaries; number of input coordinates in source 15’s ledger	phase index $0 \leq r < p$	the radius; ranks
S	static shuttle types S_j ; the channel shift in $F = S \circ \Pi$; state-code sums S_t	the padded comoving radius $S = \max(1, R + \delta)$	number of allowed rows
L	left-controller types $L_{j,x}$	the left marker coordinate (source 17)	left lane
u	controller velocity $u = d - p$; quotient offsets u_{tj}	the weight-one symbol u	incoming masses
d	the divisor in a factor p/d ; the number of increment or decrement instructions	a drift; packet widths d_r ; the initial right-marker coordinate d of the binary shuttle	comparison slack; modulus
D	$D = 6qm$ data triples; tick durations D_j	drift of a mass-two profile (source 16); marker gap $D = R - L$ (source 17)	diameter

Letter	Part V (sources 14, 15)	Part VI (sources 16, 17)	Part IV
N	the gap integer, $N = 2^a 3^b$ or raw $N = C 2^a 3^b$, with $n = 12N$	numbers N_2, N_3 of weight-two and weight-three labels (source 16); the threshold $N = 4B + 12S$ (source 17)	—
B	number of residue-expanded branches	Boolean gate count of source 16's quartic (Part IV's G); seed-prefix bound (source 17)	control baseline
C	completed pair-support count; raw cofactor C (source 15: c); new-value forms C_j ; a loader bound	number of congruence atoms (source 16); support constant C_1 and a time coefficient (source 17)	centre lane; congruence atoms
E	selector sums E_t	residual count of the quartic (source 16, Part IV's E_Φ); first-hit predicate E_q ; seed library $\mathcal{E}$ (source 17)	residual count E_Φ
G	finite gates G_I, G_q, G_H	the comoving automaton $G = \tau_{-\delta} F$	gate count
H	incoming-instruction table H ; the terminal state H of the wrapped source (source 15)	span predicate H_q (source 16)	height bound
K	loader selectors $K = (A + 1)(C + 1)$; $K_{\text{types}}, K_{\text{prescribed}}, K_{\text{completed}}$	witness count of the quartic (Part IV's K_Φ)	alphabet cap; K_Φ
M	a source machine (F_M, A_M)	total mass $M(c)$; the gap-residue modulus (source 17)	the conserved mass
P	prescribed pair-row count; the certificate P_h	de Bruijn potential (source 17)	—
p	the prime 2 or 3, factors p/d ; a period	a period	period of a type cycle
q	number of states; a source state	a normalized pair shape (source 16); a finite mode (source 17)	$q = K + 1$
T	translation T_a ; physical times T_h, T_*, T_{cl}	times in Proposition 77.1; section times T_n (source 17)	the horizon
τ	a source-step duration	translation τ_a	—
α	the singleton type map	label tuples α, β (source 16); a positive rational slope (source 17)	singleton type map
δ	a tail translation	the displacement of the isolated unit	a tail translation
h	the source horizon	first core time (source 16); edge duration (source 17)	a time; flags

Several words need the same care.

- **Mass, unit, particle.** In Part V a symbol is a set of Boolean channels and its mass is the number of channels, so a canonical configuration has *three mass units in two occupied cells*. In an ordinary integer-state number-conserving automaton (NCCA) the state is its own mass, and a “particle” of the literature is one unit of numerical mass. The tempting false reading of Part V as “three particles suffice for an NCCA” is refuted by Part VI: with a single weight-one symbol, three and four units are decidable.
- **Source and companion.** “Source 14” to “source 17” name manuscripts; inside Part V “the

source” is the simulated reversible two-counter machine. Source 15’s “companion construction” is source 14, source 16’s “companion report” is Part IV (source 13), and source 17’s “companion” is source 16; bracketed notes say so where they occur.

- **Halt and target.** Source 14 observes the clean symbol $\text{Ready}(q_f)$ *somewhere*; source 15 asks for equality of the *whole* configuration with an input-dependent target y_N ; Part VI’s observations are exact targets and finite patterns, anchored or anywhere. Part IV’s pulse is an anchored first-pulse observation. None is global halting.
- **Timed and untimed.** Source 16’s relation includes the time and is uniform in all positions; source 17’s mass-four description forgets the time and is for a fixed input, and the binary shuttle shows that the timed relation cannot be Presburger.
- **Horizon.** Sources 14 and 15 fix an external source horizon h (source steps, not CA ticks); sources 16 and 17 need no horizon. None of them is a fixed-arity polynomial for unbounded halting.

1.14 Relation of Parts V and VI to the repository

[write] The four sources cite none of the material below except Part IV, which source 16 cites by title; these cross-references are added. Results of other reports are cited by label name.

- **Re-derived results, printed as pointers.** Source 14’s Section 59 proves Part IV’s Theorem 44.1 again by the same argument; its proof is replaced by a pointer, and so are its bounded-diameter proposition (Proposition 43.1) and its finite-ring paragraph (Part IV’s Section 43); its two-mass test program is source 13’s audit minus one line, and its receipt is byte-identical to source 13’s, which is not shipped twice. Source 15’s summary of the mass-two theorem is a pointer too, and its compiler interface restates source 14’s results in its own words (Section 65). Source 16’s Lemma 75.2 and Sections 76–79 follow the architecture of Part IV’s Sections 44–45, one mass higher, and its quartic is Part IV’s compiler (Section 46); source 17’s Section 84 restates source 16. Each place carries a bracketed note; no novelty is claimed for any of them. New are source 16’s no-split lemma (Lemma 75.1), its mass-three core and uniform first-entry composition, and all of source 17 after its Section 84.
- **The certificate gate.** The product $(E_t - e_{tj})(e_{tj} + u_{tj})$ of sources 14 and 15 is the inactive-product gate of Lemma `qoc:rn:lem:gates` in Part VI of *quadratic-orthant-certificates*, and a form of Part III’s selector/copy device (note after Theorem 36.1); source 14’s natural-domain counterexample is consistent with that report’s real-orthant one-hot lemma: one-hot selection survives over the nonnegative reals, divisibility does not.
- **canonical-diophantine-certificates, Part XI [11].** For source 16’s semilinear relation, that Part’s Theorem `cdc:of:thm:singlefolds1` gives an orthant-nonnegative single-fold representation of degree two, as it does for Part IV’s (note after Corollary 80.1). Its Theorem `cdc:of:thm:classification` ($D_2^+ = D_3^+ = \text{SL}$) shows that source 17’s non-semilinear hit set has no orthant-nonnegative representation of degree at most three, so the degree four of source 17’s one-witness quartic is the least possible in that class (note after (90.1)).
- **The programme’s reviews.** The Hilbert-tenth-problem programme reviewed all five archives of this cluster before this write. Its review of sources 14 and 15 [45] reports “PASS within the stated model and interfaces”, with no repair, after both complete author replays and independent checks (360 literal core cases, 2,400 natural assignments, 45 compact/full

zero-fibre correspondences, 7,968 microscopic cleanup ticks); it checked Morita and Imai’s Proposition 3.4 and Lemma 3.5 in the primary paper (pp. 245–246) and records the off-zero and rational limitations of the compact certificate as documented limitations, not defects. Its review of sources 16 and 17 and of manuscript 06 [46] reports “PASS within the stated mathematical and executable scopes”, with no correction; it confirms that manuscript 05 changes only the attribution of manuscript 06 (the mathematical body is byte-identical), reconstructs all 8,192 local windows of the binary shuttle, checked Kong’s thesis, could not certify the Japanese text of the KAKEN report or obtain the full Alhazov–Imai paper, and notes the five-witness refinement of source 16’s quartic. Its index [47] lists the reviews, and its placement audit [48] authenticates the 116 files placed by 345a9e44e. Those reviews concern the archives and the placement; this write, which assembles the text, has not been reviewed by the programme. ([write] Added 3 October 2026: the programme has since reviewed this write too, in `review_signal_batch80_typesetting_ef114b0bb.md` with the independent challenge `review_signal_batch80_quantifiers_ef114b0bb.md`, both in `Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/` (commit f787251a2). They account for every formal statement, proof and display of sources 14–17 (71 statements and proofs retained and 4 printed as pointers; 98 displays retained and 7 declared deduplications) and confirm the degree-four minimality note after Proposition 89.2 in its stated class. Their patch `signal_batch80_quantifiers_ef114b0bb.patch` corrected two summaries written for this merge, and both corrections were applied on 3 October 2026: the last cell of the threshold table above had called the Alhazov–Imai construction “not reversible”, whereas its abstract merely does not assert reversibility; and the README’s summary of Part V had one fixed automaton simulate every separated reversible two-counter machine, whereas Theorem 49.1 compiles each machine M into its own automaton F_M and the one fixed rule of Theorem 49.2 comes from fixing a universal source. The abstract’s sentence on Part V, which had a fixed automaton simulate reversible two-counter machines, now says, with source 14’s own abstract, that it simulates a universal one.) Sections 61, 72, 82 and 92 give the details.

- **The repository’s universal bounds.** The 75- and 87-operation bounds of the project README [1] are untouched by Parts V and VI. Sources 14 and 15 give degree-two certificates for each external horizon only, and sources 16 and 17 give horizon-free quartics only for decidable classes; all four say so, and the programme’s reviews record the same conclusion.
- **Formal status.** Nothing in Parts V and VI is formalized in Lean or Rocq; the Hilbert-tenth-problem Lean development has no cellular-automaton module. Cooper’s quantifier elimination, which source 16 imports, is the Lean development `Logic/PresburgerArithmetic` (Section 1.9); that formalizes the imported step only.
- **The literature.** Sources 14 and 15 credit Brandt, Portmann and Uitto [37] (three finite-state agents, reflected-shuttle arithmetic, prime exponents in one gap), Dobrev et al. [38], the abstract of Dobrev, Královič and Pardubská [39] (three synchronous agents, full paper not obtained), Durand-Lose [5], Morita and Imai’s partitioned automaton of mass $m + 19$ [19], and Bennett’s uncomputation [40]. Sources 16 and 17 credit Kong’s 2021 thesis [41] (binary three-particle nonuniversality, with the four-particle case open in 2021), Imai’s 2021 KAKEN report [42] (decidability at most three particles, the problem left implicit), the Alhazov–Imai abstract [24] (five particles, not asserted reversible) and a 2021 RIMS session [43, 44]. Source 14 does not cite Kong or Imai’s report, which concern the numerical model of Part VI. No source claims priority, and this report claims none.

1.15 Status and what Parts V and VI do not claim

[write] The four sources give conventional proofs and finite exact-arithmetic checks, which are regression evidence, not proofs; all four are AI-assisted and unrefereed. Every limitation they state is kept in its Part. Collected here, they do not claim

- for source 14: an expanded universal source table, a numerical universal alphabet or row count, or a frozen universal transition table (universality is imported from Morita and Imai’s Proposition 3.4); anchored halt observation or intrinsic universality; an efficiency bound (the gap is exponential in the counters); a three-numerical-particle theorem for integer-state NCCA; real or rational exactness of the certificate; a fixed-arity polynomial for an unknown horizon, or a finite-fold or single-fold representation ([write] added 3 October 2026: still true of source 14; Part VIII composes source 15’s wrapper with the research tree’s residue-history interface into fixed-arity polynomials without a horizon for fixed sources, which are not finite-fold; Section 1.26); an exponentiation gadget (the loader is a bounded table); that the syntactic slot counts are lower bounds; novelty of the radius reductions; optimal overhead, a smallest alphabet, priority or literature-wide novelty; global halting, freezing or an eventual fixed point;
- for source 15: one target independent of the input; that the target is stationary or reached once; microscopic reversal of the forward paths; a bound on the overhead of the fresh-entry normalization; a map of the whole natural orthant, an identity off the zero sets, or rational or real exactness of the lift; novelty or priority for forward-and-reverse computation; a numerical universal alphabet;
- for source 16: one formula with the rule as a parameter; anything about intrinsic universality, signed masses, infinite alphabets, nonuniform or time-dependent rules, or active zero-weight backgrounds; a universal bound on the formula-dependent arity K , efficient preprocessing or optimal arity; a general rule-to-formula or quantifier-elimination implementation (the code validates the orbit accelerator); real or rational fibre exactness; a finite-fold conclusion for c.e. sets; novelty with respect to Kong’s thesis and Imai’s report;
- for source 17: an efficiency bound; infinite backgrounds, signed weights, infinite alphabets, active zero-weight backgrounds, several unit labels, or time-dependent or nonuniform rules; a five-particle upper bound for binary automata ([write] added 3 October 2026, batch 82: still not claimed by source 17; Reports 14 and 15, Part I of **five-particle-binary-automata**, prove it, also for globally reversible binary automata, so with source 17 five is exact for both); an implemented rule-to-section compiler or a formalization; reversibility of the binary shuttle (it is not injective; [write] added 4 October 2026, batch 91: still true of source 17’s rule; source 24 (Part IX) realizes the same timing phenomenon and the same quartic by a different, globally reversible binary rule of radius at most 90, Theorem 159.1); a general fixed-arity single-fold quartic compiler; uniform timed Presburger definability at mass four; priority, or any statement on the present status of the four-particle question;
- a proof-assistant verification; none of the sources’ code is a verified compiler.

The report claims no novelty for the results that the four sources prove again (Section 1.14).

1.16 Research questions answered or touched by Parts V and VI

[write] Dated notes at the questions themselves record the following; Table 3 carries the same information in its last rows.

- **Part IV’s remark after Theorem 44.1** (“It does not show that three suffices, or identify a least universal mass”) is *answered, model by model*. With several weight-one symbols three suffices, for pattern occurrence (Theorem 49.2) and for exact pairs (Theorem 64.1); with a single weight-one symbol neither three nor four does (Theorems 74.1 and 83.1), so that the least universal numerical mass is at least five, and exactly five if the Alhazov–Imai construction holds as its abstract states (its full paper has not been checked, and it is not asserted reversible). ([write] Added 3 October 2026, batch 82: the lower bound five holds in every finite dimension (Part VII, Theorem 101.1); source 19 states an inherited sharp four-versus-five boundary in every dimension, conditional on batch-82 Reports 16 and 26, which this report does not print: Report 16 is Part II of `five-particle-binary-automata`, whose Part I (Reports 14 and 15) proves the binary one-dimensional upper bound five, and Report 26 is printed there as Part VI (commit d2043bbd0).) ([write] Added 3 October 2026, batch 82, reciprocal note: a binary automaton has a single weight-one symbol, so in one dimension the least universal numerical mass is exactly five without the Alhazov–Imai paper: Reports 14 and 15 prove the upper bound for binary and for globally reversible binary automata (Corollaries 27.1 and 16.1 of `five-particle-binary-automata`, `fpa:bc:cor:sharp` and `fpa:cor:threshold`), and source 17 the lower bound. This report does not print Reports 14 and 15.)
- **Part IV’s fourth question** (“Resolve small-particle model comparisons”) is *answered in substance*: the number of weight-one symbols is the decisive convention (Section 1.13). Still open: the full Alhazov–Imai construction, a reversible numerical threshold, a binary upper bound, and the least number of weight-one symbols that permits a universal three-mass construction. ([write] Added 3 October 2026, batch 82: the binary upper bound and the reversible numerical threshold are now proved outside this report, as the preceding note says; the full Alhazov–Imai construction and the least number of weight-one symbols stay open.)
- **Part IV’s fifth question** (“Keep exact targets separate from halt signals”) is *answered for the three-mass weighted automaton* by source 15, whose output normalization is inverse execution from a fresh entry state; it is not answered for Part IV’s own Morita–Imai pulse interface.
- **Part IV’s non-claims** “exact-target undecidability, ... a minimum universal mass, or that three particles suffice” (Section 1.10) remain true of Part IV; Parts V and VI supply those statements in the models stated above.
- **Part IV’s first question** (“Print one safe universal source”) is *not* answered: sources 14 and 15 also leave the universal source as an existence construction (their first questions). ([write] Added 3 October 2026, batch 82: answered in substance outside this report by Part II of `five-particle-binary-automata` (Report 16): a literal universal reversible two-counter source with 122,622 controls and 141,561 rows, a clean loader and all-input proofs, for Report 15’s guarded syntax. A universal table in source 13’s pulse interface is still not printed, and sources 14 and 15’s first questions are answered only in part; notes there.)
- **Source 14’s third question** (exact finite targets need explicit treatment of the final counters) is answered by source 15; its anchored halt observer stays open. **Source 14’s and source 15’s fourth questions** (the integer-state mass convention) are answered in part by Part VI (at least five units for undecidable observations); the sharp reversible threshold stays open. ([write] Added 3 October 2026, batch 82: answered outside this report: five, since a globally reversible binary automaton is an integer-state one; note at Question 4.) **Source 16’s “four-versus-five boundary”** is answered by source 17 within its class.

- All other questions of Parts V and VI (Sections 63, 73 and 92, and source 16’s two directions in Section 81) are open. ([write] Added 3 October 2026: source 17’s seventh question is now answered for a fixed input by source 18, and its second, fourth and sixth are touched by sources 18, 20 and 21; Section 1.21.)

1.17 Source 18 and Part VII: mass at most four in every dimension (batch 82)

[write] This subsection and the four that follow were written on 3 October 2026, when four manuscripts of batch 82 of ProveIt’s incoming reports were added to the report: one as a third source of Part VI, the other three as Part VII. They are called *sources 18–21* after the file prefixes of their shipped programs and data, which continue this report’s local sequence after 17. Their batch-82 manuscript numbers are different, and so are the report numbers of the research pipeline that wrote them: **source 18 is batch-82 manuscript 19 (the pipeline’s addendum 13), source 19 is manuscript 06 (Report 29), source 20 is manuscript 17 (Report 30) and source 21 is manuscript 01 (Report 31)**; source 17 is the pipeline’s Report 12. All four arrived in commit db37d18c8 and are dated 3 October 2026; Appendix E gives the full provenance. None names a repository commit.

Source	Batch 82; pipeline	Archive and title	Printed as
18	manuscript 19; addendum 13	<code>Timed_Four_Mass_Quartic_Certificates.zip</code> ; <i>Canonical quartic certificates for timed four mass orbits</i> , a declared addendum to source 17	Part VI, Sections 93–100
19	manuscript 06; Report 29	<code>Dimension_Independent_Particle_Thresholds_Package.zip</code> ; <i>Dimension independent particle thresholds</i>	Part VII, Sections 101–117, Appendices I–J
20	manuscript 17; Report 30	<code>Sparse_Orbit_Geometry_and_Exact_Counting_Package.zip</code> ; <i>Sparse orbit geometry and exact counting</i>	Part VII, Sections 118–128, Appendix K
21	manuscript 01; Report 31	<code>Binary_Planar_Four_Particle_Shuttle_Package.zip</code> ; <i>A binary planar four particle shuttle</i>	Part VII, Sections 129–140

Citations among them, all pinned by SHA-256 and verified at placement: source 18 cites source 17; source 19 cites sources 17 and 18 and batch-82 Reports 16 and 26; source 20 cites source 19; source 21 cites sources 17, 18 and 20. Reports 16 and 26 belong to the binary five-particle line of the same batch and were placed in the collection’s report **five-particle-binary-automata** (commits bca6383e9 and 7d2b1b245); Report 16 is printed there as Part II, and Report 26 as Part VI (commit d2043bbd0). Neither is printed here.

- **Source 18 (Part VI), *timed quartics for a fixed input*.** Conditional on source 17’s effective normal form (Proposition 87.1), the complete timed orbit of a fixed input of mass at most four is a disjoint union of half-open charts with at most two natural parameters, affine domains and quadratic outputs (Lemma 94.1), and a canonical quartic with $B+K+M$ natural witnesses has exactly one witness tuple at each complete timed configuration (Theorem 93.1, Proposition 96.1). For source 17’s binary shuttle it writes the two charts explicitly and specializes the certificate to four witnesses and seven squares, exact even for nonnegative real

witnesses and uniform in the gap (Corollary 99.1). It answers source 17’s last question for a fixed input (Question 7).

- **Source 19 (Part VII), *every dimension*.** For a conservative cellular automaton on $\mathbb{Z}^d$ with a finite positively weighted alphabet, a unique vacuum and at most one weight-one symbol, exact and translated reachability and anchored and translated pattern occurrence are decidable from every input of mass at most four, uniformly in d and without reversibility (Theorem 101.1); its appendix lifts source 18’s timed quartics to $\mathbb{Z}^d$ (Theorem I.1). Its sharp four-versus-five boundary (Corollary 101.2) is conditional on Reports 16 and 26.
- **Source 20, *orbit geometry and counting*.** For a fixed input of mass at most four the stationary-frame visited set is semilinear, lies in finitely many translates of one rational plane of rank at most two, and has eventually quasipolynomial box counts of sharp degree $\text{rank}_{\mathbb{Q}}\{E, \nu\}$ (Theorem 118.2); an expanding shuttle discovers sites at the rate $T^{r/2}$ and has piecewise quadratic first arrivals (Theorem 118.3); a nonzero drift can destroy semilinearity and quasipolynomiality (Theorem 118.4), as an explicit weighted radius-six planar automaton shows with the exact count $3N - 3\sqrt{N} + O(1)$ (Theorem 125.1); canonical first-visit quartics follow (Theorem K.1).
- **Source 21, *a binary planar shuttle*.** An ordinary binary, quiescent, conservative radius-six automaton on $\mathbb{Z}^2$, obtained by lifting two outputs of source 17’s binary shuttle by one row, has a four-particle orbit with source 17’s clock (Theorem 132.1), visited set a wedge minus two lines (Theorem 133.1), exact stationary and drifted counts $4N - 4\sqrt{N} + O_k(1)$ (Theorems 134.1 and 136.2), Boolean degree exactly 45 with exactly 78 essential cells (Theorem 137.1), and a one-step quartic with 614 unique auxiliaries (Theorem 138.1).

The threshold table of Section 1.12 gains a row for every finite dimension ([write], 3 October 2026). Its entry at mass five records only what is true now: the inherited five-particle upper bound in every dimension rests on source 19’s linewise embedding and on Reports 16 and 26; Report 16 is Part II of `five-particle-binary-automata`, Report 26 is printed there as Part VI (commit `d2043bbd0`), and the programme’s reviews of both leave published theorems as explicit dependencies (note after Corollary 101.2).

1.18 Notions and letters of source 18 and Part VII

[write] The four sources keep their letters and their numbering of statements, by section; equations are numbered by section. Their macro for the natural numbers is written `\NN` here and prints $\mathbb{N} = \{0, 1, 2, \dots\}$: source 18 wrote $\mathbb{N}$, and sources 19–21 wrote $\mathbb{Z}_{\geq 0}$ for the same set (a change of rendering only). Sources 19–21’s maximum norm $\|\cdot\|_{\infty}$ is typeset by a renamed macro, because Part IV’s `\norm` prints $\mathcal{N}(\cdot)$; no other symbol was renamed and no normalization changed. The table lists the letters that carry different meanings among the new sources or against Part VI; inside a source the meaning is always that source’s.

Letter	Source 18	Part VII (sources 19, 20, 21)	Part VI (sources 16, 17)
d	the initial right-marker coordinate of the binary shuttle, $d \geq 7$, with $x = d - 7$	the dimension of $\mathbb{Z}^d$ (sources 19, 20); the current gap $d = k + n$ (source 21)	the initial right-marker coordinate (source 17)
k	k_i , the number of parameters of chart i	a period count (source 19); the initial gap, $k \geq 3$ (source 20) or $k \geq 7$ (source 21)	the witness of (90.1); the index of (89.3)
S	the common squared part; transition starts $S_\ell(n)$	the padded comoving radius (source 19, as in Part VI); the visited set S , S_F (source 20); an occupied set $S \subseteq \mathbb{Z}^2$ (source 21)	the padded comoving radius
B	the number of charts	the seed-prefix bound B , a marker B and the chart count B_{ch} (source 19); a clause count B , the count B_n , the function $B(N)$ (source 20); a rule output B and $B_k(N)$ (source 21)	the seed-prefix bound (source 17)
C	the certificates C_a , C_{prod} , C_{SOS} , C_d	a time coefficient and the quartic $C(y; w)$ (source 19); the box counts $C(N)$, $C_F(N)$ and a congruence count C (source 20); $C_{G,k}$, $C_{F,k}$ (source 21)	a time coefficient (source 17)
E	the selector sum	the anchor-update vector $E \in \mathbb{Z}^d$ (sources 19, 20); the selector sum (source 19's appendix); a head state of weight two (source 20); the eastward pair rule (source 21)	the anchor update (source 17)
W	—	the ordinary threshold (source 19); a witness count and a head state (source 20); the westward pair rule (source 21)	the threshold W (source 17)
N	the witness count	the threshold (source 19); a box radius and a shared cycle variable (source 20); a box radius (source 21)	the threshold (source 17)
τ	transition durations $\tau_\ell(n)$; translation	translation τ_v (sources 19, 20); first visit time $\tau(y)$ (source 20); unit vertical translation (source 21)	translation τ_a
D	the marker gap D_n	the drift vector of a packet (source 19); the discovery count $D(K)$, drifts D_i , a denominator (source 20)	the marker gap $D = R - L$ (source 17)
R	the relation R_a ; residuals R_0 , $R_{ir}^\geq$, R_q^{out}	the radius (source 19); the inverse $R(M)$ (source 20); a rule and $R_{G,k}$, $R_{F,k}$ (source 21)	the radius
M	the inequality count	$M = 10^6$, M_0 and an inequality count (source 19); a level in an inverse and a number of square lifts (source 20); a level (source 21)	the gap-residue modulus (source 17)
H	the equality count	a Cramer vector and an equality count (source 19); the inverse $H(M)$ and an equality count (source 20); the halo $H(A + a)$ (source 21)	a span predicate (source 16)
P	chart outputs P_i	a packet profile (source 19); a polynomial $P(y; w)$ (source 20); a polynomial and a predicate $P_{A,a}$ (source 21)	the de Bruijn potential (source 17)
α	coefficients α_ℓ of flight lengths	a rational slope (source 19); the leading coefficient of $D(K)$ (source 20); $2k - 10$ (source 21)	a rational slope (source 17)

Letter	Source 18	Part VII (sources 19, 20, 21)	Part VI (sources 16, 17)
λ, ν	—	a Bezout functional with $\lambda(\nu) = 1$ for the primitive flight direction ν (sources 19, 20)	—
V	—	the discovery count $V(T)$ (source 20); the visited set V_k (source 21)	—

Several readings need the same care.

- **The gap and the dimension.** In sources 17 and 18 the letter d is the initial right-marker coordinate of the binary shuttle; in sources 19 and 20 it is the dimension. Source 21 calls the initial gap $k \geq 7$; its k is source 17's and source 18's d , and its $d = k + n$ is source 18's $r = d + n$. The tempting reading of source 18's " $d \geq 7$ " as a dimension bound is false.
- **Particles and mass.** In source 21 every occupied site has weight one, so "four particles" is mass four. Source 20's explicit automata have mass four in three occupied sites: two unit markers and one head of weight two. Its letters E, W are head states; source 21's E, W, R, L are names of rules of a binary automaton.
- **Timed and untimed, fixed and uniform.** Source 18's relation keeps the time and the complete configuration; source 20's first-visit certificates forget the time and keep one witness per site. Every certificate of the four sources is for one fixed rule and one fixed input, with an arity that depends on the input; source 16's uniform Presburger description (Theorem 74.1) has no analogue here at mass four.
- **One step and one orbit.** Source 21's 614-auxiliary quartic certifies one local transition of its rule; source 18's and source 20's quartics certify one orbit.

1.19 Relation of source 18 and Part VII to the repository

[write] The four sources cite only Part VI's source 17, each other and the batch-82 Reports 16 and 26; the cross-references below are added. Results of other reports are cited by label name.

- **Re-derived results, printed as pointers or second routes.** Source 18 restates source 17's binary rule and its conservation certificate (a pointer replaces that subsection, Section 98), its return quartic (90.1) and its normal form (notes in Sections 94 and 99), and ships three of source 17's files byte for byte (shipped once). Its compiler is a second route to Part I's Theorem 10.2, with Part V's inactive gate and Lemma `qoc:rn:lem:gates` of *quadratic-orthant-certificates* (note in Section 96). Source 19 proves source 17's Theorem 83.1 again at $d = 1$ by the same architecture while extending it to $\mathbb{Z}^d$ (notes from Section 101 on); its Lemma 104.1 is a fixed-input, $\mathbb{Z}^d$ form of source 16's Theorem 74.1; its mass-two case meets Part IV's Theorem 44.1; its appendix restates source 18's chart partition and compiler (printed in full, with notes). Source 20's canonical membership quartic is a second route to Part IV's compiler (Section 46) and to Theorem `cdc:of:thm:singlefolds1` of *canonical-diophantine-certificates* (note in Appendix K); its Lemma 122.1 is classical. Source 21's orbit theorem is the planar lift of source 17's Proposition 89.2, and its one-step quartic uses Part IV's device of gate residuals. Each place carries a bracketed note, and no novelty is claimed for any of them.

- **New in the four sources.** Source 18’s complete-orbit chart partition, constant-term lifting, ledgers, selector elimination, signed external pairs and the four-witness binary certificate; source 19’s actual-site contact, exact contact rays, finite nonparallel switch set, compact triples that miss, the $\mathbb{Z}^d$ dispatch and drift cutoff, and the $\mathbb{Z}^d$ chart refinements; all of source 20’s geometry, counting, explicit automata and first-visit certificates beyond the membership compiler; source 21’s binary planar rule with its full-shift conservation proof, wedge, counts, exact dependency set and one-step quartic.
- **Degree four is least for source 18’s binary certificates.** The note after (90.1) transfers. The complete timed relation of the binary shuttle meets the semilinear condition $(x_0, x_1, x_2) = (0, 3, 4)$ exactly at the return times $T(n)$ of Section 98, which form a set that is not semilinear (Proposition 89.2); so the relation is not semilinear, and by Theorem `cdc:of:thm:classification` it has no natural-witness representation of degree at most three by a polynomial nonnegative on the real nonnegative orthant. This observation is [write], made at this merge; the programme has not reviewed it.
- **The programme’s reviews.** The Hilbert’s-tenth programme reviewed three of the four sources on the day of their placement, in `Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/`. Source 21: `review_binary_planar_four_particle_31.md` (commit 16f50dcc6) [56], “PASS within the stated scope; no error found”, with an independent reconstruction of the whole local quartic from the four printed rules (Section 139). Source 19: `review_dimension_independent_four_mass.md` (commit 7b547f686) [58]: the proof of Theorem 101.1 passes, a second reviewer found no gap, and the inherited five-particle upper bound is not reproved (Section 116). Source 20: a scoped intake, `review_original_frame_first_hit30_intake.md` (commit c875bad40) [59], which finds no defect in the whole-orbit-injectivity step of Corollary K.4 but is not a review of the compiler, and names source 19’s timed companion proof as the next audit; a follow-up (`original_frame_first_hit30_fixture.md` with its independent review `review_original_frame_first_hit30_fixture.md`, commit 5b4c091e6) [60] then read that proof in full, found it sound as a conditional statement, and emitted a complete 35-operation first-hit quartic for one drifted orbit of source 20’s weighted planar automaton (note in Appendix K). Source 18 has not been reviewed: the programme’s archive inventory `reviewed_report_archive_replay.md` lists its archive as relocation only. ([write] Added 4 October 2026: it has been since, in a scoped intake made after this write, (`review_timed_four_mass_source18_intake.md`, commit 39f398985) [112]: “PASS on the conditional chart/compiler argument and the explicit binary certificate”, conditional on source 17’s normal form.) The programme has also reviewed Reports 16 and 26, on which source 19’s corollary rests (commits 69730d3e8 [61] and fb7e3cb47 [57]; note after Corollary 101.2). The batch-80 review of source 17 [46], whose normal form sources 18–20 import, still applies to that import. This write has not been reviewed by the programme.
- **The repository’s universal bounds.** The project README’s fixed-universal operation bounds [1] are untouched: all quartics of the four sources are for one fixed rule and one fixed input, with arity depending on the input, or for one local transition (source 21); the programme’s review of source 21 says the same for it.
- **Formal status.** Nothing in source 18 or Part VII is formalized in Lean or Rocq, and the Hilbert-tenth-problem Lean development has no cellular-automaton module. The Presburger decision step that source 19 applies to its explicit formulas is Cooper’s quantifier elimination, formalized in `Logic/PresburgerArithmetic` (Section 1.9); that formalizes the imported step only. Woods’ counting theorems, which source 20 uses, are not formalized in the repository.

1.20 Status and what source 18 and Part VII do not claim

[write] The four sources give conventional proofs and finite exact-arithmetic checks, which are regression evidence, not proofs; all four are AI-assisted and unrefereed. Every limitation they state is kept in its place. Collected here, they do not claim

- for source 18: a proof of the imported normal form (the theorem is conditional on it); an implemented rule-to-chart preprocessing, a decision of the semantic bijectivity of a chart family, or a complexity bound; a bound uniform over inputs, uniform timed Presburger definability ([write] added 4 October 2026: still true of source 18; source 25 builds a conditional chart family uniform over inputs from source 18’s charts, Theorem 169.1, and its timed relation is not Presburger), or a single-fold or finite-fold Diophantine conclusion, including for MRDP; real or rational exactness of the generic compiler (only the four-witness binary certificate is exact for nonnegative real witnesses, and unrestricted real witnesses give spurious zeros); uniqueness after forgetting the time; witness minimality; novelty, priority or a literature status;
- for source 19: uniform timed Presburger definability; a useful complexity bound or small libraries; a general rule compiler or an end-to-end simulation of every input; reversibility; the inherited five-particle upper bound, which it does not reconstruct (Corollary 101.2 is conditional); several unit labels, negative or zero weights, infinite alphabets, an active vacuum, infinite scenery or time- or position-dependent rules; untimed finite-foldness, a single-fold MRDP theorem, a uniform arity bound or real or rational exactness for its appendix; a formalization; novelty or priority;
- for source 20: a binary or reversible realization; new theorems about Presburger arithmetic (its counting steps apply Woods’ theorems); practical algorithms or size bounds; an exhaustive asymptotic taxonomy in the original frame; a general original-frame first-hit certificate for sites or patterns under drift; a single-fold MRDP theorem, a uniform arity bound or real or rational exactness; an efficient general compiler, a formalization or a universality threshold; novelty with respect to the pebble-automaton literature;
- for source 21: reversibility or universality of either rule; a smallest mass, radius, number of essential cells, Boolean degree or auxiliary count; a classification of malformed orbits; an unbounded-time or finite-fold representation from its local quartic; a real-witness conclusion inherited from source 20;
- a proof-assistant verification; none of the sources’ code is a verified compiler.

The report claims no novelty for what the four sources prove again (Section 1.19).

1.21 Research questions answered or touched by source 18 and Part VII

[write] Dated notes at the questions themselves record the following; Table 3 carries the same information in its last rows.

- **Source 17’s seventh question** (Question 7, a fixed-input timed quartic compiler from disjoint phases) is *answered for a fixed input* by source 18, conditional on Proposition 87.1 and with natural witnesses; source 19’s Appendix I lifts the answer to $\mathbb{Z}^d$. The rule-to-chart preprocessing is not implemented.
- **Source 17’s second and fourth questions** (uniform untimed spatial definability; a stronger normal form for original-frame timed relations) are *touched*: for a fixed input the visited

set is semilinear with quasipolynomial counts (source 20), the complete timed orbit is a disjoint union of quadratic charts (source 18), and with drift original-frame site sets can be nonsemilinear (Theorem 125.1). The uniform versions stay open. ([write] Added 4 October 2026: answered conditionally by source 25; Section 1.31.)

- **Source 17’s sixth question** (minimal radius for the binary timing example) is *touched* by source 21: its planar analogue has exact radius six and exactly 78 essential cells, with no minimality claimed; the one-dimensional question stays open.
- **Source 17’s first, third and fifth questions** stay open; source 19 adds literature (Section 115), shows by an example that naive geometric bounds fail (Section 107), and leaves several unit labels open (its Question 4).
- **The numerical particle threshold** of Section 1.16 (at least five units, exactly five if the Alhazov–Imai construction holds) *extends to every finite dimension* for the lower bound (Theorem 101.1); the sharp upper bound in every dimension is conditional on Reports 16 and 26 (Corollary 101.2). ([write] Added 3 October 2026, batch 82: in one dimension the threshold is exactly five without the Alhazov–Imai paper, by Reports 14 and 15 (Part I of five-particle-binary-automata); note in Section 1.16.)
- **Source 20’s first question** (binary or reversible realizations) is *answered for binary rules* by source 21; reversible realizations stay open.
- **Source 19’s fifth question** (optimizing the timed certificates) is touched by source 18’s selector elimination and four-witness certificate, without a general answer. All other questions of the four sources (Sections 100, 117, 127 and 140) are open.

1.22 Part VIII: unbounded-time certificates for exact three-mass targets (batch 82)

[write] This subsection and the four that follow were written on 3 October 2026, when two further manuscripts of batch 82 of ProveIt’s incoming reports (cluster M2) were added to the report as Part VIII. They are called *sources 22 and 23* after the file prefixes of their shipped programs and data, which continue this report’s local sequence after 21. Their batch-82 manuscript numbers and their numbers in the research pipeline that wrote them are different: **source 22 is batch-82 manuscript 21 (the pipeline’s Report 21), and source 23 is manuscript 02 (Report 22)**. Both arrived in commit db37d18c8, were placed by commit 7d2b1b245 and are dated 3 October 2026; both pin the ProveIt commit ad634b2d1. Appendix E gives the full provenance.

Source	Batch 82; pipeline	Archive and title	Printed as
22	manuscript 21; Report 21	Unbounded_Compact_Clean_Clocks_Package.zip; <i>Unbounded compact clean clocks for exact three mass targets</i>	Part VIII, Sections 141–149
23	manuscript 02; Report 22	Canonical_Histories_and_Infinite_Fibers_Package.zip; <i>Canonical histories and infinite native fibers</i> , a declared sequel of source 22	Part VIII, Sections 150–158

- **Source 22, *unbounded compact clean clocks***. For a fixed reversible separated two-counter source with fresh entry and terminal halt, the raw unbounded residue-history polynomial of

the programme’s research tree is composed with source 15’s clean return wrapper: the raw native time becomes a positive witness θ , the redundant output-copy comparison is dropped, and one square imposes $U = 2\theta + 192(F + x + 1) + 16$. The resulting polynomial $Q_R(x, U; z, \theta)$ has no horizon, and its positive zeros at (x, U) are exactly the first exact whole-configuration targets of the cleaned three-mass automaton at time U (Theorem 141.1); every accepted nonempty fibre is infinite (Theorem 146.1). For four nonuniversal fixtures that halt in one or two source steps the complete circuits are emitted and audited: 604, 479, 477 and 480 operations, 60, 58, 58 and 58 positive witnesses, exact degrees 2344, 1192, 1192 and 1192; separately folded schedules have 603, 478, 476 and 479 operations. Initially halted sources have witness-free four-gate certificates.

- **Source 23, *canonical histories and infinite native fibers*.** One positive coordinate κ and the square $(\eta + \kappa - S - 1)^2$, with $S = n_{\text{initial}} + n_{\text{target}} + \theta$, force the outer height h to be the least power of two above S ; completeness survives because every true residue quotient is below θ . At every accepted pair the outer coordinates are unique and projection onto the twenty-two native coordinates is a bijection onto the entire native extension fibre (Theorems 150.1 and 154.1); that fibre is infinite by an elementary matrix argument in which only three native coordinates vary (Theorem 155.1). The circuits cost 608, 483, 481 and 484 operations.

How they relate to Parts I–VII. Part VIII continues Part V. It uses source 15’s wrapper, target and clock as inherited theorems and restates them for the reader (note in Section 143), and it answers, for fixed sources, the unbounded-time questions of sources 14 and 15 (Section 1.26). Its circuits are not fixed-horizon certificates like those of Parts I–VII: the arity is fixed by the source alone, and the price is infinite witness fibres. It shares no lemma with Parts I–IV, VI or VII. Its arithmetic comes from the Hilbert’s-tenth research tree, not from the collection: the residue-history packing, the native prescribed-scale AND and the Pell block are research-wip notes there (Section 1.24).

1.23 Notions and letters of Part VIII

[write] The two sources keep their letters and their numbering of statements, by section; equations are numbered by section (the sources numbered theirs consecutively). Their macro for the natural numbers is written `\NN` here and prints $\mathbb{N} = \{0, 1, 2, \dots\}$, as both sources define it; $\mathbb{Z}_{>0} = \{1, 2, \dots\}$ is theirs. Source 22’s `\file` prints delivered file names in typewriter type, as in the rest of the report, and both sources’ hash display `\sha` sets a hash as a breakable path rather than through the `seqsplit` package. No symbol was renamed and no normalization changed. Source 23 renames one of its own symbols (the native integer that its inherited block calls R is written ρ , to avoid confusion with the source program; Section 155.1). The table lists the letters that carry different meanings in the two sources or against Part V; inside a source the meaning is always that source’s.

Letter	Source 22	Source 23	Part V (sources 14, 15)
h	the outer height $h = n_0 + n_f + \theta + \eta$, “distinct from its source step count ℓ ”	the outer height $h = S + \eta$; also a native coordinate h	the source horizon, a number of source instructions (source 15’s h is source 22’s ℓ)
ℓ	the number of source instructions of the trace	the number $m + g + 1$ of AND lanes (source 22’s L)	—

Letter	Source 22	Source 23	Part V (sources 14, 15)
t	the raw native time of $P_R(x, y, t; z)$, replaced by θ	the trace length, $P = B^t$ (source 22's ℓ)	—
L	the number $m + g + 1$ of lanes	a dyadic multiplier $L \geq \max\{4, m + 1, 1 + \max_r(a_r + d_r)\}$ (source 22's K_0)	—
θ	the forward native duration, a positive witness	the same	source 15's Θ
F	the final raw payload, a retained positive witness	the same	source 15's N_h
U, T_{clean}	the requested clean time; the cleaned physical time	the same	source 15's T_{cl}
Q_R	the clean-clock polynomial (144.1)	the five-gate-folded form of the same polynomial; Q_R^{can} adds the height square	—
z, w	z the witness tuple (with F), θ separate	w the witness tuple (with θ); z_i the residue quotients (source 22's q_i)	—
q, s, r	control code q ; quotient q and residue s in $f(mq + s)$	control code q ; quotient z and residue r in $f(mz + r)$; the native scale $q = 16Q_{\text{native}}$; native coordinates r, s	a source state; number of states
B, P, J	radix $B = Ch^2$, $P = B^\ell$, $J = 1 + B + \dots$	the same, with $P = B^t$	B : number of branches
C	the dyadic constant $C = 131072$; $\mathcal{C}, \mathcal{C}_\tau$ packed words	the same C ; C_τ the packed clock word; a native coordinate c	completed pair-support count; cofactor
K	$K = 5$ controls; K_0 the inherited dyadic multiplier	$K = 5$ controls	loader selectors $(A + 1)(C + 1)$
M, A	numbers of multiplications and of additions or subtractions; ports M_*, A_*	the same counts; joined words M, A	a source machine M
H	the new sole halt of R_{clean} ; port H_*	a joined word H	the same terminal state H (source 15); an instruction table (source 14)
Q_*	the scale BP^v	Q_{native}	—
$\widehat{k}$	the clock quotient hat	written $\widehat{c}_\tau$	—
S	—	$S = n_{\text{initial}} + n_{\text{target}} + \theta$	static shuttle types S_j
D, ρ	—	$D = \rho^2 - 1$, $\rho = ic^2$ (native)	$D = 6qm$ data triples
η, τ, y	outer height slack η ; tick durations τ_i ; raw output y	the same; also native coordinates η, τ and y_{aux}	τ a source-step duration; y_N the target
E	selector words E_s	selector words E_r ; $E = (wq)(sq)$ in Section 155.1	selector sums E_t

Several readings need the same care.

- **Unique and infinite.** Part V's unique natural witness is at a fixed external source horizon; Part VIII's polynomials have no horizon and infinite witness fibres. Source 23's “canonical” outer history is unique, its witness tuples are not: reading “canonical” as “single-fold” is false.
- **Native, twice.** “Native” names the unblocked radius-four clock model, against the spatially

blocked and the four-phase models, and also the inherited native AND and Pell block with its twenty-two coordinates. “Native time” is the former, “native fibre” the latter.

- **Compact.** “Compact clean clocks” means a compressed history certificate without reverse witnesses, as in source 15’s compact certificate; the domains are unbounded integers, not a compact real set (the programme’s intake says the same).
- **Unbounded.** Unbounded running time for one fixed source, with the raw payload $x + 1$ as input. On that interface halting depends only on $\nu_2(x + 1)$ and $\nu_3(x + 1)$ (Section 148.3), so it is not an ordinary-input loader, and no universal source is instantiated.
- **Report numbers.** The pipeline’s Report 21 and Report 22 are sources 22 and 23, one apart; its Report 10 is source 15.

1.24 Relation of Part VIII to the repository

[write] The two sources cite the research tree’s pinned files and source 15’s theorem, and source 23 cites source 22; the cross-references below are added. Results of other reports are cited by label name.

- **Re-derived and repeated material, printed with notes.** Source 22’s Section 143 restates source 15’s wrapper, target and clock (Theorems 66.2, 67.1 and 68.1), and its Section 145 restates the research tree’s residue-history and exact-clock arguments for the time-substituted circuit; both are printed in full, as the sources’ own accounts. Source 23’s Section 153.1 restates source 22’s no-wrap argument, and its Sections 156.1, 157.1, 157.2 and 157.3 repeat source 22’s fixtures, zero-step circuits, sanity values and raw-input limits; its two repeated tables are printed once, in source 22. Each place carries a bracketed note, and no novelty is claimed for any of them.
- **The research tree’s sketch, completed.** The pinned note `three_mass_unbounded_interface.md` describes the clean-time bridge and its incremental cost but “does not claim a complete cleaned-time total”; source 22 emits and audits that total. The credit is in the opening of Part VIII.
- **Part V.** Sources 14 and 15 have a unique natural witness at each fixed source horizon (Theorems 58.1 and 70.2); Part VIII removes the horizon and pays with infinite fibres (Theorem 146.1), and canonicalization restores the uniqueness of the outer history only (Theorem 150.1). Source 15’s target y_N depends on the input, and so does Part VIII’s.
- **fixed-universal-polynomials, Part III.** Report 25, the sequel of source 23, classifies the entire native fibre (Theorem `fup:nf:thm:class`: seventeen coordinates fixed, five parametrized by two integer indices), which implies Theorem 155.1; source 23’s matrix argument is kept as a second route (note after that theorem). That Part’s transport corollary (its Section `fup:nf:sec:transport`) uses Theorem 154.1 and is conditional on it.
- **five-particle-binary-automata.** Source 22 contrasts itself with Reports 19 and 20 there (Parts IV and V), which certify a different automaton at a fixed horizon. That report’s Part VI prints Report 28, whose remark on “earlier observations about infinite native Pell witness fibers” is uncited in its source; a note there identifies them as source 23’s.
- **Later work of the programme on the same circuits.** After the pin `ad634b2d1` the research tree improved the native-time circuits (x, y, T) of the pinned interface: a factored first Pell

coefficient (`native_pell_factored_first_coefficient.md`, commit `bbfe44d90`; 598 to 597 operations), a paid endpoint projection that removes the output alias F (`three_mass_unbounded_endpoint_projection.md`, `06a9631b1`), a target-free height (`three_mass_target_free_height.md`, `9393b0f29`; 592, 467, 465 and 468 operations), and a refutation of removing the time from that height (`three_mass_time_free_height_obstruction.md`, `c8f08eb6e`), with its one-step exceptions (`three_mass_one_step_height.md`, `d400c01ee`). For clean time it built the direct clean clock (`three_mass_direct_clean_clock.md`, `8aabb1453`; 597, 472, 470 and 473 operations for source 22’s four sources, independently reviewed) [71] and refuted one further deletion from its height (`three_mass_input_free_height_obstruction.md`, `735f62a6a`) [72]. These are later repository results, not printed here. None supersedes a statement of Part VIII: the programme states that the direct clean clock’s counts are not to be attributed to source 22 and do not revise its counts, the endpoint projection removes the payload F that the clean formula needs, and the time-free obstruction is consistent with source 23’s keeping θ in S . None of them makes a fibre claim.

- **The programme’s intakes.** Source 22 was taken in with Report 20 in `review_event_budget_clean_clock_intake.md` (commit `8aabb1453`) [69], and source 23 with Report 25 in `review_incoming_negative_index_fibers.md` (commit `0055e1c4d`) [70]. Neither is a full review: the first finds no defect in the stated theorem scopes, and the second does not certify source 23’s canonical-height circuit costs. The pinned raw interface itself was reviewed by the programme before the sources were written (`review_three_mass_unbounded_interface.md`, commit `ad634b2d1`: “PASS within the stated fixed-source, natural-input scope”). Notes at the end of the two sources give the details. This write has not been reviewed by the programme.
- **The repository’s universal bounds.** The project README’s fixed-universal operation bounds [1] are untouched. Part VIII’s polynomials are for four fixed nonuniversal sources, with the raw payload as input; both sources and the programme’s intake say that no universal numerical bound and no ordinary-input loader follow.
- **Formal status.** Nothing in Part VIII is formalized in Lean or Rocq. The Hilbert-tenth-problem Lean development has no cellular-automaton module, and the residue-history, native AND and Pell results that Part VIII imports are research-wip notes with checked receipts, not Lean declarations. The project’s formal endpoint, the MRDP theorem `Diophantine.mrdp` of `Computability/HilbertTenthProblem/Lean/Diophantine/MRDP.lean`, is not used by either source.

1.25 Status and what Part VIII does not claim

[write] The two sources give conventional proofs, completely emitted circuits and finite exact-arithmetic checks, which are regression evidence, not proofs; both are AI-assisted and unrefereed, and both theorems are conditional on the inherited results listed in the opening of Part VIII. Every limitation they state is kept in its place. Collected here, they do not claim

- for source 22: an arbitrary c.e. ordinary-input loader (halting on the raw interface depends only on two valuations of $x + 1$), a universal source or a numerical universal bound, optimality or minimality of the fixture circuits, a rational-zero or real-zero theorem (real sum-of-squares nonnegativity does not extend the native theorem), a materialized complete positive native Pell witness, a fixed universal target, a stationary halt or later target recurrences, lower bounds from the degrees, a re-proof of the native Pell representation theorem or a regeneration of

the automaton’s local rules; that bounded tests prove the unbounded interpretation; that the emitted representations are finite-fold (they are not), or any multiplicity limitation for another representation of the same relation; that its first-halt argument applies to endpoints that permit looping;

- for source 23: a general finite-fold MRDP claim or impossibility, an ordinary-input universal loader, optimality, or a new Pell classification; a materialized positive native Pell seed (the matrix family is relative to a seed supplied by the inherited completeness theorem); a total polynomial map $(x, U) \mapsto o(x, U)$ or an all-tuple normalization from old to new witnesses; that the matrix family enumerates the whole fibre; a rational or real zero-set theorem, an arbitrary signed-witness interpretation, a fixed universal target, a stationary halt, a later-recurrence theorem, a minimal circuit bound or a new universal operation bound; a free fresh-entry transformation for every reversible program; that finite test totals or a pass under `python -O` replace the proofs;
- a proof-assistant verification; neither source’s code is a verified compiler.

The report claims no novelty for what the two sources restate or inherit (Section 1.24).

1.26 Research questions answered or touched by Part VIII

[write] Dated notes at the questions themselves record the following; Table 3 carries the same information in its last row.

- **Source 14’s Question 5 and source 15’s Question 5** (a costed unbounded-time certificate for the three-mass frontend, with degree, variable ledger and fibre properties) are *answered for fixed eligible sources*, conditional on the inherited theorems. Composition with the research tree’s residue-history interface gives horizon-free polynomials of exact degrees 2344 and 1192, with complete ledgers of 604, 479, 477 and 480 operations (603, 478, 476 and 479 folded; 608, 483, 481 and 484 canonical) and 60, 58, 58 and 58 positive witnesses for the four fixtures. Every accepted nonempty fibre is infinite (Theorem 146.1); after canonicalization the outer history is unique and the native fibre, onto which the fibre projects bijectively, is infinite (Theorems 154.1 and 155.1). So the unique witnesses of Part V do not survive, as source 14 anticipated, and the emitted representations are not finite-fold.
- **The universal-source questions** (source 14’s Question 1 and source 15’s Question 1) are not answered by Part VIII, which instantiates four nonuniversal sources and whose raw interface is not an ordinary-input loader. Nor is source 15’s Question 2 (one target independent of the input): Part VIII’s target depends on the input, as source 15’s does.
- **Part II’s Question 2** is *touched*: Part VIII encodes unbounded histories in a fixed arity for fixed sources of the conservative three-mass automaton, through arithmetic on source-level histories. The question as posed, for signal machines and histories of rational distances, stays open.
- Neither source states a list of questions of its own.

1.27 Part IX: a reversible four-particle clock and uniform charts (batch 91)

[write] This subsection and the four that follow were written on 4 October 2026, when two manuscripts of batch 91 of ProveIt’s incoming reports were added to the report as Part IX. They are called

sources 24 and 25 after the file prefixes of their shipped programs and data. Their batch-91 manuscript numbers and their numbers in the research pipeline that also wrote sources 18–23 are different: **source 24 is batch-91 manuscript 01 (the pipeline’s Report 49) and source 25 is manuscript 03 (Report 51)**. Both arrived in commit 0d7f51c44, were placed in d750d98dd, and are dated 4 October 2026; both pin this article itself (blob ec10c7a5, the text of commit 0f95145cb), and source 25 also pins source 24. Appendix E gives the full provenance.

Source	Batch 91; pipeline	Archive and title	Printed as
24	manuscript 01; Report 49	<code>A_Sharp_Reversible_Four_Particle_Timing_Threshold_Package.zip</code> ; <i>A sharp reversible four particle timing threshold</i>	Part IX, Sections 159–168
25	manuscript 03; Report 51	<code>Uniform_Timed_Charts_for_Four_Mass_Units_Package.zip</code> ; <i>Uniform timed charts for four mass units</i>	Part IX, Sections 169–180

- **Source 24, a reversible clock.** A binary number-conserving automaton, globally reversible on the full shift with forward and inverse radius at most 90, built from six guarded endpoint swaps in two involutive blocks, has from $\{0, 5, 6, d\}$, $d \geq 13$, the anchored hits $k^2 + (2d - 23)k$ of the word 1000011 (Theorem 159.1), with every intermediate state described (Proposition 162.1) and a density-one observation of the same orbit (Corollary 163.2). With the binary case of source 16’s lower bound, four particles are the least count for a non-eventually-periodic anchored occurrence set, also under global reversibility. Its quartic is source 17’s (90.1) (Remark 163.1).
- **Source 25, uniform charts.** Conditional on source 17’s scattering data and source 18’s fixed-input charts, one finite family of complete timed-orbit charts covers every ordered input of mass at most four of a fixed single-unit rule: finitely many Presburger input cells, at most two natural evolution parameters, affine stationary-frame positions and time of degree at most two, plus at most three gap quotients for literal affine domains (Theorem 169.1); hence uniform untimed stationary-frame Presburger reachability and a uniform single-fold quartic with copied inputs (Corollary 169.2, Proposition 177.1). Source 24’s clock with one site of drift per step has non-Presburger original-frame untimed reachability from one fixed input (Proposition 176.1).

1.28 Notions and letters of Part IX

[write] The two sources keep their letters and their numbering of statements, by section; equations are numbered by section. Their $\mathbb{N} = \{0, 1, 2, \dots\}$ is written $\mathbb{N}$ here, and no other symbol was renamed; no normalization changed. Source 24’s 1000011 and AR, ..., BC are set by two new macros. The table lists the letters that carry different meanings in the two sources or against Part VI; inside a source the meaning is always that source’s.

Letter	Source 24	Source 25	Part VI (sources 16–18)
H	the occurrence-time set $H(F, c; w, a)$; the isolation distance $H = 2(b + r) = 30$	the common modulus of (176.2); the equality count H_0 ; equality forms H_{hs}	a span predicate (16); the equality count (18)
b, r	write radius $b = 7$, read radius $r = 8$ (Section 161); a singleton's site and a phase index (Section 164)	a target unit's site (Section 170); the least partial gap change b of (173.1); residues r, r_i	a singleton's site b (16)
d	the initial right marker, $d \geq 13$; a pair diameter (Section 164)	the cycle-entry gap; a drift in (171.1); a denominator in (176.1); a pair separation	the initial right marker, $d \geq 7$ (17, 18); a drift (17)
D	the current clock gap of C_D ; a pair displacement in (164.1)	the marker gap $D = R - L > N$; section gaps $D_{n,s}$	the marker gap (17)
F, G	F the reversible clock; G any binary rule in Theorem 159.1; G_0 and $G = \tau_{-\delta}G_0$ in Section 164	F the rule and $G = \tau_{-\delta}F$; in Proposition 176.1, G is source 24's F and $F = \tau_1 \circ G$	F the rule, G its stationary frame (16)
A, B	the two blocks of the rule	$A = \sum \alpha_i$ (cycle slope), $A(\mathbf{x})$ an affine form; B the safety bound; $\mathbf{b}$ output labels	B the seed bound (17); A the coefficient of $\frac{n(n-1)}{2}$ in (87.2) (17), which is $A\Delta$ of source 25
C	$C_d = \{0, 5, 6, d\}$	the cycle time constant C, C_s ; the quartic $\mathcal{C}$	a time coefficient (17)
E	row endpoints E_0, E_1	the anchor increment of a cycle; the selector sum $\sum_h e_h$	the anchor update (17); the selector sum (18)
K	raw-key sets $K(c), K_A, K_B$	the number of completed contracting cycles	—
L	the cycle length $L_D = 2D - 22$	the left marker L ; output denominators L_q	the left marker (17)
S	the padded radius (Section 164); the swap S_κ	the padded radius; edge start times $S_s(n)$	the padded radius
W	a row's read interval	the ordinary threshold $W = N + 2B$; the witness count W_{nat}	the threshold (17)
Q	the quartic $Q(x, t; k)$, equal to source 17's $\mathcal{P}$	the first-reset time $Q(\mathbf{x})$	—
M	—	the mass $M(c)$; a residue modulus	the mass (16); the gap-residue modulus (17)
J	—	the number of retained charts	—
x	an anchor; $x = d - 13$ in Remark 163.1	the input coordinates $\mathbf{x}$	$x = d - 7$ (17, 18)

Several readings need the same care.

- **One quartic, two clocks.** Source 24's $Q(x, t; k)$ and source 17's $\mathcal{P}(x, t; k)$ are the same polynomial, but x is $d - 13$ for source 24's clock and $d - 7$ for source 17's: the same x belongs to different initial markers, and the hit sets of the two clocks in terms of d differ $(k^2 + (2d - 23)k)$ and $k^2 + (2d - 11)k$.
- **Timing threshold and universality threshold.** Source 24's threshold four is the least

particle count for a non-eventually-periodic anchored occurrence set; the binary universality threshold is five (Reports 14 and 15, Section 1.16). The tempting reading of source 24 as a four-particle universality result is false: its rule is covered by source 17’s decision theorem.

- **Uniform and fixed, untimed and timed, stationary and original.** Source 25’s Presburger conclusion is for the *untimed* relation in the *stationary* frame, uniform in the input; its timed relation is not Presburger (source 17’s shuttle), and its original-frame untimed relation need not be (Proposition 176.1). Source 18’s charts are for one fixed input; source 25’s family is uniform.
- **Two parameters and the witness count.** Source 25’s “at most two” counts free evolution parameters of a chart. A literal affine chart has up to five natural parameters, a chart of the quartic up to $3m + 1 \leq 13$ private variables before slacks, and the quartic has $W_{\text{nat}} = 3mJ + \sum_h k_h + I$ witnesses (Table 15).

1.29 Relation of Part IX to the repository

[write] Source 24 cites Part VI by line numbers, source 17’s binary audit, the programme’s parallel-particle review and **five-particle-binary-automata**; source 25 cites Part VI by line numbers and source 24. The other cross-references below are added. Results of other reports are cited by label name.

- **Re-derived results, printed as notes, pointers or second routes.** Source 24’s quartic is source 17’s (90.1), printed there; a numbered remark quotes source 24’s statement and proof (Remark 163.1). Its prospective-isolation lemma (Lemma 161.1) is Lemma `fpa:pi:lem:parallel` of **five-particle-binary-automata** (Report 26), and its lower bound (Theorem 164.1) is the binary, fixed-input case of source 16’s Theorem 74.1 and Corollary 74.2 by the same proof; both are printed as marked second routes. Source 25’s Section 170 restates source 16’s and source 17’s ingredients (notes name each counterpart), its first-contact lemma and dispatch make source 16’s and source 17’s fixed-input computations uniform, its clock (174.2) extends source 17’s (87.2), its quartic adapts source 18’s compiler (and so is again a second route to Part I’s Theorem 10.2), and its obstruction repeats source 24’s orbit formula. No novelty is claimed for any of these.
- **New in the two sources.** Source 24’s six-row rule with its full-shift reversibility, its complete orbit with all half-steps, all-time exclusivity and the density-one corollary; the reversible form of the sharp timing threshold. Source 25’s uniform dispatch, complete endpoint guard, symbolic contraction, joint quadratic clock, first-reset composition, congruence pullback and quotient lift, the copied-input quartic with its ledger, and the one-dimensional reversible obstruction.
- **The obstruction and Part VII.** Source 25 does not cite Part VII. Source 20’s Theorems 118.4 and 125.1 and Proposition 124.1 already give nonsemilinear original-frame site sets under drift for weighted planar rules, and source 21 a binary planar rule in the same regime. Proposition 176.1 is new in dimension one, for a binary globally reversible rule, and for complete-output reachability (note after it).
- **The programme’s review.** The Hilbert’s-tenth programme made a bounded routing intake of both archives with ten others of the same arrival, `Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/review_new_signals_0d7f51c44.md`

(commit 49b100cfd) [99]: full delivery READMEs and selected spans, “a routing and scope review, not a full proof”, with no statement needing an author correction. It reads source 24 as a decidable quadratic timing set whose certificate is not real-exact, whose lower bound is inherited and from which no inference from timing irregularity to universality is available, and source 25 as decidable low-mass dynamics with an essential copied-input, slack and selector ledger (“ ≤ 2 evolution parameters is not ≤ 2 total Diophantine witnesses”). It does not certify source 17’s and 18’s dynamics, which source 25 imports; source 17’s proofs were reviewed in [46], and source 18’s chart and compiler argument passed a scoped intake conditional on source 17’s normal form (review_timed_four_mass_source18_intake.md, commit 39f398985) [112]. This write has not been reviewed by the programme.

- **The repository’s universal bounds.** The project README’s fixed-universal operation bounds [1], now 84 operations, are untouched: source 24’s quartic is one timing relation of one rule, and source 25’s quartic is for one fixed rule and one pair of label cases.
- **Formal status.** Nothing in Part IX is formalized in Lean or Rocq, and the Hilbert-tenth-problem Lean development has no cellular-automaton module. The Presburger steps rest on the classical semilinearity theory [7]; Cooper’s quantifier elimination, formalized in `Logic/PresburgerArithmetic` (Section 1.9), is the decision procedure that a use of source 25’s Presburger conclusion would call, and formalizes only that step.

1.30 Status and what Part IX does not claim

[write] The two sources give conventional proofs and finite exact-arithmetic checks, which are regression evidence, not proofs; both are AI-assisted and unrefereed. Every limitation they state is kept in its place. Collected here, they do not claim

- for source 24: universality, literature priority, a minimal radius (90 is not asserted optimal) or an arithmetic arity or degree record; a new lower-bound argument (it is inherited); real exactness of the quartic (the false witness $(\sqrt{13} - 3)/2$ at $x = 0, t = 1$), or singleton fibres over signed integers or rationals; anything about several weight-one labels, signed weights, active zero-weight backgrounds, time-dependent rules, infinite-support inputs or other dimensions; that the 540 overlap tables exhaust all radius-45 neighbourhoods or that the period-64 check covers all 2^{64} words; that the rule computes universally; that any fixed-input classification is open; a formalization;
- for source 25: an unconditional theorem (it is conditional on sources 17 and 18, and source 18 on source 17); a general rule-to-chart implementation, a formal proof, published priority, or generic real or rational witness exactness; a witness count of two or five; that the timed relation is Presburger; several weight-one types, signed weights, an active zero-weight background, infinite support, higher dimensions or nonuniform rules; complexity or coefficient-size bounds; that the witness boxes prove uniqueness beyond their finite ranges, or that the abstract section words are realized by automata; the general two-total-variable raw-affine claim; a reopening of the imported fixed-input chart theorem;
- a proof-assistant verification; none of the sources’ code is a verified compiler.

The report claims no novelty for what the two sources prove again (Section 1.29).

1.31 Research questions answered or touched by Part IX

[write] Dated notes at the questions and non-claims themselves record the following; Table 3 carries the same information in its last rows.

- **Source 17’s non-claim** “reversibility of the binary shuttle (it is not injective)” (Section 1.15) remains true of source 17’s rule; source 24 realizes the same timing phenomenon, with the same quartic, by a different, globally reversible rule of radius at most 90.
- **Source 17’s second question** (Question 2, uniform untimed spatial definability in the stationary frame) is *answered conditionally* by source 25 (Theorem 169.1, Corollary 169.2): conditional on source 17’s scattering mechanism and source 18’s charts, and with no implemented preprocessing.
- **Source 17’s fourth question** (Question 4, a stronger normal form for original-frame uniform timed relations) is *answered conditionally for its uniform half*: a finite union of charts with at most two evolution parameters, at most three gap quotients, affine domains and quadratic outputs, uniform in the input. Source 25’s obstruction shows that even the untimed original-frame relation need not be Presburger, in one dimension and for a reversible binary rule.
- **Source 17’s sixth question** (Question 6, minimal radius of the binary timing example) is not answered; source 24 adds a reversible example of radius at most 90, not claimed minimal, and asks for the reversible radius (its Question 1).
- **Source 20’s first question** (Question 1, binary or reversible realizations) is *touched*: source 25’s obstruction is a one-dimensional, globally reversible binary instance of the drift effect; reversible realizations of source 20’s planar site-set phenomena stay open.
- **Source 24’s second question** (uniform descriptions across initial inputs) is answered conditionally by source 25, its pipeline’s Report 51.
- All other questions of the two sources (Sections 167 and 179) are open. Section 181, added at the assembly under the standing rule of 4 October 2026, states the unproved claims of the sources, of the programme’s intake and of the write as open questions, and proves one refinement (Remark 181.1).

1.32 Parts X–XIII: five-signal returns, branching, clocks and native-gap compilers (batch 91)

[write] This subsection and the four that follow were written on 4 October 2026, when thirteen further manuscripts of batch 91 of ProveIt’s incoming reports were added to the report as Parts X–XIII, together with the two of Part IX (Sections 1.27–1.31). They are called *sources 26–38* after the file prefixes of their shipped programs and data, which continue this report’s local sequence after Part IX’s 25. They are consecutive numbered reports of the research pipeline that also wrote sources 18–25, and their batch-91 manuscript numbers, pipeline report numbers and source numbers all differ. All thirteen arrived in commit 0d7f51c44, were placed by d750d98dd and are dated 4 October 2026; none names a ProveIt commit except source 26 (e3e58a882, at which it read this report’s README) and source 29 (b64f24e59), and source 27 pinned its repository searches at 7ba8f64e6. Appendix E gives the full provenance.

Source	Batch 91; pipeline	Archive and title	Printed as
26	manuscript 08; Report 56	<code>Periodic_Signal_Machine_Collision_Certificates_Package.zip</code> ; <i>Periodic signal machine collision certificates</i>	Part X, Sections 182–195
27	manuscript 09; Report 57	<code>Five_Signal_Nonsemialgebraic_Periodic_Validity_Package.zip</code> ; <i>Five signal nonsemialgebraic periodic validity</i>	Part X, Sections 196–210
28	manuscript 10; Report 58	<code>Five_Signal_Diophantine_Certificate_Source.zip</code> ; <i>Explicit Diophantine certificate for five signal infinite validity</i>	Part X
29	manuscript 11; Report 59	<code>Five_Signal_Rational_Rotation_Family_Source.zip</code> ; <i>Five signal rational rotation family</i>	Part X, Sections 225–237
30	manuscript 12; Report 60	<code>Planar_Signal_Realization_Source.zip</code> ; <i>Planar signal realization and strict validity kernels</i>	Part XI, Sections 240–256
31	manuscript 13; Report 61	<code>Invertible_Signal_Returns_Source.zip</code> ; <i>Fixed word invertibility and rational planar returns</i>	Part XI, Sections 257–266
32	manuscript 14; Report 62	<code>Projective_signal_returns_source_and_evidence.zip</code> ; <i>Five signal projective returns and mixed Zeno clocks</i>	Part XI, Sections 267–280
33	manuscript 15; Report 63	<code>Five_signal_homogeneous_realization_source_and_evidence.zip</code> ; <i>Exact local homogeneous realization with five live signals</i>	Part XI, Sections 281–292
34	manuscript 16; Report 64	<code>Five_signal_branching_source_and_evidence.zip</code> ; <i>Nondestructive branching with five live signals</i>	Part XII, Sections 294–303
35	manuscript 17; Report 65	<code>Clocked_signals_source_and_evidence.zip</code> ; <i>Exact instruction clocks and native gap halting certificates</i>	Part XII, Sections 304–315
36	manuscript 18; Report 66	<code>Bounded_certificates_and_encoded_gap_counts_source.zip</code> ; <i>Bounded halting certificates and exact encoded gap counts</i>	Part XIII, Sections 318–331
37	manuscript 19; Report 67	<code>Positive_POWER_reductions_and_exact_compiler_costs_source.zip</code> ; <i>Positive POWER reductions and exact bounded compiler costs</i>	Part XIII, Sections 332–347
38	manuscript 20; Report 69	<code>Low_arity_Diophantine_compilers_sources.zip</code> ; <i>Low-arity finite-clipping Diophantine compilers</i>	Part XIII, Sections 348–354
—	—; Report 68	never delivered; only its delivery README survives, carried by source 38	not printed (Question 1)

- **Part X (sources 26–29), periodic collision macros and five-signal validity.** For a supplied ultimately periodic complete collision macro of a rational number-preserving signal machine with at most four live signals, infinite validity is a finite Boolean formula of integer polynomial signs of degree at most two in the initial gaps, compiled into a quartic with one natural witness (Theorem 182.2); Zeno behaviour and rational deadlines are decided the same way (Theorem 182.3), and in every finite population a valid ultimately periodic macro on rational data has a rational accumulation time and rational limit positions (Theorem 182.4). Four is sharp: a five-live-signal machine returns as a half-contracted irrational rotation, and its valid set, a disk with one dense orbit removed from its boundary, has no finite polynomial-sign

description (source 27), yet explicit degree-twelve positive-integer certificates exist (sources 28 and 29, for one machine and for a rotation family).

- **Part XI (sources 30–33), *planar and projective returns*.** With five live signals and one ordered section, every rational planar return of positive determinant, then (by a seven-event reversal) every invertible one, every fixed-centre projective lift and, locally, every compatible invertible 3×3 homogeneous return is realized; the strict iteration kernels are classified (nonsemialgebraic exactly for infinite-order elliptic maps) and certified by explicit positive-integer polynomials.
- **Part XII (sources 34 and 35), *branching and clocks*.** For every fixed deterministic two-counter program, one finite number-preserving rule table with exactly five live signals simulates it from every counter pair on one fixed ordered section, by an exact nondestructive branch test (Theorems 294.2 and 294.1); every instruction is then padded to exactly $30D$ (Theorem 304.1), and the exponential initialization is replaced by three native positive integer gaps with a paid degree-twelve certificate at each horizon (Theorem 304.3). Five-live-signal counter computation is Durand-Lose’s [100]; no universality or minimum-population claim is made.
- **Part XIII (sources 36–38), *native-gap certificates and compilers*.** At each fixed horizon, clipping the counters at the horizon gives a unique-witness compiler with three (source 36), two or one (source 38) positive witnesses; the POWER module shrinks from 26 to 22, 16, 14, 13 and 12 leaves (source 37); a zero-versus-one classification says when no witness is needed (Theorem 351.1); and source 36 counts all encoded positive gap triples exactly. All are horizon-indexed families.

How they relate to Parts I–IX. Parts X–XIII return to signal machines, the subject of Parts I–III, after the cellular-automaton Parts IV–IX. Source 26 re-proves the one-macro case of Part I’s chamber theorem and answers in part Part I’s direction on accumulation; Parts XI–XIII build on the same five-signal ordered section; Part XII’s fixed-horizon quartic uses the device of Parts III and V. The openings of the four Parts give the details, and Section 1.34 collects them.

1.33 Notions and letters of Parts X–XIII

[write] The thirteen sources keep their letters and their numbering of statements, by section; equations are numbered by section. Their macro for the natural numbers is written `\NN` here and prints $\mathbb{N} = \{0, 1, 2, \dots\}$, and their positive integers are written $\mathbb{Z}_{>0}$. Renamed symbols are disclosed in the opening of each Part: in Part X the gate count Γ , the acceptance witness J_{acc} , the Gaussian quotient γ , the Pell modulus Θ , the radix β_{rad} , $\mathcal{K}$ and $\mathbf{R}$; in Part XI the reversal matrix $\mathbf{J}$; in Part XII the branch tags $\mathbf{Z}$, $\mathbf{N}$; in Part XIII the norm $\|\cdot\|_1$. No normalization changed. Each Part has its own table of letters (Part X’s opening, Table 23, Part XII’s source headings and the table in Part XIII’s opening); the table below lists only the letters whose meanings differ *between* these Parts or against earlier ones. Inside a source the meaning is always that source’s.

Letter	Part X	Part XI	Part XII	Part XIII
J	number of contacts (source 29); J_{acc} an acceptance witness, J_p a POWER slack; source 26's gate count is Γ	number of contacts (source 30); J_0, J_1 acceptance witnesses; J the reversal matrix	conditional instructions (source 35; source 34's C); Pell leaves J_A, J_B	a module leaf; the shell $\mathcal{J}(X)$; tail positions $J(a)$
M	return matrix; the Pell modulus is written Θ	return matrix M_J	—	the counter program; source 37's POWER modulus
N	population (source 26); a diagonal scaling (sources 27, 29)	the homogeneous 3×3 return (sources 31–33)	—	$N = K^2$ clipped classes (source 37)
T	a final-gap matrix, an accumulation time (source 26); a translation (sources 27, 29); a POWER output (sources 28, 29)	a clock, translations T_z	the instruction horizon; durations $T_{\text{test}}, T_{\text{inc}}, T_{\text{dec}}$	the native horizon; $K = T + 1$
$K, \mathcal{K}$	$\mathcal{K}$ the valid set (sources 27, 29)	$\mathcal{K}(A, P)$, $K(A, P)$ the strict kernel; K a shear count or a gap matrix	—	$K = T + 1$
Q	—	Q_+ and other macro steps	control labels Q_i ; the power output $Q = 2^{B-1}$ (source 35)	$Q = (T + 1)^2$ classes (source 36); Q_S a polynomial (source 38)
P	a quartic P_Φ ; a polygon; $P = 5^n$ or c^n	a polygon	the output $P = 2^{A-1}$ (source 35)	$P_{M,T}, P_S$ the compilers
D	a fixed denominator (source 26); the total width (sources 27–29)	the anchor distance D	the anchor distance; $D = g_1 + g_2 + g_3$	a gap total; D_n, D_ℓ, D_S
γ	the Gaussian quotient (written γ for the sources' α, η, η_j)	—	time coefficients γ_c	—
E, Z	a zero-gap set; the contact set E (source 29)	—	split edges E ; zero edges Z (not $\mathbb{Z}$)	POWER residuals E_i ; the deficit $E(X)$
ℓ	—	ℓ_J	—	leaves of a POWER module, $\ell \in \{26, 22, 16, 14, 13, 12\}$ (not gates)

Several readings need the same care.

- **Zeno and the spectral radius.** “Zeno if and only if the return has spectral radius less than one” is false: source 26's spectator machine has a Zeno clock at spectral radius one (Section 189).
- **Three ledgers.** Source 28's 81 witnesses and 44 equations and source 29's $1 + 80J$ and $1 + 43J$ at $J = 1$ agree as numbers but count different polynomials (a different machine, radix and contact); Report 60's $81J$ and $1 + 81J$ count a third construction (Table 16, Remark 240.4).

- **Five signals and certificates.** “Five signals admit no Diophantine certificate” is false: source 27’s obstruction is to finite polynomial-sign formulas, and sources 28–30 give certificates with existential integer witnesses.
- **Two machines of source 27.** The 114-event non-Zeno variant is a separately reclosed machine, not a prefix of the 138-event Zeno word.
- **Fixed-word invertibility and reversibility.** Source 31’s invertible section maps are maps of one fixed word between full sections; they are not global reversibility of a rule table, and they do not contradict Part III’s “No injectivity of the integer event map is asserted” (Section 239).
- **Leaves and gates; witnesses and horizons.** Source 37’s “twelve leaves” are positive module variables, not twelve gates; the one, two or three witnesses of Part XIII are at one fixed horizon, with degree growing with the horizon, not a fixed-arity certificate for unbounded halting.

1.34 Relation of Parts X–XIII to the repository

[write] The sources cite one another by SHA-256, source 26 cites this report’s README at e3e58a882 and the programme’s finite-schema review [14], and the others cite no repository file; the cross-references below are added. Results of other reports are cited by label name.

- **Re-derived results, printed as pointers, remarks or second routes.** Source 26 re-proves the one-macro case of Part I’s Theorem 5.3 and does not cite Part I’s accumulation direction or the two shuttles of Parts I and II, which are instances of its rational-limit theorem (opening of Part X). The fifteen-equation POWER module of sources 28–30 and 35 is printed once, as source 28’s Lemma 217.1; it is the power macro of Lemma `ptr:ai:lem:exp` of *periodic-turmite-first-revisits*. The repeated-Zeno rational-accumulation criteria of sources 30–33 are special cases of source 26’s Theorem 182.4 (Remarks 244.1 and 263.1). Source 35’s trace lemma restates source 34’s Theorem 301.1 (Remark 310.1); sources 36–38 restate one another and source 35 (the opening of Part XIII lists every case). Restated statements are numbered remarks that keep the duplicate’s label. No novelty is claimed for any of them.
- **Prior work credited by the sources.** Durand-Lose’s five-live-signal counter machines [100] and conservative constructions [5, 16]; Becker et al.’s accumulations [6] and intrinsically universal family [92]; the linear-loop nonsemialgebraicity of Dai and Xia [76] and Ouaknine and Worrell [77]; the orbit problem [88, 97, 87]. None of the thirteen sources claims priority, and this report claims none.
- **Neighbouring reports.** Report 55, whose power-system files source 28 copies, is Part VI of *group-theoretic-substrates*. Source 38’s classification is restated with proof in `a196460-clipping-tables` (batch-91 manuscript 21), which counts its closed tables (note after Theorem 351.1). Reports 70 and 71 of the same pipeline are Part VII of *five-particle-binary-automata*.
- **The programme’s reviews.** The Hilbert’s-tenth programme made a bounded routing intake of the twelve signal and particle archives of the arrival, sources 24–35 among them [99] (commit 49b100cfd), and a scoped intake of sources 36–38 [111] (commit a9ab9a698). Neither is a proof certification; neither found a statement needing an author correction. Their distinctions are kept in the Parts (notes at the end of each source). This write has not been reviewed by the programme.

- **The repository’s universal bounds.** The universal ordinary-input construction now stands at 84 operations (20aafb9a5); the 75- and 87-operation figures elsewhere in this report are dated history. Parts X–XIII do not improve or bear on it: their certificates are for one fixed machine, one fixed program at one fixed horizon, or a decidable matrix property, and both reviews say that no numerical universal improvement should be entered from their counts.
- **Formal status.** Nothing in Parts X–XIII is formalized in Lean or Rocq. The mathlib theorems `Pell.matiyasevic` and `Pell.eq_pow_of_pell` [86], on which every POWER module rests, are upstream results read as text by the sources, not ProveIt declarations, and the archives’ copy of that file is not shipped. ProveIt’s MRDP theorem `Diophantine.mrdp` (`Computability/HilbertTenthProblem/Lean/Diophantine/MRDP.lean`) is used by no source.

1.35 Status and what Parts X–XIII do not claim

[write] The thirteen sources give conventional proofs and finite exact-arithmetic checks, which are regression evidence, not proofs; all are AI-assisted and unrefereed. Every limitation they state is kept in its place; the Parts collect them (Section 238 and the source sections for Part X, Section 293.1, the opening of Part XII, and the opening of Part XIII). In brief, they do not claim

- any semantics at or after an accumulation point; Part X certifies infinite pre-accumulation validity of supplied macros only, and no frontend from a machine to its macro is implemented;
- a universality, minimum-population, minimal-speed or optimal-count result: five live signals are Durand-Lose’s for counters, and every machine is fixed only after its rational parameters or its program;
- a fixed-arity polynomial for unbounded halting, a finite-fold or single-fold representation, or an improvement of a universal operation bound; every certificate of Parts XII and XIII is indexed by its horizon;
- global reversibility (source 31’s maps are fixed-word section maps), noise robustness (all geometry is exact rational), or a physical realization of the arithmetic certificates beyond source 34’s conditional interface;
- a fresh verification of the Pell theorems, or a statement of Report 68, which was never delivered;
- a proof-assistant verification; none of the sources’ code is a verified compiler.

The report claims no novelty for what the sources prove again (Section 1.34).

1.36 Research questions answered or touched by Parts X–XIII

[write] Dated notes at the questions themselves record the following; Table 3 carries the same information in its last rows.

- **Part I’s direction 15.9** (accumulation limits without fictitious infinite witnesses) is *answered in part* by Part X: for supplied ultimately periodic macros, infinite validity, the Zeno classification and the rational accumulation time are decided by finite formulas or certified with existential witnesses. Executions without a supplied macro and any continuation stay open (Question X.9).

- **Part I’s direction 15.6** (robust universality) is *touched* by Part XII, whose branch tests run on full-dimensional open chambers covering every encoded counter pair; the counter values are exact ratios and not noise robust, so the direction stays open.
- **Part I’s direction 15.4** and **Part II’s Question 4** are *touched* by Part XIII: for bounded counter traces it proves witness, degree, height and operation tradeoffs (Table 50) and compresses a fixed horizon to one, two or three unique witnesses; Part I’s intercept forms and unbounded histories are not treated.
- **Part II’s Question 2** is *answered further in part, at a fixed horizon*, by Part XII (a number-preserving five-live-signal counter compiler that excludes accumulation, with bounded-horizon certificates); no fixed-arity encoding of unbounded histories of rational distances results.
- **Part III’s non-claims** on the optimality of 18 or ten live signals and on the injectivity of the integer event map are neither refuted nor answered: Part XII’s five live signals are for per-program, not globally reversible tables (and are prior art), and source 31’s invertibility concerns fixed words between full sections (notes in Section 35 and in the opening of Part XII).
- **Inside Parts X–XIII**, several questions of one source are answered by a later one: source 33 answers source 32’s realization beyond a rational centre and source 34 source 33’s physical branching (Table 34); source 35 answers source 34’s constant-duration question; source 37 answers source 35’s questions on the decoding overhead and on witness multiplicity and source 36’s on interpolants (notes at the questions). The questions of each Part are collected in Sections 238, 293.2, 317 and 355, together with the claims moved there under the standing rule of 4 October 2026.

Part I

Collision geometry is linear (source 07)

[write] Part I prints batch-78 manuscript 07, *Collision Geometry Is Linear: Event-Sparse Quadratic Diophantine Certificates for Rational Signal Machines*, “research manuscript prepared with ChatGPT for Vladimir Reshetnikov’s ProveIt research program”, dated 2 October 2026. Its labels carry the prefix `smc:cg::`; its sections are Sections 2–16 here and its two appendices are Appendices A and B. Its abstract and keywords follow.

Abstract (source 07). A rational signal machine computes by the collisions of finitely many types of particles moving at fixed rational velocities on a line. Its continuous geometry seems to require nonlinear arithmetic certificates. We show that, once a complete chronological collision skeleton is fixed, the geometry is instead governed by homogeneous rational linear equations and strict linear inequalities in the initial gaps. A sweep construction using maximal adjacency intervals proves that these conditions exclude both unrecorded earlier events and omitted simultaneous collisions. For n initial signals, m distinct velocities, and E collision sites, at most $3(n - 1) + (3m + 1)E$ raw linear conditions suffice.

After positive denominator clearing, one globally nonnegative convex quadratic polynomial certifies the skeleton over the nonnegative integers. Every accepted integer seed has exactly one slack-witness tuple. A finite disjoint first-hit family then yields one quartic with unique selector and slack witnesses. We also prove causal-depth denominator and coefficient bounds, a small integral realization theorem, a robustness–codimension criterion, and a finite-horizon quasi-polynomial counting theorem. Explicit examples include a first-collision tournament with

an exact counting formula and inverse, and a four-speed accumulating clock whose denominator size grows exponentially in causal depth. Finally, we prove that globally nonnegative quadratic certificates cannot be compressed into a single fixed-arity universal halting representation without leaving their decidable linear zero-set class. ([write] This last result is a special case of Part XI of the collection’s report *canonical-diophantine-certificates* [11]; see Remark 13.3.)

These are manuscript-level results with executable exact-arithmetic checks, not proof-assistant-certified theorems. The generic compiler and regression suite accompany the article. No priority claim, resolution of finite-fold MRDP, or improvement to an existing fixed-universal operation bound is made.

Keywords. Diophantine representation; signal machines; collision chronology; rational polyhedral cones; Presburger arithmetic; Ehrhart quasi-polynomials; Zeno accumulation; exact computation.

2 The question and the contribution

2.1 A concrete gap between universality and arithmetic

The MRDP theorem guarantees that a computably enumerable acceptance relation has a Diophantine representation. That existence theorem does not itself give a transparent arithmetic description of a particular computational substrate. A useful substrate-specific compiler should explain what its witnesses mean, why they cannot describe a counterfeit computation, and which resource bound is actually being measured. The distinction is especially important when the substrate has a geometric, asynchronous, or apparently continuous dynamics. For background on the classical theorem and a machine-based formalization, see Matiyasevich [9] and Bayer et al. [10].

The `ProveIt` Hilbert-tenth-problem material motivates precisely this separation. Its reversible finite-history audit distinguishes cheap local relations from a complete finite-input computation certificate, and points to additional obligations concerning wiring, initialization, boundaries, and halting observations [1, 2]. This article develops an analogous complete *geometric frontend* for a different substrate: one-dimensional rational signal machines. It does not attempt another operation-count reduction for the repository’s universal polynomial.

[write] The repository has since reviewed this manuscript, and the related manuscript printed as Part II, in its Hilbert-tenth-problem research tree (Section 1.4). The reversible-history audit cited here is unchanged from this source’s pin to the write.

Signal machines are not merely collections of unrelated finite circuits. Durand-Lose gives a halting universal construction with 13 signal types and 21 collision rules, obtained through cyclic tag systems [3]. This established universality is the reason to study the substrate. We do not reproduce or independently certify that universal machine’s full rule table and input loader here. Instead we prove a generic finite-history compiler whose interface applies to the rational fixed-speed model.

The central question addressed is the following.

Can an explicitly proposed finite collision history be represented arithmetically without introducing arbitrary event coordinates, without overlooking an earlier collision, and without splitting a genuinely simultaneous event into a false sequence?

Our answer is affirmative. In fact, the required geometric residuals can be linear rather than quadratic, and their sum of squares can therefore be quadratic rather than quartic. The essential work is not squaring equations; it is showing that a small set of order constraints certifies the *complete* chronology.

2.2 Results and their logical scope

For a fixed initial label word and a fixed discrete skeleton σ , we construct integer matrices $\mathbf{E}_\sigma, \mathbf{G}_\sigma$ such that the real initial-gap vectors realizing that skeleton are exactly

$$\mathcal{C}_\sigma = \{g \in \mathbb{R}^{n-1} : \mathbf{E}_\sigma g = 0, \mathbf{G}_\sigma g > 0\}. \quad (2.1)$$

Initial gap positivity is included among the strict rows. Every event time and position is a rational linear form in g . Section 5 proves the event-sparse bound

$$R_\sigma \leq 3(n-1) + (3m+1)E \quad (2.2)$$

for the number of raw linear residuals and inequalities.

The corresponding natural-number certificate is

$$P_\sigma(g, z) = \sum_i (\mathbf{E}_{\sigma,i} g)^2 + \sum_j (\mathbf{G}_{\sigma,j} g - 1 - z_j)^2. \quad (2.3)$$

Its degree is at most two, and its entire auxiliary tuple is forced to be $z_j = \mathbf{G}_{\sigma,j} g - 1$. This uniqueness concerns a fixed externally supplied skeleton, not a fixed universal MRDP polynomial.

The same linearization gives several additional conclusions. A collision at causal depth h has coordinate denominators dividing D^h after integer speed normalization, where D is the least common multiple of the nonzero speed differences. Every nonempty chamber has a small positive integer point. A chamber's synchronization equalities determine its exact codimension and its perturbation behavior. For a fixed event budget, first-acceptance seeds form a finite disjoint union of such chambers, and counting them in growing integer boxes gives exact quasi-polynomials.

These conclusions solve specific representation, realizability, precision, and counting questions formulated here. They are not presented as the settlement of a named longstanding conjecture. The literature review establishes the underlying signal-machine model and related accumulation results; it does not establish historical priority for every combination of lemmas in this article. The proofs are intended to stand on their own and to be suitable for further independent review.

3 Rational signal machines and chronological skeletons

3.1 Particles, velocities, and collision rules

Definition 3.1 (Machine). A rational signal machine consists of a finite label set M , a velocity map $v : M \rightarrow \mathbb{Q}$, and a deterministic rule for every admissible incoming collision set under consideration. An incoming set contains at least two labels with pairwise distinct velocities. Its output is a possibly empty set of labels with pairwise distinct velocities. A signal of label a moves along a straight line of slope $v(a)$ until it participates in a collision.

We use a transparent default rule when an explicit rule is absent: the incoming labels reappear as outgoing labels, ordered by increasing velocity. Such a crossing still counts as a collision. Alternatively, one may specify all relevant rules explicitly and reject a skeleton requiring an undefined rule. The attached implementation supports both conventions.

At a collision all incoming signals are consumed and the prescribed outputs are born at that space-time point. Newly born outputs are ordered by increasing velocity in the right-hand time

limit. This makes them separate rather than immediately collide again. At each positive time, all spatially distinct collisions occurring at that time are processed together.

This is the fixed-speed straight-line model used in abstract geometrical computation [3]. Our arithmetic interface restricts initial configurations to finitely many signals at distinct positions and considers ordinary finite prefixes. An infinite sequence accumulating at a finite time has no automatically assigned continuation in this article.

3.2 Initial coordinates and normalization

Fix $n \geq 2$ initial labels $a_0, \dots, a_{n-1}$, in increasing spatial order, and write $d = n - 1$. Translation invariance lets us put

$$x_0 = 0, \quad x_i = g_1 + \dots + g_i \quad (1 \leq i \leq d), \quad g_i > 0. \quad (3.1)$$

Repeated labels at distinct positions are permitted. We call g the *seed geometry*; the label word is separate discrete input.

The principal chamber theorem allows real g . The Diophantine theorem uses integer g . A rational gap vector can be multiplied by a positive common denominator to become integral. Scaling all positions by $c > 0$ scales all event times by c without changing any collision rule or chronological skeleton. Thus integral seeds suffice for existence of a rational realization, but they need not preserve a separately specified physical length or time bound.

When all velocities are rational, choose an integer $L > 0$ such that $Lv(a)$ is integral for every label. In the time coordinate $\tau = t/L$, all speeds are integers $Lv(a)$. Denominator bounds below are stated in this normalized time coordinate. An arbitrary common translation in velocity is not needed.

3.3 Batches and stable signal identities

Definition 3.2 (Chronological skeleton). A finite skeleton is an ordered list of nonempty batches. Each batch is an ordered list of pairwise disjoint contiguous blocks in the current live signal list, each block having at least two members. The blocks are listed from left to right. Each block is a collision site, with its rule and outputs determined by the incoming labels. New signal identities are assigned by batch, then by site, then by increasing output velocity.

The initial identities are $0, \dots, n - 1$. A consumed identity cannot be used again; a noninitial identity must have been generated by an earlier batch. Within a single batch, outputs cannot serve as inputs of another site. These are discrete well-formedness conditions, not arithmetic constraints left for a witness to evade.

Let B be the number of batches and E the total number of collision sites. At site e , write p_e and q_e for the numbers of incoming and outgoing signals. Let m be the number of distinct velocities. Then

$$2 \leq p_e \leq m, \quad 0 \leq q_e \leq m, \quad B \leq E. \quad (3.2)$$

The quantities B , E , and causal depth are different. Many sites may share a batch; many batches may contain events unrelated to a given causal chain.

Definition 3.3 (Realization). A positive seed g realizes a skeleton σ when its actual execution has exactly the specified batches as its first B batches, with exactly the specified incoming blocks

and outputs at every listed site. In particular, no earlier event and no simultaneous site or extra incoming signal may be omitted. Nothing is required about events after the final listed batch.

The empty skeleton is realized by every positive seed. It states only that no initial portion has yet been requested; it does not state that the machine never collides. A malformed skeleton is assigned the empty realization set. One can encode a statically rejected skeleton by the constant polynomial 1 if a total polynomial-producing interface is desired.

4 Eliminating every event coordinate

4.1 Intercept propagation

For each signal segment s , let v_s be its fixed velocity and write its supporting line as

$$x_s(t) = b_s + v_s t. \quad (4.1)$$

Its intercept b_s need not be its birth position: the birth may occur at a positive time. Initially, the b_s are the partial sums in (3.1), hence homogeneous integer linear forms in g .

Suppose an event contains two inputs s, r with $v_s \neq v_r$. Their formal meeting time and position are

$$t_e = \frac{b_r - b_s}{v_s - v_r}, \quad x_e = b_s + v_s t_e. \quad (4.2)$$

For an outgoing label of velocity w , its intercept is

$$b_{\text{out}} = x_e - w t_e. \quad (4.3)$$

The denominator $v_s - v_r$ is fixed by the machine, not a variable. Consequently these formulas preserve rational linearity.

Lemma 4.1 (Coordinate elimination). *For every well-formed finite skeleton, all formally reconstructed intercepts, event times, and event positions are homogeneous rational linear forms in the initial gap vector. An event with more than two incoming signals requires only additional homogeneous rational linear equalities.*

Proof. Induct on the ordered batches. Initial intercepts are linear. Equations (4.2) and (4.3) use only addition and multiplication or division by fixed rational constants. An additional incoming signal u requires $b_u + v_u t_e = x_e$, again a homogeneous linear equality. Every new intercept is then available for subsequent batches. No output of the current batch is used by another event in that batch, so this induction is acyclic. $\square$

All event coordinates are therefore *computed data*. They need not appear as free Diophantine witnesses. A certificate containing unconstrained coordinate variables would obscure both the low degree and the uniqueness that result from this elimination.

4.2 Local incidence does not imply a real computation

The incidence equations alone are insufficient. Three signals starting at $0, u, u + v$ with respective speeds $2, 0, -1$ formally give the first two a meeting time $u/2$. If $u = 3$ and $v = 1$, however, the

second and third signals collide already at time 1, before $u/2 = 3/2$. An equation witnessing the later formal intersection cannot establish that the two original segments are still alive there.

There is also a boundary failure. When $u = 2v$, all three signals meet at once. A purported binary event containing only the first two signals has perfectly consistent incidence coordinates but the wrong incoming set. A compiler must distinguish strict separation from weak separation at precisely these boundaries.

For events in the same batch, choose the leftmost site's formal time as t_k . Equate all other site times in that batch to t_k , and require

$$t_k - t_{k-1} > 0, \quad t_0 = 0. \quad (4.4)$$

These conditions are necessary but still do not exclude an unlisted earlier collision. The missing ingredient is a complete spatial-order certificate.

5 A sparse certificate for complete collision chronology

5.1 Maximal adjacency intervals

Between collisions the live signals have a strict spatial order. The skeleton tells us how that order is updated: replace each incoming block by its output block, in increasing velocity order. This replacement changes only a bounded number of adjacent pairs.

Definition 5.1 (Adjacency interval). A maximal adjacency interval records an ordered pair (s, r) , left signal first, that remains adjacent in the live list from one batch boundary to a later batch boundary. The interval begins when the pair first becomes adjacent and ends when it ceases to be adjacent or when the requested prefix ends. An unchanged pair is not closed and reopened merely because an unrelated event occurs elsewhere.

Intervals created at the final batch have zero duration and need not be checked: their output order is already determined by the distinct velocities. Every recorded positive-duration interval has an affine gap

$$\Delta_{sr}(t) = b_r - b_s + (v_r - v_s)t. \quad (5.1)$$

At its initial boundary this gap is prescribed to be zero exactly when both segments are born at the same collision there. Otherwise it is prescribed to be strictly positive. At its final boundary it is prescribed to be zero exactly when both segments are consumed at the same collision there. Otherwise it is strictly positive.

Both endpoints cannot be prescribed zero for a positive-duration interval. Indeed, two distinct straight-line segments cannot separate after a common birth and then meet again without one undergoing an intervening collision. Such a proposal is rejected statically. Each surviving interval contributes exactly two linear conditions before redundant zero rows are removed.

Lemma 5.2 (Endpoint lemma). *Let $a < b$ and let f be affine on $[a, b]$. If $f(a), f(b) \geq 0$ and at least one of them is positive, then $f(t) > 0$ for every $a < t < b$.*

Proof. For $t = (1 - \lambda)a + \lambda b$ with $0 < \lambda < 1$, $f(t) = (1 - \lambda)f(a) + \lambda f(b)$ is a positive convex combination of nonnegative numbers, at least one of which is positive. $\square$

[write] Part II proves the same statement, with terminal rays added, as Lemma 21.1, and the boundary half of the soundness argument below as Lemma 21.2.

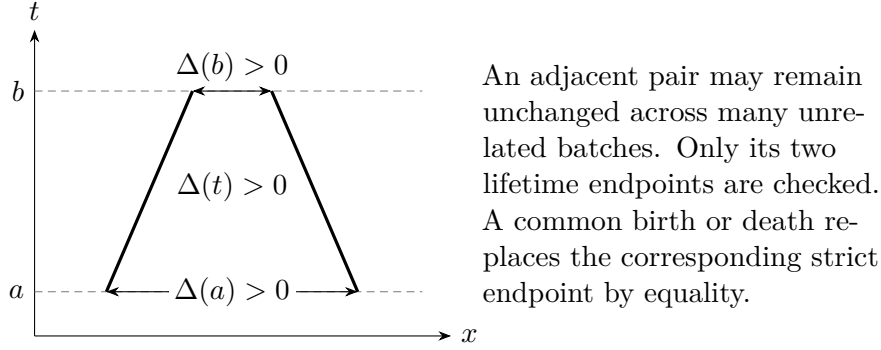


Figure 1: The affine gap between adjacent straight segments cannot become zero in the interior if its permitted endpoint conditions hold.

5.2 Soundness and completeness

Theorem 5.3 (Exact chronological chamber). *Fix a rational signal machine, an initial label word, and a finite skeleton σ . There is an effective construction of rational homogeneous linear equalities and strict inequalities whose solution set is exactly the set of real seed geometries realizing σ . All simultaneous collisions and all extra incoming signals are accounted for. Every nonempty solution set is a relatively open rational polyhedral cone in the kernel of its equality matrix.*

Proof. First reject malformed identities, overlapping or noncontiguous input blocks, undefined rules under the nontransparent convention, and incoming blocks whose velocities fail to decrease strictly from left to right. The latter condition is necessary because immediately before a multiple meeting, two incoming lines s, r ordered left to right have gap $(v_s - v_r)(t_e - t) > 0$ and hence $v_s > v_r$.

Propagate every intercept and event coordinate using (4.2)–(4.3). Add the additional incoming-incidence equalities, equal-time conditions within each batch, (4.4), initial gap positivity, and all adjacency-interval endpoint conditions described above. Every resulting expression is a homogeneous rational linear form in g .

For soundness, suppose all these conditions hold. On each positive-duration adjacency interval, the endpoint lemma gives strict separation in its interior. Hence no adjacent pair meets at an unlisted intermediate time. A meeting of nonadjacent signals would force an adjacent meeting: at the first failure of strict spatial order, at least one consecutive gap is zero. Thus no collision of any arity can occur between the listed batches.

At a batch boundary, the incidence equalities put the signals of each specified block at its reconstructed common site. Every adjacent gap not belonging to that block is strictly positive there. Consequently no additional incoming signal can join the site. The same strict conditions separate the sites of distinct blocks. They also exclude an omitted disjoint collision elsewhere in the configuration: an omitted site would have a zero gap at a boundary where strict positivity is imposed, or would lie in the interior of an unchanged adjacency interval, where positivity has just been proved.

The deterministic rule is therefore applicable to exactly the listed incoming set at each site. Its outgoing order is increasing in velocity, so outputs separate in the right-hand time limit. Static identity validation ensures that every segment has a legitimate birth and is consumed at most once. Induction on batches now proves that the reconstructed execution is exactly the actual initial prefix.

For completeness, consider an actual execution realizing σ . The coordinate propagation agrees

with its genuine intersection coordinates. The extra incidences and batch-time equalities hold, batch times strictly increase, and each endpoint gap is zero precisely at a common birth or common death of the pair. Every other endpoint gap is positive. A pair of straight segments with a common birth and a later common death cannot occur. Thus the execution satisfies every generated condition.

Finally, a finite intersection of homogeneous equality subspaces and open linear half-spaces is a relatively open convex cone in that subspace. If nonempty, its strict inequalities leave a neighborhood in the equality subspace around each point. All coefficients are rational. $\square$

5.3 The event-sparse bound

Theorem 5.4 (Linear number of chronology conditions). *Let J be the number of positive-duration maximal adjacency intervals used by the construction. Then*

$$J \leq (n - 1) + \sum_{e=1}^E (q_e + 1) \leq (n - 1) + (m + 1)E, \quad (5.2)$$

$$R_\sigma \leq 3(n - 1) + (3m + 1)E. \quad (5.3)$$

These bounds apply before deduplication or removal of identically zero equalities.

Proof. Initially there are $n - 1$ adjacent pairs. Replacing a collision block by q_e outputs creates at most $q_e - 1$ internal output pairs and two external pairs, for a total at most $q_e + 1$ when $q_e \geq 1$. If $q_e = 0$, it creates at most one bridge pair between the surviving neighbors. At a boundary or when neighboring blocks change simultaneously, fewer pairs may be created. Existing unchanged pairs need no new interval. This proves (5.2).

There are $d = n - 1$ initial gap inequalities, B positive batch-time increments, $\sum_e (p_e - 2)$ additional incoming-incidence equalities, $E - B$ equalities synchronizing the sites within batches, and $2J$ endpoint conditions. Therefore

$$\begin{aligned} R_\sigma &\leq d + B + \sum_e (p_e - 2) + (E - B) + 2J \\ &= d + \sum_e p_e - E + 2J \\ &\leq d + (m - 1)E + 2d + 2(m + 1)E \\ &= 3d + (3m + 1)E. \end{aligned}$$

$\square$

[write] The first inequality of (5.2) is Part II's Lemma 20.1; Part II's own size ledger (18.3) counts witnesses and residuals in its event-coordinate format.

The word *sparse* here refers to the number of geometric conditions, not to the number of nonzero coefficients in each eliminated linear form. A deeply propagated intercept may depend on every initial gap. Storing all rows densely can take $O(dR_\sigma)$ coefficient slots; compiling them may have additional bookkeeping cost. Neither (5.3) nor degree two is a constant arithmetic-operation bound for an unbounded computation.

An implementation may scan the live list at each batch for simplicity. That does not change the mathematical row bound. A linked-list implementation can update adjacencies locally, while the attached small implementation prioritizes clarity and exact arithmetic over this engineering optimization.

6 Quadratic Diophantine certificates with unique slacks

6.1 Positive denominator clearing

Every nonzero rational row can be multiplied by a positive common denominator and then divided by the positive greatest common divisor of its integer coefficients. For equality rows the sign may subsequently be normalized; for strict inequality rows it may not. Negating a strict row would reverse its meaning.

Delete identically zero equality rows. An identically zero strict row makes the chamber empty. Remove duplicate rows of the same kind. Denote the resulting integer matrices by $\mathbf{E}$ and $\mathbf{G}$, with r equality rows and s strict rows. Their matrix presentation is a complete, explicit description of the polynomial, even when its expanded quadratic has many monomials.

Theorem 6.1 (Unique-slack quadratic representation). *For a nonrejected skeleton, define*

$$P(g, z) = \|\mathbf{E}g\|_2^2 + \|\mathbf{G}g - \mathbf{1} - z\|_2^2, \quad (g, z) \in \mathbb{N}_0^{d+s}. \quad (6.1)$$

Then P is an integer polynomial of total degree at most two, globally nonnegative and convex on $\mathbb{R}^{d+s}$. Moreover,

$$\exists z \in \mathbb{N}_0^s \ P(g, z) = 0 \iff g \in \mathbb{Z}^d \cap \mathcal{C}_\sigma. \quad (6.2)$$

For every accepted g , the auxiliary tuple is unique:

$$z = \mathbf{G}g - \mathbf{1}. \quad (6.3)$$

Proof. Each summand is the square of an affine linear form with integer coefficients. Thus P is a quadratic sum of squares and has Hessian $2A^\top A \succeq 0$, where A is the combined coefficient matrix of these affine forms. Over the reals, and hence over the integers, a sum of squares vanishes precisely when every squared form vanishes. Its zeros therefore require $\mathbf{E}g = 0$ and $\mathbf{G}_j g = 1 + z_j \geq 1$ for every strict row. Because these expressions are integers, this is equivalent to $\mathbf{G}g > 0$. The chamber theorem proves (6.2). The same equations force (6.3), which is a nonnegative integer tuple whenever g is accepted. $\square$

[write] Theorems 5.3 and 6.1 together are the common theorem of the two Parts (Section 1.2). Part II proves it again as Theorem 18.3, with batch times and event positions kept as natural witnesses, adds halting semantics, and strengthens uniqueness to full column rank of the witness matrix (Lemma 22.1).

All event coordinates are reconstructed from g and are absent from the auxiliary tuple. This avoids arbitrary rational encodings with multiple numerator–denominator representatives. If the seed itself is given as a rational geometry in a different interface, one must separately choose and validate an encoding or a scale; the uniqueness theorem should not be silently transferred to a noncanonical rational encoding.

6.2 What uniqueness does not assert

The polynomial depends on the externally supplied skeleton. Its number of rows and slack variables is not fixed independently of the requested history. Even though each member of this family has unique auxiliary witnesses, placing an arbitrary unbounded skeleton inside one fixed Diophantine equation is a different task. An unspecified application of MRDP to a skeleton verifier does not preserve this uniqueness for free.

The present result is therefore not a proof that all computably enumerable sets admit single-fold or finite-fold Diophantine representations. Its stronger-than-usual feature is narrower and fully explicit: for each fixed collision skeleton, projection of *all* natural zeros of (6.1) onto the initial gaps is a bijection onto the actual integer realizations. There are no additional zeros corresponding to incomplete local histories.

7 Causal depth controls denominators and coefficient size

7.1 A denominator invariant

Work in the integer-speed normalization of Section 3, and set

$$V = \max(1, \max_{a \in M} |v(a)|), \quad D = \text{lcm}\{|v(a) - v(b)| : v(a) \neq v(b)\}. \quad (7.1)$$

The least common multiple of an empty set is taken as 1; in that case no collision is possible from a simple initial configuration.

Assign depth zero to initial segments. The depth of an event is one plus the maximum depth of its incoming segments. Every output segment inherits the depth of its birth event. This is causal depth, not the chronological index of the batch.

Theorem 7.1 (Causal denominator bound). *At an event of depth h , each coefficient of its time and position forms belongs to $D^{-h}\mathbb{Z}$. The same holds for every outgoing intercept. At an integer seed, the event coordinates therefore have reduced denominators dividing D^h .*

Proof. Initial intercept coefficients are integral. Suppose an event has depth h . Its incoming intercepts have denominators dividing powers of D no larger than D^{h-1} . Put them over the common denominator D^{h-1} . Formula (4.2) divides their difference by a nonzero integer velocity difference dividing D . Its time coefficients thus belong to $D^{-h}\mathbb{Z}$. The position and all output intercepts use only addition and multiplication by integer velocities, preserving that lattice. Evaluation at integer gaps preserves the same denominator divisibility. $\square$

[write] Part II proves the same divisibility for event coordinates as Lemma 19.1, with its own proof.

Only primes already present in the speed differences can appear in these coordinate denominators at an integer seed. At a rational seed, primes in its initial common denominator may also occur. Parallel collisions elsewhere do not increase the denominator exponent along an unrelated causal chain.

This is an upper bound, not an assertion that all factors of D^h occur. Cancellations and special velocity configurations can make the actual precision requirement much smaller. Section 12 exhibits an explicit case in which exponential denominator magnitude is nevertheless unavoidable.

7.2 An explicit coefficient bound

For a rational linear form, use the ℓ^1 norm of its coefficient vector. Put $K = 2 + 4V$. Initial intercepts have norm at most d .

Lemma 7.2 (Growth of propagated forms). *Every intercept of depth at most h has coefficient norm at most dK^h . Every event time and event position of depth at most h satisfies the same bound.*

Proof. Assume the input intercept norms at a depth- h event are at most $M = dK^{h-1}$. A nonzero integer speed difference has absolute value at least one, so (4.2) gives $\|t_e\|_1 \leq 2M$. Consequently

$$\|x_e\|_1 \leq (1 + 2V)M, \quad \|b_{\text{out}}\|_1 \leq (1 + 4V)M.$$

Each factor is at most K . Induction proves the claim; forms of smaller depth obey the larger bound as well. $\square$

Proposition 7.3 (Integer row height). *If the maximum event depth is h , each primitive integer chamber row can be chosen with every coefficient of absolute value at most*

$$A = 2d(1 + V)(DK)^h. \quad (7.2)$$

In particular, for a fixed machine, row coefficient bit lengths are $O(h + \log d)$.

Proof. An endpoint gap is a difference of two intercepts plus a velocity difference times a batch time. Its coefficient norm is at most $2dK^h + 2VdK^h = 2d(1 + V)K^h$. Additional incidence residuals and differences of event times satisfy the same bound. Initial gap rows also do. All these coefficients have denominators dividing D^h by Theorem 7.1. Multiplication by D^h bounds every resulting integer coefficient by (7.2). Passing to a primitive integer row can only decrease this bound. $\square$

The exponent in (7.2) bounds coefficient *magnitude*; its logarithm is linear in depth. Exact rational arithmetic can therefore remain polynomial in the externally bounded history length even when ordinary fixed-precision floating-point simulation becomes unreliable. This is not an assertion that the total number of events before acceptance has a computable bound in the initial input length.

8 Small integral realizations and finite-horizon synthesis

Theorem 8.1 (Small positive integer seed). *Let $\mathcal{C} = \{g \in \mathbb{R}^d : \mathbf{E}g = 0, \mathbf{G}g > 0\}$ be a nonempty chamber whose strict rows include $g_i > 0$. Suppose all matrix coefficients are integers of absolute value at most $A \geq 1$. Then $\mathcal{C}$ contains a positive integer point satisfying*

$$\max_i g_i \leq d^{d/2} A^d. \quad (8.1)$$

Thus every realizable collision skeleton has a realization with a seed encoding of polynomial bit length in the explicit skeleton and initial label word, for a fixed machine.

Proof. Homogeneity permits a positive scaling of any point in $\mathcal{C}$ so that every strict expression is at least 1. Hence the closed rational polyhedron

$$Q = \{g : \mathbf{E}g = 0, \mathbf{G}g \geq \mathbf{1}\}$$

is nonempty. It lies in $[1, \infty)^d$. Minimizing $\sum_i g_i$ over Q attains an optimum: intersecting with a nonempty objective sublevel gives a compact set. The optimal face is a nonempty compact polytope and has a vertex; that point is also a vertex of Q .

At such a vertex choose d independent active constraints, including sufficient equality rows. They give a nonsingular integer matrix B and a right-hand side c with entries in $\{0, 1\}$. The vertex

is $B^{-1}c$. Let $\delta = |\det B|$, a positive integer. Cramer's rule shows that $g = \delta B^{-1}c$ is integral. It is positive, satisfies the homogeneous equalities, and has every strict row at least $\delta \geq 1$.

Each coordinate of g is, up to sign, a determinant in which one column of B has been replaced by c . Hadamard's inequality bounds its magnitude by $d^{d/2}A^d$; using the sharper bound on the replaced column is unnecessary. This proves (8.1). Combining it with (7.2) gives the stated bit-length conclusion. $\square$

Corollary 8.2 (Real, rational, and integer realizability coincide). *A fixed skeleton is realizable by a positive real seed if and only if it is realizable by a positive rational seed, if and only if it is realizable by a positive integer seed. Realizability is decidable by rational linear feasibility, using $Eg = 0, Gg \geq 1$.*

The equivalence is specific to homogeneous rational chambers. It should not be transferred to arbitrary nonlinear geometry. Nor does the scaled integer seed necessarily satisfy extra nonhomogeneous requirements such as an absolute prescribed distance or a fixed deadline.

Corollary 8.3 (An NP upper bound for bounded synthesis). *Fix a rational signal machine and a distinguished output label. Given an initial label word of length n and a unary bound T , the problem of whether some positive seed produces the distinguished label within a complete prefix containing at most T collision sites is in NP.*

Proof. A skeleton of at most T sites has at most T batches and at most $n + mT$ created signal identities. Since m is fixed, its discrete description has polynomial length in $n + T$. Guess such a skeleton, check its identities and rules, and check that the distinguished label appears in its prefix.

If it is realizable, Theorem 8.1 supplies a positive integer seed with polynomial bit length: $d = n - 1$, $h \leq T$, and $\log A = O(T + \log n)$ for the fixed machine. Guess that seed as well. Propagating rational forms and verifying every chamber row then takes a polynomial number of operations on polynomial-length integers. This is a polynomial-time verifier. No search over exponentially long real numbers is required. $\square$

Only an upper bound is claimed. There is no NP-hardness result here. The budget counts total sites, including all sites in the accepting batch, not elapsed physical time. A physical-time bound can contain arbitrarily many events and even an accumulation, so it is not interchangeable with T .

9 Robustness and the rank of synchronization constraints

The chamber description distinguishes two different sources of numerical difficulty. A trace can be sensitive because it demands an exact equality among initial gaps, or its exact coordinates can require large denominators even while its combinatorial pattern is robust. The latter phenomenon will occur in the accumulating clock.

Theorem 9.1 (Certified perturbation radius and codimension). *Let $\mathcal{C} = \{g : Eg = 0, Gg > 0\}$ be nonempty, with zero strict rows removed, and let $g^* \in \mathcal{C}$. Define*

$$\rho(g^*) = \min_j \frac{G_j g^*}{\|G_j\|_1} > 0. \quad (9.1)$$

Every perturbation $\eta \in \ker E$ with $\|\eta\|_\infty < \rho(g^)$ preserves the skeleton. The chamber has relative dimension $d - \text{rank } E$. It has nonempty ambient interior if and only if $\text{rank } E = 0$; otherwise it has ambient Lebesgue measure zero.*

Proof. The equality constraints persist for $\eta \in \ker E$. For every strict row, Hölder’s elementary norm inequality gives

$$G_j(g^* + \eta) \geq G_j g^* - \|G_j\|_1 \|\eta\|_\infty > 0.$$

Hence the perturbed point stays in the chamber. This proves relative openness in the full equality kernel, whose dimension is $d - \text{rank } E$. A proper linear subspace has empty interior and measure zero in $\mathbb{R}^d$. Conversely, when the rank is zero, the finitely many strict inequalities define an open neighborhood of each accepted point. $\square$

The radius in (9.1) is a certified lower bound, not necessarily the largest possible radius in the equality subspace. Restricting perturbations to that subspace can increase the true distance to a forbidden boundary.

One must also distinguish genuinely independent synchronization equations from identities already forced by previous dynamics. Some formal coincidences yield an identically zero row and impose no additional codimension. Therefore the statement “every multiple collision is fragile” is too strong. What matters is the rank of the *nontrivial* conditions on the input geometry. When a loader or problem specification already restricts the admissible inputs to a subspace, robustness should be measured relative to that allowed input space.

The theorem provides a useful diagnostic for a proposed geometric computer. A simulator can report not merely a diagram, but also an exact equality matrix, a synchronization rank, and a rational certified tolerance at an integer seed. Those objects separate an intended robust mechanism from an accidental coincidence created by a particular numeric sample.

10 Finite-horizon acceptance, semilinearity, and counting

10.1 Disjoint first-acceptance chambers

Fix a machine, an initial label word, a marked accepting label, and a site budget T . Acceptance means the first appearance of that label in a batch such that the total number of processed sites through that entire batch is at most T . If the initial word already contains the marked label, every seed accepts with zero events.

There are only finitely many candidate skeletons with at most T sites. Indeed, at every stage there are at most $n + mT$ identities; the rule table and possible contiguous input blocks are finite. Enumerate those skeletons that end at the first accepting batch and reject all statically invalid ones. Distinct nonempty chambers in this first-acceptance list are disjoint: for a fixed seed, deterministic complete chronology has just one first accepting prefix.

Theorem 10.1 (Finite-horizon semilinearity). *The positive integer seed geometries accepted within site budget T form an effectively constructible semilinear set. More specifically, they are a finite disjoint union of the integer points of rational homogeneous chambers of the form (2.1).*

Proof. The enumeration above is finite and effective. For each skeleton, Theorem 5.3 gives a complete linear characterization. On integer points, a strict integer inequality is equivalent to being at least 1, so each chamber is definable in Presburger arithmetic. Finite union remains Presburger-definable. The effective equivalence with semilinear sets is the classical theorem of Ginsburg and Spanier [7]. $\square$

This use of the classical equivalence is not a new proof of that theorem. The new substrate-specific input to it is the exact finite union of collision chambers, including chronology and simultaneous-event boundaries.

[write] Part II proves the same statement for its fixed schemas and event bounds, prefix and halting, as Proposition 26.1.

10.2 A single quartic for an unknown bounded first-hit skeleton

The skeleton need not remain externally supplied when the event budget is fixed. Enumerate the nonrejected first-acceptance skeletons $\sigma_1, \dots, \sigma_K$ and let their matrices be $\mathbf{E}_k, \mathbf{G}_k$. Empty chambers may remain in this list without harm. Introduce natural selectors $b_1, \dots, b_K$ and a separate slack vector z_k for each chamber. Define

$$\begin{aligned} Q_T(g, b, z) = & \left(\sum_{k=1}^K b_k - 1 \right)^2 + \sum_{k=1}^K \sum_i (b_k \mathbf{E}_{k,i} g)^2 \\ & + \sum_{k=1}^K \sum_j (b_k (\mathbf{G}_{k,j} g - 1) - z_{k,j})^2. \end{aligned} \quad (10.1)$$

For an empty family, use $Q_T = 1$.

Theorem 10.2 (Canonical first-hit quartic). *Equation (10.1) is a globally nonnegative integer polynomial of degree at most four. Its natural zeros project bijectively onto the integer seeds accepted within the fixed event budget. No Boolean equations for the selectors and no additional inactive-slack constraints are needed.*

Proof. At a natural zero, $\sum_k b_k = 1$ forces exactly one selector to be 1 and all others to be 0. For the selected chamber, the remaining equations are precisely $\mathbf{E}_k g = 0$ and $\mathbf{G}_k g - \mathbf{1} = z_k \geq 0$. For an unselected chamber, each slack residual becomes $-z_{k,j}$, forcing all of its slacks to zero. Thus the polynomial accepts exactly the union of the chamber integer points.

Those chambers are disjoint by canonical first-acceptance chronology. Therefore an accepted seed determines exactly one selected skeleton, its active slacks are uniquely determined, and every inactive slack is zero. Every residual has degree at most two before squaring, proving the quartic bound. The construction is a sum of squares and hence globally nonnegative, but it is not in general convex. $\square$

The same construction works for any finite disjoint family of chambers. If chambers overlap, it still represents their union, but a seed in several chambers has several selector witnesses. First-hit canonicalization is therefore mathematically important, not cosmetic.

Remark 10.3 ([write] Relation to a single-fold quadratic of another report). The accepted set of Theorem 10.2 is a finite union of chamber integer points, hence semilinear (Theorem 10.1). The collection's report *canonical-diophantine-certificates* [11] proves in its Part XI (Theorem `cdc:of:thm:singlefolds1`) that every semilinear set has a representation by an integer polynomial of degree at most two that is nonnegative on the real nonnegative orthant, with exactly one natural witness for each member, effectively from a Presburger formula. So, if only orthant nonnegativity is required, the first-hit set has a single-fold quadratic certificate. Theorem 10.2 gives something different: an explicit construction from the chamber matrices, globally nonnegative on all real inputs, whose selector and slacks have the chamber meaning. Within the globally nonnegative class

a quadratic has a rational affine zero set (Theorem 13.1), and the selector products of (10.1) leave that class, as the paragraph after Corollary 13.2 says. Whether a globally nonnegative quadratic suffices for a given first-hit family is not decided here.

This theorem removes external skeleton choice for a *bounded* problem; it does not remove the bound. The number of candidate skeletons, and hence the number of selectors and slacks, may be very large and grows with T . The attached implementation provides the matrix-based quartic constructor, and checks a two-chamber first-hit example. It does not include an optimized exhaustive enumerator for all skeletons of an arbitrary event budget.

10.3 Exact quasi-polynomials rather than merely leading asymptotics

Let $A_T(H)$ count the accepted seeds in $\{1, \dots, H\}^d$, with $H \geq 1$. For a single chamber put

$$\Pi_\sigma = \mathcal{C}_\sigma \cap (0, 1]^d.$$

Its closure is a bounded rational polytope in its equality subspace. The original set is obtained from that closed polytope by excluding finitely many rational faces where one or more required strict expressions vanish. The upper box facets remain included.

Theorem 10.4 (Exact finite-horizon counting). *For fixed machine, initial word, accepting label, and event budget T , $A_T(H)$ is an exact quasi-polynomial in H for all positive integers H . It has degree at most d . Its ordinary generating function is rational and has poles only at roots of unity. Furthermore,*

$$\lim_{H \rightarrow \infty} \frac{A_T(H)}{H^d} = \sum_{\sigma} \text{vol}_d(\Pi_\sigma) \in \mathbb{Q}, \quad (10.2)$$

where the sum ranges over the first-acceptance chambers, and lower-dimensional terms have zero d -dimensional volume.

Proof. Homogeneity gives $\mathcal{C}_\sigma \cap (0, H]^d = H\Pi_\sigma$. For a nonempty chamber, the closure of Π_σ is obtained by replacing its strict homogeneous inequalities by weak ones. To see equality with this weak closure, take a small strictly feasible point with every coordinate less than 1 and form convex combinations with any weakly feasible point. Those combinations converge to the latter while preserving all strict conditions and the box.

A forbidden equality $G_j g = 0$ cuts out a face of the weak closure, since $G_j g \geq 0$ throughout it. Inclusion–exclusion over these faces expresses the lattice-point count in $H\Pi_\sigma$ as a finite integer combination of counts in dilates of closed rational polytopes. Ehrhart’s rational-polytope theorem makes each term a quasi-polynomial; see Beck and Robins [8]. This also covers lower-dimensional faces. Disjointness of first-acceptance chambers permits summing these exact counts.

A quasi-polynomial of degree at most d and period p has a rational generating function whose denominator divides a power of $1 - z^p$. This proves the generating-function assertion. The normalized leading coefficient of a full-dimensional rational polytope count is its volume; removing proper faces does not change it. Equivalently, the same limit follows from Riemann sums for a bounded polyhedral set. Rational polytopes have rational volume by a rational triangulation and determinant formulas. Summing over the disjoint chambers proves (10.2). $\square$

Thus a finite-horizon seed-counting problem has a particularly rigid asymptotic form: a polynomial in H with periodic coefficient functions, not an arbitrary irregular error term. Its leading

density is rational. If the initial gaps are instead sampled independently and uniformly from $(0, 1]$, the probability of finite-horizon acceptance is exactly the same rational volume sum.

The theorem does not promise an efficient procedure for enumerating all chambers or computing all their volumes. Their number can grow rapidly with T . It also does not justify interchanging an unbounded-time limit with the box-size limit. Removing the event bound can leave an infinite union of chambers, outside the finite quasi-polynomial argument.

Because each integer seed in one chamber has one slack tuple, its chamber count is also the number of complete natural zeros of its quadratic certificate with $1 \leq g_i \leq H$, even when no separate bound is placed on z . Across different skeletons this statement uses a tagged disjoint family of certificates, not a single unqualified polynomial.

11 An exact first-collision tournament

11.1 Three chambers and a complete polynomial

Consider labels A, B, C with respective speeds $2, 0, -1$, initially at $0, u, u + v$, where $u, v > 0$. Introduce two stationary output labels H, R and rules

$$\{A, B\} \longrightarrow \{H\}, \quad \{B, C\} \longrightarrow \{R\}, \quad \{A, B, C\} \longrightarrow \{R\}.$$

Use transparent rules otherwise. The question here is whether H is produced at the *first* collision batch.

The possible first events are exactly

event	condition	time
AB	$u < 2v$	$u/2$
BC	$u > 2v$	v
ABC	$u = 2v$	$v = u/2$.

(11.1)

For instance, the formal AC meeting time is $(u + v)/3$. If $u < 2v$, it is later than $u/2$; if $u > 2v$, it is later than v . It cannot be a different first event. The equality case is the triple collision.

The complete accepting chamber has the particularly small certificate

$$P_{AB}(u, v, z_1, z_2, z_3) = (u - 1 - z_1)^2 + (v - 1 - z_2)^2 + (2v - u - 1 - z_3)^2. \quad (11.2)$$

All five variables range over $\mathbb{N}_0$. At $u = v = 1$, for example, the unique auxiliary tuple is $(0, 0, 0)$. At $(u, v) = (3, 1)$, the third strict condition fails, correctly rejecting the incidence-only counterfeit. At $(2, 1)$ it also fails, correctly rejecting an omitted simultaneous third input.

For the triple-collision skeleton the compiler instead produces

$$P_{ABC}(u, v, z_1, z_2) = (u - 2v)^2 + (u - 1 - z_1)^2 + (v - 1 - z_2)^2. \quad (11.3)$$

Its equality rank is one. The two binary-event chambers have rank zero and are open in the positive quadrant; the triple-collision chamber is one-dimensional and requires exact synchronization.

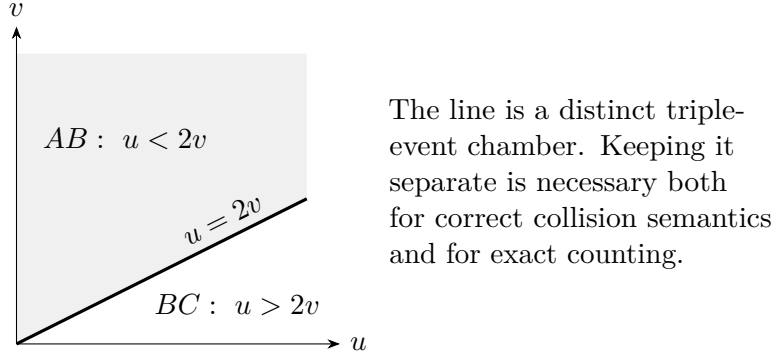


Figure 2: The first-event geometry is partitioned into two open cones and a synchronization ray. The coordinate axes are excluded.

11.2 Counting, generating function, and inversion

Let $C(H)$ count the accepting gap pairs with $1 \leq u, v \leq H$, and set $C(0) = 0$. Direct summation gives

$$\begin{aligned}
 C(H) &= \sum_{v=1}^H \min(H, 2v - 1) \\
 &= \begin{cases} 3m^2, & H = 2m, \\ 3m^2 + 3m + 1, & H = 2m + 1, \end{cases} \\
 &= \left\lfloor \frac{3H^2 + 1}{4} \right\rfloor = \frac{3H^2}{4} + \frac{1 - (-1)^H}{8}. \tag{11.4}
 \end{aligned}$$

For $H = 2m$, split the sum after $v = m$; the odd-number sum contributes m^2 and the remaining m terms contribute $2m^2$. The odd case follows by the same split after $m + 1$.

The first values are

$$0, 1, 3, 7, 12, 19, 27, 37, 48, 61, 75, 91, 108, 127, 147, 169, \dots$$

No assertion about an OEIS identification or novelty of this elementary sequence is needed. It is included as an exactly solvable counting test for the compiler and as an illustration of the general quasi-polynomial theorem. Its ordinary generating function is

$$\sum_{H \geq 0} C(H) z^H = \frac{z(1 + z + z^2)}{(1 - z)^3(1 + z)}. \tag{11.5}$$

This follows from the standard geometric-series derivatives applied to (11.4), or by expanding the displayed rational function.

There is also an exact inverse threshold. For every positive integer y ,

$$\min\{H \geq 0 : C(H) \geq y\} = \left\lceil \sqrt{\frac{4y - 1}{3}} \right\rceil. \tag{11.6}$$

Indeed, because y is integral, $\lfloor (3H^2 + 1)/4 \rfloor \geq y$ is equivalent to $3H^2 + 1 \geq 4y$. The smooth expression inside the ceiling has the expansion

$$\sqrt{\frac{4y - 1}{3}} = \frac{2}{\sqrt{3}} y^{1/2} - \frac{1}{4\sqrt{3}} y^{-1/2} - \frac{1}{64\sqrt{3}} y^{-3/2} - \frac{1}{512\sqrt{3}} y^{-5/2} + O(y^{-7/2}). \tag{11.7}$$

The binomial series supplies all further orders. The ceiling in (11.6) is essential: omitting it would incorrectly replace an integer inverse's bounded rounding oscillation by a decaying error.

11.3 Independent simultaneous sites

Take the transparent two-speed machine with $v(P) = 1$, $v(N) = -1$, and initial word P, N, P, N . With gaps $(2, 5, 2)$, the two disjoint pairs $(0, 1)$ and $(2, 3)$ collide simultaneously at time 1. A correct first batch must contain both sites. A proposed batch containing only $(0, 1)$ is rejected by the unchanged adjacency interval of $(2, 3)$, whose gap would vanish at the final boundary where positivity is required.

For general positive gaps (g_1, g_2, g_3) , this simultaneous first-batch skeleton has the single nontrivial equality $g_1 = g_3$. Changing the last gap from 2 to 3 destroys that equality. This example tests a different failure mode from the omitted third input of a multiple collision: the missing event is spatially separate, not an extra signal at the same site.

[write] Part II's hidden collision at the final cut (Section 24.2) is another instance of this failure mode, rejected by the same strict endpoint of a surviving pair (Section 1.2).

12 A four-speed clock with exponentially growing denominators

12.1 Rules and a closed-form execution

Consider the four labels

$$v(L) = 0, \quad v(P) = 2, \quad v(R) = -1, \quad v(N) = -2$$

with rules

$$\{P, R\} \longrightarrow \{N, R\}, \quad \{L, N\} \longrightarrow \{L, P\}, \quad (12.1)$$

and transparent defaults. Initially put L at 0, P at u , and R at $u + v$, with $u, v > 0$. The left boundary is stationary, the right boundary moves left at speed 1, and a particle alternately travels right at speed 2 and left at speed 2, reflecting at the boundaries via the two rules.

Write $S = u + v$ and $W = 3u + 2v$. The collision times are

$$t_{2k+1} = S - \frac{W}{3^{k+1}}, \quad k \geq 0, \quad (12.2)$$

$$t_{2k} = S - \frac{W}{2 \cdot 3^k}, \quad k \geq 1. \quad (12.3)$$

Odd collisions occur on the right boundary at position $S - t_{2k+1}$; even collisions occur at position zero.

Proposition 12.1 (Clock correctness and common chamber). *For every $u, v > 0$, equations (12.2)–(12.3) describe the actual alternating execution. The collision times strictly increase to S . Every finite alternating prefix has the whole positive (u, v) quadrant as its seed chamber.*

Proof. The first rightward meeting occurs at $t_1 = v/3$, which equals $S - W/3$. At a right-boundary collision at time t , the remaining boundary distance is $S - t$. The new left-moving particle takes $(S - t)/2$ to reach zero, leaving time-to-boundary-meeting $(S - t)/2$. From there the right-moving particle closes the remaining gap at relative speed 3, leaving two thirds of that remaining time when

it next hits the right boundary. Hence a full right-to-right bounce multiplies $S - t$ by $1/3$, with the intermediate left collision having half the previous remaining time. This proves the formulas inductively.

The first time is positive. Every subsequent difference is positive, and all times are less than S . Between collisions the moving particle lies strictly between the two boundaries. The boundaries themselves meet only at S , not at any finite indexed event. There is therefore no unlisted collision in any finite prefix. The labels alternate exactly as specified by (12.1), independently of the positive gaps. This proves both correctness and the common-chamber statement. $\square$

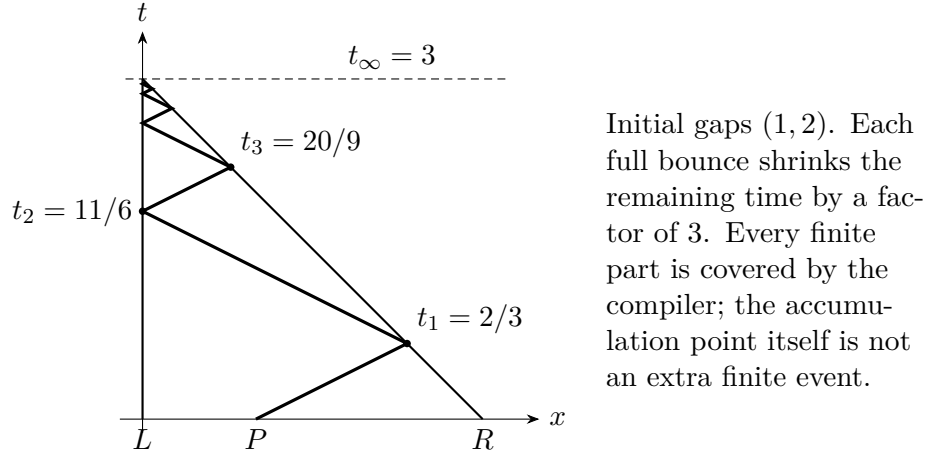


Figure 3: A four-speed accumulating clock. The drawing displays only a finite number of segments; the formulas specify all finite prefixes.

12.2 A denominator lower bound without fragile synchronization

For the particular integer seed $(u, v) = (1, 2)$, $S = 3$ and $W = 7$, giving

$$\frac{2}{3}, \frac{11}{6}, \frac{20}{9}, \frac{47}{18}, \frac{74}{27}, \frac{155}{54}, \frac{236}{81}, \frac{479}{162}, \dots$$

At event $2k$, the reduced denominator is exactly $2 \cdot 3^k$:

$$t_{2k} = \frac{6 \cdot 3^k - 7}{2 \cdot 3^k},$$

and the numerator is odd and is not divisible by 3. The causal depth of event j is j , since the reflected particle links every event to its predecessor. Thus denominator magnitude is exponential in causal depth, while its bit length is linear in depth.

Here the integer speed differences have least common multiple $D = 12$, so Theorem 7.1 correctly gives a divisible upper bound 12^j . The example does not make that upper bound sharp in its base, but it rules out a depth-independent denominator bound or a polynomial-magnitude bound for general rational signal machines.

At the same time, every finite prefix has equality rank zero and remains valid throughout the positive quadrant. Growing exact precision is therefore not evidence of a fragile collision pattern. Robust symbolic chronology and large rational denominators are compatible.

Accumulations in signal machines are established phenomena. Becker et al. [6] distinguish the behavior of small speed sets and show, in particular, that four speeds permit accumulation whereas the three-speed case with rational speed and initial-distance ratios does not. The clock is an explicit test case and a denominator-growth illustration, not a claim to have discovered accumulation. Our compiler verifies every ordinary finite prefix before S ; it assigns no transition or computational power to the limiting point at S .

[write] Part II's bounded shuttle (Proposition 25.3) uses the same mechanism on a different four-speed machine, with reduced denominators exactly 3^{k-1} (Section 1.2).

13 Why a fixed universal convex quadratic cannot follow

A natural question is whether the quadratic frontend can be compressed into one fixed polynomial with a fixed number of witnesses for arbitrary unbounded computations. Within the globally nonnegative quadratic class, there is an elementary obstruction stronger than a failure of the present construction.

Theorem 13.1 (Linear zero-set rigidity). *Let $Q(w)$ be a quadratic polynomial with rational coefficients that is nonnegative for every $w \in \mathbb{R}^N$. Its real zero set is either empty or a rational affine subspace. The projection of its nonnegative integer zeros onto any subset of coordinates is an effectively semilinear, decidable set.*

Proof. Write

$$Q(w) = w^\top A w + 2b^\top w + c$$

with A rational and symmetric. Global nonnegativity implies $A \succeq 0$. It also implies $b \perp \ker A$: otherwise the restriction to an appropriate kernel direction would be a nonconstant affine function unbounded below. Thus $b \in \text{range } A$ and the consistent rational system $Aw_0 = -b$ has a rational solution.

Completing the square gives

$$Q(w) = (w - w_0)^\top A(w - w_0) + \mu, \quad \mu = c + b^\top w_0 \geq 0.$$

If $\mu > 0$, there are no zeros. If $\mu = 0$, positive semidefiniteness gives $Q(w) = 0$ exactly when $A(w - w_0) = 0$, equivalently $Aw = -b$. This is a rational affine subspace.

Clearing denominators gives finitely many integer linear equations. Intersecting with $\mathbb{N}_0^N$ and projecting coordinates is an existential Presburger definition. Effective semilinearity and decidability follow from Ginsburg–Spanier and Presburger arithmetic [7]. $\square$

Corollary 13.2 (Convex-quadratic compression barrier). *An undecidable halting relation cannot be represented by one fixed globally nonnegative rational quadratic with a fixed finite number of natural parameters and witnesses, even after a computable input-loading map.*

Proof. The projected zero relation would be decidable by Theorem 13.1. Its preimage under a computable input-loading map would then be decidable as well, contradicting the assumed undecidability. $\square$

This theorem applies to global nonnegativity on *all real variables*, not merely nonnegativity on a special natural-number input slice. It does not classify arbitrary quadratic equations, arbitrary

nonconvex polynomials, or cubic Diophantine representations. For example, $x - y^2 = 0$ is a quadratic equation with a nonlinear zero set, but its polynomial is not globally nonnegative and is outside the theorem.

The quartic first-hit construction in (10.1) leaves the convex-quadratic class by multiplying selectors and input-dependent linear forms. Its finite-horizon relation remains decidable for other reasons. A truly unbounded compiler must additionally encode an unbounded amount of discrete chronology into finitely many integer variables. That compression, not the local collision geometry, is where the universal Diophantine problem re-enters.

Remark 13.3 ([write] Credit). Theorem 13.1 and Corollary 13.2 are a special case of results already in Part XI of the collection’s report *canonical-diophantine-certificates* [11], which source 07 does not cite. There, Lemma `cdc:of:lem:affinezero` shows that a real polynomial p of degree at most three that is nonnegative on an open convex set O and vanishes at $a \in O$ vanishes on O exactly along $a + \ker \nabla^2 p(a)$; with $O = \mathbb{R}^N$ this is the affine zero set of Theorem 13.1. Theorem `cdc:of:thm:cubicsemilinear` shows that the full natural zero set of a *rational* polynomial of degree at most three that is nonnegative only on the real nonnegative orthant is a finite union of natural solution sets of rational affine systems, and Theorem `cdc:of:thm:classification` that the coordinate projections of such zero sets are exactly the semilinear sets, $D_2^+(k) = D_3^+(k) = \text{SL}(k)$, effectively. Those results assume less (nonnegativity on the orthant only, degree up to three, no convexity or sum-of-squares hypothesis) and give the same decidable, semilinear conclusion; that report itself notes that its theorem is stronger than the observation that a globally nonnegative polynomial cannot have odd highest degree. Corollary 13.2 therefore also holds for orthant-nonnegative polynomials of degree at most three. No novelty is claimed here for either statement. Source 07’s proof by completing the square is kept as a direct second route for the globally nonnegative quadratic case, and its clause “even after a computable input-loading map” is its own: a computable preimage of a decidable set is decidable.

13.1 The classical unbounded existence result remains available

For a finite rational initial configuration, an exact event-driven simulator can find the next collision batch by computing all positive pairwise meeting times and taking the smallest. It can then apply all rules at the sites of that batch. Repeating this procedure semidecides the appearance of a marked signal in an ordinary finite prefix. An infinite accumulating run simply causes the simulator to execute indefinitely when no earlier acceptance has occurred; no post-accumulation convention is needed for semidecidability.

Consequently the pre-accumulation finite-prefix acceptance relation, with machine and rational input encoded by natural numbers, is computably enumerable. MRDP supplies a fixed-arity Diophantine representation of that encoded relation [9, 10]. This last existence step is classical. Neither MRDP nor the present frontend automatically supplies unique witnesses, an explicit small expanded universal polynomial, or the repository’s particular operation ledger.

For a fixed universal machine with an undecidable marked-halting interface, there can be no total computable function of the encoded seed that bounds the number of sites needed by every accepting run. Such a function, followed by exact bounded simulation, would decide halting. This explains why the bounded chamber theorems and the universal existence theorem coexist without contradiction.

[write] The last argument is the standard bounded-search argument, used for other statements in Proposition `cdc:bd:prop:nobound` of [11] and elsewhere in the collection.

14 Implementation, reproducible checks, and trust boundaries

14.1 What the attached code computes

The accompanying standard-library Python implementation uses exact `Fraction` arithmetic for every geometric calculation. The essential interfaces are:

```
Machine(speeds, rules, transparent_default=True)
simulate(machine, initial_labels, gaps, max_batches=20)
compile_skeleton(machine, initial_labels, skeleton)
UnionCertificate(tuple_of_chamber_certificates)
```

A skeleton is a tuple of batches; each batch is a tuple of input-ID tuples. For example, `((0, 1), (2, 3))` denotes one batch with two independent simultaneous sites. Fresh identities are generated canonically by increasing outgoing velocity.

The simulator uses an all-pairs minimum-time calculation. The symbolic compiler uses intercept propagation and adjacency intervals. Their different chronology mechanisms provide a useful cross-check: the regression suite compares exact numerical prefixes against the regions predicted by the symbolic compiler on many other initial seeds.

A compiled `Certificate` contains the complete integer matrices, exact rational event forms, causal depths, interval count, and raw row ledger. Its `witnesses(gaps)` method returns the unique natural slack tuple, while `evaluate(gaps, witnesses)` evaluates the full polynomial. The `UnionCertificate` implements (10.1). Its constructor does not prove that a supplied family is globally disjoint; that is a mathematical precondition for uniqueness. At a supplied seed, its witness routine rejects multiple accepting chambers rather than silently claiming a unique answer.

The machine constructor copies and freezes the speed and rule tables. This prevents a caller's later dictionary mutation from changing the meaning of an already constructed machine. Input validation rejects unknown labels, repeated-speed outputs, malformed groups, use of dead signals, noncontiguous blocks, overlapping sites, and inappropriate incoming speed orders. These checks are part of the semantic contract.

The exported JSON matrices specify full polynomials, not just sample zeros. For example, `tournament_left.json` contains all rows of (11.2), while `first_hit_quartic_union.json` supplies every matrix and the exact selector formula for a two-chamber union. Expanding these expressions into monomials is optional and provides no extra mathematical information.

14.2 Run and reproduce

From the extracted archive's root directory, run

```
python code/run_checks.py
latexmk -pdf -interaction=nonstopmode -halt-on-error article.tex
```

Python 3.10 or newer and a standard LaTeX installation are sufficient. No Python third-party dependencies are needed. The test command overwrites the generated JSON data with deterministic results, including `data/checks.json`.

[write] These are the delivered names. In this report the programs are shipped as `code/07-collision-geometry-run_checks.py` and `code/07-collision-geometry-signal_certificates.py` (with the delivered `Makefile` as `code/07-collision-geometry-Makefile`), and the seven

JSON files of `data/` as `data/07-collision-geometry-*.json`. The test program imports `signal_certificates` by its delivered name and rewrites all seven JSON files in place, relative to its own directory’s parent; it must therefore be run on a copy with the delivered layout (see the README). It was so run at the write, under Python 3.14.4 on Windows: it passed with 34,560 assertions, and the seven regenerated files equal the shipped ones apart from line endings. The delivered article PDF is not shipped; this report’s PDF replaces it.

The completed run accompanying this manuscript passed **34,560 exact assertions**. Its principal coverage is summarized in Table 4. These are finite regression tests, not exhaustive verification of all machines or all input geometries.

Check family	Instances
Tournament seeds $(u, v) \in \{1, \dots, 40\}^2$	1,600
Exact seed-count boxes $H = 0, \dots, 80$	81
Exact inverse thresholds $y = 1, \dots, 2000$	2,000
Four-speed clock batches at $(1, 2)$	24
Other positive clock seeds with the same 24-batch skeleton	25
Pseudorandom rule tables (fixed random seed 20261002)	6
Distinct nonempty prefix skeletons obtained from those tables	24
Independent seed–skeleton chronology comparisons	1,536
Two-chamber quartic-union seed checks	1,600

Table 4: Finite exact test coverage. Individual assertions also check witness mutations, denominators, malformed inputs, and table immutability.

The pseudorandom cases include creation, annihilation, and transparent rules. The checked skeletons are generated from exact sample executions and then tested against other seed geometries. The suite also specifically rejects an earlier omitted collision, an omitted third input, and an omitted independent simultaneous site. It tests the clock’s exact closed-form times and denominator divisibility. For the quartic union, it checks selector validity and that inactive slacks are forced to zero.

14.3 An auditable resource ledger

Table 5 separates retained conditions from raw generated rows. The generic bound is deliberately loose; redundancy elimination can be substantial. The 24-batch clock is a useful example: its entire seed chamber reduces to only three retained strict rows, even though its geometric coordinates become increasingly complicated. One of those rows is redundant with positive gaps, but the implementation does not attempt full linear redundancy minimization.

Certificate	E	B	r	s	Raw rows	Bound
Tournament AB	1	1	0	3	7	16
Tournament BC	1	1	0	3	7	16
Tournament ABC	1	1	1	2	8	16
Two simultaneous sites	2	1	1	4	11	23
Clock, 24 batches	24	24	0	3	122	318

Table 5: Exact compiler ledger: r is the retained equality count and s the retained strict-row count. The rightmost column is $3(n - 1) + (3m + 1)E$, not an arithmetic-operation count.

14.4 What has and has not been verified

The prose proofs establish general claims by mathematical argument. The code and data test finite instances with exact arithmetic. No Lean or other proof-assistant verification of this article is included, and the implementation is not a formally verified compiler. The universal 13-type machine’s complete published rule table and finite-input loader have not been reimplemented in the package. The code supplies a generic machine interface and the specific worked examples only.

The repository was inspected on 2 October 2026. The observed main-branch head was

```
f1edb38f93aa0a362a34b531a600909f6e74608a.
```

The relevant reversible-history audit was fetched as blob

```
cb31d0a108b9a19fc33da6fa89a73b604e15e260.
```

No repository-wide build was performed and no repository files were changed. The current README and the older audit have different historical operation ledgers; none is used as a theorem in the present proofs. The new row counts are not comparable to a fixed-universal arithmetic count without a separate input and history compression analysis.

[write] The pin `f1edb38f9` is an ancestor of this report’s placement commit. The audit blob `cb31d0a10` is still the current `Computability/HilbertTenthProblem/Papers/1980/REVERSIBLE_FINITE_HISTORY_MODELS.md` at the write; the project README [1] has gained text since the pin and still states the 75- and 87-operation bounds. The repository’s own review of this package [14] is described in Section 1.4.

15 Further research questions and a concrete development path

[write] Table 3 pairs these directions with the questions of Part II. The first two coincide with Part II’s Questions 5 and 1.

15.1 Formalizing the chronology theorem

The most direct formalization target is the endpoint lemma followed by the sweep invariant: the live list is in strict spatial order on every open interval between batches, and its equal-position blocks at each boundary are exactly the declared sites. Stable IDs make consumption and birth finite-map properties. A Lean development should keep the real semantic statement separate from rational coefficient computation, then prove that evaluation commutes with the intercept compiler. Finally, the sum-of-squares zero lemma and natural slack uniqueness can be linked to the repository’s Diophantine infrastructure.

A practical first milestone is a small checker for exported matrices and rational event forms. A second is a proof-producing compiler rather than a trusted Python translator. Tests that compare two unverified programs should remain auxiliary evidence, not be promoted to kernel verification.

15.2 An audited universal signal-machine frontend

Instantiate a published finite-input universal construction with its exact labels, velocities, all required collision rules, transparent defaults, input encoding, and marked acceptance convention. The 13-type

construction of Durand-Lose [3] is a concrete candidate. The task is not merely to copy a picture: simultaneous three-input cases, finite workspace supply, and the loader’s correspondence with the simulated machine must be audited. The eventual theorem should state an exact equivalence between an ordinary integer program input and the existence of a compiled accepting prefix.

This would turn the present generic geometry theorem into a fully explicit end-to-end substrate result. A fixed-arity universal polynomial would still require a separate outer compression step and its own complete resource ledger.

[write] **Added 2 October 2026 (batch 79).** This direction is answered by Part III, except for the ordinary-integer loader. Part III instantiates a published finite-input universal construction literally: Morita’s 15-state, 6-symbol reversible universal Turing machine compiled through Durand-Lose’s 2012 conservative stacks, with 114 labels, their speeds, all 445 collision rules, undefined encounters forbidden rather than completed, the input encoding, and marked acceptance distinguished from null termination (Theorem 29.1); simultaneous and triple collisions are treated exactly (Lemma 34.1; a triple collision is undefined), the workspace is a fixed population of 18 signals, and the loader’s correspondence with the simulated machine is proved (Sections 32–33). The machine is not the 13-type construction named above, and the input is a prepared cyclic-tag tape loaded as two rational distances: the exact equivalence with an *ordinary integer* program input asked for here still needs a digit-extraction and base-conversion loader, whose cost Part III states but does not pay (Section 33). As the paragraph above says, no fixed-arity universal polynomial follows; Part III’s certificates are for fixed horizons.

15.3 Sharper denominator invariants

The factor D^h treats all speed differences uniformly. Can one use a velocity-labelled dependency graph to compute exact prime-valuation bounds or a smaller denominator ideal for each event? Which speed sets force uniformly bounded denominators on all integral seeds, and which permit exponential growth? The clock supplies one lower-bound mechanism, while the small-speed accumulation results [6] suggest that the interaction between arithmetic precision and global dynamics is subtler than a least-common-multiple bound alone.

A particularly concrete project is to propagate prime-valuation vectors through (4.2) and detect systematic cancellation at each rule. Any claimed lower bound must use reduced denominators, not merely the unreduced fractions produced by a particular implementation.

15.4 From few rows to small arithmetic circuits

The event-sparse theorem controls row count but permits dense coefficient vectors. Can common subexpressions in intercept forms give a substantially smaller arithmetic circuit than dense matrix evaluation? An appropriate cost model must distinguish fixed machine numerals, input-dependent arithmetic, equality tests, and the conversion from a residual system to one polynomial. The original intercept dependency graph already provides a sharing structure; the challenge is to preserve its certified semantic meaning while optimizing its arithmetic schedule.

A useful result would be a proved tradeoff among witness count, maximum degree, coefficient height, and literal addition/multiplication count. No one-dimensional comparison such as “quadratic beats a degree-169 polynomial” is meaningful when arity and unbounded-history coding differ.

[write] [Added 4 October 2026: touched, not answered. For finite clipping tables of bounded counter programs, source 38 (Part XIII) proves such a tradeoff among witness count, degree,

coefficient size and operations (Table 50, Section 352); Part I's intercept forms are not treated.]

15.5 Avoiding exhaustive first-hit enumeration

The quartic in (10.1) is explicit but can contain an enormous number of chambers. Can a local selector circuit encode a bounded unknown skeleton with polynomially many variables while retaining canonical first-hit uniqueness, including all simultaneous events? A promising direction is to encode changes of adjacency directly rather than a rectangular array of all particle positions. The main obstruction is that a locally plausible selection must still describe one globally ordered and complete execution.

Even a polynomial-size bounded compiler would not solve fixed-arity unbounded compression. Keeping those two improvements distinct will make future operation claims interpretable.

15.6 Robust universality and minimal synchronization rank

For a given computational simulation, determine the rank of its required input-geometry equalities. Can the same computation be realized on a full-dimensional chamber, or does its interface genuinely require exact ratios among certain gaps? The answer must be relative to the legal input space rather than to coordinates the loader never permits to vary.

One possible invariant is the minimum independent synchronization rank among equivalent signal-machine implementations with a fixed speed set. The chamber theorem makes that invariant concrete for each finite trace, but extending it uniformly over a universal family needs a precise notion of simulation and allowed perturbation.

15.7 Counting complexity and asymptotic structure

Determine when first-hit chamber counts can be computed without enumerating all skeletons. Are there useful machine classes for which a recurrence on adjacency types produces the Ehrhart generating function directly? When do quasi-polynomial periods collapse, and which restrictions imply polynomial counts? The exact inverse in the tournament example is simple; in higher degree, inverse asymptotics must retain residue-class corrections and rounding effects rather than replacing them by an unjustified smooth error.

For unbounded acceptance, investigate convergence of finite-horizon rational densities. The increasing accepted sets give monotonicity, but a rational limit is not automatic. Quantitative tail estimates would be needed for a useful approximation theorem and for interchanging input-box and time limits.

15.8 Algebraic velocities and nonlinear motion

With velocities in a fixed algebraic number field, formal event coordinates remain linear over that field. However, positivity is no longer a rational half-space condition in the same elementary form, and integer realization by scaling is not automatic. A number-field sign certificate might extend part of the method, but the rational-cone, Presburger, and Ehrhart conclusions would require new arguments or additional hypotheses.

If signals accelerate or their velocities depend on variable parameters, (4.2) ceases to divide only by a fixed constant. This identifies a precise boundary of the present linearization. Studying which

broader motion laws still admit low-degree complete chronological certificates is a more informative problem than assuming every geometric model behaves like the fixed-speed case.

[write] [Added 4 October 2026: Part XII’s branch tests run on full-dimensional open chambers that cover every encoded counter pair with uniform margins (Equations (298.2)–(298.3)), so its branching needs no gap equalities; the counter values themselves are exact ratios, and source 34 states that the encoding is not noise robust. The direction stays open.]

15.9 Accumulation limits without fictitious infinite witnesses

The clock gives a finitely described accumulation time, but a generic finite prefix certificate cannot assert all events of an infinite run. A separate notion of a certified self-similar tail could perhaps prove a limit formula or a no-collision property beyond a given recursive pattern. Such a notion must specify the dynamics and limit semantics explicitly. It must not silently treat an infinite sequence of successful finite checks as one ordinary existential Diophantine witness.

[write] [Added 4 October 2026: answered in part by Part X. For a supplied ultimately periodic complete macro, source 26 decides infinite validity by a finite formula of quadratic sign tests and one unique natural witness when at most four signals are live (Theorem 182.2), classifies Zeno behaviour and computes the rational accumulation time (Theorem 182.3), and proves rational accumulation times and limit positions for every population (Theorem 182.4); the clock of Section 12 is an instance for rational seeds. With five signals no finite sign formula need exist (source 27), but existential integer certificates do (sources 28 and 29). Executions without a supplied macro and any continuation beyond the accumulation stay open (Question X.9).]

16 Conclusion

For rational fixed-speed signal machines, the continuous coordinates of a finite computation are not the difficult nonlinear part. Once a discrete chronological skeleton is fixed, they are rational linear forms in the initial gaps. The nontrivial certificate obligation is completeness of chronology: excluding unlisted earlier meetings and accounting for every signal and every site at a simultaneous event. Maximal adjacency intervals solve that obligation with a linear number of endpoint conditions.

The resulting exact chamber description yields quadratic natural-number certificates with unique slack tuples, and finite disjoint first-hit unions yield quartics with unique selector and slack tuples. It also yields useful results about denominator growth, small integral realizations, perturbation stability, and seed-counting quasi-polynomials. The executable examples test these claims against independent exact simulation and expose the boundary cases that incidence-only encodings miss.

The unresolved universal task is sharply separated: finitely encode an unbounded discrete skeleton together with the ordinary input interface. The convex-quadratic rigidity theorem explains why this cannot be achieved by merely hiding more geometry inside one fixed globally nonnegative quadratic. ([write] The rigidity theorem is a special case of Part XI of *canonical-diophantine-certificates*; Remark 13.3.) The present contribution is a complete and auditable geometric frontend, together with a clear account of which further arithmetic and formalization steps remain.

Part II

Direct Diophantine certificates (source 11)

[write] Part II prints batch-78 manuscript 11, *Direct Diophantine Certificates for Rational Signal Machines: Finite collision schemas with unique quadratic witnesses*, “research article and exact arithmetic replay”, dated 2 October 2026; its document metadata name the author as “Research report prepared with OpenAI”. Its labels carry the prefix `smc:sd:`; its sections are Sections 17–28 here and its two appendices are Appendices C and D. It proves the theorem of Part I by a second route (Section 1.2); it is printed in full, and its lemmas that coincide with lemmas of Part I carry [write] notes. Its abstract and keywords follow.

Abstract (source 11). A prescribed finite collision history of a rational signal machine admits an explicit Diophantine certificate that retains the machine’s geometry. After external normalization, we compile each finite schema into a sum of squares of integer affine residuals. Natural zeros are exactly its realizations, and every admissible input has exactly one witness tuple. The witness coefficient matrix has full column rank, so the polynomial is strictly convex in the witnesses. A sparse construction uses event coordinates, batch times, and typed endpoints of maximal adjacency lifetimes; for bounded collision arity and fanout its size is linear in the initial support and number of events. The endpoint types enforce complete collisions, simultaneous events, and the absence of a hidden collision at the final cut. We prove denominator divisibility and an explicit witness-height bound for the four-speed interface of a known universal signal machine. The construction is a uniform compiler for externally supplied finite schemas, not a fixed-arity representation of unbounded halting. Exact standard-library replays and fully worked examples accompany the proofs.

Keywords. Diophantine representation; signal machine; abstract geometrical computation; finite trace; unique witness; convex quadratic; semilinear set.

17 Purpose and relation to prior work

Turing-complete substrates can store and transform information in very different ways. Signal machines use a finite collection of particle types moving continuously on a line and interacting at collisions. A direct arithmetization of their geometry should explain both the prescribed events and the absence of all other events. Merely writing down line intersections leaves the latter obligation unresolved.

This article establishes that obligation for each fixed finite collision schema. Its contribution is an explicit integer certificate, including denominator control, canonical witnesses, sparse size bounds, and exact endpoint conditions. No general claim that signal-machine evolution is newly linearizable is intended. Durand-Lose’s linear real-register simulation already computes the next delay, updates gaps, and processes simultaneous events by finitely many linear branches [4, Section 4]. Our arithmetic constraints begin after a particular finite branch history has been fixed.

The substrate itself is already Turing universal. Durand-Lose gives a halting construction with 13 meta-signals and 21 *non-blank* collision rules; its four speeds are $-2, 0, 1, 2$ [3, Theorem 5 and Figure 4]. The construction uses a finite initial configuration and is separate from the paper’s weaker, non-halting cellular-automaton simulation with an infinite background. After a Galilean shift the speeds become $0, 2, 3, 4$. The rational normalized placements used in the halting arrangement can be scaled to integer positions, as proved generally below. We have not transcribed or executed that universal collision table. The accompanying replay uses small exact test machines.

The results are deliberately bounded in a precise sense. The machine, input length and labels, schema, and number of batches are compiler data. The polynomial's arity grows with the schema. Nothing here packs unbounded execution into one fixed-arity quadratic, resolves the single-fold Diophantine representation problem, improves a universal-polynomial arithmetic-operation bound, or formalizes the proof in a proof assistant. A targeted literature check supports the stated relation to earlier work; it does not establish exhaustive novelty.

[write] Source 11 does not know source 07, printed as Part I, which proves the same theorem with all event coordinates eliminated and allows signed rational speeds; Section 1.2 compares the two routes. The repository has since reviewed this manuscript in its Hilbert-tenth-problem research tree (Section 1.4).

18 Model and exact theorem

Throughout, $\mathbb{N} = \{0, 1, 2, \dots\}$. Inputs and witnesses are natural numbers unless explicitly stated otherwise.

Definition 18.1 (Normalized signal machine). A machine $\mathcal{M}$ consists of a finite set of labels, a fixed speed $v(\ell) \in \mathbb{N}$ for each label, and a deterministic collision table. A legal incoming set has at least two labels with pairwise distinct speeds. Its outgoing set also has pairwise distinct speeds and may be empty. An initial configuration consists of labels $\ell_0, \dots, \ell_{n-1}$ at positions

$$0 \leq p_0 < p_1 < \dots < p_{n-1}, \quad p_j \in \mathbb{N}.$$

Between collisions a signal follows a straight line of its label's speed. At a collision every incident signal is consumed and the designated outgoing set is emitted at that point.

If an incoming set has no rule, the collision is forbidden, unless the model explicitly completes the table by a transparent rule. Such a rule consumes the incident segments and emits the same labels, ordered by speed. The theorem works for either convention after the machine is fixed. The replay scripts use identity completion, so an omitted rule in their input is transparent, not forbidden.

We distinguish labels from *segment identities*. Each initial signal has its own identity, and each emitted signal receives a fresh one, even if a transparent rule reproduces the same label. A segment has one birth and at most one death. Initial signals are born at time zero. This distinction is essential for endpoint typing.

Definition 18.2 (Finite collision schema). Fix the machine and the initial label list. A schema $\mathcal{S}$ specifies K nonempty chronological batches. In batch i , a partition of the ordered pre-batch live list into consecutive blocks designates each nonsingleton block as an event and each singleton as a persisting signal. Every event uses a legal rule, and its incoming speeds strictly decrease from left to right. Its outputs are ordered by strictly increasing speed. All replacements in a batch are performed atomically. Segment births precede their deaths by at least one batch.

The schema fixes no event coordinates or durations. Its events within a batch are ordered from left to right by their incoming blocks. Zero-output events are retained as events. A schema with $K = 0$ has no events. No initial collision markers, infinite initial support, or accumulation events are included.

A *prefix realization* has exactly these first K nonempty collision batches and ends immediately after batch K . The $K = 0$ prefix is the initial configuration alone. A *halting realization* additionally

has no future collision. Halting does not require all signals to disappear or stop moving. There is no arbitrary terminal observation time.

Let E be the number of events, $I = \sum_e |\text{in}(e)|$ the number of incoming incidences, r the largest incoming arity, and s the largest fanout. Set $n_- = \max(n - 1, 0)$ and

$$D = \text{lcm}\{|v - w| : v, w \text{ are machine speeds, } v \neq w\}, \quad B = D^K, \quad (18.1)$$

with the empty least common multiple equal to one.

[write] Theorem 18.3 is, for prefix semantics, the theorem of Part I's Theorems 5.3 and 6.1, in a different witness format: natural absolute positions as input, and batch times, event positions and slacks as witnesses on the grid B . Its halting version, the full column rank and the size ledger (18.3)–(18.4) are not in Part I. Source 11's proof is printed in full as the second route.

Theorem 18.3 (Sparse canonical certificate). *There is an effective compiler which, for every such fixed schema, produces integer affine residuals $r_1, \dots, r_R$ in the input coordinates $\mathbf{p}$ and witnesses $\mathbf{y} \in \mathbb{N}^W$, and a polynomial*

$$P_S(\mathbf{p}, \mathbf{y}) = \sum_{a=1}^R r_a(\mathbf{p}, \mathbf{y})^2.$$

For prefix semantics, a natural zero exists exactly when $\mathbf{p}$ realizes the schema. For halting semantics the same statement holds after the terminal conditions below are included. For every admissible input the witness is unique. The witness coefficient matrix has full column rank; consequently, if $W > 0$, the Hessian in the witness variables is positive definite.

If L denotes the number of recorded adjacency lifetimes, including terminal rays when present, then

$$L \leq n_- + \sum_e (|\text{out}(e)| + 1), \quad (18.2)$$

$$W \leq E + 2K + 2L + n_-, \quad R \leq I + K + 2L + n_-. \quad (18.3)$$

In particular,

$$W \leq (2s + 5)E + 3n_-, \quad R \leq (r + 2s + 3)E + 3n_-. \quad (18.4)$$

The polynomial has degree at most two, and degree exactly two whenever its witness vector is nonempty.

Syntactically illegal schemas are rejected before compilation. For the halting version, a final live order with a negative adjacent relative velocity is likewise rejected as impossible. One can instead output the constant polynomial one for these cases; the stated resource ledger concerns the structurally admitted construction. At $K = 0$, the prefix compiler may use only input-order checks. For $n \leq 1$ it has no witnesses and returns zero. This harmless degenerate case is not a strictly convex function on a positive-dimensional space.

19 Rational normalization and denominator control

19.1 Normalization is an external interface

Suppose the original speeds are rational numbers $u(\ell)$. Choose a rational $c \leq \min_\ell u(\ell)$ and a positive integer m such that every $m(u(\ell) - c)$ is an integer. For a fixed rational input choose a

rational translation a and positive integer q so that every $q(x_j + a)$ is a nonnegative integer. The invertible coordinate change

$$X = q(x + a - ct), \quad T = \frac{q}{m}t \quad (19.1)$$

preserves simultaneous incidence, spatial order at each time, positive time direction, and the collision rules. Its speeds are

$$\frac{dX}{dT} = m(u(\ell) - c) \in \mathbb{N},$$

and its initial coordinates are natural numbers. Thus every finite rational configuration can be put into the theorem's interface.

The constants a, c, m, q are chosen *outside* the existential witness tuple. For a varying input family with fixed denominator and shift, the natural input $p_j = q(x_j + a)$ is the declared coordinate encoding. Allowing an arbitrary additional scale as a witness produces many proportional encodings; allowing raw variable numerators and denominators introduces products when denominators are cleared. Neither change preserves the present canonical, jointly quadratic claim without a new argument. Inputs at arbitrary real coordinates are outside this interface.

[write] Part I needs no Galilean shift: it allows signed integer speeds after the time rescaling $\tau = t/L$ (Section 3) and takes the gap vector as input. With $m = L$ the two normalizations give the same speed differences, hence the same D (Section 1.2).

19.2 Causal depth and a fixed grid

Assign depth zero to initial births. The depth of event e is one plus the maximum depth of the birth events of its incoming segments, with initial births contributing zero. Every birth lies in an earlier batch, so the maximum event depth H satisfies $H \leq K$.

Lemma 19.1 (Denominator divisibility). *([write] The same as Part I, Theorem 7.1, there stated for the coefficient forms in the initial gaps; source 11's proof is kept as a second route because it works with event-coordinate witnesses.) Every event time and position of depth at most h has reduced denominator dividing D^h . In particular D^H , and hence $B = D^K$, clears every event coordinate and time. It also clears the position of any live segment sampled at an event time.*

Proof. Initial birth positions and times are integers. Suppose an event of depth h has incoming lines

$$x = b_1 + v_1(t - a_1), \quad x = b_2 + v_2(t - a_2), \quad v_1 \neq v_2.$$

All birth coordinates have denominators dividing D^{h-1} by induction. The intersection time is

$$t = \frac{b_2 - v_2 a_2 - b_1 + v_1 a_1}{v_1 - v_2}. \quad (19.2)$$

The numerator's denominator divides D^{h-1} , while $|v_1 - v_2|$ divides D . Thus the reduced denominator of t divides D^h . Substitution into either line gives the same assertion for its position. A multi-signal event is pinned by any two incoming speeds; the remaining incidences are additional equalities. Finally a sampled position is an integer-linear combination of a birth position, birth time and sample time, all cleared by D^H . $\square$

This lemma bounds denominator divisibility, not numerator magnitude. Its practical advantage is that B is a computable constant of the schema. The compiler can use the sharper D^H if desired; the supplied implementation uses D^K .

20 The sparse construction

20.1 Geometry variables and flight equations

Use a natural variable T_i for the scaled time of batch i , with $T_0 = 0$, and a natural variable X_e for the scaled position of each event. Introduce $d_i \in \mathbb{N}$ and require

$$T_i - T_{i-1} - d_i - 1 = 0 \quad (1 \leq i \leq K). \quad (20.1)$$

Thus each batch occurs at a strictly later time on the B -grid.

For a segment σ of speed v_σ , let $b(\sigma)$ be its birth batch. Define its scaled birth-position expression A_σ to be Bp_j for initial segment j , and X_f if it is emitted at event f . Its position at any relevant batch time is the affine expression

$$Z_\sigma(T_i) = A_\sigma + v_\sigma(T_i - T_{b(\sigma)}). \quad (20.2)$$

If σ is consumed by event e in batch i , impose the flight residual

$$X_e - Z_\sigma(T_i) = 0. \quad (20.3)$$

There are I such residuals. All coefficients are integers because the speeds and B are fixed integers. No product of two unknowns appears.

Input order is encoded, rather than silently assumed, by natural slacks u_j and residuals

$$p_{j+1} - p_j - u_j - 1 = 0 \quad (0 \leq j < n - 1). \quad (20.4)$$

An explicitly restricted ordered-input domain permits these residuals and witnesses to be omitted. We keep them in every count.

20.2 Maximal adjacency lifetimes

Maintain the ordered live list combinatorially. A consecutive ordered pair of segment identities has a lifetime beginning at the first boundary at which it becomes adjacent and ending at the boundary at which it ceases to be adjacent. A pair that remains adjacent across several batches has one lifetime, not a new copy in every slab. Perform all replacements in a simultaneous batch before creating new pairs; pairs appearing only during an arbitrary sequential processing order are never recorded.

For a prefix, close each still-active positive-duration lifetime at the final cut T_K . A pair newly born at batch K has no positive-duration finite lifetime and is omitted. For halting semantics, still-active pairs receive infinite terminal lifetimes instead, including pairs born at the final batch. The initial $K = 0$ prefix needs no finite lifetimes.

For a pair (σ, τ) , left to right, its scaled gap is the affine function

$$G_{\sigma\tau}(T) = Z_\tau(T) - Z_\sigma(T), \quad (20.5)$$

whose slope is the fixed relative velocity $v_\tau - v_\sigma$.

Lemma 20.1 (Lifetime count). ([write] *The same as the first inequality of (5.2) in Part I, Theorem 5.4; source 11's proof is kept as a second route because it works with event-coordinate witnesses.*) *The number L of recorded lifetimes satisfies (18.2).*

Proof. There are at most n_- initial adjacent pairs. Replacing one consecutive event block by k outputs creates at most $k + 1$ new pairs: at most $k - 1$ internal output pairs and two outer-neighbor pairs when $k \geq 1$, or at most one bridge pair when $k = 0$. Boundary events can create fewer. A sequential accounting of the disjoint replacements in a batch therefore counts at most $\sum_e (|\text{out}(e)| + 1)$ creations. Some counted pairs may be transient and disappear within the same batch; atomic processing removes those and cannot increase the count. Every lifetime has exactly one creation, including any terminal lifetime. Closing a lifetime does not create an additional one. $\square$

20.3 Typed endpoint constraints

For a finite lifetime from batch a to batch $b > a$, impose two constraints:

1. At the lower endpoint, impose $G(T_a) = 0$ if and only if both segments were emitted by the *same event* in batch a . Otherwise impose $G(T_a) = h^- + 1$ with a fresh $h^- \in \mathbb{N}$.
2. At the upper endpoint, impose $G(T_b) = 0$ if and only if both segments are consumed by the *same event* in batch b . Otherwise impose $G(T_b) = h^+ + 1$ with a fresh $h^+ \in \mathbb{N}$.

The “if and only if” here specifies which residual the compiler emits from the schema; it is not a new logical connective in the polynomial. Permitted equality means an equality residual, not an arbitrary nonnegative gap. In particular, a surviving pair at the final cut receives a *strict* upper gap.

For a terminal ray starting at batch a , require its relative velocity to be nonnegative, a check on machine constants. Impose a zero initial gap only for two same-event outputs at that starting boundary; impose $G(T_a) = h + 1$ otherwise. Same-event outputs have strictly increasing speeds, so a zero-gap terminal pair automatically has strictly positive relative velocity. A terminal pair with zero relative velocity must therefore start with positive gap.

Every stated condition is either an integer affine residual or a finite compile-time structural check. Define P_S by squaring and adding the residuals (20.1), (20.3), (20.4), and all endpoint residuals.

21 Proof of geometric correctness

21.1 Why endpoint types suffice

Lemma 21.1 (Positive interior gap). ([write] *The same as Part I, Lemma 5.2 (the endpoint lemma) together with the static rejection of a common birth and a common death that precedes it; source 11’s proof is kept as a second route because it works with event-coordinate witnesses, and it adds terminal rays.*) *Every finite lifetime satisfying its endpoint residuals has strictly positive gap in its open time interval. Every admitted terminal lifetime has strictly positive gap at every time after its start.*

Proof. A finite gap is affine. Its endpoint values are nonnegative. If either is strictly positive, every proper convex combination of the endpoint values is strictly positive. If both are zero, the endpoint types say that the pair shares its birth event at the lower boundary and its death event at the upper boundary. Birth ordering gives $v_\sigma < v_\tau$, so

$$G(T_b) = G(T_a) + (v_\tau - v_\sigma)(T_b - T_a) > 0,$$

a contradiction. Thus at least one endpoint is strict, and no extra endpoint-sum residual is needed. On a terminal ray the nonnegative slope preserves a strict initial gap; a zero initial gap has strictly positive slope and becomes positive immediately. $\square$

It is important that this argument uses segment identities and the exact start and end boundaries. Merely allowing two nonnegative endpoint gaps, even with positive sum, does not exclude an unrecorded meeting at the upper endpoint.

Lemma 21.2 (Complete boundaries). *At every batch satisfying the residuals, each designated incoming block meets at its event point, distinct designated events have distinct ordered positions, and no persisting signal or signal of another event also meets that point. The post-batch order is valid; its only zero gaps are between outputs of the same event.*

Proof. Before the batch, every adjacent pair belongs to a lifetime that has existed for a positive duration. For a pair in one event block, the flight equations place both segments at that event. Any pair across two distinct pre-batch blocks is not consumed by the same event. If its lifetime ends at this boundary, the strict upper endpoint gives a positive gap. If it persists across the boundary and this is an interior point of its lifetime, Lemma 21.1 gives a positive gap. If the boundary is the final cut, a surviving pair has a strict upper endpoint, or lies on an admitted terminal ray. These cases exhaust the pre-batch neighboring pairs.

All positions in an incoming block are equal by the flight equations, whereas successive blocks are strictly separated. Hence the designated blocks are precisely the maximal equality classes at that time. In particular a third incident signal cannot be omitted, two designated events cannot coincide, and a completely erased event still has a position in the strict block order. Replacing each event block at its own point by the prescribed speed-ordered outputs therefore preserves weak spatial order. Across distinct surviving blocks the gap stays strict; within one output fan the gap is zero and the increasing speeds separate it immediately. If erased blocks disappear, the new bridge gap spans one or more former strict block separations, so is positive. The replacements are atomic and introduce no additional intermediate configuration. $\square$

This proof also explains why final-born prefix pairs can be omitted. Their boundary order is already certified by the incoming block partition and flight equations. Immediately after that boundary, a common output fan separates and all other gaps remain positive for a sufficiently small interval. A prefix does not assert the absence of collisions later in that interval's future.

21.2 Soundness and completeness

Proof of the existence assertion in Theorem 18.3. For soundness, suppose $P_S(\mathbf{p}, \mathbf{y}) = 0$ with natural inputs and witnesses. Each square is nonnegative, so every residual is zero. The input packet gives strict initial order. Time residuals give strictly increasing batch times. Divide all scaled times and event positions by B and draw the prescribed segments using (20.2).

On each open slab, the combinatorial live list is fixed and every neighboring pair is in an active lifetime. Lemma 21.1 excludes all adjacent meetings. A meeting of nonadjacent signals would force an adjacent equality in the ordered list at or before that meeting, so there are no other meetings either. At the next batch, Lemma 21.2 proves that the designated events are exactly all collisions and contain every incident signal. Their legal rules are the fixed schema rules. Induction across the K batches gives precisely the prescribed prefix. For halting semantics every final adjacent gap remains positive for all future time by the terminal-ray argument, so no future collision occurs.

For completeness, start with an actual realization. Lemma 19.1 puts every event time, event position and endpoint sample on the B -grid. Assign their scaled values to T_i and X_e , and set $d_i = B(t_i - t_{i-1}) - 1$. All are natural. Input gaps determine natural u_j . Each lower or upper endpoint at which the schema prescribes equality is an actual common birth or death point. Every

other endpoint is strictly separated by the definition of a complete collision history; its positive scaled integer gap minus one supplies the unique natural slack. An actual halting configuration has nonnegative adjacent relative velocities, since a negative one would eventually cause a meeting, and its terminal initial gaps have exactly the required types. All residuals vanish. $\square$

The argument covers simultaneous multi-signal collisions, disjoint events in the same batch, annihilation, transparent collisions and empty final configurations. Its dependence on complete blocks and typed boundaries is substantive, not a cosmetic choice of encoding.

22 Canonical witnesses and strict convexity

Uniqueness of integer witnesses alone would not prove strict convexity. We establish the stronger linear-algebra statement directly.

Lemma 22.1 (Full column rank). *Write the residual vector as*

$$\mathbf{r}(\mathbf{p}, \mathbf{y}) = A\mathbf{y} + C\mathbf{p} + \mathbf{c}.$$

For every structurally admitted schema, the matrix A has trivial kernel over $\mathbb{R}$, even if the schema has no realization for the chosen input.

Proof. Let $A\Delta\mathbf{y} = 0$ and hold the input fixed. Initial birth positions have zero variation. Induct on batches. Assume all earlier times and event positions have zero variation. Choose any event e in batch i and two of its incoming segments σ, τ with distinct speeds. The homogeneous flight equations are

$$\Delta X_e - v_\sigma \Delta T_i = 0, \quad \Delta X_e - v_\tau \Delta T_i = 0,$$

because both segments were born in earlier batches. Their difference gives $(v_\sigma - v_\tau)\Delta T_i = 0$, hence $\Delta T_i = 0$ and $\Delta X_e = 0$. Every other event in the batch then has zero position variation from any one of its flight equations. Equation (20.1) gives $\Delta d_i = 0$. Induction fixes all geometry variables and time slacks.

Each remaining witness is an input-order or endpoint slack with coefficient -1 in its own defining residual. With the geometry and input fixed, that residual forces its variation to be zero. Thus $\Delta\mathbf{y} = 0$. $\square$

Corollary 22.2 (Canonical zero and Hessian). *For fixed input there is at most one real solution of the residual system, and hence at most one natural zero of P_S . The Hessian in witness coordinates is $2A^\top A$, positive definite when $W > 0$.*

Proof. Two solutions differ by a vector in $\ker A$, which is zero by Lemma 22.1. Moreover $\mathbf{z}^\top(2A^\top A)\mathbf{z} = 2\|A\mathbf{z}\|^2 > 0$ for every nonzero $\mathbf{z}$. Existence for admissible inputs was proved above. $\square$

Strict convexity is asserted in the witness variables with the input held fixed. Joint strict convexity in input and witness coordinates is neither needed nor generally true. The compiler introduces no witness for a schedule choice, denominator choice or arbitrary final observation time. Adding any of those could destroy the canonical property.

Proof of the size assertions in Theorem 18.3. Geometry uses E event positions, K batch times and K time slacks. A finite lifetime needs at most two slacks and two residuals; a terminal lifetime needs at most one of each. Input order uses n_- slacks and residuals. Flight and time residual counts are I and K . This proves (18.3). Lemma 20.1, $I \leq rE$, $K \leq E$ and $|\text{out}(e)| \leq s$ give (18.4). Squaring affine residuals gives degree at most two. If $W > 0$, full column rank forces a nonzero quadratic part. $\square$

The ledger counts variables and affine residuals. With a fixed machine, every residual has bounded width: an endpoint gap uses only the birth positions and times of two segments, one endpoint time, and possibly a slack. A linear-size arithmetic circuit follows if integer constants are treated as available coefficients. This does not give a literal gate ledger under a model that charges separately for constructing large constants, input loading, multiplication conventions or natural-to-positive conversion. In particular, no improvement to previously studied 75-operation or 87-operation constructions is asserted here.

23 A full-layer reference construction

A denser construction is useful for cross-checking. In slab i , let N_i be the number of pre-batch signals and g_i the number of blocks, including singleton through-signals. Introduce natural scaled end coordinates $C_{i,j}$, a duration slack d_i , and one gap slack between successive blocks. The scaled start coordinate $A_{i,j}$ is Bp_j initially or an alias of an earlier endpoint. Require

$$\begin{aligned} C_{i,j} - A_{i,j} - v_j(d_i + 1) &= 0 && \text{for every live signal,} \\ C_{i,j} - C_{i,\text{first}(b)} &= 0 && \text{inside each collision block } b, \\ C_{i,\text{first}(b')} - C_{i,\text{first}(b)} - h - 1 &= 0 && \text{for successive blocks } b, b'. \end{aligned}$$

All blocks, including zero-output collisions, participate in the last row. The exact counts, with input-order checks, are

$$W = n_- + \sum_i (N_i + g_i), \quad R = n_- + \sum_i (2N_i - 1). \quad (23.1)$$

Indeed each slab uses $N_i + 1 + (g_i - 1)$ witnesses, and $N_i + (N_i - g_i) + (g_i - 1)$ residuals.

The two boundary lists have nonnegative adjacent gaps, with strict separation across ending blocks. A pair with zero gap at its starting boundary is a same-event output pair with increasing speeds; it cannot meet again at the ending boundary after positive time. Affineness therefore excludes all interior collisions. The full ending partition supplies completeness at each boundary. Conversely the denominator lemma gives a witness from every realization. The same two-incoming-speed induction fixes each duration, then all endpoints and gap slacks, and proves full witness rank. A halting certificate additionally checks that the final live speeds are nondecreasing; the final boundary order is already certified.

This reference compiler stores coordinates for every live signal at every layer, even if that signal survives unchanged through many layers. The sparse compiler avoids these repetitions by storing only event coordinates and maximal adjacency lifetimes. Both implementations share an exact simulator and small arithmetic utilities; agreement between them is useful evidence, but is not independent formal verification of the common simulator.

24 Worked examples

24.1 One annihilation with its complete quadratic

Consider labels R, S with speeds $2, 0$ and the rule $\{R, S\} \mapsto \emptyset$. Let the initial positions be $p_0 < p_1$, and specify one event. Here $D = B = 2$. Write T, X, d for its scaled time, position and duration slack, u for input order and h for the lower adjacency slack. The compiler gives

$$\begin{aligned} P = & (p_1 - p_0 - u - 1)^2 + (T - d - 1)^2 \\ & + (X - 2p_0 - 2T)^2 + (X - 2p_1)^2 \\ & + (2p_1 - 2p_0 - h - 1)^2 + (2p_1 - 2p_0 - 2T)^2. \end{aligned} \quad (24.1)$$

The last row is the zero upper gap of the two segments consumed by the same event. Its redundancy with flight equations is harmless. Every $p_0 < p_1$ has the unique witness

$$T = p_1 - p_0, \quad X = 2p_1, \quad d = u = p_1 - p_0 - 1, \quad h = 2(p_1 - p_0) - 1.$$

For $(p_0, p_1) = (0, 1)$ the tuple (T, X, d, u, h) is $(1, 2, 0, 0, 1)$, corresponding to physical time $1/2$ and position 1 . There are five witnesses and six residuals. The final state is empty, so the prefix is also halting. Changing any one witness while fixing the others cannot keep all residuals zero.

24.2 A hidden collision exactly at the final cut

Use the four-speed interface $\{0, 2, 3, 4\}$, with labels a, b, c, d respectively, and the annihilation rule $\{d, a\} \mapsto \emptyset$. Start with

$$(\ell_0, \ell_1, \ell_2, \ell_3) = (d, a, d, a), \quad (p_0, p_1, p_2, p_3) = (0, 4, 10, 14).$$

There are two simultaneous collisions at physical time 1 , at positions 4 and 14 . Now submit a false one-batch schema containing only the first event and marking the last two signals as through-signals. Since $B = 12$, the first event's flight equations force $T_1 = 12$ and $X_0 = 48$.

The omitted pair's gap is

$$G(T) = 12(14 - 10) - 4T = 48 - 4T.$$

It is positive before the cut, has lower value 48 , and reaches zero at the cut. Thus conditions of the form $G(0) \geq 0$, $G(T_1) \geq 0$, and $G(0) + G(T_1) > 0$ would accept its endpoints. Our upper endpoint is strict because the pair survives rather than dying in one designated event. Its residual becomes

$$G(T_1) - h^+ - 1 = -h^+ - 1,$$

which cannot vanish for $h^+ \in \mathbb{N}$. The false schema has no certificate. The correct simultaneous schema consumes both pairs atomically and has event positions 48 and 168 on the scaled grid.

[write] Part I's independent simultaneous sites (Section 11.3) are another instance of this failure mode: an omitted, spatially separate simultaneous collision, rejected by the strict endpoint of the surviving pair (Section 1.2).

24.3 An outgoing fan at a finite cut

Replace the annihilation rule in the preceding setup by $\{d, a\} \mapsto \{b, c\}$. Each collision emits speeds 2 and 3. At the final cut the two members of each fan have zero gap and immediately separate; they need no zero-duration finite lifetime. In a halting certificate they instead receive terminal rays with zero starting gap and positive relative velocity. The gap between the two different fans is strict at the cut. Its relative velocity is $2 - 3 = -1$, so that terminal ray is rejected: the two fans will interact later. This separates valid finite-prefix geometry from the stronger no-future-collision requirement.

25 Witness height for the four-speed interface

Let the normalized machine speeds lie in $\{0, 2, 3, 4\}$, and let M_j be the largest live position immediately after batch j , with $M_0 = \max_j p_j$ and $\max \emptyset = 0$. Let t_j be elapsed physical time, with $t_0 = 0$.

Lemma 25.1 (Geometric growth). *Whenever batch $j + 1$ exists, its delay satisfies $\delta_{j+1} \leq M_j$. All positions at that boundary, including erased event positions, are at most $5M_j$. Consequently*

$$M_j \leq 5^j M_0, \quad t_j \leq M_0 \frac{5^j - 1}{4}.$$

Proof. A next collision must be approached by an adjacent closing pair. The pair has a strictly positive gap g and positive integer closing speed $c \geq 1$. Because all live positions are nonnegative, $g \leq M_j$; its meeting delay is $g/c \leq M_j$. The earliest collision delay is no larger. During this interval every signal advances by at most $4M_j$, so every pre-collision position is at most $5M_j$. Outputs remain at those event positions and deletion cannot increase the maximum. Iterate this bound and sum $\delta_i \leq 5^{i-1} M_0$ to obtain the stated geometric series. $\square$

The argument concerns an existing next batch. Empty or singleton configurations have none. Nor does a valid configuration whose maximum position is zero: any coincident outgoing fan has strictly increasing speeds and separates. This avoids dividing by a nonexistent positive gap.

Theorem 25.2 (Canonical height). *For $B = 12^K$, every canonical input coordinate and witness of the sparse or full-layer compiler is at most*

$$60^K \max(1, M_0). \tag{25.1}$$

Each such integer therefore has bit length $O(K + \log(M_0 + 1))$.

Proof. For the full speed set, the nonzero absolute differences are 1, 2, 3, 4, whose least common multiple is 12. The same 12 is a valid clearing modulus for a subset. Up to batch K , every live sample and every event position is at most $5^K M_0$, including positions subsequently erased. Multiplication by 12^K bounds them by $60^K M_0$. Every endpoint gap is the nonnegative difference of two positions in this range, so its slack is smaller. Input-order slacks are at most M_0 . The time bound is

$$12^K t_j \leq 12^K M_0 \frac{5^K - 1}{4} \leq 60^K M_0,$$

and the scaled duration minus one is no larger than the corresponding scaled time. The empty and all-zero cases are covered by $\max(1, M_0)$; if $K = 0$, all remaining slacks are initial-gap slacks bounded by M_0 . Taking binary logarithms proves the bit bound. $\square$

This bounds actual zero witnesses, not arbitrary trial assignments. Degree two alone would supply no such height guarantee. Likewise, the bound is exponential in batch count K as a numerical magnitude while linear in K as a bit length. For a fixed machine with maximal speed V , the same argument gives $M_j \leq (1 + V)^j M_0$; the four-speed constant above is the concrete interface relevant here.

[write] Part I bounds the chamber's row coefficients (Proposition 7.3) and the size of *some* realizing seed (Theorem 8.1); Theorem 25.2 bounds the canonical witnesses of every realizing input. The two bounds measure different things and neither implies the other.

25.1 Exponential denominator growth can be necessary

Proposition 25.3 (A bounded geometric shuttle). *There is a fixed four-label machine with integer initial positions whose $2k$ -batch prefix contains a time of reduced denominator 3^{k-1} , although every event lies in $0 < t < 2$ and $0 < x < 6$. Thus a common integer grid for exact event geometry may require exponentially large scale in the number of batches.*

Proof. Give labels L, R, F, B speeds 3, 2, 4, 0, respectively, and use rules

$$\{F, R\} \mapsto \{B, R\}, \quad \{L, B\} \mapsto \{L, F\}.$$

Start with L, F, R at positions 0, 1, 2. The two walls follow $x_L(t) = 3t$ and $x_R(t) = 2 + 2t$, so their separation is $2 - t$ until they meet at $t = 2$. The first right collision occurs at $t_{R,1} = 1/2$ and the first left collision at $t_{L,1} = 1$. At a left collision at time $a < 2$, the outgoing speed-four shuttle catches the right wall after $(2 - a)/2$, giving right time $r = 1 + a/2$. Its speed-zero replacement is then caught by the left wall after $(2 - r)/3$, giving the next left time

$$a' = r + \frac{2 - r}{3} = \frac{4}{3} + \frac{a}{3}.$$

These positive delays place the shuttle strictly between the walls between alternating collisions; there are no other earlier meetings. The recurrence and first collision give

$$t_{L,k} = 2 - 3^{1-k}, \quad t_{R,k} = 2 - \frac{3}{2} 3^{1-k} \quad (k \geq 1).$$

The first six batch times are $1/2, 1, 3/2, 5/3, 11/6, 17/9$. At the k th left event the numerator of $(2 \cdot 3^{k-1} - 1)/3^{k-1}$ is coprime to three, so the reduced denominator is exactly 3^{k-1} . Event positions lie on a wall before time two and hence are below six. Every finite prefix is valid. $\square$

This example supplies a lower bound for common-grid geometry encodings: after $2k$ batches a clearing scale is divisible by 3^{k-1} , and its scaled last time requires $\Omega(k)$ bits. It makes the linear event-count dependence in the witness-bit bound qualitatively sharp for this type of encoding; it is not a lower bound on arbitrary Diophantine representations. The collision times approach a Zeno accumulation at $t = 2$. We assert nothing about a continuation at or beyond that accumulation, which is outside the finite-schema theorem. The replay checks 64 batches with exact fractions.

[write] Part I's four-speed clock (Section 12, Proposition 12.1) is the same shuttle mechanism on a different machine, with speeds 0, 2, -1 , -2 and reduced denominators exactly $2 \cdot 3^k$ at event $2k$ for the seed (1, 2). The worked-example receipt shipped as `data/11-signal-certificates-expected-worked_examples_receipt.json` records the last left denominator of the 64-batch prefix as $617673396283947 = 3^{31}$, which is 3^{k-1} for $k = 32$.

26 Fixed-schema arithmetic and the unbounded boundary

Proposition 26.1 (Semilinear feasible inputs). ([write] *The same as Part I, Theorem 10.1, for prefix semantics; source 11’s proof is kept as a second route because it works with event-coordinate witnesses, and it adds the halting case.*) *For a fixed normalized machine, fixed initial label list and fixed schema, the set of natural input positions admitting a prefix certificate is semilinear. The same holds for the halting certificate. For a fixed total-event bound, the union over all schemas with that bound is also semilinear, provided the initial length and labels are fixed.*

Proof. A zero of a sum of squares is exactly a common zero of its affine residuals. Existence of natural witnesses is therefore an existential formula using only integer constants, addition, order and equality. Such Presburger-definable sets are semilinear [7, Theorem 1.3]. Terminal slope checks depend only on the fixed schema; they either remove it or leave the same argument.

For a total-event bound E_0 , at most $n + sE_0$ segment identities can have been created. A finite label set and bounded finite list admit finitely many consecutive partitions, legal rule choices and batch assignments with at most E_0 events. Thus there are finitely many schemas. Finite unions of semilinear sets are semilinear. $\square$

More concretely, a semilinear set is a finite union of sets $\mathbf{a} + \sum_{j=1}^m \mathbb{N}\mathbf{b}_j$. The proposition describes a strong boundary: a fixed externally supplied finite schema reduces to integer linear feasibility despite its quadratic packaging. Strict convexity does not import undecidability into that fixed finite system.

Enumerating all finite schemas gives an enumerable family of quadratics covering finite-event halting, with varying arities and coefficients. In a deterministic machine an actually halting run has a definite finite sequence of complete nonempty batches. That fact does not encode its arbitrary length into one fixed tuple. A single fixed-arity representation of unbounded halting requires an additional history-packing construction, with its own decoder, arithmetic costs and witness-multiplicity proof. Generic Diophantine representability results do not automatically preserve this compiler’s quadratic degree, sparsity, or single witness. Nor does the construction certify continuation through a Zeno accumulation; no finite schema contains infinitely many events.

[write] For a fixed event budget, Part I’s Theorem 10.2 packs the finite union over all first-hit skeletons into one quartic with unique selector and slack witnesses; it does not remove the budget either.

26.1 Strictly positive witnesses

If a convention requires witnesses in $\mathbb{Z}_{>0}$, replace every y_j by $Y_j - 1$ and define

$$Q_S(\mathbf{p}, \mathbf{Y}) = P_S(\mathbf{p}, \mathbf{Y} - \mathbf{1}).$$

This is an integer polynomial of the same degree. The coordinatewise translation is a bijection from positive witness tuples to natural witness tuples and preserves uniqueness and the Hessian. The input-coordinate convention remains unchanged unless it is translated separately and explicitly. This substitution is exact but is not an assertion that conversion is free in an arithmetic-operation ledger.

27 Verification evidence and research questions

The package supplies two exact-arithmetic compilers, a rational event simulator, and deterministic test suites. Arithmetic uses Python integers and `fractions.Fraction`; no floating-point solver or Diophantine oracle is involved. Table 6 summarizes reproduced checks. Counts of certificates include both prefix and halting variants when both are generated; they are not counts of distinct universal-machine programs.

Check	Full layers	Sparse	Additional
Certificates	122	444	878
Exact full-rank tests	30	40	12
Single-coordinate mutations	4 229	10 102	—
One-layer candidate schedules	832	832	—
Terminal certificates	—	198	372
Height checks	—	444	878

Table 6: Finite implementation evidence. The sparse suite includes 1 114 layers, 1 186 events, 3 982 lifetimes and a dedicated hidden-final-collision regression. The additional suite covers 506 fixtures and 2 152 local growth inequalities.

The full-layer suite additionally enumerates all 625 natural tuples in a small witness box for a two-signal example and finds its unique zero. The sparse suite’s original height test uses a factor-two safety margin; the additional suite checks the sharper constant-one bound (25.1). A release-specific example check verifies the complete annihilation polynomial, the omitted-final-collision obstruction, empty inputs, positive-witness translation, simultaneous events, output fans and the denominator-growth shuttle. Appendix C describes how to reproduce every reported receipt.

These checks can catch coefficient, endpoint and bookkeeping defects. They do not prove an infinite theorem, establish exhaustive novelty, or replace an independent formalization. The mathematical proofs above establish the stated scope.

[write] The counts in Table 6 equal the four expected receipts shipped as `data/11-signal-certificates-expected-*_receipt.json`. The replay was run at the write on a copy with the delivered layout, under Python 3.14.4 on Windows: all four receipts and both coefficient-example files matched. The repository’s review [14] replayed it as well and adds independent finite checks (Section 1.4).

27.1 Questions with concrete next steps

[write] Table 3 pairs these questions with the research directions of Part I: Questions 1 and 5 coincide with Sections 15.2 and 15.1, and Questions 3 and 4 overlap Sections 15.3, 15.4 and 15.5.

1. **Instantiate the universal table and input map.** Transcribe the 13-label, 21-non-blank-rule machine with an explicit convention for every blank collision. Verify the cyclic-tag input encoding and rational halting gadget, then compile and replay complete finite examples. This would connect the present implementation to the cited universal machine rather than only to its speed interface. ([write] Added 2 October 2026: still open as stated. Part III instantiates a different universal machine, Morita’s UTM(15,6) through Durand-Lose’s 2012 stacks; the 13-label table has not been transcribed.)
2. **Exploit conservative bounded populations carefully.** A reversible conservative rational

construction is already Turing universal [5]. Its positive energy assignment bounds the number of simultaneously live signals for fixed total energy, but not the number of collisions. A direct unbounded representation would need to instantiate its finite transition branches, encode unbounded histories of rational distances, and pay for strict guards and input loading. Conservation alone neither bounds runtime nor excludes accumulation. ([write] Added 2 October 2026: answered in part, at a fixed horizon. Part III instantiates every finite transition branch of a reversible conservative universal signal machine with 18 live signals, pays for the strict guards and the loader, and excludes accumulation on encoded runs (Section 32.4); Part IV does the same for conservative lattice dynamics. Neither encodes unbounded histories of rational distances in a fixed arity.) ([write] Added 3 October 2026: Part VIII encodes unbounded histories in a fixed arity for fixed sources of Part V’s conservative three-mass automaton, with infinite witness fibres; not for signal machines or histories of rational distances, so the question stays open as posed.) ([write] Added 4 October 2026: Part XII instantiates every branch of a number-preserving, not globally reversible, five-live-signal counter compiler (a population already reached by Durand-Lose [100]), excludes accumulation on encoded runs, and certifies bounded horizons with growing arity; Part XIII bounds the witnesses at each horizon. No fixed-arity encoding of unbounded histories of rational distances results; the question stays open as posed.)

3. **Sharpen the denominator and cost ledger.** Causal depth can replace batch count in B . Further denominator cancellation may depend on the actual branch structure. A useful next result would make such improvements canonical and give a literal arithmetic-gate ledger, including the construction of coefficients and any conversion of input conventions. ([write] Causal-depth clearing itself is proved: Lemma 19.1 shows that D^H suffices, and Part I, Theorem 7.1, gives D^h event by event. What remains open is a canonical choice compatible with the witness format, and the gate ledger.)
4. **Compress histories without losing canonicity.** The sparse certificate removes repeated geometry, not the unbounded number of events. Can a restricted family of repeated collision motifs be packed with a unique decoder witness and explicit degree, height and operation bounds? Any answer should separate verification of a supplied history from existence of an outcome. ([write] Added 4 October 2026: touched for bounded counter traces. Sources 36–38 (Part XIII) replace the $T(E + 2)$ -witness trace of a fixed horizon by three, two or one unique positive witnesses with explicit degree, height and operation bounds; collision histories and unbounded horizons are not treated.)
5. **Formalize geometric completeness.** The most valuable proof-assistant target is the typed-endpoint soundness theorem, including atomic simultaneous replacement and disappearing event blocks. The replay data provide regression fixtures, but the formal objects must model the real semantics, not only the executable’s output.

28 Conclusion

A finite signal-machine schema admits a direct, canonical Diophantine encoding because its collision geometry can be expressed by affine integer constraints on a predetermined rational grid. Sparse adjacency lifetimes make the certificate linear in event count and initial support. Correct endpoint types are what turn line-incidence equations into a complete execution certificate. The resulting quadratic is strictly convex in its witness variables, and the four-speed case has an explicit linear

witness-bit bound. The remaining unbounded problem is to encode arbitrary schema length without losing the quantitative and multiplicity properties that make this finite construction useful.

Part III

Conservative signal machines as Diophantine frontends (source 12)

[write] Part III prints batch-79 manuscript 16, *Conservative signal machines as Diophantine frontends*, an AI-assisted research note dated 2 October 2026, from the archive `Conservative_Signal_Frontend_Corrected.zip`. That archive is the corrected edition of the batch’s manuscript 07, which has the same article and is not printed separately (Section 37 and Appendix E). It is called source 12 after the prefix of its shipped files. Its labels carry the prefix `smc:cs:`; its sections are Sections 29–37 here, and its three appendices are Appendices F–H, after the provenance appendix of the report. Its statements keep their numbering by section; equations are numbered by section (the source numbered them consecutively). Source 12 was written without Parts I and II and names no repository commit; [write] notes at Lemma 34.1, in Section 35 and after Theorem 36.1 relate it to Parts I–II and to the collection, and Section 1.7 describes the Part. Its abstract follows.

We instantiate a fixed reversible, number-conserving universal signal machine with 114 meta-signals, 445 explicitly enumerated binary collision rules, and exactly 18 live signals on every encoded input. The source is Morita’s 15-state, 6-symbol reversible universal Turing machine, compiled through the stack construction of Durand-Lose. Rational finite-support tape encodings occupy a fixed spatial interval; an exact replay checks the published 184-step example at every Turing-machine read. For any fixed-population, equal-cardinality signal machine, complete simultaneous-event dynamics admits the integer branch normal form $N_j = c_j I - ce_j^T$. These gap matrices satisfy $N_j^2 = c_j N_j$ and have coefficients of absolute value at most 24 in the literal machine. Encoding finite label control in one additional coordinate yields an 18-dimensional partial piecewise homogeneous integer-linear map for the literal machine. A separate binary-alphabet existence construction gives dimension and population ten. Finally, a direct branch-copy construction produces a nonnegative quadratic polynomial with a unique natural witness for each legal step, and canonical quadratic certificates for every fixed finite horizon. No fixed-arity compression of unbounded computation is claimed. The release contains the complete collision table, exact arithmetic code, receipts, and reproducible article source.

29 Results and scope

Signal machines are an unusually direct esoteric substrate for arithmetic descriptions of computation: a finite set of point particles moves at fixed velocities, and only collisions change their labels. Their geometry is continuous, but every finite collision history starting from rational data is rational. The issue is to expose that arithmetic without silently assuming generic position, constant-time tape loading, or a fixed number of witnesses for an unbounded history.

The universal computational mechanisms here are prior work. Durand-Lose supplies reversible conservative stacks and a reversible Turing-machine compiler [16]; Morita supplies the particular universal transition table [17]. Event simulation by linear real-register operations is also established background [4]. This note makes their composition literal, derives a population-dimensional integer

normal form with full tie semantics, and gives its canonical quadratic finite-step certificate. Priority of this precise synthesis has not been established by an exhaustive literature search.

Throughout, $\mathbb{N} = \{0, 1, 2, \dots\}$. A *meta-signal* is a label equipped with a speed; a *live signal* is one instance at one position. A fixed finite set of labels is compatible with arbitrarily long computations. Population and alphabet size are different resources.

Theorem 29.1 (Literal universal frontend). *There is a fixed partial signal machine $\mathcal{U}_{18}$ whose complete finite data are included with this note, with the following properties.*

- (i) *It has 114 meta-signals and 445 collision rules. Every rule has exactly two inputs and two outputs, with distinct speeds on each side. Distinct input sets have distinct output sets.*
- (ii) *Its speeds are $\{-12, -9, -4, 0, 4, 9, 12\}$. The least common multiple of all positive speed differences is $D = 65520$.*
- (iii) *Each well-formed finite input to Morita’s explicitly halting cyclic-tag interpreter is loaded into exactly 18 rational-position signals. Before halt these signals remain in $[-8, 8]$. They simulate its reversible Turing machine exactly. The signal table does not depend on the input program or word.*
- (iv) *Designated terminal pairs produce a halt label and an escape token. The token undergoes exactly eight further crossings on these encoded configurations, then no further collision occurs. Infinite encoded computations have unbounded elapsed time.*
- (v) *At event cuts, 17 nonnegative integer gaps and one integer mode coordinate define an exact partial piecewise homogeneous integer-linear map on $\mathbb{N}^{18}$. Finite reachability of its designated accepting modes is universal under the explicit input recoding described below.*

Here “universal” imports the halting-preserving simulation theorem for Morita’s source machine; the finite replay is an implementation check, not a replacement proof of universality. The source distinguishes its phase-zero halt from exhaustion of the current word. Only $(q_1, \mathbf{b})$ is accepting; $(q_2, \$)$ is nonaccepting null termination. Both outcomes are observable separately in $\mathcal{U}_{18}$. Unspecified source instructions and collisions remain undefined.

Theorem 29.2 (Binary existence bound). *There exists a fixed reversible number-conserving universal signal machine with ten live signals on every effectively encoded input. It uses speeds $\{-6, -3, -2, 0, 2, 3, 6\}$ and has a nine-gap integer lift using the current positive pivot speed c_j as its scale multiplier. Including label control gives a partial piecewise homogeneous integer-linear map on $\mathbb{N}^{10}$ with universal finite accepting reachability. Each gap branch is a rank-eight scaled projection with at most 16 nonzero entries, all of absolute value at most 12. A uniform alternative multiplier is $D = 360$.*

Theorem 29.2 is an existence-by-compiler statement using established binary reversible Turing-machine simulation [18, 17]. This release does not print or instantiate the resulting universal binary interpreter. The fully literal result is Theorem 29.1. Neither bound is claimed optimal.

Theorem 29.3 (Canonical quadratic finite-step certificate). *For either fixed integer map, one can effectively construct a polynomial of total degree at most two which vanishes over natural arguments precisely on legal one-step input/output pairs. Its full auxiliary witness is unique whenever it exists. For each externally fixed horizon K , the same construction gives a quadratic certificate for K steps with size linear in K , the machine being fixed.*

For the literal machine, the release enumerates a forward-closed family of 49,700 modes and 80,501 branches, and supplies the complete numeric mode/event data, an exact sparse coefficient exporter, and a factored polynomial circuit. The one-step packet has exactly 4,196,998 auxiliary variables. It is not expanded into a flat monomial list; that representation is much larger. The binary existence result remains specified by a compiler. The horizon K determines the arity of the certificate. It is not another free variable in a single fixed-arity quadratic polynomial.

30 Primary sources and the literal interpreter

30.1 The source machine and its terminal convention

Morita's Theorem 3.3 and Table 5 specify a 15-state, 6-symbol reversible universal Turing machine with 62 quintuples [17]. Its states are $q_0, \dots, q_{14}$ and its tape alphabet is

$$S = (\mathbf{b}, \mathbf{Y}, \mathbf{N}, *, \$, 1),$$

with $\mathbf{b}$ the blank. A quintuple (q, a, b, d, s) means: in state q scanning a , write b , move one cell in direction $d \in \{-1, +1\}$, and enter s . The complete table is reproduced as mathematical data in Appendix F and in the release JSON.

Two cells have designated terminal meaning:

$$(q_1, \mathbf{b}) = \text{halt}, \quad (q_2, \$) = \text{null}.$$

The other 26 empty cells are unspecified. In particular, q_1 and q_2 are not unconditional halting states: they also have ordinary transitions. Treating the entire states as final would change the computation.

For each target state s , all incoming transitions have one common direction d_s , and their written symbols are distinct. These local conditions imply reversibility: the target state fixes the previous head location; the symbol there identifies the unique predecessor transition, prior state, and overwritten symbol. All 62 transitions were checked against these conditions.

30.2 The finite input language

The source interpreter uses the explicitly halting cyclic-tag convention of Morita's Definition 3.3. A program has $k \geq 1$ cyclic productions

$$(\text{halt}, p_1, \dots, p_{k-1}), \quad p_i \in \{\mathbf{Y}, \mathbf{N}\}^*.$$

An empty current word gives the outcome `null`. Reading $\mathbf{N}$ deletes it and advances the phase. Reading $\mathbf{Y}$ in a nonzero phase deletes it, appends that phase's production, and advances the phase. Reading $\mathbf{Y}$ in phase zero halts. The simulation theorem applies to this convention; no unproved identification with every other cyclic-tag convention is used.

Let $\text{rev}(p)$ denote reversal. At phase zero the finite program region is

$$P = \text{rev}(p_{k-1}) * \dots * \text{rev}(p_1) * \mathbf{b}. \quad (30.1)$$

The tape is $P\$w$, surrounded by infinite blank tails, and the head initially scans the first cell of w , or a blank if w is empty, in state q_0 . Equivalently, start with a delimiter for every production

including halt, then replace the rightmost delimiter by the blank phase marker. Other initial phases replace the appropriate delimiter counted from the right. There is finite nonblank support on both sides of the head; there is no nonblank periodic background.

For (halt, YN, YYN) and $w = \text{NYY}$, the source Figure 28 starts with

$$\text{NYY} * \text{NY} * \text{b\$NYY}.$$

The printed prose following Table 5 reverses the letters of p_1 relative to the example and figure. We use the general reversal convention and the figure, which agree with the full 184-step table replay. The release documents this discrepancy rather than silently changing the sample.

30.3 What is imported and what is checked

Durand-Lose’s 2012 Sections 3–4 give the stack construction and reversible-TM translation used here. The later shrinking construction in that paper is not used. The 2006 result also establishes reversible conservative universality, but omits some compiler detail and is not an enumerated fixed universal collision table [5]. The 2007 linear-BSS comparison already treats next-event arithmetic [4].

The literal collision table is a new finite instantiation of these ingredients, not a claim to a new universal computational mechanism. Its structural checks, source-tape comparison, and integer event identities are exact. They do not constitute proof-assistant formalization or an exhaustive audit of every source reduction.

31 Finite-support stacks with a fixed number of particles

31.1 Exact rational encoding

For an alphabet of $\ell \geq 2$ symbols, number the digits $1, \dots, \ell$, with digit 1 blank, and put $b = \ell + 1$. Encode a finite digit word followed by infinitely many blanks by

$$\sigma(d_1 \cdots d_m) = \sum_{i=1}^m \frac{d_i}{b^i} + \frac{1}{\ell b^m}. \quad (31.1)$$

Let C_ℓ be the set of all such values. Trailing blank digits give the same eventually blank tape, as they should. Delimiters in a source input language determine any required finite-word boundary.

Lemma 31.1 (Stack arithmetic). *$C_\ell \subset [1/\ell, 1)$ is closed under*

$$\text{push}_v(\sigma) = \frac{v + \sigma}{b}, \quad \text{pop}(\sigma) = (v, b\sigma - v), \quad v = \lfloor b\sigma \rfloor.$$

The pop digit belongs to $\{1, \dots, \ell\}$, the remainder belongs to C_ℓ , and no admissible value lies on a digit boundary v/b .

Proof. Equation (31.1) is the base- b expansion of an infinite digit stream in $\{1, \dots, \ell\}$ eventually equal to 1. Its minimum is the all-1 stream, $1/\ell$; its value is strictly below the all- ℓ stream, which has value 1. Removing or prepending a digit leaves an eventually-1 stream. Thus $b\sigma = v + \rho$ with $\rho \in [1/\ell, 1)$, proving the floor formula and excluding boundaries. $\square$

This is an invariant on the encoded domain. Arbitrary rationals in the same interval are not substitutes: for example $2/3$ is a forbidden binary boundary. The gadget has exact rational dynamics, not a rounding convention for malformed stack values.

31.2 The complete parametric collision gadget

There are stationary marks $m_0, \dots, m_\ell$ at positions $0, \dots, \ell$, one stationary memory signal at σ , and one operation token arriving from the right. Choose a positive speed c and put $a = \ell/(\ell + 2)$. The labels and speeds are

Labels	Speeds
m_i, mem	0
$\text{ml}, \text{sl}_v, \text{gl}, \text{vl}_v$	$-c$
$\text{mr}, \text{sr}_v, \text{ack}, \text{vr}_v$	c
$\text{catch}, \text{fix}, \text{gr}, \text{set}$	$c/3, -ac, ac, -c/3$

The nontransparent rules are, for $1 \leq v \leq \ell$,

$$\{m_0, \text{ml}\} \mapsto \{m_0, \text{mr}\}, \quad \{\text{mem}, \text{sl}_v\} \mapsto \{\text{ml}, \text{sr}_v\}, \quad (31.2)$$

$$\{\text{sr}_v, m_v\} \mapsto \{m_v, \text{catch}\}, \quad \{\text{mr}, \text{catch}\} \mapsto \{\text{ml}, \text{fix}\}, \quad (31.3)$$

$$\{\text{mr}, \text{fix}\} \mapsto \{\text{mem}, \text{ack}\}, \quad \{\text{mem}, \text{gl}\} \mapsto \{\text{ml}, \text{gr}\}, \quad (31.4)$$

$$\{\text{mr}, \text{gr}\} \mapsto \{\text{ml}, \text{set}\}, \quad \{m_v, \text{set}\} \mapsto \{\text{vl}_v, m_v\}, \quad (31.5)$$

$$\{\text{mr}, \text{vl}_v\} \mapsto \{\text{mem}, \text{vr}_v\}. \quad (31.6)$$

For every $1 \leq i \leq \ell$, add identity crossings of m_i with

$$\text{ml}, \text{mr}, \text{fix}, \text{ack}, \text{gl}, \text{gr}, \text{sl}_v, \text{vr}_v \quad (1 \leq v \leq \ell),$$

and with sr_v, vl_v only when $i < v$. All other encounters are undefined. This last clause is essential: completing every unspecified pair by an identity rule can create an output-set collision with a nontransparent rule and destroy reversibility.

There are $5\ell + 10$ labels and

$$5 + 4\ell + 6\ell + 2\ell^2 + \ell(\ell - 1) = 3\ell^2 + 9\ell + 5 \quad (31.7)$$

rules. Thus one binary module has 20 labels and 35 rules, while one six-symbol module has 40 labels and 167 rules. The gadget uses only $\ell + 3$ live signals, including its operation token.

Every displayed rule is two-to-two, with distinct input and output speeds. Its output set determines which displayed case and, where applicable, which v occurred. The transparent outputs are the corresponding untouched inputs, and the listed transparent cases exclude the nontransparent outputs. This proves local rule injectivity; the release also checks it mechanically.

31.3 Geometric proof of push and pop

For push, take time zero at the encounter of **mem** with **sl_v** at $x = \sigma$. The memory moves left at speed c , reflects at zero, then moves right at speed c . Meanwhile **sr_v** goes right to mark v and produces a token of speed $c/3$. The reflected memory and that token meet at

$$x_0 = v + \sigma.$$

They produce leftward memory of speed c and a **fix** token of speed ac . After the memory reflects once more, their encounter has coordinate

$$x_0 \frac{1-a}{1+a} = \frac{v+\sigma}{\ell+1}.$$

The memory becomes stationary there and the acknowledgement leaves to the right. This is exactly push.

For pop, at the first memory/get encounter the memory moves left and reflects while the get token moves right at speed ac . Their meeting coordinate is

$$\sigma \frac{1+a}{1-a} = (\ell+1)\sigma = v + \rho, \quad \rho \in C_\ell.$$

The new set token moves left at speed $c/3$, so its first lower integer mark is exactly v . That mark turns it into the leftward value token vl_v of speed c . The reflected memory meets this token at ρ , becomes stationary, and returns vr_v to the right.

All additional marker crossings are precisely those in the transparent list. The values $\sigma, \rho \in [1/\ell, 1)$ keep the critical meetings away from integer marks. Each construction has a fixed finite number of moving segments and finitely many marker crossings, hence a uniform finite collision bound for fixed ℓ . Its largest local coordinate is less than $\ell + 1$. The memory and operation token remain the same two live resources throughout.

32 Reversible control and the fixed population

32.1 Alternating write and move form

Let a reversible Turing machine have disjoint write states Q_W and move states Q_M , with partial maps

$$\delta_W : Q_W \times S \longrightarrow Q_M \times S, \quad \delta_M : Q_M \longrightarrow Q_W \times \{L, R\}.$$

Require δ_W to be injective and the projection of δ_M onto Q_W to be injective. Thus each target write state has a unique preceding move and a fixed incoming direction. Require also that every move-state in the image of δ_W is in the domain of δ_M , and every used noninitial write or terminal state has an incoming move. The initial read side must agree with any existing incoming direction. A designated terminal *pair* (q, a) has no writing transition; other undefined pairs remain forbidden. The compiler may alternatively be given states whose every symbol is terminal, but this stronger condition is not imposed on Morita's table.

For a reversible quintuple table with nonzero movements, replace each transition (q, a, b, d_s, s) by the write $(q, a) \mapsto (r_s, b)$ followed by $r_s \mapsto (s, d_s)$, using one fresh move state for each target s . The local reversibility conditions from Section 30 give exactly the two injectivity requirements. Morita's table yields 15 move states and 62 writing transitions. Its q_0 has incoming direction right, so the initial read can use the right module consistently even though q_0 has predecessors.

32.2 Layout and tape invariant

Put $R = \ell + 2$. The left module has marks at $-R + i$ and memory at $-R + \sigma_L$; the right module is its reflected, disjointly labelled copy, with marks at $R - i$ and memory at $R - \sigma_R$. Put the current controller state at zero. There is one travelling operation/value token shared by the modules. The population is

$$2((\ell + 1) \text{ marks} + 1 \text{ memory}) + 1 \text{ controller} + 1 \text{ token} = 2\ell + 6. \quad (32.1)$$

In particular, $\ell = 6$ gives 18 and $\ell = 2$ gives ten. The available state labels depend on the fixed interpreter, but the number of live controller signals is always one.

At a controller read, the arriving value token holds the scanned symbol; the two stationary memories hold the cells strictly left and strictly right of the head, nearest cell first. Suppose the current write sends (q, a) to (r, b) and the next move is $r \mapsto (s, d)$. The controller sends a push- b token to the side opposite d . On acknowledgement it changes to s and sends a pop request to side d . The returned value is the new scanned symbol. This preserves the tape invariant and performs exactly one source transition.

Distinct writing rules cannot share an output unless they share both (r, b) , which injectivity forbids. Distinct acknowledgement rules cannot share an output unless they share the target s , also forbidden. The writing and acknowledgement output phases have disjoint controller-state sets. Gadget outputs contain no controller state. These observations prove injectivity of the compiled control rules and their separation from the gadget rules.

32.3 Terminal exits and conservation

For each designated terminal pair (q, a) , replace its arriving controller/value pair by a fresh stationary label $H_{q,a}$ and a fresh token $E_{q,a}$ moving right at maximum speed. Add transparent crossings of this escape label with every stationary marker and memory label. All these rows are two-to-two and their fresh labels preserve output injectivity.

At that time both memories and all marks are stationary. On the right there are $\ell + 1$ marks and one memory, all at distinct positions. Hence exactly $\ell + 2$ further crossings take the token to the rightmost mark; thereafter it has no future encounter. This is eight crossings for the literal machine and four for the binary construction. The larger bound of all $2\ell + 4$ stationary module signals is also immediate but unnecessary.

A terminal mode is recorded by the presence of $H_{q,a}$. Only the specified accepting subset is a reachability target: in the literal instance this is $H_{q_1,b}$, while $H_{q_2,s}$ records nonaccepting termination. Such a cut can precede the last escape crossing. If a collision-free cut is desired, additionally require the token to be rightmost and every adjacent velocity difference nonpositive. For an accepting label the two acceptance cuts differ by a fixed finite suffix. Acceptance is never a Zeno-limit convention.

Giving every signal energy one makes every rule conserve both number and this positive energy. Local reversibility refers to the partial collision table and finite well-defined flow; no uniqueness assertion about continuation through singular accumulation points is needed.

32.4 No accumulation on encoded runs

All active stack geometry remains outside $(-1, 1)$, apart from the controller token travelling between the center and a gadget. A center-to-gadget-to-center exchange covers distance at least two at speed at most $c_{\max}$, so takes at least $2/c_{\max}$ time. Each exchange contains only uniformly finitely many collisions, by Section 31.3. Thus an infinite encoded computation requires infinitely many exchanges and unbounded time. A finite number of source steps has finitely many events and finite duration. This excludes Zeno acceleration on the encoded domain without making a claim about every malformed rational configuration of the partial machine.

The outer marks remain at $-R, R$. At every event cut on an encoded run, including the final escape crossing, the span is exactly $2R$. The token later flies beyond R , but those later non-event observation times are not states of the event map.

32.5 The ten-signal existence specialization

Morita, Shirasaki and Gono give a simulation by a one-tape two-symbol reversible Turing machine [18]; the blank is included in that two-symbol alphabet [17]. Fix one universal ordinary interpreter and apply that conversion. If its reversible description uses separate write/move quadruples, writing becomes a zero-move quintuple and a movement becomes one identity-writing quintuple per symbol. Alphabet size does not change.

Factor transitions by their target state as above. A zero-move entry into s can be replaced by a right move into a fresh write state z_s , an identity write into a fresh move state t_s , and a left move into s . Both new maps satisfy the same injectivity conditions. The detour restores the head and tape. Initial entry is chosen consistently with the existing incoming direction, using the two orientations of the loader below; if there is no incoming move either entry can be chosen. Halting can be designated by the finitely many undefined terminal pairs. Applying the binary gadget proves Theorem 29.2; this is a compiled-existence argument, not a second printed universal table.

33 The exact input loader and its cost

33.1 Both finite tape halves

At the initial head position, list the cells strictly to its left in decreasing index order to form w_L . List the scanned cell and the cells to its right in increasing order to form w_R . Include every nonblank cell; omitted tails are blank. This works whether finite data lie on one side or both sides of the head, as happens in Morita's program/data layout.

Let $m = \max(|w_L|, |w_R|)$, append blank digits to the shorter word, and define

$$Q = \ell b^m, \quad A_\epsilon = \ell \sum_{i=1}^m d_{\epsilon,i} b^{m-i} + 1 \quad (\epsilon \in \{L, R\}).$$

Then $\sigma_\epsilon = A_\epsilon/Q$. Place the initial pop token at $x = 1$ pointing toward the right module. All initial labels and their order are fixed. Only two memory coordinates vary.

The ordered gap numerators, with a run of r equal entries denoted $Q^{[r]}$, are

$$h_0 = (A_L, Q - A_L, Q^{[\ell-1]}, 2Q, Q, Q, Q^{[\ell-1]}, Q - A_R, A_R). \quad (33.1)$$

There are $2\ell + 5$ entries, whose sum is $2(\ell + 2)Q$. For the binary case this is

$$h_0 = (A_L, Q - A_L, Q, 2Q, Q, Q, Q, Q - A_R, A_R). \quad (33.2)$$

For the literal machine it gives 17 gaps and span $16Q$. The initial mode coordinate, introduced below, is its fixed mode code times this sum.

If the initial state instead has left incoming direction, include the scanned cell in w_L and exclude it from w_R . Place a leftward pop token at -1 . The binary gap vector becomes

$$(A_L, Q - A_L, Q, Q, Q, 2Q, Q, Q - A_R, A_R).$$

For general ℓ , the vector is

$$(A_L, Q - A_L, Q^{[\ell-1]}, Q, Q, 2Q, Q^{[\ell-1]}, Q - A_R, A_R).$$

These are two fixed entry conventions, each with a fixed initial label order. The literal six-symbol interpreter uses right entry.

33.2 Arithmetic and representation cost

The rational loader uses $O(m)$ multiply-by- b and add-digit operations on $O(m)$ -bit integers for fixed alphabet. This is linear in the prepared tape-word length in an arithmetic-operation model; it is not a constant-cost conversion from a raw base-two integer or a claim of linear bit complexity with arbitrary multiplication primitives. Ordinary binary input integers require digit extraction, source-interpreter recoding, length/power construction, and base conversion. Those costs must be paid either in preprocessing or by a separate arithmetic representation theorem.

The source program in (30.1) has length linear in its explicit production description, and its two tape halves can be obtained effectively. The separate universal reduction into that source language is imported from the cited simulation theorem, not counted as free computation.

Keeping rational positions gives a fixed physical box. Scaling all positions and times by Q instead gives integer initial positions with the same fixed speed and rule tables, at the price of input-dependent physical scale. These two conventions have the same live population and event ordering.

34 The complete simultaneous-event normal form

The next results do not depend on stacks or Turing machines. Fix any finite partial signal machine with rational speeds and fixed population $n \geq 2$, whose rules preserve the cardinality of each collision block and whose input and output sets each have pairwise distinct speeds. Rescale the time unit once so all speeds are integers.

34.1 Post-event germs and first-collision guards

A state at an event cut is an ordered label word $a_1, \dots, a_n$ and a gap vector $g \in \mathbb{Q}_{\geq 0}^{n-1}$. Put $v_i = v(a_i)$ and $c_i = v_i - v_{i+1}$. We require

$$g_i = 0 \implies v_i < v_{i+1}. \quad (34.1)$$

Thus signals emitted at the same point are sorted by strictly increasing outgoing speed. This is a post-event *germ*, not an arbitrary coincident snapshot. It prevents zero-time recollision and equal-speed overlap. Strictly ordered initial slices satisfy the condition automatically.

If every $c_i \leq 0$, there is no future collision. Otherwise each closing edge has $g_i > 0$ by (34.1); its putative encounter time is g_i/c_i . Let J be the nonempty set attaining the smallest such time and fix the canonical pivot $j = \min J$. For a fixed mode and candidate J , the exact branch guards are

$$c_j > 0, \quad g_j > 0, \quad (34.2)$$

$$c_j g_i - c_i g_j = 0 \quad (i \in J), \quad (34.3)$$

$$c_j g_i - c_i g_j > 0 \quad (i \notin J). \quad (34.4)$$

The first condition is a compile-time test on speeds. These guards imply that every $i \in J$ is closing; equality would otherwise contradict nonnegativity and the germ condition.

Lemma 34.1 (Complete event selection). *Under the germ condition and guards (34.2)–(34.4), the next positive collision time is $\tau = g_j/c_j$. Exactly the adjacent edges in J have zero gap then. The maximal connected components of J are the complete collision blocks.*

Proof. Every gap evolves affinely as $g_i - c_i t$. At $t = \tau$, multiplication by $c_j > 0$ gives exactly the expressions in (34.3)–(34.4). The endpoint gaps are nonnegative, zero precisely for $i \in J$. An initially positive gap remains positive for $0 < t < \tau$, as a convex combination of its positive initial value and nonnegative endpoint. An initially zero gap has $c_i < 0$ and is strictly positive for every $t > 0$. Hence no collision is earlier. Zero endpoint gaps join exactly the contiguous signals occupying the same endpoint position. $\square$

[write] Lemma 34.1 is the one-batch case of the chamber theorem of Part I, Theorem 5.3, with its typed endpoints (Lemma 5.2); Part II proves the same step as Lemmas 21.1 and 21.2. The guards (34.3)–(34.4) are exactly Part I’s equality and strict rows for one batch, written after multiplication by the positive pivot speed, and a batch with several components of J is a batch of several sites in Part I’s terms. The one addition is the germ condition (34.1): Parts I and II start from strictly ordered positions, while an event cut here may carry zero gaps between diverging outputs, and the proof’s sentence on initially zero gaps disposes of them. Source 12 was written without these Parts; its proof is kept as a second route. What is new in this Part is the use made of the lemma: the mode-indexed integer map of Section 35, the pivot-scaled lift, and the literal universal machine to which they are applied.

For every nontrivial collision block, its exact incoming set must be a defined rule input. Apply all such rules atomically and sort each output by speed. An undefined block has no branch. Thus a triple arrival cannot be replaced by a preferred pairwise rule, and simultaneous collisions at distinct locations cannot be processed with an accidental temporal priority. The literal table has only two-input rules, so a genuine triple collision is undefined.

34.2 Why no coordinate permutation is needed

The endpoint gap vector before label replacement is

$$\bar{g} = A_j g, \quad A_j = I - \frac{c e_j^\top}{c_j}. \quad (34.5)$$

Each collision replaces k signals at one position by exactly k signals at that position. Therefore the ordered multiset of positions is unchanged by replacement. Reordering outgoing labels at a coincident point changes the mode, but not the gap vector. Consequently $g' = \bar{g}$: equation (34.5) is the entire branch map even when several collision blocks occur simultaneously.

The matrix is a rank-one *correction* to the identity, not a rank-one matrix. Equal-cardinality at each block is important. Merely conserving the total population across a collection of events would not by itself imply this coordinate identity.

35 Integer lifting and internalized finite control

35.1 The pivot-scaled integer lift

For a branch with canonical pivot j , the rational projection from (34.5) has only the denominator c_j . Multiplying by that very denominator gives

$$h_{k+1} = N_j h_k, \quad N_j = c_j I - c e_j^\top \in \mathbb{Z}^{(n-1) \times (n-1)}. \quad (35.1)$$

Choose Q clearing the initial gap denominators and let

$$\lambda_0 = Q, \quad \lambda_{k+1} = c_{j_k} \lambda_k, \quad h_k = \lambda_k g_k. \quad (35.2)$$

Here c_{j_k} is the positive closing speed of the uniquely selected branch at step k . All guards are homogeneous, so they apply unchanged to h_k . The multiplier is determined by the branch, and no extra scale coordinate, division, gcd reduction, or denominator witness is needed in the integer map.

Conversely, a legal integer orbit determines its branch sequence, hence all λ_k . Set $g_k = h_k / \lambda_k$ and reconstruct the event delay as

$$\tau_k = \frac{h_{k,j_k}}{\lambda_k c_{j_k}}.$$

Lemma 34.1 and induction reconstruct exactly the rational geometry, up to global translation. This proves finite-orbit equivalence in both directions, including simultaneous events.

Proposition 35.1 (Sparse scaled projections). *For every branch with $c_j > 0$,*

$$A_j^2 = A_j, \quad N_j^2 = c_j N_j, \quad \text{rank } N_j = n - 2.$$

Row j vanishes. Every other row has a diagonal entry c_j and possibly one further entry in column j . There are at most $2(n - 2)$ nonzero entries. If all speed differences have absolute value at most V , every matrix entry has absolute value at most V .

Proof. Since $e_j^\top c = c_j$, multiplication expands to $A_j^2 = A_j$. Its kernel is exactly the span of the nonzero vector c : $A_j c = 0$, while $A_j x = 0$ implies $x = c x_j / c_j$. Its rank is $(n - 1) - 1$. Since $N_j = c_j A_j$, its square and rank follow. For $i \neq j$ the only possible nonzero row entries are c_j in column i and $-c_i$ in column j , proving the sparse and coefficient bounds. $\square$

For the literal machine, the 17-dimensional gap matrices have rank 16, at most 32 nonzero entries, and entries of absolute value at most 24. The binary existence construction has nine-dimensional gap matrices of rank eight, at most 16 nonzero entries, and entries of absolute value at most 12. These bounds do not depend on the source interpreter's number of states.

35.2 A uniform alternative for comparison

It is also valid to use one fixed multiplier

$$D = \text{lcm}\{|v - v'| : v \neq v' \text{ are available integer speeds}\},$$

with $D = 1$ if the set is empty. This gives $\tilde{h}_k = Q D^k g_k$ and

$$\tilde{h}_{k+1} = M_j \tilde{h}_k, \quad M_j = D I - \frac{D}{c_j} c e_j^\top, \quad M_j^2 = D M_j. \quad (35.3)$$

The literal machine has $D = 65520$, with coefficient bound $8D = 524160$ because the minimum positive speed difference is 3. The binary speeds give $D = 360$ and coefficient bound 4320. This alternative is exact and useful for cross-checking the replay, but is unnecessary for a pure integer reachability frontend: the pivot-scaled matrices are much smaller. Both conventions have the same homogeneous guards and mode sequence.

[write] The multiplier D is the least common multiple of the nonzero speed differences used in Parts I and II ((7.1); (18.1)), and the uniform lift is their device: after k events every gap is an

integer once multiplied by QD^k , which is the event-by-event denominator bound of Theorem 7.1 (with the event count bounding the causal depth) and the grid $B = D^K$ of Part II on which Lemma 19.1 places the event coordinates. The pivot-scaled lift (35.2) sharpens that device for one branch sequence: it multiplies by the actual pivot speed c_j instead of D and needs no extra coordinate, and on the primary example it ends with a largest gap numerator of 39,320 bits against 179,449 under the uniform QD^k lift (Section 37).

35.3 Control uses one coordinate

There are finitely many ordered n -label modes; enumerate them injectively by positive integers q . This can be done lexicographically, including modes later discarded as invalid. Put

$$S = \mathbf{1}^\top h, \quad u = qS, \quad X = (h, u) \in \mathbb{N}^n. \quad (35.4)$$

Require $S > 0$. This mode encoding is used on the positive-span invariant domain of the compiled machines. An arbitrary signal machine can collapse its entire population to one point; at such a zero-span cut this particular coordinate does not distinguish outgoing modes and needs a separate convention. The generic gap theorem itself does not require positive span. In fact, positive span is invariant under every legal step of either compiled table, not only its encoded runs: zero output span would mean that all n particles form one complete collision block, requiring an n -input rule, whereas these tables have only two-input rules and $n > 2$. Each branch for mode q includes the homogeneous equality $u - qS = 0$, applies (35.1), and sets

$$u' = q'\mathbf{1}^\top N_j h, \quad (35.5)$$

where q' is the uniquely determined outgoing mode. These equations define a homogeneous integer-linear matrix $\widehat{M}_r$ on all n coordinates. Its last input column is zero; the mode is read through its guard. The small coefficient bounds 24 and 12 apply to the gap block; the added mode row contains the fixed, potentially large codes q' .

The guard also includes the germ conditions. On natural gap variables these are particularly simple: whenever $c_i \geq 0$, require $h_i > 0$; when $c_i < 0$, no extra constraint is needed beyond $h_i \geq 0$. The remaining guards are the first-collision equalities and inequalities. They are homogeneous rational polyhedral-cone conditions, with strict positivity interpreted on natural arguments as value at least one.

For the literal release, give the 114 labels their zero-based lexicographic indices $\iota(a)$ and use the explicit code

$$q(a_1, \dots, a_{18}) = 1 + \sum_{i=1}^{18} \iota(a_i) 114^{18-i}.$$

The initial and final collision-free modes have codes

$$\begin{aligned} q_{\text{initial}} &= 473631179350110573467332809270905830, \\ q_{\text{accept}} &= 473631179350110518412162422097252939, \\ q_{\text{null}} &= 473631179350110521664110943253890124. \end{aligned}$$

Their exact label orders and decimal codes are included in the machine JSON. Thus $u_0 = 16q_{\text{initial}}Q$, while the final accepting target is the single equality $u = q_{\text{accept}} \sum h$ on admissible post-event germs. The null code is a different nonaccepting target. The code values are serialized as strings to avoid loss of precision in JSON consumers using floating-point numbers.

Distinct mode codes cannot overlap when $S > 0$. For a fixed valid mode and gap vector, the set of minimizing edges is unique and its pivot is fixed. The branch guards are consequently disjoint. Enumerating every mode and every nonempty subset of edges, discarding impossible or undefined cases, is a complete finite branch compiler. This proves the pure n -coordinate claims of Theorems 29.1 and 29.2.

35.4 Limits of the normal form

No injectivity of the integer event map is asserted. Reversing a signal flow, quotienting global translation, forgetting elapsed time, taking an event section, and changing the common scale are different operations. The original partial collision table is reversible; that does not automatically make (35.1) invertible. [write] [Added 4 October 2026: still true of this Part. Source 31 (Part XI) proves that a fixed complete binary word with positive flights induces an invertible linear map between full sections (Theorem 257.1); that concerns one word between full sections, not the event map (35.1) across the branches of a universal machine, and retracts nothing here (Section 239).]

For a general configuration, summing the pivot-scaled update gives

$$\mathbf{1}^\top N_j h = c_j \mathbf{1}^\top h - (v_1 - v_n) h_j.$$

It is not valid in general to discard the second term. On encoded runs, the extreme markers are stationary at the input of every event, including the last escape crossing. Their fixed event-cut span gives

$$S_k = \sum_i h_{k,i} = 2(\ell + 2)\lambda_k, \quad S_{k+1} = c_{j_k} S_k.$$

At the last cut the token may be the rightmost outgoing germ, but there is no subsequent event requiring stationary extrema. For all these encoded cuts the geometry and time can be decoded directly from the integer state:

$$g_k = \frac{2(\ell + 2)}{S_k} h_k, \quad \tau_k = \frac{2(\ell + 2)h_{k,j_k}}{S_k c_{j_k}}.$$

For the literal machine $S_k \leq 16Q 24^k$, and for the binary specialization $S_k \leq 8Q 12^k$. Every gap is bounded by S_k , and $u_k \leq q_{\max} S_k$. Their bit length is $O(k + \log Q)$ for the fixed machine. Fixed coordinate dimension does not make integer-height growth free.

Exact points and accepting modes are different targets. In the literal mode closure of Appendix H, every active branch has stationary extreme labels, and every pivot speed is at least 3. Thus $S' \geq 3S$. Given two exact positive-span integer states $X_{\text{in}}, X_{\text{out}}$ in this domain, point-to-point reachability is decidable by deterministic simulation for at most $\lfloor \log_3(S_{\text{out}}/S_{\text{in}}) \rfloor$ steps when the ratio is at least one; a smaller target span rules out a positive-length run. For the fixed machine this costs polynomial time in the binary endpoint length. The universal question is reaching the fixed accepting mode or cone with unbounded output span, not reaching one prescribed integer point. No endpoint-size bound for acceptance is supplied.

Optional exact elapsed time. One additional natural coordinate suffices if the absolute collision timestamp is wanted. Starting at time zero, let $p_k = \lambda_k t_k$. The same branch updates it by

$$p_{k+1} = c_{j_k} p_k + h_{k,j_k}, \quad p_0 = 0.$$

Indeed, multiplying $t_{k+1} = t_k + h_{k,j_k}/(\lambda_k c_{j_k})$ by $\lambda_{k+1} = c_{j_k} \lambda_k$ gives exactly this integer-linear row. On literal encoded runs the timestamp is $t_k = 16p_k/S_k$. The timed frontend has 19 coordinates, or 11 in the binary specialization; the extra row still has coefficients bounded by 24, respectively 12. Its finite-horizon packet remains quadratic. This optional coordinate is separate from the 18-dimensional untimed reachability theorem.

The dimension retains every physical collision and is not an optimized arithmetic state dimension. For comparison, at completed Turing-machine reads a standard two-stack macro-description can use (A_L, A_R, Q, u) , with u coding the finite state and scanned-symbol pair times Q . A right move writing w and popping v from the right can use

$$Q' = 7Q, \quad A'_L = A_L + wQ, \quad A'_R = 49A_R - 7vQ,$$

with digit guard $vQ < 7A_R < (v+1)Q$; left moves are symmetric. This coarser quotient skips internal collisions and is distinct from the sparse projection frontend. Elementary stack arithmetic and the selector/copy technique are used as standard algebraic mechanisms, not claimed as new inventions.

[write] The selector/copy construction of Section 36, in which one selector per branch and one copy of the input per branch represent a disjoint union of polyhedral pieces, is the disjunctive (convex-hull) formulation of a union of polyhedra due to Balas [26, 27], here in homogeneous form with the selectors entering the strict guards; Jeroslow and Lowe [28] characterized the sets that such formulations represent with integer variables. Source 12 cites neither; the credit is added here, and source 12 claims no novelty for the technique.

36 A direct canonical quadratic Diophantine packet

36.1 Finite linear branch systems

Let F be a partial map on $\mathbb{N}^d$ with finitely many disjoint branches $r = 1, \dots, B$. Branch r has update $Y = \widehat{M}_r X$, where $\widehat{M}_r$ has integer entries, and finitely many homogeneous integer-linear guards of three kinds:

$$L(X) = 0, \quad L(X) > 0, \quad L(X) \geq 0.$$

The event map above has this form with $d = n$. Naturality of the input coordinates is part of the domain; its mode, germ, nonzero-span, first-collision, tie, and collision-legality conditions are included in the branch description. Collision legality is a finite compile-time filter.

For every branch introduce a selector $b_r \in \mathbb{N}$, a local input copy $X_r \in \mathbb{N}^d$, and one slack $s_{r,L} \in \mathbb{N}$ for each strict or weak inequality. Form the following affine residuals:

$$\sum_{r=1}^B b_r - 1, \tag{36.1}$$

$$X_i - \sum_r X_{r,i} \tag{36.2} \quad (1 \leq i \leq d),$$

$$Y_i - \sum_r (\widehat{M}_r X_r)_i \tag{36.3} \quad (1 \leq i \leq d),$$

$$L(X_r) \tag{36.4} \quad \text{for each equality guard,}$$

$$L(X_r) - b_r - s_{r,L} \tag{36.5} \quad \text{for each strict guard,}$$

$$L(X_r) - s_{r,L} \tag{36.6} \quad \text{for each weak guard.}$$

Let $\mathcal{R}$ be this list and define the explicit polynomial

$$P(X, Y; \mathbf{z}) = \sum_{R \in \mathcal{R}} R^2 + \sum_{r=1}^B \left(\sum_{h \neq r} b_h \right) \left(\sum_{i=1}^d X_{r,i} \right). \quad (36.7)$$

All sums here are finite and their coefficients are directly computable from the branch table. The second group consists of nonnegative complementarity products on natural arguments. No appeal to MRDP is involved.

Theorem 36.1 (Exactness and canonical witness). *For $X, Y \in \mathbb{N}^d$, there exists $\mathbf{z} \in \mathbb{N}^A$ with $P(X, Y; \mathbf{z}) = 0$ if and only if $Y = F(X)$. Whenever such a tuple exists, it is unique. The polynomial has total degree at most two and is nonnegative on every natural assignment.*

Proof. Every square and every displayed complementarity product is nonnegative on natural arguments. A zero therefore makes all terms vanish. Equation (36.1) forces exactly one selector, say b_{r_*} , to be 1 and all the others to be zero. For every $r \neq r_*$, the first factor of its complementarity term is 1, so all coordinates of X_r are zero. Equation (36.2) then gives $X_{r_*} = X$.

Homogeneity makes every inactive guard form zero; equations (36.5) and (36.6) consequently force its slack to zero. On the active branch an equality guard vanishes, a strict guard has value $1 + s_{r_*,L} \geq 1$, and a weak guard has value $s_{r_*,L} \geq 0$. Thus X lies on that branch and (36.3) gives $Y = \widehat{M}_{r_*} X$.

Conversely, for a legal step set its unique branch selector to 1, set its local copy to X , set its strict slack to $L(X) - 1$ and weak slack to $L(X)$, and set every inactive quantity to zero. These are natural values and every term vanishes. Disjoint branch guards fix the active selector, and all copies and slacks are then forced, proving uniqueness. The residuals are affine and the additional products are quadratic, proving the degree bound. $\square$

The product must use $\sum_{h \neq r} b_h$, as written. Replacing it by $1 - b_r$ would lose global nonnegativity on arbitrary natural assignments before the selector equation has been enforced. The polynomial is not asserted to be convex or a sum of squares over all real arguments.

[write] Two remarks place Theorem 36.1 in the collection. (i) Source 12 states nonnegativity on natural arguments only, but P is nonnegative on the whole real nonnegative orthant: the residuals enter squared, and each complementarity product $(\sum_{h \neq r} b_h)(\sum_i X_{r,i})$ is a product of two affine forms with nonnegative coefficients. So (36.7) has exactly the form $\sum_i A_i^2 + \sum_j B_j C_j$ of the spine of *quadratic-orthant-certificates* (its equation `qoc:eq:spine`), the format of that report's three Parts. This is a statement about the sign of P only; no claim is made here that the nonnegative real zeros of P are natural. (ii) The one-step graph $\{(X, F(X))\}$ of a homogeneous-guarded piecewise-linear map on $\mathbb{N}^d$ with finitely many branches is a semilinear set, so the *existence* of a single-fold representation of degree two, nonnegative on the real orthant, also follows from Theorem `cdc:of:thm:singlefolds1` of Part XI of *canonical-diophantine-certificates*, via Presburger quantifier elimination. The construction here is explicit, needs no elimination, and carries the exact ledger (36.8)–(36.9); that is what this Part contributes, not the existence statement. The fixed-horizon sum (36.10) is of the same “one natural zero, fixed horizon” kind as the trace polynomial of Theorem `pqc:pm:thm:trace` in *probabilistic-quantum-and-continuous-computation*, which handles a different dynamics with a clamp gadget; that network is total, so its trace polynomial has exactly one natural zero, whereas the event map here is partial and the natural fibre of (36.10) is empty or a singleton, “one natural zero” meaning at most one. (The text after the semicolon was added on 2 October 2026, after finding 3 of the review cited at the end of the next paragraph.)

[write] **Added 2 October 2026 (batch 79): the strong selector gates of *quadratic-orthant-certificates*.** Part V of that report (its source 16, batch-79 manuscript 09, written in commit 11abe5008) uses the same selector device for the branches of literal counter programs, with one change: its products $(\sum_{k \neq r} e_{j,k})(e_{j,r} + \sum_i x_{j,r,i})$ (its equation `qoc:um:eq:polynomial`) add the selector itself to the right factor. Two positive selectors would then make some product positive, so the gate forces one-hot selection over the nonnegative reals, and for natural inputs the nonnegative real zero fibre of that polynomial is its natural fibre (its Theorem `qoc:um:thm:poly`); its source 17 isolates the gate as Lemma `qoc:rn:lem:gates` and uses it again for reset-net traces in its Part VI. The products of (36.7) lack that term, and source 12 claims exactness over natural witnesses only. An elementary observation, not made by either source, shows how far the present gates reach. At a nonnegative real zero of (36.7), if two selectors were positive, every first factor $\sum_{h \neq r} b_h$ would be positive, so every local copy X_r would vanish and $X = 0$ by (36.2). Hence, for natural $X \neq 0$ and natural Y , exactly one selector equals one, every inactive copy and slack vanishes, the active slacks are the integers $L(X) - 1$ and $L(X)$, and the nonnegative real fibre is the natural one. At $X = 0$ all local copies vanish, and (36.5) forces $b_r = 0$ for every branch with a strict guard. Every branch with only homogeneous equality and weak guards contains zero; disjointness therefore allows at most one such branch. The selector equation then makes the whole packet's nonnegative real fibre natural at zero as well (or empty). The weak gates alone do not enforce one-hot selection at zero. Adding b_r to each right factor, as in that report, would enforce it independently of branch disjointness, without changing the counts (36.8)–(36.9). Neither report re-proves a theorem of the other. (Corrected in place on 2 October 2026 following the Hilbert's-tenth programme's review `review_batch79_j2_bbc67d225.md` of `Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/` (commit 37a829e0b, finding 4): this note had ended by calling fractional selectors at $X = 0$ an exception of the packet; they are possible for the weak gates in isolation only, and the sentences on $X = 0$ above replace that ending. For the actual event branches, whose positive-span and positive-pivot guards exclude zero, the case $X = 0$ does not arise.)

36.2 The exact witness and residual ledger

Let E, T, W be, respectively, the total counts of equality, strict, and weak guards across all B branches. There are

$$A = B(d + 1) + T + W \quad \text{auxiliary natural variables,} \quad (36.8)$$

$$R = 1 + 2d + E + T + W \quad \text{squared affine residuals,} \quad (36.9)$$

$$B \quad \text{unsquared nonnegative quadratic products.}$$

The variables are exactly B selectors, Bd local coordinates, and $T + W$ slacks. For the literal frontend, replace d by 18, so $A = 19B + T + W$ and $R = 37 + E + T + W$. For the binary existence frontend, $A = 11B + T + W$ and $R = 21 + E + T + W$.

This is a variable-and-residual ledger, not a minimal arithmetic-gate count. Expanding a squared linear form may create many monomials. The release gives the literal machine's numeric ledger and a complete factored coefficient description in Appendix H. No minimization or flat monomial expansion is claimed.

36.3 Fixed horizons and terminal reachability

For an externally fixed $K \geq 1$, introduce intermediate states $X_1, \dots, X_{K-1}$ and a separate packet witness $\mathbf{z}_k$ for each step. Define

$$P_K(X_0, X_K; \mathbf{Z}) = \sum_{k=0}^{K-1} P(X_k, X_{k+1}; \mathbf{z}_k). \quad (36.10)$$

There are exactly $KA + (K - 1)d$ witness variables with the two endpoints supplied as parameters, KR squared affine residuals, and KB complementarity products. The polynomial is quadratic and nonnegative on natural arguments. By Theorem 36.1, its natural fiber is empty or a singleton: determinism fixes every intermediate state, and the step witnesses are canonical.

A designated accepting mode is a finite disjunction of homogeneous linear mode equalities, optionally with the linear collision-free condition. Its finite disjunction can be handled by the same selector/copy construction. If the intended question is *first* acceptance at step K , disallow accepting input modes in the earlier branch packets. Nonaccepting null modes are never promoted to accepting ones. Reaching H and finishing the escape suffix are separate predicates, as discussed above.

The statement “there exists some finite K ” is an unbounded union of these fixed-horizon relations. A uniform fixed-arity Diophantine representation of that union would need an additional sequence or iteration compression theorem. None is supplied here. Universality of the dynamics and existence of a direct quadratic *step* polynomial do not remove that requirement.

37 Exact replay and the release contract

37.1 Literal machine counts

For $\ell = 6$, the two stack copies contribute 80 meta-signals and 334 rules. The controller has 15 write-state labels and 15 move-state labels. The 62 source transitions give 62 writing rules and the 15 target states give 15 acknowledgement rules. Two designated terminal pairs contribute four fresh labels, two terminal rules, and 32 transparent escape rules, one per terminal pair and each of the 16 stationary module labels. Hence

$$80 + 30 + 4 = 114, \quad 334 + 62 + 15 + 2 + 32 = 445.$$

The exported table lists every label with its integer speed and every input/output pair. Undefined encounters are explicitly forbidden; there is no implicit identity completion.

37.2 Replays and their meaning

The Python replay uses standard-library integer arithmetic and exact fractions. The core stack suite checks 1,120 operations over alphabets 2, 3 and 4, totaling 24,028 collisions and 34 simultaneous batches. A small compiled binary source machine is tested on all 63 binary words of length at most five, totaling 5,045 batches. These tests cover gadget behavior, conservation, local injectivity, complete earliest-event guards, simultaneous processing, and the rational/integer branch identity.

The source-machine audit checks all 184 transitions of Morita’s published example in both directions. The Figure 28 snapshots at times 0, 6, 59, 178 and 184 match state, head, and full displayed tape window. A separate bounded differential suite covers 2,430 source cyclic-tag cases,

7,687 macro checks and 169,019 Turing microsteps; 40 cases remain active at its 12-macrostep bound. A q_0 state alone is not a reliable macro boundary, so that test also compares the restored program, phase marker, and active word.

For the literal signal machine, each controller read is checked against an independent ordinary tape: controller state, returned symbol, and both complete rational half-stack values must all agree. The tested cases are

Program and word	TM steps	Signal batches	Outcome
(halt, YN, YYN), NYY	184	11214	halt
Same program, Y	2	154	halt
Same program, empty word	1	106	null
(halt), N	11	722	null
(halt, ε), NY	25	1602	null
Total	223	13798	

All these signal batches are single-collision batches, and each has an eight-crossing escape suffix. The primary example's final exact clock is

$$\frac{142256492930823919}{199397583417606}.$$

An additional event-timing implementation computes all pairwise intersections and groups coincident endpoints independently of the adjacent-gap selector. It matches all 11,214 primary-example batches and verifies the cumulative QD^k integer lift at every batch; the final largest numerator has 179,449 bits. Its generic suite checks 8,092 valid germs, including 1,184 simultaneous-event batches, 352 triple-or-larger events, and 3,430 initially zero-gap germs. A missing triple rule is explicitly rejected. There are also 1,296 two-half loader cases and 32 two-entry-side compiled runs totaling 2,574 batches. The pivot-scaled suite independently checks all 13,798 literal batches and 7,282 generic events, including 4,708 events whose total span changes. The primary example's final largest gap numerator has 39,320 bits under pivot scaling, compared with 179,449 under uniform scaling. It also checks exact fixed-span geometry and delay decoding, the production adaptive-lift helper, and the optional timestamp numerator at every event; the primary example's final time numerator has 39,329 bits. These are checks of the same rule data, not an independent proof of universality.

These finite comparisons substantiate the literal implementation. The proof for all inputs is the stack geometry, control invariant, and imported source universality theorem, not extrapolation from the tested cases.

37.3 Reproduction and boundaries

The release has four parts: the article source and PDF; original exact replay code; complete mathematical machine data; and public-safe replay receipts with provenance links. The run script recreates the source table and its checks, the generic stack checks, and the literal signal table and replays. A separate generic event-branch compiler and arithmetic packet demonstration exercise exact canonical witnesses and residual counts on a fully enumerated finite branch system. A separate closure compiler enumerates a forward-closed mode family for the literal initial mode and records all 80,501 numeric event branches, as described in Appendix H. This avoids enumeration of all 114^{18} possible label words.

No third-party source PDFs, source-page images, private working notes, or third-party software are bundled. Source PDFs can be obtained from the linked primary locations; their SHA-256 digests are recorded for comparison. The Python programs use only the standard library. Rebuilding the article additionally requires a standard LaTeX installation.

The strongest literal deliverable is the 18-live machine. The smaller ten-live result is derived. The general event theorem includes exact ties. The direct Diophantine result concerns a step and every fixed finite horizon. No fixed-arity polynomial for arbitrary halting histories, no fastest loader, and no optimal signal-count claim is made.

[write] **The release in this report.** “The release” is the delivered archive. Its programs are shipped as `code/12-conservative-signal-*` and its records as `data/12-conservative-signal-*`, with the delivered sub-directories `code/`, `numeric/`, `correction/`, `receipts/` and `data/` flattened into these two directories, and its `CORRECTION.md`, `SOURCE-PROVENANCE.md` and `numeric/PROOF_AND_LEDGER.md` at the report root under the same prefix; the README of this report maps every name. The delivered article source and PDF and the delivered README are not shipped: this Part prints the article, Appendix F prints its Morita table, and this report’s README replaces the delivered one. The two compact closure records `numeric/MODES.jsonl.gz` (1,054,196 bytes) and `numeric/EVENTS.jsonl.gz` (580,458 bytes) are *not* shipped either: they are regenerated by the shipped compiler in about a minute, with identical decompressed content, and the README gives the command (“Reconstructing the excluded data”); the machine file, the ledger and the circuit file named in Appendix H are shipped, and the verbose branch and residual-row files were already omitted from the delivered archive, as that appendix says. The programs need the delivered layout and must be run on a copy (README).

[write] **The corrected archive and the repository’s review.** This Part prints the corrected edition of the archive (batch 79 manuscript 16). An earlier edition (manuscript 07 of the same batch) has the same article, data and receipts, and differs only in the generic packet evaluator `code/quadratic_packet.py`. The repository’s Hilbert-tenth-problem programme reviewed that earlier edition [29]: it passed the stated construction, independently enumerated the complete outgoing successor set of all 49,700 modes and all 80,501 branches (the least mode-only closure), checked that the trillion-term monomial count of Appendix H cannot be reduced by cancellation, and found genuine input-validation defects in the generic evaluator (a padded output (1,999), a floating endpoint and a surplus slack were accepted). It wrote the repair `conservative_signal_packet_domains.patch` [29]. The corrected edition applies exactly that patch and says so in its `CORRECTION.md` (shipped as `12-conservative-signal-CORRECTION.md`); the shipped evaluator `code/12-conservative-signal-quadratic_packet.py` is the patched one, and the correction receipts are shipped. The repair does not change Theorem 36.1, the literal construction or any ledger, and it does not make arbitrary caller-supplied overlapping branch families disjoint. The programme then reviewed the corrected edition itself [31]: it reread the article and the proof notes, compared the two archives member by member, passed all ten original author entryptoints and the three correction runs, rechecked one explicit integral event for every branch, and found no new defect on the valid domain. That review also proves a reduction that source 12 does not claim, a natural-domain guard projection: deleting the germ rows, the span row and the first-event rows of unselected edges with negative closing speed (18 rows and slacks per branch) is a bijection of complete *natural* zero sets, though not of nonnegative real ones, and reduces the literal one-step packet from 4,196,998 to 2,747,980 witnesses and from 2,762,961 to 1,313,943 squared residuals, with an explicit evaluation schedule 7,489,424 operations shorter. The projection is the review’s, is not re-proved here, and changes no theorem of this Part. Neither review is a proof-assistant verification, and neither finds a new universal operation bound.

Part IV

Sparse Diophantine certificates for finite mass lattice dynamics (source 13)

[write] Part IV prints batch-79 manuscript 19, *Sparse Diophantine certificates for finite mass lattice dynamics*, “Research report prepared with AI assistance for Vladimir”, dated 2 October 2026, from the archive `Sparse_Lattice_Diophantine_Certificates.zip`. It is called source 13 after the prefix of its shipped files. The delivered article is a main file that reads five further files; they are printed here in the delivered order, without its separate table of contents. Its labels carry the prefix `smc:sl:`; its sections are Sections 38–48 here, and it has no appendices. Its statements keep their numbering by section; equations are numbered by section (the source numbered them consecutively). Source 13 names no repository commit and cites no repository file. Its comparison and sorting gadgets and its post-elimination compiler re-derive devices of *canonical-diophantine-certificates*; [write] notes at those places say so, and Section 1.7 describes the Part. Its abstract follows.

Finite conserved mass bounds the number of occupied unit records without bounding the distances between them. We exploit this distinction to compile a fixed finite horizon of one-dimensional partitioned lattice dynamics into one quartic sum of squares with a unique natural witness. The construction explicitly pays for collision detection, simultaneous redistribution, sorting, finite table lookup, and empty space. We give exact witness and residual ledgers, a coefficient-level reproducible implementation, and an optional strengthening whose nonnegative-real fiber is also exact. A source-critical interface to the reversible conservative construction of Morita and Imai includes a necessary sign correction and a reachable-legality hypothesis; a fully specified doubling example is separated from the existence of a fixed safe universal source. Complementary proofs show decidability under bounded orbit diameter and, without a diameter promise, at total mass at most two. At mass at most two, a stronger effective semilinearity argument yields horizon-free canonical quartics of fixed rule-dependent arity. No fixed-arity representation of unbounded universal computation, new classical universality result, or minimum universal mass is claimed.

38 Results and model

The geometric resource in this report is an unbounded integer gap, rather than an unbounded number of occupied cells. A conserved positive integer mass M permits a canonical description by M repeated unit records, even when the occupied positions are very far apart. The goal is to verify every lattice microstep directly on those records, including collisions and the assertion that no output has been omitted in empty space.

Three levels of claim must be kept separate. First, the main certificate theorem applies to *any supplied total conservative finite table*. Second, the corrected Morita–Imai compiler provides explicit reversible examples and an interface from suitably legal two-counter computations. Third, a fixed safe universal source exists by classical simulation constructions, but the release does not print a particular universal source machine and its numerical state count. Its fully printed worked machine computes doubling and is not universal.

The sparse construction and its exact ledgers are the arithmetic contribution considered here. The circuit-to-sum-of-squares device and classical reversible computational universality are not new claims. Dense fixed-box or ring tableaux already record bounded cellular computations; a direct sparse infinite-line argument additionally has to justify grouping, ordering, and empty space.

Priority of the particular constructions and the mass-two theorem below has not been established by an exhaustive literature review.

[write] The dense fixed-box tableau is, for example, the finite cellular-automaton compiler of Part VIII of *canonical-diophantine-certificates* (its Section `cdc:mem:sec:substrates`), whose one-hot tableau occupies a fixed box; this Part is the sparse infinite-line counterpart for one-dimensional partitioned conservative dynamics. Source 13 was written without this report and without that one.

Throughout, $\mathbb{N} = \{0, 1, 2, \dots\}$. Fix $K \geq 1$, put $q = K + 1$, and supply every row of a function

$$g : \{0, \dots, K\}^3 \longrightarrow \{0, \dots, K\}^3, \quad r + c + l = g_L(r, c, l) + g_C(r, c, l) + g_R(r, c, l).$$

A configuration c gives a triple $(L_c(x), C_c(x), R_c(x))$ at each $x \in \mathbb{Z}$. One update is

$$F(c)(x) = g(R_c(x - 1), C_c(x), L_c(x + 1)). \quad (38.1)$$

Thus the *input* order of the table is (R, C, L) and its *output* order is (L, C, R) . In physical terms the right channel streams right, the left channel streams left, and the center stays in place; g then redistributes the incoming mass. The input-output order matters even when the numerical table happens to be the identity.

Conservation forces $g(0, 0, 0) = (0, 0, 0)$. Finite support stays finite, and the sum of all channel values remains M . If g is a permutation, then F is globally bijective: apply g^{-1} at each site and reverse the two lane shifts. Both operations preserve the zero background and finite support. Reversibility is useful for the intended substrate, but is unnecessary for the forward certificate theorem.

38.1 Canonical records and fixed parameters

Let the valid initial configuration be nonempty, with minimum occupied position o and diameter D . Replace each channel mass by that many copies of a record (x, z) , where $z = 0, 1, 2$ means left, center, right. Sort lexicographically, and shift the physical position by $T - o$, where $T \geq 0$ is the externally fixed horizon. Thus a record at physical site p has

$$x = p - o + T.$$

The initial row has exactly M records. At time $t \leq T$, every position lies between $T - t$ and $D + T + t$. Repeated records are allowed and have no individual labels. The sorted repeated-unit row therefore encodes the physical configuration uniquely. There is neither zero padding nor a hidden list of empty sites.

The initial data, its translation, the local table, and the horizon are compiler parameters, not quantified witnesses. Later we explicitly pay for a loader when two initial counter positions are allowed to remain free natural parameters. The vacuum $M = 0$ is a separate trivial constant orbit.

Theorem 38.1 (Sparse finite-horizon certificate). *For a supplied total conservative g , valid nonempty initial configuration of mass M , and fixed horizon T , there is an explicitly generated integer polynomial*

$$P(\mathbf{z}) = \sum_{j=1}^R \rho_j(\mathbf{z})^2, \quad \deg \rho_j \leq 2,$$

with exactly one zero in $\mathbb{N}^V$. Its row decoder gives precisely $c, F(c), \dots, F^T(c)$. The preferred full-table construction uses exactly

$$V = T[M(q^3 + 14) + 5M(M - 1)], \quad (38.2)$$

$$R = T[17M + 5M(M - 1)]. \quad (38.3)$$

Every witness in this unique solution is at most

$$H = 3(D + 2T) + M + 3K + 3. \quad (38.4)$$

Consequently the tuple has at most $V[\log_2(H + 1)]$ bits under fixed-width coordinate encoding. A prescribed endpoint of mass M can be imposed with $2M$ additional affine residuals and no witnesses, leaving zero or one natural solution.

For $T = 0$ the witness tuple is empty, and is the unique element of $\mathbb{N}^0$. The empty sum is the zero constant. A mismatched-mass endpoint can instead be rejected by adding the constant residual 1. The degree bound is at most four, not a claim that every small instance has degree exactly four. Dimension grows with T and M ; the theorem is not a fixed-arity equation with an unknown unbounded horizon.

39 The canonical arithmetic layer

39.1 Comparison and equality without ambiguous ties

For already established nonnegative integral expressions x, y , define $\text{CMP}(x, y)$ by fresh natural witnesses b, d and residuals

$$b(b - 1), \quad y - x - (2b - 1)d - b + 1. \quad (39.1)$$

If $b = 1$, the second equation says $d = y - x$, possible exactly when $y \geq x$. If $b = 0$, it says $d = x - y - 1$, possible exactly when $x > y$. This proves unique solvability, including ties, with

$$b = \mathbf{1}_{\{y \geq x\}}, \quad d = \begin{cases} y - x, & y \geq x, \\ x - y - 1, & x > y. \end{cases}$$

The inputs can be affine expressions; there is no charge for an unnecessary alias wire. To test equality, use two comparisons, $\text{CMP}(x, y)$ and $\text{CMP}(x + 1, y)$, and take the affine difference of their flags:

$$e(x, y) = b_{\text{weak}} - b_{\text{strict}} = \mathbf{1}_{\{x=y\}}. \quad (39.2)$$

This costs four witnesses and four residuals, with no additional equality-variable witness.

[write] The comparison gadget (39.1), with the same two residuals and the same case analysis, is Lemma `cdc:mem:lem:compare` of Part V of *canonical-diophantine-certificates*. Source 13 does not cite it; the gadget is printed here as delivered and no novelty is claimed for it.

39.2 Streaming and all simultaneous collision groups

Suppose the current row is (x_i, z_i) , $0 \leq i < M$, in sorted order. Each x_i, z_i is affine in earlier variables. Introduce natural L_i, C_i, R_i with residuals

$$L_i + C_i + R_i - 1, \quad z_i - C_i - 2R_i. \quad (39.3)$$

For valid integral $z_i \in \{0, 1, 2\}$, these force the unique one-hot channel vector. A streamed position y_i is forced by

$$y_i + L_i - x_i - R_i = 0. \quad (39.4)$$

It is nonnegative at every layer $t < T$, because $x_i \geq T - t \geq 1$. This is genuine line transport; there is no clipping at zero and no wraparound.

For each unordered pair $i < j$, compute $e_{ij} = \text{EQ}(y_i, y_j)$; set $e_{ji} = e_{ij}$ and $e_{ii} = 1$ as aliases and constants. The whole collision partition is determined at once. Introduce three incoming mass variables with residuals

$$u_i - \sum_j e_{ij} R_j, \quad v_i - \sum_j e_{ij} C_j, \quad w_i - \sum_j e_{ij} L_j. \quad (39.5)$$

Their order is (R, C, L) , as required by (38.1). Equal destinations have identical incoming triples. Each component is at most K : a particular incoming lane is a translated copy of one old lane at one site. That statement uses validity of the previous row, not a guessed bound on the number of simultaneous records.

39.3 A single table-row selector simplex

Index every input table row by $\tau = (r, c, l) \in \{0, \dots, K\}^3$. For particle i , introduce q^3 natural selectors $s_{i,\tau}$, three output masses a_i, b_i, c_i , and the seven affine residuals

$$\begin{aligned} & \sum_{\tau} s_{i,\tau} - 1, \\ & u_i - \sum_{\tau} r s_{i,\tau}, \quad v_i - \sum_{\tau} c s_{i,\tau}, \quad w_i - \sum_{\tau} l s_{i,\tau}, \\ & a_i - \sum_{\tau} g_L(\tau) s_{i,\tau}, \quad b_i - \sum_{\tau} g_C(\tau) s_{i,\tau}, \quad c_i - \sum_{\tau} g_R(\tau) s_{i,\tau}. \end{aligned} \quad (39.6)$$

A natural vector summing to one has exactly one nonzero entry, equal to one. The three input moment equations select precisely the row (u_i, v_i, w_i) , because all input triples are distinct and the table is total. The output equations then force $g(u_i, v_i, w_i)$. This single simplex avoids factoring the lookup into three separate selector vectors and all their products. Its small residual count does not make the q^3 selectors or the long coefficient sums free.

39.4 Rank redistribution

Within each destination group use the old record order to define its one-based rank

$$r_i = \sum_{j \leq i} e_{ij}.$$

This is an affine expression, not a fresh witness. Compute two flags by comparisons

$$h_i = \text{CMP}(r_i, a_i), \quad k_i = \text{CMP}(r_i, a_i + b_i),$$

where the notation denotes the returned flag. The next channel is the affine expression

$$z'_i = 2 - h_i - k_i. \quad (39.7)$$

Ranks within a group are exactly $1, \dots, u_i + v_i + w_i$. Conservation makes this number $a_i + b_i + c_i$. The first a_i ranks have $(h_i, k_i) = (1, 1)$ and channel zero; the next b_i have $(0, 1)$ and channel one; the last c_i have $(0, 0)$ and channel two. Thus the output multiset at the collision site is exact. The intermediate rank convention only chooses identical physical units, and no persistent identity survives.

39.5 A fixed sorting network

Sort the M records (y_i, z'_i) by a fixed bubble sorting network of $\binom{M}{2}$ adjacent compare-exchanges. For inputs $(x, z), (y, w)$ compare the keys $3x + z$ and $3y + w$ using (39.1), giving b, d . Introduce four output witnesses and residuals

$$\begin{aligned} x_{\min} &= y - b(x - y), & z_{\min} &= w - b(z - w), \\ x_{\max} &= x - y + x_{\min}, & z_{\max} &= z - w + z_{\min}. \end{aligned} \quad (39.8)$$

For $b = 1$ the left input is the smaller or tied key; for $b = 0$ the right is smaller. Ties retain the left input, but equal keys are already the same physical record. Since channels lie in $\{0, 1, 2\}$, key order is lexicographic order. Every comparator costs six witnesses and six residuals, including its comparison. The network outputs are the next row by alias, without copy variables. All equations in this layer are of degree at most two.

[write] This compare-exchange is the two-field case of the record comparator of the same Part V (its equations `cdc:mem:eq:record-low` and `cdc:mem:eq:record-high`, with the cost count of Proposition `cdc:mem:prop:comparator-cost`); it is printed as delivered, as a re-derivation of that device.

40 Correctness and exact paid costs

Proof of Theorem 38.1. Assume inductively that the current row is the canonical true row and every earlier auxiliary witness is uniquely fixed. The channel equations and comparison lemma uniquely fix streaming and the entire equality partition. Equations (39.5) fix the incoming mass. The single natural simplex forces the unique table row and its output. Conservation and rank redistribution then give exactly the outgoing multiset. The sorting network uniquely produces the canonical next row.

The empty-space assertion is indispensable. Every nonzero incoming site appears among the y_i , since it has at least one streamed unit. A site absent from that list has incoming triple $(0, 0, 0)$ and hence zero output. A listed site has positive incoming total, which conservation keeps positive. Therefore there is no hidden output outside the represented sites, no omitted occupied site, and no ghost site introduced by an unconstrained witness.

This proves soundness and uniqueness by induction through T layers. Conversely, start with the actual physical history, expand each configuration into repeated unit records, and evaluate the stated one-hot vectors, comparisons, masses, ranks, and sorting gates. They provide nonnegative integral values satisfying every residual. Thus the unique natural solution exists. Counting records of channel z at position x and undoing the shift by $T - o$ is the decoder. The deterministic gate evaluation is its inverse on the fixed-input, fixed-horizon solution set.

If g is reversible, the final physical configuration also determines all preceding physical configurations by F^{-1} . This does not assert reversibility of the auxiliary arithmetic circuit, nor any property of arbitrary real solutions.

For the height bound, positions stay in $[0, D + 2T]$. Sorting keys are at most $3(D + 2T) + 2$; comparison slacks are bounded by the appropriate key difference, group rank, or local mass. Incoming and outgoing lane masses are at most K , ranks at most M , and selectors at most one. Bound (38.4) covers all these quantities with slack. Endpoint equations only select or reject the already unique row. $\square$

40.1 Scalar ledger

The per-layer cost is as follows. The lookup row excludes the separately listed incoming masses.

Block	Witnesses	Residuals
Channel decoding	$3M$	$2M$
Streaming	M	M
Pairwise destination equality	$4\binom{M}{2}$	$4\binom{M}{2}$
Incoming lane masses	$3M$	$3M$
Full row-selector lookup	$M(q^3 + 3)$	$7M$
Rank comparisons	$4M$	$4M$
Sorting network	$6\binom{M}{2}$	$6\binom{M}{2}$
Total	$M(q^3 + 14) + 5M(M - 1)$	$17M + 5M(M - 1)$

The exact counts include slacks, Boolean comparison flags, sorting outputs, and repeated simultaneous-collision work. Constants and affine aliases are not witnesses. The local table itself has q^3 input rows and three output entries per row; its construction, range/conservation checks, storage, and any permutation test are separate explicit costs. Expanding weighted input into M unit records requires $\Omega(M)$ output work and storage. The size bounds are polynomial in the numerical mass M , not in its binary bit length. For a fixed source substrate, $M = m + 19$ is constant while gaps and their bit lengths may grow.

Let $\mathcal{L}$ be the total number of nonzero monomial terms in the residual list, and $\mathcal{O}$ the sum of the degrees of those monomials, including repeated variable occurrences. The release declares the following literal evaluator: initialize a monomial with its signed integer coefficient, multiply by each listed variable, add its value into a residual accumulator initially zero, square each residual, and add those squares starting at zero. Under exactly this convention,

$$\#\text{multiplications} = \mathcal{O} + R, \quad \#\text{additions} = \mathcal{L} + R. \quad (40.1)$$

Multiplication by a coefficient is already included by the initialization convention; negation signs reside in signed coefficients. Even a final zero residual is squared by this evaluator. The exporter supplies these exact counts and the largest residual-coefficient bit length for each instance.

There are $O(T[M^2 + Mq^3])$ such arithmetic operations. This is scalar arithmetic, not bit complexity. Witness height, coefficient size, polynomial-description length, arithmetic cost, and witness dimension are different resources. The statement concerns a coefficient-explicit residual-list sum of squares. Fully expanding and collecting the quartic polynomial can create many more terms; no equality with the residual-list cost is asserted.

40.2 Restricted row lists

A useful optional backend replaces all q^3 rows in (39.6) by a supplied list of S distinct allowed input rows, each conserving mass. It has

$$V_S = T[M(S + 14) + 5M(M - 1)], \quad R_S = R.$$

The same proof certifies exactly histories whose every occupied incoming triple belongs to that list. If some required row is absent, its input moments cannot be matched by a natural one-hot selector, and the certificate is unsatisfiable. This backend does *not* certify all histories of an arbitrary total

extension merely because the listed rows extend to one. Coverage of the particular history or a valid-trajectory invariant is a separate proof obligation. The worked source example uses such a coverage theorem; the generic theorem always uses the supplied full table.

40.3 The domain of uniqueness and an orthant-exact option

The default simplex equations rely on natural integrality. They need not be unique even over the nonnegative real orthant. For a concrete full-table example let $K = 2$, $M = 1$, $T = 1$, and $g(r, c, l) = (r, c, l)$ numerically, with the stated input/output orders. Start with one right-channel unit. Its incoming triple is $(1, 0, 0)$ and its output is one left unit. Replace the table selector at $(1, 0, 0)$ by half at $(0, 0, 0)$ and half at $(2, 0, 0)$. All three input moments and all output moments remain unchanged. Every other witness can keep its canonical value, so the original residuals still vanish, but the selector is fractional. The release checks this with exact rational arithmetic.

For any nonnegative simplex $s_1 + \dots + s_d = 1$, add the one residual

$$\sum_{j=1}^d s_j^2 - 1. \quad (40.2)$$

If it vanishes, then

$$0 = \left(\sum_j s_j\right)^2 - \sum_j s_j^2 = 2 \sum_{i < j} s_i s_j.$$

Every summand is nonnegative; at most one selector is nonzero, and that selector equals one. Adding (40.2) to each channel simplex and each table simplex therefore restores the inductive proof over $\mathbb{R}_{\geq 0}$ for fixed natural input data. Comparisons have Boolean flags already, and their nonnegative slacks are uniquely forced once the inputs are fixed integers. All subsequent witnesses remain the canonical natural values.

Corollary 40.1 (Exact nonnegative-real fiber). *The strengthened full-table or restricted-row compiler has the same witness count and exactly $2MT$ extra residuals. Its nonnegative-real solution set is the same singleton or empty set as its natural solution set. Residual degree stays at most two and polynomial degree at most four.*

This says nothing about the unrestricted real domain, where negative selectors invalidate the nonnegativity argument. An earlier factorized lookup remains an independently tested baseline: three q -selectors, q^2 pair products, and q^3 triple products give

$$V_{\text{fact}} = T[M(q^3 + q^2 + 3q + 14) + 5M(M - 1)],$$

$$R_{\text{fact}} = T[M(q^3 + q^2 + 19) + 5M(M - 1)].$$

Its orthant strengthening costs $4MT$ residuals. These are baseline counts, not the preferred single-row-simplex ledger in (38.2)–(38.3).

41 The reversible conservative source interface

Morita and Imai [19] construct a reversible number-conserving partitioned automaton from a reversible two-counter machine. The following source audit fixes the exact contract used here. It distinguishes the published theorem, a correction forced by its encoding, and an additional reachable-legality argument needed when unspecified local rows are completed arbitrarily.

41.1 Source syntax and encoding

A transition is (i, j, op, k) , with source state q_i , target q_k , counter $j \in \{0, 1\}$, and operation increment, decrement, no-op, zero test, or positive test. There are $m \geq 2$ states, q_0 has no incoming transition, and q_{m-1} is designated final with no outgoing transition. Distinct transitions with the same source may only be complementary zero/positive tests on the same counter; the same restriction applies to distinct transitions with the same target. This is the paper's syntactic deterministic and reversible model, a stronger specification than a vague assertion of injectivity for some other instruction language.

Put

$$a_i = i + 10, \quad B = m + 16, \quad b_i = B - a_i - \gamma_i,$$

where $\gamma_i = 2, 4, 6, 7$ for increment counter zero, decrement counter zero, increment counter one, decrement counter one respectively, and zero otherwise. The alphabet cap is $K = m + 18$ and the number of ordinary cell states is $(m + 19)^3$.

The encoding $\phi(q_i, n_0, n_1)$ places (a_i, b_i, γ_i) at the origin. Counter j contributes marker mass 2^j at position n_j , normally in the center lane. If $n_j = 0$ and q_i increments that counter, its marker is instead added to the right lane at the origin. Contributions at the same site add. Hence

$$M = B + 1 + 2 = m + 19 = q, \quad (41.1)$$

independently of counter magnitude. The support occupies at most three sites, all in $\{0, n_0, n_1\}$. Coincident markers have total three, retaining their separate zero/positive information through distinct weights. Input size is paid through distances and their bit lengths, not a variable number of particles.

41.2 The printed sign defect and the repaired local injection

On printed page 249, rule (8.2) has an output-center $+\gamma_k$. When a no-op enters a state incrementing counter h , the input and repaired output are

$$\begin{aligned} & (a_i, B - a_i + c2^h + c'2^{1-h}, 0) \\ & \longmapsto (a_k, B - a_k - \gamma_k + c'2^{1-h}, \gamma_k + c2^h), \quad c, c' \in \{0, 1\}. \end{aligned} \quad (41.2)$$

The printed plus would increase the local total by $2\gamma_k$, namely four or twelve. The minus conserves mass and gives exactly the declared target encoding. This is a necessary correction to that printed formula; no author-issued erratum is asserted. The target's increment counter h must also be read from q_k , rather than borrowed from the no-op quadruple's semantically irrelevant counter tag. Opposite-tag cases are included in the executable audit.

[write] The repository's Hilbert-tenth-problem programme checked this sign against the primary paper (printed page 249 of [19]) in its review of source 13 [30]: subtracting the target opcode is forced by mass conservation and the target encoding, and the target's increment counter, not the no-op tag, controls the marker routing. Like source 13, the review records a mathematical correction to a published formula, not an author erratum. No report or README of the repository made a claim that this correction refutes.

For clarity, the structural proof behind finite row checks is as follows. The ordinary signal rules route signal mass or transfer a marker between center and signal; waiting rows are identities. After (41.2), state-update rows preserve B plus the zero-marker mask. Startup and final rows

exchange a unit signal plus a reservoir $m + 15$ with control mass B . For nonfinal i , the smallest baseline obeys

$$b_i \geq m + 6 - (m - 2) - 7 = 1;$$

the final baseline is seven. Added marker mass is at most three, all center values are at most $m + 18$, and state signals are at most $m + 9$. Thus all coordinates stay in range.

Routine signal domains differ from control domains by their nonzero lane pattern and by signal values below or above ten. At a fixed control state its unique operation type chooses the rule family. Zero masks are distinct; complementary test branches have disjoint masks. The no-outgoing convention at the final state removes domain conflicts there. This proves the function property.

For injection, different target state signals have different left components. At a fixed target, its center/right pair recovers the zero-mask bits: an increment target carries its own zero marker in the right signal and leaves the other bit in the center; otherwise both bits remain centered. A target with a movement or no-op predecessor has only that incoming transition. If it has two incoming tests, they test the same counter with complementary outcomes and hence disjoint masks. The *whole output triple* distinguishes them, even when the center alone does not. Waiting outputs have right component zero, whereas a transition into an operation state has a positive right signal. Startup targets q_0 , which has no incoming source transition, and the final unit signal with its large reservoir is distinct from an ordinary travelling unit signal. The remaining routine signal output patterns are disjoint by lane and signal type. Together these observations establish a mass-preserving partial injection.

Lemma 41.1 (Completion within mass fibers). *Every finite mass-preserving partial injection on the local triple alphabet extends effectively to a mass-preserving permutation. This applies to the repaired table.*

Proof. For each s , let Ω_s be the triples with coordinate sum s . Used inputs and used outputs have equal cardinality within Ω_s . Bijection of their unused complements, for example by pairing lexicographically sorted lists, completes this fiber. Their union is the required permutation. The unique weight-zero triple is fixed. Lane transport followed by the local permutation is a global reversible conservative CA. $\square$

Simply assigning identity to every unspecified input is generally wrong. The specified row $(2, 1, 0) \mapsto (0, 0, 3)$ can leave $(0, 0, 3)$ unspecified as an input. Its identity image would duplicate an already used output. Also, Lemma 41.1 is about a partitioned/block local permutation; it is not a completion theorem for arbitrary partial ordinary-neighborhood CA rules.

41.3 Exact source microtimes and the pulse observer

For a *legal* source step, the repaired specified rows, without using any completion row, carry $\phi(I)$ to $\phi(J)$ in the following number of microsteps:

Source operation on a counter of value n	Microsteps
Increment, including $n = 0$	$2n + 2$
Decrement, with $n > 0$	$2n$
No-op or legal zero/positive test	2

Here is the geometric reason for the clocks. The state signal travels from the origin to site -1 and back, while the control waits for its operation signal. An increment signal traverses the distance to its marker, carries that marker one step farther right, deposits it, and returns to the origin, a total of $2n + 2$ steps. When $n = 0$, the marker is already in the initial right signal, giving the same two-step formula. A decrement signal reaches its positive-position marker, reflects while carrying it one step left, deposits it, and returns in $2n$ steps. At $n = 1$ the returning marker joins the origin's zero mask directly. The other counter's marker is transparent to these routes, including coincident marker positions, because the rule families carry its distinct mass unchanged. Tests and no-ops only require the state signal's two-step return. The state-update rows then redistribute the zero mask into exactly the target encoding. These routes also explain why no constant slowdown is promised.

Let t_h be the sum of these times up to first entry into $q_f = q_{m-1}$. From $\phi(q_f, n_0, n_1)$ the state signal is in the right lane at -1 at $t_h + 1$; the final rule creates a left unit at the origin at $t_h + 2$; and

$$L_{t_h+3}(-1) = 1. \quad (41.3)$$

That unit subsequently moves left forever. Before final entry, the specified simulation trajectory never has this left-unit signal. After its departure, the fixed centers are

$$C(0) = m + 15 + \mathbf{1}_{\{n_0=0\}} + 2\mathbf{1}_{\{n_1=0\}}, \quad C(x) = \mathbf{1}_{\{x=n_0\}} + 2\mathbf{1}_{\{x=n_1\}} \quad (x > 0).$$

All right channels are zero. The results remain readable while information escapes spatially. This is an anchored pulse, not an absorbing globally halted configuration. The complete successful orbit has unbounded diameter.

The equivalence between this observer and source halting requires that *every reachable nonfinal source configuration on the input under discussion has a legal next instruction*. This is called reachable legality below. It is not asserted for every arbitrary counter pair at every internal state.

41.4 An explicit false observer on an unsafe source

Take $m = 2$ and the sole source instruction “decrement counter zero from q_0 to q_1 .” On counters $(0, 0)$ that instruction is illegal, so the designated final state is not reached. Its encoded initial CA state is $(10, 7, 4)$ at site zero, of mass 21. At time one the nonzero sites are

$$-1 : (0, 0, 10), \quad 0 : (0, 7, 0), \quad 1 : (0, 0, 4).$$

The next origin input $(10, 7, 0)$ is unspecified. Both this input and the output $(1, 16, 0)$ are unused, and both have mass 17, so adding

$$(10, 7, 0) \longmapsto (1, 16, 0)$$

is compatible with Lemma 41.1. At time two the configuration is $0 : (1, 16, 0)$ and $2 : (0, 0, 4)$; at time three it has a left unit at -1 , falsely satisfying the proposed observer. The release supplies the entire resulting 9,261-row permutation and a replay through time eight.

Thus an arbitrary partial reversible source and an arbitrary conservative permutation completion do not provide a halting equivalence on every encoded pair. Missing test branches and unnamed dead states have the same logical risk. This example does not affect the generic theorem, which always certifies the actual supplied CA dynamics, and does not refute a safe universal-source construction.

42 Safe universal sources and the paid loader

42.1 The existence argument and its exact scope

Morita's 1996 theorems [20] give designated-final-state simulation equivalences. The following preservation of reachable legality is an explicit invariant argument through those constructions, rather than an attribution of a separate totality theorem to that paper.

Start with a fixed universal deterministic Turing machine whose transition is defined at every nonhalting state/symbol pair. On its computably valid program/input encodings, choose a complete guarded counter simulation: decrement is entered only after positivity, every appropriate test branch exists, and each arithmetic routine terminates on its finite inputs. A fresh initial state and a terminal designated final state give the required endpoint conventions. This yields a safe irreversible counter simulator. Splitting high-indegree target states through finite no-op merge chains, as in Morita's preparatory reduction, preserves legal progress: one legal step only acquires a finite chain of unconditional steps.

The reversible history compiler adds a history counter H and scratch counter W . At an original-state boundary the invariant is

$$(\text{simulated counters correct}), \quad H \in \mathbb{N}, \quad W = 0.$$

An already reversible incoming relation is copied. For an overlapping incoming pair, the compiler takes the original legal operation and then records which predecessor was used. It transfers H to W through positivity-guarded decrements and reconstructs $2H$ or $2H + 1$ in the history counter while emptying W . Each decrement is legal by its positive guard; required positive back-edge tests follow increments; each transfer loop terminates by decreasing its source counter. The boundary invariant is restored at the target. Thus a legal step expands into a finite nonempty legal path, without a new nonfinal dead end. Erasing history is unnecessary for this existence argument.

The reversible multi-counter machine is then represented by two counters

$$(N, S) = \left(\prod_{i=1}^k p_i^{v_i}, 0 \right), \quad N \geq 1, \quad (42.1)$$

where the p_i are distinct fixed primes. This is a constrained encoding of all data, history, and scratch counters, not an arbitrary pair in $\mathbb{N}^2$. The prime compiler has the following legal-progress invariants.

- An increment of counter i transfers N to scratch through guarded decrements and reconstructs $p_i N$ while consuming scratch. Both loops terminate and end with scratch zero
- A legal decrement has $v_i > 0$, so $p_i \mid N$. After transfer, each positive scratch batch has at least p_i units. Its p_i successive decrements are legal even where only the first is explicitly guarded. Consuming all batches gives $(N/p_i, 0)$
- A test computes quotient and residue by a guarded finite-state division loop. Divisibility by p_i is exactly $v_i > 0$; nonzero residue is exactly $v_i = 0$. The appropriate source branch is selected, and quotient and residue restore the original N through terminating positively guarded loops
- A no-op is copied. If a source has only one test exit, the omitted exit is never needed on its legal encoded trajectory

The original state, prime encoding, and scratch-zero invariant hold again at each boundary. Fresh internal states preserve the source's required reversible incoming syntax and initial/final conventions. Induction proves that every reachable nonfinal primitive two-counter state on a valid input has a successor. If the original machine never halts, each of its steps expands into a nonempty finite legal path, so no stuck intermediate configuration can masquerade as a halt.

Fixing the universal program fixes the resulting finite two-counter source and hence, by the corrected local construction and a chosen completion, one finite reversible conservative CA. Its valid program/data encodings vary only through the initial counter positions. Eventual anchored pulse occurrence is therefore computably enumerable hard on this computable valid-input family; direct simulation makes it computably enumerable. This supplies the usual undecidable local-observation problem for a fixed finite-mass substrate.

The release does not materialize a chosen universal source table or give its numerical m . It does not claim faithful source interpretation for all $(n_0, n_1) \in \mathbb{N}^2$, intrinsic CA universality, or an efficient simulation. In particular, the often cited 96-state construction using an ultimately periodic infinite background is not a finite-mass substitute. Exact-target reachability is also a different question: the final counter positions need not be computable in advance, and the escaping halt signal is not a specified complete target. Transferring exact-target undecidability would require an additional normal form or compute/uncompute argument.

42.2 A uniform initial loader with no hidden input oracle

For fixed source data m , initial opcode γ_0 , and fixed T , leave $n_0, n_1 \in \mathbb{N}$ as free polynomial parameters. At position T form ten left records, $m + 6 - \gamma_0$ center records, and γ_0 right records. Add one counter-zero record at $n_0 + T$ and two counter-one records at $n_1 + T$. A marker's channel is right precisely when its position parameter is zero and q_0 increments that counter; otherwise it is center.

Two comparisons $\text{CMP}(n_j, 0)$ determine the zero flags, costing four witnesses and four residuals. The initial list is sorted by the same network as before. Its exact loader cost is

$$V_{\text{load}} = R_{\text{load}} = 4 + 3M(M - 1). \quad (42.2)$$

Even unused zero tests are included in this declared ledger. The exported residual coefficients are independent of the supplied values of n_0, n_1 ; only those parameter values and the canonical witnesses vary. Bound (38.4) remains valid with $D = \max(n_0, n_1)$, including loader comparison slacks.

42.3 Paying for the exact first pulse

For $T \geq 1$, site -1 is shifted to $T - 1$. For each $t = 1, \dots, T$ and each record, compute $\text{EQ}(x_{t,i}, T - 1)$. Reuse the channel one-hot vectors already present at $t < T$, and add $3M$ channel witnesses with $2M$ residuals for the final row. Add one residual at each time,

$$\sum_i e(x_{t,i}, T - 1) L_{t,i} - \mathbf{1}_{\{t=T\}}. \quad (42.3)$$

This forces detector left mass zero at every earlier observed time and one at T . The initial detector is blank because ϕ is supported on nonnegative sites. The additional cost is exactly

$$V_{\text{obs}} = 4MT + 3M, \quad (42.4)$$

$$R_{\text{obs}} = 4MT + 2M + T. \quad (42.5)$$

For the orthant-exact variant add M final decoder norm residuals, on top of the core $2MT$ extra residuals. No observation witnesses are needed for a fixed endpoint beyond the core row witnesses, but that is a different predicate.

These guards mean *zero before and one now*. For an arbitrary CA they are not the generic first occurrence of the value one, because earlier detector masses greater than one are also excluded. On valid source simulations they match the exact unique pulse in (41.3). Combining (38.2), (38.3), (42.2), (42.4), and (42.5) gives the full paid uniform-input first-pulse certificate. Safe-source legality concerns the semantic interpretation of that predicate, not the arithmetic soundness of the actual-CA certificate.

42.4 The fully specified doubling example

The worked source has seven states and the following complete transition list; all other source transitions are absent:

Source	Counter	Operation	Target
q_0	1	zero test	q_1
q_1	0	zero test	q_6
q_1	0	positive test	q_2
q_2	0	decrement	q_3
q_3	1	increment	q_4
q_4	1	increment	q_5
q_5	1	positive test	q_1

On $(q_0, n, 0)$ every reachable nonfinal instruction is legal: the first test sees zero, each decrement follows a positive counter-zero test, and the last positive test follows two increments. The machine reaches $(q_6, 0, 2n)$ after $5n + 2$ source steps. Summing the exact microclocks gives

$$t_h = 5n^2 + 7n + 4, \quad t_{\text{pulse}} = 5n^2 + 7n + 7. \quad (42.6)$$

Indeed iteration $k = 0, \dots, n - 1$ takes a positive test, a decrement of $n - k$, increments at $2k$ and $2k + 1$, and a final test, totalling $2n + 6k + 10$ microsteps. The initial and final zero tests add four; summing yields (42.6).

Here $K = 25$, $M = q = 26$. The corrected specified local table has 97 distinct rows, and its lexicographic mass-fiber completion has all $26^3 = 17,576$ rows. For $n = 2$, twelve source steps reach $\phi(q_6, 0, 4)$ at time 38, create a left unit at the origin at time 40, and produce the detector pulse at time 41. The full table and trajectory checks are shipped. This explicit example verifies the interface and its costs without representing a universal-machine claim.

43 Geometric barriers and translating periodicity

The next results use a more general model than the triple-channel compiler. Let Q be a finite alphabet, 0 a quiescent state, and $F : Q^{\mathbb{Z}} \rightarrow Q^{\mathbb{Z}}$ a deterministic translation-invariant CA of radius r . Let

$$w : Q \rightarrow \mathbb{N}, \quad w(0) = 0, \quad w(a) \geq 1 \quad (a \neq 0),$$

conserve total weight on every finite-support configuration. Thus the vacuum is the unique zero-weight symbol. A mass- M configuration has at most M occupied sites. Distinct states may have the same positive weight; this matters when comparing particle bounds in numerical-state models.

Zero-weight active symbols, signed weights allowing cancellation, and a nonzero infinite periodic background are outside these hypotheses. Quiescence and locality preserve finite support at each finite time, but do not bound its position or its diameter uniformly in time. A finite-support configuration on $\mathbb{Z}$ is not a finite-ring configuration.

Write $(\mathcal{T}_a x)(i) = x(i - a)$ for translation to the right. For nonempty x , let $\ell(x) = \min \text{supp}(x)$ and $\mathcal{N}(x) = \mathcal{T}_{-\ell(x)} x$ be its normalized shape. The vacuum is handled separately.

Proposition 43.1 (Deciding a diameter bound and its tail). *Given F , a finite configuration x , and an integer $D \geq 0$, one can decide whether $\text{diam}(F^t x) \leq D$ for every $t \geq 0$. Under this full-orbit bound, exact finite-configuration reachability and finite-word occurrence, anchored or at any translate, are decidable. Without the bound promise, the same procedure decides occurrence before the first escape and otherwise reports the escape, without concluding anything about later unrestricted occurrence.*

Proof. There are finitely many normalized nonempty shapes of diameter at most D , at most $(|Q| - 1)|Q|^D$. With mass M a sharper bound is

$$\sum_{k=1}^{\min(M, D+1)} \binom{D}{k-1} (|Q| - 1)^k. \quad (43.1)$$

Position zero is occupied, the remaining $k - 1$ positions are chosen in $\{1, \dots, D\}$, and all occupied symbols are nonzero. An exact-weight count is the coefficient of z^M in

$$\left(\sum_{a \neq 0} z^{w(a)} \right) \left(\sum_{a \in Q} z^{w(a)} \right)^D.$$

Simulate until the diameter exceeds D or a normalized shape repeats. One event must happen in finitely many steps. If

$$\mathcal{N}(F^i x) = \mathcal{N}(F^j x), \quad i < j, \quad p = j - i, \quad \delta = \ell(F^j x) - \ell(F^i x),$$

then determinism and translation equivariance give, for every $0 \leq s < p$ and $k \geq 0$,

$$F^{i+s+kp} x = \mathcal{T}_{k\delta} F^{i+s} x. \quad (43.2)$$

Consequently no future shape escapes. An escape disproves the diameter bound, while a repeat proves it.

For exact target y , inspect the finite prefix and then each phase $z = F^{i+s} x$. It can reach y exactly when its normalized shape agrees and

$$\ell(y) - \ell(z) = k\delta \quad \text{for an integer } k \geq 0.$$

This is a divisibility and sign test; when $\delta = 0$ it is simply equality of offsets.

For a full word on a fixed interval, each occupied phase site moves as $u + k\delta$. Required nonzero symbols give position equalities and symbol tests; required zeros exclude the moving occupied sites from their positions. The resulting finite Boolean combination of affine equalities and inequalities in

one integer $k \geq 0$ is decidable, including an all-zero word. For occurrence anywhere, translation of the entire phase does not change the answer, so a finite prefix and one list of phases suffice. A local predicate allowing several alphabet symbols is a finite union of these tests.

If an escape is encountered before an observed hit, it only disproves the bound. It does not reject a target that might appear after the escape. This is why the unrestricted-reachability conclusion requires the stated full-orbit promise. $\square$

If F is injective on finite configurations, a normalized repeat has no true transient. From $F^j x = \mathcal{T}_\delta F^i x$ and equivariance, cancel F^i to obtain $F^{j-i} x = \mathcal{T}_\delta x$. Thus a wholly bounded-diameter reversible orbit is translating periodic from time zero. If the existence of some diameter bound is promised but no value is supplied, simply simulate normalized shapes until a repeat; the promise ensures termination. Without that promise this is only a semidecision of boundedness.

These arguments are elementary single-orbit consequences of finite normalized shape space. Classical number-conserving CA dynamics already studies bounded-diameter classes and periodic or generalized-shift behavior; see Durand et al. [21] and Formenti–Grange [22]. No new classification theorem is claimed here.

Proposition 43.2 (No computable complete diameter cutoff). *Suppose a fixed F and computable finite input encodings $e \mapsto x_e$ reduce an undecidable predicate to eventual occurrence of an effective finite local observation, possibly anchored. There is no total computable $B(e)$ such that, whenever the observation first occurs at h , every configuration through time h has diameter at most $B(e)$.*

Proof. If such B existed, simulate until a hit, an escape from the bound, or a normalized repeat. A hit answers yes. Escape before a hit answers no by the assumed completeness of B . A normalized repeat permits the exact tail decision in Proposition 43.1, including anchored observations. One of these events occurs in finitely many bounded-width shapes, contradicting undecidability. $\square$

This does not forbid sparse witnesses with large gaps, unbounded integer values, or an externally varying horizon. It forbids a computable complete geometric cutoff through successful runs. A computable complete time or witness-length cutoff would similarly turn the relevant finite search into a decision procedure.

[write] In *canonical-diophantine-certificates* the finite-search step of this proof is the standard argument of Proposition `cdc:bd:prop:nobound` (also used at the end of Part I’s Section 13); the statement, a cutoff on orbit *diameter*, is source 13’s.

43.1 Finite rings and global halting

On a specified n -cell ring, every mass sector is finite, so orbit exploration decides reachability. Its cardinality is

$$[z^M] \left(\sum_{a \in Q} z^{w(a)} \right)^n \leq \sum_{k \leq M} \binom{n}{k} (|Q| - 1)^k.$$

For triple-channel cap K it is $[z^M](1 + \dots + z^K)^{3n}$, at most $\binom{3n+M-1}{M}$. Fixed M makes the bound polynomial in the *numerical* ring length, not necessarily in its binary encoding length. A reversible ring orbit is a cycle from its start.

A line computation from support diameter D_0 is reproduced through a specified horizon T by a ring whose periodic copies have disjoint radius- rT influence intervals; $n > D_0 + 2rT$ suffices. The zero padding and ring length depend on T . This cannot replace the infinite line for all times at once.

An injective dynamics cannot enter an existing fixed point or finite cycle from outside: the cycle already supplies the unique predecessor of each of its points. A reversible simulator therefore reports a halt by an observation while retaining information elsewhere. Stabilization of a finite window, or a departing pulse, must not be called global CA halting.

44 Decidability at total mass at most two

Theorem 44.1 (Two-unit-mass decision theorem). *Under the positive-weight, unique-vacuum hypotheses of Section 43, exact finite-configuration reachability and finite-word occurrence, anchored or anywhere, are decidable from every finite-support initial configuration of total mass at most two. Reversibility is not required. For a fixed CA F , these problems have an algorithm polynomial in the input size when finite configurations are sparse lists of nonzero coordinate/state pairs with binary coordinates, and a pattern is an explicit full finite word, whose length counts toward input size. An anchored word has a binary anchor coordinate.*

The theorem rules out universality by these observation problems at masses zero, one, and two within this model. It does not show that three suffices, or identify a least universal mass. The proof is given in detail because the absolute-position and long-excursion issues are easy to miss. ([write] Added 2 October 2026, batch 80: Part V shows that three suffices in this model when there are many weight-one symbols (Theorem 49.2, and for exact pairs Theorem 64.1); Part VI shows that with at most one weight-one symbol masses three and four remain decidable (Theorems 74.1 and 83.1). Source 14 proves the present theorem again; Section 59.) ([write] Added 3 October 2026, batch 82: with a single unit symbol, decidability extends to mass four in every finite dimension (Part VII, source 19, Theorem 101.1); at $d = 1$ with one weight-one symbol the mass-two case of this theorem is a special case of it. In one dimension the least universal mass with one weight-one symbol is five, reached by binary and by globally reversible binary automata (Reports 14 and 15, Part I of `five-particle-binary-automata`).)

44.1 Single particles are finite-state walkers

Let $Q_j = \{a \in Q : w(a) = j\}$. From a singleton of type $a \in Q_1$ at position u , positivity and conservation force the next state to be a single weight-one site. Locality puts it within r of u . Thus computable maps

$$\alpha : Q_1 \rightarrow Q_1, \quad d : Q_1 \rightarrow [-r, r] \cap \mathbb{Z}$$

give its next type $\alpha(a)$ and displacement $d(a)$. A type trajectory has a finite transient and cycle. For two hypothetical independent walkers use the product type map on Q_1^2 . Its transient μ and period p satisfy $\mu + p \leq |Q_1|^2$. At each cycle phase $0 \leq s < p$ their positions are

$$L(\mu + s + kp) = L_s + k\Delta_L, \quad R(\mu + s + kp) = R_s + k\Delta_R, \quad k \geq 0. \quad (44.1)$$

Types are fixed within a phase. The displacements over a full product cycle are constants. Reversibility would make the singleton type map a permutation, but no part of the proof needs that simplification.

44.2 Exact acceleration to the next encounter

A mass-two configuration is *near* if its diameter is at most $2r$. Every other such configuration consists of two weight-one sites $L < R$ with gap greater than $2r$. No radius- r neighborhood sees

both. Quiescence then implies that the next update is exactly the superposition of the isolated updates, as long as the configuration starts far.

Each step changes the gap by at most $2r$. Starting with gap greater than $2r$, the walkers cannot cross without entering the near region. For each affine phase of (44.1), solve

$$R_s - L_s + k(\Delta_R - \Delta_L) \leq 2r, \quad k \geq 0. \quad (44.2)$$

If the drift is negative, the first admissible k is the appropriate integer ceiling, truncated below by zero. For zero or positive drift, the phase either already satisfies the inequality at $k = 0$ or never does. Inspect the finite transient too and choose the least resulting physical time. This gives the exact first encounter or proves no encounter occurs.

At that first near time the independently calculated positions are still correct, because the preceding update began far. Stop independent motion there and use the actual CA rule for subsequent behavior. Hypothetical crossings or negative signed gaps after that first time are never used as real dynamics. This boundary is essential for collision correctness.

44.3 Only finitely many encounter shapes

There are exactly

$$N_{\text{near}} = |Q_2| + 2r|Q_1|^2 \quad (44.3)$$

normalized mass-two near shapes: one weight-two site, or an ordered pair of weight-one types at separation $1, \dots, 2r$. Record each near shape together with its time and absolute offset. From a previously unseen near configuration, perform one actual CA step. If the result is still near, continue directly. If it is far, accelerate to its first subsequent near encounter by (44.2), or conclude that this is a terminal forever-far segment.

A weight-two singleton may split, but its next diameter is at most $2r$ and is included. A near pair may merge into one weight-two site. Simultaneous collision, exchange, crossing, and coincident outputs are all handled by the actual CA update rather than a proposed particle collision law. A step starting near has diameter at most $4r$, so every newly departing far excursion starts with gap at most $4r$.

After at most $N_{\text{near}} + 1$ near visits, either a normalized near shape repeats or a terminal forever-far segment has been found. Every finite excursion is computed exactly. A repeated near shape gives the translating periodic tail (43.2). This proves termination of the compressed orbit construction. If $r = 0$, there are no separated-pair interactions, the near-pair term in (44.3) is zero, and each occupied site evolves locally; the same singleton and phase arithmetic handles the case directly.

Mass zero is the stationary vacuum. Mass one is the finite-state walker already described. These complete the orbit descriptions needed for Theorem 44.1.

44.4 Testing every intermediate time

Skipping an excursion without checking its intermediate configurations would not decide reachability. On a phase of a far segment restrict the integer k in (44.1) to its exact interval, which may have no upper bound. The two types a_s, b_s are fixed.

For an exact target with two weight-one sites (u, a) and (v, b) , $u < v$, require the type matches and solve

$$L_s + k\Delta_L = u, \quad R_s + k\Delta_R = v$$

for a common admissible integer k . Incompatible mass or occupied-site counts reject the phase. A weight-two singleton is checked at actual near states.

For a full word on an anchored interval $I = [A, B]$, there are only four possibilities relevant to a far segment:

1. More than two nonzero symbols, or a weight-two nonzero symbol, cannot match
2. Two nonzero symbols require the two matching type and position equations
3. One nonzero symbol at u requires a matching walker there and the other walker strictly outside I
4. An all-zero word requires both walkers strictly outside I

Each case is a finite union of intersections of linear equalities and inequalities in one integer k . Solve them by interval intersection, ceiling/floor operations, and divisibility. This also treats the mass-one walker by omitting the second position.

For occurrence at an arbitrary translate, anchor any nonzero word at a matching occupied site. Two nonzero word symbols require their specified relative gap and types. With one nonzero symbol the other particle must lie outside the corresponding translated interval. An all-zero finite word always occurs sufficiently far from finite support. Again the tests reduce to one-variable affine arithmetic. An empty word, if allowed, occurs everywhere by convention.

A later translated copy of a repeated near loop may hit an anchored target that the first loop missed. Therefore do not stop at a failed first copy. Expand the finite loop into its phases and apply (43.2) to every phase, including all intermediate excursion times; the offset arithmetic then searches every later copy. Unanchored queries need only one copy. This establishes correctness of every stated observation decision.

44.5 Why the fixed-rule algorithm is polynomial

Fix F . Alphabet size, radius, product-type transients and periods, and the number of near shapes are constants. Only the initial far gap may be arbitrarily large. Formula (44.2) computes its first encounter, if any, using binary arithmetic; that time is at most $O(g + 1)$ with a rule-dependent constant, but the algorithm does not simulate g steps. The resulting coordinates have input-linear bit length.

Every subsequent returning excursion begins with gap at most $4r$. During the constant-length product-type transient and one cycle, gap changes by a rule-dependent constant. If cycle drift is nonnegative and this inspected prefix has no encounter, none will occur. If drift is negative, it is an integer at most -1 , so the remaining number of cycles to encounter is also bounded by a rule-dependent constant. Thus every returning post-encounter excursion has constant duration for fixed F .

There are constantly many distinct near visits before a repeat or forever-far tail. Accordingly the compressed orbit has constantly many affine-phase pieces, and every repeating near loop has constant length. The target tests use numbers of input-linear bit length and a fixed number of arithmetic variables. Scanning the explicitly supplied word and performing those tests is polynomial time. No optimal exponent or uniform complexity bound in the description size of F is claimed. Binary encoding of a gigantic gap is crucially different from listing every empty site in that gap.

This proof resembles classical one-counter reachability: after quotienting translation, a separated pair has one unbounded gap, finite internal control, and exceptional dynamics only near a fixed

threshold. The direct argument additionally retains absolute displacement for anchored observations. Haase et al. [23] provides related one-counter context; it is not cited as stating this precise CA theorem. A targeted search did not establish priority of the exact statement.

44.6 What particle bounds from the literature do and do not say

Alhazov and Imai’s 2016 paper on particle complexity [24] is directly relevant. Its verified abstract describes finite number-conserving CA with nonnegative integer states and finite total numerical sum, and announces a universal construction with five particles and a sufficiently large finite radius. It also announces a radius-one construction based on two-register machines. The abstract does not assert reversibility. These are finite-mass infinite-line claims, not finite-ring universality.

Only the abstract-level model and upper-bound statements are relied on here; the full construction and any optimality claim have not been verified for this report. In a numerical-state NCCA with weight equal to the state value, there is only one weight-one state. The broader weighted-alphabet model of Theorem 44.1 allows multiple internal types of weight one. A small-mass construction in that broader model cannot automatically be quoted as a numerical-state particle bound. Neither the five-particle literature statement nor the mass-two decision theorem establishes a three-particle reversible universal model or a minimum universal mass.

45 Horizon-free semilinearity at mass at most two

The low-mass decision argument supports a stronger Diophantine conclusion. Unlike the generic sparse certificate, the following result has no supplied horizon and has fixed witness arity after the rule and its arithmetic description are fixed. It applies to a decidable subclass, so it creates no fixed-arity statement for arbitrary universal computation.

Theorem 45.1 (Effective timed and untimed semilinearity). *Fix a CA satisfying the hypotheses of Theorem 44.1, and fix the ordered nonzero labels of an initial configuration and a target, each of mass at most two. The numerical-coordinate relation*

$$\mathcal{R}_F(\mathbf{x}, \mathbf{y}, T) \iff T \in \mathbb{N} \text{ and } F^T(x) = y$$

is effectively Presburger-definable. Its projection $\exists T \mathcal{R}_F$ is also effectively Presburger-definable. Equivalently these are effective semilinear coordinate relations, with signed coordinates interpreted by orthants or the canonical natural encoding below. Taking a finite union allows the labels to vary over their finite allowed cases.

Here Presburger arithmetic has addition, order, integer constants, and quantification, but no multiplication of two unbounded variables. Semilinearity concerns the numerical coordinates, not a language of their binary digit strings. The proof must show why the initial unbounded gap never becomes a variable period or a variable coefficient in a later phase.

45.1 A linear guard for the entire independent prefix

Fix the initial ordered weight-one types at positions $u < v$, with $g = v - u > H_0 = 2r$. Let λ_i, ρ_i be the cumulative independent displacements at time i , starting at zero. For the product-type transient

μ and period p , put $i_s = \mu + s$ and write

$$\begin{aligned} t &= i_s + pk, \\ L(t) &= u + \lambda_{i_s} + D_L k, \quad R(t) = v + \rho_{i_s} + D_R k, \quad k \geq 0. \end{aligned} \quad (45.1)$$

All displayed coefficients except u, v, k are fixed by the rule and initial types. Set $c_i = \rho_i - \lambda_i$ and $D_* = D_R - D_L$. The gap at phase s is $g + c_{i_s} + D_* k$.

Define $\text{Safe}(g, T)$ to mean that every gap at $0 \leq h < T$ exceeds H_0 . This ensures that actual motion agrees with the independent motion at time T , including a possible first near time. It is already a Presburger statement by quantifying h and expanding its finite phase alternatives. A quantifier-free finite guard in the chosen phase is also explicit.

If $T = i < \mu$, require $g + c_h \geq H_0 + 1$ for $0 \leq h < i$. If $T = \mu + s + pk$, first impose those inequalities for all $h < \mu$. For a cycle phase $j < s$, earlier occurrences have indices $0, \dots, k$; for $j \geq s$ they have indices $0, \dots, k-1$, an empty set if $k = 0$. Minima of these arithmetic progressions give the remaining guards:

Drift	$j < s$	$j \geq s$
$D_* \geq 0$	$g + c_{i_j} \geq H_0 + 1$	$k = 0$ or $g + c_{i_j} \geq H_0 + 1$
$D_* < 0$	$g + c_{i_j} + D_* k \geq H_0 + 1$	$k = 0$ or $g + c_{i_j} + D_*(k-1) \geq H_0 + 1$

These are Boolean combinations of constant-coefficient affine inequalities. At time zero all earlier-index sets are empty, as required. Conjoin the applicable guard, the time equation, the current type matches, and target-coordinate equations from (45.1). A finite disjunction over phases describes every target time before or at first near entry, as well as all times on a forever-free initial flight.

45.2 First entry with the absolute position preserved

For $r \geq 1$ the first near entry from an initially far pair has two distinct weight-one sites and gap $e \in \{1, \dots, H_0\}$. A step from gap at least $2r + 1$ leaves gap at least one, so a merger or crossing cannot skip this boundary. There are finitely many entry shapes E .

For a transient entry time $\tau = i$, impose its Safe guard, phase labels, and

$$g + c_i = e, \quad \xi = u + \lambda_i, \quad \tau = i.$$

For a periodic entry at $\tau = \mu + s + pk_0$, impose its Safe guard, $k_0 \geq 0$, phase labels, and

$$g + c_{i_s} + D_* k_0 = e, \quad \xi = u + \lambda_{i_s} + D_L k_0, \quad \tau = \mu + s + pk_0. \quad (45.2)$$

Here ξ is the absolute leftmost entry coordinate. The guard enforces *earliest* entry, rather than an arbitrary later meeting of hypothetical walkers. Every actual first entry satisfies one such case; conversely the guard and bounded displacement prove that each satisfying case is the real first entry. Thus the full relation between initial coordinates, entry shape, time, and absolute offset is Presburger.

45.3 Finite rule-fixed orbit templates after entry

For each normalized near shape E at offset zero, preprocess its orbit by the near/far algorithm of Theorem 44.1. There are only N_{near} such shapes, each fixed by the rule. Every returning excursion

has rule-dependent constant duration; preprocess until a normalized near repeat or a terminal nonreturning free flight. Expand the finite returning flights and one complete recurrent loop when present.

This gives a finite collection of templates, each with fixed occupied-site count, fixed ordered labels, and integer constants $\theta, \pi, \eta_j, \delta_j$, representing

$$\text{elapsed time} = \theta + \pi n, \quad \text{site } j = \eta_j + \delta_j n. \quad (45.3)$$

A listed isolated time fixes $n = 0$; an infinite template has $n \geq 0$ and $\pi > 0$. A translating loop has the same displacement for every particle in a phase. A terminal free flight has the finite-state phase templates already used above, possibly with different left and right drifts. The occupied-site order holds on every represented time, by translation for a loop and by nonreturn for a free tail.

Starting E at offset ξ adds ξ to all positions. Compose (45.3) with (45.2):

$$T = \tau + \theta + \pi n, \quad y_j = \xi + \eta_j + \delta_j n.$$

Substitution gives affine expressions in the unbounded variables k_0, n . There is no product $k_0 n$: the entry shape has been chosen from a *finite* set, so every subsequent period and drift is a rule-fixed constant, independent of the initial gap. A finite union over entry cases and templates therefore remains Presburger. Targets before or at entry were already covered; these compositions cover every later target. Overlap of descriptions is harmless for definability.

An initially near input selects one of the same finite templates directly, with its initial offset and $\tau = 0$. A mass-two singleton can split and a pair can merge, so one discards mismatched *template* site counts rather than assuming the occupied-site count is conserved. A total-mass mismatch is impossible. Vacuum and mass-one cases have stationary or finite-state affine descriptions. At radius zero, each occupied coordinate is fixed and its label follows a finite map, giving a finite transient and periodic label condition. These arguments prove the timed part of Theorem 45.1; existentially quantifying T proves the untimed part.

45.4 Effective elimination and canonical signed input

Standard effective Presburger quantifier elimination converts the constructed formula into a quantifier-free Boolean combination of affine inequalities and fixed-modulus congruences; see Cooper [25]. The effective equivalence with semilinear sets is classical, as in Ginsburg–Spanier [7]. These standard results are used for the final arithmetic normal form, rather than assumed as a CA-specific reachability theorem.

Encode each signed coordinate a by natural inputs (a^+, a^-) with

$$a = a^+ - a^-, \quad a^+ = 0 \text{ or } a^- = 0.$$

This gives exactly one representation of each integer. Include this domain condition and strict ordering of occupied coordinates in the predicate before elimination. With at most two initial and two target sites, there are at most eight natural coordinate inputs, and at most nine external natural inputs when T is retained. For untimed reachability, eliminate T itself before the next compiler. Leaving it as a witness would generally destroy uniqueness when a target is visited at multiple times.

46 Canonical fixed-arity quartics for the low-mass relation

Here is a complete deterministic compiler from a fixed quantifier-free Presburger predicate $\Phi(\mathbf{z})$ on natural inputs. Normalize its atoms to $L(\mathbf{z}) \geq 0$ or $L(\mathbf{z}) \equiv 0 \pmod{d}$, where L is affine with integer

coefficients and $d \geq 1$ is fixed. Equality becomes two inequalities and strict inequality a shifted weak inequality. Express the Boolean structure by AND, OR, and NOT gates. Let I, C, G be the actual numbers of inequality atoms, congruence atoms, and Boolean gates in the chosen circuit, respectively. Identical nodes may be shared or repeated; the counts refer to the circuit actually used.

46.1 Inequalities and congruences have unique arithmetic witnesses

For $L \geq 0$ introduce natural b, s with residuals

$$b(b-1), \quad L - (2b-1)s - b + 1.$$

They force $(b, s) = (1, L)$ when $L \geq 0$, and $(b, s) = (0, -L-1)$ otherwise. Thus both witnesses exist uniquely for every input, and b is the truth bit even when the atom is false.

For $L \equiv 0 \pmod{d}$ introduce six natural witnesses q^+, q^-, a, h, b, s and five residuals

$$\begin{aligned} q^+q^-, \quad L - d(q^+ - q^-) - a, \\ a + h - (d-1), \quad b(b-1), \quad a - (2b-1)s - b. \end{aligned} \tag{46.1}$$

The first three impose the unique signed Euclidean quotient $q = q^+ - q^-$ and remainder $0 \leq a < d$. The product zero condition gives the unique sign-separated quotient representation; $h = d - 1 - a$. The last two force $b = s = 0$ if $a = 0$, and $b = 1, s = a - 1$ if $a > 0$. Therefore the congruence truth bit is the affine expression $1 - b$. All witnesses are uniquely fixed, including negative quotients and modulus one.

46.2 Boolean gates and the exact ledger

For each Boolean gate introduce one natural output z_g and its corresponding residual

$$\begin{aligned} \text{AND : } z_g - uv, \\ \text{OR : } z_g - u - v + uv, \\ \text{NOT : } z_g - 1 + u. \end{aligned}$$

Inputs are preceding outputs, atom truth bits, or constants zero and one. They are already Boolean, so each gate output is uniquely its correct Boolean value. Add one residual $o - 1$ for the final output o . Summing the squares of all residuals gives a polynomial of degree at most four with exactly

$$K_\Phi = 2I + 6C + G \quad \text{natural witnesses,} \quad E_\Phi = 2I + 5C + G + 1 \quad \text{residuals.} \tag{46.2}$$

Every input has exactly one complete assignment to the atom and gate witnesses. The final residual accepts it exactly when Φ is true. Thus the natural fiber is a singleton for true inputs and empty for false inputs. This does not rely on a product of branch equations with unconstrained inactive witnesses.

Corollary 46.1 (Horizon-free canonical low-mass certificates). *For each fixed CA and ordered label case in Theorem 45.1, timed and untimed exact reachability admit effectively constructible integer polynomials of degree at most four with zero or one natural auxiliary witness tuple. Their witness arity is the rule- and compiler-dependent constant K_Φ in (46.2), after effective quantifier elimination and a chosen Boolean circuit. No supplied time bound is required.*

These constants may be extremely large. The report neither instantiates a quantified-formula eliminator for every possible rule nor claims a small universal numerical bound, efficient uniform preprocessing, or an optimal arity. Total arity is the number of external natural inputs plus K_Φ , at most $9 + K_\Phi$ timed or $8 + K_\Phi$ untimed for fixed labels. Mass two does not mean two witnesses. Coefficients and their bits, formula length, and Boolean-circuit size remain paid parts of a concrete instantiation.

Uniqueness here comes from the *post-elimination arithmetic compiler*, not from CA determinism: an untimed target may be reached at many or infinitely many times. No corresponding real or rational witness characterization is claimed. No analogous semilinearity or finite-fold statement for arbitrary recursively enumerable sets follows from this decidable low-mass result.

[write] **A second route, at a higher degree.** Corollary 46.1 rests on the general fact that a semilinear set has a single-fold Diophantine representation, and for that fact the collection already has a sharper statement: Theorem `cdc:of:thm:singlefolds1` of Part XI of *canonical-diophantine-certificates* gives, for every semilinear set, an integer polynomial of degree at most *two*, nonnegative on the real nonnegative orthant, with exactly one natural witness for each member and none otherwise, built in the same way (Presburger quantifier elimination, then deterministic evaluation of the order and congruence atoms, with the comparison of `cdc:of:lem:unboundedcompare` and a Euclidean congruence gadget, and Boolean gates). The compiler of this section is a second route to that fact, of degree four (squared residuals) rather than an orthant-nonnegative quadratic, which carries the exact ledger (46.2). What is source 13's here is the semilinearity of the low-mass relation (Theorem 45.1), to which either compiler applies. Theorem 45.1 is related only loosely to Part I's rigidity theorem (Theorem 13.1): there a fixed globally nonnegative quadratic forces a semilinear zero set, here low-mass dynamics forces a semilinear reachability relation.

47 Reproduction and evidence

The accompanying ZIP contains editable LaTeX, the rendered PDF, standard-library Python implementations, coefficient-explicit fixtures, completed finite local tables, structured test receipts, and a SHA-256 manifest. It contains no copies of third-party papers. Source links and exact attribution limits are given in the bibliography and provenance document.

47.1 What each layer of verification establishes

The producer compiler constructs each sparse polynomial as a canonical list of integer-coefficient monomials. An export includes the public input/table descriptor, variable and residual ordering, canonical witness values, row aliases, degree and scalar ledgers, and every residual coefficient. Its bound verifier regenerates the declared circuit and compares it before evaluating the witness. This catches deletion of equations, changed input binding, changed row aliases, altered accounting, and malformed scalar types. Merely checking that a supplied witness satisfies some unbound list of equations would be weaker. A caller seeking one particular input must still compare the public descriptor with that intended input; no file can infer an unstated intention.

An independent implementation separately constructs the preferred row-selector polynomials. Its cross-check compares residual coefficient multisets under an explicit wire-name correspondence, not only decoded trajectories or final zero scores. Separate exact dynamics tests compare sparse and dense physical evolution. A third set of source checks verifies the corrected rule schemes, legal source macrosteps, mass-fiber completion, and the false-observer example. Finally, the low-mass

audits test encounter acceleration and query arithmetic; they are not exhaustive tests of all CA rules.

Finite tests support the implementations and expose edge cases; the general correctness results rest on the proofs in this report. None of the code constitutes a proof-assistant formalization, an exhaustive literature search, or a test of every natural witness.

47.2 Selected literal fixtures

Fixture	V	R	Multiplications	Additions
Binary three-way collision	288	243	1,614	1,424
Same, orthant-exact	288	261	1,830	1,559
Gap 10^{100}	108	88	540	495
Mixed ternary mass	610	370	3,694	3,287

All counts use the literal convention (40.1). The huge-gap example has maximum witness bit length 334 and residual-coefficient bit length 333, illustrating that sparse dimension can remain small while geometric values and coefficients grow. The mixed-mass fixture has largest witness bit length four. The orthant collision fixture adds eighteen residuals, exactly $2MT$.

The restricted-row doubling fixture is bound to the full completed local table but uses the declared 97-row simulation subset. It verifies an actual first pulse, rather than merely simulating a source trajectory outside the polynomial. The larger independently replayed $n = 2$, $T = 41$ case has 257,872 witnesses, 157,683 residuals, 1,263,585 nonzero residual terms, 1,694,649 literal multiplications, and 1,421,268 additions. Its largest witness is 45. Both coefficient-explicit instances are shipped. The smaller $n = 0$, $T = 7$ instance has 45,712 witnesses, 28,585 residuals, 222,845 residual terms, 299,323 multiplications, and 251,430 additions. Its largest witness is 25. The counts for each are generated and checked by the included script.

47.3 Audit coverage

The independent dense/sparse dynamics tests cover 13,200 randomized steps and 73,728 steps across all 36 binary conservative local permutations, for 86,928 comparisons. The independent preferred literal compiler checks 36 instances containing a total of 17,058 witnesses and 9,663 core residuals, and verifies the $2MT$ orthant increment. The earlier factorized independent compiler checks are retained separately with their own ledgers; those results are not relabeled as preferred-backend tests.

The preferred producer/independent coefficient cross-check matches 54 compilers, 21,096 witnesses under explicit renaming, and 11,844 residuals coefficientwise. It additionally checks 20 uniform-loader instances, 24 paid loader-plus-observer instances (12 valid first pulses and 12 correctly rejected earlier pulses), six restricted-row coefficient comparisons, and ten accounting or declaration mutations. Restricted-domain exits and a restricted table omitting the vacuum are rejected. These tests use the final preferred backend, not the older factorized compiler.

The source audit exhaustively enumerates 373 valid three-state source machines under its stated syntax and endpoint conventions, checking 8,672 legal macrosteps with both counters between zero and three. It also checks 3,000 seeded random source machines with 2–12 states for local properties, and 22,004 legal macrosteps on the first 300. Every rule family is exercised. There are 672 startup-signal predecessor tests. The doubling source is replayed for $n = 0, \dots, 10$, including

exact microtimes, final counters, and persistent result centers. The nonuniversal doubling and false-signal completions contain 17,576 and 9,261 rows respectively.

The independent mass-two arithmetic audit checks 93,663 cases: 13,656 exhaustive and 20,000 random encounter instances; 20,000 anchored-word, 20,000 unanchored-word, and 20,000 exact-configuration queries; and seven large-integer or unbounded cases. A separate encounter regression checks 6,750 far segments and a 100-digit encounter. These are tests of the affine arithmetic used in the proof, not claims about that many distinct CA rules.

The semilinearity component regression checks 1,697,593 cases: 484,788 exhaustive and 1,210,000 random free-prefix guards, four hundred-digit time cases, 81 inequality and 1,452 signed congruence bounded-fiber cases, twelve Boolean truth-table cases, 256 composed-polynomial inputs, and 1,000 single-variable mutations. An independent audit checks another 837,150 arithmetic and actual-CA cases, including 394,740 timed configurations with splitting, merging, recurrent near templates, and terminal free flights. Both audits include exact affine and negative-quotient edge cases. Neither implements arbitrary-rule quantifier elimination or constitutes formal verification.

The release runner regenerates outputs in a separate working directory and compares stable results with the shipped receipts. Its README lists commands, dependencies, and the optional larger reproduction. The manifest detects accidental modification; it is an integrity record, not a replacement for the mathematical or semantic checks.

[write] **The release in this report.** The programs are shipped as `code/13-sparse-lattice-*`, the receipts, generic fixtures and the two completed tables as `data/13-sparse-lattice-*`, and the provenance record as `13-sparse-lattice-SOURCE-PROVENANCE.md`; the delivered sub-directories `replay/`, `receipts/` and `tables/` are flattened, and the three programs named `audit.py` keep their directory in their shipped names (README of this report). The two completed tables (`data/13-sparse-lattice-doubling-complete.csv` and `data/13-sparse-lattice-false-signal-complete.csv`) are kept byte for byte with their delivered CRLF line ends. The two coefficient-explicit Morita fixtures named above (the $n = 0$, $T = 7$ and $n = 2$, $T = 41$ instances, 0.84 and 4.37 MB compressed) are *not* shipped: the shipped generator rebuilds them in about 14 s and 94 s with identical decompressed payloads, and the README gives the commands; their receipts are shipped. The delivered manuscript (six L^AT_EX files), PDF, README, checksum ledger, packager and one byte-identical copy of the Morita audit are not shipped. The programs need the delivered layout and POSIX (LF) newlines, and must be run on a copy (README). Since batch 80 the corrected code edition (next paragraph but one) adds the exactness suite `code/13-sparse-lattice-test_poly_exactness.py`, its receipt `data/13-sparse-lattice-poly_exactness.json` and the correction note `13-sparse-lattice-CORRECTION.md`.

[write] **The repository's review.** The programme reviewed this archive in full [30]: it found the generic history theorem, the loader and observer, the corrected source interface, the diameter barrier, the mass-two decision proof and the semilinearity argument sound within their stated domains, regenerated both large Morita fixtures, and ran both complete eighteen-stage replays. Its one finding is a low-level exactness defect: the public Poly constructor of `sparse_mass.py` accepts floating coefficients and mutable terms, so a hand-built floating residual can round -1 to 0. The high-level compiler and the JSON decoder already require exact coefficients, so the finding refutes no theorem and no exported certificate. The review's repair is the patch `sparse_mass_exact_polynomials.patch` in the programme's directory [30]; the shipped `code/13-sparse-lattice-sparse_mass.py` was the delivered, unpatched file until batch 80 (see the paragraph after this one). Two later reductions of the programme bear on this Part's ledgers without changing its theorems: a wire projection of the preferred compiler with a fixed initial configuration [32] removes $T[7M + M(M - 1)]$ natural

variables and as many rows, leaving $V = T[M(S + 7) + 4M(M - 1)]$ and $R = T[10M + 4M(M - 1)]$ with quadratic residuals and a bijection of natural zero sets; and a five-witness congruence atom [33] lowers the witness count (46.2) of the post-elimination compiler from $2I + 6C + G$ to $2I + 5C + G$ at the same residual count. Both are the programme's, are not re-proved here, and are not gate counts.

[write] **Corrected code edition (added 2 October 2026, batch 80).** `Sparse_Lattice_Diophantine_Certificates_corrected.zip` (arrival 4e270aa46, batch 80, manuscript 07; placed by 8a4e64732, batch 80, cluster K1; the archive dates its correction 3 October 2026) repairs the review's finding; its authors read the review's commit but did not execute its code. `Poly` now validates directly supplied terms: immutable tuples, exact integer indices and coefficients (not Booleans, floats or subclasses), nonzero coefficients, sorted indices, distinct sorted monomials; `Poly.make` remains the canonicalizing constructor, and negative (parameter) and repeated (power) indices and the zero and constant polynomials stay valid. The new suite `code/13-sparse-lattice-test_poly_exactness.py` rejects 24 malformed constructor categories and the floating false zero (the exact residual at $2^{60} + 1$ is -1 , with score 1), checks canonicalization and mutation isolation, and compares 2,700 exact expressions on 300 seeded pairs; with `-baseline-module` it reproduces the defect on the original (residual 0.0, accepted witness) and compares the same expressions with the original producer. The release runner runs it first (nineteen stages with `-regenerate-source`, eighteen before). The repair is written independently and is equivalent to the programme's patch: the patched original passes the suite and the unpatched original fails it. The patch is not to be applied to the shipped file, which already contains the repair (applied anyway, it would insert a second constructor check). The manuscript, the fixtures, both tables, every receipt value other than the producer hash and the new stage, and every theorem and ledger of this Part are unchanged; at the batch-79 write the shipped file was the original, and since batch 80 it is the corrected edition's. The correction note is `13-sparse-lattice-CORRECTION.md`, and the shipped provenance record `13-sparse-lattice-SOURCE-PROVENANCE.md` now carries the corrected edition's dated preface. The programme's correction audit [36] confirms the repair, that removing the new guard leaves the module's syntax tree identical to the original's, and the unchanged mathematics.

48 Further questions and boundaries

The strongest next steps are narrowly defined rather than general claims of universality improvement.

1. **Print one safe universal source.** Choose and export a concrete universal counter simulator, retain the history and prime-encoding invariants, and provide its actual m , complete local table, valid input encoding, and proof of reachable legality. This would replace the present existence interface with a fully numerical instance ([write] added 3 October 2026, batch 82: answered in substance outside this report by Part II of `five-particle-binary-automata` (Report 16), which prints a universal reversible two-counter source with 122,622 controls and 141,561 literal rows, makes this Part's history and prime-encoding invariants literal, and proves its clean loader and universality for all inputs, for Report 15's guarded syntax; a universal table in this Part's pulse interface is still not printed)
2. **Compress sparse microtime when structure permits it.** The certificate pays one layer per physical lattice step. Large counter gaps create long round trips. An exact arithmetic macrostep compiler for the source could remove these journeys without changing the finite-mass semantics, but would have to pay for branch tests, timing, and valid intermediate routes

3. **Separate mass, state, radius, and degree costs.** The row-selector simplification reduces residual count dramatically while retaining q^3 witnesses and long sums. Other explicit lookup or collision organizations may trade these resources. None of the displayed counts is a lower bound or minimality record
4. **Resolve small-particle model comparisons.** Check the full particle-complexity constructions and their precise observation, state-weight, radius, and reversibility conventions before claiming a sharper universal mass threshold. The mass-two theorem is a rigorous exclusion in its stated positive-weight model; it is not a proof of a matching upper bound ([write] added 2 October 2026, batch 80: answered in substance by Parts V and VI. The number of weight-one symbols is the decisive convention: with many, three units suffice for a fixed reversible automaton, Part V; with at most one, masses three and four are decidable, so integer-state automata need at least five particles, Part VI. Still open: the full Alhazov–Imai construction, a reversible numerical threshold, a binary upper bound, and the least number of weight-one symbols for universality at mass three) ([write] added 3 October 2026, batch 82: the binary upper bound and the reversible numerical threshold are now proved outside this report, by Part I of *five-particle-binary-automata*: five, with source 17’s lower bound; Section 1.16)
5. **Keep exact targets separate from halt signals.** A reduction to a specified complete target needs its own output-normalization argument. The safe-source pulse interface alone cannot supply it ([write] added 2 October 2026, batch 80: answered for the three-mass weighted automaton by source 15, Theorem 67.1, whose output normalization is inverse execution of a fresh-entry source, with a target y_N that depends on the input; the pulse interface of this Part is unchanged)

The construction gives a family of finite-horizon Diophantine certificates with explicit variable growth. Treating T as an additional unknown in a single fixed-arity universal equation, or hiding all horizons behind an unspecified MRDP representation, would be a different result and would not inherit these scalar counts. No such transfer is made for the universal higher-mass sparse family. The horizon-free fixed-arity construction in Sections 45–46 is instead for the decidable mass-at-most-two subclass and pays its actual post-elimination circuit size.

[write] Two of these questions touch questions elsewhere in the collection. The first is the lattice counterpart of Part III’s literal source: Part III prints a complete universal conservative signal machine, while source 13 deliberately leaves its safe universal source as an existence construction. The certificate of Theorem 38.1 also bears on the question “Local certificates on adaptive support geometry” of Part XVI of *canonical-diophantine-certificates*, which asks for a sparse lattice support checked canonically at a cost proportional to support and boundary, paying for sorting, distinctness and nearest-neighbour tests: source 13 pays exactly those costs (pairwise equality and a bubble-sorting network) for one-dimensional conservative partitioned automata, at size $O(T(M^2 + Mq^3))$, which is not proportional to support plus boundary, and not for sandpiles. That question stays open.

Part V

Three conserved mass units suffice (sources 14 and 15)

[write] Part V prints two AI-assisted research manuscripts of ProveIt’s incoming reports, both dated 3 October 2026 and arrived in batch 80 (commit `4e270aa46`). *Source 14* is batch-80 manuscript 10, *Three mass units in a reversible weighted cellular automaton*, author line “Mathematical construction and reproducibility report”, from the archive `Three_Mass_Reversible_Computation.zip`; its labels carry the prefix `smc:tm:`, and its sections are Sections 49–63 here. *Source 15* is batch-80 manuscript 02, *Exact targets with three conserved mass units*, author line “A clean return extension of the reversible weighted CA construction”, from the archive `Exact_Targets_Three_Mass_Units.zip`; its labels carry the prefix `smc:ct:`, and its sections are Sections 64–73. They are called sources 14 and 15 after the prefixes of their shipped files. Source 15 is a declared extension of source 14 (“a separate, 15-page mathematical extension of the frozen three-mass construction”, in its delivery README): it vendors five of source 14’s programs byte for byte and pins source 14’s L^AT_EX source, PDF and checksum ledger by SHA-256 as its “frozen companion release 9”, and all three pins match source 14’s archive. Neither manuscript names a repository commit or cites Part IV or any repository file. Statements keep their numbering by section, and equations are numbered by section. The programme has reviewed both archives [45] (Section 1.14).

[write] **Relation to Part IV.** [Editorial note: source 14’s Section 59 proves Theorem 44.1 of Part IV (source 13) again, by the same argument: the same walker maps α, d , the same near/far threshold $2r$ and encounter inequality, the same count $|Q_2| + 2r|Q_1|^2$ of near shapes, the same translating tail and the same fixed-rule polynomial bound. Its two-mass arithmetic test is source 13’s audit (the shipped `code/14-three-mass-test_two_mass_arithmetic.py` differs from `code/13-sparse-lattice-two-mass-audit.py` only by omitting the receipt-writing line), and its receipt is byte-identical to `data/13-sparse-lattice-two-mass-arithmetic.json`, which is therefore not shipped twice. The section is printed as a pointer, keeping source 14’s statement and its use in Theorem 49.2; source 14 calls that proof “self-contained” and does not cite source 13.] The certificate gate $(E_t - e_{tj})(e_{tj} + u_{tj})$ of source 14 (and of source 15) is the inactive-product gate of *quadratic-orthant-certificates*, Part VI (`qoc:rn:lem:gates`); a note after Theorem 58.1 says so. Part IV and Part V both draw on Morita and Imai’s reversible two-counter normal form [19] (source 14 cites its Proposition 3.4, source 15 adds its Lemma 3.5); Part IV also interfaces with their partitioned automaton, whose mass is $m + 19$ for an m -state source, as source 14 recalls (Section 62); source 14’s automaton keeps three units for every source, on finite-support configurations with no background. Source 14 does not cite Kong’s thesis or Imai’s KAKEN report, which concern the numerical model of Part VI; a note in Section 62 points there.

Source 14 (batch-80 manuscript 10): Three mass units in a reversible weighted cellular automaton

Mathematical construction and reproducibility report, 3 October 2026. Its abstract follows.

A fixed one-dimensional reversible cellular automaton of radius at most four can simulate a universal reversible two-counter machine using configurations of total mass exactly three. The model has a finite alphabet of Boolean channel subsets, with subset cardinality as conserved weight and a unique zero-weight vacuum. Two counters are encoded by a single gap $12 \cdot 2^a 3^b$. A reversible residue query, rational gap changes, and finite instruction-tag cleanup give an explicit

local permutation schema. Spatial blocking gives radius one at the same clock with four times as many types. A coordinate-preserving phase alternative has a fourfold clock cost. Both preserve the mass. A self-contained mass-at-most-two decidability proof makes three the sharp threshold for fixed-rule undecidable finite-pattern occurrence in this weighted model. We also give a quadratic natural-number certificate at a specified source-step horizon, with exact physical time and a fully stated resource ledger. The source compiler and literal finite examples are executable. Universality uses the classical existence of a universal reversible two-counter machine; this release does not contain an expanded universal source-machine table. The three-unit statement does not transfer to integer-state number-conserving CA, where the state itself is its numerical mass.

49 Results and scope

All configurations in the simulation live on the infinite line $\mathbb{Z}$, with finite support. A finite-support initial condition is an ordinary finite object even when its nonzero sites are very far apart. There is no finite-ring assumption and no active zero-weight background.

Theorem 49.1 (Three-unit compiler). *For every finite deterministic reversible two-counter machine M in the separated-instruction form of Section 50, there is an effective finite-alphabet CA F_M with the following properties.*

1. F_M and its inverse have radius at most four. A weight function w takes positive integer values on all non-vacuum symbols, has a unique zero-weight vacuum, and is conserved on finite configurations.
2. The canonical encoding of (q, a, b) has total mass exactly three and gap $12N$, where $N = 2^a 3^b$. It is computable from the source input.
3. Each enabled source transition produces precisely the next canonical encoding after the exact finite time specified in Section 56. Nonhalting encoded trajectories never depend on unspecified collision rows.
4. A designated one-cell symbol $\text{Ready}(q_f)$ occurs somewhere if and only if the encoded source computation reaches its halt state. The observation is taken after all finite scratch and instruction data have been reset.

Theorem 49.2 (Sharp threshold in the weighted model). *There exists one fixed reversible CA satisfying Theorem 49.1 for which occurrence somewhere of a fixed finite pattern is undecidable on a computable family of initial configurations of mass exactly three. In contrast, for every fixed deterministic conservative CA with a finite alphabet, finite radius and a unique zero-weight vacuum, exact finite-configuration reachability and finite-pattern occurrence are decidable from every initial configuration of mass at most two. Reversibility is unnecessary for the lower-mass assertion. Therefore three is the least mass allowing fixed-rule undecidable finite-pattern occurrence in this model.*

Section 57 strengthens the upper bound to a fixed rule of radius one, with the same three mass units. Its inverse also has radius one. Radius zero cannot yield this undecidable observation problem, so both the weighted mass threshold and the radius threshold are attained.

The lower-mass statement includes anchored observations. The upper-mass result here uses occurrence *somewhere*; it does not assert anchored halt observation, or exact-target configuration undecidability. The latter would require an additional construction when output counters are

unknown. Universality means computation-universality under an effective input encoder and designated observation, not intrinsic universality or an efficiency bound.

The local alphabet is $2^{\mathcal{T}}$ for a finite type set $\mathcal{T}$. Many distinct symbols have weight one. The finite type information is charged to alphabet size, while each occupied channel contributes one unit of mass. An ordinary integer-state number-conserving CA has just one symbol whose numerical state is one. Merely renumbering our symbols cannot preserve all their weights. Thus this is not a three-numerical-particle theorem for that more restrictive model.

[write] [The lower-mass half of Theorem 49.2 is Theorem 44.1 of Part IV (see Section 59). Part VI makes the model point of the last paragraph precise: with at most one weight-one symbol, masses three and four are decidable (Theorems 74.1 and 83.1).]

50 Reversible counter machines and the gap encoding

A source configuration is $(q, a, b) \in Q \times \mathbb{N}^2$, where $\mathbb{N} = \{0, 1, 2, \dots\}$. Instructions have a source state, target state, counter index in $\{0, 1\}$, and one of five operations: increment, decrement, no-op, zero test, or positive test. Tests leave both counters unchanged; decrement is enabled only for a positive counter. The final state q_f has no outgoing instructions.

The separated syntax has at most one outgoing motion instruction at a state, or zero/positive test alternatives on one counter. Dually, incoming instructions at a state consist of at most one motion instruction, or zero/positive tests on one counter. Distinct tests with the same endpoint must have disjoint guards. These are the syntactic conditions underlying the reversible normal form of Morita and Imai [19]. They are sufficient for forward determinism and backward uniqueness on counter configurations. Fused guarded arithmetic is not being substituted for this form.

Number the instructions $1, \dots, m-1$ and reserve 0 for “no enabled instruction.” Encode the two counters by

$$N = 2^a 3^b \geq 1, \quad n = 12N. \quad (50.1)$$

Increment and decrement multiply N by $2, 1/2, 3, 1/3$; a zero test on a counter is nondivisibility by its associated prime. The six-valued residue $N \bmod 6$ determines both zero/positive statuses. Prime-exponent encoding in one distance and finite-state divisibility protocols already appear in the three-agent grid-exploration construction of Brandt, Portmann and Uitto [37]; neither device is claimed as new here.

Consequently there are total finite tables

$$I : Q \times \mathbb{Z}_6 \longrightarrow \mathbb{Z}_m, \quad H : Q \times \mathbb{Z}_6 \longrightarrow \mathbb{Z}_m.$$

Here $I(q, r)$ returns the enabled outgoing instruction or 0. For H , a unique incoming motion is selected without consulting the residue, while incoming test alternatives are distinguished by their guard on the unchanged output counters. If no incoming instruction applies, set $H = 0$. On the output of any actual source step, H returns that step’s unique instruction. No search through predecessor counter values is required.

The input cost is geometric as well as finite-state. A sparse binary-coordinate encoding of $-12 \cdot 2^a 3^b$ uses $\Theta(a + b + 1)$ bits, while its numerical gap is exponential in the counter values. If a, b themselves are supplied in binary, this conversion need not be polynomial in their input bit length. The reduction requires a computable encoder, and makes no stronger efficiency claim. A fixed source table has a fixed finite channel set; allowing a larger table changes that set and its alphabet cost.

The classical existence dependency is precise: Proposition 3.4 of Morita and Imai [19], using their Definitions 3.1–3.3, states that a deterministic reversible two-counter machine can simulate any Turing machine. Apply it to a fixed universal Turing machine and fix the resulting finite source machine. The compiler theorem is effective once that finite table is supplied. This release supplies small literal test tables, rather than that expanded universal table. The earlier universal-source result is Morita [20].

[write] [Part IV’s source interface (Section 41) uses the number-conserving partitioned automaton of the same paper, with a corrected sign; source 14 uses only its two-counter normal form and Proposition 3.4. The programme’s review checked Definitions 3.1–3.3, Proposition 3.4 and Lemma 3.5 in the primary paper [45].]

51 The globally reversible channel construction

For every $t \in \mathcal{T}$, fix an integer velocity $v(t) \in [-4, 4]$. At each lattice site the symbol is a subset $A \subseteq \mathcal{T}$. Each channel is Boolean: the same type cannot occur twice at one site. Set $w(A) = |A|$ and use the empty subset as vacuum. Different channels may occupy the same cell, so number of occupied cells and mass are different quantities.

One tick is

$$F = S \circ \Pi, \tag{51.1}$$

where Π applies a single permutation π of $2^{\mathcal{T}}$ independently at each site, and S shifts channel t by $v(t)$. In particular, the output bit for channel t at site x is the t -bit of π at site $x - v(t)$ before the tick.

Lemma 51.1 (Finite completion). *Suppose the prescribed singleton rules form a permutation, and the prescribed two-element subset rules form a finite partial injection. They extend effectively to a cardinality-preserving permutation of $2^{\mathcal{T}}$, fixing the empty subset and every subset of cardinality at least three.*

Proof. The directed graph of a partial injection on the finite set of unordered pairs has in-degree and out-degree at most one. Each component with edges is a cycle or a finite directed path. Close each noncyclic path by the edge from its terminal vertex back to its initial vertex. Keep every pair outside this support fixed. Distinct components are disjoint, so the result is a permutation. Combine this pair permutation with the singleton permutation and the identities in all other cardinality sectors. $\square$

Identity on *all* unspecified pairs before path closure would generally fail: a prescribed image could collide with such an identity image. The completion procedure is part of the construction.

Lemma 51.2 (Global reversibility and conservation). *The completed rule in Lemma 51.1 defines a CA F that is a bijection on the full configuration space $(2^{\mathcal{T}})^{\mathbb{Z}}$. Both F and F^{-1} have radius at most four, and F conserves total mass on every finite-support configuration.*

Proof. Both Π and S are full-space bijections. Their inverse composition is $\Pi^{-1}S^{-1}$; the latter first shifts each channel by the negative of its velocity. Each local output depends on positions within distance four. The local permutation preserves cardinality, and shifting individual Boolean channels permutes their sites without creation or loss. Summing channel occupancies proves conservation. Vacuum is fixed by both operations. $\square$

All nontrivial rows below are in the singleton or pair sector. Use fresh labels for distinct routines, waiting modes, shuttle directions and data. Labels that are intentionally shared are identified explicitly.

52 Canonical sections and finite gates

The right-marker type R is stationary at coordinate 0. A left-controller type $L_{j,x}$ and a stationary shuttle S_j are colocated at $-n$, where

$$x = (q, i, z) \in Q \times \mathbb{Z}_m \times \mathbb{Z}_6, \quad j \in \{0, \dots, 7\}.$$

Thus a canonical configuration occupies two cells but has three units of mass. At stage zero, $i = z = 0$ defines the clean symbol

$$\text{Ready}(q) = \{L_{0,(q,0,0)}, S_0\}.$$

During flights the same three units usually occupy three cells. Finite control is inside the weight-one controller type, never in a mass-free track.

A finite data permutation G between static stages j and k is implemented by the pair rules

$$\{L_{j,x}, S_j\} \mapsto \{L_{k,G(x)}, S_k\}. \quad (52.1)$$

Both output velocities are zero, so this consumes exactly one tick. Each output stage has exactly one preceding routine. Since G is a permutation, these rules have distinct outputs; fresh internal labels separate them from unrelated routine outputs.

53 Exact reversible residue queries

Each query label ℓ has fresh shuttles

$$P_{\ell,d,r}, M_{\ell,d,r}, \quad d \in \{+, -\}, \quad r \in \mathbb{Z}_{144},$$

with velocities $+1, -1$ according to d . On singleton inputs, π increments the P phase and decrements the M phase. All waiting controllers and R are fixed as singletons. Define the finite lookup $h(r) = \lfloor r/24 \rfloor$, for representatives $0 \leq r < 144$.

For $\epsilon \in \{+1, -1\}$, the query from stage j to k uses the following pair rows; phase arithmetic is modulo 144 and scratch arithmetic modulo 6:

$$\{L_{j,x}, S_j\} \mapsto \{L_{\ell,P,x}, P_{\ell,+,1}\}, \quad (53.1)$$

$$\{R, P_{\ell,+,r}\} \mapsto \{R, P_{\ell,-,r+1}\}, \quad (53.2)$$

$$\{L_{\ell,P,(q,i,z)}, P_{\ell,-,r}\} \mapsto \{L_{\ell,M,(q,i,z+\epsilon h(r))}, M_{\ell,+,r-1}\}, \quad (53.3)$$

$$\{R, M_{\ell,+,r}\} \mapsto \{R, M_{\ell,-,r-1}\}, \quad (53.4)$$

$$\{L_{\ell,M,x}, M_{\ell,-,0}\} \mapsto \{L_{k,x}, S_k\}. \quad (53.5)$$

The exit row is prescribed *only* at phase zero. It never erases an arbitrary phase. For the stage-zero outgoing query, launch rows with source state q_f are omitted.

Lemma 53.1 (Residue query). *At gap $n = 12N$, the routine preserves the gap and q, i , maps z to $z + \epsilon(N \bmod 6)$, and exits at exactly $4n + 1$ ticks with no clock garbage.*

Proof. Set the launch time to zero. After the first tick the shuttle has moved one cell right and has phase one. At time n it reaches R with phase n . The next collision tick also updates the phase, so at time $2n$ it is back at the left controller with phase

$$r = 2n \bmod 144 = 24(N \bmod 6).$$

Therefore $h(r) = N \bmod 6$. The turnaround row performs the required scratch translation and starts the negative clock. At time $3n$ the shuttle reaches R with phase $r - n$ modulo 144. At time $4n$ it is back at the left controller with phase zero. The exit row produces the stationary stage at time $4n + 1$.

Only the shuttle moves, and $n \geq 12$, so there are no simultaneous endpoint or triple collisions. The two reflection families are injective in phase. The turnaround's output phase $r - 1$ recovers r , after which subtraction recovers the old z . Other finite data are retained. Thus every pair family is injective. The signs use separate routine labels. The minus query independently traverses the same geometry and subtracts its measured residue; it is an inverse operation on finite scratch, rather than a claim that its physical trajectory is the reversed plus trajectory. $\square$

The phase updates on collision ticks are necessary. Omitting the $+1$ or -1 in a reflection or turnaround would alter both the residue and the boundary time. The modulus $144 = 2 \cdot 12 \cdot 6$ simultaneously accounts for the round trip, the lattice scaling and the six-valued residue.

54 Exact rational gap changes

The reflected-shuttle geometric mechanism has a direct three-agent predecessor in Brandt, Portmann and Uitto [37], Section 3, which credits the factor-two case to Dobrev and coauthors [38]. Its rational-speed agents give the same ratio after time rescaling. The construction here supplies integer-lattice alignment, reversible local channel rules and cleanup.

For a moving instruction tag i with factor $p/d \in \{2, 1/2, 3, 1/3\}$, set

$$c = p + d, \quad u = d - p.$$

Use a controller B_x of velocity u and shuttles $A_{i,+}, A_{i,-}$ of velocities $c, -c$. All these types are singleton-fixed. Retain the full data $x = (q, i, z)$. The rows are

$$\{L_{4,x}, S_4\} \mapsto \{B_x, A_{i,+}\}, \tag{54.1}$$

$$\{R, A_{i,+}\} \mapsto \{R, A_{i,-}\}, \tag{54.2}$$

$$\{B_x, A_{i,-}\} \mapsto \{L_{5,x}, S_5\}. \tag{54.3}$$

No-op and test instructions use the static row from stage four to stage five, with the full tag retained.

Lemma 54.1 (Rational gap primitive). *For a valid factor p/d applied to N , the new gap is $n' = pn/d = 12N'$. The moving routine exits after $n/d + 1$ ticks. A test or no-op exits in one tick. The maximum absolute velocity is four.*

Proof. Temporarily translate the initial left endpoint to 0 and R to n . During the outward flight the shuttle is at ct and the left controller at ut . The shuttle arrives at R at $t_R = n/c$. After reflection its position is $2n - ct$. It meets the controller at

$$T = \frac{2n}{c+u} = \frac{n}{d}, \quad n' = n - uT = \frac{p}{d}n.$$

The collision is present at time T ; the next tick produces the stationary exit pair without moving it, so the total duration is $T + 1$.

For prime two, $(c, u) = (3, -1)$ or $(3, 1)$; for prime three, $(c, u) = (4, -2)$ or $(4, 2)$. The factor 12 makes n/c integral. Valid division requires $d \mid N$, hence N' is integral and remains positive. Multiplication has $d = 1$. All encounters are therefore at integer positions and times.

Since $|u| < c$, the shuttle separates immediately, visits R first and then meets the controller. When $u \leq 0$, the marker gap increases. When $u > 0$, its minimum up to the closing encounter is $n' > 0$. The markers cannot meet, and no triple collision or crossed-but-unseen encounter occurs. Distinct tags have distinct shuttles and remain in x at exit, making the prescribed rows injective. $\square$

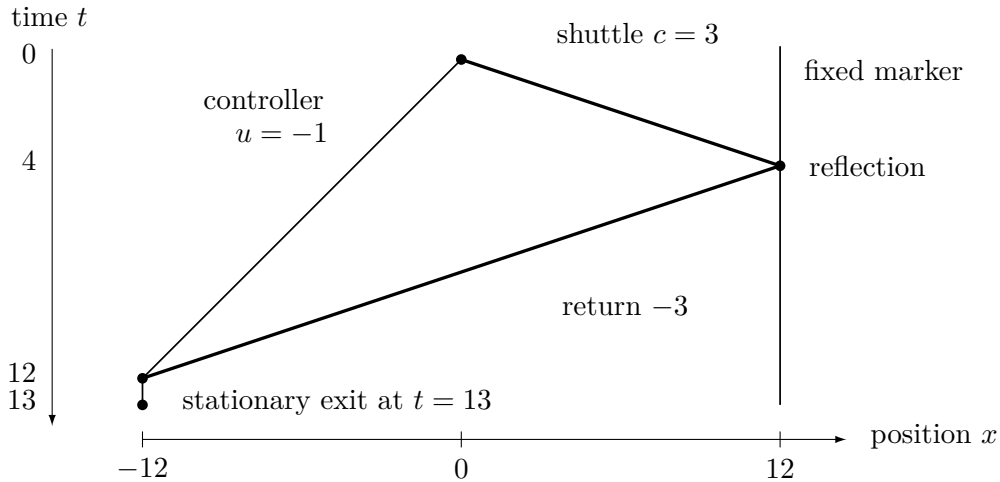


Figure 4: One exact multiplication by two at initial gap 12. The shuttle reflects at $t = 4$, meets the moving controller at $t = 12$, and the closing collision produces the stationary pair one tick later. The final gap is 24. Lines show velocities between integer-time configurations; this is the arithmetic primitive alone, not the surrounding source-step queries.

55 Instruction cleanup and the simulation proof

The fixed sequence of eight routines is

$$0 \xrightarrow{Q_+} 1 \xrightarrow{G_I} 2 \xrightarrow{Q_-} 3 \xrightarrow{G_q} 4 \xrightarrow{A} 5 \xrightarrow{Q_+} 6 \xrightarrow{G_H} 7 \xrightarrow{Q_-} 0.$$

Here the four queries have distinct labels, but each is shared by all instruction tags. The gates are

$$\begin{aligned} G_I(q, i, z) &= (q, i + I(q, z), z), \\ G_H(q, i, z) &= (q, i - H(q, z), z). \end{aligned}$$

For fixed $i \neq 0$, G_q transposes the source and target states of instruction i and fixes all other states, retaining i, z . If source equals target, that transposition is the identity; for $i = 0$ use the identity. Thus all three gates are permutations on *every* finite data value, not only valid inputs.

Start at $\text{Ready}(q)$ with $i = z = 0$. The first query loads $r = N \bmod 6$. Gate G_I writes the enabled instruction $i = I(q, r)$, and the first unquery resets $z = 0$. For $i \neq 0$, G_q changes q to the instruction's target. Arithmetic produces N' and still retains i . The output query loads $N' \bmod 6$; backward uniqueness gives

$$H(q', N' \bmod 6) = i.$$

Gate G_H therefore clears i , and the last unquery clears z . The exit is exactly $\text{Ready}(q')$ at gap $12N'$.

No hidden tag erasure. The arithmetic exit (54.3) retains i in $L_{5,x}$ while it removes the tag from the shuttle. All branches enter the same stage-five query. The instruction is cleared only by the common stage-six gate, while q, z determine its predecessor. The final unquery also has a common label. There is no later many-to-one merger of instruction-specific labels after the instruction register has become zero.

Disabled instructions. If $I = 0$, the stage-four row emits a fresh shuttle of velocity -1 and a fresh stationary trap controller carrying x . The shuttle leaves leftward, away from both stationary markers, and never collides again. Its type is singleton-fixed. The row is injective, and the resulting infinite escape creates no $\text{Ready}(q_f)$. This explicitly covers decrement at zero and any nonfinal source state with no applicable instruction.

Lemma 55.1 (Union of prescribed rows). *All prescribed singleton and pair rows satisfy Lemma 51.1 simultaneously.*

Proof. Singleton clocks are disjoint finite cycles; every other singleton is fixed. They lie in a different weight sector from the pair rows. Gate outputs are separated by static stage and their data permutations. Query reflections carry R and a fresh routine, mode and direction. Query turnarounds retain phase information; only the unique zero phase exits. Different waiting modes and routine labels separate these families. Arithmetic dispatch retains all finite data in its moving controller, and its return retains the instruction tag in the common static output. Static ratio-one branches and moving branches have disjoint instruction tags. Trap labels are fresh. Finally each static stage has one preceding routine, including the common clean exit at stage zero. These observations establish distinct domains and images for the union, not merely for each macro map in isolation. Outputs are allowed to be the next routine's inputs; this is intended and does not violate injectivity. $\square$

Proof of Theorem 49.1. Complete the pair map using Lemma 51.1. Global reversibility and conservation follow from Section 51. Lemmas 53.1 and 54.1, together with the cleanup argument, give the canonical-section simulation inductively. Each nonfinal enabled step uses only prescribed rules, with no triples; a disabled step follows the prescribed escape forever. Hence unspecified completion rows cannot manufacture a false halt on an intended nonhalting source computation.

Observe only the cell symbol $\text{Ready}(q_f)$, with both registers zero. An intermediate occurrence of controller state q_f at a different stage is not acceptance. The clean output symbol occurs exactly at a committed halt section, including time zero if the source is already final. Launch rows from the final section are deliberately unspecified. Their completed future is irrelevant after observation. A reversible dynamics cannot enter a periodic orbit from outside it; it may nevertheless make the

entire orbit, including the initial state and its first observed halt, periodic. No global freezing or eventual fixed point is asserted. $\square$

55.1 Exact type and collision counts

For this ledger let $q = |Q| \geq 1$ include the halt state, let $m - 1$ be the number of source instructions and let d be the number of increment or decrement instructions. State labels are distinct and every instruction endpoint belongs to Q . These are syntactic counts for the stated compiler, including types and rows that may be unreachable from a particular encoded input.

Put $D = 6qm$, the number of finite data triples. The disjoint type families have the following sizes:

Type family	Number
Static controllers in eight stages	$8D$
Waiting controllers in four queries	$8D$
Query shuttles with direction and clock	$4 \cdot 2 \cdot 2 \cdot 144 = 2304$
Static shuttles	8
Moving arithmetic controllers	$6qd$
Arithmetic shuttles	$2d$
Trap controllers	$6q$
Escape shuttle and right marker	2

Thus the exact channel count s and full native cell-alphabet size are

$$s = 96qm + 6qd + 2d + 6q + 2314, \quad |2^{\mathcal{T}}| = 2^s.$$

The three gates contribute $3D$ prescribed pairs. Each query contributes $146D + 288$: one launch and one return per datum, 144 turnarounds per datum and 288 right-marker reflections. The first query omits $6m$ halt launches. Arithmetic contributes D dispatches (including static and trap dispatches), $6qd$ returns and d reflections. Therefore

$$P = 3528qm + 6qd + d - 6m + 1152$$

is the exact number of prescribed pair rows.

For completeness the number of rows after path closure is also explicit. Let A, B be the domain and image sets of the prescribed partial injection. Their intersection contains only static pairs. The domain omits the $6m$ stage-zero halt pairs, and the image omits the $6q$ stage-five pairs with instruction register zero. Those omissions are disjoint, so

$$X = |A \cap B| = 8D - 6m - 6q.$$

The nonstatic outputs are not domains: their shuttle direction or fresh trap label separates them from any prescribed collision input. There are $|A \setminus B| = P - X$ path starts; closing one path adds exactly one row, while existing cycles need none. Hence the stored completed pair-support count is

$$C = 2P - X = 7008qm + 12qd + 2d + 6q - 6m + 2304.$$

Unlisted pairs are implicitly fixed; C is not the number of all possible pairs or all cell states. In either radius-one realization below, s becomes $4s$ and the full alphabet has 2^{4s} states. No analogous claim about the lifted pair-row count is needed. These formulas yield a numerical universal alphabet only after a particular universal source and its q, m, d have been supplied.

56 Exact microtime

Write $n = 12N$ and $n' = 12N'$ at consecutive clean sections. There are two queries at each gap, three one-tick finite gates, and one arithmetic routine. Thus the whole source-step duration is

$$\tau(N, N', p, d) = \begin{cases} 8n + 8n' + n/d + 8, & \text{increment or decrement,} \\ 16n + 8, & \text{test or no-op.} \end{cases} \quad (56.1)$$

For a motion instruction the four query constants contribute four ticks, the gates three and the arithmetic exit one. For a static instruction the arithmetic itself is one tick. In particular

operation	$\times 2$	$\div 2$	$\times 3$	$\div 3$	static
$\tau(N)$	$300N + 8$	$150N + 8$	$396N + 8$	$132N + 8$	$192N + 8$

with divisions restricted to enabled inputs. These formulas agree with the primitive boundary convention $F = SII$, with the ready shuttle initially colocated with the left controller.

For a valid h -step source run, with N_t the gap integer at source step t , the exact committed physical time is

$$T_h = \sum_{t=0}^{h-1} \tau_t(N_t), \quad T_0 = 0. \quad (56.2)$$

A source horizon and a CA tick horizon are different parameters. The certificate below records source horizon h and can optionally constrain T_h without expanding every physical tick.

57 Radius one realizations

Both realizations below keep the mass exactly three and use four times as many channel types as the native rule. Spatial blocking preserves every tick and groups four old positions into one new cell. Internal phase dilation preserves the old spatial coordinates and multiplies physical time by four. Neither radius reduction is a novelty claim.

57.1 Spatial blocking with the native clock

There is also a time-preserving radius-one realization by the standard operation of grouping four consecutive old sites into one new site. The new labels record positions within a block, rather than clock phases.

Theorem 57.1 (Time-preserving spatial blocking). *Let F be the native CA, with s Boolean channel types and both forward and inverse radius at most four. There is a globally reversible mass-conserving radius-one CA H on $2^{\mathcal{T} \times \{0,1,2,3\}}$ and a bijective mass-preserving spatial recoding $\mathcal{B}$ such that $H\mathcal{B} = \mathcal{B}F$. No time dilation is required.*

Proof. For every old coordinate, use its unique Euclidean decomposition $x = 4j + a$, with $j \in \mathbb{Z}$ and $a \in \{0, 1, 2, 3\}$, including when $x < 0$. Define

$$(\mathcal{B}c)_j = \{(t, a) : t \in c_{4j+a}\}.$$

This is a bijection of the two full configuration spaces. Each old occupied channel becomes exactly one new occupied channel, so it preserves total mass on finite configurations and maps vacuum to

vacuum. Set $H = \mathcal{B}F\mathcal{B}^{-1}$. The old sites influencing a new output block j lie between $4j - 4$ and $4j + 7$. They belong only to blocks $j - 1, j, j + 1$, proving radius at most one. The same argument for F^{-1} proves radius at most one for H^{-1} .

An explicit channel implementation first applies the completed old permutation π independently to each of the four offset projections. A post-collision lane (t, a) in block j is then transported to

$$\left(j + \left\lfloor \frac{a + v(t)}{4} \right\rfloor, \quad t, \quad (a + v(t)) \bmod 4 \right).$$

Its block displacement is $-1, 0$ or 1 . This transport is a permutation of all Boolean lanes; its inverse uses the opposite old displacement. The four on-site permutations and this transport preserve mass also on arbitrary mixed-offset configurations. Thus this implements the stated conjugacy on the full shift, not merely on the intended three-unit runs. $\square$

The canonical sites $-12N$ and 0 become blocks $-3N$ and 0 , with all three channel labels at offset zero. The blocks are distinct since $N \geq 1$. Observe the exact block symbol containing only the offset-zero versions of $L_{0,(q_f,0,0)}$ and S_0 . Every committed halt in the encoded native orbit has this block symbol, and its occurrence implies the old one-cell halt symbol. Hence the same first-halt observation occurs at the same tick; the whole reduction survives unchanged except for the spatial encoding. The new full alphabet has 2^{4s} states. The counts P, C above refer to the native pair dictionary, not to a flattened pair dictionary for either radius-one realization.

For example, the nine-step certificate fixture halts at tick 5340 in this spatially blocked radius-one CA. The coordinate-preserving phase realization instead uses tick 21360. Both use three mass units and $4s$ types. Native or spatially blocked time uses `clock_scale=1`; the four-phase realization uses `clock_scale=4`. The branch forms and certificate degree and slot ledger are otherwise the same.

57.2 Internal phase dilation with unchanged coordinates

Theorem 57.2 (Radius-one slowdown). *Let $F = S_v \circ \Pi$ be a Boolean-channel CA on 2^T , where the local permutation π preserves cardinality and $|v(t)| \leq r$ for an integer $r \geq 1$. There is a globally reversible, cardinality-conserving CA G of radius at most one on $2^{T \times \mathbb{Z}_r}$ and a mass-preserving phase-zero embedding E such that $G^r \circ E = E \circ F$.*

Proof. Identify a cell with $(a_0, \dots, a_{r-1}) \in (2^T)^r$ and put

$$\Pi_0(a_0, \dots, a_{r-1}) = (\pi(a_0), a_1, \dots, a_{r-1}).$$

Let P advance every occupied phase by one modulo r . For $k \in \{0, \dots, r-1\}$ define

$$\varepsilon(t, k) = \begin{cases} \text{sgn}(v(t)), & k < |v(t)|, \\ 0, & k \geq |v(t)|, \end{cases} \quad \delta(t, p) = \varepsilon(t, p - 1 \bmod r).$$

Take $G = S_\delta \circ P \circ \Pi_0$. The first two factors acting in each cell are permutations, and S_δ is an independent shift of each Boolean lane. All three preserve cardinality, including on arbitrary mixed-phase configurations. Their inverses give $G^{-1} = \Pi_0^{-1} \circ P^{-1} \circ S_\delta^{-1}$. Since $\delta \in \{-1, 0, 1\}$, both global maps have radius at most one.

On a phase-zero input, only the first of the next r ticks applies π to a nonempty phase-zero subset. The next phases are $1, \dots, r-1, 0$; cardinality preservation gives $\pi(\emptyset) = \emptyset$. For $1 \leq j \leq r$,

each post-collision type t has phase $j \bmod r$ and has moved

$$\sum_{k=0}^{j-1} \varepsilon(t, k) = \text{sgn}(v(t)) \min(j, |v(t)|).$$

At $j = r$ this displacement is $v(t)$, proving the identity. The velocity used is that of the type after π , so type-changing collisions cause no exception. Boolean capacity is respected: the three stages are permutations of cell states or of full-shift Boolean coordinates. $\square$

The operation Π_0 uses the cardinality of the phase-zero projection, not the total cell cardinality. Thus a phase-zero pair still undergoes its old collision when other-phase types share its site. This detail is essential to the stated full-alphabet implementation. The already completed original permutation supplies Π_0 directly; no new partial-rule completion is needed.

For the construction here take $r = 4$. The alphabet has $2^{4|T|}$ states, with weight equal to occupied channel count and with a unique zero-weight vacuum. All three original mass units remain charged; the phase is their finite internal label, rather than a background clock. Original durations are multiplied by four, including static and collision ticks. In particular, a query of duration $4n + 1$ takes $16n + 4$ new ticks.

The new observer requires the exact cell state consisting of the co-located phase-zero versions of $L_{0,(q_f,0,0)}$ and S_0 . On encoded orbits all mass units have phase $j \bmod 4$ at time $4k + j$, so no intermediate time $0 < j < 4$ can trigger this observer. At times $4k$ it is exactly the original committed-halt observation. Hence the undecidability reduction and the total mass three are preserved.

Finally, radius zero is impossible for this fixed-pattern occurrence undecidability. A reversible radius-zero CA is a sitewise finite permutation, so all configurations have a common finite period dividing the least common multiple of its state-cycle lengths. Finite-pattern occurrence from an explicitly presented finite-support input can therefore be checked over one period and finitely many relevant positions. Thus radius one is minimal in this model and for this observation notion; this asserts neither intrinsic universality nor a minimal alphabet.

For the certificates in Section 58, write Θ for native radius-four time, also the spatially blocked radius-one time. The phase-dilated radius-one observation time is 4Θ . The exporter and checker accept the request setting `clock_scale` equal to 1 (the default) or 4. They retain native ticks in each branch row and use `clock_scale` · Θ for the aggregate physical-time expression, extracted time coordinate and optional time constraint. Replacing $(\Theta - T)^2$ by $(4\Theta - T)^2$ changes no degree or slot count.

58 Natural number quadratic certificates

Fix a source machine, initial state q_0 , a positive raw gap integer N_0 and an externally specified source horizon $h \in \mathbb{N}$. Require that q_f have no outgoing instructions. The certificate below recognizes *first committed halt exactly at source step h* . Its variables are natural numbers. The horizon is a parameter of the compiler, not one of the polynomial's variables.

A raw integer $N \geq 1$ need not be of the form $2^a 3^b$. It uniquely factors as $C 2^a 3^b$ with $\gcd(C, 6) = 1$. The arithmetic and residue guards preserve C and correctly simulate the valuations a, b . Thus the raw-gap macro and the CA lemmas remain valid; the standard source input family is the special case $C = 1$. No smoothness test is silently asserted by accepting arbitrary positive raw inputs.

58.1 Residue-expanded branches

Expand each zero test at prime p into $p - 1$ branches, one for each nonzero residue $r \in \{1, \dots, p - 1\}$. Every other instruction gives one branch. Let B be the resulting number of branches, and let s_j, t_j be numerical codes for branch j 's source and target states, using distinct integers for distinct states.

For each step $0 \leq t < h$ and branch $1 \leq j \leq B$, introduce two natural variables e_{tj}, u_{tj} . The first selects the branch; the second is its nonnegative magnitude or quotient offset. The following affine forms describe the old value A_j , new value C_j and exact tick duration D_j , with arguments (e, u) suppressed:

Branch	A_j	C_j	D_j
Increment at p	$e + u$	$p(e + u)$	$108A_j + 96C_j + 8e$
Decrement at p	$p(e + u)$	$e + u$	$96A_j + 108C_j + 8e$
Positive test at p	$p(e + u)$	$p(e + u)$	$192A_j + 8e$
Zero test at p , residue r	$pu + re$	$pu + re$	$192A_j + 8e$
No-op	$e + u$	$e + u$	$192A_j + 8e$

For $e = 1$, the increment/no-op forms cover every positive integer; decrement and positive-test forms cover precisely positive multiples of p ; and the zero-test forms cover precisely positive integers with residue r . All forms vanish at $e = u = 0$. Tick coefficients follow directly from (56.1), including the eight constant ticks only on the active branch.

Set, using the forms at step t ,

$$E_t = \sum_j e_{tj}, \quad U_t = \sum_j A_j(e_{tj}, u_{tj}), \quad V_t = \sum_j C_j(e_{tj}, u_{tj}),$$

$$S_t = \sum_j s_j e_{tj}, \quad Q_t = \sum_j t_j e_{tj}, \quad \Theta = \sum_{t,j} D_j(e_{tj}, u_{tj}).$$

The interfaces are affine substitutions, not additional variables:

$$\hat{N}_t = \begin{cases} N_0 & t = 0, \\ V_{t-1} & t > 0, \end{cases} \quad \hat{q}_t = \begin{cases} \text{code}(q_0) & t = 0, \\ Q_{t-1} & t > 0. \end{cases}$$

Write $V_{-1} = N_0$ and $Q_{-1} = \text{code}(q_0)$ for the horizon-zero endpoint convention.

58.2 The polynomial and its exactness

Define

$$P_h = \sum_{t=0}^{h-1} \left[(E_t - 1)^2 + (S_t - \hat{q}_t)^2 + (U_t - \hat{N}_t)^2 \right] + (Q_{h-1} - \text{code}(q_f))^2 + \sum_{t=0}^{h-1} \sum_{j=1}^B (E_t - e_{tj})(e_{tj} + u_{tj}). \quad (58.1)$$

This is one integer polynomial of degree at most two. Each square has an affine argument. Every product is nonnegative on the full natural orthant, since $E_t - e_{tj} = \sum_{k \neq j} e_{tk} \geq 0$ and $e_{tj} + u_{tj} \geq 0$.

Theorem 58.1 (Unique natural witness at a fixed horizon). *For fixed $N_0 \geq 1$, q_0 and h , the natural solutions of $P_h = 0$ are in bijection with first-halt source executions of length exactly h . Since the source is deterministic, there is exactly one full natural witness if such a run exists and no witness otherwise. The affine output V_{h-1} is its final raw gap integer, and Θ is its exact physical CA time.*

Proof. All summands of (58.1) are nonnegative, so a zero forces every summand to vanish. The square $(E_t - 1)^2$ gives $E_t = 1$. Natural selectors then have exactly one value one and all others zero; no separate Boolean equations are needed. For an inactive branch, the product is $1 \cdot u_{tj}$, which forces $u_{tj} = 0$. The active branch's product vanishes automatically and permits its nonnegative magnitude.

Thus U_t, V_t, S_t, Q_t reduce to the selected branch's forms and endpoint codes. Control and input squares connect the branch to the actual preceding state and raw integer. The table enforces its guard and arithmetic exactly by natural quotients. A zero-test branch is uniquely selected by the nonzero residue. Induction from $N_0 \geq 1$ proves that every successive raw value is positive. The terminal square forces the last state to be q_f . Since q_f has no outgoing branch, it could not have occurred earlier in this h -step run. The affine tick expression is then the sum of the already proved physical durations.

Conversely, select the actual branch of a first-halt run at each step, set its e to one and all other branch variables to zero. For an increment or no-op put $u = N - 1$; for a decrement or positive test put $u = N/p - 1$; for a zero test with residue r put $u = (N - r)/p$. These are uniquely determined naturals, and every summand vanishes. Determinism determines the execution, the residue determines any expanded branch and the affine input form determines its u . Therefore every variable is unique. When $h = 0$, there are no core variables and the sole core square compares q_0 with q_f ; success has output N_0 and time zero. $\square$

The inactive-product factor $e + u$ is deliberate. A weaker gate that left an inactive branch's quotient unrestricted would destroy uniqueness and could contaminate the interface sums. Here every inactive variable is forced to its canonical zero value.

[write] [The inactive-product gate $(E_t - e_{tj})(e_{tj} + u_{tj})$ is the gate of Lemma `qoc:rn:lem:gates` of Part VI of *quadratic-orthant-certificates*, whose products $(\sum_{s \neq t} e_{js})(e_{jt} + \sum_p x_{jtp})$ force the inactive magnitudes to zero in the same way, and a form of Part III's selector/copy device (note after Theorem 36.1) and of the one-hot selectors of *canonical-diophantine-certificates*; the raw-integer branch forms, such as $pu + re$ for a residue, are source 14's own. The natural-domain counterexample below is consistent with that report's real-orthant one-hot lemma: one-hot selection survives over the nonnegative reals, divisibility does not.]

The natural domain is essential. With a prime-two zero-test branch and raw input $N_0 = 2$, the real or rational assignment $e = 1, u = 1/2$ satisfies $2u + e = 2$. For a one-step zero-test-to-halt machine it makes the entire polynomial zero although the integer guard is false. The certificate therefore does not claim real- or rational-domain exactness. Neither nonnegativity alone nor a squared input relation enforces divisibility.

58.3 Resource ledger and endpoint constraints

For the fixed-raw-input core (58.1), the exact *syntactic slot counts* are

$2Bh$ natural variables,	$3h + 1$ affine squares,	Bh nonnegative affine products.
--------------------------	--------------------------	-----------------------------------

The degree is at most two. A slot can simplify to zero for a degenerate instance; these are the counts emitted before such simplification, not lower bounds on irreducible polynomial complexity. Interface sums, state codes, raw values and clocks do not introduce auxiliary variables.

To require a fixed final raw value N_* , add $(V_{h-1} - N_*)^2$. To require a fixed physical time T_* , add $(\Theta - T_*)^2$. Each adds one affine-square slot and no variable. Instead, exporting a free natural coordinate for either endpoint adds one variable and that same square. In a successful fixed-input fiber these outputs are uniquely determined.

A free positive raw input is represented by $N_0 = x + 1$, with $x \in \mathbb{N}$. It adds one input coordinate, no square, and no hidden positive-domain convention. The theorem then states uniqueness *for each fixed input coordinate*; different accepted inputs may give different polynomial zeros.

The arity grows linearly with the externally chosen h and with the fixed source's expanded branch count B . Eventual halting is the union over all horizons. No claim is made that merely existentially quantifying an unknown horizon preserves this finite arity, this degree or this ledger.

58.4 An explicitly paid bounded counter loader

For fixed numerical counter inputs (a, b) , an external computable encoder may calculate the constant $N_0 = 2^a 3^b$ before constructing the polynomial. This is not an internal polynomial representation of exponentiation with free unbounded exponents.

A fully explicit optional loader supports free counter coordinates within supplied fixed bounds $0 \leq a \leq A, 0 \leq b \leq C$. Introduce

$$K = (A + 1)(C + 1)$$

natural selectors λ_{ij} for $0 \leq i \leq A, 0 \leq j \leq C$, and add the three affine squares

$$\left(\sum_{i,j} \lambda_{ij} - 1 \right)^2, \quad \left(a - \sum_{i,j} i \lambda_{ij} \right)^2, \quad \left(b - \sum_{i,j} j \lambda_{ij} \right)^2.$$

Substitute the affine expression

$$N_0 = \sum_{i,j} 2^i 3^j \lambda_{ij}$$

directly into the initial interface. The coefficients are precomputed integers. Natural one-hot selectors force the unique pair $(i, j) = (a, b)$, or no witness if the counters are outside the supplied bounds. This adds exactly K loader variables and three affine-square slots; each genuinely free a or b also adds its input coordinate. No separate N_0 variable is needed and the degree remains at most two. The cost is numerical in the fixed bounds, not a uniform logarithmic-size exponentiation gadget. This finite loader makes no assertion for free unbounded counter values.

59 Why masses at most two are decidable

This section gives the lower bound needed by Theorem 49.2. Let F be any deterministic translation-invariant CA of finite radius r over a finite alphabet Q . Assume a quiescent state 0, integer weights $w(0) = 0, w(q) \geq 1$ for $q \neq 0$, and conservation on finite configurations. Define $Q_j = \{q : w(q) = j\}$. Translation by a is denoted T_a . For nonempty x , let $\ell(x) = \min \text{supp}(x)$ and normalize by $\bar{x} = T_{-\ell(x)}x$.

Theorem 59.1 (Two-unit decidability). *From a finite configuration of mass at most two, one can decide exact finite-configuration reachability and occurrence of an explicitly supplied full finite word, either at a specified anchor or somewhere. For fixed F , these decision problems have polynomial complexity in the input bit length: configurations use sparse coordinate/state lists with binary coordinates, and the complete pattern length counts toward input size.*

[write] [Editorial note: this is Theorem 44.1 of Part IV (source 13), stated there for the same model with the same complexity claim, and proved there by the same argument. Source 14’s four subsections (“Free particles and the first encounter”, “Near configurations and recurrence”, “Observations at every intermediate time” and “Polynomial complexity for a fixed rule”) give that proof again step by step: the finite walker maps $\alpha : Q_1 \rightarrow Q_1$ and $d : Q_1 \rightarrow [-r, r] \cap \mathbb{Z}$; the near/far threshold $2r$, with the first encounter solved phase by phase ($k = \lceil (A - 2r)/(-D) \rceil$ for an intercept $A > 2r$ and a negative drift D); the count $K_{\text{near}} = |Q_2| + 2r|Q_1|^2$ of normalized near shapes; the translating tail $F^{i+s+kp}x = T_{k\delta}F^{i+s}x$; the tests at every intermediate time and at all later translates; and the fixed-rule polynomial bound, which is not uniform in the rule. Reversibility is not used. These subsections are not reprinted; they survive in the arrival archive.]

Proof of Theorem 49.2. The lower bound is Theorem 59.1. For the upper bound, choose a fixed universal source machine in the separated reversible form and apply Theorem 49.1. If fixed-pattern occurrence for its mass-three encodings were decidable, the computable input encoding followed by that decision procedure would decide the universal source’s halting problem. This is impossible. Combining the two assertions proves the threshold in precisely the common weighted finite-alphabet model. $\square$

60 Geometric and finite-ring boundaries

Fixed mass does not imply bounded support diameter. In the three-unit simulation the unbounded variable is the gap. A specified diameter bound would change the decision problem fundamentally.

Proposition 60.1 (Bounded diameter). *Given F , a finite initial configuration x and a bound D , it is decidable whether the orbit ever exceeds diameter D . Under the promise of staying within that diameter, exact reachability and anchored or unanchored finite-pattern occurrence are decidable.*

[write] *Pointer.* This is Proposition 43.1 of Part IV (source 13), proved there by the same finite-shape argument: simulate until the diameter escapes or a normalized shape repeats, a repeat giving the translating tail. Source 14’s proof is not reprinted. $\square$

For the undecidable mass-three family, no computable input-dependent bound can cover the whole pre-halt diameter on every successful computation. If such a bound existed, the preceding procedure could answer yes at a hit, no upon prior escape, or decide the translating tail upon a repeat, thereby deciding halting. This argument concerns a *complete* bound through every possible successful witness; it does not preclude bounded individual runs or unbounded sparse certificates. [write] [This is Proposition 43.2 of Part IV, for the mass-three family; its finite-search step is `cdc:bd:prop:nobound.`]

On a specified ring of L cells, a fixed-mass sector is finite, and orbit exploration decides

reachability. Its size is the coefficient of z^M in

$$\left(\sum_{q \in Q} z^{w(q)} \right)^L.$$

For fixed M this is polynomially bounded in the numerical ring length, which is not necessarily polynomial in its binary encoding length. A finite ring can reproduce an infinite-line run through a specified tick horizon T by sufficient padding, for example $L > D_0 + 2rT$ for initial diameter D_0 . This is only a finite-horizon replacement. [write] [Same as Part IV’s paragraphs on finite rings and global halting (Section 43).]

61 Literal examples and reproducibility

The archive contains the exact tested channel generator, standalone arithmetic and inverse checks, the certificate compiler, literal finite rule exports, machine-readable receipts, and this LaTeX source. Only source generated for this construction is packaged; third-party papers are linked in the references rather than redistributed.

The principal literal example has states q_0, q_1, q_2, q_3, q_f and the four instructions

$$q_0 \xrightarrow{\text{inc}_0} q_1, \quad q_1 \xrightarrow{\text{inc}_1} q_2, \quad q_2 \xrightarrow{\text{dec}_0} q_3, \quad q_3 \xrightarrow{\text{dec}_1} q_0.$$

The final state is unused in this cyclic example. For $N = 1$, the committed sections are

tick	0	308	1108	2016	2420
state	q_0	q_1	q_2	q_3	q_0
N	1	2	6	3	1

The full configuration returns to its initial value at tick 2420. This is a test of all four gap factors and exact cleanup, not a universal machine.

This instance has 4872 channel types, 89446 prescribed pair rows and 177752 pair-support rows after completion. Its full cell alphabet has 2^{4872} subsets. The exported sparse permutation specifies every singleton action and every pair-support action; all other pairs and all sectors of cardinality zero or at least three are identities. This is an exact finite representation of the entire local rule, without enumerating all 2^{4872} symbols. [write] [The three literal rule exports of this section (`examples/cycle_rule.json`, `merge_rule.json` and `trap_rule.json`, 17.2 MB together) are not shipped in this report: the shipped `code/14-three-mass-export_examples.py` rebuilds them byte for byte in seconds, and the shipped `data/14-three-mass-rule_index.json` records the size and SHA-256 digest of each (README, “Reconstructing the excluded data”).]

Two additional source examples test merging inverse guards and a disabled decrement. The merging example has a zero-test instruction and positive-test instruction from different states into a common final state. The output residue distinguishes the predecessor and clears its tag. At $N = 1, 3$ the zero branch halts after 200,584 ticks, and at $N = 2, 6$ the positive branch halts after 392,1160 ticks. The disabled decrement at $N = 1$ takes the explicit escape route.

61.1 A literal certificate with nine source steps

The separate certificate fixture `mixed_request.json` has the successive operations

$$\text{inc}_0, \text{inc}_1, \text{positive}_0, \text{positive}_1, \text{dec}_0, \text{dec}_1, \text{zero}_0, \text{zero}_1, \text{nop},$$

with a fresh control state after every operation and halt after the ninth. For fixed initial $N_0 = 1$, its raw values and native times are

t	0	1	2	3	4	5	6	7	8	9
N_t	1	2	6	6	6	3	1	1	1	1
T_t	0	308	1108	2268	3428	4336	4740	4940	5140	5340

The prime-three zero test contributes two residue branches, so $B = 10$. The core has 180 auxiliary variables, 28 affine-square slots and 90 product slots. The literal exported version also records two natural output coordinates, `output_N` and `physical_time`: its total arity is 182, with 30 square slots and the same 90 products. These are output coordinates, not extra free inputs. Its unique witness has `output_N = 1` and native time 5340; phase-dilated radius-one time is 21360, while spatially blocked radius-one time remains 5340. The native expanded polynomial contains 16593 nonzero monomials and maximum absolute coefficient 682112 (20 bits). These last counts refer to this one fully specified fixture, with both output coordinates. [write] [The expanded certificates of this fixture and of its phase-dilated version (`examples/mixed_certificate.json` and `mixed_radius_one_certificate.json`, about 1.8 MB each) are not shipped either; `code/14-three-mass-certificate.pyexport` rebuilds them, and their requests, witnesses and checker receipts are shipped as `data/14-three-mass-mixed*`.]

61.2 What the verification establishes

The reproducibility script reruns the following independent checks against the packaged sources and records receipts with the executed source hashes.

- The local CA audit checks all 177752 completed pair-support rows as a permutation, 3000 random global inverse cases, 144 query cases and 136 arithmetic cases with exact boundaries and stepwise inverses.
- The certificate audit checks 1125 schema and ledger instances, 2160 independently expanded polynomial evaluations and 43586 fully enumerated natural assignments in finite boxes. It constructs 246 unique witnesses and rejects 834 incompatible guards or horizons.
- A separate simulator replays 555394 original-CA microsteps, including 191 enabled macro transitions, three complete mixed runs and six disabled-decrement escape cases. It reads the serialized local maps rather than calling the producer’s step or ready predicates.
- The phase-radius-one audit checks every supported pair under the four-phase embedding, arbitrary-cardinality and mixed-phase inputs, both inverse orders, all synchronized collision boundaries and halt phases. The four cycle runs use 115136 slowed ticks in total.
- The spatial-blocking audit checks all 177752 supported native pairs at all four offsets, 19488 singleton/offset cases, 5000 random configurations and 37460 complete-trajectory ticks. The blocked and native trajectories agree at every tick, including the 5340-tick mixed halt. Its 1368268 checks also pass with optimization enabled.
- The two-unit arithmetic tests cover 93663 encounter and observation cases, including large binary coordinates and later translating phases. They validate the arithmetic components; they are not a complete implementation of the general lower-mass decision procedure. [write] [This is source 13’s mass-two audit (Section 47): the program is `code/13-sparse-lattice-two-mass-audit.py` without its receipt-writing line, and its receipt is source 13’s.]

- The size-ledger audit checks all type families, overlap and path-closure counts on 33 finite sources. The independent clock audit performs 28820 explicit checks over 1623 semantic cases and 54 CLI runs, both normally and with optimization enabled.
- The serialized-checker and strict-input tests exercise coefficient, branch, ledger, endpoint, scalar-type, ownership and witness corruptions. The final receipts give exact mutation counts for the packaged revision.

The producer also runs its own simulation, certificate and literal-export suites. None of these finite tests substitutes for the all-input proofs, and none supplies a universal source-machine table.

61.3 Replaying the release

Python 3.10 or later and its standard library suffice for every executable check. From the extracted archive root, run

```
python3 verify_manifest.py
sh replay.sh
```

The replay writes fresh evidence to `.replay/` without replacing the published examples or receipts. It checks the stored literal rules, exports new copies, compares their hashes and rebuilds the native and phase-radius-one certificate examples. The source and fixture hashes are listed in `SHA256SUMS`. Assertions are required: do not run Python with `-O` or `PYTHONOPTIMIZE` set.

To export a certificate, construct its unique witness when one exists, and run the independent checker directly:

```
mkdir -p .replay
python3 code/certificate.py export examples/mixed_request.json \
    .replay/mixed.json --expanded
python3 code/certificate.py witness .replay/mixed.json \
    .replay/mixed-witness.json
python3 code/checker.py .replay/mixed.json \
    .replay/mixed-witness.json
```

The JSON request declares the source table, initial state, source horizon, input representation, optional endpoints and physical clock convention. The checker reconstructs the expected schema independently before checking its natural witness; it does not trust a bare assertion that the submitted polynomial encodes a source run.

The PDF is rebuilt with `sh build.sh` using a normal TeX Live installation with the packages named in `tex/report.tex`. The archive includes the modular LaTeX sources and a standalone combined `three-mass-report.tex`. PDF byte identity additionally depends on the installed TeX and font versions; the script fixes the document date.

[write] **The release in this report.** These are the delivered names. Source 14's programs are shipped as `code/14-three-mass-*` (the two programs of `code/legacy/` as `code/14-three-mass-legacy-*`), its examples and receipts as `data/14-three-mass-*`, and its `SOURCE_PROVENANCE.md` as `14-three-mass-SOURCE_PROVENANCE.md`. Not shipped: the manuscript (printed here) in its standalone and modular forms, its PDF and README, the checksum ledger `SHA256SUMS` with `verify_manifest.py`, the five literal exports named above, and the two-mass receipt, which is a byte copy of source 13's. `replay.sh` and `build.sh` need the delivered layout, and `build.sh` the

unshipped `tex/report.tex`; the README of this report gives a rerun recipe in a scratch copy of the archive.

62 Provenance and limits of the claims

The mathematical statements in this report separate a compiler theorem, a classical source-existence dependency and finite implementation checks. The generator is parameterized by a source-machine table. Fixing an existing universal source establishes a fixed universal CA mathematically, but the literal example export in this archive is not a frozen universal transition table. No concrete universal alphabet cardinality is being claimed from the small examples.

The three-unit compiler was checked independently for local injectivity, full-space inverse behavior, exact collision times, finite cleanup and the committed observation. The mass-at-most-two argument was independently checked, including anchored observations in later translates and binary-coordinate complexity. The tests exercise finite instances and proof components; the proofs provide the all-input and all-source-machine conclusions.

There is substantial close prior art. Brandt, Portmann and Uitto [37] already use three finite-state agents, reflected-shuttle multiplication and division, walking-clock divisibility, and prime exponents encoded in one gap. Their algorithm is a grid-exploration protocol; the full-shift reversible CA extension and reversible information cleanup are separate obligations proved here. The factor-two geometric predecessor is credited to Dobrev and coauthors [38].

The publisher’s abstract for Dobrev, Kráľovič and Pardubská, *Busy agents on a line* [39], reports Turing-machine simulation by three synchronous distinguishable finite-state agents on $\mathbb{Z}$. That abstract’s universality statement does not assert global CA reversibility. The full journal proof was not obtained for this report. Consequently even a nonreversible three-agent universality claim would have a close published predecessor.

Durand-Lose [5] establishes reversible conservative rational signal-machine universality. Its collision geometry uses rational space and time and additional counter/history components; it is not a mass-three integer-lattice CA theorem. Morita and Imai [19] already give a reversible number-conserving partitioned CA with mass $m + 19$ for an m -state reversible source machine. A three-component numerical partitioned alphabet itself permits multiple states of the same total weight. The equal-weight-type distinction therefore separates the present claim from literal single-integer-state NCCA, not from every partitioned predecessor.

The 2016 paper by Alhazov and Imai [24] reports a universal finite number-conserving CA with five numerical particles in its primary abstract. That abstract does not claim reversibility. Its complete paper was not verified for this report, so no hidden lower bound or reversible strengthening is attributed to it. Its numerical-state mass model differs from the typed weighted model here. This report makes no literature-wide novelty, priority or smallest-alphabet claim.

[write] [For the numerical-state model, Part VI shows that three and four particles do not suffice for undecidable observations, and credits two 2021 precursors that source 14 does not cite: Kong’s thesis [41] (binary three-particle nonuniversality) and Imai’s KAKEN report [42].]

The certificate is over natural numbers, not real solutions with a silently inferred divisibility condition. The source horizon is externally specified; the arity grows with it. An unbounded existential time cannot be inserted into a fixed-arity polynomial merely by writing a variable in place of the horizon. Free exponent inputs require their own stated loader and resource payment. No unrelated numerical record or previously used variable ledger is imported into this construction.

The unique-zero and positive-weight assumptions are indispensable for the lower bound. Active zero-weight states, signed cancellation or an infinite nonzero background would invalidate the mass-to-support argument. The construction and threshold are for the infinite line, not a finite ring, and acceptance is a finite observed event rather than cessation of a reversible global evolution.

63 Questions left open by this construction

The proofs resolve the mass and radius thresholds within the stated weighted model. Several sharper or more concrete tasks remain distinct.

1. **Expand a universal source.** Supply a particular universal reversible two-counter table in the separated syntax, then freeze its generated channel names and local permutation. This would turn the existence-based instance into one explicit universal rule with an exact alphabet count. ([write] Added 2 October 2026, batch 80: open. Source 15's Question 1 and Part IV's first question ask the same.) ([write] Added 3 October 2026, batch 82: answered in part outside this report. Part II of **five-particle-binary-automata** (Report 16) prints a literal universal reversible two-counter table, in Report 15's guarded syntax, not in this source's separated syntax; the generated channel names and local permutation of this automaton are still not frozen.)
2. **Reduce the finite-state overhead.** Can fewer clock phases, channel types or collision rows implement the same cleanup and residue query? The phase implementation pays a factor four in types and time; spatial blocking pays a factor four in types while preserving time. These are constructive bounds for different spatial conventions, not optimal overhead theorems. ([write] Compare source 15's Question 3.)
3. **Develop quantitative observation extensions.** Construct and cost an anchored halt observer, for example by carrying a halt message to the fixed marker, while minimizing extra channel types and rules. Exact finite-target encodings also require explicit treatment of the final counters. Neither extension is part of the validated generator here. ([write] Added 2 October 2026, batch 80: the exact finite-target part is answered by source 15 (Theorem 67.1): its target y_N is computed from the input, and the final counters are restored by inverse execution rather than treated explicitly. The anchored halt observer is still open.)
4. **Change the mass convention.** What is the sharp reversible threshold when cell states are nonnegative integers equal to their conserved mass? The abundance of equal-weight types used here cannot be carried over by relabeling. ([write] Added 2 October 2026, batch 80: answered in part by Part VI. With at most one weight-one symbol, exact reachability and pattern occurrence are decidable at masses three and four, so undecidable observations need at least five units (Corollaries 79.1 and 83.2); no reversibility is assumed there. The sharp reversible threshold is still open. Compare source 15's Question 4.) ([write] Added 3 October 2026, batch 82: answered outside this report: five. A globally reversible binary automaton is an integer-state one, and Report 15 (Part I of **five-particle-binary-automata**, Corollary 16.1 there) builds one whose halt-pattern occurrence and exact finite-target reachability are undecidable at five particles; source 17's lower bound covers every rule with one weight-one symbol. What remains is re-scoped to the number of weight-one types, Question 5.)
5. **Cost an unbounded-time encoding for this frontend.** Instantiate known fixed-arity Diophantine coding for this particular simulation with an explicit degree and variable ledger,

and determine which witness properties survive the composition. Uniqueness at each fixed horizon does not by itself yield a finite-fold or single-fold representation after introducing such coding. This is a concrete construction and accounting task, not an open question about existence of ordinary Diophantine representations. ([write] Compare source 15’s Question 5.) ([write] Added 3 October 2026: answered for fixed eligible sources by Part VIII, conditional on its inherited theorems. Source 22 composes source 15’s wrapper with the research tree’s residue-history interface into horizon-free polynomials of exact degrees 2344 and 1192, with complete ledgers of 604, 479, 477 and 480 operations for four nonuniversal fixtures (Theorem 141.1); every accepted nonempty fibre is infinite (Theorem 146.1), so the unique witness does not survive, as anticipated here. Source 23 makes the outer history unique, and the remaining native fibre is still infinite (Theorems 154.1 and 155.1). Part VIII does not address a universal source, Question 1.)

6. **Complete the prior-art comparison.** A literature-wide priority assessment would require the uninspected full texts and a precise comparison of the update, reversibility, particle and observation conventions. No such assessment is asserted here.

Source 15 (batch-80 manuscript 02): Exact targets with three conserved mass units

A clean return extension of the reversible weighted CA construction, 3 October 2026. [write] Its “companion construction” and “companion report” are source 14, printed above; its delivered bibliography entry for it (“Companion construction report, frozen release 9”) is replaced by references to source 14’s statements. Its abstract follows.

A fresh-entry reversible two-counter program can be wrapped so that it first runs forward, then executes its logical inverse, and finally changes to a designated terminal state with its entire input restored. Applying the existing three-mass cellular-automaton compiler to this wrapped program gives undecidable exact finite-configuration reachability for one fixed reversible weighted CA. Both configurations in the reduction have mass three and the same two-site support. The target is computable from the input, without knowing whether or when the computation halts. Spatial blocking yields radius one with no clock penalty; a coordinate-preserving phase construction costs a factor four in time. Decidability at mass at most two makes three the sharp mass threshold for exact-pair reachability in this model. We prove the wrapper, the exact first-target clock, and a compact quadratic natural-number certificate, including an affine bijection of its zero fibers with those of the full wrapped computation. The universal-source existence dependency, finite implementation checks, and limits of the result are stated separately.

64 Result and scope

The previous construction established undecidable occurrence of a designated clean halt symbol somewhere on the infinite line. Its final counter values were generally unknown. That observation did not by itself provide an exact target configuration that could be computed before running the source. The extension here removes this obstacle by restoring the input before announcing completion.

The change is at the source-program level. The same local-rule generator is applied to a different finite source table. Thus the resulting CA generally has different control types and a different fixed rule from the one compiled from the original program. The earlier release is unchanged.

Theorem 64.1 (Exact-pair reachability at mass three). *There is a fixed one-dimensional reversible CA A over a finite alphabet, with a unique zero-weight vacuum and positive integer weights on all other symbols, for which the following hold.*

1. *Total weight is conserved on every finite configuration. Both A and A^{-1} have radius at most four.*
2. *There is a computable family of finite configurations (x_N, y_N) , $N \geq 1$, of total mass exactly three, such that deciding whether $A^t(x_N) = y_N$ for some $t \geq 0$ is undecidable. [write] [The undecidable family is the image of the source’s input encoding ($c = 1$, or any N for a many-one complete source), as Corollary 67.2 states.]*
3. *The input and target have the same support, at coordinates $-12N$ and 0 . Their only difference is the finite control state in the left pair. The target depends on N and is computed without knowing the halt time or the final counters.*
4. *There are also fixed radius-one rules with radius-one inverses satisfying the same exact-pair undecidability conclusion and conserving the same three mass units.*

For every fixed deterministic conservative finite-alphabet CA in the same one-dimensional weighted model, exact finite-configuration reachability from mass at most two is decidable. Thus three is the least mass permitting the stated fixed-rule undecidability.

Here and below $\mathbb{N} = \{0, 1, 2, \dots\}$. Configuration equality is equality at every absolute lattice coordinate, including vacuum everywhere else. The problem is not equality up to translation, a pattern query, a limit configuration, or an assertion that evolution stops. In fact a bijection cannot carry a distinct initial configuration into a fixed point: a fixed point already has itself as its unique predecessor.

The model matters. The alphabet is a power set of finitely many Boolean channel types, and a symbol’s weight is its cardinality. There are many different weight-one symbols. This is not the ordinary single-integer-state number-conserving CA convention in which a state’s numerical label is itself its mass. The result is for finite-support configurations on $\mathbb{Z}$, with unbounded possible separation, not for a prescribed finite ring or an active mass-free background.

What is proved anew. The new ingredients are the clean source wrapper, exact whole-configuration reduction, first-hit clock identity, and certificate composition and lift. The local residue and gap routines, global reversible completion, radius reductions, and low-mass decision theorem are dependencies from the companion construction [source 14, Part V of this report]. We state their precise interface and summarize the lower-bound argument so the reduction can be checked without reconstructing the earlier report. The archive contains the exact executable dependencies, not an expanded universal source table.

[write] [In this report the restated interface is printed with pointers to source 14’s statements (Section 65), and the summary of the lower-bound argument is replaced by a pointer to Part IV’s Theorem 44.1 (Section 67).]

65 The source model and the inherited compiler

65.1 Separated reversible instructions

A source configuration is $(q, a, b) \in Q \times \mathbb{N}^2$. Each instruction has a source state, target state, counter index in $\{0, 1\}$, and an operation among INC, positive-only DEC, NOP, ZERO, and POS. Tests preserve both counters. A zero test is enabled exactly when its counter is zero; a positive test and decrement require a positive counter.

The separated syntax imposes the following conditions on both the incoming and outgoing instructions at every state: a group is empty, a single instruction, or disjoint zero/positive tests on one and the same counter. In particular arithmetic cannot be mixed with another instruction on that side. These conditions imply forward determinism and uniqueness of a predecessor, including the guards. The designated halt state q_h has no outgoing instruction. Getting stuck elsewhere is possible and does not count as halting.

We require the additional structural condition

$$\text{no instruction has target } q_0. \quad (65.1)$$

It must hold for the whole finite table, not merely for transitions enabled on one chosen input. The necessity of this condition and its classical availability are addressed in Section 66.1.

65.2 Raw integers and exact canonical sections

Represent the two counters by a positive integer

$$N = c 2^a 3^b, \quad \gcd(c, 6) = 1. \quad (65.2)$$

The usual encoded source inputs use $c = 1$. Allowing arbitrary raw $N \geq 1$ is also sound: prime-two and prime-three valuations are the counters, and the cofactor c is invariant. An increment or decrement at prime $p \in \{2, 3\}$ changes N by p or $1/p$; zero and positive tests inspect nondivisibility or divisibility by p .

For a source table M , the inherited compiler constructs a finite set $\mathcal{T}_M$ of typed channels, velocities $v : \mathcal{T}_M \rightarrow [-4, 4] \cap \mathbb{Z}$, and a cardinality-preserving local permutation π on $2^{\mathcal{T}_M}$. One native CA tick is

$$A_M = S\Pi, \quad (65.3)$$

where Π applies π independently at all sites and S shifts each channel by its velocity. The empty subset is vacuum. Its weight is zero; all other weights are positive integers.

Let R be the stationary right-marker channel. A canonical section $C_M(q, N)$ is exactly

$$C_M(q, N)(x) = \begin{cases} \{L_{0,(q,0,0)}, S_0\}, & x = -12N, \\ \{R\}, & x = 0, \\ \emptyset, & \text{otherwise.} \end{cases} \quad (65.4)$$

The controller holds the state and finite instruction and residue registers, both zero here. The stationary shuttle S_0 is a separate unit. Consequently the section has two occupied cells and mass three, not mass two.

Proposition 65.1 (Compiler interface used in this extension). *For every separated deterministic reversible source M with terminal state q_h , the inherited compiler has the following properties.*

1. A_M is a bijection of the full configuration space. Both directions have radius at most four, and total mass is conserved on finite configurations.
2. Each enabled source instruction from (q, N) to (q', N') carries $C_M(q, N)$ to $C_M(q', N')$ after the exact duration in Table 7. The right marker stays at zero throughout.
3. Intended macro executions use only the prescribed local collision rows. Intermediate stages cannot contain an unrelated clean stage-zero final pair.
4. A stuck nonfinal source enters a prescribed escape: its shuttle moves permanently left away from the stationary markers. This continuation cannot later produce a clean terminal section and uses no arbitrary completion row.

Source operation	Raw update	Exact native ticks
INC at p	$N' = pN$	$108N + 96N' + 8$
DEC at p	$N' = N/p, p \mid N$	$96N + 108N' + 8$
ZERO, POS, NOP	$N' = N$	$192N + 8$

Table 7: The complete source-step clock, including residue queries, finite gates, and cleanup. These are not arithmetic-flight times alone.

[write] [Interface with source 14. Item 1 of Proposition 65.1 is Theorem 49.1(1), with the path-closure completion of Lemma 51.1 and Lemma 51.2; items 2 and 3 are Theorem 49.1(3), by Lemmas 53.1, 54.1 and 55.1 and the cleanup argument of Section 55; item 4 is source 14's disabled-instruction escape there. The section $C_M(q, N)$ is source 14's canonical configuration with the clean symbol Ready(q) (Section 52), and Table 7 is (56.1) at $n = 12N$: $8n + 8n' + n/d + 8$ equals the displayed entry for each of the five operation types. Source 15's two paragraphs recalling the path-closure completion and deriving the clock from the residue queries ($4n + 1$ ticks) and gap routines ($n/d + 1$ ticks) are not reprinted.]

66 A clean return source program

Let R be a source satisfying Section 65. Form two disjoint copies of its states and one new state,

$$Q_{\text{cl}} = \{F: q : q \in Q\} \dot{\cup} \{B: q : q \in Q\} \dot{\cup} \{H\}.$$

For each instruction $q \xrightarrow{\text{op}} q'$, introduce

$$F: q \xrightarrow{\text{op}} F: q', \quad B: q' \xrightarrow{\text{op}^{-1}} B: q. \quad (66.1)$$

The inverse exchanges increments with decrements, retains the counter index, and leaves both kinds of test and no-op unchanged. Add just two bridges:

$$F: q_h \xrightarrow{\text{NOP}} B: q_h, \quad B: q_0 \xrightarrow{\text{NOP}} H. \quad (66.2)$$

The new initial state is $F: q_0$ and the sole designated terminal state is H . In particular the two copies of q_h are ordinary, nonterminal states for the CA compiler.

Lemma 66.1 (Preservation of reversible syntax). *The program R_{cl} defined by (66.1)–(66.2) is separated, deterministic, and reversible. Its initial state has no incoming instruction and H has no outgoing instruction.*

Proof. The forward copy retains each original incoming and outgoing group. In the backward copy, outgoing groups are the inverses of original incoming groups, and incoming groups are inverses of original outgoing groups. A singleton arithmetic instruction remains singleton. A test group remains the same pair of disjoint tests on the same unchanged counter. Thus reversal preserves the syntactic conditions.

The first bridge uses an empty outgoing side at $F: q_h$ and an empty incoming side at $B: q_h$; both are empty because q_h had no outgoing instruction. The second uses the empty outgoing side at $B: q_0$, by freshness (65.1), and the fresh incoming side at H . Each added group is a legal singleton no-op. No other side is changed. The claimed entry and exit properties follow immediately. $\square$

Theorem 66.2 (Restoration and first source halt). *Suppose the original run from (q_0, N_0) first reaches q_h after h instructions. Write its sections as (q_t, N_t) , $0 \leq t \leq h$, with q_h also denoting its final state. Then the wrapped run is exactly*

$$\begin{aligned} & (F: q_0, N_0), \dots, (F: q_h, N_h), \\ & (B: q_h, N_h), (B: q_{h-1}, N_{h-1}), \dots, (B: q_0, N_0), (H, N_0). \end{aligned}$$

It first reaches H after $2h + 2$ instructions and restores the whole raw integer, including its cofactor. If the original run does not reach q_h , the wrapped run does not reach H .

Proof. Before the first bridge, the forward copy executes exactly the original instructions. The bridge is available precisely at $F: q_h$ and preserves N_h . Every reversed arithmetic instruction is enabled on the actual preceding output: the inverse of an increment is a decrement on a number just multiplied by the relevant prime, and the inverse of a decrement is an always-enabled increment. Each inverse test sees the same unchanged raw number. These operations restore the previous raw value exactly. Backward uniqueness gives no alternative predecessor branch.

Inducting in reverse order reconstructs $(q_{h-1}, N_{h-1}), \dots, (q_0, N_0)$. The original initial state cannot occur at a positive source time, because no instruction targets it. Therefore the second bridge cannot be encountered prematurely. It preserves N_0 and changes only the state to H . The count is $h + 1 + h + 1 = 2h + 2$. The invariant cofactor in (65.2) is unchanged by every operation and is restored along with both valuations.

If the original run is infinite, its forward copy is infinite and never takes the first bridge. If it is stuck nonfinally, the forward copy is stuck at that same section. Neither case reaches a backward-copy state or H . When $h = 0$, the hypotheses imply $q_0 = q_h$ is isolated; the run consists of the two bridges and the same proof applies. $\square$

66.1 Why the fresh entry condition matters

Consider the separated reversible source

$$q_0 \xrightarrow{\text{ZERO}_0} q_1, \quad q_1 \xrightarrow{\text{INC}_0} q_0, \quad q_0 \xrightarrow{\text{POS}_0} q_h. \quad (66.3)$$

From $N_0 = 1$ its trace is $(q_0, 1), (q_1, 1), (q_0, 2), (q_h, 2)$. A backward copy first arrives at $B: q_0$ with $N = 2$, before undoing the increment. An unconditional exit there would restore the wrong input.

It also conflicts syntactically with the inverse decrement already leaving $B: q_0$. The implementation rejects this source. Rejecting only an incoming instruction enabled on the current input would be insufficient.

Nor can an arbitrary reversible source be freshened merely by prepending a no-op into its old initial state: that new incoming motion may conflict with existing incoming instructions. The source-existence dependency must genuinely supply the needed normal form. Morita and Imai [19], Proposition 3.4, give universal reversible two-counter simulation, and Lemma 3.5 supplies a simulator with no instruction entering its initial state while preserving reversibility. Apply these results after choosing one universal source problem. The finite counts below start after this normalization; no bound on its overhead is claimed. A terminal halt may be enforced by deleting outgoing instructions from the designated halt state, preserving first-halt behavior and the separated conditions.

66.2 Relation to classical uncomputation

Forward computation followed by inverse execution to clear temporary information is a classical idea. Bennett's 1973 construction [40] restores the input after retaining the desired output and clearing intermediate information. Here the source is already reversible, so its inverse execution needs no new history tape. A terminal control label records success while both counter values are restored. This report claims neither novelty nor priority for the forward-and-reverse idea; it proves the particular source syntax, finite geometry, exact target, and resource consequences needed for this CA.

67 Exact configurations and the sharp threshold

Apply the inherited compiler to R_{cl} with H as its designated final state, and abbreviate its CA to A . Set

$$x_N = C_{R_{cl}}(F: q_0, N), \quad y_N = C_{R_{cl}}(H, N). \quad (67.1)$$

Both descriptions are computed directly from the supplied N . They have identical absolute support and differ only in the controller state inside the weight-two left symbol.

Theorem 67.1 (Whole-configuration equivalence). *For every positive raw input N ,*

$$(\exists t \in \mathbb{N} : A^t(x_N) = y_N) \iff R \text{ reaches } q_h \text{ from } (q_0, N).$$

On a first source halt at h , the target is first reached at the end of cleaned source instruction $2h + 2$.

Proof. On a halt, Theorem 66.2 gives the canonical section (H, N) after the indicated number of source instructions. Proposition 65.1 resets the finite registers at every section. The right marker has stayed at zero and the left pair returns to $-12N$. There are exactly the three prescribed channels and vacuum elsewhere, so the resulting object is the entire target in (67.1), not just a matching halt symbol.

Before this section, Theorem 66.2 lists all canonical source states, none equal to H . Intermediate macro stages have different stage or shuttle types and cannot equal the clean stage-zero target. The prescribed-row simulation therefore rules out an earlier hit.

For a nonhalting source, an infinite forward execution continues through its intended macros without an H section. A finite stuck execution takes the explicit escape continuation: the escaping channel moves forever away and never reassembles a clean pair. These are complete descriptions

of both possible nonhalting continuations, not finite-prefix checks. No arbitrary completion row supplies a later false target. The equivalence follows. $\square$

The theorem concerns the first exact hit. It makes no assertion that the target is stationary, that it occurs only once, or that the subsequent orbit remains a simulated source run. Completion may determine post-hit behavior. None of those later facts is needed for reachability.

Corollary 67.2 (One fixed rule). *Fix a universal reversible fresh-entry source as above. Then exact-pair reachability for its fixed cleaned CA is undecidable on the computable family (67.1). Exact-pair reachability is recursively enumerable: direct simulation semidecides a hit.*

Proof. Encode an instance of the fixed source’s halting problem into its initial counter values, then into N and the pair (x_N, y_N) . Both endpoints are computed without the halt time or output values. A decision procedure for this restricted CA reachability family would decide source halting by Theorem 67.1. Direct CA simulation enumerates finite times and checks finite configuration equality, proving the semidecision statement. With a standard many-one complete halting source, the same reduction gives recursively enumerable completeness. $\square$

This supplies a fixed rule, not a variable rule supplied alongside each input. It does not supply one fixed target for every N . The dependence of y_N on N is explicit and is part of the reduction.

67.1 Why mass two cannot do the same

[write] [Source 15 summarizes here, “from the companion proof”, the mass-two decision argument: finite walker maps, the near/far threshold $2r$, the $|Q_2| + 2r|Q_1|^2$ near shapes, the translating recurrent tail, exact target tests at every phase and at all later translates, and polynomial bit complexity for a fixed rule. That is Theorem 44.1 of Part IV, proved there, and Section 59 of source 14; the summary is not reprinted.]

Combining this decision theorem with Theorem 67.1 proves the mass threshold in Theorem 64.1. The assumptions exclude zero-weight active computation and signed mass cancellation. A prescribed finite ring has a finite mass sector and is decidable by orbit search, so it is a different problem.

68 Radius one and the exact clock

68.1 The first target time

Let Θ be the native microtime of the original forward source trace and N_h its final raw value. Reversal exchanges an increment $N \rightarrow N'$ with a decrement $N' \rightarrow N$. Table 7 gives

$$108N + 96N' + 8 = 96N' + 108N + 8.$$

The decrement/increment pair has the same symmetry. Inverse tests and no-ops leave the raw value unchanged and have the same duration. Hence the reversed half takes exactly Θ , in reverse order of macro durations. The two bridges cost $192N_h + 8$ and $192N_0 + 8$.

Theorem 68.1 (Exact first-target clock). *If the original first halt is after h source instructions, the native CA first reaches its exact target at*

$$T_{cl} = 2\Theta + 192(N_h + N_0) + 16. \quad (68.1)$$

For $h = 0$ this is $384N_0 + 16$, with both no-op bridges retained.

This is logical inverse execution using forward CA time and different finite controller types. It is not the microscopic inverse orbit under A^{-1} , and need not reverse particle worldlines. The time identity follows from the explicit macro formulas, not from global reversibility by itself.

68.2 Four-site spatial blocking

Group old sites $4k, 4k+1, 4k+2, 4k+3$ into blocked site k . Every old channel gets an offset 0, 1, 2, 3. This is a conjugacy of full shifts, with weight equal to the sum of the four old weights. A native radius-four update and its radius-four inverse become radius-one updates because the four-site block needs only itself and its two neighboring blocks.

The native right marker stays at blocked coordinate zero, offset zero. Since $-12N = -4(3N)$, the left pair is at blocked coordinate $-3N$, also offset zero. Conjugacy preserves exact equality at every tick, and therefore preserves the first-hit time T_{cl} . No extra particle or background is added.

68.3 Four-phase internal dilation

The alternative radius-one construction keeps native coordinates and gives every channel four phase copies. A native velocity in $[-4, 4]$ is spread over four single-cell submoves; the local collision permutation acts once per four-phase cycle. On synchronized phase-zero configurations, four new ticks implement one native tick. Each individual step remains a full-space bijection, including on configurations outside the synchronized encoded sector, as established by the inherited phase construction.

The target uses phase-zero channels. At every time all three encoded tokens have phase $t \bmod 4$, so an off-section time cannot equal that nonempty target. At time $4t$, its state equals the native state after t ticks under the phase-zero encoding. The first exact target is consequently at $4T_{\text{cl}}$, with the same coordinates $-12N$ and 0. Both radius-one alternatives have four times the native number of channel types. Spatial blocking preserves the clock; internal dilation preserves the native coordinate convention. [write] [The two constructions are source 14's Theorems 57.1 and 57.2, applied to the wrapped source.]

Radius zero cannot support the stated undecidable exact-pair problem: sites neither move nor communicate, and a finite-support configuration has a finite orbit of independent finite-state updates. Thus radius one also attains the smallest possible radius for this result.

69 Finite program and local rule costs

Let the normalized original source have $q = |Q|$ states, J instructions, instruction-register modulus $m = J + 1$, and d arithmetic instructions. Let B be its residue-expanded branch count: a zero test at prime three contributes two branches, and each other instruction contributes one. The wrapper does not simplify or remove unreachable states.

The inherited verified local-rule formulas can be applied directly to these primed quantities:

$$K_{\text{types}} = 96q'm' + 6q'd' + 2d' + 6q' + 2314, \quad (69.1)$$

$$K_{\text{prescribed}} = 3528q'm' + 6q'd' + d' - 6m' + 1152, \quad (69.2)$$

$$K_{\text{completed}} = 7008q'm' + 12q'd' + 2d' + 6q' - 6m' + 2304. \quad (69.3)$$

The latter two count native pair rows before completion and pair-support rows after completion. The full native cell alphabet has $2^{K_{\text{types}}}$ symbols. It is specified by sparse singleton and pair permutations

Quantity	Original	Cleaned
States	q	$q' = 2q + 1$
Instructions	J	$J' = 2J + 2$
Instruction modulus	m	$m' = 2m + 1$
Arithmetic instructions	d	$d' = 2d$
Expanded branches	B	$B' = 2B + 2$
First-halt source horizon	h	$h' = 2h + 2$

Table 8: Exact unsimplified wrapper counts. The first five rows count a finite program; the last counts one successful execution.

with identities elsewhere, without enumerating all subsets. [write] [These are the counts s , P and C of source 14's Section 55.1 at the primed values.]

Both radius-one alternatives use $4K_{\text{types}}$ channels. Equation (69.3) is not a flattened count of blocked local rows: spatial blocking uses a composed finite description. The counts are constructive upper descriptions for this generator, not smallest-alphabet or optimal-overhead theorems. They do not assign a numerical universal alphabet size, because no expanded universal normalized source is supplied here.

70 Compact quadratic certificates

The finite source horizon h is an external parameter. The certificate is a polynomial over natural numbers whose zeros encode first halt of the original source at that horizon. Reusing this certificate avoids independently guessing the inverse computation. We give the original encoding and the precise sense in which its witness determines a full cleaned witness.

70.1 Forward branch forms and exactness

Expand zero tests at prime p into residues $r \in \{1, \dots, p-1\}$. For every step $0 \leq t < h$ and branch $1 \leq j \leq B$, use natural variables e_{tj} and u_{tj} . The affine forms in Table 9 describe an old raw value A_j , a new raw value C_j , and duration D_j . Fix an injective integer coding of states; let s_j and t_j be branch j 's source and target codes. They may coincide for a self-loop.

Branch	$A_j(e, u)$	$C_j(e, u)$	$D_j(e, u)$
INC at p	$e + u$	$p(e + u)$	$108A_j + 96C_j + 8e$
DEC at p	$p(e + u)$	$e + u$	$96A_j + 108C_j + 8e$
POS at p	$p(e + u)$	$p(e + u)$	$192A_j + 8e$
ZERO at p , residue r	$pu + re$	$pu + re$	$192A_j + 8e$
NOP	$e + u$	$e + u$	$192A_j + 8e$

Table 9: All forms vanish when $e = u = 0$. At $e = 1$, natural u enforces the exact enabled arithmetic or residue class.

Define the affine sums, separately at every t ,

$$E_t = \sum_j e_{tj}, \quad U_t = \sum_j A_j(e_{tj}, u_{tj}), \quad V_t = \sum_j C_j(e_{tj}, u_{tj}),$$

$$S_t = \sum_j s_j e_{tj}, \quad Q_t = \sum_j t_j e_{tj}, \quad \Theta = \sum_{t,j} D_j(e_{tj}, u_{tj}).$$

Set $V_{-1} = N_0$ and $Q_{-1} = \text{code}(q_0)$. These are affine interfaces, not new variables. The core polynomial is

$$P_h = \sum_{t=0}^{h-1} [(E_t - 1)^2 + (S_t - Q_{t-1})^2 + (U_t - V_{t-1})^2] + (Q_{h-1} - \text{code}(q_h))^2 + \sum_{t=0}^{h-1} \sum_{j=1}^B (E_t - e_{tj})(e_{tj} + u_{tj}). \quad (70.1)$$

It has degree at most two. Every summand is nonnegative on the full natural orthant, since $E_t - e_{tj} = \sum_{k \neq j} e_{tk} \geq 0$. [write] [Table 9 and (70.1) are source 14's branch table and certificate (58.1), with q_h for q_f ; the gate is discussed in the note after Theorem 58.1.]

Lemma 70.1 (Forward zero fibers). *For fixed input coordinates, P_h has a natural zero exactly when the original run first halts at horizon h . If a zero exists, its full forward witness is unique. Its affine endpoints are N_0 , $N_h = V_{h-1}$, and Θ .*

[write] *Pointer*. This is source 14's Theorem 58.1, for the same polynomial. Source 15's proof repeats source 14's (the square $(E_t - 1)^2$ and natural selectors force one-hot selection, every inactive product forces its offset to zero, the active offset is $N - 1$, $N/p - 1$ or $(N - r)/p$, the terminal square forces a first halt, and $h = 0$ leaves only the constant halt square); it is not reprinted. $\square$

70.2 The compact clean certificate and roles of coordinates

Use the forward polynomial unchanged and replace its exposed physical-time form by (68.1). By Theorems 66.2 and 67.1, its zero already proves the exact cleaned target and determines every source section in both halves. The fixed compiler then determines every intervening CA microconfiguration. This is a certificate for the specified encoded source computation, not a general certificate for arbitrary CA trajectories.

To impose a supplied exact microtime T , add just

$$(\kappa[2\Theta + 192(N_h + N_0) + 16] - T)^2, \quad \kappa = \begin{cases} 1 & \text{native or spatial,} \\ 4 & \text{phase.} \end{cases} \quad (70.2)$$

For fixed T this adds no variable. To expose time as a free output coordinate, introduce one natural T and the same square; it is uniquely determined in each accepted input fiber. The target raw value is the already existing input expression N_0 . Its restoration needs neither a separately guessed output value nor another endpoint square.

The exact core ledger is

$$2Bh \text{ natural variables,} \quad 3h + 1 \text{ affine-square slots,} \quad Bh \text{ affine-product slots.} \quad (70.3)$$

These are unsimplified emitted slots; a degenerate term may vanish identically. By comparison, applying the original exporter directly to the cleaned program at horizon $2h + 2$ would give

$$8(B + 1)(h + 1) \text{ core variables,} \quad 6h + 7 \text{ squares,} \quad 4(B + 1)(h + 1) \text{ products.} \quad (70.4)$$

The smaller ledger is justified by source composition and the zero-fiber lift below, not by dropping constraints from an arbitrary reverse run.

70.3 Input modes and their paid costs

A fixed positive raw input is a constant. A free raw input uses $N_0 = x + 1$ with one genuine natural input coordinate x and no loader square. Fixed counters (a, b) can be encoded externally by the computable constant $2^a 3^b$.

For an explicitly bounded counter interface, fix bounds $0 \leq a \leq A$ and $0 \leq b \leq C$. Introduce $K = (A + 1)(C + 1)$ natural selectors λ_{ij} and three squares

$$\left(\sum \lambda_{ij} - 1\right)^2, \quad \left(a - \sum i \lambda_{ij}\right)^2, \quad \left(b - \sum j \lambda_{ij}\right)^2.$$

Use $N_0 = \sum 2^i 3^j \lambda_{ij}$ by affine substitution. The selectors are uniquely one-hot on accepted inputs. Every genuinely free a or b is an additional input coordinate; fixed values are constants. This finite table is not an unbounded exponentiation gadget.

Let r count genuine input coordinates, k loader selectors, $\ell \in \{0, 3\}$ loader squares, $s \in \{0, 1\}$ a free time output, and $\epsilon \in \{0, 1\}$ whether a time equality is included. The full compact ledger is

$$\begin{aligned} \text{natural variables} &= 2Bh + r + k + s, \\ \text{affine squares} &= 3h + 1 + \ell + \epsilon, \\ \text{affine products} &= Bh, \quad \text{degree} \leq 2. \end{aligned} \tag{70.5}$$

For $h = 0$, the time form is $384N_0 + 16$ before any phase scaling, and all optional input and time costs are still charged.

70.4 An affine lift to the full cleaned witness

Index original expanded branches by $j = 0, \dots, B - 1$ for this subsection. In the full cleaned exporter, branches j are the forward copies, $B + j$ are their inverse copies, and $2B, 2B + 1$ are the bridges. Its steps are indexed $0, \dots, 2h + 1$. Denote its branch variables by $\tilde{e}_{s,k}, \tilde{u}_{s,k}$.

The following affine substitution L maps compact witnesses to full ones:

$$\tilde{e}_{t,j} = e_{t,j}, \quad \tilde{u}_{t,j} = u_{t,j}, \quad 0 \leq t < h, \tag{70.6}$$

$$\tilde{e}_{2h-t, B+j} = e_{t,j}, \quad \tilde{u}_{2h-t, B+j} = u_{t,j}, \quad 0 \leq t < h, \tag{70.7}$$

$$\tilde{e}_{h, 2B} = 1, \quad \tilde{u}_{h, 2B} = N_h - 1, \tag{70.8}$$

$$\tilde{e}_{2h+1, 2B+1} = 1, \quad \tilde{u}_{2h+1, 2B+1} = N_0 - 1. \tag{70.9}$$

Every unlisted full-core coordinate is zero. Preserve all genuine input coordinates, loader selectors, and any time output. Coordinate names are merely labels: rename them consistently if a user-specified public name collides with a larger full-core variable name.

Theorem 70.2 (Affine bijection of natural zero fibers). *With the same fixed source, horizon, input interface, and optional time condition, L restricts to a bijection from compact natural zeros to natural zeros of the full cleaned-source certificate at horizon $2h + 2$. Projection onto its original forward variables and public coordinates is its inverse, after any consistent renaming.*

Proof. Start with a compact zero. Lemma 70.1 gives a genuine forward trace and positive endpoints N_0, N_h . Thus the two bridge magnitudes in (70.8)–(70.9) are natural, and every coordinate of L is natural.

For an original increment, $N = e + u$ and $N' = p(e + u)$. Its inverse decrement on N' uses exactly the same e, u , so exchanges the old and new affine raw values. An original decrement has the same property with increment as its inverse. Tests and no-ops have unchanged raw values and retain the same parameters, including the zero-test residue. Consequently (70.7) realizes the unique inverse trace step by step. The bridge rows are no-ops at N_h and N_0 . All unselected coordinates are zero, so their inactive products vanish. The source and raw interface squares match the concatenated trace, and the full terminal state is H .

The unchanged loader satisfies its original three squares when present. Summing the full duration forms gives (68.1); hence any time endpoint square also agrees. The lifted assignment is therefore a full natural zero.

Conversely, the forward-certificate exactness theorem applied to the full cleaned exporter turns any of its natural zeros into a first-halt cleaned execution of length $2h + 2$. The only route to H traverses a forward first halt, both bridges, and the unique inverse trace. If that forward length is k , Theorem 66.2 gives total length $2k + 2$, so $k = h$. Projecting to the first h rows and original public coordinates gives a compact zero; the clock identity supplies its time condition. Full witness uniqueness at fixed input forces all remaining rows to be precisely (70.7)–(70.9). Thus the two maps are inverses on zero fibers. $\square$

The restriction to zero fibers is essential. The bridge expression $N_h - 1$ can be negative at a general natural assignment that is not a compact zero. The claim is not a map of the entire natural orthant, nor an identity of the two polynomial values off their zero sets. The natural domain is also essential for the original encoding: fractional branch magnitudes can bypass discrete divisibility or residue guards. No assertion for nonnegative rational or real solutions is made. [write] [The programme's review [45] exhibits both limitations exactly: for a one-step increment at $N_0 = 1$ and the off-zero assignment $e = u = 0$, the compact polynomial has value 3, its lift has value 18 and a bridge coordinate -1 ; and a one-step prime-two zero test at $N_0 = 2$ has the nonnegative rational zero $e = 1, u = 1/2$ of source 14's Section 58. It records them as documented limitations, not defects, and checks 45 compact-to-full natural zero-fibre bijections.]

Finally, this is a family indexed by externally fixed h . Eventual halting is a union over all finite horizons. Merely replacing h by a variable does not give a fixed-arity unbounded-time quadratic polynomial, and fixed-horizon uniqueness does not imply a finite-fold unbounded-time representation.

71 A complete exact-target example

Take six states $q_0, \dots, q_5$, halt q_5 , and the five-instruction chain

$$q_0 \xrightarrow{\text{INC}_0} q_1 \xrightarrow{\text{INC}_1} q_2 \xrightarrow{\text{DEC}_0} q_3 \xrightarrow{\text{DEC}_1} q_4 \xrightarrow{\text{INC}_0} q_5.$$

It is fresh-entry and separated reversible. Use $N_0 = 5$, whose two counter valuations are initially zero and whose preserved cofactor is five. The original raw values are

$$5, 10, 30, 15, 5, 10.$$

The source halts after $h = 5$ with $N_h = 10 \neq N_0$ and native time $\Theta = 13\,480$. The different endpoint value makes this a genuine restoration test rather than a run whose original output happens to equal its input.

Cleaned source step	State	Raw N	Native or spatial tick
0	F: q_0	5	0
1	F: q_1	10	1 508
2	F: q_2	30	5 476
3	F: q_3	15	9 984
4	F: q_4	5	11 972
5	F: q_5	10	13 480
6	B: q_5	10	15 408
7	B: q_4	5	16 916
8	B: q_3	15	18 904
9	B: q_2	30	23 412
10	B: q_1	10	27 380
11	B: q_0	5	28 888
12	H	5	29 856

Table 10: Every clean section of the example. The inverse source half restores all intermediate raw values in reverse order. Its microscopic paths need not be reversed.

The initial configuration is the clean F: q_0 controller and shuttle at -60 , with R at 0. The exact target uses the clean H controller and shuttle at -60 , the same R at 0, and vacuum everywhere else. Its first time is

$$2(13\,480) + 192(10 + 5) + 16 = 29\,856.$$

Spatial blocking moves the left pair to -15 , offset zero, and keeps that time. The phase construction keeps coordinate -60 and first reaches its phase-zero target at 119 424 ticks.

Here $q = 6$, $J = 5$, $m = 6$, $d = B = 5$, while the cleaned values are $q' = 13$, $J' = 12$, $m' = 13$, $d' = 10$, $B' = 12$. The native generator gives 19 416 channels, 598 096 prescribed pair rows, and 1 188 236 completed pair-support rows. These are example counts, not universal-machine counts.

The compact certificate has 50 core variables, 16 core squares, and 25 products. The packaged free-time version has 51 total variables and 17 squares, with time output 29 856. By comparison, the full cleaned exporter has 288 core variables, 37 core squares, and 144 products before the same time-output charge. The affine lift reuses the forward witness, with bridge magnitudes $N_h - 1 = 9$ and $N_0 - 1 = 4$.

The archive’s request, literal expanded certificate, unique witness, and independent checker receipt reproduce Table 10. They describe a small finite chain rather than a universal program.

72 Validation and reproducibility

72.1 What the implementations check

The cleanup exporter validates structural freshness, builds the forward/inverse source table, composes the original polynomial, and exposes the cleaned clock. A separate checker reconstructs the wrapped program and affine outputs instead of trusting these claims in the exported JSON. It invokes the unchanged independent forward checker and validates the natural witness, optional literal polynomial expansion, and full resource ledger. Its decoded cleaned trace is compared with actual source execution and literal lattice evolution.

The producer tests exercise all primitive operations, both nonzero prime-three residues, guarded branch/merge behavior, a mixed chain with unequal original endpoint values, $h = 0$, immediate and delayed stalls, a growing nonhalting prefix, raw cofactors, loader interfaces, invalid freshness, clock symmetry, and maliciously altered certificates. Successful lattice checks compare the entire target and every canonical section, test conservation and inverses, and require prescribed pair rows through the first hit. Selected cases execute all native, spatial, and phase ticks.

The independent lattice harness separately builds the wrapper, executes source instructions, recognizes the full canonical configurations, and steps the generated collision and stream maps without calling the compiler’s ordinary step or ready helpers. Its certificate audit exercises valid interfaces and rejects deliberate mutations. The separate lift audit compares explicit forward/reverse variable reuse with the full cleaned exporter and its frozen independent checker.

Finite evidence and theorem evidence. Finite tests verify concrete implementations and boundary cases. The all-input restoration and nonhalting exclusion follow from the proofs, the full-space reversible behavior from the compiler, and undecidability from the classical source dependency. No finite collection of successful traces proves universality, and a growing finite prefix does not alone prove nonhalting. The archive records each test’s actual scope rather than conflating these forms of evidence.

Replay summary. The completed release receipt records the exact test counts and subject hashes. The independently checked lattice sample comprises 14 small sources and 35 runs, including 26 successful cleanups and nine stuck inputs. It covers 111 712 native ticks, 9 456 spatial ticks, and 37 824 phase ticks. The independent certificate pass checks 61 valid model/interface cases, rejects 174 mutations and 19 malformed syntax or API cases, and enumerates 1 536 natural assignments for prime-three zero-test branches. The producer checks 77 affine-lift cases; the independent lift audit checks 24 cases and 96 public-coordinate naming-collision regressions. The producer lattice suite covers 45 halting runs, 21 stalls, 197 374 native ticks, 21 120 spatial ticks, and 84 480 phase ticks; it also checks 140 clock-symmetry cases. Normal and optimized Python runs are recorded separately.

72.2 Portable replay and source provenance

The portable archive includes this standalone editable LaTeX source and PDF, the cleanup code and independent checker, literal examples, the five exact frozen Python dependencies, independent audit harnesses, receipts, and build/replay scripts. Python replay uses only the standard library. PDF rebuild uses a conventional TeX installation and the packages declared at the top of the source. No third-party paper PDFs are redistributed.

All package files are covered by a SHA-256 manifest except the manifest itself. A verifier checks both hashes and the expected file set. The archive safety check rejects absolute paths, parent traversal, duplicate members, and symbolic links. Fresh replay is performed from an extracted copy rather than from the source working directory. Temporary generated outputs remain outside the sealed file set. The final PDF is rendered and every page is visually checked.

The provenance ledger records hashes for the five frozen dependencies, the exact extension sources, source-audit scripts, and the companion report that establishes inherited results. Independent harnesses are made portable only by changing dependency/output path selection; that adaptation is recorded explicitly and the adapted copies are rerun. Receipts identify which subject code was actually tested. References link to primary publications; packaging does not silently substitute a

different local-rule generator.

[write] **The release in this report.** These are the delivered names. Source 15’s top-level programs and its `audit/` programs are shipped as `code/15-clean-targets-*`; its `examples/`, `receipts/` and `provenance/` records as `data/15-clean-targets-*`; and `CLEAN-TARGET-THE OREM.md`, `SOURCE_PROVENANCE.md` and `audit/INDEPENDENT-AUDIT.md` as `15-clean-targets-*` at the report root. “The five exact frozen Python dependencies” (`vendor/*.py`) are byte copies of source 14’s programs and are shipped once, as `code/14-three-mass-*`. Not shipped: the manuscript, PDF and README, the SHA-256 manifest and `verify_manifest.py`. The programs import `vendor/` by its delivered name, so they must be run in a restored layout; on Windows `verify_manifest.py` rejects every nested member (it compares native paths with `/`-separated names), and the replay’s byte-exact example comparison fails on CRLF output although the content agrees, so the replay is run without the manifest there (README).

73 Boundaries and precise next questions

The construction settles the mass threshold for exact-pair reachability in the stated weighted model. It also settles radius one as a sufficient radius with three conserved mass units. Several stronger or more concrete claims remain distinct.

1. **A fully expanded universal instance.** Instantiate a particular universal reversible two-counter table with fresh entry and terminal halt, provide its input encoder, and freeze the resulting cleaned local rule. This would yield an exact universal alphabet and row count. The small examples in this archive do not perform that instantiation. ([write] Compare source 14’s Question 1 and Part IV’s first question; all three are open.) ([write] Added 3 October 2026, batch 82: answered in part outside this report, as the note at Question 1 says; the cleaned universal rule of this automaton is not frozen.)
2. **One target independent of the input.** Determine whether a suitable fixed reversible weighted CA can have undecidable reachability to one chosen finite configuration from a computable input family. The present restoration construction gives y_N , not a single y . Injectivity and the backward orbit of a fixed target must be handled explicitly; no claim of impossibility follows from this report.
3. **Smaller finite-state overhead.** Reduce the duplicate state space, residue clock, or two bridge costs while retaining global reversibility and exact scratch cleanup. Compare improvements using the same target convention and all resource ledgers. A faster logical inverse cannot be assumed to reverse microscopic paths. ([write] Compare source 14’s Question 2.)
4. **The integer-state mass convention.** Find the corresponding sharp reversible threshold when each cell state’s integer value is its conserved mass. Equal-weight channel types used here cannot be transferred by relabeling. This is a different model-specific question. ([write] Compare source 14’s Question 4, answered in part by Part VI, as the note there says.) ([write] Added 3 October 2026, batch 82: answered outside this report, five, as the note at Question 4 says.)
5. **A costed unbounded-time certificate.** Compose this frontend with an explicit fixed-arity Diophantine encoding of unbounded computation, and state degree, variable costs, and the resulting fiber properties. The fixed-horizon affine lift alone supplies no finite-fold or single-fold unbounded-time conclusion. ([write] Compare source 14’s Question 5.) ([write] Added 3

October 2026: answered for fixed eligible sources by Part VIII, which composes this frontend with the research tree’s residue-history interface: exact degrees 2344 and 1192, ledgers of 604, 479, 477 and 480 operations (603, 478, 476 and 479 folded, 608, 483, 481 and 484 canonical) and 60, 58, 58 and 58 positive witnesses for four nonuniversal fixtures, and infinite witness fibres, also after canonicalization of the outer history (Theorems 146.1 and 155.1). The emitted representations are not finite-fold. Part VIII does not address Questions 1 and 2.)

6. **Mechanized compiler composition.** Formalize the separated syntax, inverse source transform, prescribed collision routes, and zero-fiber correspondence in one proof system. The key interfaces to preserve are exact absolute configurations, natural domains, and the distinction between source steps and CA ticks.

There is close geometric prior art. Brandt, Portmann and Uitto [37] use three finite-state agents, a prime-exponent gap, and reflected-shuttle arithmetic. The companion report [source 14, Section 62] compares those mechanisms with the global reversible CA and its cleanup obligations. The extension here does not claim a literature-wide priority result or a new general principle of uncomputation. Its conclusion is the explicit exact-target strengthening, under the model and dependencies stated above.

Part VI

A single unit symbol: decidability at masses three and four (sources 16, 17 and 18)

[write] Part VI prints two further AI-assisted research manuscripts of batch 80 (arrival 4e270aa46), both dated 3 October 2026. *Source 16* is batch-80 manuscript 05, *Three mass units with a single unit symbol*, author line “Decidability theorem and executable checks”, from the archive `Single_Unit_Three_Mass_Decidability (1).zip` (with a space before the parenthesis); its labels carry the prefix `smc:su:`, and its sections are Sections 74–82 here. That archive is the “attribution revision” (its `REVISION.md`, shipped as `16-single-unit-REVISION.md`) of the batch’s manuscript 06, `Single_Unit_Three_Mass_Decidability.zip`: the revision credits Kong’s 2021 thesis and Imai’s 2021 KAKEN report in its abstract, its source comparison and two bibliography entries, and leaves the mathematics, programs, receipts and audit unchanged (seven files byte-identical). Manuscript 06 is therefore superseded and not printed. *Source 17* is batch-80 manuscript 03, *Four mass units and exact reachability*, author line “Decidability proof and a binary timing certificate”, from the archive `Four_Mass_Decidability_Package.zip`; its labels carry the prefix `smc:fm:`, and its sections are Sections 83–92. It builds on source 16, which it calls “the separate 3 October 2026 companion report”, and restates the ingredients it needs (its Section 84, printed with pointers). Neither manuscript names a repository commit; source 16 cites Part IV (source 13) by its title, “Section 9”, for the post-elimination compiler. Statements keep their numbering by section; equations, which the sources tagged (1), (2), ..., are numbered by section. The programme has reviewed all three archives [46] (Section 1.14). ([write] Added 3 October 2026: a third source, batch-82 manuscript 19, follows source 17 as *source 18*; it gives canonical timed quartics for a fixed input and answers source 17’s last question. It is printed after source 17 with its own opening, and Section 1.17 describes it.)

[write] **Relation to Part IV and Part V.** [Editorial note: source 16’s mass-two case and its architecture are Part IV’s. The mass-two case is Theorems 44.1 and 45.1 of Part IV, proved

there for any number of weight-one symbols; source 16’s Lemma 75.2 and Sections 76–79 follow the architecture of Sections 44–45 (first entry with the absolute position, finite templates, no product k_0n), and its quartic is Part IV’s compiler of Section 46. Bracketed notes mark each place.] The decisive new ingredient is source 16’s no-split lemma (Lemma 75.1), which uses the single unit symbol: in the comoving frame an isolated unit is fixed, so a bounded mass-two packet never separates into two distant pieces. Part VI is consistent with Part V, which uses many weight-one types (source 14, Section 49; source 16, Section 79).

Source 16 (batch-80 manuscript 05): Three mass units with a single unit symbol

Decidability theorem and executable checks, 3 October 2026. Its abstract follows; the two bracketed notes in it are [write].

For every fixed one-dimensional deterministic conservative cellular automaton with a finite alphabet, positive integer non-vacuum weights, and at most one weight-one symbol, the complete timed evolution relation at total mass at most three is effectively Presburger-definable in all input and output positions. Exact reachability and finite-pattern occurrence, at a fixed position or anywhere, are decidable. In the stationary-unit frame, every bounded two-unit cluster remains bounded and has finite-phase motion; three-unit encounters then reduce to a finite core. Composition with a canonical arithmetic compiler gives horizon-free quartic certificates with zero or one natural witness tuple. Consequently, standard nonnegative integer-state number-conserving CA cannot realize undecidable finite halt-pattern occurrence with three numerical particles. This separates them from typed weighted three-mass constructions [such as that of Part V (source 14), which uses many weight-one types]. Kong’s June 2021 thesis already proves binary three-particle nonuniversality using a periodic-orbit decomposition. The present weighted, uniform arithmetic strengthening is stated explicitly; its novelty and the current mass-four boundary remain unresolved here. [Added 2 October 2026: the mass-four boundary is settled within the same class by source 17 (Theorem 83.1).]

74 Model and main result

Let A be a finite alphabet with a distinguished vacuum 0 and a weight $w : A \rightarrow \mathbb{N}$, where $w(0) = 0$ and $w(a) > 0$ for every $a \neq 0$. Let $F : A^{\mathbb{Z}} \rightarrow A^{\mathbb{Z}}$ be a deterministic, time-independent, translation-equivariant CA of finite radius R . The vacuum is quiescent. For every finite-support configuration c , assume

$$M(F(c)) = M(c), \quad M(c) = \sum_{i \in \mathbb{Z}} w(c_i).$$

There is at most one symbol u with $w(u) = 1$. Other symbols may share weights, including weights two and three. No reversibility assumption is made. The vacuum is the only *zero-weight* symbol; other constant configurations are allowed to be fixed.

Write $\tau_a c$ for translation of a configuration by a sites to the right. A sparse configuration is specified by a strictly increasing position tuple $x_1 < \dots < x_n$ and its nonzero label tuple $\alpha = (\alpha_1, \dots, \alpha_n)$. At mass at most three, $n \leq 3$, and only finitely many label tuples are possible.

Theorem 74.1 (Uniform timed evolution). *For each fixed F satisfying these assumptions and each pair of nonzero label tuples α, β of total weight at most three, an algorithm constructs a Presburger formula*

$$\text{Rel}_{\alpha, \beta}^F(x_1, \dots, x_n, t, y_1, \dots, y_m)$$

which holds exactly when $t \in \mathbb{N}$, both position tuples are strictly increasing, and $F^t(c_{\alpha,x}) = c_{\beta,y}$. The vacuum is handled as a separate case. The construction is effective from the finite local rule, radius, and weights, under the stated hypotheses.

“Uniform” means that the formula includes all sparse input positions, output positions, and times for one fixed rule. It does not assert a single Presburger formula in which arbitrary CA rule tables are themselves encoded as an additional numerical parameter. The effective equivalence with semilinear sets is classical [7]. Presburger arithmetic here is first-order arithmetic over $\mathbb{Z}$ with addition, order, and integer constants. Quantification over $\mathbb{N}$ is expressed by nonnegativity constraints.

Corollary 74.2. *For any such fixed F , the following problems are decidable on finite inputs of mass at most three:*

1. *Exact finite-configuration reachability, including specified-time reachability and reachability up to translation.*
2. *Occurrence of a given finite pattern at a specified position, or at some position, at some nonnegative time.*
3. *Any fixed finite collection of such observations at specified relative time offsets.*

For fixed input and fixed observation, its set of occurrence times is effectively semilinear and hence eventually periodic.

[write] [The mass-two case is Theorems 44.1 and 45.1 of Part IV, proved there for any number of weight-one symbols. Lemma 75.2 and Sections 76–79 follow the architecture of Part IV’s Sections 44–45 (first entry with the absolute position preserved, finite templates after entry, no product k_0n), one mass higher. Source 16 cites Part IV only for its compiler (Section 80).]

75 The stationary frame and two-unit clusters

If no unit symbol exists, every nonempty configuration of mass at most three has exactly one occupied site. Conservation forces its successor to have exactly one site of the same weight. There are finitely many labels and bounded one-step displacements, so this is a finite-state walker. Its motion is effectively eventually periodic up to translation, and Theorem 74.1 follows immediately. Henceforth assume u exists.

Conservation forces $F(u \text{ at } 0)$ to be exactly one copy of u , at some site $\delta \in [-R, R]$. Define

$$G = \tau_{-\delta} \circ F, \quad S = \max\{1, R + |\delta|\}.$$

Then G has radius at most S , is conservative, and fixes each isolated copy of u . Moreover $F^t = \tau_{\delta t} \circ G^t$. It suffices to prove the theorem for G ; replacing each output position y_j by $y_j - \delta t$ gives the formula for F . If the comoving rule has radius zero, each occupied coordinate stays fixed and its label follows a finite weight-preserving map; finite transient and periodic label conditions prove the result directly. The padding $S \geq 1$ also includes this case in the argument below without dividing by a zero radius.

Connect two occupied sites if their distance is at most $2S$. Components always refer to this graph. If distinct components are separated by more than $2S$, no radius- S input neighborhood sees both. Quiescence therefore gives exact independence for one step: the global successor is the disjoint union of the successors of those components evolved alone. Their radius- S output neighborhoods are disjoint as well.

Lemma 75.1 (Two-unit diameter invariance). *If a mass-two configuration has diameter at most $2S$, then every iterate under G has diameter at most $2S$.*

Proof. There are only two possibilities. A single weight-two symbol has its successor inside an interval of width $2S$ by finite propagation.

Otherwise the configuration is u at sites 0 and d , where $1 \leq d \leq 2S$. If $d > S$, each original site has exactly its isolated-unit radius- S neighborhood. Both original sites therefore output u . These two sites already exhaust the conserved mass, so the entire configuration is fixed.

If $d \leq S$, consider an output site other than 0 and d . If its neighborhood sees no unit, it outputs zero by quiescence. If it sees just one unit, its local input agrees with an isolated-unit configuration at that output site; since isolated units are fixed, it again outputs zero. A new nonzero site must see both units, and thus lies in

$$[d - S, S].$$

This interval also contains 0 and d , and has width $2S - d \leq 2S$. All output support is therefore contained in it. Induction proves the claim. $\square$

This lemma excludes a potentially misleading geometric scenario. Under a coarse propagation bound alone, a pair could appear to split, let one fragment join a third unit, and leave the other behind without a full encounter. Here that cannot happen across distinct $2S$ -components: a mass-two component never splits into two such components. The unique stationary unit label and mass conservation are essential to this conclusion. [write] [This no-split lemma is the decisive new ingredient of source 16. It has no counterpart in Part IV, where several weight-one types let a near pair separate into two walkers (Section 44).]

Lemma 75.2 (Finite phase motion). *For each normalized mass-two configuration q of diameter at most $2S$, its isolated orbit has a computable finite transient and a translated periodic tail. There exist $\mu \geq 0$, $p \geq 1$, $D \in \mathbb{Z}$, and explicitly computed finite configurations $q_0, \dots, q_{\mu+p-1}$ such that*

$$G^{\mu+r+np}(q) = \tau_{nD} q_{\mu+r} \quad (0 \leq r < p, n \geq 0).$$

Each configuration on this orbit has diameter at most $2S$.

Proof. Normalize every nonempty configuration by moving its leftmost occupied site to zero. Lemma 75.1 leaves finitely many normalized mass-two shapes: one for each weight-two symbol, and one pair u, u at each distance $1, \dots, 2S$. The successor shape and change in the leftmost position depend only on the current shape. Iterating this finite deterministic transition system reaches a repeated shape. The repeat supplies μ, p, D and the stated identity by translation equivariance. $\square$

[write] [Same argument as Proposition 43.1 and (43.2) of Part IV, applied to the finitely many shapes left by Lemma 75.1.]

If N_2 denotes the number of weight-two labels, there are exactly $N_2 + 2S$ possible normalized shapes. No speed bound on the movement of the leftmost occupied site is needed: its displacement on a shape transition is simply computed from the rule. A mass-two input of diameter greater than $2S$ is two independent unit symbols and is fixed by G .

76 The bounded encounter core

For mass exactly three, define the encounter core by the diameter condition

$$\text{Core} = \{c : M(c) = 3, \text{diam}(\text{supp } c) \leq 4S\}.$$

The core includes every connected mass-three configuration, and may include some disconnected ones. Its use requires no claim that a configuration entering it has already undergone a local interaction. Modulo translation, the core is finite. If N_3 is the number of weight-three labels, its number of normalized shapes is

$$N_3 + 8SN_2 + \binom{4S}{2}.$$

These are respectively one weight-three site, a weight-two site together with a unit site, and three unit sites. One step from the core has diameter at most $6S$, by finite propagation.

Every mass-three configuration outside the core has one of two forms:

1. Three mass-one components, each an isolated u . They are mutually separated by more than $2S$ and the entire configuration is fixed by G .
2. One mass-two component of diameter at most $2S$, and one isolated u .

A weight-three singleton is always in the core. Thus these cases exhaust all possibilities, including an initial weight-two singleton far from a unit.

Lemma 76.1 (Exact evolution before the core). *Suppose an outside-core configuration consists of a mass-two component $\tau_a q$ and a unit at b . Let*

$$z_k = \tau_a G^k(q)$$

be the isolated mass-two evolution. Consider the counterfactual union of its support with the fixed site b , and let h be the first $k \geq 0$ at which the span of that union is at most $4S$, if there is such a k . Before time h , and also at time h when finite, the actual orbit is exactly z_k together with the unit at b . The union at those times has disjoint occupied sites. If h does not exist, this describes the entire orbit.

Proof. By Lemma 75.1, $\text{diam } z_k \leq 2S$. If b were within $2S$ of any occupied site of z_k , the whole union would have span at most $4S$. Therefore, whenever the counterfactual span is greater than $4S$, the mass-two component and the unit are separated by more than $2S$. Their next step is independent and their outputs are disjoint. Induction proves equality up to and including the first core time. Although a later counterfactual superposition could have overlapping sites, no such overlap occurs in the part used here. $\square$

Lemma 76.2 (Effective first entry). *The first entry in Lemma 76.1, or the fact that no entry occurs, is computable without stepping through the initial gap.*

Proof. The finite transient from Lemma 75.2 is checked directly. On each periodic phase, write $k = \mu + r + np$ with $n \geq 0$; the leftmost site is $a + A_r + nD$, and the diameter is a constant $d_r \leq 2S$. Core membership is exactly the pair of inequalities

$$b - 4S \leq a + A_r + nD \leq b + 4S - d_r. \quad (76.1)$$

For a fixed phase this is an integer interval condition on n . If $D = 0$ it is always true or always false. If $D > 0$, compute the first nonnegative integer above the lower bound and check the upper bound; if $D < 0$, negate the inequalities. Take the earliest time among the finitely many valid phase and transient candidates. $\square$

[write] [Lemmas 76.1 and 76.2 are the exact acceleration to the next encounter of Part IV ((44.2) and the paragraph after it), with a bounded mass-two packet in place of a walker. The core and its shape count are new.]

77 Finite encounter dynamics and complete orbits

Starting from any normalized core shape, take one exact CA step. If the output is again in the core, this gives a transition with elapsed time one and a computed translation. If it is outside, apply the classification above and Lemma 76.2. Either there is no return, or the first return gives another core shape with a positive integer elapsed time and an integer translation. All these data are effective, and depend only on the starting normalized shape.

This produces a finite deterministic graph of core shapes. Every outgoing edge either reaches a core shape with a fixed time/translation increment, or enters a completely described nonreturning orbit. There are two terminal forms: a fixed configuration, or the independent combination of a stationary unit and the finite-phase mass-two walker.

Proposition 77.1 (Eventual orbit structure). *From each fixed finite configuration of mass at most three, the complete orbit under G is effectively eventually of one of the following forms:*

1. *The entire configuration is periodic up to a single translation: there are $T \geq 0$, $P \geq 1$, $d \in \mathbb{Z}$ with $G^{t+P}(c) = \tau_d G^t(c)$ for all $t \geq T$.*
2. *A stationary unit and a mass-two component evolve independently, with the latter periodic up to translation after a finite transient.*

The fixed configuration case is included in the first form with $P = 1, d = 0$.

Proof. An initially unbounded separation is resolved by Lemma 76.2, or gives an immediate nonreturning description. After a core entry, there are finitely many normalized core shapes. If a nonreturning edge is taken, its description gives the second form or a fixed configuration. Otherwise a normalized core shape repeats: at two times $t_0 < t_1$ the actual configurations satisfy $G^{t_1}(c) = \tau_d G^{t_0}(c)$. Determinism and translation equivariance imply

$$G^{t_1+n}(c) = \tau_d G^{t_0+n}(c) \quad (n \geq 0).$$

This identity covers every intermediate configuration, including every part of every subsequent excursion, not just the core-return times. $\square$

[write] [Same argument as Part IV's finite near-shape recurrence ((44.3) and the paragraphs after it), with the mass-three core in place of the near shapes.]

For a fixed starting configuration, these descriptions immediately yield a Presburger relation between time and all occupied output positions. A phase $t = T + r + nP$ gives constant labels and positions of the form $c_j + nd_j$. The drifts d_j are either all equal, or are the walker drift together with zero for the stationary unit. Any finite initial segment can be included by a finite

disjunction. Positions, labels, and finite-pattern observations at intermediate times are retained by this construction.

This proves a per-input semilinearity statement. The next section supplies the additional step needed for one formula covering arbitrarily large initial gaps.

78 Uniform Presburger construction

For every normalized core shape c , the preceding finite computation gives an effective formula $K_c(a, t, y)$ describing the whole orbit from $\tau_a c$. There are finitely many such formulas. This notation suppresses the finitely many output-label cases; all displayed formulas may be expanded separately for each label tuple.

The mass-zero, mass-one, and mass-two cases were already handled by finite phase motion and separated stationary units. Initial core configurations are covered by the formulas K_c . Initial configurations with three mass-one components are fixed in the stationary frame.

It remains to treat the outside-core pair-plus-unit case, with bounded normalized pair shape q , pair translation a , and unit position b . There are only finitely many choices of q . Its isolated profile from Lemma 75.2 is uniformly Presburger in (a, k) : each periodic phase has time $k = \mu + r + np$ and positions obtained by adding $a + nD$ to fixed constants. Its finite transient contributes finitely many cases.

Let $H_q(a, b, k)$ express that the span of the counterfactual support together with b is at most $4S$. Formula (76.1) and its finite transient cases show that H_q is Presburger. Define its first-hit predicate by

$$E_q(a, b, k) \equiv k \geq 0 \wedge H_q(a, b, k) \wedge \neg \exists j (0 \leq j < k \wedge H_q(a, b, j)).$$

Presburger arithmetic is closed under these quantifiers and Boolean operations.

For output time t , use the union of two cases:

1. *No entry through time t .* Require $t \geq 0$ and $\neg \exists j (0 \leq j \leq t \wedge H_q(a, b, j))$. Describe the output as the isolated pair profile at time t together with the unit at b .
2. *Entry by time t .* Existentially choose $0 \leq k \leq t$ satisfying $E_q(a, b, k)$. For each normalized core shape c , express that the entry configuration is $\tau_e c$, for some integer e . Then require $K_c(e, t - k, y)$.

The entry-shape condition is Presburger: its labels and relative positions belong to a finite list, while the pair-profile positions, b , and the translation e are related by linear equalities. The minimum-position choice for e , disjoint support, and ordering can all be expressed by finite conjunctions or disjunctions. The post-entry period and drift depend only on the finitely chosen core shape, not on the unbounded initial gap or first-entry time; this introduces no product of two quantified variables. Lemma 76.1 guarantees that these are valid configurations at the first entry.

Every initial label/position case reduces to one of the listed cases through finitely many comparisons and bounded-offset equalities. The result is a Presburger formula for G . Substituting $y_j - \delta t$ gives the formula for F , completing Theorem 74.1.

[write] [Same architecture as Part IV’s linear guard, first entry and templates ((45.2) and (45.3): “There is no product $k_0 n$ ”). New here is the composition of a bounded mass-two packet with a distant stationary unit.]

79 Observation and computational consequences

Exact reachability is obtained by existentially quantifying t . Reachability up to translation also quantifies one spatial shift. To test a finite pattern p at position z , quantify the at-most-three output positions and labels, and require the specified state at every pattern site. In particular, a zero in the pattern means that no occupied output position is present at that site; zeros are not ignored. The finite union over possible output label tuples is Presburger. Quantify t and, for occurrence anywhere, z . A finite collection of relative-time observations uses a conjunction of the same timed relation at $t + d_1, \dots, t + d_k$, with independent output position tuples. All these constructions are effective. [write] [Same tests as Part IV’s at every intermediate time (Section 44), written in Presburger arithmetic.]

Corollary 79.1 (Numerical particle lower bound). *Let F be a fixed one-dimensional finite-alphabet number-conserving CA with nonnegative integer states and weight $w(q) = q$. From configurations whose numerical state sum is at most three, finite-pattern occurrence and exact finite-configuration reachability are decidable. Consequently such a mass-three family cannot encode arbitrary Turing-machine halting through a computable finite-configuration encoder and a designated finite halt-pattern occurrence event.*

Proof. An integer-state alphabet has at most one state of numerical value one, and all its nonzero states have positive integer weight. The theorem applies. An effective reduction of undecidable halting to the stated decidable observation problem would be a contradiction. $\square$

The conclusion concerns ordinary effective computation with finite input encodings and finite halt observations. It is not a claim about intrinsically universal simulation, noncomputable output interpretation, signed conserved quantities, infinite alphabets, spatially nonuniform rules, time-dependent rules, or active zero-weight backgrounds. Those are different models.

Multiple distinct weight-one symbols invalidate the stationary-unit argument: isolated particles can carry finite internal states or have different motions. A three-mass construction in such a typed weighted alphabet therefore does not contradict Corollary 79.1. [write] [Part V (source 14) is such a construction.] Conversely, simply renumbering those unit symbols as distinct positive integer states cannot preserve their unit weights. The precise distinction is the number of available labels of mass one, not the number of geometrically occupied sites.

[write] [Added 4 October 2026: source 24’s Theorem 164.1 (Part IX) proves the binary, fixed-input case of Theorem 74.1 and Corollary 74.2 again, by the same steps.]

80 Horizon free canonical quartic certificates

The uniform relation has a direct Diophantine consequence. Encode a signed coordinate x by natural inputs (x^+, x^-) with

$$x = x^+ - x^-, \quad x^+ = 0 \text{ or } x^- = 0.$$

This is a unique representation of each integer. Include its domain condition, the ordering constraints, and $t \geq 0$ in the Presburger predicate. Apply effective Presburger quantifier elimination, for example Cooper’s procedure [25], to obtain a quantifier-free predicate on the chosen natural inputs. For untimed reachability, eliminate time as well. Leaving a reaching time as an auxiliary witness would generally destroy uniqueness, because a target may be visited at several or infinitely many times.

Use the canonical quantifier-free Presburger-to-quartic compiler in Section 9 of the companion report [Part IV of this report (source 13), Section 46]. Its precise interface is as follows. Normalize atoms to affine inequalities $L \geq 0$ and fixed-modulus congruences $L \equiv 0 \pmod{d}$, and choose an AND/OR/NOT circuit. Let I, C, B be its actual counts of inequality atoms, congruence atoms, and Boolean gates. The compiler gives an integer polynomial, a sum of squares of quadratic residuals, with degree at most four and

$$K = 2I + 6C + B \quad \text{natural witnesses,} \quad E = 2I + 5C + B + 1 \quad \text{residuals.} \quad (80.1)$$

(The companion report calls the gate count G ; B avoids confusion with the stationary-frame CA in this article.) Its natural witness fiber is a singleton when the predicate is true and empty when false. The atom encodings assign canonical truth bits and slack or Euclidean-division data, and the acyclic Boolean circuit has uniquely determined outputs; the final residual requires the output bit to equal one. The established compiler is reused here rather than replaced by a new implementation. [write] [The signed encoding and the elimination of time are those of Part IV’s Section 45, last subsection, and (80.1) is (46.2).]

Corollary 80.1 (Canonical quartics at mass three). *For every fixed CA and fixed input/output label case in Theorem 74.1, timed and untimed exact reachability admit effectively constructible integer polynomials of degree at most four with zero or one natural auxiliary witness tuple. Their witness arity is the formula-dependent constant K in (80.1), after quantifier elimination and a chosen Boolean circuit. No supplied time horizon is required. The same conclusion applies to each fixed finite-pattern observation predicate from Corollary 74.2.*

Proof. Apply Theorem 74.1, the stated natural input encoding, quantifier elimination, and the canonical compiler in that order. The first three steps preserve the predicate; the last step has the asserted degree, counts, and natural-fiber property. $\square$

[write] [Corollary 80.1 is Part IV’s Corollary 46.1 one mass higher, with up to twelve rather than eight external natural inputs.]

For exact reachability there are at most twelve external natural coordinate inputs, and thirteen when time is retained. Thus total arity is at most $12 + K$ untimed or $13 + K$ timed for fixed labels. The formula-dependent quantity K is not a universal numerical arity bound. Quantifier-elimination cost, formula size, circuit size, integer coefficients and their bit lengths remain paid parts of any concrete instantiation; no efficient preprocessing or optimal arity is claimed. The executable package validates the orbit accelerator, not a general-purpose rule-to-formula or quantifier-elimination implementation.

Uniqueness comes from post-elimination arithmetic, not merely from deterministic CA evolution. No corresponding exactness of the nonnegative-real or rational witness fiber is claimed. This corollary concerns the decidable single-unit-label class and gives no finite-fold conclusion for arbitrary recursively enumerable sets or for a universal typed three-mass construction.

[write] **Added 2 October 2026 (batch 80).** Two sharper compilers apply. The programme’s five-witness congruence atom [33], already recorded for Part IV (Section 47), applies verbatim after elimination and gives $K = 2I + 5C + B$ natural witnesses at the same E , keeping the single-fold property and degree four; source 16’s provenance record says that it retains the verified six-witness bound, and the programme’s review of this source [46] identifies the substitution as a formula-dependent saving, with no total-gate claim. Theorem `cdc:of:thm:singlefolds1` of Part XI of

canonical-diophantine-certificates gives the same semilinear relation an orthant-nonnegative single-fold representation of degree two, as the note after Part IV’s Corollary 46.1 says for mass two. No novelty is claimed for Corollary 80.1.

81 Primary sources and what remains unresolved

There is a direct primary-source precursor. Gil-Tak Kong’s June 2021 thesis [41], Section 2.4.1, Theorem 2, printed page 14 (PDF page 15), proves non-strong-Turing-universality with three particles. Its definitions are explicitly one-dimensional and binary: Definition 1, printed page 8 (PDF page 9), uses alphabet $\{0, 1\}$. The proof distinguishes eventual periodicity up to translation of the whole configuration from eventual decomposition into independent periodic components. Thus binary three-particle nonuniversality and this orbit-decomposition approach have an explicit 2021 precursor.

The present article states and proves a weighted extension with at most one unit label, effective uniform timed Presburger formulas in all sparse positions, exact and finite-pattern observation consequences, and the canonical quartic composition. These are the precise claims to compare with earlier work; we do not claim that the brief sources establish or exclude these stronger statements, or that their absence from a short source would establish novelty.

Imai’s KAKEN final report dated 25 June 2021 [42], PDF page 4, Section 4, item (2), also reports that motion representation was used to establish decidability when the particle number is at most three. The Japanese sentence leaves the decision problem implicit; it does not identify exact reachability, pattern occurrence, an input encoding, or a complexity bound. This establishes a numerical-NCCA decidability precursor, while leaving its relation to the fully specified weighted timed claim here undetermined. PDF page 3 distinguishes the need to encode NCCA internal state geometrically from the capabilities of pebble automata and discusses the earlier five-particle construction.

The publisher abstract of Alhazov and Imai [24] describes finite-support, nonnegative-integer-state NCCAs. It reports a radius-one universal construction via two-register machines and a universal construction with five particles when a sufficiently large finite radius is allowed. The accessible abstract does not claim minimality. The full paper was not accessible in this check, so its detailed discussion of smaller particle counts could not be verified.

A Kyoto University RIMS report [43], printed page 93, records Kong and Imai’s presentation on particle complexity on 2 February 2021; the KAKEN project record [44] places that title under academic year 2020. The thesis above supplies a directly inspectable mathematical source.

Kong’s thesis conclusion, printed page 15 (PDF page 16), describes the four-particle case as open in 2021. That is a historical statement, not evidence that the question remains open today. The current article proves insufficiency of three for the stated observation task in its weighted single-unit-label class, while the 2016 abstract supplies a five-particle numerical upper bound. It does not settle the current four-versus-five boundary or the novelty of its explicit weighted, uniform Presburger, or Diophantine strengthenings.

[write] **Added 2 October 2026 (batch 80).** The four-versus-five boundary is settled within this weighted single-unit class by source 17 (Part VI): exact reachability and pattern occurrence are decidable at mass four (Theorem 83.1), so the numerical lower bound is five (Corollary 83.2); a five-particle upper bound rests on the Alhazov–Imai abstract only. The programme’s review [46] read the cited pages of Kong’s thesis (the binary Definition 1, Theorem 2, and the four-particle case open

in 2021) and retrieved Imai’s KAKEN report, whose Japanese text it could not extract and does not certify; it obtained no full text of the Alhazov–Imai paper. [write] **Added 3 October 2026 (batch 82)**. For binary automata, and for globally reversible binary automata, the upper bound five is now proved outside this report, by Reports 14 and 15 (Part I of **five-particle-binary-automata**, Corollaries 27.1 and 16.1 there); binary automata have a single weight-one symbol, so the four-versus-five boundary of this class no longer rests on the Alhazov–Imai abstract.

Two further directions remain outside this proof: classifying the effect of two or more unit labels, including the least unit-label count permitting a universal three-mass construction; and deriving quantitative bounds for encounter acceleration, Presburger elimination, and the resulting quartic ledger. [write] [Both are open; they are also source 17’s Questions 5 and 3.]

82 Executable checks and limits of validation

The accompanying Python script implements the finite mass-two phase profile, exact first-core acceleration for positive, zero, and negative drift, and whole-orbit acceleration using the 4S core. It compares accelerated configurations against direct simulation, including all intermediate sampled times. The tests use:

- Sixteen binary conservative radius-two rules obtained from disjoint parallel swaps of selected 10 pairs, in both directions. The selection is false for an isolated unit, so its drift is zero.
- All five conservative elementary binary rules, 170, 184, 204, 226, and 240, each shifted into its unique-unit stationary frame. Their conservation is certified exactly by a potential on the four-vertex de Bruijn graph; all 256 elementary rules are tested against that finite identity. [write] (4 October 2026, batch 91.) The same potential criterion one radius up, on the sixteen-vertex de Bruijn graph of four-bit words, is proved as Lemma **fpa:tr:lem:potential** in Part VII of **five-particle-binary-automata** (Report 70); with 423 explicit periodic collisions it shows that the only injective number-conserving binary rules of radius two are the five shifts (**fpa:tr:thm:radius-two**; the count of 428 conservative tables is Fuk’s and Sullivan’s). No priority is claimed there.
- An ordinary four-state NCCA with numerical states 0, 1, 2, 3, using locally guarded translation, a $2 + 1$ merge into 3, and a reverse split. Its label values equal its weights.
- A locally guarded weighted rewrite CA with a unique unit label, several weight-two labels, and two weight-three labels. Its cases include positive and negative walkers, zero-drift phase motion, merging, splitting inside a bounded cluster, recurrent encounters, and terminal separation.

Each sample rule has a direct conservation argument: binary moves swap disjoint occupied/empty pairs; weighted rewrites have disjoint guarded supports and preserve the mass on each rewritten support. The code also tests dense, multi-center configurations as a consistency check.

The recorded run passed 7,127 orbit cases and 1,140,320 direct-versus-accelerated time comparisons across 23 rules. It checked 102 mass-two profiles, 5,103 signed linear interval cases, and 2,760 dense conservation cases. A binary pair with period two and drift one was tested against an initial unit at $10^{30} + 17$; the accelerator handled the enormous separation using 13 core visits, without iterating through the gap.

These counts describe actual CA orbit cases and direct-versus-accelerated configuration comparisons, not counts of symbolic proof templates. These finite tests exercise the implementation and the proposed normal forms. They are neither an exhaustive enumeration of all conservative

CA rules nor a machine-checked proof of the theorem. A separate executable audit passed 1,186 boundary orbits with 118,600 direct time comparisons, 4,109 independent first-core checks, and 216,160 finite-window conservation cases. It also tested reflected negative-drift gaps, zero drift, sorting and translations, with closed-form checks through time $10^{80} + 123$; its full ledger is included in the package. The mathematical argument, including the no-split lemma, uniform first-entry composition, and all-intermediate-time cycle identity, was reviewed independently as well.

[write] **The release in this report.** These are the delivered names. The script and the independent audit are shipped as `code/16-single-unit-test_single_unit_acceleration.py` and `code/16-single-unit-audit_accelerator.py`, with `code/16-single-unit-run-tests.sh` and `code/16-single-unit-build.sh`; their receipts as `data/16-single-unit-test-results.json` and `data/16-single-unit-audit-results.json`; and `REVISION.md`, `SOURCE-PROVENANCE.md` and `independent/AUDIT.md` as `16-single-unit-*` at the report root. The manuscript, PDF, README and checksum ledger are not shipped, and nothing of manuscript 06 is (its programs, receipts and audit are the same files). `run-tests.sh` copies the two programs to a temporary directory and compares the receipts; it needs the delivered layout (README). The programme’s review reran both programs and reproduced both receipts [46].

Source 17 (batch-80 manuscript 03): Four mass units and exact reachability

Decidability proof and a binary timing certificate, 3 October 2026 (its PDF metadata name the author as “Mathematical research report”). [write] Its “companion” report is source 16, printed above. Its abstract follows.

For every fixed one-dimensional finite-alphabet conservative cellular automaton with a unique zero-weight vacuum, positive integer non-vacuum weights, and at most one weight-one symbol, exact finite-configuration reachability and finite-pattern occurrence are decidable at total mass at most four. In the stationary-unit frame, recurrent separated interactions reduce to a finite-control additive gap counter. A translating three-mass gadget either escapes or brings all four masses into a bounded encounter; it cannot retain a second independent gap. The remaining expanding shuttles have spatial phases affine in their excursion number and quadratic accumulated time. This also handles anchored observations in the original moving frame.

Timed semilinearity does fail at mass four: an explicit binary radius-six NCCA has anchored return times $t_k = k^2 + (2d - 11)k$, $d \geq 7$. Its full local table has an exact de Bruijn conservation certificate. A single squared quadratic gives a horizon-free quartic timing certificate with exactly one natural witness at a hit. Primary 2021 lower-particle precursors are credited; no priority claim or present-day literature-status conclusion is inferred from an unsuccessful search.

83 Model and results

Let A be finite, with a distinguished vacuum 0 , and let $w : A \rightarrow \mathbb{N}$ satisfy $w(0) = 0$ and $w(a) > 0$ for $a \neq 0$. Let $F : A^{\mathbb{Z}} \rightarrow A^{\mathbb{Z}}$ be a deterministic, time-independent, translation-equivariant cellular automaton of finite radius R . Assume that $0^{\mathbb{Z}}$ is fixed and that, for every finite-support configuration c ,

$$M(F(c)) = M(c), \quad M(c) = \sum_{i \in \mathbb{Z}} w(c_i).$$

There is at most one symbol of weight one. Labels of higher weights may be repeated. Write $\tau_a c$ for translation to the right by a sites.

Theorem 83.1 (Four-mass decision theorem). *For every fixed F satisfying these assumptions, the following problems are decidable for finite initial configurations of mass at most four:*

1. *Exact reachability of a specified finite configuration, including reachability up to translation.*
2. *Occurrence of a specified finite pattern at a specified position, or anywhere, at some nonnegative time.*

The procedures are effective from the finite rule, radius and weights, under the stated hypotheses. No bound on their computational efficiency is asserted.

A pattern may contain zeros, and these are enforced. The theorem concerns finite inputs and finite observations. It does not cover infinite initial backgrounds, signed weights, an infinite alphabet, an active zero-weight background, multiple unit labels, or time-dependent or spatially nonuniform rules.

Corollary 83.2 (Ordinary numerical particles). *A fixed one-dimensional finite-alphabet NCCA whose states are nonnegative integers and whose weights equal those integers cannot encode arbitrary Turing-machine halting by a computable finite-configuration encoder of total numerical mass at most four and a designated finite halt-pattern observation. The same conclusion holds for exact-target halting observations.*

Proof. There is at most one numerical state equal to one. Such an encoding would reduce an undecidable problem to a decision problem in Theorem 83.1. $\square$

The 2016 publisher abstract of Alhazov and Imai reports a universal numerical NCCA with five particles [24]; the full paper was not available for this check. The present lower bound matches that reported numerical upper bound for the usual effective finite-input halting-observation interpretation. [write] [It would match, if the 2016 construction uses an effective finite input encoding and a finite halt-pattern observation; only its abstract was checked, and source 17’s delivery README says “compatible with”.] It does not establish a five-particle upper bound for the *binary* subclass. [write] [Added 3 October 2026, batch 82: still true of this source. Part I of **five-particle-binary-automata** (Reports 14 and 15) proves that bound, also for globally reversible binary automata, so with this corollary five is exact for binary and for globally reversible binary number-conserving automata.]

84 Stationary units and small packets

Assume first that the unit symbol u exists. Its isolated successor must be exactly one u , at a fixed displacement $\delta \in [-R, R]$. Put

$$G = \tau_{-\delta}F, \quad S = \max(1, R + |\delta|).$$

Then G has radius at most S , fixes each isolated u , and $F^t = \tau_{\delta t}G^t$. Components below join occupied sites at distance at most $2S$. If two components are separated by more than $2S$, their next steps evolve independently and have disjoint output support.

[write] [This section restates the ingredients of source 16 that source 17 needs (Section 75 and the next two); the mass-two case for any number of weight-one symbols is Part IV’s Theorem 44.1. The proofs of the two lemmas are pointers.]

Lemma 84.1 (Bounded mass-two packets). *A mass-two configuration of diameter at most $2S$ stays within that diameter under G . Its orbit is effectively eventually periodic up to translation.*

[write] *Pointer*. This is source 16’s Lemmas 75.1 and 75.2, proved there by the same argument (the cases $d > S$ and $d \leq S$, with all new support in $[d - S, S]$, then finitely many normalized shapes); not reprinted. $\square$

An isolated packet profile can therefore be written, after a finite transient, as

$$G^{\mu+r+kp}(q) = \tau_{kD}q_r, \quad 0 \leq r < p, \quad k \geq 0. \quad (84.1)$$

Its labels, phase shapes, period, drift and transient are explicit finite data. Two units farther than $2S$ apart are fixed.

Lemma 84.2 (Effective three-mass profiles). *Every fixed mass-three input has an effectively computable finite prefix and one of these tails:*

1. *the whole configuration is periodic up to a single translation;*
2. *a stationary unit and an independent mass-two packet with finite-phase motion.*

Every intermediate configuration is included in the resulting description.

[write] *Pointer*. This is source 16’s Proposition 77.1, with Lemmas 76.1 and 76.2 for the first entry into the core of diameter $4S$ and the finite graph of core returns; not reprinted. $\square$

For total mass at most three, these profiles already give a fixed-input timed Presburger description, and the frame translation δt preserves it. Thus all conclusions of Theorem 83.1 at mass at most three are settled here; the remaining component analysis assumes mass exactly four.

This is the small-packet ingredient of the companion mass-three report [source 16, Part VI]. The present argument does not require its stronger uniform timed-Presburger conclusion.

85 A finite library of local encounters

Let $\mathcal{E}$ be the finite set of normalized connected mass-three configurations. Every seed has diameter at most $4S$. When a bounded mass-two packet first comes within $2S$ of a unit, their union is such a seed. At the first-contact time their supports are still disjoint: they evolved from components separated by more than $2S$ at the preceding step.

Tag each entry seed by the target unit’s coordinate within it. This finite bookkeeping tag matters when the incoming packet itself consists of two unit symbols. It adds no physical label; it records which incoming coordinate was the stationary target.

For each $e \in \mathcal{E}$, compute Lemma 84.2’s orbit profile and choose a time T_e giving one of two normalized outcomes:

Compact outcome. The complete mass-three configuration has a translated-periodic tail.

Emission outcome. A stationary unit at coordinate a_e remains, and a mass-two packet has nonzero periodic drift and recedes from that unit forever. Its launch phase and coordinates are fixed relative to the seed.

If an independent packet has zero drift, its support is bounded and periodic, so reclassify the whole tail as compact. With nonzero drift, prolong the finite prefix by whole periods until every future packet phase is farther than $2S$ from the stationary unit on the drift side. Earlier detachments and

returns are part of the finite prefix. Infinitely recurrent detachments within a translated periodic three-mass orbit belong to its compact phase library.

Choose an integer $B \geq 4S$ bounding the absolute value of every occupied coordinate, relative to its normalized seed, in every finite prefix through T_e and every initial-period phase representative of either outcome. Include the residual marker coordinate a_e . Then compact phase diameters are at most $2B$, and emitted head-to-marker offsets have magnitude at most $2B$. Set

$$N = 4B + 12S, \quad W = N + 2B. \quad (85.1)$$

These constants are noncircular: B is computed from the original span- $4S$ seed family, not from all mass-three configurations of diameter W .

Lemma 85.1 (Safe local resolution). *Suppose a configuration consists of $\tau_a e$, $e \in \mathcal{E}$, and a fourth unit at b , with $|b - a| > B + 2S$. The complete isolated seed prefix through T_e evolves independently of that unit. Subsequently:*

1. *a compact outcome either stays independent forever, or first contacts the fourth unit in a complete configuration of diameter less than W ;*
2. *an emission changes the nearby residual marker by a fixed seed-dependent offset and leaves the fourth unit fixed.*

All alternatives and the first-contact times are effectively computable.

Proof. During the prefix the seed support lies in $[a - B, a + B]$, at distance greater than $2S$ from b . Exact independence applies step by step, including its terminal time. A compact phase has diameter at most $2B$. Its zero drift or drift away from b gives independence forever. If its drift points toward b , compute first proximity at most $2S$ from its affine phase profiles. Such a contact exists: finite propagation prevents a packet from crossing the stationary unit without proximity at most $2S$. At contact the total diameter is at most $2B + 2S < W$. The emission statement is the computed isolated profile. $\square$

An outside- W seed-plus-unit configuration automatically satisfies the safety condition: the seed diameter is at most $4S$, so $|b - a| > W - 4S > B + 2S$. Thus a translating mass-three gadget is not collapsed into a constant-time local reaction. Its arbitrarily long trip is either terminal or ends in a bounded full encounter.

86 The global scattering section

Call every complete mass-four configuration of diameter at most W an *ordinary state*; there are finitely many up to translation. A *live emission section* consists of two stationary units at $L < R$ and one bounded mass-two packet at a prescribed launch offset from one endpoint, with nonzero drift toward the other endpoint. Write $D = R - L > N$. The finite mode records the source side, emitted packet type and phase, and tagged seed information. The geometric ordinary and live sets can overlap. We use this deterministic representation convention: at a designated emission checkpoint, select the live tag whenever $D > N$, even if the span is at most W ; at other dispatch states apply the ordinary span test first. These tags belong to the analysis and do not add memory to the CA. A live step retains its tag through the next designated emission or an explicit ordinary reset.

Lemma 86.1 (Additive gap transitions). *Every continuing live-to-live transition, for sufficiently large gap as already ensured by (85.1), has the form*

$$q' = f(q, D \bmod M), \quad D' = D + c, \quad L' = L + e, \quad h = \alpha D + \beta, \quad (86.1)$$

where the constants are selected from a finite effective list, $c, e \in \mathbb{Z}$, $\alpha > 0$ is rational, and h is integer on the permitted residue class. M is one fixed positive integer. All intermediate configurations have effective finite-phase descriptions.

Proof. The emitted packet is permanently independent of the source unit. Follow it until first proximity at most $2S$ to the target unit at b . The resulting connected mass-three seed has left edge a' with $|a' - b| \leq 4S$. The old source marker p therefore satisfies

$$|p - a'| \geq D - 4S > N - 4S > B + 2S.$$

Lemma 85.1 applies to its complete local prefix, however long that fixed prefix is. For an emission from the new seed, the new stationary marker is $p' = a' + a_{e'}$, so

$$|p' - b| \leq 4S + B.$$

Only the contacted marker moves. Since $D > N > B + 4S$, marker order cannot reverse. The new gap and left marker thus change by bounded constants. Unit identities can switch within the encounter; the new coordinates, not identities, are what the section records.

If the outgoing packet points outward, the tail is terminal independent motion. If it points inward with new gap at most N , its complete launch configuration has diameter at most $N + 2B = W$ and is ordinary. A compact outcome is accelerated separately by Lemma 85.1; it either escapes or resets to an ordinary state. Carrying a marker inside a translating three-mass packet cannot preserve an extra unbounded gap through that reset.

For the arithmetic, reflect coordinates if necessary so the packet drift is $d > 0$, with period p . In phase r , let its left edge be $p_0 + h_r + kd$ and its width be $d_r \leq 2S$, where p_0 is the source marker. Contact with target $p_0 + D$ is exactly

$$D - 2S - d_r \leq h_r + kd \leq D + 2S. \quad (86.2)$$

The hull criterion is exact: because packet width is at most $2S$, the union of the radius- $2S$ neighborhoods of its occupied sites is the single expanded hull interval. There is no hole that falsely counts as a contact.

On each residue $\rho \bmod d$, every candidate period count is $(D - \rho)/d$ plus an integer constant. The choice of earliest valid phase therefore depends only on ρ ; all candidate times have leading coefficient p/d . The bound $D > N$ ensures nonnegative candidate counts in the periodic regime. Finite propagation ensures that one phase is valid. The entry seed and its tag are fixed relative to the target. Add the fixed local-prefix time to obtain (86.1). Take M to be a common multiple of the nonzero emission drifts. Every precontact phase and every local-prefix configuration is retained, not merely the transition endpoint. $\square$

Lemma 86.2 (Complete effective dispatch). *From an arbitrary mass-four configuration, one can effectively obtain a terminal independent profile, an ordinary state, or a live emission section, with an exact description of the intervening orbit. From an ordinary state, the next such section can be chosen at positive elapsed time.*

Proof. The possible $2S$ -component mass partitions are

$$4, \quad 3 + 1, \quad 2 + 2, \quad 2 + 1 + 1, \quad 1 + 1 + 1 + 1.$$

A connected component of mass m has diameter at most $2S(m - 1)$. The first partition is ordinary since $6S < W$; four unit components are fixed. For $3 + 1$, an outside- W state has a connected seed and a safely distant unit, so Lemma 85.1 applies.

For $2 + 2$, both packets have finite-phase independent motion until first proximity at most $2S$. Use a common phase period; each signed relative gap is affine in the period count. Finite transient checks and integer interval calculations find first contact or prove none occurs. First contact has total span at most $6S < W$.

For $2 + 1 + 1$, compute the packet's first proximity to either stationary unit by the same finite-phase calculation. If absent, the entire tail is independent. At contact, a connected three-mass seed is formed. A simultaneous contact connects all four masses and has span at most $6S$. Otherwise either the complete configuration is ordinary, or its remaining unit is safely distant and Lemma 85.1 applies. Before every first contact, independence is exact; it also holds at the first-contact time itself.

From an ordinary state take one exact step. If still ordinary, retain that step. Otherwise its span is at most $W + 2S$, and the exhaustive dispatch above applies. Its result is terminal, ordinary, or live. This supplies a positive-time outgoing section edge for each of the finitely many normalized ordinary states. Live edges include a nonempty flight. Zero-time section loops do not arise. $\square$

87 Termination and the expanding shuttle form

Enlarge the finite live mode by $D \bmod M$. A repeated finite control then has gap difference divisible by M , and consequently the same residue itinerary on each repeated cycle. Translation L is recorded as output information; it is not used as another control counter.

Proposition 87.1 (Effective orbit alternatives). *For every fixed mass-four input, its G -orbit is effectively eventually one of:*

1. *independent finite-phase objects;*
2. *a whole configuration periodic up to translation;*
3. *an expanding shuttle cycle, whose corresponding-section gap and left marker are affine in a cycle number n , whose complete spatial phases are Presburger in n and one within-flight period count, and whose accumulated section time is quadratic in n with positive leading coefficient.*

Proof. Follow the effective section machine. If an ordinary normalized state repeats, the complete configurations at those two times differ by a translation. Determinism gives alternative 2 at every subsequent intermediate time.

During an uninterrupted live excursion, store the finite controls. A control repeats after finitely many transitions. Let Δ be the change in gap between those repeats. If $\Delta = 0$, the complete launch configuration repeats up to translation, again giving alternative 2. If $\Delta > 0$, all intermediate large-gap guards remain valid forever and the same residue itinerary repeats. This gives alternative 3.

If $\Delta < 0$, compute the maximal number of cycle repetitions for which *every intermediate gap* remains greater than N . Fast-forward those repetitions, summing their constant anchor updates

and affine durations, then execute the finite remaining cycle prefix until the ordinary region is reached. This is exact finite arithmetic. There are only finitely many ordinary normalized states, so one cannot visit them indefinitely without repetition. The section analysis therefore terminates with one of the stated alternatives.

For the expanding case, write the corresponding-section coordinates as

$$D_n = D_0 + n\Delta, \quad L_n = L_0 + nE, \quad \Delta > 0. \quad (87.1)$$

There are finitely many edges per cycle. A bounded scattering-prefix phase has positions affine in n . A flight phase has marker positions affine in n and packet positions affine in (n, j) , where $j \geq 0$ is its packet-period count. Its allowable range is given by linear inequalities. The finite union of these phase formulas describes every visited configuration and is Presburger.

Each live edge duration is affine in its incoming gap, with positive leading coefficient. Consequently the complete cycle duration is affine in n and strictly increases, and its start time has the form

$$T_n = T_0 + A \frac{n(n-1)}{2} + Cn, \quad A > 0. \quad (87.2)$$

Coefficients are computable rationals yielding integer times. There is a computable C_1 for which every configuration throughout cycle n has support inside

$$[L_0 + nE - C_1, \quad L_0 + nE + D_0 + n\Delta + C_1]. \quad (87.3)$$

Local prefixes have uniformly bounded offsets; each free flight stays between its two markers up to bounded phase deviations. Thus (87.3) includes every phase, including all short local transients, rather than just sections. $\square$

Corollary 87.2 (Untimed spatial description in the stationary frame). *For each fixed input of mass at most four, the set of configurations visited by G has an effectively constructible Presburger description in their occupied coordinates, separately for each possible output-label tuple.*

Proof. Use Proposition 87.1. Independent finite-phase and translated periodic tails have affine descriptions. Expanding spatial phases were described in its proof. All earlier parts are finite, or finitely many explicitly given finite-phase segments, so can be included effectively. An enormous finite prefix may be enumerated; no efficiency claim is involved. $\square$

88 Anchored observations in the original frame

Translation-invariant questions, including exact reachability up to translation and occurrence anywhere, have the same answers for F and G . Existentially quantify the spatial shift in the preceding corollary.

For anchored questions, if the eventual orbit is independent finite-phase or translated periodic, its fixed-input timed relation is Presburger. Adding δt to all coordinates preserves that property. If the orbit is an expanding shuttle and $\delta = 0$, its untimed spatial description suffices.

Suppose now $\delta \neq 0$. For $\delta > 0$, throughout cycle n every occupied coordinate in the F -orbit is at least

$$\delta T_n + L_0 + nE - C_1,$$

which tends to $+\infty$ by (87.2). For $\delta < 0$, every occupied coordinate is at most

$$\delta T_n + L_0 + nE + D_0 + n\Delta + C_1,$$

which tends to $-\infty$. These estimates are uniform over the whole cycle because $t \geq T_n$ and (87.3) holds throughout.

Given a finite target window, compute an integer cycle index n_* beyond which the appropriate quadratic bound lies permanently outside it. Simulate or accelerate the finite prefix up to section T_{n_*} . A pattern with a nonzero symbol can occur only in this prefix; an all-zero pattern occurs afterward. A nonempty exact target has the same cutoff. The vacuum case is immediate from conservation. This proves anchored decidability without a general quadratic Diophantine solver and without claiming timed Presburger definability.

For a pattern, every zero site is enforced by absence of an occupied coordinate there. Since mass is at most four, only finitely many nonzero label tuples are possible. Exact configuration equality and these finite pattern tests are thus effective in all the described cases.

If there is no unit symbol, mass at most four occupies at most two sites. An isolated site of weight two or three is a finite-state walker; a weight-four local configuration may split into two weight-two walkers. A bounded encounter core has finitely many normalized states. Outside it there are two independent finite-phase walkers, whose first encounter is computed by common-period linear inequalities. The same finite-core return argument gives an eventually translated-periodic or independent-two-walker orbit. This handles the missing case and completes Theorem 83.1.

89 An explicit binary expanding shuttle

The loss of timed semilinearity already occurs for an ordinary binary NCCA. Join consecutive occupied sites whose distance is at most two. On every such component, apply these replacements up to translation:

$$\begin{array}{llll}
 \{0, 1\} & \mapsto & \{1, 2\} & \text{right-moving pair,} \\
 \{0, 2\} & \mapsto & \{-1, 1\} & \text{left-moving pair,} \\
 \{0, 1, 3\} & \mapsto & \{-1, 1, 4\} & \text{right bounce,} \\
 \{0, 2, 4\} & \mapsto & \{0, 3, 4\} & \text{left bounce.}
 \end{array} \tag{89.1}$$

Leave every other component unchanged.

Proposition 89.1 (A genuine conservative local rule). *Rule (89.1) defines a binary conservative CA of radius at most six.*

Proof. Each replacement preserves the number of occupied sites, and its output hull extends the input hull by at most one site. Distinct input components have separation at least three, so these expanded output hulls still have disjoint lattice sites. This proves global conservation for every finite configuration.

A recognized component is an exact pattern of span at most four, with no occupied site within two beyond either endpoint. For any output site, every recognized pattern that can change it, together with both endpoint guards, is contained in its radius-six neighborhood. All other sites retain their old value. Thus the component definition is a local rule, including on arbitrarily large unrecognized components. The vacuum and isolated units are fixed. $\square$

The rule is not injective: the distinct configurations $\{0, 1, 4, 6\}$ and $\{1, 2, 3, 5\}$ both map to $\{1, 2, 3, 5\}$. No reversibility is asserted. [write] [Added 4 October 2026: source 24 (Part IX) gives a different binary rule, globally reversible on the full shift with forward and inverse radius at most 90, whose four-particle orbit from $\{0, 5, 6, d\}$ has the hits $k^2 + (2d - 23)k$ (Theorem 159.1).]

The package exports all $2^{13} = 8192$ local outputs. Independently, a potential P on the $2^{12} = 4096$ de Bruijn vertices satisfies every identity

$$f(w_0, \dots, w_{12}) - w_6 = P(w_1, \dots, w_{12}) - P(w_0, \dots, w_{11}). \quad (89.2)$$

The complete table and potential are included as a compact JSON certificate. Summation over the lattice telescopes, yielding another exact conservation proof, not merely a dense-window test.

Proposition 89.2 (Exact quadratic hit times). *From occupied sites $\{0, 3, 4, d\}$, $d \geq 7$, the anchored pattern 10011 at sites $0, \dots, 4$ occurs exactly at*

$$t_k = k^2 + (2d - 11)k, \quad k \in \mathbb{N}. \quad (89.3)$$

Its occurrence-time set is not eventually periodic and is not Presburger.

Proof. The adjacent pair travels from $\{3, 4\}$ to $\{d - 3, d - 2\}$ in $d - 6$ steps. The component $\{d - 3, d - 2, d\}$ then becomes $\{d - 4, d - 2, d + 1\}$ in one step. The left-moving pair returns to $\{2, 4\}$ in $d - 6$ steps, after which the component $\{0, 2, 4\}$ becomes $\{0, 3, 4\}$. One cycle takes $2d - 10$ steps and increments d by one. Summing cycle durations proves (89.3). Between returns the adjacent right-moving pair is strictly farther right, or the left-moving pair has distance two and cannot occupy 3, 4 together. There are no extra pattern hits.

Successive hit gaps are $2k + 2d - 10$, which are unbounded. Every infinite ultimately periodic subset of $\mathbb{N}$ has bounded gaps after a finite prefix. The claimed failure follows; the equivalence of Presburger and semilinear sets is classical [7]. $\square$

[write] [The hypothesis $d \geq 7$ is safe but not sharp: a direct simulation of rule (89.1) at this write, like the placement check, gives exactly the times (89.3) also for $d = 6$, through $t \leq 2500$, while the formula fails for $d = 5$. Section 90 writes $d = 7 + x$ with natural x .]

90 A one-witness quartic timing certificate

Write $d = 7 + x$, where $x \in \mathbb{N}$ is an input. Define

$$\boxed{\mathcal{P}(x, t; k) = [k^2 + (2x + 3)k - t]^2}. \quad (90.1)$$

For natural inputs x, t , this degree-four polynomial has one natural auxiliary witness k , one squared quadratic residual, and three total variables. Its natural witness fiber is a singleton exactly when pattern 10011 occurs at time t , and empty otherwise. Indeed the residual must vanish, and

$$[(k + 1)^2 + (2x + 3)(k + 1)] - [k^2 + (2x + 3)k] = 2k + 2x + 4 > 0.$$

No time horizon or orbit tableau is supplied.

For natural x, t , a rational root k of the monic integer polynomial $k^2 + (2x + 3)k - t$ is integral, so the same fiber statement holds over nonnegative rationals. Over nonnegative reals the quadratic increases from zero to infinity; every natural t , including non-hit times, has a root.

Evaluating $k(k + 2x + 3) - t$ and squaring uses four additions or subtractions and two multiplications; at fixed gap the counts are two and two. These are straight-line evaluator counts, excluding input construction and bit complexity. The certificate is specific to this timing formula: mere decidability does not imply a general fixed-arity single-fold quartic compiler.

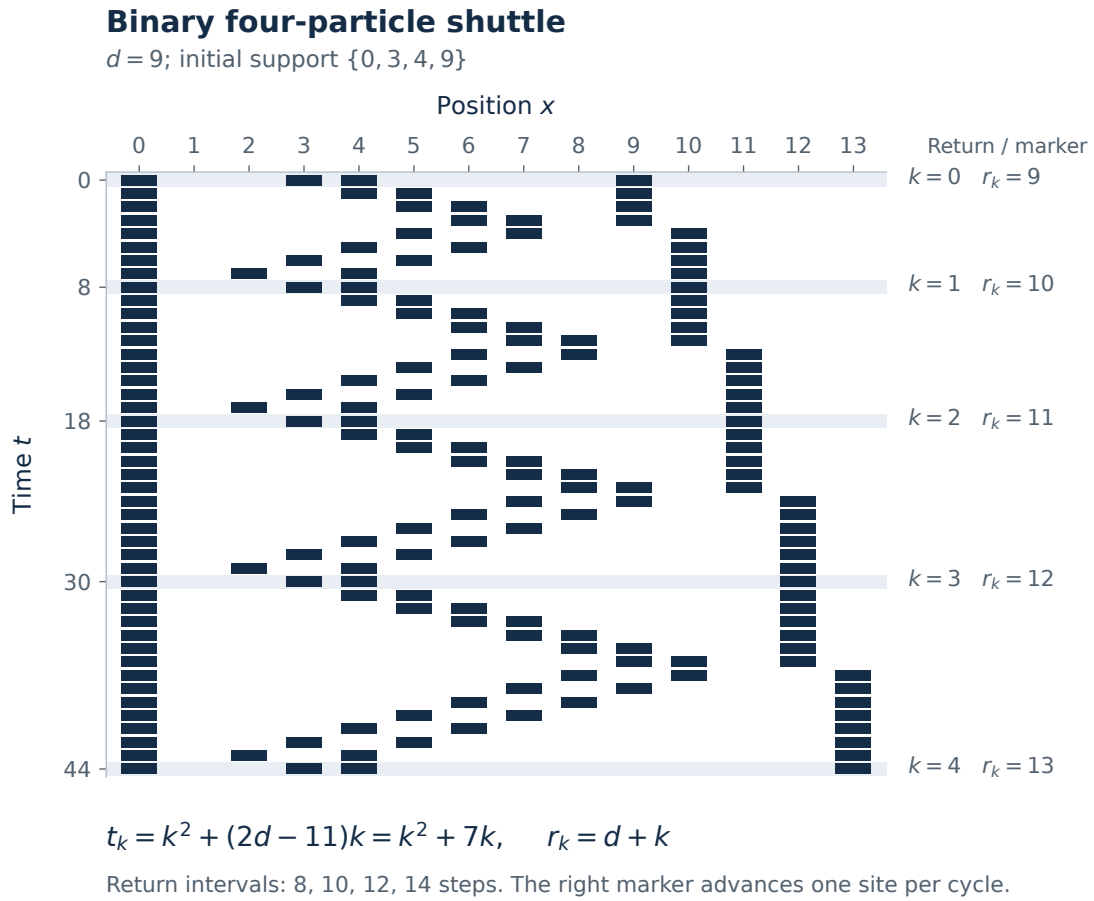


Figure 5: An actual trace of the exported binary local rule. The left marker stays fixed; the right marker moves outward by one after each cycle. The two remaining units alternate between an adjacent right-moving pair and a separated left-moving pair. The exact trace data and plotting source accompany the package.

[write] [By Theorem `cdc:of:thm:classification` of Part XI of *canonical-diophantine-certificates* ($D_2^+ = D_3^+ = \text{SL}$), the natural zero set of a polynomial of degree at most three that is nonnegative on the real nonnegative orthant has semilinear projections. The hit set $\{(x, t)\}$ of (90.1) is not semilinear (its sections are not eventually periodic, Proposition 89.2), so it has no such representation of degree at most three, single-fold or not, and the degree four of (90.1) is the least possible in that class. The programme’s review [46] confirms the uniqueness of the witness over the naturals and the nonnegative rationals, the real counterexample at $x = 0, t = 1$ (witness $(\sqrt{13} - 3)/2$), the loss of uniqueness over signed integers, and the schedule of two multiplications and four additions. ([write] Added 4 October 2026: source 24’s Corollary 4.1 is this certificate, for its reversible clock with $d = 13 + x$; Remark 163.1.)]

91 Primary attribution and scope of the literature check

Kong’s June 2021 thesis [41] studies one-dimensional *two-state* NCCAs explicitly. Its Definition 1 is on printed p. 8 (PDF p. 9). Section 2.4 proves translated eventual periodicity for two particles and three-particle non-strong-universality; Theorem 2 appears on printed p. 14 (PDF p. 15). The following conclusion, printed p. 15 (PDF p. 16), describes the four-particle case as open *at that time*. These are direct primary precursors. Their binary scope must not be silently enlarged to arbitrary weighted alphabets. [write] [Theorem 83.1 covers two-state NCCAs, whose unique unit state is 1, for its decision problems, and so bears on the four-particle case that Kong’s thesis called open in 2021, subject to the thesis’s notion of strong universality, which is not compared here.]

Imai’s final KAKEN report dated 25 June 2021 [42], PDF p. 4, reports a motion-representation decidability result at particle count at most three. Its brief statement does not identify the exact decision problem. Its methods discussion on PDF p. 3 distinguishes pebble-machine state from numerical NCCA state that must be encoded geometrically, and discusses the five-particle construction and smaller-particle analysis. The unspecified word “decidable” should not be read as a sourced statement of uniform timed Presburger definability or of the full weighted theorem here.

The Alhazov–Imai 2016 publisher abstract [24] describes finite-support nonnegative-integer-state NCCAs and reports universality with five particles at sufficiently large finite radius. This is an upper bound, not an abstract claim of minimality. Its full paper was unavailable in this check. The thesis’s nearby upper-bound reference appears mismatched, so the 2016 result is cited independently rather than inherited through that reference.

No later primary four-particle resolution was located in the targeted search. This is not evidence that the 2021 question remains open today and is not evidence of novelty of the present proof. The mathematical claim in this report rests on its argument, not on a literature-status assertion.

92 Checks and remaining questions

The general decision theorem is a mathematical proof, independently inspected; there is no implemented rule-to-section compiler and no machine-checked formalization. Finite examples cannot verify a theorem over all rules. The explicit binary rule, by contrast, has a complete exported local table and an exact finite conservation certificate.

The producer’s reproducible standard-library Python checks include all 8192 identities in (89.2), 262,144 exhaustive width-eighteen conservation cases, 8,192 component-rule/local-table comparisons, 10,000 larger random locality and translation comparisons, 302,400 direct trajectory steps, 2,424

predicted hits, and 20,000 finite unique-witness checks. A separately written replay, importing no producer evaluator, checked all 8192 local outputs, all 8192 potential identities, 49,640 direct steps over 40 initial gaps, and 840 exact pattern hits. A second independent guarded-pattern detector reconstructed every local output, checked 1,396 complete closed-form transitions and hit-exclusivity cases through section index 10^{80} , and made 25 large-integer quartic/discriminant checks. The section arithmetic audit also checked 951 first-contact instances and huge residue shifts, 272 weighted-shuttle transitions, and 27,360 original-frame support bounds. These ledgers supplement the exact rule and timing proofs; they do not establish the general theorem by testing.

The portable package contains this editable L^AT_EX source and PDF, the full rule table and potential, reproduction scripts, recorded test ledgers, independent audit materials, and exact figure data. The earlier mass-three release is not changed by this package.

[write] **The release in this report.** These are the delivered names. The programs are shipped as `code/17-four-mass-*` (those of `independent/` and `figures/` flattened), the receipts, the conservation certificate and the figure data as `data/17-four-mass-*`, the article’s figure as `figures/17-four-mass-binary-four-particle-shuttle.pdf`, and the proof-detail and audit notes (`BINARY-EXPANDING-SHUTTLE.md`, `EXPANDING-SHUTTLE.md`, `SOURCE-PROVENANCE.md`, `finite-seed-section-lemma.md`, and `independent/BINARY-INDEPENDENT-AUDIT.md`, `FINAL-REVIEW.md` and `INDEPENDENT-AUDIT.md`) as `17-four-mass-*` at the report root. The typed weighted shuttle whose 272 transitions are counted above is described in `17-four-mass-EXPANDING-SHUTTLE.md` and tested by `code/17-four-mass-test_expanding_shuttle.py`. The figure table `data/17-four-mass-binary-four-particle-shuttle.csv` keeps its delivered CRLF line ends. Not shipped: the manuscript, PDF, README, checksum ledger and the raster copy of the figure. `test_binary_expanding_shuttle.py` rewrites the sealed conservation certificate in place, with CRLF on Windows, after which `test_exact_boundaries.py` fails its hash test; run the suite on a copy and restore the certificate first (README). The programme’s review reran all nine invocations and reproduced every receipt [46].

Several questions remain useful without being asserted as new open problems:

1. Determine the current published status and priority of the four-particle boundary by obtaining the missing full papers and later primary results. ([write] Open; the programme’s review [46] did not obtain the full Alhazov–Imai paper either.) ([write] Added 3 October 2026: touched by Part VII, whose sources 19 and 20 add pebble-automaton and plane-walking literature (Sections 115 and 127); source 19 did not locate a primary theorem for its class and draws no priority conclusion.)
2. Investigate uniform *untimed* spatial Presburger definability in the stationary frame; the theorem here only needs a fixed-input construction. ([write] Added 3 October 2026: touched for a fixed input by source 20, whose visited sets are semilinear with quasipolynomial box counts in every dimension (Theorem 118.2); the uniform question stays open.) ([write] Added 4 October 2026: answered conditionally by source 25 (Part IX): conditional on this source’s scattering mechanism and source 18’s fixed-input charts, one finite chart family is uniform in all ordered inputs of mass at most four, and complete untimed reachability in the stationary frame is uniformly Presburger-definable (Theorem 169.1, Corollary 169.2). No rule-to-chart preprocessing is implemented.)
3. Give explicit complexity bounds or an implemented general section compiler, accounting for the potentially enormous finite seed library. ([write] Added 3 October 2026: open; source 19’s nearly parallel example (Section 107) shows that naive geometric bounds fail, and its Question 3

asks the same in every dimension.)

4. Determine what stronger arithmetic normal form describes original-frame uniform timed relations. The binary shuttle rules out Presburger arithmetic. ([write] Added 3 October 2026: touched. For a fixed input the complete timed orbit is a disjoint union of charts with at most two parameters and quadratic outputs (source 18, Lemma 94.1), and with drift the original-frame site set can be nonsemilinear with counts $3N - 3\sqrt{N} + O(1)$ (source 20, Theorem 125.1); the uniform question stays open.) ([write] Added 4 October 2026: answered conditionally for the uniform half by source 25: a finite union of charts with at most two evolution parameters, at most three gap quotients, affine domains and quadratic original-frame outputs (Theorem 169.1, (176.3)); its Proposition 176.1 shows that even the untimed original-frame relation need not be Presburger, for a reversible binary rule from one fixed input.)
5. Classify the effect of several weight-one labels. The stationary-unit and no-split arguments used here no longer apply. ([write] Part V shows that with many weight-one types three units suffice; the least number of weight-one symbols that permits a universal construction at mass three or four is open, as source 16 also says.) ([write] Added 3 October 2026: still open in every dimension; source 19’s Question 4.)
6. Study minimal radius or other resource bounds for the binary timing example separately from universal-computation particle bounds. ([write] Added 3 October 2026: touched by source 21, whose planar binary rule has exact radius six and exactly 78 essential cells (Theorem 137.1), with no minimality claimed; the one-dimensional question stays open.) ([write] Added 4 October 2026: not answered; source 24 gives a globally reversible binary example of radius at most 90, not claimed minimal, and asks for a smaller reversible radius, Question 1.)
7. Audit a fixed-input timed quartic compiler built from canonical disjoint phases of the orbit normal form. Quadratic time expressions and unique phase parameters suggest a route, but no such general compiler is proved here. ([write] Added 3 October 2026: answered for a fixed input by source 18 below (Theorem 93.1 and Proposition 96.1), conditional on Proposition 87.1, with natural witnesses; source 19’s Appendix I lifts it to $\mathbb{Z}^d$. The rule-to-chart preprocessing is not implemented.)

Source 18 (batch-82 manuscript 19): Canonical quartic certificates for timed four mass orbits

Research addendum 13. Complete fixed input orbits and exact witness accounting, 3 October 2026 (its PDF metadata name the author as “Mathematical research report”). [write] Source 18 is an AI-assisted research manuscript of ProveIt’s incoming reports, batch 82, manuscript 19 (arrival db37d18c8), from the archive `Timed_Four_Mass_Quartic_Certificates.zip`; it was added to Part VI on 3 October 2026. Its labels carry the prefix `smc:tq:`, and its sections are Sections 93–100 here. It is the research pipeline’s “addendum 13” and a declared addendum to source 17, which it calls “companion report 12” (source 17 is the pipeline’s Report 12); source number, manuscript number and pipeline number all differ, and bracketed notes translate them. It answers source 17’s last question (Question 7) for a fixed input, conditional on source 17’s normal form (Proposition 87.1), with natural witnesses. Its binary rule and its return quartic are source 17’s and are printed as pointers (Section 98); its chart compiler is a second route to Part I’s Theorem 10.2 (note in Section 96). It names no repository commit, and the Hilbert’s-tenth programme had not reviewed it when this was written (Section 1.19). Its abstract follows; the bracketed note in it is [write].

Fix a conservative one-dimensional cellular automaton with at most one weight-one symbol and an initial configuration of mass at most four. We construct a canonical natural-witness quartic for its complete timed orbit from the effective normal form in companion report 12 [source 17]. Disjoint half-open charts have affine domains and quadratic outputs. The baseline compiler uses $B + K + M$ witnesses for B charts, K parameters, and M inequalities. An explicit binary shuttle admits a four-witness, seven-square specialization, uniform in the natural gap parameter $x = d - 7$; at integer external inputs it is also exact for nonnegative real witnesses. The general theorem remains fixed-input and conditional on the imported normal form, with no generic real-witness, uniform-input, or Diophantine finite-fold claim.

93 The relation and its precise scope

Throughout, $\mathbb{N} = \{0, 1, 2, \dots\}$. Let A be a finite alphabet with a unique zero-weight symbol 0, with positive integer weights on all other symbols. Let $F : A^{\mathbb{Z}} \rightarrow A^{\mathbb{Z}}$ be a deterministic, translation-equivariant cellular automaton of finite radius. Assume $0^{\mathbb{Z}}$ is quiescent and total mass is conserved on every finite-support configuration. There is at most one symbol of weight one. Fix F and a finite-support configuration c with total mass at most four.

For a fixed ordered tuple $a = (a_1, \dots, a_m)$ of nonzero symbols, define

$$R_a(t, x_1, \dots, x_m) \iff \begin{cases} t \in \mathbb{N}, & x_1 < \dots < x_m, & x_i \in \mathbb{Z}, \\ F^t(c)_{x_i} = a_i & (1 \leq i \leq m), \\ F^t(c)_x = 0 & \text{for every } x \notin \{x_1, \dots, x_m\}. \end{cases}$$

The empty tuple is permitted. It describes exactly the vacuum at time t . If the requested symbols have incompatible total mass, the relation is empty.

Theorem 93.1 (Canonical timed quartic). *Assume the effective fixed-input orbit normal form in companion report 12 [source 17, Part VI]. One can effectively construct an integer-coefficient polynomial*

$$C_a(t, x_1, \dots, x_m; w_1, \dots, w_N)$$

of total degree at most four such that, for $t \in \mathbb{N}$ and $x_i \in \mathbb{Z}$,

$$\#\{w \in \mathbb{N}^N : C_a(t, x; w) = 0\} = \begin{cases} 1, & R_a(t, x), \\ 0, & \text{otherwise.} \end{cases}$$

The witness count N is fixed independently of t and x . A product-gated version is nonnegative for all real external coordinates and natural witnesses. A sum-of-squares version is nonnegative on the whole real space.

The coefficients, number of charts, witness count, and finite preprocessing may depend on both F and c . A computable but enormous finite history may be fully materialized. There is no complexity bound here, nor a bound uniform over arbitrary initial configurations. The explicit one-parameter binary family in Section 99 has a separate, stronger conclusion.

The theorem concerns natural witness fibers at integer external inputs. Global nonnegativity alone does not make rational or real witness zeros exact. Section 98.4 proves the stronger nonnegative-real conclusion only for the explicit binary specialization. Time stays external: forgetting time can destroy uniqueness, even for a stationary orbit.

94 The imported normal form and the chart interface

The only dynamical theorem imported from report 12 is its effective fixed-input eventual orbit classification, including all finite phase data. Its proof reduces separated recurrent interactions to finite control and a single additive gap counter. In a frame where the unique unit symbol is stationary, the eventual orbit is one of the following:

1. A whole configuration periodic up to translation
2. Finitely many mutually independent finite-phase objects, with a common clock after their finite transients
3. An expanding shuttle, with a finite control cycle, an affine growing marker gap, affine marker anchors, and finite-phase flights separated by fixed finite local scattering prefixes

The no-unit-symbol case and mass-at-most-three cases use the first two possibilities. Coherent translating three-mass tails are separately accelerated to a bounded full encounter or to an independent terminal tail. They are not treated as fixed-duration continuing shuttle scatterings. The classification itself is not re-proved in this addendum. [write] [The imported classification is Proposition 87.1 of source 17, with its section times (87.2) and the acceleration of Section 87; “report 12” is source 17 throughout this source.]

Lemma 94.1 (Sufficient chart interface). *It suffices to construct an effective finite chart family with the following properties. Chart i has $k_i \leq 2$ natural parameters z_i , a domain*

$$D_i = \{z_i \in \mathbb{N}^{k_i} : A_{ir}(z_i) \geq 0 \ \forall r, \quad H_{is}(z_i) = 0 \ \forall s\},$$

where A_{ir} and H_{is} are integer-affine, and a fixed output-symbol tuple. Its output $P_i(z_i)$ consists of time and sorted positions, is rational-polynomial of total degree at most two, and is integer-valued on D_i . Before filtering by output labels, the disjoint union of all chart domains maps bijectively to all timed configurations of the orbit.

The required bijection includes parameter uniqueness, not just uniqueness of a list of phase names. In particular, a finite union of overlapping parametrizations is insufficient. Once this interface is proved, retain only charts with symbol tuple a . Their outputs are exactly R_a , still without duplication. The algebraic compiler of Section 96 then proves Theorem 93.1. The supplied compiler checks exact chart syntax and degree; it does not decide this semantic bijectivity hypothesis.

95 A half open partition of every time

95.1 Finite histories and periodic tails

Let T_0 be the computable start of the eventual profile. Materialize the configuration at every integer $0 \leq t < T_0$, making each a parameter-free chart. This is allowed even if an accelerated negative-gap excursion makes T_0 impractically large. The boundary T_0 belongs only to the tail.

For a translated-periodic tail with period $p > 0$, each residue $r \in \{0, \dots, p-1\}$ has one natural parameter n and

$$t = T_0 + r + pn.$$

Its occupied coordinates are affine in n and its symbols are fixed. Euclidean division of $t - T_0$ gives exactly one r, n .

For an independent terminal tail, advance the common start beyond every object's transient and take a common multiple p of their periods. At each phase r , all positions again have affine displacements in the same n . The absorbed finite prefix remains disjoint. Giving different objects independent free clocks would be incorrect: their phases are synchronized by actual time. Stationary objects have zero displacement. The vacuum is an ordinary one-clock stationary tail, with no spatial coordinates.

95.2 Quadratic section times in an expanding shuttle

Number repetitions of one expanding control cycle by $n \geq 0$. The refined control includes the necessary gap residues. At corresponding sections,

$$D_n = D_0 + n\Delta, \quad L_n = L_0 + nE, \quad \Delta > 0.$$

Each of finitely many continuing transitions ℓ in the cycle has a nonempty remote flight and a fixed local scattering prefix. Its gap and anchor are affine in n , with the same Δ and E slopes. Its total duration is affine,

$$\tau_\ell(n) = A_\ell n + B_\ell.$$

Although the original flight calculation may use rational coefficients, $\tau_\ell(n)$ is integral for every $n \geq 0$ along this fixed itinerary. Thus $B_\ell = \tau_\ell(0)$ and $A_\ell = \tau_\ell(1) - \tau_\ell(0)$ are integers. The flight coefficient is positive. Summing the durations gives $An + B$ with $A > 0$, $B > 0$, so cycle n starts at

$$T(n) = T_0 + \frac{An(n-1)}{2} + Bn. \quad (95.1)$$

This integer-valued quadratic satisfies $T(n+1) - T(n) = An + B > 0$ and tends to infinity. Consequently the half-open intervals $[T(n), T(n+1))$ partition the entire tail. The start $S_\ell(n)$ of transition ℓ is $T(n)$ plus the sum of its preceding transition durations, hence is also quadratic and integer-valued.

95.3 Flight phases and local prefixes

Write the flight duration as $h_\ell(n) = \alpha_\ell n + \beta_\ell$ and the local-prefix length as the fixed integer $b_\ell \geq 0$. Transition ℓ owns

$$[S_\ell(n), S_\ell(n) + h_\ell(n) + b_\ell).$$

Its flight owns the first $h_\ell(n)$ times; the local prefix owns the remaining b_ℓ times. The next launch belongs to the next transition or cycle. A zero-length prefix supplies no chart.

If the flight packet has period p , then for every fixed $r \in \{0, \dots, p-1\}$ use natural parameters n, j with

$$t = S_\ell(n) + r + pj, \quad h_\ell(n) - r - pj - 1 \geq 0. \quad (95.2)$$

The domain inequality is affine. It says exactly that the integer elapsed time $r + pj$ is strictly less than $h_\ell(n)$. Euclidean division selects one r, j at every flight time. Marker positions are affine in n ; packet positions are affine in (n, j) , with symbols fixed by the phase. Empty phase domains may be retained without introducing any zero.

For each fixed $0 \leq s < b_\ell$, the local-prefix chart has just n and

$$t = S_\ell(n) + h_\ell(n) + s.$$

The finite control fixes the gap residue, contact phase, and tagged seed. Thus the contacted seed anchor is a fixed offset from an affine marker coordinate. Every occupied coordinate during its fixed local prefix is affine in n , including the distant old marker.

Uniqueness is now proved in order: first the cycle n , then its transition ℓ , then flight versus local prefix, and finally r, j or the fixed prefix index s . Every integer time is included, and no boundary is included twice. Time, phase, quotient, and prefix length have all been accounted for by the chart choice and its at most two parameters; no extra clock witness is needed.

95.4 Original coordinates and strict affine sorting

If the unique unit moves by δ sites per step under F , write $F^t = \tau_{\delta t} G^t$, where the unit is stationary for G . For any indexed occupied site v , let g_v be its stationary-frame coordinate in a chart. Its original-frame coordinate is

$$x_v = g_v(n, j) + \delta t(n, j). \quad (95.3)$$

This has degree at most two. Importantly, $x_v - x_w = g_v - g_w$ is still affine.

There is no need to assume a global ordering of moving objects. Refine each chart by every permutation π of its finitely many indexed occupied sites, adding the domain inequalities

$$g_{\pi(k+1)} - g_{\pi(k)} - 1 \geq 0 \quad (1 \leq k < m). \quad (95.4)$$

If rational coefficients arise in an affine description, clear positive denominators first. On integer sites these inequalities are exactly strict spatial order. Actual sites are distinct, so precisely one permutation passes at each original chart point. Symbols follow their indexed sites; repeated symbols do not multiply the accepted orders. Now filter by a . When a phase's order is already proved fixed, the redundant sorting refinement can be omitted. This completes Lemma 94.1.

96 An exact quartic compiler

The construction in this section applies to any family satisfying Lemma 94.1, independently of its dynamical origin. Suppose there are B charts, $K = \sum_i k_i$ private parameters, M domain inequalities, H domain equalities, and Q external output coordinates $y_1, \dots, y_Q$. For timed m -site configurations, $Q = m + 1$. First multiply each rational-affine domain constraint by its own positive common denominator. All domain forms below therefore have integer coefficients. A natural slack must be introduced *after* this step: for example, $(n - 1)/2 \geq 0$ does not mean that $(n - 1)/2$ itself is a natural integer.

96.1 Private parameters and constant term lifting

For each chart i introduce a natural selector e_i , its private natural parameters z_i , and a private natural slack u_{ir} for each inequality. Set

$$E = \sum_{i=1}^B e_i, \quad Z_i = \sum_{j=1}^{k_i} z_{ij} + \sum_r u_{ir}.$$

For a polynomial f belonging to chart i , define

$$\mathcal{L}_i f = f(z_i) - f(0) + f(0)e_i. \quad (96.1)$$

This changes only the constant term. Multiplying the quadratic monomials by e_i would instead raise an output residual to degree three and its square to degree six. Such multiplication is unnecessary because the gates below force every inactive chart's private variables to zero.

Use residuals

$$\begin{aligned} R_0 &= E - 1, \\ R_{ir}^{\geq} &= \mathcal{L}_i A_{ir} - u_{ir}, \quad R_{is}^{\leq} = \mathcal{L}_i H_{is}, \end{aligned} \quad (96.2)$$

$$R_q^{\text{out}} = L_q \left(y_q - \sum_{i=1}^B \mathcal{L}_i P_{iq} \right), \quad 1 \leq q \leq Q. \quad (96.3)$$

Here $L_q > 0$ is a common integer denominator for all rational coefficients in the *pooled* output expression. Clearing branch outputs separately and then adding the independently rescaled results would alter the equation. For example, outputs $z/2$ and $z/3$ require a single output residual such as $6y - 3z_1 - 2z_2$, not $y - z_1 - z_2$. Every residual has integer coefficients and degree at most two.

96.2 Two polynomial formats

Let the common squared part be

$$S = R_0^2 + \sum_{i,r} (R_{ir}^{\geq})^2 + \sum_{i,s} (R_{is}^{\leq})^2 + \sum_{q=1}^Q (R_q^{\text{out}})^2.$$

The product-gated and literal sum-of-squares forms are respectively

$$C_{\text{prod}} = S + \sum_{i=1}^B (E - e_i) Z_i, \quad (96.4)$$

$$C_{\text{SOS}} = S + \sum_{i=1}^B ((1 - e_i) Z_i)^2. \quad (96.5)$$

All product gates in (96.4) are nonnegative on the natural witness orthant: $E - e_i$ is the sum of the other natural selectors. They have degree at most two. The SOS gates are squares of quadratic residuals. Both forms have degree at most four; the former is nonnegative for real external coordinates and natural witnesses, the latter for all real inputs. One could use $E - e_i$ inside the squared gate as well, since it equals $1 - e_i$ at selector zeros.

Proposition 96.1 (Exact zero fibers). *The natural zeros of either polynomial over a fixed external output y are in bijection with the admissible chart points mapping to y .*

Proof. Every summand is nonnegative on the intended domain. At a zero, $R_0 = 0$ gives $E = 1$. Natural selectors therefore have exactly one value equal to one and all others equal to zero. Separate Boolean-selector equations are not needed.

Let i be the selected chart. For every inactive chart h , its gate has coefficient one, so $Z_h = 0$. Every private parameter and slack is natural; therefore every individual summand in Z_h is zero. Equation (96.1) then gives $\mathcal{L}_h f = 0$ for each inactive domain or output form. On the selected chart it gives $\mathcal{L}_i f = f(z_i)$. All surviving residuals say exactly

$$A_{ir}(z_i) = u_{ir} \geq 0, \quad H_{is}(z_i) = 0, \quad y = P_i(z_i).$$

Conversely, an admissible chart point determines a zero: select that chart, use its parameters, set its slacks to the uniquely determined integers $A_{ir}(z_i)$, and set every other private coordinate to zero. These two constructions are inverses. The chart bijection then proves the empty-or-singleton assertion of Theorem 93.1. $\square$

Naturality is essential to the inactive-zero step. Signed private variables could cancel in Z_h . Free difference pairs for private witnesses would also admit many representations and destroy the stated uniqueness. The compiler uses no such encoding.

[write] [Second route. The device of this compiler is Part I's: Theorem 10.2 multiplies the rows of disjoint chambers by natural selectors that sum to one, and Proposition 96.1 is a second route to it for chart families with private parameters. The inactive gate $(E - e_i)Z_i$ is the gate of Part V's certificate (source 14, Theorem 58.1) and the inactive-product gate of Lemma `qoc:rn:lem:gates` in Part VI of *quadratic-orthant-certificates*. Source 18's own refinements are the constant-term lifting (96.1), which keeps quadratic outputs at degree two, the pooled output denominators L_q , and the selector elimination below. Source 18 calls its proof elementary algebra and cites none of these; no novelty is claimed for the device.]

96.3 Exact witness and residual ledgers

For $B \geq 1$, both formats use precisely

$$N = B + K + M \leq 3B + M \quad (96.6)$$

natural witnesses when $k_i \leq 2$. The slots before deletion of zero terms are as follows.

Quantity	Product-gated	Sum of squares
Selectors	B	B
Private parameters	K	K
Private inequality slacks	M	M
Squared residuals	$1 + M + H + Q$	$1 + M + H + Q + B$
Nonsquared nonnegative products	B	0
Total degree upper bound	4	4

These counts do not bound coefficient sizes or the preprocessing needed to find the chart family. The polynomial may have degree less than four. If $B = 0$, use the constant polynomial 1 with zero witnesses. If $B = 1$, the product gate is identically zero because there is no inactive chart; the SOS gate can be retained. Parameter-free charts and syntactically retained empty domains cause no difficulty.

96.4 Removing one selector in the SOS form

For $B \geq 1$, the SOS construction can omit its last selector. Retain natural $e_1, \dots, e_{B-1}$, set $E_* = \sum_{i < B} e_i$, and substitute $e_B = 1 - E_*$ everywhere. Replace R_0 by $E_*(E_* - 1)$, keeping every other squared residual and every squared gate. At a zero, the natural integer E_* is either zero or one, so the implicit selector is also zero or one and exactly one of all B selectors is one. The inactive-zero and uniqueness proof is unchanged. Every residual remains quadratic; the result has $B - 1 + K + M$ natural witnesses and the same $1 + M + H + Q + B$ square slots. This refinement is for the SOS form. Its whole-space nonnegativity does not require the implicit selector to be nonnegative away from zeros. For $B = 1$ the selector residual is identically zero and may be deleted.

97 Signed external positions without extra witnesses

Signed integer positions are already permitted as external inputs in the theorem. For an entirely natural external coordinate relation, replace each x_i by the canonical pair $(x_i^+, x_i^-) \in \mathbb{N}^2$, imposing

$$x_i = x_i^+ - x_i^-, \quad x_i^+ x_i^- = 0.$$

Substitute $x_i^+ - x_i^-$ in the output residuals. Add the nonnegative product $x_i^+ x_i^-$ to C_{prod} , or its square to C_{SOS} . There are m extra product slots or m extra squared residuals, respectively, and *no additional witnesses*: these pairs are external inputs. The number of external coordinates becomes $1 + 2m$; the original $Q = m + 1$ output equations remain. Total degree is at most four.

For any signed integer x , the unique accepted pair is $(\max(x, 0), \max(-x, 0))$. Without the product condition, the value -1 would be encoded by both $(0, 1)$ and $(1, 2)$, among infinitely many pairs. Those unrestricted difference pairs describe a different external relation. This external coding must not be confused with an encoding of the private parameters or slacks, whose literal natural values were needed in the proof.

98 A complete binary shuttle orbit

98.1 The conservative radius six rule

[write] [Pointer. This subsection restates source 17's binary rule: its four component replacements up to translation, with the roles right-moving pair, left-moving pair, right bounce and marker shift, and left bounce, are rule (89.1); the proof that the component prescription conserves mass and is a genuine local binary rule of radius six is that of Proposition 89.1; and the 8192-entry rule table with its exact conservation potential on all 4096 length-twelve binary words is the certificate (89.2). Source 18 reuses source 17's three files for it byte for byte (they are shipped once, as source 17's). The omitted text survives in the arrival archive. Its closing sentence is kept:] These finite algebraic identities certify global conservation; bounded orbit simulations serve a different purpose.

98.2 Two disjoint sorted charts

Fix $d \geq 7$ and start at $\{0, 3, 4, d\}$. Set

$$T(n) = n^2 + (2d - 11)n, \quad r = d + n.$$

Cycle n begins with $\{0, 3, 4, r\}$. The right-moving pair takes $r - 6$ steps to reach $\{r - 3, r - 2\}$. Its right bounce gives $\{0, r - 4, r - 2, r + 1\}$. The left-moving pair takes another $r - 6$ steps to reach $\{2, 4\}$, and its next bounce starts cycle $n + 1$. One cycle therefore takes $2r - 10$ steps, which agrees with $T(n + 1) - T(n)$.

There are exactly two charts, both with natural n, j and the inequality

$$d + n - 6 - j \geq 0. \tag{98.1}$$

Their outputs are

$$\text{Right: } \quad t = T(n) + j, \quad (x_0, x_1, x_2, x_3) = (0, 3 + j, 4 + j, d + n), \tag{98.2}$$

$$\text{Left: } \quad t = T(n) + d + n - 5 + j, \quad (x_0, x_1, x_2, x_3) = (0, d + n - 4 - j, d + n - 2 - j, d + n + 1). \tag{98.3}$$

Both use $0 \leq j \leq r - 6$. The right chart owns $[T(n), T(n) + r - 5)$, including its right contact configuration. The next step is the bounce, owned by the left chart at $j = 0$. The left chart owns $[T(n) + r - 5, T(n + 1))$, including its left contact. The next step belongs to cycle $n + 1$. Since $r \geq 7$, both displayed coordinate lists are strictly increasing. Thus no sorting inequalities need to be added. These charts describe every state, not only returns. [write] [The hypothesis $d \geq 7$ is safe but not sharp, as for Proposition 89.2: in the placement check the two charts also reproduce every state through $t < 2500$ for $d = 6$.]

98.3 An explicit eight witness baseline certificate

Use the ordered witness tuple

$$w = (e_R, e_L, n_R, j_R, u_R, n_L, j_L, u_L).$$

There are three selector/domain residuals,

$$\begin{aligned} r_0 &= e_R + e_L - 1, \\ r_R &= (d - 6)e_R + n_R - j_R - u_R, \\ r_L &= (d - 6)e_L + n_L - j_L - u_L, \end{aligned}$$

one time residual

$$\begin{aligned} r_t &= t - (n_R^2 + (2d - 11)n_R + j_R \\ &\quad + n_L^2 + (2d - 10)n_L + (d - 5)e_L + j_L), \end{aligned}$$

and four position residuals,

$$\begin{aligned} r_{x_0} &= x_0, \\ r_{x_1} &= x_1 - (3e_R + j_R + (d - 4)e_L + n_L - j_L), \\ r_{x_2} &= x_2 - (4e_R + j_R + (d - 2)e_L + n_L - j_L), \\ r_{x_3} &= x_3 - (de_R + n_R + (d + 1)e_L + n_L). \end{aligned}$$

Let $S_d = r_0^2 + r_R^2 + r_L^2 + r_t^2 + \sum_{v=0}^3 r_{x_v}^2$. The explicit product-gated certificate is

$$C_d = S_d + e_L(n_R + j_R + u_R) + e_R(n_L + j_L + u_L). \quad (98.4)$$

The SOS variant replaces these two product terms by

$$((1 - e_R)(n_R + j_R + u_R))^2 + ((1 - e_L)(n_L + j_L + u_L))^2.$$

Its ledger is $B = 2$, $K = 4$, $M = 2$, $H = 0$, $Q = 5$: eight natural witnesses, eight squared residuals plus two product slots, or ten squared residuals and no product slots. Both polynomials have degree exactly four.

For example, the minimal gap $d = 7$ gives the following boundary checks. The selected branch parameters and its slack determine all eight witnesses; the inactive branch's coordinates are zero.

t	Occupied sites	Branch	n	j	u
0	$\{0, 3, 4, 7\}$	Right	0	0	1
1	$\{0, 4, 5, 7\}$	Right	0	1	0
2	$\{0, 3, 5, 8\}$	Left	0	0	1
3	$\{0, 2, 4, 8\}$	Left	0	1	0
4	$\{0, 3, 4, 8\}$	Right	1	0	2

At $t = 2$ the witness is $(0, 1, 0, 0, 0, 0, 1)$; at $t = 4$ it is $(1, 0, 1, 0, 2, 0, 0, 0)$. The two changes of branch exhibit the half-open ownership convention explicitly.

98.4 A four witness binary specialization

The two binary charts share their domain and their quadratic cycle time. This permits a stronger specialized SOS certificate with shared natural witnesses (e, n, j, u) . Let $e = 0$ select the right chart and $e = 1$ the left. For $d \geq 7$ define seven residuals

$$\begin{aligned} q_e &= e(e - 1), \\ q_u &= d + n - 6 - j - u, \\ q_t &= t - (n^2 + (2d - 11)n + j + e(d + n - 5)), \\ q_0 &= x_0, \\ q_1 &= x_1 - (3 + j + e(d + n - 7 - 2j)), \\ q_2 &= x_2 - (4 + j + e(d + n - 6 - 2j)), \\ q_3 &= x_3 - (d + n + e). \end{aligned}$$

Then

$$C_d^{(4)} = q_e^2 + q_u^2 + q_t^2 + q_0^2 + q_1^2 + q_2^2 + q_3^2$$

has degree four, four natural witnesses, seven squared residuals, and no nonsquared product slots. At a natural zero, $q_e = 0$ forces $e \in \{0, 1\}$, and $q_u = 0$ forces the shared domain $0 \leq j \leq d + n - 6$ with its unique slack. For $e = 0$ the output is exactly (98.2); for $e = 1$ it is exactly (98.3). Their proved timed partition therefore gives exactly one witness tuple at every valid complete timed configuration and none otherwise. In particular, the quadratic time term is shared rather than multiplied by a selector, so no cubic residual appears.

At $d = 7$, the five states in the preceding table have witnesses $(0, 0, 0, 1)$, $(0, 0, 1, 0)$, $(1, 0, 0, 1)$, $(1, 0, 1, 0)$, and $(0, 1, 0, 2)$. This is an upper bound for this binary family, not a witness-minimality claim or a generic shared-parameter optimization for arbitrary charts.

Exactness on the nonnegative real witness orthant. For this specialized certificate, integer external inputs in the stated domain force every real zero's witnesses to be integers. Indeed $q_e = 0$ gives $e \in \{0, 1\}$, and $q_3 = 0$ gives $n = x_3 - d - e \in \mathbb{Z}$. Then $q_1 = 0$ gives $j = x_1 - 3$ when $e = 0$, or $j = d + n - 4 - x_1$ when $e = 1$. Finally $q_u = 0$ gives $u = d + n - 6 - j \in \mathbb{Z}$. If the four real witnesses are nonnegative, they are therefore natural, and the proved singleton-fiber conclusion applies. This also covers nonnegative rational witnesses. It uses the spatial output equations and does not extend to the one-witness return-time certificate or arbitrary chart compilers.

Unrestricted real witnesses still fail. At $d = 7$, $t = 0$, the sorted external output $(0, 5, 7, 8)$ makes all seven residuals vanish at $(e, n, j, u) = (1, 0, -2, 3)$, although the true initial state is $\{0, 3, 4, 7\}$. In the uniform version this is $x = 0$; the negative j is essential.

99 A uniform gap specialization with the same cost

For this particular rule and initial family, set $d = 7 + x$ and make $x \in \mathbb{N}$ an additional external input. Substitute $7 + x$ for d in every residual of Section 98.3. The new monomials include xe_R , xe_L and xn ; all residuals still have total degree at most two in all external and witness variables. Consequently

both baseline polynomials still have degree four and eight natural witnesses, with unchanged slot counts. The same substitution in Section 98.4 gives degree four, four natural witnesses, and seven squares. The selector-eliminated baseline SOS likewise keeps its seven witnesses and ten squares.

Corollary 99.1 (One-parameter complete orbit family). *The substituted four-witness certificate has a singleton natural witness fiber exactly at the complete timed orbit of the initial configuration $\{0, 3, 4, 7 + x\}$, for each natural external x . The same statement holds for nonnegative real witnesses. All other integer output configurations at a natural time have no such witness.*

Proof. For each fixed $x \in \mathbb{N}$, the substituted polynomial is precisely the four-witness fixed-gap polynomial with $d = 7 + x \geq 7$, already proved correct. The simultaneous degree bound follows from the explicit residuals, rather than from an unproved parameterized version of the generic compiler. $\square$

The implementations represent x symbolically in the baseline `uniform_binary_shuttle` and the compact `compact_binary_uniform_sos`. Its tests include $x = 0$, compare the expanded polynomial with independently compiled fixed-gap polynomials, and compare the certificate with direct orbit charts. This is uniformity over one explicit one-parameter input family. It is not a generic extension to arbitrary externally parameterized chart domains or arbitrary initial configurations.

The smaller return-hit certificate in report 12 [source 17’s (90.1)], written there with the witness k and $d = 7 + x$ is

$$[n^2 + (2d - 11)n - t]^2,$$

with just one natural witness. It describes the times when the anchored pattern 10011 appears at sites $0, \dots, 4$, equivalently these cycle returns. It does not output every intermediate complete configuration. That narrower relation explains its smaller cost. The return gaps grow without bound; their nonperiodicity also separates this quartic representation from a claim of timed Presburger definability.

100 Verification and mathematical provenance

100.1 Exact implementation and reproducibility

The release contains the standalone source of this report, its PDF, exact Python code, machine-readable expanded polynomials, and deterministic test receipts. The code uses only standard-library integer and rational arithmetic. Charts and sparse polynomials snapshot mutable constructor inputs. Boolean and floating-point coefficients, malformed degrees and arities, and negative natural witnesses are rejected before arithmetic. There are no approximate-zero tests or pre-validation value-keyed caches.

The chart tests perform the following finite exact checks.

Check	Count
Direct binary CA states	5,208
Complete eight-witness cube evaluations across both forms	13,122
Wrong output rejections	1,920
Rational-coefficient single-chart evaluations	250
Malformed input rejections	40
Immutable snapshot checks	5
Strict affine sorting refinements	10
Original-frame quadratic shifts	160
Large-integer nonnegativity checks	200
Canonical signed external pairs	50
Uniform-gap orbit certificate comparisons	1,456
Uniform versus fixed expanded-polynomial comparisons	2,560

The separately frozen four-witness suite checks 5,208 direct CA states, 26,040 wrong-output rejections, 3,125 full four-witness cube evaluations, 122 valid-cube phase comparisons, and 32 malformed-input rejections. The selector-elimination suite checks 2,187 seven-witness cube points, 872 uniform-gap orbit comparisons, 10,935 uniform/fixed cube comparisons, 18 one-chart no-selector cases, two parameter-free cases, and 14 malformed inputs. The refinements were mathematically reviewed and added in separate modules; the original compiler and fixtures remain byte-identical.

A missing-gate counterexample is also exercised: deleting inactive-chart gates admits a false witness, whereas the actual compiler rejects it. The finite cube includes invalid selectors and nonzero inactive branches. The separately included binary-rule suite verifies all conservation-potential identities, compares its component and radius-table rules, and checks a larger collection of direct trajectories. These computations supplement the exact arguments; they do not prove the normal form or decide injectivity of an arbitrary chart description.

Run `./run-tests.sh` from the extracted bundle. The script checks its manifest, reruns all four suites, and requires byte-identical regenerated JSON fixtures. Run `./build.sh` to compile the standalone LaTeX source with pdfLaTeX. Detailed replay and visual-review scope is recorded in the verification files. No network access or external service is needed.

[write] **The release in this report.** These are the delivered names. The programs of `code/phase/` and the top-level `build.sh` and `run-tests.sh` are shipped as `code/18-timed-quartics-*`, the JSON fixtures, receipts and verification records as `data/18-timed-quartics-*`, and `verification/independent-audit.md` as `18-timed-quartics-independent-audit.md` at the report root. Not shipped: the manuscript (printed here), its PDF, README, checksum ledger and manifest checker `verify_manifest.py`, and the three files of `code/` that are byte copies of source 17's (`test_binary_expanding_shuttle.py`, `binary-radius6-conservation-certificate.json` and `binary-shuttle-test-results.json`, shipped once as `code/17-four-mass-test_binary_expanding_shuttle.py` and `data/17-four-mass-*`). `run-tests.sh` needs the delivered layout; the README gives the rerun recipe. At placement on Windows the binary suite and the selector-elimination suite passed and their regenerated fixtures were identical to the delivered ones apart from line ends, the two exhaustive cube suites exceeded the 170-second cap and were not run to completion, and the binary test rewrote the sealed conservation certificate with CRLF line ends, after which the manifest check failed, as for source 17 (Section 92).

100.2 Audit correction and boundaries of the result

Independent mathematical review covered the chart partition, compiler, ledgers, binary formulas, and SOS refinements, conditional on the companion normal form. Test reruns establish reproducibility, not new algebraic proofs. A preliminary small-cube expectation was corrected from twelve roots to fourteen; the compiler and theorem were unchanged.

The following boundaries remain:

- The effective bijective low-degree charts, not decidability alone, supply the quartic construction
- Forgetting time can lose uniqueness; a stationary orbit repeats forever
- No general uniform-input arity bound or general single-fold or finite-fold Diophantine conclusion, including for MRDP, is asserted
- Only the four-witness binary specialization is proved exact for nonnegative real witnesses at integer external inputs. The generic compiler makes no such claim; unrestricted real witnesses can give spurious zeros

Further directions. Implement the effective rule-to-chart preprocessing, characterize when untimed projection retains canonical witnesses, and optimize arithmetic evaluation costs. These are research directions, not literature-level open-problem assertions or witness-minimality claims.

100.3 Attribution and classical context

The normal form and binary rule come from unchanged companion report 12 [source 17, Part VI]. Its lower-particle precursors include Kong’s explicitly binary thesis [41] and Imai’s project report [42]; neither is attributed this weighted-alphabet four-mass normal form or compiler. Ginsburg and Spanier [7] supply the classical Presburger/semilinear context. The displayed compiler proof is elementary algebra. No novelty, priority, or current open-problem status is inferred from these comparisons or a literature search failure.

Part VII

Mass at most four in every dimension: orbit geometry, counting and a binary planar shuttle (sources 19, 20 and 21)

[write] Part VII prints three AI-assisted research manuscripts of ProveIt’s incoming reports, batch 82 (arrival db37d18c8), all dated 3 October 2026 and added on that day. Each is a numbered report of one research pipeline: *source 19* is batch-82 manuscript 06, the pipeline’s Report 29; *source 20* is manuscript 17, Report 30; *source 21* is manuscript 01, Report 31. Their author lines are “Research report 29”, “Research report 30” and “Research report 31”; they name no assistant and no repository commit. The three numbers of each manuscript differ; brackets translate the pipeline’s report numbers wherever they occur (Report 12 is source 17, addendum 13 is source 18, Reports 29–31 are sources 19–21; Reports 16 and 26 are batch-82 manuscripts placed in the collection’s report

`five-particle-binary-automata`, where Report 16 is printed as Part II).

The three form one chain on top of Part VI. Source 19 lifts source 17’s four-mass decision theorem from $\mathbb{Z}$ to every $\mathbb{Z}^d$ (Theorem 101.1), proving the case $d = 1$ again by the same architecture (notes in its Sections 101–111), and its Appendix I lifts source 18’s fixed-input timed quartics to $\mathbb{Z}^d$. Source 20 draws quantitative orbit geometry out of source 19’s normal form: semilinear visited sets, quasipolynomial box counts of sharp degree, quadratic clocks, first arrivals, and the nonsemilinear effect of drift, with explicit weighted planar automata and canonical first-visit quartics. Source 21 realizes source 20’s planar wedge and drift regime by an ordinary binary rule, built by lifting source 17’s binary shuttle into the plane. Each pins its predecessors by SHA-256 (source 19 pins source 17’s L^AT_EX source, source 18’s and those of Reports 16 and 26; source 20 pins source 19’s; source 21 pins those of sources 17, 18 and 20), and the pins were verified at placement. Source 19’s sharp four-versus-five boundary is conditional on Reports 16 and 26, as it says itself (note after Corollary 101.2).

[write] **Literature.** Sources 19 and 20 compare the encounter normal form with pebble automata (Delorme and Mazoyer [49], Lacalle and Gajardo [50], Blum and Sakoda [51]), plane-walking automata (Salo and Törmä [52]), two-dimensional reversible number-conserving universality (Morita, Tojima, Imai and Ogiro [53]) and Schaeffer’s physically universal automaton [54], besides Kong’s thesis [41] and the Alhazov–Imai abstract [24] already credited in Part VI; source 20 uses Woods’ Presburger counting theorems [55]. These pebble-automaton, plane-walking and counting references are new to the collection. Several were checked only through publisher abstracts, as the sources say, and none of the three claims novelty or priority.

[write] **Review status.** The Hilbert’s-tenth programme has reviewed source 19 (`review_dimension_independent_four_mass.md`, commit 7b547f686; Section 116) and source 21 (`review_binary_planar_four_particle31.md`, commit 16f50dcc6; Section 139), and made a scoped intake of source 20’s original-frame certificate, with a follow-up on its timed dependency in source 19 (`review_original_frame_first_hit30_intake.md` and `original_frame_first_hit30_fixture.md`, commits c875bad40 and 5b4c091e6; Appendix K); source 18 has not been reviewed ([write] added 4 October 2026: it has been since, in a scoped intake (`review_timed_four_mass_source18_intake.md`, commit 39f398985) [112]; Section 1.19). Section 1.19 gives the relation to the rest of the report, and Section 1.20 collects what the four batch-82 sources do not claim.

Source 19 (batch-82 manuscript 06): Dimension independent particle thresholds

Effective reachability at mass four and the inherited mass five boundary, “Research report 29”, 3 October 2026. [write] From the archive `Dimension_Independent_Particle_Thresholds_Package.zip`; its labels carry the prefix `smc:zd:`, its sections are Sections 101–117 here, and its two appendices are Appendices I and J, after Part III’s. The Hilbert’s-tenth programme has reviewed its proof (note at the end of Section 116). Its abstract follows; the bracketed notes in it are [write].

For a translation invariant finite range cellular automaton on $\mathbb{Z}^d$, with a finite positively weighted alphabet, a unique weight zero vacuum, and at most one weight one symbol, exact finite configuration reachability and finite pattern occurrence are decidable from every input of mass at most four. The result is uniform in the finite dimension and does not require reversibility. A mass two packet has finitely many normalized shapes. Its successful flights toward a stationary unit follow finitely many exact lattice rays. Two consecutive successful flights in nonparallel directions force the marker separation into an effectively computable finite set. Outside that set, the dynamics reduces to finite control and one additive scalar gap, with the vector anchor retained as output. Expanding cycles have quadratic section times; a drift cutoff restores anchored

observations in the original frame. We give the full effective proof, all weighted and no unit cases, a decision algorithm, and reproducible finite arithmetic checks. An appendix extends Report 13's [source 18's] fixed input timed quartic certificate compiler using canonical higher dimensional phase charts. The previously established one dimensional five particle universal construction embeds linewise into every finite dimension, giving the inherited sharp four versus five boundary [conditional on batch-82 Reports 16 and 26, which this report does not print; note after Corollary 101.2]. This is an ordinary mathematical proof, not a formalization, an implemented general rule compiler, or a claim of novelty or priority.

101 Statement and scope

The mass of a finite configuration $c: \mathbb{Z}^d \rightarrow A$ is

$$m(c) = \sum_{x \in \mathbb{Z}^d} w(c(x)).$$

Here A is finite, $0 \in A$ is the unique weight zero symbol, and every other symbol has a positive integer weight. A finite configuration equals 0 outside a finite support. A cellular automaton (CA) is deterministic, translation equivariant, and has a finite neighborhood. Conservation below means conservation of this finite total mass for every finite configuration. Write $\tau_v c(x) = c(x - v)$.

Theorem 101.1 (Uniform finite input decision theorem). *Let $d \geq 1$ be finite. Let F be a CA on $\mathbb{Z}^d$ with a quiescent vacuum and the weights just specified. Assume mass conservation and at most one weight one symbol. Given its finite local rule, a finite neighborhood bound, the weights, and a finite input c of mass at most four, there is an effective procedure deciding each of the following:*

1. *whether $F^t(c) = b$ for a specified finite configuration b and some $t \in \mathbb{N}$;*
2. *whether $F^t(c) = \tau_v b$ for some $t \in \mathbb{N}$ and $v \in \mathbb{Z}^d$;*
3. *whether a specified finite pattern occurs in its prescribed window at some time;*
4. *whether that pattern occurs at some translated window at some time.*

Pattern zeros are enforced. Reversibility is not required. The rule hypotheses are promises; the procedure need not certify conservation for an arbitrary submitted rule.

[write] [For $d = 1$ this is Theorem 83.1 of Part VI (source 17, the pipeline's "Report 12"), whose proof the present one generalizes step by step: the stationary-unit frame, bounded packets, the tagged seed library with B fixed first, the thresholds N before W , one additive gap, the dispatch over the five mass partitions, termination in three forms, and the drift cutoff. Brackets below mark each place; at $d = 1$ the steps there are a re-proof by the same architecture, and no novelty is claimed for them. The higher-dimensional ingredients are new: contact through actual occupied sites rather than a bounding box (Figure 6), exact contact rays with a Bezout functional and residues (Section 106), the finite nonparallel switch set (Lemma 107.1), compact triples that miss a remote unit (Section 105), and a transverse offset folded into the finite mode (Section 108). Source 19 itself cites source 17 only in Section 114 and in its lineage appendix.]

The proof constructs a fixed input orbit description. In the frame fixing isolated mass one particles, its untimed part is effectively Presburger definable. Its eventual form is independent finite phase objects, a whole configuration periodic up to translation, or an expanding one counter shuttle. In the last case exact elapsed time is retained, but is quadratic in the shuttle cycle index. No assertion of uniform timed Presburger definability is made.

The input dimension may be part of the finite rule description. Every construction below is a finite enumeration or effective integer calculation for that dimension. No useful complexity bound is asserted. In particular, “effective” does not mean that the normalized state libraries or resulting formula are small.

Corollary 101.2 (Inherited sharp boundary). *In the effective finite input and finite observation simulation setting of Reports 16 and 26 [batch-82 reports of the same pipeline; note below], the sharp conserved mass boundary is four versus five in every finite dimension: mass at most four has the decision property of Theorem 101.1, and the previously established binary five particle universal construction embeds in every $\mathbb{Z}^d$.*

The upper part inherits the source construction and compiler proofs in those reports; this report proves the higher dimensional lower part and the linewise inheritance argument. Section 114 states the distinction precisely. It does not transfer undecidability to an arbitrary observation convention lacking the inherited effective encoding and halt observation.

[write] [The upper half of Corollary 101.2 is not proved in this report. It is the linewise embedding (Section 114) of the binary five-particle universal construction of batch-82 Reports 16 and 26 of the same research pipeline, which source 19 pins by SHA-256 (Appendix J). Both were placed in the collection’s report `five-particle-binary-automata` (file prefixes `16-literal-source-` and `26-parallel-involutions-`; placement commits `bca6383e9` and `7d2b1b245`). Report 16 is printed there as Part II (labels `fpa:ls:`); Part I of that report, Reports 14 and 15, proves with source 17 that five particles are the exact one-dimensional binary threshold (its Corollary `fpa:bc:cor:sharp`). Report 26 is printed there as Part VI (commit `d2043bbd0`). The four-versus-five boundary in every dimension is therefore sharp only conditionally on those releases having their stated properties. The Hilbert’s-tenth programme has reviewed both. Its review of Report 16 (`review_literal_universal_reversible_source16.md`, commit `69730d3e8`) [61] found no source-interface or arithmetic contradiction; it records that Report 16 closes the literal-source and clean-nonblocking obligations of the earlier five-particle compiler, keeps the published Neary–Woods universality theorem and the earlier compiler theorems as explicit dependencies, and notes that Report 16’s own threshold claim inherits source 17’s mass-four theorem, which it does not recertify. Its review of Reports 26–28 (`review_parallel_particle_reports.md`, commit `fb7e3cb47`) [57] found the two-involution mechanism of Report 26 sound but states that it did not reconstruct the complete source compiler and that Report 26’s full source binding remains outside its replay. The lower half, decidability at mass at most four in every finite dimension, is proved in full here. Source 19’s own adversarial review says the same: “the sharpness statement remains conditional on the cited five-particle source releases having their stated properties” (`19-mass-four-zd-independent-review.md`).]

[write] (4 October 2026, batch 91.) Reports 70 and 71 of the same pipeline, printed as Part VII of `five-particle-binary-automata` (labels `fpa:tr:`, `fpa:sw:`), replace Report 26’s parallel rule by two further full-shift, number-conserving two-involution rules with the same admissible doubled micrograph and smaller sufficient radii, $108D + 149 + 3J$ (`fpa:tr:thm:main`) and $76D + 105 + 3J$ (`fpa:sw:thm:main`), against Report 26’s $180D + 258 + 9J$. Since the linewise embedding uses only the admissible one-dimensional dynamics, either rule can stand in for Report 26’s; the four-versus-five boundary is unchanged, and the new rules differ from Report 26’s only on malformed inputs. Neither Report is independently audited beyond a scoped programme intake (`a9ab9a698`).

102 Stationary unit frame and independent components

Use the maximum norm on $\mathbb{Z}^d$. A radius bound $R \geq 0$ means that the output at x depends only on $x + [-R, R]^d$. For a nonempty finite configuration,

$$\text{diam}(c) = \max\{\|x - y\|_\infty : x, y \in \text{supp}(c)\}.$$

Normalize it by subtracting its lexicographically first occupied site. Translation preserves lexicographic order, so this normalization is translation compatible. Bounded diameter and bounded mass leave only finitely many normalized configurations, and their successors can be computed by evaluating the rule on a finite neighborhood of their support.

Suppose the unique possible weight one symbol u exists. An isolated u must evolve to exactly one u : positivity and conservation leave no other output of mass one. Quiescence and finite range confine its displacement δ to $\|\delta\|_\infty \leq R$. Put

$$G = \tau_{-\delta}F, \quad S = \max(1, R + \|\delta\|_\infty).$$

Then G has radius at most S , fixes each isolated u , and

$$F^t = \tau_{t\delta}G^t. \tag{102.1}$$

The case with no unit symbol is treated separately in Section 112.

Join occupied sites at distance at most $2S$. A connected component of mass m has at most m occupied sites and therefore diameter at most $2S(m - 1)$. Components mutually farther than $2S$ have disjoint radius S influence neighborhoods. Their one step evolutions are independent and have disjoint output supports. By induction, the independent evolutions agree with the full evolution through the first configuration at which two such supports have distance at most $2S$. The next step is computed using the full rule. A geometric contact need not imply a nontrivial collision, but processing it exactly is harmless.

This observation is used with *actual occupied sites*. In several dimensions an expanded bounding box is generally larger than the union of the contact boxes of those sites. Replacing the latter by the former is incorrect.

103 Bounded mass two packets

[write] [At $d = 1$ the lemma below is source 17's Lemma 84.1, which restates source 16's no-split Lemma 75.1; the coordinatewise intersection of two radius- S cubes is the $\mathbb{Z}^d$ form of that argument. With one weight-one symbol and $d = 1$, the mass-two case is also a special case of Part IV's Theorem 44.1, proved there for any number of weight-one symbols.]

Lemma 103.1. *A mass two configuration of diameter at most $2S$ remains within that diameter under G . It has an effectively computable finite transient and a periodic profile up to a vector translation.*

Proof. A weight two singleton produces support in its radius S cube, of diameter at most $2S$. Otherwise the configuration is two units, translated to 0 and v , with $\|v\|_\infty \leq 2S$.

If $\|v\|_\infty > S$, each original occupied site sees only its own unit and stays occupied by u . These two sites already exhaust the total mass, so the configuration is fixed. If $\|v\|_\infty \leq S$, any newly

occupied site other than those two must see both units: a neighborhood seeing zero units gives vacuum, and a neighborhood seeing just one gives the stationary singleton output. The intersection of the two radius S cubes contains both original sites, and its coordinate widths are $2S - |v_i| \leq 2S$. The entire successor support lies in that intersection.

There are finitely many normalized mass two shapes of this diameter. Each has one next normalized shape and a translation increment. A repeated shape gives the finite transient followed, after resetting the time origin, by

$$P(r + kp) = kD + P_r, \quad 0 \leq r < p, \quad k \in \mathbb{N}, \quad (103.1)$$

where $D \in \mathbb{Z}^d$, the period $p \geq 1$, and all labeled phase shapes P_r are explicit. This finite graph can be enumerated. $\square$

A pair of units at distance greater than $2S$ is fixed. The lemma includes all weight two singleton labels and two unit shapes, so it is not restricted to binary alphabets. The word “packet” will denote such a bounded mass two object, and “dimer” will be used synonymously even when its support is a singleton.

104 Mass three profiles and a finite encounter library

Lemma 104.1. *Every fixed mass three input has an effective finite prefix followed by either a whole configuration periodic up to a vector translation, or an independent stationary unit and a finite phase mass two packet. Three stationary units are included as a constant case. Mass at most three consequently has a fixed input timed Presburger description.*

Proof. Take as the core every normalized mass three configuration of diameter at most $4S$. Outside this finite core, the $2S$ component partition is $2 + 1$ or $1 + 1 + 1$. The latter is fixed. In the former case Lemma 103.1 gives the dimer’s finite transient and phases. In an eventual phase its occupied coordinates are $p + kD$ for a finite list of offsets p , while the other unit is stationary.

Entry of the complete configuration into the diameter $4S$ core is a finite conjunction of coordinatewise linear inequalities in k . Their common integer solution set is an interval, possibly empty or unbounded. Compute its first nonnegative integer, compare the corresponding actual times over all phases, and also check the finite transient. If no core entry exists, there is no later dimer unit contact: such a contact would be connected and have diameter at most $4S$.

From each normalized core state, take one exact step and then its just computed first return when needed. The result is a finite deterministic graph of core returns, labeled by finite elapsed times and translation vectors, with possible terminal independent tails. Repetition of a normalized core state gives equality of two complete configurations up to translation. Determinism and equivariance propagate this equality to all future times, including every intermediate excursion. The corresponding finite period can be enumerated even if enormous.

In the independent case, take a common period for the finite phase objects. In either case coordinates and time become affine in one nonnegative integer after finitely many phases. Restoring δt keeps them affine. $\square$

[write] [In $d = 1$, source 16 proves the stronger uniform form, Theorem 74.1 (uniform in all positions and the time), and source 17’s Lemma 84.2 is the fixed-input form used here. Lemma 104.1 is weaker in $d = 1$ but holds in every $\mathbb{Z}^d$, and source 19 needs only the fixed-input form.]

[write] [At $d = 1$ the tagged seed library and its bound B are source 17's (Section 85).] Enumerate every normalized *connected* mass three seed. Its diameter is at most $4S$, so this is a finite effective list. A dimer's first contact with a unit is such a seed. Include a finite bookkeeping tag for the incoming target unit's coordinate. The tag records the geometric role of that coordinate; it introduces neither a new CA symbol nor a persistent physical identity.

For each tagged seed, compute the preceding lemma's profile and extend its finite prefix to one of two checkpoints:

Compact

The whole triple is periodic up to translation, with finitely many bounded phase representatives.

Emission

A stationary unit remains and the independent dimer has nonzero drift D . From the checkpoint onward it is purely periodic up to D and never comes within $2S$ of the residual unit.

If an independent dimer has $D = 0$, the complete triple is eventually periodic and belongs to Compact. If $D \neq 0$, choose a coordinate with nonzero drift. Its linear escape allows the finite prefix to be extended effectively until every later phase remains beyond the residual unit's contact interval. This produces Emission.

Use a common seed anchor frame: normalize each triple seed with occupied anchor 0, and measure every occupied coordinate through its checkpoint in that frame. Choose B to bound all these coordinates, every compact phase representative, and every emission first period representative, including residual unit offsets. On reanchoring an emission at its residual unit, include all resulting finite difference offsets as well, enlarging B if necessary. The list is finite. This B is computed solely from the original connected mass three seeds, *before* any mass four core is chosen. Compact phase diameters are at most $2B$.

105 Compact triples reset or escape

Place a tagged library seed at an anchor a and a fourth unit at b , with $\|b - a\|_\infty > B + 2S$. Its entire precomputed prefix through the checkpoint is independent of the fourth unit. At an emission, the remote unit remains fixed; the new residual unit is at a seed dependent fixed offset from the target anchor.

For a compact checkpoint, compute the first contact with the fourth unit by enumerating the compact phases and the actual occupied sites in each. For each candidate site the distance at most $2S$ conditions are coordinatewise linear inequalities in the period count. Check the finite prefix separately. A translating triple can miss the remote unit in higher dimensions. If it does, this is a terminal independent profile. If it hits, the full configuration has diameter at most $2B + 2S$.

Thus an arbitrarily long trip of a compact triple either ends in a bounded full encounter or remains independent forever. It cannot carry two independently changing unbounded gaps through a continuing interaction. This includes triples that detach and recombine during their periodic orbit: their entire period, not just a section shape, is retained in the compact profile.

106 Exact contact rays

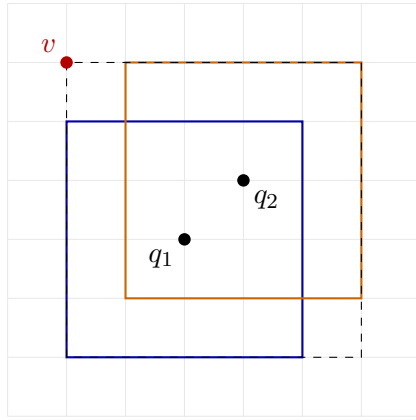
At an emission checkpoint put the source residual unit at 0 and write the dimer profile as

$$Q(r + kp) = kD + Q_r, \quad D \neq 0, \quad 0 \leq r < p, \quad k \in \mathbb{N}.$$

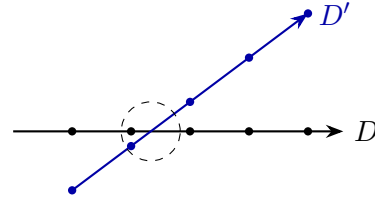
It never meets its source again in this isolated profile. For a target unit at v , contact at time $r + pk$ occurs exactly when

$$v = a + kD, \quad a \in Q_r + ([-2S, 2S]^d \cap \mathbb{Z}^d). \quad (106.1)$$

The notation denotes the union over the actual occupied sites of Q_r , with the phase r retained. All offsets a form a finite effective set. Repeated witnesses are harmless. Minimize the *actual integer time* $r + pk$ over all valid witnesses. If there is no witness with $k \geq 0$, the packet and both markers remain independent forever.



Actual contact boxes
leave holes in their bounding hull



Nonparallel flight equations
have at most one pair of counts

Figure 6: Left: for $S = 1$, sites $(0, 0)$, $(1, 1)$ do not contact $(-2, 3)$, although their expanded bounding hull contains it. Right: schematic arithmetic rays; the finite exceptional region is computed from exact determinants, not from the drawing's angle or scale.

To express the successful rays in finite control, write $D = m\nu$, where ν is the primitive integer vector on its unoriented line, with first nonzero coordinate positive, and $m \in \mathbb{Z} \setminus \{0\}$. By Bezout's identity compute an integer linear functional λ with $\lambda(\nu) = 1$. Every vector has the unique decomposition

$$v = n\nu + z, \quad n = \lambda(v), \quad z = v - \lambda(v)\nu, \quad \lambda(z) = 0.$$

For each contact offset define $\ell = \lambda(a)$ and $z_a = a - \ell\nu$. Equation (106.1) becomes

$$z = z_a, \quad n \equiv \ell \pmod{|m|}, \quad k = (n - \ell)/m \geq 0, \quad (106.2)$$

$$t = (p/m)n + r - (p/m)\ell. \quad (106.3)$$

For $|n| > \max_a |\lambda(a)|$ on the drift facing sign, all eligible counts are nonnegative. On the opposite sign there is no candidate. On the eligible sign, the transverse offset, residue, and phase identify the possible witnesses and the earliest constant term in (106.3).

Relative to the target, the entire contact dimer shape is $Q_r - a$. Hence its tagged seed is also determined by these finite data. If several witnesses attain the same actual time they describe

the same complete configuration; a fixed witness ordering removes bookkeeping ambiguity. All intermediate flight phases remain available: their coordinates are affine in the marker vector and one period variable, with a linear first contact cutoff. Every local encounter prefix phase is retained too.

107 The effective nonparallel exception

Let a dimer fly from marker A to marker B . At the next emission the contacted marker is $B + c$, with c in a finite library of marker update vectors, while the old source A remains stationary. Thus the incoming marker vector $v = B - A$ and outgoing target vector v' satisfy

$$v' = A - (B + c) = -v - c.$$

If both flights succeed, with nonzero drifts D, D' , their contact ray equations give

$$v = a + kD, \quad v' = a' + k'D', \quad kD + k'D' = -(a + a' + c), \quad k, k' \in \mathbb{N}. \quad (107.1)$$

The offsets a, a', c range over finite computable lists.

Lemma 107.1 (Finite nonparallel switch set). *For all nonparallel pairs of emission drifts, the marker vectors in (107.1) form an effectively computable finite set.*

Proof. Choose coordinates i, j with

$$\Delta_{ij} = D_i D'_j - D_j D'_i \neq 0, \quad H = -(a + a' + c).$$

The only possible rational solution of the two selected equations is

$$k = \frac{H_i D'_j - H_j D'_i}{\Delta_{ij}}, \quad k' = \frac{D_i H_j - D_j H_i}{\Delta_{ij}}. \quad (107.2)$$

Retain it only when both entries are nonnegative integers and every remaining coordinate equation in (107.1) holds. Compute $v = a + kD$. There are finitely many drift and offset inputs, so this enumerates the full finite exceptional set. $\square$

This is an exact calculation, rather than a qualitative appeal to divergent directions. In particular, a bound depending only on the small offsets is insufficient. Let $M = 10^6$ and set

$$D = (M, 1), \quad D' = (-M - 1, -1), \quad a = a' = 0, \quad c = (0, -1).$$

The determinant is 1, while (107.1) has the solution $k = M + 1, k' = M$. The exceptional vector $v = (M + 1)D$ has

$$\|v\|_\infty = M^2 + M = 1\,000\,001\,000\,000.$$

Higher dimensional examples follow by adjoining zero coordinates. This shows why nearly parallel drift vectors can create a very large but finite exception.

Outside a computable bound every continuing sequence of successful dimer exchanges uses one fixed *unoriented* rational line. Opposite orientations on that line are permitted. A nonparallel outgoing packet that misses the old marker is terminal, not another counter mode. A compact triple is already covered by Section 105.

108 Finite control and one scalar gap

For every primitive line ν in the emission library, choose one Bezout functional λ . Order the two markers by their λ coordinates, calling them L and R , and write

$$R - L = x\nu + z, \quad x = \lambda(R - L) > 0, \quad \lambda(z) = 0. \quad (108.1)$$

This is a bookkeeping order for the chosen line, not a global linear order of $\mathbb{Z}^d$. For a successful long flight the transverse offset belongs to the finite set of $\pm(a - \lambda(a)\nu)$ from contact offsets on this line. Its sign depends on the source side.

The finite mode records ν , the source side, the complete emission profile and launch shape, the transverse offset, and any needed tagged seed information. The absolute vector anchor L is retained as output; it does not affect the next rule application.

108.1 Noncircular threshold selection

[write] [At $d = 1$ these are source 17's thresholds (85.1), $N = 4B + 12S$ and $W = N + 2B$, also chosen in that order; here N must in addition exceed the Bezout and exceptional-switch quantities.] Choose an integer N larger than all required finite quantities:

1. every $|\lambda(a)|$ in the contact lists, to validate the large gap residue formulas;
2. every $|\lambda(c)|$ for marker shifts, so the marker order cannot reverse;
3. every $|\lambda(v)|$ for exceptional switch vectors, for the relevant incoming functionals;
4. a safety bound separating the remote marker from every local seed prefix.

For the fourth item it suffices, with further finite launch offsets included if needed, to take

$$N > \max_{\lambda} \|\lambda\|_1 (B + 10S + 1),$$

since $|\lambda(v)| \leq \|\lambda\|_1 \|v\|_{\infty}$. In particular $x > N$ makes the remote marker safely farther than $B + 6S$ from the incoming seed anchor. All data used here were already computed from the original seed and phase libraries.

Now choose W larger than $6S$, $B + 6S$, $2B + 2S$, and the diameter of every successful emission launch with $0 < x \leq N$. This last list is finite: the lines and transverse offsets are finite, the scalar ranges over a finite interval, and all launch shapes have fixed finite offsets. More explicitly, a successful vector satisfies $n = \lambda(a) + km$ with $k \geq 0$. If $|n| \leq N$, then

$$0 \leq k \leq \frac{N + |\lambda(a)|}{|m|}.$$

Enumerate those finitely many integers for every offset and profile, and include their full launch diameters in W . This also includes $n = 0$ and every degenerate marker order. Thus W is computed *after* N , without enlarging and recomputing the seed library.

Call a configuration of diameter at most W ordinary. At designated emission checkpoints use the live representation when the next cross marker flight succeeds and $x > N$. Otherwise use an ordinary state if its diameter is at most W , or an explicit terminal independent profile if the flight misses. At other dispatch states test ordinary first. Ordinary and tagged live states may overlap; the dispatch convention is part of the algorithm, not an extra state of the CA.

108.2 The transition formula

On a live to live edge Lemma 107.1 prevents a nonparallel line switch. Only the contacted marker changes, by one vector c . Since the order remains fixed,

$$x' = x + c_0, \quad L' = L + e, \quad z' = z + c_\perp,$$

with constants from finite effective lists. Fold z into the mode. Let M_0 be the least common multiple of the positive numbers $|m|$ in all emission drift decompositions. Equations (106.2)–(106.3), plus the fixed local prefix duration, give

$$q' = f(q, x \bmod M_0), \quad x' = x + c_0, \quad L' = L + e, \quad h = \alpha x + \beta, \quad (108.2)$$

where $\alpha > 0$ is rational and h is a positive integer on the permitted residue. Here $\alpha = p/|m|$ for the flight. Positivity follows because live targets are safely distant. [write] [At $d = 1$ this is source 17's Lemma 86.1, equation (86.1), with the marker gap D in the role of the scalar x .]

An updated successful gap at most N has bounded full diameter and resets to ordinary. An emission that misses is terminal. A compact outcome is analyzed by Section 105; it then resets or is terminal. The *choice to continue live or exit through a compact outcome or emission miss*, apart from bounded lower guard effects, depends only on mode and residue. Whether a compact outcome subsequently hits the remote marker may depend on the actual gap, and is calculated separately. It is not a live edge and must not be treated as residue determined.

There is no further control clock: the CA is time independent, the units are stationary in G , and packet phase is already recorded. Neither absolute anchor nor accumulated time is queried by the control. The vector separation is consequently not a pair of freely testable scalar counters.

109 Complete dispatch from arbitrary inputs

For total mass four the possible $2S$ component mass partitions are

$$4, \quad 3 + 1, \quad 2 + 2, \quad 2 + 1 + 1, \quad 1 + 1 + 1 + 1.$$

Lower masses were covered above. Each mass partition is handled as follows. [write] [At $d = 1$ the same five partitions are source 17's Lemma 86.2.]

Connected mass four

Its diameter is at most $6S$, so it is ordinary. This includes a weight four singleton and all weighted connected shapes.

Four unit components

All are stationary.

Mass three plus one

Outside ordinary, the connected triple has diameter at most $4S$. Since $W > B + 6S$, the remote unit is far enough from the chosen triple seed anchor for the library prefix to be independent. Apply the tagged seed library and Section 105, or enter an emission checkpoint.

Two dimers

Combine their finite transients and then take a common phase period. At each common phase the relative occupied coordinates are affine in one nonnegative integer period count. Their first distance

at most $2S$ event is the earliest solution among finitely many coordinatewise interval tests, one for each pair of occupied sites. If absent, the tail is independent. At contact, each packet has diameter at most $2S$, so the full configuration has diameter at most $6S$ and is ordinary.

A dimer and two units

Use the exact contact calculation against each unit, including any finite dimer transient. If the initial dimer has eventual drift $D = 0$, inspect its transient and one entire positioned period against both units; the nonzero drift ray formula is unnecessary. If neither is hit, the tail is independent. A first simultaneous contact with both yields a connected mass four state of diameter at most $6S$. Otherwise the contact forms a tagged triple seed. If the full state is ordinary, reset. If not, the remaining unit satisfies the safety separation and the library applies.

Actual times are compared across all sites, phases, and possible target units. Ties are retained at the level of complete configurations. No argument assumes a packet must hit a marker merely because one coordinate moves toward it.

From an ordinary state take at least one exact CA step before invoking dispatch. This yields a positive time edge from one of finitely many normalized ordinary states to another ordinary state, a live checkpoint, or an explicit terminal profile. All intervening configurations are kept as finite phase descriptions. A live edge has a positive length flight. Thus the section graph has no artificial zero time loops.

110 Termination and the complete orbit form

Enlarge the finite mode by $x \bmod M_0$. Follow the section machine from the given input. A repeat of a normalized ordinary state means two complete configurations are translates, so the whole future is periodic up to translation, at every intermediate time.

Within an uninterrupted live excursion, the unoriented line is fixed and there are finitely many enlarged modes. At a repeated mode let Δ be the net scalar change around the recorded cycle. The residue has repeated, hence Δ is divisible by M_0 and the same residue itinerary is available on later repetitions.

$$\Delta = 0$$

The whole launch configuration repeats up to the anchor translation. Determinism gives a translated periodic future.

$$\Delta > 0$$

Every intermediate lower guard valid in the recorded cycle remains valid in all later repetitions. The same itinerary persists forever, giving an expanding shuttle.

$$\Delta < 0$$

Compute the greatest number of complete repetitions for which every intermediate scalar stays above N . This is a finite collection of linear inequalities. Execute or accelerate those repetitions. The remaining finite cycle prefix exits the live region to ordinary or terminal. An above threshold mode dependent exit would already have appeared during the recorded cycle, because the enlarged control itinerary is the same.

The decreasing case can be simulated without acceleration: there are finitely many repetitions, however large that number is. Acceleration merely gives a compact exact description of the finite prefix.

There are only finitely many normalized ordinary states. Therefore the analysis terminates with a finite prefix and one of three eventual forms:

- A. independent finite phase objects;
- B. the entire configuration periodic up to translation;
- C. an expanding shuttle.

All described prefixes and phases are effective. This is an unbounded termination proof, not an inference from finite experiments. [write] [At $d = 1$ this is source 17's Proposition 87.1, with the same three cases for Δ .]

110.1 Expanding shuttle coordinates and time

Suppose the cycle in case C has $\Delta > 0$ and total anchor update $E \in \mathbb{Z}^d$. At its corresponding sections,

$$x_n = x_0 + n\Delta, \quad L_n = L_0 + nE, \quad n \in \mathbb{N}. \quad (110.1)$$

There are finitely many edges and local phases per cycle. During a local encounter prefix every occupied coordinate is affine in n . During a flight it is affine in (n, j) , where $j \in \mathbb{N}$ is the packet period count subject to linear first contact bounds. Labels are fixed within a phase. These descriptions include the whole orbit between sections. Their finite union is Presburger definable when elapsed time is forgotten.

The duration of every live edge is $\alpha x + \beta$. Its duration at corresponding cycles grows by $\alpha\Delta > 0$. Summing finitely many edges gives a positive affine cycle duration, so section times have the form

$$T_n = T_0 + A \frac{n(n-1)}{2} + Cn, \quad A > 0. \quad (110.2)$$

The effective rational coefficients take integral values on all natural cycle indices. The duration of cycle n is $An + C > 0$.

There are effective constants $K_0, K_1 \geq 0$ such that every stationary frame occupied point y in cycle n satisfies

$$\|y\|_\infty \leq K_0 + K_1 n. \quad (110.3)$$

Both markers are affine in n ; every local offset is bounded; the flight period count is bounded above by an affine function of n ; and every packet drift and phase offset is fixed finite data. Taking maxima over finitely many phases yields the bound. This reasoning requires neither coordinatewise monotonicity nor a line through the spatial origin.

111 Deciding observations and restoring translation

In cases A and B, a common period expresses all occupied coordinates and time affinely in a nonnegative integer, with finitely many phases and a finite transient. Exact targets and finite patterns are finite Presburger predicates on these coordinates. For example, a prescribed nonzero site must equal some occupied site with the prescribed label; a prescribed zero site must differ from every occupied site. There are at most four occupied sites and finitely many label possibilities. Exact configuration equality also requires that there be no additional occupied sites. Quantify an integer vector for translated observations. Restoring (102.1) preserves affine formulas here.

In case C, the finite union of the affine in (n, j) stationary frame phase formulas decides every *untimed* exact target and pattern query for G , anchored or translated. Translation invariant queries have the same answer for F and G by (102.1). When $\delta = 0$, this finishes all queries.

Suppose $\delta \neq 0$ and consider an anchored query for F . Choose a coordinate i with $\delta_i \neq 0$. Since $t \geq T_n$ throughout cycle n , equations (102.1) and (110.3) imply

$$y_i(F^t(c)) \geq \delta_i T_n - K_0 - K_1 n \quad \text{if } \delta_i > 0, \quad (111.1)$$

$$y_i(F^t(c)) \leq \delta_i T_n + K_0 + K_1 n \quad \text{if } \delta_i < 0. \quad (111.2)$$

The reversed inequality in the second line is essential. The nonzero quadratic leading term eventually dominates the linear term uniformly over every phase and every time in the cycle. [write] [At $d = 1$ this cutoff is source 17's Section 88.]

For any specified finite target window, compute n_* beyond which the appropriate bound lies permanently beyond that window. One may solve a rational quadratic inequality, or increase n until the bound is beyond the window and its forward difference points outward; discrete convexity then proves permanence. Consequently a nonempty exact target, or an anchored pattern prescribing any nonzero site, can occur only before T_{n_*} . That finite time interval can simply be simulated. Its possible enormous size does not affect decidability. A wholly zero anchored pattern occurs after the cutoff. An empty exact configuration is settled by mass conservation; mass zero input is handled immediately.

This completes the proof of Theorem 101.1 when a unit symbol exists. In particular, the proof restores absolute position without invoking unrestricted quadratic Diophantine solvability and without introducing an absolute position control counter.

112 Alphabets with no unit symbol

When no symbol has weight one, mass at most four occupies at most two sites. A singleton of weight two or three cannot split into two nonvacuum sites, so it is a finite state walker. Its label evolution and vector increments become periodic. A weight four singleton may split into two weight two walkers; at the split their separation is at most $2R$.

Take a finite normalized core of singleton and two site configurations of diameter at most $2\max(1, R)$. Outside it the only nontrivial case is two independent weight two walkers. Their finite transients and common eventual phases give coordinatewise affine inequalities for the first core encounter. Solve them exactly, or certify no encounter. A core return has a computable finite elapsed time and translation. Its finite core return graph gives an eventually translated periodic whole configuration or a terminal pair of independent walkers, exactly as in Lemma 104.1. These profiles have timed Presburger descriptions and decide all four observations.

Symbols of weight greater than four never appear at these total masses, by conservation and positivity. This also covers every missing small mass and weighted singleton case, completing the theorem.

113 The effective decision algorithm

The construction may be implemented eagerly or by computing only visited normalized states. It uses the promised rule hypotheses as follows.

1. Check mass zero and immediate label or mass impossibilities. If there is no unit symbol, use Section 112.
2. Compute the isolated unit displacement δ , the stationary unit rule G , and S .
3. Enumerate normalized mass two states of diameter at most $2S$ and compute their finite profiles.
4. Enumerate normalized mass three states of diameter at most $4S$, build their return graph, and compute the tagged connected seed library, its prefixes, compact and emission profiles, and B .
5. Enumerate actual site contact offsets and phases. Compute primitive drift directions and Bezout functionals.
6. Enumerate marker shifts and solve (107.1) for every nonparallel profile and offset combination, retaining only valid nonnegative integral solutions in all coordinates.
7. Select N and then W as in Section 108.1. Construct ordinary dispatch and finite live mode residue transitions as needed.
8. Follow Section 110 until case A, B, or C is established, retaining exact durations, vector anchors, and all intermediate phases.
9. Test the finite prefix and eventual profiles using Section 111, with the effective original frame cutoff when necessary.

Every enumeration is finite. Unbounded event searches reduce to one variable integer linear equations or inequalities, or to the proved one counter termination argument. Presburger decision is applied only after the explicit finite formulas have been constructed. No step replaces an unbounded question by an unjustified finite search.

114 The inherited mass five upper bound

Let H be the previously established one dimensional binary five particle universal CA. For $x = (x_1, \dots, x_d)$, define $\hat{H}$ by applying H independently on each line parallel to the first coordinate axis. It has the same finite alphabet and radius bound when the one dimensional neighborhood is regarded as a subset of $\mathbb{Z}^d$. A finite configuration meets finitely many such lines, and conservation on each line gives conservation of the finite total particle count.

If H is reversible, apply H^{-1} independently on the same lines. This is the inverse of $\hat{H}$. If $H = I_2 \circ I_1$ with full shift involutions I_1, I_2 , their linewise lifts remain involutions on the entire higher dimensional full shift, and preserve finite particle number when the original involutions do. An input supported on a single line has exactly its original one dimensional dynamics on that line; all other lines remain vacuum.

Therefore every effective input encoding, clean target or halt observation, exact clock, and source simulation proved for the chosen one dimensional construction is inherited on a fixed line. Reports 16 and 26 [batch 82; Report 16 is Part II of `five-particle-binary-automata`, Report 26 is placed there; note after Corollary 101.2] supply the relevant universal source and compilation statements. This paragraph does not independently reconstruct that source or materialize a universal rule array. Together with Theorem 101.1, it gives Corollary 101.2 in that inherited simulation setting.

Report 12's [source 17's, Proposition 89.2] binary four particle quadratic shuttle also embeds linewise. Hence decidability at four does *not* imply eventual timed semilinearity. Report 13

[source 18, Part VI] already treats fixed input timed quartic certificates in the one dimensional setting; it is a precursor, not a result reclassified here as a discovery.

The hypotheses delimit the conclusion. Multiple unit labels can give independent finite state walkers with different drifts, invalidating the stationary dimer argument. Negative weights, weight zero nonvacuum symbols, infinite alphabets, active vacuum, infinite input scenery, and time or position dependent rules are outside the theorem. Infinite marked axes or a free external tape cannot be silently supplied without changing the model.

115 Relationship to earlier literature

The literature check included close pebble and walker models as well as conservative cellular automata. It did not locate a primary theorem deciding reachability for the exact class in Theorem 101.1. That bounded search result supplies no novelty or priority conclusion. The accompanying literature record separates inspected full texts from publisher abstracts.

Delorme and Mazoyer study pebble automata on the unbounded plane and report a three pebble intrinsic universality threshold, with no intrinsically universal two pebble automaton [49]. Their publisher abstract was checked, but the full 2002 article was not retrieved. Their finite state head and passive movable pebbles are a close predecessor model. Intrinsic universality is distinct from the exact reachability question here. A naive unique unit encoding charges at least mass two for a stateful head and one for each pebble; this bookkeeping observation is an inference, not their theorem or an established complete encoding.

Lacalle and Gajardo analyze two distinguishable pebbles on a one symbol grid [50]. Their full preprint provides one pebble periodicity, geometric ray escape arguments, and real time two counter recognition of two pebble traces. Its conclusion explicitly extends its geometric argument to $\mathbb{Z}^d$. Trace recognition by a counter machine is not itself a reachability decision theorem. The present proof must additionally account for CA cluster splitting, fusion, control transfer, weighted labels, and synchronous intermediate phases. The transient bound in their local one pebble argument should not be applied to an arbitrary distant head and pebble start.

Blum and Sakoda report, with attribution to Sipser, that two planar finite automata and one pebble simulate a universal two counter machine [51]. This cautions against equating agent count with conserved mass: two independently stateful agents and a pebble have naive minimum unique unit mass $2 + 2 + 1 = 5$. Their discussion of marked axes likewise cannot provide a free finite mass background. Their two pebble nonuniversality statement is not substituted for our decision algorithm.

Salo and Törmä's plane walking model has independently controlled heads reading a separate supplied configuration [52]. Its sparse input and hierarchy results are relevant, but static marks carry information and must be counted in a conservative encoding. Independently controlled heads cannot automatically be represented by interchangeable weight one symbols.

Morita, Tojima, Imai, and Ogiro give genuinely two dimensional reversible number conserving universal constructions [53]. The accessible publisher abstract does not establish the present mass four or unique unit restriction. Schaeffer's number conserving physically universal CA has a diffusion theorem bounding the duration of finite configuration computation [54]; it is not a small finite seed Turing universality counterexample. Block phases and velocity tracks also need explicit encoding when comparing its model to an ordinary translation invariant CA.

Kong's thesis discusses the one dimensional binary small particle hierarchy and leaves four

particles open in its concluding discussion [41]. It neither proves the weighted higher dimensional theorem nor explicitly gives this uniform reachability procedure. The separately identifiable particle complexity paper is Alhazov and Imai [24]; its full construction was not independently reverified in this literature pass. For the inherited upper bound, the supplied Reports 16 and 26 [batch 82; Part II of, and placed in, `five-particle-binary-automata`] remain the actual proof dependencies.

116 Executable evidence and reproducibility

The package includes the unchanged arithmetic audit source and its deterministic expected receipt. It tests the formulas used in the proof, not the class of all CA rules. Normal and optimized Python runs have the same exactly typed receipt. The audit uses explicit failure checks, not optimization removable assertions.

Finite check	Count
First contact profiles in dimensions 1 to 3	600
Exact contact ray shifts exceeding 10^{60}	600
Tied earliest contact witness cases	30
Common clock two packet comparisons in dimensions 1 to 5	500
Nonparallel switch equations in dimensions 2 to 6	500
Two unit intersection box checks in dimensions 1 to 4	892
Original frame escape cutoffs	1,000
Expanded hull false positive regression	1
Nearly parallel exceptional separation	1 000 001 000 000

The final row is a norm, not a number of trials. Randomized finite cases use the fixed seed 20261003. Bounded direct traces are compared only inside their tested windows; they are not used to infer nonoccurrence beyond those windows. Huge ray shifts are tested by exact integer arithmetic.

The replay entrypoint verifies an independently fixed required inventory and the pinned scientific source digest before running any packaged Python helper. The full manifest checks every payload byte and rejects extra files and symlinks. Tests execute in fresh external temporary copies, in both normal and optimized modes. Their receipts are checked recursively with exact value *and type* equality against the expected schema, so booleans cannot masquerade as integer counts. The original scientific payload is checked before and after for identical bytes, permission modes, and modification times. Outputs are written outside the release directory.

A tamper regression exercises changed and resealed scientific code, missing or extra payloads, and symlink substitution. This is accidental corruption and dependency substitution protection around a trusted entrypoint, not cryptographic authentication against an adversary who replaces the entire package and all trust anchors. The ZIP hash must be obtained from a trusted distribution if authenticity matters.

Run `./reproduce.sh -output-dir ../report29-replay` from the extracted release, after inspecting the script if desired. Run `./build.sh ../report29-build` to rebuild the PDF into an external directory using pdfLaTeX. The README gives dependencies, scope, and exact output semantics. Copyrighted literature reading copies are excluded.

A fresh independent arithmetic audit additionally checks 4,335 exact contact comparisons (including zero drift), 2,172 finite control seed and time invariances, 7,040 nonparallel solution recoveries, 58,800 small launch bounds, and 28,800 indices beyond quadratic cutoffs. It also rejects

a remaining coordinate inconsistency and a bounding hull false positive, and exercises a distinct nearly parallel exception of norm 1 000 002 000 000. Its delivered copy changes only the proof binding hash to that of the sanitized proof copy; the arithmetic source and expected values are otherwise unchanged.

The proof was independently inspected as ordinary mathematics. The arithmetic evidence strengthens confidence in the contact and cutoff calculations; it does not constitute a formal proof, a general rule to section compiler, or a complete end to end simulation test for every promised input.

[write] **The release in this report.** These are the delivered names. `build.sh`, `reproduce.sh`, `replay.py`, `tamper_regression.py`, `scientific/audit_arithmetic.py`, `scientific/check_independent_arithmetic.py` and `scientific/timed/check_corollary.py` are shipped as `code/19-mass-four-zd-*` (the last as `code/19-mass-four-zd-timed-check_corollary.py`); the four `verification/*.json` records as `data/19-mass-four-zd-*`; and `scientific/PROOF.md`, `audit.md`, `independent-review.md`, `literature.md`, `timed/PROOF.md` and `timed/REVIEW.md` as `19-mass-four-zd-*.md` at the report root. Not shipped: the manuscript (printed here), its PDF, README and checksum ledger. The replay entry point checks an exact inventory and pinned hashes under the delivered names, so it runs only in the delivered layout (README). At placement on Windows, `replay.py` passed in normal and optimized mode (receipts typed-equal to the expected ones) and `tamper_regression.py` rejected every tamper case.

[write] **The programme’s review.** The Hilbert’s-tenth programme reviewed this archive’s placed proof in `review_dimension_independent_four_mass.md`, with its checker and receipt, in `Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/` (commit 7b547f686, 3 October 2026) [58]: the proof “passes this mathematical review”. It read the complete proof, including the no-unit case and the absolute-frame observation cutoff, and a separate reviewer of the proof and the new checker found no gap. Its new checks are arithmetic fixtures, not a cellular-automaton analyzer: support contact against 216 targets of a three-dimensional packet profile, 125 ray translations, the residue and transverse decomposition, a marker missed despite movement toward it, six nonparallel pairs on 1,470 right sides against a bounded search (with a parallel opposing pair rejected), and three counter cycles. It executed none of the archive’s Python. It states that the hypotheses are necessary to the argument, that the five-particle upper bound is inherited from separate source and compiler proofs which it does not reprove, and that no arithmetic-circuit lower bound or smaller universal polynomial follows.

117 Further questions

Several next steps would add information without changing the present theorem.

1. Give a precise comparison theorem between the CA encounter normal form and classical two pebble machines. The reduction must preserve splitting, fusion, geometric tags, and observation semantics, rather than merely count objects.
2. Implement and verify the finite rule to encounter library compiler for bounded small rules. The present arithmetic audit does not implement that construction. A small verified compiler would test the integration points between exact local evolution, first contacts, and section dispatch.
3. Quantify state library sizes, determinant exceptions, and Presburger formula growth as functions of dimension, radius, alphabet, and weights. The proof makes no complexity claim; the nearly parallel example excludes especially naive small geometric bounds.

4. Classify which relaxations of unit uniqueness already allow undecidable fixed mass dynamics. Multiple labels can provide independent walker control, but that observation alone is not a complete conservative CA construction at a specified mass. ([write] Added 3 October 2026: open. In $d = 1$ Part V shows that many weight-one types already allow undecidable pattern occurrence at mass three (Theorem 49.2); the least relaxation is open, as source 17’s Question 5 also says.)
5. Optimize the input dependent canonical timed certificates of Appendix I, particularly the spatial sorting and fixed pattern refinement costs, without confusing those bounds with a uniform arity theorem over arbitrary inputs. ([write] Added 3 October 2026: open. Source 18’s selector elimination (Section 96) saves one witness per chart family, and its four-witness binary certificate (Section 98.4) shows what shared parameters can save in one explicit family; neither is a general optimization.)
6. Formalize the dimer invariance, first contact arithmetic, nonparallel exception, and section termination lemmas in a proof assistant. The currently supplied evidence is mathematical exposition and finite executable checking.

Source 20 (batch-82 manuscript 17): Sparse orbit geometry and exact counting

Semilinear spatial sets, quadratic clocks, and the effects of drift, “Research report 30”, 3 October 2026. [write] From the archive `Sparse_Orbit_Geometry_and_Exact_Counting_Package.zip`; its labels carry the prefix `smc:og:`, its sections are Sections 118–128 here, and its appendix is Appendix K, after source 19’s. It builds on source 19 (“Report 29”), which it pins by SHA-256. The Hilbert’s-tenth programme made a scoped intake of its original-frame certificate (note in Appendix K). Its abstract follows; the bracketed note in it is [write].

For a fixed finite input of conserved mass at most four, the orbit classification in Report 29 [source 19] has quantitative geometric consequences. In the frame fixing the unique possible unit symbol, the set of distinct occupied sites is effectively semilinear and lies in finitely many translates of one rational subspace of rank at most two. Centered box counts are eventually quasipolynomial of degree zero, one, or two. An expanding shuttle has sharp degree $r = \text{rank}_{\mathbb{Q}}\{E, \nu\}$, discovery count $\alpha c^{-r/2} T^{r/2} + O(T^{(r-1)/2})$, and first arrival times piecewise rational quadratic in the site coordinates. We retain the exact quadratic clock and its floor oscillations, prove inverse estimates, and give a stronger exact rank one inverse theorem. A fully specified radius six weighted cellular automaton realizes a planar wedge. Restoring a nonzero unit drift can destroy both semilinearity and eventual box quasipolynomiality: the same example has an exact floor root count $3N - 3\sqrt{N} + O(1)$. A third direction is genuinely needed for another drifted realization, even against finite unions of bounded width planes. A constructive appendix gives ordinary degree at most four Diophantine certificates with exactly one natural witness tuple for each stationary frame visited site and each reachable canonical complete configuration in either frame. The stationary first time uses one shared cycle variable; full original frame targets use whole orbit injectivity. The mathematical proofs are separated from finite computational checks. The report uses existing Presburger counting theorems and states its close relationship to classical two pebble work; no novelty, binary realization, or reversibility claim is made.

118 Scope and the inherited classification

Let F be a deterministic, translation equivariant, finite radius cellular automaton on $\mathbb{Z}^d$, for a finite $d \geq 1$. Its finite alphabet A has a unique weight zero vacuum 0, positive integer weights on every other symbol, and at most one symbol u of weight one. Assume that F preserves the total weight

of every finite configuration. Fix a finite input c_0 of mass at most four. The vacuum is quiescent. These are exactly the promise hypotheses of the lower mass classification in Report 29 [source 19, Part VII].

If u exists, its isolated evolution is a single unit translated by a fixed vector $\delta \in \mathbb{Z}^d$. With the convention $\tau_v c(x) = c(x - v)$, put

$$G = \tau_{-\delta} F, \quad F^t = \tau_{t\delta} G^t. \quad (118.1)$$

Then G fixes isolated units. If no unit symbol exists, use $G = F$ and $\delta = 0$. All unlabeled counts below count the union of occupied sites, regardless of the state occupying a site on a particular visit.

Definition 118.1. The stationary frame spatial set, centered box count, and time discovery count are

$$\begin{aligned} S &= \bigcup_{t \geq 0} \text{supp}(G^t(c_0)), \\ C(N) &= \#(S \cap [-N, N]^d), \quad N \in \mathbb{N}, \\ V(T) &= \# \bigcup_{0 \leq t \leq T} \text{supp}(G^t(c_0)), \quad T \in \mathbb{N}. \end{aligned}$$

Each site contributes once. These are not counts of visits, occupied site and time pairs, or phase witnesses. Write S_F and C_F for the corresponding original frame quantities.

118.1 Exactly what is used from Report 29

The present report does not repeat the decision theorem or the full proof that a sparse CA has the following classification. It uses the effective orbit construction of Report 29, Sections 2–12 [source 19, Sections 102–112], particularly the fixed line reduction, additive scalar gap, and complete intermediate phase description. After a computable finite prefix, precisely one of the following forms is supplied:

- A. Independent finite phase objects. Taking a common period, each occupied coordinate is one of finitely many $p + kD_i$, $k \in \mathbb{N}$. In the stationary unit frame there are at most two nonzero drift directions: a genuinely mobile object costs mass at least two.
- B. A whole configuration periodic up to translation. Its occupied coordinates are finitely many rays $p + kD$ with one common drift.
- C. An expanding shuttle. A repeated finite control cycle has scalar gap, anchor, and exact section times

$$x_n = x_0 + n\Delta, \quad L_n = L_0 + nE, \quad T_n = T_0 + cn^2 + bn, \quad (118.2)$$

where $\Delta > 0$, $c > 0$, and all data are effective. Section times are integral and strictly increasing; b, c may be rational. A fixed primitive integer vector $\nu \neq 0$ gives the common unoriented flight line. Every edge, encounter prefix, packet period phase, label, and within cycle time offset is retained.

The source writes $T_n = T_0 + An(n-1)/2 + Cn$, so here $c = A/2$ and $b = C - A/2$. This is a change of notation, not an extra timing assumption. The exact affine within cycle offsets used later also come from those retained phases. Knowledge only of an unordered spatial union and $T_n \sim cn^2$ would not suffice for the first arrival assertions.

The entire dependency is fixed input. “Effective” means that a finite description can be constructed from the promised rule and this input. It does not mean a small description, a polynomial time algorithm, an implemented general classifier, or a single uniform timed Presburger description covering arbitrary inputs.

118.2 Main conclusions

A rational affine plane is a translate of a real subspace spanned by integer vectors. Its rank is the dimension of that subspace. A set has finite width in such a plane if it is contained in the plane plus a fixed bounded set.

Theorem 118.2 (Stationary frame geometry and counting). *Under the stated hypotheses, the following descriptions are effective.*

1. *S is semilinear, equivalently Presburger definable. Its characteristic generating function is rational after partitioning $\mathbb{Z}^d$ into disjoint sign orthants and reflecting each orthant into $\mathbb{N}^d$.*
2. *S lies in finitely many exact affine translates of one rational subspace of rank at most two, hence in a finite width rational affine plane of rank at most two.*
3. *For a nonempty orbit, $C(N)$ is eventually quasipolynomial of degree zero, one, or two. Degree zero is equivalent to S being finite. If S is infinite, then*

$$C(N) = \kappa N^r + O(N^{r-1}), \quad \kappa \in \mathbb{Q}_{>0}, \quad r \in \{1, 2\}. \quad (118.3)$$

4. *Counts in boxes with independently variable integer lower and upper endpoints are piecewise quasipolynomial. Any fixed affine one parameter family of finite integer boxes consequently has an eventual quasipolynomial count of degree at most two, possibly zero or of smaller degree.*

For independent or translated periodic tails the degree is one if any occupied ray has nonzero drift, and zero otherwise. For an expanding shuttle it is exactly $r = \text{rank}_{\mathbb{Q}}\{E, \nu\}$. The empty orbit has identically zero count.

A rank two containing plane does not force degree two: two nonparallel independent rays already need such a plane but have only linearly many sites in a box.

Theorem 118.3 (Expanding shuttle timing). *For an expanding shuttle, let $D(K)$ count the distinct sites in the fixed prefix and the first K complete eventual cycles. Use the half open cycle intervals $[T_n, T_{n+1})$. Then D is effectively eventually quasipolynomial and*

$$D(K) = \alpha K^r + O(K^{r-1}), \quad \alpha \in \mathbb{Q}_{>0}, \quad r = \text{rank}_{\mathbb{Q}}\{E, \nu\}. \quad (118.4)$$

Consequently

$$V(T) = \alpha c^{-r/2} T^{r/2} + O(T^{(r-1)/2}). \quad (118.5)$$

The first visit time $\tau(y)$ on S is effectively piecewise rational polynomial of degree at most two on a finite semilinear partition. For visited sites escaping every bounded set,

$$\tau(y) = \Theta(\|y\|_{\infty}^2). \quad (118.6)$$

The largest first visit time of a site in $S \cap [-N, N]^d$ is $\Theta(N^2)$.

The last assertion is coverage of the sites this particular orbit eventually visits, not coverage of the entire lattice box. The count $V(T)$ need not be eventually quasipolynomial in ordinary time.

Theorem 118.4 (Drift restoration). *For an expanding shuttle, S_F is contained in finitely many affine translates of $\text{span}_{\mathbb{R}}\{E, \nu, \delta\}$, of rank at most three. If $\delta \neq 0$, then $C_F(N) = \Theta(N)$. Nevertheless S_F need not be semilinear and C_F need not be eventually quasipolynomial. These failures and rank three sharpness occur for explicit finite range, globally weight conserving CAs with one unit symbol and a mass four input. Independent and whole translated periodic tails remain semilinear in the original frame.*

119 Complete phase coordinates and the common plane

At edge i of cycle n , the two geometrically designated markers have coordinates

$$L_0 + nE + e_i, \quad L_0 + nE + e_i + (x_0 + n\Delta + d_i)\nu + z_i.$$

Here e_i, d_i, z_i are fixed data in the finite mode. The symbols need not distinguish the markers: their names are bookkeeping roles. A packet emitted from either marker begins at a bounded offset from it and has a fixed period drift $m_i\nu$, with nonzero integer m_i . Its retained phase sites have the form

$$y = p_i + n(E + \varepsilon_i\Delta\nu) + jm_i\nu, \quad \varepsilon_i \in \{0, 1\}. \quad (119.1)$$

The integer domain is a finite conjunction or finite union of linear inequalities and congruences in n, j . After splitting finitely many residues and boundary phases, it has $0 \leq j \leq a_in + b_i$. Each genuine growing full flight phase has $a_i > 0$. Constant offsets and initial gap contributions are incorporated in p_i . Encounter prefixes and marker positions have the same affine form without j .

Thus all these points belong to $p_i + \text{span}_{\mathbb{R}}\{E, \nu\}$. There are finitely many p_i . Any finite prefix is included by adding one translate per exceptional point. This proves the common subspace assertion, including exact finite translates rather than only a tube estimate.

Proposition 119.1 (Segment confinement and pointed cone). *Every support in cycle n is within a fixed distance of the segment joining*

$$L_0 + nE \quad \text{and} \quad L_0 + n(E + \Delta\nu).$$

If E and ν are independent, the eventual site set lies in a fixed width neighborhood of the pointed rational cone generated by E and $E + \Delta\nu$. Its opening in their plane is strictly less than 180 degrees.

Proof. A successful long flight begins within bounded distance of one marker and terminates within bounded distance of the other. Its period samples progress on the common line by a fixed nonzero multiple of ν . The retained within period offsets are bounded. These facts confine the whole flight to bounded distance from the joining segment, even if individual microscopic steps do not move monotonically. Every local encounter prefix has bounded duration and bounded relative support. Marker positions at the finitely many edges differ from the section markers by bounded vectors. These bounds are uniform in n .

If E, ν are independent, so are $E, E + \Delta\nu$. A real linear functional can take value one on both generators. Their positive cone is therefore pointed and lies in an open half plane of their span. If E is parallel to ν , the cone can degenerate to a ray or a line; only rank one is asserted there. $\square$

For an independent tail, stationary units add offsets but no drift. Every genuinely mobile independent component costs at least mass two, so at most two drift vectors occur. Taking their span proves the common rank two bound. If there is no unit symbol, at most two occupied sites exist and the classification supplies the same independent alternative. For a whole translated periodic tail, all phases use one drift vector. These observations cover terminal missed flights, independent dimers, zero drift periodic objects, and stationary configurations.

120 Semilinearity and exact distinct site counts

A subset of $\mathbb{Z}^d$ is semilinear if it is a finite union of sets

$$a + \mathbb{N}v_1 + \cdots + \mathbb{N}v_s, \quad a, v_i \in \mathbb{Z}^d.$$

Presburger arithmetic permits integer addition, order, congruences, Boolean operations, and quantifiers, but not products of variables. Its definable sets are precisely the semilinear sets.

120.1 Membership before enumeration

For each phase, existentially quantify n, j in its integer domain and impose equation (119.1). Disjoin all phases, marker and encounter lists, and finite prefix points. This gives an effective Presburger formula for $y \in S$. Independent rays have the same property.

The counted free variable must be y . Different cycles, periods, states, or phases can describe the same site. Summing the sizes of their parameter domains would count multiplicities and can change both coefficients and growth degree. Existential site membership removes them before counting.

Woods' Theorem 1.5 [55] supplies the standard equivalence with rational characteristic generating functions. For signed coordinates, use a disjoint orthant partition and reflection into $\mathbb{N}^d$, or the injective encoding

$$y \mapsto (y_1^+, y_1^-, \dots, y_d^+, y_d^-), \quad y_i^+ = \max(y_i, 0), \quad y_i^- = \max(-y_i, 0).$$

An unrestricted two sided Laurent sum may have incompatible expansions or no common domain of convergence. No such unspecified formal series is needed.

120.2 Existing Presburger counting theorem

The formula

$$y \in S \quad \wedge \quad -N \leq y_i \leq N \quad (1 \leq i \leq d)$$

has finite fibers in the spatial variable y . Woods' Theorem 1.10 gives an effective piecewise quasipolynomial count in N . In one nonnegative parameter, finitely many rational polyhedral pieces leave finitely many exceptional bounded intervals and eventual residue polynomials. This is eventual quasipolynomiality. With independently variable endpoints, the same theorem gives a multivariate piecewise quasipolynomial.

These are applications of existing counting results, not new general theorems about Presburger arithmetic. Their use is effective without a complexity promise. The special work here is extracting the correct finite fiber family and proving its sharp degree.

120.3 Sharp degree and common leading coefficient

Proof of the counting assertions in Theorem 118.2. A rational rank r affine plane has $O(N^r)$ lattice points in $[-N, N]^d$. Choose r coordinate projections independent on its direction space; on an exact affine plane these determine at most one point. Finitely many parallel translates still give $O(N^r)$. For finite width, bounded integer transverse offsets give the same estimate. Thus an expanding tail has $C(N) = O(N^r)$ for $r = \text{rank}_{\mathbb{Q}}\{E, \nu\}$.

Fix one genuine growing flight phase, present on all sufficiently large cycles or on one fixed positive density residue class. Its period parameter has at least $an - b$ values, with $a > 0$. For a sufficiently small constant $\eta > 0$, the sites with $n \leq \eta N$ all lie in the radius N box once finite exceptions are removed.

If $r = 2$, the vectors $E + \varepsilon \Delta \nu$ and $m\nu$ are independent. Hence $(n, j) \mapsto y$ is injective within this single phase. There are $\Theta(N)$ admissible cycle values and $\Theta(n)$ positions in cycle n , yielding $\Omega(N^2)$ distinct sites.

If $r = 1$, that two parameter map need not be injective. Instead choose just one admissible cycle n within a bounded distance of ηN . Its $\Theta(N)$ period values give distinct sites because $m\nu \neq 0$. This proves $\Omega(N)$ without intercycle injectivity.

Every eventual residue polynomial therefore has exact degree r . Its leading coefficient is common across residues: monotonicity $C(N) \leq C(N+1)$ orders consecutive coefficients, and following the whole residue cycle forces equality. The common coefficient κ is positive and rational. Lower order terms are uniformly $O(N^{r-1})$.

For independent and translated periodic tails, each nonconstant ray contributes $\Theta(N)$ sites, whereas constant rays contribute finitely many. This proves their degree claims and the finite set characterization of degree zero. $\square$

In particular, the ordinary generating function $\sum_{N \geq 0} C(N)z^N$ is rational. For some effective period M , after accounting for finitely many initial terms its denominator divides $(1 - z^M)^{r+1}$.

120.4 Why multibox chambers matter

For the basic wedge $\{(n, j) : n \geq 0, 0 \leq j \leq n\}$, the centered rectangle count equals

$$\begin{cases} (N_1 + 1)(N_1 + 2)/2, & N_1 \leq N_2, \\ (N_2 + 1)(N_2 + 2)/2 + (N_1 - N_2)(N_2 + 1), & N_1 \geq N_2. \end{cases}$$

The chamber boundary is real; one global multivariate quasipolynomial is not generally available. Along $N_2 = 0$ the degree drops to one. Translated or degenerate boxes can miss every unbounded component. A positive centered rational dilation in every coordinate preserves the sharp degree, after resolving floors by residues. A general affine family guarantees only the upper degree bound.

121 Quadratic clocks and discovery inverses

Constrain the phase witnesses by $0 \leq n < K$ to count all sites through complete cycles. The free variable remains y , so $D(K)$ is a finite fiber Presburger count. All its positions have norm $O(K)$ in the common rank r subspace. The same growing phase lower bounds prove $D(K) = \Theta(K^r)$. Monotonicity again forces one positive rational leading coefficient α across residues, proving (118.4).

In particular,

$$D(K+1) - D(K) = O(K^{r-1}). \quad (121.1)$$

For rank one, an entire late cycle discovers only boundedly many new sites even though its shuttle makes $\Theta(K)$ visits along a long flight.

121.1 Retaining the clock exactly

For sufficiently large T , let $n(T)$ be the unique cycle with $T_{n(T)} \leq T < T_{n(T)+1}$. Then

$$n(T) = \left\lfloor \frac{-b + \sqrt{b^2 + 4c(T - T_0)}}{2c} \right\rfloor = \sqrt{T/c} + O(1). \quad (121.2)$$

The discriminant is nonnegative in the eventual range. The increasing quadratic on nonnegative cycle indices makes the indicated root the appropriate one.

Since $D(n(T)) \leq V(T) \leq D(n(T) + 1)$, substitution in (118.4) gives (118.5). Thus rank one has square root discovery with bounded error, and rank two has linear discovery with error $O(\sqrt{T})$. Merely knowing $T_n \sim cn^2$ would prove the leading asymptotic equivalence, but the displayed error requires $T_n = cn^2 + O(n)$.

There is an exact description stronger than this asymptotic. For current cycle n and time offset s , count the distinct sites from the prefix, earlier cycles, and current phase occurrences with their affine elapsed time at most s . This is a Presburger finite fiber count $Q(n, s)$, hence effectively piecewise quasipolynomial. Consequently

$$V(T) = Q(n(T), T - T_{n(T)}). \quad (121.3)$$

The assertion of piecewise quasipolynomiality belongs to Q , not to its composition with the quadratic floor root. Section boundaries and within cycle oscillations have not been discarded.

121.2 Spatial and cycle inverses

For an infinite spatial set define

$$R(M) = \min\{N : C(N) \geq M\}, \quad J(M) = \min\{K : D(K) \geq M\}.$$

Comparing $\kappa N^r + O(N^{r-1})$ with the level M gives

$$R(M) = (M/\kappa)^{1/r} + O(1), \quad J(M) = (M/\alpha)^{1/r} + O(1). \quad (121.4)$$

Indeed a fixed sufficiently large shift of $(M/\kappa)^{1/r}$ changes the leading polynomial by enough to dominate the uniform $O(N^{r-1})$ error.

These inverses are also exactly effective. On each eventual residue $N = u + Pq$ for a fixed period P , substitute into the corresponding degree one or two polynomial for $C(N)$, solve the eventual increasing linear or quadratic inequality, and minimize over the finitely many residues and finite exceptional values. This produces finitely many rational affine ceiling formulas in rank one or quadratic root ceiling formulas in rank two. In rank one R is eventually quasi affine. In rank two its square root growth rules out eventual quasipolynomiality.

For discovery of M distinct sites, let

$$H(M) = \min\{T : V(T) \geq M\}.$$

Monotonicity and (118.5) give

$$H(M) = \begin{cases} (c/\alpha^2)M^2 + O(M), & r = 1, \\ (c/\alpha)M + O(\sqrt{M}), & r = 2. \end{cases} \quad (121.5)$$

An exact rank one improvement is available because of the bounded number of new sites per cycle.

Proposition 121.1 (Quadratic rank one discovery inverse). *If $r = 1$, then $H(M)$ is effectively eventually quasipolynomial of degree exactly two, with common leading coefficient c/α^2 . Accordingly $V(T)$ has an eventual exact description using a finite maximum of quadratic floor root formulas, without becoming a quasipolynomial in T .*

Proof. By (121.1), some fixed integer B bounds the number of genuinely new sites in every sufficiently late cycle. The eventual graph of $D(n)$ is Presburger because D is quasi affine. For each fixed $1 \leq \ell \leq B$, “at least ℓ new sites in cycle n have appeared by offset s ” is Presburger: use ℓ pairwise distinct site variables, occurrence by s , and absence from the fixed prefix and every earlier cycle. A finite disjunction over ℓ defines the earliest cycle $n(M)$ and earliest offset $s(M)$ at which the count reaches M . This also handles cycles that introduce no new site.

These are one parameter Presburger functions, hence eventually quasi affine. An elementary proof of the underlying piecewise affine fact appears in Lemma 122.1. Substitution in

$$H(M) = T_0 + cn(M)^2 + bn(M) + s(M)$$

gives an eventual quadratic quasipolynomial. Equation (121.5) supplies its common leading coefficient. Solving each eventual residue polynomial inequality $H(M) \leq T$ and taking the largest admissible M gives the stated floor root description of V . $\square$

122 First arrival times and coverage

The spatial union alone does not tell us when a site first appears. The complete phase form supplies the missing timing information. Let $R(y, n, h)$ be the Presburger relation asserting that site y is occupied at offset h in the half open cycle n . The offset is affine in the phase parameters: preceding edge durations are affine in n , and a packet period contributes its fixed period length times j plus a retained phase offset.

Exclude the finite set of sites first seen before T_0 , treating those separately. Define $n_*(y)$ as the least cycle containing y , then $h_*(y)$ as the least offset in that cycle. Their graphs are Presburger: impose occurrence and negate any occurrence at an earlier cycle, or at an earlier offset in the same least cycle. Minimization is legitimate here because the cycle and time domains are nonnegative integers and every y in the domain is eventually visited.

[write] [The next lemma is classical: Presburger-definable functions have semilinear graphs [7] and are therefore piecewise rational affine. Source 20 proves it directly; no novelty is claimed for it.]

Lemma 122.1 (Presburger functions are piecewise rational affine). *A Presburger definable function with integer input and output is rational affine on each part of an effective finite semilinear partition of its domain.*

Proof. Decompose its graph into finitely many linear sets. On one such component write the generators as (v_i, w_i) , with v_i in the input coordinates and w_i in the output coordinates. Every

integer relation $\sum_i a_i v_i = 0$ also satisfies $\sum_i a_i w_i = 0$: separate positive and negative coefficients to obtain two points of this graph component with the same input, then use functionality. Therefore $v_i \mapsto w_i$ extends to a rational linear map on the rational span of the v_i . Adding the base point makes the function rational affine on the component's domain. A finite Boolean refinement of the semilinear domains yields a partition. All steps are effective. $\square$

This argument allows lower dimensional input pieces and several output coordinates. It asserts no uniqueness of the semilinear parameter witnesses.

By the lemma, both $n_*(y)$ and $h_*(y)$ are piecewise rational affine. Refining their partitions jointly gives

$$\tau(y) = T_0 + cn_*(y)^2 + bn_*(y) + h_*(y) \quad (122.1)$$

as a rational polynomial of degree at most two on every cell. Earlier prefix sites contribute singleton cells. This proves the exact effective statement in Theorem 118.3.

The source support bound $\|y\|_\infty \leq K_0 + K_1 n$ implies that the earliest cycle of a distant site is at least a positive constant times $\|y\|_\infty$ minus a constant. Since section times are quadratic, $\tau(y) = \Omega(\|y\|_\infty^2)$. Conversely, finitely many affine formulas for $n_*(y)$ give $n_*(y) = O(1 + \|y\|_\infty)$; the within cycle duration is $O(1 + n)$. Hence $\tau(y) = O(1 + \|y\|_\infty^2)$. This proves (118.6).

To obtain box coverage time, the upper bound applies uniformly to sites of norm at most N . For the lower bound, choose fixed $0 < \varepsilon < 1$. Equation (118.3) gives $C(N) - C(\lfloor \varepsilon N \rfloor) > 0$ for all sufficiently large N , so some visited site in the box has norm at least εN . Its first arrival takes $\Omega(N^2)$ time.

Remark 122.2. Piecewise quadratic first arrival does not imply that the full space and time orbit is Presburger. The relation $t = cn^2 + bn + T_0$ contains a genuine quadratic clock. A piecewise quadratic formula by itself is not yet a canonical Diophantine certificate. Appendix K supplies the additional disjoint clause construction, signed quotient encoding, and uniqueness proof.

123 A globally conservative planar wedge automaton

We now give actual CAs realizing the geometric phenomena. Their unrestricted conservation is proved directly, rather than inferred from an abstract shuttle schedule or from finite simulations.

Work on $\mathbb{Z}^2$ with alphabet $\{0, u, E, W\}$ and weights

$$w(0) = 0, \quad w(u) = 1, \quad w(E) = w(W) = 2.$$

The letters E, W denote internal head states. There is only one unit symbol. Let $e_1 = (1, 0)$, $e_2 = (0, 1)$. A head at a is eligible if there is no other head within maximum norm distance four. Its action reads the old configuration. Any failed guard gives no rewrite. Every site not explicitly written stays unchanged.

Proposition 123.1 (Full rule validity). *Simultaneously applying Table 11 at eligible heads defines a translation equivariant radius six CA. It preserves total weight on every finite configuration, including configurations with many heads, obstructed destinations, and extra unit symbols. Its vacuum is quiescent and its isolated unit is stationary.*

Proof. Rules 1 and 2 have mutually exclusive forward guards; so do rules 3 and 4. All writes lie in $a + [-2, 2]^2$. Distinct eligible heads are more than four apart, so their write cubes are disjoint. No

Rule	Required old symbols	New symbols at written sites
1	E at a ; 0 at $a + e_1$	0 at a ; E at $a + e_1$
2	E at a ; u at $a + e_1$; 0 at $a + 2e_1 + e_2$	W at a ; 0 at $a + e_1$; u at $a + 2e_1 + e_2$
3	W at a ; 0 at $a - e_1$	0 at a ; W at $a - e_1$
4	W at a ; u at $a - e_1$; 0 at $a + e_2$ and $a - e_1 + e_2$	0 at a and $a - e_1$; E at $a + e_2$; u at $a - e_1 + e_2$

Table 11: The complete planar rewrite rule, in addition to the head isolation condition. All vacancy tests are part of the definition.

eligible rewrite can overwrite an ineligible head: a head at a written site would be within distance two and would itself disqualify the candidate. Each listed rewrite preserves the exact weight sum on its written sites. Every newly filled site is explicitly required to be vacant unless it is the head's own rewritten site. In particular no additional unit is silently erased.

For locality, the output at x only needs to consider heads within distance two of x and their radius four isolation tests. All this lies within distance six of x . The listed destination guards are within the same radius six neighborhood. If no action writes x , retain its old value. This is a well defined finite radius local rule. Disjoint, weight preserving rewrites prove conservation for all finite inputs. Vacuum and an isolated unit have no eligible head action and remain unchanged. $\square$

123.1 Sections and the exact spatial wedge

Fix $k \geq 3$. Start with units at $(0, 0)$ and $(k, 0)$ and an E head at $(1, 0)$. There is exactly one head forever, so its eligibility test always passes. Inductively the section at cycle n is

$$L_n = (0, n), \quad R_n = (k + n, n), \quad E \text{ at } (1, n). \quad (123.1)$$

Put $K = k + n$. There are $K - 2$ eastward free steps, one right bounce by rule 2, $K - 2$ westward free steps, and one left bounce by rule 4. The duration is $2(K - 1)$, so

$$T_n = n^2 + (2k - 3)n. \quad (123.2)$$

The right bounce moves the right unit to $(k + n + 1, n + 1)$. The left bounce moves the left unit and head up by one, yielding the next section. All guards hold along this intended orbit.

Every row n eventually has exactly the visited horizontal coordinates $0, 1, \dots, k + n$. The early arrival of the right unit on row $n + 1$ adds only that next row's endpoint. Therefore the entire occupied site union is exactly

$$S = \{(x, y) \in \mathbb{Z}^2 : y \geq 0, 0 \leq x \leq k + y\}. \quad (123.3)$$

This realizes $E = e_2$, $\nu = e_1$, $\Delta = 1$, $c = 1$, and rank two. No special rare label or section subsequence is being counted.

For $N \geq k$, direct summation gives

$$\begin{aligned} C(N) &= \sum_{y=0}^N (\min(N, k + y) + 1) \\ &= \frac{N^2 + (2k + 3)N + 2 + k - k^2}{2}. \end{aligned} \quad (123.4)$$

Thus $\kappa = 1/2$ and the density in centered square boxes is $1/8$. A finite union of bounded width lines has only $O(N)$ lattice sites there, so rank two is sharp even against finitely many lines. The wedge does not explore a half plane.

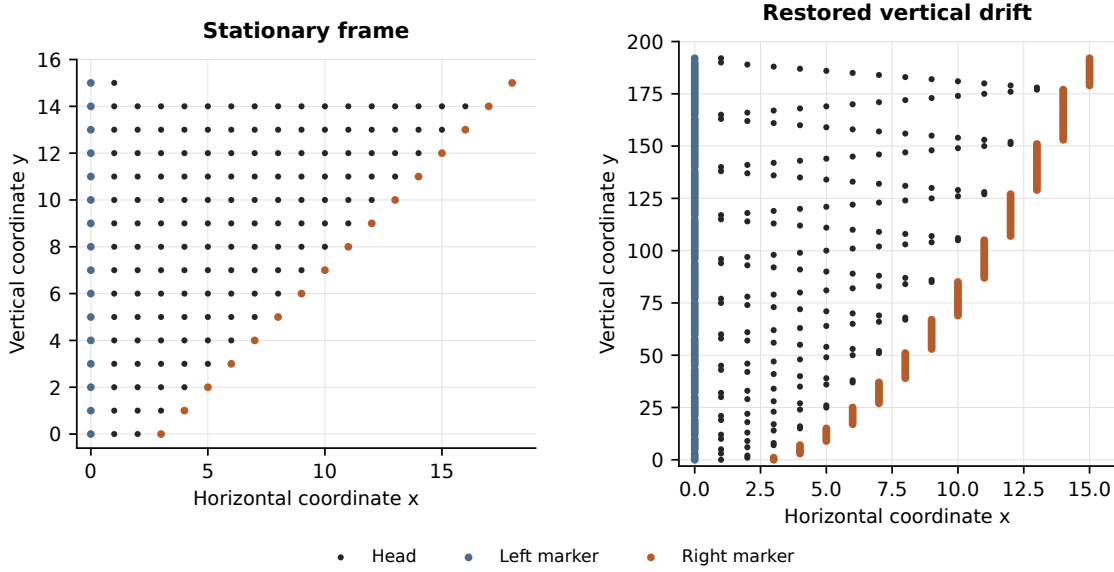


Figure 7: Actual occupied site unions from the authored local rule simulator, with $k = 3$. Left: the stationary frame wedge through section $n = 15$. Right: the same planar rule with a vertical shift after every step, through time 180. Colors distinguish the persistent left marker, head, and right marker in this intended one head orbit; the displayed sets are not an abstract flight schedule.

123.2 Exact discovery and first arrival for this example

At the section time T_n , the number of sites already visited is

$$B_n = \frac{n(n-1)}{2} + (k+1)n + 3. \quad (123.5)$$

Rows below n are complete and row n contributes its two left positions and right endpoint. For $0 \leq h < 2(k+n-1)$,

$$V(T_n + h) = B_n + \min(h, k+n-2) + \mathbf{1}_{\{h \geq k+n-1\}}. \quad (123.6)$$

The eastward flight discovers its interior one site per step. The right bounce discovers one next row endpoint. The westward flight revisits old sites, and the final left bounce creates two sites counted at the next section. Formula (123.6) exposes the within cycle oscillation hidden in $V(T) = T/2 + O(\sqrt{T})$.

For row $n \geq 1$, first arrival is

$$\tau(x, n) = \begin{cases} T_n, & x = 0 \text{ or } 1, \\ T_n + x - 1, & 2 \leq x \leq k+n-1, \\ T_n - (k+n-2), & x = k+n. \end{cases} \quad (123.7)$$

The last line is the preceding cycle's right bounce. On row zero, the three initial sites have time zero and each other $2 \leq x \leq k-1$ has time $x-1$. These formulas show explicitly the finite semilinear partition and rational quadratic first arrival description.

124 Rank one behavior and nonsemilinear drift

Degree zero occurs for one stationary unit. Degree one occurs for an isolated E head moving east under the preceding rule. These inputs have mass one and two, respectively.

For an expanding rank one example, change the planar rule in exactly two places:

1. In rule 2, change both the vacant destination guard and the new unit destination from $a + 2e_1 + e_2$ to $a + 2e_1$.
2. Replace rule 4 by: if W is at a and u is at $a - e_1$, change the head in place to E and leave that neighboring unit unchanged. Remove the former upward destination vacancy tests, since no new site is filled in this action.

The isolation convention and rules 1 and 3 are unchanged. This is a complete guard and write specification. The same disjoint rewrite proof establishes a radius six globally conservative CA.

From the same input, sections become $L_n = (0, 0)$, $R_n = (k + n, 0)$, with E at $(1, 0)$, and retain clock (123.2). Its full visited set is the nonnegative horizontal ray. The exact time count, after the initial interior has been visited, is

$$V(T) = \left\lfloor \sqrt{T + (k-1)(k-2)} \right\rfloor + 3, \quad T \geq k-2. \quad (124.1)$$

Indeed site $x \geq k+1$ is first reached by the right marker at time

$$(x-2)^2 - (k-1)(k-2).$$

Consequently

$$H(M) = (M-3)^2 - (k-1)(k-2), \quad M \geq k+2. \quad (124.2)$$

This directly realizes the rank one inverse theorem, with $\alpha = c = 1$. It also proves that an actual permitted CA can have nonquasipolynomial $V(T)$: an unbounded eventual quasipolynomial cannot grow like $\sqrt{T}$.

Now call this horizontal rule G and set $F = \tau_{e_2} G$. Translation commutes with G , so the isolated unit drift is e_2 and normalization recovers G . The full original frame occupied site set is

$$S_F = \{(x, t) : x \in \text{supp}(G^t(c_0)), t \geq 0\},$$

where the horizontal support is read as a set of x coordinates. Every row contains exactly three sites; all satisfy $x \geq 0$ and $x = O(\sqrt{t})$, yet x is unbounded.

Proposition 124.1. *The full set S_F in this example is not semilinear.*

Proof. Suppose it is a finite union of linear sets, and let (a, b) be a period vector of a nonempty component. Negative b is impossible because arbitrarily large period multiples would enter negative rows. If $b = 0$, finite rows force $a = 0$. If $b > 0$, the uniform sublinear bound on x forces $a = 0$: otherwise along its multiples $|x|/t$ tends to the nonzero ratio $|a|/b$. Hence every period is vertical, and finitely many component base points bound x globally, a contradiction. $\square$

This proof concerns every occupied site, not just special labels or sparse section times. Its nonsemilinearity alone does not rule out an eventual polynomial box count. The next example supplies that separate conclusion.

125 An exact nonquasipolynomial box count

Return to the planar wedge rule of Section 123 and again set $F = \tau_{e_2}G$, now shifting inside the wedge's plane. At time t in cycle n , let $\sigma(t) = 0$ before that cycle's right bounce finishes, and 1 afterward. The heights of the three objects in the original frame are

$$\text{left: } t + n, \quad \text{head: } t + n, \quad \text{right: } t + n + \sigma(t).$$

Every trajectory has strictly increasing height. Usually height increases by one; at its one upward move in each cycle it increases by two. The left marker and head have the same height sequence and different horizontal coordinates.

There are no cross time overlaps with the right trajectory. Before the right bounce it has the head's height at the same time but strictly larger x . After the bounce its height is either the head's next height or, at the left bounce, a missing head height. In the former case the right coordinate is $k + n + 1$, while the head's next coordinate is at most $k + n - 1$. The three entire trajectories are therefore pairwise disjoint, not merely distinct at each instant.

The left marker and head skip the heights

$$a_n = T_n + n - 1 = n^2 + (2k - 2)n - 1, \quad n \geq 1. \quad (125.1)$$

The right bounce finishes at $t = T_n + k + n - 1$, so the right trajectory skips

$$b_n = T_n + k + 2n - 1 = n^2 + (2k - 1)n + k - 1, \quad n \geq 0. \quad (125.2)$$

Each height sequence starts at zero and contains exactly the nonnegative integers outside its displayed skipped sequence. The number of its holes through height N is thus a quadratic inequality count.

For $N \geq 2k + 4$, the horizontal box constraint is inactive at all heights at most N . One way to see this is that the first right marker occurrence at coordinate $k + j$, $j \geq 1$, has height

$$(j - 1)^2 + (2k - 1)(j - 1) + k.$$

If its horizontal coordinate exceeds $N \geq k + 1$, this height exceeds N as well. The head is never farther right than the marker. We retain the convenient conservative threshold $2k + 4$ in the formula and audits.

Define

$$A(N) = \max \left(0, \left\lfloor \sqrt{(k - 1)^2 + N + 1} \right\rfloor - (k - 1) \right), \quad (125.3)$$

$$B(N) = \max \left(0, 1 + \left\lfloor \frac{-(2k - 1) + \sqrt{(2k - 1)^2 + 4(N - k + 1)}}{2} \right\rfloor \right). \quad (125.4)$$

These count (125.1) and (125.2), respectively, with their different starting indices.

Theorem 125.1 (Exact drifted wedge count). *For every integer $k \geq 3$ and $N \geq 2k + 4$, the original frame full occupied site count is*

$$C_F(N) = 3(N + 1) - 2A(N) - B(N). \quad (125.5)$$

In particular,

$$C_F(N) = 3N - 3\sqrt{N} + O(1), \quad (125.6)$$

and C_F is not eventually quasipolynomial. Thus S_F is not semilinear either.

Proof. The three trajectories are disjoint, and their heights are precisely the nonnegative integers minus the displayed holes. For N in the stated range, counting their points in the box is exactly counting these height sequences through N . This gives (125.5). Both $A(N)$ and $B(N)$ equal $\sqrt{N} + O(1)$, proving (125.6).

If C_F were eventually quasipolynomial, its linear growth and $C_F(N)/N \rightarrow 3$ would force every residue polynomial to be $3N$ plus a constant. Then $C_F(N) - 3N$ would be bounded, contradicting the negative square root term. A semilinear S_F would have an eventual quasipolynomial box count by the already cited Presburger theorem, so it too is excluded. $\square$

Formula (125.5) is the exact statement. Its bounded floor oscillations are retained; equation (125.6) is not a floor free expansion to all orders. The inverse box radius satisfies the useful two term estimate

$$\min\{N : C_F(N) \geq M\} = M/3 + \sqrt{M/3} + O(1). \quad (125.7)$$

Substitute $N = M/3 + \sqrt{M/3} + L$ into (125.6); a fixed sufficiently large positive or negative L brackets the level M . Exact inversion remains governed by the floor formula and monotonicity.

126 General drift bounds and rank three sharpness

For any expanding shuttle, add δ times its actual time to (119.1). Actual time is T_n plus an affine function of n, j . Every resulting point therefore lies in a fixed affine translate of

$$\text{span}_{\mathbb{R}}\{E, \nu, \delta\}.$$

There are only finitely many phase offsets. The quadratic degree in n is the source of the extra geometry. Drift restoration is not an affine change of the untimed spatial parameters alone.

In independent and translated periodic tails, actual time is affine in the common period index. Drift restoration simply changes each ray vector, preserving semilinearity.

126.1 Every nonzero drifting shuttle has linear box order

The stationary support norm bound and quadratic clock imply, uniformly over the at most four occupied sites at time t ,

$$y_F(t) = t\delta + O(\sqrt{t}). \quad (126.1)$$

If $\delta \neq 0$, any original frame site in the radius N box must occur at $t = O(N)$. There are at most four occupied sites at each time, so $C_F(N) = O(N)$.

For the matching lower bound, choose fixed small $\eta > 0$ and $T = \lfloor \eta N \rfloor$, so all support through T lies in the radius N box once N is large. Pick an occupied site at time T . Quiescence and finite radius imply that it has an occupied predecessor within a fixed radius at time $T - 1$; otherwise its entire input neighborhood would be vacuum. Repeat to time zero. This produces a bounded jump path of occupied sites contained in the box. A coordinate with $\delta_i \neq 0$ has endpoint displacement $\Theta(N)$ by (126.1). A bounded jump path spanning that displacement contains $\Omega(N)$ distinct sites. Thus $C_F(N) = \Theta(N)$ without assuming labeled particle identities or mistaking occurrences for distinct sites.

126.2 A three dimensional realization

Embed the planar wedge rule in $\mathbb{Z}^3$ using its e_1, e_2 directions and the same radius four head isolation test in the three dimensional maximum norm. The disjoint write and radius six proof is unchanged. Compose with the shift $\delta = e_3$.

At section times the original frame support contains

$$(0, n, T_n), \quad (k + n, n, T_n).$$

If all visited sites stayed within bounded distance of one affine plane of rank at most two, some nonzero real linear functional

$$\ell(x, y, z) = ax + by + cz$$

would stay bounded on them. Its value on the first sequence is $bn + c[n^2 + (2k - 3)n]$, so boundedness forces $b = c = 0$. Its value on the second then forces $a = 0$, a contradiction.

The stronger finite union assertion requires the flights, rather than only those two section sequences. The eastward flight contains

$$(1 + j, n, T_n + j), \quad 0 \leq j \leq k + n - 2. \quad (126.2)$$

On these points,

$$\ell = cn^2 + (b + c(2k - 3))n + (a + c)j + a.$$

If $a + c \neq 0$, a fixed bounded strip for ℓ contains only $O(1)$ of the integer j values in each cycle. If $a + c = 0$, this is a nonconstant unbounded polynomial in n unless $a = b = c = 0$, so the strip contains none of the flight for every sufficiently large n . A finite collection of strips therefore covers only $O(1)$ of its $\Theta(n)$ sites. Every bounded width affine plane of rank at most two lies inside one such strip.

Corollary 126.1. *The rank three upper bound is sharp even if finitely many bounded width affine planes of rank at most two are allowed, with arbitrary orientations.*

The original frame example still has only linear centered box order. Containing rank, growth degree, and semilinearity are distinct properties.

127 Prior work and mathematical interpretation

The geometric picture is close to classical two pebble automata. Lacalle and Gajardo [50], printed pages 2 and 5, describe conical confinement, zigzag coverage, and expanding oscillations versus periodic behavior, attributing core predecessors to Delorme and Mazoyer. Their conclusion on printed page 17 extends their geometric arguments to $\mathbb{Z}^d$. Their principal theorem concerns real time trace recognition, a different statement from the present fixed input site enumeration and weighted CA classification.

Blum and Sakoda [51], printed page 148, discuss sector exploration below a half plane and the inability of one finite automaton with two pebbles to explore an entire half plane. The pointed cone statement here should be read alongside that close predecessor geometry. Delorme and Mazoyer [49] remains a central reference. Its publisher abstract and the explicit account in Lacalle and Gajardo were checked; the complete 2002 text was not retrieved for this report. That limitation supplies no basis for a priority inference.

The explicit CAs use a mass two stateful head and two identical mass one markers. Report 29’s [source 19’s] classification also covers splitting, fusion, weighted symbols, and independent packet tails. Its proof is the required bridge; a general sparse CA should not silently be identified with a pebble machine. Conversely, the semilinearity to counting implication is existing Presburger mathematics [55].

For Diophantine computational substrates, the useful distinction is between untimed spatial definability and the extra quadratic clock. The spatial set admits effective linear arithmetic descriptions and rational generating functions, while first arrival and inverse discovery can be genuinely quadratic. Drift couples that clock into spatial coordinates, creating nonsemilinear sets and square root corrections despite a very small finite conserved mass. This is a concrete separation of descriptive resources, not an undecidability or universality claim.

127.1 What remains open here

1. Determine which of these sharp geometric and drift phenomena admit binary or reversible realizations under the same small mass restrictions. The weighted examples do not answer this. ([write] Added 3 October 2026: answered for binary rules by source 21, whose ordinary binary radius-six rule realizes a planar wedge, with two missing lines, and the drifted square-root regime (Theorems 133.1 and 136.2). Reversible realizations stay open: source 21’s rules are not asserted reversible.) ([write] Added 4 October 2026: touched in one dimension by source 25’s Proposition 176.1, whose globally reversible binary rule with one site of drift per step has non-Presburger original-frame untimed reachability from one fixed input; reversible realizations of the planar site-set phenomena stay open.)
2. Extract practical algorithms and size bounds for the effective semilinear partitions, counting chambers, and first arrival formulas. General effectivity alone need not be computationally useful.
3. Optimize the canonical first arrival Diophantine encoding of Appendix K, especially the cell count, quotient variables, and shared quadratic lifts, without mistaking input dependent arity for a uniform bound. ([write] Open; source 21’s 614-auxiliary quartic certifies one local transition of its binary rule, not a first-arrival relation.)
4. Classify possible subleading terms and exact inverse formulas in the original frame. The present example exhibits a square root correction; the general $\Theta(N)$ theorem does not assert an exhaustive asymptotic taxonomy.
5. Clarify the exact correspondence with each classical pebble model, including indistinguishable markers, collision semantics, and its chosen observation convention, before making any novelty comparison.

128 Reproducibility and proof boundaries

The portable release contains this editable source and PDF, authored figures with their simulator, the proof and independent counting review, finite scientific checkers, expected JSON receipts, and a sealed replay entry point. No external paper cache is included. Bibliographic links are sufficient to locate the primary literature.

The main checker implements both explicit rule variants, the section clock, stationary box

formula, and exact drifted count. It also samples malformed configurations and compares global rewrites with cellwise radius six evaluation. That cellwise strategy reuses the same action predicate: it is an independently evaluated local output route, not an independently reimplemented transition rule. A separate included example checker independently transcribes the transition rules and checks first arrival and boundary arithmetic.

Every scientific correctness check uses explicit exceptions rather than removable Python assertions. A deliberately failing guard self test confirms enforcement under optimization. The replay runs fresh isolated normal and optimized copies, compares receipts recursively by exact type and value, and confirms that packaged file bytes, modes, and modification times remain unchanged. Booleans cannot masquerade as integer counts. An independently fixed inventory and pins are checked before invoking any packaged helper; a manifest by itself is insufficient when edited together with its payload.

The bundled tamper regressions test ordinary payload changes, removal paired with manifest editing, coordinated checker and manifest alteration, a type only receipt mutation, an extra file, and a symlink payload. Their aim is narrowly to exercise the stated verification boundary. This is not an adversarial security system against an attacker who replaces the trusted bootstrap and its reference pins together.

A PDF build writes only to an external directory. The release is not modified by replay or rebuild. Rebuilding requires a conventional L^AT_EX distribution; the geometry and example checkers use the Python standard library, while the optional first visit algebra fixtures use SymPy. Figure regeneration additionally uses Matplotlib. All pages of the final PDF are rendered and visually inspected for clipping, overlap, missing symbols, and reference errors.

Finite checks audit transcription, formula arithmetic, and selected boundary cases. They do not establish conservation on all configurations, classification termination, semilinearity, or asymptotic claims. Those conclusions rest on the unrestricted mathematical arguments in this report and the precisely stated Report 29 [source 19] dependency. No binary CA construction, reversibility proof, formal proof assistant certification, efficient general compiler, or new universality threshold is claimed.

[write] **The release in this report.** These are the delivered names. `build.sh`, `reproduce.sh`, `replay.py`, `tamper_regression.py`, `figures/make_figures.py` and the checkers `scientific/geometry/audit.py`, `scientific/examples/independent-audit.py` and `scientific/first-visit/audit.py` are shipped as `code/20-orbit-geometry-*` (sub-directories joined to the name by hyphens, for example `code/20-orbit-geometry-geometry-audit.py`); the normal and optimized receipts of the geometry and first-visit checkers and the four `verification/*.json` records as `data/20-orbit-geometry-*`; the figure as `figures/20-orbit-geometry-visited-geometry.pdf`; and the nine proof, review and supplement notes of `scientific/` as `20-orbit-geometry-*.md` at the report root. Not shipped: the manuscript (printed here), its PDF, README and checksum ledger, and the two supplement checksum manifests `scientific/first-visit/MANIFEST.sha256` and `scientific/geometry/MANIFEST.sha256`. The replay entry point needs the delivered layout (README). At placement on Windows, `replay.py` passed in normal and optimized mode and `tamper_regression.py` rejected every tamper case.

Source 21 (batch-82 manuscript 01): A binary planar four particle shuttle

Exact orbit geometry and a one step quartic certificate, “Research report 31”, 3 October 2026. [write] From the archive `Binary_Planar_Four_Particle_Shuttle_Package.zip`; its labels carry the prefix `smc:bp:`, and its sections are Sections 129–140 here. It builds on sources 17, 18 and 20

(“Reports 12, 13 and 30”), which it pins by SHA-256, and modifies source 17’s binary shuttle; no code is shared. The Hilbert’s-tenth programme has reviewed it (Section 139). Its abstract follows; the bracketed notes in it are [write].

We give an ordinary binary, quiescent, conservative cellular automaton on $\mathbb{Z}^2$ with an explicit four-particle expanding shuttle. Its stationary-frame visited set is a wedge with exactly two missing lines, its centered-box count has leading coefficient $1/2$, and its first-arrival times and generalized inverse are explicit. Composing the rule with unit vertical translation gives four disjoint traces and the exact floor-root count $4N - 4\sqrt{N} + O_k(1)$, with inverse $M/4 + \sqrt{M/4} + O_k(1)$. Both specified rules have exact centered Chebyshev radius six and precisely 78 essential input cells. Their unique multilinear Boolean polynomial has degree 45. A separately constructed one-step sum-of-squares quartic has 614 canonical natural auxiliaries, 693 residuals, 79 external variables, and exactly 3403 collected terms. Its auxiliary tuple is unique even over the reals. The construction modifies the two bounce outputs of the one-dimensional binary shuttle in Report 12 [source 17] and retains its moving pairs and quadratic clock. It answers the binary-realization question left by Report 30’s weighted planar example [more precisely: it gives a binary realization of the planar wedge and of the drifted square-root regime of source 20’s weighted example, answering source 20’s first question for binary rules but not for reversible ones]. Full-shift correctness is proved, rather than inferred from finite tests. No reversibility, universality, optimal-radius, or general finite-fold Diophantine claim is made.

129 Scope and relation to the earlier reports

Report 30 established quantitative geometry for promised conservative cellular automata of fixed mass at most four and supplied a weighted planar realization [source 20, Part VII]. The present report supplies a binary realization directly. Every occupied site has weight one, and the selected orbit contains exactly four occupied sites at every time. The proof of this example is self-contained and does not require a general orbit-classification theorem. [write] [Its counts are exact instances of source 20’s Theorems 118.2 and 118.3 (an expanding shuttle of rank two, with leading coefficient $1/2$) and, after the vertical shift, of Theorem 118.4 (linear box order with a square-root deficit, as in source 20’s Theorem 125.1).]

The starting mechanism is the one-dimensional binary shuttle in Report 12, Section 7 [Section 89 here, rule (89.1)], “An explicit binary expanding shuttle” [source 17, Part VI]. Its replacements, up to translation, are

$$\{0, 1\} \mapsto \{1, 2\}, \quad \{0, 2\} \mapsto \{-1, 1\}, \quad \{0, 1, 3\} \mapsto \{-1, 1, 4\}, \quad \{0, 2, 4\} \mapsto \{0, 3, 4\}.$$

We keep both moving-pair replacements and all four input shapes. At the right bounce, only the outgoing right marker moves up one row. At the left bounce, the entire output triple moves up one row. These two transverse changes make successive sweeps occupy successive rows. Thus the four horizontal patterns are not presented as new patterns. The quadratic section clock is inherited from the same flight lengths; the planar visited set and vertical-drift geometry are proved below.

Report 13 distinguishes local or fixed-formula certificates from more general timed constructions [source 18, Part VI]. We preserve that distinction. The quartic in Section 138 describes *one local transition*, with a literal expanded polynomial and exact resource ledger. It is not an unbounded orbit tableau. Report 30’s [source 20’s Corollary K.4] separate complete-target theorem is discussed only in Section 140; it does not follow from the local quartic.

Throughout, $k \geq 7$ is an integer, $\mathbb{N} = \{0, 1, 2, \dots\}$, and a “natural witness” means a tuple of nonnegative integers. The notation $O_k(1)$ permits a bound depending on the fixed initial gap k .

Spatial counts count distinct lattice sites ever occupied, not visits, particles with multiplicity, or space-time events.

130 The exact component rule

Identify a binary configuration with its occupied set $S \subseteq \mathbb{Z}^2$, which may be infinite. Join distinct occupied sites whenever their Chebyshev distance is at most two. A connected component that is a translate of an input A in Table 12 is replaced by the corresponding translate of B . All other components remain unchanged. Only translates are recognized; neither reflections nor rotations are implicitly included. Call the resulting map G .

Rule	Input A	Output B
E	$\{(0, 0), (1, 0)\}$	$\{(1, 0), (2, 0)\}$
W	$\{(0, 0), (2, 0)\}$	$\{(-1, 0), (1, 0)\}$
R	$\{(0, 0), (1, 0), (3, 0)\}$	$\{(-1, 0), (1, 0), (4, 1)\}$
L	$\{(0, 0), (2, 0), (4, 0)\}$	$\{(0, 1), (3, 1), (4, 1)\}$

Table 12: The four exact component replacements. A surrounding empty halo is part of recognition, not an optional condition.

Lemma 130.1 (Recognition and bounded matching). *The component prescription is unambiguous. On every finite or infinite configuration it gives a bijection from old occupied sites to new occupied sites whose displacements have max norm at most one. In particular, outputs from different components do not overlap.*

Proof. Each input shape is connected in the distance-two graph. An exact translate has a unique anchor, namely its minimum x coordinate and common y coordinate. The pair shapes have different gaps; the triple shapes have different gap lists. Therefore distinct rules cannot recognize the same component. Isolated sites, nonhorizontal components, larger components, and infinite components are unrecognized and stay fixed.

Use the following source-to-target bijections within recognized components:

$$\begin{aligned}
 E : & \quad (0, 0) \mapsto (1, 0), \quad (1, 0) \mapsto (2, 0), \\
 W : & \quad (0, 0) \mapsto (-1, 0), \quad (2, 0) \mapsto (1, 0), \\
 R : & \quad (0, 0) \mapsto (-1, 0), \quad (1, 0) \mapsto (1, 0), \quad (3, 0) \mapsto (4, 1), \\
 L : & \quad (0, 0) \mapsto (0, 1), \quad (2, 0) \mapsto (3, 1), \quad (4, 0) \mapsto (4, 1).
 \end{aligned}$$

Use the identity on every frozen component. Every displacement has norm at most one. Distinct components have all mutual distances at least three. If matched outputs from two components coincided, their source sites would have distance at most two by the triangle inequality, a contradiction. This includes a recognized component next to a frozen finite or infinite component. Outputs within each component are distinct by the displayed bijection. The combined matching is therefore a bijection onto the new occupied set. $\square$

For finite S , the lemma proves exact particle conservation. For infinite S , it proves a well-defined bounded-displacement matching without subtracting infinite cardinalities. Two output components may join after a step; the next step simply recognizes or freezes the newly formed component. Such joining does not invalidate the one-step argument.

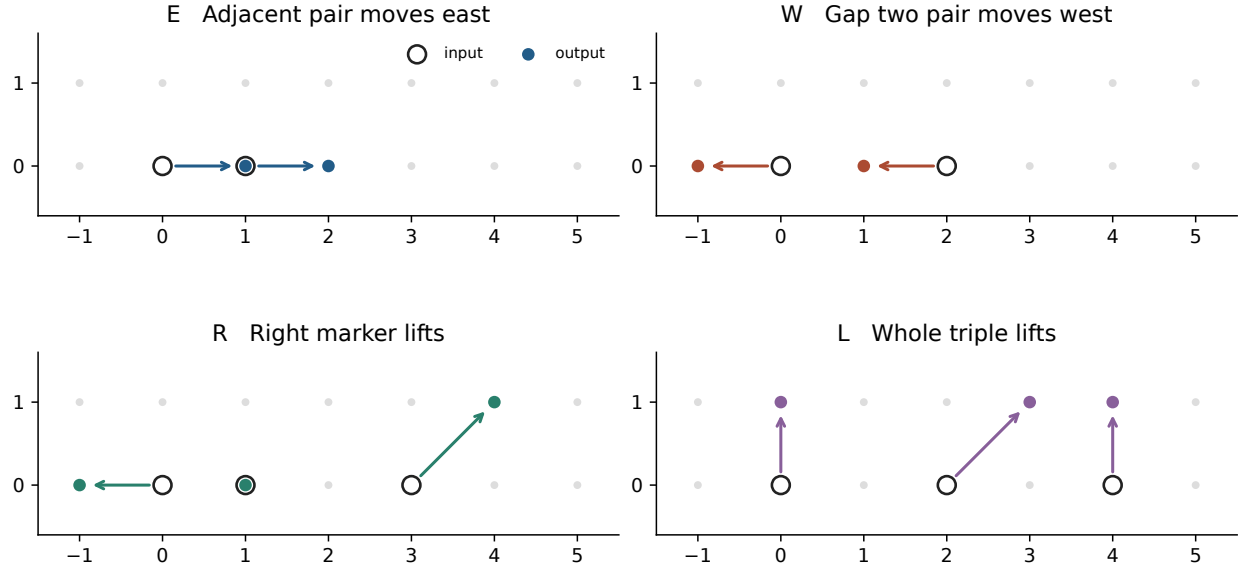


Figure 8: Exact primitive replacements and the source-to-target matching. Gray grid points are unoccupied reference sites. The two transverse bounce changes are visible in R and L . The required empty radius-two halo around each input is not drawn.

131 Locality on the full shift

For a translated input $A + a$, define its halo and recognition predicate by

$$H(A + a) = (A + a) + [-2, 2]^2, \quad P_{A,a}(S) \iff S \cap H(A + a) = A + a,$$

where intervals denote integer lattice intervals. This finite predicate is equivalent to saying that $A + a$ is exactly an occupied connected component. Indeed, the shape is connected, and the first site of any occupied path leaving it must lie in its halo. Conversely, an empty surrounding halo excludes every such path. This works on infinite configurations as well: a fragment of a larger or infinite component cannot be falsely rewritten because a window truncates the rest.

To evaluate the new bit at z , consider only anchors for which z is in a primitive input or output. If the old bit is zero, a successful predicate turns it on exactly when the corresponding replacement creates z . If the old bit is one, a successful predicate turns it off exactly when the replacement removes z . Otherwise the bit is preserved. Lemma 130.1 ensures that these local decisions agree with the component prescription on the entire binary full shift.

For every candidate and each input site p , one has $\|p - z\|_\infty \leq 4$. Its halo is therefore within radius six of z . More precisely, the changed-site cylinders read only

$$D_G = [-6, 6] \times [-3, 2], \quad |D_G| = 78. \quad (131.1)$$

The central unchanged bit lies in the same rectangle. All formulas commute with translations. No input shape is recognized in the vacuum, so G is quiescent. This proves the upper radius bound on all 2^{169} square radius-six windows without enumerating them. Section 137 proves that the smaller 78-cell dependency rectangle is exact.

Let τ be vertical translation by $(0, 1)$, with $\tau S = S + (0, 1)$, and set

$$F = \tau \circ G. \quad (131.2)$$

Translation equivariance gives $F^t(S) = \tau^t G^t(S)$. Evaluating F at z evaluates G at $z - (0, 1)$, so its read rectangle is

$$D_F = [-6, 6] \times [-4, 1]. \quad (131.3)$$

Thus F also has radius at most six, fixes the vacuum, and preserves every finite mass. Although the component matching for G moves sites by at most one, no such displacement bound is needed to reprove conservation after the global translation.

132 Four particles and the exact section clock

Start from

$$S_0 = \{(0, 0), (3, 0), (4, 0), (k, 0)\}, \quad k \geq 7.$$

Put

$$T_n = n^2 + (2k - 11)n \quad (n \in \mathbb{N}). \quad (132.1)$$

Theorem 132.1 (Complete phase description). *For every $n \geq 0$,*

$$G^{T_n}(S_0) = \{(0, n), (3, n), (4, n), (k + n, n)\}.$$

Writing $d = k + n$, the cycle $[T_n, T_{n+1})$ consists of these two inclusive phase lists:

$$G^{T_n+j}(S_0) = \{(0, n), (3 + j, n), (4 + j, n), (d, n)\}, \quad 0 \leq j \leq d - 6, \quad (132.2)$$

$$G^{T_n+d-5+q}(S_0) = \{(0, n), (d - 4 - q, n), (d - 2 - q, n), (d + 1, n + 1)\}, \quad 0 \leq q \leq d - 6. \quad (132.3)$$

In particular, each displayed support contains four distinct particles.

Proof. At the section, the adjacent pair is isolated from both markers. For $j < d - 6$, its separation from each marker is at least three, so E advances it by one. At $j = d - 6$, the right component is exactly the R triple $\{d - 3, d - 2, d\}$ in row n ; the left marker is separated because $d - 3 \geq 4$. The next state is the $q = 0$ state of (132.3).

The returning gap-two pair is separated from the lifted marker by three columns: $(d+1) - (d-2) = 3$. A row difference of one does not reduce this max-norm distance. The separation only grows during the return. For $q < d - 6$, the pair's leftmost site is at least three, so W applies and decreases both horizontal coordinates by one. At $q = d - 6$, the lower component is the exact L triple $\{0, 2, 4\}$, separated from the upper marker by at least $d - 3 \geq 4$. One L step gives the next section.

The duration is $(d - 6) + 1 + (d - 6) + 1 = 2d - 10$. Summing these durations for $d = k, k + 1, \dots, k + n - 1$ gives (132.1). This proves induction from $n = 0$ [read: this completes the induction on n]. The endpoints in the two lists cover exactly $2d - 10$ states and exclude the next section; they remain valid for $k = 7$. $\square$

[write] [Theorem 132.1 is the planar lift of source 17's Proposition 89.2: the four input shapes, the two moving pairs, the cycle of $2d - 10$ steps and the clock are source 17's, and so are the steps of the proof; only the transverse lifts of the right marker and of the left-bounce output are new, as source 21 says above.]

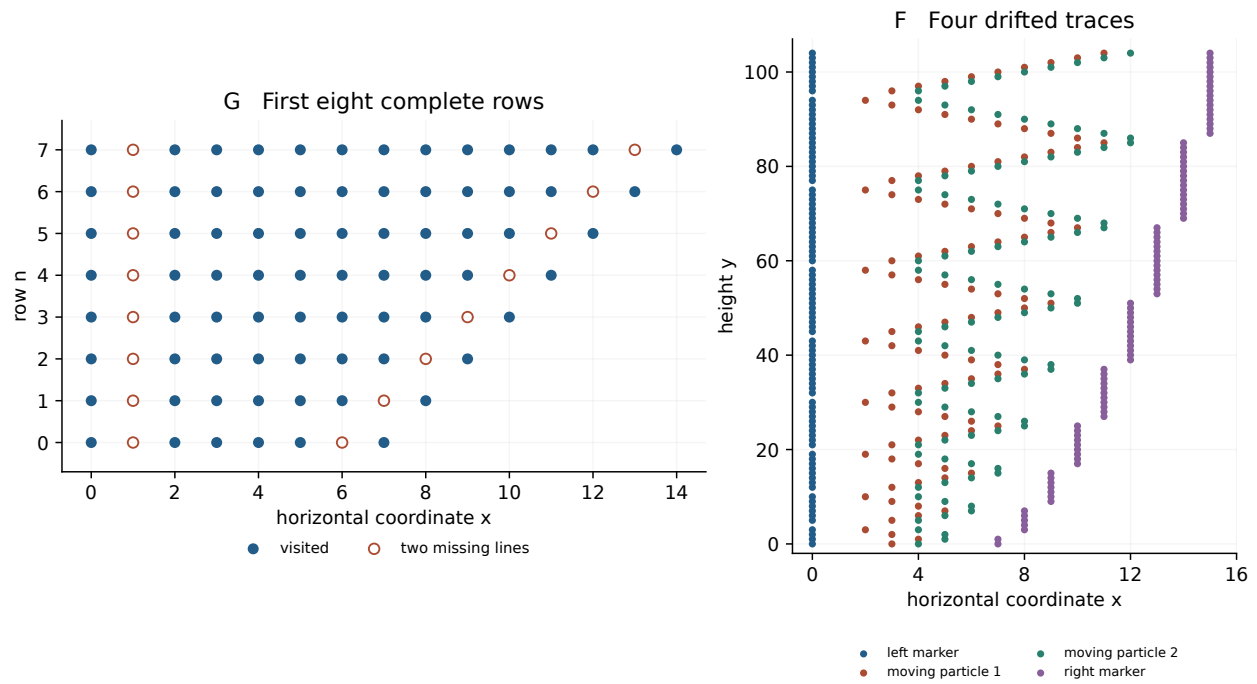


Figure 9: Actual simulation of the component rule from $k = 7$. Left: the first eight completed rows visited by G , simulated through T_8 , with the two missing families marked by open circles. Right: every occupied site of F through time 96, colored by the four particles using the bounded matching. The drifted traces have no repeated site. Both panels are generated from the rule, not drawn from the count formulas.

133 Visited sites and first arrivals

Theorem 133.1 (The wedge and its two missing lines). *The complete stationary-frame visited set is*

$$V_k = \{(0, n) : n \geq 0\} \cup \{(x, n) : n \geq 0, 2 \leq x \leq k + n - 2\} \cup \{(k + n, n) : n \geq 0\}. \quad (133.1)$$

It is the integer wedge $n \geq 0, 0 \leq x \leq k + n$ with precisely the two families $x = 1$ and $x = k + n - 1$ removed.

Proof. On row n , the east phase visits $[3, k + n - 2]$ and the west phase additionally reaches 2. It never reaches 1. The markers occupy 0 and $k + n$. The right marker may first occupy its row during the preceding cycle, but no extra sites result. Future cycles use higher rows. Hence the union is exactly (133.1). $\square$

Let $\theta_k(x, n)$ be the first time $G^t(S_0)$ occupies (x, n) , and leave it undefined at an unvisited site. For $n \geq 1$, put $d = k + n$. Then the exact disjoint branches are

$$\theta_k(x, n) = \begin{cases} T_n, & x \in \{0, 3, 4\}, \\ T_n + 2d - 11, & x = 2, \\ T_n + x - 4, & 5 \leq x \leq d - 2, \\ T_n - (d - 6), & x = d. \end{cases} \quad (133.2)$$

The same formula holds on row zero except that $\theta_k(k, 0) = 0$. For the right-marker branch, its creation time is $T_{n-1} + d - 6$, and $T_n - T_{n-1} = 2d - 12$, giving the displayed value. The interior branch is the leading east-moving member's first visit. The site 2 first appears in the last pre- L return state, whose relative time is $d - 5 + d - 6 = 2d - 11$. These branches partition every visited row even at the smallest gap. They establish piecewise-quadratic first-arrival data directly, without using a certificate compiler.

134 Exact centered counts and stationary inverse

For integer $N \geq 0$, define

$$C_{G,k}(N) = \#(V_k \cap [-N, N]^2), \quad R_{G,k}(M) = \min\{N \in \mathbb{N} : C_{G,k}(N) \geq M\}, \quad M \in \mathbb{N}.$$

Theorem 134.1 (Stationary count and complete inverse). *One has*

$$C_{G,k}(N) = \begin{cases} 1, & N = 0, \\ N(N + 1), & 1 \leq N \leq k - 2, \\ \frac{N^2 + (2k - 1)N - k^2 + 3k - 4}{2}, & N \geq k - 1. \end{cases} \quad (134.1)$$

The exact generalized inverse is

$$R_{G,k}(M) = \begin{cases} 0, & 0 \leq M \leq 1, \\ \left\lceil \frac{\sqrt{1 + 4M} - 1}{2} \right\rceil, & 2 \leq M \leq (k - 2)(k - 1), \\ \left\lceil \frac{\sqrt{8M + 8k^2 - 16k + 17} - (2k - 1)}{2} \right\rceil, & M > (k - 2)(k - 1). \end{cases} \quad (134.2)$$

Thus $C_{G,k}(N) = \frac{1}{2}N^2 + (k - \frac{1}{2})N + O_k(1)$ and $R_{G,k}(M) = \sqrt{2M} - k + \frac{1}{2} + O_k(1)$. [write] [The terms $-k + \frac{1}{2}$ are absorbed by $O_k(1)$; more precisely $R_{G,k}(M) = \sqrt{2M} - k + \frac{1}{2} + \vartheta_M + o(1)$ with $0 \leq \vartheta_M < 1$, the ceiling's rounding.]

Proof. All visited coordinates are nonnegative. For $1 \leq N \leq k - 2$, every included row contributes exactly N sites: 0 and $2, \dots, N$. At $N = 0$ only the origin contributes.

For $N \geq k$, the full wedge contributes

$$(N + 1)^2 - \frac{(N - k)(N - k + 1)}{2}.$$

Subtract the $N + 1$ holes on $x = 1$ and the $N - k + 2$ holes on $x = k + n - 1$. Simplification gives the final branch. At $N = k - 1$, the complete square loses $N + 1$ sites on $x = 1$ and one upper-edge hole; $N(N + 1) - 1$ agrees with that branch. Omitting either family of holes would produce the wrong polynomial.

The count is strictly increasing, and solving the corresponding quadratic inequality gives the two ceiling formulas. The threshold in (134.2) places the last inverse in the last count branch. The asymptotics follow by expanding its positive root; integer rounding contributes a bounded error. $\square$

135 Vertical drift and four disjoint traces

For each time t , let $n(t)$ be the unique integer satisfying $T_{n(t)} \leq t < T_{n(t)+1}$. Call the left marker and two matched moving particles the three bulk trajectories. Their F height is

$$h(t) = t + n(t).$$

The right marker has G row $r(t) \in \{n(t), n(t) + 1\}$, so its F position is $(k + r(t), v(t))$, where $v(t) = t + r(t)$. The functions $n(t)$ and $r(t)$ are nondecreasing, so $h(t)$ and $v(t)$ are strictly increasing.

Lemma 135.1 (Disjointness of all four traces). *Each particle's F trace has no self-intersection, and the four traces are pairwise disjoint. Every time contributes four new sites to the all-time visited set.*

Proof. The strictly increasing heights exclude self-repetition. All three bulk sites at a given time have the same height and distinct horizontal coordinates. Their heights agree across two times only when the times agree, so the three bulk traces are pairwise disjoint.

Suppose a marker site at time t equals a bulk site at time u . Write $r = r(t)$ and $n = n(u)$. Every bulk horizontal coordinate is at most $k + n - 2$, while the marker coordinate is $k + r$. Equality forces $n \geq r + 2$. Height equality gives $u = t + r - n \leq t - 2$. Monotonicity then implies $n(u) \leq n(t) \leq r(t) = r$, contradicting $n \geq r + 2$. Thus no marker site lies on a bulk trace. $\square$

The three bulk trajectories have one site at every nonnegative height except the skipped levels

$$a_n = T_n + n - 1 = n^2 + (2k - 10)n - 1, \quad n \geq 1. \quad (135.1)$$

Indeed $n(t)$ rises by one at T_n , so h jumps by two there. The right marker rises to row $n + 1$ at

$$\rho_n = T_n + k + n - 5.$$

It therefore skips precisely

$$b_n = \rho_n + n = n^2 + (2k - 9)n + k - 5, \quad n \geq 0. \quad (135.2)$$

Both sequences are strictly increasing. They interlace: $b_0 < a_1$, and for $n \geq 1$,

$$a_n < b_n < a_{n+1}, \quad b_n - a_n = n + k - 4, \quad a_{n+1} - b_n = n + k - 5.$$

No height is skipped by both the bulk and marker families. Notice that “four new sites per time” alone does not count sites in a spatial box: the four sites may enter that box at different heights and horizontal positions.

136 The exact drift count and its inverse

Write $V_{F,k} = \bigcup_{t \geq 0} F^t(S_0)$ and define $C_{F,k}(N)$ and $R_{F,k}(M)$ by the same centered-box and generalized-inverse conventions.

Lemma 136.1 (The horizontal cutoff). *Every F -visited site (x, y) satisfies $0 \leq x \leq \max(k + 1, y)$ and $y \geq 0$. Consequently, for $N \geq k + 1$, horizontal truncation removes no visited site of height at most N .*

Proof. For bulk sites in rounds $n = 0, 1$, $x \leq k + n - 2 \leq k - 1$. For $n \geq 2$, $T_n \geq T_2 = 4k - 18 \geq k - 2$, and hence $x \leq k + n - 2 \leq T_n + n \leq y$.

For the right marker, rows $r \leq 1$ give $x = k + r \leq k + 1$. When $r \geq 2$, its first time in that row is $\rho_{r-1} = T_{r-1} + k + r - 6$, so

$$y - x \geq T_{r-1} + r - 6 \geq T_1 - 4 = 2k - 14 \geq 0.$$

All coordinates are nonnegative by the phase formulas and the drift. The first lifted marker has $x = k + 1$, so the claimed threshold cannot simply be replaced by $N \geq k$. $\square$

Theorem 136.2 (Floor-root count and inverse). *For every integer $N \geq k + 1$, let $\alpha = 2k - 10$ and $\beta = 2k - 9$. Then*

$$A_k(N) = \#\{n \geq 1 : a_n \leq N\} = \left\lfloor \frac{\sqrt{\alpha^2 + 4(N+1)} - \alpha}{2} \right\rfloor, \quad (136.1)$$

$$B_k(N) = \#\{n \geq 0 : b_n \leq N\} = 1 + \left\lfloor \frac{\sqrt{\beta^2 + 4(N - k + 5)} - \beta}{2} \right\rfloor, \quad (136.2)$$

$$\begin{aligned} C_{F,k}(N) &= 4(N + 1) - 3A_k(N) - B_k(N) \\ &= 4N - 4\sqrt{N} + O_k(1). \end{aligned} \quad (136.3)$$

As $M \rightarrow \infty$,

$$R_{F,k}(M) = \frac{M}{4} + \sqrt{\frac{M}{4}} + O_k(1). \quad (136.4)$$

Proof. Lemma 136.1 reduces the box count to a height count. The three disjoint bulk traces each omit $A_k(N)$ of the $N + 1$ heights, and the marker trace omits $B_k(N)$. Lemma 135.1 allows these counts to be added without overlap. Solving the two quadratic inequalities gives (136.1)–(136.2); the stated cutoff ensures that their expressions are nonnegative. Each is $\sqrt{N} + O_k(1)$.

For the inverse, set $m = M/4$ and $u = m + \sqrt{m}$. Then $u - \sqrt{u} = m + O(1)$. Rounding u to an integer changes the count by $O_k(1)$. Beyond the transient, increasing N by one adds four sites at a nonskipped height, one at a bulk skip, or three at a marker skip. The interlacing excludes a simultaneous skip. Thus an $O_k(1)$ count discrepancy can be corrected by $O_k(1)$ further radius, proving (136.4). $\square$

For completeness, the finite transient also has an exact formula requiring no limiting convention:

$$C_{F,k}(N) = \sum_{t=0}^N \#\{(x, y) \in G^t(S_0) : x \leq N, t + y \leq N\} \quad (N \geq 0). \quad (136.5)$$

All coordinates are nonnegative and the traces are disjoint; times $t > N$ have height greater than N . Thus (136.5) follows directly and can use the complete phase formulas. Together with (136.3), it gives an exact algorithm for every inverse value: check the finite transient, then find the least integer satisfying the monotone floor inequality. The implementation evaluates roots with integer square roots, including exact-square boundaries, rather than floating-point approximations.

137 Exact Boolean degree and all essential inputs

For input shape A at anchor a , define the exact-component indicator

$$M_{A,a}(c) = \prod_{p \in A+a} c_p \prod_{q \in H(A+a) \setminus (A+a)} (1 - c_q). \quad (137.1)$$

On binary inputs it equals one precisely for an isolated recognized component. The local rule is the following real multilinear polynomial on binary inputs:

$$g_z(c) = c_z + \sum_{(A,B)} \sum_{b \in B \setminus A} M_{A,z-b}(c) - \sum_{(A,B)} \sum_{a \in A \setminus B} M_{A,z-a}(c). \quad (137.2)$$

Its validity on the whole Boolean cube follows from exact recognition and disjoint outputs, not merely from checking the selected orbit. There are respectively 2, 4, 4, 6 signed translated indicators of types E, W, R, L . Each halo is a rectangle of height five and width 6, 7, 8, 9, respectively. The indicator degrees are consequently 30, 35, 40, 45.

Summing (137.2) over sites gives a second finite-input conservation proof: each translated-indicator sum is unchanged by the offset, and $|B \setminus A| = |A \setminus B|$. Only finitely many indicators can be nonzero on a finite support, so these sum rearrangements are legitimate.

Theorem 137.1 (Exact degree and exact dependency set). *The unique real multilinear polynomial for the local Boolean rule has degree exactly 45. Every cell of D_G is essential and no other cell is; F has exactly the translated dependency set D_F . Both specified rules therefore have exact centered Chebyshev radius six.*

Proof. The polynomial (137.2) is multilinear because each indicator uses every input bit at most once. The six L indicators have distinct rectangular halos of size 45. Each has three occupied factors and 42 complementary factors, so its degree-45 top coefficient is $(-1)^{42} = 1$, multiplied by the birth or removal sign. Their top monomials are distinct and cannot cancel one another; every other indicator has degree at most 40. This proves exact degree 45 by uniqueness of multilinear interpolation on the Boolean cube.

For the three births, the L halos have horizontal intervals $[-2, 6]$, $[-5, 3]$, $[-6, 2]$ and vertical interval $[-3, 1]$. For the three removals, the horizontal intervals are $[-2, 6]$, $[-4, 4]$, $[-6, 2]$ and the vertical interval is $[-2, 2]$. Their union is exactly D_G . Every one of its 78 variables therefore occurs in a nonzero degree-45 monomial.

If a Boolean function were independent of one such variable, the difference between the polynomial's one and zero restrictions would vanish on the Boolean cube. Uniqueness of multilinear interpolation would force this derivative polynomial to vanish identically, contradicting the nonzero monomial containing that variable. Thus all 78 variables are essential. Section 131 excludes any other dependency. Translating the evaluation point gives D_F and its essentiality. Both rectangles contain a horizontal coordinate of absolute value six, which proves the exact centered radius. $\square$

The release also supplies an explicit finite witness for each cell: two configurations differing at that cell only, with different central outputs. The file `essential-input-witnesses.json` in `scientific/audit/` records all 78 pairs. They are checked with both the direct component simulator and the independently encoded cylinder evaluator. Translating both inputs down one row gives the corresponding F witness. This is a concrete flip-witness confirmation independent of relying on the top-monomial argument. Exact radius here is a property of these two rules, not a least-radius claim over all possible binary constructions.

138 A literal one step quartic with unique auxiliaries

Enumerate the 78 cells of D_G in row-major order, increasing y and then x , and add one external output variable y_{out} . There are 79 external variables. For each of the sixteen indicators, use its own unshared multiplication chain, listing occupied factors first and complementary factors afterward. Within each group use row-major order. Start the chain with its first external occupied factor. At every subsequent factor introduce one new auxiliary z and one residual

$$z - uv \quad \text{or} \quad z - u(1 - v), \quad (138.1)$$

where u is the preceding chain value and v is the next external input. Indicator order and all variable indices are recorded canonically in the certificate.

There are

$$2 \cdot 29 + 4 \cdot 34 + 4 \cdot 39 + 6 \cdot 44 = 614 \quad (138.2)$$

auxiliaries and the same number of gate residuals. Add the 78 bit residuals $c_i(c_i - 1)$ and the output residual

$$y_{\text{out}} - c_{(0,0)} - \sum_{\text{births}} z_{\text{last}} + \sum_{\text{removals}} z_{\text{last}}. \quad (138.3)$$

Let P be the sum of squares of these $614 + 78 + 1 = 693$ residuals. The circuit gives an ordinary integer polynomial, not a compressed truth predicate or a symbolic placeholder for an unspecified computation. [write] [Arithmetizing a Boolean circuit by bit equations and product gates is the standard device of Part IV's gate residuals (Section 46); the counts 614, 693 and 3403 are source 21's own.]

Theorem 138.1 (The local zero set and canonical witnesses). *The total degree of P is exactly four. For any fixed external input-output tuple, its natural auxiliary fiber has exactly one element if that tuple is a valid binary local transition, and has no element otherwise. The same statement holds when the auxiliaries range over all real numbers.*

Proof. Every residual has degree at most two, so P has degree at most four. The squares of bit residuals supply positive fourth-power monomials of the external input variables, which cannot be canceled by the other squared residuals here; hence the degree is exactly four.

Since P is a sum of real squares, $P = 0$ is equivalent to simultaneous vanishing of every residual. The bit residuals force each external input into $\{0, 1\}$. Each acyclic chain equation then forces its next auxiliary uniquely, starting from an external input. All forced values are themselves zero or one. Finally (138.3) forces the output to equal (137.2), hence the actual binary local output. Conversely, for a valid transition these forced chain values make every residual zero.

Therefore the unique real auxiliary tuple is natural, and invalid external tuples have no real or natural auxiliary solution. This argument proves uniqueness by the specific deterministic chain structure. It is not an inference from finite perturbation tests or from decidability. $\square$

138.1 The exact algebra ledger

There are $2 \cdot 1 + 4 \cdot 1 + 4 \cdot 2 + 6 \cdot 2 = 26$ plain-product gates, each with two residual monomials, and $2 \cdot 28 + 4 \cdot 33 + 4 \cdot 37 + 6 \cdot 42 = 588$ complementary-factor gates, each with three. The bit residuals each have two monomials and the output residual has 18. Thus the uncollected ledgers are

$$\begin{aligned} \text{residual-monomial occurrences} &= 26 \cdot 2 + 588 \cdot 3 + 78 \cdot 2 + 18 = 1990, \\ \text{ordered square-expansion occurrences} &= 26 \cdot 4 + 588 \cdot 9 + 78 \cdot 4 + 18^2 = 6032. \end{aligned}$$

Collecting equal monomials in the literal exported polynomial gives Table 13. The last three values were independently reconstructed and compared exactly, not assumed from the circuit description.

Resource	Exact value
External input bits	78
External variables including output	79
Canonical natural auxiliaries	614
Total variables	693
Squared residuals	693
Residual-monomial occurrences	1990
Ordered square-expansion occurrences	6032
Distinct collected monomials	3403
Largest absolute collected coefficient	2
Total degree	4

Table 13: Literal resource ledger for the deliberately unshared one-step circuit. No auxiliary-minimality claim is made.

The machine-readable `scientific/local-quartic-certificate.json` contains all 3403 collected terms, the full variable order, all 614 gates, all 693 residuals, and all sixteen indicators. A monomial is a list of variable indices with repetitions representing powers; coefficients are exact integers. The independent artifact reader re-expands squared residuals using unordered pairs and compares the resulting sparse polynomial exactly. A second reconstruction begins from the geometric primitive replacements and independently rebuilds indicators, gates, residuals, and every coefficient. Thus the expanded artifact is checked against both its circuit and its intended geometry.

139 Verification and portable reproduction

The release separates proofs from executable evidence. The reviewed source proofs `PROOF.md` and `LOCAL_ALGEBRA.md` are preserved under `scientific/`. This report supplies a connected mathematical presentation. Independent reviews and explicit flip witnesses accompany the source files. No finite regression is offered as an enumeration of all full-shift configurations.

139.1 Independent routes and finite coverage

The direct simulator `component_rule.py` builds occupied components and applies the four replacements. The pointwise evaluator `local_rule.py` imports neither that simulator nor its rule dictionary; it encodes the sixteen occupied-and-empty cylinders independently and exports exact integer masks. Both G and F are evaluated.

The primary suite checks all 4096 supports in a 4×3 rectangle, all 8192 supports in a 13-site line, arbitrary exterior contexts, one-site halo interference, interacting active and malformed templates, births over a full-radius candidate region, large translations, exact orbit phases, first arrivals, both box counts, disjoint traces, and integer-root boundary cases. Its deterministic receipt reports 76680 direct orbit phase steps, 46980 first-arrival checks, and 53601 drift count checks. These are finite checks of the displayed formulas, not substitutes for the all-input proofs.

The quartic artifact reader checks 350 binary local fixtures and their incorrect binary outputs. At the vacuum witness it tests all 614 upward unit perturbations, one per auxiliary. It also compares the sparse identity at 60 nonbinary integer points and obtains all 78 essentiality witnesses. These tests do not exhaust all auxiliary perturbations on all inputs; uniqueness is proved in Theorem [138.1](#). Independent component, drift, cross-implementation, and geometry-to-quartic reconstruction checks provide separate routes. Every scientific test used by the release replay uses explicit exception checks, so optimization does not disable its checks. Normal and `-O` runs must agree with the exact expected receipts, including JSON value types.

139.2 Integrity before execution and isolated outputs

The portable verifier first compares the exact required payload file set and independently pinned SHA-256 values before executing any packaged scientific code. It then runs in fresh external copies. Adjacent-output scripts are executed only in those copies. Expected receipts are compared recursively with exact types, so a Boolean is not accepted in place of an integer. The verifier checks that release bytes, file modes, and modification times are unchanged after the run. The release checksum list and archive checksum provide an additional external integrity layer; executing the verifier still presumes trust in that bootstrap file.

The README supplies exact commands and requirements. Scientific replay needs Python 3.10 or later and only the standard library. PDF rebuild needs `pdflatex` and the listed TeX packages; figure regeneration additionally needs Matplotlib. Supplied vector PDF figures allow a report rebuild without Matplotlib. Build products and replay receipts belong outside the release. Bounded tamper regressions test changed code, missing required files, altered certificate content, and strict typed comparisons; they do not run the scientific suite on tampered payloads.

The portable package contains no Python bytecode caches, private absolute paths, third-party source caches, or files from earlier releases. Those earlier releases are cited, not modified. The exact source and receipt lineage is recorded in `verification/source-lineage.json`.

[write] **The release in this report.** These are the delivered names. `build.sh`, `reproduce.sh`, `replay.py`, `tamper_regression.py` and `figures/make_figures.py` are shipped as `code/21-planar-shuttle-*`, the programs of `scientific/code/` as `code/21-planar-shuttle-code-*` and those of `scientific/audit/` as `code/21-planar-shuttle-audit-*`; the receipts, review bindings and certificates of `scientific/audit/` as `data/21-planar-shuttle-audit-*`, the two certificates of `scientific/` (among them `local-quartic-certificate.json`), the figure data and the `verification/` records as `data/21-planar-shuttle-*`; the two figures as `figures/21-planar-shuttle-rule.pdf` and `figures/21-planar-shuttle-orbits.pdf`; and `scientific/PROOF.md`, `LOCAL_ALGEBRA.md` and the four audit notes as `21-planar-shuttle-*.md` at the report root. Not shipped: the manuscript (printed here), its PDF, README and checksum ledger. Several delivered receipts are byte copies of one another (normal, optimized and expected copies); they are shipped as delivered, because the replay compares them by name. At placement on Windows every replay step passed in normal and optimized mode after two workarounds for harness defects, which the README states: `replay.py` starts its children with `-I`, which ignores `PYTHONUTF8`, so two audits fail to print non-ASCII text on a cp1252 console unless run with `-X utf8`; and several writers emit CRLF line ends on Windows, so the byte comparison of the rebuilt local quartic certificate fails although its content equals the delivered file.

[write] **The programme’s review.** The Hilbert’s-tenth programme reviewed this archive in `review_binary_planar_four_particle31.md`, with its checker and receipt, in `Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/` (commit 16f50dcc6, 3 October 2026): “PASS within the stated scope; no error found”. It read the proof notes in full, authenticated the archive and seven members, and, from the four printed replacement rules alone, rebuilt the sixteen indicators, the 32 recognition cylinders of G and F , the 614 product-chain gates and the 693 residuals; their expansion matches all 3403 saved terms, of degree four. It confirms the six noncancelling degree-45 leaders, the 78 essential cells and exact radius six for these rules, the phase intervals including $k = 7$, both missing lines, the cutoff $N \geq k + 1$, and both drift asymptotics, and adds 483 phase states, 183 first arrivals and 261 box counts for $k = 7, 8, 11$. It executed none of the archive’s Python, and it notes that the 614 auxiliaries are not a complete evaluation cost of the quartic and that the rule gives no universal simulation; the repository’s universal operation bounds are unchanged.

140 Consequences limits and next questions

This example realizes two distinct counting regimes with an ordinary binary mass-four orbit: a quadratic stationary-frame wedge count with sharp coefficient $1/2$, and a linear drifted count with a square-root deficit of coefficient four. The deficit records skipped heights of four disjoint traces, rather than weighted-site multiplicities. The two removed wedge lines and the finite horizontal cutoff are essential to the exact formulas.

The construction proves existence for the specified initial family, not a classification of arbitrary malformed orbits. Malformed inputs are covered by the conservation and locality proof; they are not claimed to enter a shuttle. Neither G nor F is asserted reversible or universal. No smallest mass, smallest radius among constructions, minimal number of essential cells, optimal Boolean degree, or optimal auxiliary count is asserted.

The local quartic relates 78 input bits to one next-step output. Its unique real auxiliary tuple follows from acyclic multiplication chains and bit equations. It does not imply an unbounded-time finite-fold representation for arbitrary recursively enumerable sets. Report 30 separately proves that,

for a fixed promised conservative CA and fixed mass-at-most-four input, untimed reachability of a canonical *complete configuration* has an effective ordinary quartic with one natural witness tuple, optionally including first occurrence time [source 20, Part VII]. That theorem uses the full classified timed charts and whole-orbit injectivity in the expanding case. Its hypotheses include this example, but its general certificate is not the 614-auxiliary local polynomial here. Site or partial-pattern recurrence cannot be substituted for complete-target injectivity, and no real-witness conclusion is inherited from that theorem.

Natural next questions remain tightly connected to this construction:

1. Can a binary planar four-particle wedge be realized with a smaller centered radius or fewer essential cells while retaining a transparent all-input conservation proof? ([write] Open. Compare source 17’s Question 6 for the one-dimensional rule, which stays open too.)
2. Can shared products reduce the 614 local auxiliaries without introducing noncanonical inactive values, and what exact collected polynomial results?
3. Which changes to the two transverse bounces preserve four-trace disjointness after drift, and which change the coefficient of the square-root deficit?

These are questions, not claims settled by the present finite checks or by the existing general classification.

Part VIII

Unbounded-time certificates for exact three-mass targets (sources 22 and 23)

[write] Part VIII prints two AI-assisted research manuscripts of ProveIt’s incoming reports, batch 82 (arrival db37d18c8; cluster M2, placed by 7d2b1b245), both dated 3 October 2026 and added on that day. They are numbered reports of the research pipeline that also wrote sources 18–21: *source 22* is batch-82 manuscript 21, the pipeline’s Report 21; *source 23* is manuscript 02, Report 22, a declared sequel of Report 21. Their author lines are “Report 21 Mathematical construction and portable reproducibility package” and the same with “Report 22”; they name no assistant. Both pin the ProveIt commit ad634b2d1 of the research files they build on. Brackets translate the pipeline’s numbers where they occur: Report 10, which source 22’s provenance record names as “Report10 construction”, is source 15; Reports 19 and 20 are Parts IV and V of the collection’s report *five-particle-binary-automata*; Reports 21 and 22 are sources 22 and 23; and Report 25, the sequel of Report 22, is Part III of *fixed-universal-polynomials*.

[write] **What the two sources do.** Part V certifies the three-mass automaton at a fixed external *source horizon*: one polynomial for each horizon, with a unique natural witness (Theorems 58.1 and 70.2). Its two sources end by asking for a costed *unbounded-time* encoding, with its degree, its variable ledger and the fibre properties that survive (source 14’s Question 5 and source 15’s Question 5). Source 22 gives one for each fixed eligible source (fresh entry, terminal halt, $q_0 \neq q_h$): a polynomial $Q_R(x, U; z, \theta)$ with no horizon, whose positive zeros at a natural pair (x, U) exist exactly when source 15’s cleaned automaton first reaches its exact whole-configuration target at time U from the raw payload $x + 1$ (Theorem 141.1). For four fixed nonuniversal sources it emits the complete circuits, of 604, 479, 477 and 480 operations and exact degrees 2344, 1192, 1192 and 1192, and it

proves that every accepted nonempty witness fibre is infinite (Theorem 146.1). Source 23 appends one positive coordinate and one square, which fix the outer history height at the least power of two above $S = n_{\text{initial}} + n_{\text{target}} + \theta$. The outer coordinates then become unique, projection onto the twenty-two native coordinates is a bijection onto the entire native extension fibre (Theorem 154.1), and that fibre is still infinite (Theorem 155.1); the circuits cost 608, 483, 481 and 484 operations. Unbounded time thus costs Part V’s unique natural witness, and canonicalization restores uniqueness of the outer history only.

[write] **Dependencies.** Both main theorems are composition theorems. They are conditional, exactly as the sources state, on results that this report does not print:

- (i) the raw unbounded residue-history interface `three_mass_unbounded_interface.md` of the Hilbert’s-tenth research tree, with its literal arithmetic receipt and its independent review (“PASS within the stated fixed-source, natural-input scope”), at ProveIt commit `ad634b2d1` [62], and the residue-affine packed-history proof [63];
- (ii) source 15’s clean-target theorem (Theorems 66.2, 67.1 and 68.1; its audited statement is shipped as `15-clean-targets-CLEAN-TARGET-THEOREM.md`);
- (iii) the native prescribed-scale AND theorem and the Pell notes of the same tree [65, 66, 67, 68].

The files of (i) and (iii) are research-wip notes in `Computability/HilbertTenthProblem/Papers/`, not printed reports of the collection, and the native Pell representation theorem behind them is not re-proved by either source. The two archives carried byte copies of these files and of source 15’s theorem; they are not shipped twice. At this write the pinned files are unchanged in the repository.

[write] **Credit.** The clean-time bridge itself is in the research-wip note, at its lines 96–98: the formula $T_{\text{clean}} = 2T_{\text{native}} + 192(x + 1) + 192F + 16$, its literal cost $2M + 4A$ and one more square, with the remark “This is an incremental schedule only; the prototype emits the native-time relation and does not claim a complete cleaned-time total”. Its four raw circuits have 598, 473, 471 and 474 operations (lines 106–109), and $598 + 6 = 604$ and so on. The clock identity is source 15’s Theorem 68.1. New in source 22 are the complete emitted, audited and costed composition, its exact degrees, the separately folded schedules and the infinite-fibre theorem; new in source 23 are the canonical height, the entire-fibre projection and the fixed-scale infinitude by a matrix argument. Neither source claims novelty for what it inherits.

[write] **Review status.** The Hilbert’s-tenth programme has made a scoped intake of each source, not a full review: of source 22 in `review_event_budget_clean_clock_intake.md` (commit `8aabb1453`) [69], and of source 23 in `review_incoming_negative_index_fibers.md` (commit `0055e1c4d`) [70]; notes at the end of the two sources give the details. Section 1.24 relates Part VIII to the rest of the report and to later work of the programme on the same circuits, and Section 1.25 collects what the two sources do not claim.

Source 22 (batch-82 manuscript 21): Unbounded compact clean clocks for exact three mass targets

Fully emitted fixed source Diophantine certificates, “Report 21 Mathematical construction and portable reproducibility package”, 3 October 2026. [write] From the archive `Unbounded_Compact_Clean_Clocks_Package.zip`; its labels carry the prefix `smc:uc:`, its sections are Sections 141–149 here, and its equations are numbered by section. Its four bibliography entries are this report’s [62], [63] and [64] and, for the clean-target theorem, source 15 (Part V). Its abstract follows.

We close the clean-time output interface of the existing fixed-source unbounded residue-history certificate. For a fresh-entry, terminal-halt reversible source, retain its positive final payload

F , promote its nonempty native duration θ to a positive witness, remove only the redundant output-copy comparison, and impose

$$T_{\text{clean}} = 2\theta + 192(F + x + 1) + 16.$$

All chronology, native AND, endpoint, clock and no-wrap guards remain. Four complete native or spatially blocked circuits have respectively 604, 479, 477 and 480 arithmetic operations, 60, 58, 58 and 58 positive witnesses, and exact degrees 2344, 1192, 1192 and 1192. Each includes twenty comparison squares and the entire finalizer. The original four-phase variant pays one additional multiplication. A separately audited constant-folded schedule reduces both native and phase-four totals to 603, 478, 476 and 479, with unchanged witness counts and degrees. Exact degrees have deterministic modular leading-component certificates. Both directions of the first-target interpretation follow from explicitly identified inherited theorems. Every accepted nonempty fiber has infinitely many positive witnesses, as every sufficiently large dyadic height admits a fresh native extension. The four fixtures themselves halt in one or two source steps and are nonuniversal demonstrations. No arbitrary c.e. ordinary-input loader, universal numerical bound, optimality claim, rational-zero theorem or materialized full positive Pell witness is supplied. Initially halted sources have separate four-gate, witness-free quadratic certificates.

141 Result and scope

Throughout, $\mathbb{N} = \{0, 1, 2, \dots\}$ and every existential coordinate of a nonempty packet belongs to $\mathbb{Z}_{>0}$. Fix a reversible separated two-counter source R whose entry q_0 has no incoming instruction and whose terminal halt q_h has no outgoing instruction. For the nonempty construction assume $q_0 \neq q_h$. The raw initial payload is $N_0 = x + 1$, with $x \in \mathbb{N}$; its prime-two and prime-three valuations encode the two counters. Factors coprime to six are part of the payload and are preserved.

The inherited three-mass compiler uses a finite alphabet of Boolean channel subsets, with mass equal to the number of occupied channels and vacuum of mass zero. The resulting cellular automaton (CA) preserves total mass three on the configurations here. This is a weighted finite-alphabet model, not the model in which an integer state label itself equals its mass.

Theorem 141.1 (Fixed-source clean-time composition). *Assume the inherited complete raw-history/native theorem and the inherited exact clean-target CA theorem described in Section 142. For an eligible fixed source R , the construction below gives an integer polynomial $Q_R(x, U; z, \theta)$ with no external history horizon, such that for every $x, U \in \mathbb{N}$,*

$$\exists(z, \theta) \in \mathbb{Z}_{>0}^{d_R+1} \quad Q_R(x, U; z, \theta) = 0$$

if and only if the cleaned CA first reaches its exact canonical whole-configuration target at time U from its canonical input at raw payload $x + 1$. The terminal payload F is a retained coordinate of z and

$$U = 2\theta + 192(F + x + 1) + 16.$$

Spatial blocking preserves this time. The four-phase construction replaces the right side by four times the entire expression. The dimension depends on the fixed source, not its runtime.

This is a composition theorem with explicit dependencies. The portable packet emits the construction completely for these four fixed sources:

Fixture	Literal source	Accepted raw inputs	Steps
INC2;DEC2	$s \xrightarrow{\text{INC2}} a \xrightarrow{\text{DEC2}} h$	all $N_0 > 0$	2
ZERO3	$s \xrightarrow{\text{ZERO3}} h$	$3 \nmid N_0$	1
NOP	$s \xrightarrow{\text{NOP}} h$	all $N_0 > 0$	1
POSITIVE3	$s \xrightarrow{\text{POSITIVE3}} h$	$3 \mid N_0$	1

A disabled test is a nonfinal stuck state. Each fixture has singleton incoming and outgoing instruction groups, fresh entry and terminal halt. Its generic certificate has no horizon parameter, but these short executions do not demonstrate a long or universal execution. Their simple accepted relations could have much smaller specialized descriptions; the measured costs concern the fully retained generic compiler, not minimal representations.

The present unbounded packet differs from Reports 19–20 [Parts IV and V of **five-particle-binary-automata**, on the binary five-particle automaton]: those reports concern finite-horizon CA schemas and canonical natural witness tuples. Here the native positive witnesses have unbounded height freedom, and accepted nonempty fibers are provably infinite.

142 Authenticated dependencies

The public raw interface, its arithmetic receipt, independent review and residue-history proof are pinned to ProveIt commit

ad634b2d10ad666260f9fdff04ec94b75169ee4b

The local **reproducibility/source/** directory contains their exact bytes, including two Python files retained solely as inert provenance. None of those third-party Python files is imported or executed by the new emitter or checkers. The JSON gate lists are read as arithmetic data.

The raw receipt **three_mass_unbounded_interface.json** has SHA-256

fe69f0504c8f68c4130c4b21fed5c817596ab5c0e2fdf9621b0bf5f45c12ac0e

The unchanged earlier clean-target proof, **inherited_clean_target_theorem.md**, has SHA-256

3d85f8cde66837d8916fb3273c2c57cc1a91ce37de1f3d6c6fe28a4db9103168

The complete source hashes, public URLs, Git-blob hashes and byte counts are in **source/provenance.json**. The outer release records original and delivered hashes separately; no emitted circuit or inherited source file was adapted. The folding auditor has a documented portable path-guard adaptation, with its self-hash receipt regenerated.

Write the inherited raw polynomial as

$$P_R(x, y, t; z), \quad x, y, t \in \mathbb{N}, \quad z \in \mathbb{Z}_{>0}^{d_R}.$$

Its positive coordinates include the unshifted terminal payload F , twenty-two native auxiliaries, all thirty residue-selector hats in the fixtures, quotient and selected-quotient hats, height and global slacks, and the shifted clock quotient. Its twenty comparisons are two outer rows, sixteen native rows, the exact native-clock row and $F = y$. The inherited assertion is

$$\exists z > 0 \ P_R(x, y, t; z) = 0 \iff \begin{array}{l} R \text{ first halts from } (q_0, x + 1) \\ \text{with raw output } y \text{ and native time } t. \end{array}$$

In particular every such zero has $F = y > 0$ and $t > 0$ when $q_0 \neq q_h$.

The raw input port is paid as $5x + q_0$, and its target port as $5(F - 1) + q_h$ for these fixtures. Their subtraction, multiplication and code addition remain in the emitted clean circuits. F is neither $F - 1$ nor a valuation. The existential dimension and literal loaders therefore preserve the inherited domain contract.

The second dependency is the earlier clean-wrapper and canonical-section theorem [source 15, Theorems 66.2, 67.1 and 68.1], including the prescribed escape of nonfinal stuck configurations. This report does not regenerate the CA local rule tables, reprove the historical native/Pell theorem or replay its third-party implementation. The new portable arithmetic replays have no network or external repository dependency; this computational self-containment is separate from foundational independence of the inherited theorems.

[write] [The pinned commit is ProveIt’s `ad634b2d1`; the pinned interface, its receipt and review, and the residue-history proof are the files `three_mass_unbounded_interface.md`, `.json`, `review_three_mass_unbounded_interface.md` and `residue_affine_packed_history.md` of `Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/` [62, 63], unchanged there at this write. The “unchanged earlier clean-target proof” is source 15’s file shipped as `15-clean-targets-CLEAN-TARGET-THEOREM.md`, byte for byte. The directory `reproducibility/source/` held byte copies of these files and is not shipped; its `provenance.json` is shipped as `data/22-clean-clocks-source-provenance.json`.]

143 The clean source and exact physical target

[write] [This section restates source 15’s clean return wrapper and exact target for the reader: the wrapper and its restoration theorem are Section 66 and Theorem 66.2, the freshness discussion is Section 66.1, the target is Theorem 67.1, and the clock identity (143.1) is Theorem 68.1, in which source 15 writes Θ for θ , N_h for F and T_{cl} for T_{clean} . It is printed in full as source 22’s own account; no novelty is claimed for it.]

143.1 Forward and reverse source traces

Construct R_{clean} with controls $F : q$, $B : q$ for each source control q , and a new sole halt H . Copy every source instruction forward, copy its inverse backward, and add

$$F : q_h \xrightarrow{\text{NOP}} B : q_h, \quad B : q_0 \xrightarrow{\text{NOP}} H.$$

INC and positive-only DEC exchange; ZERO, POSITIVE and NOP are self-inverse. In the backward copy, incoming and outgoing source groups interchange. Singleton arithmetic groups remain singletons, and a complementary pair of tests remains disjoint. Terminality makes the first bridge legal; entry freshness makes the second bridge legal.

Suppose the original first-halting trace has ℓ instructions and payloads $N_0, \dots, N_\ell = F$. The cleaned trace is its forward copy, the halt bridge, its uniquely determined inverse trace, and the entry bridge. Each reverse step restores its actual forward input. No original instruction enters q_0 , so the reverse copy cannot reach $B : q_0$ prematurely. Thus the first H occurs after $2\ell + 2$ source instructions and restores the entire integer N_0 , including its coprime cofactor. A nonhalting or nonfinal stuck forward run never reaches the first bridge.

Freshness is a genuine hypothesis. An arbitrary source cannot be made eligible merely by prepending a NOP: it can conflict with existing incoming groups. If q_0 recurs in the original trace,

an unguarded reverse exit at B : q_0 could leave before the input is restored. The packet includes a rejecting regression for such a nonfresh source.

143.2 Geometry and first target

At a canonical native section, the fixed right marker is at absolute coordinate 0, and the clean stage-zero controller with its stationary shuttle is at $-12N_0$. All other channels and sites are vacuum. The input controller is F : q_0 ; the exact target replaces it by H at the same positions. Equality includes the entire finite configuration, not just an observed control label or a translate.

The inherited section theorem excludes an earlier target at a macro boundary or inside a macro. A stuck source follows its prescribed shuttle escape and cannot form that target. Arbitrary completion rows are not used before the first target or along these nonhalting and stuck continuations. No claim is made about later target recurrences or stationary posthalt behavior.

Blocking four sites gives radius one, moves the left pair to $-3N_0$ with offset zero, and preserves every time. The alternative four-phase radius-one simulator keeps the native positions; its exact target is phase zero. Its first time is four times the native time, since off-section phases cannot equal this target. Both preserve mass three without a background.

143.3 The clock identity

A forward step $N \rightarrow N'$ has native duration

$$\begin{aligned} \text{INC} : & 108N + 96N' + 8, \\ \text{DEC} : & 96N + 108N' + 8, \\ \text{ZERO, POSITIVE, NOP} : & 192N + 8 \quad (N' = N). \end{aligned}$$

Interchanging N, N' and inverting the instruction preserves the duration. If θ is the forward duration, the reverse half also takes θ . The two NOP bridges take $192F + 8$ and $192N_0 + 8$. Therefore

$$T_{\text{clean}} = 2\theta + 192(F + N_0) + 16 = 2\theta + 192(F + x + 1) + 16. \quad (143.1)$$

This is forward execution of a logically reversed source with different control states. It does not follow from microscopic CA reversibility alone.

144 The complete polynomial and both directions

The only exact raw coordinate y occurs in the final output-copy residual $F - y$. It occurs in neither the raw core nor the other nineteen comparisons. The differently named native auxiliary `native__y_aux` is retained. Replace every occurrence of the raw natural time t by a fresh positive coordinate θ , including its occurrence in the height and clock row. Drop only $F = y$, retain positive F and its encoded endpoint, and append (143.1).

Let $E_1, \dots, E_{19}$ be the retained residuals after the time substitution, with F among z . The complete emitted polynomial is

$$Q_R(x, U; z, \theta) = \sum_{i=1}^{19} E_i(x, z, \theta)^2 + (2\theta + 192(F + x + 1) + 16 - U)^2. \quad (144.1)$$

Equivalently, as an identity of integer polynomials on all tuples,

$$Q_R(x, U; z, \theta) = P_R(x, F, \theta; z) + (2\theta + 192(F + x + 1) + 16 - U)^2. \quad (144.2)$$

The old last square is identically zero after the explicit substitution $y = F$. This identity does not compare the polynomials while leaving y independently free. For phase four, multiply the entire clean affine expression by four inside the new square.

Proof of Theorem 141.1. For soundness, an integer zero of (144.1) makes every residual zero, since a sum of real squares is zero only when each square is zero. Set the forgotten natural coordinate $y = F$ and the original natural time $t = \theta$. Positivity makes both substitutions admissible. This reconstructs a zero of the complete inherited raw certificate, hence an actual first-halting source trace with its exact terminal payload and native time. The clean-wrapper theorem and (143.1) give its first exact whole-configuration target at the requested U .

Conversely, if the cleaned CA first reaches its target, the inherited clean theorem gives an original first halt. Its F is positive and its θ is positive because the source starts away from halt and every genuine instruction has positive duration. The raw theorem supplies positive witnesses at $y = F, t = \theta$. These witnesses satisfy the nineteen retained rows, while (143.1) supplies the twentieth. Thus (144.1) vanishes. The spatial and phase statements follow from their inherited section correspondence. $\square$

Promoting t directly to a positive variable costs one witness and no arithmetic gate; a separate $W - 1$ shift is unnecessary for nonempty runs. No reverse history packet, reverse native witnesses, new selector lane, hidden external horizon or unpaid exponentiation is introduced. Removing F itself would invalidate the positive target port and the halt-bridge duration; removing chronological or native rows is also unjustified.

145 Retained chronology and nonwrapping time

[write] [This section restates, for the circuit with t replaced by θ , the residue-history proof [63] and the exact-clock argument of the pinned interface and of its independent review [62], with the radix Ch^2 in place of the former linear one; it is printed in full as source 22's own account.]

145.1 Pretyping and the actual history

The general encoded source configuration is $n = K(N - 1) + q$. Here $K = 5$ and $m = 6K = 30$. The residue fixes the control and payload modulo six. The source becomes a total positive residue-affine map

$$f(mq + s) = a_s q + d_s, \quad q \geq 0, \quad 1 \leq s \leq m,$$

with integer $a_s \geq 0, d_s > 0$. Missing guards, the halt control, trap and unused labels go to a permanent rejecting trap with unchanged payload. A nonempty orbit ending at the original halt is consequently a genuine first-halting source orbit.

In the general construction choose a source-dependent dyadic C at least the inherited radix multiplier and $2384m + 2$. For the four fixtures $C = 131072$ suffices. For the packet's height h , distinct from its source step count ℓ , retain

$$h = n_0 + n_f + \theta + \eta, \quad B = Ch^2, \quad C = 131072, \quad \eta > 0.$$

Write selector hats as $\widehat{E}_s = E_s + 1$, quotient hat as $\widehat{W} = W + 1$, and exceptional-slope hats as $\widehat{Z}_a = Z_a + 1$. Set

$$J = \sum_s E_s, \quad P = (B - 1)J + 1, \quad G_a = \sum_{s:a_s=a} E_s, \quad R_h = (h - 1)J.$$

The two retained outer equations are

$$J + (W + 1) + \sum_a (Z_a + 1) + \beta = P, \quad \beta > 0, \quad (145.1)$$

$$B\mathcal{N} + n_0 = \mathcal{C} + Pn_f, \quad (145.2)$$

where the paid chronological words are

$$\mathcal{C} = mW + \sum_s sE_s, \quad \mathcal{N} = a_0W + \sum_a (a - a_0)Z_a + \sum_s d_sE_s.$$

Negative slope differences are allowed in this algebraic expression; nonnegativity is not presumed before typing.

Equation (145.1) first excludes $J = 0$: then $P = 1$ while $W + 1 + \beta \geq 2$. It bounds $W, Z_a, J < P$, with $E_s, G_a \leq J$. The selector, class and range AND lanes are

$$(E_s, J, E_s), \quad (W, (B - 1)G_a, Z_a), \quad (W, R_h, W).$$

All lane coefficients lie in $[0, P)$. Concatenate them in base P to ports H_*, M_*, A_* . If $L = m + g + 1$ and v is the least power of two at least L , then

$$0 \leq H_*, M_*, A_* < P^L \leq P^v < Q_* = BP^v.$$

These bounds are obtained before assuming a history or digit typing. The sixteen retained native comparisons, on the paid padded ports

$$16H_* + 12, \quad 16M_* + 10, \quad 16A_* + 8, \quad 16Q_*,$$

give the prescribed-scale native AND theorem. They force Q_* dyadic and H_* AND $M_* = A_*$. Hence B, P are dyadic, and so is the integer h , since $B = Ch^2$ and C is dyadic.

The identity $P - 1 = (B - 1)J$ now forces $P = B^\ell$ for an integer $\ell \geq 1$: writing $B = 2^b, P = 2^p$ and reducing p modulo b proves $b \mid p$. Thus $J = 1 + B + \dots + B^{\ell-1}$. Selectors are subsets of these unit positions. Their sum is J ; the bound $m < B$ prevents a carry, so exactly one residue is selected at each time. The range lane gives quotient digits $0 \leq q_i < h$, and class lanes select their correct subsets. The words in (145.2) consequently have genuine positive base- B digits. Their bounds and the endpoint bounds yield

$$n_0 = \mathcal{C}_0, \quad \mathcal{C}_{i+1} = \mathcal{N}_i, \quad \mathcal{N}_{\ell-1} = n_f.$$

This is actual chronological transport, not an unordered flow or a bare endpoint comparison. The inherited proof supplies the exact digit bounds; replacing its former linear radix by Ch^2 preserves them because the new radix dominates the old multiplier times h .

145.2 Why the clock congruence is exact

Each current encoded configuration lies in $[1, mh]$. No current configuration repeats in a deterministic first-halting orbit: repetition would force a cycle and exclude the later first halt. Therefore

$$1 \leq \ell \leq mh.$$

Each current payload satisfies $N \leq 6h$, and every genuine native tick satisfies

$$0 < \tau \leq 396N + 8 \leq 2384h.$$

It follows that

$$\sum_{i < \ell} \tau_i \leq 2384mh^2 < Ch^2 - 1 = B - 1, \quad (145.3)$$

since $C \geq 2384m + 2$. Independently the positive height identity gives $0 < \theta < h < B - 1$.

The paid affine packed clock $\mathcal{C}_\tau = \sum_i \tau_i B^i$ uses exactly the same selected-quotient and selector hats. Its retained comparison is

$$\mathcal{C}_\tau = (B - 1)(\hat{k} - 1) + \theta, \quad \hat{k} > 0.$$

Reducing modulo $B - 1$ gives $\sum_i \tau_i \equiv \theta \pmod{B - 1}$. Both numbers lie in $[0, B - 2]$ by (145.3); hence they are equal. Hiding θ existentially changes none of these paid guards. The new cleanup square is applied only after exact native time is established.

This argument requires first-halt rejecting totalization. It does not apply automatically to arbitrary endpoints that permit looping after an earlier acceptance. Nor is a bounded sample of nonhalting behavior proof of nonhalting.

146 Every accepted nonempty witness fiber is infinite

Theorem 146.1 (Infinite positive fibers). *For any accepted fixed pair (x, U) of an emitted nonempty native, spatial or phase-four circuit, its complete positive witness fiber is infinite. Rejected fibers are empty. This assertion excludes the separate initially halted circuits.*

Proof. Fix the actual first-halting source trace, its payload F , native time θ , encoded endpoints n_0, n_f , length $\ell \geq 1$ and residue quotients q_i . For every sufficiently large dyadic integer

$$h > \max\{n_0 + n_f + \theta, q_0, \dots, q_{\ell-1}\},$$

set $\eta = h - n_0 - n_f - \theta > 0$, $B = Ch^2$, $P = B^\ell$ and $J = 1 + B + \dots + B^{\ell-1}$. Pack the same residue indicators, quotients and selected quotients in this new base. Zero words have positive hat one and are allowed.

Let $g \leq m - 1$ be the number of exceptional slope classes. They are disjoint, so $\sum_a Z_a \leq W \leq (h - 1)J$. The exact remaining global slack is

$$\begin{aligned} \beta &= P - J - (W + 1) - \sum_a (Z_a + 1) \\ &= (B - 2)J - W - \sum_a Z_a - g \\ &\geq (B - 2h)J - g > 0. \end{aligned}$$

For the last inequality, the new radix satisfies $B \geq K_0 h$, where the inherited dyadic multiplier K_0 is at least 4 and $m + 1$; also $h \geq 3$ and $J \geq 1$. Thus $(B - 2h)J \geq (K_0 - 2)hJ > g$. These are strict positive supplied slacks, not merely nonnegative residuals.

All actual lane equalities hold. Selector words are at most J , quotient and selected-quotient words are at most $(h - 1)J < P$, class masks are at most $(B - 1)J = P - 1$, and the range mask is below P . Therefore the concatenated ports are below the dyadic prescribed scale $Q_* = BP^v$. Their padded positive versions are strictly below $16Q_*$. The inherited prescribed-scale completeness theorem supplies fresh positive native witnesses at every such scale. No old native tuple is assumed to remain valid when h changes.

Chronological transport is the actual fixed path, and

$$\hat{k} = 1 + \frac{\mathcal{C}_\tau - \theta}{B - 1} \in \mathbb{Z}_{>0},$$

since $\mathcal{C}_\tau - \theta = \sum_i \tau_i(B^i - 1)$ is a nonnegative multiple of $B - 1$. For $\ell = 1$ the quotient is zero and the hat is one. The terminal payload and cleanup equation preserve the same fixed external (x, U) . Thus every permitted height extends to a full positive zero.

There are infinitely many such dyadic heights; distinct heights give distinct retained η . The resulting complete tuples are distinct regardless of native multiplicity at a fixed scale. $\square$

The result is an all-height consequence of the inherited completeness theorem. The checker verifies 54 larger-height outer extensions as implementation evidence; it does not numerically construct huge full native/Pell witnesses. A deterministic source trace and unique reverse lift therefore coexist with infinitely many arithmetic witnesses. These emitted representations are not finite-fold, without implying any multiplicity limitation for a different representation of the same simple relation.

[write] [Compare Part V: at a fixed source horizon sources 14 and 15 have exactly one natural witness (Theorems 58.1 and 70.2), and source 14 warns that uniqueness at each fixed horizon does not by itself yield a finite-fold representation after unbounded coding is introduced (Question 5). This theorem shows that for the present coding it does not. Source 23 (Theorems 154.1 and 155.1) shows that infinitude survives even at the least admissible height.]

147 Complete arithmetic ledgers and exact degrees

147.1 Literal cost model

Each binary $+$, $-$ or $\times$ gate costs one operation. Constant multiplications and squarings count as multiplications; the integer literals are explicitly listed in each ledger. This model does not synthesize those constants from a restricted initial constant set. There is no gate for exponentiation, division, bitwise AND or an unspecified predicate. Powers such as P^v are realized by their fixed squaring chains. Bitwise AND appears only in the semantic checker of the native encoding, not as a free polynomial gate.

The new affine bridge consists of these six paid gates:

$$F + x, \quad F + x + 1, \quad 192(F + x + 1), \quad 2\theta, \quad 192(F + x + 1) + 2\theta, \quad 192(F + x + 1) + 2\theta + 16.$$

They cost $2M + 4A$, where A includes addition and subtraction. Deleting the old $F = y$ square and its accumulation removes $1M + 2A$; adding the clean-time square and its accumulation restores

$1M + 2A$. Every final twenty-row SOS uses twenty subtractions, twenty squarings and nineteen additions: $20M + 39A = 59$ gates.

Fixture	Raw core	M	A	Total	Witnesses	Degree
INC2;DEC2	539	239	365	604	60	2344
ZERO3	414	184	295	479	58	1192
NOP	412	182	295	477	58	1192
POSITIVE3	415	189	291	480	58	1192

These totals include the core, six bridge gates and the full finalizer. Each has two natural parameters x, U and twenty comparisons. Spatial blocking uses precisely the same polynomials. Phase four adds one multiplication by literal four before the final clean comparison, giving totals 605, 480, 478, 481 with the same A , witnesses and degrees.

147.2 Separately emitted constant folded schedules

The original freeze and its counts above remain unchanged. A separate addendum folds the affine input shift and scale into literal constants. The native and phase-four clocks are identically

$$192(F + x) + 2\theta + 208, \quad 768(F + x) + 8\theta + 832,$$

respectively, since $192 + 16 = 208$ and $4 \cdot 208 = 832$. In either model, compute $F + x$, the first scalar product, the θ product, their sum, and the final constant addition. This uses five bridge gates, $2M + 3A$, with no changed coordinate domain. The complete twenty-row finalizer is rebuilt unchanged in size.

Folded native or phase four	M	A	Total	Witnesses	Degree
INC2;DEC2	239	364	603	60	2344
ZERO3	184	294	478	58	1192
NOP	182	294	476	58	1192
POSITIVE3	189	290	479	58	1192

This saves one addition from the original native schedule and one multiplication plus one addition from the original unrolled phase-four schedule. The equality holds on every integer tuple, so all semantic and fiber statements are unchanged. The addendum supplies eight complete circuits, their full ledgers, degree certificates and an independent data-only audit. The separate audit checks all 4072 folded gates and, in addition to formal degree bounds, evaluates complete univariate polynomial specializations modulo 1000003. Its pinned reference files are byte-identical copies of the original eight circuits. This is an elementary schedule improvement within the stated literal-constant model, not an optimality claim. The unchanged first freeze and its original audits are retained separately.

147.3 Witnesses and literal inheritance

The witnesses are the raw positive list followed by `theta_positive`: twenty-two native coordinates, thirty selector hats, one quotient hat, g selected-quotient hats, two slacks, F , the shifted clock quotient and θ . Thus the count is $58 + g$, with $g = 2$ for INC2;DEC2 and zero for the other fixtures. The exact gate and coordinate names are in every JSON circuit and matching textual DAG.

The independent auditor checks 3560 inherited-gate identities across all eight nonempty variants. Every raw core gate is unchanged except the operand substitution $t \mapsto \theta$. It checks all 152 retained comparison-pair identities and the complete twenty-row finalizers. The complete ancestor set of the output equals all declared gates and ports: there are no undeclared operands, forward references, dead charged gates or dead supplied coordinates.

147.4 Deterministic exact degree certificates

The claimed degrees are exact total polynomial degrees in all supplied coordinates, before restricting to the zero set or quantifying witnesses. Give each coordinate v_i degree one and each integer literal degree zero. Propagate a formal bound d by addition of bounds at multiplication and maximum at addition or subtraction. Simultaneously evaluate the degree- d homogeneous component under the integer ray

$$v_i = (i + 2)z, \quad i = 0, 1, \dots,$$

using the exact emitted order `parameters + auxiliaries`, modulo $p = 1000003$.

At a sum or difference of bound d , an operand contributes its stored coefficient only if its formal bound equals d . At a product, multiply the stored top coefficients. A zero coefficient never lowers the bound: cancellation on this chosen ray cannot justify discarding lower components for subsequent operations. Induction on the gate list proves both the degree upper bound and the claimed component evaluation. A nonzero output value modulo p implies a genuinely nonzero integer homogeneous component, so the lower bound matches the upper bound. Primality is not needed for this nonzero-modulus implication.

Circuit family	Degree	Top value modulo 1000003
INC2;DEC2 native and phase four	2344	135347
ZERO3 NOP POSITIVE3 native and phase four	1192	977370
Initially halted native	2	585225
Initially halted phase four	2	418734

The full weight map and every per-gate degree/top row are in the JSON artifacts. Two independently implemented data-only checkers recompute the certificates. Nonempty traces contain two zero-top entries, and an explicit cancellation regression verifies retention of their bounds. Equality of the high-degree native and phase-four coefficients is consistent with changing only a degree-one time residual. No degree claim is a lower bound on circuit size or on the intrinsic complexity of the represented relation.

148 Initially halted sources and raw input limits

148.1 The separate zero step circuit

If $q_0 = q_h$, freshness and terminality make it isolated. The cleaned trace consists of two NOP bridges, with $\theta = 0$ and $F = N_0$. Hence

$$T_{\text{clean}} = 384N_0 + 16 = 384x + 400.$$

The separate polynomial is

$$(384x + 400 - U)^2,$$

with four gates, $2M + 2A$, one comparison, zero witnesses and degree two. Its phase-four counterpart is $(1536x + 1600 - U)^2$, with the same ledger by constant folding. Neither is forced through the positive-time nonempty packet. At an accepted external pair the witness fiber is the singleton empty tuple, so the infinite-fiber theorem does not apply.

148.2 All input clock sanity checks

On accepted paths of the four fixtures, $F = N_0$. Direct source arithmetic gives

Fixture	Native forward duration θ	Clean duration
INC2;DEC2	$600N_0 + 16$	$1584x + 1632$
NOP	$192N_0 + 8$	$768x + 800$
ZERO3 when $3 \nmid x + 1$	$192N_0 + 8$	$768x + 800$
POSITIVE3 when $3 \mid x + 1$	$192N_0 + 8$	$768x + 800$

All phase-four durations are multiplied by four. The generic circuits deliberately retain F , rather than specializing these coincidences away. For example a fresh singleton INC2 source has $F = 2N_0$, $\theta = 300N_0 + 8$, and clean time $1176N_0 + 32$. Additional wrapper tests cover this unequal-payload behavior and coprime factors five and seven, without claiming a full emitted generic circuit count for these extra sources.

A fake ZERO3 halt at $N_0 = 3$ and fake POSITIVE3 halt at $N_0 = 1$ fail the retained chronological endpoint row: their totalized successors are the rejecting trap, not halt. This construction removes no selector and no encoded halt endpoint, and does not pass through a weakened fixed-horizon selector interface.

148.3 What the raw coordinate cannot represent

Write $N_0 = 2^a 3^b c$, with $\gcd(c, 6) = 1$. Scaling the raw payload by a positive factor coprime to six preserves guards and the control sequence, and scales the payload throughout. For each corresponding tick,

$$\tau(cN) - 8 = c(\tau(N) - 8).$$

When output and time are existentially hidden, halting thus depends only on $\nu_2(x + 1), \nu_3(x + 1)$. In particular $x = 0$ and 4 cannot be distinguished for halting; among positive coordinates, neither can $x = 4$ and 6 . This excludes interpreting the raw $x + 1$ interface as an arbitrary c.e. ordinary-input loader.

The general fixed-source theorem is compatible with encoded input contracts, but a numerical universal claim would require a concrete fixed universal separated reversible source table, its proved ordinary-input or decoder interface, and a fully charged emission and audit. None is supplied here. The inherited clean theorem's abstract computable reduction is not such a paid loader.

All zero-set interpretations require natural external coordinates and strictly positive integer witnesses. Real SOS nonnegativity does not extend the native theorem to rational or real witnesses. The target depends on the input; no fixed universal target, stationary halt, arbitrary later recurrence, optimality or new universal operation bound is asserted.

149 Independent checks and reproducible delivery

149.1 What was checked

The new emitter authenticates the pinned raw JSON, parses its literal arithmetic, verifies the old finalizers and emits complete clean circuits. It does not execute the upstream author scripts. The independent literal auditor checks all ten JSON circuits and all eight text DAGs: 4092 gates, 162 comparison rows, inheritance, closure, liveness, literal constants, counts, degree certificates and complete SOS output identity.

The literal auditor also checks 120 signed assignments, 320 modular assignments and 606 exact or adjacent-time zero-step evaluations. These are finite arithmetic identities. No implication from modular SOS zero to individual modular residual zero is used. Eight deliberately corrupted circuit copies are rejected: omitted witness, undeclared operand, changed inherited gate, deleted comparison, wrong cleanup constant, extra dead gate, undercounted ledger and false degree.

A separate semantic checker records:

- 19200 residue-state and clock evaluations, including rejecting controls;
- 576 accepted full outer packed histories, with all four outer equations and the actual joined AND checked;
- 576 each of strict no-wrap and coprime-scaled trace checks;
- 192 false prime-three halts rejected by chronological transport;
- 576 rejections each for clean-time, native-time and final-payload mutations;
- 384 complete signed/modular SOS evaluations and 390 zero-step exact evaluations;
- 29 additional physical-wrapper checks, including 13 unequal-payload cases, and one nonfresh-entry rejection;
- 54 larger-height outer extensions at unchanged input and clean time.

Native and phase-four cases are counted separately. These are not hundreds of distinct source programs. Bounded tests and all-tuple arithmetic identities support implementation correctness; the inherited theorems and the proofs above establish the unbounded semantic interpretation.

No complete positive native/Pell zero tuple has been numerically materialized. Outer-history witnesses stop at the native interface. Signed or modular evaluations of the entire DAG are not evaluations at a full positive satisfying native witness. The package preserves this distinction explicitly in its receipts.

149.2 Portable commands and integrity boundary

The release contains this editable `report21.tex`, its rendered `report21.pdf`, the unchanged mathematical freeze under `reproducibility/`, the separate constant-folded alternatives under `folded-clocks/`, and standard-library integrity and replay tools. From the release root:

```
python -B verify_release.py
python -B replay.py --output /tmp/new-report21-replay
```

The output directory must be new and outside the release. Replay verifies the sealed release first, copies the mathematical packet into fresh normal and optimized work areas, and runs all own emitters and independent checks in isolated standard-library Python processes. Each mode authenticates the freeze before and after; deterministic re-emission and receipt bytes must agree. The assertion-dependent secondary degree checker runs normally and is explicitly required to reject optimized execution; the full independent circuit auditor recomputes every degree certificate successfully in both modes. The original release is left unchanged. Exact-inventory attack tests are run in both modes.

For direct mathematical replay, enter `reproducibility/` and run:

```
python -B emit_clean_clocks.py --check
python -B check_semantics.py --check
python -B audit/check_emitted_circuits.py \
    --expect audit/independent_emitted_circuit_audit.json
python -B audit/check_degree_certificate.py
python -B verify_manifest.py
```

The constant-folding addendum is also checked in both modes by the main replay. To replay it directly, enter `folded-clocks/` and run:

```
python -B emit_folded_clocks.py --check
python -B audit/check_folded_clocks.py \
    --expect audit/independent_folded_clock_audit.json
python -B verify_manifest.py
```

The degree checker rewrites only its deterministic receipt; the main replay runs it in the disposable copy. No third-party executable source is invoked. To rebuild the PDF with a local TeX installation, run `sh build.sh`; temporary TeX products remain outside the sealed tree. The distributed PDF is already built and is not needed for arithmetic replay.

The release manifest enumerates every delivered payload file by exact relative path, byte length and SHA-256. Its checker rejects missing files, extra files or directories, symlinks, nonregular objects, duplicate JSON keys, noncanonical paths, altered pins and altered bytes, including under Python optimization. A companion digest protects the manifest's bytes against accidental change. These are consistency checks, not a digital signature; authenticity requires a separately trusted archive digest. The outer inventory also authenticates the untouched inner freeze and its own manifest.

All pages of the final PDF were rendered and visually reviewed; its text and mathematical claims were independently reviewed against the frozen source. The packaged quality and replay receipts distinguish those review results from foundational theorem dependencies. A content scan found no private credentials or personal conversation material in the release. Public author attribution and public mathematical provenance remain intentional.

[write] **The release in this report.** These are the delivered names. Of `reproducibility/`, the emitter `emit_clean_clocks.py`, the semantic checker `check_semantics.py` and the two checkers of `audit/` are shipped as `code/22-clean-clocks-*`; the receipts of `receipts/` and `audit/` and the source record `source/provenance.json` as `data/22-clean-clocks-*` (the last as `data/22-clean-clocks-source-provenance.json`); `THEOREM.md`, `README.md` and the two audit notes as `22-clean-clocks-THEOREM.md`, `22-clean-clocks-REPRODUCIBILITY-README.md`, `22-clean-clocks-PROOF-AUDIT.md` and `22-clean-clocks-EMITTED-CIRCUIT-AUDIT.md`. Of `folded-clocks/`, the emitter and its checker are `code/22-clean-clocks-folded-*`, its receipts

and logs `data/22-clean-clocks-folded-*`, and `ADDENDUM.md`, `PORTABILITY.md`, `README.md` and the folding audit `22-clean-clocks-FOLDED-ADDENDUM.md`, `-FOLDED-PORTABILITY.md`, `-FOLDED-README.md` and `22-clean-clocks-FIVE-GATE-FOLDING-AUDIT.md`. The release-level `build.sh`, `delivery-provenance.json`, `privacy-scan.json`, `report21-qa.json`, `report21-replay.json` and `report21-math-review.md` are shipped under the prefix too. Not shipped: the manuscript (printed here) and its PDF, the delivery README, the release and packet checksum manifests with their checkers and the release tools (`verify_release.py`, `replay.py`, `release_checks.py`, `write_manifest.py`), the byte copies of repository files in `reproducibility/source/`, three log files that are byte copies of shipped ones, and the 42 circuit files (34 emitted files and eight reference copies), which the shipped emitters regenerate byte for byte from the repository’s raw receipt in a few seconds (README, “Reconstructing the excluded data”); the shipped emission receipts list the SHA-256 of every one. The replay above needs the delivered layout, and on Windows it fails: two checkers key their hash tables by native path strings (`audit/check_emitted_circuits.py`, lines 382, 393 and 394, and `audit/check_folded_clocks.py`, lines 423 and 425), and `replay.py` starts its children isolated with `-I -S`. With POSIX path strings every delivered command passed at placement, in normal and optimized mode, on unchanged copies; the secondary degree checker rejects optimized mode, as designed (README).

[write] **The programme’s intake.** The Hilbert’s-tenth programme took this source in, together with Report 20 (Part V of `five-particle-binary-automata`), in `review_event_budget_clean_clock_intake.md` (commit `8aabb1453`, 3 October 2026) [69]. It read the theorem and the folding addendum against the inherited clean-wrapper theorem, executed no archived Python, and found “no newly identified defect in their stated theorem scopes”. It confirms that Theorem 141.1 has fixed arity for each fixed source and no external horizon because it inherits the already unbounded native/Pell packed-history theorem, whose replacement it is not; that the new content is the clean wrapper and the exact affine first-hit clock; that accepted nonempty fibres are infinite, all domains integral and the raw input $x + 1$; and that the folded identity $192(F + x) + 2\theta + 208$ is a genuine paid saving. It stresses that “compact” means a compressed history certificate without reverse witnesses, not a compact real domain. The same commit adds a separate construction, `three_mass_direct_clean_clock.md` with an independent review (PASS) [71], which imposes the clean clock directly on the packed clock word for the same four sources: 597, 472, 470 and 473 operations, 59, 57, 57 and 57 positive witnesses and 19 comparisons, six operations and one witness fewer than the folded circuits of Section 147.2. The programme states that these counts must not be attributed to source 22 and do not revise its historical counts. A later note [72] proves that the further deletion of the input from that construction’s height (596, 471, 469 and 472 operations) is unsound, with false zeros at infinitely many input pairs.

Source 23 (batch-82 manuscript 02): Canonical histories and infinite native fibers

Exact fixed source Diophantine certificates for three mass computation, “Report 22 Mathematical construction and portable reproducibility package”, 3 October 2026. [write] From the archive `Canonical_Histories_and_Infinite_Fibers_Package.zip`; its labels carry the prefix `smc:ch:`, its sections are Sections 150–158 here, and its equations are numbered by section. It is a declared sequel of source 22 (“Report 21”), whose frozen base and folded packets it carried byte for byte (shipped once, as source 22’s). Its bibliography entries are this report’s [65], [66], [63], [62], [67] and [68]. Its abstract follows; the bracketed note in it is [write].

The unbounded clean-clock certificates of Report 21 [source 22] admit a canonical outer history without acquiring finite full witness fibers. Put $S = n_{\text{initial}} + n_{\text{target}} + \theta$ and retain $h = S + \eta$. One new positive coordinate κ and the comparison $\eta + \kappa = S + 1$ force the least dyadic $h > S$.

Completeness survives at this exact height because every genuine residue quotient is below the physical native duration θ . At every accepted external pair, all supplied outer coordinates are unique, and projection onto the twenty-two native coordinates is a bijection onto the entire native extension fiber at its fixed ports and scale. A separate elementary matrix argument proves that every nonempty such native fiber is infinite: only three native coordinates need vary, while the other nineteen stay fixed. Thus full accepted nonempty fibers remain infinite even after outer canonicalization. Eight complete circuits have 608, 483, 481 and 484 operations in each clock model, 61, 59, 59 and 59 positive coordinates, and exact degrees 2344, 1192, 1192 and 1192. These are structural results for the retained representations. They imply no general finite-fold MRDP claim, no ordinary-input universal loader, no optimality result and no new Pell classification. No complete positive native Pell tuple is numerically materialized.

150 The two results and their common scope

We use $\mathbb{N} = \{0, 1, 2, \dots\}$ and $\mathbb{Z}_{>0} = \{1, 2, \dots\}$. External coordinates are natural integers. Every supplied witness coordinate in a nonempty circuit is strictly positive. Computed arithmetic registers may be signed; they are not extra witnesses.

Fix an eligible deterministic reversible separated source R , using INC2, INC3, enabled DEC2 and DEC3, ZERO2, ZERO3, POSITIVE2, POSITIVE3 and NOP. Its entry q_0 has no incoming instruction, its designated halt q_h has no outgoing instruction, and initially $q_0 \neq q_h$. Eligibility means that the inherited complete raw residue-history/native interface and exact clean-target wrapper apply. It does not assert a free fresh-entry transformation for every reversible program.

The positive raw input is $N_0 = x + 1$. The valuations at two and three encode the counters, and the cofactor coprime to six remains part of the payload. Disabled guards are nonfinal stuck states. The cleaned cellular automaton (CA) has total mass three on the configurations considered, using a finite alphabet of Boolean channel subsets whose mass is the number of occupied channels. This is not the convention that a state's integer label equals its mass.

Let $Q_R(x, U; w)$ denote the complete five-gate-folded clean-clock polynomial of Report 21 [source 22, Section 147.2; as a polynomial it equals source 22's Q_R of (144.1), whose circuits it computes with one gate fewer, and its witness tuple w is source 22's (z, θ)]. For a nonempty first-halting source history let F be its final raw payload and $\theta > 0$ its forward physical native duration. Native and spatially blocked time satisfy

$$U = 2\theta + 192(F + x + 1) + 16. \quad (150.1)$$

The phase-four time is four times the whole right side. The exact target restores the entire raw input and the prescribed absolute marker/controller geometry; phase four has a phase-zero target. The semantic claim is a first-hit claim about this whole finite configuration, not a claim about arbitrary microscopic states or later recurrence.

Theorem 150.1 (Canonical outer history and infinite full fibers). *Assume the inherited native completeness, residue-history and clean-target theorems described in Section 151. Set*

$$S = n_{\text{initial}} + n_{\text{target}} + \theta, \quad h = S + \eta,$$

where η is the existing positive height slack. Add $\kappa \in \mathbb{Z}_{>0}$ and define

$$Q_R^{\text{can}}(x, U; w, \kappa) = Q_R(x, U; w) + (\eta + \kappa - S - 1)^2. \quad (150.2)$$

Then for each $x, U \in \mathbb{N}$:

1. *A positive zero exists exactly when the inherited cleaned CA first reaches its specified target at U in the chosen clock model.*
2. *At an accepted pair, h is the least power of two strictly greater than S , and every supplied outer coordinate is unique.*
3. *Projection onto the twenty-two native coordinates is a bijection onto the entire positive native extension fiber at its uniquely fixed padded ports and prescribed scale.*
4. *Every such nonempty full fiber is infinite. Starting from any one witness, only native j, o, y_{aux} need vary; all supplied outer coordinates and the other nineteen native coordinates can stay fixed.*

The initially halted, witness-free construction is separate and excluded from items 2–4.

Equation (150.2) is an identity of integer polynomials on all tuples. Items 2–4 are statements about accepted positive fibers. In particular the fiberwise inverse in item 3 inserts fixed outer constants; it is not an all-input polynomial procedure for discovering a halting trace or a global polynomial coordinate change from old witnesses to new witnesses.

The outer-canonical packet alone does not decide native cardinality. Its statement of that limitation is retained in the archive. The separate fixed-scale theorem in Section 155 supplies the additional fact needed for item 4. Report 21’s infinite-fiber conclusion [Theorem 146.1] remains correct; the new result shows that enlarging height is not necessary for infinitude.

151 Dependencies and source ledger

The public native and residue-history sources are pinned to ProveIt commit

ad634b2d10ad666260f9fdff04ec94b75169ee4b

The raw complete arithmetic receipt has SHA-256

fe69f0504c8f68c4130c4b21fed5c817596ab5c0e2fdf9621b0bf5f45c12ac0e

It is read as literal JSON data. No upstream Python implementation is imported or executed by the new proofs’ checkers or replay harness. Local copies, their provenance and pinned public links are included in the archive.

Dependency	Role in the present argument
Native prescribed AND64 [65, 66]	Dyadic typing, soundness and positive extension at the prescribed fixed scale; the sixteen native equations
Residue-affine packing [63]	Typed chronological histories, positive packed hats, lane bounds and native input interface
Raw three-mass interface [62]	Literal gates, coordinate order, physical clock, first-halt rejection and public arithmetic provenance
Inherited clean-target theorem [source 15]	Forward/reverse wrapper, full-payload restoration, exact absolute target and equation (150.1)
Report 21 [source 22] base and folding proofs	Complete cleaned arithmetic, exact no-wrap argument, paid ports, five-gate time bridge and frozen parent circuits
Outer canonicalization audit	Quotient bound at the least height, outer uniqueness, native-fiber bijection and literal extra-square implementation
Fixed-scale native audit	Three-coordinate dependency isolation, positivity and the matrix family of Section 155

The unchanged clean-target proof snapshot has SHA-256

3d85f8cde66837d8916fb3273c2c57cc1a91ce37de1f3d6c6fe28a4db9103168

The outer packet pins its original base and folded manifests respectively by

1bf1225aa950ad1d4f842c8bf098e1935925cd1d52c90453b7696c1321648f95

774a4498984ebccf8216897e21bf597084e70b148557ba4fa12b597bba44ba91

The separately frozen outer-canonical packet manifest has SHA-256

3546ec269df10808a2067cea9df325709df11966b346857271e66ab4855cadce

The separately frozen native-fiber packet manifest has SHA-256

d716d6dc668547ee5d64dfe77620b66f93ab839f95e0261250c2a387b5338dca

Release provenance distinguishes original snapshots from any documented portability edits and records delivered hashes. Earlier reports and source packets are left unchanged.

This report does not reprove the historical native/Pell representation theorem or regenerate the CA local rules. It proves the new outer canonicalization and fixed-scale multiplicity statements from those explicit dependencies. The portable replay is computationally self-contained; that fact is distinct from foundational independence of the inherited mathematics.

[write] [The pinned commit and files are those of source 22 (note at the end of Section 142); the clean-target proof snapshot is source 15’s theorem. The outer packet is `canonical-height/` and the native-fibre packet `native-fibers/`, shipped as `23-canonical-fibers-*`; the base and folded manifests, which the outer packet pins and which describe source 22’s packets, and the two later packet manifests are not shipped, and their SHA-256 values above are the record of them. The “outer canonicalization audit” and “fixed-scale native audit” of the table are `23-canonical-fibers-CANONICAL-HEIGHT-PROOF-AUDIT.md` and `23-canonical-fibers-NATIVE-FIBERS-REVIEW.md`.]

152 Why the physical clock pays for the canonical height

152.1 Encoded states and exact instruction durations

Assign controls in $\{1, \dots, K\}$, including a rejecting trap, and put $m = 6K$. Encode a positive raw payload and control by

$$n = K(N - 1) + q, \quad 1 \leq q \leq K.$$

Missing guards, original halt, trap and unused labels transition to the trap with unchanged payload, and the trap never exits. Thus a nonempty path ending at the original halt represents its first genuine halt. The retained residue-affine table is

$$f(mz + r) = a_r z + d_r, \quad 1 \leq r \leq m, \quad z \geq 0, \quad a_r \geq 0, \quad d_r \geq 1.$$

For a current encoded configuration,

$$\left\lfloor \frac{n-1}{6K} \right\rfloor = \left\lfloor \frac{N-1}{6} \right\rfloor. \quad (152.1)$$

Indeed, writing $N - 1 = 6a + b$, $0 \leq b \leq 5$, leaves remainder $Kb + q - 1$ between zero and $6K - 1$.

The literal native tick durations, measured at the current payload, are

Enabled instruction	Native duration τ
INC2	$300N + 8$
INC3	$396N + 8$
DEC2	$150N + 8$
DEC3	$132N + 8$
ZERO POSITIVE NOP	$192N + 8$

Therefore, in any genuine nonempty first-halting trace with $\theta = \sum_i \tau_i$,

$$z_i = \left\lfloor \frac{N_i - 1}{6} \right\rfloor < N_i \leq \tau_i \leq \theta. \quad (152.2)$$

The weaker uniform inequality $\tau_i \geq 132N_i + 8$ is enough. The argument does not require monotone payloads or an endpoint bound on intermediate growth. A large temporary payload costs physical time at the very step where it appears.

With

$$n_{\text{initial}} = Kx + q_0, \quad n_{\text{target}} = K(F - 1) + q_h, \quad S = n_{\text{initial}} + n_{\text{target}} + \theta,$$

every $h > S$ already exceeds every true quotient. This is the completeness point that an arbitrary endpoint-only canonical-height proposal would lack.

152.2 The strict dyadic interval

Lemma 152.1. *For $S \in \mathbb{Z}_{>0}$, $h = S + \eta$, and $\eta, \kappa \in \mathbb{Z}_{>0}$, the new comparison $\eta + \kappa = S + 1$ is equivalent to*

$$S < h \leq 2S, \quad \eta = h - S, \quad \kappa = 2S + 1 - h.$$

Exactly one power of two lies in $(S, 2S]$, the least one strictly greater than S .

Proof. The two inequalities are precisely positivity of the displayed slacks. If $2^a \leq S < 2^{a+1}$, then $2^{a+1} \in (S, 2S]$ and adjacent powers lie outside. When $S = 2^a$, the chosen height is $2S$, with $\eta = S$ and $\kappa = 1$; the inadmissible choice $h = S$ would make $\eta = 0$. $\square$

The old radix is $B = Ch^2$, where C is a fixed dyadic integer satisfying

$$C \geq L, \quad C \geq 2384m + 2, \quad L \geq \max\{4, m + 1, 1 + \max_r(a_r + d_r)\},$$

and L is dyadic. The native theorem forces B and the length power P to be dyadic. Since C is dyadic and h is a positive integer, $B = Ch^2$ dyadic forces h dyadic: an odd prime divisor of h would divide B . The new row is not an independent dyadicity oracle. In the emitted fixtures $K = 5$, $m = 30$, $C = 131072$.

153 Soundness and exact canonical completeness

153.1 The old no-wrap argument remains necessary

[write] [This subsection restates source 22's argument of Section 145.2 (with t for source 22's ℓ , C_τ for its $\mathcal{C}_\tau$ and $\widehat{c}_\tau$ for its $\widehat{k}$); it is printed as source 23's account of why that argument is still needed.]

A positive integer zero of (150.2) annihilates all old comparison squares and the new square. Old soundness supplies the first-halting history, F and its clock. Lemma 152.1 then forces the stated height. No old comparison is deleted.

For clarity, the inherited exact-clock proof does not obtain the clock merely from a congruence. Before digit typing, its global equation implies $J \geq 1$, $P \geq B$ and that all lane coefficients lie in $[0, P)$. The native block establishes actual packed AND and dyadic B, P . Then

$$P - 1 = (B - 1)J$$

forces $P = B^t$, $J = 1 + B + \cdots + B^{t-1}$ for $t \geq 1$. Selector lanes type one residue per time and the range mask $(h - 1)J$ types $0 \leq z_i < h$. Chronological transport gives the deterministic orbit from the initial endpoint to the halt, with each current encoded state in $[1, mh]$.

A current state cannot repeat before a deterministic first halt, so $t \leq mh$. Its payload satisfies $N_i \leq 6h$, giving $\tau_i \leq 396N_i + 8 \leq 2384h$. Consequently

$$0 < \sum_i \tau_i \leq 2384mh^2 < Ch^2 - 1 = B - 1. \quad (153.1)$$

Independently $0 < \theta < h < B - 1$. If $C_\tau = \sum_i \tau_i B^i$ is the typed clock word, the retained paid comparison is

$$C_\tau = (B - 1)(\hat{c}_\tau - 1) + \theta.$$

Reduction modulo $B - 1$ gives $\sum_i \tau_i \equiv \theta$, and the strict interval bounds give equality. First-halt totalization is essential here. The new cap neither replaces the proof nor permits loops or posthalt continuation.

153.2 Positive witnesses at the exact selected height

Conversely take the genuine first-halting trace and its $\theta = \sum_i \tau_i$. Choose the unique dyadic $h \in (S, 2S]$, set $\eta = h - S$, $\kappa = 2S + 1 - h$, and use (152.2) to fit all true quotient digits. Set

$$B = Ch^2, \quad P = B^t, \quad J = \sum_{i=0}^{t-1} B^i.$$

Pack the chronological selector words E_r , quotient word W and selected-quotient words Z_a for the g exceptional slope classes. Their supplied hats are $E_r + 1, W + 1, Z_a + 1$. Zero words therefore have positive hat one and are allowed.

Disjoint exceptional classes give $\sum_a Z_a \leq W \leq (h - 1)J$. The global comparison fixes the remaining slack as

$$\begin{aligned} \beta &= P - J - (W + 1) - \sum_a (Z_a + 1) \\ &= (B - 2)J - W - \sum_a Z_a - g \\ &\geq (B - 2h)J - g \geq (L - 2)hJ - g > 0. \end{aligned} \quad (153.2)$$

For the final strict bound use $L \geq 4$, $L \geq m + 1$, $g \leq m - 1$, $h \geq 3$ and $J \geq 1$. The physical nonempty setting has $S > 2$, so its selected height satisfies the required lower bound.

All scalar AND lane coefficients lie in $[0, P)$. Selectors and class selectors are at most J ; W and every Z_a are at most $(h-1)J < P$; class masks are at most $(B-1)J = P-1$. Let $\ell = m + g + 1$ be the number of lanes and v the least power of two at least ℓ . For the joined words H, M, A ,

$$0 \leq H, M, A < P^\ell \leq P^v < Q_{\text{native}} = BP^v.$$

The genuine AND holds lane by lane. The actual native ports and scale are

$$\mathbf{a} = 16H + 12, \quad \mathbf{b} = 16M + 10, \quad \mathbf{c} = 16A + 8, \quad q = 16Q_{\text{native}}. \quad (153.3)$$

The three ports are positive and below q , q is dyadic, and the low padding satisfies $12 \text{ AND } 10 = 8$. Prescribed-scale completeness supplies all twenty-two positive native coordinates at this exact scale. It is neither necessary nor generally correct to reuse a native tuple from another height.

Finally

$$\hat{c}_\tau = 1 + \frac{C_\tau - \theta}{B-1} \in \mathbb{Z}_{>0},$$

since $C_\tau - \theta = \sum_i \tau_i(B^i - 1)$ is a nonnegative multiple of $B-1$. For one step this hat equals one. Transport, physical time, clean time and the new cap all hold, so every one of the twenty-one comparisons vanishes. This proves completeness at the canonical height, rather than at an unspecified sufficiently large height.

154 The precise canonical object

Fix an accepted (x, U) and clock model. Determinism and first-halt semantics fix the entire trace, its length t , final payload F , native duration θ , and every residue and quotient. These fix the endpoints and S , hence h, η, κ , then B, P, J , every packed word and hat. Equation (153.2) fixes β and the clock comparison fixes $\hat{c}_\tau$. All outer registers and the padded ports (153.3) are fixed as well.

The supplied outer coordinates are the selector hats, quotient hat, exceptional selected-quotient hats, $\eta, \beta, F, \hat{c}_\tau, \theta, \kappa$. In these fixtures their number is 39 for INC2;DEC2 and 37 for the other three. Quantities S, h, B, P, J are computed registers, not additional coordinates.

Write $o(x, U)$ for this unique outer tuple and $\mathcal{E}(\mathbf{a}, \mathbf{b}, \mathbf{c}, q) \subseteq \mathbb{Z}_{>0}^{22}$ for the set of *all* positive native assignments satisfying the complete sixteen-comparison block at the fixed boundary. The native coordinate order, with the literal prefix `native__` understood, is

$$F_0, F_1, F_2, a, c, d, f, h, i, j, k, o, r, s, w, \\ \tau, \eta, \zeta, ga, y_{\text{aux}}, \text{odd_half}, \text{bound_beta}.$$

Native h, η, τ are different coordinates from the outer height, outer slack and physical clock. Native F_3 and q are computed ports, not supplied coordinates.

Theorem 154.1 (Entire-fiber projection). *Projection that retains the twenty-two native coordinates is a bijection*

$$\{(w, \kappa) > 0 : Q_R^{\text{can}}(x, U; w, \kappa) = 0\} \longrightarrow \mathcal{E}(\mathbf{a}, \mathbf{b}, \mathbf{c}, q).$$

Proof. A complete zero satisfies the native block at the fixed outer boundary, so projection is well defined. If two witnesses project to the same native tuple, their outer tuples both equal $o(x, U)$; this proves injectivity. Conversely, adjoin $o(x, U)$ to *any* member of $\mathcal{E}$. The sixteen native comparisons hold by definition, and the five nonnative comparisons depend only on the fixed outer data and external ports. All twenty-one comparisons therefore hold. This proves surjectivity onto the entire native fiber, not merely a selected family of native extensions. $\square$

The comparison-independence statement is checked transitively on all eight complete emitted DAGs; no native witness enters an outer comparison through a hidden intermediate. All native gate rows, the ordered coordinate list and the sixteen comparison pairs are retained literally from the folded parents.

This is accepted-fiber canonicity. It does not give a total polynomial function $(x, U) \mapsto o(x, U)$, nor an all-tuple polynomial normalization map from an old witness to a new witness. The extra square restricts the old zero set's outer projection, even though the old-polynomial portion remains identical on all tuples.

155 Infinitely many native extensions at fixed scale

155.1 The two affected equations

Fix the four ports (153.3) and any positive native solution. All symbols in this section except the four ports refer to native coordinates. Define

$$p = 2r + 1, \quad \rho = ic^2, \quad D = \rho^2 - 1, \quad u = jc - p = of - c, \quad y = y_{\text{aux}}.$$

We write ρ instead of R to avoid confusing this auxiliary integer with the fixed source program. The exact native comparisons involving j, o, y are

$$u = jc - p = of - c, \quad \rho^2(u^2 - y^2) = 1 - y^2. \quad (155.1)$$

The second is equivalently $(\rho u)^2 - Dy^2 = 1$. The distinct unchanged auxiliary norm is

$$\rho^2 = (a^2 + 4a + 3)(f^2 - 1).$$

It remains valid because all its supplied coordinates are fixed.

Positivity is available directly from the source equations. Since $q \geq 16$ and $s = 2\text{odd_half} + 1 \geq 3$, let $E = (wq)(sq) > 0$. The retained equations give

$$k = r + 1 + hE > r + 1, \quad c = (sq)k + \eta > 48(r + 1) > p.$$

As $j \geq 1$, $u \geq c - p > 0$. Also $\rho = ic^2 > 1$, so $D > 0$. The optional nonsquare fact $(\rho - 1)^2 < D < \rho^2$ is true but not needed below. No deep classification of the auxiliary Pell index is used to establish these inequalities.

The only direct gate uses of the three changing coordinates are $of = o \cdot f$, $jc = j \cdot c$, and $\text{aux_}y2 = y^2$. Their transitive comparison descendants are exactly native comparisons 13 and 14, numbered from one. These are full comparisons 15 and 16. All other fourteen native comparisons and all five outer comparisons are independent of them. Derived gate values along those paths may change; the claim fixes the other *supplied coordinates*, not every arithmetic register.

155.2 A congruence preserving norm family

Theorem 155.1 (Fixed-scale native infinitude). *Every nonempty positive fiber of this prescribed native block is infinite. From any positive seed, the other nineteen supplied native coordinates can remain unchanged.*

Proof. Put $M_0 = \rho cf$ and

$$B_0 = \begin{pmatrix} \rho & D \\ 1 & \rho \end{pmatrix}, \quad \begin{pmatrix} V_0 \\ Y_0 \end{pmatrix} = \begin{pmatrix} \rho u \\ y \end{pmatrix}.$$

The determinant of B_0 is $\rho^2 - D = 1$, so its reduction modulo M_0 belongs to the finite group $\text{GL}_2(\mathbb{Z}/M_0\mathbb{Z})$. Let $L_0 \geq 1$ be its order. A completely specified alternative is $L_0 = (M_0^4)!$: there are at most M_0^4 residue matrices, so the order is at most M_0^4 and divides this factorial. The order is not claimed to divide M_0^4 itself.

For every $n \geq 0$ define

$$\begin{pmatrix} V_n \\ Y_n \end{pmatrix} = B_0^{nL_0} \begin{pmatrix} V_0 \\ Y_0 \end{pmatrix}, \quad u_n = \frac{V_n}{\rho}, \quad j_n = \frac{u_n + p}{c}, \quad o_n = \frac{u_n + c}{f}. \quad (155.2)$$

Set $y_{\text{aux},n} = Y_n$ and leave all other native coordinates unchanged.

Since $B_0^{nL_0} \equiv I \pmod{M_0}$,

$$V_n \equiv \rho u \pmod{\rho cf}.$$

Thus u_n is an integer and $u_n \equiv u \pmod{cf}$. The seed identities $u + p = jc$ and $u + c = of$ then make j_n, o_n integers. No coprimality assumption on c, f is required.

For arbitrary V, Y ,

$$\begin{aligned} (\rho V + DY)^2 - D(V + \rho Y)^2 &= (\rho^2 - D)(V^2 - DY^2) \\ &= V^2 - DY^2. \end{aligned}$$

Therefore the seed norm $V_0^2 - DY_0^2 = 1$ is preserved. Substituting $V_n = \rho u_n$ gives exactly

$$\rho^2(u_n^2 - Y_n^2) = 1 - Y_n^2, \quad j_n c - p = u_n = o_n f - c.$$

These are both changed comparisons in (155.1). The other fourteen native comparisons keep their seed values by the dependency statement above.

Finally $V_0, Y_0 > 0$ and one multiplication by B_0 sends a positive vector to

$$V' = \rho V + DY > V, \quad Y' = V + \rho Y > Y.$$

Since $L_0 \geq 1$, V_n, Y_n strictly increase with n . All reconstructed coordinates are positive, and distinct n give distinct Y_n . This yields infinitely many positive native tuples at the same ports and scale. $\square$

These matrix powers and the factorial describe witnesses mathematically. They introduce no new circuit gates, comparisons or existential coordinates. The result is seed-relative: it does not assert that inadmissible ports possess a seed. At a valid canonical boundary, the inherited prescribed-scale theorem supplies one; Theorem 154.1 lifts the whole resulting family to the complete canonical circuit.

The family need not enumerate the entire native fiber. It proves a cardinality lower bound sufficient here. The pinned auxiliary-index notes already retain noncanonical indices and exhibit related congruence behavior [67, 68]; the present matrix argument needs no novel index-classification or divisibility theorem.

[write] [Report 25, the pipeline's sequel to this source (batch-82 manuscript 07), classifies the entire positive native fibre at fixed ports and scale: exactly seventeen of the twenty-two coordinates are fixed, and the other five are parametrized without repetition by two integer indices. It is printed

as Part III of the collection’s report `fixed-universal-polynomials` (Theorem `fup:nf:thm:class`), which implies Theorem 155.1: there a seed also has infinitely many native extensions, with the seventeen fixed coordinates among the nineteen that the matrix family keeps fixed. The elementary matrix argument above is kept as a second route; the classification is the complete answer. That Part also prints a corollary that transports the classification to the complete canonical circuits through Theorem 154.1 (its Section `fup:nf:sec:transport`), conditional on this source.]

156 Literal circuits and full operation ledgers

156.1 Four fixtures in two clock models

The archive emits eight complete nonempty circuits and two unchanged zero-step circuits. The fixed sources are [write] [those of source 22, whose table in Section 141 is the same as source 23’s and is printed once: $\text{INC2;DEC2}, s \xrightarrow{\text{INC2}} a \xrightarrow{\text{DEC2}} h$, accepting all $N_0 > 0$ in two steps; and ZERO3, NOP and $\text{POSITIVE3}, s \rightarrow h$ in one step, accepting $3 \nmid N_0$, all N_0 and $3 \mid N_0$ respectively]. They are nonuniversal demonstrations of the generic unbounded mechanism. The generic eligible-source proof does not mean that a literal circuit for every eligible source, or a long universal computation, has been emitted. These simple accepted relations admit smaller specialized descriptions; no minimality claim is made.

An operation is one binary multiplication, addition or subtraction in the literal straight-line DAG. Reuse is allowed; integer constants are literal; signs in intermediate registers are unrestricted. Every comparison’s subtraction and square, and every final accumulation, are charged. Both native/spatial and phase-four models have the same complete ledger:

Fixture	M	A	Total	Outer	Native	All	Degree
INC2;DEC2	240	368	608	39	22	61	2344
ZERO3	185	298	483	37	22	59	1192
NOP	183	298	481	37	22	59	1192
POSITIVE3	190	294	484	37	22	59	1192

Here A includes additions and subtractions; Outer, Native and All count supplied positive coordinates. Every nonempty circuit has two natural external ports, twenty-one comparisons and the complete 62-gate SOS finalizer. Across both clock models this is 4112 nonempty gates; the two zero-step circuits bring the delivered total to 4120.

156.2 Materializing S without a hidden free gate

The folded parent’s graph did not already contain S . Its height schedule was

$$d = n_{\text{initial}} + n_{\text{target}}, \quad b = d + \eta, \quad h = b + \theta.$$

In each of the eight actual parent DAGs, b , named `bridge_height_without_time`, has exactly one consumer, the height addition, and occurs in no comparison. Replace the last two additions by

$$S = d + \theta, \quad h = S + \eta.$$

The same two gates now materialize S ; associativity preserves h on every integer tuple. The single-consumer graph condition ensures every other old register and all twenty old residual polynomials

remain unchanged. This is a checked graph reassociation, not an assumption that a useful affine register is free.

Two additional producers compute $\eta + \kappa$ and $S + 1$. Rebuild the entire finalizer from all twenty-one comparisons: twenty-one residual subtractions, twenty-one self-products and twenty accumulation additions give

$$21M + 41A = 62 \text{ operations.}$$

The parent's twenty-row finalizer cost $20M + 39A = 59$. Thus the fully paid net change is $1M + 4A = 5$ gates and one positive coordinate. The folded-parent totals were 603, 478, 476, 479, producing the displayed 608, 483, 481, 484.

The resulting output satisfies (150.2) on arbitrary signed integer assignments, including away from the old zero set. The natural ports and all old coordinate positions retain their order; κ is appended. The folded time bridge stays

$$192(F + x) + 2\theta + 208 \quad \text{or} \quad 768(F + x) + 8\theta + 832.$$

In both models the canonical S uses forward native θ , never the phase-four time or a source-step count.

156.3 Exact polynomial degree

Degree means total degree of the complete output polynomial in all supplied ports and witnesses, before restricting to zeros. Formal upper bounds propagate by maximum at addition/subtraction and sum at multiplication. Along the integer ray $v_i = (i + 2)z$, with v_i ordered as parameters followed by auxiliaries, propagate coefficients at those formal bounds modulo 1000003. A vanishing leading coefficient never lowers its stored bound.

The nonzero output coefficients at the formal upper bounds are 135347 at degree 2344 for INC2;DEC2 and 977370 at degree 1192 for each other fixture. Thus the lower and upper bounds match. The appended coordinate leaves old weights unchanged, reassociation preserves old polynomials, and the new square has only degree two. Independently computed full univariate specializations agree with the certificates, including all gate coefficients. These degrees are properties of the emitted polynomials; they are not lower bounds on every representation of the same accepted relation.

157 Boundary cases and limits

157.1 Initially halted and rejected inputs

[write] [The zero-step polynomials are source 22's, Section 148.1, unchanged.] If $q_0 = q_h$, freshness and terminality make the control isolated. The cleaned computation consists of two NOP bridges, $\theta = 0$ and $F = x + 1$. The separate polynomials are

$$(384x + 400 - U)^2, \quad (1536x + 1600 - U)^2.$$

Each has four operations, $2M + 2A$, one comparison, no witnesses and exact degree two. The files are byte-identical to the frozen originals. An accepted zero-step pair has the single empty witness tuple. It has no twenty-two-coordinate native fiber, and is not covered by the nonempty infinitude conclusion.

For stuck or nonhalting inputs the old positive fiber is empty, so adding a square cannot create a zero. Rejecting totalization and inherited soundness, rather than a bounded simulation prefix, justify the unbounded statement. Wrong requested clean times remain rejected; in phase four a time not divisible by four cannot satisfy the clean-time row. Quotient words equal to zero are valid via hat one, including the one-step clock quotient hat.

157.2 Sanity checks on all accepted fixture inputs

Let $N_0 = x + 1$. On the accepted paths of these four fixtures $F = N_0$: [write] [the table is source 22's, in Section 148.2, printed once: forward native time $600N_0 + 16$ and native clean time $1584x + 1632$ for INC2;DEC2, and $192N_0 + 8$ and $768x + 800$ for NOP, for ZERO3 if $3 \nmid N_0$ and for POSITIVE3 if $3 \mid N_0$]. Phase-four times are four times these values. The generic circuits retain F and its paid ports rather than specializing the equal-input/output coincidence away.

157.3 What does not follow

[write] [The first paragraph repeats source 22's Section 148.3.] The raw input contract still preserves cofactors coprime to six. Writing $N_0 = 2^a 3^b c$ with $\gcd(c, 6) = 1$, such scaling preserves guards and source control sequence; each corresponding duration obeys $\tau(cN) - 8 = c(\tau(N) - 8)$. When output and time are hidden, halting depends only on the two valuations. For example $x = 0$ and $x = 4$ cannot be distinguished for halting by this raw interface. It is not a paid arbitrary ordinary-input/program loader.

No general finite-fold MRDP impossibility follows from an infinite fiber in this particular representation. Nor does deterministic computation force uniqueness of arithmetical auxiliaries. The established conclusions are exactly unique outer history, a fixed-port native-fiber bijection, and infinitely many remaining native extensions. No rational or real zero-set theorem, arbitrary signed-witness interpretation, fixed universal target, stationary halt, later-recurrence theorem, minimal circuit bound, or new universal numerical operation bound is asserted.

The matrix formula is explicit relative to a positive seed, but neither this report nor its finite replay materializes a complete positive native Pell seed. Existence comes from the inherited completeness theorem; infinitude then follows from the mathematical matrix argument.

158 Independent evidence and portable replay

158.1 Mathematical checks and finite evidence

The outer packet has independent proof and complete-circuit audits. Its circuit audit checks the exact inherited graph, liveness, coordinate order, all-tuple extra-square identity, complete ledgers, degree and zero-step identity. It records 96 signed and 256 modular identity evaluations and 29 rejected circuit corruptions. Finite modular evaluations are supplemental arithmetic tests; no inference from a modular SOS zero to separate modular residual zeros is made.

The independent outer semantic checker records, among other tests:

- 32768 scalar strict-next-dyadic cases, including 16 exact dyadic boundaries;
- 6656 residue/control quotient identities and 1749 enabled primitive duration bounds;
- 3072 accepted full *outer* histories, comprising 1536 parent and 1536 canonical candidates;

- 1536 larger-height repackings that retain the old outer constraints and AND but fail the new cap;
- 1024 false-halt rejections, 9216 wrong-clean-time rejections and 3072 cap-slack mutation rejections;
- 60 accepted all-zero quotient-word histories and 390 exact zero-step evaluations.

These are tests across the four small fixtures and both models, not counts of distinct universal programs. Accepted outer histories check positive outer slacks, padded native-port bounds, genuine joined AND and strict no-wrap. They do not construct complete positive native zeros.

The fixed-scale audit separately checks the source inequalities, the exact two affected native comparisons, the full transitive dependency graph and the norm/congruence proof. Its local numerical family examples exercise only the auxiliary recurrence system. They are deliberately labeled as such, not as complete positive native witnesses. The combined release checker connects the native coordinate order and dependencies directly to the eight actual canonical DAGs.

Normal and optimized Python runs are reported separately. Passing under `python -0` is not, by itself, proof that assertions executed; the delivered replay records the actual coverage and uses explicit failures for release-integrity checks. Neither finite test totals nor the optimized interpreter replace the general proofs in Sections 152–155.

158.2 Package and reproduction contract

The portable archive includes this LaTeX source and PDF, exact complete arithmetic JSON/text DAGs, the two theorem packets, pinned inherited data and prose, independent checkers, original/delivered provenance, and a release-level exact inventory. All computation is local standard-library Python; public source URLs are provenance links and are not needed for replay.

The principal directories are `canonical-height/`, `native-fibers/` and `inherited-source/`. The outer packet includes authenticated base/folded reference data. The release inventory rejects missing, changed and extra files, unexpected directories, hidden extras and symlinks; its negative tests also exercise malformed manifests and tampered pinned dependencies. A checksum is an integrity check, not a digital signature: archive authenticity requires a trusted separately obtained digest.

From an extracted release directory, the documented entry points are

```
python -B verify_release.py
python -B release_checks.py
python -B replay.py --output ../report22-replay
```

Choose a new output directory outside the sealed release. The replay must not write generated receipts, bytecode or build products into the sealed inventory. Optional PDF rebuilding uses `build.sh`, writes temporary TeX files outside the package, and replaces only the PDF; it is therefore an intentional release edit and requires resealing if retained.

The final delivery process includes a fresh ZIP extraction, read-only normal and optimized replay from that extracted tree, adversarial inventory checks, independent article review, PDF text/build checks and visual inspection of every page. Separate machine-readable receipts record the exact executed checks and outcome. Replaying the mathematical packet authenticates and recomputes its data; it does not claim to solve an unbounded witness search.

[write] **The release in this report.** These are the delivered names. Of `canonical-height/`, the emitter `emit_canonical_clocks.py` and the two checkers of `audit/` are shipped as `code/23-canonical-fibers-*`, its emission receipt as `data/23-canonical-fibers-canonical-emission.json` and the audit receipts as `data/23-canonical-fibers-*`, and `README.md`, `THEOREM.md` and the two audit notes as `23-canonical-fibers-CANONICAL-HEIGHT-*.md`. Of `native-fibers/`, the checker is `code/23-canonical-fibers-native-check_native_fibers.py`, its receipt `data/23-canonical-fibers-native-CHECK-RECEIPT.json`, and `README.md`, `THEOREM.md` and `REVIEW.md` are `23-canonical-fibers-NATIVE-FIBERS-*.md`. The release-level `build.sh`, the checkers `check_combined_fibers.py`, `check_privacy.py` and `check_provenance.py`, the records `combined-checks.json`, `delivery-provenance.json` and `report22-qa.json`, and `report22-math-review.md` (as `23-canonical-fibers-MATH-REVIEW.md`) are shipped under the prefix too. Not shipped: the manuscript (printed here) and its PDF, the delivery `README`, the checksum manifests with their checkers and the release tools, the directory `inherited-source/` (byte copies of source 22's and of repository files; its five-gate audit copy differs from source 22's only in a sanitized example path at its line 179), the reference copies in `canonical-height/reference/` (of source 22's notes, two manifests and ten circuits, and of three repository files), the six files of `native-fibers/source/` (four byte copies of repository notes, and two byte copies of files shipped by `fixed-universal-polynomials` as `data/07-native-fiber-source-native_blocks.json` and `data/07-native-fiber-source-provenance.json`), and the 18 emitted circuit files, which the shipped emitter regenerates byte for byte from source 22's regenerated circuits (`README`). On Windows the emitter fails when it checks its reference manifest, because it keys the manifest by a native path string (line 16 of `emit_canonical_clocks.py`); with POSIX path strings all eleven commands of the delivered replay passed at placement in normal and optimized mode, on unchanged copies (`README`).

[write] **The programme's intake.** The Hilbert's-tenth programme took this source in within its intake of the negative-index and native-fibre archives, `review_incoming_negative_index_fibers.md` (commit 0055e1c4d, 3 October 2026) [70], under the heading "Report25 and Report22: fibers, not a competing soundness result". It reads this source as supplying the earlier fixed-slice infinitude (Theorem 155.1), whose native-fibre statement explains why canonicalizing outer histories does not remove auxiliary multiplicity; it notes that neither this archive nor Report 25's asserts finite-fold universality, removes a positivity constraint or reduces the programme's current universal circuit, and that infinite fibres are compatible with that circuit's full-zero bijection. It states that it does not certify source 23's canonical-height circuit costs. The programme's archive inventory `reviewed_report_archive_replay.md` (commit ceb7222e9) lists this archive as a reviewed intake and source 22's as relocation only; the scoped intake of source 22 described at the end of Section 149 came later.

Part IX

A reversible four-particle clock and uniform timed charts at mass four (sources 24 and 25)

[write] Part IX prints two AI-assisted research manuscripts of ProveIt's incoming reports, batch 91 (arrival 0d7f51c44, placement d750d98dd), both dated 4 October 2026 and added on that day. They are numbered reports of the research pipeline that wrote sources 18–23: *source 24* is batch-91

manuscript 01, the pipeline’s Report 49, *A sharp reversible four particle timing threshold*, from the archive `A_Sharp_Reversible_Four_Particle_Timing_Threshold_Package.zip`; source 25 is batch-91 manuscript 03, the pipeline’s Report 51, *Uniform timed charts for four mass units*, from `Uniform_Timed_Charts_for_Four_Mass_Units_Package.zip`. The source numbers are the file prefixes 24-reversible- and 25-uniform- of their shipped files and continue this report’s sequence; manuscript, source and pipeline numbers all differ, and brackets translate them where the sources use them (source 25 calls source 24 “Report 49”). Their labels carry the prefixes `smc:rt:` and `smc:ut:`; labels added in this write carry `smc:ix:`. Neither manuscript has an author line beyond its title and report number, and neither names an assistant.

Both sources were written against this report itself. Each pins its `article.tex` as Git blob `ec10c7a579e04d3d27a92035d9a180dfcb695ff2` (SHA-256 `17d3c0d9...`), the text of commit `0f95145cb`, which is Parts I–VIII as printed here, and cites Part VI by line numbers of that blob; brackets translate every line range into the labels printed here, since the line numbers of this article change as it grows. Source 24 also pins the programme’s `review_parallel_particle_reports.md` (blob `30c7366b`, commit `fb7e3cb47` [57]) and this report’s `17-four-mass-BINARY-INDEPENDENT-AUDIT.md` (blob `0aa7b970`); source 25 pins source 24’s L^AT_EX source by SHA-256 (`c7813317...`). All these pins equal the repository and delivery bytes; they were verified at placement.

[write] **Two continuations of Part VI.** Source 24 supplies what source 17’s binary shuttle lacks: source 17 states that its radius-six rule “is not injective” and asserts no reversibility (after Proposition 89.1). Source 24 gives a different binary rule, globally reversible on the whole full shift, with forward and inverse radius at most 90, whose four-particle orbit from $\{0, 5, 6, d\}$ has the anchored hits $k^2 + (2d - 23)k$ (Theorem 159.1). Writing $d = 13 + x$ gives exactly source 17’s timing polynomial with $d = 7 + x$, so source 24’s quartic is source 17’s certificate (90.1) word for word; it is printed once there, and a numbered remark keeps source 24’s statement (Remark 163.1). New in source 24 are the six-row reversible rule, its full-shift involution proof, the complete orbit with every half-step, all-time exclusivity and the density-one observation (Corollary 163.2). Its two other ingredients are credited by the source itself and are printed as marked second routes: the prospective-isolation lemma is Lemma `fpa:pi:lem:parallel` of `five-particle-binary-automata` (Report 26, Part VI there), which source 24 cites through the programme’s review, and its three-particle lower bound is the binary, fixed-input case of source 16’s Theorem 74.1 and Corollary 74.2. The non-reversible sharp threshold four was therefore already implied by sources 16 and 17; only its reversible form is new. Source 25 answers source 17’s second question (Question 2, uniform untimed spatial definability in the stationary frame) and the uniform half of its fourth (Question 4), conditionally: it imports source 17’s scattering mechanism and source 18’s fixed-input charts, and source 18 is itself conditional on source 17’s normal form (Proposition 87.1), so the theorem is a conditional on a conditional (Theorem 169.1). Its non-Presburger obstruction (Proposition 176.1) is source 24’s clock with a drift of one site per step. Source 25 does not cite Part VII: source 20’s planar weighted automata already have nonsemilinear original-frame site sets under drift (Proposition 124.1, Theorem 125.1), so a non-Presburger original-frame phenomenon was known here in dimension two; source 25’s obstruction is new in dimension one, for a binary globally reversible rule, and for complete-output reachability from one fixed input (note after Proposition 176.1).

[write] **Review status.** The Hilbert’s-tenth programme made a bounded routing intake of both archives together with ten other signal and particle archives of the same arrival [99] (`review_new_signals_0d7f51c44.md`, commit `49b100cfd`). It read each delivery README in full and selected spans (source 24’s manuscript lines 50–64, 202–228 and 308–319; source 25’s proof note lines 1–39 and 130–185); it is “a routing and scope review, not a full proof”, and it found “no specific contradictory theorem/interface statement requiring an author correction”. Its findings: source 24

is “a decidable quadratic timing set, not a universal substrate improvement. The certificate is not real-exact. The three-particle lower bound is inherited; no timing irregularity $\Rightarrow$ universality inference is available”; for source 25, “ ≤ 2 evolution parameters is not ≤ 2 total Diophantine witnesses; a per-chart pre-slack bound is $3m + 1 \leq 13$ ”, the copied-input, slack and selector ledger is essential, and there is no general rule-to-chart implementation. It does not certify the inherited source 17 and 18 dynamics. The universal-polynomial bound of the project, 84 operations, is untouched by both sources.

[write] **Notation.** Each source keeps its letters, and inside a source a letter has that source’s meaning. The sources’ $\mathbb{N} = \{0, 1, 2, \dots\}$ is written $\mathbb{N}$ here (a change of rendering only); their `\supp` and `\diam` are this report’s operators, and source 25’s $\mathcal{L}$ is source 18’s. No symbol was renamed and no normalization changed. Letters that need care: in source 24, $H(F, c; w, a)$ with arguments is an occurrence-time set, while the bare $H = 2(b + r) = 30$ is the isolation distance; $b = 7$ and $r = 8$ are the write and read radii in Section 161, but in Section 164 b is a singleton’s site and r a phase index; d is the initial right marker in Theorem 159.1 and a pair diameter in Section 164; D is the current clock gap in Proposition 162.1 and a packet displacement in (164.1). Source 25’s $S, B, N = 4B + 12S, W = N + 2B, D, L, R$ are source 17’s (Section 85); its C is a cycle time constant, chosen by the source to avoid the safety bound B ; its H is a common modulus, H_0 an equality count and H_{hs} the equality forms; its J counts retained charts (other Parts use J differently); its K counts completed contracting cycles (Part II’s K is a batch count); its E is both a cycle anchor increment and the selector sum $\sum_h e_h$, as in source 18; its $G = \tau_{-\delta}F$ is the stationary-frame rule, while in Proposition 176.1 G is source 24’s clock (source 24’s F) and $F = \tau_1 \circ G$. The table of Section 1.28 collects the letters.

Source 24 (batch-91 manuscript 01): A sharp reversible four particle timing threshold

“Report 49”, 4 October 2026. [write] From the archive `A_Sharp_Reversible_Four_Particle_Timing_Threshold_Package.zip` (main file `Research_Report49/manuscript/report49.tex`, 14-page PDF; neither is shipped). Its labels carry the prefix `smc:rt:`, and its sections are Sections 159–168 here; its appendix is printed as the last of them. Statements keep their numbering by section, and equations are numbered by section. Its abstract follows.

We specify a one-dimensional, binary, globally reversible, number-conserving cellular automaton of radius at most 90. From the four-particle input $\{0, 5, 6, d\}$, $d \geq 13$, the exact word 1000011 at the fixed coordinates $[0, 6]$ occurs precisely at times $k^2 + (2d - 23)k$, $k \in \mathbb{N}$. Six literal endpoint swaps, protected by isolation and prospective raw-key preservation, define the automaton on the entire binary full shift. We prove reversibility also for malformed and infinite configurations and derive every intermediate state of the clock orbit. An explicitly credited adaptation of the inherited mass-at-most-three argument proves that every fixed-input exact anchored observation below four particles is eventually periodic. Thus four is the least particle count for this timing phenomenon, also when global reversibility is required. The relation has the elementary quartic certificate $[k^2 + (2x + 3)k - t]^2 = 0$, with empty or singleton fibers over $\mathbb{N}$ and over $\mathbb{Q}_{\geq 0}$, but not an exact interpretation over $\mathbb{R}_{\geq 0}$. This is a reversible realization and a sharp timing result, not a universality, arithmetic-record, or literature-priority claim.

Conventions. $\mathbb{N} = \{0, 1, 2, \dots\}$. Intervals of sites are inclusive integer intervals. Composition BA means apply A first. A finite set denotes the binary configuration with exactly those occupied sites.

159 Statement and scope

A binary cellular automaton (CA) is a deterministic, time-independent, translation-equivariant map $F : \{0, 1\}^{\mathbb{Z}} \rightarrow \{0, 1\}^{\mathbb{Z}}$ whose output at a site depends on a fixed finite neighborhood. Radius R means the neighborhood is contained in $[-R, R]$. The vacuum is the all-zero configuration. We use finite number conservation:

$$|\text{supp } F(c)| = |\text{supp } c| \quad \text{for every finite-support } c.$$

This is the standard convention used in Morita’s definition of an RNCCA [73]. Global reversibility here means a two-sided inverse on *all* bi-infinite binary configurations, not merely on an invariant family of finite inputs.

For a word $w \in \{0, 1\}^{m+1}$ anchored at a , define

$$H(F, c; w, a) = \{t \in \mathbb{N} : (F^t(c))_{a+i} = w_i \ (0 \leq i \leq m)\}.$$

Zeros are part of the observation. The configuration may contain other particles outside the observed interval. A set $H \subseteq \mathbb{N}$ is eventually periodic if some $T \geq 0$, $P \geq 1$ satisfy $t \in H \iff t + P \in H$ for every $t \geq T$.

Theorem 159.1 (Sharp reversible timing threshold). *There exists a fixed binary number-conserving CA F such that:*

1. F and its inverse are CAs of radius at most 90 on $\{0, 1\}^{\mathbb{Z}}$.
2. For every $d \geq 13$, with $C_d = \{0, 5, 6, d\}$,

$$H(F, C_d; 1000011, 0) = \{k^2 + (2d - 23)k : k \in \mathbb{N}\}.$$

This set is infinite and not eventually periodic.

3. For every binary number-conserving CA G , every finite input with at most three occupied sites, and every fixed exact anchored finite word, the occurrence-time set is effectively eventually periodic.

Consequently four is the least particle count permitting a non-eventually-periodic occurrence-time set in this model, both without a reversibility restriction and with global reversibility required.

The lower-bound argument in item 3 is inherited, not newly claimed: Section 164 gives a self-contained binary, per-input adaptation of the stronger weighted uniform timed-semilinearity theorem in the inspected source [source 16, Theorem 74.1]. The generic guarded-swap mechanism likewise comes from the repository’s parallel-particle review [57]. The contribution examined here is the six-row reversible clock, its exact anchored timing, and the resulting sharp threshold within this model.

[write] [Without item 1 the theorem is already in this report: item 3 is the binary case of source 16’s Corollary 74.2, since a binary automaton has the single weight-one symbol 1 (its last sentence: occurrence times of a fixed observation from a fixed input of mass at most three are effectively eventually periodic), and source 17’s noninjective radius-six rule realizes item 2’s phenomenon with the hits $k^2 + (2d - 11)k$ from $\{0, 3, 4, d\}$ (Proposition 89.2). The threshold four without reversibility thus follows from sources 16 and 17. New here is that the upper example is globally reversible. In one dimension the least particle count for binary *universality*, reversible or not, is five (Reports 14 and 15, Part I of **five-particle-binary-automata**; Section 1.16); the timing threshold four is a different and smaller threshold, and the two are compatible because source 17’s four-mass decision theorem covers this rule (Section 165).]

160 The literal local rule

In each of two blocks, a row consists of two distinct finite endpoints E_0, E_1 and a read interval W containing them. Table 14 is the complete row data; no unlisted legal-state convention is imposed.

Block	Row	E_0	E_1	W
A	AR	$\{0, 1\}$	$\{0, 2\}$	$[-4, 6]$
A	AL	$\{0, 4\}$	$\{0, 3\}$	$[-4, 8]$
A	AC	$\{0, 1, 6\}$	$\{-1, 2, 7\}$	$[-5, 8]$
B	BR	$\{0, 2\}$	$\{1, 2\}$	$[-4, 6]$
B	BL	$\{0, 3\}$	$\{-1, 3\}$	$[-5, 7]$
B	BC	$\{-5, 0, 3\}$	$\{-5, 0, 1\}$	$[-5, 8]$

Table 14: The complete six-row specification of source 24’s rule. Coordinates are offsets from the row’s anchor; every unspecified site in its read interval must be zero.

A *raw key* is a pair $\kappa = (\text{row}, x)$, where $x \in \mathbb{Z}$ is an anchor. For configuration c , the key is present exactly when

$$\text{supp}(c) \cap (x + W) = x + E_0 \quad \text{or} \quad x + E_1.$$

The key does *not* include the endpoint side. This omission is essential: swapping an endpoint must preserve the key rather than replace it with a differently named key. Let $K_A(c)$ and $K_B(c)$ denote the raw-key sets for the respective blocks.

For a present key κ , let $S_\kappa c$ replace the occupied endpoint with the other endpoint and leave every other site unchanged. Set

$$b = 7, \quad r = 8, \quad H = 2(b + r) = 30.$$

For a fixed block, select $\kappa = (\text{row}, x)$ precisely when both conditions hold:

- (i) **Isolation:** no other raw key has anchor within distance 30 of x . Another row at the same anchor counts as another key.
- (ii) **Prospective preservation:** after the hypothetical swap, the raw-key set with anchors in $[x - 15, x + 15]$ is exactly the same as before it.

All selected swaps act simultaneously. Keys failing either test do nothing; sites in no selected write support retain their state. Finally define $F = B \circ A$.

This is a finite local algorithm, including on malformed input. In particular, prospectivity must search for *new* keys after the hypothetical swap, not merely test whether previously present keys survive. The next section proves that simultaneous writes are unambiguous and local.

161 Full shift reversibility

We give the complete guarded-swap proof, including preservation of rejected keys. It uses no particle-count bound.

[write] [Second route. Lemma 161.1 is the prospective-isolation lemma of Report 26, Lemma fpa:pi:lem:parallel of five-particle-binary-automata (Part VI there), with the same isolation distance $H = 2(b + r)$ and the same radius bound $3(b + r)$; that report applies it to a binary five-particle compiler. Source 24 credits the mechanism to the programme's review of Reports 26–28 [57] and gives the proof again for its own row format, in which a key does not record its endpoint side. The proof is printed because it is self-contained; no novelty is claimed for the lemma.]

Lemma 161.1 (Prospective isolation). *Consider a finite family of endpoint swaps with distinct, nonempty endpoints of equal weight. Every endpoint is contained in its own read window; all writes lie in $[-b, b]$ and all reads in $[-r, r]$, where $r \geq b$. Suppose each swap exchanges the two sides of its own raw-key predicate. Select a raw key at x by isolation within $H = 2(b + r)$ and preservation of all raw keys within $b + r$ under its hypothetical swap. The parallel block T is a translation-equivariant full-shift involution of radius at most $3(b + r)$. It fixes vacuum and conserves particle number on finite configurations.*

Proof. Write $K(c)$ for all raw keys in the block. A hypothetical write at x can change a raw predicate at y only if the radius- b write window and radius- r read window meet, so only if $|x - y| \leq b + r$. Therefore the local prospective test $P_\kappa(c)$ is equivalent to the global equality

$$K(S_\kappa c) = K(c). \quad (161.1)$$

This equality is about key identity, not endpoint side.

Two selected anchors have distance greater than H . Their write windows are disjoint. Moreover, a single raw-predicate read window cannot meet two selected write windows: if it met those at x and y , then $|x - y| \leq 2(b + r) = H$, a contradiction. Every raw predicate therefore sees either no write or exactly the change produced by one selected hypothetical swap. In the latter case (161.1) preserves its truth value. Thus

$$K(Tc) = K(c) \quad (161.2)$$

for arbitrary finite or infinite c .

Isolation consequently has the same truth value before and after T . To check selection itself, the prospective test at a present key anchored at x depends only on

$$[x - (b + 2r), x + (b + 2r)] :$$

its tested anchors are within $b + r$ and their reads extend another r . Any *other selected* key at y has $|x - y| > H$, because its own isolation test sees the raw key at x , whether or not x is selected. Its radius- b write window misses this entire comparison window.

If κ is selected, its own swap exchanges the two configurations whose raw sets are compared in (161.1); the other selected writes cannot affect that comparison. Hence its prospective test remains true. If κ is isolated but prospectively rejected, no own write occurs and no other selected write reaches the comparison window; it remains rejected. If it is nonisolated, it remains nonisolated by (161.2), hence rejected. Indeed, the same window argument preserves its prospective status too. No new raw key appears by (161.2). The entire selected-key set is therefore unchanged. A second block application performs the same disjoint endpoint involutions and returns c .

For locality at an output site z , only anchors within b of z can write there. At any such anchor, isolation reads radius $H + r$, and prospectivity reads radius $b + 2r$. Thus an adequate output radius is

$$b + \max\{H + r, b + 2r\} = 3(b + r).$$

All conditions use relative coordinates, so the rule is translation-equivariant. Nonempty endpoints imply no key on vacuum and only finitely many present keys on finite support. Disjoint equal-weight swaps then preserve the finite particle count. The proof and the involution identity apply without a finite-support assumption. $\square$

In Table 14, the endpoint sizes agree row by row, the endpoints lie inside their read intervals, their supports lie in $[-7, 7]$, and the read intervals lie in $[-8, 8]$. Exact replacement exchanges the two own-key alternatives. Lemma 161.1 gives

$$A^2 = B^2 = \text{id}, \quad F^{-1} = A \circ B,$$

on the full shift, and radii at most 45 for each block and 90 for F and F^{-1} . These are certified upper bounds, not minimal-radius claims. Although a time step factors into two blocks, the same fixed composite rule is used at every time; this is not a spatially partitioned or time-alternating model with an extra hidden clock state.

161.1 Why isolation alone is insufficient

Take

$$X = \{-8, -5, 0, 1\}, \quad Y = \{-8, -5, 0, 2\}.$$

The only A -key of X is $(AR, 0)$. Its hypothetical swap yields Y , which has both $(AR, 0)$ and $(AC, -7)$. Thus the actual rule rejects the swap. An isolation-only variant sends X to Y and fixes Y , and is therefore noninjective.

The example also tests simultaneous accepted and rejected keys: add $\{70, 71\}$ to X . The key at 0 is still isolated but prospectively rejected; $(AR, 70)$ is selected. The actual output is $X \cup \{70, 72\}$, and both statuses are preserved. This is a concrete malformed-input check, not an assumption that malformed configurations are all fixed.

[write] [Report 26 gives a counterexample of the same kind for its own rule family, on a binary ring of period 12 (Section `fpa:pi:sec:lemma` of `five-particle-binary-automata`); the example above is source 24's own, for its six rows.]

162 The complete four particle orbit

For $D \geq 13$, write

$$C_D = \{0, 5, 6, D\}, \quad L_D = 2D - 22.$$

Proposition 162.1 (Every intermediate state). *For $0 \leq s < L_D$,*

$$F^s(C_D) = \begin{cases} \{0, 5 + s, 6 + s, D\}, & 0 \leq s \leq D - 11, \\ \{0, 2D - 18 - s, 2D - 14 - s, D + 1\}, & D - 10 \leq s \leq 2D - 23. \end{cases} \quad (162.1)$$

At the endpoint, $F^{L_D}(C_D) = C_{D+1}$.

Proof. The two phase intervals are adjacent and disjoint and cover the whole cycle. We first compute

all four possible kinds of complete step, retaining the intermediate half-step:

$$\{0, p, p+1, D\} \xrightarrow{\text{AR}(p)} \{0, p, p+2, D\} \xrightarrow{\text{BR}(p)} \{0, p+1, p+2, D\}, \quad (162.2)$$

$$\begin{aligned} & \{0, D-6, D-5, D\} \xrightarrow{\text{AC}(D-6)} \{0, D-7, D-4, D+1\} \\ & \xrightarrow{\text{BL}(D-7)} \{0, D-8, D-4, D+1\}; \end{aligned} \quad (162.3)$$

$$\{0, q, q+4, D+1\} \xrightarrow{\text{AL}(q)} \{0, q, q+3, D+1\} \xrightarrow{\text{BL}(q)} \{0, q-1, q+3, D+1\}, \quad (162.4)$$

$$\{0, 5, 9, D+1\} \xrightarrow{\text{AL}(5)} \{0, 5, 8, D+1\} \xrightarrow{\text{BC}(5)} \{0, 5, 6, D+1\}. \quad (162.5)$$

Equations (162.2) and (162.4) are flights; (162.3) and (162.5) are the two contacts. At $D = 13$, the left-flight range is empty and the left phase is the single state $s = 3$. Thus the smallest allowed parameter is included.

It remains to prove that these proposed arrows are the actual guarded moves. At every displayed state and half-state exactly one particle pair has distance at most four; all other pairs have distance at least five. Every endpoint in Table 14 contains such a close pair. Consequently every raw key must use this unique pair.

For block A , gaps 1, 2 permit AR; the only competing endpoint with gap 1 is the input of AC. It requires a third particle six sites to the right of the pair's left site, precisely where the AR guard also reads. At right contact it creates AC and excludes AR; during right flight the right marker is farther away. Gaps 4, 3 permit AL; the only competing gap-3 endpoint is the output of AC. Its third particle is eight sites right of the pair's left site, precisely inside the AL guard. Thus on the reverse side of the contact it excludes AL. That marker distance does not occur in the free left-flight half-states.

For block B , gaps 2, 1 permit BR, with only the output of BC competing at gap 1. Gaps 3, 4 permit BL, with only the input of BC competing at gap 3. Both BC endpoints require a third particle five sites left of the close pair's left site. That site is in the relevant BR or BL read interval, so the contact and free move cannot coexist. The marker occurs exactly at left contact or its reverse half-step. At the right contact the pair's left site is $D - 7 \geq 6$, so the origin cannot accidentally trigger BC.

The unused origin at right contact is outside $[D - 11, D + 2]$, since $D \geq 13$. The unused right marker at left contact is $D + 1 \geq 14$, outside $[0, 13]$. The displayed flight bounds likewise keep spectator particles out of the relevant read intervals. Hence each proposed half-step has exactly one raw key before and after its hypothetical swap, and it is the *same row-anchor key*. Isolation and prospectivity both pass. These arrows prove (162.1) and its endpoint by induction. $\square$

[write] [Compare source 17's shuttle (Proposition 89.2 and Figure 5): there the moving pair has separations one and two and the rule is a component prescription of radius six; here the separations are one and four, the bounces are two guarded half-steps each, and every move is an involutive swap. Both clocks lengthen by two steps per cycle, as the right marker advances by one: a cycle takes $2D - 22$ steps here and $2d - 10$ there.]

163 Exact timing and an elementary quartic

At $s = 0$ in (162.1), the sites 0, 5, 6 are occupied and sites 1, 2, 3, 4 are empty. During the right phase the adjacent pair is $\{5, 6\}$ only when $s = 0$. During the left phase the moving particles have

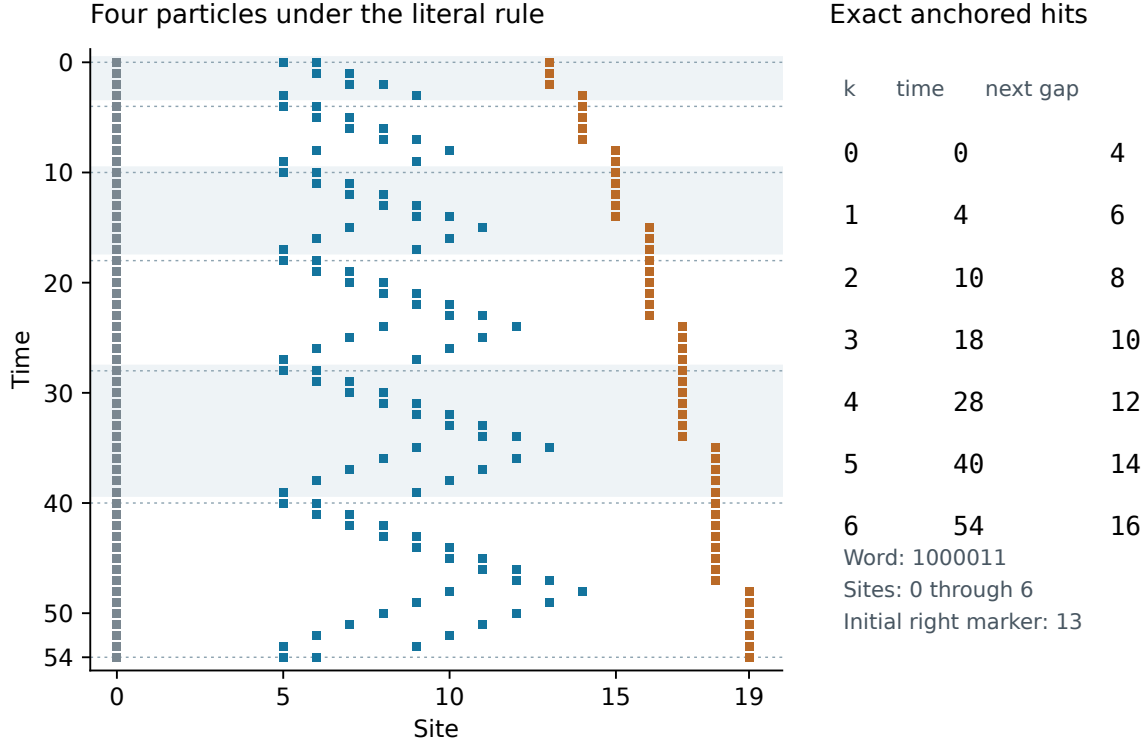


Figure 10: The first six cycles of source 24’s clock from C_{13} , obtained by iterating the literal guarded rule, not by plotting the phase formula. Each square is one occupied site; time increases downward. Dashed lines mark exactly the seven observed hits 0, 4, 10, 18, 28, 40, 54. The moving pair changes its separation from one to four at the right contact, and the right marker advances one site per cycle. The plotted coordinates and observations are saved as a CSV, and independently checked against (162.1). [write] [Shipped as `figures/24-reversible-clock-spacetime.pdf`; the CSV is `data/24-reversible-figures-clock-spacetime.csv`.]

separation four, while the other two particles occupy 0 and $D + 1 \geq 14$; hence the exact word cannot occur. This proves *all-time exclusivity*, not just the existence of section returns.

Starting at C_d , the k th hit is the configuration C_{d+k} at time

$$t_k = \sum_{j=0}^{k-1} L_{d+j} = \sum_{j=0}^{k-1} (2(d+j) - 22) = k^2 + (2d - 23)k. \quad (163.1)$$

The empty sum gives $t_0 = 0$. Successive gaps are

$$t_{k+1} - t_k = 2k + 2d - 22,$$

which tend to infinity. Every infinite eventually periodic subset of $\mathbb{N}$ has bounded successive gaps: beyond its threshold, a nonempty periodic residue class supplies a point at least once per period. Thus (163.1) is not eventually periodic. In one variable Presburger-definability is equivalent to eventual periodicity, so the hit set is not Presburger-definable [7].

Remark 163.1 ([write] Source 24’s quartic is source 17’s). Source 24 states as its Corollary 4.1, *Exact nonnegative rational fibers*: “For $x, t \in \mathbb{N}$, put $d = 13 + x$ and

$$Q(x, t; k) = [k^2 + (2x + 3)k - t]^2. \quad (163.2)$$

The polynomial has total degree four and a single natural witness coordinate k . Its zero fiber over $\mathbb{N}$ is empty when t is not a hit, and a singleton when t is a hit. The zero fiber over $\mathbb{Q}_{\geq 0}$ is exactly the same fiber.” Its proof: “The timing polynomial is strictly increasing for $k \geq 0$: its consecutive differences are $2k + 2x + 4 > 0$. Its values on $\mathbb{N}$ are exactly (163.1). Squaring introduces no additional zeros. If a rational k is a root of the monic integer polynomial $k^2 + (2x + 3)k - t$, write $k = a/b$ in lowest terms with $b > 0$. Multiplication by b^2 implies $b \mid a^2$, hence $b = 1$. A nonnegative rational root is therefore natural. This is the elementary monic rational-root argument, not an inference from finite rational testing.”

This is the certificate $\mathcal{P}(x, t; k)$ of (90.1), polynomial for polynomial: source 17 writes $d = 7 + x$ for its shuttle, whose hits $k^2 + (2d - 11)k$ become $k^2 + (2x + 3)k$, and source 24 writes $d = 13 + x$, whose hits $k^2 + (2d - 23)k$ become the same expression. The statement, the strict monotonicity, the monic rational-root argument and the failure over nonnegative reals (every natural t has a nonnegative real root) are all in Section 90, where the result is printed, together with its least-degree note and the programme’s confirmation [46]; source 24 says the certificate has “the same arithmetic form” as the earlier shuttle’s. Only the automaton behind the polynomial differs: for source 24 it is globally reversible.

The domain restrictions matter. At a genuine hit, the second root is $-k - (2x + 3)$, so unrestricted integer or rational fibers are not singletons. Over nonnegative reals every $t \geq 0$ has a nonnegative root; for example $x = 0, t = 1$ has the false witness $(\sqrt{13} - 3)/2$, although 1 is not a clock hit. Thus the certificate is not real-exact. It is a direct elementary encoding of this one timing relation, already of the same arithmetic form as the earlier nonreversible shuttle’s certificate [17-four-mass-BINARY-INDEPENDENT-AUDIT.md; source 17, Section 90]; no universal-polynomial degree, arity, or single-fold record is claimed. [write] [No certificate with (x, t) as its only external data is real-exact: Remark 181.1.]

163.1 A dense observation of the same orbit

The complete orbit also illustrates why quadratic *section times* do not force every finite-word hit set to be a union of quadratic sequences.

Corollary 163.2 (A density one hit set). *For $d \geq 13$ and t_k from (163.1), the exact word 100000 at $[0, 5]$ has occurrence set*

$$H(F, C_d; 100000, 0) = \bigcup_{k \geq 0} [t_k + 1, t_{k+1} - 2] \cap \mathbb{N}.$$

This set has natural density one and an infinite complement. It cannot be a finite union of eventually arithmetic progressions and positive-leading-coefficient quadratic polynomial sequences, even when the latter are restricted to eventual arithmetic progressions of their parameter.

Proof. The origin is always occupied and sites 1, 2, 3, 4 are always empty in (162.1). Site 5 is occupied exactly at the first and last state of each cycle: $s = 0$ in the right phase and $s = L_D - 1$ in the left phase. Hence the only omitted times in cycle k are t_k and $t_{k+1} - 1$, proving the interval formula. Since $t_k = k^2 + O(k)$, there are only $O(\sqrt{T})$ omissions through time T , but infinitely many altogether. Thus the density is one and the set is not cofinite.

A quadratic sequence with positive leading coefficient has only $O(\sqrt{T})$ values in $[0, T]$, so any finite union of such sequences has density zero. A finite union of eventual arithmetic progressions is eventually periodic. If its union with a density-zero set has density one, that eventually periodic part

must itself have density one, and therefore be cofinite. Its union would then be cofinite, contradicting the infinitely many omissions above. $\square$

This is an elementary additional observation of the same proved orbit. It does not contradict the inherited fixed-input quadratic *chart* description, which allows a second within-flight parameter and hence growing intervals of times; see Section 165.

[write] [The “inherited fixed-input quadratic chart description” is source 18’s Lemma 94.1: in its charts the within-flight parameter j ranges over an interval whose length grows with the cycle, which produces exactly such bands of times.]

164 Why three particles cannot suffice

This section adapts the stationary-unit, no-split, finite-encounter argument from the inherited source [source 16], specifically its model and timed-evolution theorem at lines 5406–5459 and proof at lines 5464–5743 of the pinned source [write] [Section 74 with Theorem 74.1 and Corollary 74.2, and Sections 75–79]. The source treats weighted alphabets with at most one weight-one label and establishes uniform Presburger evolution. We reprove only the binary, fixed-input statement needed here. This narrower presentation is included for logical completeness, not as a claim of a new lower-bound argument.

[write] [Second route, step by step. The stationary frame below is that of Section 75; the no-split argument is Lemma 75.1; the pair tail (164.1) is Lemma 75.2; the core of diameter at most $4S$ and the exact evolution before it are Section 76 with Lemma 76.1; the first-entry interval (164.2) is (76.1) of Lemma 76.2 verbatim; the return graph is Proposition 77.1; and the observation step is Section 79. Source 24’s version is shorter because it is binary and per input: it needs no uniform construction (Section 78), and it concludes eventual periodicity of each observation directly. Source 19’s Lemma 104.1 is the same fixed-input statement on $\mathbb{Z}^d$.]

Theorem 164.1 (Inherited binary per-input bound). *Fix a one-dimensional binary, quiescent, number-conserving CA G_0 of finite radius. For every fixed finite-support input with at most three particles, the relation between time and its occupied output coordinates is effectively Presburger-definable. In particular, every fixed exact anchored finite-word occurrence set is effectively eventually periodic. Reversibility is unnecessary.*

Proof. We give the orbit decomposition and then the observation argument.

A stationary singleton frame. Conservation forces the successor of a single particle at 0 to be a single particle at some δ with $|\delta| \leq R$, where R is a radius of G_0 . Let τ_a denote translation to the right by a , and set

$$G = \tau_{-\delta}G_0, \quad S = \max\{1, R + |\delta|\}.$$

Every singleton is fixed by G , whose radius is at most S . Also $G_0^t = \tau_{\delta t}G^t$. The padding $S \geq 1$ includes radius-zero rules.

Join occupied sites at distance at most $2S$ into components. Distinct components are more than $2S$ apart, so no radius- S read neighborhood meets two of them. Quiescence and finite propagation imply exact one-step independence: their separately computed successor supports are disjoint and their union is the global successor.

A bounded pair cannot split into distant components. Suppose a two-particle configuration has diameter $d \leq 2S$, translated to $\{0, d\}$. If $d > S$, each original occupied site's neighborhood is exactly its isolated-singleton neighborhood, so both sites remain occupied. They already exhaust the conserved mass, and the pair is fixed. If $d \leq S$, any new occupied output site must see both particles: a neighborhood seeing zero or just one would give the vacuum or stationary-singleton output there. Such a site lies in $[d - S, S]$, which also contains $0, d$ and has width $2S - d \leq 2S$. The successor pair thus still has diameter at most $2S$. Induction proves this bound at all times.

Normalize a bounded pair by putting its leftmost site at zero. There are only $2S$ possible normalized pair shapes. The next shape and the displacement of its leftmost site depend only on that shape. A repeated shape yields a computable finite transient and a translated periodic tail:

$$G^{\mu+r+np}(q) = \tau_{nD} q_{\mu+r} \quad (0 \leq r < p, n \geq 0), \quad (164.1)$$

for constants μ, p, D and finitely many explicit configurations. A pair initially farther than $2S$ apart consists of two fixed singletons. This settles zero, one, and two particles.

The finite three-particle encounter core. For mass three take the core to be the configurations of diameter at most $4S$. It has $\binom{4S}{2}$ normalized shapes. Every connected three-particle configuration is in the core, since its two adjacent gaps are at most $2S$. Outside the core there are therefore either three isolated fixed singleton components, or a bounded pair and one stationary singleton. A single step from the core has diameter at most $6S$ by finite propagation.

In the pair-plus-singleton case, let z_k be the isolated pair orbit and let b be the singleton's site. Until the first k at which the span of $\text{supp}(z_k) \cup \{b\}$ is at most $4S$, the actual orbit is exactly this union, also at the first such time if it exists. Indeed, if the singleton were within $2S$ of the pair, the pair's diameter bound $2S$ would put the whole span within $4S$. Before entry the two components are more than $2S$ apart, so one-step independence applies inductively. In particular, the counterfactual union is disjoint throughout the portion used.

First entry is effective. Inspect the finite transient in (164.1). On a periodic phase the pair's leftmost site has the form $a + A_r + nD$ and its diameter is a constant $d_r \leq 2S$. Core entry on that phase is exactly

$$b - 4S \leq a + A_r + nD \leq b + 4S - d_r, \quad n \in \mathbb{N}. \quad (164.2)$$

This is an integer interval problem. For $D = 0$ it is always or never satisfied; for $D > 0$ use the least nonnegative integer above the lower bound and test the upper bound; for $D < 0$ reverse the inequalities. Comparing the finitely many phase and transient candidates determines the first entry or certifies no entry.

A finite return graph describes all intermediate times. For each normalized core shape, take one exact step. A core successor gives a return edge of duration one and a fixed spatial displacement. If the successor is outside, the preceding classification and first-entry procedure either give a later first return, with a fixed positive duration and displacement, or a completely described nonreturning orbit. These data depend only on the normalized starting shape, so this is a finite deterministic graph.

If a nonreturning edge occurs, the tail is fixed or is a stationary singleton plus an independent bounded pair with translated periodic tail. Otherwise a normalized core shape repeats. At two actual times $t_0 < t_1$ we then have $G^{t_1}(c) = \tau_a G^{t_0}(c)$, and determinism and translation equivariance give

$$G^{t_1+n}(c) = \tau_a G^{t_0+n}(c) \quad (n \geq 0).$$

This identity includes *every intermediate time*, not just returns to the core. Thus, after a finite transient, the entire configuration is periodic up to one translation, or is an independent stationary singleton and a bounded translated-periodic pair.

Exact observations in the original frame. On each of finitely many time phases $t = T + r + nP$, every occupied coordinate has the form $a_j + nd_j$ in the stationary frame. In the original frame it becomes

$$a_j + \delta(T + r) + n(d_j + \delta P).$$

All coefficients are fixed integers. Together with the finite transient, these formulas give a finite disjunction of linear existential formulas for time and occupied coordinates; particle ordering, if needed, can also be imposed by linear inequalities. Hence the timed relation is Presburger-definable.

For an exact word anchored at z , each required one means at least one of these at-most-three coordinates equals the corresponding site; each required zero means *none* equals that site. On a fixed time phase, each equality of an affine coordinate to a fixed site is either always true, always false, or true at at most one integer n . Consequently the full observation is eventually constant on each phase. A finite union of those phase tails and finitely many exceptions is effectively eventually periodic. The interval computations, finite-state iterations, and finite equality tests above make the threshold and period computable. This proves the theorem without requiring the source's stronger uniform-in-input-position construction. $\square$

Applying Theorem 164.1 to the reversible subclass supplies the lower bound in Theorem 159.1; Sections 161–163 supply the four-particle upper example. The lower bound relies on binary particles having a single internal type. It makes no assertion about several weight-one labels, signed weights, active zero-weight backgrounds, time-dependent rules, infinite-support inputs, or other dimensions.

165 Relation to the existing results

The scope separates three different questions: timing regularity, decidability of finite observations, and computational universality. The present example separates eventual periodicity from the other two: a simple decidable quadratic set can have unbounded gaps.

The earlier four-particle shuttle. The inspected repository audit of source 17 describes a different radius-six binary four-particle shuttle and explicitly proves that rule noninjective [17-four-r-mass-BINARY-INDEPENDENT-AUDIT.md; the rule (89.1) and the note after Proposition 89.1]. Its arithmetic timing corollary already has the elementary squared-polynomial form used here. It does not supply global reversibility. The six-row rule above addresses that missing property using the inherited prospective-isolation mechanism [57]. Its reversibility proof deliberately handles candidate births, rejected keys, and arbitrary full-shift configurations.

Low-particle work and universality. The inherited source records Kong's 2021 binary three-particle nonuniversality argument and its periodic-orbit decomposition [[41]; source 16, Section 81]. The source then gives the stronger weighted single-unit-label timed theorem used for our lower bound. We rely on the pinned source proof, reproduced in the binary specialization above; we do not attribute its full weighted or uniform conclusion to Kong.

The same inspected article records already available binary and globally reversible binary five-particle universality constructions in the repository’s *five-particle-binary-automata* collection, Reports 14 and 15, and its four-mass decidability theorem [**five-particle-binary-automata**, Part I; source 17, Theorem 83.1; the dated note in Section 81]. Those are separate results. We do not rerun their source programs or use their universality claims in a proof here. Four-particle nonperiodic timing is compatible with four-particle decidability, and the present rule is not claimed to compute universally. The inherited source 17 already proves fixed-input alternatives consisting of independent finite-phase objects, whole configurations periodic up to translation, or expanding shuttles with affine section gaps and anchors and positive-quadratic accumulated section times (lines 6269–6330 [write] [Proposition 87.1 and Corollary 87.2]). Source 18 refines this to complete disjoint fixed-input timed charts with at most two natural parameters, affine domains, and degree-at-most-two coordinate and time polynomials [write] [Lemma 94.1]. These include within-flight time bands, not only isolated quadratic return sequences; Corollary 163.2 is consistent with that established description.

The standard multistate setting. Morita proves a universal 96-state, four-neighbor one-dimensional RNCCA by simulating a reversible partitioned CA [73]. That primary result establishes the expressive power of reversible number conservation with many states; it does not give this binary fixed-four-particle timing statement. No literature-priority claim follows from the present source comparison. The radius bound 90 is also not asserted optimal.

[write] [Further context in this collection. Report 26 (Part VI of **five-particle-binary-automata**) applies the same prospective-isolation lemma to make a binary five-particle universal compiler a product of two full-shift involutions; source 24 applies it to a four-particle clock, which is not universal. The programme’s intake reads source 24 as “a decidable quadratic timing set, not a universal substrate improvement” [99].]

166 Independent checks and reproducibility

The unrestricted conclusions above are mathematical proofs, not mechanically formalized theorems. Finite computation checks the literal implementation and boundary arithmetic; it cannot replace the full-shift argument. The frozen independent audit records a final mathematical PASS and an additional complete UndefinedBehaviorSanitizer replay [24-**reversible-frozen-audit-AUDIT.md**].

166.1 Three separately inspectable checkers

The candidate checker was newly written from the stated rule. The independent sparse checker was separately written from the six rows and imports no candidate implementation. It discovers every possible raw key by aligning occupied sites with endpoint offsets, tests exact read windows, and rediscovers the complete raw-key set after hypothetical swaps. The independent C++ implementation uses rotated masks on 64-periodic words rather than sparse alignment code.

The independent audit records:

- All 540 potentially changing source-side, target-row, relative-anchor interactions, comprising 49,840 complete local assignments. Forty-eight assignments change a raw predicate, so prospective rejection is nonvacuous. At most 12 free bits occur in any interaction table.
- Every step of 108 cycles, $D = 13, \dots, 120$: 11,988 complete steps and unique selected raw keys at every half-step; plus 36 large-coordinate phase and contact tests through $D = 10^{100}$.

- A continuous 3,780-step orbit with exactly the 61 hits $k^2 + 3k$, $0 \leq k \leq 60$.
- 4,736 arbitrary or malformed finite configurations, yielding 9,472 checked blocks: 4,179 changed blocks, 2,962 with multiple selected keys, and 21 with isolated prospectively rejected keys. Both inverse identities and translation covariance were checked.
- All 262,144 binary words of width 18 embedded in period 64, and 252 additional two-island words: 524,792 checked blocks, including 80 isolated-rejection blocks and 216 multiple-selection blocks. Both inverse identities passed on every enumerated width-18 word.
- The compact candidate-birth example and simultaneous accepted/rejected example; 42 large-integer quartic tests and 1,516,800 rational trials.

The periodic tests are exact induced tests on infinite periodic configurations. Period 64 is greater than $2H = 60$, avoiding duplicate lifts in an isolation neighborhood, and greater than $2(b + r) = 30$, separating the raw-predicate influence neighborhoods of copies of a hypothetical write. Consequently the ring prospective test agrees with the specified single-anchor test. The enumeration is *not* all 2^{64} periodic words. Likewise the local interaction tables exhaust each pairwise raw-predicate interaction, not every radius-45 input neighborhood. These distinctions are part of the audit scope.

The C++ source uses explicit failure checks rather than assertions disabled by optimization. Its optimized and UndefinedBehaviorSanitizer runs produced byte-identical JSON receipts. Python checks likewise use explicit failure tests. The recorded audit environment was Python 3.12.14 and Debian g++ 14.2.0.

166.2 The offline release packet

The archive includes the editable manuscript and PDF, figure and numerical trace, inspected checker sources, read-only frozen candidate and audit files, and SHA-256 manifests. The inherited article is preserved as inert evidence with its exact excerpt and provenance; it is never compiled, imported, or executed. No upstream executable, automation, or schedule is run.

The release helper `code/replay.py` first verifies the manifest, then copies only the inspected checker sources into a fresh temporary directory. All checker output and newly compiled binaries stay in that scratch directory. It compares deterministic results to the frozen receipts, ignoring only the recorded sparse checker's elapsed-time field. It also verifies the source hashes and excerpt identity, regenerates the numerical spacetime trace from the literal rule, and rebuilds this manuscript with shell escape disabled. No frozen input is overwritten and no network access is required. The optional sanitizer mode recompiles only the independently authored C++ checker with UndefinedBehaviorSanitizer. See `README.txt` for exact commands and prerequisites.

SHA-256 manifests authenticate correspondence with the inspected packet once their digest is obtained through a trusted channel; they are integrity checks, not a claim of digital signatures or third-party certification. The independent audit's original manifest is retained separately. Figure 10 is supplementary illustration; its seven hit rows and every plotted occupied coordinate are reproduced by the checker-based trace generator.

[write] **The release in this report.** These are the delivered names, below `Research_Report49/`. The programs are shipped as `code/24-reversible-*` (for example `frozen/candidate/check_reversible_clock.py` as `code/24-reversible-frozen-candidate-check_reversible_clock.py` and `frozen/audit/exhaustive_periodic.cpp` as `code/24-reversible-frozen-audit-exhaustive_periodic.cpp`), the receipts, interaction tables, figure trace and provenance records as `data/24-reversible-*`, the figure as `figures/24-reversible-clock-spacetime.pdf`, and the

audit notes and the audited specification as `24-reversible-frozen-*.md` at the report root. Not shipped: the manuscript and its PDF (printed here), `README.txt`, the three checksum manifests (`MANIFEST.sha256` and two `SHA256SUMS`), the byte copy of this report’s `article.tex` and its line excerpt (the repository file at blob `ec10c7a5` is the original), the byte-identical sanitizer receipt and a byte-identical copy of the candidate specification; all survive in the arrival archive. `code/replay.py` verifies `MANIFEST.sha256` first and therefore cannot run in this report; the checkers import or write by their delivered names, so they are rerun on a re-extracted copy of the archive (`README`). The programme’s intake reports these checks as historical assertions; ProveIt reran them on copies at placement: every receipt was obtained again up to Windows line ends, except that the sparse checker’s receipt also differs in its elapsed time and in the hash of its tables (written with CRLF line ends; the LF form has the frozen hash), and the sanitizer run, which needs a non-Windows compiler, was not repeated (its frozen receipt equals the optimized one byte for byte).

167 Further questions

These are directions for investigation, not consequences claimed by this report.

1. **Radius and guard efficiency.** Can the radius-90 reversible example be reduced substantially while retaining a transparent full-shift proof? The conservative bounds $b = 7$, $r = 8$, and global isolation $H = 30$ are sufficient rather than optimized. Any reduction must still preserve rejected-key status and exclude candidate births. ([write] Open. It is the reversible form of source 17’s Question 6, whose noninjective binary example has radius six.)
2. **Uniform descriptions across initial inputs.** The inherited sources already give effective fixed-input four-particle orbit alternatives and quadratic phase charts. A different question is whether a suitable exact timed normal form can be made uniform in the initial particle positions, with precisely stated syntax and parameter bounds. The present family gives one uniform quadratic timing formula, but does not settle that general problem. No fixed-input classification is asserted open here. ([write] Added 4 October 2026: answered conditionally by source 25, the same pipeline’s next report but one, for the single-unit class at mass at most four: one finite chart family uniform in all ordered initial coordinates, with at most two evolution parameters and at most three gap quotients (Theorem 169.1), conditional on source 17’s scattering data and source 18’s fixed-input charts.)
3. **A useful criterion for safe parallel swaps.** Can prospectivity be replaced by a simpler finite syntactic criterion for suitable endpoint families, while retaining the full-shift conclusion? The compact counterexample above rules out isolation alone, but does not preclude stronger row-specific compatibility conditions.
4. **Proof-assistant verification.** A formalization could connect the finite predicate algorithm to the global involution theorem, the exact orbit, and the inherited low-particle lower bound. Exhaustive local interaction records provide useful test data, but a full proof must also cover arbitrary infinite supports and all intermediate times.

168 Dependency and provenance ledger

[write] [Source 24’s Appendix A.] The proof dependencies are short and explicit. Lemma 161.1 establishes full-shift involutions from finite row data; Table 14 instantiates it. Proposition 162.1

establishes the complete legal orbit using the guards rather than bypassing them. Equation (163.1) and the exclusivity argument yield exact nonperiodic timing. Theorem 164.1, adapted from the inherited proof, supplies the only lower-bound input. Corollary 4.1 [write] [Remark 163.1] uses only the timing formula, monotonicity, and the monic rational-root argument. The earlier shuttle, five-particle universality constructions, and Morita’s theorem provide context, not logical inputs to these proofs.

The inherited source is the ProveIt repository’s combined `signal-machine-collision-certificates/article.tex`, retrieved read-only on 4 October 2026. Its Git blob object is `ec10c7a579e04d3d27a92035d9a180dfcb695ff2`; its SHA-256 and the exact line-excerpt hash are in `frozen/audit/source-provenance.json` [write] [shipped as `data/24-reversible-frozen-audit-source-provenance.json`]. The excerpt is lines 5400–5774 inclusive [write] [from the end of source 16’s abstract through Section 79]. The model, per-input orbit argument, and treatment of exact zeros are preserved in that excerpt. The full inert source preserves the wider context, including the five-particle cross-reference at lines 5897 onward [write] [the dated note of 2 October 2026 in Section 81].

The audited candidate specification has SHA-256

505f6d8b6be6e226bd8128785913d2e8b30e0da0e328f44cf99d7b06872053ab.

The original independent-audit manifest has SHA-256

47e602d8d935baad3eba4dd7ec54c2902a56e4e3eda4d847b9b2c88ccaab7ff7.

The release manifest covers the separate frozen copies rather than mutable working files. Read-only filesystem modes guard against accidental editing; the hashes detect alteration. They do not make the files cryptographically immutable. [write] [The candidate specification is shipped as `24-reversible-frozen-audit-audited-candidate-specification.md` (its byte-identical copy `frozen/candidate/candidate-specification.md` is not shipped); the original audit manifest `frozen/audit/SHA256SUMS` and the release manifest are not shipped and survive in the arrival archive. Both hashes above were verified at this write.]

[write] **Source 24’s bibliography.** Its nine entries are cited here as follows. *inherited*: this article at blob `ec10c7a5`, Part VI, source 16 (“the main-branch link may change”; the packet held the inspected bytes). *parallel*: the programme’s review [57], inspected at blob `30c7366b`, “source of the prospective-isolation mechanism; attribution recorded in the frozen candidate and independent audit”. *oldclock*: `17-four-mass-BINARY-INDEPENDENT-AUDIT.md`, inspected at blob `0aa7b970` (“the earlier noninjective radius-six shuttle is distinct from the present rule”). *kong*: Kong’s thesis [41]; the source’s attribution follows this article, because its own fresh retrieval from the university server was rate-limited. *five*: `five-particle-binary-automata`, Reports 14 and 15, Part I, “cited for contextual distinction only; not an independently replayed input”. *morita*: Morita [73], definitions in its Section 2 and the 96-state four-neighbor construction in its Section 3. *ginsburg*: Ginsburg and Spanier [7]; the per-input eventual-periodicity argument of Section 164 is also given directly. *audit*: the frozen independent audit of 4 October 2026, final mathematical and UBSan PASS, shipped as `24-reversible-frozen-audit-AUDIT.md`.

Source 25 (batch-91 manuscript 03): Uniform timed charts for four mass units

“Report 51”, 4 October 2026. [write] From the archive `Uniform_Timed_Charts_for_Four_Mass_Units_Package.zip` (main file `Research_Report51/manuscript/report51.tex`, 21-page PDF; neither is shipped). Its labels carry the prefix `smc:ut:`, and its sections are Sections 169–180 here;

its appendix is printed as the last of them. Its main theorem is conditional on source 17’s scattering data and source 18’s fixed-input chart theorem, which it imports from this article by line number (brackets translate the lines into labels); its obstruction uses source 24, which it calls “Report 49”. Its abstract follows.

Fix a one-dimensional finite-radius conservative cellular automaton with a unique zero-weight vacuum, positive integer nonvacuum weights, and at most one weight-one symbol. Conditional on the pinned source-17 scattering mechanism and source-18 fixed-input chart theorem, we construct one finite effective family of complete timed-orbit charts, uniform in all ordered initial coordinates at total mass at most four. On finitely many disjoint Presburger input cells, at most two natural evolution parameters suffice; stationary-frame positions are affine and time has joint total degree at most two. Literal affine domains require at most three additional, uniquely determined initial-gap quotients, giving at most five natural chart parameters. The decisive step is a fully guarded symbolic contraction up to the first bounded ordinary reset, after which only fixed templates are added. This proves uniform untimed stationary-frame Presburger reachability. A copied-input, selector-safe compilation gives a finite-arity quartic with empty or singleton natural witness fibers for the uniform complete timed relation. Its witness count includes copies, quotients, selectors and slacks, and is not two or five. A shifted version of Report 49’s [source 24’s] reversible clock shows that original-frame untimed reachability can fail to be Presburger even from one fixed four-particle input. We give the full arithmetic construction, endpoint ownership proof, conditional dependency ledger and independent adversarial checks. No general compiler implementation, formal verification, or literature-priority claim is asserted.

Conventions. $\mathbb{N} = \{0, 1, 2, \dots\}$. Coordinates are integers. Translation τ_a moves a configuration to the right by a . A complete configuration lists every occupied site and its label; all unlisted sites are vacuum. “Affine” allows rational coefficients until positive denominators are cleared.

169 The uniform statement and its scope

Let A be a finite alphabet with distinguished vacuum 0. Let $w : A \rightarrow \mathbb{N}$ have $w(0) = 0$ and $w(a) > 0$ for $a \neq 0$. Let $F : A^{\mathbb{Z}} \rightarrow A^{\mathbb{Z}}$ be a deterministic, time-independent, translation-equivariant cellular automaton of finite radius R , with quiescent vacuum and

$$M(F(c)) = M(c), \quad M(c) = \sum_{i \in \mathbb{Z}} w(c_i)$$

for every finite-support configuration c . There is at most one symbol of weight one; labels of higher weights need not be unique. Fix a tuple $\mathbf{a} = (a_1, \dots, a_m)$ of nonzero input labels with total weight at most four. An input is determined by $\mathbf{x} = (x_1, \dots, x_m) \in \mathbb{Z}^m$, $x_1 < \dots < x_m$, with these labels in order. Thus $m \leq 4$. Empty input is allowed.

[write] [This is the model of source 16 (Section 74) and source 17 (Section 83) at mass at most four.]

If a unit symbol u exists, conservation implies that an isolated u becomes exactly one u , displaced by some $\delta \in [-R, R]$. Put

$$G = \tau_{-\delta} F, \quad S = \max\{1, R + |\delta|\}. \quad (169.1)$$

Then G has radius at most S and fixes every isolated u . If no unit exists, put $G = F$, $\delta = 0$ and $S = \max\{1, R\}$. In either case $F^t = \tau_{\delta t} G^t$.

Theorem 169.1 (Conditional uniform timed chart theorem). *Assume the finite scattering data of source 17 and the complete fixed-input chart theorem of source 18, in the precise forms stated in*

Section 170. For fixed F and $\mathbf{a}$, one can effectively construct a finite family, independent of the magnitudes of $\mathbf{x}$, with these properties.

1. The ordered input space is partitioned into finitely many disjoint cells defined by integer-affine equalities and inequalities and fixed-modulus congruences in the initial gaps $g_i = x_{i+1} - x_i$.
2. On each input cell, a chart has a fixed ordered output-label tuple and at most two additional natural evolution parameters $\mathbf{z}$. Its remaining domain constraints are rational-affine equalities and inequalities in $(\mathbf{x}, \mathbf{z})$. Its time $T(\mathbf{x}, \mathbf{z})$ is a rational polynomial of total degree at most two jointly in all these variables. Every occupied coordinate $V_j(\mathbf{x}, \mathbf{z})$ in the G frame is affine. All outputs are integral on the domain.
3. For each fixed valid input $\mathbf{x}$, the disjoint union of the chart domains maps bijectively to the complete timed orbit

$$\{(t, G^t(c_{\mathbf{a}, \mathbf{x}})) : t \in \mathbb{N}\}.$$

The bijection includes uniqueness of the chart and its natural parameters, not merely coverage by finitely many formulas.

A literal integer-affine-domain version is obtained by adding at most $m - 1 \leq 3$ uniquely input-determined natural gap quotients, hence at most $m + 1 \leq 5$ natural chart parameters for nonempty input. The empty input has a one-clock vacuum chart.

The word *uniform* is essential: the same finite family handles arbitrarily large input gaps. Source 18 by itself fixes the entire initial configuration and permits an input-dependent finite history to be materialized. That alone does not imply this theorem. Here all preprocessing depends only on the fixed rule, radius, weights and input-label case. No practical complexity or coefficient-size bound is claimed.

[write] [The two imports are source 17's Lemmas 85.1, 86.1 and 86.2 with Proposition 87.1, and source 18's Theorem 93.1 with Lemma 94.1. Source 18 is itself conditional on Proposition 87.1, so Theorem 169.1 rests on source 17 twice; source 17's proofs have been reviewed by the programme [46], source 18's chart and compiler argument passed a scoped intake conditional on source 17's normal form (review_timed_four_mass_source18_intake.md, commit 39f398985) [112], and the intake of source 25 [99] does not certify either import. The theorem answers source 17's Question 2 and the uniform half of its Question 4 under these conditions, and the corresponding remarks in Section 1.21; source 18's statement that it gives no bound uniform over inputs remains true of source 18. With the uniform family the timed relation is still not Presburger: source 17's binary shuttle already excludes that for one fixed input (Proposition 89.2).]

Corollary 169.2. *Under the same assumptions, complete untimed reachability in the stationary frame is uniformly Presburger-definable in input and output coordinates. In the original frame, complete timed outputs have degree at most two. For each fixed input-label and output-label case, the uniform complete timed relation has an integer polynomial of degree at most four with exactly one natural witness tuple at a member and none otherwise, using canonical natural pairs for signed external coordinates.*

The Presburger conclusion concerns *untimed stationary-frame* outputs. The timed relation is not claimed Presburger, and Proposition 176.1 shows that original-frame untimed reachability can genuinely fail to be Presburger. The single-fold statement concerns natural witnesses and keeps time external; it is not a generic rational- or real-witness theorem. No theorem here covers several weight-one types, signed weights, an active zero-weight background, infinite support, higher dimensions, or nonuniform rules.

170 Imported finite dynamics and the dependency boundary

The inert, pinned combined article [sources 16, 17 and 18 of this report, at blob [ec10c7a5](#)] is included in the release. Its source numbering is internal to that article: source 16 is *Three mass units with a single unit symbol*; source 17 is *Four mass units and exact reachability*; source 18 is *Canonical quartic certificates for timed four mass orbits*. The article’s repository blob and exact content hash are recorded in Section 180. The following are the imported dynamical ingredients, not newly claimed classifications.

[write] [Restatement. This section restates source 16’s Sections 75–77, source 17’s Sections 84–86 and source 18’s chart interface, in places word for word; brackets name each counterpart. It is printed because it fixes the exact form in which source 25 uses them and adds two explicit bounds (the remark that the section mechanism applies at every current gap $D > N$, and the inequality after (170.5)); no novelty is claimed for it.]

170.1 Independent components and the no split packet lemma

Join occupied sites at distance at most $2S$. If distinct components are farther than $2S$ apart, no radius- S read neighborhood sees both. Quiescence and finite propagation imply exact one-step independence; the separately evolved output supports are disjoint and their union is the actual successor.

Source 16 proves that a mass-two packet of diameter at most $2S$ never exceeds that diameter under G [write] [Lemma 75.1]. The short argument also explains why a coarse finite-speed estimate would be insufficient. A single weight-two label has its next support within a width- $2S$ interval. For two units at $0, d$, $1 \leq d \leq 2S$, if $d > S$, each original site sees precisely its isolated-unit neighborhood. Both units remain, exhausting all mass. If $d \leq S$, a new nonzero site must see both units: seeing zero or just one would give the vacuum or stationary-singleton output there. Every new site is in $[d - S, S]$, which also contains $0, d$ and has width at most $2S$. Induction proves the claim.

There are finitely many normalized mass-two shapes: the weight-two labels and two units with separation $1, \dots, 2S$. Each has a finite deterministic shape orbit and a computable finite-phase description

$$G^{\mu+r+p^k}(q) = \tau_{k\eta} q_r, \quad 0 \leq r < p, \quad k \in \mathbb{N}, \quad (170.1)$$

with fixed integers μ, p, η and explicitly known phase configurations. A bounded pair cannot emit one independent fragment and retain another across distinct $2S$ components. [write] [This is (84.1) of source 17’s Lemma 84.1, and source 16’s Lemma 75.2.]

Source 16 also supplies the finite mass-three core of diameter at most $4S$, its first-return computation and complete fixed-input profiles [write] [Section 76 and Proposition 77.1; source 17’s Lemma 84.2]. A core return repeats a normalized configuration; determinism then repeats the *whole* intervening orbit up to translation, rather than merely its section points. These data yield, for each connected mass-three seed, a fixed prefix followed by a compact translated-periodic configuration or by a stationary unit and an independently receding mass-two packet. Zero packet drift is absorbed into the compact case.

170.2 The noncircular seed library and safety constants

Let $\mathcal{E}$ be the finite family of normalized connected mass-three seeds, each of diameter at most $4S$. A seed may carry a finite analytical tag recording the target unit’s coordinate at entry; this is not a

physical label or extra memory in the automaton. For every seed select the complete finite prefix and eventual compact or emission outcome supplied above. Prolong an emission prefix until the nonzero-drift packet stays farther than $2S$ from its residual unit in every future phase.

Choose a single integer $B \geq 4S$ bounding the absolute coordinates, relative to the seed's left edge, of every occupied site throughout every chosen prefix and every initial-period representative. Include each residual marker. Compact phase diameters and emitted head-to-marker offsets are then at most $2B$. Source 17 fixes

$$N = 4B + 12S, \quad W = N + 2B. \quad (170.2)$$

The seeds defining B are the original span- $4S$ seeds. They are not all shapes of span W , and B does not depend on the arbitrary gap being analyzed. This noncircular choice is part of the uniformity input. [write] [Section 85; (170.2) is (85.1).]

A seed translated to left edge a , together with a fourth unit at b , is safely independent throughout its complete fixed prefix whenever $|b - a| > B + 2S$. For an outside- W connected-seed-plus-unit input, its seed diameter gives

$$|b - a| > W - 4S > B + 2S. \quad (170.3)$$

A compact outcome can then have a very long flight. Its first actual proximity to the fourth unit, if any, has total diameter at most $2B + 2S < W$. It therefore either escapes independently or enters a bounded full encounter. It is never replaced by a purported constant-time scattering reaction. [write] [Lemma 85.1 and the paragraph after it.]

170.3 The large gap section interface

An *ordinary* mass-four state has complete diameter at most W . A *live emission section* has two stationary units at $L < R$, gap $D = R - L > N$, and one bounded mass-two head at a prescribed finite-phase launch offset from one marker, drifting toward the other. Its finite control records the launch side, packet phase and type, and finite seed tags.

The geometric ordinary and live sets can overlap. At a designated emission checkpoint, use the live tag whenever $D > N$, even if the span is at most W . At other dispatch states apply the ordinary span test first. Once entered, a live step retains its tag through the next designated emission or explicit reset. Accordingly, *first ordinary reset* below means the first ordinary dispatch under this convention, not the first incidental intermediate time of span at most W . [write] [This is source 17's representation convention, Section 86.]

After adjoining a fixed gap residue to the control, a continuing live edge has fixed data

$$D' = D + c_q, \quad L' = L + e_q, \quad h_q(D) = \alpha_q D + \beta_q, \quad (170.4)$$

with $c_q, e_q \in \mathbb{Z}$, $\alpha_q > 0$ rational, and an integer positive duration on its admissible residue. Every intermediate flight phase and local-prefix state is retained. The source construction applies at *every current* gap $D > N$, not just beyond an unrecorded input-dependent threshold. [write] [(170.4) is (86.1) of Lemma 86.1, whose “sufficiently large gap” is the bound $D > N$ of (85.1), as source 17 says there.]

Here are the source bounds behind that assertion. At contact with target marker b , the seed's left edge a' satisfies $|a' - b| \leq 4S$. The other marker p then obeys

$$|p - a'| \geq D - 4S > N - 4S > B + 2S.$$

The entire seed prefix is safe. The new residual marker moves from b by at most $4S + B$, and $D > N > B + 4S$ prevents marker order reversal. Thus both gap and anchor change by fixed additive constants.

For the flight, reflect if necessary so the packet drift $\eta > 0$. Let its phase- r left edge relative to the launch marker be $h_r + \eta k$ and its width be $d_r \leq 2S$. First proximity in that phase is exactly

$$D - 2S - d_r \leq h_r + \eta k \leq D + 2S. \quad (170.5)$$

The hull test is exact here because a width- $2S$ mass-two support has no gap between the radius- $2S$ neighborhoods of its occupied sites. The launch bound $|h_r| \leq 2B$ gives

$$D - 2S - d_r - h_r > N - 4S - 2B = 2B + 8S > 0.$$

No hidden negative period count or new transient occurs. On each $D \bmod \eta$, candidate counts are D/η plus fixed offsets, and candidate times have the same leading coefficient p/η . Eligibility and the earliest phase depend only on this residue. Finite propagation prevents crossing the target without proximity. These facts give the finite control and affine duration in (170.4). [write] [(170.5) is (86.2), with source 17's drift d written η ; the displayed lower bound makes explicit source 17's statement that $D > N$ ensures nonnegative candidate counts.]

The noncontinuing alternatives are retained: an outward head escapes independently; an inward emission with next gap at most N has span at most $N + 2B = W$; a compact outcome either escapes or reaches an ordinary configuration after its separately accelerated flight. The large-gap mechanism does not retain a second independent gap through a compact reset.

170.4 Exactly what the fixed templates supply

For *each fixed* complete input, source 18 converts source 17's normal form into finitely many disjoint half-open charts, with at most two natural parameters, integer-affine domains, quadratic time, affine stationary-frame positions, and complete timed-orbit bijectivity [write] [Lemma 94.1, Section 95]. It may explicitly materialize a large finite history depending on that fixed input. We use this only for the finitely many normalized ordinary shapes of span at most W , after the first ordinary reset. The new uniform deduction is initial symbolic dispatch plus symbolic live-cycle contraction and additive reset composition. Neither the fixed-input theorem alone nor a collection of separately computed input orbits would justify that deduction.

171 A uniform first contact calculation

Lemma 171.1 (First contact on finitely many input cells). *Consider finitely many fixed finite-phase objects with anchors affine in the external input coordinates. Their independent counterfactual motion has a finite disjoint input partition on which either no monitored proximity ever occurs or the first proximity time and all occupied positions at that time are affine. Every earlier complete configuration has half-open affine charts using at most one natural evolution parameter. All residue divisors are fixed constants.*

Proof. Take a common fixed transient μ and common phase period P . At elapsed time $\mu + r + Pk$, each occupied site has a form

$$a_v(\mathbf{x}) + b_{v,r} + d_{v,r}k, \quad 0 \leq r < P, \quad k \in \mathbb{N},$$

where the phase constants and drifts are fixed. For any monitored pair of occupied sites, proximity at most $2S$ is exactly

$$-2S \leq A(\mathbf{x}) + dk \leq 2S. \quad (171.1)$$

A finite union over site pairs is necessary for compact shapes with holes. A hull-only test for an arbitrary compact mass-three phase is not valid.

If $d = 0$, the earliest phase count is $k = 0$ when $|A(\mathbf{x})| \leq 2S$ and no candidate exists otherwise. If $d > 0$, the earliest possible candidate is

$$k_* = \max \left\{ 0, \left\lceil \frac{-2S - A(\mathbf{x})}{d} \right\rceil \right\}; \quad (171.2)$$

it is retained only if the upper inequality in (171.1) holds there. For $d < 0$, replace (A, d) by $(-A, -d)$. Split the numerator modulo the positive fixed divisor, then split the maximum at zero. The resulting count and candidate time are affine, with affine eligibility tests. Each transient time is one further constant-time candidate with affine eligibility.

Among the finitely many eligible candidates, choose the least actual time, breaking ties by a fixed index. Comparisons between affine candidate times remain affine. This gives disjoint input cells, including a cell with no candidate. Ties select an arithmetic description only: the actual configuration retains *all* simultaneous physical contacts.

The precontact chart in phase r has guard

$$0 \leq \mu + r + Pk < h(\mathbf{x}),$$

where h is the selected first-contact elapsed time. In a no-contact cell omit the upper bound. The finite transient is handled separately. All guards and positions are affine. Before the first proximity, the objects are farther than $2S$ apart, so one-step independence holds. The preceding independent step therefore also gives the actual complete configuration *at* first contact, with disjoint occupied sites. The next piece owns that endpoint; the precontact piece does not.

If entry time and anchors already depend affinely on $\mathbf{x}$, add the entry time and substitute the anchors. This does not change degree. Every ceiling divides by a fixed integer, never by a variable gap. Rational-affine integrality and congruence pullbacks are treated explicitly in Section 176. $\square$

[write] [For one fixed input this is the first-entry computation of source 16 (Lemma 76.2, interval (76.1)) and of source 17 (Lemma 85.1); the lemma makes it uniform in the input by the residue splitting of (171.2). The warning against hull tests for compact shapes complements (86.2), where the hull test is exact because the moving object is a mass-two packet.]

172 Uniform initial dispatch

Partition the initial coordinates by $2S$ -component structure, labels and bounded internal gaps. A component of mass b has at most b occupied sites and diameter at most $2S(b - 1)$. Its normalized shape therefore lies in a fixed finite list, chosen with affine equality and inequality tests.

At total mass four, the component mass partitions and their dispatch are exhaustive.

4 The initial complete configuration has diameter at most $6S < W$ and is ordinary.

1 + 1 + 1 + 1

All components are stationary units under G and the complete configuration is fixed.

2 + 2

Apply Lemma 171.1 to the two packet profiles. If no contact occurs, their synchronized motion is a terminal independent profile. At first contact, the two widths at most $2S$ and the contact distance at most $2S$ give complete span at most $6S < W$, hence an ordinary state.

3 + 1

If already ordinary, dispatch there. Otherwise (170.3) ensures the seed's complete fixed prefix is independent of the fourth unit. An emission yields an independent outward tail or a live section, with the small-gap branch ordinary. A compact outcome is followed by Lemma 171.1 until escape or ordinary first contact. That flight can be arbitrarily long in the original input gap.

2 + 1 + 1

Apply Lemma 171.1 to the packet and both units. A simultaneous contact connects all four masses, so its span is at most $6S$. A single contact yields a connected mass-three seed plus the remaining unit. This is either ordinary or safely distant and follows the 3 + 1 case. Arithmetic tie-breaking must not discard the simultaneous-contact branch.

[write] [These are the five mass partitions of source 17's Lemma 86.2, now dispatched uniformly in the input coordinates.]

Every path has a bounded number of symbolic stages: at most one first packet contact, one fixed seed prefix, and one compact flight. This bound depends on the component cases, not on any initial gap. At a designated inward emission checkpoint apply the live-tag convention of Section 170. Every necessary comparison involves affine positions. Consequently each initial cell ends in an independent terminal profile, an ordinary state, or a live section with entry time, anchor and gap affine in $\mathbf{x}$. All earlier times already have half-open affine-output charts with at most one evolution parameter.

An independent terminal profile uses a *common* phase period and one common clock. Assigning a separate free clock to each object would describe configurations that never coexist. Once phases are synchronized, all object positions are affine in the same natural period count.

172.1 Mass at most three

With a unit symbol, source 16's finite normalized core of span at most $4S$ supplies the same construction. Before first core entry there are independent finite-phase packets and units. The first-contact or first-core-entry arithmetic is affine after residue refinement. Once a bounded core shape is selected, its full fixed template has affine time and positions. A repeated normalized core shape repeats the complete intervening orbit up to translation. Thus no four-mass shuttle or quadratic live-cycle clock is needed at these masses.

[write] [At mass at most three, source 16's Theorem 74.1 already gives a uniform Presburger timed relation, which is stronger; this subsection only fits that case into the chart format.]

172.2 The case without a unit symbol

Every occupied site then has weight at least two, so at mass at most four there are at most two occupied sites. A separated two-site input must have weights 2 + 2. An isolated weight-two site cannot split: its successor must be one weight-two site. Its label and bounded one-step displacement form a finite-state walker with eventual finite-phase motion. A weight-three singleton also cannot split; a weight-four singleton *can* split, and is not silently treated as a walker.

Take all complete configurations of diameter at most $2S$, including singletons, as a finite normalized core. Outside it there are precisely two independent weight-two walkers. Lemma 171.1 describes their first proximity uniformly in the input gap. From each fixed core shape, take one exact step and compute the next core return or an independent escaping profile. Each return has a fixed positive elapsed time and translation increment. The finite deterministic graph of normalized core states either escapes or repeats. Repetition gives a complete translated-periodic template, including intermediate states. This handles possible weight-four splits and proves uniform affine charts without introducing fictitious unit markers. Vacuum has empty support and $t = z$, $z \in \mathbb{N}$; padding $S \geq 1$ handles radius zero without division by zero.

173 Finite control and every live cycle endpoint

Absorb D modulo a common fixed modulus into live control q . Include all emission drifts, duration denominators and residues needed by compact-outcome contacts. The unguarded large-gap continuation map is now deterministic on a finite set. Its actual use is subject to $D > N$ at the current section and $D' > N$ for the next live section.

Follow control until a terminal branch, a failed small-gap guard or a repeated control. Before a repeat there are only finitely many edges. Their gap and anchor updates and accumulated times are affine in the entry data. A compact outcome uses the first-contact lemma and either escapes or resets. These are finite affine branches. If control repeats, the intervening cycle word is fixed. Because control includes residues, the net gap change is a multiple of the control modulus, so its residue itinerary repeats.

Let the cycle have p edges indexed $i = 0, \dots, p-1$. Write the fixed gap and anchor increments as γ_i, ε_i , and put

$$\begin{aligned} c_0 &= 0, & c_s &= \sum_{i < s} \gamma_i, & c_p &= \Delta, \\ e_0 &= 0, & e_s &= \sum_{i < s} \varepsilon_i, & e_p &= E, \\ b &= \min_{0 \leq s \leq p} c_s. \end{aligned} \tag{173.1}$$

For cycle-entry gap d and left marker ℓ , edge s in cycle n starts with

$$D_{n,s} = d + n\Delta + c_s, \quad L_{n,s} = \ell + nE + e_s. \tag{173.2}$$

Lemma 173.1 (Complete endpoint guard). *Cycle n is completed live-to-live exactly when*

$$d + n\Delta + c_s > N \quad (0 \leq s \leq p), \quad \text{equivalently} \quad d + n\Delta + b > N. \tag{173.3}$$

This condition includes the final endpoint as well as every interior endpoint.

Proof. At each live section, the fixed edge data apply whenever the current gap exceeds N . If the next endpoint also exceeds N , the designated inward emission remains live. Induction over the cycle therefore proves sufficiency. If any endpoint first fails, its arriving edge has been traversed, but its terminal launch is an ordinary reset. The cycle is then not completed live-to-live. This proves necessity and explains why its final endpoint cannot be omitted. $\square$

If $\Delta \geq 0$ and $d + b > N$, every later cycle is valid. If $d + b \leq N$, the first cycle fails, and a finite affine split chooses its first failed endpoint. The case $\Delta = 0$ is not replaced by an input-uniform fixed period: the physical period can be $Ad + \beta$ and depends on the arbitrary entry gap. Its clock must retain the product nd .

Lemma 173.2 (Exact symbolic contraction). *Suppose $\Delta = -a < 0$. The number of wholly completed live-to-live cycles before the failed cycle is*

$$K = \max \left\{ 0, \left\lceil \frac{d + b - N}{a} \right\rceil \right\}. \quad (173.4)$$

On a finite sign and residue refinement of the input cells, K is affine in $\mathbf{x}$. The first failed endpoint $s \in \{1, \dots, p\}$ is uniquely selected by the affine guards

$$d - Ka + c_r > N \quad (0 \leq r < s), \quad d - Ka + c_s \leq N. \quad (173.5)$$

Its complete normalized launch shape belongs to a fixed finite ordinary family, and its anchor is affine in $\mathbf{x}$.

Proof. By (173.3), cycle n completes precisely when $an < d + b - N$. Counting the natural integers satisfying this strict inequality yields (173.4), including $K = 0$ and exact divisibility. If $K > 0$, the final-endpoint guard for cycle $K - 1$ proves that cycle K starts live. If $K = 0$, that is the entry hypothesis. Some endpoint in the next cycle fails, so its first failure is indeed in $1, \dots, p$. The edge arriving there is traversed under a live start, and the endpoint belongs to the reset rather than to a live successor.

For $v = d + b - N > 0$, fix $v \equiv r \pmod{a}$, $0 \leq r < a$. Then

$$K = (v - r)/a + \mathbf{1}_{r>0};$$

for $v \leq 0$, $K = 0$. Every divisor is fixed, hence these are affine input formulas after the finite refinement. No new free reset-count witness is needed.

At failure, the preceding gap is an integer greater than N and the last fixed decrement γ_{s-1} crosses N . The endpoint gap is thus at least $N + 1 + \gamma_{s-1}$ and at most N ; source 17 additionally preserves positive marker order. The launch span is at most W . Finitely many normalized shapes suffice, selected by affine equalities. By (173.2), its anchor is $\ell + KE + e_s$, affine after substitution. $\square$

[write] [For one fixed input, source 17's Proposition 87.1 needs only the expanding case $\Delta > 0$ of a repeated control, since a contracting cycle reaches an ordinary state after finitely many cycles; uniformly in the input that number K is unbounded, and Lemma 173.2 computes it symbolically. Note that b in (173.1) is the least partial gap change, not a coordinate as in Section 170.]

174 Quadratic clocks and all intermediate times

Write the edge duration at gap z as $h_i(z) = \alpha_i z + \beta_i$, and define

$$\begin{aligned} A &= \sum_{i=0}^{p-1} \alpha_i, & C &= \sum_{i=0}^{p-1} (\alpha_i c_i + \beta_i), \\ A_s &= \sum_{i<s} \alpha_i, & C_s &= \sum_{i<s} (\alpha_i c_i + \beta_i). \end{aligned} \quad (174.1)$$

We use C here so as not to confuse this cycle constant with the safety bound B . A full cycle entered at gap z lasts $Az + C$. If its initial global time is T_* , the start time of edge s in cycle n is

$$S_s(n) = T_* + n(Ad + C) + \frac{A\Delta n(n-1)}{2} + A_s(d + n\Delta) + C_s. \quad (174.2)$$

Indeed the preceding cycles have durations $A(d + j\Delta) + C$, $0 \leq j < n$, and the preceding edges of cycle n contribute the final two terms. Formula (174.2) also holds at $s = p$, the endpoint of a whole word.

[write] [At $s = 0$ this is source 17's section clock (87.2), $T_n = T_0 + A' \frac{n(n-1)}{2} + C'n$, with $A' = A\Delta$ and $C' = Ad + C$: the letters A and C differ between the two sources. Here the entry gap d is an affine function of the input, which produces the bilinear term nd .]

Initial dispatch and the finite control preperiod make T_*, d, ℓ affine in the input coordinates on each cell. Hence (174.2) has joint total degree at most two in $(\mathbf{x}, n)$. Its non-affine terms are nd and $n(n-1)$; no coefficient depending on $\mathbf{x}$ multiplies n^2 . In particular $\Delta = 0$ retains the generally bilinear term nAd .

174.1 Flight and local prefix charts

At edge s , let the packet period be P_s and the affine flight duration be $f_s(z)$, with any fixed edge offsets absorbed. For each phase $0 \leq r < P_s$, use $n, j \in \mathbb{N}$ and

$$\begin{aligned} t &= S_s(n) + r + P_s j, \\ 0 &\leq r + P_s j < f_s(d + n\Delta). \end{aligned} \quad (174.3)$$

Every occupied stationary-frame coordinate is affine in $(\mathbf{x}, n, j)$. Fixed scattering-prefix phase v has time

$$t = S_s(n) + f_s(d + n\Delta) + v$$

and affine positions in $(\mathbf{x}, n)$. A finite refinement fixes labels and phase choices. The flight and prefix own consecutive half-open intervals; the endpoint launch belongs to the next piece. The two natural parameters have distinct meanings: cycle number and within-flight period quotient.

For complete contracting cycles impose $0 \leq n < K(\mathbf{x})$. This is an affine guard after the residue refinement. In the final partial cycle substitute $n = K(\mathbf{x})$, select s by (173.5), and retain every edge before its first failed endpoint, including the entire arriving edge strictly before its endpoint. At most j remains free on these charts. Substitution of an affine K into a quadratic time preserves total degree two; coordinates stay affine.

For a failing first nonnegative cycle use $n = 0$. For a valid nonnegative cycle let n be unbounded. A noncontinuing terminal or compact branch occurs in the finite control preperiod; a repeated deterministic continuing control cannot later acquire a new outcome except through a failed gap guard.

The reset timestamp at endpoint s is $S_s(K)$, a quadratic input function. One must *not* put a nonlinear guard $t < Q(\mathbf{x})$ into a chart domain. Chronology is enforced by the affine cycle bounds and affine within-phase bounds above; time is a quadratic *output*. Positive actual edge durations and Euclidean division inside each packet period then give unique ownership of every pre-reset time.

174.2 Sorting and labels

The finitely many indexed occupied sites may change spatial order between phases or input cells. Refine a chart by each candidate permutation π , adding

$$V_{\pi(j+1)} - V_{\pi(j)} - 1 \geq 0.$$

Stationary-frame differences are affine, so these are affine constraints after denominator clearing. Actual occupied sites are distinct, and exactly one strict permutation passes. Labels follow their sites; repeated labels do not create additional accepted orders. The same ordering applies in the original frame because the common shift δt cancels in every difference.

[write] [For one fixed input the flight charts and the sorting refinement are source 18's ((95.2) and (95.4)); here the cycle number and the within-flight quotient keep their meaning, and the input enters affinely.]

175 The first ordinary reset removes unbounded geometry

Proposition 175.1 (Additive template composition). *Suppose a disjoint input cell reaches its first ordinary state, under the designated live-tag convention, at time $Q(\mathbf{x})$ with left anchor $a(\mathbf{x})$. Refine it to fix the normalized complete shape o . If Q is quadratic and a affine, then its entire suffix has charts of the form*

$$t = Q(\mathbf{x}) + t_o(\mathbf{z}), \quad V_j = a(\mathbf{x}) + v_{o,j}(\mathbf{z}), \quad (175.1)$$

with at most two natural evolution parameters, affine domains, quadratic time, and affine stationary-frame positions.

Proof. There are finitely many normalized complete configurations of mass at most four and diameter at most W . Precompute the source-18 fixed-input chart family for each such o , in the stationary-frame rule G . Translation equivariance and determinism say that the suffix from the actual reset is exactly the translate by $a(\mathbf{x})$ of the orbit from o , with time shifted by $Q(\mathbf{x})$. This gives (175.1).

This operation adds two quadratic expressions; it does not insert the quadratic arrival time as a variable argument of another quadratic clock. The template domain depends only on its own evolution parameters; combine it with the affine input cell. The normalized shape has erased every unbounded relative geometric coordinate. Only the affine anchor and quadratic elapsed time survive. No cycle-count parameter from the initial contraction persists as an independent witness.

The fixed template may contain a huge explicitly materialized prefix. Its size depends on G and one of finitely many bounded shapes, not on the arbitrary original gaps. Thus this finite preprocessing is compatible with input uniformity. The reset time itself belongs to the template at relative time zero, and all pre-reset charts exclude it. $\square$

Proof of Theorem 169.1, except for the quotient syntax. Initial dispatch covers every possible component case and all intervening times. It reaches a terminal independent profile, a bounded ordinary state, or a live section. The finite-control preperiod and cycle construction cover every live excursion until its first reset, or forever if it remains live. Compact exits use the first-contact construction, so they leave no unaccounted long flight. A first ordinary reset invokes Proposition 175.1, which covers the entire remaining orbit without symbolically concatenating later input-dependent excursions.

Every branch choice is a finite affine or fixed-residue refinement. Earliest-contact tie-breaking, the first failed endpoint, half-open stage boundaries, cycle counts, synchronized phases and Euclidean

within-flight quotients give exactly one description per physical time for each fixed input. Sorting preserves uniqueness. Small masses and the no-unit case were handled in Section 172. The family is finite because the rule, phase libraries, controls, cycles and ordinary shapes are finite. The only unbounded free evolution coordinates used are the cycle number and within-flight quotient, or the at-most-two parameters of a fixed suffix template. Formulae (174.2) and (175.1) give the asserted degree bounds.

The bijection is *fiberwise in the input*. Forgetting the input itself need not leave an injective parametrization: a noninjective rule can merge different initial configurations. This is not an additional global injectivity claim. $\square$

176 Input residues and the honest parameter ledger

All residue tests arise from relative coordinates; a common translation of the input leaves them unchanged. An integral affine combination of input positions invariant under such translations can be rewritten as an affine combination of the gaps $\mathbf{g} = (g_1, \dots, g_{m-1})$.

One must pull rational-affine residues back *before* choosing the final modulus. If $(\mathbf{a} \cdot \mathbf{g} + c)/d$ is integer-valued on the current cell, with $d > 0$, its residue $\rho \bmod M$ is equivalent there to

$$\mathbf{a} \cdot \mathbf{g} + c \equiv d\rho \pmod{dM}. \quad (176.1)$$

The modulus is dM , not merely $\text{lcm}(d, M)$. For example, the parity of $D/2$ requires $D \bmod 4$ on even D ; $D \bmod 2$ cannot distinguish $D = 2$ from $D = 4$.

Only finitely many branches, fixed divisors, cycles, normalized resets and fixed templates occur. Therefore finitely many integer-affine input congruences suffice. Choose a common positive multiple H of their pulled-back moduli. For every gap residue vector $0 \leq r_i < H$, introduce the equations

$$x_{i+1} - x_i = Hu_i + r_i, \quad u_i \in \mathbb{N}, \quad 1 \leq i < m. \quad (176.2)$$

Since $g_i \geq 1$, these quotients are natural; if $r_i = 0$, ordered input forces $u_i \geq 1$. Each input has exactly one residue vector and exactly one quotient vector. After these substitutions and positive denominator clearing, every remaining domain constraint is integer-affine. This completes the syntax part of Theorem 169.1.

Quantity for one nonempty-input chart	Bound	Role
Free evolution parameters	2	Cycle and phase, or template
Canonical initial-gap quotients	$m - 1 \leq 3$	Unique from input
Total natural affine-chart parameters	$m + 1 \leq 5$	Sum of the preceding rows
Private signed-input copies for quartic	$2m \leq 8$	Selected chart only
Private quartic variables before slacks	$3m + 1 \leq 13$	Copies, quotients, evolution

Table 15: Source 25’s ledgers. Different ledgers answer different questions. Selectors and inequality slacks are additional quartic witnesses. The free-evolution bound is not a raw-affine auxiliary-variable bound or a total quartic witness bound.

Dropping the time output leaves a finite union of affine-domain affine-output relations, with fixed congruences or their canonical affine lifts. Existential projection over natural parameters

is Presburger-definable. This gives the first assertion of Corollary 169.2; the classical connection with effective semilinearity is due to Ginsburg and Spanier [7]. Fixed finite-pattern tests, including required zeros, and translation queries can also be formed within Presburger arithmetic from these complete stationary-frame configurations. The argument does not make a quadratic time graph Presburger, and forgetting time can destroy the timed chart's single-fold property.

Finally, original-frame coordinates are

$$Y_j = V_j + \delta T. \quad (176.3)$$

They have total degree at most two. This is sufficient for the quartic compiler below, but is not an untimed original-frame Presburger deduction.

176.1 A genuine original frame obstruction

The frame restriction is substantive. The following direct corollary uses the already audited reversible clock of Report 49 [source 24, Proposition 162.1]; it does not depend on the conditional uniform-chart imports.

Proposition 176.1 (Non Presburger absolute untimed reachability). *There is a globally reversible conservative binary CA F , with F and F^{-1} of radius at most 91, whose forward untimed set of complete outputs from the single fixed input $\{0, 5, 6, 13\}$ is not Presburger-definable in its four sorted absolute occupied coordinates. Its stationary-singleton frame has a Presburger-definable untimed orbit from that input.*

Proof. Call the radius-at-most-90 clock in Report 49 G (that report calls it F). For $D \geq 13$, put $C_D = \{0, 5, 6, D\}$ and $L_D = 2D - 22$. The report's complete-orbit proposition gives

$$G^s(C_D) = \begin{cases} \{0, 5 + s, 6 + s, D\}, & 0 \leq s \leq D - 11, \\ \{0, 2D - 18 - s, 2D - 14 - s, D + 1\}, & D - 10 \leq s \leq 2D - 23, \end{cases} \quad (176.4)$$

and $G^{L_D}(C_D) = C_{D+1}$. [write] [(176.4) is (162.1), repeated with source 24's F written G .] The displayed coordinates are strictly increasing in each range and their minimum is always zero. An isolated singleton is fixed, since every literal rule endpoint requires at least two particles.

Set $F = \tau_1 \circ G$. Translation commutes with G , so $F^t = \tau_t \circ G^t$ and $F^{-1} = G^{-1} \circ \tau_{-1}$. Both maps have radius at most 91, preserve particle number and are inverse on the full shift. The singleton drift of F is 1, so its stationary frame is precisely G .

From C_{13} , cycle k starts at $t_k = k^2 + 3k$ and has gap $D = 13 + k$. Let $\mathcal{R}_F \subseteq \mathbb{Z}^4$ be the complete forward untimed output set, in sorted coordinates. Its minimum coordinate at physical time t is t . Intersect $\mathcal{R}_F$ with the affine slice

$$y_2 - y_1 = 5, \quad y_3 - y_1 = 6.$$

The right phase of (176.4) meets this slice exactly at $s = 0$. The left phase has middle-pair separation four and cannot meet a slice requiring separation one. The full phase bounds and sorting exclude spectator substitutions, including at the smallest gap $D = 13$. Thus the exact intersection is

$$\{(t_k, t_k + 5, t_k + 6, t_k + 13 + k) : k \in \mathbb{N}\}. \quad (176.5)$$

If $\mathcal{R}_F$ were Presburger, intersection with this slice and projection onto y_1 would make $\{k^2 + 3k : k \in \mathbb{N}\}$ Presburger. A unary Presburger subset of $\mathbb{N}$ is eventually periodic [7]. Here successive gaps

$2k + 4$ tend to infinity: an eventual period $P \geq 1$ would put $t_k + P$ strictly between consecutive terms for sufficiently large k , a contradiction.

By contrast the complete stationary-frame untimed orbit is explicitly

$$\begin{aligned} \mathcal{R}_G = & \{(0, p, p + 1, D) : D \geq 13, 5 \leq p \leq D - 6\} \\ & \cup \{(0, q, q + 4, E) : E \geq 14, 5 \leq q \leq E - 9\}. \end{aligned}$$

These are precisely the two phase ranges with $D = 13 + k$, and their union is Presburger. This proves both claims. $\square$

Report 49's [source 24's] inherited binary theorem also proves that complete original-frame timed evolution is Presburger for every fixed input of at most three particles [write] [Theorem 164.1; for weighted single-unit rules, source 16's Theorem 74.1]. Projecting away time preserves Presburger definability. Thus four is the least particle count for failure of fixed-input original-frame untimed reachability in the binary conservative model, including its globally reversible subclass. This sharpness deduction uses that explicitly credited lower-bound theorem in addition to the clock orbit.

The failure already holds for one fixed anchored input, so a uniform original-frame untimed relation cannot always be Presburger either: specializing its input coordinates would give $\mathcal{R}_F$. These statements concern forward time and absolute coordinates. Quotienting outputs by arbitrary translation removes this particular obstruction. The remaining classification question is which additional rules or drift conditions preserve original-frame untimed regularity.

[write] [Relation to Part VII, which source 25 does not cite. Source 20's Theorem 118.4 states that a nonzero drift can destroy semilinearity in the original frame, and its explicit weighted radius-six planar automata have original-frame visited site sets that are not semilinear (Proposition 124.1, Theorem 125.1); source 21 realizes the planar drift regime by a binary rule. A non-Presburger original-frame phenomenon of a fixed four-mass input was therefore known in this report in dimension two. Proposition 176.1 differs in three respects: it is one-dimensional, its rule is binary and globally reversible, and the non-Presburger set is the set of complete outputs (a relation in $\mathbb{Z}^4$), not a set of visited sites. Its argument, a drift of one site per step added to a clock whose stationary orbit is Presburger, is the one-dimensional, reversible instance of the mechanism of Theorem 118.4. In one dimension the obstruction is also close to source 17's remark (Question 4) that "the binary shuttle rules out Presburger arithmetic" for original-frame timed relations; the new point is that it survives forgetting the time.]

177 A selector safe uniform quartic

We now give the complete algebraic construction. This is a uniform adaptation of source 18's fixed-input compiler, not the unmodified use of that compiler on external-variable coefficients.

[write] [Source 18's compiler is Section 96 with Proposition 96.1: selectors, constant lifting (96.1), pooled output denominators and natural slacks, and its signed external pairs are Section 97. It was already noted as a second route to Part I's Theorem 10.2 (Section 1.19). The new ingredient here is the private copy of the external input (177.1), gated with the other private variables, which keeps the degree at four when chart formulas are quadratic in the input.]

Fix an ordered output-label tuple $\mathbf{b} = (b_1, \dots, b_\ell)$. Keep only the charts with that tuple and use original-frame outputs (176.3), or stationary-frame outputs if that is the desired relation. There are J retained charts. If $J = 0$, the constant polynomial 1 with no witnesses represents the empty

relation. Assume $J \geq 1$ and nonempty input for the following ledger; vacuum is treated by its one-clock chart.

177.1 Canonical external coordinates and private input copies

Encode each signed external input coordinate uniquely as

$$x_i = x_i^+ - x_i^-, \quad x_i^+, x_i^- \in \mathbb{N}, \quad x_i^+ x_i^- = 0.$$

Use the same canonical convention for external output coordinates $y_j = y_j^+ - y_j^-$. Time t remains a natural external variable. These are external encodings, not freely chosen difference-pair witnesses.

For each chart h , introduce a natural selector e_h , private natural copies X_{hi}^+, X_{hi}^- , its natural gap quotients $\mathbf{u}_h$, its $k_h \leq 2$ natural evolution coordinates $\mathbf{z}_h$, and later one natural slack per inequality. The copy residuals are

$$R_{hi}^\pm = X_{hi}^\pm - e_h x_i^\pm. \quad (177.1)$$

They are quadratic in all variables. Replace every occurrence of x_i within chart h by $X_{hi}^+ - X_{hi}^-$. Denote its private vector before slacks by $v_h = (X_h^+, X_h^-, \mathbf{u}_h, \mathbf{z}_h)$. Now its domain forms $A_{hr}(v_h) \geq 0$, $H_{hs}(v_h) = 0$ are affine and its time/coordinate outputs $P_{hq}(v_h)$ are quadratic polynomials with genuinely numerical coefficients.

Clear the positive denominator of each domain form *before* introducing a natural slack. An inequality such as $(n-1)/2 \geq 0$ requires the integral form $n-1 \geq 0$, not a natural slack equated to a potentially half-integral left-hand side. Include the quotient equations (176.2), input order tests and all chart guards among the domain forms. The private quotient variables are determined by the selected external input, not free additional choices at a polynomial zero.

177.2 Constant lifting, gates and pooled outputs

For a polynomial in the private variables of chart h , define

$$\mathcal{L}_h f = f(v_h) - f(0) + e_h f(0). \quad (177.2)$$

Only the numerical constant is multiplied by the selector. With $E = \sum_h e_h$, introduce natural slacks s_{hr} and set

$$\begin{aligned} R_0 &= E - 1, \\ R_{hr}^{\geq} &= \mathcal{L}_h A_{hr} - s_{hr}, & R_{hs}^{\leq} &= \mathcal{L}_h H_{hs}, \end{aligned} \quad (177.3)$$

$$\begin{aligned} Z_h &= \sum_i (X_{hi}^+ + X_{hi}^-) + \sum_i u_{hi} + \sum_j z_{hj} + \sum_r s_{hr}, \\ R_h^{\text{off}} &= (1 - e_h) Z_h. \end{aligned} \quad (177.4)$$

Let $Y_0 = t$ and $Y_q = y_q^+ - y_q^-$ for $1 \leq q \leq \ell$. Choose one positive common denominator L_q for the *pooled* rational output expression in coordinate q , and set

$$R_q^{\text{out}} = L_q \left(Y_q - \sum_{h=1}^J \mathcal{L}_h P_{hq} \right), \quad 0 \leq q \leq \ell. \quad (177.5)$$

Clearing each chart separately and then summing the separately rescaled values would change the relation. For outputs $z_1/2$ and $z_2/3$, the pooled equation is $6Y - 3z_1 - 2z_2 = 0$, not $Y - z_1 - z_2 = 0$.

Define the integer polynomial

$$\begin{aligned} \mathcal{C} = & R_0^2 + \sum_{h,i,\pm} (R_{hi}^\pm)^2 + \sum_{h,r} (R_{hr}^\geq)^2 + \sum_{h,s} (R_{hs}^\leq)^2 \\ & + \sum_{q=0}^{\ell} (R_q^{\text{out}})^2 + \sum_h (R_h^{\text{off}})^2. \end{aligned} \quad (177.6)$$

Every residual has total degree at most two; hence $\deg \mathcal{C} \leq 4$. It is a literal sum of squares, globally nonnegative on real values, although its exactness assertion below concerns natural witnesses only.

Proposition 177.1 (Empty or singleton natural fibers). *For valid canonical external input and output pairs, the natural zero fiber of (177.6) is in bijection with the chart points mapping from that external input to that complete timed output. It is therefore empty or a singleton, and is a singleton exactly for the desired uniform timed relation.*

Proof. At a zero every squared residual vanishes. $R_0 = 0$ and natural selectors imply that exactly one selector is 1 and the others are 0, without extra Boolean equations. On an inactive chart, (177.4) gives $Z_h = 0$. Since every summand is natural, all its private copies, quotients, evolution parameters and slacks vanish individually. Its lifted domain and output forms consequently vanish.

On the active chart, (177.1) sets the private copies equal to the fixed canonical external coordinates. Constant lifting returns the original forms. The surviving domain equations say exactly $A_{hr} = s_{hr} \geq 0$ and $H_{hs} = 0$, and the pooled outputs say $Y_q = P_{hq}$. The quotient equations determine their unique natural values. By Theorem 169.1, there is exactly one admissible selected chart point if the complete timed output is reached, and none otherwise. Each active slack is the uniquely determined integer A_{hr} .

Conversely, a chart point fixes its selector, copies, quotients, evolution parameters and slacks uniquely; set all other private coordinates to zero. This is a zero of (177.6), and the two constructions are inverse. Naturality is indispensable: signed private coordinates could cancel in Z_h , and arbitrary signed difference-pair witnesses would admit multiple representations. $\square$

If noncanonical external natural pairs must be rejected rather than excluded by the interface, append the squared quadratic residuals

$$(x_i^+ x_i^-)^2 \quad (1 \leq i \leq m), \quad (y_j^+ y_j^-)^2 \quad (1 \leq j \leq \ell). \quad (177.7)$$

They add no witnesses and preserve degree four. Invalid ordering or incompatible labels is rejected by the retained chart guards or by the empty family. Finitely many label cases may alternatively be combined using a tagged, padded external encoding; we make no optimized arity claim for that optional packaging.

177.3 Why private copies are necessary

Source 18's constant lift is degree-safe when the entire input is fixed. Applying it directly to $f(\mathbf{x}, \mathbf{z})$ can fail: $f(\mathbf{x}, 0)$ may be quadratic in the external input. The naive term $e_h x^2$ has degree three, and its squared output residual can have degree six. Private copies convert such a term to X_h^2 ; the copying equation $X_h - e_h x$ has degree two, so its square has degree four. Including the copies in the inactive gate makes this repair exact rather than merely degree-preserving.

For a concrete two-chart arithmetic fixture, let $x \in \mathbb{N}$ be external and take the private natural vector

$$(e_0, e_1, X_0, X_1, u_0, u_1, n_0, n_1) \in \mathbb{N}^8.$$

Use the residuals

$$\begin{aligned} e_0 + e_1 - 1, \quad X_0 - e_0x, \quad X_1 - e_1x, \\ X_0 - 2u_0, \quad X_1 - 2u_1 - e_1, \\ (1 - e_0)(X_0 + u_0 + n_0), \quad (1 - e_1)(X_1 + u_1 + n_1), \\ t - (X_0^2 + X_0n_0 + n_0^2 + X_1^2 + X_1n_1 + n_1^2 + e_1), \\ v - (X_0 + n_0 + X_1 - n_1). \end{aligned}$$

The charts select even or odd input with a unique parity quotient and use a genuinely quadratic external-input term after copying. Their sum of squares has total degree four. The naive alternative $[t - (xn + n^2 + ex^2)]^2$ has total degree six. This fixture checks the algebraic architecture only; it is not claimed to be the orbit chart of a particular cellular automaton.

177.4 The full witness and residual ledger

Let I be the total number of inequalities and H_0 the total number of equalities across retained charts, after denominator clearing. Equalities include all quotient equations. The number of natural witnesses is exactly

$$\begin{aligned} W_{\text{nat}} &= J + \sum_{h=1}^J (2m + (m-1) + k_h) + I \\ &= 3mJ + \sum_h k_h + I \leq (3m+2)J + I. \end{aligned} \tag{177.8}$$

Here the first J are selectors. Each nonempty-input chart has at most $3m+1 \leq 13$ private variables before selector and slacks. The displayed sum-of-squares construction has

$$1 + 2mJ + I + H_0 + (\ell + 1) + J \tag{177.9}$$

squared residual slots, with a further $m + \ell$ if (177.7) is included. Zero or redundant residuals may be removed, but these are the explicit unoptimized counts. The total is fixed for the rule and label cases, independent of input gap sizes and requested time. It need not be small. This completes Corollary 169.2.

[write] [The programme’s intake confirms the reading of this ledger: “ ≤ 2 evolution parameters is not ≤ 2 total Diophantine witnesses; a per-chart pre-slack bound is $3m+1 \leq 13$ ”, and there is no “optimized numerical gate ledger” [99]. The quartic is for one fixed rule and one pair of label cases, so the project’s universal operation bounds [1] are untouched.]

178 Adversarial boundaries and verification

The frozen proof note received a scoped structural PASS; a separate adversarial audit then examined the imported source arguments, uniform initial dispatch, every cycle endpoint, template composition, congruence pullback and the copied-input compiler [25-uniform-frozen-science-PROOF-NOTE.md with 25-uniform-frozen-science-REVIEW.md; 25-uniform-frozen-audit-AUDIT.md]. Both conclusions are conditional on the stated pinned dynamics and fixed-template theorem. Mathematical proofs carry the unrestricted claims; finite computations validate explicit arithmetic fixtures.

178.1 Failures of tempting shortcuts

The following examples were retained as negative controls.

1. **Omitting the final endpoint.** With $N = 10$, initial gap 12 and cycle word (-3) , the start is live but the endpoint gap is 9. A start-only check wrongly accepts a whole live-to-live cycle.
2. **Checking only start and finish.** With $N = 10$, initial gap 13 and increments $(-5, +6)$, the start and final gap 14 are live and net drift is positive, but the interior gap is 8. Positive net drift alone does not give an infinite live cycle.
3. **Replacing a strict count by the wrong rounding.** Completed cycles satisfy $an < v$. If $v > 0$ is a multiple of a , $\lfloor v/a \rfloor + 1$ overcounts; the correct count is $\lceil v/a \rceil$.
4. **A hull test for arbitrary compact shapes.** For support $\{0, 20, 21\}$, probe 10 and proximity radius 2, the probe is inside the expanded hull but is farther than 2 from every occupied site. The general first-contact calculation must use occupied-site pairs.
5. **An insufficient congruence modulus.** On even D , knowing $D \bmod 2$ does not determine whether $D/2$ is even. Pulling back a modulus through a denominator requires the product in (176.1).
6. **Uncopied external quadratic terms.** The naive selector-times- x^2 lift creates an actual sextic squared residual. Copying inputs, and gating all copies, is necessary for this quartic architecture.
7. **A false fixed period at zero gap drift.** $\Delta = 0$ does not remove the input dependence from $n(Ad + C)$. Deleting nd destroys uniform timing even though section geometry has constant gap.

These refute weakened arguments, not the stated theorem. The arithmetic section examples do not assert realizability of every abstract increment word by a CA.

178.2 Fresh arithmetic checks and their limits

The frozen candidate checker uses only Python's standard library, exact integers and rational arithmetic. It recorded:

- 26,733 first-contact interval cases, including zero and signed drift
- 69,800 direct contracting-cycle comparisons across 1,396 words
- 105 large-integer contraction cases with gaps through 10^{100}
- 1,538 nonnegative-cycle guard cases and 225 directional degree checks

The separate audit checker uses Python and SymPy. Its direct section oracle advances one edge and every elapsed time; the independently constructed charts use the closed clock and accelerated reset formula. It recorded:

- 7,020 abstract section/input cases and 7,353,761 timed phase descriptions, with no missing or duplicate time
- 3,242 bounded-reset cases, including additive composition with a fixed prefix and periodic reset template

- 2,196 synchronized first-contact cases with multiple phases, multiple site pairs, and signed or zero drift
- 703,125 natural witness tuples in finite boxes for the copied-input two-chart quartic, plus symbolic degrees four and six for the repaired and naive constructions
- 10,374 denominator-inside-congruence comparisons and explicit rejection of incomplete end-point, hull and modulus variants

The independent manuscript review also checked the explicit signed-input copying, quotient and residual ledgers with separate algebra fixtures. A separate literal-row check of the Report 49 consequence compared 10,301 complete states through 100 cycles, ending at time 10,300, and found exactly the 101 predicted section hits. It imports no clock implementation; the all-time obstruction follows from the proof above.

The witness boxes test their finite ranges only; they do not prove all-natural-fiber uniqueness. The section words are abstract arithmetic fixtures, not a generated library of CA rules. Neither checker implements the general rule-to-chart construction. These limits are recorded in the frozen receipts and retained in the release.

178.3 An isolated reproducibility packet

The release contains the editable TeX, rendered PDF, byte-preserved frozen proof and audit packets, the pinned inherited article as inert text, and new release helpers. The replay helper verifies all frozen hashes, copies only the inspected fresh arithmetic checkers into new scratch directories, runs them with Python assertions enabled, and compares their deterministic receipts exactly. It does not execute the inherited article, upstream programs, saved schedules, or any general CA compiler. All checker output is directed to the copied scratch files, leaving frozen evidence unchanged.

The PDF helper copies only this manuscript into a new external build directory, uses installed TeX with shell escape disabled, and fixes the source date and PDF metadata for deterministic rebuilding. The archive helper uses a fixed member order, timestamps and permissions. The release manifest verifies content correspondence; hashes are not digital signatures or third-party certification. Exact commands, prerequisites and path-safety rules appear in the release README. Mathematical audit scope and release integrity are distinct checks.

[write] **The release in this report.** These are the delivered names, below `Research_Report51/`. The five checkers are shipped as `code/25-uniform-frozen-*` (`science/check_arithmetic.py`, `audit/check_adversarial.py`, `manuscript-audit/check_manuscript_algebra.py`, `frame-obstruction/independent_check.py`, `manuscript-audit-corollary/check_frame_obstruction.py`), the four release helpers as `code/25-uniform-code-*`, the review patch of the corollary audit as `code/25-uniform-frozen-manuscript-audit-corollary-manuscript-diff.patch`, the receipts, manifests and release evidence as `data/25-uniform-*`, and the proof note, its review, the audits, the frame-obstruction proof, the scope note and the packet READMEs as `25-uniform-frozen-*.md` at the report root. Not shipped: the manuscript and its PDF (printed here), the delivery `README.txt`, five checksum manifests (`MANIFEST.sha256` and four further `SHA256SUMS` or `MANIFEST.sha256` files), the 40 page renders and two text dumps of the manuscript audits, a $\text{\LaTeX}$ log, the byte copy of this report's `article.tex` and the byte copy of source 24's manuscript; all survive in the arrival archive. The release helpers verify the unshipped manifests first and so cannot run in this report; each checker writes its receipt into its working directory and is rerun on a copy (README). ProveIt reran all five checkers on copies at placement: every receipt

was obtained again, identical up to Windows line ends.

179 Relation to earlier work and further questions

The source dependency is deliberately narrower than a claim that this report independently establishes a new low-mass CA classification. Source 16 already supplies uniform timed Presburger behavior through mass three. Source 17 supplies four-mass decidability and its finite-control, single-gap scattering construction. Source 18 already supplies complete fixed-input quadratic charts and their natural-single-fold quartic compiler. The additional deduction here is one uniform family across arbitrary initial positions, with the exact residue and quotient syntax, first-reset argument and copied-input repair made explicit.

The inherited source credits Kong’s 2021 thesis for a binary three-particle nonuniversality precursor [41]. This is historical context, not an attribution of the present weighted uniform theorem to that thesis. The fresh university PDF retrieval for this report was rate-limited; the detailed precursor attribution follows the inspected pinned source and scope note. Morita’s primary 2012 construction gives a universal 96-state four-neighbor reversible number-conserving CA by a multistate simulation [73]; it does not settle this fixed-four-mass unique-unit-label timing statement. No priority conclusion follows from the literature comparison or from a failed search for a matching theorem.

Meaningful remaining questions include the following.

1. **Remove the conditional packaging.** Can the entire finite-library construction, source-18 template theorem and present uniform deduction be joined into one independently formalized result, with the live-tag convention and all intermediate-time ownership represented explicitly?
2. **Implement the effective construction.** A rule-to-chart implementation would need to enumerate actual seeds, produce validated period and contact data, track congruence pullbacks, emit guarded cycle charts and verify fixed templates. Arithmetic fixture checkers do not perform these tasks. What certificate interface makes such preprocessing independently checkable without trusting the generator? ([write] The same gap is source 17’s Question 3, source 18’s further directions (Section 100) and source 19’s Question 2.)
3. **Optimize auxiliary variables and modulus size.** The at-most-three input quotient cost is transparent but not proved minimal. Can literal raw-coordinate affine domains use fewer total natural auxiliaries for this class? Which geometric or arithmetic restrictions allow a smaller common modulus or copied-input ledger?
4. **Characterize original frame regularity.** Proposition 176.1 settles the existence of a genuine non-Presburger obstruction, already for a reversible binary four-particle rule and one fixed input. Which additional conditions on the rule, phase drifts or observation encoding ensure original-frame untimed Presburger reachability? ([write] For planar weighted rules, source 20 already exhibits nonsemilinear original-frame site sets under drift (Theorem 125.1); the existence of an obstruction is new here only in one dimension and for reversible rules. The characterization is open.)
5. **Separate natural from real exactness.** The generic compiler is exact over natural witnesses only. Which restricted chart families have enough integrality structure to force all nonnegative rational or real zeros to represent physical integer-time states, and what extra conditions are

genuinely necessary? ([write] Source 18’s four-witness binary certificate is one such family: it is exact for nonnegative real witnesses (Corollary 99.1).)

6. **Quantify preprocessing.** Effectiveness gives no useful bound on chart count, phase periods, coefficients or the materialized prefixes of bounded templates. Explicit rule-dependent complexity bounds would clarify when the construction is practical. They must account for fixed but potentially enormous reset-template histories.

None of these questions reopens the already imported fixed-input chart theorem. Nor is the general two-total-variable raw-affine claim asserted as a consequence of the present proof.

180 Pinned source and proof dependency ledger

[write] [Source 25’s Appendix A.] The frozen inherited article is `frozen/science/pinned/inherited-article-source.tex` [write] [not shipped: it is a byte copy of this report’s `article.tex` at blob `ec10c7a5`]. It was retrieved read-only from the ProveIt repository’s combined `signal-machine-collision-certificates/article.tex`; the inspected Git blob is `ec10c7a579e04d3d27a92035d9a180dfcb695ff2`. Its exact SHA-256 is

17d3c0d9b449c689f88c1dc082ffee9b116993e4f5585b65d52c4fe591b2d962.

Source line ranges refer to those pinned bytes, not to a later main-branch view.

- Source 16: stationary frame, two-unit diameter invariance, finite normalized mass-two shapes and mass-three encounter-core profiles; see especially lines 5470–5550 and the subsequent core-return argument. [write] [Section 75 with Lemmas 75.1 and 75.2; then Sections 76–77.]
- Source 17: model and hypotheses, lines 5979 onward; original connected-seed library and fixed B, N, W , lines 6083–6121; safe resolution and the tagged section convention, lines 6123–6164; all-current- $D > N$ additive section proof, lines 6166–6223; exhaustive dispatch and fixed-input normal forms through line 6340. [write] [Section 83; Section 85 through (85.1); Lemma 85.1 and the opening of Section 86; Lemma 86.1; Lemma 86.2, Proposition 87.1 and Corollary 87.2.]
- Source 18: complete fixed-input chart interface and half-open partition, lines 6660–6797; selector, constant lifting, pooled denominator and natural-fiber compiler proof, lines 6798–6910. Its input is fixed; uniformity is not imported from its theorem statement. [write] [Lemma 94.1 and Section 95; Section 96 with Proposition 96.1.]

For the separate frame obstruction, the exact Report 49 manuscript is also retained as inert text at `frozen/report49/report49.tex`; its source and content pins are in `PROVENANCE.json` in that directory. The new short deduction uses its complete-orbit formula and full-shift reversibility, not its executable checkers. [write] [The manuscript copy is not shipped; it is source 24, printed above, and its SHA-256 `c7813317...` equals that of batch-91 manuscript 01. `PROVENANCE.json` is shipped as `data/25-uniform-frozen-report49-PROVENANCE.json`.]

The new mathematical route is: affine first contact and initial dispatch; complete endpoint guard; exact piecewise-affine contracting count; joint quadratic clock; finite normalized first reset and additive fixed-template suffix; honest congruence lift; selector-safe private copies. The frozen independent audit checks this route and the sufficiency of the actual imported section mechanism.

The frozen science manifest hash is

0fb634cca9f665b44c3601509a6cef49391599064a9715164227fc475260be4c.

The frozen independent-audit manifest hash is

7719ee3d73a9811975218309d61db5ec6e622977f3cc6197c76771f7c7edb085.

The proof note hash is

2e6484f4a2dbc0419e9662bf1ca02d4154469a1f4dcf93f4320f9951c9124962.

These frozen packets are included byte-for-byte. Their statements retain their original review dates and scope. This manuscript reorganizes and expands their exposition without modifying those scientific inputs.

[write] [The three hashes are those of the files shipped as `data/25-uniform-frozen-science-MANIFEST.json`, `data/25-uniform-frozen-audit-MANIFEST.json` and `25-uniform-frozen-science-PROOF-NOTE.md`, verified at this write.]

[write] **Source 25’s bibliography.** Its nine entries are cited here as follows. *source16*, *source17*, *source18*: sources 16, 17 and 18 of this article at blob `ec10c7a5` (source 17 at lines 5979–6340 for the imported model, library, section mechanism and normal form, “conditional dynamical input, with all phase data retained”; source 18 from line 6570, fixed-input charts and compiler at lines 6660–6910, “the source calls itself Research addendum 13; its companion Report 12 is source 17”); the source adds that the main-branch links may change and that the release pins the inspected bytes. *proofnote*: `25-uniform-frozen-science-PROOF-NOTE.md`, with its scoped structural review, arithmetic checker, receipt and pinned source. *audit*: `25-uniform-frozen-audit-AUDIT.md`, “PASS for the residue-cell theorem and canonical-quotient lift, conditional on the stated source-17 and source-18 inputs”. *report49*: source 24, “delivered and independently audited research report, Sections 2–6: the literal reversible rule, complete four-particle orbit and inherited lower-particle theorem”; the radius-91 shifted consequence is proved in source 25, and a separate check is shipped as `code/25-uniform-frozen-frame-obstruction-independent_check.py`. *ginsburg*: Ginsburg and Spanier [7], used through closure of Presburger formulas under finite union and existential quantification. *kong*: Kong’s thesis [41], “precursor attribution follows the inspected source 16 and frozen scope note”, a fresh full-text retrieval having been rate-limited. *morita*: Morita [73], “multistate universality context, not a logical input to the uniform four-mass theorem”.

181 Further questions and research (Part IX)

[write] This section was added at the assembly of batch 91 under the standing rule of 4 October 2026: an unproved claim of a source, of a review or of this write is not dropped but stated here as an open question, with its source, its argument sketch and what is missing; a claim shown false stays on record with its refutation. The two sources’ own questions are in Sections 167 and 179. The items below come from source 25’s frozen scope note, from the programme’s intake [99] and from this write.

1. **Quadratic arithmetic bands for fixed-input hit sets** (source 25’s frozen scope note, `25-uniform-frozen-science-pinned-frontier-scope-check.md`, section “Correct fixed-input arithmetic envelope”). *Claim*: for a fixed rule of source 17’s class and a fixed input of mass at most four, every exact anchored finite-pattern occurrence set is a finite union of *quadratic arithmetic bands*

$$\{Q_i(n) + p_i j : n \geq n_i, n \equiv \rho_i, l_i(n) \leq j \leq u_i(n), j \equiv \sigma_i\},$$

with Q_i a rational quadratic that is integral on its domain, l_i, u_i rational affine and the congruences of fixed moduli, together with finitely many exceptions and an eventually periodic set. *Sketch given:* in an expanding orbit with zero drift every anchored test, including each required zero, is a Boolean combination of affine conditions on the chart parameters (n, j) of source 18’s Lemma 94.1; split into disjoint cases, refine residues and take the eventually dominant affine bounds; finitely many exceptional cycles give finite intervals. With nonzero drift source 17’s cutoff (Section 88) leaves finitely many hits of a pattern with a nonzero symbol, and an all-zero pattern holds from some time on; nonexpanding tails have Presburger timed relations. Corollary 163.2 is a single band ($Q(k) = t_k + 1$, $p = 1$, $0 \leq j \leq 2k + 2d - 25$), and it is the note’s reason why isolated quadratic sequences do not suffice. *Missing:* the note itself calls this an unproved corollary: a self-contained proof of the band refinement “would still need to be written and checked”. A density or counting theorem for such hit sets is not proved, and neither is a version uniform in the input, which Theorem 169.1 would suggest under its two conditions.

2. **Which relations admit certificates exact over nonnegative real witnesses?** (The programme’s intake: source 24’s certificate “is not real-exact” [99]; source 25’s Question 5.) Remark 181.1 below proves more than the intake’s finding: no polynomial certificate of any degree and with any number of witnesses represents the hit relation of Theorem 159.1, or the set of Corollary 163.2, with external (x, t) alone and nonnegative real witnesses. Membership exactness over nonnegative reals is possible exactly for traces on $\mathbb{N}^m$ of semialgebraic sets; source 18’s real-exact four-witness certificate (Corollary 99.1) works because its external data include the integer output positions. *Open:* source 25’s question proper, which asks for *witness-level* exactness (every nonnegative real or rational zero is a physical chart point): which chart families of Theorem 169.1, with which external outputs, have it? For nonnegative rational witnesses the monic argument of Remark 163.1 shows that the trace obstruction does not apply, and no general criterion is known.
3. **A fixed-input criterion for original-frame untimed regularity** ([write], proposed at this assembly; it refines source 25’s Question 4). *Proposed statement:* for a fixed rule of source 17’s class with singleton drift δ and a fixed input of mass at most four, the original-frame untimed reachability set is Presburger if and only if $\delta = 0$ or the eventual orbit (Proposition 87.1) is not an expanding shuttle; the condition is decidable by source 17’s Theorem 83.1. *Sketch:* if $\delta = 0$ the original frame is the stationary one and Corollary 87.2 applies; for independent finite-phase or translated-periodic tails the timed relation is Presburger (Section 88) and so is its projection. For an expanding shuttle with $\delta \neq 0$ the section at cycle n has gap $D_0 + n\Delta$, left marker $L_0 + nE$ and time T_n of (87.2) with positive quadratic coefficient, so its leftmost original-frame coordinate is quadratic in n with leading coefficient $\delta A/2 \neq 0$. A Presburger slice fixing the launch shape relative to the left marker should meet the orbit in finitely many such sequences, and their projection, infinite and of density zero, is not eventually periodic. Proposition 176.1 is the case of source 24’s clock with $\delta = 1$. *Missing:* a proof that the slice meets the orbit only in finitely many quadratic sequences (other phases of a cycle can repeat the launch shape relative to the marker; each such phase occurs at T_n plus an affine function of n , which keeps the leading coefficient, but labels and the identity of the leftmost particle need checking), and any statement uniform in the input.
4. **Timing irregularity and universality** (the programme’s intake: for source 24 “no timing irregularity $\Rightarrow$ universality inference is available” [99]; source 24’s non-claim that its rule is not claimed to compute universally, Section 165). At four particles the inference fails for source 24’s

own rule: its anchored hit set is not eventually periodic (Theorem 159.1), while every exact and anchored mass-four question for it is decidable by source 17’s Theorem 83.1, since it is a binary number-conserving rule. *Open*: whether the six-row rule computes universally with five or more particles, or with unboundedly many. No argument is given in either direction. One easy observation: every endpoint of Table 14 contains a pair at distance at most four, so a configuration with all pairwise distances at least five is fixed. What is missing is any analysis beyond the four-particle orbit.

5. **Programme review of the unread spans** (the programme’s intake [99]). The intake read source 24’s manuscript only at lines 50–64, 202–228 and 308–319: the statement of Theorem 159.1, the timing section with Remark 163.1, and Section 165. It did not read the proofs of Lemma 161.1, Proposition 162.1, Corollary 163.2 or Theorem 164.1; the first and last are second routes to reviewed results (Report 26 through [57], and source 16). For source 25 it read the proof note at lines 1–39 and 130–185, so the uniform first contact, dispatch and guarded cycle acceleration (Sections 171–173) and Proposition 176.1 with its frozen proof note are not read by the programme. Source 18, which source 25 imports, passed a scoped intake on 3 October 2026, conditional on source 17’s normal form (`review_timed_four_mass_source18_intake.md`, commit 39f398985); that intake did not re-establish any uniform bound. *Missing*: a programme review of these proofs. The frozen audits (PASS) and the reruns at placement are regression evidence for the arithmetic fixtures and the literal rule, not proofs.

Remark 181.1 ([write]) No polynomial certificate of a time-only clock relation is exact over nonnegative reals). Let $T \subseteq \mathbb{N}^m$ and let $P(\mathbf{v}, \mathbf{w})$ be a real polynomial with n witnesses such that, for every $\mathbf{v} \in \mathbb{N}^m$, $P(\mathbf{v}, \mathbf{w}) = 0$ has a solution $\mathbf{w} \in \mathbb{R}_{\geq 0}^n$ exactly when $\mathbf{v} \in T$. Then $S = \{\mathbf{v} \in \mathbb{R}^m : \exists \mathbf{w} \in \mathbb{R}_{\geq 0}^n, P(\mathbf{v}, \mathbf{w}) = 0\}$ is semialgebraic by the Tarski–Seidenberg theorem, and $T = S \cap \mathbb{N}^m$. Conversely every such trace is represented: a basic set $\{f = 0, g_1 > 0, \dots, g_r > 0\}$ is the projection of the zeros of $f^2 + \sum_i (g_i w_i^2 - 1)^2$, a finite union is the projection of the zeros of the product of these polynomials in disjoint witnesses, and a real witness w is written $a - b$ with $a, b \geq 0$.

For $m = 1$ a semialgebraic subset of $\mathbb{R}$ is a finite union of points and intervals, so its trace on $\mathbb{N}$ is finite or cofinite. Fix $x \in \mathbb{N}$ in the hit relation $t = k^2 + (2x + 3)k$ of (163.1) with $d = 13 + x$. Its section $\{k^2 + (2x + 3)k : k \in \mathbb{N}\}$ is infinite and misses $t_k + 1$ for every k , because consecutive hits differ by $2k + 2x + 4 \geq 4$. So no polynomial certificate $P(x, t; \mathbf{w})$, of any degree and with any number of witnesses, is exact for nonnegative real witnesses with (x, t) as its only external data. This is sharper than the false witness $(\sqrt{13} - 3)/2$ printed after Remark 163.1, which refutes real exactness for one certificate only. The same argument applies to the set of Corollary 163.2 (infinite, with infinite complement) and to source 17’s hits (Proposition 89.2). Over $\mathbb{Q}_{\geq 0}$ the obstruction disappears, since a projection over rational witnesses need not be semialgebraic, and source 24’s quartic is exact there (Remark 163.1).

Part X

Periodic collision macros and five-signal validity (sources 26–29)

[write] Part X prints four AI-assisted research manuscripts of ProveIt’s incoming reports, batch 91 (arrival 0d7f51c44; cluster W2, placed by d750d98dd), all dated 4 October 2026 and added on

that day. They are consecutive numbered reports of one research pipeline. *Source 26* is batch-91 manuscript 08, the pipeline’s Report 56; *source 27* is manuscript 09, Report 57; *source 28* is manuscript 10, Report 58; *source 29* is manuscript 11, Report 59. The source numbers are the file prefixes of their shipped programs, data and records (26-periodic-, 27-fivesignal-, 28-diophantine-, 29-rotations-), continuing Part IX’s sources 24 and 25; their labels carry the prefixes `smc:pm:`, `smc:fv:`, `smc:fd:` and `smc:rf:`. Their author lines are “Report56”, “Report57”, “Report 58” and “REPORT 59”; they name no assistant. Source 26 pins the ProveIt commit `e3e58a882`, at which it read this report’s README and the programme’s finite-schema signal review; source 27 pinned its repository searches at `7ba8f64e6` and inherits source 26’s copies; source 28 inherits source 27 and pins the mathlib4 commit `ac77769f`; source 29 pins `b64f24e59`. Brackets translate the pipeline’s numbers where they occur: Reports 56–59 are sources 26–29; Report 55, whose power-system files source 28 copies, is Part VI of the neighbouring report *group-theoretic-substrates* (batch-91 manuscript 07); Reports 60–63 are printed in Part XI of this report and Reports 64 and 65 in Part XII.

[write] **What the four sources do.** Source 26 fixes a rational number-preserving signal machine and a *supplied* complete finite collision macro that returns to its own outgoing section. With at most four live signals the translation-reduced section has dimension at most two, and validity of every finite prefix of the endlessly repeated macro is a finite Boolean formula of integer polynomial signs of degree at most two in the initial gaps, compiled into a quartic with exactly one natural witness (Theorem 182.2); Zeno behaviour and rational deadlines are decided the same way (Theorem 182.3); and for any finite population a valid ultimately periodic complete macro on rational data has rational accumulation time and rational limit positions (Theorem 182.4). Source 27 shows that four is sharp for the sign classification: an explicit five-live-signal machine with a complete 138-event word returns as a half-contracted rotation of cosine $3/5$, and its valid set is a disk with one dense orbit deleted from its boundary, which no finite polynomial-sign formula describes, even on rational or on positive integer gaps (Theorem 196.2). Source 28 gives that very predicate an explicit existential positive-integer polynomial: 3 inputs, 81 witnesses, 44 equations, 344 gates, exact degree 12 (Theorem 211.1). Source 29 realizes every rational rotation of infinite order, at every positive rational scale, with five live signals, and proves the corresponding classification, a real-input obstruction, an $O(L^3)$ rational decision and a certificate with $1 + 80J$ witnesses for J contacts (Theorems 225.1, 225.2 and 231.1).

[write] **Relation to Parts I and II.** Source 26 compared itself with this report’s README and with the programme’s finite-schema review only: “The complete merged repository article exceeded the connector’s readable-file limit and was not obtained” (Section 193). It therefore does not cite three things printed here. (i) Its one-step exactness lemma and one-macro chamber (Lemma 184.1 and (184.3)) are the one-macro case of Part I’s chamber theorem (Theorem 5.3), re-proved; the source says it “re-proves the needed conservative one-macro chamber”. (ii) Its infinite criterion, clock classification and rational-limit theorem answer in part Part I’s research direction on accumulation (Section 15.9), which asks for a certified self-similar tail that “must not silently treat an infinite sequence of successful finite checks as one ordinary existential Diophantine witness”. (iii) Part I’s four-speed clock (Proposition 12.1, Zeno time $S = u + v$) and Part II’s bounded shuttle (Proposition 25.3, Zeno time 2) are valid ultimately periodic complete macros of population three, and for rational seeds their rational accumulation times are instances of Theorem 182.4; the shrinking-wall shuttle in source 26’s spectator example is the same classical mechanism. Notes at these places, and a dated note at Part I’s direction, record this; there is no conflict. Nothing in Part X defines a continuation at or after an accumulation point, so the report’s front non-claim on accumulation semantics (Section 1.5) stays true.

[write] **One object, three ledgers.** Sources 28 and 29 and Part XI’s Report 60 [batch-91 manuscript 12] certify orbit avoidance on the critical circle with the same pattern, a primitive Gaussian or companion fraction, a canonical valuation of its denominator and two fifteen-equation power systems, but they count it differently (Table 16). Every row adds 2×26 witnesses and

Table 16: [write] Witness and equation counts of the critical-circle certificates of sources 28 and 29 and of source 30 (Report 60, Part XI).

Source	Object	Witnesses	Equations	Per contact
28 (Report 58)	source 27’s machine, one contact	81	44	14 outer equations including the radius equation; 29 outer leaves including the radius slack δ
29 (Report 59)	the rotation family, J contacts	$1 + 80J$	$1 + 43J$	13 outer equations, 28 outer leaves; one shared radius witness and equation
Part XI, Re- port 60	elliptic rational maps, J contacts	$81J$; strict kernel $1 + 81J$	$44J$; strict kernel $1 + 44J$	14 outer equations, 29 outer leaves (one more positive ac- ceptance witness and equation); one shared radius witness and equation

2×15 equations for the two power systems. At $J = 1$ the counts of sources 28 and 29 coincide (81 and 44), but the polynomials do not: source 29’s machine for the rotation (3, 4, 5) at scale 1/2 has exactly source 27’s return matrix (its formula (229.2) gives source 27’s (196.1); checked in exact arithmetic at this write) but is not source 27’s machine (it has 174 events, and its critical radius is $\rho^2 = 1/65$, not $4/1845$; Section 235), its radix is $\beta_{\text{rad}} = 4P + 2c + 1$ rather than source 28’s $4P + 1$, and source 29 claims no gate count. Report 60’s strict kernel has one witness and one equation more than source 29’s ledger *per contact*, hence J more of each at the same J (163 against 161 witnesses at $J = 2$): it excludes the time-zero match and the positive-time matches by two outer equations with positive witnesses J_0 and J_1 , where source 29’s single equation (O13) also covers exponent 0 (Remark 240.4). [Corrected at the assembly: the draft of this paragraph said “one witness and one equation more” in all, which is false for $J \geq 2$.] The counts are not contradictory; they are three constructions, and the programme’s review reports the first two as “reported source ledgers, not independently rebuilt”.

[write] **Renamed symbols.** The letter J has three meanings in the sources: source 26’s number of logic gates is written Γ here (its ledger $4A + J$ is printed $4A + \Gamma$), source 28’s strict acceptance witness is written J_{acc} (source 29’s name), and J is source 29’s number of contacts throughout. The Gaussian quotient that sources 27, 28 and 29 call α , η and η_j is written γ and γ_j (source 27 uses η for a centred coordinate and source 26 uses α for an eigenvalue); source 27’s components A, B of that quotient are written γ_1, γ_2 , and its normalized coordinates (u, v) in Section 204 and Figure 12 are written (w_1, w_2) (source 28’s u, v, A, B are integers). Source 27’s valid set K and rotation R are written $\mathcal{K}$ and $\mathcal{R}$, as source 29 writes them (Part II’s K is a number of batches). The Pell modulus that sources 28 and 29 call M is written Θ , as the turmite report prints the same system (the sources’ M is also their return matrix), and source 28’s radix $b = 4P + 1$ is written

β_{rad} , source 29’s name for the corresponding radix (b is a leg of source 29’s Pythagorean triple). Equation tags (Ok) and (Pk) are local to each source. Every other letter keeps its source’s meaning inside that source; the front matter lists the remaining collisions.

[write] **Dependencies.** Source 27’s sharp-boundary corollary uses source 26’s theorem as an explicit, separately audited dependency (source 27 “does not re-prove” it). Source 28 inherits source 27’s validity criterion (its Theorem 212.1 is Proposition 203.2 in integer coordinates) and specializes two theorems of mathlib’s `PellMatijasevic.lean`; its fifteen-equation power system is the construction of the turmite report’s Lemma `ptr:ai:lem:exp` (*periodic-turmite-first-revisits*, Report 42) and of Report 55, from which source 28 copied the builder and its notes; these three files are shipped with *group-theoretic-substrates* as `code/09-chrono-matrix-science-build_certificate.py`, `09-chrono-matrix-science-SOURCE_NOTES.md` and `data/09-chrono-matrix-independent_audit-pell-dependency.json`, byte for byte the archive’s `science/sources/prior-build_certificate.py.txt`, `POWER_SOURCE_NOTES.md` and `pell-dependency.json`. Source 29 is separate from sources 27 and 28 (a different machine, no reuse of their certificate) and copies source 28’s proof as an inert dependency. None of the four is formalized.

[write] **Review status.** The Hilbert’s-tenth programme made a bounded routing intake of all four, with eight other signal and particle archives of the arrival, in `review_new_signals_0d7f51c44.md` (commit 49b100cfd, 4 October 2026) [99]: full READMEs and selected spans, not a proof audit, and “No specific contradictory theorem/interface statement requiring an author correction was found in the selected passages”. Its scope notes are recorded at the end of each source. This write has not been reviewed by the programme.

Source 26 (batch-91 manuscript 08): Periodic signal machine collision certificates

Exact infinite validity and observable clock arithmetic, “Report56”, 4 October 2026. [write] From the archive `Periodic_Signal_Machine_Collision_Certificates_Package.zip`; its labels carry the prefix `smc:pm:`, its sections are Sections 182–195 here (its two appendices are the last two), and its equations are numbered by section. Its bibliography entries are this report’s [6], [75], [74] and [14]. Its abstract follows.

Fix a rational, number-preserving signal machine, a complete finite collision macro, and an outgoing collision section to which the macro returns with the same ordered labels and co-location pattern. With at most four live signals, the translation-reduced section has dimension at most two. We prove that validity of every finite prefix of the endlessly repeated macro is equivalent to an effectively constructed Boolean formula of integer polynomial signs of degree at most two in the initial gap numerators. A literal sign-and-logic compiler produces a sum-of-squares polynomial of degree at most four, with exactly one natural witness tuple on each accepted encoded input. The arity depends on the fixed macro, not on the number of repetitions.

The same construction classifies finite-time accumulation from the observed clock sequence, including exact cancellation of invisible unstable or neutral modes, and handles rational deadline queries. A four-signal spectator example has a strict one-cycle chamber, a weak limiting boundary, and a Zeno clock despite return spectral radius one. In any fixed finite population, rational initial data and a valid ultimately periodic complete macro imply rational finite accumulation time and rational limiting positions.

The proofs cover simultaneous collisions, all real spectral degeneracies, and the nonreal case. They concern finite pre-accumulation prefixes; no continuation through an accumulation is defined. The implemented emitter starts at a supplied quadratic sign formula. A general machine-and-macro frontend is specified mathematically but is not implemented. No literature-priority claim, general Positivity algorithm, or new general MRDP theorem is asserted.

Reading guide. Sections 182–184 specify the model and exact chronological chamber. Sections 185–188 prove the sign, Diophantine, and clock claims. Sections 189–190 give the spectator example and rational-limit obstruction. Sections 191–193 separate implementation, evidence, reproducibility, and prior work.

182 The certificate class and its three conclusions

We use $\mathbb{N} = \{0, 1, 2, \dots\}$. All witnesses below are natural numbers. Signal positions and time are real; every speed and local linear coefficient is rational. A rational initial gap vector is encoded as $x = p/q$ with $p \in \mathbb{N}^d$ and $q \in \mathbb{Z}_{>0}$. Positivity conditions reject zero gap numerators where the section requires strict separation. An overall rational spatial translation may be supplied independently.

Definition 182.1 (Fixed data and variable input). A *complete collision batch* contains every collision site at one timestamp, every incident signal at each site, and the selected local rule there. Consecutive batches have strictly positive time separation. Fix a rational signal machine whose local rules preserve the number of signals, an outgoing collision section, and a nonempty finite word W of complete batches. The final section must have exactly the initial ordered label word and outgoing co-location pattern. The machine, rules, speeds, word, and sections are compiler data. Only the initial inter-site gaps and stated arithmetic query parameters are polynomial inputs.

The standard collision convention requires pairwise distinct speeds within each incoming set and within each outgoing set [6]. It excludes equal-speed co-located outputs. Equal-speed signals at different positions remain permitted. Completeness allows several disjoint collision sites at the same timestamp and collisions involving more than two signals. Zero-duration cascades are excluded, rather than silently merged into a permitted macro.

Theorem 182.2 (Infinite periodic validity). *For the fixed data above, with at most four live signals, one can effectively construct a Boolean combination of rational polynomial sign tests, each of degree at most two in the initial gaps, that is true exactly when every finite prefix of W^ω is the genuine complete collision history. After positive denominator clearing, the tests have integer coefficients. For integer gap numerators, an explicitly constructed ordinary integer sum-of-squares polynomial of degree at most four has exactly one natural witness tuple on each accepted encoded input and no witness on each rejected encoded input.*

The number of witnesses is independent of the repetition count. It can grow with the finite supplied word. Thus the theorem does not provide one fixed arity over all macros. Matrix and macro coefficients must be fixed before introducing the gap variables: allowing them as additional polynomial inputs would require different degree accounting.

Theorem 182.3 (Observable physical clock). *On the valid inputs of Theorem 182.2, finite total elapsed time is characterized by another degree-at-most-two sign formula. Validity together with Zeno behavior, and validity together with non-Zeno behavior, therefore admit the same unique-witness quartic construction. For rational initial gaps and a finite accumulation time, that time is rational and effectively computable. Equality to, and strict comparison with, a supplied rational deadline at section time zero also use degree-at-most-two integer sign tests.*

Theorem 182.4 (Rational limits for every fixed finite population). *For any fixed finite number of signals, a valid ultimately periodic complete collision macro with rational speeds and rational initial positions can have only rational finite accumulation times and rational limiting positions of its periodically named collision sites.*

The last theorem is conditional on validity and finite accumulation. It does not claim an effective validity test in arbitrary dimension. In particular, an execution with irrational accumulation time or irrational limiting location cannot admit a valid ultimately periodic complete macro of the specified kind. None of the three theorems specifies an execution at or after the accumulation instant.

[write] [Theorem 182.4 is the general repeated-Zeno rational-accumulation statement of this report. Part I's clock (Proposition 12.1: accumulation at $S = u + v$) and Part II's shuttle (Proposition 25.3: accumulation at time 2) are instances for rational seeds, each, after a one-batch prefix, a two-batch macro of three signals that returns to its section after one bounce at each wall (the clock's return multiplies the remaining time $S - t$ by $1/3$, Part I's proof; the shuttle's left times satisfy $a' = 4/3 + a/3$). The Zeno criteria and rational accumulation times of Part XI's Reports 61–63 are special cases of it and of Theorem 182.3; notes there point here. Together with Theorems 182.2 and 182.3 it answers in part Part I's research direction on accumulation limits (Section 15.9): for supplied ultimately periodic complete macros the infinite repetition is certified by a finite formula and one witness tuple, not by an infinite sequence of finite checks, and the limit time is a rational formula; the direction stays open for executions without a supplied periodic macro and for any continuation beyond the accumulation.]

183 The dimension of the outgoing collision section

Immediately after a complete batch, list the N outgoing signals by increasing position. Within each co-located group, order the signals by strictly increasing speed. Let g_i be the gap from signal i to signal $i + 1$, for $1 \leq i < N$, and let Z be the indices of the zero gaps inside the co-located groups. All other gaps are strictly positive.

An outgoing collision section has at least one co-located pair, so $|Z| \geq 1$. Remove the overall translation and set

$$d = N - 1 - |Z|.$$

Let $L : \mathbb{R}^d \rightarrow \mathbb{R}^{N-1}$ insert zeros at the indices in Z , and let R select the complementary coordinates. Then $RL = I$, and every section configuration has the form $g = Lx$ with $x > 0$ componentwise. Consequently

$$d \leq N - 2 \leq 2 \quad (N \leq 4). \quad (183.1)$$

Two simultaneous disjoint binary collisions at population four have two zero gaps and $d = 1$. A triple collision also has $d = 1$, and a four-signal collision has $d = 0$. At $d = 0$, homogeneity forces every proposed flight time to be zero; hence a nonempty repeated macro with positive flight times is impossible.

This reduction is anchored *immediately after a collision*. Retaining an arbitrary later observation phase would add a phase coordinate and invalidate (183.1). Sorting each co-located output group by speed is also essential: these groups then separate immediately into the stated order. An arbitrary within-site order would not represent the actual outgoing mode.

184 A complete one-macro chronological chamber

In a fixed outgoing mode, let v_i be signal i 's speed and put $c_i = v_i - v_{i+1}$. During a collision-free interval of duration τ ,

$$g'_i = g_i - c_i \tau. \quad (184.1)$$

Only an adjacent pair with $c_i > 0$ can cause the next collision. Such a closing pair has $g_i > 0$, since an outgoing co-located pair has the opposite strict speed order. Its candidate time is g_i/c_i . If no adjacent pair is closing, a proposed next batch is impossible.

Let S be the specified incoming zero-gap indices at the next batch, and choose a pivot $j \in S$. Require $c_j > 0$ and define $\tau = g_j/c_j$. Require $\tau > 0$. For every other closing index i , impose

$$\frac{g_i}{c_i} - \tau \begin{cases} = 0, & i \in S, \\ > 0, & i \notin S. \end{cases} \quad (184.2)$$

Reject structurally if an index of S is not closing. Add all specified zero and strict-positive gap conditions in the initial and resulting outgoing sections. Every maximal consecutive block in S is one collision site: check its entire incoming label set and rule, replace all sites simultaneously, and sort each output block by increasing speed. Number preservation keeps the number of gaps fixed. Within each output block the gaps are zero; outside those blocks the ordered inter-site gap indices retain their meaning.

Lemma 184.1 (One-step exactness). *The structural conditions, section conditions, positive flight time, and candidate comparisons above are necessary and sufficient for the indicated complete next batch.*

Proof. Necessity is first-collision semantics. Conversely, every adjacent gap remains nonnegative until the pivot time. A nonadjacent crossing would require a prior adjacent crossing. The strict comparisons exclude an earlier competitor and exclude any omitted collision site at the pivot time. The tied indices include every zero gap of every collision block, hence every incoming signal in a multiple collision. Strict gaps between the maximal blocks keep distinct purported sites distinct. The stated rules replace these blocks at once; distinct, speed-sorted outputs establish the next outgoing section. No collision is hidden by the representation. $\square$

Each new gap, position increment, and flight time is a homogeneous rational linear form in the starting gaps. Symbolically compose these fixed finite linear operations over the m batches of W . No physical trajectory has to be executed to perform this construction. It gives rational matrices E, G, T and rows ℓ, r such that

$$\begin{aligned} \text{one macro is legal} &\iff Ex = 0 \text{ and } Gx > 0, \\ \text{final full gaps} &= TLx, \\ \text{macro elapsed time} &= \ell x, \\ \text{leftmost position increment} &= rx. \end{aligned} \quad (184.3)$$

The equality matrix includes the final zero-gap conditions of the initial section. The strict matrix includes every positive flight time and every required strict gap. A mismatch of final labels or co-location pattern is structural rejection.

[write] [Lemma 184.1 and the chamber (184.3) are the one-macro case of Part I's Theorem 5.3, whose typed endpoint conditions (Lemma 5.2) play the role of the candidate comparisons (184.2), specialized to number-preserving rules and to a word that ends on its own initial section. Part II proves the same step for its normalized schemas (Lemma 21.2), and Part III's Lemma 34.1 is another one-batch case. Source 26 did not read those Parts (Section 193); its proof is printed as its own account and claims no novelty for the chamber (Section 193). New in source 26 are the return matrix of the closed word and the all-repetition criterion that follows.]

Set $M = RTL$. On the chamber, $TLx = Lmx$. The condition need not hold at every point of the ambient vector space. Keeping redundant rows is harmless. A direct construction uses at most

$$(2m + 1)(N - 1) + m \leq 7m + 3 \quad (184.4)$$

zero/strict rows: $(m + 1)(N - 1)$ section conditions, at most $m(N - 1)$ candidate comparisons, and m positive-time conditions. This is a loose row bound, not an optimized arithmetic-operation count.

Proposition 184.2 (Exact infinite criterion). *Every finite prefix of W^ω is the unique actual complete history if and only if*

$$EM^n x = 0, \quad GM^n x > 0 \quad \text{for every } n \geq 0. \quad (184.5)$$

Proof. An actual history satisfies the one-macro conditions at every repetition, giving necessity. For sufficiency, apply Lemma 184.1 through the first macro. Its final section conditions give $TLx = Lmx$ with the prescribed labels and co-locations. Apply the same argument to Mx , and continue by induction. The final section equalities at each stage justify each next invocation. This establishes every finite prefix without assuming that M preserves the entire physical section off the legal orbit. $\square$

185 All cases of the order two sign problem

Pad dimension one to two with a zero coordinate and a zero row and column of M . This leaves every physical scalar observation unchanged. For a row a , Cayley–Hamilton implies

$$aM^n x = 0 \text{ for all } n \geq 0 \iff ax = 0 \text{ and } aMx = 0. \quad (185.1)$$

One equality suffices in dimension one, but the padding causes no difficulty.

185.1 The nonreal branch must be rejected before squaring

Let $\delta_M = (\text{tr } M)^2 - 4 \det M$. If $\delta_M < 0$, a real scalar observation has the form

$$aM^n x = \rho^n (A \cos(n\theta) + B \sin(n\theta)), \quad 0 < \theta < \pi.$$

No nonzero such sequence remains nonnegative. If $\theta/(2\pi)$ is rational, the normalized values over a full period sum to zero, so nonnegativity would force them all to be zero. If it is irrational, the powers of $e^{i\theta}$ are dense in the unit circle; a nonzero real linear functional is negative on an open arc, reached by that orbit. The elementary density argument follows by applying the pigeonhole principle to fractional parts to obtain arbitrarily small nonzero rotation steps.

Thus weak nonnegativity holds only for the zero observation, and strict positivity never holds. A nonempty legal macro has a strictly positive flight-time guard. Therefore (184.5) is empty in this branch. This test precedes squaring: a pure-imaginary conjugate pair squares to a *negative* scalar matrix and must not be mistaken for nonnegative real spectrum.

185.2 Real spectrum and the parity reduction

Assume $\delta_M \geq 0$ and put $B = M^2$. Its eigenvalues satisfy $\alpha \geq \beta \geq 0$. For each guard row a and parity $\varepsilon \in \{0, 1\}$, set

$$A_0 = aM^\varepsilon x, \quad A_1 = aM^{\varepsilon+2} x.$$

The parity subsequence $v_n = aM^{2n+\varepsilon}x$ has these initial values and satisfies the characteristic recurrence of B . Write

$$t = \operatorname{tr} B, \quad h = \det B, \quad \Delta = t^2 - 4h.$$

The following exhaustive list supplies exact infinite tests.

Proposition 185.1 (Nonnegative-root recurrence tests). *The sequence v_n is strictly positive for every $n \geq 0$ precisely under the conditions below:*

1. If $\alpha > \beta > 0$, then $A_0 > 0$ and $A_1 - \beta A_0 \geq 0$.
2. If $\alpha > \beta = 0$, then $A_0 > 0$ and $A_1 > 0$.
3. If $\alpha = \beta = \lambda > 0$, then $A_0 > 0$ and $A_1 - \lambda A_0 \geq 0$.
4. If $\alpha = \beta = 0$, strict positivity is impossible.

For weak nonnegativity, replace the first $A_0 > 0$ by $A_0 \geq 0$ in cases 1 and 3. In cases 2 and 4 the condition is $A_0 \geq 0$ and $A_1 \geq 0$.

Proof. In case 1, put $D = (A_1 - \beta A_0)/(\alpha - \beta)$. Then

$$\frac{v_n}{\alpha^n} = D + (A_0 - D)(\beta/\alpha)^n. \quad (185.2)$$

This is a convex interpolation between the initial value A_0 and the limiting value D . At each finite n , the coefficient of A_0 is positive. Hence $D = 0$ is allowed even for strict positivity. Conversely the initial sign and limiting sign are necessary.

In case 2, the first term is A_0 , and all later terms are $A_1\alpha^{n-1}$; the strict tail inequality cannot be weakened. In case 3, Cayley–Hamilton gives the Jordan formula

$$v_n = \lambda^n \left[A_0 + n \left(\frac{A_1}{\lambda} - A_0 \right) \right]. \quad (185.3)$$

The initial value must have the stated sign and the slope must be nonnegative. This includes the scalar case with zero slope and makes no diagonalizability assumption. In case 4, $B^2 = 0$, so the terms from index two onward vanish. (For $B = M^2$ here, $M^2 = 0$ already.) The weak conditions follow from the same formulas. $\square$

Apply Proposition 185.1 to both parities. This covers positive and negative eigenvalues of M , distinct opposite eigenvalues of equal modulus, repeated negative eigenvalues with a Jordan block, and zero roots. For example, when the eigenvalues of M are $\lambda, -\lambda$, B is scalar but *both* parity initial values must still be positive. A nonzero alternating observation fails.

185.3 Removing the only square root without raising the degree

Only case 1 may have a nonrational coefficient. Define

$$C = 2A_1 - tA_0, \quad H = \Delta A_0^2 - C^2. \quad (185.4)$$

Under $A_0 \geq 0$ and $\Delta > 0$,

$$A_1 - \beta A_0 \geq 0 \iff C \geq 0 \vee (C < 0 \wedge H \geq 0). \quad (185.5)$$

Indeed, twice the expression on the left is $C + A_0\sqrt{\Delta}$. If $C \geq 0$ it is nonnegative. Otherwise compare two nonnegative quantities before squaring. This also proves the equivalence at $A_0 = 0$. In the repeated-root case $\lambda = t/2$ is rational. Every fixed rational denominator may be cleared by a positive integer.

Consequently all infinite strict or weak guards are Boolean formulas in integer linear or quadratic polynomials in x . One guard requires at most six distinct polynomials, namely A_0, C, H for each parity; the other branches need fewer. An infinite equality requires at most two linear polynomials by (185.1). This proves the sign-formula part of Theorem 182.2. It is an explicit order-two calculation, not a proposed finite spectral test for arbitrary-order linear recurrences.

186 A literal quartic with exactly one natural witness

Let $\Phi(p)$ be a Boolean formula of sign tests of distinct integer polynomials $Q_1(p), \dots, Q_A(p)$, each of degree at most two. Here p denotes the chosen integer inputs; common positive denominator scaling preserves the homogeneous validity signs.

For each Q , introduce natural witnesses e_-, e_0, e_+, z and the following six residuals:

$$\begin{aligned} e_-(e_- - 1), \quad e_0(e_0 - 1), \quad e_+(e_+ - 1), \\ e_- + e_0 + e_+ - 1, \\ Q - (e_+ - e_-)(z + 1), \quad e_0 z. \end{aligned} \tag{186.1}$$

Every residual has total degree at most two, including when Q is quadratic.

Lemma 186.1 (Canonical natural sign fiber). *For every encoded input p , the simultaneous zero set of (186.1) in natural witnesses consists of exactly one tuple. Its three flags indicate the sign of $Q(p)$; if $Q(p) \neq 0$, then $z = |Q(p)| - 1$, while if $Q(p) = 0$, then $e_0 = 1$ and $z = 0$.*

Proof. The first three residuals force Boolean flags. Their one-hot sum leaves one flag equal to one. The sign equation forces the selected sign and, when nonzero, the displayed magnitude. At zero, the final equation $e_0 z = 0$ eliminates the otherwise free slack. The displayed choices satisfy all six equations, proving existence and uniqueness. $\square$

Read $Q > 0$ as e_+ , $Q = 0$ as e_0 , and $Q \geq 0$ as the Boolean negation of e_- . Translate the formula to an acyclic AND/OR/NOT circuit, topologically ordered. For each of its Γ gates introduce one natural output b and one defining residual:

$$\begin{aligned} \text{AND: } b - uv, \quad \text{OR: } b - u - v + uv, \\ \text{NOT: } b - 1 + u. \end{aligned} \tag{186.2}$$

The inputs of the first gates are uniquely Boolean by Lemma 186.1. Inductively every gate output is also uniquely Boolean, so separate gate-Boolean equations are unnecessary. Add the output residual $b_{\text{out}} - 1$ and take the sum of the squares of all residuals.

Theorem 186.2 (Explicit arithmetic ledger). *This construction yields an ordinary integer polynomial $P_\Phi(p, w)$ of degree at most four. For every encoded input, $\Phi(p)$ is true if and only if $P_\Phi(p, w) = 0$ for a natural witness tuple w , and that tuple is unique when it exists. Before optional constant-output simplifications, the ledger is*

$$\#w = 4A + \Gamma, \quad \#\text{residuals} = 6A + \Gamma + 1. \tag{186.3}$$

Proof. A sum of integer squares vanishes exactly when all its residuals vanish. Every residual has degree at most two, hence the sum has degree at most four. The sign witnesses and gate outputs are uniquely determined, and the final residual accepts exactly the true formula. Counting four witnesses and six residuals per atom, one of each per gate, and the final residual gives (186.3). A constant-false formula can instead be represented by 1, and constant true by 0, with an empty witness tuple. $\square$

[write] [The source and the programme’s review write the gate count Γ as J : “ $4A + J$ witnesses and $6A + J + 1$ residuals for A sign atoms/ J logic gates”. It is not the contact count J of source 29 or the witness J_{acc} of sources 28 and 29.]

For S infinite strict/weak rows and E_0 infinite equality rows, the preceding sign construction gives $A \leq 6S + 2E_0$ before clock/deadline predicates are added. This is an explicit arity accounting, not an optimal count. Shifting each natural witness by one provides a strictly positive witness convention without changing degree or uniqueness.

No quadratic guard is multiplied by a selector: doing so could create a cubic residual. The sign equation in (186.1) is what preserves degree two. No MRDP theorem, Pell equation, exponentiation atom, square-root witness, or unbounded history code is used. “Exactly one” means per supplied input encoding. Two numerator/denominator encodings of the same rational geometry are distinct inputs; no canonical rational representation is asserted. This completes Theorem 182.2.

[write] [The one-hot sign gadget (186.1) with a unique slack is the same device as Part I’s unique-slack quadratic (Theorem 6.1) and Part IV’s canonical quartics with a unique slack for each inequality (Section 46), applied here to quadratic sign atoms with a one-hot sign selector; it is printed as the source’s own construction, and the source claims no novelty for its Boolean arithmetic (Section 193).]

187 Summability is a property of the observable clock

Under (184.5), the macro durations

$$u_n = \ell M^n x$$

are strictly positive, being sums of strictly positive flight times. Total elapsed time is $\sum_{n \geq 0} u_n$. Test its even and odd subsequences using the same $(A_0, A_1, \alpha, \beta)$ notation as Section 185, now with $a = \ell$.

Proposition 187.1 (Exact clock classification). *For a valid strictly positive parity subsequence, summability is determined as follows.*

1. If $\alpha > \beta > 0$ and $\alpha < 1$, it is summable.
2. If $\alpha \geq 1 > \beta > 0$, it is summable exactly when $A_1 - \beta A_0 = 0$.
3. If $\alpha > \beta \geq 1$, it is not summable.
4. If $\alpha > \beta = 0$, it is summable exactly when $\alpha < 1$.
5. If $\alpha = \beta = \lambda > 0$, it is summable exactly when $\lambda < 1$.

The nilpotent case has no valid strictly positive infinite clock. The full clock is summable exactly when both parity subsequences are summable.

Proof. In the distinct positive-root case use (185.2). Both modes decay when $\alpha < 1$. If $\alpha \geq 1 > \beta$, summability requires and is implied by vanishing of the dominant coefficient D . If both roots are at least one, a strictly positive sequence cannot be summable, including the case $D = 0$. When $\beta = 0$, strict validity forces $A_1 > 0$; its geometric tail converges exactly when $\alpha < 1$. For a repeated positive root, (185.3) has a positive initial value and nonnegative slope. The geometric factor, with its possible linear factor n , is summable exactly when $\lambda < 1$. $\square$

In the cancellation branch, $\Delta > 0$ and $A_0 > 0$, and

$$A_1 - \beta A_0 = 0 \iff C < 0 \wedge H = 0. \quad (187.1)$$

This is the exact equality version of the sign-separated squaring in (185.5). Comparisons of the fixed algebraic constants α, β with one are decidable exactly while compiling. They do not introduce algebraic coefficients into the emitted integer polynomial.

Conjoin validity with this Zeno formula, or with its negation for non-Zeno behavior, and apply Theorem 186.2. The observable $\ell M^n x$ may lose a neutral or unstable whole-matrix mode through exact cancellation along the input. Therefore $\rho(M) < 1$ is not a necessary condition for a summable clock. Conversely a decaying visible component does not itself certify legal collisions: the full strict guard system remains essential.

188 Rational clock values and exact deadline queries

For rational input $x = p/q$, define numerator observations

$$a_e = \ell p, \quad b_e = \ell M^2 p, \quad a_o = \ell M p, \quad b_o = \ell M^3 p.$$

Retain $t = \text{tr}(M^2)$ and $h = \det(M^2)$. If the valid clock converges and $1 - t + h \neq 0$, its sum is

$$T = \frac{b_e + b_o + (1 - t)(a_e + a_o)}{q(1 - t + h)}. \quad (188.1)$$

To obtain this formula for one parity recurrence, sum its characteristic recurrence, or evaluate its generating function

$$\frac{A_0 + (A_1 - tA_0)z}{1 - tz + hz^2}$$

at $z = 1$ after any cancellations. Summing the two parity expressions gives (188.1). A denominator of either sign is allowed in this intermediate formula, including when an unstable mode has been canceled.

If $1 - t + h = 0$ and a valid convergent clock exists, M^2 has a simple root one and another root $\beta < 1$. Both parity coefficients of the root-one mode vanish. The sum is then

$$T = \frac{a_e + a_o}{q(2 - t)}. \quad (188.2)$$

A repeated root one cannot support a strictly positive summable clock. Formula (188.2) is the remaining geometric sum, since $\beta = t - 1$. These cases cover all valid convergent clocks and establish their rationality and effective evaluation in dimension at most two.

Clear the fixed rational coefficients and normalize the denominator sign to write $T = L(p)/(Dq)$ with L an integer linear form and $D \in \mathbb{Z}_{>0}$ fixed. For a supplied deadline H/K , where $K > 0$,

$$T = H/K \iff DqH - KL(p) = 0, \quad (188.3)$$

$$T < H/K \iff DqH - KL(p) > 0. \quad (188.4)$$

These polynomials have total degree at most two in the integer inputs p, q, H, K . Conditions $q > 0$ and $K > 0$ may be encoded with the same sign compiler rather than promised as domains. Signed deadline numerators may be handled as signed integer inputs or by a separately specified natural encoding.

The stated degree accounting fixes the section time to zero. An initial rational time offset can be added by ordinary cross-multiplication, but if its numerator and denominator are new variables the degree ledger must be reconsidered. No uncounted variable-denominator offset is included here. Together with Section 187, this completes Theorem 182.3.

189 A spectator makes the limiting boundary visible

Consider five meta-signals with speeds

$$L : +1, \quad R : -1, \quad P : +3, \quad Q : -3, \quad S : -2.$$

Use the two nontransparent collision rules

$$\{P, R\} \longrightarrow \{Q, R\}, \quad \{L, Q\} \longrightarrow \{L, P\}. \quad (189.1)$$

Every other allowed incoming set may use its transparent rule $C \rightarrow C$. Each rule preserves population and has distinct incoming/outgoing speeds. The example has four simultaneously live signals and five speed values across its meta-signals; these are different counts.

Immediately after a left bounce, put L, P at zero, R at d , and S at $d + y$, with $d, y > 0$. The outgoing order is L, P, R, S , and the full gap vector is $(0, d, y)$. The proposed macro is one P – R bounce followed by one Q – L bounce.

[write] [Without the spectator S this is the shrinking-wall shuttle of Part I's clock (rules (12.1), speeds $0, 2, -1, -2$) and of Part II's bounded shuttle (speeds $3, 2, 4, 0$), here with walls of speeds ± 1 ; Section 1.2 compares the two. The source credits the mechanism to the literature [6]; the spectator and the weak limiting boundary are what the example adds.]

189.1 The exact complete two-bounce chamber

The first bounce occurs after $d/4$, provided $y > d/4$: the competing R – S candidate is at time y . The outgoing order after that bounce is L, Q, R, S with gaps

$$(d/2, 0, y - d/4).$$

The second bounce takes a further $d/8$, provided $y - d/4 > d/8$. Combining the conditions gives exactly

$$d > 0, \quad y > 3d/8. \quad (189.2)$$

Throughout a legal prefix the spectator stays to the right of R , while the shuttle remains between L and R until each bounce. Thus no spectator–shuttle or spectator–left-wall collision precedes the

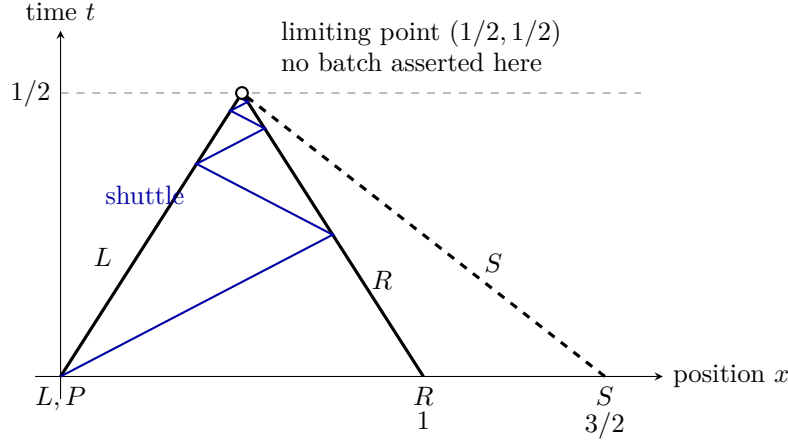


Figure 11: (Source 26.) Analytic spacetime diagram of the included limiting boundary $d = 1$, $y = 1/2$. The walls are $x_L = t$ and $x_R = 1 - t$; the spectator is $x_S = 3/2 - 2t$. The blue shuttle joins the exact bounce points from (189.4); its final short segment denotes convergence of the displayed sequence. The open circle marks a limit, not a prescribed collision or post-accumulation continuation. This drawing uses derived formulas, not a physical simulation.

R – S candidate; these are complete chronology checks, not just a selection of desired bounces. At equality in (189.2), an additional remote spectator collision belongs to the endpoint batch, so the proposed batch is incomplete and must be rejected.

The return map, time, and leftmost displacement are

$$M\begin{pmatrix} d \\ y \end{pmatrix} = \begin{pmatrix} d/4 \\ y - 3d/8 \end{pmatrix}, \quad \ell(d, y) = 3d/8, \quad r(d, y) = 3d/8. \quad (189.3)$$

After k legal cycles, direct induction yields

$$d_k = \frac{d}{4^k}, \quad y_k = y - \frac{d}{2}(1 - 4^{-k}), \quad t_k = \frac{d}{2}(1 - 4^{-k}). \quad (189.4)$$

For $K \geq 1$, the first K complete strict cycles are valid exactly when

$$y > \frac{d}{2}(1 - 4^{-K}). \quad (189.5)$$

Equality gives an omitted spectator collision in that finite endpoint batch and is invalid. Intersecting (189.5) over every finite K gives the different boundary condition

$$d > 0, \quad y \geq d/2. \quad (189.6)$$

At $y = d/2$ every finite prescribed batch is legal. The spectator reaches the common limiting position only at the accumulation time, which is outside the finite-prefix semantics.

All three core signals tend to $(x, t) = (d/2, d/2)$. The spectator has limiting position y , equal to $d/2$ exactly on the boundary. Figure 11 illustrates that included boundary without supplying any dynamics beyond it.

189.2 Two failures of tempting shortcuts

The full return matrix and clock row are

$$M = \begin{pmatrix} 1/4 & 0 \\ -3/8 & 1 \end{pmatrix}, \quad \ell = \begin{pmatrix} 3/8 & 0 \end{pmatrix}.$$

Thus $\rho(M) = 1$, but every input in (189.6) is Zeno, with

$$T = \sum_{n \geq 0} \frac{3d}{8} 4^{-n} = d/2.$$

The neutral spectator-gap coordinate is invisible to the clock. This is a signal-realizable counterexample to “Zeno if and only if the whole return matrix has spectral radius below one.”

There is also no finite repetition horizon depending only on this fixed machine and macro that certifies infinite validity for every rational or integer seed. For any $K \geq 0$, take

$$d = 4^{K+1}, \quad y = d/2 - 1.$$

For $K \geq 1$, the threshold for K cycles is $d/2 - 2$, which is smaller than y . The next threshold is $d/2 - 1/2$, which is larger than y . Thus the first K cycles succeed and the next fails by a genuine extra R - S collision. For $K = 0$, the empty prefix is valid and the first proposed bounce has a competing tie.

For positive integer d, y , the infinite predicate itself has the particularly small unique-witness quadratic

$$(2y - d - z)^2 = 0, \quad z \in \mathbb{N}. \quad (189.7)$$

This example does not replace the general quartic theorem. The exported quartic deliberately passes the same predicate through the general sign compiler, making the ledger and canonical-zero branch inspectable. The shrinking-wall shuttle is classical [6]; the added spectator isolates the weak boundary and observable-clock warning used here.

190 Rational accumulation limits in every finite dimension

We now retain any fixed finite population, assuming that the repeated complete macro is valid. Its rational gap return matrix M may have dimension greater than two. Let x be rational and $f_n = \ell M^n x$ be its positive macro durations. The generating function is

$$F(z) = \sum_{n \geq 0} f_n z^n = \ell(I - zM)^{-1} x \in \mathbb{Q}(z) \quad (190.1)$$

as a formal rational generating function, and as an analytic identity near zero.

If $\sum f_n$ converges, Abel’s theorem gives

$$\lim_{z \rightarrow 1^-} F(z) = \sum_{n \geq 0} f_n.$$

The rational function agrees with the convergent series on the interval where the latter is defined, with removable cancellations understood. A rational function over $\mathbb{Q}$ having a finite limit at the rational point one has a rational limit: cancel common powers of $z - 1$ and evaluate the remaining

quotient. This value is effectively computable by polynomial arithmetic. The accumulation time is therefore rational, conditional on validity and convergence.

For positions, number preservation lets the k th *spatially ordered* signal position $X_k(t)$ be followed continuously through every collision. All replaced signals at a site have the same position, and they are replaced by the same number of outgoing signals at that position. Let $V = \max |\text{speed}|$. Each ordered position is V -Lipschitz on every finite prefix. If the accumulation time T is finite, it follows that every $X_k(t)$ has a limit as $t \rightarrow T^-$; the gaps have limits too.

The displacement of the leftmost position during macro n is $rM^n x$, with

$$|rM^n x| \leq V f_n.$$

Its series converges absolutely. The same rational-generating-function argument makes its total displacement rational. For a convergent component sequence h_n of $M^n x$, Abel summation gives

$$\lim_{n \rightarrow \infty} h_n = \lim_{z \rightarrow 1^-} (1 - z) \sum_{n \geq 0} h_n z^n. \quad (190.2)$$

The expression on the right is again a rational function over $\mathbb{Q}$ with a finite value at one. Hence all limiting gaps are rational, and adding the rational leftmost limit makes every limiting ordered position rational.

A fixed named collision site at a specified phase of the macro differs from an appropriate macro-boundary position by at most V times that macro's duration. These durations tend to zero in a finite-time accumulation. Its limit is therefore one of the same rational ordered-position limits. A finite rational prefix before the repeated part contributes only rational event times and positions, so an ultimately periodic macro has the same conclusion. This proves Theorem 182.4.

General signal-machine accumulations can have more complicated coordinates [6]. The theorem is an obstruction to their description by the present exact finite periodic macros, not a contradiction of that broader literature. It supplies no arbitrary-dimensional recognition algorithm.

191 Implemented objects and exact evidence

The mathematical construction above has two distinct implementation layers. The one-macro chamber frontend is explicit in Section 184, but no general parser from an arbitrary machine and macro was implemented in this work. The new implemented emitter starts with an integer sign formula of degree at most two. It builds sparse polynomial residuals, expands their sum of squares, and writes complete coefficient data for two concrete certificates.

Export	Atoms	Gates	Witnesses	Residuals	Monomials
Four-signal predicate	2	2	10	15	65
Quadratic-sign diagnostic	3	5	17	24	117

Both expanded polynomials have exact degree four. The first recognizes $d > 0$ and $2y \geq d$ on natural inputs. The second exercises AND, OR, and NOT using the nonsquare-spectrum recurrence matrix

$$A = \begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix}.$$

It recognizes strict positivity of the first coordinate of $A^n(x, y)$ for every $n \in \mathbb{N}$, on natural x, y . This is a recurrence diagnostic. No physical complete collision macro realizing this particular matrix is asserted.

The source emitter receipt records 8 774 comparisons of the expanded polynomial with its residual sum, including 7 936 rejected one-coordinate mutations of canonical witnesses. These finite regressions support the implementation; they do not prove the infinite equivalences or uniqueness on untested inputs. Those are proved in Sections 185–186.

The author-side exact arithmetic receipt includes 7 938 rational diagonal-mode comparisons, 882 Jordan/nilpotent comparisons, 45 000 integer-matrix prefix probes, 15 147 sign-gadget assignments, 6 912 return-map identities, 64 strict-endpoint/late-failure checks, 66 clock-sum identities, and 12 Boolean-gate cases. The matrix probes used a finite prefix; their observed agreement is not a finite-cutoff theorem.

An independent mathematical audit read the entire proof and found no substantive correction required. Its separately authored exact arithmetic checks cover strict and weak recurrence branches, radical signs, canonical witnesses, cancellation, clock sums, and spectator identities. One deliberately late failure has first nonpositive term at index 6 932, underscoring the distinction between a proof of infinite validity and a short experimental prefix. The proof audit is conventional mathematical review, not proof-assistant verification or journal peer review.

The frozen source proof is preserved byte-for-byte in the package. Its SHA-256 is

`df7cefb472a76e6f34ce7fff0b1e8548782a82f4863fac2ecd2d682c69941a77`

A separate independent artifact audit reconstructed every sign gadget, gate residual, and atom polynomial, then independently expanded both sums of squares. The complete coefficient dictionaries agree exactly with both exported ordinary polynomials. This is an all-assignment polynomial identity, stronger than finite evaluation agreement. The complete sign truth tables contain 9 and 27 cases. Further regressions cover 882 natural input pairs and 21 188 one-coordinate witness mutations. The canonical-fiber proof therefore transfers to these literal exports through exact residual and coefficient identities. The audit neither imported nor executed the author emitter; it treated that source as inert and the exports as data. It did not invoke an upstream signal-machine program or simulate collisions.

[write] [Shipped names: the frozen proof is `26-periodic-science-PROOF.md` (the hash above is its SHA-256), its notes `26-periodic-science-SOURCE_NOTES.md`; the emitter and the author checker are `code/26-periodic-science-emit_sign_certificate.py` and `code/26-periodic-science-check_arithmetic.py`, with the two exports and their receipt as `data/26-periodic-science-exports-four_signal_quartic.json`, `data/26-periodic-science-exports-quadratic_sign_quartic.json` and `data/26-periodic-science-exports-sign_compiler_receipt.json` and the arithmetic receipt as `data/26-periodic-science-arithmetic_receipt.json`; the independent proof audit is `26-periodic-independent_audit-AUDIT.md` with `code/26-periodic-independent_audit-check_exact_algebra.py`, and the artifact audit is `26-periodic-independent_audit-emitted-artifact-audit-AUDIT-ADDENDUM.md` with `code/26-periodic-independent_audit-emitted-artifact-audit-check_exported_quartics.py`. On copies, all four stages reproduce their frozen outputs up to line endings; the README gives the route.]

192 Reproducibility and execution boundary

The release separates the editable article, frozen science, frozen independent audits, and newly authored release/replay tools. Exact path and hash inventories bind the artifacts to their evidence. The source and audit packets remain inert inputs except for explicitly inspected new arithmetic-only checker programs named by the replay contract.

The release utility verifies the exact file inventory, bytes, and POSIX modes against a manifest whose trusted SHA-256 is supplied separately. A PDF rebuild uses a fresh external directory, deterministic date settings, and `pdflatex` with shell escape disabled. Archive creation uses sorted members and fixed timestamps. Build and archive operations check preservation of the release inputs. The reproducibility instructions and machine-readable receipts state the tool versions, byte identities, and output requirements actually checked.

The arithmetic replay also uses a fresh external output location. It validates the frozen input pins before executing any permitted checker, copies the needed data into a separate working area, and compares regenerated evidence with the frozen receipts or exports. The assertion-based checks must run without Python optimization; removing assertions is not an independent successful check. The replay tools reject unsafe path overlap, reused outputs, and symlink-based output redirection, and their adversarial tests are included.

No upstream program, saved computation schedule, downloaded executable, physical simulator, or post-accumulation dynamics is executed. Evaluating a newly authored rational recurrence, expanding a polynomial, and checking the displayed return-map identities are arithmetic checks of the proof objects. They are not a simulation of the signal machine. The analytic diagram is drawn directly from its stated equations.

What successful replay establishes. It establishes agreement of the pinned finite exact arithmetic artifacts and the independently checked exports, preservation of the supplied sources, and reproducibility of the packaged build under the recorded environment. It does not convert finite regression tests into an infinite proof, remove the fixed-word hypothesis, or implement the missing general chamber frontend.

[write] [In this report the release’s manifest (`MANIFEST.json`) and its `SHA256SUMS` files are not shipped (they were verified at placement and survive in the arrival commit), the manuscript, PDF and page renders are not shipped, and the release and replay tools (`code/26-periodic-release.py`, `code/26-periodic-audit_replay-replay_audit.py`) check the delivered layout and POSIX modes, so they do not run on the shipped names. The README gives a rerun route on a copy of the archive.]

193 Prior work and remaining boundaries

The repository comparison is pinned to commit

`e3e58a8820593857650bb7a0193a11ec32db67d6`

The signal-machine volume’s README and its independent finite-schema review describe rational linear chambers for finite complete schemas, including simultaneous-event chronology, and explicitly exclude accumulation semantics [74, 14]. The reviewed source notes emphasize that a physical-time bound is not an event bound. The present report re-proves the needed conservative one-macro chamber and adds an exact all-repetition criterion for the supplied word.

The complete merged repository article exceeded the connector’s readable-file limit and was not obtained. No theorem in this report depends on an unread result from that article. The packet retains the actual inert text sources used for comparison, with separate byte hashes; these are saved-text hashes, not claimed Git blob identifiers. Targeted searches for “periodic collision” and “collision word” found no matches. This is a bounded overlap check, not an exhaustive novelty result. Neighboring canonical-certificate, liveness, continuous-computation, and stochastic/thermal descriptions were considered as existing work.

[write] [The unread article is this report. It contains the chamber theorem that source 26 re-proves (Part I’s Theorem 5.3; note after (184.3)), the accumulation direction that Theorems 182.2–182.4 answer in part (Section 15.9), and two shuttles that are instances of Theorem 182.4 (note after it). The statement that no theorem depends on it remains true. The two searches had no match in the collection at the pin, and the phrases now occur in this Part. The README and review copies that the packet keeps are not byte copies of the repository files (each has one more trailing line feed); they are not shipped, and the files themselves are unchanged since the pin.]

Becker, Chapelle, Durand-Lose, Levorato, and Senot give the standard distinct-speed convention and classical shrinking-wall shuttle [6]. Four speeds in that literature must not be confused with four live signals here. Ouaknine and Worrell study low-order positivity, including the distinction between strict and non-strict versions [75]. The elementary order-two formulas used here are proved in full; deciding order-two recurrence signs is not claimed as new.

The proposed contribution is the explicit composition of collision-anchored dimension reduction, complete chronological guards, all-cases order-two positivity, deterministic natural sign witnesses, and observable-clock summability. The finite-schema chamber, elementary recurrence facts, Boolean arithmetic, rational generating functions, and contracting shuttle are not individually claimed new. No priority claim is made for the composition either.

The main unfinished engineering task is a general audited parser/emitter from a finite machine-and-macro description to its chamber rows and final quartic DAG. Other natural questions include which larger-population macro classes admit comparable low-degree formulas, and how additional geometric restrictions might control arity. No maximal decidable population, optimal arity, constant operation budget, faithful reduction from arbitrary Positivity instances, or arbitrary-run liveness theorem is established. The result remains a certificate class for supplied periodic collision words and their finite pre-accumulation prefixes.

[write] [Of the questions in the last paragraph, the population one is answered by source 27: with five live signals no finite polynomial-sign formula exists for some supplied macro (Corollary 206.1), although an existential integer certificate still does (source 28). The parser/emitter and arity questions stay open.]

194 A compact compiler checklist (source 26, Appendix A)

For a fixed supplied machine, section, and word, the proof gives the following finite procedure.

1. Verify number preservation, distinct-speed collision sets, complete batches, and the exact return of ordered labels and outgoing co-locations. Reject structurally invalid data.
2. Remove translation and zero section gaps. If the remaining dimension is zero, reject any nonempty positive-flight macro. For population at most four, otherwise retain one or two gap coordinates.

3. Compose the finite symbolic gap updates, retaining every tie equality, strict competing-time comparison, section condition, and positive flight time. Produce E, G, M, ℓ, r .
4. Replace each infinite equality by its first two observations. Check δ_M before squaring. Reject the nonreal branch using its impossible positive clock guard.
5. For real spectrum, square M , split each strict guard into even and odd subsequences, and use all four cases of Proposition 185.1. Eliminate square roots using the sign-separated formula (185.5).
6. If requested, add both parity clock-summability tests, including exact dominant-mode cancellation. For valid Zeno inputs, select (188.1) or (188.2) and add the rational deadline predicates.
7. Clear fixed denominators positively; deduplicate sign polynomials. Introduce the six-residual canonical gadget for each atom and deterministic Boolean gates. Require the circuit output to be one and sum all residual squares.

Only the sign-formula-to-polynomial part of this checklist is implemented generically in the supplied emitter. Earlier steps are proved mathematical constructions, not claimed software features.

195 Boundary cases that a certificate must preserve (source 26, Appendix B)

Case	Required treatment
Simultaneous disjoint sites	Include all sites; strict comparisons exclude omitted ties
Multiple collision	Include the maximal zero block and every incident signal
Co-located output order	Distinct speeds sorted increasingly; no zero-duration cascade
One-dimensional section	Zero padding is harmless; no added phase coordinate
Nonreal spectrum	Reject before squaring using the impossible positive clock
Zero smaller root	Require $A_1 > 0$ for a strict positive tail
Repeated positive root	Allow zero Jordan slope; use the exact linear-in- n formula
Negative real roots	Test both parity subsequences
Vanishing dominant mode	Permit it if the smaller positive mode supplies all strict terms
Radical elimination	Check signs before squaring; retain $A_0 = 0$ for weak tests
Zero sign atom	Force the magnitude slack to zero as well as the zero flag
Clock eigenvalue one	Cancel it observably; use the exceptional clock sum
Spectator limiting tie	Include $y = d/2$ for finite prefixes; no batch at the limit
Equivalent rational encodings	Uniqueness holds separately for each encoded input

[write] **The programme’s intake of source 26.** The review [99] reads it as “A supplied lasso class, not arbitrary execution/nontermination. Arity is fixed per machine/word, not uniform over words. The general machine+macro→chamber→DAG frontend is not implemented. Rational accumulation conclusions are conditional on validity.” Each point is a limitation the source states itself; no correction is asked for.

Source 27 (batch-91 manuscript 09): Five signal nonsemialgebraic periodic validity

An explicit physical obstruction to finite polynomial sign classification, “Report57”, 4 October 2026. [write] From the archive `Five_Signal_Nonsemialgebraic_Periodic_Validity_Package.zip`; its labels carry the prefix `smc:fv:`, its sections are Sections 196–210 here (its two appendices, the 36 guards and the 138 rules, are the last two; the archive set them from two `\input` files that are not shipped, and the same data are shipped as `data/27-fivesignal-science-construction-GUA.RDS.txt` and `data/27-fivesignal-science-construction-RULES.json`), and its equations are numbered by section. “Report56” in it is source 26 (Part X); its other bibliography entries are this report’s [6], [76], [77], [78], [79] and [80]. Its valid set K and rotation R are written $\mathcal{K}$ and $\mathcal{R}$, and its Gaussian quotient $\alpha = A + iB$ and normalized coordinates (u, v) of Section 204 are written $\gamma = \gamma_1 + i\gamma_2$ and (w_1, w_2) (opening of Part X). Its abstract follows.

We give a finite rational-speed number-preserving signal machine with exactly five live signals and a supplied complete periodic macro of 138 binary collisions. The machine uses 169 meta-signal types and eight speeds. Its return on three positive gaps is rationally conjugate to a contraction by one half combined with the rotation having cosine $3/5$ and sine $4/5$. We prove the entire physical chronology, including every transparent crossing, competing collision, tie exclusion, outgoing order, and phase-label return. Thirty-six strict rational linear guards describe exactly one legal macro.

The scale-normalized set of infinitely valid inputs is the disk of squared radius $4/1845$ with precisely one oriented countable dense orbit removed from its boundary. It is nonsemialgebraic. More strongly, no finite Boolean formula in polynomial signs, even with real coefficients, agrees with the predicate on all rational inputs or on all positive integer gap triples. The integer claim uses an eventual homogeneous-ray argument; it does not assume that the proposed formula is homogeneous.

The rational-input predicate nevertheless has an elementary exact decision procedure. Every valid run of this machine is Zeno, with an explicitly computed rational linear accumulation time. A separately reclosed 114-event machine gives the same normalized validity set with non-Zeno executions. Combined with the separately audited at-most-four-signal theorem of Report56, the example establishes a sharp four-versus-five boundary for universal finite polynomial-sign classification of supplied complete periodic macros. It implies neither undecidability nor impossibility of existential-natural Diophantine certificates. Abstract linear-loop nonsemialgebraicity is prior work; no literature-priority or phase-, rule-, or speed-minimality claim is made.

Evidence. The underlying construction and all 138 local rules passed an independent conventional proof and inert-data audit. The portable packet includes frozen sources, independently authored checkers and receipts, all guards, and all rules. No physical trajectory, saved collision program, or author checker is executed by the release replay.

196 The exact theorem and its scope

We write $\mathbb{N} = \{0, 1, 2, \dots\}$. A signal machine has finitely many meta-signals, each with a fixed real velocity, and deterministic local collision rules. Every incoming and outgoing collision set has pairwise distinct velocities. Different meta-signals may share a velocity at distinct spatial positions. A number-preserving rule emits as many signals as it consumes. These are the standard conventions of [6].

Here all velocities are rational. Initial positions may be real for the geometric theorem; rational and integer initial-gap statements are proved separately. Some literature includes rational initial positions in the phrase “rational signal machine,” so we use “rational-speed machine” when that distinction matters. A live-signal count counts strands, including the two strands sharing a collision point, rather than occupied spatial points, speeds, or available labels.

Definition 196.1 (The supplied macro problem). Fix a machine, an outgoing collision section, and a nonempty finite word W of complete collision batches. Completeness means that the batch contains every simultaneous collision site and every incident signal at each site. Successive batches have strictly positive time separation. The final section must restore the initial ordered labels and co-location pattern. The variable input is the initial vector of positive inter-site gaps. Its validity means that every finite prefix of W^ω is the genuine complete physical history.

Our word has one binary site in each batch. Rules, speeds, and the entire word are fixed data. We do not search for a periodic itinerary, decide unrestricted machine behavior, or define a continuation through an accumulation.

Theorem 196.2 (Explicit physical obstruction). *There is a rational-speed number-preserving machine with five live signals, 169 meta-signals, and exactly eight speeds, with a complete 138-binary-event word W , whose return on positive gaps is*

$$M = \frac{1}{30} \begin{pmatrix} 7 & -2 & 10 \\ 14 & 11 & -10 \\ -6 & 6 & 15 \end{pmatrix}. \quad (196.1)$$

The exact one-macro chamber is the intersection of the 36 strict rational linear guards in Section 209. Put

$$D = g_1 + g_2 + g_3, \quad x = g_1, \quad y = g_1 + g_2, \quad \xi = x - D/3, \quad \eta = y - 2D/3. \quad (196.2)$$

For $D > 0$, the infinitely valid inputs are exactly those for which $w = (\xi/D, \eta/D)$ belongs to

$$\mathcal{K} = \{w : \|w\| < \rho\} \cup (\{w : \|w\| = \rho\} \setminus \{R^{-n}p : n \in \mathbb{N}\}), \quad (196.3)$$

where

$$R = \begin{pmatrix} 3/5 & -4/5 \\ 4/5 & 3/5 \end{pmatrix}, \quad \rho^2 = \frac{4}{1845}, \quad p = \left(\frac{4}{205}, -\frac{26}{615} \right). \quad (196.4)$$

The real validity cone is nonsemialgebraic. No finite polynomial-sign formula agrees with it on every positive rational gap triple, or even on every positive integer gap triple. These assertions allow arbitrary real coefficients in the alleged formula.

Theorem 196.3 (Arithmetic and time). *The rational-input validity predicate of Theorem 196.2 is decidable by a finite elementary exact-arithmetic procedure. Every valid input is Zeno, with accumulation position 0 and time, measured from the initial outgoing section,*

$$T(D, \xi, \eta) = \frac{74528}{855}D + \frac{53102}{3705}\xi - \frac{144302}{3705}\eta. \quad (196.5)$$

For rational initial data this time is rational. A separate 114-event variant has the same normalized chamber and set $\mathcal{K}$, but every valid infinite execution is non-Zeno.

The proofs occupy Sections 198–205. The real nonsemialgebraicity claim also excludes a finite description using quantifiers over real polynomial variables, by real quantifier elimination. Quantifiers over natural numbers are a different matter: neither that claim nor the integer sign obstruction rules out Diophantine representations. Section 206 states exactly how Report56 enters.

[write] [The validity set (196.3) has the shape of source 29's general formula (225.2) with one contact ($J = 1$), but Theorem 196.2 is not a literal instance of Theorem 225.1: for the same rotation and the scale $1/2$, source 29's machine has 174 events and $\rho^2 = 1/65$ (Section 235), and source 29 says so. Part XI's Report 60 [batch-91 manuscript 12] extends the realization and the classification to every orientation-preserving rational planar map. Source 28 (Theorem 211.1) gives the predicate of Theorem 196.2 an existential positive-integer certificate.]

197 Five strands and a collision section

At the outgoing section, place four stationary marker identities at

$$L = 0 < X = x < Y = y < D,$$

and a messenger at L with velocity $+1$. The stationary labels are $L0, X0, Y0, D0$, all with speed zero. The outgoing label order is

$$(L0, Q0, X0, Y0, D0),$$

with the first two labels co-located. The free positive gaps are $g = (x, y - x, D - y)$. Overall translation and the fixed zero gap have been removed. Throughout the macro the messenger has speed $+1$ or -1 . At most one marker moves at a time, with speed strictly between -1 and 1 , and it is restored to its stationary label before the current primitive ends.

The phrase “marker identity” tracks one strand as its phase label changes. The finite collection of temporary labels supplies a fixed controller for this particular finite word, not an unbounded program store. Every explicit collision consumes a messenger and one marker and emits a messenger and that same marker identity. Thus all executions of the supplied word have exactly five live signals.

For any binary collision at position q and time t , two incoming velocities $v_1 > v_2$ have positions $q - v_1\epsilon < q - v_2\epsilon$ just before it. Incoming local order is therefore decreasing velocity. Outgoing local order is increasing velocity. The construction respects both orders at every rule; the exhaustive local order receipt is provided with the packet. Since $|a| < 1$ for a moving marker and $|v_Q| = 1$, no output pair has equal speeds or remains co-located for a positive interval.

198 Primitive geometry with complete chronology

All times in this section start at the current primitive’s initial section. Reflection across the right anchor means using the coordinate $q \mapsto D - q$ and negating every nonzero physical velocity. It preserves all strict-order and timing arguments below.

198.1 Scaling a marker about the anchor

Let a stationary target be at $z > 0$, the messenger at 0 with speed $+1$, and all other markers stationary. For $u > 0$, define

$$a = \frac{u - 1}{u + 1} \in (-1, 1).$$

The primitive $L_z(u)$ launches the target at the first messenger hit, reflects the messenger to -1 , reflects it again at the anchor, restores the target at the next hit while reflecting the messenger to -1 , then reflects the messenger at the anchor back to $+1$. Every stationary spectator on each traversed segment is crossed transparently.

Lemma 198.1 (Exact scaling primitive). *The target ends at uz . The prescribed complete binary word is valid exactly when z and uz lie strictly between the target’s unchanged nearest stationary neighbors, with no upper-neighbor condition if there is no such neighbor. Its duration is $2(z + uz)$.*

Proof. The principal events have time-position pairs

$$(z, z), \quad (2z, 0), \quad (2z + uz, uz), \quad (2z + 2uz, 0). \quad (198.1)$$

Indeed the second target hit solves

$$t - 2z = z + a(t - z),$$

and $(1 + a)/(1 - a) = u$. The target moves monotonically on the segment from z to uz . Strict endpoint separation from its fixed nearest neighbors therefore excludes every target–stationary–marker encounter, including one remote from the messenger.

Every inner stationary spectator at $s < \min(z, uz)$ is crossed by the messenger at the four times

$$s, \quad 2z - s, \quad 2z + s, \quad 2z + 2uz - s. \quad (198.2)$$

The first and third traversals encounter spectators in increasing spatial order; the second and fourth encounter them in decreasing order. The strict inequalities $0 < s < \min(z, uz)$ place every crossing strictly between the relevant principal events, and distinct spectator positions give distinct crossing times. No outer spectator is reached.

Immediately after target launch, the messenger’s relative velocity is $-1 - a < 0$, so it separates toward the anchor. On its outward return the relative velocity is $1 - a > 0$ and the next meeting is the unique solution already computed. The last departure separates from the restored stationary marker. Thus the listed events are complete, binary, and strictly time-separated. No unlisted messenger or marker collision can occur. Conversely, reaching a neighbor gives a coincident forbidden contact; passing it forces an earlier target–neighbor contact. Either contradicts this complete word. The last time in (198.1) gives the duration. $\square$

198.2 Homothety toward a reflector

In every use of the next primitive, its target x is the nearest marker to the active anchor. This restriction is part of the statement. Let a stationary reflector be at $z > x$, let the nearest unchanged neighbor beyond the target be $y \leq z$, and choose $v > 0$. Put

$$a = \frac{1 - v}{1 + v} \in (-1, 1).$$

The primitive $H_z(v)$ launches the target when the outward messenger hits it, without changing the messenger’s speed $+1$. The messenger crosses any intervening stationary markers, reflects at z , crosses the same spectators inward, restores the target without changing its speed -1 , and reflects at the anchor to $+1$.

Lemma 198.2 (Exact reflector primitive). *The target ends at $x' = vx + (1 - v)z$. The prescribed complete binary word is valid exactly when $0 < x, x' < y$. Its duration is $2z$.*

Proof. The outward and inward messenger lines are $q = t$ and $q = 2z - t$. Target restoration solves

$$2z - t = x + a(t - x),$$

giving $q = x'$ and $t = 2z - x'$. The target is monotone between its endpoints, so the endpoint inequalities exclude every fixed-marker collision. A spectator s strictly between the target’s neighbor and reflector, or equal to that neighbor when it is not itself the reflector, is crossed at s and $2z - s$. Since $x' < s$, its inward crossing precedes restoration; because $0 < x' < z$, restoration is after the reflector hit and before the anchor hit. All spatial separations are strict, so all times are distinct. The messenger separates from the target after launch with relative speed $1 - a > 0$ and catches it inward with relative speed $-1 - a < 0$. There is no stationary marker below the target whose unlisted crossing could invalidate this chronology. Necessity follows from the same endpoint contact argument as Lemma 198.1. The messenger returns to the anchor at time $2z$. $\square$

198.3 A translation from two primitives

For $c < 1$, apply $L_x(1/(1-c))$ and then $H_z(1-c)$ to the nearest-anchor target. Denote the composite by $T_z(c)$. Both target phases have velocity $a = c/(2-c)$, although their labels are distinct. Algebra gives

$$x \mapsto \frac{x}{1-c} \mapsto x + cz. \quad (198.3)$$

If $0 < x, x + cz < y \leq z$, the middle point lies between the two endpoints. For $c > 0$, the inequalities $x < x/(1-c) < x + cz$ follow from $x + cz < z$; for $c < 0$, the reversed inequalities follow from $x < z$. The $c = 0$ case is immediate. Thus Lemmas 198.1 and 198.2 prove necessity and sufficiency of just the two endpoint conditions, along with unchanged marker order. This is an exact guard reduction, not a sufficient safety margin. The target moves monotonically in the direction of c throughout the composite.

199 Three shears and all finite phases

Two translations $T_y(c), T_D(-2c/3)$ send the left-based target to $x + c(y - 2D/3)$. Repeating this pair twice with $c = -1/4$ gives

$$A: \quad x \mapsto x - y/2 + D/3, \quad y \mapsto y. \quad (199.1)$$

The two physical target velocities are $-1/9$ and $1/11$.

Transfer the messenger from L to D by transparently crossing X, Y and reflecting at D . In the reflected coordinate, set $r = D - y$ and $s = D - x$. Repeat $T_s(2/5), T_D(-4/15)$ twice with r the target. This gives $r \mapsto r + 4s/5 - 8D/15$, hence

$$B: \quad y \mapsto y + 4x/5 - 4D/15, \quad x \mapsto x. \quad (199.2)$$

The reflected-coordinate target velocities are $1/4, -2/17$, so the physical velocities are $-1/4, 2/17$. Transfer back by crossing Y, X and reflecting at L . In every H use, X is nearest L , or Y is nearest D , exactly as required in Section 198.2.

Apply A , then B , then A . In the centered coordinates (196.2),

$$\begin{pmatrix} \xi' \\ \eta' \end{pmatrix} = \begin{pmatrix} 1 & -1/2 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 4/5 & 1 \end{pmatrix} \begin{pmatrix} 1 & -1/2 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \xi \\ \eta \end{pmatrix} = R \begin{pmatrix} \xi \\ \eta \end{pmatrix}. \quad (199.3)$$

Equivalently $x' = 3x/5 - 4y/5 + 2D/3$, $y' = 4x/5 + 3y/5$, and $D' = D$. Each shear has two translations using the nearer reflector with eight events apiece, and two using the farther reflector with ten apiece: 36 events. Each anchor transfer has three events. The rotation word therefore has $3 \cdot 36 + 2 \cdot 3 = 114$ events.

Table 18 is the complete marker-identity chronology, including spectators. Indices are zero-based. Letters stand for the messenger's collision with that identity; the full labels and velocities appear in Section 210. Every L principal event reverses the messenger. H launch/restoration events preserve its direction; the reflector and anchor reverse it. Transparent crossings preserve direction.

Table 18: (Source 27.) All finite phases of the contracted machine

Phase	Events	Marker word
L01	0–3	XLXL
H02	4–7	XYXL
L03	8–11	XLXL
H04	12–17	XYDYXL
L05	18–21	XLXL
H06	22–25	XYXL
L07	26–29	XLXL
H08	30–35	XYDYXL
transfer right	36–38	XYD
L09	39–42	YDYG
H10	43–46	YXYD
L11	47–50	YDYG
H12	51–56	YXLXYD
L13	57–60	YDYG
H14	61–64	YXYD
L15	65–68	YDYG
H16	69–74	YXLXYD
transfer left	75–77	YXL
L17	78–81	XLXL
H18	82–85	XYXL
L19	86–89	XLXL
H20	90–95	XYDYXL
L21	96–99	XLXL
H22	100–103	XYXL
L23	104–107	XLXL
H24	108–113	XYDYXL
L25	114–117	XLXL
L26	118–125	XYXLXYXL
L27	126–137	XYDYXLXYDYXL

200 Exact chamber and all competitor exclusions

For an A block with incoming coordinates (x, y, D) , define

$$\begin{aligned}
 a_0 &= x, & a_1 &= x - y/4, & a_2 &= x - y/4 + D/6, \\
 a_3 &= x - y/2 + D/6, & a_4 &= x - y/2 + D/3.
 \end{aligned}$$

Require $0 < y < D$ and $0 < a_i < y$ for $0 \leq i \leq 4$, then replace x by a_4 . These twelve rows are precisely the initial/final endpoint conditions of its four translations and its fixed marker order.

For the B block, form $r = D - y$, $s = D - x$ at its own incoming section and define

$$\begin{aligned} b_0 &= r, & b_1 &= r + 2s/5, & b_2 &= r + 2s/5 - 4D/15, \\ b_3 &= r + 4s/5 - 4D/15, & b_4 &= r + 4s/5 - 8D/15. \end{aligned}$$

Require $0 < s < D$ and $0 < b_i < s$ for $0 \leq i \leq 4$, then replace y by $D - b_4$. Again there are twelve rows. Both transfers are legal under the strict anchor/data order at their endpoints.

Pull back the first A's twelve rows, B's twelve rows through the first A, and the last A's twelve rows through the preceding blocks. The resulting 36 homogeneous rational linear rows are listed in Section 209. They are an exact system, not claimed to be irredundant. Let C be their open cone and let P be its $D = 1$ slice in (ξ, η) coordinates.

Proposition 200.1 (Complete one-macro equivalence). *The complete 114-event rotation word is legal if and only if all 36 rows hold. There are no additional equality conditions, positive-flight conditions, earlier-competitor conditions, or omitted-tie conditions to impose.*

Proof. Every legal word must have the stated ordered markers at each translation endpoint, so all rows are necessary. For sufficiency, the translation lemma places every internal scaling endpoint between checked endpoints. The active target is confined throughout each primitive to its open nearest-neighbor interval. Consequently no moving marker can collide with a stationary marker, and no two stationary markers meet. This excludes collisions remote from the messenger as well as adjacent ones.

The messenger has speed ± 1 , faster in absolute value than the target. Lemmas 198.1 and 198.2 give its unique principal meetings and every intervening spectator crossing, with positive separation of successive times. At each such meeting all uninvolved markers are strictly elsewhere. Hence no additional earlier collision or simultaneous site exists, no third signal joins a binary event, and each intended output immediately separates in increasing-speed order. Inducting over primitives and transfers proves the complete word. A failed strict endpoint order either gives a forbidden coincident contact or forces a prior marker crossing; it cannot leave this word valid. This establishes exact equivalence, rather than only sufficient legality on a smaller region. $\square$

[write] [This is source 26's one-macro chamber (184.3) for this word, with no equality rows, proved here directly from the primitive lemmas rather than through source 26's candidate comparisons.]

At the center $x = D/3$, $y = 2D/3$, every A target endpoint is $D/3$ or $D/6$, and every reflected B endpoint is $D/3$ or $3D/5$, between 0 and its fixed neighbor $2D/3$. Thus P contains the origin in its interior. The initial rows $0 < x < y < 1$ make P bounded.

201 Contraction and a finite deterministic rule table

After the rotation, perform $L_x(1/2)$, then $L_y(1/2)$, then $L_D(1/2)$, each with target velocity $-1/3$. These phases contain respectively 4, 8, and 12 binary events, as shown in Table 18.

There is no new guard. When a target is halved, all its inner stationary markers have already been halved and remain strictly below its endpoint; all outer markers are unscaled and strictly above its start. Its monotone motion therefore obeys Lemma 198.1. The complete word has 138 events, and its return is M in (196.1). The rational coordinate matrix is

$$\begin{pmatrix} D \\ \xi \\ \eta \end{pmatrix} = Qg, \quad Q = \begin{pmatrix} 1 & 1 & 1 \\ 2/3 & -1/3 & -1/3 \\ 1/3 & 1/3 & -2/3 \end{pmatrix}. \quad (201.1)$$

Direct composition gives

$$QMQ^{-1} = N = \text{diag}(1/2, R/2). \quad (201.2)$$

The eigenvalues are $1/2, (3 + 4i)/10, (3 - 4i)/10$. Therefore neither an all-real-spectrum claim nor a claim that all peripheral phases must be roots of unity holds for these physical periodic returns. The normalized coordinate w rotates by R while D halves.

For $j = 0, \dots, 137$ introduce a messenger label Q_j whose velocity is the incoming messenger velocity at event j ; set $Q_{138} = Q_0$, with Q_0 of speed $+1$. Give each of the 27 target primitives its own temporary moving-marker label. The four stationary labels are unchanged. At event j , with the prescribed incoming and outgoing marker labels $Z_{\text{in}}, Z_{\text{out}}$, include the binary rule

$$\{Q_j, Z_{\text{in}}\} \longrightarrow \{Q_{(j+1) \bmod 138}, Z_{\text{out}}\}. \quad (201.3)$$

Transparent crossings have $Z_{\text{in}} = Z_{\text{out}}$ and equal messenger velocities; anchor/reflector bounces have equal marker labels and opposite messenger velocities. Target launches and restorations exchange stationary and temporary labels exactly as the primitive proof prescribes.

Every explicit input contains its unique Q_j , so no two explicit input sets conflict. Each input and output has distinct speeds, because the messenger speed is ± 1 and the marker speed lies in $(-1, 1)$. Complete the finite machine by the identity rule $S \mapsto S$ on every other allowed collision input set with pairwise distinct speeds. There are finitely many such sets. This completion is deterministic and globally number-preserving; it does not make an omitted triple contact part of the prescribed binary word.

There are $138 + 27 + 4 = 169$ meta-signals, and the exact speed set is

$$\left\{-1, -\frac{1}{3}, -\frac{1}{4}, -\frac{1}{9}, 0, \frac{1}{11}, \frac{2}{17}, 1\right\}. \quad (201.4)$$

Event 137 emits $\{Q_0, L0\}$ at L , with Q_0 moving right. The other markers are stationary and strictly ordered. The original labels, output order, and zero-gap pattern are therefore restored. Section 210 prints all 138 rules in compact form; the JSON supplies every label definition and every explicit input/output set without compression.

202 The exact tangent and a real forbidden collision

Write a guard as $aD + b\xi + c\eta > 0$. Every a is positive. On $D = 1$, the squared distance of its supporting line from the origin is $a^2/(b^2 + c^2)$. Exact rational comparison of the 36 rows gives block minima

$$\frac{1}{45} \quad \text{for the first A,} \quad \frac{4}{1845} \quad \text{for B,} \quad \frac{4}{153} \quad \text{for the last A.}$$

The unique global minimizer is row 18:

$$\frac{D}{15} - \frac{3}{5}\xi + \frac{13}{10}\eta > 0. \quad (202.1)$$

For $a = 1/15$ and normal $n = (-3/5, 13/10)$, the tangent point is $-an/\|n\|^2 = p$, and $\|p\|^2 = 4/1845$. All other rows have strictly larger distance; each is strictly positive on the entire closed radius- ρ disk. Row 18 is nonnegative there and vanishes only at p . Thus this disk touches the closed chamber at exactly one point, and its entire boundary except p belongs to the open chamber.

This contact is physically identifiable. At $D = 1$ and $w = p$ the initial positions are

$$x = \frac{217}{615}, \quad y = \frac{128}{205}.$$

After the first A, $x = 46/123$. In the first reflected L phase, $r = 1 - y = 77/205$ and $s = 1 - x = 77/123$, so $(5/3)r = s$. The moving Y marker reaches X exactly at its prescribed restoration. Event 41 expects $\{Q_{41}, Y_{L09}\}$, but the true contact also contains $X0$, with three distinct incoming physical velocities $-1, -1/4, 0$. Hence the complete binary word fails. The missing point is an actual omitted triple contact, not an arbitrary exclusion imposed by an overcautious guard.

Two exact integer examples are

$$g_{\text{reject}} = (217, 167, 231), \quad g_{\text{accept}} = (233, 171, 211). \quad (202.2)$$

Both have sum 615; they normalize respectively to p and $Rp = (28/615, -2/205)$. The second is valid by the oriented-orbit argument below. These examples distinguish the two directions of rotation at the boundary.

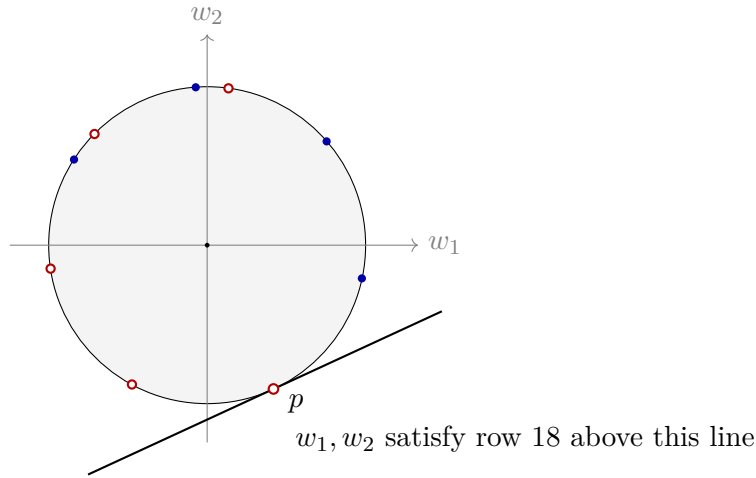


Figure 12: (Source 27.) Analytic geometry of the critical disk. Filled blue points show four accepted forward powers $R^k p$; open red points show p and four rejected inverse powers. Each full one-sided orbit is dense. Only the nearest supporting line is drawn; this is not a trajectory simulation or a drawing of the whole chamber. All conclusions use the exact formulas, not this finite illustration.

203 Infinite validity and three sign obstructions

Lemma 203.1 (Infinite order and density). *The rotation R has infinite order, and both one-sided orbits of every nonzero point are dense on its circle.*

Proof. Its complex eigenvalue is $\zeta = (3 + 4i)/5$. If it were a root of unity, $\zeta + \zeta^{-1} = 6/5$ would be a rational algebraic integer, hence an integer, a contradiction. Thus its angle divided by 2π is irrational. The standard elementary irrational-rotation argument now gives density: pigeonholing multiples of the angle modulo 2π produces arbitrarily small nonzero rotation steps, whose multiples approximate every circular position. Equivalently, the closure of its cyclic subgroup is a closed subgroup of the circle; a proper closed subgroup is finite, contrary to infinite order. Positive and negative one-sided powers have the same full-circle closure. $\square$

Proposition 203.2 (Exact infinitely legal set). *The set of normalized inputs satisfying every finite macro prefix is exactly $\mathcal{K}$ in (196.3).*

Proof. By Proposition 200.1 and cyclic label restoration, every finite prefix is valid exactly when $R^n w \in P$ for all $n \geq 0$. Induction justifies applying the same guard system after each return; contraction cancels in normalization.

If $\|w\| < \rho$, each guard is strictly positive at every point of that smaller circle, so all iterates are legal. If $\|w\| = \rho$, the only failure point on the circle in one macro is p . Thus a future macro fails exactly when $R^n w = p$ for some $n \geq 0$, or $w = R^{-n}p$. If $\|w\| > \rho$, the nearest guard is strictly negative on a nonempty open arc of that circle. Lemma 203.1 forces the forward orbit into that arc. These three cases prove the formula, including its inverse orientation. $\square$

[write] [The first sentence of this proof is source 26's infinite criterion (Proposition 184.2) for this word, which has no equality rows; the reduction to it is the source's own.]

Proposition 203.3 (Real nonsemialgebraicity). *The set $\mathcal{K}$ and its positive homogeneous gap cone are not semialgebraic.*

Proof. The excluded set $E = \{R^{-n}p : n \geq 0\}$ is countably infinite, since R has infinite order and $p \neq 0$. If $\mathcal{K}$ were semialgebraic, its complement on the algebraic circle $S = \{\|w\|^2 = \rho^2\}$ would be semialgebraic. A semialgebraic subset of a circle is a finite union of isolated points and arcs. It cannot be countably infinite: an arc is uncountable, and a finite union of points is finite. Therefore $\mathcal{K}$ is not semialgebraic. A semialgebraic gap cone would give a semialgebraic $D = 1$ section under the invertible rational coordinates (201.1), contradicting this result. $\square$

Proposition 203.4 (Rational-input sign obstruction). *No finite Boolean formula of polynomial signs with real coefficients agrees with infinite validity on all positive rational gap triples.*

Proof. All points $R^k p$, for integer k , are rational. The rejected set E and the accepted set $A = \{R^k p : k \geq 1\}$ are disjoint: an equality would imply $R^{n+k}p = p$ with $n + k > 0$. Both are dense in S . Their associated $D = 1$ gap triples are strictly positive, because the entire critical disk lies inside the strict initial-order rows; only the different row 18 touches it.

Restrict an alleged formula to $D = 1$ and then to S . Its truth set is semialgebraic. Away from finitely many partition points, it has constant membership on each remaining open arc. Every such arc contains both a point of A and a point of E , which require opposite answers. This contradiction proves the claim. Real, rather than rational, coefficients do not evade the finite-arc property. $\square$

Proposition 203.5 (Integer-input sign obstruction). *No finite Boolean formula of polynomial signs with real coefficients agrees with infinite validity on all $g \in \mathbb{Z}_{>0}^3$.*

Proof. Suppose F did. The physical validity predicate is constant on each positive ray. For each polynomial $f = \sum_{j=0}^d f_j$, with homogeneous components f_j , the eventual sign of $f(tg)$ as $t \rightarrow +\infty$ is the sign of the highest-degree component $f_j(g)$ that is nonzero. If all components vanish, its sign is zero. For example, eventual positivity is the finite formula

$$\bigvee_{j=0}^d \left(f_j(g) > 0 \wedge \bigwedge_{k=j+1}^d f_k(g) = 0 \right).$$

There are analogous formulas for negative and zero sign. Replace every sign atom of F by its eventual-sign formula to obtain a semialgebraic F_∞ . It is invariant under positive scaling, and for each fixed g agrees with $F(tg)$ for all sufficiently large t .

For a positive integer vector g , all integer multiples mg have the same correct physical answer. Hence their eventual answer $F_\infty(g)$ agrees with validity. Every positive rational vector has a positive integer multiple; scale invariance extends agreement to every positive rational vector, contradicting Proposition 203.4. The original F was never assumed homogeneous. This eventual-ray bridge is the necessary additional argument for the integer claim. $\square$

Together these results prove the nonsemialgebraic part of Theorem 196.2. They obstruct finite polynomial-sign descriptions without natural-number witnesses. Even allowing real quantifiers still yields a semialgebraic set by quantifier elimination, but natural-number quantification does not share that property.

204 Elementary exact membership on rational inputs

First reject a gap triple that is not strictly positive. For positive rational gaps, compute D, ξ, η and $w = (w_1, w_2) = (\xi/D, \eta/D)$ exactly. Compare $w_1^2 + w_2^2$ with $4/1845$. A smaller value is accepted and a larger value rejected. Only equality needs an orbit test.

On the critical circle form the Gaussian rational

$$\gamma = \frac{w_1 + iw_2}{p_x + ip_y} = \gamma_1 + i\gamma_2 \in \mathbb{Q}(i). \quad (204.1)$$

By (196.3), rejection is exactly

$$\gamma = \left(\frac{3 - 4i}{5} \right)^n \quad \text{for some } n \in \mathbb{N}. \quad (204.2)$$

Let q be the least common positive denominator of the reduced rational components γ_1, γ_2 . This denominator determines the only possible exponent.

Lemma 204.1 (Exact inverse-power denominators). *The least common denominator of the components of $((3 - 4i)/5)^n$ is exactly 5^n for every $n \geq 0$.*

Proof. The $n = 0$ case is 1. For $n \geq 1$, work modulo 5 in the quotient ring with $i^2 = -1$. One has $3 - 4i = 3 + i$ and $(3 + i)^2 = 3 + i$ there. Therefore the integer coefficient pair of $(3 - 4i)^n$ is congruent to $(3, 1)$ modulo 5. Neither coefficient is divisible by 5, so the denominator 5^n cannot cancel in either component. The quotient ring need not be a field; only this multiplication identity is used. $\square$

[write] [This is the case $(a, b, c) = (3, 4, 5)$ of source 29's Lemma 232.1; source 28 proves it again inside its recognition lemma (Lemma 216.1).]

The algorithm is now finite:

1. If q is not a power of 5, accept.
2. Otherwise divide by 5 repeatedly to find the forced exponent n with $q = 5^n$.
3. Compute $(3 - 4i)^n$ by exact integer arithmetic and compare (204.2). Reject exactly on equality.

For $q = 1$ the forced exponent is zero, so only $\gamma = 1$ is excluded. The forward point $\gamma = (3 + 4i)/5$ is accepted even though its denominator is 5: equality, not denominator alone, decides the answer. No unbounded orbit search or numerical density test appears in this algorithm.

Under a signed-integer/positive-denominator encoding, this establishes a recursive rational predicate. The MRDP theorem then implies the existence of an existential-natural Diophantine representation [80]. That observation does not provide a small polynomial, a fixed small witness count, a canonical or unique witness, or an efficient compiler. In particular, this report proves no quartic impossibility theorem. The failure of a finite sign classifier cannot be promoted to impossibility of a polynomial equation with integer-quantified auxiliary variables.

[write] [Source 28 supplies such a representation explicitly, with 81 positive witnesses, 44 equations and exact degree 12 (Theorem 211.1); it does not give a unique witness or a quartic.]

205 The exact Zeno clock and the separate non Zeno machine

Primitive times are computed from the proofs, without trajectory execution:

$$t_L(z, u) = 2(1 + u)z, \quad t_H(z, v) = 2z, \quad t_T(x, z, c) = \frac{2(2 - c)}{1 - c}x + 2z. \quad (205.1)$$

Each anchor transfer takes D . Substituting the successive target positions in the four translations of an A block yields

$$t_A(x, y, D) = 16x - y/5 + 16D/3.$$

For the reflected B block in its own incoming coordinates (r, s, D) , it yields

$$t_B(r, s, D) = \frac{1016}{57}r + \frac{2972}{285}s + \frac{1388}{855}D.$$

Substitute the preceding shear outputs into the next block's time and add the two transfers. The 114-event rotation duration is

$$\begin{aligned} \ell_R &= \frac{222}{5}D + \frac{30512}{1425}x - \frac{9942}{475}y \\ &= \frac{32134}{855}D + \frac{30512}{1425}\xi - \frac{9942}{475}\eta. \end{aligned} \quad (205.2)$$

The final three half-scalings take $3(x_{\text{rot}} + y_{\text{rot}} + D)$, equal to $6D + (21/5)\xi - (3/5)\eta$. Thus the contracted macro duration is

$$\ell(D, \xi, \eta) = \frac{37264}{855}D + \frac{36497}{1425}\xi - \frac{10227}{475}\eta. \quad (205.3)$$

It is positive on the legal chamber, because it is a sum of strictly positive physical flights; negative coefficients in this coordinate expression do not mean negative times.

Every moving point remains in the interval $[0, D]$ of its macro's initial anchors. The twelve translation L primitives each take less than $4D$, the twelve H primitives each take at most $2D$, transfers take $2D$, and the three final half-scalings take less than $9D$ together. Hence $\ell < 83D$. Since $D_n = 2^{-n}D_0$, every valid infinite run has total elapsed time less than $166D_0$. All five strand positions converge to the fixed left anchor 0.

For the exact sum, let ℓ also denote its row vector and use N from (201.2). All its eigenvalues have modulus $1/2$, so

$$\sum_{n \geq 0} \ell N^n = \ell(I - N)^{-1}.$$

Solving, or verifying the row identity $T - TN = \ell$, gives (196.5). In uncentered coordinates the same time is

$$T(D, x, y) = \frac{1204366}{11115}D + \frac{53102}{3705}x - \frac{144302}{3705}y. \quad (205.4)$$

The formula is conditional on infinite validity. Evaluating it at an invalid seed does not establish a physical accumulation time. Rational inputs give rational time, in agreement with Report56’s arbitrary-fixed-population rational-limit theorem [Theorem 182.4].

205.1 The 114 event variant must be reclosed

The supplied JSON is the 138-event contracted machine. Its zero-based event 113 emits Q_{114} , beginning contraction. Repeating its first 114 events is not a periodic run of that unchanged rule table.

For a separate machine, retain rules 0 through 113, change the messenger output of rule 113 from Q_{114} to Q_0 , and discard unused later messenger and temporary labels. This makes the last reflection restore the original Q_0 outgoing section. Its return is $2M$, conjugate to $\text{diag}(1, R)$, and its exact 36-row normalized chamber is unchanged. Every macro includes two full transfers between anchors a fixed distance $D > 0$ apart, so its duration is at least $2D$. Its valid infinite executions are consequently non-Zeno. This proves the variant assertion of Theorem 196.3 without treating a prefix as a cyclic machine.

[write] [The reclosing is also recorded in the shipped 27-fivesignal-science-companion-VARIANT_CLARIFICATION.md; the programme’s review stresses the same point (“truncating the 138-event table is not the same periodic machine”).]

206 The sharp population boundary and what it does not say

Report56 [source 26] proves, for any fixed rational number-preserving machine and supplied complete periodic macro with at most four live signals, an effective Boolean formula of rational polynomial signs of degree at most two in the initial gaps. After denominator clearing, its explicit compiler yields a degree-at-most-four sum-of-squares polynomial with exactly one natural witness tuple on each accepted encoded input. Its hypotheses include the same outgoing collision section, distinct-speed conventions, restored ordered labels and co-location pattern, complete simultaneous batches, and positive time between batches used here.

The dimension calculation explains the threshold. For N signals and $z \geq 1$ zero gaps inside outgoing co-location groups, translation reduction gives $N - 1 - z \leq N - 2$ free coordinates. At population at most four this is at most two. Our binary outgoing section at population five has three positive gaps. Dividing by scale leaves two affine coordinates, but their constant baseline is the third homogeneous coordinate. Therefore a planar normalized rotation with an axial offset does not contradict Report56’s two-dimensional homogeneous classification.

Corollary 206.1 (Sharp universal sign-classification boundary). *Under the fixed-complete-macro hypotheses of Report56, all machines with at most four live signals have finite polynomial-sign validity classifiers, whereas some five-live-signal machine does not. The latter failure already holds if agreement is required only on rational gaps or only on positive integer gaps.*

Proof. The lower-population direction is the separately audited Report56 theorem [Theorem 182.2], whose frozen proof and independent audit are included as an explicit dependency. The upper

counterexample is Theorem 196.2. No reproof of the former is claimed by the new five-signal audit. $\square$

This is a boundary for a particular universal representation property of supplied complete macros. It is not a boundary for decidability of arbitrary signal machines, nor a claim that every five-signal macro is nonsemialgebraic. The rational predicate of this very counterexample is decidable. It does not invalidate Report56's rational accumulation theorem in larger populations. It also does not prove that five signals prevent unique-witness or quartic Diophantine certificates by other methods; a finite-sign frontend and existential-natural polynomial representation are different resources.

[write] [The dependency on source 26 is printed in this Part (Section 182); the archive's byte copies of source 26's proof, audit and manuscript are not shipped twice, and its pins of source 26's PDF, manifest and archive (`dependency_report56/DELIVERY_HASHES.json`, not shipped) match the delivered archive of source 26 exactly.]

207 Prior work and useful next questions

Dai and Xia already prove that nontermination sets of general three-variable linear loops need not be semialgebraic; their Section 4 uses a diagonal positive-spectrum update [76]. Ouaknine and Worrell discuss a closely related irrational-rotation cone with a non-strict guard in Example 3.1 [77]. Thus neither abstract three-variable nonsemialgebraicity nor the general rotation-cone mechanism is claimed as new here. The strict guard is essential to the precise disk-minus-oriented-orbit set in this report.

Signal-machine definitions, distance encodings, geometric multiplication/division, shrinking, and acceleration are established in the literature [6, 78, 79]. The physical claim documented here is a particular fully specified five-live-signal realization with an exact complete-word chamber and the stated arithmetic behavior. Bounded literature and repository overlap checks found no matching construction; they do not establish priority. The eight-speed, 169-type, and 138-event counts are certified counts, not optimality claims.

[write] [The repository searches (`signal rotation, non-semialgebraic, periodic collision, conservative stacks`) were pinned at `7ba8f64e6`, as the shipped `27-fivesignal-science-companion-BOUNDARY_AND_ARITHMETIC.md` records; at that pin the collection had no five-signal or nonsemialgebraic validity result, and this Part now has them.]

Several next questions stay within the demonstrated result:

1. Can the number of phases, types, or distinct speeds be reduced while preserving a complete guarded physical realization? A smaller count needs its own full chronology audit.
2. Can a general machine-and-macro frontend automatically derive an exact chamber and produce a checkable completeness certificate? The present release replays exact algebra and static rule verification, not physical event selection.
3. Which other rotations, contractions, or guard geometries are realizable with five live signals? The example disproves universal spectral restrictions, but is not a realization theorem for arbitrary rational matrices. ([write] Answered for every rational rotation of infinite order and every positive rational scale by source 29 (Theorem 225.1), and for every orientation-preserving rational planar map by Part XI's Report 60; other guard geometries stay open.)
4. Can the elementary inverse-power test here be converted into a useful small Diophantine

certificate, with explicit witness or degree bounds? MRDP alone supplies no economical answer. ([write] Answered by source 28: 81 positive witnesses, 44 equations, 344 gates and exact degree 12 (Theorem 211.1), and for the family by source 29's $1 + 80J$ witnesses and $1 + 43J$ equations; whether smaller witness or degree bounds exist stays open.)

5. How do strict versus non-strict guards change representability for other supplied macros? Boundary orbit orientation and actual multiple-collision semantics must remain explicit.

None of these questions is needed to certify the result above. No universal Positivity-to-signal-macro reduction is asserted.

208 Evidence and portable reproducibility

The construction is a conventional proof, not proof-assistant checked. Its independent audit rederived the two primitives, every endpoint map and guard row, both returns, all supporting-line distances, the tangent and actual triple contact, the rational-membership criterion, and the clocks. It checked each of the 138 rules against separately written phase templates, including exact velocities, temporary-label lifetimes, local output order, transparent crossings, and cyclic labels. It returned PASS for that stated scope.

The release distinguishes three kinds of evidence. Mathematical arguments establish the all-input claims. Exact independent algebra and static checks verify the displayed finite data. Rendering, hash, replay, and archive checks establish artifact integrity. Finite checks are not substitutes for the proofs, and an artifact hash is not a mathematical correctness certificate.

The packet's top-level README gives executable commands. Frozen construction, companion, and audit inputs are preserved under `science/` and `independent_audit/`; the separate lower-population proof and audit are under `dependency_report56/`. The machine-readable rule table is `science/construction/RULES.json`; the guards are `science/construction/GUARDS.txt`. The exhaustive local-order receipt is `independent_audit/all_138_local_rule_checks.tsv`.

The portable adapter runs only the two newly authored independent checkers after verifying their exact approved hashes. It rebinds their input/output path constants in memory, leaving the frozen checker bytes unchanged, and sends every output to an exclusive fresh directory. It compares regenerated substantive receipts and captured stdout with pinned expected values. It does not execute the construction's emitter, its algebra or membership scripts, the saved JSON rule program, or a physical simulator. Its operations are exact rational algebra and inert structural validation. Release tools likewise refuse existing output targets and build deterministic PDFs and ZIPs in fresh locations; the manifest binds every included file. The README states the local tool prerequisites and the trusted external manifest pin required for integrity verification.

[write] [Shipped names: `science/construction/PROOF.md`, `RULES.json` and `GUARDS.txt` are `27-fivesignal-science-construction-PROOF.md`, `data/27-fivesignal-science-construction-RULES.json` and `data/27-fivesignal-science-construction-GUARDS.txt`; the companion is `27-fivesignal-science-companion-BOUNDARY_AND_ARITHMETIC.md`; the audit is `27-fivesignal-independent_audit-INDEPENDENT_AUDIT.md` with `data/27-fivesignal-independent_audit-all_138_local_rule_checks.tsv` and the checkers `code/27-fivesignal-independent_audit-audit_exact_algebra.py` and `code/27-fivesignal-independent_audit-audit_inert_rules.py`; the adapter is `code/27-fivesignal-audit_replay-portable_audit_replay.py`. The archive's `dependency_report56/` copies, its manifests and its top-level README are not shipped (source 26 is printed above). The adapter and release tools expect the delivered layout; the

README gives a rerun route on a copy.]

208.1 Principal frozen source pins

Construction proof `science/construction/PROOF.md`:

1bea81f8f2e6693d4b7958fb44763eb644663d208fe70c577e0137184e17a725

Explicit rule table `science/construction/RULES.json`:

0e145aecc4f60685570c41be57d04905eab5b3bfa1a25b185d0b7bcf27e2f2db

Exact guards `science/construction/GUARDS.txt`:

d8c1dec4945dd4f3dedfdf469d3b6e51ea9991e8234cccb1daf49c0d8cbc6ce9

Boundary and arithmetic companion:

9642699a09f3ebc558fb1268c61370fb514839404e834acf288f9504e3e81b77

Independent five-signal audit:

b119df1c0be8685078b011c14e3c71decabce343f9ba9041ee6f97b4c58a5226

Separate Report56 proof and audit, respectively:

df7cefb472a76e6f34ce7fff0b1e8548782a82f4863fac2ecd2d682c69941a77

2ad7bcd43a2926b0403a1df9cdfa35544add3aea98094b5ea128feace86c59c2

209 All 36 exact strict guards (source 27, Appendix A)

Every row below means $aD + b\xi + c\eta > 0$ in the initial centered coordinates of (196.2). Blocks A, B, C denote the first, middle, and last shear. Repeated or redundant rows are retained. The squared distance column is $a^2/(b^2 + c^2)$ on $D = 1$. Row 18 is the unique minimum; the comparison can be performed entirely in rational arithmetic.

Row	Block	a	b	c	Distance squared
1	A	$\frac{2}{3}$	0	1	$\frac{4}{9}$
2	A	$\frac{1}{3}$	0	-1	$\frac{1}{9}$
3	A	$\frac{1}{3}$	1	0	$\frac{1}{9}$
4	A	$\frac{1}{3}$	-1	1	$\frac{1}{18}$
5	A	$\frac{1}{6}$	1	$-\frac{1}{4}$	$\frac{4}{153}$
6	A	$\frac{1}{2}$	-1	$\frac{5}{4}$	$\frac{4}{41}$
7	A	$\frac{1}{3}$	1	$-\frac{1}{4}$	$\frac{16}{153}$
8	A	$\frac{1}{3}$	-1	$\frac{5}{4}$	$\frac{16}{369}$

Row	Block	a	b	c	Distance squared
9	A	$\frac{1}{6}$	1	$-\frac{1}{2}$	$\frac{1}{45}$
10	A	$\frac{1}{2}$	-1	$\frac{3}{2}$	$\frac{1}{13}$
11	A	$\frac{1}{3}$	1	$-\frac{1}{2}$	$\frac{4}{45}$
12	A	$\frac{1}{3}$	-1	$\frac{3}{2}$	$\frac{4}{117}$
13	B	$\frac{2}{3}$	-1	$\frac{1}{2}$	$\frac{16}{45}$
14	B	$\frac{1}{3}$	1	$-\frac{1}{2}$	$\frac{4}{45}$
15	B	$\frac{1}{3}$	0	-1	$\frac{1}{9}$
16	B	$\frac{1}{3}$	-1	$\frac{3}{2}$	$\frac{4}{117}$
17	B	$\frac{3}{5}$	$-\frac{2}{5}$	$-\frac{4}{5}$	$\frac{9}{20}$
18	B	$\frac{1}{15}$	$-\frac{3}{5}$	$\frac{13}{10}$	$\frac{4}{1845}$
19	B	$\frac{1}{3}$	$-\frac{2}{5}$	$-\frac{4}{5}$	$\frac{5}{36}$
20	B	$\frac{1}{3}$	$-\frac{3}{5}$	$\frac{13}{10}$	$\frac{20}{369}$
21	B	$\frac{3}{5}$	$-\frac{4}{5}$	$-\frac{3}{5}$	$\frac{9}{25}$
22	B	$\frac{1}{15}$	$-\frac{1}{5}$	$\frac{11}{10}$	$\frac{4}{1125}$
23	B	$\frac{1}{3}$	$-\frac{4}{5}$	$-\frac{3}{5}$	$\frac{1}{9}$
24	B	$\frac{1}{3}$	$-\frac{1}{5}$	$\frac{11}{10}$	$\frac{4}{45}$
25	C	$\frac{2}{3}$	$\frac{4}{5}$	$\frac{3}{5}$	$\frac{4}{9}$
26	C	$\frac{1}{3}$	$-\frac{4}{5}$	$-\frac{3}{5}$	$\frac{1}{9}$
27	C	$\frac{1}{3}$	1	$-\frac{1}{2}$	$\frac{4}{45}$
28	C	$\frac{1}{3}$	$-\frac{1}{5}$	$\frac{11}{10}$	$\frac{4}{45}$
29	C	$\frac{1}{6}$	$\frac{4}{5}$	$-\frac{13}{20}$	$\frac{4}{153}$
30	C	$\frac{1}{2}$	0	$\frac{5}{4}$	$\frac{4}{25}$
31	C	$\frac{1}{3}$	$\frac{4}{5}$	$-\frac{13}{20}$	$\frac{16}{153}$
32	C	$\frac{1}{3}$	0	$\frac{5}{4}$	$\frac{16}{225}$
33	C	$\frac{1}{6}$	$\frac{3}{5}$	$-\frac{4}{5}$	$\frac{1}{36}$
34	C	$\frac{1}{2}$	$\frac{1}{5}$	$\frac{7}{5}$	$\frac{1}{8}$
35	C	$\frac{1}{3}$	$\frac{3}{5}$	$-\frac{4}{5}$	$\frac{1}{9}$
36	C	$\frac{1}{3}$	$\frac{1}{5}$	$\frac{7}{5}$	$\frac{1}{18}$

210 All 138 explicit binary rules (source 27, Appendix B)

In the following table, row j means

$$\{Q_j, Z_{\text{in}}\} \longrightarrow \{Q_{(j+1) \bmod 138}, Z_{\text{out}}\}.$$

The v_Q^- and v_Q^+ columns define the incoming and outgoing messenger speeds. Stationary labels $L0, X0, Y0, D0$ have speed zero. Every temporary label's speed is printed in the separate label table

below. Inputs and outputs are sets; their spatial orders are determined by decreasing incoming and increasing outgoing velocity, respectively. The TSV receipt lists those orders for every row. Unused allowed input sets use identity rules. This convention plus the tables specifies the whole finite machine.

j	Phase	Z_{in}	Z_{out}	Q_{out}	v_Q^-	v_Q^+
0	L01	X0	X_L01	Q1	1	-1
1	L01	L0	L0	Q2	-1	1
2	L01	X_L01	X0	Q3	1	-1
3	L01	L0	L0	Q4	-1	1
4	H02	X0	X_H02	Q5	1	1
5	H02	Y0	Y0	Q6	1	-1
6	H02	X_H02	X0	Q7	-1	-1
7	H02	L0	L0	Q8	-1	1
8	L03	X0	X_L03	Q9	1	-1
9	L03	L0	L0	Q10	-1	1
10	L03	X_L03	X0	Q11	1	-1
11	L03	L0	L0	Q12	-1	1
12	H04	X0	X_H04	Q13	1	1
13	H04	Y0	Y0	Q14	1	1
14	H04	D0	D0	Q15	1	-1
15	H04	Y0	Y0	Q16	-1	-1
16	H04	X_H04	X0	Q17	-1	-1
17	H04	L0	L0	Q18	-1	1
18	L05	X0	X_L05	Q19	1	-1
19	L05	L0	L0	Q20	-1	1
20	L05	X_L05	X0	Q21	1	-1
21	L05	L0	L0	Q22	-1	1
22	H06	X0	X_H06	Q23	1	1
23	H06	Y0	Y0	Q24	1	-1
24	H06	X_H06	X0	Q25	-1	-1
25	H06	L0	L0	Q26	-1	1
26	L07	X0	X_L07	Q27	1	-1
27	L07	L0	L0	Q28	-1	1
28	L07	X_L07	X0	Q29	1	-1
29	L07	L0	L0	Q30	-1	1
30	H08	X0	X_H08	Q31	1	1
31	H08	Y0	Y0	Q32	1	1
32	H08	D0	D0	Q33	1	-1
33	H08	Y0	Y0	Q34	-1	-1
34	H08	X_H08	X0	Q35	-1	-1
35	H08	L0	L0	Q36	-1	1
36	right	X0	X0	Q37	1	1
37	right	Y0	Y0	Q38	1	1
38	right	D0	D0	Q39	1	-1
39	L09	Y0	Y_L09	Q40	-1	1
40	L09	D0	D0	Q41	1	-1
41	L09	Y_L09	Y0	Q42	-1	1
42	L09	D0	D0	Q43	1	-1
43	H10	Y0	Y_H10	Q44	-1	-1
44	H10	X0	X0	Q45	-1	1
45	H10	Y_H10	Y0	Q46	1	1
46	H10	D0	D0	Q47	1	-1

j	Phase	Z_{in}	Z_{out}	Q_{out}	v_Q^-	v_Q^+
47	L11	Y0	Y_L11	Q48	-1	1
48	L11	D0	D0	Q49	1	-1
49	L11	Y_L11	Y0	Q50	-1	1
50	L11	D0	D0	Q51	1	-1
51	H12	Y0	Y_H12	Q52	-1	-1
52	H12	X0	X0	Q53	-1	-1
53	H12	L0	L0	Q54	-1	1
54	H12	X0	X0	Q55	1	1
55	H12	Y_H12	Y0	Q56	1	1
56	H12	D0	D0	Q57	1	-1
57	L13	Y0	Y_L13	Q58	-1	1
58	L13	D0	D0	Q59	1	-1
59	L13	Y_L13	Y0	Q60	-1	1
60	L13	D0	D0	Q61	1	-1
61	H14	Y0	Y_H14	Q62	-1	-1
62	H14	X0	X0	Q63	-1	1
63	H14	Y_H14	Y0	Q64	1	1
64	H14	D0	D0	Q65	1	-1
65	L15	Y0	Y_L15	Q66	-1	1
66	L15	D0	D0	Q67	1	-1
67	L15	Y_L15	Y0	Q68	-1	1
68	L15	D0	D0	Q69	1	-1
69	H16	Y0	Y_H16	Q70	-1	-1
70	H16	X0	X0	Q71	-1	-1
71	H16	L0	L0	Q72	-1	1
72	H16	X0	X0	Q73	1	1
73	H16	Y_H16	Y0	Q74	1	1
74	H16	D0	D0	Q75	1	-1
75	left	Y0	Y0	Q76	-1	-1
76	left	X0	X0	Q77	-1	-1
77	left	L0	L0	Q78	-1	1
78	L17	X0	X_L17	Q79	1	-1
79	L17	L0	L0	Q80	-1	1
80	L17	X_L17	X0	Q81	1	-1
81	L17	L0	L0	Q82	-1	1
82	H18	X0	X_H18	Q83	1	1
83	H18	Y0	Y0	Q84	1	-1
84	H18	X_H18	X0	Q85	-1	-1
85	H18	L0	L0	Q86	-1	1
86	L19	X0	X_L19	Q87	1	-1
87	L19	L0	L0	Q88	-1	1
88	L19	X_L19	X0	Q89	1	-1
89	L19	L0	L0	Q90	-1	1
90	H20	X0	X_H20	Q91	1	1
91	H20	Y0	Y0	Q92	1	1
92	H20	D0	D0	Q93	1	-1
93	H20	Y0	Y0	Q94	-1	-1
94	H20	X_H20	X0	Q95	-1	-1
95	H20	L0	L0	Q96	-1	1
96	L21	X0	X_L21	Q97	1	-1
97	L21	L0	L0	Q98	-1	1
98	L21	X_L21	X0	Q99	1	-1

j	Phase	Z_{in}	Z_{out}	Q_{out}	v_Q^-	v_Q^+
99	L21	L0	L0	Q100	-1	1
100	H22	X0	X_H22	Q101	1	1
101	H22	Y0	Y0	Q102	1	-1
102	H22	X_H22	X0	Q103	-1	-1
103	H22	L0	L0	Q104	-1	1
104	L23	X0	X_L23	Q105	1	-1
105	L23	L0	L0	Q106	-1	1
106	L23	X_L23	X0	Q107	1	-1
107	L23	L0	L0	Q108	-1	1
108	H24	X0	X_H24	Q109	1	1
109	H24	Y0	Y0	Q110	1	1
110	H24	D0	D0	Q111	1	-1
111	H24	Y0	Y0	Q112	-1	-1
112	H24	X_H24	X0	Q113	-1	-1
113	H24	L0	L0	Q114	-1	1
114	L25	X0	X_L25	Q115	1	-1
115	L25	L0	L0	Q116	-1	1
116	L25	X_L25	X0	Q117	1	-1
117	L25	L0	L0	Q118	-1	1
118	L26	X0	X0	Q119	1	1
119	L26	Y0	Y_L26	Q120	1	-1
120	L26	X0	X0	Q121	-1	-1
121	L26	L0	L0	Q122	-1	1
122	L26	X0	X0	Q123	1	1
123	L26	Y_L26	Y0	Q124	1	-1
124	L26	X0	X0	Q125	-1	-1
125	L26	L0	L0	Q126	-1	1
126	L27	X0	X0	Q127	1	1
127	L27	Y0	Y0	Q128	1	1
128	L27	D0	D_L27	Q129	1	-1
129	L27	Y0	Y0	Q130	-1	-1
130	L27	X0	X0	Q131	-1	-1
131	L27	L0	L0	Q132	-1	1
132	L27	X0	X0	Q133	1	1
133	L27	Y0	Y0	Q134	1	1
134	L27	D_L27	D0	Q135	1	-1
135	L27	Y0	Y0	Q136	-1	-1
136	L27	X0	X0	Q137	-1	-1
137	L27	L0	L0	Q0	-1	1

Temporary marker labels and their physical speeds

Label	Speed
X_L01	$-\frac{1}{9}$
X_H02	$-\frac{1}{9}$
X_L03	$\frac{1}{11}$
X_H04	$\frac{1}{11}$
X_L05	$-\frac{1}{9}$

Label	Speed
X_H06	$-\frac{1}{9}$
X_L07	$\frac{1}{11}$
X_H08	$\frac{1}{11}$
Y_L09	$-\frac{1}{4}$
Y_H10	$-\frac{1}{4}$
Y_L11	$\frac{2}{17}$
Y_H12	$\frac{2}{17}$
Y_L13	$-\frac{1}{4}$
Y_H14	$-\frac{1}{4}$
Y_L15	$\frac{2}{17}$
Y_H16	$\frac{2}{17}$
X_L17	$-\frac{1}{9}$
X_H18	$-\frac{1}{9}$
X_L19	$\frac{1}{11}$
X_H20	$\frac{1}{11}$
X_L21	$-\frac{1}{9}$
X_H22	$-\frac{1}{9}$
X_L23	$\frac{1}{11}$
X_H24	$\frac{1}{11}$
X_L25	$-\frac{1}{3}$
Y_L26	$-\frac{1}{3}$
D_L27	$-\frac{1}{3}$

[write] **The programme’s intake of source 27.** The review [99] reads it as “A sign-classification obstruction, explicitly not an obstruction to existential integer Diophantine representation or an undecidability claim”, and notes that the 114-event non-Zeno variant “needs a reclosed phase cycle; truncating the 138-event table is not the same periodic machine” (Section 205). Both are statements of the source itself; no correction is asked for.

Source 28 (batch-91 manuscript 10): Explicit Diophantine certificate for five signal infinite validity

Exact positive integer certificate and reproducibility audit, “Report 58”, 4 October 2026. [write] From the archive `Five_Signal_Diophantine_Certificate_Source.zip`; its labels carry the prefix `smc:fd:`, its sections are Sections 211–224 here (its two appendices are the last two), and its equations are numbered by section; the tags (O1)–(O14), (D1)–(D2), (P1)–(P15), (5.1)–(5.15) and (c.1)–(c.15) are the source’s. Its bibliography entries are this report’s [81], [82], [83], [84], [85] and [86]; the five-signal machine it certifies is source 27’s. Its Gaussian quotient η , radix $b = 4P + 1$, acceptance witness J and Pell modulus M are written γ , β_{rad} , J_{acc} and Θ (opening of Part X). Its abstract follows.

For a fixed, independently audited five-live-signal machine, we give an explicit integer polynomial whose positive-integer zeros characterize infinite validity of its complete collision macro. The native representation has three positive gap inputs, 81 positive integer witnesses, 44 equations,

and exact total degree 12. Its literal binary arithmetic circuit has 344 gates: 136 multiplications, 110 additions, and 98 subtractions. A separately emitted one-input representation includes a fully paid Cantor decoder and has 85 positive witnesses, 46 equations, and 367 gates, again of exact degree 12.

The construction reduces the inherited disk-minus-inverse-orbit validity criterion to primitive Gaussian fractions, an exact denominator valuation, and bounded extraction of a Gaussian power from two ordinary powers. Each ordinary power is represented by fifteen explicit polynomial equations, specialized from pinned constructive Pell theorems. All domain adapters, residuals, squares, and decoding operations are charged. The report proves both directions and treats zero exponents, zero numerator components, critical-circle equality, orientation, and strict slacks. Independent sparse-polynomial audits match every residual to the frozen arithmetic source and compute its exact leading form. Complete positive Pell witnesses are supplied for twelve exponent-zero evaluations. The physical theorem and Pell theorems are explicit dependencies; finite tests supplement these proofs and do not replace them.

211 The result and the objects being represented

Throughout, $\mathbb{N} = \{0, 1, 2, \dots\}$ and $\mathbb{Z}_{>0} = \{1, 2, 3, \dots\}$. A positive witness is an ordinary independent integer variable ranging over $\mathbb{Z}_{>0}$. The input domain is also $\mathbb{Z}_{>0}$; there is no promise that an input satisfies a separate code-validation test.

The machine is the fixed five-live-signal, rational, number-preserving machine of the pinned physical proof [81]. Its outgoing section consists of four markers at $0 < x < y < D$ and the right-moving messenger co-located with the left marker. The three positive section gaps are

$$(g_1, g_2, g_3) = (x, y - x, D - y).$$

Its complete contracted macro has 138 strictly time-separated binary collisions and return matrix

$$M = \frac{1}{30} \begin{pmatrix} 7 & -2 & 10 \\ 14 & 11 & -10 \\ -6 & 6 & 15 \end{pmatrix}.$$

Normalized centered coordinates rotate by $(3 + 4i)/5$. We write $\text{Valid}(g_1, g_2, g_3)$ when that specified complete macro can be iterated indefinitely. The inherited theorem includes legality of the complete physical collision word and its strict tie exclusions. It is not only an assertion about iterating a matrix. Report 57 [source 27] provides the physical proof and its independent audit; the exact proof and audit bytes used here are included in this package.

Theorem 211.1 (Native positive-input certificate). *The frozen arithmetic source `science/evidence/three-input-linear.dag.js` on defines an explicit polynomial*

$$F_{\text{nat}} \in \mathbb{Z}[g_1, g_2, g_3, w_1, \dots, w_{81}]$$

of exact total degree 12 such that, for every $(g_1, g_2, g_3) \in \mathbb{Z}_{>0}^3$,

$$\text{Valid}(g_1, g_2, g_3) \iff \exists (w_1, \dots, w_{81}) \in \mathbb{Z}_{>0}^{81} : F_{\text{nat}}(g_1, g_2, g_3, w) = 0.$$

The source has 44 equation residuals and 344 binary arithmetic gates, consisting of 136 multiplications, 110 additions, and 98 subtractions.

The polynomial is given explicitly by the sum of the squares of the fourteen outer residuals in Section 213 and the two fifteen-equation power systems in Section 223, after the positive-domain substitutions of Section 213.1. The frozen DAG fixes the literal expression sharing and hence the gate count. There are no exponentiation, division, comparison, or logical primitives in that source.

For naturals a, b , put

$$\text{pair}(a, b) = \frac{(a+b)(a+b+1)}{2} + b, \quad \text{code}(g_1, g_2, g_3) = 1 + \text{pair}(g_1 - 1, \text{pair}(g_2 - 1, g_3 - 1)).$$

The half in this mathematical definition is eliminated from the certificate by the two integer equations of Section 218.

Theorem 211.2 (Paid one-input certificate). *The frozen source `science/evidence/one-input-linear.dag.json` defines a polynomial*

$$F_{\text{one}} \in \mathbb{Z}[z, w_1, \dots, w_{85}]$$

of exact total degree 12 with the following property. Every $z \in \mathbb{Z}_{>0}$ has a unique decoded positive gap triple g , and

$$\text{Valid}(g) \iff \exists w \in \mathbb{Z}_{>0}^{85} : F_{\text{one}}(z, w) = 0.$$

Its 46 residuals and 367 binary gates include all decoding costs: 144 multiplications, 118 additions, and 105 subtractions.

What is and is not claimed. These are explicit existential positive-integer representations. The inherited impossibility of a finite polynomial-sign classifier does not preclude them. Neither theorem is a real-witness equivalence or a claim of finite-fold, singlefold, or unique witnesses. Signed-pair adapters in fact give infinitely many witnesses for each solvable input. No optimality, circuit lower bound, novelty priority, undecidability, or new physical population threshold is established. The construction does not use generic MRDP representability as its final representation, a variable-length conjunction, an unexpanded power oracle, or a hidden witness array.

[write] [Shipped names: the four frozen DAGs are `data/28-diophantine-science-evidence-three-input-linear.dag.json`, `data/28-diophantine-science-evidence-one-input-linear.dag.json` and the two `...-quartic.dag.json` files beside them, each with its `.receipt.json`. The counts 81 and 44 are the case of one contact of source 29's ledger $1 + 80J$, $1 + 43J$, for a different polynomial; the opening of Part X compares the three ledgers of Part X and Part XI.]

212 The inherited physical criterion in integer coordinates

Define the following arithmetic expressions:

$$D = g_1 + g_2 + g_3, \quad x = g_1, \quad y = g_1 + g_2, \quad (212.1)$$

$$A = 3x - D, \quad B = 3y - 2D, \quad (212.2)$$

$$\Delta(g) = 4D^2 - 205(A^2 + B^2), \quad (212.3)$$

$$U = 6A - 13B, \quad V = 13A + 6B. \quad (212.4)$$

We reserve $\Delta(g)$ for the determined integer deficit and δ for its nonnegative witness below. The normalized centered point is

$$\omega = \frac{A + iB}{3D}.$$

The critical squared radius and tangency point from the physical theorem are

$$\rho^2 = \frac{4}{1845}, \quad p = \frac{4}{205} - \frac{26}{615}i = \frac{2}{615}(6 - 13i).$$

Direct rational calculation gives

$$\rho^2 - |\omega|^2 = \frac{\Delta(g)}{1845D^2}, \quad (212.5)$$

$$\gamma := \frac{\omega}{p} = \frac{U + iV}{2D}, \quad (212.6)$$

$$U^2 + V^2 = 205(A^2 + B^2). \quad (212.7)$$

Because $D > 0$, the deficit has precisely the radius-deficit sign. On $\Delta(g) = 0$, the number γ has modulus one.

Theorem 212.1 (Inherited complete-macro validity criterion). *For the fixed machine and every positive gap triple,*

$$\text{Valid}(g) \iff \Delta(g) > 0 \quad \text{or} \quad \left[\Delta(g) = 0 \text{ and } \gamma \neq \left(\frac{3 - 4i}{5} \right)^m \text{ for every } m \in \mathbb{N} \right]. \quad (212.8)$$

Inputs with $\Delta(g) < 0$ are invalid. The $m = 0$ contact is excluded.

Theorem 212.1 is the precise inherited dependency, independently checked in the pinned physical audit [82]. Its proof uses the complete strict chamber, the unique tangency point, and the infinite-order normalized rotation. It excludes an omitted triple contact at the tangency. The contraction cancels upon normalization, so the corresponding explicitly closed uncontracted machine has the same geometric criterion; our primary predicate remains the fixed contracted macro above. We do not reconstruct or execute its collision program here.

[write] [Theorem 212.1 is source 27's Proposition 203.2 in the integer coordinates $A = 3\xi$, $B = 3\eta$, with its exclusion test (204.2); it is printed and proved there, and is not re-proved here.]

The inherited real, rational, and integer finite-polynomial-sign obstructions concern formulas with finitely many sign atoms in the input coordinates alone. The new certificate instead quantifies over 81 or 85 positive integer auxiliaries. The scopes are different and compatible. Section 221 adds an elementary sign obstruction specific to the paid integer encoding.

213 The complete outer certificate and its positive domains

In addition to the input gaps, use the following conceptual integer quantities:

Domain	Quantities
Natural	$\delta, n, k, L_C, H_C, L_S, H_S$
Positive	$h, q, r, s, t, J_{\text{acc}}$
Signed	$u, v, a_1, a_2, a_3, C, S, \kappa$

Let the two explicit power systems supply positive outputs P, T , using the common natural exponent n , with bases 5 and $3 + 4\beta_{\text{rad}}$, respectively, where

$$\beta_{\text{rad}} = 4P + 1. \quad (213.1)$$

The statements $P = 5^n$ and $T = (3 + 4\beta_{\text{rad}})^n$ are conclusions of those systems, not power gates or extra defining equations. All names such as $D, A, B, U, V, \beta_{\text{rad}}$ denote expressions expanded into the DAG, not uncharged independent leaves.

The fourteen outer equations are

$$4D^2 = 205(A^2 + B^2) + \delta, \quad (\text{O1})$$

$$U = hu, \quad (\text{O2})$$

$$V = hv, \quad (\text{O3})$$

$$2D = hq, \quad (\text{O4})$$

$$a_1u + a_2v + a_3q = 1, \quad (\text{O5})$$

$$q = Pr, \quad (\text{O6})$$

$$r = 5k + s, \quad (\text{O7})$$

$$s + t = 5, \quad (\text{O8})$$

$$C + P = L_C, \quad (\text{O9})$$

$$P - C = H_C, \quad (\text{O10})$$

$$S + P = L_S, \quad (\text{O11})$$

$$P - S = H_S, \quad (\text{O12})$$

$$T - C - \beta_{\text{rad}}S = \kappa(\beta_{\text{rad}}^2 + 1), \quad (\text{O13})$$

$$\delta^2 + (r - 1)^2 + (u - C)^2 + (v + S)^2 = J_{\text{acc}}. \quad (\text{O14})$$

The domains distinguish three essential cases. First, $\delta \in \mathbb{N}$ allows the critical circle while excluding every exterior point. Second, the four bound slacks are natural, so the coefficient bounds are closed. Third, $J_{\text{acc}} > 0$ makes the final acceptance test strict. Also, both s and t must be positive.

213.1 Every final quantifier is positive

Every natural z is replaced by one positive leaf z^+ through

$$z = z^+ - 1.$$

Every signed z is replaced by two positive leaves $z^{\text{pos}}, z^{\text{neg}}$ through

$$z = z^{\text{pos}} - z^{\text{neg}}.$$

For any integer z , one possible pair is $(\max(z, 0) + 1, \max(-z, 0) + 1)$. Thus this replacement is onto $\mathbb{Z}$. It uses one subtraction gate, rather than two shifts followed by a subtraction. The source names the leaves `.Plus`, `.Positive`, and `.Negative`.

Each directly positive quantity is one positive leaf. Thus the outer system contributes

$$7 + 6 + 2 \cdot 8 = 29$$

positive leaves. Each power system contributes 26, including its output. The native total is $29 + 26 + 26 = 81$; P and T are not counted twice.

For each named equation, form its left side minus its right side. Let $\mathcal{R}$ be the fixed list of 44 resulting residuals, including the thirty power residuals. The certificate is the integer polynomial

$$F_{\text{nat}} = \sum_{R \in \mathcal{R}} R^2. \quad (213.2)$$

Over integers, $F_{\text{nat}} = 0$ is equivalent to every residual being zero, since each square is nonnegative. All loops used to describe this source range over fixed lists, never over an input or witness value. For the one-input source, the two decoder residuals are included in the same fixed sum.

214 Primitive Gaussian fractions and the canonical exponent

Lemma 214.1 (Exact common denominator). *Equations (O2)–(O5), with the stated domains, force*

$$h = \gcd(U, V, 2D), \quad (u, v, q) = (U/h, V/h, 2D/h), \quad \gcd(u, v, q) = 1.$$

The positive integer q is the least common positive denominator of both rational coordinates u/q and v/q . Conversely these values admit signed Bézout coefficients satisfying (O5).

Proof. Any common divisor of u, v, q divides the left side of (O5), so divides 1. Since $h > 0$, equations (O2)–(O4) then give

$$\gcd(U, V, 2D) = h \gcd(u, v, q) = h.$$

Conversely, $2D > 0$ makes the gcd positive, and its primitive quotient admits an ordinary three-term integer Bézout identity.

If a positive integer d clears both rational coordinates, then $q \mid du$ and $q \mid dv$. Multiply (O5) by d . Every term on the left is divisible by q , including $a_3 dq$, so $q \mid d$. Since q itself clears both coordinates, it is their exact common denominator. This proof does not assume that either coordinate is individually in lowest terms. $\square$

Negative values of U or V cause no difficulty. If both vanish, $\gcd(0, 0, 2D) = 2D$, so $u = v = 0$ and $q = 1$; choosing $a_3 = 1$ suffices. Thus the center has no exceptional or singular branch. The signed adapters preserve all Bézout coefficients.

Lemma 214.2 (Canonical valuation). *Assuming $P = 5^n$, equations (O6)–(O8) force*

$$n = v_5(q), \quad q = 5^n r, \quad 5 \nmid r,$$

and conversely these values always supply the required witnesses.

Proof. Positivity of s, t and $s + t = 5$ give $s \in \{1, 2, 3, 4\}$. Hence $r = 5k + s$ is positive and not divisible by 5, with $k \in \mathbb{N}$. Equation (O6) gives $q = 5^n r$, which uniquely fixes n as the 5-adic valuation. Conversely write $q = 5^n r$ with $5 \nmid r$ and use the division algorithm $r = 5k + s$, $1 \leq s \leq 4$, then set $t = 5 - s > 0$. $\square$

In particular $q = 1$ gives $n = 0, r = 1, k = 0, s = 1, t = 4$. These clauses are imposed at every input, including interior points. There is no boundary selector and no uncharged test for whether the input lies on the critical circle.

215 Bounded extraction of the Gaussian power

For each $n \in \mathbb{N}$, define ordinary integers C_n, S_n by

$$C_n + iS_n = (3 + 4i)^n.$$

The modulus identity gives $C_n^2 + S_n^2 = 25^n$, hence

$$|C_n|, |S_n| \leq 5^n = P.$$

These names are used in the proof; they are not additional witness variables or source operations.

Lemma 215.1 (Exact bounded residue extraction). *If $P = 5^n$, $\beta_{\text{rad}} = 4P + 1$, and $T = (3 + 4\beta_{\text{rad}})^n$, then (O9)–(O13) are solvable and force $(C, S) = (C_n, S_n)$.*

Proof. Reducing the binomial expansion in $\mathbb{Z}[X]/(X^2 + 1)$ gives

$$(3 + 4X)^n \equiv C_n + S_n X \pmod{X^2 + 1}.$$

Substitute the integer $X = \beta_{\text{rad}}$. Then

$$T \equiv C_n + \beta_{\text{rad}} S_n \pmod{\beta_{\text{rad}}^2 + 1}.$$

Consequently the true pair satisfies (O13) with an integer κ , and its four closed bounds give natural slacks in (O9)–(O12).

Conversely those four equations impose $-P \leq C, S \leq P$. Put $d_C = C - C_n$ and $d_S = S - S_n$. The remainder congruence gives

$$\beta_{\text{rad}}^2 + 1 \mid d_C + \beta_{\text{rad}} d_S,$$

while

$$\begin{aligned} |d_C + \beta_{\text{rad}} d_S| &\leq 2P(1 + \beta_{\text{rad}}) = 8P^2 + 4P \\ &< 16P^2 + 8P + 2 = \beta_{\text{rad}}^2 + 1. \end{aligned}$$

Thus $d_C + \beta_{\text{rad}} d_S = 0$. If $d_S \neq 0$, integrality gives $|\beta_{\text{rad}} d_S| \geq \beta_{\text{rad}}$, whereas $|d_C| \leq 2P < \beta_{\text{rad}}$. This contradicts $d_C = -\beta_{\text{rad}} d_S$, so $d_S = 0$ and then $d_C = 0$. $\square$

There is no restriction on the size or sign of κ . The bound argument controls the remainder pair, not the quotient. At $n = 0$ the values are

$$P = 1, \quad \beta_{\text{rad}} = 5, \quad T = 1, \quad C = 1, \quad S = 0.$$

The value $C = P$ lies on a closed bound; making every bound slack positive would incorrectly exclude it. The strict inequality used in the uniqueness proof follows from $P \geq 1$ and requires no extra witness or comparison gate.

216 The strict final equation and the two directions

Lemma 216.1 (Recognition of the forbidden inverse orbit). *For the primitive fraction and canonical exponent above,*

$$\gamma = \left(\frac{3 - 4i}{5} \right)^m \text{ for some } m \in \mathbb{N} \iff r = 1, \quad u = C_n, \quad v = -S_n.$$

Proof. For $m \geq 1$, reduce the Gaussian numerator modulo 5. Since

$$3 - 4i \equiv 3 + i \pmod{5}, \quad (3 + i)^2 \equiv 3 + i \pmod{5},$$

we have $(3 - 4i)^m \equiv 3 + i \pmod{5}$. Both numerator coordinates are nonzero modulo 5. Thus their joint denominator after division by 5^m is exactly 5^m . This calculation occurs in the quadratic quotient ring over $\mathbb{Z}/5\mathbb{Z}$; it does not assume that this ring is a field. At $m = 0$, the point is 1 and its denominator is 1.

If γ equals such a point, its exact common denominator q is 5^m . The canonical factorization $q = 5^n r$ therefore gives $m = n, r = 1$. Equality of the primitive rational coordinates gives $(u, v) = (C_n, -S_n)$. Conversely these equalities and $q = 5^n$ imply $\gamma = (C_n - iS_n)/5^n = ((3 - 4i)/5)^n$. $\square$

[write] [The first paragraph of this proof is source 27's Lemma 204.1, proved again; source 29's Lemma 232.1 is the general version.]

The sign in (O14) is deliberately $v + S$. For example, the forward boundary point $(3 + 4i)/5$ has $u = 3, v = 4, q = 5, n = 1$, so $v + S = 8$. Replacing the plus by a minus would confuse the accepted forward orbit with the excluded inverse orbit.

Proof of Theorem 211.1, assuming the power lemma proved next. Soundness. Suppose a positive-integer assignment makes $F_{\text{nat}} = 0$. By the sum-of-squares identity every residual vanishes. Equation (O1) gives $\Delta(g) = \delta \geq 0$. The two power systems give $P = 5^n$ and $T = (3 + 4\beta_{\text{rad}})^n$. Sections 214 and 215 supply the primitive fraction, canonical exponent, and exact coefficients C_n, S_n .

Because $J_{\text{acc}} > 0$, equation (O14) says that not all of $\delta, r - 1, u - C_n$, and $v + S_n$ vanish. If $\delta > 0$, the first alternative of the inherited criterion holds. If $\delta = 0$, the preceding lemma says that the input is not a forbidden inverse-orbit point. Theorem 212.1 therefore gives infinite complete-macro validity.

Completeness. Suppose $\text{Valid}(g)$. Set $\delta = \Delta(g) \in \mathbb{N}$. Choose the positive gcd and primitive quotients, and choose signed Bézout coefficients. Factor the positive denominator as $q = 5^n r$, $5 \nmid r$, and choose k, s, t by the division algorithm. The constructive power lemma supplies all auxiliaries for the two systems at exponent n . Set $C = C_n, S = S_n$; choose their four natural bound slacks and the integer quotient κ from Section 215.

If $\delta > 0$, the left side of (O14) is positive. If $\delta = 0$, validity excludes the forbidden inverse orbit, so at least one of $r - 1, u - C_n, v + S_n$ is nonzero. In either case set J_{acc} equal to the positive sum of squares. Replace every natural and signed quantity by positive leaves using Section 213.1. All 44 equations then hold simultaneously and $F_{\text{nat}} = 0$. $\square$

This is a finite mathematical argument in both directions. It performs no unbounded orbit search and needs no density argument after the physical theorem has been supplied.

217 A fully explicit positive integer power system

We now specify and justify the only number-theoretic subsystem. Its exact dependency is the pinned source `Mathlib/NumberTheory/PellMatiyasevic.lean`, identified with `mathlib4` commit

`ac77769fabe23cb237559e7f56578dbead91499f`

The actual dependency checked here is the included local file, SHA-256

993760c797ad0ff66fa77064a616fce779550bc745cbf6c44e04f392fd0bed0a

The theorem statements and constructive proof text were read as inert source. This work is not a new Lean kernel build or a new authentication of the remote commit. The independent specialization review [85] records exact local line references and domain checks.

[write] [The pinned file is cited as [86]; the archive’s copy of it and of its Apache 2.0 licence are not shipped. The fifteen-equation system below is the power macro of the turmite report (*periodic-turmite-first-revisits*, Lemma `ptr:ai:lem:exp`, Report 42: the same fifteen equations, “31 multiplications and 39 additions or subtractions, for 70 operations, with 25 positive internal witnesses”, i.e. 26 leaves with the output) and of Report 55 [*group-theoretic-substrates*, Part VI]; source 28’s archive carried Report 55’s builder, its source notes and its Pell-dependency record, which are shipped there as `code/09-chrono-matrix-science-build_certificate.py`, `09-chrono-matrix-science-SOURCE_NOTES.md` and `data/09-chrono-matrix-independent_audit-pell-dependency.json`. Source 29 prints the same system again in its own names (Section 234.3); it is printed once, here, and source 29’s account of the specialization is kept as a second account. No novelty is claimed for the system; it is a conventional specialization of the two cited theorems.]

217.1 Fifteen equations and twenty six leaves

Fix an integer base $B_0 \geq 2$, a natural exponent e , and a positive output o . Introduce directly positive quantities

$$o, w, \Theta, g, x, y, u, v, s, t, q_b, q_v, J_p,$$

two positive leaves a_{-1}, β_{-1} and the expressions

$$a = a_{-1} + 1, \quad \beta = \beta_{-1} + 1,$$

and eleven natural quantities

$$d_{wb}, d_{wk}, d_{yk}, \alpha_1, \alpha_2, \sigma_1, \sigma_2, \tau_1, \tau_2, \rho_1, \rho_2.$$

The natural quantities each use one positive leaf minus one. Set $k_0 = e + 1$ and $m_0 = B_0 o$ as expressions. The complete system is

$$x^2 = 1 + (a^2 - 1)y^2, \quad (\text{P1})$$

$$u^2 = 1 + (a^2 - 1)v^2, \quad (\text{P2})$$

$$s^2 = 1 + (\beta^2 - 1)t^2, \quad (\text{P3})$$

$$\beta = 1 + 4yq_b, \quad (\text{P4})$$

$$\beta + u\alpha_1 = a + u\alpha_2, \quad (\text{P5})$$

$$v = y^2 q_v, \quad (\text{P6})$$

$$s + u\sigma_1 = x + u\sigma_2, \quad (\text{P7})$$

$$t + 4y\tau_1 = k_0 + 4y\tau_2, \quad (\text{P8})$$

$$y = k_0 + d_{yk}, \quad (\text{P9})$$

$$w = B_0 + d_{wb}, \quad (\text{P10})$$

$$w = k_0 + d_{wk}, \quad (\text{P11})$$

$$\Theta = m_0 + J_p, \quad (\text{P12})$$

$$a^2 = 1 + ((w + 1)^2 - 1)(wg)^2, \quad (\text{P13})$$

$$2aB_0 = \Theta + (B_0^2 + 1), \quad (\text{P14})$$

$$x + \Theta\rho_1 = y(a - B_0) + m_0 + \Theta\rho_2. \quad (\text{P15})$$

These are polynomial equations. The label POWER is only a name for this finite system. It is not an additional symbol in the resulting polynomial. Section 223 prints both actual instances with their bases and output expressions substituted.

There are 13 directly positive leaves, 2 shifted-positive leaves, and 11 natural-adaptor leaves, for a total of 26 including o . In the literal frozen source each instance has 70 body gates: 31 multiplications, 24 additions, and 15 subtractions. This includes all parameter shifts, natural adaptors, the exponent shift, and multiplication $B_0 o$; the global residual-square assembly is counted separately.

Lemma 217.1 (Power-system equivalence). *For every $B_0 \geq 2$, $e \in \mathbb{N}$, and $o \in \mathbb{Z}_{>0}$, the fifteen equations have the other 25 positive leaves if and only if $o = B_0^e$.*

217.2 The precise constructive dependencies

The pinned source defines natural Pell sequences $(x_j(a), y_j(a))$ with initial pair $(1, 0)$ and parameter $a > 1$. Its theorem `Pell.matiyasevic`, lines 760–766, identifies the pair at index k by three Pell equations, $k \leq y$, and either the zero-index pair $(1, 0)$ or natural witnesses with

$$\begin{aligned} \beta > 1, \quad \beta &\equiv 1 \pmod{4y}, \quad \beta \equiv a \pmod{u}, \quad v > 0, \quad y^2 \mid v, \\ s &\equiv x \pmod{u}, \quad t \equiv k \pmod{4y}. \end{aligned}$$

Its constructive direction is at lines 767–806 and reverse direction at 807–847. Equations (P1)–(P9) are precisely the nonzero branch with $k = k_0 = e + 1$.

The theorem `Pell.eq_pow_of_pell`, lines 860–864, states in its nonzero-base, nonzero-index

branch that $B_0^{k_0} = m_0$ is equivalent to natural witnesses w, a, Θ, g , with $a > 1$, satisfying

$$\begin{aligned} x_{k_0}(a) &\equiv y_{k_0}(a)(a - B_0) + m_0 \pmod{\Theta}, \\ 2aB_0 &= \Theta + (B_0^2 + 1), \quad m_0 < \Theta, \quad B_0 \leq w, \quad k_0 \leq w, \\ a^2 - ((w + 1)^2 - 1)(wg)^2 &= 1. \end{aligned}$$

The source writes these expressions in natural arithmetic. Its constructive proof is at lines 865–897 and reverse proof at 898–928. Equations (P10)–(P15) encode these conditions. The source theorem’s modulus is here named Θ ; it must not be confused with the Pell coordinate t in (P3) and (P8).

217.3 Integer subtraction and congruence adapters

For naturals X, Y , the truncated-natural equation $X - Y = 1$ is equivalent to the integer equation $X = Y + 1$, since the positive result forces $X > Y$. This converts all three ordinary Pell equations and (P13) in both directions. The inner quantities $a^2 - 1$, $\beta^2 - 1$, and $(w + 1)^2 - 1$ are nonnegative under their domains.

The remaining difference $a - B_0$ must also be checked. Equations (P10) and (P13), together with $g > 0$, give $w \geq B_0 \geq 2$ and

$$a^2 = 1 + (w^2 + 2w)w^2g^2 > w^2.$$

Thus $a > w \geq B_0$, so integer subtraction agrees with the theorem’s natural subtraction before the power conclusion is invoked. Conversely, a natural witness $g = 0$ from the constructive theorem would force $a^2 = 1$, contradicting $a > 1$. Hence restricting g to be positive loses no witness.

For any positive modulus d , the equation

$$p + dr_1 = q + dr_2, \quad r_1, r_2 \in \mathbb{N},$$

is equivalent to $p \equiv q \pmod{d}$. If $p - q = dz$, choose $r_1 = \max(-z, 0)$ and $r_2 = \max(z, 0)$. This encodes either quotient sign, including zero. The four pairs in (P5), (P7), (P8), and (P15) are all retained. In particular $\beta - a = u(\alpha_2 - \alpha_1)$, with the same orientation as the corresponding source congruence.

Equation (P12) has a positive J_p , not a natural slack. It therefore supplies the strict modulus bound $m_0 < \Theta$. The source reverse proof uses this strictness to identify congruent natural numbers; replacing it by a weak inequality is not justified.

217.4 Both directions and exponent zero

Proof of Lemma 217.1. Soundness. From (P1)–(P9), the domains, and the exact subtraction conversion, the nonzero branch of `Pell.matiyasevic` applies at $k_0 = e + 1 \geq 1$. It yields

$$x = x_{k_0}(a), \quad y = y_{k_0}(a).$$

Equations (P10)–(P15) then provide all conditions of the positive branch of `Pell.eq_pow_of_pell`, including $m_0 < \Theta$ and the justified natural subtraction $a - B_0$. Thus

$$B_0^{e+1} = m_0 = B_0 o.$$

Cancel positive B_0 to obtain $o = B_0^e$.

Completeness. If $o = B_0^e$, set $k_0 = e + 1$ and $m_0 = B_0 o = B_0^{k_0}$. The constructive power theorem supplies natural w, a, Θ, g with its conditions. Here $w \geq B_0 \geq 2$, $\Theta > m_0 > 0$, and $g > 0$ as shown above. Choose the positive leaf $a_{-1} = a - 1$, positive $J_p = \Theta - m_0$, and natural $d_{wb} = w - B_0, d_{wk} = w - k_0$.

Take $x = x_{k_0}(a)$ and $y = y_{k_0}(a)$ and apply the constructive Pell-pair theorem. The zero branch is impossible because $k_0 \leq y$ and $k_0 \geq 1$. It supplies natural u, v, s, t, β in the nonzero branch. Their stronger positive requirements follow: $y \geq k_0 \geq 1$; the Pell equations make $x, u, s > 0$; $v > 0$ is explicit in the theorem; and $\beta > 1$ gives a positive β_{-1} . Divisibility $y^2 \mid v$ gives a positive q_v , and $\beta \equiv 1 \pmod{4y}$ with $\beta > 1$ gives a positive q_b . Finally $t \equiv k_0 \pmod{4y}$ excludes $t = 0$, because $1 \leq k_0 \leq y < 4y$.

Set $d_{yk} = y - k_0 \in \mathbb{N}$ and choose the four natural quotient pairs as above. Replace all natural quantities by their positive-adaptor leaves. This gives all required positive leaves simultaneously. $\square$

At exponent $e = 0$, the shifted index is $k_0 = 1$. Both proofs still use only the positive-index theorem branches; cancellation yields $o = 1$. No selector or separate exponent-zero case is introduced in the source.

In the actual certificate, the first base is 5. The second base is

$$3 + 4\beta_{\text{rad}} = 3 + 4(4P + 1) = 16P + 7 \geq 23,$$

because P is a positive leaf even before the first module's semantics is used. Both modules use the same natural exponent n . Thus every hypothesis is justified without circular reasoning.

218 Paid decoding to one positive input

For an external input $z > 0$, introduce positive gap witnesses g_1, g_2, g_3 and one natural witness j . Put $a = g_1 - 1, b_0 = g_2 - 1, c = g_3 - 1$; the temporary name b_0 here is unrelated to the extraction radix β_{rad} . Add

$$2j = (b_0 + c)(b_0 + c + 1) + 2c, \tag{D1}$$

$$2(z - 1) = (a + j)(a + j + 1) + 2j. \tag{D2}$$

All four quantities are represented by positive leaves, with $j = j^+ - 1$. These are two ordinary integer polynomial equations, not a pairing-function primitive or a promise that the code is legal.

For any $N \in \mathbb{N}$, there is a unique diagonal $d \in \mathbb{N}$ such that

$$\frac{d(d+1)}{2} \leq N < \frac{(d+1)(d+2)}{2}.$$

The offset $s = N - d(d+1)/2$ satisfies $0 \leq s \leq d$, giving the unique pair $(d-s, s)$ with Cantor value N . Apply this first to $N = z - 1$, obtaining (a, j) , and then to j , obtaining (b_0, c) . Consequently every positive input decodes to exactly one positive gap triple. Conversely every positive triple gives a positive code satisfying both equations.

Adding these residuals to the sum of squares and using Theorem 211.1 proves Theorem 211.2. The uniqueness of decoded gaps is only a property of the transport. It does not imply uniqueness of the other existential witnesses.

The decoder contributes 17 body gates: 6 multiplications, 6 additions, and 5 subtractions. Two extra residual differences, their two squares, and two additions to the existing sum add 6 gates. Thus the exact increase is four witnesses, two equations, and 23 gates:

$$8 \text{ multiplications} + 8 \text{ additions} + 7 \text{ subtractions.}$$

The decoder residuals have degree two, so they do not alter the degree-twelve leading form.

219 Literal arithmetic ledger and exact degree

The cost convention is literal: every gate is binary addition, subtraction, or multiplication; fixed integer coefficients are free leaves; multiplication by a fixed coefficient is charged. Named aliases only point to existing expressions. The count concerns the pinned expression DAG, not a minimum among equivalent circuits.

Native linear region	Leaves	Eq.	$\times$	$+$	$-$	Gates
Radius and U, V arithmetic	1	1	12	6	4	22
Primitive Gaussian fraction	12	4	7	2	5	14
Natural valuation exponent	1	0	0	0	1	1
First power system	26	15	31	24	15	70
Valuation and positive residue	4	3	2	2	1	5
Extraction bases	0	0	2	2	0	4
Second power system	26	15	31	24	15	70
Bounded complex extraction	10	5	3	3	11	17
Strict acceptance	1	1	4	4	2	10
Residuals, squares, and sum	0	0	44	43	44	131
Total	81	44	136	110	98	344

The last row of operations uses one subtraction per residual, one multiplication per square, and $E - 1$ additions to sum E squares. Its size is $3E - 1$, giving 131 gates at $E = 44$. The independent audit checked this literal tail topology, in addition to polynomial equality, so no residual is silently omitted, duplicated, or given an unchecked weight.

Proposition 219.1 (Exact leading homogeneous form). *For both primary sources, the total degree is exactly 12. In the actual independent positive leaves, its homogeneous degree-twelve part is*

$$w_5^8 g_5^4 + w_c^8 g_c^4, \quad (219.1)$$

where the subscripts distinguish the two power systems.

Proof. Every outer residual has degree at most three, and every power residual has degree at most six. Each (P13) residual has leading term $-w^4 g^2$, since

$$a^2 - 1 - ((w + 1)^2 - 1)(wg)^2$$

has that unique degree-six term. The other residuals have smaller degree. Squaring and summing therefore gives exactly (219.1), with both coefficients equal to +1. The two monomials use distinct power-module leaves, so this homogeneous part is nonzero. The decoder residuals have degree two. Hence the joint degree in inputs and positive witness leaves is exactly twelve in both cases. $\square$

The independent exact expansion confirmed the entire leading form, not only a syntactic upper bound. It also found every declared input and witness in the collected polynomial support. Graph liveness and algebraic liveness both hold; these are different checks.

219.1 An optional witness and gate tradeoff

The optional quartic source replaces (O8) with

$$(s - 1)(s - 2)(s - 3)(s - 4) = 0$$

and drops the positive leaf t . Over integer witnesses it enforces precisely the same four possible values of s . It saves one witness but costs six gates: three more multiplications, one fewer addition, and four more subtractions. The exact degree and equation count are unchanged.

Source	Inputs	Wit.	Eq.	$\times$	$+$	$-$	Gates	Terms
Native linear	3	81	44	136	110	98	344	772
One-input linear	1	85	46	144	118	105	367	802
Native quartic	3	80	44	139	109	102	350	775
One-input quartic	1	84	46	147	117	109	373	805

The linear source is the primary presentation because its range proof is simpler. The quartic source is a separately audited tradeoff, not a new record or a minimality assertion. “Terms” counts nonzero monomials after exact coefficient collection, not gates.

220 Independent audit evidence and adversarial checks

The fresh independent audit [84] returned PASS with no required change to the frozen certificate. It independently reconstructed every equation residual as a sparse multivariate integer polynomial and compared it with every raw DAG. It also expanded the final output and compared it with the sum of the independently specified residual squares. The author emitter, author semantic checker, and prior source checker were not used as the independent specification and were not executed during that audit or this release preparation.

The raw-source audit checked leaf declarations, integer constants, permitted operations, topological references, all named residuals, the literal assembly tail, all graph dependencies, and collected algebraic support. Its receipts contain canonical expanded-polynomial hashes, each residual’s degree and hash, region ledgers, and the exact degree-twelve form. Source-copy hashes and byte lengths were checked against the supplied pin manifest.

220.1 Exact arithmetic tests and their limits

The independent arithmetic audit uses only integer arithmetic and exact rational fractions. Its finite challenge set includes

- 15,625 positive gap triples with coordinates from 1 through 25;
- both orbit orientations through exponent 100 at scales 1, 2, and 7, giving 606 cases;
- 820 rational critical-circle parameter cases;
- 12,500 primitive-denominator and three-term Bézout cases, including zero and negative components;
- exhaustive bounded extraction at $n = 0, 1, 2, 3$, with $P = 1, 5, 25, 125$;

- 20,000 positive-code round trips and 8,000 positive-gap round trips.

The comparisons use the already-proved physical disk-minus-inverse-orbit formula. They are not physical trajectory simulations and do not independently replace the physical proof.

Two orientation fixtures are especially useful:

Gaps	δ	u	v	q	n	J_{acc}	Outcome
(217, 167, 231)	0	1	0	1	0	0	Rejected contact
(233, 171, 211)	0	3	4	5	1	64	Accepted forward point

220.2 Malformed candidates identify essential guards

At the rejected contact, take $n = 0$ but falsely choose $(C, S) = (27, 0)$. With $P = 1, \beta_{\text{rad}} = 5, T = 1$, equation (O13) still holds for $\kappa = -1$, and the final square sum is 676. The upper bound blocks it because $P - C = -26$. The independent audit constructed full positive leaves for all other equations and evaluated each raw DAG: the only failed residual was `complex.C_upper`, equal to -26 , with total output 676.

If primitive Bézout were omitted, the contact could be represented by $(u, v, q) = (5, 0, 5)$, with $n = 1, r = 1$ and $(C, S) = (3, 4)$. The false acceptance sum would be 20. Every Bézout combination of $(5, 0, 5)$ is divisible by 5, so the actual primitive equation forbids it.

At the forbidden point $(3 - 4i)/5$, undercounting n as 0 would give $r = 5$ and a false acceptance sum 36. It would require $s = 5, t = 0$, or $s = 0$ after carrying, each violating positivity. Allowing a natural J_{acc} would admit the contact with $J_{\text{acc}} = 0$. Allowing a signed radius slack would admit exterior examples such as $(1, 1, 10)$, where $\Delta = -82449$, despite a positive squared acceptance sum. These examples show why the actual domains cannot be weakened.

220.3 Complete positive Pell witnesses at exponent zero

The independent audit did not merely substitute a claimed power output for these fixtures. It constructed all positive Pell auxiliaries for bases $B_0 \in \{5, 23\}$ and exponent zero. Put $w = B_0$ and choose

$$a = x_w(w + 1), \quad g = y_w(w + 1)/w.$$

The elementary Pell recurrence gives $x_j \equiv 1$ and $y_j \equiv j \pmod{w}$, proving this quotient integral. At shifted index $k_0 = 1$, take

$$x = a, \quad y = 1, \quad u = 2a^2 - 1, \quad v = 2a.$$

Choose $\beta \equiv a \pmod{u}$, $\beta \equiv 1 \pmod{4}$, with $\beta > 1$, then $s = \beta, t = 1$. The remaining quotients and slacks follow from the displayed equations. Here u is odd, so the two congruences for β are compatible. The complete stored witness list has largest entry 78 decimal digits.

For each of the four DAG variants, full assignments were evaluated at the center $(1, 1, 1)$, the accepted boundary point $\gamma = -1$ with gaps $(193, 243, 179)$, and the accepted boundary point $\gamma = i$ with gaps $(231, 191, 193)$. These are twelve complete positive zero evaluations, stored in `independent_audit/full-positive-witnesses.json`. At the rejected contact, setting $J_{\text{acc}} = 1$ leaves only the final residual -1 and output 1. Changing a paid code while holding its decoded gaps fixed gives the expected decoder failure.

Every positive witness was separately incremented by one. All non-neutral mutations failed; the only neutral mutations were Bézout leaves multiplying a zero component. Neutrality in those cases is mathematically expected and was not misreported as a failed test.

The nonzero-exponent orbit and outer-arithmetic tests compute ordinary power outputs directly. They do not instantiate the potentially enormous full Pell auxiliary tuples for all such exponents. The all-exponent equivalence is proved by the pinned constructive theorems and the specialization in Section 217, not by extrapolating finite tests.

[write] [Shipped names: 28-diophantine-independent_audit-INDEPENDENT_AUDIT.md, 28-diophantine-independent_audit-POWER_AUDIT.md, data/28-diophantine-independent_audit-full-positive-witnesses.json, the receipts data/28-diophantine-independent_audit-exact-source-receipt.json, ...-arithmetic-receipt.json and ...-POWER_DAG_RECEIPT.json, and the checkers code/28-diophantine-independent_audit-audit_exact_source.py, ...-audit_arithmetic.py and ...-check_power_dag_independent.py. On copies with their path constants mapped, all three reproduce their frozen receipts up to line endings and path separators.]

221 An elementary obstruction for the paid integer encoding

The following encoding corollary is independent of the certificate’s witness count or degree. It was checked in the independent audit and is included for clarity about the different representations.

Corollary 221.1. *The transported predicate on positive codes cannot agree everywhere with a finite Boolean combination of univariate real-polynomial sign tests.*

Proof. Each univariate real polynomial is either identically zero or has the sign of its leading coefficient at every sufficiently large positive argument. A finite Boolean combination of its sign atoms is therefore eventually constant.

For every positive integer t , the center gaps (t, t, t) are accepted. The contact gaps $(217t, 167t, 231t)$ are rejected by homogeneity and Theorem 212.1. Their respective Cantor codes are unbounded as t grows. Thus arbitrarily large positive codes occur in both outcome classes, contradicting eventual constancy. $\square$

This is an elementary statement about finite sign classifiers under this particular encoding. It establishes no arithmetic-circuit lower bound, undecidability theorem, physical threshold, or obstruction to existential positive-integer witnesses. Theorem 211.2 explicitly supplies such witnesses.

222 Reproducibility and provenance

The release includes the complete frozen science packet and independent audit, byte-for-byte, alongside this editable LaTeX source and its rendered PDF. The physical proof and physical audit are the exact dependencies copied into `science/sources/`; the whole physical paper need not be duplicated in this manuscript. The pinned Pell source and its Apache 2.0 license are also included inertly.

Proof and source checks. The reusable checks concern exact source equality, explicit finite arithmetic challenges, positive witness evaluations, integrity of frozen inputs, and deterministic

rendering. The audit scripts are independent of the author emitter and semantic checker. The included `portable_replay/replay_report58.py` authenticates the complete 33-file science tree and 11-file independent-audit tree before running the three pinned independent checkers. Their source bytes are compiled unchanged in isolated Python with assertions enabled. A narrow logical-path adapter redirects only approved reads and writes, preserving historical path strings in the receipts. All four regenerated receipt and witness files, and the arithmetic checker's stdout, are required to match the frozen references byte-for-byte. The release-copy replay passed those exact comparisons with the original and staged inputs preserved.

The portable adapter's historical external physical-audit lookup is served by the included authenticated physical-audit copy. It therefore rechecks those frozen bytes, rather than independently rereading an external original. The root audit had established the original byte identity. The included portability suite passed 33 cases, including relocated paths with spaces, read-only inputs, malformed and overlapping output paths, symlinks, altered inputs, unexpected inventory entries, and optimized-Python refusal. Python startup isolation and the logical-path adapter are not an operating-system or hostile-code sandbox. The package README and `portable_replay/REPLAY_README.md` give exact commands and further limits.

Output discipline. Release tools use explicit output directories, reject unsafe overwrites or overlap with frozen inputs, and keep temporary caches outside the packet. They must not execute any author emitter, upstream or saved program, physical simulator, trajectory replay, or Lean executable. Byte integrity and successful finite replay are evidence about the delivered files, not a fresh formal verification of the inherited mathematical dependencies.

Determinism. The PDF build uses fixed source-date metadata, disabled shell escape, and clean temporary build directories. The distributable archive uses a sorted file inventory, fixed timestamps, normalized modes, and a manifest of exact file sizes and SHA-256 digests. The archive's own digest is supplied separately to avoid a self-referential manifest. Render inventories and review receipts describe the actual reviewed PDF bytes. Final release status and package hashes are supplied by the accompanying manifest and README rather than being recursively embedded into the PDF.

Assurance boundaries. No generic MRDP theorem is being used as a black-box final representation. No real-witness, finite-fold, unique-witness, minimality, or universal lower-bound claim follows from this package. The complete geometric theorem is inherited; the Pell specialization is conventional mathematics checked against a pinned source, not a new proof-assistant build. Exact finite tests support the explicit all-input proofs.

[write] [In this report: the archive's `science/sources/` copies of source 27's proof, companion and audit are shipped once, with source 27 (`27-fivesignal-science-construction-PROOF.md`, `27-fivesignal-science-companion-BOUNDARY_AND_ARITHMETIC.md`, `27-fivesignal-independent_audit-INDEPENDENT_AUDIT.md`); its three Report 55 files are shipped with *group-theoretic-substrates* (note in Section 217); the Pell source, its licence, the manuscript, PDF and the inventory manifests are not shipped. The portable replay (`code/28-diophantine-portable_replay-replay_report58.py`) and the release tools require explicit canonical absolute POSIX paths and the delivered layout, so they refuse to run here; the README gives a rerun route on a copy.]

223 Both instantiated power systems (source 28, Appendix A)

This appendix removes the generic base and output names from the two fifteen-equation systems. Every symbol with subscript 5 or c denotes a distinct local quantity. The shared exponent is $n \in \mathbb{N}$. In each system $o, w, \Theta, g, x, y, u, v, s, t, q_b, q_v, J_p$ are positive, the parameters a, β are their own positive leaves plus one, and each of the eleven $d, \alpha, \sigma, \tau, \rho$ quantities is a positive leaf minus one. The output is P in the first system and T in the second. These domain substitutions are part of the polynomial specification; no auxiliary domain oracle is used.

223.1 Base five and output P

$$x_5^2 = 1 + (a_5^2 - 1)y_5^2, \quad (5.1)$$

$$u_5^2 = 1 + (a_5^2 - 1)v_5^2, \quad (5.2)$$

$$s_5^2 = 1 + (\beta_5^2 - 1)t_5^2, \quad (5.3)$$

$$\beta_5 = 1 + 4y_5q_{b,5}, \quad (5.4)$$

$$\beta_5 + u_5\alpha_{1,5} = a_5 + u_5\alpha_{2,5}, \quad (5.5)$$

$$v_5 = y_5^2q_{v,5}, \quad (5.6)$$

$$s_5 + u_5\sigma_{1,5} = x_5 + u_5\sigma_{2,5}, \quad (5.7)$$

$$t_5 + 4y_5\tau_{1,5} = (n + 1) + 4y_5\tau_{2,5}, \quad (5.8)$$

$$y_5 = (n + 1) + d_{yk,5}, \quad (5.9)$$

$$w_5 = 5 + d_{wb,5}, \quad (5.10)$$

$$w_5 = (n + 1) + d_{wk,5}, \quad (5.11)$$

$$\Theta_5 = 5P + J_{p,5}, \quad (5.12)$$

$$a_5^2 = 1 + ((w_5 + 1)^2 - 1)(w_5g_5)^2, \quad (5.13)$$

$$2a_5 \cdot 5 = \Theta_5 + (5^2 + 1), \quad (5.14)$$

$$x_5 + \Theta_5\rho_{1,5} = y_5(a_5 - 5) + 5P + \Theta_5\rho_{2,5}. \quad (5.15)$$

The source's local prefix is `power5..` In particular $J_{p,5}$ is its positive `strict` leaf, and $a_5 = \text{aMinus1} + 1$, $\beta_5 = \text{betaMinus1} + 1$ within that namespace.

223.2 The second ordinary power and output T

Here $\beta_{\text{rad}} = 4P + 1$ is the outer expression, not an independent leaf. The second base is $3 + 4\beta_{\text{rad}}$. The complete instance is

$$x_c^2 = 1 + (a_c^2 - 1)y_c^2, \quad (\text{c.1})$$

$$u_c^2 = 1 + (a_c^2 - 1)v_c^2, \quad (\text{c.2})$$

$$s_c^2 = 1 + (\beta_c^2 - 1)t_c^2, \quad (\text{c.3})$$

$$\beta_c = 1 + 4y_c q_{b,c}, \quad (\text{c.4})$$

$$\beta_c + u_c \alpha_{1,c} = a_c + u_c \alpha_{2,c}, \quad (\text{c.5})$$

$$v_c = y_c^2 q_{v,c}, \quad (\text{c.6})$$

$$s_c + u_c \sigma_{1,c} = x_c + u_c \sigma_{2,c}, \quad (\text{c.7})$$

$$t_c + 4y_c \tau_{1,c} = (n + 1) + 4y_c \tau_{2,c}, \quad (\text{c.8})$$

$$y_c = (n + 1) + d_{yk,c}, \quad (\text{c.9})$$

$$w_c = (3 + 4\beta_{\text{rad}}) + d_{wb,c}, \quad (\text{c.10})$$

$$w_c = (n + 1) + d_{wk,c}, \quad (\text{c.11})$$

$$\Theta_c = (3 + 4\beta_{\text{rad}})T + J_{p,c}, \quad (\text{c.12})$$

$$a_c^2 = 1 + ((w_c + 1)^2 - 1)(w_c g_c)^2, \quad (\text{c.13})$$

$$2a_c(3 + 4\beta_{\text{rad}}) = \Theta_c + ((3 + 4\beta_{\text{rad}})^2 + 1), \quad (\text{c.14})$$

$$x_c + \Theta_c \rho_{1,c} = y_c(a_c - (3 + 4\beta_{\text{rad}})) + (3 + 4\beta_{\text{rad}})T + \Theta_c \rho_{2,c}. \quad (\text{c.15})$$

The source's local prefix is `power_complex.`; it remains a system for an ordinary integer power. Complex-number notation appears only in the proof of bounded extraction. There are no complex-valued source leaves or complex arithmetic primitives.

224 Selected exact byte pins (source 28, Appendix B)

These pins identify the frozen scientific objects. Paths are relative to the release root. The complete source provenance is in `science/sources/SOURCE_PINS.json`, the science inventory in `science/PACKET_MANIFEST.json`, and the independent audit inventory in `independent_audit/AUDIT_MANIFEST.json`.

Object	SHA-256
<code>science/PROOF.md</code>	<code>b55f301d2b1531f348163f026b3c1f7f827669d266afb77d854eef7ceb21c482</code>
<code>science/PACKET_MANIFEST.json</code>	<code>8e2d450010e9029d500bfd2dec67995ec61d54ab08665d449c3b9d18143c965b</code>
<code>science/sources/five-signal-PROOF.md</code>	<code>1bea81f8f2e6693d4b7958fb44763eb644663d208fe70c577e0137184e17a725</code>
<code>science/sources/INDEPENDENT_AUDIT.md</code>	<code>b119df1c0be8685078b011c14e3c71decabce343f9ba9041ee6f97b4c58a5226</code>
<code>science/sources/BOUNDARY_AND_ARITHMETIC.md</code>	<code>9642699a09f3ebc558fb1268c61370fb514839404e834acf288f9504e3e81b77</code>

Object	SHA-256
science/sources/pell-source.lean	993760c797ad0ff66fa77064a616fce779550bc745cbf6c44e04f392fd0bed0a
science/sources/SOURCE_PINS.json	11d2372f087c5fb75fd23473d09839b595fe7ec5484cbbb9cb55bd202f203a48
science/evidence/three-input-linear.dag.json	707721d1a8dca21b29acda57df8df6e763a1df89761f513122bec5edd9261ccc
science/evidence/one-input-linear.dag.json	b64a268cc03033c13c020abc0cb12c1c81f9ba630aa75812ac84b08e2971efc3
science/evidence/three-input-quartic.dag.json	11c1a3d6eca23af2a9ede232e1a5ab36ba395a3235a424ef76ff14d5e19f14d0
science/evidence/one-input-quartic.dag.json	3d2fe384ea6564950e0ca4b392396e3fa4489d88e1974dfd269a42610428ed24
independent_audit/INDEPENDENT_AUDIT.md	e790c49af8d56190740520637af65d436e75f7af2a30856590646f3bf6372dbd
independent_audit/POWER_AUDIT.md	90eab5e5b8f1fc369805e4722a0bc6cbc6534cb9997a2038095ae44157be1df3
independent_audit/AUDIT_MANIFEST.json	23e0751b30845d4925cd4ebb884ed537465715a0d184456d775a76ebfcac30ba
independent_audit/exact-source-receipt.json	abf4b13afb1d2b839255f9fc49fdda26c7cfbadb51042d99bdf657cc3adb7f88
independent_audit/full-positive-witnesses.json	6347a9f792c9f0cf198cef8cd0d4c5de6acfa036fa2917e7a3c05889ab47138c

[write] [Every pinned object except `pell-source.lean` is shipped under the prefix `28-diophantine-` (or, for the three `science/sources/` copies, under `27-fivesignal-`); the receipt `exact-source-receipt.json` was regenerated on a copy, identical up to line endings and the separator in four displayed paths.]

[write] **The programme’s intake of source 28.** The review [99] calls it “Genuine explicit ordinary-integer arithmetic and paid decoding, but the represented predicate is decidable rotation validity, not universal halting”, writes the native ledger as “344 gates = 136M+208A” (additions and subtractions together) and the paid one as “367 gates = 144M+223A”, and adds that “The counts are reported source ledgers, not independently rebuilt in this intake.” No correction is asked for, and no universal operation bound follows: the project’s 84-operation universal bound is untouched.

Source 29 (batch-91 manuscript 11): Five signal rational rotation family

Exact collision chronology, boundary geometry, rational decision and integer certificates, “REPORT 59”, 4 October 2026. [write] From the archive `Five_Signal_Rational_Rotation_Family_Source.zip` (manuscript `manuscript/report59.tex`, the final reviewed edition; the review’s three earlier editions are not shipped, and its edit history is shipped as `code/29-rotations-manuscript-review-SOURCE_CHANGES.diff`); its labels carry the prefix `smc:rf:`, its sections are Sections 225–237 here, and its equations are numbered by section. Its fixture table and geometry figure were `\input` files generated by `code/29-rotations-tools-derive_presentation.py`; they are printed in place (the generator reproduces them byte for byte up to line endings). Its bibliography entries are this report’s [87], [88], [16], [6] (its `agc8`), [79], [76], [77], [86], [89], [90] and [91]. Its Euclidean norm $\|\cdot\|$ is written with `\enorm`, its Gaussian quotients η_j are written γ_j , and its Pell modulus M is written Θ (opening of Part X); J is its number of contacts throughout. Its abstract follows.

For every rational planar rotation of infinite order and every positive rational scale factor, we give an explicitly constructible finite rational-speed, number-preserving signal machine with exactly five live signals. A fixed complete collision word realizes the scaled rotation. Its exact finite-word domain is a rational open polygon; infinite validity is an open disk together with its critical circle after finitely many backward rotation orbits are deleted. This yields real, rational and positive-integer finite polynomial-sign obstructions, while rational membership for each fixed machine has a conservative deterministic $O(L^3)$ bit bound. Every valid infinite run is Zeno exactly when the scale factor is below one, with a rational linear clock on rational inputs. A literal integer certificate has $1 + 80J$ positive witnesses, $1 + 43J$ equations and exact sum-of-squares degree twelve, where J is the number of distinct nearest contacts. All fifteen equations of each POWER module are displayed and its constructive Pell dependency is identified.

Verification status. The frozen conventional proof and static physical constructor passed a separate adversarial audit. Its independent reconstruction matched all fields of 45 fixtures: 13,572 rules, 3,336 guards and 2,976 phases. The general theorem rests on the parameterized chronology and loop invariants, not on extrapolation from these fixtures.

Scope. The implemented object is a static physical constructor. The arithmetic result is an explicit finite formula, with no emitted general arithmetic DAG or exact gate-ledger claim. No author constructor, author checker, upstream program, physical simulator, saved collision schedule or Lean process is run by this release. This report is separate from Reports 57 and 58 [sources 27 and 28] and makes no priority, optimality, finite-fold or post-accumulation claim.

Full source and evidence accompany the report. The rules remain in machine-readable fixtures rather than being printed as thousands of rows.

225 Theorems and quantifiers

Let $a, b, c \in \mathbb{Z}$ satisfy

$$c > 1, \quad a^2 + b^2 = c^2, \quad \gcd(a, b, c) = 1, \quad R = \frac{1}{c} \begin{pmatrix} a & -b \\ b & a \end{pmatrix}. \quad (225.1)$$

These are exactly the rational planar rotations of infinite order. Fix any $\lambda \in \mathbb{Q}_{>0}$. The machine and its complete word are fixed *after* these parameters are chosen; only the live population is uniformly five. Word length, alphabet and rule table need not be uniform in the rotation.

Theorem 225.1 (Physical realization and exact domain). *There is an explicitly constructible finite rational-speed number-preserving one-dimensional signal machine with exactly five live signals and a fixed complete binary-collision macro whose return in rational homogeneous coordinates is*

$$(D, X, Y) \mapsto N(D, X, Y), \quad N = \text{diag}(\lambda, \lambda R).$$

Its exact one-macro domain on the normalized section $D = 1$ is a bounded open rational polygon P containing zero. If ρ is the least distance from zero to its supplied nonconstant guard lines, and $E = \{p_1, \dots, p_J\}$ is the set of distinct perpendicular contacts of all minimizing lines, then $J \geq 1$, $\rho^2 \in \mathbb{Q}_{>0}$, $E \subset \mathbb{Q}^2$, and the exact normalized infinite validity set is

$$\mathcal{K} = \bigcap_{n \geq 0} R^{-n} P = \{w : \|w\| < \rho\} \cup \left(\{w : \|w\| = \rho\} \setminus \bigcup_{j=1}^J \{R^{-n} p_j : n \geq 0\} \right). \quad (225.2)$$

No uniqueness, disjoint-orbit or noninitial-face assumption is imposed on the contacts.

The input section has four stationary markers at $0, x, y, D$ with $0 < x < y < D$ and a right-moving messenger at the left anchor. Its three free positive gaps are

$$g = (g_1, g_2, g_3) = (x, y - x, D - y), \quad X = x - D/3, \quad Y = y - 2D/3, \quad w = (X/D, Y/D).$$

The messenger and left anchor are the two outgoing strands at their contact; they count as separate live signals.

[write] [Source 29's (X, Y) are source 27's centred coordinates (ξ, η) , and the section is source 27's. Theorem 225.1 contains the shape of source 27's validity set (Theorem 196.2) as the case of one contact, for a different machine (Section 235). Part XI's Report 60 [batch-91 manuscript 12] extends the realization to every orientation-preserving rational planar map and carries this source's frozen proof as a dependency.]

Theorem 225.2 (Classification, decision and certificate). *The positive homogeneous gap cone of infinite validity is not semialgebraic. No finite Boolean formula in polynomial-sign atoms, even with arbitrary real coefficients, agrees with validity on every positive rational gap triple or every positive integer gap triple. Nevertheless, for each fixed compiled machine, rational membership is decidable in deterministic $O(L^3)$ bit operations under elementary arithmetic, where L is the total input encoding length.*

For ordinary positive integer gaps, validity has an explicit existential-positive-integer certificate with exactly $1 + 80J$ witnesses and $1 + 43J$ polynomial equations. Their sum of squares has exact total degree twelve. An optional paid one-input Cantor transport uses $5 + 80J$ witnesses and $3 + 43J$ equations, still of exact degree twelve.

On all positive real gaps, validity is G_δ but not F_σ , excludes finite existential mixed integer/real polynomial-sign certificates, and is properly co-semidecidable in the finite-time field-operation BSS model of Section 231.

Every infinitely valid run is Zeno if and only if $0 < \lambda < 1$. Then all strands approach the fixed left anchor and the accumulation time is rational on rational input.

The finite-sign obstruction has no quantified integer witnesses. It therefore does not contradict the displayed Diophantine representation. The rational decision bound fixes the machine and its contacts; it is neither a uniform compilation bound nor a bound on finding Pell witnesses.

226 Rotation parameters and shear synthesis

Primitivity implies $a, b \neq 0$, c odd and $\gcd(a, c) = \gcd(b, c) = 1$. Indeed, a common prime divisor of a leg and c divides both legs. If c were even, the parity equation would either make both legs even or give $2 \equiv 0 \pmod{4}$.

Put $\zeta = (a + ib)/c$. If it were a root of unity, then $\zeta + \zeta^{-1} = 2a/c$ would be a rational algebraic integer and therefore an integer. Thus $c \mid 2$, contrary to odd $c > 1$. Both the forward and backward powers of R are dense in the planar rotation group. Conversely, the rational finite-order rotations have entries 0 or ± 1 and primitive denominator one. This proves the parameter coverage in (225.1).

Define

$$\begin{aligned} t_x &= -\frac{b}{c+a}, & t_y &= \frac{b}{c}, & (226.1) \\ N_x &= \max(1, \lceil 4|t_x| \rceil), & N_y &= \max(1, \lceil 4|t_y| \rceil), \\ \delta_x &= t_x/N_x, & \delta_y &= t_y/N_y, & K_0 &= 2N_x + N_y. \end{aligned}$$

Since $|a| < c$, $c + a > 0$ even when $a < 0$. The step sizes are nonzero and have absolute value at most $1/4$. For

$$S_x(t) = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}, \quad S_y(s) = \begin{pmatrix} 1 & 0 \\ s & 1 \end{pmatrix},$$

one obtains

$$S_x(t_x)S_y(t_y)S_x(t_x) = \begin{pmatrix} 1 + t_x t_y & 2t_x + t_x^2 t_y \\ t_y & 1 + t_x t_y \end{pmatrix} = R. \quad (226.2)$$

Chronologically performing x , then y , then x shears gives this product; its equal outer factors remove any apparent convention reversal. Total shear size need not be small. The number of micro-shears grows while each step remains small.

227 Five signals and complete primitive chronology

During a primitive only one marker moves, with speed strictly between -1 and 1 . The messenger has speed ± 1 ; the remaining markers are stationary. Every spectator crossing belongs to the prescribed word. A complete word records all collisions, including any collision assigned the default identity rule.

[write] [The two primitives and the translation below are source 27's (Lemmas 198.1 and 198.2 and (198.3)), proved again in source 29's words; source 29 credits translation and scaling to the literature (Section 236).]

227.1 Single marker scaling

For $u > 0$, the primitive $L_z(u)$ begins with a target at anchor-distance $z > 0$. At the first messenger hit give it speed

$$h = \frac{u - 1}{u + 1}.$$

Reflect the messenger left, reflect it at the anchor, restore the target to speed zero on the next hit, reflect left again, and return to the anchor. The four principal time–position pairs are

$$(z, z), \quad (2z, 0), \quad (2z + uz, uz), \quad (2z + 2uz, 0). \quad (227.1)$$

The target line $q = z + h(t - z)$ meets the restoration point because $h(1 + u) = u - 1$. Thus the endpoint is uz and the duration is $2(z + uz)$.

The exact guard requires both z and uz to lie strictly between the unchanged nearest stationary neighbors, with no upper-neighbor test for an outermost target. To prove sufficiency, the target moves monotonically between these endpoints, so cannot collide with a stationary marker. An inner marker $s < \min(z, uz)$ is met on the four legs at

$$s, \quad 2z - s, \quad 2z + s, \quad 2z + 2uz - s. \quad (227.2)$$

These are ordered by increasing, decreasing, increasing and decreasing spatial order, respectively. The target-end gaps are $z - s_{\max}$ and $uz - s_{\max}$; the anchor-end gaps are positive marker coordinates. No outer marker is reached by the messenger. Since $|h| < 1$, messenger–target relative speeds have fixed strict signs on each leg, so there is no unlisted recollision. All other marker pairs are stationary and distinct. This also rules out a remote simultaneous collision.

For necessity, equality at a checked neighbor gives an extra or coincident contact by that checkpoint; passage beyond the neighbor forces one earlier by continuity. Equality with the anchor collapses required timing. Even an identity-default contact violates the selected complete word.

227.2 Reflector homothety

The target at t is the nearest marker to the anchor, and a stationary reflector lies at $z > t$. For $v > 0$, $H_z(v)$ launches the target at speed $h = (1 - v)/(1 + v)$ while keeping the messenger outward. The messenger crosses every intervening spectator, reflects at z , crosses the spectators in reverse order, restores the target while continuing inward, then reflects at the anchor. Solving the two world lines gives

$$t' = vt + (1 - v)z, \quad t_{\text{restore}} = 2z - t', \quad \text{duration} = 2z. \quad (227.3)$$

The exact guard is $0 < t, t' < s$, where $s \leq z$ is the unchanged nearest outer marker. A spectator q is met at times q and $2z - q$. The inequalities $t, t' < q$ place its crossings strictly after launch and before restoration; $0 < t' < z$ places restoration strictly between reflector and final anchor bounces. With no spectator the same argument uses $q = z$ at the endpoints. Monotonicity excludes target-spectator contacts, and the fixed relative speeds exclude extra messenger hits. Failure yields an extra or tied contact or collapsed chronology as above.

The nearest-target hypothesis is essential. It holds in every use below: X is targeted from L , or Y from the reflected anchor D . No H call omits a stationary marker between its anchor and target.

227.3 Translation and reflection

For $e < 1$ compose

$$T_z(e) = L_t(1/(1 - e)) ; H_z(1 - e).$$

The target positions and common moving speed are

$$t \mapsto \frac{t}{1 - e} \mapsto t + ez, \quad h = \frac{e}{2 - e}. \quad (227.4)$$

Suppose $0 < t, t + ez < s \leq z$. If $e > 0$, $t + ez < z$ implies $t < (1 - e)z$, and

$$\frac{t}{1 - e} - t = \frac{et}{1 - e} > 0, \quad t + ez - \frac{t}{1 - e} = \frac{e((1 - e)z - t)}{1 - e} > 0.$$

If $e < 0$, $t < z$ makes both differences negative. At $e = 0$ all positions agree. The intermediate endpoint is therefore between the checked start and finish, and the target is monotone across both moving phases and its intervening pause. The translation has precisely the start/end ordering guard, with no hidden extra inequality.

Reflection in D changes a coordinate q to $D - q$, takes D as anchor and negates every physical velocity. The chronology and exact-guard results are unchanged.

228 The macro and its exact finite chamber

A left-based pair

$$T_y(\delta) ; T_D(-2\delta/3) \quad (228.1)$$

updates x to $x + \delta(y - 2D/3)$, leaving y, D fixed. The near translation uses eight events: four in L and four in H with reflector Y . The far translation uses ten: four in L and six in H with spectator Y and reflector D . Thus each micro-shear has eighteen events.

Repeat (228.1) N_x times with δ_x . Transfer the messenger to D by crossing X , crossing Y , then reflecting at D . In reflected coordinates $r = D - y$, $s = D - x$, repeat $T_s(\delta_y); T_D(-2\delta_y/3)$ N_y times. Since

$$r' = r + t_y(s - 2D/3), \quad y' = y + t_y(x - D/3),$$

this gives the centered $S_y(t_y)$ shear with the correct sign. Transfer back by crossing Y , crossing X , reflecting at L , then repeat the first x block. The rotation word has

$$m_{\text{rot}} = 18K_0 + 6. \quad (228.2)$$

All translation parameters satisfy $e < 1$: $|\delta| \leq 1/4$ and $|2\delta/3| \leq 1/6$.

228.1 Endpoint rows and necessity

For an x block with n pairs and incoming x, y, D , put

$$\begin{aligned} a_{2k} &= x + k\delta(y - 2D/3) & (0 \leq k \leq n), \\ a_{2k+1} &= x + k\delta(y - 2D/3) + \delta y & (0 \leq k < n). \end{aligned} \quad (228.3)$$

Its exact guards are

$$0 < y < D, \quad 0 < a_i < y \quad (0 \leq i \leq 2n). \quad (228.4)$$

For the reflected block use the same expressions with (x, y) replaced by (r, s) , obtaining $0 < s < D$ and $0 < b_i < s$. Each block supplies $4n + 4$ strict homogeneous rational linear rows. Pull back the three sets through their preceding rational block maps. There are exactly

$$4K_0 + 12 \quad (228.5)$$

supplied rows in (D, X, Y) . They may include redundant or repeated rows. The ordered initial rows imply $D > 0$, also understood as the ambient section assumption.

The primitive lemmas prove sufficiency by induction. For necessity on a positive initial section, take the first failed endpoint checkpoint. Previous primitives have their prescribed order. By the checkpoint the target must reach or cross a fixed neighbor, or two required event times collapse, forcing an extra or simultaneous contact. A failed initial-order row means the input is outside the required section. Thus these rows describe the whole complete-word domain, not merely a convenient sufficient neighborhood.

At the center $x = D/3$, $y = 2D/3$, every pair returns its target to anchor-distance $D/3$. Its odd endpoint is

$$D/3 + 2\delta D/3 \in [D/6, D/2] \subset (0, 2D/3).$$

Every row is therefore strictly positive at the center, for arbitrarily many steps and either sign of δ . The normalized polygon P is open, convex and rational with zero in its interior. Initial-order rows place it inside $0 < x < y < 1$, so it is bounded.

228.2 The loop invariant for every parameter

This proof links the arbitrary-parameter construction to the implemented static constructor. It is not a finite-fixture inference. At entry to pair k of an anchor-based block, the invariant is:

1. The target row equals $t_0 + k\delta(z - 2D/3)$; the fixed reflector row z and outer anchor-distance D are unchanged
2. All four markers are stationary, and the messenger is at the same anchor in the same outgoing direction
3. The accumulated phase word includes every principal and spectator event, and its current target endpoints satisfy all preceding strict checkpoints
4. The duration row is the sum of the rational homogeneous primitive duration rows already appended

The near translation appends $t_k + \delta z$; the far translation appends t_{k+1} . Their exact endpoint guards and primitive restoration prove the invariant for $k + 1$. It holds at $k = 0$ by the incoming section. This proves all block maps, rows, durations and counts for arbitrary finite n .

The transfers are valid by the terminal strict order, take duration D each and reverse direction at the opposite anchor. Combining the invariant for the first x , reflected y and second x blocks proves (226.2) as an exact return. The center positivity argument proves the constructor's guard assertions mathematically; the assertions are not used as a substitute for that argument.

229 Scaling, finite rules and the five live population

For $0 < \lambda < 1$, append $L_x(\lambda)$, $L_y(\lambda)$, $L_D(\lambda)$. Each inner neighbor has already been scaled and each outer neighbor is unscaled. If $s < z$, then $\lambda s < \lambda z$, so a target moving from z to λz stays above its scaled inner neighbors and below unscaled outer neighbors. For $\lambda > 1$ reverse the order to D, Y, X : every outer neighbor is already scaled and each inner neighbor remains unscaled, giving the analogous strict order proof. For $\lambda = 1$ omit scaling.

The X, Y, D scaling primitives use respectively 4, 8, 12 events regardless of execution order. Their speed is $(\lambda - 1)/(\lambda + 1) \in (-1, 1)$; they introduce no new guard and restore the complete outgoing section. If $x_{\text{rot}}, y_{\text{rot}}$ denote postrotation coordinates, their duration is

$$2(1 + \lambda)(x_{\text{rot}} + y_{\text{rot}} + D).$$

Consequently

$$m = 18K_0 + 6 + 24\mathbf{1}_{\{\lambda \neq 1\}}, \quad N = \text{diag}(\lambda, \lambda R). \quad (229.1)$$

In ordinary gaps the return matrix is

$$M = \frac{\lambda}{3c} \begin{pmatrix} c + 2a - b & c - a - b & c - a + 2b \\ c - a + 3b & c + 2a & c - a - 3b \\ c - a - 2b & c - a + b & c + 2a + b \end{pmatrix}. \quad (229.2)$$

With

$$Q = \begin{pmatrix} 1 & 1 & 1 \\ 2/3 & -1/3 & -1/3 \\ 1/3 & 1/3 & -2/3 \end{pmatrix}$$

one has $QM = NQ$. Negative entries of M outside the chamber do not assert a physically executable return there.

229.1 Explicit finite grammar

Give every pre-event messenger state a distinct label Q_j for $0 \leq j < m$, with its specified speed ± 1 , and identify $Q_m = Q_0$. Each of the $4K_0$ elementary L/H pieces in the translations gets a unique temporary target label; add three temporary labels when $\lambda \neq 1$. Keep stationary labels L_0, X_0, Y_0, D_0 . At event j emit

$$\{Q_j, Z_{\text{in}}\} \longrightarrow \{Q_{j+1 \bmod m}, Z_{\text{out}}\}. \quad (229.3)$$

Launch and restoration switch between stationary and temporary target labels; spectator and bounce events keep the stationary identity. In L , reverse messenger direction at all four principal events. In H , preserve direction at launch/restoration and reverse at reflector/anchor. Include all spectator crossings. Each transfer reverses direction only at its final anchor.

Every input and every output has pairwise distinct speeds because marker speeds have modulus less than one and messenger speeds are ± 1 . Each Q_j occurs in only one explicit input set, so prescribed rules cannot conflict. At a binary contact the incoming local order is decreasing speed and outgoing local order increasing speed; strands separate immediately. The strict endpoint proof keeps all other markers away and excludes remote collisions.

Assign identity to every other collision input set with pairwise distinct speeds. The alphabet is finite, hence this is a finite deterministic number-preserving completion. Default rules cannot hide extra contacts from the complete word. Each explicit rule consumes and emits one messenger and one marker of the same identity. Thus the live population is always five, although the meta-signal count is

$$\#\text{meta-signals} = 22K_0 + 10 + 27 \mathbf{1}_{\{\lambda \neq 1\}}. \quad (229.4)$$

A meta-signal is a type; a live signal is an instance. Neither is the count of distinct speeds.

229.2 Source-level invariant coverage

The audited constructor is 158 lines. Its exact rational linear-row representation and finite loops have the following all-parameter interpretation:

- L expands increasing inner-marker order, launch, decreasing order, anchor bounce, increasing order, restoration, decreasing order, anchor bounce. It adds $4 + 4k$ events for k inner spectators and one temporary label
- H expands launch, forward spectators, reflector, reverse spectators, restoration, anchor. It adds $4 + 2k$ events and one temporary label; reflection negates all velocities
- Every translation targets the anchor-nearest marker, so its L inner list is empty. Near H has no spectator; far H has exactly the near reflector as spectator. Its duration row is $2(1 + 1/(1 - e))t + 2z$
- The shear loop is governed by the invariant in Section 228.2. The scaling branch uses X, Y, D below one and D, Y, X above one, with inner lists respectively empty, $[X]$, and $[X, Y]$

- The rule loop advances distinct messenger labels cyclically, changes a target only at its paired launch/restoration, and preserves all four marker identities. It produces m messenger labels, $4K_0 + 3 \mathbf{1}_{\{\lambda \neq 1\}}$ temporary labels and four stationary labels
- The contact scan uses exact squared distances and exact equality, then deduplicates exact rational contacts. Serialization uses integer or rational strings, never floating-point approximations

The mathematical domain is the stated integer/rational parameter set. The constructor's assertions are development checks, not a security-hardened validator under Python optimization. There is no event selector or physical trajectory solver in the static constructor.

[write] [The constructor is shipped as `code/29-rotations-science-static_family.py`; run on a copy it regenerates all 45 fixtures byte for byte (`data/29-rotations-science-evidence-*.json`) in about a second.]

230 Infinite validity and all tangency cases

Write a normalized supplied row as $\alpha + v \cdot w > 0$, with $\alpha > 0$. Ignore constant-only rows. For each remaining row define

$$d_{\text{row}}^2 = \frac{\alpha^2}{v \cdot v}, \quad p_{\text{row}} = -\frac{\alpha v}{v \cdot v}. \quad (230.1)$$

There is a nonconstant row because P is bounded. Let ρ^2 be the positive rational minimum of the squared distances, and let E deduplicate the contacts at this minimum. Every contact is nonzero, rational and has norm ρ .

For $\|w\| < \rho$, Cauchy–Schwarz makes every row strictly positive. On $\|w\| = \rho$, equality can occur only for a minimizing row at its perpendicular contact; every other row is strict. Thus the circle complement of P is exactly E , including redundant, proportional and repeated guard rows. Above radius ρ , a nearest row is negative on a nonempty open arc. Every forward irrational rotation orbit meets that arc.

Normalization removes λ , so at each macro return w updates by R . Exact finite-word equivalence now proves (225.2). If a contact has positive initial gaps, its failed guard is a physical extra or tied contact. If it lies on an initial-order face, it is not an admissible initial section. Its shifted inverse-orbit points are still excluded by the same finite-chamber criterion.

The excluded critical-circle set is countably infinite, since a nonzero contact already has an infinite orbit. It is dense, as is a backward orbit. Its complement is uncountable. A semialgebraic subset of a circle is a finite union of points and arcs, which cannot realize this pattern. Hence $\mathcal{K}$ is not semialgebraic. Slicing the homogeneous gap cone at $D = 1$ gives the real-gap assertion. Real quantifier elimination also rules out a finite existential-real polynomial description of the exact real set.

[write] [This and the next two subsections give, for the whole family and for J contacts, the arguments of source 27's Propositions 203.2–203.5, which treat one machine with one contact. Both accounts are printed: source 29's must handle co-orbital contacts and contacts on an initial-order face, which source 27's machine does not have.]

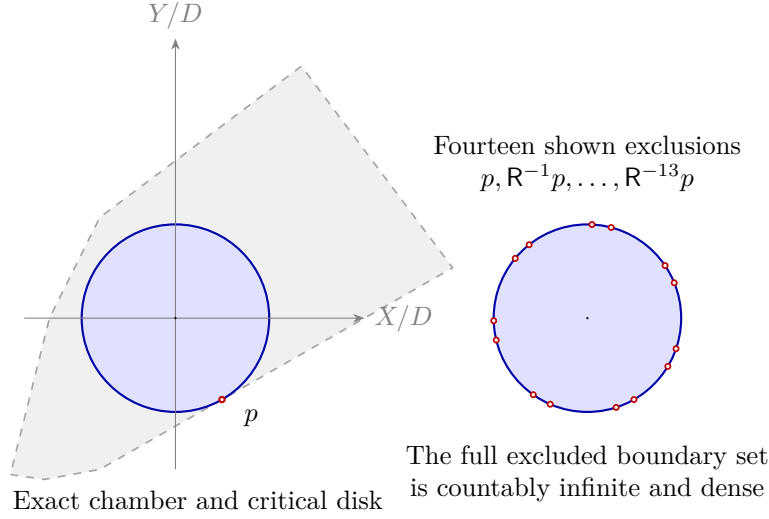


Figure 13: (Source 29.) Analytic normalized geometry for the new $(3, 4, 5)$ family, independent of λ . Dashed polygon edges are strict guards. The open disk is valid; on its critical circle exactly the entire backward orbit of $p = (4/65, -7/65)$ is excluded. The right panel shows only fourteen members of that infinite excluded set, not a complete plot of it. Vertices and orbit points are computed by exact rational formulas from the frozen fixture; this is no trajectory simulation.

230.1 Rational sign obstruction with co-orbital contacts

Fix $p \in E$ and its full orbit $O = \{R^k p : k \in \mathbb{Z}\}$. Infinite order makes each orbit index unique. The set

$$I = \{k \in \mathbb{Z} : R^k p \in E\}$$

is finite, nonempty and contains zero. Put $k_* = \max I$. Restricted to O , the forbidden union of backward orbits consists exactly of indices $k \leq k_*$. Thus the accepted forward tail $\{R^k p : k > k_*\}$ and rejected backward tail $\{R^k p : k \leq k_*\}$ are both dense and rational. No algorithm for deciding whether contacts are co-orbital is needed in this proof.

The positive-gap restriction needs an additional check. The closed critical disk lies in the closed initial-order triangle. Each strict initial-order line can vanish on the circle at only one perpendicular contact. There are therefore only finitely many zero-gap circle points. Removing these from the rejected tail leaves it dense; the accepted tail is already in P and has positive gaps.

A finite polynomial-sign formula restricts on the algebraic circle to a semialgebraic set, whose membership is constant on each of finitely many open arcs after removal of finitely many endpoints. It cannot accept one dense rational tail and reject the other. This works even for arbitrary real polynomial coefficients.

230.2 Positive integer obstruction without homogeneous input formulas

Suppose a finite Boolean polynomial-sign formula F were correct on all positive integer gap triples. Decompose each polynomial as $f = \sum_d f_d$ into homogeneous components. For a fixed g , the eventual sign of $f(tg)$ as $t \rightarrow +\infty$ is the sign of the highest-degree component nonzero at g , or zero if all components vanish. This is a finite semialgebraic case distinction, invariant under positive rescaling.

Replace each sign atom by this eventual-sign distinction, giving a homogeneous semialgebraic

formula F_∞ . The physical predicate is constant on positive rays. Correctness of F at every positive integer multiple forces correctness of F_∞ on each positive integer ray. Clearing denominators and rescaling extends this correctness to all positive rational gap triples, contradicting the preceding subsection. No homogeneity was assumed for the original F .

231 All real inputs and finite certificate obstructions

The following strengthening concerns correctness on *all positive real gaps*. It is a separate, independently audited application of classical topology, real algebraic geometry and the finite-path analysis of exact real computation. It leaves the encoded-rational and integer-input conclusions intact.

Let $G = (0, \infty)^3$, and let $V \subset G$ be the infinite-validity set for any one fixed compiled family member. Define the critical circle and its full excluded set by

$$S = \{w : \|w\| = \rho\}, \quad \mathcal{B} = \bigcup_{j=1}^J \{\mathbb{R}^{-n} p_j : n \geq 0\}.$$

The set $\mathcal{B}$ is countably infinite and dense in S , irrespective of duplicated or co-orbital representatives. Equation (225.2) says that $\mathcal{K}$ is the closed disk with $\mathcal{B}$ deleted. Recall that F_σ means a countable union of closed sets and G_δ a countable intersection of open sets in the stated topology.

Theorem 231.1 (Proper topological class and real-input obstruction). *For every fixed family member, V is G_δ but not F_σ , both relative to G and in $\mathbb{R}^3$. Its relative complement is F_σ but not G_δ . There are no finite k, m and finite Boolean polynomial-sign formula Φ , with arbitrary fixed real coefficients, such that on every $g \in G$,*

$$g \in V \iff \exists n \in \mathbb{N}^k \exists y \in \mathbb{R}^m \Phi(g, n, y). \quad (231.1)$$

The same holds for signed integer witnesses. In the finite-time exact real field-operation BSS model specified below, V is not semidecidable, even with finitely many arbitrary fixed real constants, whereas $G \setminus V$ is semidecidable using rational constants alone.

231.1 The Baire argument and admissible closed arc

An explicit open-set representation is

$$\mathcal{K} = \bigcap_{m \geq 1} \{w : \|w\|^2 < \rho^2 + 1/m\} \cap \bigcap_{b \in \mathcal{B}} (\mathbb{R}^2 \setminus \{b\}).$$

Thus $\mathcal{K}$ is G_δ . If $S \setminus \mathcal{B}$ were F_σ relative to S , write it as a countable union of relatively closed sets. Each has empty interior because $\mathcal{B}$ is dense, so each is nowhere dense. The singleton members of $\mathcal{B}$ are also nowhere dense. This would cover the compact nonempty circle by countably many nowhere dense closed sets, contrary to the Baire category theorem. Consequently $\mathcal{K}$ is not F_σ .

The same proof works on any nondegenerate closed arc $A \subset S$. The intersection $\mathcal{B} \cap A$ is countable and dense in this perfect compact metric space. The elementary compact-arc Baire statement can also be proved directly: successively choose nested nonempty closed subarcs avoiding the successive closed nowhere dense sets, with diameters tending to zero. Compactness gives an intersection point outside the proposed cover.

To transfer to actual positive gaps, use the normalized initial-order triangle

$$\mathcal{D} = \{(u, v) : 1/3 + u > 0, 1/3 + v - u > 0, 1/3 - v > 0\}.$$

The map

$$(d, (u, v)) \mapsto d(1/3 + u, 1/3 + v - u, 1/3 - v)$$

is a homeomorphism from $(0, \infty) \times \mathcal{D}$ onto G . The closed critical disk lies in the closed initial-order triangle. Each of the three affine gap forms is positive at zero and can vanish on its circle only at its single perpendicular contact. Hence at most finitely many points of S fail the positive-gap condition, and some nondegenerate closed arc A lies wholly inside $\mathcal{D}$.

Normalization is continuous on G , so V is G_δ relative to G . As G is ambient open, it is also G_δ in $\mathbb{R}^3$. Equivalently, the exact strict guard description expresses V as $G \cap \bigcap_{n \geq 0} (M^n)^{-1}C$, where C is the open homogeneous chamber in gap coordinates, obtained by pulling the supplied guard rows back through Q . If V were relatively F_σ , its preimage under the continuous fixed-scale embedding

$$\iota : A \longrightarrow G, \quad \iota(u, v) = (1/3 + u, 1/3 + v - u, 1/3 - v)$$

would make $A \setminus \mathcal{B} F_\sigma$ in A , a contradiction. Ambient F_σ would imply relative F_σ , so is also impossible. Taking complements proves the asserted proper class for invalidity. No Borel-completeness claim is made.

231.2 Why finitely many mixed witnesses cannot suffice

Every semialgebraic subset of $\mathbb{R}^a$ is F_σ . For example,

$$\{f > 0\} = \bigcup_{j \geq 1} \{f \geq 1/j\}$$

for a real polynomial f ; analogous formulas handle $f < 0$, while zero and weak-sign sets are closed. Resolve a finite Boolean sign formula into a finite union of finite intersections of these atoms. The class F_σ is closed under finite intersections and countable unions. This argument does not assume an arbitrary semialgebraic set is itself locally closed.

For a fixed integer tuple n in a hypothetical (231.1), the set

$$A_n = \{g \in \mathbb{R}^3 : \exists y \in \mathbb{R}^m \Phi(g, n, y)\}$$

is semialgebraic by Tarski–Seidenberg real projection, with no compactness restriction on the witnesses and allowing arbitrary fixed real coefficients [91]. The union over countably many n is F_σ . Agreement on all G would therefore make V relatively F_σ , impossible. This also rules out countable disjunctions of such certificates. In the special case $\exists n P(g, n) = 0$, fixed- n zero loci are already closed, so real projection is unnecessary.

This conclusion does not cover arbitrary alternating integer/real quantifiers or nonpolynomial oracle predicates. A formula required to agree only at integer inputs, or at discrete encodings of rational inputs, may behave differently at every other real input. Thus Section 234’s existential-positive-integer theorem is fully compatible with this obstruction.

231.3 The exact real computation model and semidecision

Use the ordinary finite-time Blum–Shub–Smale model with finite control, finitely many fixed real constants, real registers, field operations $+$, $-$, $\times$, $/$ where defined, copying/register shifts and exact equality and order tests. Ordinary loops and discrete control are allowed. There is no digit-extraction or floor primitive, one-step real-to-integer-word cast, oracle or completed-infinite-computation instruction. Semidecidability relative to G means halting exactly on the members among inputs in G , with no condition outside that promise. The model and relative convention, and the F_σ upper bound, are established in [89, 90].

For completeness, unroll a finite program into its countable family of finite accepting paths. A path fixes its branch outcomes and register accesses and uses finitely many instructions and registers. By induction, every encountered register is a rational function of the input, with coefficients in the field generated by the program’s fixed real constants. Its path conditions consist of finitely many rational sign tests and every nonzero-divisor restriction. They are semialgebraic: for example, when $b \neq 0$, $a/b > 0$ is equivalent to $ab > 0$, and $a/b = 0$ to $a = 0$. Retain intermediate division-domain restrictions even when a later expression cancels a denominator.

Each finite accepting-path domain is therefore semialgebraic; their countable union is F_σ . Intersecting with G gives the same relative upper bound. Finitely many arbitrary real constants only change coefficients, not this argument. The proper class proved above rules out semidecision of V , and hence decision as well.

Invalidity has a direct rational semidecision. Maintain $q = g$. Evaluate every supplied (D, X, Y) guard row at Qq ; halt when one is nonpositive; otherwise replace q by Mq and repeat. Exact chamber equivalence guarantees that an invalid input has a finite failed iterate, including an equality at an excluded boundary orbit. Valid inputs never halt. This procedure uses only finitely many fixed rational rows and the rational return matrix; it neither simulates physical collisions nor inspects an infinite-time limit. Alternatively, normalize, halt above the disk and enumerate inverse iterates of all contacts, halting at an exact match. To classify nonpositive inputs as invalid too, add their initial sign test.

Thus validity is *properly co-semidecidable* in this specified model. This is no Type-2 or oracle classification, BSS-degree completeness statement, physical hypercomputation claim, or Turing undecidability claim for encoded rationals. Encoded rational inputs supply numerator/denominator data that an arbitrary exact real register does not supply, and the $O(L^3)$ procedure below need not cover irrational inputs. Positive scaling cancels in normalization, so this real-input conclusion is independent of contraction and Zeno behavior.

[write] [The real-input companion is shipped as `29-rotations-real-input-PROOF.md`, with its notes and its independent audit (`29-rotations-real-input-SOURCE_NOTES.md`, `29-rotations-real-input-audit-AUDIT.md`, `29-rotations-real-input-audit-SOURCE_VERIFICATION.md`). The companion was written for source 27’s instance and is specialized to this family through the admissible-arc argument (Section 237). The programme’s review: the theorem “excludes finite existential mixed real/integer polynomial certificates on **all real gaps** and gives a model-specific BSS obstruction; it does not prohibit integer-input certificates or establish rational-input undecidability”.]

232 Exact rational membership and its bit bound

For rational positive gaps compute $w = (X/D, Y/D)$ and compare $\|w\|^2$ to ρ^2 exactly. Accept below, reject above. At equality, for every contact form the rational Gaussian quotient

$$\gamma_j = \frac{w_x + iw_y}{p_{jx} + ip_{jy}}. \quad (232.1)$$

Invalidity is equivalent to one equality $\gamma_j = ((a - ib)/c)^n$ for $n \geq 0$.

Lemma 232.1 (Exact power denominator). *The joint reduced denominator of $((a - ib)/c)^n$ is exactly c^n for every $n \geq 0$.*

Proof. For each prime $p \mid c$, p is odd and divides neither a nor b . In the algebra $(\mathbb{Z}/p\mathbb{Z})[i]$, set $z = a - ib$. Since $a^2 + b^2 \equiv 0$,

$$z^2 = 2az, \quad z^n = (2a)^{n-1}z \quad (n \geq 1).$$

Both coordinates on the right are nonzero modulo p . This computation does not presume the algebra is a field. Each integer numerator coordinate of $(a - ib)^n$ is consequently coprime to c , so the denominator is exactly c^n . The case $n = 0$ has numerator pair $(1, 0)$ and denominator one. $\square$

[write] [For $(a, b, c) = (3, 4, 5)$ this is source 27's Lemma 204.1, there proved by the identity $(3 + i)^2 = 3 + i$ modulo 5.]

For each γ_j , let q_j be its least common positive denominator. Repeated division by the whole integer c decides whether $q_j = c^n$. If not, no inverse-power match is possible. If yes, n is forced; compare with that one Gaussian power exactly. Repeating this for the finite contact list decides membership. There is no unbounded orbit search and density is not used as an algorithm.

232.1 Fixed machine polynomial time

Fix all machine parameters, J , ρ^2 and contact coordinates. Encode input rational gaps by ordinary binary signed numerators and positive denominators, with total length L . Malformed or nonpositive inputs are rejected by exact checks. Initial lowest-term reduction is optional.

The radius test and each quotient formation use a constant number of rational operations. Products of a constant number of input numerators or denominators have $O(L)$ bits, so all intermediate fractions and reduced joint denominators do too. Schoolbook division and the Euclidean algorithm give a conservative $O(L^3)$ bound for this fixed number of reductions, comparisons and gcd operations.

The whole-integer power test uses at most $\log_c q_j = O(L)$ divisions of $O(L)$ -bit integers by fixed c . If successful, compute $(a - ib)^n$ by $n = O(L)$ successive multiplications by the fixed Gaussian integer. At step $k \leq n$ each coefficient has magnitude at most $c^k \leq q_j$, hence $O(L)$ bits. These multiplications and the final comparison take polynomial time, indeed $O(L^2)$ with elementary fixed-coefficient arithmetic. Since J is fixed, $O(L^3)$ bounds the whole procedure conservatively.

This is not an optimality claim, a uniform parameter-size or compilation bound, or a bound on finding the potentially enormous Diophantine witnesses. The decision algorithm never constructs the auxiliary ordinary powers used in the certificate.

233 The exact Zeno classification and rational clock

There are $2K_0$ translation gadgets. Each L portion takes less than $4D$ and each H portion at most $2D$, so the rotation duration is less than $(12K_0 + 2)D$, including the two length- D transfers. With scaling, the extra duration is less than $6(1 + \lambda)D$. Thus a valid macro has a rational homogeneous linear duration row ℓ satisfying

$$0 < \ell(D, X, Y) < C_\lambda D, \quad C_\lambda = 12K_0 + 8 + 6\lambda \quad (233.1)$$

when $\lambda \neq 1$; the smaller rotation-only bound applies at one. Rational linearity follows from the primitive durations and endpoint maps.

At repetition n , $D_n = \lambda^n D_0$. For $\lambda < 1$, the geometric upper bound gives finite total time, and every strand in macro n lies in $[0, D_n]$, so all approach the fixed left anchor. Since all eigenvalues of N have modulus $\lambda < 1$,

$$T(z) = \sum_{n \geq 0} \ell N^n z = \ell(I - N)^{-1} z, \quad z = (D, X, Y). \quad (233.2)$$

This row has rational coefficients and gives rational accumulation time on rational gaps. For $\lambda \geq 1$, the transfers alone contribute $2D_n$ per macro, whose sum diverges. This proves the exact iff. No continuation through an accumulation is asserted.

[write] [Rational accumulation time on rational gaps is also the conclusion of source 26's Theorem 182.4, which holds for every valid ultimately periodic complete macro; here the time is an explicit row. Source 27's clock (196.5) is the corresponding row for its machine at $\lambda = 1/2$.]

The independent static check also obtains the duration using the closed sum

$$\sum_{k=0}^{n-1} t_k = nt_0 + \frac{\delta n(n-1)}{2} (z - 2D/3),$$

rather than the author's incremental loop, and matched every fixture duration row.

234 A literal fixed arity degree twelve certificate

This section concerns three ordinary positive integer input gaps and one fixed machine. Its contact set E is computed first; all machine and contact quantities are fixed coefficients, not witnesses. Write $\rho^2 = H/K$ in positive integer lowest terms and put

$$\begin{aligned} D &= g_1 + g_2 + g_3, & x &= g_1, & y &= g_1 + g_2, \\ A &= 3x - D, & B &= 3y - 2D, & \Delta &= 9HD^2 - K(A^2 + B^2). \end{aligned} \quad (234.1)$$

Introduce one natural witness $d \in \mathbb{N}$ and impose

$$\Delta = d. \quad (234.2)$$

For each p_j fix integers r_j, s_j and $t_j > 0$ with $p_j = (r_j/t_j, s_j/t_j)$. The contact is nonzero. Define fixed linear input expressions

$$U_j = t_j(r_j A + s_j B), \quad V_j = t_j(r_j B - s_j A), \quad Q_j = 3(r_j^2 + s_j^2)D > 0. \quad (234.3)$$

Then $\gamma_j = (U_j + iV_j)/Q_j$, also at the center. These aliases are expressions, not additional leaves.

[write] [This is source 28's construction (Sections 213–216) carried out for each contact of a general family member, with the radius equation shared; source 28 treats source 27's machine and gives a literal DAG and gate ledger, which source 29 does not. For $(a, b, c) = (3, 4, 5)$ the radix (234.4) is $4P + 11$, not source 28's $4P + 1$. The opening of Part X compares the ledgers.]

234.1 The thirteen outer equations per contact

For each j independently introduce

$$\begin{aligned} n, k, L_C, H_C, L_S, H_S &\in \mathbb{N}, \\ h, q, r, s, t, J_{\text{acc}} &\in \mathbb{Z}_{>0}, \\ u, v, e_1, e_2, e_3, C, S, \kappa &\in \mathbb{Z}. \end{aligned}$$

Let $\epsilon = \text{sgn } b$ and use the explicit POWER module of the next subsection twice, with distinct internal witnesses, to define

$$P = \text{POWER}(c, n), \quad \beta_{\text{rad}} = 4P + 2c + 1, \quad T = \text{POWER}(a + |b|\beta_{\text{rad}}, n). \quad (234.4)$$

The radix β_{rad} is a linear expression in P , not a module Pell parameter. Suppressing the contact index on witnesses, impose all thirteen equations:

$$U_j = hu, \quad (\text{O1})$$

$$V_j = hv, \quad (\text{O2})$$

$$Q_j = hq, \quad (\text{O3})$$

$$e_1u + e_2v + e_3q = 1, \quad (\text{O4})$$

$$q = Pr, \quad (\text{O5})$$

$$r = ck + s, \quad (\text{O6})$$

$$s + t = c, \quad (\text{O7})$$

$$C + P = L_C, \quad (\text{O8})$$

$$P - C = H_C, \quad (\text{O9})$$

$$S + P = L_S, \quad (\text{O10})$$

$$P - S = H_S, \quad (\text{O11})$$

$$T - C - \beta_{\text{rad}}S = \kappa(\beta_{\text{rad}}^2 + 1), \quad (\text{O12})$$

$$d^2 + (r - 1)^2 + (u - C)^2 + (v + \epsilon S)^2 = J_{\text{acc}}. \quad (\text{O13})$$

All domains are essential. Replace a natural z by $z_+ - 1$ with $z_+ \in \mathbb{Z}_{>0}$. Replace a signed z by $z_+ - z_-$ with both leaves in $\mathbb{Z}_{>0}$. Positive quantities remain individual leaves. These are explicit affine substitutions throughout the displayed system.

234.2 Exactness of the outer system

Equations (O1)–(O4) say that (u, v, q) is the primitive joint quotient of (U_j, V_j, Q_j) . Bézout forces $\text{gcd}(u, v, q) = 1$; with $h > 0$ this forces h to be the positive gcd of the original triple. Since $Q_j > 0$, q is exactly the least common positive denominator of u/q and v/q . Zero and negative numerator components are allowed. At the center, $U_j = V_j = 0$ and $q = 1$.

POWER gives $P = c^n$. Equations (O6)–(O7) imply $1 \leq s \leq c - 1$, $k \geq 0$, and $c \nmid r$. Every positive q has a unique factorization $q = c^n r$ with this condition, by repeated division by the *whole* integer c . For composite c this is not an additive prime-adic valuation and does not assert $\gcd(r, c) = 1$.

Let $C_n + iS_n = (a + i|b|)^n$. Its modulus is $c^n = P$, so $-P \leq C_n, S_n \leq P$. Substitution of i by β_{rad} modulo $\beta_{\text{rad}}^2 + 1$ gives

$$(a + |b|\beta_{\text{rad}})^n \equiv C_n + \beta_{\text{rad}}S_n \pmod{\beta_{\text{rad}}^2 + 1}.$$

Thus the true pair satisfies extraction and bounds. Conversely, any bounded solution C, S has

$$|(C - C_n) + \beta_{\text{rad}}(S - S_n)| \leq 2P(1 + \beta_{\text{rad}}) < \beta_{\text{rad}}^2 + 1.$$

Indeed $\beta_{\text{rad}} > 4P$ gives $\beta_{\text{rad}}^2 + 1 - 2P(1 + \beta_{\text{rad}}) > 8P^2 - 2P + 1 > 0$ for $P \geq 1$. Divisibility forces this difference to zero; then $|C - C_n| \leq 2P < \beta_{\text{rad}}$ forces both coefficient differences to zero. Extraction is exact.

Negative a creates no POWER-base problem. Already from positivity of the independent leaf P ,

$$a + |b|\beta_{\text{rad}} \geq -(c - 1) + (4P + 2c + 1) = 4P + c + 2 > 2.$$

The first base is $c \geq 5$, so both module hypotheses hold before invoking their semantic conclusion.

Lemma 232.1 now says that γ_j is a forbidden inverse power precisely when

$$r = 1, \quad u = C, \quad v = -\epsilon S.$$

The sign in (O13) is therefore $v + \epsilon S$. At $n = 0$ one gets $P = 1, C = 1, S = 0$, correctly excluding $\gamma_j = 1$ on the boundary.

Equation (234.2) forces $\Delta \geq 0$. For $\Delta > 0$, a positive acceptance witness exists for each contact because the sum in (O13) is already positive. At $\Delta = 0$, each (O13) is solvable exactly when its contact is not a forbidden match. Conjoining all contacts yields precisely (225.2). Conversely, choose gcds, Bézout coefficients, canonical c -power factors, true Gaussian coefficients, bounded slacks, congruence quotients and the POWER witnesses. Every valid input then supplies all witnesses, with J_{acc} equal to its nonzero displayed sum. This proves both directions without a boundary selector or a generic MRDP appeal.

234.3 All fifteen POWER equations

Remark 234.1 ([write] restatement of source 28’s system (P1)–(P15): source 29’s POWER display). The module is printed once, as source 28’s system (P1)–(P15) of Section 217, whose Lemma 217.1 states and proves its equivalence. Source 29 displays the same fifteen equations, for “an integer base $B_0 \geq 2$ and a natural exponent e ”, in its own names: its output out , Pell parameter α , Pell coordinates $x_p, y_p, u_p, v_p, s_p, t_p$ and congruence quotients r_1, r_2 are source 28’s o, a, x, y, u, v, s, t and ρ_1, ρ_2 ; $w, g, q_b, q_v, J_p, \beta$, the eleven natural quantities and the expressions $k_0 = e + 1, m_0 = B_0 \text{out}$ are named alike in both, and the modulus is Θ in both (the sources’ M). Its equations (P1), (P13) and (P15), whose labels `smc:rf:eq:P1`, `smc:rf:eq:P13` and `smc:rf:eq:P15` this remark carries, are source 28’s (P1), (P13) and (P15) in these names.

After the same natural-to-positive adapters this is 26 positive leaves, including its output, and fifteen equations. The exponent and base are expressions supplied by the caller, not additional module leaves.

The precise mathematical dependency is the constructive characterization `Pell.matiyasevic` and power characterization `Pell.eq_pow_of_pell` in the pinned mathlib source [86]. They are specialized theorems, not unexpanded arithmetic oracle symbols. The source is read as inert mathematical text; no Lean execution is claimed.

234.4 Specializing the constructive Pell theorems

[write] [Second account (source 29) of the proof of Lemma 217.1, in source 29's names; it adds the explicit substitution dictionary from the mathlib statements.]

For the index characterization substitute source a, k, x, y by α, k_0, x_p, y_p , and source auxiliaries u, v, s, t, b by $u_p, v_p, s_p, t_p, \beta$. Equations (P1)–(P3) are the three Pell identities. Equation (P4) gives $\beta \equiv 1 \pmod{4y_p}$; (P5) gives $\beta \equiv \alpha \pmod{u_p}$; (P6) gives $y_p^2 \mid v_p$; (P7)–(P8) give the two remaining congruences; (P9) gives $k_0 \leq y_p$. These are the positive-index branch, since $k_0 = e + 1 \geq 1$ and $y_p \geq k_0$ rule out the source's zero-index pair $(1, 0)$. Thus (x_p, y_p) is the Pell pair at index k_0 for α .

The source uses natural truncated subtraction. For naturals, $X - Y = 1$ in truncated subtraction is equivalent to the ordinary integer identity $X = Y + 1$, precisely the form of the displayed Pell equations. No subtraction convention is changed without this equivalence.

For the power characterization substitute source n, k, m by B_0, k_0, m_0 , and its w, a, t, z by w, α, Θ, g . Equations (P10)–(P11) give $w \geq B_0, k_0$; (P12) gives the required strict bound $\Theta > m_0$; (P13) is the auxiliary Pell identity; (P14) is the required relation $2\alpha B_0 = \Theta + B_0^2 + 1$; (P15) is the congruence modulo Θ after identifying the Pell pair.

Since $g \geq 1$ and $w \geq B_0 \geq 2$, (P13) forces $\alpha > w \geq B_0$. Its ordinary subtraction $\alpha - B_0$ therefore agrees with the source's natural subtraction. The theorem gives $B_0^{k_0} = m_0 = B_0$ out. Cancelling the positive B_0 gives out $= B_0^e$, including $e = 0$ through the index shift.

Conversely, the constructive theorems supply witnesses for $B_0^{k_0}$. Their auxiliary g is nonzero because $\alpha \geq 2$. Pell x coordinates are positive; $y_p \geq k_0 \geq 1$ and $v_p > 0$ give $q_v > 0$. The conditions $\beta > 1$ and $\beta \equiv 1 \pmod{4y_p}$ give $q_b > 0$. Since $t_p \equiv k_0 \pmod{4y_p}$ and $1 \leq k_0 \leq y_p < 4y_p$, $t_p \neq 0$. The strict modulus bound gives $J_p > 0$. Every signed congruence difference is represented by the difference of two natural multiples, exactly as the displayed paired variables. Thus the positive domains lose no solutions, proving both directions of the literal module.

234.5 Witness ledger and exact degree

The per-contact outer domains have six naturals, six positive quantities and eight signed quantities represented by sixteen positive leaves. Each POWER module contributes 26 leaves, with P and T counted only as their respective outputs. Hence

$$W = 1 + J(6 + 6 + 16 + 26 + 26) = 1 + 80J, \quad E_q = 1 + J(13 + 15 + 15) = 1 + 43J. \quad (234.5)$$

The first 1 is the shared radius witness or radius equation. Define F literally as the sum of squares of all residual polynomials after substitution of every alias and affine domain adapter. This is a finite formula for each fixed chamber: J is a machine-dependent fixed integer, not a witness-indexed array.

Outer residuals have total degree at most three in input and independent witness leaves. POWER residuals have degree at most six, even when the second base is affine in P . Residual (P13) has degree-six leading term $-w^4 g^2$. Its square supplies a nonzero degree-twelve contribution. The

degree-twelve homogeneous part of F is a sum of squares of real homogeneous polynomials, which cannot cancel identically. As $J \geq 1$, $\deg F = 12$ exactly.

For a paid one-input transport, introduce positive gap witnesses g_1, g_2, g_3 and one natural witness j , and append

$$2j = (g_2 + g_3 - 2)(g_2 + g_3 - 1) + 2(g_3 - 1), \quad (234.6)$$

$$2(z - 1) = (g_1 - 1 + j)(g_1 + j) + 2j. \quad (234.7)$$

These encode the bijection $z = 1 + \text{pair}(g_1 - 1, \text{pair}(g_2 - 1, g_3 - 1))$. They add four positive leaves and two quadratic residuals, giving $5 + 80J$ witnesses and $3 + 43J$ equations, still of exact degree twelve. This has a different input ledger from the native three-gap theorem.

There is no exact gate count or emitted general arithmetic DAG claim. Positive-pair encodings of signed values alone admit infinitely many presentations, so neither uniqueness nor finite-foldness follows.

[write] [At $J = 1$ the ledger (234.5) gives 81 witnesses and 44 equations, source 28's counts (Theorem 211.1); the paid transport gives 85 and 46, again source 28's (Theorem 211.2). The decoder (234.6)–(234.7) is source 28's (D1)–(D2). Part XI's Report 60 counts the same pattern as $81J$ and $44J$ for the orbit complement and $1 + 81J$, $1 + 44J$ for the strict kernel: it has one more outer equation and one more positive witness per contact (opening of Part X).]

235 Concrete family data and independent evidence

For $(a, b, c) = (3, 4, 5)$, the construction has $N_x = 2$, $N_y = 4$, $K_0 = 8$. There are 150 events without scaling and 174 with $\lambda \neq 1$. At $\lambda = 1/2$ there are 213 meta-signals and 44 supplied guards. The unique nearest row and contact are

$$D/5 - 4X/5 + 7Y/5 > 0, \quad \rho^2 = 1/65, \quad p = (4/65, -7/65). \quad (235.1)$$

The radius itself is $1/\sqrt{65}$. The earlier Report 57 machine [source 27] has larger shear steps, 138 half-scaled events and squared radius $4/1845$. The same return matrix does not determine the same finite chamber or infinite-validity predicate. This report does not replace that machine or silently reuse its certificate.

235.1 Summary of all physical fixtures

The frozen evidence contains fifteen signed and coordinate-ordered triples at each $\lambda \in \{1, 1/2, 2\}$, totaling 45 outputs. The following table groups the three scales per rotation. The guard count and squared radius are scale-independent. Columns m_1 and $m_{\neq 1}$ give the unscaled and either scaled event count. All physical fixtures have $J = 1$; the theorem does not assume it.

(a, b, c)	N_x	N_y	m_1	$m_{\neq 1}$	Guards	ρ^2
(3, 4, 5)	2	4	150	174	44	1/65
(3, -4, 5)	2	4	150	174	44	4/369
(-3, 4, 5)	8	4	366	390	92	1/185
(-3, -4, 5)	8	4	366	390	92	4/1233
(4, 3, 5)	2	3	132	156	40	9/505
(4, -3, 5)	2	3	132	156	40	16/765
(-4, 3, 5)	12	3	492	516	120	1/360
(-4, -3, 5)	12	3	492	516	120	4/2169
(5, 12, 13)	3	4	186	210	52	49/4381
(-5, 12, 13)	6	4	294	318	76	49/7956
(12, 5, 13)	1	2	78	102	28	3200/152217
(-12, 5, 13)	20	2	762	786	180	1/936
(119, 120, 169)	2	3	132	156	40	126736/9541233
(-119, 120, 169)	10	3	420	444	104	198025/55282266
(120, -119, 169)	2	3	132	156	40	400/25533

The static constructor output contains parameter fractions, counts, gap return matrix, rational duration row, every named guard row, exact squared inradius, nearest-row records, deduplicated contacts, primitive phase ranges, complete speed/label map, every indexed binary rule and the identity-default declaration. Rational values are serialized as integer or numerator/denominator strings.

The author fixture checks test center positivity, return and conjugacy, phase continuity, counts, distinct collision speeds, rational contacts and nonnegativity at each nearest contact. The separate auditor read the entire proof and constructor inertly and reconstructed all output fields by independent closed block formulas and arithmetic-series duration sums. Label-occupancy checks update only finite sets of labels; they do not compute positions or times or select collisions.

Independent evidence	Count
Physical fixtures with complete field equality	45
Indexed binary rules	13,572
Supplied guard rows	3,336
Primitive phase ranges	2,976
Rejected deliberate static mutations	12
Exact symbolic identities	13
Synthetic boundary index checks	101
Signed/composite denominator checks	672
Bounded radix extraction checks	672

The twelve negative controls delete a spectator, reverse contraction order, change a reflected speed sign, mutate a guard coefficient, delete an initial row, change a tangent sign, alter a duration, reverse the return orientation, remove the default, stale a messenger label, collapse a temporary label or move a phase endpoint. The auditor verified the exact 60-file author packet, all 59 nonself manifest entries and seven source dependencies, with preservation snapshots.

235.2 The separate degeneracy diagnostic

To test multiple, duplicate, co-orbital and initial-order tangencies together, the independent geometry checker uses a synthetic rational polygon with $\rho^2 = 1/18$ and contacts

$$p = (1/6, -1/6), \quad R^{-3}p, \quad R^2p, \quad -p,$$

with proportional duplicate rows. It has eight supplied rows, six nearest rows and four distinct contacts; three nearest rows represent p . The first three distinct contacts share one rotation orbit. The point p lies on the initial face $y = x$.

The denominator-based test agrees with acceptance of $R^k p$ exactly for $k > 2$ over 101 checked indices. The sole zero-gap point in that checked index range is $k = 0$. This is a separate geometric diagnostic, not a polygon emitted by any of the 45 physical fixtures. The general proof of Section 230 supplies the result beyond those finite tests.

The additional arithmetic cases use 32 signed/coordinate-ordered triples over $c = 5, 13, 65, 169$ and exponents zero through twenty. Thus both a distinct-prime composite denominator and a prime-square denominator are covered. Finite checks support, rather than replace, the prime-divisor and bounded-radix proofs. General large Pell witnesses are not instantiated.

[write] [Shipped names: the audit `29-rotations-independent-audit-AUDIT.md` and its arithmetic part `29-rotations-independent-audit-ARITHMETIC_AUDIT.md`, the checkers `code/29-rotations-independent-audit-audit_static.py` and `code/29-rotations-independent-audit-audit_geometry.py` with their receipts `data/29-rotations-independent-audit-STATIC_INDEPENDENT_RECEIPT.json` and `data/29-rotations-independent-audit-GEOMETRY_INDEPENDENT_RECEIPT.json`, and the arithmetic review `29-rotations-science-arithmetic_review-ARITHMETIC_REVIEW.md` with `code/29-rotations-science-arithmetic_review-check_arithmetic.py`. On copies all reproduce their frozen receipts up to line endings (the static audit after one path constant is mapped).]

236 Prior work and claim boundaries

Conservative rational signal arithmetic, translation, scaling and shrinking are established techniques. Durand-Lose’s AGC6 [16] explicitly equates incoming and outgoing counts, implements rational stack operations through translation and scaling, and gives conservative shrinking. AGC8 [6] distinguishes meta-signals, live instances and speed counts and studies accumulations and rationality. The appearance of five reference markers or five meta-signals in AGC5 [79] is not an exactly-five-total-live number-preserving result.

Dense rational-rotation semialgebraic obstructions also precede this work. Fijalkow–Ohlmann–Ouaknine–Pouly–Worrell, STACS 2017, Example 1 [87], use the rational rotation $\frac{1}{5}(\begin{smallmatrix} 4 & -3 \\ 3 & 4 \end{smallmatrix})$ and show that an unreachable circle target may lack a semialgebraic separating invariant. Their introduction recalls the polynomial-time Kannan–Lipton orbit decision theorem [88]. Our fixed-machine denominator procedure is a direct special-case algorithm, not a new general orbit-decision phenomenon. The strict-boundary input classifier and complete physical word are distinct objects from an inductive nonreachability invariant.

Earlier three-variable linear-loop nonsemialgebraicity is established by Dai–Xia [76]; Ouaknine–Worrell [77] give an irrational-rotation loop with a non-strict guard whose nontermination set is a closed cone. Strictness is precisely what creates the boundary deletions here. The finite rational-tail argument and eventual-sign passage to integer rays still require their own proofs.

The contribution documented here is the explicit five-live-signal physical realization for each rational infinite-order rotation, with its exact complete-word chamber, exact strict-boundary classifier and literal fixed-arity arithmetic specialization. The bounded comparison found no matching complete realization in the inspected sources, but it is not an exhaustive novelty search or a priority certificate. No unchanged universal rotation table, arbitrary-matrix compilation, minimal population or word length, general undecidability, arithmetic gate optimality or continuation after accumulation is claimed.

[write] [The repository comparison of the archive’s literature notes was pinned at `b64f24e59` (this report’s README, blob `572efe05`, unchanged until this Part was written); the notes are shipped as `29-rotations-science-PRIOR_WORK.md`. Within this report the closest results are sources 26–28, which source 29 cites by Report number, and Part XI’s Report 60, which generalizes it. The machine depends on the rotation, as the source says; the programme’s review makes the same point for the generalization in Report 60 (“without a generic emitted gate ledger or one unchanged universal table”, and “Physical compilation cost/word length is not claimed polynomial in matrix bit length”), and it applies here too. Source 29’s counts (228.2), (229.1) and (229.4) are exact but grow with $|t_x| = |b|/(c + a)$: for the primitive triples $(-(c - 1), b, c)$ with b odd and $c = (b^2 + 1)/2$ one has $c + a = 1$, $N_x = 4|b|$ and $m \geq 18 \cdot 8|b| = 144|b|$ events, exponential in the bit length of the parameters (for $(-12, 5, 13)$, $N_x = 20$ and $m_1 = 762$, as in the fixture table below).]

237 Release verification and reproducibility

This manuscript transcribes and expands the final frozen proof and incorporates the independent arbitrary-parameter source-invariant audit. The frozen science and audit files are preserved byte-for-byte in the accompanying archive. Section 231 incorporates the separately audited real-input companion, specialized through the admissible-arc argument to this family; its proof, source notes and fresh independent audit are also preserved. The author packet’s original README predates the completed physical audit; the separate final audit and its receipt supply the later PASS verdict.

The release tools use explicit input and fresh external output paths, verify exact file inventories and SHA-256 pins, reject unsafe destinations and preserve all frozen inputs. Builds are isolated and deterministic under a fixed epoch and environment, and use no shell escape. Replay executes only independently authored inspected checkers and release tools. Author constructor/checker files, saved physical schedules, upstream code and the Lean source remain inert evidence. A full page render is inspected before sealing. Exact build and render receipts accompany the archive rather than being treated as mathematical proof.

Primary bindings

The following full SHA-256 values identify the source of this report. File paths are relative to the archive root.

Final proof `science/PROOF.md`

`14c3d694d9e0c21f3ad3b125c0d1f2bbf47793e1283c9314aa3855ef60c89fc5`

Static physical constructor `science/static_family.py`

`c6ce6ae3873da154c4b7f200f762408f36583d20e424ac6fca4661bd0bd9ce41`

Science packet manifest `science/PACKET_MANIFEST.json`

`646de47472089e9909fafbf0ed1e1a7851a114d862e4df9b31d0802815772044`

Independent audit `independent-audit/AUDIT.md`

b645448d86f3041df3c406f88640cd7f9c73aee11715b6b9eb43310a352a5799

Independent receipt `independent-audit/AUDIT_RECEIPT.json`

327ac64f5df5f8dbd2929cdcbabb147a19198bdf9b23291edb948b726514bbbe

Real input companion `real-input/PROOF.md`

58e75fba06c017fb8e2e59925b03a3b96e389e268ea257db79d1f5201b3a7b1d

Real input independent audit `real-input-audit/AUDIT.md`

34d1c28ad1bfc4f21a96bb28aed20f23e52f5c636999f255cc61a40e75d6589f

Pell source `science/inert_sources/pell-source.lean`

993760c797ad0ff66fa77064a616fce779550bc745cbf6c44e04f392fd0bed0a

[write] [Every binding except the Pell source is shipped under the prefix 29-rotations- (29-rotations-science-PROOF.md, code/29-rotations-science-static_family.py, data/29-rotations-science-PACKET_MANIFEST.json, 29-rotations-independent-audit-AUDIT.md, data/29-rotations-independent-audit-AUDIT_RECEIPT.json, 29-rotations-real-input-PROOF.md, 29-rotations-real-input-audit-AUDIT.md). The archive’s copies of sources 27 and 28 (its dependencies/ folder) are shipped once, with those sources. The release and replay tools (code/29-rotations-tools-release59.py, code/29-rotations-tools-portable_tools-replay_family59.py) need POSIX paths and the delivered layout; the README gives a rerun route on a copy.]

[write] **The programme’s intake of source 29.** The review [99] records the family, the polynomial-time fixed-machine membership and the ledger $1 + 80J$, $1 + 43J$, and notes “No emitted generic arithmetic DAG or gate ledger”, which is the source’s own scope statement. No correction is asked for.

238 Further questions and research (Part X)

[write] This section collects what the four sources state as possible, expected or not established without proving it, and the questions they leave, each with its origin, the argument that exists and what is missing. Source 27’s own five questions are printed in its Section 207, with notes on the two that sources 28 and 29 answer in part; they are not repeated. No claim of the sources was found to be false; the exact-arithmetic identities of sources 27 and 28 that were recomputed at this write (source 27’s clock rows (205.2), (205.3), (196.5) and (205.4), its two boundary examples (202.2), source 28’s identities (212.5)–(212.7) on random gaps, and the equality of source 29’s return matrix (229.2) at $(3, 4, 5)$, $\lambda = 1/2$ with source 27’s) all hold.

X.1 An audited machine-and-macro frontend (source 26, Sections 191 and 193; source 27, Question 2). The chamber, the infinite criterion and the sign formula are proved constructions (Sections 184–186 and the checklist of Section 194), but only the last step, from a quadratic sign formula to the quartic, is implemented. Missing: a parser from a finite machine and a supplied word to the rows E, G, M, ℓ, r , the sign formula and the quartic DAG, with an independent audit; source 27 adds a checkable completeness certificate for the derived chamber.

X.2 Low-degree formulas beyond four live signals (source 26, Section 193: “which larger-population macro classes admit comparable low-degree formulas, and how additional geometric restrictions might control arity”). Source 27 shows that a finite sign formula need not exist at

five signals (Corollary 206.1); sources 28 and 29 give existential integer certificates of degree 12 for the rotation macros, and Part XI's Report 60 extends them to elliptic planar returns. Missing: a statement for every supplied macro with five or more live signals (a bounded-degree existential certificate, or an obstruction to one), and any arity bound uniform over macros; source 26 states that its arity grows with the word.

- X.3 Variable time offsets and signed deadlines** (source 26, Section 188). The source says that an initial rational time offset “can be added by ordinary cross-multiplication, but if its numerator and denominator are new variables the degree ledger must be reconsidered”, and that signed deadline numerators may be handled “by a separately specified natural encoding”. Neither encoding is given. Missing: the encodings and their degree and witness ledgers; the degree-at-most-two claim is proved only for section time zero and fixed coefficients.
- X.4 An effective validity test in higher dimension** (source 26, Theorem 182.4 and the remark after it). The rational-limit theorem is conditional on validity and supplies “no arbitrary-dimensional recognition algorithm”. For the return matrices of order at most two the test is Proposition 185.1; in general the infinite criterion (184.5) is a conjunction of positivity problems for linear recurrences of higher order, whose general decidability is not known; source 26 asserts no general Positivity algorithm (it cites [75] for the low-order case) and no reduction from arbitrary Positivity instances. Missing: a decision procedure for the infinite criterion of a supplied macro with more than four live signals, or a reduction showing that one would decide Positivity.
- X.5 Unique-witness, finite-fold or quartic certificates at five signals** (source 27, Section 206: its obstruction “does not prove that five signals prevent unique-witness or quartic Diophantine certificates by other methods”; sources 28 and 29). The certificates of sources 28 and 29 have infinitely many witnesses for each accepted input (signed-pair adapters and the Pell witnesses), and degree 12. Missing: either a certificate for source 27's predicate with a unique (or finitely many) positive witnesses, or with degree four as at four signals, or a proof that none exists. A canonical choice of every signed pair and of the Pell witnesses would be one route, as Part VIII's canonical height is for its outer history (Theorem 150.1); it is not attempted by the sources.
- X.6 Smaller certificates** (source 27, Question 4; sources 28 and 29). Sources 28 and 29 state that their counts are neither optimal nor lower bounds, and source 28's quartic variant trades one witness for six gates. Missing: lower bounds, or smaller representations of the same predicates; for source 29's family an emitted arithmetic DAG with a literal gate ledger (“No emitted generic arithmetic DAG or gate ledger”, the programme's review).
- X.7 Shorter words and fewer types** (source 27, Question 1; source 29, Section 236). Source 27's counts of eight speeds, 169 types and 138 events are not claimed minimal, and source 29's word length grows with $|b|/(c+a)$ (note at the end of Section 236). Missing: a realization whose word length is polynomial in the bit length of (a, b, c) , or a lower bound; any reduction needs its own complete chronology proof.
- X.8 Finer classification of the real validity sets** (source 29, Section 231). The real validity set is proved G_δ and not F_σ , and properly co-semidecidable in the finite-time BSS model; source 29 makes “No Borel-completeness claim” and no Type-2, oracle or BSS-degree statement. Missing: those classifications.
- X.9 Accumulation without a supplied macro, and beyond it** (Part I's direction, Section 15.9).

Sources 26–29 certify infinite repetition of a supplied ultimately periodic macro and compute its accumulation time; none of them finds a periodic itinerary, treats executions that are not ultimately periodic, or defines any dynamics at or after the accumulation. Those parts of Part I’s direction stay open.

Part XI

Planar and projective returns with five live signals (sources 30, 31, 32 and 33)

239 Sources, scope and conventions of Part XI

[write] Part XI prints four AI-assisted research manuscripts of ProveIt’s incoming reports, batch 91 (arrival 0d7f51c44; cluster W3, placed by d750d98dd), all dated 4 October 2026 and added on that day. They are consecutive numbered reports of the research pipeline that also wrote the sources of Parts X and XII. *Source 30* is batch-91 manuscript 12, the pipeline’s Report 60, from the archive `Planar_Signal_Realization_Source.zip`; *source 31* is manuscript 13, Report 61 (`Invertible_Signal>Returns_Source.zip`); *source 32* is manuscript 14, Report 62 (`Projective_signal_returns_source_and_evidence.zip`); and *source 33* is manuscript 15, Report 63 (`Five_signal_homogeneous_realization_source_and_evidence.zip`). Their title pages read “Research Report 60” to “Research Report 63” and their author lines “Research report”; they name no assistant, no repository and no ProveIt commit, and the run logs they carry show the runtime paths of an AI pipeline. The source numbers are the file prefixes `30-planar-`, `31-invertible-`, `32-projective-` and `33-homogeneous-` of their shipped files; their labels carry the prefixes `smc:pl:`, `smc:iv:`, `smc:pj:` and `smc:hl:`, and sentences written in the merge carry `smc:xi:` labels.

[write] **Pins.** The four sources pin their predecessors by SHA-256, never by commit. Source 30 pins source 29’s proof (the pipeline’s Report 59; 14c3d694..., shipped once, as `29-rotations-science-PROOF.md`) and the file `Mathlib/NumberTheory/PellMatiyasevic.lean` of `mathlib4` commit `ac77769fabe2` (993760c7..., not shipped, Section 293.3), besides its own four frozen proofs. Sources 31, 32 and 33 pin source 30’s physical proof (`e0ddd64c...`, shipped as `30-planar-science-physical-PROOF.md`); sources 32 and 33 pin source 31’s proof (`dc97e30c...`, `31-invertible-science-frozen-proof-PROOF.md`); and source 33 pins source 32’s proof (`502e90e0...`, `32-projective-science-frozen-proof-PROOF.md`) and its independent review. Each archive carried byte copies of the proofs it pins; they are shipped once, at the earliest source, and the copies are not shipped again.

[write] **Pipeline numbers.** The sources keep the pipeline’s numbers in their text. Reports 56–59 are sources 26–29 (Part X), Reports 60–63 are sources 30–33 (this Part), Reports 64 and 65 are sources 34 and 35 (Part XII), and Reports 66, 67 and 69 are sources 36–38 (Part XIII). Brackets translate the numbers where a reference matters.

[write] **The common section.** All four sources work on one outgoing section of a finite deterministic rational-speed number-preserving signal machine: stationary markers $L = 0 < X = x < Y = y < R = D$ and a messenger leaving L at speed $+1$. The messenger and the anchor are two outgoing strands at one contact, so the section has exactly five live signals. The centred coordinates

are $\xi = x - D/3$, $\eta = y - 2D/3$, $z = (D, \xi, \eta)^\top$ and $w = (\xi/D, \eta/D)^\top$, and the native gaps are $g = (x, y - x, D - y)$. A *complete word* lists every contact, transparent crossings and identity-rule contacts included. Every machine is fixed only after its rational parameters (a matrix, a scale) are chosen: none of the four sources gives one unchanged table for all matrices.

[write] **What the four sources do.** They form one chain of realization theorems, each inheriting the previous one by a pinned byte copy.

- (i) Source 30 realizes every $A \in \text{GL}_2(\mathbb{Q})$ with $\det A > 0$, at every positive rational scale λ , by a complete binary word returning $z \mapsto \lambda \text{diag}(1, A)z$ on an exact bounded open rational polygon (Theorem 240.1). It classifies the strict infinite-iteration kernel $\mathcal{K}(A, P)$ of every rational planar map on every bounded open convex rational polygon: it is nonsemialgebraic exactly in the infinite-order elliptic case (Theorem 240.2). It certifies that case with an explicit degree-twelve polynomial in $81J$ positive witnesses for J contacts (Theorem 240.3), and every other case by a unique-slack quadratic (Corollary 253.1).
- (ii) Source 31 proves that a fixed complete binary word with positive flights induces an invertible rational linear map between full sections (Theorem 257.1), with determinant $(-1)^m \prod_j s_j^+ / s_j^-$ (Theorem 257.2). A seven-event reversal (Theorem 257.3) extends source 30 to all of $\text{GL}_2(\mathbb{Q})$, and a fixed-scale return $\lambda \text{diag}(1, A)$ exists exactly when $\det A \neq 0$ (Corollary 257.4).
- (iii) Source 32 realizes every lift $N = \begin{pmatrix} s_0 & b \\ 0 & A \end{pmatrix}$ with $s_0 > 0$ and $A \in \text{GL}_2(\mathbb{Q})$, whose normalized action is the projective map $w \mapsto Aw / (s_0 + bw)$ fixing the centre (Theorem 267.1). It exhibits one 44-event word with both clock behaviours (Theorem 267.2) and proves a spectral Zeno criterion on the active cyclic subspace (Theorem 273.1).
- (iv) Source 33 proves that $N \in \mathbb{Q}^{3 \times 3}$ is realized on some full-dimensional open set exactly when $\det N \neq 0$ and $\mathcal{C} \cap N^{-1}\mathcal{C} \neq \emptyset$ (Theorem 281.1). Physical shape translations replace the rational eigenray that source 32 needs. Source 33 adds a rational linear-programming decision, a degree-six certificate with eight positive witnesses (Corollary 287.1) with native arithmetic graphs of 48 and 57 operations, and a compatible matrix with no infinitely valid input.

Each source answers a question of its predecessor; Section 293 gives the status of every question of the four.

[write] **Relation to Part X.** Source 30 generalizes sources 27 and 29 of Part X (Reports 57 and 59). Their rotation machines are the cases $A \in \text{SO}_2(\mathbb{Q})$ of Theorem 240.1, and their kernels (a disk minus one or finitely many dense backward orbits on its boundary) are cases of Theorem 240.2. Source 29's all-real (non- F_σ) obstruction (Theorem 231.1) is not restated by source 30. Source 30's certificate follows the pattern of sources 28 and 29 (bounded radix extraction of a Gaussian or companion power and two POWER modules); its fifteen-equation POWER module is printed once, in Part X (Section 217), and Section 251 here is a pointer (Remark 251.1). The witness counts of sources 28, 29 and 30 are compared in Table 16 of Part X; Remark 240.4 says why they differ. The rational accumulation time that all four sources prove for Zeno runs is a special case of source 26's Theorem 182.4, which covers every valid ultimately periodic complete macro of any fixed finite population. Source 30's cyclic-subspace proof is printed as a second route (Remark 244.1). Source 32's Theorem 273.1 adds a spectral criterion and is printed in full, and the restatements of sources 31 and 33 are Remarks 263.1 and 288.1.

[write] **Relation to Parts I and III.** The linearity half of source 31's Theorem 257.1 is a case of Part I's Theorem 5.3: once a complete chronology is fixed, every event coordinate is a rational linear form in the initial gaps. Only the invertibility is new here. The unique-slack quadratic

certificates of source 30 (Corollary 253.1), source 31 (Corollary 261.1) and source 32 (Section 274.2) use the device of Part I’s Theorem 6.1, applied to a new target, the infinite-validity sets. Source 31’s invertibility is compatible with Part III. There the integer gap matrices of one event have rank $n - 2$ (Proposition 35.1), and Section 35 states “No injectivity of the integer event map is asserted”. Part III’s matrices map a collision section into the whole gap space with the colliding gap set to zero, and its non-claim concerns the event map across branches of a universal machine. Source 31’s map goes between the full sections of one fixed word. Source 31 itself says that the images of different words may overlap, and that number preservation is not global reversibility. Nothing in Part III is retracted. Part III’s reversible universal machine with ten live signals (Theorem 29.2) is a different resource point. The machines of this Part are number-preserving, not globally reversible, and none of them is universal.

[write] **Review status.** The Hilbert’s-tenth programme made a bounded routing intake of the twelve signal and particle archives of arrival 0d7f51c44, these four included, in `review_new_signals_0d7f51c44.md` (commit 49b100cfd) [99]. It read every delivery README and selected spans of each source’s proofs. It found “No specific contradictory theorem/interface statement requiring an author correction” in those passages, and it does not certify the omitted chronology, the Pell proofs or the source graphs. It asks readers to keep apart local and invariant realization, full-section and projected maps, fixed-word invertibility and global reversibility, and real and integer inputs. It also records that source 33’s 48/57-operation certificates “concern a decidable matrix property; they do not compete with the universal 84-operation construction”. It ran none of the sources’ programs. This Part keeps those distinctions; Section 293 collects what the four sources do not claim.

[write] **Notation.** Part XI keeps each source’s notation inside that source. Table 23 lists the letters that carry several meanings across the four sources; the tempting false reading is the other source’s meaning. One symbol is renamed: source 31’s orientation-reversing matrix, which sources 31 and 32 call J , is written J here, because source 30 (and source 29 in Part X) use J for the number of contacts. Its subscripted forms P_J , T_J , M_J , ℓ_J and r_J become P_J , T_J , M_J , ℓ_J and r_J ; the matrix itself is unchanged. As in the rest of this report, $\mathbb{N} = \{0, 1, 2, \dots\}$ (source 30 defines its $\mathbb{N}_0$ the same way but never uses it), and source 30’s closure `cl` is printed with this report’s macro. Source 30’s figure colours, which its preamble defined by redefining the standard colour names `blue` and `red`, and the messenger and target colours of sources 31–33, use colour names of their own here. They do not recolour the figures of earlier Parts.

[write] **How the sources are printed.** The four sources follow in pipeline order, each with its title, its subtitle and its abstract. Their sections are this report’s sections, with equations numbered by section. The appendices of sources 32 and 33 become the last sections of those sources, and their “Appendix” cross-references read “Section”. A restated result is printed once, where the theme is owned. Every other occurrence is a numbered remark that quotes the statement, keeps the duplicate’s label and points to the printed proof. Source 32’s restatement of the inherited compiler (its Appendix A) and of the anchor scaling primitive are such remarks, and so are source 30’s POWER module and the accumulation-time restatements of sources 31 and 33. Each source’s own literature discussion, evidence section and questions are kept in full. Each source’s bibliography is merged into this report’s; Section 293.3 gives the details.

Source 30 (batch-91 manuscript 12): Planar signal realization and strict validity kernels

Exactly five live signals and explicit positive integer certificates, “Research Report 60”, 4 October 2026. [write] From the archive `Planar_Signal_Realization_Source.zip` (main file `Report60/Report60.tex`, 28-page PDF); its labels carry the prefix `smc:pl:`. Its sections are Sections 240–256

Table 23: [write] Letters with several meanings in Part XI. Each meaning holds only inside the named source.

Symbol	Meaning
$J, \mathbf{J}$	J : the number of fixed contacts and, as J_0, J_1, J_p , three positive witnesses (source 30; source 29 in Part X counts <i>distinct nearest</i> contacts with J). $\mathbf{J}$: the seven-event reversal matrix $\begin{pmatrix} -2/3 & 5/6 \\ 0 & 1 \end{pmatrix}$ (sources 31, 32; their J)
K	the total shear subdivision count (sources 30, 31, 32); the integer gap-coordinate matrix and K_{esc} (source 33); source 30's kernel is $\mathcal{K}$, not K , and source 31's kernel is $K(A, P)$
N	a denominator, a micro-step count and a horizon (source 30, which writes the return matrix N_0); the homogeneous 3×3 return matrix (sources 31, 32, 33)
H	the dilation factor count, a radix bound and a 2×2 metric matrix (source 30); the dilation count (sources 31, 32); the gap matrix with $H\mathbf{z} = (x, y - x, D - y)$ (source 33)
C	the companion matrix, C_q and C_A (source 30); the inherited compiler argument $F(C, \lambda)$, the ordered cone and C_v (source 32); source 33's ordered cone is $\mathcal{C}$ (Part I's $\mathcal{C}_\sigma$ is the chamber of a skeleton)
P	the guard polygon (all sources); $P_0 = q^m$ (source 30); the positive matrix P_g (source 32)
F	the certificate polynomial (sources 30, 33); the final target position $-\frac{2}{3}x + \frac{5}{6}y$ (source 31); the inherited planar compiler $F(C, \lambda)$ (source 32)
T	the transfer count, the translation $T_z(e)$ and the power output T_0 (source 30); clocks T_N, T_∞, T_J (source 31); a duration $T(z)$ (source 32); the shape translation T_a and the primitive $T_D(e)$ (source 33)
s, v, h	speeds, spectator positions and parameters, defined locally in each primitive

here. Its nine bibliography entries are this report's [6], [16], [96], [97], [87], [76], [98], [75] and [86]. Its abstract follows.

Every rational orientation-preserving planar map admits an exactly five-live-signal realization with a finite complete collision word, rational speeds, full outgoing-phase closure and an exact bounded open rational guard polygon. The construction includes a centered determinant dilation, so it does not require a rational square root of the determinant. We then classify the strict infinite-iteration kernel of every rational planar map on every bounded open convex rational polygon containing the origin. Its sole nonsemialgebraic case is infinite-order elliptic dynamics: an open rational ellipse together with its boundary minus finitely many backward contact orbits. An elementary denominator test gives uniform polynomial-time rational membership in that branch. Finally, those orbit complements have an explicit positive-integer polynomial certificate with $81J$ witnesses, $44J$ residuals and exact degree twelve; the strict kernel adds one witness and one residual. All remaining rational-input branches have explicit quadratic certificates with uniquely forced positive slack tuples. Every equation, positivity adapter and inherited Pell dependency is displayed. Conventional proofs, frozen source packets, and fresh independent exact/static audits are kept distinct from physical simulation and formal proof certification.

240 The three results and their common interface

An outgoing signal section has stationary markers at $0 < x < y < D$ and a messenger at the left anchor moving at speed $+1$. Put

$$\xi = x - D/3, \quad \eta = y - 2D/3, \quad w = (\xi/D, \eta/D)^T.$$

The messenger and left anchor are distinct outgoing strands at their shared contact. Live-signal count is a strand count, not a count of occupied spatial sites. A complete word records *all* collisions, including spectator crossings and any unprescribed contact receiving an identity rule.

Theorem 240.1 (Physical realization). *For every $A \in \text{GL}_2(\mathbb{Q})$ with $\det A > 0$ and every $\lambda \in \mathbb{Q}_{>0}$, an effective finite deterministic number-preserving rational-speed signal machine has exactly five live signals and a fixed complete binary-collision word returning*

$$(D, \xi, \eta)^T \mapsto \lambda \text{diag}(1, A)(D, \xi, \eta)^T.$$

The word restores every marker label and the messenger's outgoing left-anchor phase. Its exact normalized domain is a bounded open rational polygon P with $0 \in \text{int } P$. Infinite repetition is valid precisely when

$$w \in \mathcal{K}(A, P) := \bigcap_{n \geq 0} A^{-n}P. \quad (240.1)$$

Every valid infinite run is Zeno exactly when $0 < \lambda < 1$; then all strands approach the left anchor, and the accumulation time is rational for rational initial data.

Here $A^{-n}P$ means preimage, even in the later abstract classification where singular A is allowed. “Fixed” means fixed after A, λ have been selected. This is not one unchanged finite machine accepting arbitrary matrix inputs.

[write] [As existence statements, the physical realizations of Part X are cases of Theorem 240.1: source 29's (Theorem 225.1) is the case of an infinite-order rotation $A \in \text{SO}_2(\mathbb{Q})$ at a positive rational scale, and source 27's contracted rotation (Theorem 196.2) is one such case; their words are different constructions. The last sentence of the theorem, the rational accumulation time, is a special case of source 26's Theorem 182.4 (Remark 244.1).]

Theorem 240.2 (Exact planar geometry and decision). *Let $A \in \mathbb{Q}^{2 \times 2}$ and $P = \{w : h_i w < b_i, 1 \leq i \leq m\}$ be bounded, open, convex, rational and contain 0. Then $\mathcal{K}(A, P)$ is nonsemialgebraic exactly when*

$$\det A = 1, \quad -2 < \text{tr } A < 2, \quad \text{tr } A \notin \{-1, 0, 1\}. \quad (240.2)$$

In that case there are effective rational $Q = Q^T > 0$, $r^2 > 0$, and a nonempty finite set $E \subset \mathbb{Q}^2 \setminus \{0\}$ such that

$$\mathcal{K} = \{w : w^T Q w < r^2\} \cup \left(\{w : w^T Q w = r^2\} \setminus \bigcup_{z \in E, n \geq 0} \{A^{-n} z\} \right). \quad (240.3)$$

No finite real-coefficient polynomial-sign formula agrees with this predicate on every rational point. In all other cases the real kernel is a finite conjunction of real-algebraic linear equalities and rational strict or weak affine guards restricted to the bounded-orbit subspace; rational-input membership has a finite rational linear-sign formula.

For each fixed A , rational membership is polynomial-time in the encoding of P, w . The elliptic branch is polynomial-time uniformly in A, P, w . Uniform explicit affine-sign output can require exponentially many atoms. A credited low-order recurrence result also gives a uniform counting-hierarchy upper bound; no uniform polynomial-time result is inferred from it.

[write] [The kernels of source 27 (Theorem 196.2) and source 29 (Theorem 225.2) are instances of the elliptic case. Source 29 also proves an obstruction on all positive real inputs (Theorem 231.1), which source 30 does not restate; source 30's sign obstructions are for rational and positive integer inputs. The rotational-density mechanism is prior work of Fijalkow et al. [87], as source 30 says in Section 255.]

Theorem 240.3 (Explicit positive-integer certificate). *Fix an infinite-order elliptic rational A and $J \geq 1$ nonzero rational contacts. For a rational point encoded by integer linear expressions $(X/N, Y/N)$ with $N > 0$, avoidance of their backward orbits has an explicit integer-coefficient sum-of-squares polynomial of exact degree 12, existentially quantified over $81J$ ordinary positive integer witnesses. There are $44J$ residual equations before squaring.*

For the strict kernel (240.3), the ledger is $1 + 81J$ witnesses and $1 + 44J$ residuals. Three positive physical gaps can be used directly as inputs. An optional paid one-positive-input Cantor transport adds four witnesses and two quadratic residuals, giving $5 + 81J$ and $3 + 44J$ respectively, still of degree 12.

Remark 240.4 ([write] Three ledgers). Part X prints two earlier ledgers for related objects. Source 28 certifies one fixed rotation machine with 81 positive witnesses and 44 equations (85 and 46 with paid one-input transport). Source 29 certifies the rotation family with $1 + 80J$ witnesses and $1 + 43J$ equations, where J counts the distinct nearest contacts. Source 30's $81J$ and $44J$ (and $1 + 81J$, $1 + 44J$ for the strict kernel) count a different construction, for an arbitrary infinite-order elliptic map and any J fixed contacts. Per contact it has fourteen outer equations (O1)–(O14) and seven strictly positive variables, where source 29 has thirteen equations and six. Source 30 excludes the time-zero match and the positive-time matches by two separate positive witnesses J_0 and J_1 , and it notes that for a general elliptic map the times zero and one both have denominator one and must be distinguished (Section 249). Source 29's single gate (O13) also excludes the time-zero match of a rotation, whose Gaussian data at exponent 0 are $P = 1$, $C = 1$, $S = 0$. Whether source 30's two gates can be merged is open (Question 1). The three counts are explicit upper bounds for three constructions, not competing measurements of one object, and none is claimed minimal; Table 16 in Part X tabulates them.

Outside the infinite-order elliptic case, the finite rational linear description instead gives a quadratic sum-of-squares certificate with a unique positive slack tuple for each valid fixed integer-gap input (Corollary 253.1).

The elliptic certificate is a fully displayed finite polynomial schema. It is not a generic emitted arithmetic DAG, an optimized gate count, a small-witness algorithm, or a finite-fold representation. The sign obstruction has no quantified integer witnesses and so does not contradict this certificate. No universal Turing-completeness theorem, priority claim, minimal live-signal lower bound, or continuation through a Zeno point is asserted.

241 Complete collision primitives

Distances below are measured from the current stationary anchor. The messenger points into the interval at primitive entrance and has speed ± 1 . Only one target marker can move, always with speed of modulus less than one. Reflecting about the right anchor reverses physical velocities and preserves every distance inequality and duration.

241.1 Anchor scaling

For $u > 0$, launch a target initially at distance z with speed

$$h = \frac{u - 1}{u + 1} \in (-1, 1)$$

when the outward messenger reaches it. Reverse the messenger, bounce it at the anchor, restore the target on the next outward meeting, and reverse and bounce the messenger once more. The four principal time–position pairs are

$$(z, z), \quad (2z, 0), \quad (2z + uz, uz), \quad (2z + 2uz, 0). \quad (241.1)$$

Indeed $z + h((2z + uz) - z) = uz$. Denote this operation by $L_z(u)$; its duration is $2(z + uz)$.

The exact guards require z and uz to lie strictly between the same unchanged nearest stationary neighbors. The lower endpoint is the anchor for the nearest target; the outermost target has no upper neighbor. Monotonicity of the target between these endpoints prevents all target–neighbor contacts. For each stationary inner spectator at distance s , all four messenger crossings are

$$s, \quad 2z - s, \quad 2z + s, \quad 2z + 2uz - s. \quad (241.2)$$

The inequalities $s < z$ and $s < uz$ put these groups in the prescribed outward, reversed, outward and reversed orders between the principal events. Outer markers are never reached. The strict speed bound separates the launch and restoration strands and gives a unique intersection. Thus there are exactly $4 + 4k$ events with k inner spectators.

Necessity is equally important. If a prospective endpoint reaches or crosses a neighbor, continuity forces a target–neighbor contact by that checkpoint. Equality may instead give a simultaneous contact or collapse a prescribed interval. None preserves the selected globally time-separated complete binary word. The argument stops at the first obstruction and does not assume an affine continuation remains physical after it.

241.2 Reflector homothety

Let the target at t be nearest to the anchor, and let a stationary reflector be at $z > t$. For $v > 0$, its first outward hit launches the target at $h = (1 - v)/(1 + v)$ while the messenger continues outward, crosses every spectator, reflects at z , crosses those spectators in reverse, restores the target on its inward leg, and bounces at the anchor. Solving the two lines gives

$$t' = vt + (1 - v)z, \quad T_{\text{restore}} = 2z - t', \quad T_{\text{finish}} = 2z. \quad (241.3)$$

Call the operation $H_z(v)$. Its exact guards are $0 < t, t' < s$, where $s \leq z$ is the nearest unchanged outer marker. The target remains in this interval by monotonicity. A spectator q is crossed at q and $2z - q$; $t' < q$ places its returning crossing before restoration. Also $0 < t' < z$ places restoration between the reflector and final anchor bounces. There are $4 + 2k$ events for k intermediate spectators. Failed endpoints force the same first-contact or collapsed-interval obstruction as for scaling.

241.3 Translation and its complete guard reduction

For $e < 1$, concatenate

$$T_z(e) := L_t\left(\frac{1}{1 - e}\right) ; H_z(1 - e), \quad t \mapsto \frac{t}{1 - e} \mapsto t + ez. \quad (241.4)$$

Both target-moving phases have speed $e/(2 - e) \in (-1, 1)$. The exact guard is simply $0 < t, t + ez < s \leq z$. To justify the omitted internal row, put $k = t/(1 - e)$ and $f = t + ez$. Then

$$k - f = \frac{e(f - z)}{1 - e}.$$

For $0 < e < 1$, $t < k < f$ because $f < z$; for $e < 0$, $f < k < t$ because $t < z$; for $e = 0$ all three coincide. Thus the internal endpoint lies between the retained endpoints. Conversely an invalid internal endpoint already breaks the scale; if it is valid and the final endpoint fails, the homothety breaks the word. This proves necessity and sufficiency for the grouped translation.

Zero parameters are retained. When $e = 0$ or $u = v = 1$, the temporary target label has speed zero, but its messenger partner has speed ± 1 . These are nonempty positive-duration collision sequences, not empty-word conventions.

241.4 Full anchor transfers

With every marker stationary and $0 < x < y < D$, a left-to-right transfer crosses x , crosses y and reflects at D , at relative times x, y, D . The final messenger points left. The reverse transfer crosses y , crosses x and reflects at 0 , at times $D - y, D - x, D$, and ends pointing right. Each costs time D and three events. Its guard is exactly the existing strict ordering; it is never a zero-time phase switch.

242 Signed shears and determinant dilation

Write

$$U(t) = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}, \quad V(t) = \begin{pmatrix} 1 & 0 \\ t & 1 \end{pmatrix}.$$

For rational t , let $N = \max(1, \lceil 4|t| \rceil)$ and $\delta = t/N$, so $|\delta| \leq 1/4$, including $\delta = 0$.

242.1 Exact centered shear blocks

A left-anchor micro-shear is $T_y(\delta)$ followed by $T_D(-2\delta/3)$. It sends x to $x + \delta(y - 2D/3)$ and fixes y, D , hence acts by $U(\delta)$ on (ξ, η) . Its first translation has $4 + 4$ events. The second has $4 + 6$, including both crossings of the y spectator. The total is 18 events and four temporary target labels.

For all N steps, define the incoming-coordinate endpoints

$$a_{2k} = x + k\delta(y - 2D/3) \quad (0 \leq k \leq N), \quad (242.1)$$

$$a_{2k+1} = a_{2k} + \delta y \quad (0 \leq k < N). \quad (242.2)$$

The exact supplied rows are

$$y > 0, \quad D - y > 0, \quad a_i > 0, \quad y - a_i > 0 \quad (0 \leq i \leq 2N). \quad (242.3)$$

There are $4N + 4$ rows. The translation lemma supplies all hidden internal checkpoints, so these rows characterize all $18N$ events, including spectator crossings. The return is $U(t)$.

For a right-anchor block put $r = D - y$, $s = D - x$, and use the same construction with $(x, y) = (r, s)$. It sends r to $r + t(s - 2D/3)$, so

$$y' = y + t(x - D/3),$$

which is $V(t)$ with the *same* sign. Its $4N + 4$ rows are $s > 0, D - s > 0, b_i > 0, s - b_i > 0$, using the reflected endpoint formulas. It also has $18N$ events and four temporary labels per micro-step.

At the center $x = D/3, y = 2D/3$, every completed micro-shear fixes its target. The intermediate translation endpoint is

$$D/3 + 2\delta D/3 \in [D/6, D/2],$$

strictly inside the interval ending at $2D/3$. Thus every retained row is center-strict, for arbitrarily large total shear: large parameters are realized by finitely many small legal steps.

242.2 Every determinant-one factorization and zero pivot

Let $B = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ with $ad - bc = 1$. Products act on column vectors; the rightmost factor is performed first. Direct multiplication gives the following exhaustive branches:

$$c \neq 0 : B = U((a-1)/c)V(c)U((d-1)/c), \quad (242.4)$$

$$c = 0, b \neq 0 : B = V((d-1)/b)U(b)V((a-1)/b), \quad (242.5)$$

$$b = c = 0 : B = U(a-1)V(1)U(a^{-1}-1)V(-a). \quad (242.6)$$

For (242.4), the upper-right entry is $(ad-1)/c = b$. For (242.5), the lower-left entry is $(ad-1)/b = c$. The diagonal branch first establishes $a \neq 0$ and $d = a^{-1}$, and works for negative a . No other branch divides by a or d , so zero diagonal pivots are covered.

The chronological axis lists are respectively $x, y, x; y, x, y$; and y, x, y, x , with reversed parameter order from the displayed products. Start at the left anchor, insert a full transfer whenever the next axis changes anchor, and end with a return to the left if required. For B_0 blocks and indicators $\sigma_j = 0$ for x , 1 for y , the transfer count is

$$T = \sigma_1 + \sum_{j=1}^{B_0-1} |\sigma_{j+1} - \sigma_j| + \sigma_{B_0}.$$

The branches give $T = 2, 4, 4$ respectively, with $B_0 = 3, 3, 4$. The initial and final transfers in the second branch cannot be omitted. If K is the sum of the subdivision counts, the shear word has $18K + 3T$ events, $4K + 4B_0$ supplied guards and $4K$ temporary labels. All markers and the messenger phase close.

For $B = I$ the retained list is $y(-1), x(0), y(1), x(0)$. It has $K = 10, T = 4$ and 192 events; it is a genuine positive-duration closed word. No identity simplification is used in these counts.

242.3 A centered one-axis dilation

For $q \in [3/4, 5/4]$, perform

$$L_x(q) ; T_D((1-q)/3).$$

The target endpoints are $x, qx, x' = qx + (1-q)D/3$, hence $\xi' = q\xi, \eta' = \eta, D' = D$. Its translation parameter is in $[-1/12, 1/12]$. The exact eight rows are

$$\begin{aligned} y > 0, \quad D - y > 0, \quad x > 0, \quad y - x > 0, \\ qx > 0, \quad y - qx > 0, \quad x' > 0, \quad y - x' > 0. \end{aligned} \quad (242.7)$$

The endpoint qx must remain. For example $q = 5/4, x = 1/2, y = 3/5, D = 1$ gives $x' = 13/24 \in (0, y)$ but $qx = 5/8 > y$. The translation's own internal endpoint needs no extra row by (241.4). At the

center the guarded positions are $D/3, qD/3, D/3$, with $qD/3 \in [D/4, 5D/12]$. The block has 14 events, three temporary labels and a center-strict exact chamber.

For arbitrary $q > 0$, $q \neq 1$, set

$$H = \max \left(1, \left\lceil \frac{4|q-1|}{\min(1,q)} \right\rceil \right), \quad u_j = 1 + \frac{j(q-1)}{H}, \quad q_j = \frac{u_j}{u_{j-1}}. \quad (242.8)$$

All $u_j \geq \min(1, q) > 0$, and $|q_j - 1| \leq 1/4$. Their product telescopes to q . Compose these H small blocks to realize $C_q = \text{diag}(q, 1)$, using $14H$ events, $3H$ temporary labels and $8H$ guards. For $q = 1$, put $H = 0$ and omit only this suffix.

Given A with $q = \det A > 0$, define $B = C_q^{-1}A \in \text{SL}_2(\mathbb{Q})$. Perform the shear word for B , complete its left-anchor return, and then perform C_q . The return is $C_q B = A$. In particular nonsquare determinants such as 2 and $3/2$ require no irrational square root and no extra live signal.

242.4 Arbitrary positive global scale and full ledger

If $\lambda < 1$, scale targets x, y, D in that order by λ , always about the left anchor. Inner neighbors have already been scaled and outer neighbors have not. Both target endpoints therefore stay in the unchanged neighbor interval. For $\lambda > 1$ use D, y, x in reverse order, so outer neighbors have already been scaled. The same monotonicity argument applies. The event counts 4, 8, 12 sum to 24; there are three temporary labels and no additional guard. For $\lambda = 1$ omit this suffix.

Writing $\varepsilon = \mathbf{1}_{\lambda \neq 1}$, the complete counts are

$$\begin{aligned} m_{\text{events}} &= 18K + 3T + 14H + 24\varepsilon, \\ m_{\text{guards}} &= 4K + 4B_0 + 8H, \\ m_{\text{temporary}} &= 4K + 3H + 3\varepsilon, \\ m_{\text{metasignals}} &= 22K + 3T + 17H + 27\varepsilon + 4, \\ m_{\text{live}} &= 5. \end{aligned} \quad (242.9)$$

The alphabet consists of one messenger phase for each event, the temporary target labels, and the four permanent stationary labels. These counts are not minimized and need not be polynomial in the bit length of A .

243 Exact chamber and finite deterministic completion

Pull every supplied row back through earlier homogeneous rational returns in coordinates $z = (D, \xi, \eta)^T$. Each has form $\alpha D + \beta \xi + \gamma \eta > 0$ with rational coefficients. Transfers and global scaling introduce no extra rows. On $D = 1$ their conjunction defines P .

Sufficiency follows by induction over the exact primitive chronology. Every nonmoving marker is stationary, the sole moving marker stays in its strict neighbor interval, and every messenger leg lists all stationary points it crosses. The strict speed inequalities exclude hidden target intersections and ensure all binary contacts are separated in time.

For necessity, take the first primitive or grouped translation with a failed prospective endpoint, all previous checkpoints being ordered. Its preceding word prefix is valid. A scale or homothety then forces an additional, simultaneous, collapsed or reordered contact by the failed checkpoint. For a translation, the internal scale either already fails, or its endpoint is valid and the final homothety

fails. Initial row failure means the section itself was not allowed. This uses affine equations only until the first physical obstruction. Default rules cannot erase extra contacts from a *complete* collision word.

Every completed shear and determinant dilation fixes the center, and all internal center check-points were verified. Therefore every pulled-back row is strict at 0. The first block's rows imply $0 < x < y < D$; for a right-based first block, $0 < r < s < D$ is the equivalent condition. Hence P is a full-dimensional bounded open convex rational polygon. Supplied rows may be repeated or redundant: exactness is a statement about their conjunction, not facet irredundancy.

Give the messenger a fresh label Q_j before event j , for $0 \leq j < m_{\text{events}}$, and identify the final outgoing Q_m with Q_0 . Temporary target labels carry their rational speeds in $(-1, 1)$. At the prescribed event emit

$$\{Q_j, Z_{\text{in}}\} \longrightarrow \{Q_{(j+1) \bmod m}, Z_{\text{out}}\}.$$

Only launch and restoration change the marker label. Spectators and anchor/reflector bounces preserve the stationary marker. Input keys are distinct because Q_j is distinct. Incoming and outgoing pairs have distinct speeds, since the messenger speed is ± 1 . Incoming order is decreasing speed and outgoing order increasing speed, so outgoing strands separate. At the final event every permanent marker label is restored and Q_0 points right at the left anchor.

Complete the finite rule table with the identity output on every other legal input set of pairwise distinct speeds, including larger sets. This is deterministic and preserves cardinality. The prescribed rules consume and emit one messenger and one marker, and the default completion also preserves cardinality. There are exactly five live strands throughout every finite part of a valid run. The standard outgoing-collision convention is consistent with the definition and initial-configuration discussion in Becker et al. [6].

244 Iteration and the rational Zeno clock

At a completed word $D' = \lambda D$ and $w' = Aw$. Full label and phase restoration allows literal word repetition; the exact chamber then proves (240.1). Its center is always valid, so the infinite domain is nonempty. The compiler's P is its actual exact chamber; no claim permits arbitrary replacement of P while retaining the physical counts.

Every primitive duration is a rational homogeneous linear form in its incoming positions. Thus the full duration is a fixed rational row ℓz . Before the global suffix all positions lie in $[0, D]$. A translation takes less than $6D$, a micro-shear less than $12D$, a centered dilation less than $10D$, and a transfer exactly D . The global suffix takes

$$2(1 + \lambda)(x + y + D) < 6(1 + \lambda)D.$$

Every retained micro-shear contains a far-reflector H_D phase of duration $2D$, and at least one such step remains even for the identity. Consequently, throughout the exact chamber,

$$2D \leq \ell z < (12K + T + 10H + 6(1 + \lambda)\varepsilon)D. \quad (244.1)$$

At iteration n , $D_n = \lambda^n D_0$. The two-sided comparison with a geometric series proves Zeno behavior exactly for $0 < \lambda < 1$. In that case every point of the n th word lies in $[0, \max(1, \lambda)D_n]$, hence all strands tend to the fixed left anchor. For $\lambda \geq 1$ the lower bound forces divergent total time.

For rational data and $\lambda < 1$, write $N_0 = \lambda \text{diag}(1, A)$ and take the rational cyclic subspace

$$V = \text{span}_{\mathbb{Q}}\{z, N_0 z, N_0^2 z\}.$$

It is invariant by Cayley–Hamilton. A valid normalized orbit stays in bounded P , so $N_0^n z \rightarrow 0$. Since z is cyclic on V , the restriction has no eigenvalue of modulus at least one. Thus $I - N_0$ is invertible on V over $\mathbb{Q}$, and

$$T_\infty = \ell((I - N_0)|_V)^{-1} z \in \mathbb{Q}. \quad (244.2)$$

A rational basis of V reduces this to rational linear algebra. The full-space inverse need not exist: an unused expanding eigenmode of A can have eigenvalue $1/\lambda$. Formula (244.2) avoids that singularity. No state or continuation after accumulation is defined.

Remark 244.1 ([write] A second route to a Part X theorem). The rationality of T_∞ is a special case of source 26’s Theorem 182.4 (Part X): a valid ultimately periodic complete macro of any fixed finite population, with rational speeds and rational initial positions, has only rational finite accumulation times. The cyclic-subspace computation (244.2), which avoids an unused expanding mode, is source 30’s own route and is printed here once. Sources 31, 32 and 33 repeat it for their compositions (Remark 263.1, equation (273.4) and Remark 288.1); source 32’s Theorem 273.1 adds a spectral criterion.

245 The bounded-orbit subspace and every spectral case

This part applies to *every* rational 2×2 matrix, including singular and negative-determinant matrices, independently of the physical realization theorem. Let

$$P = \{w \in \mathbb{R}^2 : h_i w < b_i, 1 \leq i \leq m\}$$

be bounded, open, convex and rational with $0 \in P$. All $b_i > 0$. Discard constant rows; redundant nonconstant rows are allowed. Define

$$\mathcal{B}(A) = \{w : \sup_{n \geq 0} \|A^n w\| < \infty\}.$$

Because P is bounded, $\mathcal{K}(A, P) \subseteq \mathcal{B}(A)$.

In real Jordan coordinates, $\mathcal{B}(A)$ is the sum of the entire generalized eigenspaces for $|\mu| < 1$ and the ordinary eigenspaces for $|\mu| = 1$. Expanding modes contribute only zero. A stable Jordan block has a polynomial times geometric decay; a nontrivial Jordan block at ± 1 grows linearly unless the generalized coordinate vanishes. An expanding block grows for every nonzero component. An invertible eigenbasis prevents unbounded components from canceling in norm. These observations give an effective real-algebraic basis, using quadratic roots and linear equations.

Repeated eigenvalues are rational, equal to $\text{tr}(A)/2$. A repeated stable eigenvalue, including a nontrivial block or $A^2 = 0$, gives $\mathcal{B} = \mathbb{R}^2$. At ± 1 , the scalar case gives $\mathbb{R}^2$ and a nontrivial block gives the ordinary eigenline. A repeated expanding eigenvalue gives $\{0\}$. Distinct real eigenvalues contribute their stable and unit eigenspaces. Nonreal conjugates have common modulus $\sqrt{\det A}$ and yield $\mathbb{R}^2$ when that modulus is at most one, or $\{0\}$ when it is greater than one. This exhausts the planar possibilities.

Lemma 245.1 (Closure, without replacing strict validity). *For*

$$C = \bigcap_{n \geq 0} \{w : h_i A^n w \leq b_i \text{ for every } i\}, \quad \text{cl}(\mathcal{K}) = C.$$

Proof. The forward inclusion follows from continuity. For $w \in C$ and $0 \leq s < 1$, $h_i A^n(sw) \leq sb_i < b_i$, so $sw \in \mathcal{K}$ and $sw \rightarrow w$. The assertion concerns the closure only, not the membership of every point on its boundary. $\square$

245.1 Zero and one bounded dimension

If $\mathcal{B} = \{0\}$, then $\mathcal{K} = \{0\}$. If $\mathcal{B}$ is a line, A acts on it by a real μ with $|\mu| \leq 1$, and

$$\mathcal{K} = \begin{cases} \mathcal{B} \cap P, & 0 \leq \mu \leq 1, \\ \mathcal{B} \cap P \cap A^{-1}P, & -1 \leq \mu < 0. \end{cases} \quad (245.1)$$

For nonnegative μ the orbit lies in $[0, w]$; for negative μ its even and odd subsequences lie in $[0, w]$ and $[0, Aw]$. Convexity and $0 \in P$ prove both formulas, including $0, \pm 1$. The guard rows are rational in ambient coordinates, but the line itself need not be rational.

For example,

$$A = \begin{pmatrix} 0 & 1 \\ -1 & 3 \end{pmatrix}, \quad P = (-1, 1)^2, \quad \mu = (3 - \sqrt{5})/2$$

gives the nontrivial kernel segment on $y = \mu x$. No finite Boolean combination of rational affine signs describes this *real* segment. Otherwise, at every point at least one nonconstant atomic form would vanish, since nonzero signs persist in an ambient disk. Finitely many rational lines cannot cover an irrational-slope segment: each meets it in at most one point.

The same line contains no nonzero rational point. Thus its rational-input predicate is simply $w = 0$. More generally, a rational nonzero eigenvector makes its eigenvalue rational by dividing a nonzero coordinate of Aw by the corresponding coordinate of w , and a rational eigenvalue has a rational eigenspace. This proves the coefficient-field distinction required in Theorem 240.2.

245.2 Two stable dimensions and a finite strict horizon

Suppose every eigenvalue has modulus less than one. Enumerate exact rational powers until

$$\|A^k\|_\infty \leq 1/2, \quad k \geq 1, \quad C_A = \max_{0 \leq j < k} \|A^j\|_\infty.$$

Termination follows from $A^n \rightarrow 0$. Compute rational $R, \epsilon > 0$ with

$$P \subset [-R, R]^2, \quad [-\epsilon, \epsilon]^2 \subset P.$$

An outer R follows by enumerating the closed polygon's vertices. One may take $\epsilon = \min_i b_i / (2\|h_i\|_1)$. Choose $q_0 \geq 0$ with $2^{-q_0} C_A R < \epsilon$ and put $N = q_0 k$. For $n = lk + j \geq N$, $l \geq q_0$, $0 \leq j < k$, every $w \in P$ satisfies

$$\|A^n w\|_\infty \leq 2^{-l} C_A R < \epsilon.$$

All later guards are automatically strict, so

$$\mathcal{K} = \bigcap_{0 \leq n \leq N} A^{-n} P. \quad (245.2)$$

The redundant time N makes the convention valid also when $N = 0$. This is a finite open rational polygon. Neither diagonalizability, positivity, invertibility nor a rational eigenbasis was used.

For fixed A , k, C_A are constants. In dimension two, candidate vertices come from pairs of supplied lines; exact feasibility and Cramer's rule have polynomial bit cost. Their coordinates, R , and ϵ have polynomial bit length in the input length L . Thus $\log(R/\epsilon) = O(L)$ and $N = O_A(L)$. The powers through N and all pulled-back rows have polynomial bit length, proving fixed- A polynomial membership. If A is input, the enumeration is only an effective procedure; it does not prove a uniform polynomial horizon or output size.

245.3 Real unit spectra and mixed unit–stable spectra

If $\mathcal{B} = \mathbb{R}^2$ and both real eigenvalues have modulus one, semisimplicity gives $A^2 = I$, so $\mathcal{K} = P \cap A^{-1}P$. Otherwise consider distinct eigenvalues $\varepsilon \in \{1, -1\}$ and μ with $|\mu| < 1$. Since the trace is rational, μ is rational. Let

$$E = \frac{A - \mu I}{\varepsilon - \mu}, \quad F = I - E, \quad A = \varepsilon E + \mu F.$$

If $0 < |\mu| < 1$, write $r = \mu^2 \in (0, 1)$. For $j = 0, 1$,

$$A^{2k+j}w = \varepsilon^j Ew + r^k \mu^j Fw.$$

Each scalar guard is a convex interpolation of its initial parity value $h_i A^j w$ and limiting value $h_i \varepsilon^j Ew$. Since $r^k > 0$, all finite values are strict exactly when the first is strict and the second weak. Hence

$$\mathcal{K} = P \cap A^{-1}P \cap \{w : Ew \in \text{cl}(P), \varepsilon Ew \in \text{cl}(P)\}. \quad (245.3)$$

Necessity follows from $k = 0$ and the limit; sufficiency follows from positive weight on the strict endpoint. Both signs of both eigenvalues are covered.

If $\mu = 0$, the limit is actually reached at times one and two. The correct formula instead is

$$\mathcal{K} = P \cap A^{-1}P \cap A^{-2}P, \quad (245.4)$$

with every guard strict. For $\varepsilon = 1$ the last factor is redundant. Weak limiting inequalities cannot be used at an attained limit.

A concrete half-open example is

$$A = \text{diag}(1, 1/2), \quad P = \{(x, y) : -1 < x < 2, -1 < y < 1, x + y < 1\}.$$

The rectangle guards persist. The remaining guards $x + 2^{-n}y < 1$ for all n are equivalent to $x + y < 1$ and $x \leq 1$, so $\mathcal{K} = P \cap \{x \leq 1\}$. It contains $(1, -1/2)$ even though the limiting point $(1, 0)$ is outside P . No finite strict time prefix gives this set, because such an intersection is open. Finite linear description and finite strict-prefix determination are different properties.

246 The rational critical ellipse and its missing orbits

For a nonreal unit spectrum, $\det A = 1$ and $t = \text{tr } A \in (-2, 2)$. Choose nonzero $v \in \mathbb{Q}^2$. The columns of $S = [v, Av]$ are independent, since A has no real eigenvector. Cayley–Hamilton yields

$$AS = SC, \quad C = \begin{pmatrix} 0 & -1 \\ 1 & t \end{pmatrix}.$$

For $H = \begin{pmatrix} 1 & t/2 \\ t/2 & 1 \end{pmatrix}$, direct multiplication gives $C^T H C = H$. Since $1 - t^2/4 > 0$, H is positive definite. Therefore

$$Q = S^{-T} H S^{-1} \in \mathbb{Q}^{2 \times 2}, \quad Q > 0, \quad A^T Q A = Q. \quad (246.1)$$

A real change of basis makes the map a Euclidean rotation. A rational Euclidean conjugacy is neither needed nor automatic.

If an eigenvalue is a root of unity, its sum with its inverse is the rational algebraic integer t , hence an integer. The only possibilities are $-1, 0, 1$, giving orders $3, 4, 6$. Conversely each such trace gives that order. These cases have the finite open polygon

$$\mathcal{K} = \bigcap_{0 \leq n < \text{ord}(A)} A^{-n}P.$$

Every other rational trace in $(-2, 2)$ is an irrational-angle rotation, and every nonzero orbit is dense in its Q -ellipse.

For each nonzero supplied row put

$$d_i = h_i Q^{-1} h_i^T > 0, \quad r_i^2 = \frac{b_i^2}{d_i}, \quad z_i = \frac{b_i}{d_i} Q^{-1} h_i^T. \quad (246.2)$$

All these data are rational and $z_i^T Q z_i = r_i^2$. Set $r^2 = \min_i r_i^2$ and deduplicate the contacts z_i attaining that minimum to obtain nonempty E . The Q -inner-product Cauchy–Schwarz inequality gives

$$h_i w \leq \sqrt{d_i} \sqrt{w^T Q w}.$$

Every point below radius r^2 therefore satisfies every guard strictly. On the critical ellipse a guard can fail only by equality, exactly at its minimizing contact. On a larger ellipse a minimizing guard is strictly violated on a nonempty open arc, reached by orbit density. This proves (240.3), including all strict-boundary exceptions.

For example $A = \begin{pmatrix} 0 & -1 \\ 1 & 1/2 \end{pmatrix}$ preserves $Q = \begin{pmatrix} 1 & 1/4 \\ 1/4 & 1 \end{pmatrix}$. Its Euclidean rotation has cosine $1/4$ and sine $\sqrt{15}/4$. Rational conjugation of a rational matrix stays rational, so no rational basis can turn this map into that Euclidean rotation matrix. The metric formulation is the correct general one.

246.1 Dense rational tails and finite-sign impossibility

Choose $z \in E$. The rational orbit $O = \{A^k z : k \in \mathbb{Z}\}$ has no repetitions. Let

$$I_z = \{k \in \mathbb{Z} : A^k z \in E\}, \quad k_* = \max I_z.$$

The set is nonempty and finite. Restricted to O , the union of backward contact orbits is exactly the indices $k \leq k_*$; all $k > k_*$ are accepted. Both tails are dense on the critical ellipse and consist of rational points.

A finite set of real polynomials partitions an ellipse into finitely many open arcs with constant signs, together with finitely many exceptional points; restrictions vanishing identically contribute a constant zero sign. Both tails meet each open arc. Thus no finite polynomial-sign Boolean formula separates them, even only on rational inputs. In particular the real kernel is nonsemialgebraic. All previous cases had finite algebraic linear descriptions, completing the dichotomy.

For the physical machine, let positive gaps be $g_1 = x$, $g_2 = y - x$, $g_3 = D - y$. The closed critical ellipse lies in the closed initial-order triangle. Each of the triangle's three boundary lines meets the nondegenerate ellipse in finitely many points. Removing those finitely many from the rejected tail leaves it dense; the accepted tail already belongs to P . Affine normalization therefore transfers the impossibility to every positive rational gap triple.

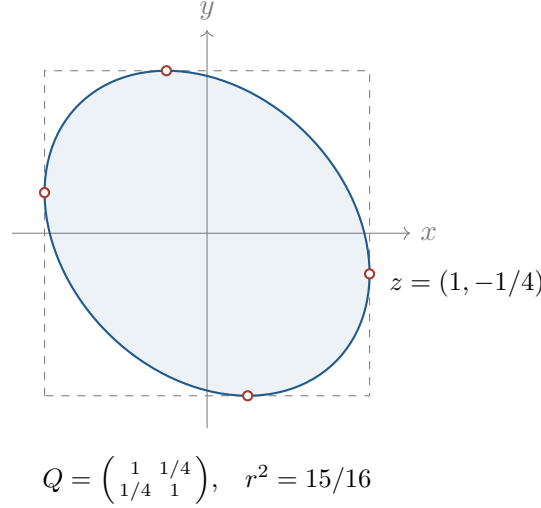


Figure 14: Analytic geometry for $A = \begin{pmatrix} 0 & -1 \\ 1 & 1/2 \end{pmatrix}$ and $P = (-1, 1)^2$. The four open circles mark the rational contact set. The interior is accepted; the boundary additionally loses every backward iterate of each contact, a dense countable set not individually drawn. This is an exact formula illustration, not a particle simulation or a complete plot of the accepted boundary.

Suppose a finite polynomial-sign formula agreed on every positive *integer* gap triple. Decompose each atomic polynomial into homogeneous parts. Along a positive ray, its eventual sign is the sign of its highest nonzero homogeneous component, or zero if all vanish. This eventual-sign rule is itself a finite semialgebraic formula. For any positive rational gap triple, clearing denominators gives arbitrarily large integer points on the same ray, and physical validity is homogeneous. The eventual-sign formula would therefore agree on every positive rational ray, contradicting the dense-tail argument. This excludes unquantified finite sign formulas, not existential integer certificates.

247 An elementary exact elliptic decision algorithm

Construct Q, r^2, E by (246.1)–(246.2). For rational w , accept when $w^T Q w < r^2$ and reject when it is greater. On equality, reject exactly if $A^n w = z$ for some $z \in E, n \geq 0$. Although the general rational Orbit Problem already supplies a polynomial algorithm [97], here a denominator forces a unique positive candidate time.

Write $\text{tr } A = p/q$ in lowest terms, with $q > 0$. Infinite order implies $q \geq 2$. For a contact z put

$$S_z = [z, A^{-1}z], \quad C = \begin{pmatrix} 0 & -1 \\ 1 & p/q \end{pmatrix}, \quad y = S_z^{-1}w.$$

The columns are independent and $A^{-1}S_z = S_z C$. The forbidden condition is $y = C^n e_1$. Define

$$V_0 = 1, \quad V_1 = p, \quad V_k = pV_{k-1} - q^2V_{k-2}.$$

Then $Ce_1 = e_2$, and for $n \geq 2$,

$$C^n e_1 = \left(-\frac{V_{n-2}}{q^{n-2}}, \frac{V_{n-1}}{q^{n-1}} \right)^T. \quad (247.1)$$

For every prime dividing q , $V_k \equiv p^k \not\equiv 0$ modulo that prime. Thus $\gcd(V_k, q) = 1$, and the joint least common positive denominator is *exactly* q^{n-1} for every $n \geq 1$, including $n = 1$.

First test $y = e_1$, the exceptional exponent $n = 0$. Otherwise compute the common reduced denominator b . If b is not a pure whole-integer power q^m , there is no hit. If $b = q^m$, compute only $n = m + 1$ by exact recurrence or powering and compare with y . In particular $b = 1$ supplies the candidate e_2 at time one, after the separate e_1 test. Repeated exact division by q tests a pure power; no prime factorization is needed. Negative p , composite q , repeated prime factors and zero components cause no exception.

All basis changes and contact data have polynomial bit length. The candidate time obeys

$$n = m + 1 \leq 1 + \lfloor \log_2 b \rfloor.$$

This includes $b = 1$; the additive one is necessary. Exact recurrence values have polynomial bit length at this candidate time, and the number of contacts is at most the number of supplied guard rows. Hence the branch is polynomial-time uniformly in A, P, w . Finite-order elliptic cases need at most six checks. The bound concerns point membership, not the construction of Pell witnesses below.

247.1 Credited uniform counting-hierarchy context

For any guard define $u_n = b - hA^n w$, let $s = \text{tr } A$ and $d = \det A$. Cayley–Hamilton gives

$$u_{n+3} = (s + 1)u_{n+2} - (s + d)u_{n+1} + du_n. \quad (247.2)$$

Thus each guard is strict positivity of a rational recurrence of order at most three. Ouaknine and Worrell [75], Theorem 4.1 and Section 6, give the counting-hierarchy upper bound

$$\text{coNP}^{\text{P}^{\text{P}^{\text{P}^{\text{P}}}}}$$

for this low-order strict-positivity problem. Their rational-to-integer scaling $v_n = \ell^{n+1}u_n$ preserves sign, and their degeneracy treatment covers repeated and unit roots. Their recurrence convention excludes a zero last coefficient. If A is singular, checking the first two terms removes every possible planar zero-eigenvalue transient; then zero characteristic factors are dropped before applying the theorem. Apply it to each supplied guard and conjoin the answers. Polynomially many such conjunctions retain this upper bound by the usual nondeterministic complement/guard-index choice.

This is an application of an existing complexity theorem, not a new uniform polynomial bound and not an optimality assertion. It coexists with the elementary fixed- A polynomial proof, the uniformly polynomial elliptic branch, and the explicit-output lower bound next.

248 Exponential explicit affine output in a stable family

Let $M \geq 4$, $\lambda = 1 - 1/M$, and

$$A_M = \begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix}, \quad P = (-1, 1)^2.$$

The input has $O(\log M)$ bits and is stable. For $n \geq 1$,

$$A_M^n(x, y) = (\lambda^n x + n\lambda^{n-1}y, \lambda^n y).$$

In the positive quadrant the first-coordinate upper guard is $y < f_n(x)$, where

$$f_n(x) = \frac{\lambda^{1-n}}{n} - \frac{\lambda x}{n}.$$

Adjacent lines meet at

$$x_n = \lambda^{-n} \left(1 - \frac{n(1-\lambda)}{\lambda} \right), \quad y_n = (1-\lambda)\lambda^{-n}. \quad (248.1)$$

Put $x_0 = 1$. Direct subtraction gives

$$x_n - x_{n-1} = -n(1-\lambda)^2 \lambda^{-n-1} < 0.$$

The adjacent difference $f_{j+1} - f_j$ has positive slope and zero at x_j . For $x_n < x < x_{n-1}$ these ordered signs telescope toward n , giving $f_n(x) < f_j(x)$ for every $j \neq n$, including arbitrarily large j .

Let $N = \lfloor (M-1)/2 \rfloor$. For $1 \leq n \leq N$, the displayed segment lies in $0 < x < 1$ and $0 < y < 1$: the factor in x_n is nonnegative in this range, monotonicity gives $x_n < 1$, and Bernoulli's inequality gives $\lambda^n \geq 1 - n/M \geq 1/2$, so $y_n \leq 2/M \leq 1/2$. The first endpoint satisfies $f_1(1) = 1/M$. Both coordinates of every iterate are nonnegative on the segment, so all lower guards are automatic. All second-coordinate upper guards are strict, as is the time-zero first-coordinate guard. Therefore

$$\{(x, f_n(x)) : x_n < x < x_{n-1}\}$$

is a genuine exposed boundary segment of $\text{cl}(\mathcal{K})$, activating only the first-coordinate upper guard at time n .

These N facet lines have distinct slopes $-\lambda/n$. Any finite affine-sign Boolean formula for $\mathcal{K}$ must include each line among its atomic zero sets: otherwise a generic point on the segment avoids all atomic zeros and the formula is locally constant across a true boundary. Thus $\Omega(M)$ atoms are necessary. Taking $M = 2^L$ gives exponential output size in the encoded input length. This does not lower-bound succinct membership algorithms, unrestricted nonlinear atom languages, or quantified auxiliary descriptions.

249 Integer coefficient extraction for arbitrary rational elliptic maps

Fix an infinite-order elliptic $A \in \text{SL}_2(\mathbb{Q})$ and nonzero rational contacts $z_1, \dots, z_J$. Write $\text{tr } A = p/q$ in lowest terms, with $q \geq 2$ and $|p| < 2q$. Matrix, trace, contact and metric data are fixed coefficients, not witnesses. An input point is

$$w = (X/N, Y/N)^T, \quad X, Y \in \mathbb{Z}, \quad N \in \mathbb{Z}_{>0},$$

without any coprimality requirement. Five positive inputs suffice by taking $X = a_1 - a_2$, $Y = a_3 - a_4$, $N = a_5$. More generally X, Y, N may be fixed integer linear expressions in positive inputs, provided $N > 0$ on that domain.

For physical positive gaps g_1, g_2, g_3 , use exactly

$$X = 2g_1 - g_2 - g_3, \quad Y = g_1 + g_2 - 2g_3, \quad N = 3(g_1 + g_2 + g_3). \quad (249.1)$$

Then $(X/N, Y/N) = (\xi/D, \eta/D)$. No extra witnesses are introduced by these input aliases.

249.1 Companion coordinates and integer power coefficients

For one contact z , set $B = A^{-1}$, $S = [z, Bz]$ and $C = \begin{pmatrix} 0 & -1 \\ 1 & p/q \end{pmatrix}$. Then $BS = SC$. With $U_{-1} = 0$, $U_0 = 1$ and $U_k = (p/q)U_{k-1} - U_{k-2}$, induction gives

$$C^0 e_1 = e_1, \quad C^n e_1 = (-U_{n-2}, U_{n-1})^T \quad (n \geq 1).$$

Writing $V_k = q^k U_k$ recovers the integer recurrence and exact denominator proof in Section 247. Both $n = 0$ and $n = 1$ have denominator one and must be distinguished.

Let β be a root of $f(Z) = Z^2 - pZ + q^2$. The roots are distinct, of modulus q , with $\delta = 4q^2 - p^2 > 0$. Reduction modulo the monic f gives unique integers

$$\beta^n = C_n + D_n \beta, \quad C_0 = 1, D_0 = 0, \quad C_{n+1} = -q^2 D_n, \quad D_{n+1} = C_n + p D_n. \quad (249.2)$$

For $n \geq 1$, $D_n = V_{n-1}$; for $n \geq 2$, $C_n = -q^2 V_{n-2}$. Thus $q \mid C_n$ for every $n \geq 1$ and

$$C^n e_1 = \frac{(C_n/q, D_n)^T}{q^{n-1}} \quad (n \geq 1). \quad (249.3)$$

Since $|\beta - \bar{\beta}| = \sqrt{\delta}$,

$$D_n = \frac{\beta^n - \bar{\beta}^n}{\beta - \bar{\beta}}, \quad |D_n| \leq 2q^n / \sqrt{\delta} \leq 2q^n, \quad |C_n| \leq 2q^{n+1}.$$

The last inequality also holds at $n = 1$, where $C_1 = 0$.

For natural m , set $P_0 = q^m$, $n = m + 1$ and define expressions

$$H = 2q^2 P_0, \quad R = 4H + |p| + 1. \quad (249.4)$$

The true coefficients obey $|C_n|, |D_n| \leq H$, and $R \geq 2$. Divisibility of $Z^n - C_n - D_n Z$ by $f(Z)$ gives

$$R^n \equiv C_n + R D_n \pmod{R^2 - pR + q^2}.$$

If another bounded integer pair C, D has the same congruence, then

$$|(C - C_n) + R(D - D_n)| \leq 2H(1 + R) < R^2 - pR + q^2.$$

For the strict inequality,

$$\begin{aligned} R^2 - pR + q^2 - 2H(1 + R) &\geq R(R - |p| - 2H) + q^2 - 2H \\ &= R(2H + 1) + q^2 - 2H > 0. \end{aligned}$$

The difference is a multiple of a positive modulus with smaller absolute value, so it is zero. Since $|C - C_n| \leq 2H < R$, $C - C_n = -R(D - D_n)$ forces $D = D_n$ and $C = C_n$. One ordinary power R^{m+1} therefore extracts the coefficient pair exactly within the inclusive bounds.

250 The outer positive-integer equations

For each contact choose a positive integer t_j clearing S_j^{-1} , and let the fixed integer matrix be $M_j = t_j S_j^{-1}$. Define integer linear input expressions

$$U_j = M_{j,11}X + M_{j,12}Y, \quad V_j = M_{j,21}X + M_{j,22}Y, \quad W_j = t_j N > 0.$$

Then $S_j^{-1}w = (U_j/W_j, V_j/W_j)^T$, including $w = 0$.

For a kernel with fixed Q, r^2 , choose a positive integer L clearing their denominators, and set $a = Lr^2 > 0$, $G = LQ \in \mathbb{Z}^{2 \times 2}$. Introduce one natural d and the residual equation

$$\Delta := aN^2 - G_{11}X^2 - 2G_{12}XY - G_{22}Y^2 = d. \quad (250.1)$$

It enforces exactly $\Delta \geq 0$. For pure orbit avoidance, omit d and this equation and replace d^2 by zero below.

For each contact independently introduce six natural variables

$$m, k, L_C, H_C, L_D, H_D,$$

seven strictly positive variables

$$h, b, r, s, t, J_0, J_1,$$

and eight unrestricted signed integer variables

$$u, v, e_1, e_2, e_3, C, D, \kappa.$$

Use two independent copies of the full POWER module in Section 251:

$$P_0 = \text{POWER}(q, m), \quad H = 2q^2 P_0, \quad R = 4H + |p| + 1, \quad T_0 = \text{POWER}(R, m + 1).$$

The outputs P_0, T_0 are already witness leaves of their modules; $H, R, m + 1$ are expressions, not extra witnesses. Add the following fourteen equations per contact:

$$\begin{aligned} U_j &= hu, & (O1) \\ V_j &= hv, & (O2) \\ W_j &= hb, & (O3) \\ e_1 u + e_2 v + e_3 b &= 1, & (O4) \\ b &= P_0 r, & (O5) \\ r &= qk + s, & (O6) \\ s + t &= q, & (O7) \\ C + H &= L_C, & (O8) \\ H - C &= H_C, & (O9) \\ D + H &= L_D, & (O10) \\ H - D &= H_D, & (O11) \\ T_0 - C - RD &= \kappa(R^2 - pR + q^2), & (O12) \\ d^2 + (u - b)^2 + v^2 &= J_0, & (O13) \\ d^2 + (r - 1)^2 + (qu - C)^2 + (v - D)^2 &= J_1. & (O14) \end{aligned} \quad (250.2) \quad (250.3)$$

250.1 Ordinary positive domains without unpaid adapters

Every natural variable is replaced by $z = z_+ - 1$ with one ordinary positive leaf z_+ . Every signed variable is replaced by $z = z_+ - z_-$ with two ordinary positive leaves. Strictly positive variables remain single leaves. These affine substitutions require no equations and permit zero natural values and arbitrary signed values. All final inputs and witnesses are ordinary positive integers; no zero-valued variable is silently put in a positive domain.

250.2 Soundness

Equations (O1)–(O4) force $\gcd(u, v, b) = 1$ and $h = \gcd(U_j, V_j, W_j) > 0$. Hence b is the common reduced denominator. To see this directly, if $a' > 0$ clears u/b and v/b , then b divides $a'u, a'v$; multiplying the Bezout equality by a' proves $b \mid a'$. Conversely b clears the pair. At zero input the primitive representation is $u = v = 0, b = 1$, so the argument still applies.

The first POWER module forces $P_0 = q^m$. Equations (O6)–(O7) say $k \geq 0$ and $1 \leq s \leq q - 1$, so q does not divide r . Consequently (O5) expresses the unique decomposition

$$b = q^m r, \quad m \geq 0, \quad r > 0, \quad q \nmid r$$

obtained by repeatedly dividing by the *whole* integer q . For composite q this is not an additive prime-adic valuation; r may share some prime factors with q .

Equations (O8)–(O11) impose the inclusive coefficient bounds. The second POWER module and (O12), together with Section 249, force $(C, D) = (C_{m+1}, D_{m+1})$. The modulus is positive and its quotient κ is allowed either sign.

Time-zero equality with e_1 is exactly $u = b, v = 0$. For $d = 0$, positive J_0 in (O13) excludes exactly that equality. At positive time, the denominator lemma and (249.3) give

$$S_j^{-1}w = C^n e_1 \text{ for some } n \geq 1 \iff r = 1, qu = C_{m+1}, v = D_{m+1}. \quad (250.4)$$

Indeed a hit at n forces $b = q^{n-1}$ and $m = n - 1$; conversely these numerator equalities give the orbit point at $n = m + 1$. Thus positive J_1 in (O14) excludes exactly all positive-time matches when $d = 0$.

For the kernel, (250.1) rejects exterior points. For interior points $d > 0$, both acceptance sums are automatically positive. On the boundary $d = 0$, the two gates for every contact are solvable exactly when all backward orbits are avoided. This proves soundness without a selector variable, an orbit-search bound, or a generic MRDP invocation.

250.3 Completeness

For an accepted input, choose its positive $\gcd$ h , primitive quotients u, v, b , and signed Bezout coefficients. Choose the unique whole- q decomposition, its quotient and positive remainder split $s + t = q$. Choose the true coefficients of β^{m+1} , their four nonnegative bound slacks, and the integer extraction quotient κ . The POWER theorem supplies all internal witnesses. The acceptance sums are positive integers precisely under the accepted conditions, so choose those values for J_0, J_1 . For the kernel, take $d = \Delta \geq 0$. Finally apply the affine domain adapters. This constructs witnesses in every accepted case. Repeated contacts and overlapping backward orbits merely repeat conjuncts; no contact-orbit equivalence classification is required.

251 The complete fifteen-equation POWER module

Remark 251.1 ([write] Printed once, in Part X). Source 30 makes explicit “the constructive Pell specialization used in Report 59” [source 29], “rather than treated as an exponentiation oracle”. Its statement reads as follows. For an integer expression $B_0 \geq 2$ and a natural exponent expression e , take thirteen directly positive leaves out, $w, M, g, x_p, y_p, u_p, v_p, s_p, t_p, q_b, q_v, J_p$, two positive leaves α_+, β_+ with $\alpha = \alpha_+ + 1$ and $\beta = \beta_+ + 1$, and eleven natural variables $d_{wb}, d_{wk}, d_{yk}, \alpha_1, \alpha_2, \sigma_1, \sigma_2, \tau_1, \tau_2, r_1, r_2$,

each one shifted positive leaf. That makes 26 positive leaves, the output included. With $k_0 = e + 1$ and $m_0 = B_0$ out, the fifteen equations (P1)–(P15) have a solution in the other 25 leaves exactly when out = B_0^e .

This is the system of source 28’s Section 217 and Lemma 217.1, equation for equation. Source 30’s $x_p, y_p, u_p, v_p, s_p, t_p, \alpha, \alpha_+, \beta_+, r_1, r_2$, out are source 28’s $x, y, u, v, s, t, a, a_{-1}, \beta_{-1}, \rho_1, \rho_2, o$. Both sources specialize the same two theorems of the pinned mathlib file [86]. The first is `Pell.matiyasevic` (lines 760–766); its zero-index alternative $x = 1, y = 0$ is excluded by $k_0 \geq 1$ and $y_p \geq k_0$. The second is `Pell.eq_pow_of_pell` (lines 860–864). Both sources convert the natural-number subtractions exactly: $a - b = 1 \iff a = b + 1$ for naturals, and $\alpha > w \geq B_0 \geq 2$ makes $\alpha - B_0$ an ordinary difference. Both prove soundness and completeness, the exponent $e = 0$ included, by the same argument. The equations, the theorem interface and the proof are printed in Part X. As source 30 says, these are theorem specializations of the pinned source, “not a new proof of the constructive Pell theorems and not a locally executed Lean build”. Source 30’s archive preserved the source file with its licence; they are not shipped here (Section 293.3).

Source 30’s own addition is the instantiation: “In the certificate the first base is $q \geq 2$; the second is the affine expression $R = 8q^2P_0 + |p| + 1 \geq 2$ before either semantic conclusion is invoked. The second exponent $m + 1$ is natural. There is no unpaid hypothesis for an input-dependent base.”

252 The exact witness and polynomial ledger

For each contact the six natural outer variables cost six positive leaves, the seven positive variables cost seven, and the eight signed variables cost sixteen. The two independent POWER modules cost 52, including both outputs. Therefore

$$6 + 7 + 16 + 52 = 81$$

positive witnesses occur per contact. The 14 outer residuals and 30 POWER residuals give 44 equations. The strict-kernel radius adds exactly one natural witness and one equation.

Predicate and input convention	Positive witnesses	Residuals	Degree
J -contact orbit complement	$81J$	$44J$	12
Strict kernel, native inputs	$1 + 81J$	$1 + 44J$	12
Strict kernel, paid one-input transport	$5 + 81J$	$3 + 44J$	12

Define F literally as the sum of squares of every displayed residual after substituting all input aliases, H, R, k_0, m_0 , the two Pell-parameter shifts, natural shifts and signed differences. Coefficients are integers after the stated fixed denominator clearing. Over ordinary positive integers, $F = 0$ is equivalent to all residuals vanishing.

Outer residual degree is at most three; the only cubic type is κR^2 . Each POWER residual has degree at most six, even when its base is affine in the other module’s independent output. Equation (P13) has degree-six leading monomial $-w^4g^2$. Its square contributes w^8g^4 of degree twelve. The degree-twelve part is a nonempty sum of squares of nonzero real homogeneous polynomials and cannot vanish identically. Since $J \geq 1$, the exact total degree is 12.

For the paid one-input version, introduce positive gap witnesses g_1, g_2, g_3 and one natural witness

j , and add

$$2j = (g_2 + g_3 - 2)(g_2 + g_3 - 1) + 2(g_3 - 1), \quad (252.1)$$

$$2(z - 1) = (g_1 - 1 + j)(g_1 + j) + 2j. \quad (252.2)$$

These express the Cantor bijection

$$z = 1 + \text{pair}(g_1 - 1, \text{pair}(g_2 - 1, g_3 - 1)), \quad \text{pair}(a, b) = \frac{(a + b)(a + b + 1)}{2} + b.$$

They add four positive leaves after $j = j_+ - 1$ and two quadratic residuals, preserving degree twelve. This is a different input ledger from the native three-gap statement.

No uniqueness or finite-fold claim follows: even a signed difference has infinitely many positive-pair encodings. No witness-height bound, efficient Pell-witness construction, or source-specific arithmetic gate count is given. The elementary polynomial-time decision procedure of Section 247 does not compute the enormous radix powers or Pell witnesses used in this existential proof.

252.1 A compiled nonorthogonal elliptic example

An independently checked physical fixture takes

$$A = \begin{pmatrix} 3/5 & -8/5 \\ 2/5 & 3/5 \end{pmatrix}, \quad \lambda = 1/2.$$

Its chronological shear parameters are $x(-1), y(2/5), x(-1)$, giving $K = 10, T = 2, H = 0$. The exact count is 210 events, 52 supplied guards, 43 temporary labels and 257 meta-signals, with five live strands. It preserves $Q = \text{diag}(1, 4)$. Independent contact computation on its *actual compiled chamber* gives

$$r^2 = 1/45, \quad E = \{(-2/15, 1/30)^T\}.$$

Thus the three-gap validity certificate has 82 positive witnesses, 45 residuals and exact degree 12. This example connects the physical, geometric and arithmetic interfaces; it does not substitute a convenient square for the compiler's exact chamber. The analytic ellipse in Figure 14 is a separate geometry illustration.

253 Quadratic certificates outside the exceptional case

The remaining spectral branches admit a much simpler fixed-input certificate. This is an elementary corollary of the rational-input part of the classification, not a lower bound on the degree or witness count of elliptic certificates.

Corollary 253.1 (Unique positive slack tuple). *Fix rational A that is not infinite-order elliptic, and bounded open convex rational P containing 0. For the three positive integer inputs in (249.1), there is an explicit integer-coefficient sum of squares $F(g, w)$ of affine linear residuals such that normalized membership in $\mathcal{K}(A, P)$ holds exactly when one positive witness tuple solves $F = 0$. Invalid inputs have no such tuple. The stated retained-row construction has exact total degree two, possibly with zero witnesses. No POWER module is required.*

[write] [This is the unique-slack device of Part I's Theorem 6.1, applied to a finite rational description of an infinite-validity kernel.]

253.1 The finite rational rows and their exact ledger

On rational points, Sections 245–246 give a finite conjunction

$$e_j w = 0 \quad (1 \leq j \leq r), \quad \alpha_i + v_i w > 0 \quad (1 \leq i \leq s), \quad \beta_k + z_k w \geq 0 \quad (1 \leq k \leq t),$$

with rational coefficients. For $\mathcal{B} = \{0\}$ or an irrational line, replace this by the actual rational-point kernel $w = 0$, rather than trying to encode the irrational line by rational coefficients. For a rational bounded-orbit line use its rational equation and the one- or two-time formula (245.1). The full-dimensional branches are the stable finite horizon, real semisimple unit spectrum, finite elliptic order, mixed nonzero stable mode with weak limiting guards, and mixed zero mode with strict guards. These are all the nonexceptional cases.

Clear each row with its own fixed positive integer factor and multiply the normalized inequality by $N > 0$. The resulting integer homogeneous linear input forms are

$$\begin{aligned} E_j(g) &= d_j e_j(X, Y)^T, \\ L_i(g) &= c_i(\alpha_i N + v_i(X, Y)^T), \\ M_k(g) &= a_k(\beta_k N + z_k(X, Y)^T), \end{aligned}$$

where every clearing factor is positive. Equality, strictness and weakness are therefore preserved. Use one positive witness p_i for each strict row, one positive witness q_k for each weak row, and define literally

$$F(g, p, q) = \sum_{j=1}^r E_j(g)^2 + \sum_{i=1}^s (L_i(g) - p_i)^2 + \sum_{k=1}^t (M_k(g) - q_k + 1)^2. \quad (253.1)$$

There are exactly $s + t$ witnesses and $r + s + t$ residuals. Input aliases add no leaves or equations. With m supplied nonconstant polygon rows, a convenient unreduced ledger is

Branch	r	s	t
Zero or irrational bounded-orbit subspace	2	0	0
Rational line, nonnegative scalar	1	m	0
Rational line, negative scalar	1	$2m$	0
Stable plane, finite horizon H_0	0	$(H_0 + 1)m$	0
Real semisimple unit spectrum	0	$2m$	0
Nonreal finite order $d_0 \in \{3, 4, 6\}$	0	$d_0 m$	0
Mixed unit and nonzero stable eigenvalue	0	$2m$	$2m$
Mixed unit and zero stable eigenvalue	0	$3m$	0

Repeated or redundant rows remain as separate residuals; their uniquely forced slack coordinates do not create choices. Except in the two zero-kernel branches, retain the original time-zero guards, so at least one inequality remains.

253.2 Correctness, uniqueness and degree

Each square in (253.1) is nonnegative. Thus $F = 0$ exactly when all residuals vanish, forcing

$$p_i = L_i(g), \quad q_k = M_k(g) + 1.$$

These forced values are positive precisely when the corresponding integer rows are strict or weak as required. Every valid fixed input consequently has one tuple, and every invalid input has none. Equality residuals introduce no witness choices.

All residuals are affine linear, hence total degree is at most two. If any inequality remains, its witness appears with coefficient -1 in its own residual only, so its square has coefficient one in F and the degree is exactly two. In the zero-witness branches take $F = X^2 + Y^2$, also nonconstant of degree two. The empty product $\mathbb{Z}_{>0}^0$ contains exactly the empty tuple: it is the unique witness precisely when $X = Y = 0$, equivalently $g_1 = g_2 = g_3$ on positive inputs. There is no hidden witness there.

For completeness, the input map is invertible over $\mathbb{Q}$:

$$g_1 = (N + 3X)/9, \quad g_2 = (N - 3X + 3Y)/9, \quad g_3 = (N - 3Y)/9.$$

Thus a nonzero equality form does not become identically zero under the aliases. Exact degree two describes this retained construction, not a minimum: if the entire positive-gap triangle is valid, a deliberately simplified predicate could instead have the constant-zero polynomial. Different integer triples can encode the same rational point, but witness uniqueness is for each fixed triple and fixed residual list. The construction size depends on fixed (A, P) , and a stable horizon can be large.

253.3 A half-open normalized example

This is an *abstract* kernel example, not a claimed physical compiler fixture. Let

$$A = \text{diag}(1, 1/2), \quad P = \{(u, v) : -1 < u < 1, -1 < v < 1, u + v < 1/6\}.$$

The parity-free interpolation gives $\mathcal{K} = P \cap \{u \leq 1/6\}$. With the aliases (249.1), the five strict forms and one weak form are

$$(L_1, L_2, L_3, L_4, L_5) = (N + X, N - X, N + Y, N - Y, N - 6X - 6Y), \\ M = N - 6X.$$

A complete six-witness formula is $F = \sum_{i=1}^5 (L_i - w_i)^2 + (M - w_6 + 1)^2$. For $g = (6, 1, 5)$, $(N, X, Y) = (36, 6, -3)$ and $(u, v) = (1/6, -1/12)$, with the unique positive tuple

$$(w_1, \dots, w_6) = (42, 30, 33, 39, 18, 1).$$

The orbit stays strictly in P at every finite time but tends to $(1/6, 0) \notin P$. The final slack 1 encodes the weak equality $M = 0$ exactly. The point $g = (7, 1, 5)$ satisfies all time-zero guards but has $M = -9$, forcing the last slack to -8 and thus correctly admitting no positive witness.

Combining this corollary with Theorem 240.3 completes the positive-integer certificate story for every member of the positive-determinant physical family. The abstract corollary itself also applies to singular and negative-determinant rational A under the geometric hypotheses; that does not supply a physical realization for those matrices.

254 Independent evidence and reproducibility boundaries

The report integrates three main separately frozen source packets, each with a fresh independent conventional review, and the separately reviewed elementary quadratic-certificate corollary. The

archive preserves both the original packets and their later reviews. Original constructors and checkers are included as inert source, not executed by the report builder. The accompanying portable replay runs separately authored, inspected independent checkers against authenticated relocated data; it never advances physical particles or selects the next collision.

254.1 Physical static reconstruction

The physical source has 25 rational matrices at each $\lambda \in \{1, 1/2, 2\}$, totaling 75 fixtures. The independent reconstruction matches every field: factorization, endpoint returns, primitive phases, rational duration row, all rules and label speeds, guard rows, counts and completion declaration. It checks all three pivot branches, positive and negative diagonal factors, either shear sign, zero diagonal pivots, finite and infinite elliptic order, nonorthogonal elliptic conjugacy, Jordan blocks, irrational real eigenvalues, stable and unstable complex spectra, unit-contracting modes, and nonsquare determinants.

Independent rational half-plane clipping verifies every closed chamber vertex against every row and positive exact area for every fixture; the largest chamber has eight vertices. Duration rows satisfy the claimed bounds across each chamber closure, and strictly at the center. A separate exact algebra sweep covers all 1,056 integer matrices with entries in $[-3, 3]$ and positive determinant. The intermediate-dilation guard counterexample in Section 242 is also checked.

Independent physical evidence	Count
Full fixture objects matched	75
Complete indexed binary event/rule records	24,678
Supplied guard records	7,020
Additional integer matrix factorizations	1,056

The independent checker was run normally and with Python optimization and gave identical output. These are static grammar and rational-algebra checks. Universal exact chronology rests on the monotonicity, speed separation and first-obstruction proof, not on the finite fixture suite.

254.2 Geometric and arithmetic reconstruction

The geometry audit independently checks 1,212 preserved-metric identities, 1,212 contact identities, 7,676 denominator instances and 8,484 forward/backward decisions. It also tests 490 mixed-spectrum grid points, the attained-zero-stable-mode exception, four stable contraction fixtures, and 180 interior samples on 90 facet lines. Its normal and optimized runs agree. The infinite set classification and universal facet argument remain conventional mathematical proofs.

The arithmetic audit checks 2,508 reduced signed traces with $2 \leq q \leq 45$, 92,796 exact backward powers and 90,288 forward powers for direction, 499,092 whole-base factorizations, 24,276 rational-grid comparisons, 350 rational contact bases, and 24 exhaustive bounded-extraction searches. It independently constructs actual positive POWER-module witnesses for $(B, e) = (2, 0), (2, 1), (3, 0), (3, 1), (4, 0), (6, 0)$; all 26 leaves and 15 equations check, with the largest sample leaf having 34,887 bits. These small cases support the translation but do not replace the all-exponent constructive Pell dependency.

Its independent symbolic expansion reconstructs four adapted schemas:

Schema	Inputs	Witnesses	Residuals	Monomials
One-contact kernel, trace $-5/6$	5	82	45	847
One-contact avoidance, trace $-5/6$	5	81	44	725
Two-contact kernel, trace $1/2$	3	163	89	1,590
Compiled example of Section 252.1	3	82	45	812

All four have exact degree twelve. Monomial counts describe these particular expanded test schemas, not a generic arithmetic-DAG size or a universal gate count.

254.3 Source pins and the minor wording erratum

The source and review bindings below are SHA-256 digests, not signatures. They identify the inspected bytes. A trusted externally supplied manifest pin authenticates the released archive inventory.

Frozen component	SHA-256
Physical proof	e0ddd64cdbbb5c7f448266ab58f1dfefb2 868892ac0f230632b2337e4d0bed065
Physical independent review	1867f384f9c44ca0055d29ce782664b8 efdda49a953a3b81591148fdda34688c
Geometric proof	70eb8f8398d474c3343493fc88c006ce 91951a2eb747c9b4133c5c481b666231
Geometric independent review	2009679b3e04d769d989440963dbd74f 96aa20be61a552db471f5cad2c7f0b97
Arithmetic proof	e93fdbefc3ba72e34c44e4f9d8a9b97e 72027f72dc3ce53d54570113b7db5e5a
Arithmetic independent review	a7d3b6585503e5953e020be0580620914 b1dea4f520af74643aba65d45ad11b5
Quadratic corollary proof	faaf236df7e6da3f05a38ea75ab91f0aa 91e37db581fa00bd6116215bd306346
Quadratic independent review	5d35de325076b7c8b07bcd805347a2bcef 9dd81c26715a214391afdb9583cccb
Pell source	993760c797ad0ff66fa77064a616fce779 550bc745cbf6c44e04f392fd0bed0a

All three main components and the quadratic corollary passed their source-bound independent reviews. The arithmetic audit records one non-substantive wording erratum, E1: the original sentence bounding both divisions and recurrence steps by $\log_2 b$ omitted the candidate's additive one. Section 247 uses the corrected $1 + \lfloor \log_2 b \rfloor$ recurrence bound; successful whole-base divisions remain at most $\lfloor \log_2 b \rfloor$, with a final divisibility test if needed. The frozen source is unchanged. This wording change affects no theorem, polynomial, domain, ledger or complexity class.

The archive contains full proof texts, inert source programs, exact fixture JSON, review receipts, the pinned Pell source and license, editable LaTeX, safe deterministic release/build tools, and independent portable replay instructions. Rebuilding typesets the manuscript and renders every page in a fresh external workspace. It does not run the scientific constructors, source checkers, upstream programs, collision schedules or Lean. The rendered ellipse is analytic geometry; it is not physical simulation evidence.

[write] [In this report the four proofs, the reviews and the replay notes are shipped as `30-planar-r-*.md` at the report root, the programs as `code/30-planar-*`, and the 75 fixtures, the receipts

and the other evidence as `data/30-planar-*`. The PDF, the editable LaTeX, the page renders, the build transcripts and the pinned Pell source with its licence are not shipped; they survive in the arrival archive. The README gives the command that regenerates the fixtures byte for byte.]

255 Literature positioning and precise claim boundaries

Signal-machine definitions, finite meta-signal alphabets, rational velocities and the outgoing-collision convention are established in abstract geometrical computation [6]. Small-speed accumulation results concern *speed* counts, a different resource from five simultaneously live signals. Rational stack translation and scaling, together with energy-conservative shrinking, occur in Durand-Lose’s earlier work [16]. The present physical proof does not require that paper as an unproved primitive lemma; its chronology is supplied directly. The fresh manuscript review verified the stack displacement and scaling passages in Section 3 (PDF pages 4–6), the introduction and the later shrinking construction. The frozen companion packet’s earlier failed retrieval is a historical access note; it does not negate this later verification.

Maximal output-admissible sets and stable finite determination are established control-theory tools. Gilbert and Tan [96], Section IV and Theorem 4.1, treat asymptotic stability with observability, bounded output constraints and an interior-origin condition. With output matrix I , the contraction-tail proof above is a strict-guard specialization. Their Section V concerns a semisimple $+1$ /stable setting and finitely determined inner approximations; it should not be described as a treatment of arbitrary unit-circle spectra. The audit verified the primary archived paper’s indexed theorem and scope excerpts; direct PDF retrieval returned an error. No broader scope is attributed to it.

Kannan–Lipton point-to-point rational orbit membership is polynomial-time. The later primary treatment by Chonev, Ouaknine and Worrell [97] states that result and distinguishes point targets from more general targets. The rational contacts here make it applicable directly. The denominator algorithm is an elementary specialization, not a new general orbit decidability theorem.

Fijalkow and coauthors [87], Example 1, already use a rational irrational-angle rotation to obstruct semialgebraic invariant separation. The mechanism of rotational density is prior work. Here it is applied to the full strict polygon kernel, with explicit rational tangent contacts and dense accepted and rejected tails on one rational orbit.

Dai and Xia [76] analyze two-variable loops with homogeneous guards $Bw > 0$. Affine polygon bounds introduce an additional coordinate when homogenized; their two-variable theorem cannot simply be relabeled as this exact affine planar classification. Almagor, Ouaknine and Worrell [98] prove semialgebraic source-to-target orbit decidability in dimension at most three. A singleton rational source and the complement of P already imply uniform decidability here. The credited low-order strict-positivity transfer in Section 247 gives stronger complexity context without asserting uniform polynomial time.

The claim is an explicit integrated realization, strict-boundary classification and positive-integer certificate with transparent interfaces. No exhaustive literature search, priority, novelty of each elementary lemma, optimum witness degree, minimum population, or unrestricted universality is certified. Negative-determinant and singular maps are covered by the abstract classification, but not by the stated physical compiler; no physical impossibility is inferred for them. The real irrational-stable-line set may require algebraic coefficients even when its rational-input predicate is trivial. Encoded rational and integer inputs are not interchangeable with all-real inputs. No end-to-end formal verification, arbitrary-matrix fixed alphabet, uniform polynomial physical compiler, or post-Zeno semantics is claimed.

256 Further questions with concrete targets

1. **Two independent rotation blocks.** For rational $A = \text{diag}(R_1, R_2)$ with multiplicatively independent infinite-order complex multipliers and a bounded open convex rational polytope, determine the exact strict kernel on invariant two-tori. A useful target includes zero-radius strata, active guard contact sets, simultaneous backward hits and rational-input complexity. Closed support bounds alone omit the strict-boundary arithmetic. General torus/orbit methods are already known; the question asks for a transparent exact specialization and literature comparison.
2. **Uniform stable decision complexity.** Compare the credited counting-hierarchy bound with sharper low-order recurrence results to determine whether this bounded convex planar setting admits a uniform polynomial decision algorithm. The exponential facet example rules out small explicit affine output, not such a decision algorithm.
3. **Compiler size and alternative dilations.** Seek bit-efficient rational shear and dilation schedules while preserving every spectator crossing and exact phase closure. Any improvement should state its encoded input model and account for temporary labels, events and guards separately. The present near-one subdivisions establish existence but can be large.
4. **Orientation reversal and rank loss.** Determine which negative-determinant or singular rational planar returns admit a comparably small number-preserving complete-word realization. One must specify the section and collision semantics; the present theorem’s exclusion is not an obstruction result.
5. **Certificate resources beyond degree.** Analyze constructive witness height, modular extraction size and possible reductions of the 81J ledger. Distinguish fixed contacts from contacts supplied as input, and native gap arity from paid pairing. Existing degree and witness counts are explicit upper bounds, not lower bounds.
6. **Formal interface certification.** A future formal development could certify the universal first-obstruction chronology, exact spectral case split and polynomial adapters together. The present inert Pell source is only one dependency; its existence does not make the whole physical-to-arithmetic chain formally verified.

[write] [Status at this write: Question 4 is answered for negative determinant by source 31. Its Corollary 257.4 says that a fixed-scale full-section return $\lambda \text{diag}(1, A)$ exists exactly when $\det A \neq 0$, and its reversing primitive has seven events (Theorem 257.3). Singular planar returns are excluded only under source 31’s full-section, open-chamber, binary, positive-flight and fixed-scale hypotheses (Section 260). The question stays open for other sections and observation models, which source 31’s Question 5 asks about. The other five questions are open.]

Source 31 (batch-91 manuscript 13): Fixed word invertibility and rational planar returns

Collision parity and an explicit seven event reversal, “Research Report 61”, 4 October 2026. [write] From the archive `Invertible_Signal_Returns_Source.zip` (main file `Report61/Report61.tex`, 19-page PDF); its labels carry the prefix `smc:iv:`. Its sections are Sections 257–266 here. Its four bibliography entries are source 30 (its frozen physical proof) and this report’s [92], [5] and [93]. Its abstract follows.

A fixed complete binary-collision word with positive flights induces a rational linear isomorphism between its full outgoing positional sections. After quotienting common translation, a literal return to the same sorted collision face has determinant equal to $(-1)^m$ times the product of outgoing-to-incoming relative speed ratios, where m is its event count. An explicit seven-event machine realizes the orientation-reversing matrix $J = \begin{pmatrix} -2/3 & 5/6 \\ 0 & 1 \end{pmatrix}$ on the stationary-marker section with five live signals. Its exact chamber is $0 < x < y < D, y < 4x$; its infinite-validity predicate on positive integer gaps has an explicit quadratic certificate with two uniquely forced positive witnesses. Composing this primitive with the positive-determinant compiler of Report 60 gives every rational invertible planar matrix, with literal phase closure, an exact center-containing rational polygon, and any prescribed positive rational scale factor. Singular planar matrices are excluded only under the full-section, open-chamber, binary positive-flight and fixed-scale hypotheses stated here. Full proofs, rule and gap tables, an analytic spacetime diagram, the Zeno-time extension, and independent static evidence are supplied. Number preservation does not imply global reversibility, and no universality, novelty, optimality or unrestricted impossibility claim is made.

257 The result and the model

The missing distinction in the planar realization problem is invertibility, not orientation preservation. An odd complete word can reverse orientation. The seven-event example below supplies precisely the extra primitive needed to extend Report 60 from positive determinant to all of $GL_2(\mathbb{Q})$.

A *meta-signal* is a label carrying a fixed rational speed. A *live signal* is an occurrence of such a label on the line. A collision rule replaces its incoming signals at their common position by outgoing signals there. The speeds within each incoming set and within each outgoing set are distinct; different labels elsewhere may have equal speeds. A *complete word* records every event in its interval, including identity-rule contacts. It is not a selected subsequence of interesting contacts.

Theorem 257.1 (Full outgoing section invertibility). *Fix a finite complete collision word on a nonempty full-dimensional open chamber of an outgoing binary-collision section. Assume:*

1. *The live population n is constant, and each prescribed event consumes two signals and emits exactly two at the same position.*
2. *The two incoming speeds and the two outgoing speeds are distinct at every event.*
3. *Initially exactly one outgoing pair coincides and every other adjacent gap is positive. Every subsequent event follows a strictly positive flight, with no intervening or simultaneous remote event.*
4. *The word fixes all phases, partners and spatial ordering, hence every flight's velocity vector. A literal return restores the outgoing phase, label pattern and section identification.*
5. *The section coordinates retain every positional degree of freedom, except possibly the legitimate quotient by common spatial translation.*

The positional section map is the restriction of a rational invertible homogeneous linear map between the full section spaces. The induced map after quotienting common translation is rational and invertible too.

This is a fixed-itinerary geometric statement. Its inverse need not be implemented by the machine, and images of different words can overlap. It imposes no injectivity condition on the global map from incoming label sets to outgoing label sets.

[write] [The linearity is a case of Part I's Theorem 5.3; the invertibility is new to this report. It is compatible with Part III's gap matrices of rank $n - 2$ and with Part III's non-claim on injectivity (Section 239). Source 31 credits the inverse of a fixed-itinerary section map to Alishah, Duarte and Peixe [93] (Section 264).]

Theorem 257.2 (Exact determinant and parity). *For a literal m -event return to the same sorted collision face and outgoing phase, use increasing-gap coordinates at both ends. Then*

$$\det M = (-1)^m \prod_{j=1}^m \frac{s_j^+}{s_j^-}, \quad \text{sgn } \det M = (-1)^m, \quad (257.1)$$

where $s_j^- > 0$ is the incoming pair's closing speed and $s_j^+ > 0$ its outgoing separation speed. Here M acts on the full translation-quotiented section. A different end identification requires its own coordinate determinant factor.

Theorem 257.3 (Seven event stationary section reversal). *There is a finite deterministic rational-speed number-preserving machine with five live signals, eleven standalone meta-signals and seven specified rules. Its outgoing section consists of stationary markers L, X_0, Y, R at $0, x, y, D$, respectively, and a messenger Q_+ at L moving at speed $+1$. Its seven-event complete binary return has exactly the chamber*

$$0 < x < y < D, \quad y < 4x, \quad (257.2)$$

and restores all section labels and phases with

$$D' = D, \quad x' = -\frac{2}{3}x + \frac{5}{6}y, \quad y' = y. \quad (257.3)$$

Put $X_c = x - D/3$ and $Y_c = y - 2D/3$. The centered planar action is

$$J = \begin{pmatrix} -2/3 & 5/6 \\ 0 & 1 \end{pmatrix}, \quad \det J = -2/3. \quad (257.4)$$

The standard center $(x, y) = (D/3, 2D/3)$ is strictly inside the chamber.

Corollary 257.4 (Exact rational planar realization criterion). *Let $A \in \mathbb{Q}^{2 \times 2}$ and $\lambda \in \mathbb{Q}_{>0}$. In the five-live-signal full-section model of Theorem 257.1, a fixed complete word returning the preceding stationary-marker section by*

$$(D, X_c, Y_c)^T \mapsto \lambda \text{diag}(1, A)(D, X_c, Y_c)^T \quad (257.5)$$

on a nonempty full-dimensional open chamber exists if and only if $\det A \neq 0$. When it exists, the machine can be finite, deterministic and number-preserving; the phase closes literally; and its exact normalized chamber is a bounded open rational polygon containing the center.

The constructive direction imports only the positive-determinant realization theorem from the frozen Report 60 proof [[write] source 30, Theorem 240.1; shipped as 30-planar-science-physics-1-PROOF.md]. The invertibility obstruction, determinant calculation and seven-event construction are proved directly here. The prior proof is included as a pinned, inert source dependency.

The obstruction does not concern a projection that forgets hidden continuous coordinates, a lower-dimensional boundary or invariant slice, population-changing or nonbinary collisions, or a new operation declared at an accumulation limit. Those changes alter the hypotheses. No universal Turing-completeness statement follows from the ability to compile a selected rational linear return.

258 Transverse flight and the full section inverse

Give the n live positional slots temporary names and write their coordinates as $q \in \mathbb{R}^n$. If a is a collision pair, let h_a subtract its two positions and set

$$H_a = \ker h_a = \{q : h_a q = 0\}.$$

This is the full positional hyperplane at that collision, before any translation quotient. The other strict ordering inequalities select an open part of it; they introduce no further linear equation.

Lemma 258.1 (Transverse constant flight). *Let $v \in \mathbb{Q}^n$ be the fixed velocity vector of a flight from H_a to H_b , with $h_a v \neq 0$ and $h_b v \neq 0$. The landing map and its inverse are*

$$F_{ab}(q) = q - v \frac{h_b q}{h_b v}, \quad q \in H_a, \quad (258.1)$$

$$F_{ab}^{-1}(r) = r - v \frac{h_a r}{h_a v}, \quad r \in H_b. \quad (258.2)$$

They are rational homogeneous linear isomorphisms between H_a and H_b .

Proof. The collision time solves $h_b(q + \tau v) = 0$, so $\tau = -h_b q / (h_b v)$ and gives (258.1). The assumptions make each affine line parallel to v intersect each hyperplane exactly once. For an explicit verification, put $r = F_{ab}(q)$ with $h_a q = 0$. Then

$$h_a r = -\frac{(h_a v)(h_b q)}{h_b v}, \quad r - v \frac{h_a r}{h_a v} = q.$$

Interchanging a, b proves the opposite composition. All coefficients are rational. The maps are homogeneous because the contact equations have no external spatial offset or prescribed clock time. $\square$

As an ambient map on $\mathbb{R}^n$, $I - v h_b / (h_b v)$ has kernel $\mathbb{R}v$. That fact does not make the section map singular: v is transverse to H_a , so the restricted kernel is zero. This is the precise point at which the apparent projection obstruction disappears.

For a physical flight, distinct source outgoing speeds give $h_a v \neq 0$, while distinct target incoming speeds give $h_b v \neq 0$. Positivity of τ and absence of earlier contacts restrict its domain but do not change its ambient linear extension. The inverse formula does not assert a positive reverse flight in the given machine. On an actual open chamber it is a bijection onto the open image of that chamber in H_b , not necessarily onto the entire physical target chamber.

258.1 Collisions and positional identifications

At a binary event the two incoming coordinates and two outgoing coordinates all have the same value. Choose either fixed matching between the input slots and output slots. The induced positional identification is invertible; exchanging the matching merely permutes coincident coordinates. The entire fixed word is a product of these collision identifications and the maps (258.1), proving the full-section assertion in Theorem 257.1.

The coordinates need not represent material particles. Labels change at a collision, and no persistent particle identity is assumed. Sorted positional slots provide a canonical chart: at a binary event the pair is adjacent; just before contact the left incoming strand has greater speed; just after

contact assign the two slots in increasing outgoing-speed order. At the contact itself reassignment changes no positional vector because the two entries coincide.

One meta-signal can have several occurrences at different positions. Sorting distinguishes them, so repeated labels discard no continuous information. Two occurrences of the same label have the same speed and cannot themselves form one of the stipulated distinct-speed pairs. If a proposed word leaves partner occurrences ambiguous, refine it into its finitely many ordered charts. The theorem is chartwise and does not claim that an unspecified union of charts has one linear formula.

The open full-dimensional chamber is also essential. Two linear maps that agree on such a chamber agree on the whole section space. Agreement on an invariant line, a boundary or another proper slice cannot identify the full linear extension and cannot support the singular-map obstruction.

258.2 Common translation and dimension

Let $e = (1, \dots, 1)^T$. Every collision row has $h_a e = 0$; every flight fixes e ; and every slot permutation fixes e . Hence the full isomorphism preserves $\mathbb{R}e$ and induces an isomorphism

$$H_{\text{in}}/\mathbb{R}e \longrightarrow H_{\text{out}}/\mathbb{R}e.$$

An anchor fixed to zero is a choice of representative of this quotient. Even if that anchor moved during the word, subtracting its final position is still a legitimate quotient chart. The general theorem needs no stationary-anchor hypothesis.

For n signals, a single binary section has dimension $n - 1$, and quotienting translation leaves $n - 2$. When a nonzero positive positional scale is available (in particular, for $n \geq 3$ with a positive remaining gap), locally quotienting that scale leaves $n - 3$. For $n = 2$ the translation quotient is already zero-dimensional and there is no nontrivial scale quotient. For five signals the dimensions are 4, 3, 2. The full homogeneous quotient coordinates in the stationary-marker model are exactly (D, x, y) or (D, X_c, Y_c) .

The initial anchor/messenger contact is an outgoing section, not an extra zero-time collision added to the word. The final anchor contact is the last counted event and emits the restored outgoing section. No label switch is being used as a substitute for an actual positive-duration contact.

259 Determinants in sorted gap coordinates

Put $d = n - 1$ and define the adjacent gaps

$$g_i = q_{i+1} - q_i \quad (1 \leq i \leq d).$$

These are full coordinates after quotienting translation. Collision rank a is the face $E_a = \{g : g_a = 0\}$, with the chart obtained by omitting g_a and keeping every other coordinate in increasing index order.

On a flight write $w_i = v_{i+1} - v_i$. For a flight from E_a to E_b the map is

$$P_{ab}g = g - w \frac{g_b}{w_b}. \quad (259.1)$$

The source opens, so $w_a > 0$; the target closes, so $w_b < 0$.

Lemma 259.1 (Complementary minor formula). *In the omitted-coordinate charts,*

$$\det(P_{ab} : E_a \rightarrow E_b) = (-1)^{a+b} \frac{w_a}{w_b}. \quad (259.2)$$

Proof. Let $P = I - we_b^T/w_b$. Its restriction matrix deletes row b and column a . Set $u = w/w_b$, so $e_b^T u = 1$. The rank-one adjugate identity

$$\text{adj}(I - uv^T) = (1 - v^T u)I + uv^T$$

with $v = e_b$ gives $\text{adj } P = we_b^T/w_b$. The (a, b) adjugate entry is the cofactor of the (b, a) entry, hence is $(-1)^{a+b}$ times the desired minor. This proves (259.2). The adjugate identity follows, for example, first when $1 - v^T u \neq 0$ by the rank-one inverse and determinant formulas, and then everywhere by polynomial continuation. If $a = b$, the restricted map is the identity and the formula gives 1; such a flight cannot be a nontrivial positive flight with both $w_a > 0$ and $w_a < 0$. $\square$

For an independent geometric derivation, let $\Omega = dg_1 \wedge \cdots \wedge dg_d$ and $\alpha = \iota_w \Omega$. Replacing a tangent vector u_i by $P_{ab}u_i$ adds a multiple of w , so the alternating expression $\Omega(w, u_1, \dots, u_{d-1})$ is unchanged. On E_a the induced form is

$$\alpha|_{E_a} = (-1)^{a-1} w_a dg_1 \wedge \cdots \wedge \widehat{dg_a} \wedge \cdots \wedge dg_d.$$

Comparing source and target forms again gives (259.2).

259.1 Telescoping the signs and speeds

Let the successive section faces be $a_0, a_1, \dots, a_m$. Multiplying the one-flight determinants telescopes the chart signs:

$$\text{sgn det } M = (-1)^{m + \sum_{j=1}^m (a_{j-1} + a_j)} = (-1)^{m + a_0 + a_m}.$$

For a literal return $a_m = a_0$, this becomes $(-1)^m$. On flight j , the absolute determinant is its source opening speed divided by its target closing speed. The list of source opening speeds is the initial opening followed by the openings after events $1, \dots, m-1$. Literal outgoing-phase closure makes the initial opening equal to the opening after event m . Thus the numerator list is a cyclic rearrangement of all event outgoing separation speeds. This proves (257.1) and Theorem 257.2.

If the final phase differs, retain the initial opening in place of the final one. If a separate coordinate identification C maps the target face to the source face, its determinant must also be included. More generally, replacing the input and output charts by C_{in} and C_{out} multiplies the displayed determinant by $\det C_{\text{out}} / \det C_{\text{in}}$. A same-chart literal return changes by conjugacy, so its determinant is unchanged. An ambient label-permutation sign cannot simply be substituted for the determinant induced on the actual face quotient.

For the stationary-marker section the first gap is zero and the other gaps are $(x, y - x, D - y)$. Their invertible linear relation to (D, x, y) changes a literal return by conjugacy. Centering is another conjugacy. The parity formula therefore applies in both of those coordinate systems.

260 What the singular obstruction actually excludes

The full translation-quotiented five-signal return has a rational invertible 3×3 homogeneous matrix N . If its scale law is the fixed $D' = \lambda D$ and its centered planar law is (257.5), then

$$N = \lambda \text{diag}(1, A), \quad \det N = \lambda^3 \det A. \quad (260.1)$$

Since $\lambda > 0$, $\det A$ cannot vanish. This proves the necessary direction of Corollary 257.4. It also gives $\text{sgn det } A = (-1)^m$ for a literal m -event return. Neither odd length nor negative determinant is forbidden.

The normalized affine variant $w' = Aw + b$ has homogeneous matrix

$$N = \lambda \begin{pmatrix} 1 & 0 \\ b & A \end{pmatrix},$$

with the same determinant $\lambda^3 \det A$. It cannot be singular either under these hypotheses. No information is lost merely by representing a positive-scale quotient in the chart $D = 1$.

For comparison, let a general invertible homogeneous matrix be

$$N = \begin{pmatrix} a & \beta \\ c & B \end{pmatrix}, \quad \rho(w) = a + \beta w > 0.$$

Its normalized action is the projective map

$$f(w) = \frac{c + Bw}{\rho(w)}, \quad Df(w) = \frac{B - f(w)\beta}{\rho(w)}, \quad \det Df(w) = \frac{\det N}{\rho(w)^3} \neq 0.$$

For the determinant identity, multiply N on the right by $\begin{pmatrix} 1 & 0 \\ w & 1 \end{pmatrix}$ and on the left by $\begin{pmatrix} 1 & 0 \\ -f(w) & 1 \end{pmatrix}$; the result has first column $(\rho(w), 0, 0)^T$ and lower right block $B - f(w)\beta$. This proves the displayed Jacobian formula without assuming B invertible. A nonconstant scale row changes the normalized map from affine to projective, but does not introduce a local singularity where the chart is defined.

Discarding an independent continuous variable is instead a projection. Likewise, a formula valid only on a lower-dimensional set cannot determine the full section matrix. Population change, multicollisions, simultaneous remote contacts and a declared continuation at an accumulation point are outside the transverse single-face argument. This report makes no singular-map impossibility claim for those alternative models.

261 The seven event reversal machine

The construction retains four stationary section markers and one messenger. Only the marker initially called X_0 moves during the word. The right marker R is untouched but supplies the independent scale coordinate D of the normalized planar section.

261.1 All labels and all specified rules

Table 25 lists all eleven standalone meta-signals. Subscripts s, m, c mean setup, middle and cleanup. The distinct labels X_{pre} and X_{post} deliberately have the same speed; their phases must remain distinguishable.

The seven specified binary rules are in Table 26. The last three columns give their incoming closing speed, outgoing separation speed and ratio in sorted spatial order. Every incoming pair and every outgoing pair has distinct speeds, and all seven input sets are different. Rule 4 is a marker-only contact. It leaves the messenger's position, velocity and label Q_m unchanged. Assigning a new messenger phase at that event would not be the stated machine.

Every other admissible input set with pairwise distinct speeds receives the identity output. There are finitely many subsets of the finite label set, so this is a finite deterministic completion, preserving cardinality even for unprescribed larger contacts. Such contacts, if they occur, still count

Table 25: Complete speed table for the standalone reversal

Role	Meta-signals	Assigned speed
Stationary section markers	L, X_0, Y, R	0 for each
Outgoing messenger	Q_+	1
Setup messenger	Q_s	$-3/2$
Middle messenger	Q_m	$-1/4$
Cleanup messenger	Q_c	-1
First moving target phase	X_{pre}	$-1/2$
Fast target phase	X_{fast}	2
Final moving target phase	X_{post}	$-1/2$

Table 26: Rules and exact relative speeds

j	Input	Output	s_j^-	s_j^+	s_j^+/s_j^-
1	$\{Q_+, X_0\}$	$\{Q_s, X_{\text{pre}}\}$	1	1	1
2	$\{Q_s, L\}$	$\{Q_+, L\}$	$3/2$	1	$2/3$
3	$\{Q_+, X_{\text{pre}}\}$	$\{Q_m, X_{\text{fast}}\}$	$3/2$	$9/4$	$3/2$
4	$\{X_{\text{fast}}, Y\}$	$\{X_{\text{post}}, Y\}$	2	$1/2$	$1/4$
5	$\{Q_m, L\}$	$\{Q_+, L\}$	$1/4$	1	4
6	$\{Q_+, X_{\text{post}}\}$	$\{Q_c, X_0\}$	$3/2$	1	$2/3$
7	$\{Q_c, L\}$	$\{Q_+, L\}$	1	1	1

as extra events and invalidate the selected complete binary word. The completion does not make the global label-rule map injective. In fact rules 2, 5 and 7 already share the output set $\{Q_+, L\}$.

Initially the live signals are L, Q_+ together at 0, X_0 at x , Y at y and R at D . They are five outgoing strands at four occupied sites. Every specified rule consumes and emits two strands; every finite prefix retains population five.

261.2 Ballistic lines and exact contact data

Write $q(t)$ for the messenger's position and $z(t)$ for the moving target's position. The markers L, Y, R remain at 0, y, D . Define

$$\xi = \frac{5}{4}y - x, \quad F = \frac{2}{3}\xi = -\frac{2}{3}x + \frac{5}{6}y. \quad (261.1)$$

The following equations are obtained from the fixed speeds and continuity at each contact. They are global-time line equations, not a numerical trajectory or an event-selection algorithm. For example, $q = z$ at the first contact gives $t_1 = x$. Thereafter $q = 0$ on the setup line gives $t_2 = 5x/3$. Solving $t - 5x/3 = 3x/2 - t/2$ gives $t_3 = 19x/9$ and contact position $4x/9$. The line $2t - 34x/9$ reaches y at $t_4 = 17x/9 + y/2$. The unchanged messenger line reaches zero at $t_5 = 35x/9$, where $z = \xi$. The last two flights have duration F each.

At sections 1, 3, 6 the contact is messenger/target; at 2, 5, 7 it is messenger/ L ; at 4 it is target/ Y .

Table 27: Exact affine lines on the seven open flights

Flight	$q(t)$	$z(t)$
$0 \rightarrow 1$	t	x
$1 \rightarrow 2$	$5x/2 - 3t/2$	$3x/2 - t/2$
$2 \rightarrow 3$	$t - 5x/3$	$3x/2 - t/2$
$3 \rightarrow 4$	$35x/36 - t/4$	$2t - 34x/9$
$4 \rightarrow 5$	$35x/36 - t/4$	$5y/4 + 17x/18 - t/2$
$5 \rightarrow 6$	$t - 35x/9$	$5y/4 + 17x/18 - t/2$
$6 \rightarrow 7$	$23x/9 + 5y/3 - t$	F

Table 28: All section times and positions

Section	Absolute time	Messenger position	Target position
Initial	0	0	x
1	x	x	x
2	$5x/3$	0	$2x/3$
3	$19x/9$	$4x/9$	$4x/9$
4	$17x/9 + y/2$	$(4x - y)/8$	y
5	$35x/9$	0	ξ
6	$35x/9 + F$	F	F
7	$35x/9 + 2F$	0	F

The seven positive-flight candidates and total duration are

$$(\Delta t_1, \dots, \Delta t_7) = \left(x, \frac{2x}{3}, \frac{4x}{9}, \frac{9y-4x}{18}, \frac{4x-y}{2}, F, F \right), \quad (261.2)$$

$$T_J = \frac{23}{9}x + \frac{5}{3}y. \quad (261.3)$$

These formulas follow by subtraction from Table 28. Each line in Table 27 has exactly the corresponding messenger and target speeds in Table 25.

261.3 The whole chamber has no omitted contacts

Assume (257.2). Then $0 < \xi < y$ and $0 < F < y$: indeed $\xi > y/4 > 0$ from $x < y$, while $\xi < y$ is precisely $x > y/4$. Every duration in (261.2) is positive. The fixed spatial order on each open flight is

$$L, \quad q, \quad z, \quad Y, \quad R.$$

Its four gaps at every endpoint are listed in Table 29. Exactly the prescribed contact gap is zero at each section. Every nonzero entry is strictly positive. For the only less immediate row, section 4

Table 29: Complete endpoint gap table in sorted order

Section	$q - L$	$z - q$	$Y - z$	$R - Y$
Initial	0	x	$y - x$	$D - y$
1	x	0	$y - x$	$D - y$
2	0	$2x/3$	$y - 2x/3$	$D - y$
3	$4x/9$	0	$y - 4x/9$	$D - y$
4	$(4x - y)/8$	$(9y - 4x)/8$	0	$D - y$
5	0	ξ	$y - \xi$	$D - y$
6	F	0	$y - F$	$D - y$
7	0	F	$y - F$	$D - y$

has $4x - y > 0$ by the guard and $9y - 4x > 5y > 0$ by $x < y$. For all other rows positivity follows directly from $0 < x < y$, $0 < \xi$, $F < y$, and $D > y$.

On an open flight each gap is affine in time. If it is positive at both endpoints it remains positive between them. The departure gap has value zero only initially and positive outgoing derivative, and is positive at the next endpoint. The arrival gap is positive initially and zero only at the final endpoint. Thus no adjacent gap vanishes during an open flight. Nonadjacent signals cannot meet without an intervening adjacent gap vanishing. The endpoint table also excludes simultaneous remote contacts.

In particular, after event 4 the target moves left at $-1/2$, faster leftward than the messenger's $-1/4$. A target/messenger collision is a genuine competing possibility, not an ignored interaction. Their gap is positive at event 4, positive with value ξ at event 5, and affine throughout that interval. It cannot vanish before event 5. The right marker R remains strictly beyond Y at all times. This proves sufficiency on the entire open chamber, rather than on a set of test inputs.

An alternative whole-chamber certificate puts $k = x/y$ and $d = D/y - 1$. The chamber is $1/4 < k < 1$, $d > 0$, $y > 0$. Each noncontact gap divided by y is an affine function of k nonnegative

at $1/4$ and 1 , positive at at least one endpoint, or is d . The same endpoint test works for all seven normalized flight durations. This elementary interval certificate is used by the independent arithmetic review.

261.4 Necessity from the first wrong event

Begin with only $0 < x < y < D$. The messenger first reaches X_0 at $t_1 = x$. Until event 2 it moves left faster than X_{pre} , and reaches L at $5x/3$, when the target is still $2x/3 > 0$. After the bounce it meets the left-moving target at $19x/9$, position $4x/9 > 0$, before the target could reach L . No marker to the right is approached on these first three flights. Thus events 1–3 have exactly the stated order without using $y < 4x$.

Immediately after event 3 the messenger moves left and the target moves right. They separate. There are exactly two possible next contacts: the target reaches Y , or the messenger reaches L . Their prospective absolute times are

$$t_{X/Y} = \frac{17x}{9} + \frac{y}{2}, \quad t_{q/L} = \frac{35x}{9}, \quad t_{q/L} - t_{X/Y} = \frac{4x - y}{2}.$$

Both prospective flights start from the already valid prefix. If $y = 4x$, the two contacts are simultaneous at different positions. If $y > 4x$, messenger/ L occurs first. Both violate the selected complete, globally separated word before its stipulated fourth event. There is no need to continue a fictitious trajectory beyond that first obstruction. Together with sufficiency, this proves that the exact chamber is precisely (257.2).

On $D = 1$ the chamber is the bounded open rational triangle

$$0 < y < 1, \quad y/4 < x < y.$$

Writing $u = X_c/D$, $v = Y_c/D$, a convenient exact strict guard list is

$$P_J = \left\{ (u, v) : \frac{1}{3} + u > 0, \frac{1}{3} - u + v > 0, \frac{1}{3} - v > 0, \frac{2}{3} + 4u - v > 0 \right\}. \quad (261.4)$$

Redundancy in this four-row description is harmless. Each row is strict at $(u, v) = (0, 0)$.

261.5 An analytic spacetime view

Figure 15 substitutes $D = 1$, $x = 1/3$, $y = 2/3$ into the exact lines and endpoints. This input is in the interior, and $F = 1/3$. The picture illustrates a proved word; it is not evidence from a simulated run.

261.6 Return matrix and literal closure

At event 6 the target is restored to X_0 with speed zero, and the messenger becomes Q_c with speed -1 . At event 7 the messenger reaches the stationary L and returns to label Q_+ and speed $+1$. Neither Y nor R changed label or position. Thus the final outgoing section is literally the initial one, with x replaced by F .

The homogeneous return in (D, x, y) is

$$M_J = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -2/3 & 5/6 \\ 0 & 0 & 1 \end{pmatrix}.$$

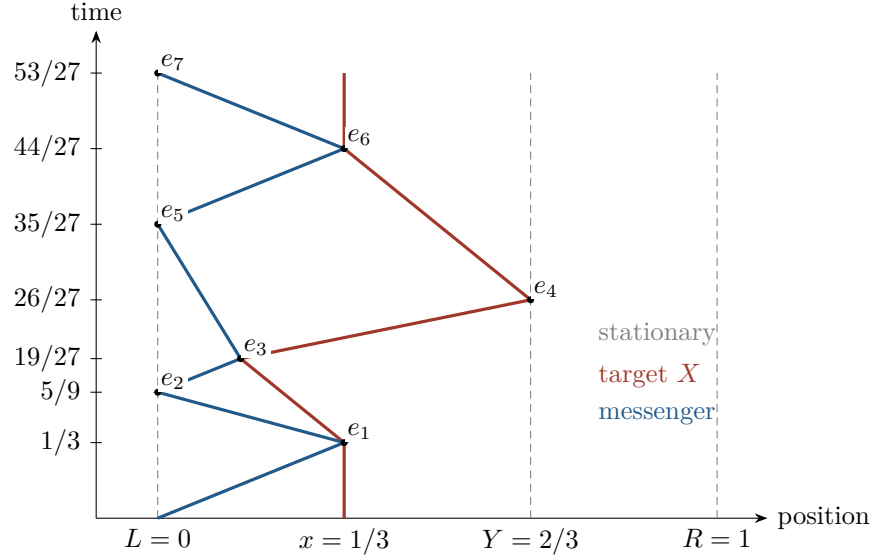


Figure 15: Exact analytic spacetime diagram at the standard center, drawn directly from Tables 27–28. Time increases upward. Event 4 is the target’s contact with Y ; the messenger continues along the same blue line. Straight segments and event times are formula substitutions, not output of a trajectory simulator.

It fixes $(D, D/3, 2D/3)^T$, because $-2D/9 + 5D/9 = D/3$. Centering therefore leaves exactly $\text{diag}(1, J)$, with no affine remainder, and

$$J^{-1} = \begin{pmatrix} -3/2 & 5/4 \\ 0 & 1 \end{pmatrix}.$$

The seven ratios in Table 26 multiply to $2/3$. The word has seven events, so Theorem 257.2 gives determinant $-2/3$, agreeing with the displayed matrix. The sorted collision faces are 1, 2, 1, 2, 3, 1, 2, 1, including the initial face, which makes the closed-face convention explicit. This completes the proof of Theorem 257.3.

Deleting the unused R strand preserves the primitive and reduces its live population to four. It also removes D as an independent coordinate: the translation quotient then has dimension two and its positive-scale normalization has dimension one. This observation is not a four-live-signal normalized planar compiler, nor a lower bound for other encoding models.

261.7 Exact infinite validity and a two witness certificate

The standalone primitive has a particularly simple repetition problem. Both D and y remain fixed, and the ratio $t = x/y$ evolves by

$$f(t) = \frac{5}{6} - \frac{2}{3}t, \quad P_t = (1/4, 1).$$

The first two word requirements give

$$P_t \cap f^{-1}P_t = (1/4, 7/8).$$

Indeed $f(t) > 1/4$ is equivalent to $t < 7/8$, and $f(t) < 1$ imposes only $t > -1/4$, already implied by $t \in P_t$. Moreover,

$$f((1/4, 7/8)) = (1/4, 2/3) \subset (1/4, 7/8).$$

Thus the two-word condition is forward invariant and is already the exact infinite-validity set. The lower endpoint $t = 1/4$ fails by simultaneous contacts in the first word. The upper endpoint $t = 7/8$ completes the first word, then lands at $1/4$ and fails in the second. Neither boundary is admitted.

Corollary 261.1 (Two positive integer witnesses). *For positive integer physical gaps*

$$g_1 = x, \quad g_2 = y - x, \quad g_3 = D - y,$$

infinite repetition of this standalone seven-event word is valid if and only if there exist positive integers u, v such that

$$(3g_1 - g_2 - u)^2 + (7g_2 - g_1 - v)^2 = 0. \quad (261.5)$$

This is a quadratic integer-coefficient certificate with exactly two positive integer witnesses, uniquely forced whenever it holds.

Proof. The ratio conditions $1/4 < x/y < 7/8$ are exactly $g_2 < 3g_1$ and $g_1 < 7g_2$. For integer gaps their positive differences are positive integers. A sum of real squares is zero precisely when each summand is zero, forcing $u = 3g_1 - g_2$ and $v = 7g_2 - g_1$. Conversely positive witnesses force both strict inequalities. The gap g_3 is deliberately absent: the far marker is untouched and imposes only its already assumed positivity. $\square$

The witness count is for the displayed certificate; no minimality is claimed. For real positive gaps the two strict linear inequalities still describe the exact infinite-validity set, but the integer witness formulation is stated only for integer inputs.

[write] [The two forced witnesses are the unique slacks of Part I's Theorem 6.1 for the two strict rows of the interval $(1/4, 7/8)$.]

There is also an exact finite-macro clock. For an infinitely valid input,

$$x_n = \frac{y}{2} + \left(-\frac{2}{3}\right)^n \left(x - \frac{y}{2}\right).$$

Summing $23x_n/9 + 5y/3$ from $n = 0$ to $N - 1$ gives, for every integer $N \geq 0$,

$$T_N = \frac{53}{18}yN + \frac{23}{15} \left(x - \frac{y}{2}\right) \left(1 - \left(-\frac{2}{3}\right)^N\right). \quad (261.6)$$

The correction term is bounded, while $y > 0$ makes the linear term diverge. Thus infinite standalone repetition is not Zeno; acceleration requires an additional shrinking scale mechanism such as the one in Section 263.

262 From positive determinant to all invertible planar maps

262.1 The precise imported theorem

The positive-determinant compiler of Report 60 [[write] source 30, Theorem 240.1; shipped as 30-planar-science-physical-PROOF.md] gives the following statement: for each $B \in \text{GL}_2(\mathbb{Q})$

with $\det B > 0$ and $\lambda \in \mathbb{Q}_{>0}$, a finite deterministic rational-speed number-preserving machine on exactly five live signals has a complete binary word returning

$$(D, X_c, Y_c)^T \mapsto \lambda \operatorname{diag}(1, B)(D, X_c, Y_c)^T.$$

It uses the same stationary-marker section and initial messenger speed $+1$. Every marker label and the messenger phase close literally. Its exact chamber on $D = 1$ is a bounded open rational polygon P_B containing 0 . It supplies all guards as strict rational affine rows and proves their necessity by the earliest failed primitive, not by continuing an invalid prospective word.

Only this physical realization theorem and its duration bound are imported here. Report 60's separate planar spectral classification and arithmetic-certificate results are not needed for the realization criterion of this report. The frozen positive-compiler proof is included in the evidence package so the dependency can be inspected without executing any prior constructor.

For orientation, its construction has four ingredients. Anchor scaling moves a target distance z to uz using target speed $(u - 1)/(u + 1)$ and four principal messenger contacts. A reflector homothety sends a nearest target t to $vt + (1 - v)z$ with duration $2z$. Their composition supplies guarded rational translations. Subdivided signed shear blocks realize

$$U(a) = \begin{pmatrix} 1 & a \\ 0 & 1 \end{pmatrix}, \quad V(a) = \begin{pmatrix} 1 & 0 \\ a & 1 \end{pmatrix},$$

using exact neighbor-interval guards. Rational shear factorizations, including zero-pivot and diagonal branches, produce every element of $\operatorname{SL}_2(\mathbb{Q})$. A positive centered one-axis dilation changes determinant without taking an irrational square root. Finally ordered anchor scales supply the common factor λ . Full anchor transfers restore the messenger's side and direction.

With K the total shear subdivision count, T the number of full transfers, H the number of small centered-dilation factors and $\varepsilon = \mathbf{1}_{\lambda \neq 1}$, the positive word has

$$m_B = 18K + 3T + 14H + 24\varepsilon. \quad (262.1)$$

Here T is even, so m_B is even. The construction retains genuine positive-duration zero-parameter factors where necessary, including for the identity matrix. The complete argument, factors, guards and label counts belong to the pinned dependency; these counts are not presented as optimized.

[write] [The imported theorem is source 30's Theorem 240.1, proved in Sections 241–244. The counts (262.1) are source 30's (242.9), and the duration bound used below is its (244.1). The summary is kept as source 31's statement of its interface.]

262.2 Matrix composition and the exact guard pullback

If $\det A > 0$, the imported compiler already proves existence. If $\det A < 0$, set

$$B = J^{-1}A. \quad (262.2)$$

Since $\det J < 0$, $\det B > 0$. Run the positive compiler for (B, λ) and then the seven-event reversal. The chronological composition is

$$\operatorname{diag}(1, J) [\lambda \operatorname{diag}(1, B)] = \lambda \operatorname{diag}(1, JB) = \lambda \operatorname{diag}(1, A). \quad (262.3)$$

The first block restores the stationary-marker geometry exactly; its final positive scale λ does not change the normalized guards of the second block.

If $w = (X_c/D, Y_c/D)^T$, the shape entering the reversal is Bw . Consequently the complete composite word has exactly the chamber

$$P = P_B \cap B^{-1}P_J = \{w : w \in P_B, Bw \in P_J\}. \quad (262.4)$$

Each condition is a strict rational affine row. Both sets contain 0 as an interior point, so their intersection is full-dimensional and center-containing. It is bounded because it is contained in the bounded P_B ; it is an open convex rational polygon.

Sufficiency follows by executing two valid word blocks in order. For necessity, if a P_B row fails, the first block fails by its own first-failure proof. After a successful first block, failure of a pulled-back P_J row makes the second block fail by Section 261.4. This blockwise argument establishes exactness. Replacing either exact chamber by a safe neighborhood or an arbitrarily chosen invariant polygon would not establish the same statement.

262.3 A literal finite label interface

The geometric composition must be accompanied by a consistent finite rule table. Blindly identifying the standalone Q_+ with the positive compiler's initial messenger label is not valid: the input set involving X_0 could then demand different outputs at the wrong phase.

Let the positive block's initial messenger be P_0 , and let its internal messenger labels be private to that block. Give the reversal fresh labels Q_+, Q_s, Q_m, Q_c and fresh moving-target labels $X_{\text{pre}}, X_{\text{fast}}, X_{\text{post}}$; identify only the stationary section roles L, X_0, Y, R . Make two interface replacements:

1. At the positive block's final event, replace its usual output messenger P_0 by the reversal entry label Q_+ , which has the same required speed +1.
2. At the reversal's final event, replace its output Q_+ by P_0 , again with speed +1.

Keep the reversal's intermediate outputs Q_+ at events 2 and 5 private to the reversal. Its marker-only event 4 has the fresh X_{fast} in its input and cannot conflict with a positive-block rule. It changes no messenger label. The whole word therefore begins and ends with exactly P_0 at L , while restoring all marker labels. No extra live signal or zero-time event is inserted.

All positive-block explicit rules involve a messenger; all reversal rules have the finite labels and distinct speeds already checked. Fresh labels separate potential duplicate input sets across the blocks. Give every remaining admissible input set its identity output. This completes the finite deterministic number-preserving machine.

The selected negative-determinant word has $m_B + 7$ events, which is odd by (262.1), as Theorem 257.2 requires. Eleven meta-signals is the standalone reversal count only; it is not the count of the whole composite machine. The finite rule table may depend on A and λ . No single unchanged universal table, bounded word length or efficient bit-complexity bound is asserted.

Together with the obstruction (260.1), this proves both directions of Corollary 257.4.

263 Repetition and accumulation time

Literal phase closure means the next macro begins in the same section and uses the same complete word. At macro n , the scale is $D_n = \lambda^n D_0$ and the normalized shape is $A^n w$. The exact one-word

chamber therefore gives

$$\text{infinite complete-word validity} \iff w \in K(A, P) := \bigcap_{n \geq 0} A^{-n}P. \quad (263.1)$$

This set contains 0, since $0 \in P$ and $A0 = 0$. The polygon is the actual word chamber, with strict boundaries retained. Formula (263.1) is not a claim that the compiler can realize an arbitrary prescribed polygon at the same cost.

263.1 Uniform duration bounds and the Zeno criterion

Each event time is a rational homogeneous linear form on its fixed word, so the macro duration is a fixed rational row ℓz , where $z = (D, X_c, Y_c)^T$. The positive compiler supplies

$$2D \leq \ell_B z < C_B D, \quad C_B = 12K + T + 10H + 6(1 + \lambda)\varepsilon. \quad (263.2)$$

Its lower bound comes from a retained far-reflector phase, including in its identity case. On the reversal's entrance chamber, (261.3) and $x, y < D$ give

$$0 < T_J < \frac{38}{9}D.$$

After the positive block, its entrance scale is λD . Thus for the composite word

$$2D \leq \ell z < \left(C_B + \frac{38\lambda}{9}\right) D. \quad (263.3)$$

For positive determinant use (263.2) directly. In centered coordinates the reversal duration row itself is

$$\ell_J = \left(\frac{53}{27}, \frac{23}{9}, \frac{5}{3}\right),$$

so for the negative branch $\ell = \ell_B + \lambda \ell_J \text{diag}(1, B)$.

Along an infinitely valid run the bounds are constant multiples of $D_n = \lambda^n D_0$. The sum of macro durations is finite exactly when $0 < \lambda < 1$. Each macro contains a finite positive number of positive-flight events, so this is precisely the Zeno criterion. For $\lambda \geq 1$, the lower bound alone forces divergent total time. For $\lambda < 1$, all strands approach the fixed left anchor: during macro n they remain in $[0, \max(1, \lambda)D_n]$, both through the imported compiler and the appended reversal.

No operation or continuation through the accumulation point is specified. Number preservation applies to every finite stage; it does not by itself select an evolution after infinitely many collisions.

263.2 Rational accumulation time on the valid cyclic subspace

Remark 263.1 ([write] Printed once, as source 30's cyclic-subspace route). Source 31 states the following. Assume rational initial data, infinite validity and $0 < \lambda < 1$, and put $N = \lambda \text{diag}(1, A)$ and $V = \text{span}_{\mathbb{Q}}\{z, Nz, N^2z\}$. Then V is a rational N -invariant subspace containing the forward orbit, every eigenvalue of $N|_V$ has modulus less than one, and

$$T_\infty = \sum_{n \geq 0} \ell N^n z = \ell[(I - N)|_V]^{-1} z \in \mathbb{Q}. \quad (263.4)$$

The unrestricted expression $\ell(I - N)^{-1} z$ can fail to exist: A may have the eigenvalue $1/\lambda$ while no infinitely valid orbit uses the corresponding mode. The proof is source 30's argument for (244.2) in Section 244, now for every $A \in \text{GL}_2(\mathbb{Q})$ and for the duration row ℓ of the composite word above. The rationality is a special case of source 26's Theorem 182.4 (Remark 244.1).

264 Primary literature and the boundaries of the claim

The transverse-flight inverse is a signal-machine specialization of standard constant-flow section geometry. The following primary sources provide definitions and close precedents; they are not substitutes for the direct proofs above.

Becker et al. [92], Definitions 1–2, manuscript PDF pp. 4–5, assign speeds to meta-signals and require distinct speeds inside collision input and output sets. The configuration framework allows repeated occurrences of a label. Example 5 on PDF p. 6 explicitly uses distinct labels at equal speed. These points justify keeping the counts of live occurrences, label types and speeds separate.

Durand-Lose [5], Definitions 1, 3 and 4, manuscript PDF pp. 3–5, distinguishes rational signal machines, positive-energy conservativeness and reversible collision rules. Equal energies make number preservation a special conservative case. Reversibility there requires injectivity of the rule map and has an accumulation qualification. Its rational convention also restricts positions, and its time framework, to rationals. The open real chambers in this report are a rational-coefficient geometric extension; rational points in them still give rational configurations. None of the results here implies that the identity-completed machine is globally reversible.

Alishah, Duarte and Peixe [93], Section 5, manuscript PDF pp. 21–24, study constant-direction oblique section maps in polytope skeleton flows. Equation (5.2) gives the projection, Proposition 5.4 its trajectory sector, and Definition 5.7 with (5.5) composes fixed itineraries. The paragraph immediately after (5.3), PDF p. 22, identifies the inverse with the reversed-flow section map; Proposition 5.10 also invokes local linear isomorphisms. This is a direct geometric antecedent for the inverse used here. Its setting does not itself provide the present signal-machine parity calculation, seven-event rule table or compiler interface.

The source packet and independent review made a bounded primary-source comparison, not an exhaustive literature search. Failure to locate the exact combined statement does not establish novelty or priority. No minimality claim is attached to seven events, eleven labels, the set of speeds or the imported factorization. Five live signals here is the full-coordinate normalized-section dimension count, not a prohibition on alternative lower-population encodings with different sections or observations.

265 Proof evidence and reproducibility

The evidence has three logically separate layers.

1. **Conventional proof.** Sections 258–263 give the inverse, cofactor identity, chronology, exact first failure, phase interface and realization argument. General claims do not follow from finite fixtures.
2. **Frozen author arithmetic.** The source packet contains its original standard-library rational-row checker and receipt. It verifies line increments, gap and flight positivity certificates, centering, inverse and speed products. Its ninety complementary-minor fixtures are examples supporting a separately proved identity. This program is included as inert source evidence; manuscript production did not run it or any prior compiler.
3. **Independent arithmetic and review.** The separate review rederived all global affine lines and used $k = x/y \in (1/4, 1)$ for a whole-interval positivity certificate. Its newly written exact checker verifies static contact, velocity, continuity and endpoint identities. It does not choose

the next event, advance a machine state, sample trajectories or execute a saved collision schedule. The review is a conventional mathematical PASS with exact arithmetic support, not a proof-assistant certificate.

The author’s cone certificate is also explicit. With

$$r = x - y/4 > 0, \quad s = y - x > 0, \quad t = D - y > 0,$$

one has

$$(D, x, y) = \left(\frac{4r + 4s}{3} + t, \frac{4r + s}{3}, \frac{4r + 4s}{3} \right).$$

Every required noncontact gap and duration becomes a nonzero nonnegative rational combination of r, s, t , while contact rows vanish identically. This certifies the full chamber, rather than inferring it from numerical samples. The independent endpoint method proves the same inequalities by a different elementary certificate.

The package includes a standalone editable L^AT_EX file, the equivalent modular manuscript, the compiled PDF, the frozen proof and JSON rule/guard tables, the pinned positive-compiler source, independent review and deterministic arithmetic receipts, and release-only build and integrity tools. The build tools typeset the owned manuscript and render its pages; they do not execute the frozen scientific programs. The analytic diagram is written directly in L^AT_EX from the exact event lines. Any portable independent replay has its own explicitly reviewed tool and provenance; it is separate from manuscript typesetting.

[write] [Shipped here: the frozen proof, its README and review guide, the reviews and the replay notes as `31-invertible-*.md`; the programs as `code/31-invertible-*`; the JSON tables, receipts and pins as `data/31-invertible-*`. The pinned positive-compiler proof is shipped once, as `30-planar-science-physical-PROOF.md`. The PDF, the LaTeX and the page renders are not shipped.]

265.1 Pinned mathematical inputs

The main source is the frozen proof packet dated 4 October 2026. The elementary infinite-validity, two-witness and finite-clock corollaries in Section 261.7 are additional direct manuscript deductions, independently checked during manuscript review; they are not represented as contents of the earlier frozen packet. The following SHA-256 pins identify the mathematical input and the reviewed dependency, independently of any later release packaging.

Report 61 proof text

dc97e30c56817370138d6be760d44a5df031743a0ee1011cca39381b67bb0cec

Report 61 source manifest

715bbbc817188a1a4d90ea7ad673fe4c69a9df105e202828c1835ae308e6f7eb

Positive compiler proof from Report 60

e0ddd64cdbbb5c7f448266ab58f1dfeb2868892ac0f230632b2337e4d0bed065

Independent scientific audit manifest

dbbc49723fad5b5ab34cd8dab1b8824d66d22b50e384d948e1de0576b7e16aef

The dependency’s own broader claims and old fixture outputs are not reclassified as newly executed evidence. No prior delivered source is altered. Fresh manuscript and release audits, when included, are separate from the frozen scientific review.

266 Further questions with a precise scope

The fixed-scale planar criterion is complete under its stated assumptions. It leaves several concrete questions.

1. **General homogeneous returns.** Which rational $N \in \mathrm{GL}_3(\mathbb{Q})$ can be realized on (D, X_c, Y_c) when the final positive scale depends on shape? Any proposal must specify an open compatible ordered-section domain. Its normalized dynamics is projective, so the fixed-scale result alone does not answer it.
2. **Compilation efficiency.** What word length, label count and speed complexity suffice as functions of the bit length of A and λ ? The current exact factorization is constructive, but no polynomial-size or optimality theorem is proved.
3. **Minimal reversing primitives.** Can an orientation-reversing literal return be realized with fewer than seven events on this exact stationary-marker section? Parity requires odd length, but it gives no lower bound of seven. A meaningful answer must keep the section, complete-word convention and permitted rules fixed.
4. **Globally reversible completion.** Can one arrange an injective global collision-rule map while retaining the desired section return and comparable resources? The present number-preserving identity completion and its repeated output sets do not answer that different question.
5. **Alternative observations and sections.** What singular observed maps become possible when independent coordinates are deliberately forgotten or when the section has lower dimension? Such questions require a new observation model, rather than an extension of the present impossibility claim by analogy.

[write] [Status at this write: Question 1 is answered at a rational fixed centre by source 32 (Theorem 267.1). It is answered locally, on some full-dimensional open set, by source 33 (Theorem 281.1): a rational homogeneous return is locally realizable exactly when it is invertible and sends some ordered input to an ordered output. It stays open for a prescribed, invariant or maximal chamber (source 33’s Question 1). The other four questions are open.] A continuation law through a Zeno point would likewise be an additional model choice. The finite-word proofs and rational accumulation time specify a limit and its time, not what happens after it.

Source 32 (batch-91 manuscript 14): Five signal projective returns and mixed Zeno clocks

Exact fixed center realization and a 44 event example, “Research Report 62”, 4 October 2026. [write] From the archive `Projective_signal_returns_source_and_evidence.zip` (main file `Report62/Report62.tex`, 22-page PDF); its labels carry the prefix `smc:pj:`. Its sections are Sections 267–280 here; the last three are its Appendices A–C. Its six bibliography entries are this report’s [94], [93], [4], [92], [6] and [95]. Its abstract follows.

Exactly five live signals suffice to realize every rational homogeneous return $N = \begin{pmatrix} s & b \\ 0 & A \end{pmatrix}$ with $s > 0$ and $A \in \mathrm{GL}_2(\mathbb{Q})$ on the stationary four-marker section. The fixed complete word has a bounded, center-strict, exact rational chamber, rational label speeds, number-preserving binary rules and literal phase closure. Its normalized return is $w \mapsto Aw/(s + bw)$ with positive denominator. An eight-event outer-marker operation and twelve-event recentering supply a scale shear; compensation and rational conjugation produce an arbitrary scale row. The inherited

planar compiler is stated precisely and yields an explicit 780-event coefficient-one ledger. A separate 44-event machine has infinite validity exactly when $D \geq 3y/2$: equality is Zeno and strict inequality is non-Zeno. Full chronology and first-failure proofs, all 44 rules, exact clocks, a cyclic-subspace spectral criterion, and native integer-gap quadratic certificates with witness counts 1, 0, 1 are included. The result concerns a fixed rational center, not general three-dimensional homogeneous realizability or Turing completeness. Analytic figures and independent static evidence accompany the proof.

267 Results and the physical section

The outgoing section has stationary markers $L = 0$, $X = x$, $Y = y$, $R = D$ with $0 < x < y < D$, and one messenger at L outgoing at speed $+1$. The anchor and messenger are two outgoing strands at one contact. Counting them separately gives exactly five live signals. Put

$$\xi = x - D/3, \quad \eta = y - 2D/3, \quad u = (\xi, \eta)^\top, \quad z = (D, u)^\top.$$

The standard centered ray is $u = 0$, $D > 0$. A complete word lists every contact in order, including transparent crossings and remote contacts. Its chamber consists of exactly those initial section points that realize that word with strictly positive flights and no simultaneous or omitted event.

Theorem 267.1 (Fixed center projective realization). *For every rational $s_0 > 0$, row $b \in \mathbb{Q}^{1 \times 2}$ and $A \in \text{GL}_2(\mathbb{Q})$, there is an effectively constructible finite deterministic rational-speed number-preserving signal machine and a fixed complete binary word whose literal section return is*

$$z \mapsto Nz, \quad N = \begin{pmatrix} s_0 & b \\ 0 & A \end{pmatrix}. \quad (267.1)$$

Its exact chamber is a finite intersection $Q = \{z : Gz > 0\}$ of strict rational homogeneous rows, supplied by an explicit finite pullback construction. The normalized chamber P on $D = 1$ is a bounded open rational polygon containing the centered origin strictly. All section labels and the outgoing messenger phase are restored. On P the normalized return is

$$f(w) = \frac{Aw}{s_0 + bw}, \quad w = u/D, \quad s_0 + bw > 0. \quad (267.2)$$

There are exactly five live signals throughout each finite portion of the execution.

These are precisely the rational projective planar transformations fixing this center that admit the indicated invertible lift positive on its ray. A common scalar multiple of N gives the same projective transformation, but a different homogeneous return and potentially a different clock. The theorem realizes the specified lift itself.

Theorem 267.2 (One machine with two clock behaviors). *For every rational $v > 1$ an explicit 44-event word on the same five-signal section has infinite complete-word validity exactly on*

$$0 < x < y < D, \quad D \geq \frac{3}{2}y.$$

Equality gives Zeno runs, and strict inequality gives non-Zeno runs. On positive integer native gaps $g = (x, y - x, D - y)$ the validity, Zeno-validity and non-Zeno-validity predicates have quadratic single-polynomial certificates with respectively one, zero and one positive-integer witnesses, uniquely determined on each accepted input.

The exact 44-rule declaration is in Section 279; the proof and clocks are in Section 274. A center-strict one-word chamber can have its center on the boundary of the infinite-validity set.

[write] [Source 27 in Part X needs two machines for its two clock behaviours: a 138-event Zeno word and a separately reclosed 114-event non-Zeno variant with the same chamber and the same set K (Theorem 196.3). Here a single word has both behaviours. Question 2 asks how far this extends.]

Scope. The construction is fixed after its rational parameters are chosen. It does not give one unchanged machine for all matrices, arbitrary prescribed guard polygons, polynomial compilation size, optimal event/speed counts, global rule reversibility, or evolution after an accumulation. Singular A is excluded by the inherited full-section invertibility theorem under its binary, positive-flight, open-chamber hypotheses. A nonpositive s_0 cannot give this literal return on a chamber containing the standard center, whose final D would be nonpositive. No general $\text{GL}_3(\mathbb{Q})$ or fixed-word Turing-completeness claim is made.

268 Exact elementary chronology

268.1 The inherited compiler interface

Write $F(C, \lambda)$ for the preserved planar compiler of Reports 60 and 61. For every $C \in \text{GL}_2(\mathbb{Q})$ and rational $\lambda > 0$, it supplies a nonempty positive-duration complete word with return

$$F(C, \lambda) : \quad z \mapsto \lambda \text{diag}(1, C)z. \quad (268.1)$$

It supplies its finite exact homogeneous guard rows, a center-strict chamber, and literal section closure. For $\det C > 0$, every event involves the messenger. For $\det C < 0$, a positive compiler is followed by a seven-event orientation-reversing block J containing one marker-only event. Its messenger label does not change at that remote event. Fresh phase copies are required at interfaces.

Section 278 gives the complete arithmetic factorization, primitive endpoint lists, orientation-reversing interface and count convention used here. The two unchanged dependency proofs are included in the evidence package. The present new proof invokes this interface rather than silently treating an arbitrary planar change of coordinates as a physical operation.

[write] [$F(C, \lambda)$ is source 30's Theorem 240.1 for $\det C > 0$ and source 31's Corollary 257.4 for $\det C < 0$. Source 32's restatement of their constructions (its Appendix A) is the pointer Remark 278.1 in Section 278. The two dependency proofs are shipped once, as 30-planar-science-physical-PROOF.md and 31-invertible-science-frozen-proof-PROOF.md.]

268.2 Anchor scaling and global homotheties

Remark 268.1 ([write] Source 30's primitive, restated). Source 32 restates source 30's anchor scaling $L_z(u)$ and its ordered global scaling (Section 241, equations (241.1)–(241.2), and Section 242), with the same chronology and first-obstruction proof. In source 32's notation, a target at distance $q > 0$ from a stationary anchor is scaled by a rational $a > 0$ with launch speed $h_a = (a - 1)/(a + 1)$, $|h_a| < 1$. Relative to the entrance, the principal time-position pairs are

$$(q, q), \quad (2q, 0), \quad (2q + aq, aq), \quad (2q + 2aq, 0), \quad (268.2)$$

the duration is $2(1 + a)q$, and a stationary inner spectator at p is crossed at

$$p, \quad 2q - p, \quad 2q + p, \quad 2q + 2aq - p. \quad (268.3)$$

The exact condition is that q and aq stay strictly between the same unchanged nearest neighbours. The word has $4 + 4k$ events with k inner spectators, and $a = 1$ gives a legitimate timed identity. A global homothety by $\lambda < 1$ scales X, Y, R in that order, and one by $\lambda > 1$ scales R, Y, X . It has $4 + 8 + 12 = 24$ events, needs no new condition, has homogeneous return λI_3 and takes

$$T_{\text{global}} = 2(1 + \lambda)(x + y + D), \quad (268.4)$$

because each marker's position when its own scale starts is its original incoming position.

269 An eight event outer marker operation

Fix $v \in \mathbb{Q}_{>0}$ and set

$$h = \frac{v-1}{v+1}, \quad d = vD + (1-v)y = y + v(D-y), \quad t_* = (v+2)D - (v+1)y. \quad (269.1)$$

The messenger crosses X, Y transparently, launches R at speed h and reflects left, bounces at Y , restores R and reflects left, crosses Y, X , then bounces at L . Distinct phases distinguish a transparent Y crossing from its middle bounce.

Event	Time	Messenger position	Contact and action
1	x	x	Cross X
2	y	y	Cross Y
3	D	D	Launch R at h ; reflect
4	$2D - y$	y	Bounce at Y
5	t_*	d	Restore R ; reflect
6	$t_* + d - y$	y	Cross Y
7	$t_* + d - x$	x	Cross X
8	$t_* + d$	0	Bounce at L

The eight flight durations, including the initial outgoing flight, are

$$x, \quad y - x, \quad D - y, \quad D - y, \quad v(D - y), \quad v(D - y), \quad y - x, \quad x. \quad (269.2)$$

They are all positive exactly on the prescribed ordered section. The restoration identity and the moving R/Y gap at the middle bounce are

$$d - D = h(t_* - D), \quad R(2D - y) - y = \frac{2v(D - y)}{v + 1} > 0.$$

Both target endpoints D, d lie strictly beyond y , so monotonicity excludes every target-spectator contact. Between Y and R there is no unlisted spectator. All inner-marker crossings before and after that interval occur in the table. Relative messenger/target speeds are nonzero; after a contact the stipulated directions separate immediately. Thus there is neither an omitted contact nor a simultaneous remote event. Conversely, there is no additional guard in (269.2) once the initial order holds. The exact return and duration are

$$(D, x, y) \mapsto (d, x, y), \quad T_{\text{outer}} = 2[(1 + v)D - vy] = 2[D + v(D - y)]. \quad (269.3)$$

This includes $v = 1$, for which the temporary R label has speed zero and all eight events are retained.

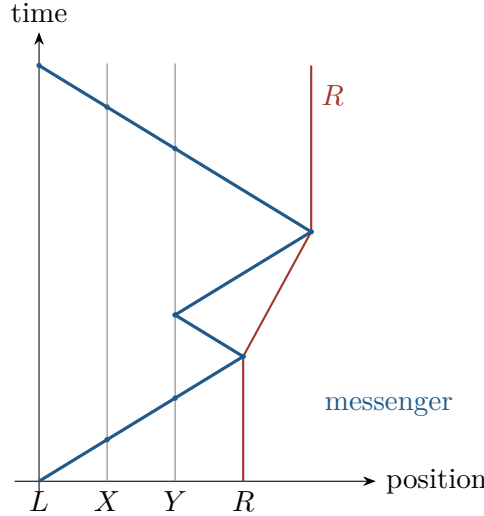


Figure 16: Analytic outer-operation illustration, from the exact table with $D = 1$, $x = 1/3$, $y = 2/3$, $v = 2$. The eight event points are drawn from affine formulas, not a collision simulation. Time increases upward.

270 Twenty event recentering

Put $s = (v + 2)/3$. After the outer operation, scale both x and y about L by s , keeping d fixed. For $s < 1$ use X then Y ; for $s > 1$ use Y then X ; at $s = 1$ retain both identity words in the first order. The two scales always use $4 + 8 = 12$ events and two fresh temporary marker labels.

For contraction, $sx < x < y < d$, and the later Y endpoints satisfy $sx < sy \leq y < d$. For expansion, Y moves first. Its only new obstruction is $sy \geq d$; if $sy < d$, its endpoints stay between x and d . The later X endpoints stay between 0 and sy . Consequently the exact raw chamber is

$$x > 0, \quad y - x > 0, \quad D - y > 0, \quad d - sy = vD + \frac{1 - 4v}{3}y > 0. \quad (270.1)$$

The fourth row is redundant for $v \leq 1$. For $v > 1$ it is essential: if it fails, the outward-moving Y reaches or crosses stationary R during its first scale. Equality gives simultaneous contact at the proposed restored endpoint; strict failure gives a prior extra contact. Thus (270.1) is exact, not just a safe sufficient subchamber.

Each scale starts and ends at L with outgoing messenger. The Y word is “cross X , launch Y , cross X , bounce L , cross X , restore Y , cross X , bounce L ,” at the then-current stationary X . Equations (268.2)–(268.3) prove strict ordering of every crossing. There is no zero-time phase transfer.

The resulting raw return and duration are

$$(D, x, y) \mapsto (d, sx, sy), \quad (270.2)$$

$$T_{\text{raw}} = 2[(1 + v)D - vy] + 2(1 + s)(x + y). \quad (270.3)$$

In centered coordinates its matrix is

$$N_v = \begin{pmatrix} s & 0 & 1 - v \\ 0 & s & (v - 1)/3 \\ 0 & 0 & v \end{pmatrix}. \quad (270.4)$$

At $x = D/3, y = 2D/3, d = sD$ and $d - sy = sD/3 > 0$. The endpoint proof also checks all internal scale positions. Thus this 20-event, three-temporary-label, four-supplied-row word is center-strict for every rational $v > 0$. Its determinant $vs^2 > 0$ agrees with even collision parity.

271 Scale shears and arbitrary fixed center returns

271.1 Compensation of the planar action

The lower-right block of (270.4) and its compensation are

$$A_v = \begin{pmatrix} s & (v-1)/3 \\ 0 & v \end{pmatrix}, \quad C_v = sA_v^{-1} = \begin{pmatrix} 1 & (1-v)/(3v) \\ 0 & s/v \end{pmatrix}.$$

Since $\det C_v = s/v > 0$, the positive inherited compiler applies. Follow the raw word by $F(C_v, 1/s)$, equivalently by $F(C_v, 1)$ and a global scale $1/s$. Direct multiplication gives

$$\text{diag}(1/s, A_v^{-1})N_v = E_{(0,k)} := \begin{pmatrix} 1 & 0 & k \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad k = \frac{3(1-v)}{v+2}. \quad (271.1)$$

This changes only the outer scale coordinate: $D \mapsto D + k\eta$. Its exact chamber includes the raw rows and *all* compiler rows pulled back by N_v . Keeping only the raw endpoint row after compensation would lose necessary chronology conditions. Because N_v sends the central ray to a positive multiple of itself, all these rows remain center-strict.

The parameter correspondence is a rational bijection

$$v > 0 \quad \longleftrightarrow \quad -3 < k < 3/2, \quad v = \frac{3-2k}{3+k}. \quad (271.2)$$

For arbitrary $k \in \mathbb{Q}$, let $n = \max(1, \lceil |k| \rceil)$ and $\delta = k/n$. Since $|\delta| \leq 1$, it lies strictly in this range. Concatenate n copies of $E_{(0,\delta)}$. Their coefficients add, yielding $E_{(0,k)}$. Every completed factor fixes the center exactly, so the finite pullback intersection remains center-strict. No large-coefficient empty-domain obstruction appears. A zero requested shear can be omitted when another nonempty block supplies the timed return.

271.2 One coefficient suffices for every scale row

For nonzero $r = (a, b)$, set

$$B = \begin{pmatrix} b & -a \\ a & b \end{pmatrix}, \quad \det B = a^2 + b^2 > 0. \quad (271.3)$$

Chronologically perform $F(B, 1)$, then $E_{(0,1)}$, then $F(B^{-1}, 1)$. Their product is

$$\text{diag}(1, B^{-1})E_{(0,1)}\text{diag}(1, B) = E_r := \begin{pmatrix} 1 & r \\ 0 & I_2 \end{pmatrix}. \quad (271.4)$$

Indeed the added row is $(0, 1)B = (a, b)$, while the planar product is $B^{-1}B = I_2$. Both physical conjugating compilers have positive determinant and exact center-strict chambers. This uses one

fixed coefficient-one shear, at the cost of two parameter-dependent planar compilers. For $r = 0$ omit this prefix.

For (267.1), first compile E_{b/s_0} , then $F(A/s_0, s_0)$. The product is exactly

$$[s_0 \operatorname{diag}(1, A/s_0)]E_{b/s_0} = \begin{pmatrix} s_0 & b \\ 0 & A \end{pmatrix}. \quad (271.5)$$

If $\det A < 0$, the last factor uses the inherited seven-event J extension. No additional live signal is introduced.

271.3 Exact chambers and first failure

Suppose a chronological list of blocks has full return matrices $M_1, \dots, M_q$ and exact guard matrices $G_1, \dots, G_q$. Its exact rows, in the initial coordinates, are

$$G_1 z > 0, \quad G_2 M_1 z > 0, \quad \dots, \quad G_q M_{q-1} \cdots M_1 z > 0. \quad (271.6)$$

The inherited endpoint lists are finite rational expressions, the new raw block uses (270.1), and global homotheties need no extra row. Thus this is an effective finite rational-arithmetic procedure, not a trajectory search.

Sufficiency follows by induction through the complete chronology of each block. For necessity, consider the earliest block whose guard fails, after a valid prefix. Its first-failure proof forces an extra collision, simultaneous contact, nonpositive flight or wrong order, so the selected complete word fails. Prospective affine formulas are invoked only until that first obstruction, never as a supposed physical continuation after it.

Every completed block sends the center to a positive multiple of itself, and every internal chamber is center-strict. Hence (271.6) contains an open neighborhood of the center on $D = 1$. The first block's section rows imply $0 < x < y < 1$, making this open rational polygon bounded. Redundant supplied rows are retained. The chamber is not replaced by its closure or by a smaller convenient neighborhood.

The ordered cone, in centered coordinates, is

$$C = \{D > 0, \quad D/3 + \xi > 0, \quad D/3 + \eta - \xi > 0, \quad D/3 - \eta > 0\}. \quad (271.7)$$

Endpoint compatibility requires $Q \subset C \cap N^{-1}C$. On normalized input put $h = s_0 + bw$ and $q = Aw$. The output rows are

$$h > 0, \quad q_1 + h/3 > 0, \quad q_2 - q_1 + h/3 > 0, \quad h/3 - q_2 > 0. \quad (271.8)$$

These are strict affine rows and are strict at $w = 0$. They imply the positive denominator in (267.2), but cannot replace intermediate collision guards. Together with the next phase argument, this completes the proof of Theorem 267.1.

271.4 Finite deterministic rules and literal closure

Give each incoming messenger flight a fresh label at its prescribed rational speed; emit the successor phase at its contact, identifying the final outgoing phase with the whole word's initial label. Give each launch and restoration its own temporary marker labels, including speed-zero identity scales. Section marker labels are shared by physical role.

All new raw and positive-compiler events involve the messenger. Their input sets are distinct because their phase labels are distinct. Their incoming and outgoing speeds are distinct: the messenger speed is ± 1 and every moving target speed has modulus less than one. The possible J suffix has its separately checked speeds and one marker-only collision. That collision must leave the messenger phase unchanged. Its fresh temporary fast- X label makes its input set unique, and interface phase copies prevent an accidental rule shared with the standalone J entry phase.

Assign identity output to every other finite input set whose speeds are pairwise distinct. There are finitely many labels and hence finitely many such sets. This is deterministic and cardinality-preserving. Every explicit event consumes and emits two signals; the completion preserves cardinality too. An extra collision remains an extra collision, even when its rule is identity, so completion does not enlarge the selected word chamber. Exactly five live signals persist throughout each finite portion. This establishes literal phase closure without asserting global injectivity of collision rules.

272 Explicit compilation counts

For the selected inherited $F(C, \lambda)$ write m, t, g for event, temporary-marker-label and supplied-guard counts. In the positive case its factorization has shear subdivision sum K , full-transfer count T , shear-block count $B_{\text{num}} \in \{3, 4\}$, determinant subdivision count H , and $\varepsilon = \mathbf{1}_{\lambda \neq 1}$. Section 278 specifies every parameter. Then

$$m(C, \lambda) = 18K + 3T + 14H + 24\varepsilon, \quad (272.1)$$

$$t(C, \lambda) = 4K + 3H + 3\varepsilon, \quad (272.2)$$

$$g(C, \lambda) = 4K + 4B_{\text{num}} + 8H. \quad (272.3)$$

The raw word always has $(m, t, g) = (20, 3, 4)$, retaining both identity inner scales at $v = 1$. The corrected shear therefore has counts

$$20 + m(C_v, 1/s), \quad 3 + t(C_v, 1/s), \quad 4 + g(C_v, 1/s).$$

All its events involve the messenger, so the phase-isolated number of meta-signals is $m + t + 4$. This counts label types, not simultaneous occurrences. No minimization is asserted.

272.1 A 780 event coefficient one ledger

For $k = 1$, equations (271.2) and (271.1) give

$$v = \frac{1}{4}, \quad s = \frac{3}{4}, \quad C_v = \begin{pmatrix} 1 & 1 \\ 0 & 3 \end{pmatrix}.$$

The positive compiler first factors

$$\text{diag}(1/3, 1)C_v = \begin{pmatrix} 1/3 & 1/3 \\ 0 & 3 \end{pmatrix} \in \text{SL}_2(\mathbb{Q}).$$

Its chronological shear list is $y(-2), x(1/3), y(6)$, with subdivision counts 8, 2, 24. Hence $K = 34$, $T = 4$, $B_{\text{num}} = 3$. The determinant factor 3 has $H = 8$. The exact ledger is

Operation	Events	Temporary labels	Supplied rows
Raw outer and recentering word	20	3	4
Signed shear steps	612	136	148
Four full transfers	12	0	0
Eight centered dilation steps	112	24	64
Global compensation 4/3	24	3	0
Total	780	166	216

The fixed-scale compensation uses $612 + 12 + 112 = 736$ events, and $780 + 166 + 4 = 950$ meta-signals suffice. For the main theorem with $b \neq 0$, put $r = b/s_0$ and use (271.3). An explicit total event count is

$$m(B, 1) + 780 + m(B^{-1}, 1) + m(A/s_0, s_0). \quad (272.4)$$

The analogous temporary-label and guard counts add with 166 and 216 for the middle kick. For $b = 0$ only the final nonempty compiler is used.

When the final planar determinant is negative, its selected positive component acquires seven events, three temporary X labels, one marker-only event and the four pulled-back supplied J guard rows. If r_J counts these marker-only events, phase-isolated messenger labels can number $m - r_J$; the total meta-signal count is $m - r_J + t + 4$. Here $r_J \leq 1$. The full determinant is $s_0 \det A$. Positive components have even length and a negative final component adds one odd J block, agreeing with $\operatorname{sgn} \det N = (-1)^m$.

Speeds. All parameters in the anchor, reflector, signed-shear and dilation formulas are rational, so every speed is rational. For the main construction all messenger speeds outside J are ± 1 and all target speeds there lie in $(-1, 1)$. The J extension adds messenger speeds $-3/2, -1/4$ and target speeds $-1/2, 2$. Repeated values on different labels are allowed. There is no uniform or optimal distinct-speed count claimed for arbitrary N . For the concrete 44-event family, Section 274 gives exactly six distinct speed values.

273 Repeated validity and exact clocks

Let $Q = \{z : Gz > 0\}$ be the exact cone of Theorem 267.1. Literal closure gives

$$\text{infinite complete-word validity} \iff GN^n z > 0 \text{ for every } n \geq 0. \quad (273.1)$$

Equivalently, the normalized survivor set is

$$K = \{w : f^n(w) \text{ is defined and belongs to } P \text{ for every } n \geq 0\}.$$

Every valid denominator is positive. On $D = 1$, (273.1) is an intersection of countably many strict rational affine half-planes in the initial w . Thus K is convex and contains the center. A countable intersection of open sets need not be open. No semialgebraicity classification is established here, and the fixed-scale planar classification cannot simply be applied to the nonlinear normalized map f .

Every prescribed flight time is a rational homogeneous linear form on the exact chamber; so is the full duration $T(z) = \ell z$ for a rational row ℓ . For this compiler there are constants $c_1, c_2 > 0$ with

$$c_1 D \leq \ell z \leq c_2 D \quad (z \in Q). \quad (273.2)$$

One can take $c_1 = 2$: if $b \neq 0$, the first $F(B, 1)$ already takes at least $2D$; if $b = 0$, the retained final planar compiler has that bound. For the upper bound, all finitely many intermediate times and positions are fixed linear forms in the initial coordinates. The closure of the normalized ordered input triangle is compact, so their absolute values have finite maxima. No shape denominator enters before normalization. This also bounds all positions during a valid macro by a constant times its entrance scale.

Theorem 273.1 (Active cyclic spectrum). *For an infinitely valid input z , put $z_n = N^n z$ and $D_n = (z_n)_0 > 0$. Then*

$$\text{Zeno} \iff \sum_{n \geq 0} D_n < \infty \iff N^n z \longrightarrow 0 \iff |\mu| < 1 \text{ for every } \mu \in \text{spec}(N|_{V_z}), \quad (273.3)$$

where $V_z = \text{span}\{z, Nz, N^2 z\}$, with dependent vectors removed to choose a basis. This space is invariant by Cayley–Hamilton.

Proof. The first equivalence is (273.2). Valid normalized marker positions are bounded, so each coordinate of z_n is $O(D_n)$; summability forces $z_n \rightarrow 0$. Conversely, decompose the cyclic subspace into its real or complex generalized eigenspaces. A component with eigenvalue μ contributes a polynomial in n times μ^n . Since z is cyclic in V_z , every eigenvalue of $N|_{V_z}$ is active. Terms from any $|\mu| \geq 1$ prevent convergence to zero; distinct spectral projections make cancellation between different generalized eigenspaces irrelevant. When all moduli are below one, polynomial-geometric decay is absolutely summable. This proves both remaining equivalences and the converse summability. $\square$

At the center the criterion reduces to $s_0 < 1$. It generally does not reduce to that scalar condition for every input.

If N, z are rational and the run is Zeno, V_z has a rational basis and $I - N$ restricted to it is invertible. Therefore the total accumulation time is rational and computable by a rational linear system:

$$T_{\text{total}} = \ell[(I - N)|_{V_z}]^{-1} z. \quad (273.4)$$

The inverse means an operator on V_z , embedded back into the full space. The full $I - N$ can be singular because of an inactive unit mode; its unrestricted inverse must not be used without checking. Since positions during macro n are $O(D_n)$, all five signals approach the fixed left anchor during a Zeno run. No continuation through that accumulation is specified.

[write] [The rationality of (273.4) is a special case of source 26’s Theorem 182.4, by source 30’s cyclic-subspace argument (Remark 244.1). The spectral equivalences of Theorem 273.1 are printed here once; source 33 restates them as Remark 288.1.]

274 A 44 event word with both clock behaviors

Fix rational $v > 1$. Append to the raw 20-event word the 24-event global contraction by $1/v$. Set

$$s = \frac{v+2}{3}, \quad r = \frac{s}{v} = \frac{v+2}{3v}, \quad a = \frac{v-1}{v}, \quad \frac{1}{3} < r < 1. \quad (274.1)$$

Its physical-coordinate return and exact one-word chamber are

$$D' = D - ay, \quad x' = rx, \quad y' = ry, \quad (274.2)$$

$$x > 0, \quad y - x > 0, \quad D - y > 0, \quad 3vD + (1 - 4v)y > 0. \quad (274.3)$$

The suffix introduces no additional guard. The last row is equivalent to $D' > y'$. This word has 44 messenger-involving events, six temporary marker labels, four supplied rows, 44 messenger phases and 54 meta-signals. The explicit labels and rules in Section 279 restore all section labels and phase Q_0 at event 44.

The six distinct speed values are

$$-1, \quad -h, \quad 0, \quad p, \quad h, \quad 1, \quad \text{where } h = \frac{v-1}{v+1}, \quad p = \frac{v-1}{v+5}. \quad (274.4)$$

Here $0 < p < h < 1$, so the listed six values are pairwise distinct. (The label count is larger because phases at equal speed must remain distinct.)

The exact duration is

$$\begin{aligned} T(z) = & 2[(1+v)D - vy] + 2(1+s)(x+y) \\ & + 2(1+1/v)[s(x+y) + vD + (1-v)y]. \end{aligned} \quad (274.5)$$

These summands are respectively the outer operation, the two inner scales, and the three global scales. In particular $T(z) > 2D$. A convenient bound on (274.3) is

$$T(z) < [2(1+v) + 4(1+s) + 2(1+1/v)(2s+v)]D. \quad (274.6)$$

Indeed $x, y < D$, and for $v > 1$ the last bracketed position sum is less than $(2s+v)D$.

274.1 Exact infinite validity including equality

Define $c = D - 3y/2$. Since $1 - r = 2a/3$, (274.2) leaves c invariant. Algebraic powers, before any physical first failure, are

$$x_n = r^n x, \quad y_n = r^n y, \quad D_n = c + \frac{3}{2}yr^n. \quad (274.7)$$

The extra one-word guard at macro n is equivalent to

$$D_{n+1} - y_{n+1} = c + \frac{1}{2}yr^{n+1} > 0. \quad (274.8)$$

If $c \geq 0$, this is strictly positive at every finite n . So are the section rows $x_n > 0$, $y_n - x_n > 0$ and $D_n - y_n = c + yr^n/2 > 0$. Induction through the exact one-word chamber proves that every macro is physically valid. If $c < 0$, (274.8) eventually becomes negative; hence a finite first guard failure occurs. This uses a prospective algebraic formula only to locate a failing checkpoint, and does not suppose that physical dynamics follows it after a wrong collision. Infinite validity is exactly $c \geq 0$, proving the first part of Theorem 267.2.

For $c = 0$, all section coordinates contract by r . Homogeneity of duration gives $T_n = r^n T_0$, hence

$$T_{\text{total}} = \frac{T_0}{1-r} < \infty. \quad (274.9)$$

This is rational on rational input. For $c > 0$, the outer phase alone is longer than $2D_n \geq 2c$ at every macro, so total time diverges. Both behaviors belong to the same finite machine and the same repeated word.

On $D = 1$ the infinite-validity set is $0 < x < y \leq 2/3$. Its accepted equality boundary $y = 2/3$ is entirely Zeno. The center $(1/3, 2/3)$ lies on that boundary, while strictly inside the one-word chamber. The return matrix in (274.2) has eigenvalues $1, r, r$. The invariant c detects its unit mode: equality uses only contracting modes, while strict validity uses the unit mode. Its determinant $r^2 > 0$ matches 44-event parity.

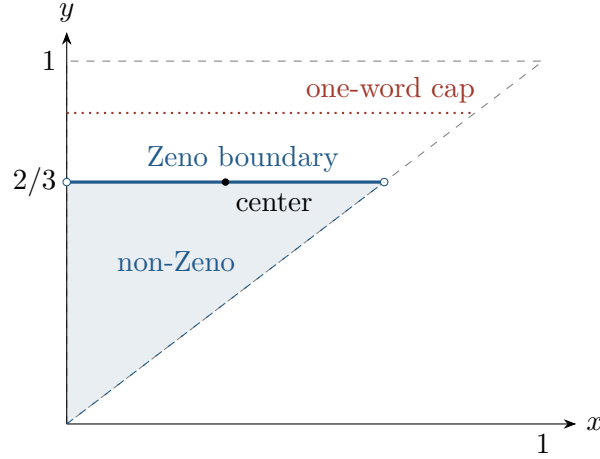


Figure 17: Exact normalized validity geometry. The interior $0 < x < y < 2/3$ is non-Zeno; the horizontal boundary $0 < x < y = 2/3$ is accepted and Zeno. The dashed order edges are excluded. The dotted one-word cap shown for $v = 2$ is $y < 6/7$, from the exact fourth guard, and is strictly above the repeated-validity boundary. This is an inequality diagram, not a simulation.

Two exact inputs in the same machine. For $v = 2$, $r = 2/3$ and direct simplification of (274.5) gives

$$T(D, x, y) = 12D + \frac{26}{3}x + \frac{5}{3}y.$$

At $(D, x, y) = (1, 1/3, 2/3)$, $c = 0$, $T_0 = 16$ and the accumulation time is 48. At $(1, 1/4, 1/2)$, $c = 1/4$ and the macro durations are $T_n = 3 + 12(2/3)^n$. The first n macros take $3n + 36(1 - (2/3)^n)$, which diverges. These values are direct evaluations of the displayed formulas.

274.2 Native positive integer gap certificates

Restrict $g_1 = x$, $g_2 = y - x$, $g_3 = D - y$ to $\mathbb{Z}_{>0}$. Positivity of these original inputs already supplies the full initial order. Since

$$c = g_3 - \frac{g_1 + g_2}{2},$$

the following single-polynomial equations give the promised exact predicates:

$$\text{infinite validity: } (2g_3 - g_1 - g_2 - w + 1)^2 = 0, \quad w \in \mathbb{Z}_{>0}, \quad (274.10)$$

$$\text{Zeno validity: } (2g_3 - g_1 - g_2)^2 = 0, \quad \text{no witnesses}, \quad (274.11)$$

$$\text{non-Zeno validity: } (2g_3 - g_1 - g_2 - w)^2 = 0, \quad w \in \mathbb{Z}_{>0}. \quad (274.12)$$

All have total degree two and use the native input gaps directly, without auxiliary aliases. A square of an integer vanishes precisely when its base does. Writing $L = 2g_3 - g_1 - g_2$ only as a displayed abbreviation, (274.10) uniquely forces $w = L + 1$, positive exactly for integer $L \geq 0$. Equation (274.12) uniquely forces $w = L$, positive exactly for $L > 0$. Equation (274.11) accepts exactly $L = 0$ and has one empty witness tuple on an accepted input, none on a rejected input. This proves the remaining part of Theorem 267.2.

The integer-domain qualification is essential: replacing positive integer witnesses by positive real or rational witnesses would make the first formula test $L > -1$, not $L \geq 0$, for arbitrary real or

rational gaps. These elementary certificates establish no broader arithmetic classification for general projective survivor sets.

[write] [They are the unique-slack certificates of Part I's Theorem 6.1 for one weak row, one equality and one strict row in L .]

275 The fixed ray boundary of the result

For an arbitrary rational homogeneous matrix M , a necessary map-level condition for an open positive ordered-section domain is

$$C \cap M^{-1}C \neq \emptyset.$$

The intersection is open; nonemptiness gives an open set of compatible input/output pairs, but no signal-machine construction. Theorem 267.1 provides a construction near its specified rational central eigenray, not a proof that this necessary condition suffices in general.

275.1 Rational fixed rays and changed encodings

Write $M = \begin{pmatrix} a & c \\ d & B \end{pmatrix}$ and suppose a rational finite point $p \in \mathbb{Q}^2$ obeys

$$M(1, p)^\top = \lambda(1, p)^\top, \quad \lambda > 0.$$

For $T = \begin{pmatrix} 1 & 0 \\ p & I_2 \end{pmatrix}$, direct multiplication gives

$$T^{-1}MT = \begin{pmatrix} \lambda & c \\ 0 & B - pc \end{pmatrix}. \quad (275.1)$$

If M is invertible, then $\det(B - pc) = \det M / \lambda \neq 0$. The fixed-center compiler therefore applies in this explicitly changed rational encoding. This algebra does not implement T or T^{-1} physically on the original marker coordinates. It must not be described as a literal original-section realization of M without an additional conversion construction.

An affine chart change preserving D cannot move a ray at infinity to a finite center. A general rational basis change can re-encode any rational eigenray, but then the normalization functional, sign and domain must be specified anew. If the ray is to retain the same positive physical marker meaning, it must lie inside C and have positive multiplier. A nonzero rational eigenvector has a rational eigenvalue, by dividing any nonzero coordinate of its image by that coordinate.

275.2 A positive compatible matrix without a rational eigenray

In positive gap coordinates consider

$$P_g = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 3 \end{pmatrix}, \quad \det P_g = 2, \quad \chi_{P_g}(t) = t^3 - 6t^2 + 8t - 2. \quad (275.2)$$

All entries are positive, so every positive gap vector maps to a positive one. A rational root of this monic cubic could only be $\pm 1, \pm 2$, and the respective values are 1, -17 , -2 , -50 , all nonzero. Thus the cubic has no rational root and the matrix has no rational eigenray.

The change to centered coordinates is

$$D = g_1 + g_2 + g_3, \quad \xi = (2g_1 - g_2 - g_3)/3, \quad \eta = (g_1 + g_2 - 2g_3)/3.$$

In these coordinates the same matrix is

$$M = \begin{pmatrix} 4 & -1 & -1 \\ -1/3 & 1/3 & 1/3 \\ -1/3 & -1/3 & 5/3 \end{pmatrix}. \quad (275.3)$$

It preserves C globally, yet no rational conjugacy can put it into a form fixing a rational ray. This is an explicit uncovered case of the present construction and rational-conjugacy argument, not an impossibility theorem for some other signal construction.

[write] [Answered by source 33 (Section 286). There the same matrix, written K in gap coordinates and N in centred coordinates, is realized on the original section, on some full-dimensional open set. The realization runs source 32's fixed-centre compiler for $N_0 = T_{-q}NT_p$ and then one 26-event physical translation; it needs no rational eigenray and no change of encoding. Source 32's sentence stays true: P_g is not covered by source 32's own construction and rational-conjugacy argument.]

276 Primary literature and attribution

Fixed-itinerary fractional-linear algebra predates this construction. Mestl, Lemay and Glass [94] derive wall maps $x \mapsto Cx/(1 + c^T x)$ and their cycle compositions in Glass networks, with exit-dependent domains. Their equations (2.3)–(2.4), printed page 35, and (4.2), printed page 40, give the closest algebraic precedent. Their flow and clock assumptions are different; they do not establish this five-live-signal realization.

Alishah, Duarte and Peixe [93], Section 5, give oblique constant-flow projections, their trajectory domains and inverses, and pulled-back itinerary domains. Their equations (5.2)–(5.5), Proposition 5.4 and Definition 5.7 situate the underlying section-map geometry. This is a polytope-flow analysis, not the finite signal compiler proved here.

Durand-Lose [4], Sections 3.1–3.2, uses scale-relative distances to represent registers and implements addition and constant multiplication by signals. Those arithmetic constructions have their own register and population resources. They cannot be cited as if they already implied the exact five-live-signal number-preserving full-section theorem.

Becker and coauthors [92], Section 2, Definitions 1–2, specify label-dependent constant speeds, deterministic rules and distinct-speed incoming and outgoing collision sets. Multiple labels may share a speed, and five live signals is not five meta-signals. The affine transformations treated in [6], Section 2.3, act on all speeds and on the common spatial coordinate; they do not implement arbitrary two-dimensional marker-shape conjugacies.

Belgacem, Edwards and Farcot [95], Section III, equations (6)–(12), explicitly track positive denominators and strict itinerary cones for fractional-linear Glass-network maps. This is a useful domain-audit precedent. No accompanying code was executed for this report.

The construction-specific content is the exact five-live-signal realization of the positive fixed-center lift, its complete chronology, exact chambers, literal phase closure and the mixed-clock example. No novelty, priority, exhaustive-search, general universality or minimality conclusion follows from the bounded source review.

277 Further questions

1. **Realization beyond a rational center.** Can a finite five-live-signal construction realize (275.3) in the original coordinates on a nonempty open chamber? A positive answer needs a new physical mechanism, not just rational fixed-ray conjugacy.
2. **Exact projective survivor geometry.** Classify the set $\{z : GN^n z > 0 \ (n \geq 0)\}$ on $D = 1$ for the present rational triangular lifts. Which such sets are semialgebraic, and which limiting strict boundaries are accepted? The mixed example shows why closure loses clock information.
3. **Efficient compilation.** Reduce the 780-event coefficient-one implementation or the size dependence of its planar conjugators, while preserving the same exact guard and literal phase accounting. No current count is a lower bound.
4. **Uniformity and speed resources.** Determine what can be shared among parameter-dependent machines, and what distinct-speed or label resources suffice under a precisely fixed encoding. A finite universal rule table is a separate claim requiring its own proof.
5. **Arithmetic descriptions on native gaps.** Extend the elementary unique-witness certificates beyond the half-space example, without hidden aliases, unrestricted auxiliary dimensions or a transfer from a different dynamical model.

Each question is open within this report's scope. None is answered by the standard full-coordinate dimension count, by collision number preservation, or by familiar projective algebra alone.

[write] [Status at this write: Question 1 is answered by source 33 (Theorem 281.1 and Section 286). The matrix (275.3) is realized in the original coordinates on a nonempty open chamber. Questions 2–5 are open. Source 33's escaping-gap matrix bears on Question 2: it is a compatible map with an empty survivor set, but it does not classify survivor sets.]

278 The precise inherited planar compiler (source 32, Appendix A)

Remark 278.1 ([write] Printed once, in sources 30 and 31). Source 32's Appendix A “states the dependency interface in an executable-free mathematical form”. It restates, without new content, the constructions of sources 30 and 31 that Theorem 267.1 uses, and they are printed there:

- the reflector homothety $H_z(v)$, the translation $T_z(e) = H_z(1 - e) \circ L_t(1/(1 - e))$ with its exact guard $0 < t, t + ez < s \leq z$, and the full anchor transfer: source 30, Section 241, equations (241.3) and (241.4);
- the signed centred shear blocks of $N_t = \max(1, \lceil 4|t| \rceil)$ micro-steps of 18 events each, with their $4N_t + 4$ exact rows and their centre-strictness: source 30, Section 242, equation (242.3);
- the factorization of $SL_2(\mathbb{Q})$ that covers every pivot, with its chronological block lists and transfer count T : equations (242.4)–(242.6);
- the centred dilation $\text{diag}(q, 1)$ with its eight exact rows, and its telescoping into H small factors: equations (242.7) and (242.8), with $B = \text{diag}(q^{-1}, 1)C$;
- the orientation-reversing block J , with its chamber $0 < x < y < D$, $y < 4x$, its speeds, its seven flight durations, the composite chamber $P_B \cap B^{-1}P_J$ and the fresh interface labels: source 31, Theorem 257.3 and Sections 261–262.3;

- the duration bounds $2D \leq T_F < (12K + T + 10H + 6(1 + \lambda)\varepsilon)D$ and the extra $(38\lambda/9)D$ of the reversal: (244.1) and (263.3);
- the invertibility theorem and the determinant formula $\det N = (-1)^m \prod_j s_j^+ / s_j^-$: source 31, Theorems 257.1 and 257.2. As source 32 repeats, neither the inverse flight formula nor number preservation establishes global machine reversibility.

Source 32's counts (272.1) and their companions are source 30's (242.9).

279 All 44 rules and their speeds (source 32, Appendix B)

The following table is the complete selected word for rational $v > 1$. Set $h = (v - 1)/(v + 1)$ and $p = (v - 1)/(v + 5)$. The stationary labels L, X, Y, R have speed zero. Temporary labels have speeds

$$R_o : h, \quad Y_i : p, \quad X_i : p, \quad X_g : -h, \quad Y_g : -h, \quad R_g : -h.$$

The outgoing phase column gives the speed of the messenger after each rule. The entrance label is Q_0 with speed +1, so the preceding row also determines every incoming phase speed. All 44 messenger phases $Q_0, \dots, Q_{43}$ are distinct. The final rule emits the original Q_0 ; all four markers have their stationary labels again. Braces denote collision input/output sets, not ordered particle identities.

Table 30: Explicit finite rule declaration for the mixed-clock word.

Event	Input	Output	Outgoing speed
1	$\{Q_0, X\}$	$\{Q_1, X\}$	+1
2	$\{Q_1, Y\}$	$\{Q_2, Y\}$	+1
3	$\{Q_2, R\}$	$\{Q_3, R_o\}$	-1
4	$\{Q_3, Y\}$	$\{Q_4, Y\}$	+1
5	$\{Q_4, R_o\}$	$\{Q_5, R\}$	-1
6	$\{Q_5, Y\}$	$\{Q_6, Y\}$	-1
7	$\{Q_6, X\}$	$\{Q_7, X\}$	-1
8	$\{Q_7, L\}$	$\{Q_8, L\}$	+1
9	$\{Q_8, X\}$	$\{Q_9, X\}$	+1
10	$\{Q_9, Y\}$	$\{Q_{10}, Y_i\}$	-1
11	$\{Q_{10}, X\}$	$\{Q_{11}, X\}$	-1
12	$\{Q_{11}, L\}$	$\{Q_{12}, L\}$	+1
13	$\{Q_{12}, X\}$	$\{Q_{13}, X\}$	+1
14	$\{Q_{13}, Y_i\}$	$\{Q_{14}, Y\}$	-1
15	$\{Q_{14}, X\}$	$\{Q_{15}, X\}$	-1
16	$\{Q_{15}, L\}$	$\{Q_{16}, L\}$	+1
17	$\{Q_{16}, X\}$	$\{Q_{17}, X_i\}$	-1
18	$\{Q_{17}, L\}$	$\{Q_{18}, L\}$	+1
19	$\{Q_{18}, X_i\}$	$\{Q_{19}, X\}$	-1
20	$\{Q_{19}, L\}$	$\{Q_{20}, L\}$	+1
21	$\{Q_{20}, X\}$	$\{Q_{21}, X_g\}$	-1
22	$\{Q_{21}, L\}$	$\{Q_{22}, L\}$	+1
23	$\{Q_{22}, X_g\}$	$\{Q_{23}, X\}$	-1
24	$\{Q_{23}, L\}$	$\{Q_{24}, L\}$	+1
25	$\{Q_{24}, X\}$	$\{Q_{25}, X\}$	+1

Continued on the next page

Event	Input	Output	Outgoing speed
26	$\{Q_{25}, Y\}$	$\{Q_{26}, Y_g\}$	-1
27	$\{Q_{26}, X\}$	$\{Q_{27}, X\}$	-1
28	$\{Q_{27}, L\}$	$\{Q_{28}, L\}$	+1
29	$\{Q_{28}, X\}$	$\{Q_{29}, X\}$	+1
30	$\{Q_{29}, Y_g\}$	$\{Q_{30}, Y\}$	-1
31	$\{Q_{30}, X\}$	$\{Q_{31}, X\}$	-1
32	$\{Q_{31}, L\}$	$\{Q_{32}, L\}$	+1
33	$\{Q_{32}, X\}$	$\{Q_{33}, X\}$	+1
34	$\{Q_{33}, Y\}$	$\{Q_{34}, Y\}$	+1
35	$\{Q_{34}, R\}$	$\{Q_{35}, R_g\}$	-1
36	$\{Q_{35}, Y\}$	$\{Q_{36}, Y\}$	-1
37	$\{Q_{36}, X\}$	$\{Q_{37}, X\}$	-1
38	$\{Q_{37}, L\}$	$\{Q_{38}, L\}$	+1
39	$\{Q_{38}, X\}$	$\{Q_{39}, X\}$	+1
40	$\{Q_{39}, Y\}$	$\{Q_{40}, Y\}$	+1
41	$\{Q_{40}, R_g\}$	$\{Q_{41}, R\}$	-1
42	$\{Q_{41}, Y\}$	$\{Q_{42}, Y\}$	-1
43	$\{Q_{42}, X\}$	$\{Q_{43}, X\}$	-1
44	$\{Q_{43}, L\}$	$\{Q_0, L\}$	+1

Events 1–8 are the outer operation. Events 9–16 scale Y by s , and events 17–20 scale X by s . This is the required expansion order for $v > 1$. Events 21–24, 25–32 and 33–44 contract X, Y, R respectively by $1/v$. The speed identities are $(s - 1)/(s + 1) = p$ and $((1/v) - 1)/((1/v) + 1) = -h$.

Every input set is distinct by its incoming phase. All incoming and outgoing pairs have distinct speeds because messenger speeds are ± 1 and target speeds lie strictly between them. At every event the outgoing directions separate. Complete the machine by identity output on every other finite input set with pairwise distinct speeds. The declaration is deterministic and number-preserving; its selected complete word has the exact chamber proved in Section 274. This static table is not an executed collision schedule.

280 Evidence and reproducibility (source 32, Appendix C)

280.1 Source binding and independent review

The new proof is bound to the frozen packet `projective-signal-shears62-20261004`. The SHA-256 of its complete main proof is

```
502e90e091eabc0a6c8eb6092e73f4407
ae84f143d337be8a4a633a3f85a7c47
```

The supplied source-evidence package preserves every frozen file, including the author's static checker, inert. The positive planar dependency is bound to SHA-256

```
e0ddd64cdbbb5c7f448266ab58f1dfef2
868892ac0f230632b2337e4d0bed065
```

and the invertibility/negative-determinant dependency to

```
dc97e30c56817370138d6be760d44a5d
f031743a0ee1011cca39381b67bb0cec
```

The line breaks are typographical; each pin is one continuous 64-character hexadecimal digest.

The original evidence records 36 named static checks, including 34 symbolic equalities and two finite check groups, plus rational parameter fixtures. Its elementary polynomial arithmetic supports the direct proof; it is not itself a proof-assistant formalization. The original checker was not re-executed for this report.

A separate fresh independent audit accepted the frozen mathematics without a required correction. It inspected the full proof, both dependencies, all 44 rules, the certificates and the original checker as text. Its freshly authored, inspected exact algebra/static-rule checker passed 157 named algebraic, structural and authentication checks. That count includes 44 literal-rule comparisons and 44 outgoing-phase-speed comparisons; it is not a count of independent mathematical theorems. Its independent review reconstructs the generic chronology, exact guards, compensation, counts, active-spectrum clock argument and mixed-clock boundary. Finite fixtures do not replace those all-parameter proofs.

The audit recorded all 17 source objects, including directories, before and after its work; bytes, object set, modes and nanosecond modification times were unchanged. Access times were not claimed invariant. The independent audit checker uses SymPy for rational-function identities and matrix algebra and reads rules as inert JSON; it neither advances physical state nor chooses a next collision.

280.2 What the package makes reproducible

The PDF, standalone LaTeX and equivalent modular LaTeX are supplied with the unchanged science and audit evidence. The fresh release tools authenticate the full frozen input inventories, enforce path and output safeguards, and pin the build's executable and TeX dependencies. A locked build checks its dependencies before execution and checks equality afterward, disables shell escape, renders every page and verifies the expected PNG count and image integrity. The complete release inventory and deterministic archive bind the final delivered bytes.

[write] [Shipped here: the frozen proof with its README, review guide, literature notes and independent audit, the reviews and the replay notes as `32-projective-*.md`; the programs and one patch as `code/32-projective-*`; evidence, receipts and pins as `data/32-projective-*`. The dependency proofs are shipped once, under the prefixes `30-planar-` and `31-invertible-`. The PDF, the LaTeX, the page renders and the build transcripts are not shipped.]

The independent portable replay re-runs only the authenticated independent checker in a fresh external workspace. Frozen author programs and prior constructors remain inert. It supports read-only relocated inputs and checks deterministic evidence against the sealed reference. The accompanying README and review define the exact command, safeguards and trusted-runtime limitations; isolation flags do not constitute an operating-system or network sandbox.

No author scientific checker, upstream mathematical program, prior constructor, physical simulator, saved collision schedule or Lean program was executed to produce this manuscript. Both figures use displayed analytic formulas only. The final manuscript and tool reviews are separate from the frozen mathematical review; the package retains their receipts rather than silently folding later work into an earlier source.

280.3 Reading order

For the central proof, read Sections [269–271](#), then the guard and phase argument in Section [271.3](#). Section [273](#) proves the general conditional clock criterion, and Section [274](#) gives the fully explicit

mixed behavior and certificates. Section 278 specifies the inherited compiler assumptions and guard production. Section 279 records every event rule. Section 275 states exactly why the general rational homogeneous problem remains open here.

Source 33 (batch-91 manuscript 15): Exact local homogeneous realization with five live signals

Physical translations and a complete compatibility criterion, “Research Report 63”, 4 October 2026. [write] From the archive `Five_signal_homogeneous_realization_source_and_evidence.zip` (main file `Report63/Report63.tex`, 18-page PDF); its labels carry the prefix `smc:hl:`. Its sections are Sections 281–292 here; the last two are its Appendices A and B. Its three bibliography entries are this report’s [92], [93] and [4]. Its abstract follows.

A rational 3×3 homogeneous matrix is realized by a positive-duration literal return of exactly five live signals on some full-dimensional open part of the standard four-marker section if and only if it is nonsingular and sends at least one ordered input to an ordered output. The sufficient construction physically translates any rational interior shape to any other, using 26-event steps with exact intermediate-corner guards. Left and right translations reduce an arbitrary compatible matrix to the preserved fixed-center compiler without requiring a rational eigenray. The full return realizes the specified homogeneous representative, including scale. We prove the ten-event translation chronology and its exact guard, give a rational polynomial-time compatibility decision, and work out a positive-gap matrix with no rational eigenray. Local realizability has an explicit degree-six Diophantine representation with eight positive-integer witnesses and literal native gap-input arithmetic DAGs of 48 and 57 binary operations. A conditional cyclic-subspace clock criterion is separated from local existence; a compatible escaping-gap map has no infinitely valid input. The claims are conventional proofs relative to stated preserved dependencies, supported by static arithmetic evidence, with no universality, optimality, or invariant-chamber conclusion.

281 The exact local criterion

Consider four stationary markers

$$L = 0, \quad X = x, \quad Y = y, \quad R = D, \quad 0 < x < y < D,$$

and a messenger at L outgoing to the right with speed $+1$. There are exactly five live signals, counting the two outgoing strands at the initial L /messenger contact. Translation of the entire configuration has been quotiented out, but scale remains a coordinate. Write

$$z = (D, \xi, \eta)^\top, \quad \xi = x - D/3, \quad \eta = y - 2D/3. \quad (281.1)$$

The ordered cone and its normalized shape triangle are

$$\mathcal{C} = \{z : D/3 + \xi > 0, \ D/3 + \eta - \xi > 0, \ D/3 - \eta > 0\}, \quad (281.2)$$

$$\Omega = \{w \in \mathbb{R}^2 : (1, w)^\top \in \mathcal{C}\}. \quad (281.3)$$

Adding the three inequalities proves $D > 0$. In particular $0 \in \Omega$.

Theorem 281.1 (Full homogeneous local realization). *For $N \in \mathbb{Q}^{3 \times 3}$ the following conditions are equivalent.*

1. *A finite deterministic rational-speed, number-preserving signal machine has a nonempty positive-duration complete binary word that returns literally to the stated section and acts by $z \mapsto Nz$ on a nonempty full-dimensional open set.*

2. $\det N \neq 0$ and $\mathcal{C} \cap N^{-1}\mathcal{C} \neq \emptyset$.

Incoming and outgoing speeds at each prescribed binary collision are distinct. Under condition 2, a machine, word, and finite list of strict rational homogeneous guard rows $Gz > 0$ can be constructed effectively. Its exact chamber

$$\mathcal{Q} = \{z : Gz > 0\}$$

contains a rational ray $(1, p)^\top$ with $p \in \Omega$. On $D = 1$ it is a bounded nonempty open rational polygon. The return restores all stationary labels and the original outgoing messenger phase, and realizes N itself, not only its projective class.

The theorem is a local one-pass statement. It does not require the chosen word to work on every compatible input, and it does not prescribe its chamber or make that chamber invariant. In particular it need not supply any infinitely valid input. The matrix-dependent finite rule table and word need not have polynomial description size. Global reversibility, continuation after accumulation, Turing completeness, novelty, and optimal counts are not claimed.

281.1 The model and inherited interfaces

A meta-signal is a label carrying a fixed rational velocity. The finite rule table assigns an outgoing set to each admissible incoming set, with pairwise distinct incoming speeds and pairwise distinct outgoing speeds. Equal velocities on labels used in different phases are allowed. A complete binary word lists every contact, including messenger crossings of a stationary marker; it may not silently discard a contact. All valid words here preserve the population at five.

Two preserved results are used with explicit scope.

1. The fixed-center compiler from Report 62 realizes every rational matrix

$$M = \begin{pmatrix} s & b \\ 0 & A \end{pmatrix}, \quad s > 0, \quad b \in \mathbb{Q}^{1 \times 2}, \quad A \in \text{GL}_2(\mathbb{Q}), \quad (281.4)$$

on an exact chamber whose complete homogeneous guard list is strict at $e_0 = (1, 0, 0)^\top$. Its positive-duration complete word is nonempty, literal on the present section, and supports fresh phase copies at interfaces. The orientation-reversing case $\det A < 0$ is included. We retain its full guard list, not merely endpoint inequalities.

2. The fixed-word full-section theorem from Report 61 makes every fixed complete binary return invertible in the three translation-quotiented positional coordinates. In the present literal section chart, an m -event return obeys

$$\text{sgn}(\det N) = (-1)^m. \quad (281.5)$$

The relevant proof and accepted fixed-center review are copied unchanged into the source archive. The translation chronology below is re-established directly, using the nearest-target geometry of the preserved positive-planar proof. No earlier executable constructor or saved trajectory is needed.

[write] [Report 62 is source 32 (Theorem 267.1) and Report 61 is source 31 (Theorems 257.1 and 257.2). Their copies are shipped once, as 32-projective-science-frozen-proof-PROOF.md, 32-projective-audits-scientific-REVIEW.md and the proofs of sources 30 and 31. The ten-event primitive of the next section is source 30's translation $T_z(e)$ with reflector $z = D$ and one spectator (equation (241.4)). Source 33 proves it again with all ten contacts listed, and it is kept as a marked second presentation.]

282 A physical nearest-target translation

Use distance coordinates from a stationary anchor: a nearest target at t , a stationary spectator at s , and the far stationary reflector at D . Thus these are precisely the four markers, with one spectator between target and reflector. Fix a rational parameter $e < 1$, and define

$$u = \frac{t}{1-e}, \quad t' = t + eD, \quad h = \frac{e}{2-e}. \quad (282.1)$$

Then $-1 < h < 1$. The proposed word consists of a homothety about the anchor by $1/(1-e)$ followed by a homothety about the far reflector by $1-e$. Both use fresh temporary target labels, even when $e = 0$.

Lemma 282.1 (Exact ten-event primitive). *The word described below takes the target from t to $t + eD$, fixes the other markers, and returns the messenger to the anchor pointing outward. Its exact guard is*

$$0 < t < s < D, \quad 0 < t' < s. \quad (282.2)$$

Every collision is binary with distinct incoming and outgoing speeds. The duration is

$$2 \left(t + \frac{t}{1-e} \right) + 2D, \quad (282.3)$$

strictly between $2D$ and $6D$ on this chamber.

282.1 All ten contacts and their chronology

The first four event time/position pairs, starting with the messenger at the anchor pointing outward, are

$$(t, t), \quad (2t, 0), \quad (2t + u, u), \quad (2t + 2u, 0). \quad (282.4)$$

At event 1 the target is launched with distance velocity h and the messenger reverses. Event 2 bounces the messenger at the anchor. Event 3 restores the target and reverses the messenger; event 4 bounces at the anchor. The target restoration equation is

$$t + h(t + u) = u.$$

No spectator lies between the target and its anchor.

Set $\alpha = 2t + 2u$. The remaining six event pairs are

$$\begin{aligned} &(\alpha + u, u), \quad (\alpha + s, s), \quad (\alpha + D, D), \\ &(\alpha + 2D - s, s), \quad (\alpha + 2D - t', t'), \quad (\alpha + 2D, 0). \end{aligned} \quad (282.5)$$

At event 5, launch the target with velocity h while the messenger continues outward. Cross the spectator at event 6, bounce at the reflector at event 7, cross the spectator inward at event 8, restore the target while continuing inward at event 9, and bounce at the anchor at event 10. Here

$$u + h((2D - t') - u) = t'$$

is the restoration identity. Figure 18 shows both moving target phases and every messenger contact.

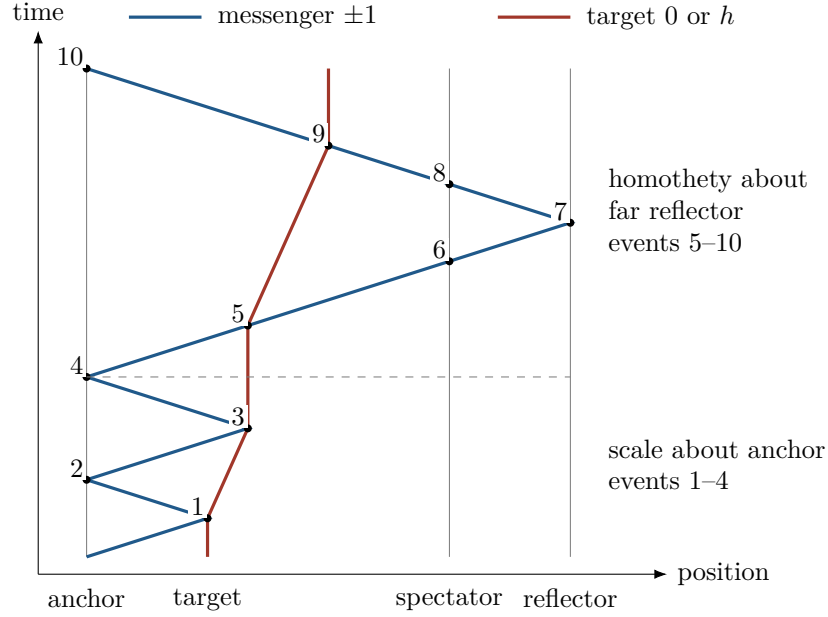


Figure 18: Physical ten-event translation, drawn from the exact formulas with $D = 4$, $t = 1$, $s = 3$, $e = 1/4$, hence $u = 4/3$, $t' = 2$, $h = 1/7$, and total duration $38/3$. Time increases upward. The dashed line is the four/six-event interface, not an extra collision. The red target remains strictly between anchor and spectator. Event numbers refer to the complete word; both spectator crossings are counted. This is an analytic illustration, not a simulated trajectory.

282.2 Why endpoint guards are exact for this primitive

The internal restored target position u lies between t and t' because

$$u - t = \frac{et}{1 - e}, \quad t' - u = \frac{e(D - t')}{1 - e}. \quad (282.6)$$

Both differences have the sign of e under (282.2); at $e = 0$ both vanish. Thus $0 < u < s$. The first four flight lengths are t, t, u, u . The last six are

$$u, \quad s - u, \quad D - s, \quad D - s, \quad s - t', \quad t'. \quad (282.7)$$

Every flight is strictly positive. During each moving phase the target moves monotonically between two points of $(0, s)$, so it cannot collide with anchor, spectator, or reflector. The messenger has speed ± 1 , while the target has speed $h \in (-1, 1)$; prescribed departures and returns are transverse. The two spectator contacts listed above exhaust all such contacts. There is no other moving marker. This proves complete chronology, not only correct endpoints.

Conversely, start with a valid ordered section and consider the first failed required endpoint. If an endpoint leaves $(0, s)$, either the continuously moving target reaches an adjacent stationary marker or a prescribed event interval collapses or reorders, no later than the proposed restoration. If the first scale is obstructed, the argument stops there; otherwise it applies to the second homothety. It never continues a prospective trajectory through an earlier wrong collision. This first-failure argument proves necessity as well as sufficiency of (282.2).

Summing the flights gives (282.3). Since $0 < t, u < D$, the duration is strictly between $2D$ and $6D$. At $e = 0$ the word is still a genuine ten-event timed identity. Temporary labels of speed zero

are legitimate: their collision partners have speed ± 1 . Reflection about R negates physical velocities and preserves all distance-coordinate times, inequalities, and counts. We denote the primitive by $T_D(e)$ with its chosen target understood.

283 Translation between arbitrary rational interior shapes

For $a \in \mathbb{Q}^2$ define the homogeneous shape translation

$$T_a = \begin{pmatrix} 1 & 0 & 0 \\ a_1 & 1 & 0 \\ a_2 & 0 & 1 \end{pmatrix}. \quad (283.1)$$

It fixes D and acts by $w \mapsto w + a$. We now implement it with the same five signals and original marker roles.

Define the normalized gap forms

$$\gamma_1(w) = \frac{1}{3} + w_1, \quad \gamma_2(w) = \frac{1}{3} + w_2 - w_1, \quad \gamma_3(w) = \frac{1}{3} - w_2. \quad (283.2)$$

For $p, q \in \Omega \cap \mathbb{Q}^2$ let

$$\begin{aligned} m &= \min\{\gamma_i(p), \gamma_i(q) : 1 \leq i \leq 3\} > 0, \\ \Delta &= q - p, \quad M = \max(|\Delta_1|, |\Delta_2|), \quad n = \max\{1, \lceil 2M/m \rceil\}, \quad \delta = \Delta/n. \end{aligned} \quad (283.3)$$

Every point on the segment from p to q has gaps at least m , because the γ_i are affine. Also $m \leq 1/3$ and $|\delta_j| \leq m/2 \leq 1/6$, so both primitive parameters δ_1 and $-\delta_2$ are less than one.

Lemma 283.1 (Literal physical shape translation). *For every rational $p, q \in \Omega$, a complete word realizes T_{q-p} on an exact rational chamber containing $(1, p)^\top$. The selected construction has $26n$ events and $4n$ temporary target labels, with n as in (283.3).*

Proof. For $k = 0, \dots, n-1$ execute these four blocks.

1. At L , apply $T_D(\delta_1)$ to nearest target X . This uses ten events and changes x to $x + \delta_1 D$.
2. Transfer $L \rightarrow R$: cross X , cross Y , and reflect at R . These are three events of total duration D .
3. At R , apply reflected $T_D(-\delta_2)$ to nearest target Y . Its distance $D - y$ decreases by $\delta_2 D$, so y increases by $\delta_2 D$. This uses ten events.
4. Transfer $R \rightarrow L$: cross Y , cross X , and reflect at L . These are three events of total duration D .

At nominal input p , the completed-step shapes are $p + k\delta$. The X-first intermediate shape is

$$c_k = p + k\delta + (\delta_1, 0).$$

Its gaps differ from those at $p + k\delta$ by $\delta_1, -\delta_1, 0$; hence all are at least $m/2 > 0$. Thus both translations have strictly ordered initial and final endpoints. Lemma 282.1 supplies their entire chronology, including their hidden scale endpoints. Transfers need only the retained section ordering. Every nominal contact is legitimate. Each step restores all stationary labels and the outgoing messenger at L , and acts by T_δ . The product is $T_\delta^n = T_\Delta$. $\square$

283.1 The exact chamber includes the intermediate corners

Set

$$H = \begin{pmatrix} 1/3 & 1 & 0 \\ 1/3 & -1 & 1 \\ 1/3 & 0 & -1 \end{pmatrix}, \quad Hz = (x, y - x, D - y)^\top. \quad (283.4)$$

An exact guard list for this translation is

$$\begin{aligned} HT_{k\delta}z &> 0 & (k = 0, \dots, n), \\ HT_{((k+1)\delta_1, k\delta_2)}z &> 0 & (k = 0, \dots, n-1). \end{aligned} \quad (283.5)$$

There are $6n + 3$ supplied scalar rows, allowing redundancy: three rows at each of $2n + 1$ checkpoints. Necessity and sufficiency follow from the exact primitive guards, applied to the first invalid checkpoint after a valid prefix. Merely testing the original and final endpoints would not justify discarding intermediate X-first corners. All supplied rows are strict at $(1, p)^\top$, so the finite intersection contains a full-dimensional open neighborhood of that ray.

Each step uses $10 + 3 + 10 + 3 = 26$ events and four temporary marker labels. With a fresh messenger phase at each event and the final phase closed to the initial one, a standalone translation uses $26n$ messenger labels, $4n$ temporary target labels, and four stationary labels: $30n + 4$ meta-signals. These are construction counts, not minima. For $p = q$ retain $n = 1$, $\delta = 0$ to obtain a nonempty 26-event identity. Such an identity may be omitted only if another nonempty macro supplies the timed return.

The duration of a step starting at (D, x, y) is

$$6D + 2 \left(1 + \frac{1}{1 - \delta_1}\right) x + 2 \left(1 + \frac{1}{1 + \delta_2}\right) (D - y). \quad (283.6)$$

The X-first operation does not change the second target's incoming distance $D - y$. Both scale parts are positive and less than $4D$, giving

$$6D < T_{\text{step}} < 14D, \quad 6nD < T_{\text{translation}} < 14nD.$$

The latter uses that D is unchanged during the translation.

284 Left and right normalization proves sufficiency

Assume $\det N \neq 0$ and $\mathcal{C} \cap N^{-1}\mathcal{C} \neq \emptyset$. This intersection is open and contains a rational point. Normalize its positive first coordinate to obtain

$$e_p = (1, p)^\top, \quad p \in \Omega \cap \mathbb{Q}^2.$$

Write

$$Ne_p = s(1, q)^\top, \quad s > 0, \quad q \in \Omega \cap \mathbb{Q}^2. \quad (284.1)$$

Set

$$N_0 = T_{-q}NT_p. \quad (284.2)$$

Then $N_0e_0 = se_0$, so

$$N_0 = \begin{pmatrix} s & b \\ 0 & A_0 \end{pmatrix}, \quad \det A_0 = \det N/s \neq 0. \quad (284.3)$$

More explicitly, for $N = \begin{pmatrix} a & b \\ c & B \end{pmatrix}$,

$$s = a + bp, \quad q = (c + Bp)/s, \quad A_0 = B - qb.$$

All quantities are rational. There is no eigenray requirement: p and q are generally different.

Execute in chronological order the physical translation $p \rightarrow 0$, the inherited fixed-center compiler for N_0 , and the physical translation $0 \rightarrow q$. Their actual matrix product is

$$T_q N_0 T_{-p} = N. \quad (284.4)$$

Each factor is a literal five-signal complete word on the original section. Thus the algebra is implemented physically, rather than by relabeling an encoding.

284.1 Pulling back every exact guard

Let G_- , G_0 , and G_+ be the exact guard matrices of the three words. Their composite chamber is exactly

$$G_- z > 0, \quad G_0 T_{-p} z > 0, \quad G_+ N_0 T_{-p} z > 0. \quad (284.5)$$

The first list is strict at e_p by Lemma 283.1. The middle word starts there at e_0 , and every one of its guards is strict. The last word starts at se_0 ; its guards are homogeneous and strict at e_0 , hence at se_0 . Every row in (284.5) is consequently strict at e_p .

Sufficiency follows by completing the three exact words. For necessity, take the first word whose exact guard fails after a successful prefix; no prospective formula is used past that failure. The first input ordering bounds the normalized chamber by Ω , so the $D = 1$ section is a bounded nonempty open rational polygon containing p . Final order implies

$$\mathcal{Q} \subseteq \mathcal{C} \cap N^{-1}\mathcal{C}.$$

Equality is not asserted and is not required.

284.2 Deterministic phase closure and the population count

Give each messenger-involving event a fresh incoming messenger phase with its prescribed rational speed, and output the next phase. Identify only the last outgoing phase of the full composite with the first. Temporary target labels are fresh at every launch/restoration, including speed-zero identities. Stationary marker labels retain their roles. Thus interfaces never merge unrelated standalone messenger rules and do not require instantaneous phase-switch events.

Every new translation contact has a messenger of speed ± 1 and a target of speed in $(-1, 1)$ or a stationary partner. Both incoming and outgoing pairs have distinct speeds. The optional seven-event orientation-reversing middle component has one marker-only event. Its messenger phase is unchanged there, and a fresh copy of its temporary marker label isolates that input set, as in the inherited compiler.

Each prescribed rule consumes two signals and emits two. Complete the table by identity rules on all other admissible finite label sets with pairwise distinct speeds. There are finitely many such sets. This gives a finite deterministic number-preserving machine, but an unlisted contact still does not belong to the selected complete word. The live population remains exactly five throughout valid finite execution; global injectivity of the rule map is not claimed.

If the physical translations use n_- , n_+ steps and the middle compiler uses m_0 events, t_0 temporary marker labels, and g_0 guard rows, then

$$\begin{aligned} m &= 26(n_- + n_+) + m_0, \\ t &= 4(n_- + n_+) + t_0, \\ g &= (6n_- + 3) + g_0 + (6n_+ + 3). \end{aligned} \tag{284.6}$$

With r marker-only events, fresh labels total $m - r + t + 4$. The chosen inherited compiler has $r \leq 1$. Both translations have determinant one and even length, so the determinant parity (281.5) is inherited from the middle component.

284.3 Necessity completes the criterion

Suppose a realization agrees with N on a nonempty full-dimensional open chamber. The full-section fixed-word theorem makes its ambient homogeneous positional map invertible. Agreement on an open set identifies that ambient linear map with N , hence $\det N \neq 0$. Both an input in the chamber and its output are ordered, so $\mathcal{C} \cap N^{-1}\mathcal{C} \neq \emptyset$. This proves the converse and completes Theorem 281.1.

285 An exact rational compatibility decision

The inverse of (283.4) is

$$H^{-1} = \begin{pmatrix} 1 & 1 & 1 \\ 2/3 & -1/3 & -1/3 \\ 1/3 & 1/3 & -2/3 \end{pmatrix}. \tag{285.1}$$

Set $K = HNH^{-1}$. Cone compatibility is equivalent to the existence of g with $g > 0$ and $Kg > 0$. By positive scaling, impose $\mathbf{1}^\top g = 1$. Solve the rational linear program

$$\begin{aligned} &\text{maximize} && \varepsilon \\ &\text{subject to} && \mathbf{1}^\top g = 1, \\ & && g_i \geq \varepsilon, \quad (Kg)_i \geq \varepsilon \quad (i = 1, 2, 3), \\ & && 0 \leq \varepsilon \leq 1. \end{aligned} \tag{285.2}$$

A strictly positive g gives a positive feasible ε after normalization, and any feasible $\varepsilon > 0$ supplies a strict witness. Thus compatibility holds exactly when the program is feasible with positive optimum. Infeasibility or optimum zero means incompatibility. Its feasible set, when nonempty, is compact: g belongs to the closed standard simplex and $\varepsilon \in [0, 1]$.

No numerical tolerance is required. There are four variables, one equality, and eight inequalities. Enumerate the at most

$$\binom{8}{3} = 56$$

triples of inequality boundaries. Solve each together with the equality whenever the four rows have full rank, reject candidates violating any weak inequality, and select the greatest ε . Every nonempty compact polytope has a vertex, including lower-dimensional feasible sets, so this finds the exact optimum or proves infeasibility. Exact Gaussian elimination or Cramer's rule uses only a fixed

number of rational operations in fixed dimension and has polynomial bit complexity in the rational input length. Computing $\det N$ is polynomial as well.

A positive rational optimum provides all data for the compiler:

$$e_p = H^{-1}g, \quad s = \mathbf{1}^\top K g > 0, \quad q = \frac{((H^{-1}Kg)_2, (H^{-1}Kg)_3)}{s}. \quad (285.3)$$

The first coordinate of $H^{-1}g$ is one, so the remaining coordinates give p .

This proves a polynomial-time endpoint-feasibility and local-realizability *decision*. It does not bound the size of the output collision word polynomially. The translation count can grow like the reciprocal interior margin, and the inherited compiler can have large subdivision counts. Small binary matrix descriptions can produce very large explicit words in this construction. Decidability is not a Turing-universality assertion.

286 A compatible matrix with no rational eigenray

Consider the strictly positive integer gap matrix

$$K = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 3 \end{pmatrix}, \quad N = H^{-1}KH = \begin{pmatrix} 4 & -1 & -1 \\ -1/3 & 1/3 & 1/3 \\ -1/3 & -1/3 & 5/3 \end{pmatrix}. \quad (286.1)$$

It has $\det K = \det N = 2$ and maps every strictly positive gap vector to a strictly positive vector. Its characteristic polynomial is

$$t^3 - 6t^2 + 8t - 2.$$

None of the possible rational roots $\pm 1, \pm 2$ is a root. A rational matrix acting on a nonzero rational eigenvector would have a rational eigenvalue, so neither K nor N has a rational eigenray.

Nevertheless choose $p = 0$. Then

$$Ne_0 = (4, -1/3, -1/3)^\top, \quad s = 4, \quad q = (-1/12, -1/12),$$

and

$$N_0 = \begin{pmatrix} 4 & -1 & -1 \\ 0 & 1/4 & 1/4 \\ 0 & -5/12 & 19/12 \end{pmatrix}. \quad (286.2)$$

Its lower block has determinant $1/2$, so the inherited fixed-center compiler applies. For the suffix $0 \rightarrow q$, the endpoint gap margin is $m = 1/4$ and $M = 1/12$, giving $n = 1$. Thus one 26-event physical suffix after the N_0 compiler realizes N exactly. The identity prefix may be omitted because the middle compiler is nonempty, or retained for uniform bookkeeping.

At the nominal input $D = 1$, the middle word finishes with $D = 4$, $x = 4/3$, $y = 8/3$. The suffix first moves X to 1, then Y to $7/3$, retaining $R = 4$. Every intermediate order is strict. Final normalized gaps are

$$(1/4, 1/3, 5/12).$$

This example closes the rational-eigenray gap without changing the section or the specified homogeneous representative.

[write] [This is the matrix P_g of source 32's Section 275, equations (275.2) and (275.3), which source 32 left uncovered; the example answers source 32's Question 1.]

287 A degree six positive integer certificate

This section represents local physical realizability, not infinite repetition. Let $K = (k_{ij})$ be an *integer* matrix in gap coordinates, and let $q > 0$ be a fixed or externally supplied integer denominator. The gap return is K/q , and the corresponding centered return is $H^{-1}(K/q)H$. Positive common scaling affects neither nonsingularity nor positive-cone compatibility.

Use eight positive-integer witnesses

$$g_1, g_2, g_3, h_1, h_2, h_3, b, d$$

and the single integer polynomial

$$F(K; g, h, b, d) = \sum_{i=1}^3 ((Kg)_i - h_i)^2 + [\det K - (2b - 3)d]^2 + [(b - 1)(b - 2)]^2. \quad (287.1)$$

Corollary 287.1. *Equation $F = 0$ has a positive-integer witness tuple if and only if the represented rational return is locally physically realizable on the standard five-signal section.*

Proof. A sum of nonnegative integer squares vanishes exactly when every residual vanishes. Therefore $Kg = h > 0$, $b \in \{1, 2\}$, and $\det K = (2b - 3)d \neq 0$. These are the two conditions of Theorem 281.1 in gap coordinates.

Conversely, strict real feasibility of rational linear inequalities gives a positive rational vector g by openness. Clear a common positive denominator to obtain positive integer g and $h = Kg$. Choose $b = 1$ if $\det K < 0$, $b = 2$ if $\det K > 0$, and $d = |\det K|$. All five residuals vanish. $\square$

The interface has nine signed integer matrix-entry inputs and eight positive-integer witnesses. There are five squared residuals: three vector coordinates, one determinant/sign residual, and one two-value gate. The exact total degree of the polynomial in all its input and witness variables is six, since the cubic determinant is squared; all other displayed terms have degree at most four. An optional positive denominator is an extra input that does not occur in the polynomial.

Representing $k_{ij} = a_{ij} - c_{ij}$ externally with positive input leaves changes nine signed matrix-entry inputs to eighteen positive external inputs, adds no witnesses, and preserves exact degree six. There is no finite-fold statement. Whenever a witness tuple exists, replacing (g, h) by (rg, rh) for every positive integer r gives infinitely many distinct witness tuples with b, d unchanged.

287.1 Centered inputs require an explicit conversion

For a native signed integer centered numerator M with $N = M/q$, put

$$S = 3H = \begin{pmatrix} 1 & 3 & 0 \\ 1 & -3 & 3 \\ 1 & 0 & -3 \end{pmatrix}, \quad B = 3H^{-1} = \begin{pmatrix} 3 & 3 & 3 \\ 2 & -1 & -1 \\ 1 & 1 & -2 \end{pmatrix}. \quad (287.2)$$

Then $SB = BS = 9I$, and the integer matrix $\widetilde{K} = SMB$ satisfies

$$H(M/q)H^{-1} = \frac{\widetilde{K}}{9q}, \quad \det \widetilde{K} = 729 \det M. \quad (287.3)$$

Thus the centered-input polynomial is explicitly

$$F_{\text{center}}(M; g, h, b, d) = \sum_{i=1}^3 (((SMB)g)_i - h_i)^2 + [729 \det M - (2b - 3)d]^2 + [(b - 1)(b - 2)]^2. \quad (287.4)$$

This has nine signed matrix-entry inputs, eight positive witnesses, five squared residuals, and exact total degree six. The fixed linear substitution creates no hidden witness or alias variable. Eighteen positive external leaves can encode the entries of M by differences. The positive denominator factor 9 adds no condition. No centered-input arithmetic operation count is asserted here.

287.2 Literal native gap input DAGs and their operation ledger

The frozen packet emits two complete ordered binary arithmetic DAGs, including every node and final output identifier. The signed-input graph has leaves for the nine entries, eight witnesses, and constants 1, 2, 3. Gate outputs are unrestricted integers. A square is one multiplication, and multiplication by the constant 2 is counted. These are arithmetic expression graphs, not collision schedules or additional Diophantine witnesses.

Part of the selected graph	$\times$	$+$	$-$
Three row dot products	9	6	0
Three row residual differences	0	0	3
Cofactor determinant	9	1	4
Sign value $2b - 3$	1	0	1
Sign times d , then determinant residual	1	0	1
Binary-gate residual $(b - 1)(b - 2)$	1	0	2
Five residual squares	5	0	0
Sum of five squares	0	4	0
Signed-input total	26	11	11
Nine external positive-input differences	0	0	9
Positive-input total	26	11	20

Table 31: Native gap-input operation ledger. The totals are 48 and 57 binary operations. The fixed centered-coordinate conversion SMB is not included.

The determinant is evaluated as

$$k_{11}(k_{22}k_{33} - k_{23}k_{32}) - k_{12}(k_{21}k_{33} - k_{23}k_{31}) + k_{13}(k_{21}k_{32} - k_{22}k_{31}).$$

The positive-input graph literally prepends all nine differences $k_{ij} = a_{ij} - c_{ij}$ to the same graph. It has eighteen positive external matrix-entry leaves and the same eight positive witnesses; the denominator remains optional and unused.

Neither native count is a minimum, a record, a general Diophantine representation comparison, or a measure of the much larger physical word compiler. The representation is for this particular decidable matrix property. It implies neither a general Diophantine normal form nor an undecidability theorem.

[write] [The Hilbert's-tenth programme's review [99] makes the same point: these certificates concern a decidable matrix property and do not compete with its universal 84-operation construction.]

288 Repeated validity and clocks are conditional

Let $\mathcal{Q} = \{z : Gz > 0\}$ denote the actual full chamber (284.5). Literal phase closure gives the exact equivalence

$$\text{the complete word is infinitely valid at } z \iff GN^n z > 0 \text{ for every integer } n \geq 0. \quad (288.1)$$

The chamber need not be forward invariant. The set in (288.1) may be empty even when the local criterion holds.

288.1 A local realization with no infinitely ordered input

In gap coordinates take

$$K_{\text{esc}} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{pmatrix}. \quad (288.2)$$

It has determinant one and is compatible for $g_3 > g_1 > 0, g_2 > 0$. Yet

$$K_{\text{esc}}^n g = (g_1, g_2, g_3 - ng_1)^\top,$$

so every strictly positive input eventually leaves the ordered cone. Theorem 281.1 locally realizes $H^{-1}K_{\text{esc}}H$, while no repeatedly valid realization with this same full return has an infinitely valid input. Figure 19 makes the distinction explicit.

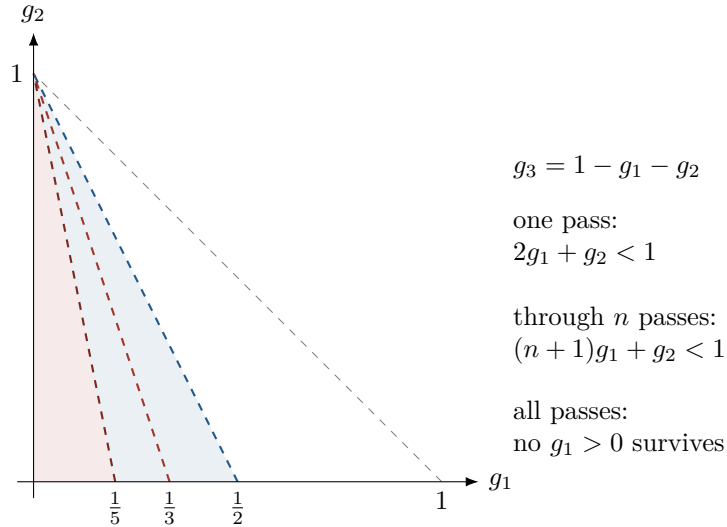


Figure 19: Local versus repeated endpoint feasibility for the escaping-gap matrix. With initial gaps normalized by $g_1 + g_2 + g_3 = 1$, the open blue triangle is compatible for one pass. Dashed lines show the boundaries after one, two, and four passes. The intersection over every n has no strictly positive input; the $g_1 = 0$ boundary is excluded. An actual word chamber is a subset of the one-pass endpoint region, and its repeated validity additionally requires every pulled-back guard $GN^n z > 0$.

288.2 A duration bound for the retained-translation construction

Retain both physical translations, including an identity when needed. The first translation alone takes more than $6D$. On the actual chamber every event time and endpoint coordinate is a fixed rational homogeneous linear form in z . The closure of the normalized triangle Ω is compact. Consequently the finitely many such forms yield an effective finite upper bound on total duration and intermediate positions proportional to incoming D . Thus the full duration is ℓz for a rational row ℓ , and there are constants $c_1, c_2 > 0$ with

$$c_1 D \leq \ell z \leq c_2 D \quad (z \in \mathcal{Q}), \quad c_1 = 6 \text{ available.} \quad (288.3)$$

The lower bound here is for this retained-translation construction; omitting a timed identity is harmless to local realization but does not invoke this particular argument unchanged.

Remark 288.1 ([write] Conditional spectral clock criterion, printed once as Theorem 273.1). Source 33's Corollary 8.1 states the following. For an infinitely valid input z , write $z_n = N^n z$ and $D_n = (z_n)_1 > 0$. For the construction retaining both translations,

$$\begin{aligned} \text{Zeno} &\iff \sum_{n \geq 0} D_n < \infty \iff N^n z \longrightarrow 0 \\ &\iff \text{every eigenvalue of } N|_{V_z} \text{ has modulus } < 1, \quad V_z = \text{span}\{z, Nz, N^2 z\}. \end{aligned} \quad (288.4)$$

If z is rational and Zeno, its total time

$$T_\infty = \ell (I - N|_{V_z})^{-1} z \in \mathbb{Q} \quad (288.5)$$

is rational, the inverse being taken within V_z . No ambient inverse is required: unused unit eigenmodes may make $I - N$ singular. This is source 32's Theorem 273.1 with its formula (273.4), for source 33's construction: the bound (288.3), with $c_1 = 6$, replaces source 32's (273.2). Source 33's proof is source 32's: sum the bound, bound ξ_n and η_n by the ordering, apply the Jordan estimates on the cyclic subspace and Cayley–Hamilton. The rationality is a special case of source 26's Theorem 182.4 (Remark 244.1).

All clock claims assume (288.1). They do not decide the general infinite-validity set, provide such an input, or continue a trajectory beyond an accumulation. The local theorem does not depend on this optional corollary.

289 Relation to the literature and scope

The signal-machine model used here is consistent with Becker et al. [92]: finitely many labels carry constant velocities; collision input and output sets have distinct velocities; and the operational next event occurs at the smallest positive meeting time. Their intrinsic universality construction concerns macro-signals and simulations of entire machines. It does not establish the present five-live-signal local criterion.

The oblique constant-flow section maps, itinerary domains, and path compositions in Alishah, Duarte, and Peixe [93] give related geometric precedents. Their setting is flows on polytopes. Durand-Lose [4] gives scale-relative signal-distance arithmetic in the linear Blum–Shub–Smale setting. That arithmetic does not alone imply the fixed-population, full-section realization proved here.

The construction-specific contribution presented in this report is the exact local criterion, obtained from the physical translation chronology and left/right reduction relative to the preserved fixed-center compiler. The explicit degree-six certificate describes that decidable criterion. The literature is positioning rather than a priority or exhaustive novelty search.

Established conclusion	Boundary of that conclusion
Some full-dimensional open chamber realizes the specified N	Not all compatible inputs or a prescribed chamber
Exact finite one-pass guards	Not forward invariance or an infinitely valid input
A polynomial-time realizability decision	Not a polynomial-size physical compiler
Eight positive witnesses and joint degree six	Not finite fibers or optimal witness/degree counts
48/57 native gap-input DAG operations	Not the centered-input conversion or collision count
A spectral clock criterion on an active cyclic space	Conditional on every repeated-word guard
A finite deterministic number-preserving rule table	Not global reversible dynamics or Turing universality

Table 32: Quantifiers and interfaces that must not be conflated.

290 Further research directions

The construction suggests several precise next questions. They are questions left open by this report, not claims that the wider literature has no answer.

1. **Guard coverage.** Can alternative physical factorizations enlarge the realized chamber toward the full endpoint-compatible region? Characterizing which prescribed polygons can be exact word chambers would strengthen the current existential result.
2. **Compilation size.** The selected rational subdivision can be large when the interior margin is small. Bounds on event counts, temporary labels, and arithmetic description length should be studied separately from the fixed-dimensional polynomial-time decision.
3. **Repeated validity.** For which matrix and guard classes can the set $\{z : GN^n z > 0 \text{ for all } n \geq 0\}$ be decided or described effectively? The escaping-gap example shows that endpoint compatibility alone cannot answer this.
4. **Physical branching.** A finite family of compatible linear returns is not yet a physical piecewise program. A further construction would need explicit branch detection, mutually consistent collision rules, complete chronology, and phase interfaces before a universality claim could follow.

[write] [Status at this write: Question 4 is answered by source 34 in Part XII. A finite table tests two branches on the full section without destroying it (Theorem 294.2), and every fixed two-counter program is compiled with five live signals (Theorem 294.1). Five-live-signal counter computation itself is Durand-Lose’s prior work, credited there. Questions 1–3 are open.]

291 The emitted arithmetic expression graph (source 33, Appendix A)

The following is a literal typeset transcription of the signed native gap-input DAG. Entries in the left column precede those in the right column. The final output is $v_{48} = F$. Symbols k_{ij} are signed inputs and g_i, h_i, b, d are the eight positive witnesses. Intermediate v_j are arithmetic gates, not quantified witnesses.

Nodes 1–24	Nodes 25–48
$v_1 = k_{11} \cdot g_1$	$v_{25} = v_{23} - v_{24}$
$v_2 = k_{12} \cdot g_2$	$v_{26} = k_{12} \cdot v_{25}$
$v_3 = k_{13} \cdot g_3$	$v_{27} = k_{21} \cdot k_{32}$
$v_4 = v_1 + v_2$	$v_{28} = k_{22} \cdot k_{31}$
$v_5 = v_4 + v_3$	$v_{29} = v_{27} - v_{28}$
$v_6 = v_5 - h_1$	$v_{30} = k_{13} \cdot v_{29}$
$v_7 = k_{21} \cdot g_1$	$v_{31} = v_{22} - v_{26}$
$v_8 = k_{22} \cdot g_2$	$v_{32} = v_{31} + v_{30}$
$v_9 = k_{23} \cdot g_3$	$v_{33} = 2 \cdot b$
$v_{10} = v_7 + v_8$	$v_{34} = v_{33} - 3$
$v_{11} = v_{10} + v_9$	$v_{35} = v_{34} \cdot d$
$v_{12} = v_{11} - h_2$	$v_{36} = v_{32} - v_{35}$
$v_{13} = k_{31} \cdot g_1$	$v_{37} = b - 1$
$v_{14} = k_{32} \cdot g_2$	$v_{38} = b - 2$
$v_{15} = k_{33} \cdot g_3$	$v_{39} = v_{37} \cdot v_{38}$
$v_{16} = v_{13} + v_{14}$	$v_{40} = v_6 \cdot v_6$
$v_{17} = v_{16} + v_{15}$	$v_{41} = v_{12} \cdot v_{12}$
$v_{18} = v_{17} - h_3$	$v_{42} = v_{18} \cdot v_{18}$
$v_{19} = k_{22} \cdot k_{33}$	$v_{43} = v_{36} \cdot v_{36}$
$v_{20} = k_{23} \cdot k_{32}$	$v_{44} = v_{39} \cdot v_{39}$
$v_{21} = v_{19} - v_{20}$	$v_{45} = v_{40} + v_{41}$
$v_{22} = k_{11} \cdot v_{21}$	$v_{46} = v_{45} + v_{42}$
$v_{23} = k_{21} \cdot k_{33}$	$v_{47} = v_{46} + v_{43}$
$v_{24} = k_{23} \cdot k_{31}$	$v_{48} = v_{47} + v_{44}$

The 57-operation positive-input graph prepends exactly these nine subtraction nodes, then uses the same 48-node graph with these k_{ij} values:

$$\begin{aligned}
 k_{11} &= a_{11} - c_{11} & k_{12} &= a_{12} - c_{12} & k_{13} &= a_{13} - c_{13} \\
 k_{21} &= a_{21} - c_{21} & k_{22} &= a_{22} - c_{22} & k_{23} &= a_{23} - c_{23} \\
 k_{31} &= a_{31} - c_{31} & k_{32} &= a_{32} - c_{32} & k_{33} &= a_{33} - c_{33}
 \end{aligned}$$

All eighteen a_{ij}, c_{ij} leaves are positive external inputs. The archived JSON files retain their native node identifiers, schema, operation maps, leaf declarations, and final output identifiers. Neither graph includes the centered-input conversion *SMB*.

292 Evidence and reproducibility boundary (source 33, Appendix B)

The source archive preserves the original Report 63 scientific packet without modification, including its proof, dependency proof/review copies, two literal arithmetic DAGs, inert author checker, and

recorded evidence. These are separately distinguished from the fresh independent audit and the owned manuscript/build tools. A successful finite check does not replace an all-parameter proof or amount to proof-assistant formalization.

[write] [Shipped here: the frozen proof and its README, the reviews and the tool notes as `33-homogeneous-*.md`; the programs as `code/33-homogeneous-*`; the two literal DAGs, the evidence, receipts and pins as `data/33-homogeneous-*`. The dependency proofs and the review of source 32 are shipped once, under the prefixes 30-, 31- and 32-. The PDF, the LaTeX and the page renders are not shipped.]

292.1 Frozen scientific inputs

The source proof is `science/frozen-proof/PROOF.md`. Its SHA-256 is

31f0db773daa015075e74dabb8955c110288e1c0f9ed2121c647df034773aa67.

The included manifest binds all scientific files and the original locations of the preserved dependencies. The author evidence records 98 named static checks, seven exact endpoint-LP fixtures, 36 translation endpoint/margin fixtures, and Diophantine fixtures. The author checker is included as inert source; it is not invoked by report building or packaging. No author or upstream mathematical program, simulator, saved collision schedule, or Lean source is executed in producing this report.

The seven endpoint fixtures illustrate why both parts of the criterion matter:

Gap matrix fixture	det K	LP optimum
Identity	1	1/3
Positive matrix of Section 286	2	1/3
Swap of first two coordinates	-1	1/3
Negative identity	-1	infeasible
diag(1, 1, -1)	-1	0
All-ones matrix	0	1/3
Escaping-gap matrix of Section 288	1	1/4

Table 33: Recorded exact fixture values. Positive compatibility does not remove the independent nonsingularity requirement, and one-pass compatibility does not imply infinite validity.

292.2 Fresh independent static audit

The fresh independent audit accepts the mathematical criterion and its stated limitations without a required correction. Its core source-bound run records 840 named assertions covering chronology formulas, endpoint and corner guards, physical translation rule declarations, normalization, exact rational feasibility, and source authentication. An archived-source replay records 832 assertions: it authenticates all included files and copied dependency bindings but does not reopen the four historical original files, each of which contributes two authentication assertions in the source-bound run. Its separate sparse-polynomial and literal-DAG audit records 737 named assertions. The latter checks the full emitted expressions and their native interfaces rather than merely trusting operation totals.

These counts have overlapping purposes and include structural and authentication checks. They are not counts of independent theorems and should not be summed into a measure of mathematical

strength. The accepted all-parameter argument remains the proof in the body of the report and its preserved dependencies. The independent generic 26-event rule declaration is an inert symbolic object, not an executed collision schedule.

292.3 Manuscript and release tooling

The editable source is supplied both as modular L^AT_EX and as a deterministic standalone flattening. Owned release tools authenticate the frozen files, flatten the manuscript, typeset it in an isolated fresh TeX format with shell escape disabled, record all format and document-pass inputs, render every PDF page, and create a deterministic evidence archive. They do not run scientific checkers as part of a typesetting or packaging command.

The build lock records the invoked interpreter/executable bytes and the union of system TeX and font-map inputs, with a preflight comparison and a post-build comparison. It does not inventory operating-system libraries or Python’s standard library and is not a sandbox for hostile source. Separate reviewer evidence covers the current owned tooling and the final manuscript. Direct model all-page visual inspection remains distinct from PDF-text, page-count, PNG, and dependency authentication checks.

A release manifest authenticates exact file inventories, bytes, modes, and nanosecond modification times. The supplied extraction path restores those metadata so that relocated verification can use the same trusted external manifest pin. Ordinary ZIP extraction is not claimed to retain all authenticated metadata. Integrity hashes are commitments, not digital signatures; a digest acquired from the same untrusted object does not constitute independent authentication.

293 What Part XI does not claim, its questions, and its provenance

293.1 Non-claims of the four sources

[write] The four sources state the following limits; none of them is lifted by printing them together.

- *All four.* The proofs are conventional proofs supported by finite exact and static checks. None is a proof-assistant formalization, and no Lean or Rocq development certifies any statement of this Part. No physical simulation, next-event search or saved collision schedule was run, and finite fixtures do not replace the all-parameter proofs. No priority, novelty, exhaustive literature search, minimality or optimality (events, labels, speeds, population, degree, witnesses, gates) is claimed. Each machine is fixed after its rational parameters: there is no unchanged universal table and no polynomial-size compiler. Number preservation is not global reversibility. No run is continued through a Zeno accumulation point. Hashes are integrity commitments, not signatures, and overlapping check counts must not be summed.
- *Source 30.* The physical compiler covers $GL_2^+(\mathbb{Q})$ at a fixed positive rational scale only. The abstract classification also covers singular and negative-determinant matrices, but without a physical realization and without a physical impossibility. There is no uniform polynomial physical compiler, no generic emitted arithmetic graph, no gate count and no Turing completeness. A real kernel on an irrational stable line may need algebraic coefficients. A finite sign obstruction is not an obstruction to existential integer certificates. Fixed- A polynomial membership, uniform elliptic polynomial membership, the credited counting-hierarchy bound and the exponential explicit output are four different claims. Erratum E1 changes only the

wording of a bound: at most $\lfloor \log_2 b \rfloor$ divisions and $1 + \lfloor \log_2 b \rfloor$ recurrence steps; no theorem changes. Figure 14 is analytic geometry, not a trajectory.

- *Source 31.* The singular obstruction holds only under the full-section, open-chamber, binary, positive-flight and fixed-scale hypotheses. It says nothing about projected, lower-dimensional, nonbinary, population-changing or limit models. The two-witness corollary and the finite clock are manuscript additions, not contents of the frozen packet. Seven events are not claimed minimal, and the inverse map need not be implemented by the machine.
- *Source 32.* The realization is for a fixed rational centre only. There is no realization of an arbitrary GL_3 matrix in the original coordinates, and rational conjugacy is not a physical change of chart. The native-gap certificates are for the stated half-space example only; the integer-domain qualification is essential.
- *Source 33.* The theorem gives some local chamber: not every compatible input, not a prescribed or invariant chamber, and no infinitely valid input. A polynomial-time decision is not a polynomial-size compiler. There is no finite-fold statement. The 48/57 graph counts exclude the conversion from centred inputs, and the clock criterion is conditional on repeated validity.

293.2 Status of the questions

[write] Table 34 lists every question of the four sources, and the two questions of Section 293.4, with their status at this write. A question answered only in part stays open and is re-scoped; the note after each question list gives the details.

293.3 Provenance of Part XI

[write] Part XI was built from four manuscripts of batch 91, all arriving in commit 0d7f51c44 (4 October 2026) and placed by commit d750d98dd, which also placed the sources of Parts IX, X, XII and XIII.

- *Source 30* (manuscript 12, Report 60): `Planar_Signal_Realization_Source.zip`, main file `Report60/Report60.tex` (905 lines; also as five modular files), 28-page PDF. It contributed the physical realization of $\mathrm{GL}_2^+(\mathbb{Q})$, the classification of planar strict kernels and their certificates. It pins its own frozen proofs (physical e0ddd64c..., geometric 70eb8f83..., arithmetic e93fdbef..., quadratic faaf236d...), their four independent reviews, source 29's proof (14c3d694...) and the mathlib Pell file (993760c7..., mathlib4 commit ac77769fabe23cb237559e7f56578dbead91499f).
- *Source 31* (manuscript 13, Report 61): `Invertible_Signal_Returns_Source.zip`, main file `Report61/Report61.tex` (649 lines; also six modular files), 19-page PDF. It contributed full-section invertibility, the parity formula, the seven-event reversal and the $\mathrm{GL}_2(\mathbb{Q})$ criterion. It pins its proof (dc97e30c...), its source manifest (715bbbc8...), its audit manifest (dbbc4972...) and source 30's physical proof.
- *Source 32* (manuscript 14, Report 62): `Projective_signal_returns_source_and_evidence.zip`, main file `Report62/Report62.tex` (774 lines; also nine modular files), 22-page PDF. It contributed fixed-centre projective realization, the 44-event mixed clock and the spectral clock criterion. It pins its proof (502e90e0...) and the proofs of sources 30 and 31.

Table 34: [write] The questions of sources 30–33.

Source	Question	Topic	Status
30	1	two independent rotation blocks	open
30	2	uniform stable decision complexity	open
30	3	compiler size, alternative dilations	open
30	4	orientation reversal and rank loss	answered for $\det A < 0$ by source 31 (Corollary 257.4); singular returns excluded under source 31’s hypotheses; open for other models
30	5	certificate resources beyond degree	open
30	6	formal interface certification	open
31	1	general homogeneous returns	answered at a rational fixed centre by source 32 and locally by source 33 (Theorem 281.1); open for prescribed or invariant chambers
31	2	compilation efficiency	open
31	3	reversing primitives with fewer than seven events	open
31	4	globally reversible completion	open
31	5	alternative observations and sections	open
32	1	realization beyond a rational centre	answered by source 33 (Section 286)
32	2	projective survivor geometry	open (source 33’s escaping-gap matrix is an example with an empty survivor set)
32	3	efficient compilation	open
32	4	uniformity and speed resources	open
32	5	arithmetic on native gaps	open
33	1	guard coverage	open
33	2	compilation size	open
33	3	repeated validity	open
33	4	physical branching	answered by source 34 in Part XII (Theorems 294.2 and 294.1)
write	1	merging source 30’s two gates	open (Section 293.4)
write	2	one word with both clock behaviours	open (Section 293.4)

- *Source 33* (manuscript 15, Report 63): `Five_signal_homogeneous_realization_source_and_evidence.zip`, main file `Report63/Report63.tex` (633 lines; also ten modular files), 18-page PDF. It contributed the local homogeneous criterion, its decision, its certificate and the escaping-gap example. It pins its proof (31f0db77...), the proofs of sources 30, 31 and 32 and source 32's review.

None of the four names a ProveIt commit. This Part prints the standalone main files.

[write] **Where the merge had to choose.** (1) Source 30's POWER module, already printed in Part X, became Remark 251.1. (2) The restated accumulation-time criterion of sources 31 and 33 became Remarks 263.1 and 288.1. Source 30's version is printed as a second route to source 26's theorem, and source 32's spectral theorem is printed in full. (3) Source 32's restatements of the inherited compiler and of anchor scaling became Remarks 278.1 and 268.1. The equation labels of its Appendix A (`eq:xblock`, `eq:dilation-guards`), which only that appendix cited, and those of source 30's POWER display (`eq:powerfirst`, `eq:powerlast`) are kept on the pointer remarks. (4) Source 33's ten-event primitive (Lemma 282.1) is a special case of source 30's translation with a fuller chronology, and it is kept as a marked second presentation. (5) The appendices of sources 32 and 33 became their last sections. (6) The reversal matrix is renamed J , the colours are renamed, and the bibliographies are merged: source 30's `agc8` and source 32's `becker8` are this report's [6], source 30's `agc6` is [16], source 31's `durand` is [5] (this report's own key `durand` is a different paper), and the `durand2007` of source 32 and `durand` of source 33 are [4]. Source 31's entry `report60` became a cross-reference to source 30. (7) Each source's title page, table of contents and float settings were dropped; its abstract is quoted under its heading.

[write] **What is shipped.** The shipped files of the four sources carry the prefixes `30-planar-`, `31-invertible-`, `32-projective-` and `33-homogeneous-`: proofs, reviews and notes at the report root, programs under `code/`, and evidence, receipts and pins under `data/`. They are byte-identical to the delivery, and the README maps every shipped name to its delivered path. Not shipped, but surviving in the arrival archive: the four PDFs and their page renders, the standalone and modular LaTeX, the TeX build transcripts, the checksum manifests and release byte inventories, empty files, in-batch byte copies (each shipped once at its earliest source), and the third-party mathlib Pell source with its licence, which is cited by commit and hash. Several delivered texts still name these files or use delivery paths; the README lists them and gives rerun routes on copies. The packaged release verifiers and portable replays refuse to run on Windows paths, and some checkers hard-code `/workspace/...` paths, so they run only on patched copies (README).

293.4 Further questions and research

[write] The merge raised two questions of its own. They are recorded here instead of being asserted or dropped.

1. *Can source 30's two acceptance gates be merged?* Source 29's single outer equation (O13) also excludes the time-zero match of a rotation: at exponent 0 its Gaussian data are $P = 1$, $C = 1$, $S = 0$. Source 30 uses two gates, (O13) for time zero and (O14) for positive times, and notes that for a general elliptic map the times zero and one both have denominator one (Section 249). A merged gate would bring source 30's ledger to $80J$ witnesses and $43J$ equations, matching source 29's per-contact count. Argument sketch: choose the companion basis $S = [z, A^{-1}z]$ so that the time-zero point e_1 is the image of the exponent $m = -1$ of the same coefficient recurrence. Missing: a positive-domain encoding of that shifted exponent within one POWER pair, and a proof that the extraction bound of Section 249 still separates the two candidates.

Neither source claims that such a merge is possible.

2. *A uniform clock family.* Source 27 needs two machines for its two clock behaviours, a 138-event Zeno word and a separately reclosed 114-event non-Zeno variant (Theorem 196.3). Source 32's 44-event word has both behaviours (Theorem 267.2), because its return matrix has the unit mode that the invariant $c = D - 3y/2$ detects. Does every rational fixed-centre lift with a unit eigenvalue and a contracting complement admit one word that is Zeno on the contracting face and non-Zeno inside? Missing: a general invariant, which source 32 has only for its example, and control of the exact one-word chamber along the face.

Part XII

Nondestructive branching and exact instruction clocks with five live signals (sources 34 and 35)

[write] Part XII prints two AI-assisted research manuscripts of ProveIt's incoming reports, batch 91 (arrival 0d7f51c44; cluster W3, placed by d750d98dd), both dated 4 October 2026 and added on that day. They are numbered reports of the research pipeline that also wrote sources 24–33 (Parts IX–XI) and 36–38 (Part XIII): *source 34* is batch-91 manuscript 16, the pipeline's Report 64, from the archive `Five_signal_branching_source_and_evidence.zip`; *source 35* is batch-91 manuscript 17, the pipeline's Report 65, from `Clocked_signals_source_and_evidence.zip`, a declared continuation of source 34. Both carry the author line “Research report” and name no assistant; their recorded run logs show the runtime of an AI pipeline. Neither names a ProveIt commit, path or gap. Source 35 pins source 34, and together they pin sources 28, 30 and 33 and the mathlib Pell source by SHA-256; and every pin resolves to a shipped file or to a file of the arrival archive (the README lists them). Their labels carry the prefixes `smc:br:` and `smc:ck:`; their equations are numbered within this report's sections, and where the pipeline's report numbers appear in their text, the bracketed source numbers after them were added in the merge.

[write] **What the two sources do.** Source 34 builds, for every fixed deterministic two-counter program, one finite rational-speed signal machine with number-preserving (cardinality-preserving) binary rules and exactly five live signals—four stationary markers and one messenger—that simulates the program from every initial counter pair on one fixed ordered full section $L < X < Y < R$. Its devices are an exact *nondestructive* branch test, whose two complete words have the exact open chambers $4x < y < 8x$ and $8x < y$ and are undone by a physically implemented, branch-tagged inverse, and a powers-of-two encoding that every guard covers uniformly. It adds, independently of the physics, a quartic with a unique positive witness for halting within a fixed instruction horizon. Source 35 pads every instruction of that compiler to exactly $30D$ (at most 54 contacts, one common set of 39 speeds), and replaces the externally supplied exponential initialization by three native positive integer gaps (still of numeric height $\Theta(2^m)$) whose decoding it pays for inside two displayed 15-residual POWER modules: joint degree 12, $T(E + 2) + 54$ positive witnesses at horizon T , a unique decoded trace, and infinite full witness fibres. It also computes the exact primitive integer initialization and its bit height.

[write] **What is not new.** Counter computation with five live signals is Durand-Lose's MCU 2004/2005 construction [100] (two fixed scale signals, two counter-position signals and one instruction

signal, with cardinality-preserving rules), which both sources credit prominently. Part XII is therefore *not* a new universality result and not a minimum-population result; neither source claims either. What the sources add is the fixed ordered full-section interface with literal label restoration, the exact nondestructive branching, the uniform guard coverage, the exact clock and the costed native-gap certificate. Opposite-speed reversal is standard [5], and bouncing delays are an established mechanism, used explicitly by Durand-Lose and Emmanuel [102]. No universal counter program is constructed, so no universal machine with an explicit table is printed here.

[write] **Relation to the other Parts.**

- *Part III.* Part III writes out a reversible, number-conserving universal signal machine with 18 live signals and derives one with ten (Theorems 29.1 and 29.2); among its non-claims is the optimality of 18 or ten live signals (Section 1.10). Part XII’s five live signals concern number-preserving tables that are *not* claimed globally reversible, compiled per fixed counter program rather than one universal table, and the count is prior art. Part XII therefore neither refutes nor answers that non-claim; it is a different resource point.
- *Part XI.* The “local-realization proof” that source 34 retains is source 33’s Theorem 281.1; source 34 re-establishes source 33’s ten-event nearest-target primitive (Lemma 282.1) as its translation block (Section 299). The determinant $-8/9$ of its seven-event word is an instance of source 31’s parity theorem (Theorem 257.2).
- *Parts X and XI, the POWER module.* Source 35’s two modules are the base-two instance of the 15-equation Pell module of source 28 (Lemma 217.1) and source 30 (Section 251); its Lemma 309.1 is a second route for that instance with positive adapters.
- *Fixed-horizon certificates.* Source 34’s quartic (Theorem 301.1) uses the device of Part V’s fixed-horizon certificate (Theorem 58.1) and of Part III’s packet (Theorem 36.1): one-hot step selectors, affine updates, a unique witness, and arity growing with the horizon; only the object is new. Part VIII removes the horizon for fixed sources of a different frontend; nothing in Part XII does.
- *Part XIII.* Sources 36–38 (the pipeline’s Reports 66, 67 and 69) continue source 35. They replace its growing trace by a bounded number of witnesses at each horizon—57 positive witnesses in Theorem 318.2, $2\ell + 5$ (29 for $\ell = 12$) in Theorem 332.2, 28 in Theorem 353.2—but remain horizon-indexed families.

[write] **Review status.** The Hilbert’s-tenth programme made a bounded routing intake of the twelve signal and particle archives of this arrival, these two included [99]. It read both delivery READMEs in full and selected spans of the four proofs (source 34: physical proof lines 1–32, 214–249, 327–390, arithmetic proof 1–26, 136–165, 183–226; source 35: native-gap proof 1–58, 162–197, 231–261, clock proof 1–20, 170–244). It calls the two reports the most useful computation interface of the arrival, and records the boundaries kept in this Part: the arithmetic encodes a trace of a fixed horizon T with arity growing in T ; the fixed-arity verification of an unbounded trace remains unpaid; five live signals, 19 or 39 speeds, 32 or 54 contacts and the 30D clock do not bound the arithmetic work of such a representation, and the rule alphabet still depends on the program, so no numerical universal improvement should be entered from these counts; and source 35’s primitive gap height is $\Theta(2^m)$, a bit height $\Theta(m)$ exponential in the binary length of the larger counter m . It found no specific contradictory theorem or interface statement requiring an author correction in the selected passages, and it is not a proof certification: omitted chronology, the Pell proofs and the archived tests were not certified. No patch was issued.

[write] **Evidence.** Each archive carried frozen proof packets, inert author checkers, independent audits with their own checkers, release tools and build records. They are shipped under the prefixes **34-branching-** and **35-clocked-** (proofs and audit reports at the report root, programs in **code/**, records in **data/**); byte copies of earlier packets are shipped once, at the source that first delivered them. The placement dossier reran every independent checker and the author checkers on copies, with only path and file-access adaptations, and reproduced the recorded verdicts; the README gives tested commands. This is not a formalization: no statement of this Part has a Lean or Rocq counterpart in the repository.

Source 34 (batch-91 manuscript 16): Nondestructive branching with five live signals

A uniform counter compiler on one fixed ordered section, “Research Report 64”, 4 October 2026. [write] From the archive **Five_signal_branching_source_and_evidence.zip**, main file **Report 64/Report64.tex** (a byte-exact flattening of the eleven modules in its **manuscript/**). Its labels carry the prefix **smc:br**; its sections are Sections 294–303 here, its Appendix A being printed as Section 303; its table of contents and page breaks are omitted. Its two branch labels, which it wrote $\backslash\mathbb{Z}$ and $\backslash\mathbb{N}$, are written **Z** and **N** here (sans serif, as in the source): **Z** marks the zero branch and **N** the nonzero branch, and neither is the set $\mathbb{Z}$ or $\mathbb{N}$. Its natural numbers include 0 and are written **N**. Its diagram colours carry the prefix **smcbr**. Its bibliography is merged into the report’s.

We construct a finite physical test that returns four stationary markers and one messenger to exactly the same ordered section while retaining one of two branch labels. The two complete collision words have exact open chambers $4x < y < 8x$ and $8x < y$ inside $0 < x < y < D$; their nondestructive versions use 14 and 12 events. A separated powers-of-two encoding makes these guards valid for every nonnegative pair of counters. Explicit increment and positive-decrement gadgets then give one finite rational-speed rule table per fixed finite two-counter program, independent of its initial counters. Exactly five signals remain live, every nonhalting instruction has at most 32 binary events and duration $D < T < 10D$, and all instruction sections restore the same marker ordering, labels, scale and messenger speed. Infinite encoded computations have no finite-time collision accumulation. A companion finite-trace construction gives a unique positive-integer witness for each fixed instruction horizon: one quartic equation with $T(E + 2)$ witnesses and $T(E + Z + 4) + 1$ residuals before bundling, where E counts split edges including a virtual halt loop and Z counts zero edges. Its arity grows with the horizon. Five-live-signal counter computation already appears in Durand-Lose’s MCU 2004/2005 work; the result here is the fixed full-section interface, exact nondestructive branching and uniform guard coverage, not a new universality or minimum-population claim.

294 Main result and scope

A *meta-signal* is a finite label with a fixed rational speed. A *live signal* is an individual occurrence of such a label. Our count of five concerns live signals, not the number of labels or speeds. At an outgoing section the five strands are

$$L = 0, \quad X = x, \quad Y = y, \quad R = D, \quad Q \text{ at } L \text{ outgoing at } +1, \quad 0 < x < y < D. \quad (294.1)$$

The four marker labels have speed zero. The messenger and L are counted as two outgoing strands at their common section contact. All later contacts on the encoded executions are binary, have distinct incoming and distinct outgoing speeds, and are separated by positive time. Deterministic collision rules consume and produce the same number of signals.

Theorem 294.1 (Uniform fixed-program compiler). *For every fixed finite deterministic two-counter program consisting of increment, zero-test/positive-decrement, and halt instructions, there is an effectively specified finite rational-speed collision table that simulates that program for every initial pair $(a, b) \in \mathbb{N}^2$ on the single section (294.1). The encoding is*

$$x = D \left(\frac{1}{20} + \frac{1}{10} 2^{-a} \right), \quad y = D \left(\frac{19}{20} - \frac{1}{10} 2^{-b} \right), \quad D > 0. \quad (294.2)$$

Every nonhalting simulated transition restores the four stationary marker labels, the scale D , and the speed+1 outgoing messenger, whose control label identifies the next instruction. It uses at most 32 binary collisions and has duration strictly between D and $10D$. The rule table uses only the 19 rational speeds in (300.1). Exactly five signals remain live, and an infinite encoded computation has no finite-time collision accumulation.

The finite table depends on the *fixed program*, not on a , b , D , or the number of executed instructions. A separately chosen fixed universal counter program would give one table through this compiler. That conditional corollary is different from saying that every counter program runs on one table without an interpreter or program encoding. No particular universal counter program is constructed here. [write] [Compare Part III, which writes out one fixed reversible universal table with 18 live signals (Theorem 29.1). A universal counter program compiled by Theorem 294.1 would give one five-signal table that is number preserving but not claimed reversible; none is written out in this report.]

Theorem 294.2 (Nondestructive full-section test). *On either of the full-dimensional open chambers*

$$\mathcal{P}_Z = \{(D, x, y) : x > 0, y - 4x > 0, 8x - y > 0, D - y > 0\}, \quad (294.3)$$

$$\mathcal{P}_N = \{(D, x, y) : x > 0, y - 8x > 0, D - y > 0\}, \quad (294.4)$$

a finite rule table returns exactly the same (D, x, y) and stationary marker labels, while outputting distinct messenger phases $Q_{\text{out},Z}$ and $Q_{\text{out},N}$. It uses respectively 14 and 12 events. Branch acquisition is a real choice of local collision partners. No marker-only collision changes a remote messenger label.

294.1 Historical boundary of the claim

Durand-Lose’s MCU 2004/2005 construction already encodes two counters by two position signals, uses two fixed scale signals and one instruction signal, and preserves signal cardinality [100]. The five-live-signal computational precedent is therefore prior work. There the counter markers can exchange order, and a zero counter lies beyond the unit marker. This report supplies a different certified interface: every instruction begins and ends in the fixed order $L < X < Y < R$, with literal label restoration and one scale. Its specific additions are the two exact branching chambers, a nondestructive test and guard coverage for the entire encoded state set. No novelty, priority, minimum-population or exhaustive-literature claim is made.

The preceding local-realization proof (retained with the evidence) establishes physical map gadgets around suitable points. Local existence alone does not imply a single encoding covered by every future instruction guard, nor does it implement a branch. Those are precisely the issues treated here. The argument below is self-contained for the test and the update primitives used by the compiler. [write] [The local-realization proof is source 33’s (Part XI, Theorem 281.1); the copy that source 34 retained is shipped once, as 33-homogeneous-science-frozen-proof-PROOF.md.]

295 An explicit changing test

We first permit X to change position. Section 297 undoes that change physically while preserving the outcome in finite labels. Only L , X , Y and the messenger participate; R is a stationary spectator.

295.1 Labels, speeds and local rules

[write] [Z and N are finite branch tags (zero and nonzero tested counter), not the sets $\mathbb{Z}$ and $\mathbb{N}$. The messenger labels C_Z, C_N are unrelated to the conditional-state count C and the control residuals C_t of Section 301.]

The stationary target label is X_0 . Its moving labels and speeds are

$$X_{\text{pre}} : -\frac{1}{2}, \quad X_{\text{fast}} : +2, \quad X_{\text{post}} : -\frac{1}{2}.$$

The messenger labels are distinct, including those with equal speeds:

label	Q_0	Q_1	Q_2	Q_3	Q_4	C_Z	C_N
speed	1	$-3/2$	1	$-1/4$	4	-1	-1

The rules preceding the final anchor contact are

$$\begin{aligned}
 \{Q_0, X_0\} &\longrightarrow \{Q_1, X_{\text{pre}}\}, \\
 \{Q_1, L\} &\longrightarrow \{Q_2, L\}, \\
 \{Q_2, X_{\text{pre}}\} &\longrightarrow \{Q_3, X_{\text{fast}}\}, \\
 \{Q_3, L\} &\longrightarrow \{Q_4, L\}, \\
 \{X_{\text{fast}}, Y\} &\longrightarrow \{X_{\text{post}}, Y\}, \\
 \{Q_4, X_{\text{post}}\} &\longrightarrow \{C_Z, X_0\}, \\
 \{Q_4, X_{\text{fast}}\} &\longrightarrow \{C_N, X_0\}.
 \end{aligned} \tag{295.1}$$

At L , each C_c emits a distinct speed-+1 phase. In the nondestructive gadget this is the reverse-entry phase in Section 297; the anchor contact is a single interface event, not two instantaneous events.

The rule with input $\{X_{\text{fast}}, Y\}$ changes only those two participants. The branch is learned later, when Q_4 meets either X_{post} or X_{fast} . Thus branch acquisition is local even though its geometric cause is an earlier marker-only contact.

295.2 The common prefix

The messenger begins at L with speed +1. Table 35 specifies all positions needed for the first three events. The other markers stay at $L = 0$, $Y = y$, $R = D$. After event 3, the messenger travels left at $-1/4$ and X right at $+2$. The messenger reaches L at

$$B = \frac{35x}{9}; \quad A = \frac{17x}{9} + \frac{y}{2} \tag{295.2}$$

is the prospective time at which this fast X reaches Y . Since

$$A - B = \frac{y - 4x}{2},$$

both selected chambers have the anchor bounce first. At B , the target is at $4x$ and the messenger acquires speed +4. If the target were to remain fast, they would meet at time $53x/9$ and position $8x$. Consequently $y < 8x$ gives the X/Y contact first, whereas $y > 8x$ gives the messenger/fast-target contact first. This comparison is made only after the common prefix and anchor bounce are valid.

Event	Time	Messenger	X	Contact
Section	0	0	x	outgoing L, Q_0
1	x	x	x	Q_0, X_0
2	$5x/3$	0	$2x/3$	Q_1, L
3	$19x/9$	$4x/9$	$4x/9$	Q_2, X_{pre}

Table 35: Common prefix of the two changing-test words. The initial section is not counted as a new event.

295.3 The Z word: $4x < y < 8x$

Set

$$F = \frac{10y - 8x}{9}. \quad (295.3)$$

The word continues as in Table 36. Its seven positive flight durations are

$$x, \quad \frac{2x}{3}, \quad \frac{4x}{9}, \quad \frac{16x}{9}, \quad \frac{y - 4x}{2}, \quad \frac{2(8x - y)}{9}, \quad F. \quad (295.4)$$

The return map is $(D, x, y) \mapsto (D, F, y)$. Its determinant is $-8/9$, as expected for the odd-length seven-event word; this determinant is a consistency check, not a substitute for chronology. [write] [The sign $(-1)^7$ is the parity rule of source 31 (Theorem 257.2); the six-event N word below has determinant $+8$ accordingly.]

Event	Time	Messenger	X	Contact
4	$35x/9$	0	$4x$	Q_3, L
5	$17x/9 + y/2$	$2y - 8x$	y	X_{fast}, Y
6	$11x/3 + 5y/18$	F	F	Q_4, X_{post}
7	$25(2x + y)/18$	0	F	C_Z, L

Table 36: Remainder of the changing-test word on $\mathcal{P}_Z$.

295.4 The N word: $8x < y$

There is no X/Y contact on this word. Table 37 completes the prefix. The six positive flight durations are

$$x, \quad \frac{2x}{3}, \quad \frac{4x}{9}, \quad \frac{16x}{9}, \quad 2x, \quad 8x. \quad (295.5)$$

The return map is $(D, x, y) \mapsto (D, 8x, y)$, with determinant $+8$ and six events.

Event	Time	Messenger	X	Contact
4	$35x/9$	0	$4x$	Q_3, L
5	$53x/9$	$8x$	$8x$	Q_4, X_{fast}
6	$125x/9$	0	$8x$	C_N, L

Table 37: Remainder of the changing-test word on $\mathcal{P}_N$.

296 Complete chronology and exact chambers

296.1 Why the endpoint checks suffice

Throughout either changing-test word, the strands have weak spatial order

$$L, Q, X, Y, R,$$

with exactly the indicated adjacent contact at an event and every other gap strictly positive. This is an order of live strands, not of finite phase labels. At common events 1–3 the positive target positions x , $2x/3$, $4x/9$ lie below y ; at event 4, $X = 4x < y$.

At the marker-only Z event 5,

$$0 < 2y - 8x < y$$

because $4x < y < 8x$. At event 6,

$$F > 0, \quad y - F = \frac{8x - y}{9} > 0. \quad (296.1)$$

At the final anchor bounce $X = F$ stays strictly between L and Y . For N , the inequality $0 < 8x < y$ certifies the final two rows. In both cases R stays beyond Y .

Every adjacent gap is affine during a flight. If positive at both endpoints it is positive throughout. A departure gap has positive outgoing derivative; an arrival gap has negative incoming derivative. Since collision partners have distinct speeds, a gap zero at just one endpoint is strictly positive in the open flight. Hence no additional adjacent gap can vanish, and a nonadjacent meeting is impossible without an adjacent meeting. Together with (295.4) and (295.5), this proves the *complete* collision chronology, including the absence of simultaneous remote contacts.

296.2 Positive coefficient certificates

The chamber coordinates make the strict inequalities checkable without numerical samples. On $\mathcal{P}_Z$ put

$$\begin{aligned} \alpha &= y - 4x > 0, & \beta &= 8x - y > 0, & \gamma &= D - y > 0, \\ x &= (\alpha + \beta)/4, & y &= 2\alpha + \beta, & D &= 2\alpha + \beta + \gamma. \end{aligned} \quad (296.2)$$

On $\mathcal{P}_N$ put

$$\begin{aligned} \alpha &= x > 0, & \beta &= y - 8x > 0, & \gamma &= D - y > 0, \\ x &= \alpha, & y &= 8\alpha + \beta, & D &= 8\alpha + \beta + \gamma. \end{aligned} \quad (296.3)$$

Every noncontact event gap and every flight duration is a nonzero linear combination of these three positive quantities with nonnegative rational coefficients. For example, in (296.2), the fourth-event target distance is $4x = \alpha + \beta$, the fifth-event messenger distance is 2α , and

$$F = 2\alpha + \frac{8\beta}{9}, \quad y - F = \frac{\beta}{9}.$$

The arithmetic evidence records coefficient rows for every gap, rather than testing a selection of counter values.

296.3 Necessity and the first forbidden event

The stated chambers are exact for the two selected complete words. Once the common prefix is valid, having event 4 precede the marker-only contact requires $y > 4x$. Having that contact precede the fast messenger catch then requires $y < 8x$. Together with the initial ordering these give precisely $\mathcal{P}_Z$. The $\mathbf{N}$ word requires the fast messenger catch before any X/Y contact, hence $y > 8x$, giving precisely $\mathcal{P}_N$.

At $y = 4x$ the messenger/ L contact and the X/Y contact are simultaneous at different locations. At $y = 8x$ the messenger, X and Y meet together. Both boundaries are excluded. There is a lower post-contact chamber $x < y < 4x$ for the underlying machine, but its word is neither claimed as part of this test interface nor needed for the counter encoding. [write] (See item 2 of Section 317.) Necessity is proved at the *first* ordering failure; no invalid trajectory is continued through an earlier wrong event.

297 The physical inverse and the retained branch

297.1 The anchor interface is a real collision

Negating velocities at an arbitrary outgoing section would be an externally imposed operation. Instead, the last forward anchor collision is used as the phase interface. Its incoming messenger speed is -1 , so it can emit a reverse-entry messenger at $+1$. The reverse phase first undoes the preceding *non-anchor* event. At the other end, a new L collision emits the desired forward output phase. There is no zero-time duplicate event at either join.

For $c \in \{Z, N\}$ introduce fresh reverse labels. A bar denotes the negative of the corresponding forward speed, and the subscript c preserves the branch. Stationary base marker labels are retained. In particular $\bar{C}_c$ has speed $+1$. The final forward anchor rule is

$$\{C_c, L\} \longrightarrow \{\bar{C}_c, L\}. \quad (297.1)$$

The two reverse-entry rules are

$$\begin{aligned} \{\bar{C}_Z, X_0\} &\longrightarrow \{\bar{Q}_{4,Z}, \bar{X}_{\text{post},Z}\}, \\ \{\bar{C}_N, X_0\} &\longrightarrow \{\bar{Q}_{4,N}, \bar{X}_{\text{fast},N}\}. \end{aligned} \quad (297.2)$$

The additional marker-only rule for Z is

$$\{\bar{X}_{\text{post},Z}, Y\} \longrightarrow \{\bar{X}_{\text{fast},Z}, Y\}. \quad (297.3)$$

For either branch c , use

$$\begin{aligned}
 \{\overline{Q}_{4,c}, L\} &\longrightarrow \{\overline{Q}_{3,c}, L\}, \\
 \{\overline{Q}_{3,c}, \overline{X}_{\text{fast},c}\} &\longrightarrow \{\overline{Q}_{2,c}, \overline{X}_{\text{pre},c}\}, \\
 \{\overline{Q}_{2,c}, L\} &\longrightarrow \{\overline{Q}_{1,c}, L\}, \\
 \{\overline{Q}_{1,c}, \overline{X}_{\text{pre},c}\} &\longrightarrow \{\overline{Q}_{0,c}, X_0\}, \\
 \{\overline{Q}_{0,c}, L\} &\longrightarrow \{Q_{\text{out},c}, L\}.
 \end{aligned} \tag{297.4}$$

Both output phases have speed $+1$. They may be different next-instruction labels or entry phases of later blocks. All rules are binary and have distinct outgoing speeds.

297.2 A literal reflected chronology

Write the forward event times as

$$0 = t_0 < t_1 < \dots < t_m = T.$$

After the outgoing interface at time T , the reverse events occur at elapsed times

$$T - t_{m-1}, T - t_{m-2}, \dots, T - t_1, T. \tag{297.5}$$

These are elapsed *event times*, not successive flight durations. The reverse flights themselves have lengths

$$t_m - t_{m-1}, t_{m-1} - t_{m-2}, \dots, t_1 - t_0.$$

They are all positive. The first $m - 1$ inverse contacts occupy the forward event positions in reverse order. The last is the fresh terminal L bounce. Between contacts the complete five-strand geometry is the forward geometry at the reflected time, with all moving velocities negated. Strict separations are therefore inherited exactly.

At the terminal section, X is again at x , Y at y , and the anchors at $0, D$. Every stationary marker has its original label. Branch tags remain in the messenger despite the common reverse prefix geometry. The output coordinate map is the identity on all three homogeneous coordinates, with no residual moving marker, shifted scale or hidden marker phase.

The changing word and its inverse use $2m$ events in total: 14 for Z and 12 for N . Their durations are

$$T_{\text{test},Z} = \frac{25(2x + y)}{9}, \quad T_{\text{test},N} = \frac{250x}{9}. \tag{297.6}$$

This proves Theorem 294.2. The method uses the standard opposite-speed reversal principle [5]; the explicit boundary rules and branch tags are essential here. It constructs an inverse *gadget* using fresh labels. It does not claim that the entire collision table, after completion on unused inputs, is globally reversible. [write] [Source 31 (Part XI) draws the same line between the invertibility of a fixed word and global reversibility.]

297.3 A readable spacetime view

Figure 20 draws one point of each chamber from the proved tables, followed by its reflected geometry. The figure is illustrative; the symbolic tables and gap inequalities, not the drawing, certify all inputs.

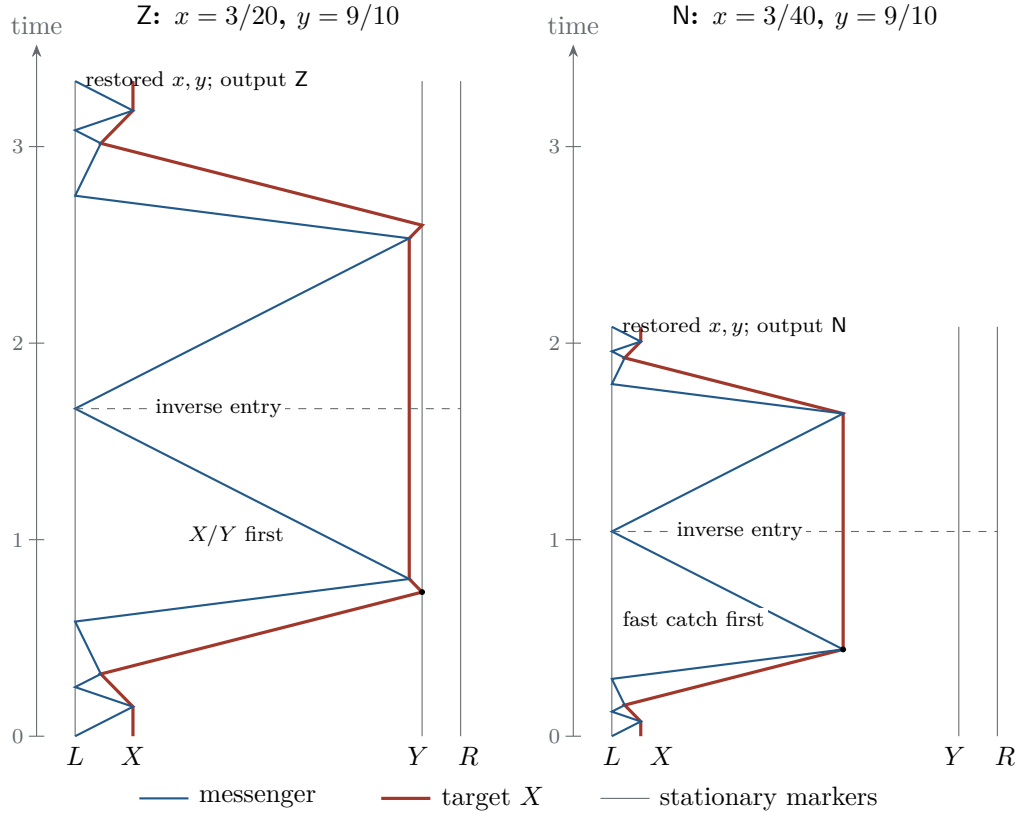


Figure 20: Two nondestructive tests at $D = 1$, with time increasing upward on the same vertical scale. The dashed line is the existing forward terminal anchor event, which emits the inverse-entry phase. The final anchor event restores the original ordered section and emits the branch output. Exact rational coordinates are substituted into the tables only for this analytic illustration; no collision scheduler or simulator generates it. The noncontact bend on a messenger trace at the marker-only event simply marks the same global event time.

298 An encoding covered by every test guard

Local gadget validity is not enough for a computation: every state reached later must stay in a certified guard. The encoding (294.2) makes this invariant explicit.

298.1 Two uniformly separated bands

For every finite nonnegative pair (a, b) ,

$$\frac{D}{20} < x \leq \frac{3D}{20}, \quad \frac{17D}{20} \leq y < \frac{19D}{20}. \quad (298.1)$$

Even the closure of this rectangle lies strictly inside $0 < x < y < D$. As a counter tends to infinity, its encoding tends to an ordinary interior shape; the limiting section does not merge markers. This does not make the encoding noise robust: resolving unbounded counters still requires arbitrarily fine exact distances.

When $a = 0$, $x = 3D/20$, so

$$y - 4x \geq \frac{D}{4}, \quad 8x - y > \frac{D}{4}, \quad x \geq \frac{D}{20}, \quad D - y > \frac{D}{20}. \quad (298.2)$$

Every zero- A state lies in $\mathcal{P}_Z$, uniformly over b . When $a > 0$, $x \leq D/10$, so

$$y - 8x \geq \frac{D}{20}, \quad x > \frac{D}{20}, \quad D - y > \frac{D}{20}. \quad (298.3)$$

Every positive- A state lies in $\mathcal{P}_N$. The secondary remote-tie boundary $y = 4x$ and the primary branch boundary $y = 8x$ are both separated from the encoded inputs. The intermediate changing-test section is also nondegenerate: on the zero branch,

$$y - F = \frac{8x - y}{9} > \frac{D}{36};$$

on the positive branch, $y - 8x \geq D/20$. Hence the physical reverse does not conceal a shrinking gap at its entry section.

298.2 Testing the other counter by reflection

Use distance from R as a reflected coordinate. The nearest target and the spectator then have distances

$$r = D - y = D \left(\frac{1}{20} + \frac{1}{10} 2^{-b} \right), \quad s = D - x = D \left(\frac{19}{20} - \frac{1}{10} 2^{-a} \right). \quad (298.4)$$

They satisfy exactly the same inequalities as (x, y) . Negating all physical speeds and exchanging the left/right roles realizes the same test for b . Reflection changes neither flight durations nor distance-coordinate guards.

The transfer from the standard L section to the reflected R section has three contacts: cross X , cross Y , bounce at R . It takes time D . The return transfer crosses Y , crosses X , then bounces at L , also taking three events and time D . Transfer crossing rules preserve the marker and use fresh messenger phases. Their endpoint bounce selects the next block's entry phase. All six crossings and bounces are counted; none is a remote relabeling.

Thus one encoding simultaneously covers every A and B test, independently of counter magnitudes. No finite-cover argument is inferred from local realization, and no initial counter value is used to choose a different compiled table.

299 Explicit uniformly guarded counter updates

Use distance coordinates from an anchor $A = 0$, with a nearest target M_0 at t , a stationary spectator S at s , and a far reflector $B = D$, where $0 < t < s < D$. These are roles occupied by the existing four markers; no extra marker is introduced. [write] [In this section A and B name the anchor and the far reflector, not the two counters A, B of Section 300, and h is a speed, not the halt edge of Section 301.] Reflection gives the right-anchor versions.

299.1 Four-event anchored scaling

For fixed rational $k > 0$, put

$$h = \frac{k-1}{k+1}, \quad -1 < h < 1.$$

Give the target label M_h speed h . Fresh messenger labels K_0, K_1, K_2, K_3 have respective speeds $+1, -1, +1, -1$. The complete rule schema is

$$\begin{aligned} \{K_0, M_0\} &\longrightarrow \{K_1, M_h\}, \\ \{K_1, A\} &\longrightarrow \{K_2, A\}, \\ \{K_2, M_h\} &\longrightarrow \{K_3, M_0\}, \\ \{K_3, A\} &\longrightarrow \{K_{\text{out}}, A\}. \end{aligned} \tag{299.1}$$

Here K_{out} has speed $+1$. The collision times and positions are

$$(t, t), \quad (2t, 0), \quad (2t + kt, kt), \quad (2t + 2kt, 0). \tag{299.2}$$

The target identity is $t + h(t + kt) = kt$. Its moving segment is monotone between t and kt . The exact guard is

$$0 < t < s < D, \quad 0 < kt < s. \tag{299.3}$$

These endpoint inequalities make every flight positive and exclude every spectator contact. If an endpoint is obstructed, an adjacent contact or a collapsed/reordered flight occurs no later than the proposed restoration. The operation fixes s, D , restores the base labels and has duration $2(1+k)t$.

299.2 Ten-event translation

For rational $e < 1$, the translation $t \mapsto t + eD$ has exact guard

$$0 < t < s < D, \quad 0 < t + eD < s. \tag{299.4}$$

Put

$$u = \frac{t}{1-e}, \quad h = \frac{e}{2-e}. \tag{299.5}$$

First scale t to u using (299.1) with $k = 1/(1-e)$; this uses target speed h . Its final anchor bounce emits V_0 at speed $+1$.

For the remaining six-event part, take messenger speeds

$$V_0, V_1, V_2 : +1, \quad V_3, V_4, V_5 : -1.$$

Use a fresh moving target label M'_h of speed h , with rules

$$\begin{aligned}
 \{V_0, M_0\} &\longrightarrow \{V_1, M'_h\}, \\
 \{V_1, S\} &\longrightarrow \{V_2, S\}, \\
 \{V_2, B\} &\longrightarrow \{V_3, B\}, \\
 \{V_3, S\} &\longrightarrow \{V_4, S\}, \\
 \{V_4, M'_h\} &\longrightarrow \{V_5, M_0\}, \\
 \{V_5, A\} &\longrightarrow \{V_{\text{out}}, A\}.
 \end{aligned} \tag{299.6}$$

The output speed is $+1$. The six messenger positions are

$$u, \quad s, \quad D, \quad s, \quad t + eD, \quad 0.$$

The six flight lengths (and times, because the messenger speed has magnitude one) are

$$u, \quad s - u, \quad D - s, \quad D - s, \quad s - (t + eD), \quad t + eD. \tag{299.7}$$

The target restoration identity is

$$u + h[2D - (t + eD) - u] = t + eD. \tag{299.8}$$

The hidden intermediate position is guarded by the same endpoint inequalities, since

$$u - t = \frac{et}{1 - e}, \quad (t + eD) - u = \frac{e[D - (t + eD)]}{1 - e}.$$

Both differences have the sign of e , so u lies between t and $t + eD$. The target then stays strictly inside $(0, s)$. Every spectator crossing is explicitly included, and the target cannot meet either anchor. The complete translation takes ten events and duration

$$2 \left[t + \frac{t}{1 - e} \right] + 2D. \tag{299.9}$$

This is the needed nearest-target primitive from the retained Report 63 [source 33] proof, with its rules and guard re-established here. [write] [It is source 33's Lemma 282.1.] Only the rational values used below enter the final compiler.

299.3 Increment and positive decrement

For an encoded distance

$$t = \frac{D}{20} + \frac{D}{10} 2^{-n},$$

increment $n \mapsto n + 1$ is the two-block operation

$$t \mapsto \frac{t}{2} \mapsto \frac{t}{2} + \frac{D}{40}. \tag{299.10}$$

It uses scaling $k = 1/2$ and translation $e = 1/40$. Since $D/20 < t \leq 3D/20$, its scale endpoint is at most $3D/40$ and its final endpoint belongs to $(D/20, D/10]$. All remain positive and strictly below the spectator distance $s \geq 17D/20$.

For $n \geq 1$, positive decrement $n \mapsto n - 1$ is

$$t \mapsto 2t \mapsto 2t - \frac{D}{20}. \quad (299.11)$$

It uses scaling $k = 2$ and translation $e = -1/20$. The positive branch gives $D/20 < t \leq D/10$. Its scale endpoint is in $(D/10, D/5]$, and its final endpoint in $(D/20, 3D/20]$. Again both guards hold uniformly, including the intermediate scale inside the translation. No positive-decrement claim is made for a zero input.

Each update has $4 + 10 = 14$ events, preserves the other counter, and restores the full anchor section. The target speeds in distance coordinates are

$$-\frac{1}{3}, \quad +\frac{1}{3}, \quad \frac{1}{79}, \quad -\frac{1}{41}. \quad (299.12)$$

For clarity, substituting into the block durations yields

$$\begin{aligned} T_{\text{inc}} &= 2D + \frac{196}{39}t, \\ T_{\text{dec}} &= 2D + \frac{290}{21}t. \end{aligned} \quad (299.13)$$

The admitted ranges prove $2D < T_{\text{update}} < 4D$. For example, the increment coefficient multiplies at most $3D/20$ and the decrement coefficient at most $D/10$. The strict lower bound follows from $t > 0$. Reflecting the same blocks updates counter B .

300 One finite table and unbounded computation

300.1 The instruction semantics

The deterministic instruction set is

INC A ; goto j , INC B ; goto j ,
 if $A = 0$ then goto j_0 , else $A := A - 1$ and goto j_1 ,
 if $B = 0$ then goto j_0 , else $B := B - 1$ and goto j_1 ,
 HALT .

An instruction section uses the base stationary labels and messenger Q_i at L outgoing at speed $+1$. The index i is the finite control state. For each instruction and branch make fresh copies of all temporary phases and moving-marker labels needed by its blocks. The copies enlarge the finite alphabet but do not create additional live signals.

An interface is implemented by selecting the outgoing messenger label at the existing final anchor contact of the preceding block. It does not add another event at the same spacetime point. Table 38 gives the complete block compilation and its event counts.

Every marker-only rule has an instruction-specific moving-marker input, and the inverse marker labels are branch tagged. Thus different contexts do not force conflicting outputs on the same input. No collision changes a label at a distance. The explicit rules either have distinct inputs or agree on their prescribed output; fresh phase copies ensure distinctness whenever contexts differ.

Instruction branch	Physical blocks	Events
INC A	Left update; final bounce emits Q_j	14
INC B	Transfer to R , reflected update, transfer to L	20
Conditional A , zero	Left nondestructive Z test; emit Q_{j_0}	14
Conditional A , positive	Left N test, positive decrement; emit Q_{j_1}	26
Conditional B , zero	Transfer to R , reflected Z test, transfer to L	20
Conditional B , positive	Transfer to R , reflected N test, positive decrement, transfer to L	32

Table 38: Event counts include all transfer crossings and bounces. The nondestructive test already counts its forward/inverse interface once.

300.2 Completion of the finite table

Complete the table by identity rules on every otherwise unspecified finite label set with pairwise distinct speeds. The alphabet is finite, so this completion is finite and deterministic. It preserves cardinality. The complete chronology proofs ensure that these default rules insert no extra event along an encoded execution: there is no additional meeting at which one would be used.

A single common speed set for every compiled program is

$$\{0\} \cup \pm \left\{ 1, \frac{1}{3}, \frac{1}{79}, \frac{1}{41}, \frac{3}{2}, \frac{1}{4}, 4, \frac{1}{2}, 2 \right\}. \quad (300.1)$$

Its nine positive magnitudes, their negatives and zero give 19 speeds. Forward testing, reverse testing and reflection account for the signs; the update speeds are (299.12). The number of finite labels depends on the program, but these 19 speed values do not. This is an upper bound, not an optimization result.

300.3 Inductive simulation proof

The initial section satisfies (294.2). At each nonhalting instruction, Section 298 selects the correct zero or positive chamber for every encoded counter pair. The test restores the same coordinates. Section 299 implements precisely the instructed update, with every intermediate guard valid. The block interfaces restore the standard section and emit exactly the correct next control label. Induction on instruction count proves correctness for every finite computation prefix and hence for any unbounded run.

For rational D , every initial position is rational and every finite event time and collision position is rational: all speeds and all displayed affine formulas are rational. The theorem also permits real positive scale as a geometric rescaling; it makes no arbitrary-real-input universality claim.

300.4 Uniform time and the no-accumulation conclusion

Equations (297.6) and (298.1) imply

$$D < T_{A\text{-test}} < 4D.$$

For the zero branch this follows from $x = 3D/20$ and $17D/20 \leq y < 19D/20$. For the positive branch, $D/20 < x \leq D/10$ gives $25D/18 < T < 25D/9$. [write] (The upper bound is attained

at $a = 1$; read $\leq$. See Remark 317.1.) Adding two transfers gives $3D < T_{B\text{-test with transfers}} < 6D$. Updates have duration strictly between $2D$ and $4D$, and each transfer takes exactly D . Therefore every nonhalting instruction satisfies

$$D < T_{\text{instruction}} < 10D. \quad (300.2)$$

The largest block count in Table 38 is 32.

Before halt every position remains in $[0, D]$: the changing-test tables and reflected inverse do; an update's moving target stays between its anchor and spectator; transfers traverse only that interval. The scale D never changes. If s_n is the time of the n th completed instruction, then $s_n > nD$ for $n > 0$. Thus any bounded time interval contains only finitely many instructions, each having finitely many positively separated events. An infinite encoded computation has no finite-time collision accumulation, including no accumulation hidden inside an instruction. No post-accumulation continuation is used.

300.5 What halting means

The principal convention is reachability of the full section with a designated control label Q_H . This is a precise observable section event, not a claim that all five physical strands stop moving.

A finite-collision convention is also available: assign Q_H speed $+1$ and identity crossings with X, Y, R . The messenger then leaves to the right, with exactly three subsequent crossings and no later collision. The four markers remain, so the live population is still five. Spatial confinement after that halt is not claimed. A separately specified harmless bounce cycle could retain confinement, but then finite-collision halting would no longer be the convention. These alternatives are distinct and must not be silently combined.

301 A quartic certificate for a fixed instruction horizon

This arithmetic result is independent of the physical implementation. Its final corollary transports a native finite-trace certificate through Theorem 294.1. The horizon indexes a family of polynomials with growing witness arity; it is not an additional input to one fixed-arity polynomial.

301.1 Fixed transition data and positive inputs

Fix a deterministic two-counter program M , an initial control state q_0 and one designated halt state H . Assign distinct integer codes to the states. Split each conditional into a zero edge and a positive-decrement edge, and add exactly one absorbing halt edge h . This edge is *virtual padding* of a terminated native trace. It does not describe the post-halt motion of the physical escape convention.

Let the E edges have fixed data

$$(u_e, v_e, d_e, f_e), \quad e = 1, \dots, E,$$

where u_e, v_e are source/target state codes, and $d_e, f_e \in \{-1, 0, 1\}$ are the counter changes. An increment state has one edge. A conditional has exactly two edges: its zero edge has both changes zero and a zero guard on the tested counter; its positive edge decrements that counter. The halt edge has $u_h = v_h = H$ and $d_h = f_h = 0$. Even if conditional targets coincide, the two guarded

edges remain distinct. Duplicate copies of an enabled edge are not allowed, since witnesses identify individual edges.

Let E_{A0} and E_{B0} be the zero-edge sets, and put $Z = |E_{A0}| + |E_{B0}|$. If I and C count increment and conditional states, excluding the halt state, then

$$E = I + 2C + 1, \quad Z = C. \quad (301.1)$$

The native input counters are $a = A - 1$, $b = B - 1$, where A, B are positive integers.

Theorem 301.1 (Unique bounded-horizon witness). *For each fixed $T \geq 1$ there is an explicit integer polynomial*

$$P_{M,T}(A, B; W)$$

of total degree exactly four, including its two native input variables, such that $P_{M,T} = 0$ has a positive-integer witness if and only if M reaches H within T transitions from $(q_0, A - 1, B - 1)$. Whenever a witness exists it is unique. The construction has exactly

$$\begin{array}{ll} T(E + 2) & \text{positive witness variables,} \\ T(E + Z + 4) + 1 & \text{residual equations before bundling,} \\ 1 & \text{final polynomial equation.} \end{array} \quad (301.2)$$

A first-halt-exactly- T variant has the same witnesses and degree, with one additional residual. The $T = 0$ case is constant and is treated separately below.

[write] [The device—one-hot step selectors, affine counter updates, positivity of the next shifted counter as the decrement guard, a sum of squares with a unique witness, and arity growing with the horizon—is that of Part V’s Theorem 58.1 and Part III’s Theorem 36.1; the object, a two-counter program with native shifted counters, is new here. In this section E and Z are counts, not sets, and C counts conditional states (Equation (301.1)); source 35 writes the same count J . Part XIII replaces the $T(E + 2)$ witnesses by three, at a degree that grows with the horizon (Theorem 318.1).]

301.2 Every witness and residual explicitly

For $t = 0, \dots, T - 1$ and $e = 1, \dots, E$, introduce a positive integer $w_{t,e}$ and abbreviate

$$s_{t,e} = w_{t,e} - 1.$$

For $t = 1, \dots, T$, introduce positive integer witnesses A_t, B_t . Set $A_0 = A$, $B_0 = B$ as abbreviations for the inputs, not new witnesses. There are $TE + 2T$ witnesses in total.

The selector and one-hot residuals are

$$S_{t,e} = (w_{t,e} - 1)(w_{t,e} - 2), \quad 0 \leq t < T, \quad 1 \leq e \leq E, \quad (301.3)$$

$$O_t = \sum_e s_{t,e} - 1, \quad 0 \leq t < T. \quad (301.4)$$

Vanishing forces $w_{t,e} \in \{1, 2\}$, hence $s_{t,e} \in \{0, 1\}$, with exactly one selected edge per step. Inactive selectors are uniquely $w = 1$; they introduce no free padding values.

The exact counter-update residuals are affine:

$$U_t = A_{t+1} - A_t - \sum_e d_e s_{t,e}, \quad (301.5)$$

$$V_t = B_{t+1} - B_t - \sum_e f_e s_{t,e}, \quad 0 \leq t < T.$$

If the selected edge decrements A , then $A_{t+1} = A_t - 1 > 0$, so $A_t \geq 2$. Thus the native entering counter is positive. The next-counter positivity already implements the positive-decrement guard, with no extra residual or witness. The same applies to B . One-hot selection is essential; it prevents sums of displacements from disguising an invalid decrement.

For every step and each zero edge include

$$Z_{t,e} = \begin{cases} s_{t,e}(A_t - 1), & e \in E_{A0}, \\ s_{t,e}(B_t - 1), & e \in E_{B0}. \end{cases} \quad (301.6)$$

A selected zero edge forces the native counter to vanish. An inactive zero edge contributes the zero residual without constraining that counter.

Finally include control residuals

$$\begin{aligned} C_{\text{initial}} &= \sum_e u_e s_{0,e} - q_0, \\ C_t &= \sum_e v_e s_{t,e} - \sum_e u_e s_{t+1,e}, & 0 \leq t < T - 1, \\ C_{\text{final}} &= \sum_e v_e s_{T-1,e} - H. \end{aligned} \quad (301.7)$$

There are $1 + (T - 1) + 1 = T + 1$ such equations, also when $T = 1$. Distinct state codes and one-hot selection ensure these compare actual states, not weighted averages of codes. State codes are fixed coefficients, not additional witnesses.

301.3 Correctness and uniqueness

Suppose the native computation first reaches H after $k \leq T$ transitions. Follow its unique edge sequence for k steps and the unique halt loop for the remaining $T - k$ steps. Set each selected edge's w to 2, every other w to 1, and A_t, B_t to the native counter values after t padded steps, plus one. All are positive. Each displayed residual vanishes. If $k = 0$, the whole padded sequence consists of halt edges.

Conversely, vanishing selectors and one-hot equations determine a single edge e_t at every step. The control residuals start its path at q_0 , join consecutive source and target states, and end it at H . The counter equations implement exactly that path's updates. A selected zero edge is valid by (301.6); a selected decrement is valid by positivity of the next shifted counter. Increment and halt edges are unconditionally valid. Hence the witnesses define a genuine padded native computation, and deleting its halt-loop suffix gives a first halt no later than T .

At an increment state one edge is possible. At a conditional with shifted tested counter 1, the zero edge is valid and a decrement would make the next shifted counter zero. At shifted counter at least 2, the zero edge is impossible and exactly the decrement edge is valid. At H there is one padding edge. Induction from the fixed input therefore fixes every edge and next counter pair uniquely. It then fixes every w uniquely to 1 or 2. This proves uniqueness of the entire positive witness tuple, including the inactive selectors.

301.4 One polynomial, exact degree and exact counts

Let $\mathcal{R}$ be the explicitly displayed list of residuals (301.3)–(301.7), and define

$$P_{M,T} = \sum_{r \in \mathcal{R}} r^2. \quad (301.8)$$

This specifies the integer polynomial without needing a monomial expansion. A sum of squares of integers is zero exactly when all residuals are zero.

Residual group	Count	Degree before squaring
Selectors	TE	2
One-hot selection	T	1
Two counter updates	$2T$	1
Zero guards	TZ	2
Initial, linking, final control	$T + 1$	1
Total	$T(E + Z + 4) + 1$	at most 2

Table 39: Exact ledger for the displayed positive- T construction. Counts refer to displayed equation slots, including redundant or identically zero residuals for degenerate programs or state codes. They are not irredundancy, rank or minimality bounds.

The total degree is at most four, jointly in inputs and witnesses. It is exactly four: $E \geq 1$ because the halt edge exists, and every squared selector residual contributes $w_{t,e}^4$ with coefficient 1. Other residuals are at most linear in that selector, so their squares cannot cancel its pure fourth power. This proves Theorem 301.1.

There are no state witnesses, sign witnesses, unconstrained padding witnesses or hidden existential bounds. In instruction counts, (301.1) gives

$$\begin{aligned}
 \#W &= T(I + 2C + 3), \\
 \#\mathcal{R} &= T(I + 3C + 5) + 1, \\
 \#\text{variables including inputs} &= 2 + T(I + 2C + 3).
 \end{aligned} \tag{301.9}$$

301.5 Exact first halt, zero horizon and the family boundary

For first halt exactly at $T \geq 1$, add the single linear residual

$$H_{\text{early}} = \sum_{t=0}^{T-1} s_{t,h}. \tag{301.10}$$

Every selector is nonnegative, so $H_{\text{early}} = 0$ forbids any padded halt edge before final arrival. With $C_{\text{final}} = 0$, the first arrival at H must then be the T th transition. Conversely every such trace satisfies (301.10). Thus

$$P_{M,T}^{\text{exact}} = P_{M,T} + H_{\text{early}}^2$$

has the same unique witness property, degree four and $T(E + 2)$ witnesses, with $T(E + Z + 4) + 2$ residuals. If $q_0 = H$ and $T \geq 1$, the by-horizon form has its unique all-loop witness, while the exact-first-halt form has none.

For $T = 0$ use the constant polynomial

$$P_{M,0} = (q_0 - H)^2. \tag{301.11}$$

There are zero witnesses. The empty tuple is a solution precisely when the initial control state is already H , also the first-halt-at-zero condition. The positive-horizon quartic claim and ledger do not apply.

For fixed M , these are effectively specified polynomials indexed by T . Their arity grows with T . It is true that the native program halts if and only if *some member* of the family has a witness. This does not create one fixed-arity Diophantine representation of unbounded halting, nor a finite-fold or single-fold result for such a representation. A packing construction would need to encode and verify a variable-length trace with finitely many integers and account for its own arithmetic modules. No packing, exponentiation, Pell or MRDP module is assumed here. [write] [Source 35 (Section 309) adds two Pell POWER modules that pay for the exponential initialization inside the polynomial (the gaps themselves still have numeric height $\Theta(2^m)$, Theorem 312.1), not to pack the trace, and Part XIII bounds the number of witnesses at each horizon; neither gives one fixed-arity representation of unbounded halting.]

301.6 Transport to physical instruction sections

For positive input A, B initialize the fixed physical compiler by

$$x = D \left(\frac{1}{20} + \frac{1}{10} 2^{-(A-1)} \right), \quad y = D \left(\frac{19}{20} - \frac{1}{10} 2^{-(B-1)} \right). \quad (301.12)$$

This exponential initialization is separate from the polynomial. Its powers are not concealed polynomial terms or witness constraints. Positive rational D gives rational initial coordinates.

Theorem 294.1 now implies that $P_{M,T} = 0$ has its unique positive witness exactly when this encoded physical run reaches the designated halt section within T simulated native instructions. The exact variant characterizes first arrival at the T th instruction section. Only the physical prefix ending at first halt is used; the virtual padding edge is never asserted to be physically executed.

If first halt occurs at $k > 0$, elapsed time lies strictly between kD and $10kD$, with at most $32k$ binary events. If $k = 0$, the initial section already carries the halt label. An instruction horizon is not a fixed collision count, because instruction branches have different event counts. This transport is a bounded native reachability statement through a proved encoding; it is not a polynomial predicate on arbitrary real or rational physical coordinates.

302 Interpretation and further questions

302.1 What the construction establishes

The local changing-test maps are nontrivial affine maps of the homogeneous section data; their physical inverse erases that coordinate change while preserving the finite branch phase. This resolves the data-restoration problem at the level of complete configurations, not just one coordinate. The separated encoding then resolves a different problem: both counters can be updated forever with the same finite rule table, because every encoded state satisfies every needed guard.

The proof uses exact rational geometry and discrete instruction semantics. It is not a physical finite-precision implementation, a noise-robust coding theorem, a proof of minimum population or speed count, a globally reversible completion, or an optimal collision compiler. The five-live-signal count is a prior computational precedent [100]; the inverse principle is also standard [5]. The finite-label/live-occurrence distinction follows the usual model terminology [92]. Any discrete universality corollary must use the precise instruction semantics, a point emphasized in the counter-machine literature [101].

302.2 Questions left open by this proof

1. **Can the same full-section interface use fewer speeds?** The set in (300.1) is chosen for explicit rational arithmetic, not minimality. It is open here whether forward/reverse test phases and both update signs can share a smaller speed alphabet without losing deterministic binary chronology. [write] (Open. Source 35's clock adds ten speed magnitudes, Equation (307.1), and asks the same question for its 39 speeds.)
2. **Can the test avoid undoing its whole geometry?** The current nondestructive primitive pays for a complete physical inverse. A direct data-restoring branch might reduce the 14/12 event counts while keeping both exact chambers and literal stationary-label restoration.
3. **Can all instruction transitions have one constant duration?** The present uniform bounds suffice for no accumulation, but the duration still depends on the encoded shape and branch. An exact constant-time padding block must itself preserve the same full section and use no additional live strand; no such construction is supplied. [write] (Added 4 October 2026: answered for this compiler by source 35, Theorem 304.1. Identity shuttles of the messenger between the four unmoved markers make every instruction last exactly $30D$, at the cost of 39 speeds and at most 54 contacts; smaller clocks stay open, Section 315.)
4. **Can a completed table be globally reversible?** Fresh inverse phases make each selected return invertible as a gadget. Global reversibility of a total finite collision table is a stronger compatibility problem, especially where distinct control paths merge. It is not resolved by adding identity rules. [write] (Open. Part III's machine is globally reversible, but it is a different, universal construction with 18 live signals.)
5. **What quantitative precision is needed for a finite computation?** The guard margins for branching are uniform, yet adjacent counter values converge geometrically. A useful finite-horizon perturbation analysis would separately quantify branch safety, counter distinguishability and accumulated timing error; no robustness statement follows just from the margins.
6. **How should collision and label complexity trade off?** Instruction-specific copying makes deterministic compilation transparent. A smaller finite control alphabet might reuse phases, but only after proving that marker-only rules and returning messengers cannot become ambiguous. No optimality result is known from this argument.

A separate arithmetic question is whether a rigorously costed packing mechanism can replace the growing-arity horizon family by a fixed-arity representation while retaining useful degree or witness bounds. The unique witnesses proved for each horizon alone do not answer it. [write] (Open as posed. Part XIII reduces the witnesses at each horizon to a bounded number—Theorems 318.2, 332.2 and 353.2—but its polynomials are still indexed by the horizon.)

These questions concern stronger properties than the theorems proved here. They are not claims that the properties are impossible.

303 Evidence, preservation and verification boundary

[write] [This is source 34's Appendix A. "The release" is the delivered archive. Its two packets are shipped as 34-branching-science-frozen-proof-* (physical) and 34-branching-science-arithmetic-* (bounded horizon), with their checkers in `code/` and their evidence in `data/`;

the two audits as `34-branching-audits-*`, the manuscript review as `34-branching-manuscript-review-*`, the release-tool review and build records as `34-branching-qa-*`, and the tools as `code/34-branching-tools-*`. The release README, the release manifest, the PDF and its page renders are not shipped; the README of this report records the pins that the release README gave.]

303.1 Two unchanged scientific packets

The physical source is the frozen five-signal branching packet. The bounded-horizon arithmetic companion retains its original development identifier (65), but is integrated into this single Report 64. [write] [The identifier is the packet's directory name `bounded-horizon-diophantine65-20261004`; it does not denote source 35, the pipeline's Report 65, as source 35 also says (Section 313).] Both original proofs, their retained dependencies, author checkers and recorded arithmetic evidence are preserved byte-for-byte. The release README supplies complete SHA-256 pins and exact paths. Hashes authenticate bytes relative to a trusted external pin; they are not digital signatures or mathematical proofs. Historical audit-pending wording in the originals remains unchanged; separate completed dossiers record later acceptance.

303.2 Static arithmetic versus simulation

The frozen physical checker declares rational event rows. Its retained evidence checks displacement identities against prescribed velocities, every adjacent endpoint gap and duration in positive chamber coordinates, designated zero gaps, endpoint maps, inverse identities and uniform update guards. These are finite algebraic certificates. The checker neither chooses a next collision nor evolves an arbitrary configuration. The complete chronology is proved in Section 296.

The physical independent audit accepted the frozen core without mathematical correction. Its 4,424 checks include 56 primitive local rules and a 118-rule/142-label example table, assembled and syntax-checked but never executed. The all-program conclusion remains the namespace and induction argument. [write] [Rerun on a copy at the placement, under Windows with the Linux-only no-access-time read flag removed: 4,424 assertions and the same verdict; the arithmetic audit also passes, with 54 schema instances, 201 finite-polynomial instances and 33,640 candidate words. The README gives the commands.] The arithmetic audit independently verifies the residual schema, quartic coefficients, guards and finite-polynomial candidate words. Its finite checks support, but do not replace, the general proof of positive-witness correctness and uniqueness.

No author scientific program, upstream constructor, native counter interpreter, physical simulator, saved physical collision schedule or proof-assistant program was executed during manuscript preparation. Scientific source files are inert preserved evidence. [write] [The placement ran them only on throw-away copies; their outputs agree with the recorded evidence up to checker self-hashes, line endings and path separators.] The diagram substitutes exact rational coordinates into the proved formulas; it is not generated by a collision scheduler.

303.3 Release and review records

The archive contains the editable LaTeX modules and their deterministic standalone flattening, final PDF, scientific sources, independent audit dossiers and release-only tools. Its README gives review outcomes and portable replay commands. Mathematical audit, manuscript transcription checks and integrity checks are distinct; their overlapping counts are not additive theorem counts.

The typesetting build uses a fresh TeX format with no shell escape and records the dependency union from every pass. A locked rebuild authenticates external manuscript and toolchain pins, checks packaged PDF equality, and renders and validates every page. The dependency lock covers executable bytes and recorded system inputs, not shared libraries, Python’s standard library or the operating system. The deterministic archive and its extractor preserve manifest-bound file and directory metadata.

Independent tool review and all-page manuscript review bind the exact final versions identified in the README and external handoff. All original scientific bytes, modes and nanosecond mtimes are preserved. A final independent terminal audit precedes publication. [write] (No record of it is in the archive; see Section 317, item 3.)

Source 35 (batch-91 manuscript 17): Exact instruction clocks and native gap halting certificates

Two continuations of the fixed ordered five signal compiler, “Research Report 65”, 4 October 2026. [write] From the archive `Clocked_signals_source_and_evidence.zip`, main file `Report65/Report65.tex` (the flattening of the five modules in its `manuscript/`, plus three PDF-metadata commands omitted here). Its labels carry the prefix `smc:ck::`; its sections are Sections 304–315 and a conclusion here; its table of contents is omitted. Two numbered equations without a label in the source carry the labels `smc:ck:n7` and `smc:ck:n10`, after their numbers (7) and (10) in the source’s PDF. Its natural numbers include 0 and are written $\mathbb{N}$; its `\Pos` is this report’s $\mathbb{Z}_{>0} = \mathbb{Z}_{>0}$. Its Lemma 7.1 restates Theorem 301.1 of source 34 and is printed as Remark 310.1, with source 35’s own short proof. Its bibliography item for Report 64 is replaced by references to source 34, and the rest of its bibliography is merged into the report’s. Its diagram colours carry the prefix `smcck`.

We add two explicit refinements to the five-live-signal counter compiler of Report 64 [source 34]. First, stationary identity shuttles pad each instruction to exactly $30D$ time, where D is the preserved anchor distance. The same five signals suffice; at most 54 binary contacts occur per instruction, including identity crossings, and one common set of 39 rational speeds works for every finite source program. A general output-linear padding lemma implements every phase change at a real contact. Second, a bounded-halting polynomial accepts three native positive integer gaps, paying for their exponential decoding inside two fully displayed 15-residual POWER modules. For a fixed program and horizon T , the formula has exactly $T(E + 2) + 54$ positive witnesses, $T(E + Z + 4) + 33$ residuals, and joint degree 12. Here E includes split conditional edges and one virtual halt loop, while Z counts zero edges. Accepted gaps decode uniquely and determine a unique padded trace, but every nonempty full witness fiber is infinite. We also derive the exact primitive integer initialization, its gcd and minimum scale, and its bit-height cost. The predicate includes being encoded, and the horizon indexes a growing-arity family. Neither arbitrary unencoded physical halting nor a fixed-arity unbounded-halting representation is claimed. Five-signal computation and bouncing delays have prior precedents; no priority or optimality claim is made.

304 Results and their precise scope

A finite signal-machine label has one fixed rational speed. A live signal is an occurrence of a label. Our population count is five live signals, regardless of how many finite phase labels a program needs. At an instruction section the four stationary markers and the outgoing messenger are

$$L = 0, \quad X = x, \quad Y = y, \quad R = D, \quad Q_i \text{ at } L \text{ with speed } +1, \quad 0 < x < y < D. \quad (304.1)$$

The literal stationary labels are L, X_0, Y, R . The two outgoing strands at L are counted separately. An instruction is a transition from a nonhalting control state; its destination may be the halt state. All contacts used below are binary, with distinct incoming and outgoing speeds. Identity crossings count as contacts.

Theorem 304.1 (Exactly clocked compiler). *For each fixed finite deterministic two-counter program with increments, zero-test/positive-decrement instructions and a designated halt state, there is a finite rational-speed collision table which simulates every pair $(a, b) \in \mathbb{N}^2$ from the encoding*

$$x = D \left(\frac{1}{20} + \frac{1}{10} 2^{-a} \right), \quad y = D \left(\frac{19}{20} - \frac{1}{10} 2^{-b} \right), \quad D > 0. \quad (304.2)$$

Each instruction takes exactly $30D$, uses at most 54 contacts, and restores the full section with the prescribed new counters and next control label. Exactly five signals remain live. A single set of 39 rational speeds suffices for all finite programs. First halt after k instructions occurs at exactly $30kD$ and after at most $54k$ contacts. An infinite computation has no finite-time collision accumulation.

The clock is constant across counter values, branches and instructions within each run. It scales with the unchanged D ; it is 30 in normalized units $D = 1$. The finite label and rule counts can depend on the program. No instruction is performed at the halt state. Three optional post-halt escape crossings are separate from the $54k$ prefix bound and do not change the first-halt time.

For the arithmetic theorem, fix integer codes for the program states, initial state q_0 and halt state H . Split each conditional into a zero edge and a decrement edge, and add one virtual absorbing edge at H . Let E be the total number of edges and Z the number of zero edges. If there are I increment instructions and J conditional instructions, then $E = I + 2J + 1$ and $Z = J$. [write] [Source 34 writes this count C (Equation (301.1)). Here J and Z are counts, unrelated to the Pell leaves J_A, J_B of Section 309 and to the integers $\mathbb{Z}$.]

Definition 304.2 (Native encoded halting). For external inputs $(g_1, g_2, g_3) \in \mathbb{Z}_{>0}^3$, set

$$D = g_1 + g_2 + g_3, \quad x = g_1, \quad y = g_1 + g_2. \quad (304.3)$$

A triple is encoded when there are $a, b \in \mathbb{N}$ such that

$$\frac{g_1}{D} = \frac{1}{20} + \frac{1}{10} 2^{-a}, \quad \frac{g_3}{D} = \frac{1}{20} + \frac{1}{10} 2^{-b}. \quad (304.4)$$

For a fixed integer $T \geq 0$, $\text{EncHalt}_{M,T}(g)$ means that g is encoded and its decoded computation reaches H in at most T instructions.

Theorem 304.3 (Native positive gap certificate). *The explicit integer polynomial $F_{M,T}$ in Section 311 satisfies*

$$\text{EncHalt}_{M,T}(g_1, g_2, g_3) \iff \exists W \in \mathbb{Z}_{>0}^{T(E+2)+54} F_{M,T}(g; W) = 0.$$

It has three external inputs, exactly $T(E + Z + 4) + 33$ residual equations before sum-of-squares bundling, and joint total degree exactly 12 . The counts include $T = 0$. For $T \geq 1$, first halt exactly at T requires one additional residual, no additional witnesses and no degree increase. Decoded counters and the trace projection are unique, but each accepted input has infinitely many full witnesses.

The three external inputs are unrestricted positive integers; the *predicate itself* includes encoding. An unencoded triple has no witness, independently of any physical behavior outside the encoded simulation. D is the expression in (304.3), not another input. There is no externally supplied exponential initialization. The horizon fixes the number of variables and residuals: this is a growing-arity family, not one fixed-arity polynomial with a free unbounded time variable.

305 The inherited core and its output coordinate costs

Report 64 [source 34] proves a single fixed-ordered-section compiler with uniform guards on (304.2). Its core trajectories, all-input guard proof, literal marker restoration and finite-rule construction are retained unchanged as an explicit dependency. We summarize exactly what is used, then derive the new elapsed-time expressions independently. [write] [The statements summarized in this subsection are source 34's: the scaling and translation guards (299.3) and (299.4), the updates (299.10) and (299.11), the test chambers and durations (Theorem 294.2 and Equation (297.6)), the encoding margins (298.2) and (298.3), and the transfers of Section 298.]

305.1 Core operations and uniform guards

In nearest-target coordinates let t be the target distance, s the spectator distance and D the far anchor distance. The inherited four-contact scaling $t \mapsto kt$ has guard

$$0 < t < s < D, \quad 0 < kt < s,$$

duration $2(1+k)t$, and moving-target speed $(k-1)/(k+1)$. The ten-contact translation $t \mapsto t + eD$ for $e < 1$ has guard

$$0 < t < s < D, \quad 0 < t + eD < s,$$

and duration $2[t + t/(1-e)] + 2D$. Its intermediate target $t/(1-e)$ lies between t and $t + eD$. Each operation restores its anchor section and all stationary labels. Every spectator crossing is included in its count.

For $t = D/20 + (D/10)2^{-n}$, increment is

$$t \mapsto t/2 \mapsto t/2 + D/40,$$

using $k = 1/2, e = 1/40$. Positive decrement is

$$t \mapsto 2t \mapsto 2t - D/20,$$

using $k = 2, e = -1/20$. For all $n \geq 0$, $D/20 < t \leq 3D/20$. On a positive decrement $n \geq 1$ gives $t \leq D/10$, so even the doubled intermediate target is at most $D/5 < s$, since $s \geq 17D/20$. These inequalities guard all endpoints and hidden translation stages.

The inherited nondestructive A test restores its input section with branch-specific outgoing labels. Its zero word has 14 contacts and duration $25(2x + y)/9$, with exact chamber $4x < y < 8x$ inside the ordered section. Its positive word has 12 contacts and duration $250x/9$, with exact chamber $8x < y$. If $a = 0$, then $x = 3D/20$ and $17D/20 \leq y < 19D/20$, which lies in the zero chamber. If $a > 0$, $x \leq D/10$, whence $y - 8x \geq D/20 > 0$. Thus all encoded states satisfy the appropriate strict guard. The B operation reflects the construction in R , using target $D - y$, spectator $D - x$, and two unit-speed transfers with three contacts each.

This report does not replace the core's chronology proof by a timing identity. The complete frozen core proof, including the changing test, its physically implemented inverse and all primitive rules, accompanies the release. [write] [It is source 34's physical packet, shipped once as **34-bran** **ching-science-frozen-proof-PROOF.md**; its arguments are printed in Sections 295–300.] The newly proved padding below has the larger guard $0 < x < y < D$.

305.2 Elapsed time in the available output geometry

Let (D, x', y') be the section *after* a core completes. Padding sees these values. We need

$$T_c = \alpha_c D + \beta_c x' + \gamma_c y'. \quad (305.1)$$

The complete coefficient ledger is Table 40. The nearest-target update durations are

Branch	α_c	β_c	γ_c	Core contacts
A increment	341/195	392/39	0	14
A zero	0	50/9	25/9	14
A positive	383/126	1310/63	0	26
B increment	69/5	0	-392/39	20
B zero	31/3	-25/9	-50/9	20
B positive	155/6	0	-1310/63	32

Table 40: Core time expressed in the output coordinates available at padding entry. A positive branch includes the nondestructive test and decrement.

$$T_{\text{inc}}(t) = 3t + 2 \left[\frac{t}{2} + \frac{t/2}{1 - 1/40} \right] + 2D = 2D + \frac{196}{39}t, \quad (305.2)$$

$$T_{\text{dec}}(t) = 6t + 2 \left[2t + \frac{2t}{1 + 1/20} \right] + 2D = 2D + \frac{290}{21}t. \quad (305.3)$$

Adding the positive-test time gives

$$T_{\text{pos}}(t) = \frac{250}{9}t + T_{\text{dec}}(t) = 2D + \frac{2620}{63}t. \quad (305.4)$$

For A increment, $x' = t/2 + D/40$, hence $t = 2x' - D/20$; substitution gives the first row. For A positive, $x' = 2t - D/20$, hence $t = x'/2 + D/40$; substitution gives the third. The nondestructive zero time is already $25(2x' + y')/9$.

For B write r for the old distance to the B target from R . Two transfers add $2D$. Increment gives

$$r = 2(D - y') - D/20 = 39D/20 - 2y', \quad T = 4D + (196/39)r = (69/5)D - (392/39)y'.$$

Positive decrement gives

$$r = (D - y')/2 + D/40 = 21D/40 - y'/2, \quad T = 4D + (2620/63)r = (155/6)D - (1310/63)y'.$$

The reflected zero test yields

$$T = 2D + \frac{25}{9}[2(D - y') + (D - x')] = \frac{31}{3}D - \frac{25}{9}x' - \frac{50}{9}y'.$$

These identities hold on the admitted domain of the respective core. They do not assert that every branch is meaningful on every ordered section.

306 Stationary shuttles and a general clock padding lemma

Drop primes temporarily and let $a_0 = x$, $b_0 = y - x$, $c_0 = D - y$. All three are strictly positive. The four markers stay at their literal labels and positions throughout padding. Only the messenger moves. For a fixed positive rational v , every listed flight has duration equal to its length divided by v .

Primitive	Contact word	Flight lengths	Duration	Contacts
LXL	X, L	a_0, a_0	$2x/v$	2
LYL	X, Y, X, L	a_0, b_0, b_0, a_0	$2y/v$	4
RXR	Y, X, Y, R	c_0, b_0, b_0, c_0	$2(D - x)/v$	4
RYR	Y, R	c_0, c_0	$2(D - y)/v$	2
LR	X, Y, R	a_0, b_0, c_0	D/v	3
RL	Y, X, L	c_0, b_0, a_0	D/v	3
LRL	X, Y, R, Y, X, L	$a_0, b_0, c_0, c_0, b_0, a_0$	$2D/v$	6

Table 41: Complete padding words. Departure is not a new contact; the terminal contact is included. All identity crossings are counted.

306.1 Every phase change occurs at a contact

Give every occurrence of every primitive fresh messenger labels, tagged by instruction, branch and stage. Let S be an anchor, U its chosen target, and B_0 any intervening marker. A round trip has an outward label p_{out} of speed εv and inward label p_{in} of speed $-\varepsilon v$, where $\varepsilon = 1$ from L and -1 from R . Its rules are

$$\{p_{\text{out}}, U\} \longrightarrow \{p_{\text{in}}, U\}, \quad \{p_{\text{in}}, S\} \longrightarrow \{q_{\text{next}}, S\}, \quad (306.1)$$

$$\{p_{\text{out}}, B_0\} \longrightarrow \{p_{\text{out}}, B_0\}, \quad \{p_{\text{in}}, B_0\} \longrightarrow \{p_{\text{in}}, B_0\}. \quad (306.2)$$

The label q_{next} has the already specified speed of the next primitive. For the full-span loop $S = L, U = R$ and both X, Y are intervening markers. A transfer label p_{tr} has speed $+1$ or -1 , identity crossings at X, Y , and destination rule

$$\{p_{\text{tr}}, U\} \longrightarrow \{q_{\text{next}}, U\}. \quad (306.3)$$

There is no separate departure rule. The preceding terminal contact emits the next departure label. All inputs and outputs consist of one speed-zero marker and a nonzero-speed messenger. Fresh occurrence labels prevent conflicting input rules. Unused finite distinct-speed inputs may receive cardinality-preserving identity rules as in the core.

306.2 Exact chronology on the whole ordered section

Every flight in Table 41 crosses one open interval between consecutive stationary markers, with positive length a_0 , b_0 or c_0 . There is no other marker in its interior. At a crossing the messenger continues in the same direction; at a target it reverses; at a terminal anchor it leaves in the next

specified direction. Consequently the next listed contact is the next physical contact. At a contact with L, X, Y, R , the respective distances to the other markers are

L	$a_0, a_0 + b_0, a_0 + b_0 + c_0$
X	$a_0, b_0, b_0 + c_0$
Y	$a_0 + b_0, b_0, c_0$
R	$a_0 + b_0 + c_0, b_0 + c_0, c_0$

and all are positive. There is only one moving signal, so neither a remote simultaneous contact nor a triple contact can occur. No extra marker or temporary marker phase is introduced. The common guard is precisely the ordered section: its boundary identifies stationary markers or collapses a mandatory full-span flight, violating the strict binary-section convention.

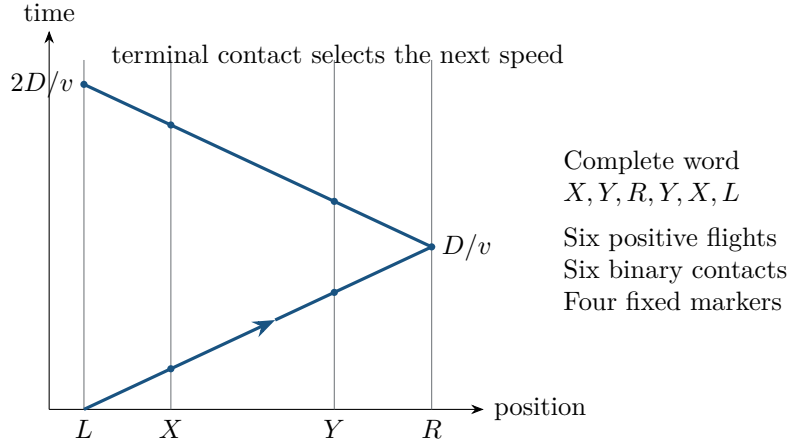


Figure 21: An analytic full-span identity loop. The drawing uses illustrative marker positions; its coordinates follow the exact formulas $t = z/v$ outward and $t = (2D - z)/v$ inward. It is not a simulated or stored event schedule.

Lemma 306.1 (Output-linear clock padding). *Suppose finitely many implemented cores return to (304.1) with an outgoing +1 messenger at L , and branch c has time (305.1) in its output coordinates. If the rational constant $C > 0$ satisfies*

$$C > 4 + \alpha_c + \max(\beta_c, 0) + \max(\gamma_c, 0) \quad (306.4)$$

for every branch, each core admits five-live-signal identity padding to total time CD . Padding uses at most 26 contacts. For a finite core family its rational speeds and labels are finite and independent of the coordinates.

Proof. Fix a branch and put

$$\begin{aligned} \ell_X &= \max(-\beta, 0), & r_X &= \max(\beta, 0), \\ \ell_Y &= \max(-\gamma, 0), & r_Y &= \max(\gamma, 0), \\ k &= C - \alpha - r_X - r_Y - 4 > 0. \end{aligned}$$

The required positive-distance decomposition is

$$CD - T_c = (4 + k)D + \ell_X x + \ell_Y y + r_X(D - x) + r_Y(D - y). \quad (306.5)$$

Its coefficients are correct since $\ell_X - r_X = -\beta$, $\ell_Y - r_Y = -\gamma$, and $4 + k + r_X + r_Y = C - \alpha$.

Execute the following order, omitting a target loop only when its coefficient is zero:

1. A unit-speed full LRL loop, taking $2D$ and six contacts
2. LXL at magnitude $2/\ell_X$, then LYL at $2/\ell_Y$
3. The unit-speed LR transfer, taking D and three contacts
4. RYY at magnitude $2/r_Y$, then RXX at $2/r_X$
5. The unit-speed RL transfer, taking D and three contacts
6. A full LRL loop at magnitude $2/k$, taking kD and six contacts

The first loop is essential: the inherited entry speed is $+1$. Its fresh terminal L contact chooses the first different speed. The construction never silently replaces that outgoing speed at entry, and never inserts a zero-time extra event. Every later selection occurs at the terminal contact in (306.1) or (306.3). If a target loop is omitted, the preceding actual contact emits the next actual phase.

The fixed loops and transfers total $4D + kD$ and 18 contacts. At most one of ℓ_X, r_X and one of ℓ_Y, r_Y is nonzero. Thus at most two target loops occur, each with at most four contacts, giving at most 26. Their times add to (306.5). The positive-flight argument applies to every stage and at every interface; all coordinates and stationary labels are preserved. The final contact emits the next instruction's speed- $+1$ label. $\square$

The bound (306.4) is sufficient, not necessary or optimal. Any finite rational output-linear family admits such a rational C . If the core templates are common to every program, the enlarged speed set is common too; only phase copies grow with program size.

307 Six instruction padding formulas

Apply Lemma 306.1 with $C = 30$. Write $LX(q)$ for an LXL loop of duration qx' , and analogously $LY(q)$, $RX(q)$, $RY(q)$ for durations qy' , $q(D - x')$, $q(D - y')$. Each has speed magnitude $2/q$. Table 42 lists all nonzero target loops. Every row also includes the initial unit-speed full loop, both unit-speed transfers and the final full loop.

Branch	Left loops	Right loops	k	Final speed $2/k$
A increment	none	$RX(392/39)$	71/5	10/71
A zero	none	$RY(25/9), RX(50/9)$	53/3	6/53
A positive	none	$RX(1310/63)$	13/6	12/13
B increment	$LY(392/39)$	none	61/5	10/61
B zero	$LX(25/9), LY(50/9)$	none	47/3	6/47
B positive	$LY(1310/63)$	none	1/6	12

Table 42: Six explicit padding words, with positive final coefficients. Coordinates are the frozen output of the core.

The four target-loop magnitudes are $39/196$, $18/25$, $9/25$, $63/655$ for coefficients $392/39$, $25/9$, $50/9$, $1310/63$ respectively. All six values of k are positive, including $1/6$ for B positive. Thus the precise output-coordinate calculation proves $30D$ without relying only on the inherited bound $T_c < 10D$.

Branch	Core	Targets	Fixed	Padding	Total
A increment	14	4	18	22	36
A zero	14	6	18	24	38
A positive	26	4	18	22	48
B increment	20	4	18	22	42
B zero	20	6	18	24	44
B positive	32	4	18	22	54

Table 43: Contact ledger. Fixed padding has two full-span loops and two transfers. Interfaces are counted exactly once.

The generic estimate $32 + 26 = 58$ is valid, but the six actual words improve it to 54. The core’s terminal L contact belongs to the core; padding first reaches X after a positive flight. A padding primitive’s terminal event emits the next departure label, and the next primitive counts only subsequent contacts. Hence no interface contact is lost or double-counted.

307.1 A common rational speed set

The inherited core uses the sufficient 19-speed set

$$\{0\} \cup \pm \left\{ 1, \frac{1}{3}, \frac{1}{79}, \frac{1}{41}, \frac{3}{2}, \frac{1}{4}, 4, \frac{1}{2}, 2 \right\}.$$

Padding adds

$$\pm \left\{ \frac{39}{196}, \frac{18}{25}, \frac{9}{25}, \frac{63}{655}, \frac{10}{71}, \frac{6}{53}, \frac{12}{13}, \frac{10}{61}, \frac{6}{47}, 12 \right\}. \quad (307.1)$$

These ten positive magnitudes are pairwise distinct and disjoint from the nine core magnitudes, so the total set has $1 + 2(9 + 10) = 39$ elements. A label never changes its assigned speed. New phase labels emitted at actual contacts implement all speed changes. Neither the set of speed values nor its size depends on counter values, scale or the particular finite program.

307.2 Finite control and physical composition

For every instruction and realized branch, make a fresh copy of its padding word. Retarget only the core’s existing last L rule: it now emits the fresh padding-entry label at the same old speed $+1$. Thus no core trajectory, duration, contact or counter output changes. The branch-specific core label chooses the branch’s fixed padding copy. Its finite phase labels carry the chosen successor j until the final loop emits Q_j .

Every padding rule keeps the stationary markers at the exact positions $(0, x', y', D)$ with literal labels L, X_0, Y, R . No data is hidden in a relabeled marker or altered scale. Two signals are consumed and two produced at every new contact. Fresh instruction, branch and stage labels give a finite deterministic rule table for a fixed finite program. Identity completion preserves cardinality on unused distinct-speed inputs. Encoded executions use only the proved words, so exactly five live signals persist.

Induction now proves Theorem 304.1. The core is guarded at every encoded input, returns the correct output, and the padding restores that output with the correct successor label. Starting at

time zero, the n th completed instruction section is at $30nD$. Each instruction contains at most 54 strictly separated contacts. An infinite run therefore cannot have finite-time accumulation, either between sections or inside a finite word. Before halt all positions remain in $[0, D]$.

Halt is designated full-section reachability, not the extinction of motion. Under the optional escape convention Q_H makes identity crossings with X, Y, R and then leaves to the right. Those exactly three post-halt contacts are not an additional source instruction. At $k = 0$ the initial label is already Q_H , and the first-halt time is zero.

308 Native positive gaps and paid decoding

From now on the three external variables are $g_1, g_2, g_3 \in \mathbb{Z}_{>0}$, with D, x, y defined by (304.3). This ensures the ordered section even before the encoding test. Introduce positive shifted counters A, B and outputs P, Q . The two POWER modules displayed in Section 309 enforce

$$P = 2^{A-1}, \quad Q = 2^{B-1}. \quad (308.1)$$

Add the two residuals

$$G_1 = (20g_1 - D)P - 2D, \quad (308.2)$$

$$G_3 = (20g_3 - D)Q - 2D. \quad (308.3)$$

When both vanish, positivity of D, P, Q forces $20g_1 - D > 0$ and $20g_3 - D > 0$. Division only by positive nonzero integers gives

$$\frac{g_1}{D} = \frac{1}{20} + \frac{1}{10P}, \quad \frac{g_3}{D} = \frac{1}{20} + \frac{1}{10Q},$$

which is exactly (304.4) with $a = A - 1, b = B - 1$. Conversely an encoded triple supplies A, B, P, Q and satisfies these two equations. The POWER modules supply all remaining decoding witnesses. Thus signs, denominators and powers are enforced internally, without a free exponential input.

Whenever decoding exists, it is unique:

$$P = \frac{2D}{20g_1 - D}, \quad Q = \frac{2D}{20g_3 - D}. \quad (308.4)$$

A positive integral power of two has a unique nonnegative exponent, hence unique A, B . An elementary exact decoder can reject a nonpositive denominator, a nonintegral quotient, or a quotient that is not a power of two. This algorithmic observation does not replace the polynomial equations. Every accepted endpoint gap is in $(D/20, 3D/20]$, so the middle gap lies in $[7D/10, 9D/10)$.

309 Two fully displayed POWER modules

The two module copies use disjoint leaves. For $i = A, B$, let $(C_i, o_i) = (A, P)$ or (B, Q) respectively. The thirteen directly positive leaves per copy are

$$o_i, w_i, M_i, \eta_i, x_i, y_i, u_i, v_i, s_i, t_i, q_{b,i}, q_{v,i}, J_i.$$

Here M_i is a modulus witness, η_i is an auxiliary Pell witness, and x_i, y_i are Pell coordinates; they are unrelated to the physical x, y or native g_j . Two further positive leaves $\alpha_{+,i}, \beta_{+,i}$ define

$$\alpha_i = \alpha_{+,i} + 1, \quad \beta_i = \beta_{+,i} + 1.$$

The eleven natural aliases in each copy are

$$d_{w2,i}, d_{wC,i}, d_{yC,i}, \alpha_{1,i}, \alpha_{2,i}, \sigma_{1,i}, \sigma_{2,i}, \tau_{1,i}, \tau_{2,i}, r_{1,i}, r_{2,i}. \quad (309.1)$$

Each alias is, by definition, its own positive leaf minus one. These are substitutions, not extra variables or equations. Thus each copy has exactly $13 + 2 + 11 = 26$ positive leaves *including its output*, or 25 auxiliary leaves beyond that output. The index C_i is the already counted shifted initial counter, not a new module leaf.

309.1 The A module

The fifteen residuals $R_{A,1}, \dots, R_{A,15}$ are the left sides below, with the stated positive substitutions fully understood:

$$\begin{aligned} x_A^2 - 1 - (\alpha_A^2 - 1)y_A^2 &= 0, \\ u_A^2 - 1 - (\alpha_A^2 - 1)v_A^2 &= 0, \\ s_A^2 - 1 - (\beta_A^2 - 1)t_A^2 &= 0, \\ \beta_A - 1 - 4y_A q_{b,A} &= 0, \\ \beta_A + u_A \alpha_{1,A} - \alpha_A - u_A \alpha_{2,A} &= 0, \\ v_A - y_A^2 q_{v,A} &= 0, \\ s_A + u_A \sigma_{1,A} - x_A - u_A \sigma_{2,A} &= 0, \\ t_A + 4y_A \tau_{1,A} - A - 4y_A \tau_{2,A} &= 0, \\ y_A - A - d_{yC,A} &= 0, \\ w_A - 2 - d_{w2,A} &= 0, \\ w_A - A - d_{wC,A} &= 0, \\ M_A - 2P - J_A &= 0, \\ \alpha_A^2 - 1 - ((w_A + 1)^2 - 1)(w_A \eta_A)^2 &= 0, \\ 4\alpha_A - M_A - 5 &= 0, \\ x_A + M_A r_{1,A} - y_A(\alpha_A - 2) - 2P - M_A r_{2,A} &= 0. \end{aligned} \quad (309.2)$$

309.2 The B module

The independent fifteen residuals $R_{B,1}, \dots, R_{B,15}$ are

$$\begin{aligned}
 x_B^2 - 1 - (\alpha_B^2 - 1)y_B^2 &= 0, \\
 u_B^2 - 1 - (\alpha_B^2 - 1)v_B^2 &= 0, \\
 s_B^2 - 1 - (\beta_B^2 - 1)t_B^2 &= 0, \\
 \beta_B - 1 - 4y_B q_{b,B} &= 0, \\
 \beta_B + u_B \alpha_{1,B} - \alpha_B - u_B \alpha_{2,B} &= 0, \\
 v_B - y_B^2 q_{v,B} &= 0, \\
 s_B + u_B \sigma_{1,B} - x_B - u_B \sigma_{2,B} &= 0, \\
 t_B + 4y_B \tau_{1,B} - B - 4y_B \tau_{2,B} &= 0, \\
 y_B - B - d_{yC,B} &= 0, \\
 w_B - 2 - d_{w2,B} &= 0, \\
 w_B - B - d_{wC,B} &= 0, \\
 M_B - 2Q - J_B &= 0, \\
 \alpha_B^2 - 1 - ((w_B + 1)^2 - 1)(w_B \eta_B)^2 &= 0, \\
 4\alpha_B - M_B - 5 &= 0, \\
 x_B + M_B r_{1,B} - y_B(\alpha_B - 2) - 2Q - M_B r_{2,B} &= 0.
 \end{aligned} \tag{309.3}$$

These displays define ordinary integer polynomials after replacing every natural alias and α_i, β_i by its specified positive-leaf expression. No POWER predicate remains uninterpreted.

309.3 The exact Pell dependency

The substantive arithmetic input is the pinned mathlib4 theorem pair `Pell.matiyasevic` and `Pell.eq_pow_of_pell` in `Mathlib/NumberTheory/PellMatiyasevic.lean`, commit

`ac77769fabe23cb237559e7f56578dbead91499f`.

The source, its license and attribution are retained inertly. [write] [In this report the Lean source file and its Apache-2.0 licence are not shipped; they are cited by commit and SHA-256, and the arrival archive keeps them. The delivered NOTICE.md is shipped as `28-diophantine-science-sources-NOTICE.md`, and the dependency record `pell-dependency.json` as `data/09-chrono-matrix-independent_audit-pell-dependency.json` of the report `group-theoretic-substrates`; both are byte-identical to source 35's copies.] The two theorem statements are at lines 760–766 and 860–864 of the pinned source. This report uses their stated constructive equivalences; it does not claim a fresh Lean verification. The source SHA-256 is

`993760c797ad0ff66fa77064a616fce779550bc745cbf6c44e04f392fd0bed0a`.

The earlier general module used base $B_0 \geq 2$, semantic exponent $e \geq 0$, Pell index $e + 1$, and target $B_0 o$. [write] [This is the module of source 28 (Lemma 217.1) and source 30 (Section 251), whose proofs source 35 carried as `POWER58_PROOF.md` and `POWER60_PROOF.md`; they are shipped once, as `28-diophantine-science-PROOF.md` and `30-planar-science-arithmetic-PROOF.md`. Here B_0 is that module's base; in Section 310 B_0 is the input counter B .] Here substitute

$$B_0 = 2, \quad e = C_i - 1, \quad k_0 = C_i, \quad m_0 = 2o_i.$$

In particular the modulus identity becomes $4\alpha_i = M_i + 5$. The semantic exponent can be zero even though the Pell index C_i is strictly positive.

Lemma 309.1 (Positive base two POWER equivalence). *For $C_i, o_i \in \mathbb{Z}_{>0}$, the 15 residuals in one module have a solution in their 25 remaining positive leaves if and only if $o_i = 2^{C_i-1}$.*

Proof. For soundness, equations 1–9 instantiate the nonzero-index branch of `Pell.matiyasevic`. They enforce $\alpha_i > 1$, $y_i \geq C_i \geq 1$, three Pell identities, $\beta_i > 1$, and the four congruences

$$\beta_i \equiv 1 \pmod{4y_i}, \quad \beta_i \equiv \alpha_i \pmod{u_i}, \quad s_i \equiv x_i \pmod{u_i}, \quad t_i \equiv C_i \pmod{4y_i},$$

together with $v_i > 0$ and $y_i^2 \mid v_i$. Paired natural aliases express signed congruence quotients without a sign restriction. The theorem identifies (x_i, y_i) with the Pell pair of index C_i .

Equations 10–15 impose $w_i \geq 2$, $w_i \geq C_i$, $M_i > 2o_i$, the auxiliary Pell identity, $4\alpha_i = M_i + 5$, and the final congruence

$$x_i \equiv y_i(\alpha_i - 2) + 2o_i \pmod{M_i}.$$

Since $w_i \geq 2$ and $\eta_i \geq 1$, equation 13 implies

$$\alpha_i^2 = 1 + ((w_i + 1)^2 - 1)(w_i \eta_i)^2 > w_i^2,$$

so $\alpha_i > w_i \geq 2$. In particular $\alpha_i - 2$ agrees with the source's natural subtraction. The theorem `Pell.eq_pow_of_pell` gives $2^{C_i} = 2o_i$, and cancellation proves the required power.

For completeness start from $2^{C_i} = 2o_i$ and use the constructive power theorem. Its auxiliary η_i cannot be zero, because that would force $\alpha_i = 1$. Apply the constructive nonzero-index Matiyasevic equivalence to the resulting Pell pair. The Pell coordinates x_i, u_i, s_i are positive. Since $y_i \geq C_i \geq 1$ and $v_i > 0$, the divisibility quotient $q_{v,i}$ is positive. Since $\beta_i > 1$ and $\beta_i \equiv 1 \pmod{4y_i}$, $q_{b,i}$ is positive. The congruence $t_i \equiv C_i \pmod{4y_i}$ and $1 \leq C_i \leq y_i < 4y_i$ exclude $t_i = 0$. The strict inequality $M_i > 2o_i$ supplies positive J_i . Every signed quotient can be written as the difference of two naturals, and all inequality slacks are nonnegative. Their shifted leaves are positive. This includes $C_i = 1$.

For natural u, v , the source equality $u - v = 1$ is equivalent to $u = v + 1$. The interior differences $\alpha_i^2 - 1$, $\beta_i^2 - 1$ and $(w_i + 1)^2 - 1$ are nonnegative. Thus replacing the source's truncated subtraction by the displayed integer equations neither loses nor adds solutions. $\square$

[write] [Lemma 309.1 is source 28's Lemma 217.1 at $B_0 = 2$ and $e = C_i - 1$, with renamed leaves; the proof above is a second route for this instance, written for positive adapters and the Pell index $C_i \geq 1$.]

309.4 An exact exponent zero fixture

For either module, $C_i = o_i = 1$ has the following assignment, with natural aliases given before their positive shifts:

$$\begin{aligned} w_i &= 2, & M_i &= 63, & \eta_i &= 3, & \alpha_i &= \beta_i = 17, \\ x_i &= 17, & y_i &= 1, & u_i &= 577, & v_i &= 34, & s_i &= 17, & t_i &= 1, \\ q_{b,i} &= 4, & q_{v,i} &= 34, & J_i &= 61, & d_{wC,i} &= 1. \end{aligned}$$

All other natural aliases are zero. Thus $\alpha_{+,i} = \beta_{+,i} = 16$, the positive leaf for $d_{wC,i}$ is 2, and the other natural-alias leaves are 1. Direct substitution checks all 15 equations. This finite fixture checks the exponent-zero boundary; Lemma 309.1, not the fixture, proves all exponents.

310 Complete bounded trace equations

Assign distinct fixed integer codes to states. Each edge e has fixed source u_e , target v_e , and changes $d_e, f_e \in \{-1, 0, 1\}$. Increment has one edge. A conditional has a zero edge with both changes zero and a positive edge decrementing its tested counter. Add exactly one virtual edge h from H to H with changes zero. There are no duplicate edges or other outgoing halt edges.

For $T \geq 1$ introduce positive leaves

$$w_{t,e} \quad (0 \leq t < T, \ 1 \leq e \leq E), \quad A_t, B_t \quad (1 \leq t \leq T).$$

Define $s_{t,e} = w_{t,e} - 1$, $A_0 = A$, $B_0 = B$. The full residual list $\mathcal{R}_{\text{trace}}$ is [write] [source 34's residuals (301.3)–(301.7), collected in one display]

$$\begin{aligned} S_{t,e} &= (w_{t,e} - 1)(w_{t,e} - 2), & 0 \leq t < T, \ 1 \leq e \leq E, & \quad (310.1) \\ O_t &= \sum_e s_{t,e} - 1, & 0 \leq t < T, \\ U_t &= A_{t+1} - A_t - \sum_e d_e s_{t,e}, & 0 \leq t < T, \\ V_t &= B_{t+1} - B_t - \sum_e f_e s_{t,e}, & 0 \leq t < T, \\ Z_{t,e} &= s_{t,e}(A_t - 1), & \text{each A-zero edge and } 0 \leq t < T, \\ Z_{t,e} &= s_{t,e}(B_t - 1), & \text{each B-zero edge and } 0 \leq t < T, \\ C_{\text{initial}} &= \sum_e u_e s_{0,e} - q_0, \\ C_t &= \sum_e v_e s_{t,e} - \sum_e u_e s_{t+1,e}, & 0 \leq t < T - 1, \\ C_{\text{final}} &= \sum_e v_e s_{T-1,e} - H. \end{aligned}$$

At $T = 0$, replace the whole list by the single constant residual

$$\mathcal{R}_{\text{trace}} = (q_0 - H), \quad (310.2)$$

and introduce no trace witnesses. [write] [As in source 34, Equation (301.11).] Count residual *slots*, retaining any that specialize to the zero polynomial for a particular program. The exact trace ledger in both cases is

$$\#W_{\text{trace}} = T(E + 2), \quad \#\mathcal{R}_{\text{trace}} = T(E + Z + 4) + 1. \quad (310.3)$$

Indeed there are TE selector equations, $3T$ one-hot/update equations, TZ zero guards and $T + 1$ control equations for $T \geq 1$.

Remark 310.1 ([write] restatement of source 34's Theorem 301.1: trace correctness and uniqueness). [write] Source 35's Lemma 7.1 states: *For fixed positive A, B , the trace equations have a positive solution exactly when the computation from $(q_0, A - 1, B - 1)$ reaches H by T . When a solution exists its trace witness tuple is unique.* This is the residual-level content of source 34's Theorem 301.1, proved in Section 301; source 35's own proof, a condensed form of that argument, follows.

Proof. The selector equations give $w_{t,e} \in \{1, 2\}$, so $s_{t,e} \in \{0, 1\}$. The one-hot equation selects exactly one edge per time. The control equations force its source to be q_0 initially, link each target

to the next source, and give final target H . Distinct state codes make equality of codes equality of states.

The update equations give the exact shifted-counter transitions. A selected zero edge forces its tested counter to equal 1. A decrement selected at value 1 would force its next positive leaf to be zero, which is impossible. Therefore decrement edges automatically require a positive unshifted counter; no guard witness is missing. At a nonhalting state exactly the appropriate deterministic edge can be selected. At H only h remains. Induction gives the real computation followed, if necessary, by virtual halt steps. It determines every selector and counter leaf uniquely. Conversely the computation padded through time T supplies these leaves and zeros every residual. For $T = 0$, (310.2) is exactly initial halt; the trace is the empty tuple. $\square$

For $T \geq 1$ define the additional residual

$$H_{\text{early}} = \sum_{t=0}^{T-1} s_{t,h}. \quad (310.4)$$

Its summands are nonnegative, so its vanishing excludes every virtual halt-loop use. With the final control equation it enforces first arrival at H exactly at T . For $T = 0$ the by-horizon and exact-horizon predicates coincide and use (310.2). The halt loop pads an arithmetic trace only; it is not a physical continuation after the designated halt section.

311 One polynomial and its exact resource ledger

After all aliases have been replaced by their defining expressions, set

$$F_{M,T} = G_1^2 + G_3^2 + \sum_{j=1}^{15} R_{A,j}^2 + \sum_{j=1}^{15} R_{B,j}^2 + \sum_{r \in \mathcal{R}_{\text{trace}}} r^2. \quad (311.1)$$

This finite sum defines every coefficient of an ordinary integer polynomial; monomial expansion is not needed to specify it. Program data are fixed coefficients and T fixes the finite index sets. There is no division or exponentiation left in (311.1). There are no undeclared variables, auxiliary predicates or variable-length conjunctions for a fixed M, T .

Over integers a sum of squares vanishes exactly when each residual vanishes. Lemma 309.1 gives (308.1), and (308.2)–(308.3) then give the unique encoded input. Remark 310.1 gives the bounded halting trace. This proves soundness of Theorem 304.3. For completeness take an encoded computation halting by T , use its shifted counters and power outputs, choose the two constructive POWER witnesses and the unique padded trace, and substitute. Every residual vanishes.

Equivalently one may count A, B, P, Q as four and the two modules' remaining 25 leaves as fifty. Counting four plus twice 26 would count the two outputs twice. Including the three native inputs, there are $T(E + 2) + 57$ total variables. In instruction counts, witnesses and residual slots are respectively

$$T(I + 2J + 3) + 54, \quad T(I + 3J + 5) + 33.$$

The exact-halt version adds H_{early}^2 at $T \geq 1$, increasing only the residual count by one.

311.1 Degree exactly twelve

The two gap residuals have degree two and trace residuals have degree at most two. In either POWER module residuals 1–3 have degree at most four; residual 6 has degree three; residuals 4–5,

Component	Positive witnesses	Residual slots
Initial shifted counters A, B	2	0
A module including P	26	15
B module including Q	26	15
Native gap constraints	0	2
Bounded trace	$T(E + 2)$	$T(E + Z + 4) + 1$
Total	$T(E + 2) + 54$	$T(E + Z + 4) + 33$

Table 44: The exact construction ledger. P, Q are already counted inside their respective modules.

7–8 and 15 have degree at most two; residuals 9–12 and 14 have degree at most one. Residual 13 has degree six, with highest homogeneous part

$$-w_i^4 \eta_i^2.$$

No other residual has degree six. Consequently the complete degree-12 homogeneous part of $F_{M,T}$ is exactly

$$w_A^8 \eta_A^4 + w_B^8 \eta_B^4. \quad (311.2)$$

The two distinct monomials have coefficient one and cannot cancel. Degree is joint in all independent positive leaves and native inputs, before imposing any equations on those leaves. It remains exactly 12 at $T = 0$ and in the exact-halt version.

311.2 Unique decoding does not mean unique full witnesses

For fixed gaps, (308.4) and the uniqueness of powers of two fix A, B, P, Q . Remark 310.1 then fixes the trace leaves. Nevertheless a nonempty full fiber is infinite. Given any full witness and any $n \in \mathbb{N}$, increase both $\alpha_{1,A}$ and $\alpha_{2,A}$ by n . These aliases appear only in residual 5 of the A module, where their equal increments cancel. All positive adapters remain positive and no other residual changes. Distinct n give distinct witnesses. This specific polynomial is neither single-fold nor finite-fold. That loss of uniqueness is an exact property of the paid decoding construction.

311.3 Transport to the exactly clocked machine

For encoded native inputs the physical and arithmetic initial states agree. If the first halt is after $k \leq T$ source instructions, Theorem 304.1 gives first physical halt at

$$30kD = 30k(g_1 + g_2 + g_3), \quad (311.3)$$

with at most $54k$ contacts before that section. For exact T , the time is precisely $30T(g_1 + g_2 + g_3)$. The by-horizon formula asserts some such $k \leq T$, not necessarily arrival at time $30TD$. Virtual arithmetic padding does not force physical instructions at H . At $T = 0$, $q_0 = H$ is necessary and sufficient after successful decoding, and the physical first-halt time is zero.

312 Primitive integer initialization and its exact cost

Every nonnegative counter pair admits native positive integer gaps. The scale needed by this particular encoding can be found exactly, including its exceptional small cases and the factor of five.

Theorem 312.1 (Exact primitive initialization). *Let $a, b \geq 0$, $m = \max(a, b)$, and define*

$$U = 2^m + 2^{m-a+1}, \quad W = 2^m + 2^{m-b+1}, \quad V = 20 \cdot 2^m - U - W. \quad (312.1)$$

These are positive integers. Their greatest common divisor $c = \gcd(U, V, W)$ is

$$c = \begin{cases} 1, & (a, b) = (0, 0), \\ 4, & (a, b) = (1, 1), \\ 2, & \{a, b\} = \{0, 1\}, \\ 10, & m \geq 2 \text{ and } a \equiv b \equiv 3 \pmod{4}, \\ 2, & m \geq 2 \text{ otherwise.} \end{cases} \quad (312.2)$$

The primitive triple and exact minimum positive integer total scale are

$$g^* = \left(\frac{U}{c}, \frac{V}{c}, \frac{W}{c} \right), \quad D_{\min} = \frac{20 \cdot 2^m}{c}. \quad (312.3)$$

Every positive integer gap triple encoding the same (a, b) is a unique positive integer multiple of g^ .*

Proof. Division by $20 \cdot 2^m$ gives exactly (304.4). Since $U, W \leq 3 \cdot 2^m$, $V \geq 14 \cdot 2^m > 0$.

At $m = 0$, $(U, V, W) = (3, 14, 3)$, so $c = 1$. At $m = 1$, the equal case gives $(4, 32, 4)$ and the mixed cases give $(6, 30, 4)$ or its reflection; their gcds are 4 and 2.

For $m \geq 2$, at least one of a, b equals m . Its endpoint gap is $2^m + 2$, with exact 2-adic valuation one. Both endpoint gaps and V are even, so c has exact 2-adic valuation one. Also c divides $U + V + W = 20 \cdot 2^m$. Thus the only possible odd prime factor is 5, with exponent at most one. Since

$$U = 2^{m-a}(2^a + 2),$$

we have $5 \mid U$ exactly when $2^a \equiv -2 \pmod{5}$, namely $a \equiv 3 \pmod{4}$ by the order four of 2 modulo 5. The same criterion holds for W . When both endpoint gaps are divisible by 5, so is V because their sum is divisible by 5. This proves (312.2).

Normalization makes g^* primitive. Every other integer realization has the same normalized coordinates and therefore equals λg^* for a positive rational λ . Since $\gcd(g^*) = 1$, integer Bézout coefficients combine the entries of g^* to 1. Applying the same combination to the integer entries of λg^* gives $\lambda \in \mathbb{Z}$. It is positive and unique. Thus (312.3) has the smallest possible positive integer total scale for this chosen encoding. $\square$

312.1 Bit height and binary counter input

For a positive integer n write $\text{bit}(n) = \lfloor \log_2 n \rfloor + 1$. Equations (312.2)–(312.3) give the exact formula

$$\text{bit}(D_{\min}) = \begin{cases} 5, & (a, b) = (0, 0), \\ 4, & (a, b) = (1, 1), \\ 5, & \{a, b\} = \{0, 1\}, \\ m + 2, & m \geq 2 \text{ and } a \equiv b \equiv 3 \pmod{4}, \\ m + 4, & m \geq 2 \text{ otherwise.} \end{cases} \quad (312.4)$$

a	b	c	Primitive gaps g^*	$D_{\min}$	$\text{bit}(D_{\min})$
0	0	1	(3, 14, 3)	20	5
1	1	4	(1, 8, 1)	10	4
0	1	2	(3, 15, 2)	20	5
2	2	2	(3, 34, 3)	40	6
3	3	10	(1, 14, 1)	16	5
3	7	10	(16, 227, 13)	256	9

Table 45: Exact initialization examples. The congruence class modulo four causes the factor-of-five reduction. These examples illustrate the all-pair proof.

For $m \geq 2$, $D_{\min}$ equals either 2^{m+1} or $10 \cdot 2^m$, proving the last two cases and the bounds

$$2^{m+1} \leq D_{\min} \leq 10 \cdot 2^m.$$

The middle gap is always largest: $g_2^* \geq 7D_{\min}/10$, whereas $g_1^*, g_3^* \leq 3D_{\min}/20$. Since $D_{\min}/2 < g_2^* < D_{\min}$,

$$\text{bit}(D_{\min}) - 1 \leq \max_j \text{bit}(g_j^*) \leq \text{bit}(D_{\min}). \quad (312.5)$$

Therefore the primitive description has bit height $m + O(1)$ and numeric height $\Theta(2^m)$. If $m \geq 1$ itself has binary length $\ell = \lfloor \log_2 m \rfloor + 1$, then $2^{\ell-1} \leq m < 2^\ell$. Its gap bit height is $\Theta(2^\ell)$ as a function of the largest counter's binary input length. In particular it is $\Theta(m)$, not $\Theta(\log m)$.

The exact primitive initialization is a theorem about this powers-of-two coordinate representation. It is not a universal lower bound for all simulations, encodings, counter algorithms or Diophantine witnesses. No height bound for the potentially much larger Pell auxiliaries is proved here.

Conversely the positive integer denominators in (308.4) imply $P, Q \leq 2D$, and hence

$$a, b \leq \lfloor \log_2(2D) \rfloor.$$

One can decode an accepted input by polynomial-time integer arithmetic in the binary length of its three gaps: form D , test the denominator signs, divide exactly, and check the binary power-of-two condition. This does not make producing the exponentially longer gap description polynomial-time in the binary counter input length, and it does not decide unbounded halting.

313 Evidence and the boundary of the proof

Report 64 [source 34] already contains the bounded-counter trace construction whose working source directory happened to end in `diophantine65`. That historical directory name does not designate an earlier Report 65. The two additions here are the clocked physical interface and the native positive-gap decoding, together with their synthesis and exact initialization cost.

The release separates the mathematical statements from the evidence used to check their transcription and arithmetic. The two candidate continuations and two independent audits are retained as inert, hash-pinned files. Their previously recorded outputs are included without rerunning their scientific code during this report assembly. The complete inherited physical and trace proofs are included with the native-gap dependency packet, as are the earlier POWER derivations and

pinned theorem source. [write] [Here they are shipped once, at their first sources: `34-branching-science-frozen-proof-PROOF.md`, `34-branching-science-arithmetic-PROOF.md`, `28-diophantine-science-PROOF.md` and `30-planar-science-arithmetic-PROOF.md`; the pinned Lean source is not shipped (Section 309).]

The clock continuation’s exact rational row checks derive all six output-coordinate costs, verify the primitive flight lengths and positive separations, compute the six positive k values, and check $T_c + P_c = 30D$, the maximum 54 and the 39-speed sufficient set. Its independent audit additionally instantiates all seven primitive words and the local rules across all six padding branches, checking phase-entry speed, distinct-speed pairs and joins. These checks support the positive-flight chronology proof; they do not replace it with sampled trajectories. [write] [Rerun on copies at the placement: the clock audit passes with 54 contacts, 136 rules and 39 speeds, and the native-gap audit passes; their records agree with the shipped ones up to path separators and file-metadata snapshots. The README gives the commands.]

The native-gap continuation’s sparse-polynomial checks count positive leaves and residual slots, verify the exact degree-12 terms, evaluate an exponent-zero fixture and finite trace fixtures, and test normalization formulas on a finite grid. Its independent audit checks the positive adapters against the pinned Pell theorem statements, independently recovers the resource ledger and degree, and checks an additional exponent-one fixture. Finite arithmetic fixtures cannot establish the all-exponent POWER equivalence; that part depends on the two named constructive theorems. Likewise physical trace transport depends on the inherited all-encoded-state compiler theorem.

No author or upstream mathematical implementation, physical simulator, next-collision search, saved accepting schedule, native machine interpreter or Lean build is executed for this report. Fresh report-owned release tools only assemble, compile, hash, inventory and package the article and inert evidence. PDF rendering is a layout check, not mathematical execution. The analytic diagram is generated directly from its displayed coordinates.

The exact source identities and independent audit verdicts are in the accompanying manifest and pin records. [write] [The pin records are shipped as `data/35-clocked-INPUT_PINS.json` and the packets’ manifests `data/35-clocked-science-*-MANIFEST.json`; the release byte inventory `MANIFEST.json` was verified at placement (298 of 298 entries) and is not shipped. The audits are shipped as `35-clocked-audits-clock-AUDIT.md` and `35-clocked-audits-native-gaps-INDEPENDENT_AUDIT.md`, with their checkers in `code/` and records in `data/`.] The release preserves the input sources’ bytes, modes and recorded modification timestamps; it creates a new report rather than editing Report 64 [source 34]. The archive includes editable LaTeX, the final PDF, the inert source packets, the independent audit dossiers and the report-owned release tools.

314 Prior work and limits of interpretation

Durand-Lose’s MCU 2004/2005 construction already uses two fixed scale signals, two counter-position signals and one instruction signal with cardinality-preserving rules [100]. It is substantive prior art for five-live-signal counter computation. Report 64 [source 34] supplied the particular fixed ordered full-section interface used here, with explicit guard coverage and nondestructive branching. Report 65 does not newly establish five-signal universality.

Bouncing signals are also an established delay mechanism. Durand-Lose and Emmanuel explicitly choose speeds for a border-and-back motion to obtain a delay proportional to width [102]. The present clock contribution is the six-template exact output-linear compensation at this five-live-signal interface, including its inherited +1 entry, complete contact accounting and literal restoration. It is

not a claim to invent waiting by bouncing. Reversing speeds and collision rules is standard in the reversible signal-machine literature [5]; no global reversibility theorem for the completed compiler is asserted.

Finite counter traces and sum-of-squares conjunctions are elementary encodings. Dudenhefner provides a precise modern treatment of the relevant counter instruction model and emphasizes that computational claims depend on the instruction set [101]. Our source language and edge semantics have therefore been stated explicitly. The substantive number-theoretic dependency is the pinned constructive Pell pair [86]. Its license and attribution travel with the copied source. [write] [Not in this report: the Lean source and its licence are not shipped [86].]

The literature check was targeted to these mechanisms and dependencies, not exhaustive. No firstness, priority or absence-of-equivalent-prior-construction claim is made. Nor are 30, 54, 39, degree 12, witness count or residual count asserted to be optimal. The exact counts describe this construction. Robustness, finite precision, minimality and efficient Pell witnesses are not established.

315 Questions opened by the construction

The following are research questions, not established conclusions.

1. **Smaller exact clocks.** What is the smallest sufficient C for the six core branches if one varies the physically valid padding words? The present output-linear lemma deliberately pays a $2D$ entry loop and two transfers; removing a cost requires a new contact-level interface proof, not only a rearranged linear identity.
2. **Fewer speeds or contacts.** Can different rational loop combinations reduce the common speed set or the worst-case contacts without changing the five-live-signal population or fixed section? Tradeoffs between clock size, speed magnitudes and word length remain open here.
3. **More general elapsed-time functions.** Which functions of the output section admit finite identity padding with fixed rational speeds and no extra live signal? The present lemma covers homogeneous linear functions. It does not handle arbitrary nonlinear costs or input-dependent speed assignment.
4. **Alternative native arithmetic interfaces.** Can the exact 54-witness decoding overhead or degree 12 be reduced with a different explicit POWER representation? A fair comparison must count positive adapters, index shifts, outputs and sign conventions consistently, and must pay for initialization internally. [write] (Added 4 October 2026: answered in part by source 37 in Part XIII. Its Theorem 332.1 gives base-two POWER presentations with 22, 16, 14, 13 and 12 leaves including the output, the 22- and 16-leaf ones at degree 12, and its Theorem 332.2 composes two of them with paid gap decoding and source 36's three-witness compiler in place of the trace, into $2\ell + 5$ positive witnesses at every horizon, at a degree that grows with T (source 35's degree is 12 at every horizon). None of those presentations has degree below 12, so the degree half stays open.)
5. **Control of witness multiplicity.** Can the positive congruence-quotient redundancy be removed constructively while maintaining soundness and completeness? The current common-shift freedom proves infinite fibers. It gives no finite-fold claim for another formula. [write] (Added 4 October 2026: the redundancy itself is removed by source 37 in Part XIII: its 22-leaf normalization drops the paired congruence quotients with soundness and completeness kept (Section 333), and its Section 335 proves that every accepted fibre nevertheless stays infinite,

by genuine parameter variation. No finite-fold representation results, and the compositions of Part XIII also have infinite full fibres.)

6. **Height bounds.** What explicit bounds can be given for the Pell witnesses as a function of the native input gaps and fixed horizon? Primitive gap height is settled here for this encoding; auxiliary certificate height is a separate issue.
7. **An independent formalization.** A future machine-checked proof could formalize the clock's local rules, the all-section chronology, the exact native formula and their composition. Reading a pinned formal theorem and evaluating finite static certificates are not substitutes for such a new formalization. [write] (Open. The repository has no Lean or Rocq development of these constructions.)

316 Conclusion

A fixed ordered-section counter compiler can be clocked exactly with its existing five live signals. The explicit six words give $30D$ per instruction, at most 54 contacts and a common sufficient set of 39 rational speeds. The native arithmetic continuation removes free exponential initialization by enforcing unique powers-of-two decoding inside two explicit POWER modules, then attaches a complete bounded trace. This yields exactly $T(E + 2) + 54$ positive witnesses, $T(E + Z + 4) + 33$ residual slots and joint degree 12. The combination identifies first physical halt after k instructions with exact time $30kD$. Its scope remains encoded inputs and a growing-arity bounded-horizon family, with unique decoded traces but infinite full witness fibers. Exact primitive initialization quantifies the representation cost rather than hiding it.

317 Further questions and research (Part XII)

[write] This section was added at the assembly under Vladimir Reshetnikov's standing rule of 4 October 2026: no unproved claim of a source, of the review or of the write is dropped silently. Part XII was drafted before the rule, and both manuscripts, their proof packets and audits, the review [99] and the write's own notes were scanned again against it. The sources' own questions stay where they are printed, in Sections 302 and 315, with dated notes where a later source answers them. The items below are the claims that are asserted without proof, that a review left unchecked, or that are wrong as printed. One claim is wrong as printed, and its failure is proved in Remark 317.1. The other notes of the write were checked against Part XIII and against the sources, and no further unproved assertion was found in them.

1. **The positive A -test time bound (source 34; refuted).** Section 300 of source 34 states that on the positive branch " $D/20 < x \leq D/10$ gives $25D/18 < T < 25D/9$ ". The upper inequality fails at $a = 1$, where the bound $25D/9$ is attained (Remark 317.1). The correct form is $25D/18 < T \leq 25D/9$. The theorems that use the bound are unaffected, because they use only $T < 4D$.
2. **The lower post-contact chamber (source 34).** *Claim.* Section 296 asserts, with no word or proof, that "there is a lower post-contact chamber $x < y < 4x$ for the underlying machine". *Sketch (the write, at the assembly).* Take $0 < x < y < 4x$. The common prefix is unchanged, since its target positions x , $2x/3$ and $4x/9$ lie below y . By Equation (295.2), $A - B = (y - 4x)/2 < 0$, so the X_{fast}/Y contact now comes first, at time A and position y .

At the anchor bounce (time B), X_{post} (speed $-1/2$) is at $5y/4 - x > 0$. It is still to the right of the slower left-moving messenger, because their gap is affine and positive at both ends. The messenger, now at speed 4, meets it at $\frac{8}{9}(5y/4 - x) = (10y - 8x)/9 = F \in (0, y)$ and emits C_Z , which returns to L . So the rules (295.1) produce the same labels and the same return map $(D, x, y) \mapsto (D, F, y)$, with events 4 and 5 interchanged. Time reversal applies the reverse rules (297.3) and (297.4) in the interchanged order. If this holds, the compiled zero test of source 34 acts on all of $0 < x < y < 8x$ except $y = 4x$. The source neither claims this nor needs it, because encoded states have $y - 4x \geq D/4$. *Missing.* The complete gap table and event-time list of the interchanged word and of its inverse (the analogue of Section 296), and an independent check of them.

3. **The terminal audit of source 34.** *Claim.* The last sentence of Section 303: “A final independent terminal audit precedes publication.” *Status.* The archive has no record of such an audit. Its delivery README states that the final terminal audit is performed after this assembly of sources and evidence, and that the pre-terminal README does not claim it was completed. Source 34’s acceptance therefore rests on the physical and arithmetic audits and on the manuscript review that it ships. *Missing.* The terminal-audit dossier, or confirmation that the audit took place.
4. **What no independent party has checked (the review; sources 34 and 35).** *Claims.* The review [99] read only the spans listed at the opening of this Part. It says that it does not certify the omitted chronology, the Pell proofs or the archived tests. The audits are the pipeline’s own, and source 35’s manuscript review calls itself model-performed. The following therefore rest on the sources’ written proofs alone:

- the complete chronology of the test words and their inverses (Sections 296 and 297);
- the argument that fresh phase copies make the compiled table of every program consistent and deterministic, and that identity completion inserts no event (Section 300). The physical audit assembled a 118-rule example table but never executed it;
- the trace-uniqueness proofs (Theorem 301.1);
- the reduction of Lemma 309.1 to the two pinned mathlib theorems.

Checked at the assembly. The write rechecked the following in exact rational arithmetic:

- the update speeds and durations (299.12) and (299.13), and the encoding margins;
- all six rows of Table 40, the six values of k in Table 42, and the count of 39 speeds;
- the exponent-zero fixture of Section 309;
- Table 45, and the 43 encoded triples with $D \leq 96$ that source 35’s native-gap audit records;
- the transcription of the fifteen residuals against the statements of `Pell.matiyasevic` and `Pell.eq_pow_of_pell` at lines 760–766 and 860–864 of the pinned source, which is kept in the arrival archive.

All agree. *Missing.* An independent proof-level review of the chronology and of per-program determinism, and a formalization (source 35’s last question, Section 315).

5. **A fully charged fixed-arity history (the review; source 34).** *Claim.* The review states that the step still missing is “an actual fixed-arity integer history representation with all

initialization, control, chronology and branch checks charged”. It adds that the counts of five live signals, 19 or 39 speeds, 32 or 54 contacts and $30D$ do not by themselves bound the arithmetic work of such a representation. *Sketch.* Source 34’s closing question (end of Section 302) asks for an arithmetic packing of the horizon family. Source 35 pays for the initialization inside the polynomial (Theorem 304.3). Part XIII bounds the witnesses at each horizon. *Missing.* A single fixed-arity representation that verifies an unbounded trace and charges all of these checks. None of the sources, and no other Part, supplies one. No universal resource bound may be entered from the counts above.

Remark 317.1 ([write] The positive A -test bound is attained). Section 300 of source 34 states that on the positive branch “ $D/20 < x \leq D/10$ gives $25D/18 < T < 25D/9$ ”. The upper inequality is not strict. Take $a = 1$ and any b . Equation (294.2) gives $x = D(1/20 + 1/20) = D/10$, and Equation (297.6) then gives

$$T_{\text{test},N} = \frac{250x}{9} = \frac{25D}{9}.$$

Since $x = D/20 + (D/10)2^{-a}$ decreases strictly in a , the correct statement is $25D/18 < T_{\text{test},N} \leq 25D/9$ on the positive branch, with equality exactly at $a = 1$. By reflection the same holds for the B test, with equality at $b = 1$. Nothing downstream changes. Since $25D/9 < 4D$, the bounds $D < T_{A\text{-test}} < 4D$ and $3D < T_{B\text{-test with transfers}} < 6D$ and Equation (300.2) hold as printed. Source 34’s frozen physical proof states only $D < T_{A\text{-test}} < 4D$. The strict upper bound first appears in the manuscript, and the manuscript review did not flag it.

Part XIII

Native-gap certificates and low-arity finite-clipping compilers (sources 36, 37 and 38)

[write] Part XIII prints three AI-assisted research manuscripts of ProveIt’s incoming reports, batch 91 (arrival 0d7f51c44; read by cluster W4 and placed by d750d98dd), all dated 4 October 2026 and added on that day. They are numbered reports of the research pipeline that also wrote sources 24–35 of Parts IX–XII. *Source 36* is batch-91 manuscript 18, the pipeline’s Report 66, *Bounded halting certificates and exact encoded gap counts* (archive `Bounded_certificates_and_encoded_gap_counts_source.zip`); *source 37* is manuscript 19, Report 67, *Positive POWER reductions and exact bounded compiler costs* (archive `Positive_POWER_reductions_and_exact_compiler_costs_source.zip`); *source 38* is manuscript 20, Report 69, *Low-arity finite-clipping Diophantine compilers* (archive `Low_arity_Diophantine_compilers_sources.zip`). The source numbers are the prefixes 36-bounded-cert-, 37-pos-power- and 38-low-arity- of their shipped files. Their title pages read “RESEARCH REPORT 66”, “67” and “69”, their PDF metadata name the author “Research report”, and they name no assistant and no ProveIt commit; the only external pin they share is the mathlib4 commit `ac77769f` of the Pell theorems [86]. Brackets translate the pipeline’s numbers where they occur: Reports 58 and 60 are sources 28 and 30 (Parts X and XI), whose POWER modules the pipeline calls “POWER58” and “POWER60”; Report 64 is source 34 and Report 65 is source 35 (Part XII); Reports 66, 67 and 69 are sources 36, 37 and 38. Report 68, which source 38 names as a predecessor, was never delivered; see below.

[write] **What the three sources do.** Source 35 (Part XII) ends with a native positive gap certificate for bounded halting of a two-counter program: an encoded gap triple, two fully displayed 26-leaf POWER modules for 2^a and 2^b , and a bounded execution *trace* whose witness count grows with the horizon T (Theorem 304.3, using source 34’s trace certificate, Theorem 301.1). The three sources of this Part replace that growing trace. Source 36 proves that clipping each native counter at T preserves every control decision through T transitions (Lemma 319.1), interpolates the $(T + 1)^2$ resulting classes, and obtains a sum of six squares with exactly *three* positive witnesses, unique when it exists, of degree at most $4(T + 1)^2 - 4$ (Theorem 318.1); composed with two 26-leaf POWER modules it gives 57 positive witnesses and 38 residual slots (Theorem 318.2). Its second half counts *all* encoded positive gap triples by their total, with slope 743/1125, density 3/20 of the represented totals, a shell formula, sharp $(\log_2 X)^2$ error envelopes and the inverse law (Theorem 318.3). Source 37 shrinks the POWER module to 22, 16, 14, 13 and 12 leaves (Theorem 332.1), proves the *exact* degree of source 36’s compiler, $4(T + 1)^2 - 4$ at positive odd and $4(T + 1)^2 - 8$ at positive even horizons (Theorem 341.1), and composes 29 positive witnesses with 18 residual slots (Theorem 332.2). Source 38 replaces the three-witness compiler by a tensor compiler with *two* positive witnesses at degree at most $4T$ (Proposition 349.1) and by a product of disjoint-cell factors with *one* positive witness (Proposition 350.1), classifies exactly when *zero* witnesses suffice (Theorem 351.1), and composes 28 or 27 positive witnesses with the 12-leaf module (Theorem 353.2). Every certificate of the Part is a *horizon-indexed family*: each T gives a separately compiled polynomial with its own finite acceptance table, degree and coefficients. Source 37 thereby answers two of source 35’s closing questions: another explicit POWER representation lowers source 35’s decoding overhead of 54 positive witnesses to 34 at the same degree 12 (16-leaf modules) or to 26 at degree 20 (12-leaf modules), and the positive congruence-quotient redundancy can be removed constructively, although every full fibre stays infinite (Sections 333, 335 and 343).

[write] **Dependencies.** (i) The all-exponent POWER equivalences rest on the constructive theorem pair `Pell.matiyasevic` and `Pell.eq_pow_of_pell` of `mathlib4` at `ac77769f`, read as text; no source runs Lean [86]. (ii) The physical reading of the certificates (bounded instruction-section reachability of a five-signal machine) is conditional on source 34’s retained five-live-signal compiler (Theorem 294.1) and is not re-audited by any of the three; source 37’s Section 345 states the interface. (iii) The bounded compiler, the clipping lemma, the interpolation degree, the zero-witness classification and the counting theorems need neither Pell theory nor a physical theorem. The archives carried byte copies of the proofs of sources 28, 30, 34 and 35 that they depend on; they are shipped once, under those sources’ prefixes (`28-diophantine-science-PROOF.md`, `30-planar-science-arithmetic-PROOF.md`, `34-branching-science-arithmetic-PROOF.md`, `35-clocked-science-native-gaps-PROOF.md`), and the README maps every delivered name.

[write] **Printed once.** The three sources restate one another and source 35, and this Part prints each such result once. A restated statement becomes a numbered remark, marked [write], that quotes it, keeps its label and points to the proof printed elsewhere; references to it say “Remark”. The clipping lemma is source 36’s Lemma 319.1; source 37’s Lemma 9.1 and source 38’s Lemma 1.2 are Remarks 340.1 and 348.2. Source 36’s three-witness compiler (Theorem 318.1), restated in source 37’s letters as its Proposition 9.2, is Remark 340.2; the surrounding displays are kept, because source 37 computes with them. Source 36’s Proposition 7.1, the primitive-scale classification, is source 35’s Theorem 312.1 with the same proof (Remark 324.1), and source 36’s 26-leaf POWER module with its Pell adapters is source 35’s Section 309 and Lemma 309.1 (Remark 323.1; the module’s display is kept). Source 38’s Proposition 6.1, the twelve-leaf module, is source 37’s at base two (Remark 353.1); its displays are kept, because source 38’s ledgers are stated in its letters, and its recovery argument is summarized with its labelled equations. Successive improvements are

not duplicates and are all printed: 57 witnesses (source 36), 29 (source 37), 28 and 27 (source 38); the degree bound $4(T+1)^2 - 4$ (source 36), its exact values (source 37) and the linear bound $4T$ (source 38). Each restatement carries a bracketed note, and no novelty is claimed for any of them.

[write] **Report 68.** Source 38 names “Reports 66–68” as its predecessors. Report 68, *Gaussian fluctuations of encoded gap counts and inverse ranks*, is in neither batch 91 nor the repository; only its delivery README survives, as a predecessor record of source 38, shipped as `38-low-arity-qa-pred-tools-README68.md`. According to that README it proves a Gaussian limit law for the deficit E of source 36’s counting theorem, with leading mean $\frac{1}{2}(\log_2 X)^2$ and variance $\frac{1637}{5400}(\log_2 X)^3$, and a corresponding law for the inverse ranks. Its article is not available here and none of its statements is proved here; they are recorded as an open question, with the README’s argument sketch and what is missing, in Section 355 (Question 1), and source 36’s question on finer error statistics (Section 331, item 5) stays open in this report. Report 70 of the same pipeline (Part VII of `five-particle-binary-automata`) also cites Report 68: its presentation tools “adapt the fully inspected accepted Report 68 presentation contract”. Reversing Report 70’s shipped `data/70-two-scale-qa-pred-contract-ADAPTATION.diff` on its `build_report70.py`, `release70.py` and `selftest70.py` gives exactly this report’s `code/38-low-arity-qa-pred-tools-build_report68.py`, `release68.py` and `selftest68.py` with “68” renamed “70” (checked 4 October 2026 by the write of that report); Report 68’s acceptance itself stays unverifiable.

[write] **A neighbouring report.** Source 38’s zero-versus-one classification (Theorem 351.1) is restated with proof in batch-91 manuscript 21, the base of the collection’s report `SetTheory/Cardinals/docs/reports/generating-functions-and-asymptotics/oeis-sequence-asymptotics/a196460-clipping-tables`, which counts the closed $n \times n$ tables that need no auxiliary variable as $C_n = 1 + a_n$, with a_n the OEIS sequence A196460. Manuscript 21 carried four of source 38’s files byte for byte; they are shipped once, here, under the prefix `38-low-arity-`.

[write] **Review status.** The Hilbert’s-tenth programme reviewed all three archives in a scoped intake, `review_new_arithmetic_0d7f51c44.md` (commit `a9ab9a698`) [111]. It read the three delivery READMEs completely and selected spans of the articles (source 36: clipping, interpolation, compiler, costs and the gap composition without the module’s adapter proof; source 37: the twelve-leaf section and the composition, trace, physical and evidence sections; source 38: everything up to the classification, the costs, and the gap compositions), ran none of the archived programs, and found no defect. It calls sources 36 and 38 “genuine finite-table arity reductions” with a sound clipping proof, stresses that “these are finite-horizon predicates”, calls source 37’s result “twelve leaves, not twelve gates” with a strict-positive-quotient scope and no optimized arithmetic circuit, and finds “no new paid circuit below the current 84-operation unbounded ordinary-input construction”. Notes at the end of each source give the details. The same programme’s review of the batch’s twelve signal reports (commit `49b100cfd`) [99] calls Report 64 [source 34] the “strongest computation lead” and Report 65 [source 35] the “best clock/interface lead”, and stresses that their horizon certificates are growing-arity families and that the native-gap initialization has gap height $\Theta(2^m)$; the three sources of this Part compress the arity of that interface at each fixed horizon and do not remove the horizon.

[write] **Letters.** Each source keeps its own letters, and the same letter means different things in different sources. The table fixes the readings; where a reading is tempting and false, the second column says so.

Symbol	Meaning in Part XIII	Not to be confused with
$\mathbb{N}$	$\{0, 1, 2, \dots\}$ in all three sources (their $\backslash\mathbb{N}$)	the same set as Part I's $\mathbb{N}_0$, which is printed with a subscript 0
$\mathbb{Z}_{>0}$	the positive integers, the domain of every witness	–
T, K	horizon (native transitions) and $K = T+1$	T of earlier Parts (time, horizon of another model)
Q	$(T+1)^2$, the number of clipped classes (source 36)	$\mathbb{Q}$, the rationals (source 37); Q_S , the one-witness polynomial (source 38); Q , the power output of source 35
N $d; c$	K^2 , the same class count in source 37 source 36: $d = (Q-1)!$ and the primitive scale $d(a, b)$, with c a gcd; source 37: $c = (N-1)!$ and $d = 2b$ in the module; source 38: $c = (K-1)!$	$\mathbb{N}$ and $\mathbb{N}_0$ each other
M	the counter program (sources 36, 38)	source 37 writes $\mathbf{M}$ for the program and M for the POWER modulus; source 36's modulus is M_0 , source 38's is M_p
D	$g_1 + g_2 + g_3$, a gap total	D_n , the n th total (source 36); D_ℓ , a module degree (source 37); D_S , a degree (source 38); D_i residuals (source 38)
$E(X), \mathcal{J}(X)$	the deficit and the shell count of source 36's counting	E_i , POWER residuals; J , a module leaf; $J(a)$, tail positions (source 38)
$\ f\ _1$ $\rho; \rho_1, \rho_2$	sum of absolute coefficients the slope 743/1125 (source 36's counting)	Part IV's $\mathcal{N}(\cdot)$ (the norm notation $\mathcal{N}$) two natural aliases of source 36's POWER module
leaf	a positive variable of a POWER module, <i>including</i> its output and <i>excluding</i> its index	a gate or arithmetic operation: “twelve leaves, not twelve gates” [111]

The three sources' macros $\backslash\mathbb{N}$, $\backslash\text{Pos}$ and $\backslash\text{pos}$, and their norm $\backslash\text{norm}$, are written $\mathbb{N}$, $\mathbb{Z}_{>0}$ and $\|\cdot\|_1$ here; nothing else is renamed. Statement numbers follow this report's sections; equations are numbered by section (the sources numbered them consecutively).

[write] **What Part XIII does not claim.** None of the three sources gives one fixed polynomial for unbounded halting, a single-fold or finite-fold MRDP representation, an optimal or minimum witness count, degree, coefficient height or arithmetic circuit, a universal-polynomial record, an efficient bound on Pell witness heights, a faster bounded decision than simulation, novelty or priority, a Lean or other formal verification, an executed counter interpreter or physical simulator, or a new physical result; the acceptance tables are paid finite preprocessing, not free advice; every full composed fibre is infinite; and source 36's counts concern all encoded initializations, not halting inputs. Each source's own statement of its limits is printed in full.

Source 36 (batch-91 manuscript 18): Bounded halting certificates and exact encoded gap counts

Two arithmetic continuations of native integer initialization, “Research Report 66”, 4 October 2026. [write] From the archive `Bounded_certificates_and_encoded_gap_counts_source.zip`, main file `Report66/Report66.tex` (770 lines; a 21-page PDF); its labels carry the prefix `smc:bc:`, its sections are Sections 318–331 here, and its equations are numbered by section. Its six bibliography entries are this report's [103], [104], [105], [86], [106] and [107]. Its table of contents and page breaks are omitted. Its abstract follows.

A finite counter horizon admits a finite classification even when the initial counters are unbounded. We prove that clipping each native counter at T preserves every control decision through T transitions. Integer interpolation of the resulting $(T + 1)^2$ classes yields an explicit sum of six squares with exactly three positive witnesses, uniquely determined whenever a witness exists. For $T \geq 1$ its joint degree is at most $4(T + 1)^2 - 4$ and its expanded support has at most $16(T + 1)^2 - 5$ monomials. The finite acceptance table, coefficients and storage costs are paid explicitly. Two fully displayed POWER modules compose this certificate with three native integer gaps, giving 57 positive witnesses and 38 residual slots; the full composed fibers are infinite.

Independently of any program or halting predicate, we count all encoded positive integer gap triples by their total. Exact floor and digit formulas give slope 743/1125, while distinct represented totals have density 3/20. We prove a shell formula, an exact residue-class error law on a power-of-two subsequence, sharp normalized logarithm-squared error envelopes, and the inverse law with multiplicity, of slope 1125/743. These are separately compiled bounded-horizon polynomial families and arithmetic counts of all encoded initializations. No fixed-polynomial unbounded-halting result, halting-input density, optimality or priority claim follows.

318 The two questions and the main conclusions

The first question is how many existential variables a *fixed-horizon* counter certificate needs when its coefficients may depend effectively on the horizon. The answer below is three for a displayed construction. This is a trade between witness count and the finite table, degree and coefficients. The second question is how many positive integer triples realize the native gap encoding up to a given total. That count concerns geometry and divisibility alone; no counter program is run or selected by it.

The two parts share the encoding

$$\frac{g_1}{D} = \frac{1}{20} + \frac{1}{10}2^{-a}, \quad \frac{g_3}{D} = \frac{1}{20} + \frac{1}{10}2^{-b}, \quad D = g_1 + g_2 + g_3, \quad (318.1)$$

where $g_1, g_2, g_3 > 0$ are integers and $a, b \in \mathbb{N} = \{0, 1, \dots\}$. Its ordered counter pair is unique because $a \mapsto 2^{-a}$ is injective. We distinguish the interpolation size $Q = (T + 1)^2$, an individual gap total D , and the counting height X throughout.

Theorem 318.1 (Three positive witnesses). *Fix a finite deterministic two-counter program M , its initial state q_0 and its halt state H . For every fixed integer $T \geq 0$, the construction in Sections 319–322 gives an integer polynomial*

$$P_{M,T}(A, B; j, r, s)$$

with positive inputs A, B and exactly three positive witnesses j, r, s , such that a witness exists if and only if M reaches H within T transitions from counters $(A - 1, B - 1)$. The witness is unique. The polynomial is a sum of six specified residual squares, has degree at most $4Q - 4$ and support at most $16Q - 5$ for $T \geq 1$, and has degree two at $T = 0$. First halt exactly at T has the same counts and bounds.

Theorem 318.2 (Paid native gap composition). *The construction in Section 323 gives one integer polynomial in three external positive gaps and 57 positive witnesses. It is a sum of 38 residual squares. Its zeros assert that the gaps are encoded and that their decoded counters satisfy the chosen bounded-halting predicate. Its joint degree is $\max(12, \deg P_{M,T})$. The decoded counters and the three compressed witnesses are unique, but every nonempty complete witness fiber is infinite. The all-exponent POWER equivalence uses the explicitly pinned constructive Pell theorem pair.*

Theorem 318.3 (All encoded initializations). *Let $\mathcal{A}(X)$ be the number of encoded positive integer triples with total at most X , and set $\rho = 743/1125$. Then*

$$\mathcal{A}(X) = \rho X - E(X), \quad E(X) \geq 0,$$

with exact floor and digit formulas below. For integer $X \rightarrow \infty$,

$$\liminf \frac{E(X)}{(\log_2 X)^2} = 0, \quad \limsup \frac{E(X)}{(\log_2 X)^2} = 1.$$

The represented totals are exactly the multiples of 10 or 16. If D_n is the total of the n th triple, counting multiplicity, then

$$D_n = \frac{1125}{743}n + O(\log^2 n),$$

and the normalized nonnegative inverse deviations have $\liminf$ zero and $\limsup$ 1125/743.

The three-witness theorem is independent of the physical compiler and of Pell theory. The counting theorem needs only the elementary primitive-scale classification proved in Section 324. Physical transport is conditional on the retained encoded-input compiler theorem. None of these statements classifies arbitrary unencoded physical executions.

[write] [The “retained encoded-input compiler theorem” is source 34’s Theorem 294.1 (Part XII). The primitive-scale classification is source 35’s Theorem 312.1, with the same proof; it is quoted as a note in Section 324. Theorem 318.1’s degree bound is sharpened to exact values by source 37 (Theorem 332.2), its three witnesses are reduced to two and one by source 38 (Propositions 349.1 and 350.1), and Theorem 318.2’s 57 witnesses become 29 (source 37) and 28 or 27 (source 38). Source 36’s own statements are unchanged by these later improvements.]

319 Exact finite horizon clipping

A native instruction either increments one counter and moves to a specified successor, or tests one counter for zero, moving to one successor at zero and otherwise decrementing that counter and moving to another successor. The other counter is unchanged. The control graph is finite and may contain loops. We count transitions, so the initial state occurs at time zero. For bookkeeping only, extend the computation after H by leaving both counters and the state unchanged.

Lemma 319.1 (Exact clipping). *For a fixed $T \geq 0$, replace an initial pair (a, b) by*

$$(a^*, b^*) = (\min(a, T), \min(b, T)). \quad (319.1)$$

The original and clipped runs have identical states $q_0, \dots, q_T$ and choose identical branches at entry times $0, \dots, T-1$. In particular their first arrival at H , when it is at most T , agrees.

Proof. The initial states agree. Suppose the branches through the first t transitions agree, where $t < T$. Each corresponding counter has then received the same changes. If its original initial value was below T , the two values are identical. Otherwise its representative started at T and can have decreased by at most t , so its current value is at least $T - t > 0$. The original equals it plus the nonnegative initial difference and is also positive. Every tested counter therefore has the same zero/nonzero outcome in both runs. The next branch and changes agree. Increment instructions and virtual halt steps also agree. Induction proves the claim through transition T ; the control states determine first arrival at H . $\square$

The strict inequality is needed at instruction *entry*, not after the last transition. A representative counter can reach zero at time T while the original is still positive. Their final counters, final zero status and behavior beyond the horizon need not agree. At $T = 0$ there are no branches to compare, and the only halting question is whether $q_0 = H$.

The threshold is sharp for this uniform clipping argument when $T \geq 1$. Consider the one-instruction loop that moves to H if $a = 0$ and otherwise decrements a and repeats. Initial $a = T - 1$ first halts on transition T , whereas initial $a = T$ first halts on transition $T + 1$. Clipping at $T - 1$ would merge different answers. This establishes no lower bound on polynomial degree or witness arity.

[write] [Sources 37 and 38 restate this lemma with the same proof, source 37 also with the same sharpness example; their statements are quoted in Remarks 340.1 and 348.2. The Hilbert’s-tenth programme’s review checked this proof: “through transition T , a counter initially at least T and its clipped representative make the same zero tests, since tests occur before the final decrement” [111].]

With shifted positive inputs, put

$$A = a + 1, \quad B = b + 1, \quad K = T + 1, \quad Q = K^2, \quad u = \min(A, K), \quad v = \min(B, K). \quad (319.2)$$

Thus the exact shifted threshold is $T + 1$, rather than the more conservative $T + 2$.

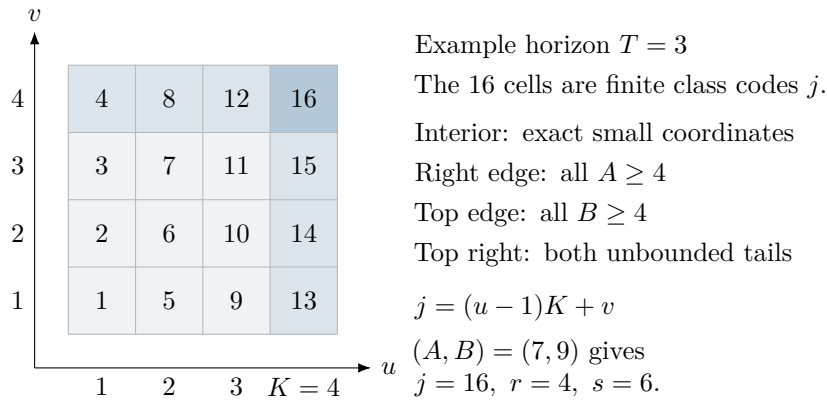


Figure 22: Finite classification with two deterministic tail slacks. The figure encodes classes only; it shows neither an acceptance table nor a simulated computation.

320 A finite table and integer interpolation

Index the Q representative pairs by

$$i = (u - 1)K + v, \quad u_i = 1 + \left\lfloor \frac{i - 1}{K} \right\rfloor, \quad v_i = 1 + ((i - 1) \bmod K), \quad 1 \leq i \leq Q. \quad (320.1)$$

In this formula the symbol u_i denotes the first coordinate; we use u for its selected value below. Let $S_{M,T}$ contain the indices whose representative inputs $(u_i - 1, v_i - 1)$ first reach H at some time at most T . For the exact-halt variant use $S_{M,T}^=$, containing precisely those with first arrival at time T .

Both sets are effectively computable finite tables. In principle, each of the Q representatives is evaluated for at most T transitions, checking q_0 first. At most QT native transitions suffice, and representative counters remain at most $2T$. These evaluations are a cost of constructing coefficients,

not a free oracle. No native interpreter is executed in this report. If $q_0 = H$, the by-horizon table is full for every T , and the exact-horizon table is empty for $T > 0$.

Fix either table and call it S . Let $d = (Q - 1)!$, with $0! = 1$. Define integer polynomials

$$\ell_i(z) = (-1)^{Q-i} \binom{Q-1}{i-1} \prod_{\substack{1 \leq h \leq Q \\ h \neq i}} (z - h), \quad (320.2)$$

$$U_0(z) = \sum_{i=1}^Q u_i \ell_i(z), \quad V_0(z) = \sum_{i=1}^Q v_i \ell_i(z), \quad (320.3)$$

$$R(z) = \prod_{i=1}^Q (z - i), \quad H_S(z) = \prod_{i \in S} (z - i). \quad (320.4)$$

An empty product is one. At the i th node,

$$\prod_{h \neq i} (i - h) = (-1)^{Q-i} (i - 1)! (Q - i)!,$$

so $\ell_i(i) = d$ and $\ell_i(h) = 0$ at all other nodes. Hence

$$U_0(i) = du_i, \quad V_0(i) = dv_i, \quad 1 \leq i \leq Q. \quad (320.5)$$

This directly proves the interpolation identities. Every coefficient is an integer; the only denominator clearing is by the fixed integer d , which depends on the horizon and not on any input or witness. The degrees of U_0, V_0 are at most $Q - 1$. The roots of H_S are exactly the accepted codes, with no roots when S is empty.

321 Six residuals and a unique witness triple

Use exactly three positive integer witnesses j, r, s . Define

$$\begin{aligned} R_0 &= R(j), \\ R_A &= (dA - U_0(j))(U_0(j) - dK), & L_A &= dA - U_0(j) - d(r - 1), \\ R_B &= (dB - V_0(j))(V_0(j) - dK), & L_B &= dB - V_0(j) - d(s - 1), \\ R_H &= H_S(j), \end{aligned} \quad (321.1)$$

and set

$$P_{M,T} = R_0^2 + R_A^2 + L_A^2 + R_B^2 + L_B^2 + R_H^2. \quad (321.2)$$

These finite formulas completely specify the polynomial. Program and horizon data determine coefficients; they are not additional polynomial variables. Six is a literal presentation count. Some slots can coincide or be identically zero.

Proof of Theorem 318.1 apart from its size bounds. An integer sum of squares vanishes exactly when each square vanishes. If $P_{M,T} = 0$, then $R(j) = 0$ forces $j \in \{1, \dots, Q\}$. Write $u = u_j$ and $v = v_j$. Using (320.5) and dividing only by the fixed positive d , the two A residuals imply

$$(A - u)(u - K) = 0, \quad A - u = r - 1 \geq 0. \quad (321.3)$$

If $u < K$, the first equation forces $A = u$. If $u = K$, the second forces $A \geq K$. These are exactly the cases $u = \min(A, K)$. The same argument proves $v = \min(B, K)$. Therefore j is the unique clipped class. The equation $H_S(j) = 0$ puts that class in the chosen acceptance table, and Lemma 319.1 gives the asserted halting predicate.

Conversely, if that predicate holds, take

$$u = \min(A, K), \quad v = \min(B, K), \quad j = (u - 1)K + v, \quad r = A - u + 1, \quad s = B - v + 1. \quad (321.4)$$

All three witnesses are positive. The range and acceptance residuals vanish. Each classification product vanishes because its coordinate either equals its input or equals K . The two slack residuals vanish by definition. Thus $P_{M,T} = 0$.

Finally the soundness argument forces u, v and then j . Each linear residual forces its slack. This proves uniqueness, including the bounds

$$1 \leq j \leq Q, \quad 1 \leq r \leq A, \quad 1 \leq s \leq B.$$

No trace values, sign choices, inactive selectors or interpolation denominators remain as free witnesses. $\square$

321.1 Horizon zero

When $T = 0$, we have $K = Q = d = 1$ and $U_0 = V_0 = 1$. Both classification products vanish identically, while $L_A = A - r$ and $L_B = B - s$. If $q_0 = H$ then

$$P_{M,0} = 2(j - 1)^2 + (A - r)^2 + (B - s)^2. \quad (321.5)$$

Otherwise

$$P_{M,0} = (j - 1)^2 + (A - r)^2 + (B - s)^2 + 1. \quad (321.6)$$

The accepted case has the unique tuple $(1, A, B)$ and the rejected case has none. Both displayed polynomials have degree two and nine nonzero monomials. The fixed count of three witnesses at this trivial horizon is a presentation convention, not a necessity claim. The by-horizon and exact-horizon questions coincide here.

322 Degree coefficient support and preprocessing costs

All degree statements are joint in the five variables A, B, j, r, s . For $T \geq 1$, $Q \geq 4$. The degrees of R_0, R_H are at most Q , those of L_A, L_B at most $Q - 1$, and those of R_A, R_B at most $2Q - 2$. Squaring gives

$$\deg P_{M,T} \leq \max(2Q, 4Q - 4, 2Q - 2) = 4Q - 4. \quad (322.1)$$

Cancellations can lower this bound; equality is not asserted or enforced artificially.

322.1 Coefficient height

For an integer polynomial f , let $\|f\|_1$ be the sum of the absolute values of its coefficients. Set

$$\Lambda = K 2^{Q-1} (Q + 1)!.$$

The norm of a product of factors $z - h$ is at most the product of $1 + h$. From (320.2), the binomial sum 2^{Q-1} , and $u_i, v_i \leq K$, we obtain

$$\|U_0\|_1, \|V_0\|_1 \leq \Lambda, \quad \|R\|_1, \|H_S\|_1 \leq (Q+1)! \leq \Lambda, \quad dK, d \leq \Lambda. \quad (322.2)$$

Coefficient norms are subadditive and submultiplicative. Each classification residual has norm at most $4\Lambda^2$; each linear slack residual has norm at most 4Λ . Consequently

$$\|P_{M,T}\|_1 \leq 2\Lambda^2 + 2(4\Lambda^2)^2 + 2(4\Lambda)^2 = 32\Lambda^4 + 34\Lambda^2 \leq 66\Lambda^4. \quad (322.3)$$

This also holds at $T = 0$. A nonzero coefficient therefore needs at most $7 + 4\lceil \log_2 \Lambda \rceil$ unsigned magnitude bits, plus a sign bit if required. The resulting bound is

$$O(Q \log(Q+1)) = O(T^2 \log(T+2))$$

for positive horizons. It is uniform in the program because the program changes only a subset of the same finite node set.

322.2 Sparse expanded support

A dense five-variable monomial bound would conceal the special structure. Put $m = Q - 1$. The first classification residual has the form

$$R_A = Af_A(j) + g_A(j), \quad f_A = d(U_0 - dK), \quad g_A = -U_0(U_0 - dK),$$

with degrees at most m and $2m$. Its square has a pure- j part of degree at most $4m$, an A part of j -degree at most $3m$, and an A^2 part of j -degree at most $2m$. The B classification has the same structure. Moreover

$$L_A = dA - dr + (d - U_0(j)),$$

so its square adds only pure- j , A , and r terms and the monomials A^2, r^2, Ar ; the coefficient of r has j -degree at most m . The B, s counterpart is identical. The remaining two squares are pure- j and have degree at most $2Q \leq 4m$.

Grouping disjoint monomial types gives the explicit ceiling in Table 46. No monomial contains

Monomial type	Maximum number
Pure powers of j	$4m + 1$
A^2 or B^2 times powers of j	$2(2m + 1)$
A or B times powers of j	$2(3m + 1)$
r or s times powers of j	$2(m + 1)$
r^2, s^2, Ar, Bs	4
Total	$16m + 11 = 16Q - 5$

Table 46: An expanded-support bound for $T \geq 1$. Shared terms are grouped before counting, and cancellations can only decrease the support.

both A and B , both r and s , or any undeclared combination of these four variables. Storing each nonzero coefficient in signed binary and its five exponents in binary costs at most

$$O(Q^2 \log(Q+1)) = O((T+1)^4 \log(T+2)) \quad (322.4)$$

bits. This is an upper bound for that expanded storage convention, not an optimality result.

322.3 What the effective generator pays

For the selected predicate the acceptance table consists of exactly Q Boolean values. In principle constructing it takes at most QT native transitions. The table is independent of the eventual large inputs A, B , but determining it is substantive finite work.

A direct coefficient generator can build each of the Q basis products by repeated multiplication by a linear factor, requiring $O(Q^2)$ integer arithmetic operations per product and $O(Q^3)$ overall. Adding, scaling and ordinary convolution to expand the six squares fit within $O(Q^3)$ further integer arithmetic operations. Product and triangle bounds as above keep intermediate magnitudes within $O(Q \log(Q + 1))$ bits. This is an arithmetic-operation bound, not a unit-cost bit-complexity claim.

There is one uniform effective procedure which takes the finite program and horizon and produces the appropriate polynomial. Nevertheless, each T selects a separately compiled polynomial with its own degree and coefficients. When T is written in binary, these costs need not be polynomial in its input length. The theorem supplies no faster method for deciding one bounded instance than its ordinary simulation, and no free access to its acceptance table.

322.4 Why an unbounded horizon does not give one polynomial

For a fixed program, halting is of course equivalent to acceptance at some finite horizon. But the expression $P_{M,T}$ changes its table, factorial, products, degrees and coefficients with T . Placing an existential phrase “there is a T ” outside this family does not express those changes inside a single fixed polynomial. This construction contains no fixed arithmetic module encoding the map from T to its coefficient table. Its unique bounded witness triples therefore imply no single-fold or finite-fold fixed-polynomial representation of unbounded halting. Using first-halt tables changes that conclusion in no way.

323 Composition with paid native gap decoding

The external inputs are arbitrary positive integers g_1, g_2, g_3 , and $D = g_1 + g_2 + g_3$ remains an expression, not a new variable. The predicate now includes both encoding (318.1) and bounded halting of the decoded counters. Introduce positive shifted-counter witnesses A, B and positive power outputs p, q . Two independent copies of the following module enforce

$$p = 2^{A-1}, \quad q = 2^{B-1}. \quad (323.1)$$

The construction replaces the growing trace in the retained native-gap certificate by the compressed polynomial (321.2); it leaves the original source packet unchanged.

323.1 All leaves and all residuals of one POWER module

For a positive input index C and output o , one copy has thirteen directly positive leaves

$$o, w, M_0, g, x_p, y_p, u_p, v_p, s_p, t_p, q_b, q_v, J_p.$$

Two more positive leaves α_+, β_+ define expressions $\alpha = \alpha_+ + 1$ and $\beta = \beta_+ + 1$. Eleven natural aliases

$$d_{wb}, d_{wk}, d_{yk}, \alpha_1, \alpha_2, \sigma_1, \sigma_2, \tau_1, \tau_2, \rho_1, \rho_2$$

are each represented by their own distinct positive leaf minus one. These are substitutions, not new equations or extra uncounted variables. Thus a copy contains $13 + 2 + 11 = 26$ positive leaves including o , or 25 auxiliaries beyond its output. The already counted C is not a module leaf. M_0 is unrelated to the source program M , and the module leaf g is unrelated to the three external gaps.

After all these substitutions, the fifteen residual polynomials are

$$\begin{aligned}
E_1 &= x_p^2 - 1 - (\alpha^2 - 1)y_p^2, \\
E_2 &= u_p^2 - 1 - (\alpha^2 - 1)v_p^2, \\
E_3 &= s_p^2 - 1 - (\beta^2 - 1)t_p^2, \\
E_4 &= \beta - 1 - 4y_p q_b, \\
E_5 &= \beta + u_p \alpha_1 - \alpha - u_p \alpha_2, \\
E_6 &= v_p - y_p^2 q_v, \\
E_7 &= s_p + u_p \sigma_1 - x_p - u_p \sigma_2, \\
E_8 &= t_p + 4y_p \tau_1 - C - 4y_p \tau_2, \\
E_9 &= y_p - C - d_{yk}, \\
E_{10} &= w - 2 - d_{wb}, \\
E_{11} &= w - C - d_{wk}, \\
E_{12} &= M_0 - 2o - J_p, \\
E_{13} &= \alpha^2 - 1 - ((w + 1)^2 - 1)(wg)^2, \\
E_{14} &= 4\alpha - M_0 - 5, \\
E_{15} &= x_p + M_0 \rho_1 - y_p(\alpha - 2) - 2o - M_0 \rho_2.
\end{aligned} \tag{323.2}$$

Every leaf in one copy is distinct from every leaf in the other copy, the external gaps, A, B , and j, r, s .

323.2 The precise imported theorem and positive adapters

Remark 323.1 ([write] restatement of source 35's Lemma 309.1). Source 36 states: “The sole substantive number-theoretic dependency is the constructive theorem pair `Pell.matiyasevic` and `Pell.eq_pow_of_pell` in the pinned mathlib source [86]. Their statements were read as inert text; no Lean execution is part of this report. The specialization yields

$$(\exists \text{ the 25 positive auxiliary leaves : } E_1 = \dots = E_{15} = 0) \iff o = 2^{C-1}, \quad C, o > 0.” \tag{323.3}$$

This module and this equivalence are source 35's (Section 309 and Lemma 309.1 of Part XII), with the letters renamed from source 35 to source 36 as $\eta_i \mapsto g$, $M_i \mapsto M_0$, $(x_i, y_i, u_i, v_i, s_i, t_i) \mapsto (x_p, y_p, u_p, v_p, s_p, t_p)$, $J_i \mapsto J_p$, $(d_{w2,i}, d_{wC,i}, d_{yC,i}) \mapsto (d_{wb}, d_{wk}, d_{yk})$, $(r_{1,i}, r_{2,i}) \mapsto (\rho_1, \rho_2)$ and $(C_i, o_i) \mapsto (C, o)$; the fifteen residuals (323.2) are source 35's, in the same order. Source 36's proof of (323.3) (soundness through the nonzero-index branch of `Pell.matiyasevic` and $\alpha > w \geq 2$; completeness through the constructive power theorem with every positive adapter, including $C = 1$; and the remark that natural subtraction adds no loophole) and its exponent-zero fixture ($C = o = 1$, $w = 2$, $M_0 = 63$, $g = 3$, $\alpha = \beta = 17$, $x_p = s_p = 17$, $y_p = t_p = 1$, $u_p = 577$, $v_p = q_v = 34$, $q_b = 4$, $J_p = 61$, $d_{wk} = 1$ and every other natural alias zero) are source 35's proof and fixture, step for step, and are not printed twice. Source 36 adds one warning: “The exponent is $C - 1$, but the *Pell index* is C ; using index $C - 1$ would mishandle the boundary.” The same module is the 26-leaf predecessor

of source 37 (Section 333) and the base-two case of the general-base POWER system of Report 58 [source 28, Part X], itself a re-derivation of the fifteen-equation Pell macro `ptr:ai:lem:exp` of the collection's report *periodic-turmite-first-revisits*. The module letters ρ_1, ρ_2 have nothing to do with the slope $\rho = 743/1125$ of Section 327.

323.3 The full composed polynomial and its ledger

Add the two gap residuals

$$G_1 = (20g_1 - D)p - 2D, \quad G_3 = (20g_3 - D)q - 2D. \quad (323.4)$$

Let E_i^A and E_i^B be the independent copies of (323.2) with $(C, o) = (A, p)$ and (B, q) . Define

$$F_{M,T} = G_1^2 + G_3^2 + \sum_{i=1}^{15} (E_i^A)^2 + \sum_{i=1}^{15} (E_i^B)^2 + P_{M,T}(A, B; j, r, s). \quad (323.5)$$

Equations (321.1), (323.2) and (323.4) specify every residual in the final polynomial. No power relation remains uninterpreted.

Correctness. A zero of $F_{M,T}$ makes every residual vanish. Equation (323.3) forces (323.1). Since $D, p, q > 0$, the gap equations force positive denominators $20g_1 - D$ and $20g_3 - D$. Dividing by positive quantities gives exactly (318.1), with $a = A - 1, b = B - 1$. Theorem 318.1 then gives the required bounded-halting predicate. Conversely, an encoded input satisfying that predicate supplies its unique A, B, p, q , constructive POWER auxiliaries and the triple (321.4). Every residual vanishes. $\square$

Resource in the displayed native gap formula	Count
External positive inputs	3
Shifted-counter witnesses A, B	2
Two POWER copies including outputs p, q	$2 \cdot 26$
Compressed witnesses j, r, s	3
All positive witnesses	57
All variables including the inputs	60
Residual slots	$2 + 30 + 6 = 38$
Final polynomial equations	1

Table 47: Literal counts for every $T \geq 0$, for either by-horizon or exact-horizon acceptance. Identically zero or duplicate slots at $T = 0$ are retained.

The gap residuals have degree two. In each POWER copy, E_{13} has degree six with leading monomial $-w^4g^2$, and every other residual has degree at most four. Its square therefore contributes w^8g^4 with coefficient one. The two such monomials use separate module leaves and cannot cancel with each other or with P , which uses no module leaf. Hence

$$\begin{aligned} \deg F_{M,T} &= \max(12, \deg P_{M,T}), \\ \deg F_{M,T} &\leq \max(12, 4(T+1)^2 - 4) \quad (T \geq 1), \\ \deg F_{M,0} &= 12. \end{aligned} \quad (323.6)$$

If $\deg P > 12$, its top terms cannot be affected by the fixed degree-twelve base. This proves the equality in the first line without claiming equality in the second. The POWER and gap base has coefficients independent of M, T , so its addition preserves the stated coefficient-bit asymptotic, up to a fixed contribution.

323.4 Unique projection and infinite full fibers

For a given accepted input, the gap equations force

$$p = \frac{2D}{20g_1 - D}, \quad q = \frac{2D}{20g_3 - D}.$$

Their powers-of-two exponents determine A, B uniquely, and the compressed theorem fixes j, r, s . But the complete 57-tuple is never unique when it exists. In the first module, increase α_1 and α_2 by the same nonnegative integer. They appear only together in E_5 ; every residual is preserved, and both positive adapters remain positive. Arbitrarily many such shifts yield infinitely many distinct full witnesses. Unique decoding and unique compressed projection must therefore not be described as a finite-fold full gap representation.

323.5 Conditional physical interpretation

The retained unlocked five-signal theorem identifies each native transition with one ordered instruction section on encoded inputs. Conditional on that theorem, (323.5) certifies bounded instruction-section reachability from those inputs. If first halt is at $k > 0$, the retained physical prefix takes strictly between kD and $10kD$ time and at most $32k$ binary contacts. At $k = 0$ the initial section is already halted. These are the retained theorem's bounds; the present arithmetic result does not re-audit the physical construction or depend on the clock refinement of Report 65 [source 35, Theorem 304.1]. A physical collision horizon is not identified with a native instruction horizon, and virtual absorbing halt steps are only arithmetic bookkeeping.

[write] [The “retained unlocked five-signal theorem” is source 34’s Theorem 294.1 (Part XII), at most 32 binary collisions per instruction with duration strictly between D and $10D$. With source 35’s clock every instruction takes exactly $30D$; source 36 does not use that refinement.]

324 Primitive integer realization and the counting object

The remainder of the report counts *all* encoded initializations. Neither M nor T occurs in the counting definitions. An initialization is a positive integer triple, counted once; its Diophantine witness tuples are not counted.

Remark 324.1 ([write] restatement of source 35’s Theorem 312.1: exact primitive scale). Source 36 states, as its Proposition 7.1: “For each ordered pair $(a, b) \in \mathbb{N}^2$, put $m = \max(a, b)$ and

$$U = 2^m + 2^{m-a+1}, \quad W = 2^m + 2^{m-b+1}, \quad V = 20 \cdot 2^m - U - W. \quad (324.1)$$

If $c = \gcd(U, V, W)$, then the primitive triple is $(U, V, W)/c$, and every encoded integer triple for

this pair is a unique positive integer multiple of it. Its total is

$$d(a, b) = \begin{cases} 20, & (a, b) = (0, 0), \\ 10, & (a, b) = (1, 1), \\ 20, & \{a, b\} = \{0, 1\}, \\ 2^{m+1}, & m \geq 2 \text{ and } a \equiv b \equiv 3 \pmod{4}, \\ 10 \cdot 2^m, & m \geq 2 \text{ otherwise.} \end{cases} \quad (324.2)$$

In every case $d(a, b) \geq 2^{m+1}$."

This is source 35's Theorem 312.1 (Part XII), whose gcd list $c \in \{1, 4, 2, 10, 2\}$ gives $d(a, b) = 20 \cdot 2^m / c$, the minimum total $D_{\min}$ there. Source 36's proof is source 35's: the small cases $(3, 14, 3)$, $(4, 32, 4)$ and $(6, 30, 4)$ directly; for $m \geq 2$ exact 2-adic valuation one and the only possible odd factor 5, which divides U exactly when $a \equiv 3 \pmod{4}$ because 2 has order four modulo 5; and a Bézout argument for the integer multiple. It is not printed twice. The added lower bound holds because $c \leq 10$, so $d(a, b) \geq 2 \cdot 2^m$. Source 36 cites this classification as inherited, from the retained native-gap proof (its bibliography entry [105], Sections 3 and 6), and its independent counting review re-derived it through least common multiples of reduced denominators (Section 330).

Here $a \equiv b \equiv 3 \pmod{4}$ means *both* counters have residue 3, and does not require $a = b$. Each counter pair supplies a ray of integer multiples; distinct ordered pairs have distinct ratios and therefore disjoint rays.

324.1 Counter index and numeric height are different

For $m \geq 2$, the primitive total lies between 2^{m+1} and $10 \cdot 2^m$, so its bit length is $m + O(1)$ and its numeric size is $\Theta(2^m)$. At a counting height X , a ray can contribute only if $m \leq \log_2 X - 1$. Thus only logarithmically many counter layers occur, but the early rays contribute linearly many multiples. The counter-layer index m is not the counting height, the primitive total, or the rank of a triple.

If the largest native counter is itself given in binary with length ℓ , then m can be of order 2^ℓ . The chosen gap description's bit height is correspondingly exponential in ℓ . This is a cost of this particular encoding, not a general encoding lower bound and not a bound on the larger Pell witnesses.

Definition 324.2 (Triple count and shell count). For integer $X \geq 0$, let $\mathcal{A}(X)$ count encoded positive integer triples with $D \leq X$, and set $\mathcal{A}(0) = 0$. For $X \geq 1$, put

$$\mathcal{J}(X) = \mathcal{A}(X) - \mathcal{A}(X - 1).$$

Thus $\mathcal{J}(X)$ counts triples at the one total X .

By Remark 324.1 and disjointness of the rays,

$$\mathcal{A}(X) = \sum_{a, b \geq 0} \left\lfloor \frac{X}{d(a, b)} \right\rfloor. \quad (324.3)$$

Only finitely many summands are nonzero. This counts neither distinct counter pairs, distinct totals, witness tuples nor halting inputs.

325 Exact floor formulas and a digital identity

For integers $b \geq 2$ and $n \geq 0$, define

$$F_b(n) = \sum_{k \geq 0} (2k+1) \left\lfloor \frac{n}{b^k} \right\rfloor. \quad (325.1)$$

Theorem 325.1 (Exact cumulative formula). *For every integer $X \geq 0$,*

$$\mathcal{A}(X) = F_2\left(\left\lfloor \frac{X}{10} \right\rfloor\right) + F_{16}\left(\left\lfloor \frac{X}{16} \right\rfloor\right) - F_{16}\left(\left\lfloor \frac{X}{80} \right\rfloor\right). \quad (325.2)$$

Proof. There are $2m+1$ ordered pairs with maximum m . Assign all of them the baseline scale $10 \cdot 2^m$. The pair $(0,0)$ actually changes a contribution $\lfloor X/10 \rfloor$ to $\lfloor X/20 \rfloor$, whereas $(1,1)$ makes the inverse change. These two low-counter corrections cancel exactly.

The remaining exceptional pairs have maximum $m = 4k+3$. Among the $k+1$ possible counter values $3, 7, \dots, 4k+3$, exactly $(k+1)^2 - k^2 = 2k+1$ ordered pairs have that maximum. For each, replace the old scale $80 \cdot 16^k$ by 16^{k+1} . Summing the baseline and corrections gives (325.2), using $\lfloor \lfloor X/c \rfloor / b^k \rfloor = \lfloor X/(cb^k) \rfloor$. $\square$

Write the base- b digits of n as $n = \sum_{j \geq 0} \delta_j b^j$, and define

$$s_b(n) = \sum_j \delta_j, \quad w_b(n) = \sum_j j \delta_j.$$

Both sums are zero at $n = 0$.

Proposition 325.2 (Digit form). *For every integer base $b \geq 2$ and integer $n \geq 0$,*

$$F_b(n) = \frac{b(b+1)n - (3b-1)s_b(n)}{(b-1)^2} - \frac{2w_b(n)}{b-1}. \quad (325.3)$$

Proof. Expanding each floor and reversing the finite sums gives

$$F_b(n) = \sum_j \delta_j \sum_{k=0}^j (2k+1) b^{j-k}.$$

The inner sum is

$$\frac{b(b+1)b^j - (3b-1)}{(b-1)^2} - \frac{2j}{b-1}.$$

To verify this without importing a summation identity, observe that its value at $j = 0$ is 1, and that both this expression and the inner sum satisfy the recurrence $S_j = bS_{j-1} + (2j+1)$. Substitution establishes the displayed value and then (325.3). $\square$

Equations (325.2) and (325.3) express the complete cumulative count by floors, digit sums and weighted digit sums. No external sequence identification is needed.

326 Shell multiplicities and represented total scales

Let $v_2(X)$ be the exponent of 2 dividing $X \geq 1$, including value zero for odd X .

Theorem 326.1 (Exact shell). *Writing $v = v_2(X)$,*

$$\mathcal{J}(X) = \begin{cases} v^2, & 5 \mid X, \\ \lfloor v/4 \rfloor^2, & 5 \nmid X. \end{cases} \quad (326.1)$$

Consequently the represented positive totals are exactly the multiples of 10 or 16. Their distinct-scale count is

$$\mathcal{B}(X) = \left\lfloor \frac{X}{10} \right\rfloor + \left\lfloor \frac{X}{16} \right\rfloor - \left\lfloor \frac{X}{80} \right\rfloor = \frac{3}{20}X + O(1). \quad (326.2)$$

Proof. A primitive scale d contributes exactly one triple at total X if and only if $d \mid X$. The low-counter exchange in the proof of Theorem 325.1 also cancels for these divisor indicators.

If $5 \mid X$, a baseline scale $10 \cdot 2^m$ divides X precisely for $0 \leq m \leq v - 1$. Their multiplicities sum to $1 + 3 + \cdots + (2v - 1) = v^2$, with an empty sum at $v = 0$. Each exceptional old scale $80 \cdot 16^k$ divides X exactly when its replacement 16^{k+1} does, namely when $v \geq 4k + 4$. All special corrections therefore cancel.

If $5 \nmid X$, no baseline scale divides X . Only the replacement scales contribute, with $0 \leq k < \lfloor v/4 \rfloor$. Their multiplicities sum to $\lfloor v/4 \rfloor^2$. This proves (326.1). Positivity of that formula is equivalent to $10 \mid X$ or $16 \mid X$. Inclusion-exclusion, with common multiples 80, gives (326.2). $\square$

The two counts answer different questions: $\mathcal{B}$ counts a represented total once, whereas $\mathcal{A}$ counts it $\mathcal{J}(X)$ times. For example, total 20 supports four triples, and total 80 supports sixteen. The density $3/20$ is a density among positive integer totals, not a triple-count slope.

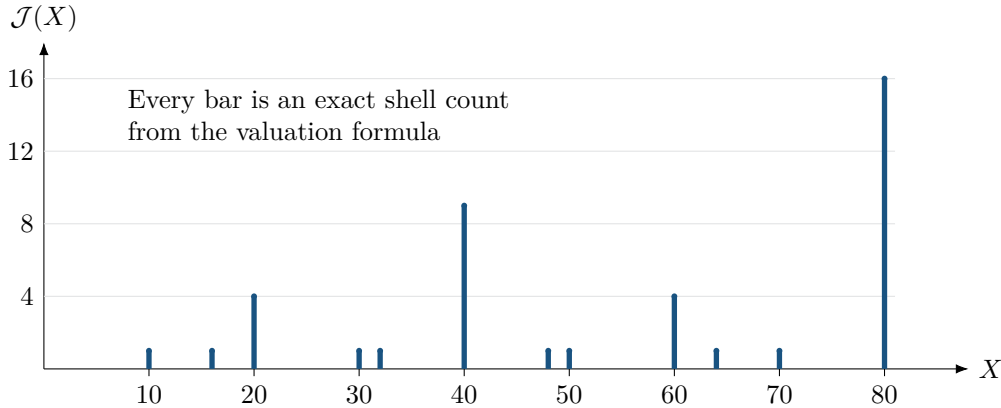


Figure 23: The represented totals through 80. The low bars at 16, 32, 48 and 64 come from the factor-5-free exceptional rays. At 80 both divisibility patterns coincide, and the shell contains $4^2 = 16$ triples. This vector plot uses the explicit formula, not a machine execution.

327 Linear growth and the sharp cumulative error

The reciprocal primitive-scale sum is absolutely convergent. Indeed, the bound $d(a, b) \geq 2^{m+1}$ and $2m + 1$ pairs per layer dominate it by a convergent geometric series with a linear weight. The same

baseline-and-correction argument gives

$$\begin{aligned}\rho &:= \sum_{a,b \geq 0} \frac{1}{d(a,b)} = \frac{1}{10} \sum_{m \geq 0} \frac{2m+1}{2^m} + \left(\frac{1}{16} - \frac{1}{80}\right) \sum_{k \geq 0} \frac{2k+1}{16^k} \\ &= \frac{3}{5} + \frac{68}{1125} = \frac{743}{1125}.\end{aligned}\tag{327.1}$$

For completeness, $\sum_{k \geq 0} (2k+1)x^k = (1+x)/(1-x)^2$ for $|x| < 1$: multiplication by $1-x$ yields $1 + 2\sum_{k \geq 1} x^k$, which is a geometric series. The low-counter reciprocal corrections cancel as before.

Define the deficit

$$E(X) = \rho X - \mathcal{A}(X) = \sum_{a,b \geq 0} \left(\frac{X}{d(a,b)} - \left\lfloor \frac{X}{d(a,b)} \right\rfloor \right) \geq 0.\tag{327.2}$$

The series converges by domination by $X \sum 1/d(a,b)$.

Proposition 327.1 (Explicit uniform envelope). *For $X \geq 1$, put $L = \lfloor \log_2 X \rfloor$. Then*

$$0 \leq E(X) < L^2 + 4L + 6.\tag{327.3}$$

In particular $\mathcal{A}(X)/X \rightarrow \rho$.

Proof. There are $(L+1)^2$ pairs with maximum counter at most L , each contributing strictly less than one in (327.2). The remaining tail is at most

$$X \sum_{m=L+1}^{\infty} \frac{2m+1}{2^{m+1}} = X \frac{2L+5}{2^{L+1}} < 2L+5.$$

The tail identity follows by writing $m = L+1+h$ and summing the constant and linear geometric terms. Adding the two bounds gives (327.3). Dividing by X proves the limit. $\square$

The slope ρ is not a density among all positive triples with bounded total. Those triples number $\binom{X}{3}$ for $X \geq 3$, so the fraction that are encoded tends to zero. The linear law describes the much thinner encoded family on its own.

327.1 A divisibility subsequence

For integers $M \geq 1$, write

$$X_M = 10 \cdot 2^M.\tag{327.4}$$

Every primitive scale with maximum counter at most M divides X_M , and none with larger maximum divides it. Normal scales have the required factor 5 and exponent of 2, and exceptional scales have exponent $m+1$, while $v_2(X_M) = M+1$. The low-counter cases are all covered by $M \geq 1$. Thus

$$\mathcal{J}(X_M) = (M+1)^2.\tag{327.5}$$

This cutoff is a statement about *divisibility*, not size. A special scale with maximum larger than M can still be below X_M : for instance $(a,b) = (3,3)$ has primitive total 16, below $X_1 = 20$, but $16 \nmid 20$.

All terms through maximum M vanish in (327.2); the tail bound with cutoff M gives $E(X_M) = O(M)$. From the shell jump,

$$E(X_M - 1) = E(X_M) + (M+1)^2 - \rho = M^2 + O(M).\tag{327.6}$$

Because $\log_2 X_M = M + O(1)$, Proposition 327.1 and these two subsequences prove

$$\liminf_{X \rightarrow \infty} \frac{E(X)}{(\log_2 X)^2} = 0, \quad \limsup_{X \rightarrow \infty} \frac{E(X)}{(\log_2 X)^2} = 1. \quad (327.7)$$

In particular the uniform $O(\log^2 X)$ error cannot be replaced by $o(\log^2 X)$. Nor is there a single constant c for which $E(X) = c(\log_2 X)^2 + o(\log^2 X)$ along all integers. This is a precise oscillation statement, rather than an undefined assertion about a smooth correction.

328 Exact residue class errors on the subsequence

The $O(M)$ error on (327.4) has an exact formula.

Theorem 328.1 (Exact linear error). *For every $M \geq 1$, with $r = M \bmod 4$,*

$$E(X_M) = \alpha_r M + \beta_r, \quad (328.1)$$

where the constants are

r	α_r	β_r
0	34/15	1261/225
1	61/30	2329/450
2	31/15	1189/225
3	32/15	1223/225

(328.2)

The exact left-neighbor deficit is this expression plus $(M + 1)^2 - \rho$.

Proof. The binary digit identity gives $F_2(2^M) = 6 \cdot 2^M - 2M - 5$. For $M \geq 3$, write $M - 3 = 4q + t$ with $0 \leq t \leq 3$, and put $c = 2^t$ and $h = \lfloor 5c/16 \rfloor$. The two base-16 arguments in (325.2) are then $5c16^q$ and $c16^q$.

Their digit-sum difference is $\delta = 4c - 15h$, because $5c$ has low digit $5c - 16h$ and next digit h , while c is a single base-16 digit. Their weighted-digit-sum difference is $q\delta + h$. For $t = 0, 1, 2, 3$ the pairs (δ, h) are respectively

$$(4, 0), \quad (8, 0), \quad (1, 1), \quad (2, 2).$$

Substituting the digit identity with $b = 16$ into (325.2), the linear terms give exactly ρX_M and the remaining terms give

$$E(X_M) = 2M + 5 + \left(\frac{47}{225} + \frac{2q}{15} \right) \delta + \frac{2h}{15}. \quad (328.3)$$

Inserting $q = (M - 3 - t)/4$ and the four pairs above yields (328.2), with $r \equiv t + 3 \pmod{4}$. The two remaining cases are direct: $\mathcal{A}(20) = 6$ and $\mathcal{A}(40) = 17$. Their deficits agree with the stated $r = 1, 2$ formulas. The left-neighbor assertion follows from (327.6). $\square$

These exact constants depend on the residue of the *exponent* M , not on a residue class of an arbitrary counting height X . No claim is made that the same linear formula describes all heights.

329 The inverse law with multiplicity

For integer $n \geq 1$, define

$$D_n = \min\{X \geq 0 : \mathcal{A}(X) \geq n\}. \quad (329.1)$$

This is the total of the n th encoded triple in a nondecreasing-total ordering. Ties may be ordered arbitrarily; the value X is repeated exactly $\mathcal{J}(X)$ times. Existence follows already from the pair $(1, 1)$, whose primitive scale is 10.

Theorem 329.1 (Inverse asymptotic and sharp constants). *The inverse deviation is nonnegative and*

$$D_n = \frac{n}{\rho} + O(\log^2 n), \quad \rho = \frac{743}{1125}. \quad (329.2)$$

Moreover

$$\liminf_{n \rightarrow \infty} \frac{D_n - n/\rho}{(\log_2 n)^2} = 0, \quad \limsup_{n \rightarrow \infty} \frac{D_n - n/\rho}{(\log_2 n)^2} = \frac{1}{\rho} = \frac{1125}{743}. \quad (329.3)$$

Proof. Because $\mathcal{A}(D_n) \leq \rho D_n$ and $\mathcal{A}(D_n) \geq n$, we have $D_n \geq n/\rho$. The scale-10 ray gives $D_n \leq 10n$. The defining inequality $\mathcal{A}(D_n - 1) < n$ implies

$$0 \leq \rho D_n - n < \rho + E(D_n - 1). \quad (329.4)$$

Using (327.3) and $D_n \leq 10n$ proves (329.2).

For the sharp upper coefficient, let $L_n = \lfloor \log_2(D_n - 1) \rfloor$. It is at most $\log_2 n + O(1)$. Equations (329.4) and (327.3) therefore give

$$D_n - \frac{n}{\rho} \leq \frac{1}{\rho}(\log_2 n)^2 + O(\log n).$$

This keeps the coefficient one from the cumulative envelope; merely inverting an unspecified $O(\log^2 X)$ error would not establish the sharp constant.

For matching subsequences, define the first and last ranks of the X_M block by

$$n_M^{\text{first}} = \mathcal{A}(X_M - 1) + 1, \quad n_M^{\text{last}} = \mathcal{A}(X_M).$$

Both inverse totals equal X_M , and both ranks are asymptotic to ρX_M . Using (327.5) and Theorem 328.1, with $r = M \bmod 4$, yields the exact formulas

$$\begin{aligned} D_{n_M^{\text{first}}} - \frac{n_M^{\text{first}}}{\rho} &= \frac{E(X_M) + (M+1)^2 - 1}{\rho} = \frac{M^2 + (2 + \alpha_r)M + \beta_r}{\rho}, \\ D_{n_M^{\text{last}}} - \frac{n_M^{\text{last}}}{\rho} &= \frac{\alpha_r M + \beta_r}{\rho}. \end{aligned} \quad (329.5)$$

Since $\log_2 n_M^{\text{first}}$ and $\log_2 n_M^{\text{last}}$ are $M + O(1)$, the first sequence attains normalized limit $1/\rho$ and the last attains zero. Together with nonnegativity and the preceding upper bound, this proves (329.3). $\square$

For comparison, the inverse of the *distinct*-scale function $\mathcal{B}$ is $(20/3)n + O(1)$. Its slope differs because every represented total is counted only once. Neither inverse indexes primitive counter pairs by their maximum, and neither has a halting interpretation.

330 Evidence dependency boundaries and reproducibility

The article assembles two frozen arithmetic continuations [103, 104] without changing them. Its primary mathematical arguments are the English proofs above. Retained numerical and symbolic checks provide additional finite evidence with distinct scopes.

330.1 The original static evidence

The three-witness packet records exact sparse-integer interpolation checks for horizons $T = 0, \dots, 6$, including every interpolation node and finite grids of positive inputs. It tests literal acceptance tables, not tables discovered by interpreting a counter program. For $Q = 1$ and $Q = 4$ it also exhausts bounded positive witness boxes for every acceptance subset, respectively two and sixteen subsets. Its stored checks account for 1,600 and 72,000 polynomial assignments in those tests.

The same evidence records the 57-witness, 38-slot gap ledger and the independent degree-twelve module monomials. It evaluates the full exponent-zero fixture at the encoded triple $(3, 14, 3)$ and checks that a common quotient-pair shift preserves the complete assignment. These tests support the displayed construction; they do not prove exponentiation for every index. For example, the recorded degree at $T = 2$ is 28, below the general bound 32, illustrating why the degree statement is an upper bound.

The counting packet records exact checks through total 200,000, 95,019 digit-identity cases, the four-class formula for $M = 1, \dots, 300$, and inverse blocks covering 131,968 ranks. These are the original packet's recorded results. Their source programs are retained inertly and are not executed or imported by the report's build or release tools.

330.2 Independent review

The independent counting review [106] accepts the frozen proof without mathematical correction. Its separate primitive-scale path uses least common multiples of reduced coordinate denominators, rather than the original gcd construction. A second path enumerates all 669,920 ordinary positive triples with total at most 160 and decodes their endpoint ratios. Its completed evidence also checks all totals through 250,000, all 164,979 ranks represented there, 4,225 counter pairs, 63,519 digital-identity cases, the four-class identities through $M = 400$, and independent denominator-derived subsequence counts through $M = 48$.

The independent bounded-certificate audit [107] accepts the main theorem, the final sparse-support refinement and the stated resource bounds without correction. It reconstructs rational Lagrange interpolation by synthetic division and then clears denominators, independently of the packet's polynomial implementation. Its completed evidence includes 140 node identities, 28 expanded native instances, 91,800 literal positive witness triples for all acceptance subsets at $Q = 1, 4$, all 512 tables at $Q = 9$, and 12 complete gap compositions. It also supplies a data-only external coefficient checker: this authenticates a supplied polynomial against a declared table, but does not discover or certify a program's acceptance table by execution. The gap result retains its pinned Pell dependency, and physical transport retains its separate conditional status.

Report assembly does not treat either packet's own checker as an independent oracle, and no finite test replaces the all-input clipping and uniqueness proofs.

[write] [In this report the two science packets are shipped as `36-bounded-cert-bc-*` and `36-bounded-cert-cnt-*` (proofs `36-bounded-cert-bc-PROOF.md` and `36-bounded-cert-c`

nt-PROOF.md, programs `code/36-bounded-cert-bc-static_algebra.py` and `code/36-bounded-cert-cnt-check_counting.py`, records under `data/`), and the two audit dossiers as `36-bounded-cert-audit-bc-*` and `36-bounded-cert-audit-cnt-*`; the README maps every delivered name. Both reviews were written by the same pipeline as the article; they are not external reviews. At placement every checker of the two packets and both audits reproduced its frozen records on copies, up to CRLF line ends (README). The Hilbert’s-tenth programme’s review read this Part’s clipping, compiler and gap sections and calls the result a “genuine finite-table arity reduction” [111].]

330.3 The pinned POWER dependency

The retained mathlib file is `Mathlib/NumberTheory/PellMatiyasevic.lean`, at commit

`ac77769fabe23cb237559e7f56578dbead91499f`.

Its SHA-256 is

`993760c797ad0ff66fa77064a616fce779550bc745cbf6c44e04f392fd0bed0a`.

The relevant theorem statements are `Pell.matiyasevic` and `Pell.eq_pow_of_pell`, at the retained source’s lines 760–766 and 860–864. The file, copyright header, notice and Apache 2.0 license are included unchanged. [write] [They are third-party files and are not shipped in this report; they are cited by commit and SHA-256 [86], and survive in the arrival archive.] The report inspects the theorem statements and their specialization; it does not claim a new Lean build, a fresh formalization or an elementary proof of the entire imported number-theoretic theorem.

330.4 What is and is not replayed

Fresh report-owned tools perform only static assembly, isolated LaTeX typesetting, rendering, source authentication and deterministic packaging. They neither import nor run the science or audit checkers, author/upstream mathematical programs, a native counter interpreter, a physical simulator, a collision schedule search or Lean. The two vector diagrams are drawn from the displayed class and shell formulas.

The release separates unchanged source packets, independent audit dossiers, editable manuscript modules, a flattened LaTeX file, the rendered PDF, source pins and build/release receipts. Source bytes, modes and modification times are retained. Dependency locking covers the recorded format and all TeX-pass input files and the named binaries; it is a typesetting reproducibility boundary, not formal verification of the mathematics or an inventory of the whole operating system. The accompanying README gives the exact commands and the external digest requirements. An independently supplied manifest pin is necessary to authenticate a received archive; a manifest inside an untrusted archive cannot authenticate itself.

[write] [The delivery README, article source and PDF and the editable manuscript modules are not shipped: this Part prints the article and this report’s README replaces the delivery README. The release tools are shipped as `code/36-bounded-cert-tools-*.py` with the release manifest and input pins as `data/36-bounded-cert-RELEASE_MANIFEST.json` and `data/36-bounded-cert-INPUT_PINS.json`; they expect the delivered layout and absolute POSIX paths and refuse to verify on Windows (README). The flattened shipped names break the packets’ relative layout, so reruns start from the arrival archive.]

331 Further questions within the established scope

The results suggest several concrete investigations. They are directions for future work, not claims of external open-problem status.

1. **Exact interpolation degree.** The stated degree bound is sufficient but sometimes not attained. The examples show parity-related cancellations in the univariate coordinate interpolants. A closed form for their exact degree and leading coefficients could sharpen the bound for this particular ordering of classes without changing witness arity. [write] [Answered by source 37: the interpolants have exact degree $K^2 - 1$ for even K and $K^2 - 2$ for odd $K \geq 3$, with explicit leading coefficients (Theorem 341.1), so the compiler has exact degree $4(T+1)^2 - 4$ at positive odd and $4(T+1)^2 - 8$ at positive even horizons (Theorem 332.2), for every acceptance table.]
2. **Description cost versus witness count.** Compare other explicit node orderings, integer-valued bases and representations of the acceptance set. The relevant costs include table construction, arithmetic bit complexity, degree and expanded storage, rather than witness count alone. No lower bound is known from the present argument. [write] [Partly addressed, for other representations rather than other orderings: source 38's tensor compiler has two witnesses, degree at most $4T$ and worse expanded support, its disjoint-cell product one witness and higher degree, both with all costs paid (Section 352, Table 50). No lower bound is proved there either, so the question stays open.]
3. **Restricted instruction classes.** A program with a smaller set of zero-sensitive states or a known invariant might admit fewer equivalence classes at a given horizon. Any such reduction must prove that its classes preserve the required control prefix, including the horizon boundary.
4. **Canonical POWER auxiliaries.** The paired quotient variables explain infinite complete fibers immediately. Choosing a canonical signed quotient representation would remove that particular redundancy, but it would not by itself prove uniqueness or even finiteness of all Pell auxiliaries. A complete finite-fiber statement would need a new argument for every remaining choice. [write] [Settled negatively for the modules of this Part: source 37 removes the paired quotients (the 22-leaf normalization, Section 333) and proves that every accepted fibre of the 22-, 16-, 14-, 13- and 12-leaf modules is still infinite, through an explicit Pell progression in the auxiliary parameter β (Section 335). A finite-fibre POWER presentation would need a different construction.]
5. **Finer error statistics.** The exact digit identity permits study of average deficits, distributions across dyadic intervals, or other explicitly specified subsequences. The liminf and limsup proved here constrain such refinements but do not provide a universal logarithm-squared correction term. [write] [Open in this report. The undelivered Report 68 is described by its surviving README as proving Gaussian fluctuations of E for a uniform cutoff; its article is not available, and its claims are recorded, unproved, as Question 1 of Section 355.]
6. **Other rational endpoint encodings.** Replacing the constants in (318.1) may change the reduced denominators, exceptional residue classes and shell formulas. Repeating the primitive-scale analysis could identify which features of the linear count and oscillatory error persist.

None of these directions identifies an OEIS entry, resolves an external conjecture, establishes a priority claim, or converts counting all encoded initializations into counting halting inputs. The

central conclusions remain exact bounded certificates with explicit costs, and exact arithmetic counts with their correct multiplicities.

[write] **Review by the Hilbert’s-tenth programme.** The programme’s scoped intake of the batch [111] read source 36’s delivery README completely and, of its article, Sections 319–322, the opening of Section 323 with the module’s leaf list, and that section’s composed polynomial, ledger, fibre and physical subsections; it did not read the module’s residuals and adapter proof or the counting half (Sections 324–329), and it ran none of the archived programs. Its findings: the clipping proof is sound (tests occur before the final decrement, so final counters need not agree); the finite interpolation gives exactly three positive witnesses and six residual slots; the range and slack residuals force the unique clipped cell with no encoded-input promise; and the polynomials are compiled separately for each program and horizon, so that neither an outer existential quantifier over T nor unique bounded witnesses gives one fixed unbounded polynomial. It reports no defect, and its universal 84-operation construction is unchanged.

Source 37 (batch-91 manuscript 19): Positive POWER reductions and exact bounded compiler costs

Explicit domains and witness degree tradeoffs, “Research Report 67”, 4 October 2026. [write] From the archive `Positive_POWER_reductions_and_exact_compiler_costs_source.zip`, main file `Report67/Report67.tex` (822 lines; a 24-page PDF); its labels carry the prefix `smc:pp:`, its sections are Sections 332–347 here, and its equations are numbered by section. It continues source 36, whose compiler and clipping lemma it restates (notes in Section 340), and the 26-leaf POWER module of sources 28, 30, 35 and 36. Its four bibliography entries are this report’s [86], [101], [108] and [109]. Its table of contents and page breaks are omitted. Its abstract follows.

We give five explicit positive-integer POWER modules with 22, 16, 14, 13 and 12 leaves, including the output. Their sums of squares have respectively 15, 9, 7, 6 and 5 residual slots and fixed-base degrees 12, 12, 16, 16 and 20. For a variable base the last two degrees are 20 and 24. The reductions distinguish sign normalization, bijective elimination, and restriction to a strict quotient domain followed by rational integer recovery. Every restored variable is proved integral and positive. All nonempty complete fibers remain infinite.

For the retained three-witness bounded counter compiler, we prove an exact degree law: degree two at horizon zero, $4(T+1)^2 - 4$ at positive odd horizons, and $4(T+1)^2 - 8$ at positive even horizons. We identify the complete top homogeneous term and sharpen the support ceiling. Composing the 12-leaf module twice with native gap decoding and the bounded compiler gives 29 positive witnesses, three external positive inputs, 18 residual slots and degree equal to the maximum of 20 and the compiler degree. The other variants give explicit alternative resource ledgers.

The all-exponent POWER theorem depends on a precisely pinned constructive Pell theorem pair. The finite-horizon compiler is separately compiled for each horizon and pays for its finite acceptance table. No new Lean verification, finite-fold representation, optimality, priority or fixed-polynomial unbounded-halting claim is made.

332 Conventions and principal results

All existential variables in this report range over $\mathbb{Z}_{>0}$. We write $\mathbb{N} = \{0, 1, \dots\}$. A *natural alias* z is the expression $z_+ - 1$ in its own distinct positive leaf z_+ . Thus a zero alias is represented by leaf 1. Aliases are expanded substitutions, never additional variables or defining equations. A shifted

Pell parameter is $\alpha = \alpha_+ + 1$, so it is at least 2. Every final equation is literally a sum of the listed squared integer-polynomial residuals.

Fix $b \in \mathbb{Z}$, $b \geq 2$, and $C, o \in \mathbb{Z}_{>0}$. The intended POWER relation is $o = b^{C-1}$. Its Pell index is C , including when the semantic exponent is zero. A module leaf count includes o and excludes the supplied index C . A fixed base is a coefficient. If the base is variable, supply an additional positive input B and put $b = B + 1$; that external input is not concealed in a module count. Degree is joint in all independent inputs and positive leaves, after adapter expansion and before imposing equations. The results cover independent or affine base conventions as stated, not arbitrary nonlinear substitutions.

Theorem 332.1 (Explicit POWER presentations). *The five residual systems displayed below satisfy*

$$o = b^{C-1} \iff \exists \text{ positive auxiliary leaves with } \mathcal{P}_\ell = 0, \quad \ell \in \{22, 16, 14, 13, 12\}.$$

Their exact literal resources are Table 48. Every accepted fixed (b, C, o) has infinitely many complete auxiliary tuples in each presentation. The theorem imports the constructive Pell statements identified in Section 334; the reductions and domain recovery are proved here.

Presentation	Leaves	Auxiliaries	Residuals	Degree fixed / variable
One-sided quotients	22	21	15	12 / 12
Polynomial elimination	16	15	9	12 / 12
Substitute v, t	14	13	7	16 / 16
Recover α	13	12	6	16 / 20
Recover u on strict domain	12	11	5	20 / 24

Table 48: Output o is included in leaves and excluded from auxiliaries. The index and any base input are external. Each row defines one final polynomial equation.

Theorem 332.2 (Exact bounded compiler and gap ledger). *For a fixed deterministic two-counter program M and horizon $T \geq 0$, let $P_{M,T}(A, B; j, r, s)$ be the six-square compiler in Section 340. Its three positive witnesses are unique when they exist. Writing $K = T + 1$, its exact joint degree is*

$$d_T = \begin{cases} 2 & T = 0, \\ 4(T+1)^2 - 4 & T > 0 \text{ odd}, \\ 4(T+1)^2 - 8 & T > 0 \text{ even}. \end{cases} \quad (332.1)$$

Two independent base-two ℓ -leaf modules with r_ℓ residuals and degree D_ℓ , composed with paid gap decoding, yield exactly $2\ell+5$ positive witnesses, $2r_\ell+8$ residual slots, three external positive gap inputs, and degree $\max(D_\ell, d_T)$. The zeros assert both that the inputs are encoded and that the decoded counters satisfy the selected bounded-halting predicate. Both by-horizon and first-halt-exactly-horizon versions have these counts. Every nonempty complete composed fiber is infinite.

In particular the last row gives 29 witnesses, 32 total variables, 18 residual slots and degree $\max(20, d_T)$. These replace the modules in a new formula; they do not retroactively recount a frozen predecessor. Report 66's [source 36's] accepted compiler, its 26-leaf POWER baseline, and the separately accepted reduction and exact-degree packets are preserved as provenance. Variable count, residual count, degree, coefficient height, expanded storage and arithmetic gates are distinct resources. No gate improvement or minimum among all formulas is claimed.

333 The positive 22 leaf module

Use thirteen directly positive leaves

$$o, w, M, g, x, y, u, v, s, t, q_b, q_v, J,$$

two positive leaves α_+, β_+ defining $\alpha = \alpha_+ + 1$ and $\beta = \beta_+ + 1$, and seven natural aliases

$$d_{wb}, d_{wC}, d_{yC}, q_\alpha, q_\sigma, q_\tau, q_r.$$

Each natural alias has its own positive adapter. There are $13 + 2 + 7 = 22$ positive leaves. The complete residual list is

$$\begin{aligned} R_1 &= x^2 - 1 - (\alpha^2 - 1)y^2, & R_2 &= u^2 - 1 - (\alpha^2 - 1)v^2, \\ R_3 &= s^2 - 1 - (\beta^2 - 1)t^2, & R_4 &= \beta - 1 - 4yq_b, \\ R_5 &= \beta - \alpha - uq_\alpha, & R_6 &= v - y^2q_v, \\ R_7 &= s - x - uq_\sigma, & R_8 &= t - C - 4yq_\tau, \\ R_9 &= y - C - d_{yC}, & R_{10} &= w - b - d_{wb}, \\ R_{11} &= w - C - d_{wC}, & R_{12} &= M - bo - J, \\ R_{13} &= \alpha^2 - 1 - ((w + 1)^2 - 1)(wg)^2, \\ R_{14} &= 2\alpha b - M - (b^2 + 1), \\ R_{15} &= x - y(\alpha - b) - bo - Mq_r. \end{aligned} \tag{333.1}$$

Define $\mathcal{P}_{22} = \sum_{i=1}^{15} R_i^2$ after every shift. For $b = 2$, the modulus residuals are $M - 2o - J$ and $4\alpha - M - 5$.

333.1 Why four signed quotients are nonnegative

The 26-leaf predecessor [write] [source 36's module, Section 323, at $b = 2$; source 35's, Section 309] has the same equations except that slots 5, 7, 8 and 15 use the four paired-natural residuals

$$\begin{aligned} \beta + u\alpha_1 - \alpha - u\alpha_2, & \quad s + u\sigma_1 - x - u\sigma_2, \\ t + 4y\tau_1 - C - 4y\tau_2, & \quad x + Mr_1 - y(\alpha - b) - bo - Mr_2. \end{aligned} \tag{333.2}$$

All eight aliases in this display are natural. They initially allow signed differences. We prove their signs from the other equations, without assuming the conclusion.

Lemma 333.1 (Positive residue). *If $L > 0$, $0 < r \leq L$, $z > 0$ and $z = r + Lq$ for an integer q , then $q \geq 0$.*

Proof. If $q \leq -1$, then $z \leq r - L \leq 0$, a contradiction. The endpoint $r = L$ is allowed. $\square$

We have $y \geq C \geq 1$ and $v = y^2q_v \geq y$. The first two Pell equations give

$$u^2 - \alpha^2 = (\alpha^2 - 1)(v^2 - 1) \geq 0, \quad u^2 - x^2 = (\alpha^2 - 1)(v^2 - y^2) \geq 0.$$

Positivity yields $0 < \alpha \leq u$ and $0 < x \leq u$. Applying Lemma 333.1 to the first three congruences, with residues α, x, C and moduli $u, u, 4y$, proves

$$\frac{\beta - \alpha}{u}, \quad \frac{s - x}{u}, \quad \frac{t - C}{4y} \in \mathbb{N}. \tag{333.3}$$

For the last congruence, $w \geq b \geq 2$, $g \geq 1$ and $R_{13} = 0$ imply $\alpha^2 > w^2$, hence $\alpha > w \geq b$. Put $h = x - y(\alpha - b)$. The two quantities being compared are nonnegative and

$$x^2 - [y(\alpha - b)]^2 = (2\alpha b - b^2 - 1)y^2 + 1 > 0,$$

so $h > 0$. Moreover $0 < bo < M$ by $R_{12} = 0$. The final paired congruence gives $h = bo + M(r_2 - r_1)$. The same lemma proves

$$q_r = \frac{x - y(\alpha - b) - bo}{M} = r_2 - r_1 \in \mathbb{N}. \quad (333.4)$$

Thus every old positive solution has the four required nonnegative quotients, not merely a special constructive solution.

333.2 The exact normalization maps

From an old tuple preserve all other leaves and define

$$q_\alpha = \alpha_2 - \alpha_1, \quad q_\sigma = \sigma_2 - \sigma_1, \quad q_\tau = \tau_2 - \tau_1, \quad q_r = r_2 - r_1.$$

The new leaves are $q_\alpha + 1, q_\sigma + 1, q_\tau + 1, q_r + 1$. Their positivity follows from the preceding proof. Conversely set the first member of every old natural pair to zero and the second to its new quotient. The old positive leaves are therefore 1 and $q + 1$, never zero. These maps prove existential equivalence, and the reverse map is a section. They are not inverse on all old tuples: adding an arbitrary common natural shift to both members of each pair changes the old tuple without changing its quotient. The old ledger $13 + 2 + 11 = 26$ becomes $13 + 2 + 3 + 4 = 22$, retaining all fifteen residual slots.

334 The precise constructive Pell dependency

The substantive number-theoretic input is Mario Carneiro's constructive pair `Pell.matiyasevic` and `Pell.eq_pow_of_pell` in `mathlib4` [86]. The pinned commit is

`ac77769fabe23cb237559e7f56578dbead91499f`

and the retained file `Mathlib/NumberTheory/PellMatiyasevic.lean` has SHA-256

`993760c797ad0ff66fa77064a616fce779550bc745cbf6c44e04f392fd0bed0a.`

Its source header, Apache 2.0 license and notice are retained. We inspect these source statements and constructive branches as text; there is no new Lean execution or proof-assistant certification in this report.

Define the integer Pell polynomials by

$$\begin{aligned} X_0(z) &= 1, & Y_0(z) &= 0, \\ X_{n+1}(z) &= zX_n(z) + (z^2 - 1)Y_n(z), & Y_{n+1}(z) &= X_n(z) + zY_n(z). \end{aligned} \quad (334.1)$$

The first imported theorem says that (x, y) is the index- k Pell pair for $\alpha > 1$ exactly when $k \leq y$ and either $(x, y) = (1, 0)$ or there exist natural u, v, s, t, β satisfying

$$\begin{aligned} x^2 - (\alpha^2 - 1)y^2 &= 1, & u^2 - (\alpha^2 - 1)v^2 &= 1, & s^2 - (\beta^2 - 1)t^2 &= 1, \\ \beta > 1, & \beta \equiv 1 \pmod{4y}, & \beta \equiv \alpha \pmod{u}, & v > 0, & y^2 \mid v, \\ s \equiv x \pmod{u}, & & t \equiv k \pmod{4y}. \end{aligned} \quad (334.2)$$

The positive-base, positive-index branch of the second theorem characterizes $n^k = m$ by natural $w, \alpha, M, g, \alpha > 1$, and

$$\begin{aligned} X_k(\alpha) &\equiv Y_k(\alpha)(\alpha - n) + m \pmod{M}, \\ 2\alpha n &= M + n^2 + 1, \quad m < M, \quad n \leq w, \quad k \leq w, \\ \alpha^2 - ((w+1)^2 - 1)(wg)^2 &= 1. \end{aligned} \tag{334.3}$$

The source uses natural subtraction. For natural a, c , the statement $a - c = 1$ with truncated subtraction is equivalent to $a = c + 1$. The inner square-minus-one differences here are nonnegative, and the proof above establishes $\alpha > b$. Thus the ordinary subtraction $\alpha - b$ agrees with its natural counterpart. No domain strengthening or rational-witness replacement is hidden in this translation.

All-exponent equivalence for the 22-leaf module. Slots 1–9 in (333.1) imply (334.2) with $k = C$. The zero branch is impossible because $y \geq C \geq 1$. Hence $(x, y) = (X_C(\alpha), Y_C(\alpha))$. Slots 10–15 supply (334.3) with $n = b, k = C, m = bo$. They imply $b^C = bo$, and cancellation of $b > 0$ gives $o = b^{C-1}$.

Conversely suppose $o = b^{C-1}$. The constructive power theorem supplies the auxiliaries for $b^C = bo$, and the constructive positive-index Pell theorem supplies the inner norms and congruences. The auxiliary g cannot vanish because that would force $\alpha = 1$. The Pell first coordinates x, u, s are positive. We have $y \geq C \geq 1, v > 0$ and $y^2 \mid v$, so $q_v > 0$. Since $\beta > 1$ and $\beta \equiv 1 \pmod{4y}$, $q_b > 0$. The congruence $t \equiv C \pmod{4y}$ and $1 \leq C \leq y < 4y$ exclude $t = 0$. Strict $M > bo$ supplies $J > 0$. The other inequalities supply the three natural slacks. Represent signed congruence quotients by pairs of naturals, then apply the normalization maps just proved. Every positive adapter exists, including at $C = 1$. $\square$

[write] [At $b = 2$ and before the normalization of Section 333, this is source 35's Lemma 309.1 and source 36's (323.3). New here are the arbitrary fixed or variable base $b \geq 2$, the explicit statements (334.2) and (334.3) of the imported theorems, and the proof that every old solution, not merely a constructive one, has nonnegative quotients.]

335 A genuine infinite family in each accepted fiber

The recurrence (334.1) proves by induction

$$X_n(z)^2 - (z^2 - 1)Y_n(z)^2 = 1, \quad X_n(1) = 1, \quad Y_n(1) = n.$$

For the norm, substitution into one recurrence step multiplies the previous norm by $z^2 - (z^2 - 1) = 1$. For $z \geq 2, n \geq 1$, both coordinates are positive by the same recurrence. Integer polynomials preserve congruences of their arguments.

Fix an accepted (b, C, o) and one 22-leaf solution. Hold $\alpha, w, M, g, x, y, u, v, q_v, J$, the three slacks and q_r fixed. For every $k \in \mathbb{N}$, put

$$\begin{aligned} \beta_k &= \beta_0 + 4yuk, & s_k &= X_C(\beta_k), & t_k &= Y_C(\beta_k), \\ q_{b,k} &= q_{b,0} + uk, & q_{\alpha,k} &= q_{\alpha,0} + 4yk, \\ q_{\sigma,k} &= (s_k - x)/u, & q_{\tau,k} &= (t_k - C)/(4y). \end{aligned} \tag{335.1}$$

Here $\beta_k \equiv \alpha \pmod{u}$ and $\beta_k \equiv 1 \pmod{4y}$, so $s_k \equiv x \pmod{u}$ and $t_k \equiv C \pmod{4y}$. Lemma 333.1, using $0 < x \leq u$ and $1 \leq C < 4y$, makes the two quotients natural. All fifteen

equations hold. The new pair at $k = 0$ need not equal the originally chosen (s, t) ; only the preserved data and congruences are required. The leaf $q_{b,k}$ is strictly increasing because $u > 0$, proving infinitely many distinct tuples. This is genuine parameter variation after the four common-shift redundancies have been removed. [write] [This settles source 36's question on canonical POWER auxiliaries (Section 331, item 4) negatively for these modules: the canonical quotients of the 22-leaf module still leave every accepted fibre infinite.]

For an explicit boundary example use $b = 2, C = o = 1$ and

$$\begin{aligned} w = 2, \quad M = 63, \quad g = 3, \quad \alpha = 17, \quad x = 17, \quad y = 1, \quad u = 577, \quad v = 34, \\ q_v = 34, \quad J = 61, \quad d_{wb} = d_{yC} = q_r = 0, \quad d_{wC} = 1, \\ \beta = s = 17 + 2308k, \quad t = 1, \quad q_b = 4 + 577k, \\ q_\alpha = q_\sigma = 4k, \quad q_\tau = 0. \end{aligned} \tag{335.2}$$

All equations are identities in $k \in \mathbb{N}$. Zero aliases use adapter 1. Later eliminations retain q_b , so this infinitude cannot disappear by removing β_+ .

336 Bijective polynomial elimination to sixteen and fourteen leaves

336.1 Sixteen leaves and nine residuals

Retain the eight positive leaves $o, g, u, v, t, q_b, q_v, J$, the positive α_+ , and the same seven natural adapters as before. Define expressions

$$\begin{aligned} \alpha &= \alpha_+ + 1, & w &= b + d_{wb}, & y &= C + d_{yC}, \\ \beta &= 1 + 4yq_b, & M &= 2b\alpha - b^2 - 1, \\ x &= y(\alpha - b) + bo + Mq_r, & s &= x + uq_\sigma. \end{aligned} \tag{336.1}$$

The nine retained residuals, in their complete order, are

$$\begin{aligned} E_1 &= x^2 - 1 - (\alpha^2 - 1)y^2, & E_2 &= u^2 - 1 - (\alpha^2 - 1)v^2, \\ E_3 &= s^2 - 1 - (\beta^2 - 1)t^2, & E_4 &= \beta - \alpha - uq_\alpha, \\ E_5 &= v - y^2q_v, & E_6 &= t - C - 4yq_\tau, \\ E_7 &= w - C - d_{wC}, & E_8 &= M - bo - J, \\ E_9 &= \alpha^2 - 1 - ((w + 1)^2 - 1)(wg)^2. \end{aligned} \tag{336.2}$$

They are respectively old slots 1, 2, 3, 5, 6, 8, 11, 12, 13. The other six old residuals become identities under (336.1). Define $\mathcal{P}_{16} = \sum_{i=1}^9 E_i^2$ after these substitutions.

Projection of an old solution gives these sixteen leaves; its six defining equations force exactly the expressions (336.1). Conversely, a positive solution of (336.2) has $w \geq b \geq 2$ and $y \geq C \geq 1$. Equation $E_9 = 0$ gives $\alpha > w \geq b$. Equation $E_8 = 0$ gives $M = bo + J > 0$; positivity of that unconstrained polynomial expression was not assumed in advance. It follows that $x > 0$ and $s > 0$. Also $\beta - 1 = 4yq_b > 0$, giving the restored positive leaf β_+ . Thus every removed domain is recovered. The two maps are inverse on complete solution tuples for fixed b, C, o .

336.2 Fourteen leaves and seven residuals

Additionally define

$$v = y^2q_v, \quad t = C + 4yq_\tau. \tag{336.3}$$

Remove leaves v, t and residuals E_5, E_6 . The remaining leaves are the six positive o, g, u, q_b, q_v, J , α_+ , and the seven natural adapters, totaling fourteen. The full ordered residual list is

$$E_1, E_2, E_3, E_4, E_7, E_8, E_9 \quad \text{with (336.1) and (336.3) expanded.} \quad (336.4)$$

Define $\mathcal{P}_{14}$ as the sum of these seven squares. The restored values in (336.3) are positive integers because $y, q_v, C \geq 1$ and $q_\tau \geq 0$. Therefore restriction and reconstruction are again mutually inverse, with no lost positive-domain condition.

337 Thirteen leaves by integral parameter recovery

Retain the six directly positive o, g, u, q_b, q_v, J and the seven natural aliases; remove α_+ . Define the following integer polynomial expressions:

$$\begin{aligned} w &= b + d_{wb}, & y &= C + d_{yC}, & \beta &= 1 + 4yq_b, \\ v &= y^2 q_v, & t &= C + 4yq_\tau, & M &= bo + J, \\ d &= 2b, & A_* &= M + b^2 + 1, \\ X &= y(A_* - db) + dbo + dMq_r, & S_* &= X + duq_\sigma. \end{aligned} \quad (337.1)$$

The notation A_* avoids collision with the outer compiler input A . It is an expression, not a leaf. The six residuals are

$$\begin{aligned} F_1 &= X^2 - d^2 - (A_*^2 - d^2)y^2, \\ F_2 &= d^2 u^2 - d^2 - (A_*^2 - d^2)v^2, \\ F_3 &= S_*^2 - d^2 - d^2(\beta^2 - 1)t^2, \\ F_4 &= d\beta - A_* - duq_\alpha, \\ F_5 &= w - C - d_{wC}, \\ F_6 &= A_*^2 - d^2 - d^2((w+1)^2 - 1)(wg)^2. \end{aligned} \quad (337.2)$$

Define $\mathcal{P}_{13} = \sum_{i=1}^6 F_i^2$. No division or rational coefficient occurs in this polynomial.

Lemma 337.1 (Rational square recovery). *If $z \in \mathbb{Q}$ and $z^2 \in \mathbb{Z}$, then $z \in \mathbb{Z}$.*

Proof. Write $z = p/q$ in lowest terms with $q > 0$. Since q^2 divides p^2 and p, q are coprime, $q = 1$. $\square$

From a fourteen-leaf solution, its equations imply $M = bo + J$ and $A_* = d\alpha$. Consequently $X = dx$ and $S_* = ds$. The six residuals above are respectively

$$d^2 E_1, d^2 E_2, d^2 E_3, dE_4, E_7, d^2 E_9;$$

the removed E_8 becomes an identity. Restriction gives a thirteen-leaf solution.

Conversely, from such a solution, set $\alpha = A_*/d$ in $\mathbb{Q}$ for the proof only. Both numerator and denominator are positive, and $F_6 = 0$ gives

$$\alpha^2 = 1 + ((w+1)^2 - 1)(wg)^2 \in \mathbb{Z}. \quad (337.3)$$

Lemma 337.1 makes α an integer. Because $w \geq b \geq 2$ and $g \geq 1$, its square exceeds w^2 , so $\alpha > w \geq b$ and the restored $\alpha_+ = \alpha - 1$ is positive. The equation $A_* = d\alpha$ recovers the old expression $M = 2b\alpha - b^2 - 1$. Define

$$x = y(\alpha - b) + bo + Mq_r, \quad s = x + uq_\sigma.$$

Both are positive integers, and $X = dx, S_* = ds$. Dividing the zero residuals by d or d^2 , which are nonzero, gives all surviving fourteen-leaf equations. Thus the maps are inverse and clearing denominators has introduced no spurious rational solution.

Here integrality also follows directly from $F_4 = 0$, since $\alpha = \beta - uq_\alpha$ is an integer. The square-recovery proof is valid but redundant in this presentation. That shortcut disappears when u is removed next; this distinction matters in the twelve-leaf proof.

At fixed base two the six residuals can be read with

$$\begin{aligned} d = 4, \quad A_* = 2o + J + 5, \quad M = 2o + J, \quad w = 2 + d_{wb}, \quad y = C + d_{yC}, \\ X = y(A_* - 8) + 8o + 4Mq_r, \quad S_* = X + 4uq_\sigma, \quad \beta = 1 + 4yq_b, \quad v = y^2q_v, \quad t = C + 4yq_r. \end{aligned}$$

In particular they are exactly $X^2 - 16 - (A_*^2 - 16)y^2$, $16u^2 - 16 - (A_*^2 - 16)v^2$, $S_*^2 - 16 - 16(\beta^2 - 1)t^2$, $4\beta - A_* - 4uq_\alpha$, $w - C - d_{wC}$ and $A_*^2 - 16 - 16((w + 1)^2 - 1)(wg)^2$.

338 Exact degrees of the first four presentations

The exact residual degrees after all shifts are as follows. Their orders are precisely (333.1), (336.2), (336.4) and (337.2).

Leaves	Fixed base	Independent base or $B + 1$
22	4, 4, 4, 2, 2, 3, 2, 2, 1, 1, 1, 1, 6, 1, 2	4, 4, 4, 2, 2, 3, 2, 2, 1, 1, 1, 2, 6, 2, 2
16	4, 4, 6, 2, 3, 2, 1, 1, 6	6, 4, 6, 2, 3, 2, 1, 2, 6
14	4, 8, 8, 2, 1, 1, 6	6, 8, 8, 2, 1, 2, 6
13	4, 8, 8, 2, 1, 6	8, 10, 10, 3, 1, 8

These upper degrees follow by expanding the displayed definitions. For example, in the sixteen-leaf form y, w have degree one, β degree two, and M degree one at fixed base or two at variable base. Thus x, s have degree at most two or three respectively. After substituting $v = y^2q_v$, the term $(\alpha^2 - 1)v^2$ has degree eight. In the thirteen-leaf variable-base form A_*, M, β have degree two, d, w, y degree one, X, S_* degree four, v degree three and t degree two.

To prove exactness, the following monomials have coefficient one in the final sums of squares:

$$\begin{aligned} \mathcal{P}_{22} : \quad & w^8 g^4, \\ \mathcal{P}_{16} : \quad & d_{wb,+}^8 g^4, \\ \mathcal{P}_{14} : \quad & \alpha_+^4 d_{yC,+}^8 q_v^4, \\ \mathcal{P}_{13} \text{ fixed} : \quad & J^4 d_{yC,+}^8 q_v^4, \\ \mathcal{P}_{13} \text{ variable} : \quad & B^8 d_{yC,+}^8 q_v^4. \end{aligned} \tag{338.1}$$

In the first two rows, only the auxiliary Pell residual contains g and its leading g term is $-w^4 g^2$ or $-d_{wb,+}^4 g^2$. In the last three rows, only the second Pell residual contains q_v ; its indicated source term

is respectively $-\alpha_+^2 d_{yC,+}^4 q_v^2$, $-J^2 d_{yC,+}^4 q_v^2$, or $-B^4 d_{yC,+}^4 q_v^2$. To produce a displayed target monomial, neither factor in that residual square can contain another variable, leaving exactly this source. The coefficients are therefore one and cannot be cancelled by another square. These terms attain degrees 12, 12, 16, 16, 20 as claimed. Fixed bases $b \geq 2$ change none of these coefficient-one arguments.

The 22-to-16-to-14-to-13 maps are bijections for fixed arguments. More concretely, they all retain the strictly increasing leaf $q_{b,k}$ in (335.1). Every accepted fiber in these four presentations is therefore infinite even though several explicit Pell coordinates have disappeared.

339 Twelve leaves on a strict quotient domain

The next step removes u but changes the domain of q_α . It is not a bijection with the entire thirteen-leaf solution set.

Retain six directly positive leaves

$$o, g, q_b, q_v, J, q_\alpha$$

and six natural aliases

$$d_{wb}, d_{wC}, d_{yC}, q_\sigma, q_\tau, q_r.$$

Each of the latter still uses one positive adapter. The new q_α is itself positive, with no minus-one shift. Define $w, y, \beta, v, t, M, d, A_*, X$ as in (337.1), and now set

$$U = d\beta - A_*, \quad S = q_\alpha X + Uq_\sigma. \quad (339.1)$$

The full five residuals are

$$\begin{aligned} H_1 &= X^2 - d^2 - (A_*^2 - d^2)y^2, \\ H_2 &= U^2 - d^2 q_\alpha^2 - (A_*^2 - d^2)q_\alpha^2 v^2, \\ H_3 &= S^2 - d^2 q_\alpha^2 - d^2 q_\alpha^2 (\beta^2 - 1)t^2, \\ H_4 &= w - C - d_{wC}, \\ H_5 &= A_*^2 - d^2 - d^2((w+1)^2 - 1)(wg)^2. \end{aligned} \quad (339.2)$$

Let $\mathcal{P}_{12} = \sum_{i=1}^5 H_i^2$ after every substitution and shift. The first residual deliberately has no extra q_α^2 factor. There are no rational coefficients, hidden congruence residuals, or rational witnesses.

For $b = 2$ the complete expressions are $d = 4$, $M = 2o + J$, $A_* = 2o + J + 5$, $w = 2 + d_{wb}$, $y = C + d_{yC}$, $\beta = 1 + 4yq_b$, $v = y^2 q_v$, $t = C + 4yq_\tau$, and

$$X = y(A_* - 8) + 8o + 4Mq_r, \quad U = 4\beta - A_*, \quad S = q_\alpha X + Uq_\sigma.$$

Thus the five residuals specialize explicitly to

$$\begin{aligned} &X^2 - 16 - (A_*^2 - 16)y^2, \\ &U^2 - 16q_\alpha^2 - (A_*^2 - 16)q_\alpha^2(y^2 q_v)^2, \\ &S^2 - 16q_\alpha^2 - 16q_\alpha^2(\beta^2 - 1)(C + 4yq_\tau)^2, \\ &w - C - d_{wC}, \quad A_*^2 - 16 - 16((w+1)^2 - 1)(wg)^2. \end{aligned}$$

339.1 Two rational recoveries and all restored domains

Suppose all twelve leaves are positive integers and all H_i vanish. Directly from the definitions,

$$w \geq b \geq 2, \quad y \geq C \geq 1, \quad \beta \geq 5, \quad v = y^2 q_v \geq y, \quad t \geq C, \quad M = bo + J > bo > 0.$$

Also d, A_*, q_α are positive. Define $\alpha = A_*/d$ in $\mathbb{Q}$ only for the proof. Equation $H_5 = 0$ gives (337.3); Lemma 337.1 makes α integral and the strict square bound gives $\alpha > w \geq b$. In particular $\alpha_+ = \alpha - 1 > 0$ and $M = 2b\alpha - b^2 - 1$.

Next define $u = U/(dq_\alpha)$ in $\mathbb{Q}$. This division is legitimate because $d, q_\alpha > 0$. Equation $H_2 = 0$ gives

$$u^2 = 1 + (\alpha^2 - 1)v^2 \in \mathbb{Z}. \quad (339.3)$$

The rational-square lemma makes u an integer. Since $v \geq 1$, we have $|u| \geq \alpha > 0$. The definition of U gives

$$\beta = \alpha + uq_\alpha. \quad (339.4)$$

If $u < 0$, integrality and the preceding bound give $u \leq -\alpha$, whence $\beta \leq \alpha - \alpha q_\alpha \leq 0$, contradicting $\beta \geq 5$. Thus $u > 0$. The proof crucially uses the positive-integer inequality $q_\alpha \geq 1$, not just nonvanishing of a rational quotient.

Define the integers $x = y(\alpha - b) + bo + Mq_r$ and $s = x + uq_\sigma$. They are positive because $\alpha > b$, $bo > 0$ and the aliases are natural. Now $X = dx$ and $S = dq_\alpha s$. Dividing $H_1 = 0$ by d^2 and $H_3 = 0$ by $d^2 q_\alpha^2$ recovers the first and third Pell equations. Equation (339.3) recovers the second. The remaining equations are $H_4 = 0$, (337.3), (339.4) and the definitions. In full they are

$$\begin{aligned} x^2 &= 1 + (\alpha^2 - 1)y^2, & u^2 &= 1 + (\alpha^2 - 1)v^2, & s^2 &= 1 + (\beta^2 - 1)t^2, \\ \beta &= 1 + 4yq_b, & \beta &= \alpha + uq_\alpha, & v &= y^2 q_v, & s &= x + uq_\sigma, \\ t &= C + 4yq_\tau, & y &= C + d_y C, & w &= b + d_{wb} = C + d_{wC}, \\ M &= bo + J, & \alpha^2 &= 1 + ((w + 1)^2 - 1)(wg)^2, \\ 2\alpha b &= M + b^2 + 1, & x &= y(\alpha - b) + bo + Mq_r. \end{aligned}$$

All restored w, M, x, y, u, v, s, t are positive. The restored $\beta_+ = \beta - 1$ is positive, as is α_+ . In the old domain the adapter for its natural quotient is now $q_\alpha + 1$; all other natural aliases retain their positive adapters. This is a genuine 22-leaf solution, so the pinned theorem proves $o = b^{C-1}$.

339.2 Completeness and the exact image of the witness map

Let $W_{13}(b, C, o)$ be the thirteen-leaf solution set, and let W_{13}^+ be its subset with natural $q_\alpha \geq 1$. For any old solution, reconstruct the 22-leaf data and use (335.1) with $k \geq 1$. Then

$$q_{\alpha,k} = q_{\alpha,0} + 4yk > 0.$$

The family preserves all equations and all positive domains. Hence an old solution always gives a strict-domain old solution, including when the initial quotient was zero.

From a strict solution, delete u and replace the old adapter $q_{\alpha,+}$ by the direct leaf $q_\alpha = q_{\alpha,+} - 1$. On that solution $U = duq_\alpha$ and $S = dq_\alpha s$, so the five residuals become

$$d^2 R_1, \quad d^2 q_\alpha^2 R_2, \quad d^2 q_\alpha^2 R_3, \quad R_{11}, \quad d^2 R_{13},$$

and vanish. The inverse keeps the other eleven leaves, restores $q_{\alpha,+} = q_\alpha + 1$ and restores the unique $u = U/(dq_\alpha)$. Its integrality and positivity were proved above. We have exactly a bijection

$$W_{13}^+(b, C, o) \longleftrightarrow W_{12}(b, C, o). \quad (339.5)$$

The inverse image inside W_{13} is the subset $\beta > \alpha$, equivalent to $q_\alpha \geq 1$ because $u > 0$. This is not literal projection because an adapter is translated, and it is not a bijection with all W_{13} . For instance the old fixture at $b = 2, C = o = 1, k = 0$ in (335.2) has $q_\alpha = 0$ and is excluded. Passing an arbitrary old tuple through the family at $k = 1$ proves existential equivalence; it need not be injective or inverse to reconstruction. Together with the old completeness theorem, this proves Theorem 332.1 for twelve leaves.

The substitution identities can also be stated without division. Let F_i be (337.2), so $F_4 = U - duq_\alpha$, and let its old $S_* = X + duq_\sigma$. As identities in polynomial indeterminates,

$$\begin{aligned} H_1 &= F_1, & H_4 &= F_5, & H_5 &= F_6, \\ H_2 - q_\alpha^2 F_2 &= F_4(U + duq_\alpha), \\ H_3 - q_\alpha^2 F_3 &= F_4 q_\sigma (2q_\alpha X + (U + duq_\alpha)q_\sigma). \end{aligned} \quad (339.6)$$

The missing congruence equation is recovered through integral u , not silently ignored.

339.3 Exact degree and infinite fibers

At fixed base, the degrees of w, y, M, A_* are one, those of β, X, U are two, those of v, S are three, and that of t is two. Thus the exact degrees of $H_1, \dots, H_5$ are

$$4, 10, 10, 1, 6.$$

Only H_2 contains q_v , and its term $-J^2 q_\alpha^2 d_{yC,+}^4 q_v^2$ yields the coefficient-one monomial

$$J^4 q_\alpha^4 d_{yC,+}^8 q_v^4 \quad (339.7)$$

in $\mathcal{P}_{12}$. No other variable can occur in either factor producing that monomial. It has degree twenty, proving exactness for every fixed $b \geq 2$.

At $b = B + 1$, the degrees of d, w, y are one, those of M, A_*, β, t are two, those of v, U are three, that of X is four and that of S is five. The exact residual degrees are

$$8, 12, 12, 1, 8.$$

The B^2 term of A_* supplies $-B^4 q_\alpha^2 d_{yC,+}^4 q_v^2$ in H_2 . Its square gives the coefficient-one monomial

$$B^8 q_\alpha^4 d_{yC,+}^8 q_v^4 \quad (339.8)$$

of degree twenty-four. The independent- b convention is the same affine degree calculation.

For every accepted argument, (335.1) with $k \geq 1$ leaves infinitely many distinct $q_{b,k}$ in the new tuples. Restricting the starting endpoint does not remove infinitude. In the explicit zero-exponent family the retained data are

$$g = 3, \quad J = 61, \quad q_v = 34, \quad d_{wb} = d_{yC} = q_r = q_\tau = 0, \quad d_{wC} = 1,$$

and $q_b = 4 + 577k$, $q_\alpha = q_\sigma = 4k$ for $k \geq 1$. Reconstruction gives $\alpha = 17, u = 577, \beta = s = 17 + 2308k, t = 1$. At $k = 0$ the polynomial identities still vanish but the new direct leaf is zero, outside the domain.

The independent twelve-leaf audit supplies a sharper boundary check. At that excluded endpoint, change q_v to 2. Every new residual still vanishes, because the terms carrying v are multiplied by $q_\alpha^2 = 0$, but old reconstruction would require $u^2 = 1 + 288 \cdot 4 = 1153$, which lies strictly between 33^2 and 34^2 . Allowing zero for the new direct quotient therefore breaks integer recovery. This is a counterexample to that enlarged-domain reconstruction argument, not a false-output example: it still has the accepted output $o = 1$.

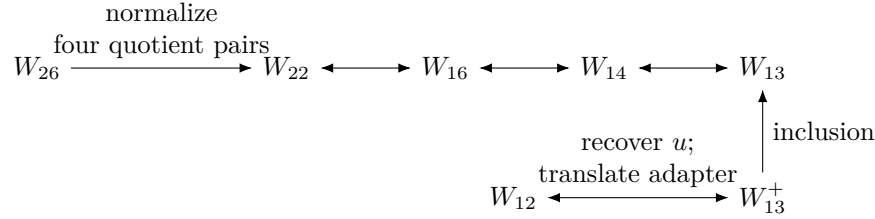


Figure 24: Solution-set maps at fixed b, C, o . The top three double arrows are bijections. The first arrow has a section but is not injective. Every nonempty W_{13} has a strict witness in W_{13}^+ by the Pell progression. The last double arrow identifies W_{12} only with this strict subset. All displayed nonempty fibers are infinite.

340 The retained three witness bounded compiler

Fix a finite deterministic two-counter program M , initial state q_0 and halt state H . Each instruction increments one counter with a specified successor, or tests it for zero with a specified zero successor and otherwise decrements it with a specified positive successor. The other counter is unchanged. States may repeat. After H use virtual absorbing steps only for native bookkeeping. These semantics are explicit; related standard instruction conventions and their importance are discussed by Dudenhefner [101].

340.1 Exact clipping on all input sizes

Remark 340.1 ([write] restatement of source 36’s Lemma 319.1: clipping). Source 37 states, as its Lemma 9.1: “Replacing native initial counters (a, b) by $(\min(a, T), \min(b, T))$ preserves all control states through time T and all selected branches at entry times $0, \dots, T - 1$. Therefore it preserves first halt by T and first halt exactly at T .” This is source 36’s Lemma 319.1. Source 37’s proof (induction on the common branches: a clipped counter that began at T is still at least $T - t > 0$ at entry time $t < T$) and its two remarks (the strict bound is needed before the last transition, not after it; the one-instruction loop from $a = T - 1$ and $a = T$ shows that clipping at $T - 1$ would merge different answers, a clipping-threshold fact and not a polynomial lower bound) are source 36’s, and are not printed twice.

For positive inputs $A = a + 1, B = b + 1$, put

$$K = T + 1, \quad N = K^2, \quad u = \min(A, K), \quad v = \min(B, K).$$

Index the finite representative pairs by $i = (u - 1)K + v$ and define

$$u_i = 1 + \left\lfloor \frac{i-1}{K} \right\rfloor = \left\lceil \frac{i}{K} \right\rceil, \quad v_i = 1 + ((i-1) \bmod K) = i + K - Ku_i. \quad (340.1)$$

Let $S \subseteq \{1, \dots, N\}$ contain the indices whose representative inputs $(u_i - 1, v_i - 1)$ satisfy the selected by-horizon or exact-horizon predicate. In principle the table can be constructed by checking the initial state and evaluating each of its N configurations for at most T transitions. This costs at most NT native transitions; the representative counters stay at most $2T$. It invokes no unbounded decision. No such interpreter is run in this report. If $q_0 = H$, the by-horizon table is full, while the exact table is empty when $T > 0$.

340.2 Integer interpolation and every residual

[write] [This subsection, Remark 340.2 and (340.6) restate source 36's Sections 320–321 in source 37's letters: N , c and M for source 36's Q , d and M . The displays (340.2), (340.3) and (340.4) are source 36's (320.2)–(320.4), (321.1) and (321.2); they are kept because Sections 341–343 compute with them. The node formulas (340.1) add the equivalent forms $\lceil i/K \rceil$ and $i + K - Ku_i$, which Section 341 uses.]

Let $c = (N - 1)!$, with $0! = 1$, and define

$$\begin{aligned} \ell_i(z) &= (-1)^{N-i} \binom{N-1}{i-1} \prod_{\substack{1 \leq h \leq N \\ h \neq i}} (z - h), \\ U_0(z) &= \sum_{i=1}^N u_i \ell_i(z), & V_0(z) &= \sum_{i=1}^N v_i \ell_i(z), \\ R(z) &= \prod_{i=1}^N (z - i), & H_S(z) &= \prod_{i \in S} (z - i). \end{aligned} \quad (340.2)$$

Empty products equal one. Since $\prod_{h \neq i} (i - h) = (-1)^{N-i} (i - 1)! (N - i)!$, the basis has $\ell_i(i) = c$ and vanishes at the other nodes. Hence

$$U_0(i) = cu_i, \quad V_0(i) = cv_i.$$

These are the usual Lagrange interpolants cleared by the fixed positive integer c ; the elementary general interpolation formula is recorded in DLMF [108]. All final coefficients are integers. No denominator depends on an input or witness.

Use exactly three positive witnesses j, r, s and the six residuals

$$\begin{aligned} R_0 &= R(j), & L_A &= cA - U_0(j) - c(r - 1), \\ R_A &= (cA - U_0(j))(U_0(j) - cK), & L_B &= cB - V_0(j) - c(s - 1), \\ R_B &= (cB - V_0(j))(V_0(j) - cK), & R_H &= H_S(j). \end{aligned} \quad (340.3)$$

The polynomial is

$$P_{M,T} = R_0^2 + R_A^2 + L_A^2 + R_B^2 + L_B^2 + R_H^2. \quad (340.4)$$

Program and horizon determine its coefficients, not new polynomial inputs. Six is a literal residual-slot count, retaining zero and duplicate slots.

Remark 340.2 ([write] restatement of source 36’s Theorem 318.1: correctness and uniqueness). Source 37 states, as its Proposition 9.2: “For every positive input pair (A, B) , a positive triple (j, r, s) solving (340.4) exists exactly when the selected bounded-halting assertion holds. When it exists it is unique.” This is the witness part of source 36’s Theorem 318.1, and source 37’s proof is source 36’s (Section 321): the range residual forces $j \leq N$, the two A residuals give $(A - u)(u - K) = 0$ and $A - u = r - 1 \geq 0$, hence $u = \min(A, K)$, likewise $v = \min(B, K)$, and the clipping lemma gives the native assertion; conversely the unique witness is

$$j = (u - 1)K + v, \quad r = A - u + 1, \quad s = B - v + 1, \quad (340.5)$$

with $1 \leq j \leq N$, $1 \leq r \leq A$ and $1 \leq s \leq B$ (source 36’s (321.4)). The proof is not printed twice.

At $T = 0$ we have $K = N = c = 1$, $U_0 = V_0 = 1$. Both classification residuals vanish and

$$P = (j - 1)^2 + (A - r)^2 + (B - s)^2 + H_S(j)^2, \quad (340.6)$$

where $H_S = 1$ or $j - 1$. Both possibilities have degree two and nine nonzero monomials. The accepted case has tuple $(1, A, B)$; the rejected case has no tuple. Keeping three witnesses at this trivial horizon is only a uniform presentation convention. [write] [Source 36’s (321.5) and (321.6), written as one formula.]

341 Exact interpolation degree at every horizon

[write] [This section answers source 36’s question on the exact interpolation degree (Section 331, item 1). It agrees with source 36’s recorded evidence: at $T = 2$, $K = 3$, the exact degree $4 \cdot 9 - 8 = 28$ is the value recorded there (Section 330), below source 36’s bound 32.]

Let $U = U_0/c$ and $V = V_0/c$ be the rational interpolants in (340.2). For $K \geq 2$ define

$$\delta_K = \begin{cases} K^2 - 1 & K \text{ even,} \\ K^2 - 2 & K \text{ odd.} \end{cases} \quad (341.1)$$

Theorem 341.1 (Interpolation degree and leading coefficient). *For every $K \geq 2$, all four polynomials U, V, U_0, V_0 have exact degree δ_K . The leading coefficient a_K of U_0 is*

$$a_K = \begin{cases} -\sum_{h=1}^{K-1} \binom{K^2-2}{hK-1}, & K \text{ even,} \\ (K^2-1) \sum_{h=1}^{K-1} (-1)^{h+1} \binom{K^2-3}{hK-1}, & K \text{ odd, } K \geq 3. \end{cases} \quad (341.2)$$

It is negative for even K , and has sign $(-1)^{(K+1)/2}$ for odd $K \geq 3$. The leading coefficient of V_0 is $-Ka_K$. At $K = 1$, both cleared interpolants are the constant 1.

341.1 Reflection and finite differences

At each node $u_{N+1-i} = K + 1 - u_i$. By uniqueness of degree-at-most- $N - 1$ interpolation,

$$U(N + 1 - z) = K + 1 - U(z).$$

Thus $U((N+1)/2+x) - (K+1)/2$ is odd in x . If K is odd and $K \geq 3$, then $N-1$ is a positive even integer and its coefficient vanishes, giving $\deg U \leq N-2$. This qualification excludes $K=1$, where $U=1$; no negative degree bound is asserted there.

For $K \geq 2$, the degree-at-most- $N-1$ polynomial $z+K-KU(z)$ interpolates the v_i . Consequently

$$V(z) = z + K - KU(z), \quad V_0(z) = c(z+K) - KU_0(z). \quad (341.3)$$

Once $\deg U \geq 2$ is established, this proves the claims for V . The identity is not asserted at $K=1$.

Let $\Delta f_i = f_{i+1} - f_i$. Repeated expansion gives

$$\Delta^q f_1 = \sum_{k=0}^q (-1)^{q-k} \binom{q}{k} f_{k+1}.$$

For a polynomial of degree at most q , this is $q!$ times its coefficient of z^q : each difference reduces a leading term az^q to qaz^{q-1} , and induction proves the statement. The step values are

$$u_i = 1 + \sum_{h=1}^{K-1} \mathbf{1}_{i > hK}.$$

For $1 \leq b \leq q$, Pascal's identity telescopes to

$$\sum_{k=b}^q (-1)^{q-k} \binom{q}{k} = (-1)^{q-b} \binom{q-1}{b-1}. \quad (341.4)$$

Therefore, whenever all boundaries $hK \leq q$ and $q \geq 1$,

$$\Delta^q u_1 = \sum_{h=1}^{K-1} (-1)^{q-hK} \binom{q-1}{hK-1}. \quad (341.5)$$

For even K , take $q = N-1$: it is odd and every hK is even. This is a strictly negative sum of positive integers, so $\deg U = N-1$ and, because $c = (N-1)!$, it proves the first coefficient formula.

For odd $K \geq 3$ the symmetry bound allows $q = N-2$. All boundaries fit since $(K-1)K \leq N-2$. Now q, K are odd, so

$$\Delta^{N-2} u_1 = S_K, \quad S_K = \sum_{h=1}^{K-1} (-1)^{h+1} \binom{K^2-3}{hK-1}. \quad (341.6)$$

It remains to prove $S_K \neq 0$ with its claimed sign. Then $\deg U = N-2$ and $a_K = cS_K/(N-2)! = (N-1)S_K$.

341.2 The strict sign for every odd K

Fix odd $K \geq 3$ and put $n = K^2 - 3$. The K complex roots of $r^K = -1$ satisfy, for every integer m ,

$$\frac{1}{K} \sum_{r^K=-1} r^m = \begin{cases} 0 & K \nmid m, \\ (-1)^{m/K} & K \mid m. \end{cases}$$

This follows by expressing them as $e^{i\pi/K}$ times the K th roots of unity and summing a finite geometric progression. Expanding $(1+r)^n$ gives

$$\frac{1}{K} \sum_{r^K=-1} r(1+r)^n = \sum_{h=1}^{K-1} (-1)^h \binom{n}{hK-1} = -S_K. \quad (341.7)$$

The bounds $0 \leq hK - 1 \leq n$ give precisely $1 \leq h \leq K - 1$. The real root -1 contributes zero since $n > 0$. Pair the other roots as $e^{\pm 2i\theta_l}$, where

$$\theta_l = \frac{(2l+1)\pi}{2K}, \quad 0 \leq l \leq \frac{K-3}{2}.$$

All these angles lie strictly between zero and $\pi/2$. Since $1 + e^{2i\theta} = 2 \cos \theta e^{i\theta}$, the real part of $r(1+r)^n$ is $2^n \cos^n \theta_l \cos((n+2)\theta_l)$. Here

$$\cos((n+2)\theta_l) = \cos\left(\frac{K(2l+1)\pi}{2} - \theta_l\right) = (-1)^{(K-1)/2+l} \sin \theta_l.$$

The paired sum in (341.7) therefore yields

$$S_K = (-1)^{(K+1)/2} \frac{2^{n+1}}{K} \sum_{l=0}^{(K-3)/2} (-1)^l \sin \theta_l \cos^n \theta_l. \quad (341.8)$$

To prove that the last sum is positive, put $f(\theta) = \sin \theta \cos^n \theta$. Its derivative is

$$f'(\theta) = \cos^{n-1} \theta (\cos^2 \theta - n \sin^2 \theta),$$

so it is strictly decreasing once $\tan \theta > 1/\sqrt{n}$. For $K \geq 3$,

$$n \geq \frac{2}{3}K^2, \quad \frac{1}{\sqrt{n}} \leq \frac{\sqrt{3/2}}{K} < \frac{3}{2K} < \frac{\pi}{2K} = \theta_0.$$

The middle inequality follows by squaring, and the last uses $\pi > 3$. Also $\tan \theta > \theta$ on $(0, \pi/2)$ because the derivative of $\tan \theta - \theta$ is $\tan^2 \theta > 0$. Thus $f(\theta_0) > f(\theta_1) > \dots > 0$. Pairing successive terms of its alternating sum gives positive differences, with a positive final term if unpaired. This includes the single term for $K = 3$. Hence (341.8) proves $(-1)^{(K+1)/2} S_K > 0$. The interpolation theorem follows. Its smallest degree at $K \geq 2$ is three, attained at $K = 2$, validating the leading-coefficient deduction from (341.3).

341.3 The complete highest homogeneous term

Let $K \geq 2$, $\delta = \delta_K$ and $a = a_K$. The unique highest-degree terms of the two classification residuals are

$$-a^2 j^{2\delta}, \quad -K^2 a^2 j^{2\delta}.$$

Indeed the competing input terms AU_0 and BV_0 have degree $\delta + 1 < 2\delta$. Their squares therefore contribute the entire degree- 4δ expression

$$(1 + K^4) a_K^4 j^{4\delta_K}. \quad (341.9)$$

The other four squared residuals have degrees at most $2N, 2\delta, 2\delta, 2N$. For even $K \geq 2$, $4\delta - 2N = 2N - 4 > 0$; for odd $K \geq 3$, $4\delta - 2N = 2N - 8 > 0$. Hence none contributes at this top degree. Its coefficient is strictly positive for every acceptance subset, including empty and full tables. Together with (340.6), this proves the exact degree law (332.1). The earlier ceiling $4(T+1)^2 - 4$ is attained at positive odd T and exceeds the true degree by exactly four at positive even T . No artificial degree-padding term is added.

342 Support coefficients and the cost of finite preprocessing

The small witness count does not give a small fixed polynomial for all horizons. Here are explicit costs of the retained compiler, sharpened only where the exact degree permits it.

[write] [This section is source 36's Section 322 in source 37's letters. The support ceiling $16\delta + 11$ replaces source 36's $16Q - 5$ (equal for even K , smaller by 16 for odd $K \geq 3$); the height bound (342.1), the storage bound and the generator costs are source 36's (322.3), (322.4) and Section 322 unchanged, kept here as the text of the sharpened section.]

342.1 Expanded support

For $K \geq 2$ put $\delta = \delta_K$. Write

$$R_A = Af_A(j) + g_A(j), \quad f_A = c(U_0 - cK), \quad g_A = -U_0(U_0 - cK).$$

The two coefficient degrees are at most $\delta, 2\delta$. Its square has only a pure- j part of degree at most 4δ , an A part of j -degree at most 3δ , and an A^2 part of j -degree at most 2δ . The B classification is analogous. Since $L_A = cA - cr + (c - U_0(j))$, its square contributes only pure- j , A and r terms, plus A^2, r^2, Ar ; its r coefficient has degree at most δ . The B, s counterpart is the same. The root-product squares have pure- j degree at most $2N < 4\delta$. Thus disjoint monomial types give

Type	Ceiling
Pure j powers	$4\delta + 1$
A^2, B^2 times j powers	$2(2\delta + 1)$
A, B times j powers	$2(3\delta + 1)$
r, s times j powers	$2(\delta + 1)$
r^2, s^2, Ar, Bs	4
Total	$16\delta + 11$

This is a ceiling, not an exact support count. It is $16N - 5$ for even K and $16N - 21$ for odd $K \geq 3$. Cancellations only reduce it. At $K = 1$ the exact support is nine as already stated. No monomial mixes A with B , r with s , or another undeclared combination of nonselector variables.

342.2 Coefficient height and storage

Let $\|f\|_1$ denote the sum of absolute coefficient values and set

$$\Lambda = K2^{N-1}(N+1)!.$$

The product of factors $z - h$ has norm at most $\prod(1+h)$. Using (340.2), the binomial sum 2^{N-1} , and $u_i, v_i \leq K$, we get

$$\|U_0\|_1, \|V_0\|_1 \leq \Lambda, \quad \|R\|_1, \|H_S\|_1 \leq (N+1)! \leq \Lambda, \quad cK, c \leq \Lambda.$$

Subadditivity and submultiplicativity imply classification-residual norm at most $4\Lambda^2$ and slack-residual norm at most 4Λ . Therefore, also at horizon zero,

$$\|P\|_1 \leq 2\Lambda^2 + 2(4\Lambda^2)^2 + 2(4\Lambda)^2 = 32\Lambda^4 + 34\Lambda^2 \leq 66\Lambda^4. \quad (342.1)$$

A nonzero coefficient needs at most $7 + 4\lceil \log_2 \Lambda \rceil$ magnitude bits, plus a sign bit if needed. This is $O(N \log(N + 1))$. Storing the $O(N)$ supported coefficients and all five exponents in binary gives an expanded-description ceiling

$$O(N^2 \log(N + 1)) = O((T + 1)^4 \log(T + 2)).$$

Neither is an optimal storage or coefficient bound.

342.3 Effective construction and its limits

The table has exactly N Boolean entries for the selected predicate. Constructing it in principle costs at most NT native transitions. A straightforward algebraic generator builds each basis product by repeated linear-factor multiplication using $O(N^2)$ integer operations per product, hence $O(N^3)$ overall. Scaling, adding and ordinary convolution to form the six squares fit within another $O(N^3)$ integer operations. Product and triangle bounds as above keep intermediate magnitudes within $O(N \log(N + 1))$ bits. These are arithmetic-operation bounds, not unit-cost bit-complexity assertions.

The generator is uniform and effective in program and horizon, but each horizon selects different coefficients and a different degree. If T is supplied in binary, these costs need not be polynomial in its bit length. The construction does not give a faster single-instance bounded-halting decision procedure than ordinary simulation. Existentially saying that some horizon accepts does not internalize varying factorials, products and acceptance tables in a single fixed polynomial. No such uniform arithmetic encoding is supplied, and no unbounded single-fold consequence follows from unique bounded triples.

343 Paid composition on three positive native gaps

The external inputs are arbitrary $g_1, g_2, g_3 \in \mathbb{Z}_{>0}$ and $D = g_1 + g_2 + g_3$ is an expression. Call the triple encoded when there are natural counters a, b with

$$\frac{g_1}{D} = \frac{1}{20} + \frac{1}{10}2^{-a}, \quad \frac{g_3}{D} = \frac{1}{20} + \frac{1}{10}2^{-b}. \quad (343.1)$$

The predicate explicitly includes this encoding. An arbitrary unencoded positive triple has no certificate, irrespective of any physical behavior from it.

Choose one of the five module presentations, with ℓ leaves, r_ℓ residuals and fixed-base degree D_ℓ . Introduce positive shifted counters A, B and two independent base-two copies with $(C, o) = (A, p)$ and (B, q) . Their outputs p, q are counted inside their modules, once each. Add

$$G_1 = (20g_1 - D)p - 2D, \quad G_3 = (20g_3 - D)q - 2D. \quad (343.2)$$

If $Q_i^{(A)}$ and $Q_i^{(B)}$ denote the fully substituted residuals from the selected display, define the entire polynomial

$$\mathcal{F}_{M,T}^{(\ell)} = G_1^2 + G_3^2 + \sum_{i=1}^{r_\ell} (Q_i^{(A)})^2 + \sum_{i=1}^{r_\ell} (Q_i^{(B)})^2 + P_{M,T}(A, B; j, r, s). \quad (343.3)$$

Every leaf in one module is distinct from the other module, the three gaps and A, B, j, r, s . Equations (333.1), (336.2), (336.4), (337.2), or (339.2), together with (340.3) and (343.2), specify every term. No unexpanded power predicate remains.

All-input correctness. A zero of (343.3) makes each residual vanish. The module theorem gives $p = 2^{A-1}$ and $q = 2^{B-1}$. Positivity of D, p, q forces $20g_1 - D > 0$ and $20g_3 - D > 0$ through the gap equations. Division by these positive quantities yields exactly (343.1), with $a = A - 1, b = B - 1$. Remark 340.2 gives the selected bounded-halting assertion on those counters.

Conversely, an encoded triple and accepted counters supply $A = a + 1, B = b + 1, p = 2^a, q = 2^b$. Both gap residuals vanish. The all-exponent module theorem supplies both complete positive POWER tuples, and (340.5) supplies the compiler witnesses. Therefore all squares vanish. There is no encoded-input promise on the quantified domain: encoding is part of the proved predicate on all positive triples. $\square$

343.1 Exact ledgers and the degree tradeoff

Decoding alone costs $2 + 2\ell$ positive witnesses and $2 + 2r_\ell$ residuals. Adding the three compiler witnesses and six residuals gives Table 49.

Module leaves	Decoding witnesses	Decoding residuals	Full witnesses	All variables	Full residuals	Fixed degree
22	46	32	49	52	38	$\max(12, d_T)$
16	34	20	37	40	26	$\max(12, d_T)$
14	30	16	33	36	22	$\max(16, d_T)$
13	28	14	31	34	20	$\max(16, d_T)$
12	26	12	29	32	18	$\max(20, d_T)$

Table 49: Paid native-gap compositions. All rows have three external positive inputs and one final polynomial equation. All variables means witnesses plus inputs. Counts hold at every horizon for both acceptance conventions.

For the degree, gap squares have degree four. The module sums have degree D_ℓ , and their explicit coefficient-one monomials in (338.1) or (339.7) involve independent module variables absent from P . If $d_T > D_\ell$, its top part (341.9) is unchanged; otherwise a module's degree witness survives. At a tie both survive. Hence

$$\deg \mathcal{F}_{M,T}^{(\ell)} = \max(D_\ell, d_T)$$

is exact, not just an upper bound. In particular, $d_0 = 2, d_1 = 12, d_2 = 28, d_3 = 60$. Thus the composed degrees at $T = 0$ and $T = 1$ are respectively 12, 12, 16, 16, 20 across the five rows, while for every $T \geq 2$ all five have the same exact compiler degree d_T . The smaller module can save witnesses and residuals without raising the composed degree once the interpolation part dominates. This compares only these displayed constructions, not all possible certificates. [write] [With source 36's 26-leaf module, $\ell = 26, r_\ell = 15$ and $D_\ell = 12$, the same formulas give $2\ell + 5 = 57$ witnesses and $2r_\ell + 8 = 38$ residual slots: source 36's Theorem 318.2. Source 38 lowers the 29 of the last row to 28 or 27 by changing the native compiler, not the module (Theorem 353.2).]

The fixed base-two module and gap part has a constant coefficient norm and finite support independent of M, T . Adding it preserves the coefficient-bit and expanded-description asymptotic bounds of Section 342, with a fixed additive contribution. It does not imply small Pell witnesses or a smaller arithmetic circuit. The leaf reduction can substantially increase expanded module support and degree; no optimized DAG is supplied.

343.2 Unique projection and infinite complete fibers

For an accepted fixed gap input the equations force

$$p = \frac{2D}{20g_1 - D}, \quad q = \frac{2D}{20g_3 - D}.$$

Injectivity of powers of two fixes A, B , and the compressed theorem fixes j, r, s . Thus the projection (A, B, p, q, j, r, s) is unique. Full tuples are infinite: vary (335.1) in one module while keeping its index and output, the other module and all outer witnesses fixed. For twelve leaves restrict to $k \geq 1$ as required. Since q_b increases, the full tuple changes. Unique decoding and a unique bounded triple therefore never give finite full fibers in these formulas.

344 The finite trace alternative and its exact paid counts

For completeness the growing trace alternative can use any same module. Give states distinct fixed integer codes, split each conditional into its zero and positive edges and add exactly one absorbing halt edge h . Let E be the number of edges and Z the number of zero edges. Each edge e has fixed source u_e , target v_e and counter changes $d_e, f_e \in \{-1, 0, 1\}$. If there are I increment and C_0 conditional instructions, then $E = I + 2C_0 + 1$ and $Z = C_0$.

For $T \geq 1$ use positive $w_{t,e}$ for $0 \leq t < T$, $1 \leq e \leq E$, and positive A_t, B_t for $1 \leq t \leq T$. Put $s_{t,e} = w_{t,e} - 1$, $A_0 = A$, $B_0 = B$. The complete trace residual list is

$$\begin{aligned} (w_{t,e} - 1)(w_{t,e} - 2) & \quad (TE \text{ slots}), \\ \sum_e s_{t,e} - 1 & \quad (T), \\ A_{t+1} - A_t - \sum_e d_e s_{t,e}, \quad B_{t+1} - B_t - \sum_e f_e s_{t,e} & \quad (2T), \\ s_{t,e}(A_t - 1) \text{ or } s_{t,e}(B_t - 1) \text{ on zero edges} & \quad (TZ), \\ \sum_e u_e s_{0,e} - q_0 & \quad (1), \\ \sum_e v_e s_{t,e} - \sum_e u_e s_{t+1,e} \quad (0 \leq t < T - 1) & \quad (T - 1), \\ \sum_e v_e s_{T-1,e} - H & \quad (1). \end{aligned} \tag{344.1}$$

Their squared sum has $T(E + 2)$ witnesses and $T(E + Z + 4) + 1$ residual slots. At $T = 0$ use the single constant residual $q_0 - H$ and no witnesses, so the same ledger holds. This constant polynomial is not called quartic.

The selector and one-hot equations choose exactly one edge at each time. Updates fix both next counters. A selected zero edge forces its tested shifted counter to be 1; a selected decrement from 1 would force a nonpositive next counter, forbidden by the domain. The state equations therefore give exactly the genuine deterministic native path padded at H . Conversely that path supplies every positive selector and next counter. Induction fixes all witnesses uniquely, including inactive selectors equal to 1. This argument needs no acyclicity, and duplicate enabled edges are not inserted.

For first halt exactly at $T \geq 1$, add the single residual

$$\sum_{t=0}^{T-1} s_{t,h}.$$

Its nonnegative terms exclude all early halt-loop uses, while the final equation still forces arrival. No new witness is added. At $T = 0$ exact and by-horizon predicates agree.

Every trace residual has degree at most two; for $T \geq 1$ the selector square gives a coefficient-one $w_{t,e}^4$, proving trace degree four. Adding the paid decoding part yields exactly

$$\begin{aligned} \text{positive witnesses} &= T(E + 2) + 2 + 2\ell, \\ \text{residual slots} &= T(E + Z + 4) + 3 + 2r_\ell, \\ \text{all variables} &= T(E + 2) + 5 + 2\ell, \\ \text{degree} &= D_\ell. \end{aligned} \tag{344.2}$$

The exact-halt version adds one residual for $T \geq 1$. For twelve leaves this is $T(E + 2) + 26$ witnesses, $T(E + Z + 4) + 13$ residuals and degree twenty. The module degree witnesses survive, also at $T = 0$. This is an alternative tradeoff: growing trace arity with constant module-dominated degree, versus constant witness arity with growing interpolation degree and finite table costs.

[write] [The trace residuals (344.1) and their correctness argument are source 34's Theorem 301.1, as used in source 35's Theorem 304.3; with source 35's 26-leaf modules ($\ell = 26$, $r_\ell = 15$, $D_\ell = 12$) the ledger (344.2) gives source 35's $T(E + 2) + 54$ witnesses, $T(E + Z + 4) + 33$ residual slots and degree 12. No novelty is claimed for the trace itself; the smaller-module counts are source 37's.]

345 Physical interpretation and semantic boundaries

The arithmetic POWER and clipping arguments do not need a physical simulator. The separately retained five-live-signal compiler theorem [source 34's Theorem 294.1] uses stationary markers $L = 0$, $X = g_1$, $Y = g_1 + g_2$, $R = D$ and an instruction messenger. On precisely the encoded inputs (343.1), it identifies each native transition with one ordered instruction section, preserves D , and uses at most 32 binary collisions and duration strictly between D and $10D$ per nonhalting transition.

Conditional on that retained theorem, the certificates here also express encoded-input bounded instruction-section reachability. If first halt occurs at $k > 0$, the physical prefix has elapsed time strictly between kD and $10kD$ and at most $32k$ collisions. At $k = 0$ it is already at the halt section. A collision-count horizon is not an instruction horizon. The arithmetic absorbing halt edge is padding only, not an assertion that the physical messenger keeps executing after halt.

Neither the three-witness compiler nor these module replacements classify arbitrary unencoded physical executions. The present work does not re-audit the physical theorem, create a universal counter program, or establish a new five-signal universality result. Its finite-table coefficients depend on the fixed program and horizon; its physical interpretation remains limited to the explicitly inherited interface.

346 Dependencies evidence and reproducibility

The article synthesizes three separately preserved mathematical continuations: the 22-leaf normalization and 16/14/13 eliminations, the strict-domain twelve-leaf continuation, and the exact interpolation-degree continuation. The complete proofs are retained, together with the earlier bounded compiler and its native-gap, trace, POWER and physical dependencies. Earlier reports are not edited or renumbered. The new notation A_* for a module expression does not change any polynomial.

The reductions supply explicit algebra and domain recovery. Their all-exponent semantic conclusion still uses the pinned mathlib constructive Pell theorem pair. The bounded compiler and exact interpolation theorem need neither Pell theory nor the physical theorem. Physical transport alone uses the retained encoded-input physical compiler. These dependencies are deliberately separated: exact expansion of a polynomial verifies neither its all-input semantics nor a physical simulation theorem.

The baseline independent audit reconstructs all eight fixed/variable-base expansions and verifies the maps, degrees, domains and infinite families. It records 31 elimination identities, 192 complete module evaluations, 96 reconstruction round trips and 28 composed examples. The exact-degree independent audit checks leading coefficients for $K = 1, \dots, 101$, all 385 node pairs for $K \leq 10$, and 565 complete sum-of-squares expansions, including every acceptance subset at $K = 1, 2, 3$. It identifies the local odd- K qualification applied explicitly in Section 341. These finite checks support the written all-horizon proofs; they do not replace them.

The independent twelve-leaf audit verifies five full module expansions, complete coefficient agreement in both retained conventions, 33 reconstruction fixtures, 675 rejected single-leaf perturbations and eight complete compositions with 777 clipping/interpolation fixtures. It separately checks the strict-domain image and the rational integrality and sign arguments; no mathematical correction was required. Its domain challenges illustrate failure of the reconstruction after changing domains, not false-output solutions within the stated theorem.

The supplied scientific source files and checker programs are inert evidence in this release. The article build imports or executes none of them. There is no native counter interpreter, physical simulator, saved schedule, upstream scientific execution or Lean build. Stored author evidence, independent mathematical evidence, manuscript review and presentation-tool checks are different evidence categories. The independent dossiers and the final release README state their exact scope and status without turning one category into another.

[write] [In this report the three science packets are shipped under the prefixes **37-pos-power-pb-** (the 22-, 16-, 14- and 13-leaf reductions), **37-pos-power-p12-** (the twelve-leaf continuation) and **37-pos-power-ed-** (the exact degree), and the three independent audits under **37-pos-power-audit-pb-**, **37-pos-power-audit-p12-** and **37-pos-power-audit-ed-**; the archive's copy of source 36's bounded-certificate packet is shipped once, under source 36's prefix, and its copies of the proofs of sources 28, 30, 34 and 35 under theirs. Nine evidence files over 1 MB (five distinct polynomial expansions and one independent expansion, with copies) are not shipped; the packets' own programs regenerate them in seconds (README, "Reconstructing the excluded data"). The audits were written by the same pipeline as the article. At placement every science and audit checker reproduced its frozen records on copies up to CRLF line ends and, for two of them, a one-line path-separator patch in the scratch copy; the twelve-leaf audit hash-pins its packet's **MANIFEST.sha256**, which is not shipped, so its rerun restores the packet from the arrival archive (README).]

The report-owned presentation tools adapt the completely inspected accepted Report 66 release contract. They flatten only the fixed manuscript modules, authenticate their reviewed pin map, build a fresh TeX format in an external directory with shell escape disabled, and retain the union of the format and all three compilation passes' recorder inputs. A dependency lock pins the named executables, interpreter and recorded TeX/font files. A locked rebuild must reproduce the packaged PDF byte for byte. Every page is rendered and inspected; PNG raster/CRC validation is additional to visual review. The lock does not inventory the operating system, shared libraries or Python standard library.

Exact manifests authenticate the release’s listed file and directory descendants, bytes, modes and nanosecond modification times. Frozen input copies preserve their dependency layout and are separately pinned. A metadata-preserving extraction operation restores the authenticated descendant metadata; ordinary ZIP extraction may not. The manifest’s own hash must be obtained outside the archive. Read-access times and transported filesystem creation/change times are not integrity claims. The README gives the exact commands, pin values and completed independent review receipts. [write] [The delivery README is not shipped; this report’s README replaces it. The release tools are shipped as `code/37-pos-power-tools-*.py`; they expect the delivered layout and refuse Windows paths.]

347 Context and questions left open

Pell-based Diophantine exponentiation is established prior mathematics. Pāk and Kaliszyk [109] describe formalizations and carefully distinguish free parameters, explicit unknowns and additional unknowns hidden in subrelations. That distinction is essential here: twelve module leaves include the output, exclude the supplied index, and cost eleven auxiliaries after the output is fixed. The native-gap composition must pay for both shifted counters, both complete modules and the compiler witnesses.

The comparison in Table 48 is between specified presentations of one relation. It is not a comparison with arbitrary Diophantine equations, prime-representing polynomials or universal polynomials. No exhaustive literature search, novelty, priority, minimum degree, minimum witness count, arithmetic-gate improvement or small-height Pell-witness result is claimed. Every displayed full POWER and native-gap fiber is infinite; eliminating obvious common-shift freedom does not make these formulas finite-fold.

Several bounded questions remain meaningful. Can another explicitly justified elimination improve a listed witness/degree pair? Can the coefficient and support ceilings be tightened without changing the literal formula? Can useful height bounds be proved for reconstructed positive Pell witnesses? Can finite-table construction exploit a particular program without treating acceptance as free data? Each would require a separately stated construction or proof. None is settled by the exact counts in this report, and none turns the horizon-indexed family into one fixed polynomial for unbounded halting.

[write] [Still open in this report. Source 38 lowers the composed witness count further by a different native compiler, not by a smaller module (Section 353), and states the same coefficient and finite-table questions as open for its own formulas.]

[write] **Review by the Hilbert’s-tenth programme.** The programme’s scoped intake [111] read source 37’s delivery README completely and, of its article, the twelve-leaf section (Section 339), the last subsection of Section 342, Sections 343–346 and the first two paragraphs of Section 347; it did not read the 22- to 13-leaf reductions or the exact-degree proof, and it ran none of the archived programs and did not newly certify the inherited Pell theorems. Its findings: the reduction makes the quotient q_α strictly positive, reconstructs the eliminated Pell data rationally and then integrally, and uses that strict sign to recover the positive branch; the converse restores only the strict-quotient subset of the older presentation; the module has twelve leaves including its output and five residual slots, degrees 20 and 24, and “No optimized arithmetic DAG for it is supplied” (“twelve leaves, not twelve gates”); the composition has 29 positive witnesses, 32 variables and 18 residual slots, and full gap fibres stay infinite. It reports no defect.

Source 38 (batch-91 manuscript 20): Low-arity finite-clipping Diophantine compilers

Two witnesses at linear degree; one witness by disjoint cells, “Research Report 69”, 4 October 2026. [write] From the archive `Low_arity_Diophantine_compilers_sources.zip`, main file `Report69/Report69.tex` (652 lines; a 17-page PDF); its labels carry the prefix `smc:la:`, its sections are Sections 348–354 here, and its equations are numbered by section. It continues sources 36 and 37 and the undelivered Report 68; its clipping lemma restates source 36’s and its POWER module is source 37’s twelve-leaf module at base two (notes in Sections 348 and 353). Its three bibliography entries are this report’s [108], [110] and [86]. Its abstract follows.

A predicate on positive integer pairs that depends only on clipping each coordinate at a fixed threshold has a direct Diophantine representation with a unique positive witness pair. Cleared tensor interpolation gives five residual squares and, at horizon $T \geq 1$, exact degree $\max\{2(T+1), 4, 2 \deg F_S\} \leq 4T$. One fixed two-zero-test program attains $4T$ for every by-horizon $T \geq 2$ in this presentation. A separate product of disjoint-cell sums of squares has one unique positive witness and exact degree $2n_I + 4Tn_E + 8Tn_C$. We prove an exact zero-versus-one native auxiliary-variable classification for every fixed finite clipping table, with no degree restriction: zero variables suffice exactly when each accepted tail cell’s full coordinate-flat closure is accepted.

All domains, empty and full tables, horizon zero, coefficient heights, expanded support, storage, factored gates and table preprocessing are paid explicitly. Composing with the accepted twelve-leaf POWER module gives 28 positive witnesses and 17 squares, or 27 witnesses with a nonnegative product block; replacing that block by its square gives 13 squares and doubles its compiler-degree contribution. Native witnesses and decoded projections are unique, but the full composed fibers are infinite. The results concern horizon-indexed families, not a fixed polynomial for unbounded halting, and make no global minimality or priority claim.

348 Finite tables, domains, and the main tradeoff

Write $\mathbb{N} = \{0, 1, 2, \dots\}$ and $\mathbb{Z}_{>0} = \{1, 2, \dots\}$. Fix $T \in \mathbb{N}$, put $K = T + 1$, and define

$$c_K(x) = \min(x, K), \quad U_S = \{(A, B) \in \mathbb{Z}_{>0}^2 : (c_K(A), c_K(B)) \in S\}, \quad (348.1)$$

where $S \subseteq \{1, \dots, K\}^2$ is fixed coefficient data. The table is never a free input or an existentially selected table. Degrees below are joint total degrees in every actual input and witness, before imposing any equations. We set $\deg 0 = -\infty$; a nonzero constant has degree zero. A residual-square count refers to the literal displayed slots, even if some are zero or repeated.

Theorem 348.1 (Native compilers, stated together). *For every fixed table S there are explicit integer polynomials with the following properties.*

1. A sum $P_S(A, B; r, s)$ of five residual squares represents U_S with exactly two declared positive witnesses, unique whenever they exist. For $T \geq 1$,

$$\deg P_S = \max\{2K, 4, 2 \deg F_S\} \leq 4T,$$

where F_S is the rejection interpolant in (349.3). A separate $T = 0$ definition has exact degree two.

2. A globally nonnegative polynomial $Q_S(A, B, w)$ represents U_S with one declared positive witness, again unique whenever it exists. For $T \geq 1$,

$$\deg Q_S = D_S := 2n_I + 4Tn_E + 8Tn_C \leq 10T^2 + 8T, \quad (348.2)$$

where n_I, n_E, n_C count accepted cells with zero, one and two tail coordinates. The empty product is 1, with $D_S = 0$.

3. With no degree restriction, the minimum number of positive existential auxiliary variables in a single integer-polynomial equality defining U_S is zero or one. It is zero exactly when the coordinate-flat closure condition of Theorem 351.1 holds.

Each construction is given and proved below. In particular, the arity classification concerns a *fixed finite table* and unrestricted degree. It does not assert that the two-witness linear-degree formula is arity-minimal, or that any degree is minimal. The one-witness full-table product reaches its displayed worst-case degree even though the full predicate itself has the zero polynomial as a zero-witness representation.

348.1 Why bounded counter programs give such tables

Fix a finite deterministic two-counter program M , initial state q_0 and halt state H . A nonhalt instruction either increments one counter and takes its specified successor, or tests a counter for zero, taking its zero successor when zero and otherwise decrementing it and taking its positive successor. The other counter is unchanged. Both counters are natural integers; arbitrary loops are allowed. Make H virtually absorbing with no counter change solely to describe a finite prefix.

Remark 348.2 ([write] restatement of source 36’s Lemma 319.1: clipping through the last tested entry). Source 38 states, as its Lemma 1.2: “The runs from (a, b) and $(\min(a, T), \min(b, T))$ have the same control states $q_0, \dots, q_T$ and the same branches in their first T transitions. Thus first arrival at H at or before T is preserved.” This is source 36’s Lemma 319.1 (source 38 states only the by- T conclusion here and treats first halt exactly at T in the next paragraph). Its proof, by induction on the common branches with the remark that a representative may become zero only *after* transition T , is source 36’s and is not printed twice.

Positive inputs A, B represent native counters $A - 1, B - 1$. The shift converts the native threshold T to $K = T + 1$. Define S by testing representatives $(i - 1, j - 1)$ either for reaching H by T or for first reaching H exactly at T . Remark 348.2 proves (348.1) for all positive input sizes in both cases. If $q_0 = H$, the by- T table is always full, and the exact- T table is empty for $T > 0$. At $T = 0$ both predicates are full exactly when $q_0 = H$.

There are K^2 table bits. A direct finite construction examines the initial state and at most T transitions from each representative: at most $K^2 T$ native transitions, with each representative counter at most $2T$. This is an effective finite preprocessing bound, not an executed interpreter in this report. It is not free advice and does not imply a faster bounded-halting algorithm. The horizon indexes different polynomials, with different products, factorials and table coefficients.

349 A two-witness tensor compiler

Assume $T \geq 1$, so $K \geq 2$, and put $c = (K - 1)!$. For $1 \leq i \leq K$ define

$$\ell_i(z) = (-1)^{K-i} \binom{K-1}{i-1} \prod_{\substack{1 \leq h \leq K \\ h \neq i}} (z - h). \quad (349.1)$$

The difference product at i is $(-1)^{K-i}(i-1)!(K-i)!$. Hence

$$\ell_i(h) = c \mathbf{1}_{i=h} \quad (1 \leq h \leq K), \quad \sum_{i=1}^K \ell_i(z) = c. \quad (349.2)$$

The second identity follows because the two sides have degree at most $K-1$ and agree at K distinct nodes. Every coefficient is integral. Define

$$F_S(x, y) = \sum_{(i,j) \notin S} \ell_i(x) \ell_j(y), \quad (i, j) \in \{1, \dots, K\}^2. \quad (349.3)$$

It has degree at most $K-1$ in each coordinate, at most $2K-2$ jointly, and at a grid node equals zero on S and c^2 off S . In particular the full table gives $F_S = 0$ and the empty table gives $F_S = c^2$. The notation x, y defines this polynomial; it introduces no witnesses in the compiler.

Set the expressions

$$u = A - r + 1, \quad v = B - s + 1, \quad R_K(z) = \prod_{i=1}^K (z - i). \quad (349.4)$$

Use precisely the five residuals

$$\begin{aligned} D_1 &= R_K(u), & D_2 &= R_K(v), \\ D_3 &= (r-1)(u-K), & D_4 &= (s-1)(v-K), \\ D_5 &= F_S(u, v), & P_S &= \sum_{i=1}^5 D_i^2. \end{aligned} \quad (349.5)$$

Here A, B, r, s are ordinary positive integer variables. Before the equations hold, u, v are not assumed positive. There are no implicit range constraints and no input-dependent denominators.

Proposition 349.1 (Both directions and the whole native fiber). *For positive A, B , there exist positive r, s with $P_S = 0$ exactly when $(A, B) \in U_S$. Every such zero has*

$$r = A - c_K(A) + 1, \quad s = B - c_K(B) + 1. \quad (349.6)$$

In particular the native witness fiber is a singleton or empty, and $1 \leq r \leq A, 1 \leq s \leq B$.

Proof. A sum of real squares vanishes only when every residual vanishes. The first two force $u, v \in \{1, \dots, K\}$. From (349.4), $A - u = r - 1 \geq 0$. If $u < K$, then $D_3 = 0$ forces $r = 1$ and $A = u$. If $u = K$, positivity of r forces $A \geq K$. These alternatives give exactly $u = c_K(A)$ and the first equality in (349.6); similarly $v = c_K(B)$. The fifth residual now asserts table membership, proving soundness and uniqueness.

Conversely, if $(A, B) \in U_S$, the positive pair in (349.6) gives those clipped representatives. The range residuals vanish. Each classification product vanishes because either its slack is one or its representative is K . The fifth residual vanishes at the accepted node. Thus $P_S = 0$. $\square$

Strict positivity is essential. If $r = 0$ were allowed, an input $A = K - 1$ could produce $u = A - r + 1 = K$; the classification product would vanish via $u - K$, selecting an incorrect tail class. Positivity is part of the theorem's domain, not an inequality that the displayed polynomial supplies on its own.

349.1 Horizon zero is a separate definition

When $T = 0$, put $\epsilon = 0$ for the full one-cell table and $\epsilon = 1$ for the empty table. Use the five residual slots

$$A - r, B - s, 0, 0, \epsilon, \quad P_S = (A - r)^2 + (B - s)^2 + \epsilon. \quad (349.7)$$

This has exact degree two, six nonzero monomials if accepted and seven if rejected. Accepted inputs have the unique positive pair $(r, s) = (A, B)$. Inserting $K = 1$ in the unchanged generic formula (349.5) would retain unnecessary quadratic classification residuals and degree-four squares. Equation (349.7) explicitly replaces that formula; it does not silently simplify its degree claim.

349.2 Exact degree and a fixed sharpness example

Lemma 349.2 (No cancellation at the highest square degree). *For real polynomials $h_1, \dots, h_m$, not all zero, $\deg \sum h_i^2 = 2 \max_i \deg h_i$.*

Proof. The top homogeneous component is the sum of squares of the top components of maximal degree. If it vanished identically, evaluation at every real point and nonnegativity would force each such component to vanish identically, a contradiction. $\square$

The substitution $x = A - r + 1$, $y = B - s + 1$ preserves the degree of every nonzero polynomial $f(x, y)$. Its top part becomes $f_{\text{top}}(A - r, B - s)$, nonzero because setting $r = s = 0$ recovers $f_{\text{top}}(A, B)$. The range residuals have exact degree K and the classification residuals exact degree two. Therefore

$$\deg P_S = \max\{2K, 4, 2 \deg F_S\} \leq 4K - 4 = 4T. \quad (349.8)$$

Empty and full tables have degree $2K$. At $T = 1$, every table has degree four.

For a fixed program attaining the upper bound, use states Q_A, Q_B, L, H , initial state Q_A , and the instructions

State	Zero branch	Positive branch
Q_A : test a	go to Q_B	decrement a , go to L
Q_B : test b	go to H	decrement b , go to L

At L , increment a and return to L . The only accepted input is $(a, b) = (0, 0)$, first reaching H on transition two. For every by-horizon $T \geq 2$, the table is $S = \{(1, 1)\}$. Then

$$F_S(x, y) = c^2 - \ell_1(x)\ell_1(y),$$

whose $x^{K-1}y^{K-1}$ coefficient is -1 . It has degree $2K - 2$, and (349.8) is exactly $4T$ for this one fixed program at every such horizon.

This is sharpness of the *displayed construction*, not a lower bound against other polynomials. For the same fixed program, first-exactly- T acceptance is empty when $T > 2$. At each individual exact horizon $T \geq 2$, a chain of $T - 2$ increments on the successful path after the second test yields the same singleton table; that delayed program depends on T .

350 One witness from disjoint clipped cells

For $K \geq 2$, define

$$p_K(x) = \prod_{h=1}^T (x - h)^2. \quad (350.1)$$

This is nonnegative on $\mathbb{R}$. On positive integers it is zero exactly for $x < K$, and strictly positive exactly for $x \geq K$. The second assertion uses the discrete domain: every positive integer smaller than $K = T + 1$ is one of its roots.

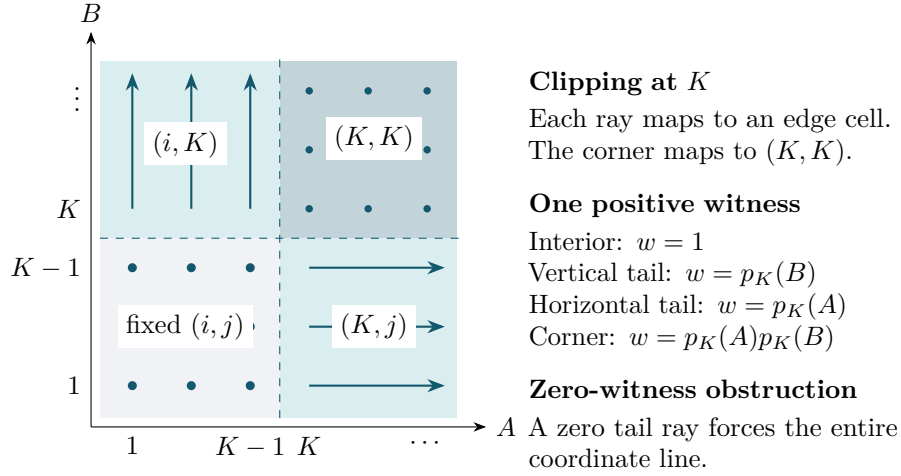


Figure 25: The positive-integer partition, drawn schematically at $K = 4$. Shading indicates regions of lattice points, not additional real-domain acceptance. The cells are disjoint; the tail labels retain the fixed threshold K rather than the actual unbounded coordinate.

For each table cell define a globally nonnegative factor:

$$\begin{aligned} f_{ij} &= (A - i)^2 + (B - j)^2 + (w - 1)^2 & (i, j < K), \\ f_{iK} &= (A - i)^2 + (w - p_K(B))^2 & (i < K), \\ f_{Kj} &= (B - j)^2 + (w - p_K(A))^2 & (j < K), \\ f_{KK} &= (w - p_K(A)p_K(B))^2. \end{aligned} \quad (350.2)$$

Set

$$Q_S(A, B, w) = \prod_{(i,j) \in S} f_{ij}(A, B, w), \quad (350.3)$$

with empty product 1. All coefficients are integers, and $Q_S \geq 0$ on all of $\mathbb{R}^3$.

Proposition 350.1 (One witness, with no inactive freedom). *For positive A, B, w , a cell factor in (350.2) vanishes exactly on its own clipped cell and at the following witness value:*

$$w = \begin{cases} 1, & A, B < K, \\ p_K(B), & A < K \leq B, \\ p_K(A), & B < K \leq A, \\ p_K(A)p_K(B), & A, B \geq K. \end{cases} \quad (350.4)$$

Consequently $\exists w \in \mathbb{Z}_{>0} : Q_S(A, B, w) = 0$ is equivalent to $(A, B) \in U_S$, and its positive witness is unique.

Proof. An interior factor is zero exactly when $A = i$, $B = j$ and $w = 1$. An edge factor f_{iK} is zero exactly when $A = i$ and $w = p_K(B) > 0$, which forces $B \geq K$ by (350.1); the other edge is symmetric. At the corner, $w = p_K(A)p_K(B) > 0$ forces both factors positive, since they are nonnegative, hence $A, B \geq K$. The displayed choices prove every converse.

A real product is zero exactly when some factor is zero. On positive integer inputs the clipped cells partition the domain, so at most one factor can vanish for a positive witness. Its cell is accepted if and only if it occurs in (350.3). Equation (350.4) then proves existence and uniqueness. Every listed value is a positive integer on its stated cell. $\square$

Allowing $w = 0$ would break this proof: in an edge factor, $p_K(B) = 0$ at a below-threshold positive integer could mimic the tail. No additional inequality is hidden in the equation; strict positivity is an explicit domain obligation.

At $K = 1$, the separate one-witness definitions are $(w - 1)^2$ for the full table and 1 for the empty table. Their degrees are two and zero. The accepted witness is $w = 1$; the rejected polynomial has no solution and its formally declared witness is unused.

350.1 Exact degree and the distinction between a block and a square

Each factor is nonzero. Their exact degrees, in the order of (350.2), are $2, 4T, 4T, 8T$: p_K has leading term x^{2T} , and its squared corner product has leading term $A^{4T}B^{4T}$. Total degree is additive under products of nonzero polynomials. Thus (348.2) is exact. The ceiling uses $n_I \leq T^2$, $n_E \leq 2T$ and $n_C \leq 1$, and is attained by the full-table *displayed product*. For nonempty S , the coefficient of $w^{2|S|}$ is one, so w genuinely occurs.

Each interior factor has three residual-square slots, each edge has two, and the corner has one. Fully distributing their product gives precisely

$$L_S = 3^{n_I} 2^{n_E} \quad (350.5)$$

literal square slots: pick one squared residual from each factor and square the product of those residuals. The empty-product convention gives the one square 1^2 . Coincident or zero terms need not be combined; this is a presentation count, never a minimum SOS length.

The compact Q_S is a product of sums of squares and is not generally one residual square. Replacing it by Q_S^2 preserves its zero set and witnesses, costs one additional multiplication, and changes its degree from D_S to $2D_S$. Exponential distributed SOS length does not imply exponential collected monomial support; degree and the fixed number of variables give the polynomial bounds in Section 352.

351 Exactly when zero native witnesses suffice

For a clipped cell $a = (a_1, a_2)$ define its tail positions $J(a) = \{\ell : a_\ell = K\}$. Its *coordinate-flat closure in the table* consists of all cells obtained by replacing any or all coordinates in $J(a)$ by arbitrary indices in $\{1, \dots, K\}$, while keeping the other coordinates fixed.

Theorem 351.1 (Exact zero-versus-one native classification). *Fix $K \geq 1$ and $S \subseteq \{1, \dots, K\}^2$. The following are equivalent:*

1. *An integer polynomial $R(A, B)$, with no witnesses, vanishes on positive integer inputs exactly on U_S .*
2. *Every accepted cell's coordinate-flat closure is contained in S .*
3. *If (i, K) is accepted then every (i, j) is accepted; if (K, j) is accepted then every (i, j) is accepted; an accepted (K, K) forces the full table.*

Within one polynomial equality, with fixed table-dependent coefficients and no degree bound, the minimum number of positive existential auxiliary variables is zero when these conditions hold and one otherwise.

The quantified indices in the last condition range from 1 to K . Joint replacement at the corner is essential. This is not the usual coordinatewise order-ideal condition: fixed coordinates smaller than K need not be replaced.

Necessity. Suppose R vanishes on all positive integer points of an accepted cell. Fix its non-tail coordinates at that cell's exact values. The resulting polynomial in the tail coordinates vanishes on the entire Cartesian product $\{K, K+1, \dots\}^{|J(a)|}$.

A polynomial zero on a product of infinite real sets is identically zero. For one variable this is the finite-root theorem. Inductively, view it as a polynomial in its last variable; fixing all earlier coordinates in their infinite sets forces every coefficient to vanish there, and the induction hypothesis makes those coefficient polynomials zero. Therefore the restriction of R to the entire coordinate flat is identically zero. In particular, each former tail coordinate may take any positive value below K as well. Exact representation requires the corresponding jointly replaced clipped cells to be accepted. This argument even allows real coefficients and makes no SOS assumption on R . $\square$

Sufficiency and the minimum. Let $X_1 = A$, $X_2 = B$ and define

$$Z_a(X) = \sum_{\ell \notin J(a)} (X_\ell - a_\ell)^2, \quad R_S(X) = \prod_{a \in S} Z_a(X). \quad (351.1)$$

An empty sum is zero and an empty product is one. A factor vanishes exactly on its coordinate flat. If the closure condition holds, every positive point on that flat is accepted; conversely an accepted point makes its own cell's factor zero. This proves exact representation. For the empty table, $R_S = 1$. An accepted all-tail cell forces a full table and contributes the zero factor, yielding $R_S = 0$, as required.

If S is nonempty and has no all-tail cell, each factor is a nonzero degree-two polynomial, so the displayed R_S has degree exactly $2|S|$. That degree need not be minimal. Necessity rules out zero variables if closure fails, while Proposition 350.1 supplies one. These two observations establish the exact minimum in the stated class. $\square$

At $K = 1$, both possible tables have minimum zero, represented by 0 and 1. Finite interior-only tables also satisfy closure. For example, a single interior point admits $(A - i)^2 + (B - j)^2 = 0$ with no witness. A single edge cell at $K \geq 2$ fails closure: its tail ray cannot be cut off below K by a witness-free polynomial equality. These examples explain why a uniform one-witness construction is not minimal for every table.

[write] [Batch-91 manuscript 21 restates this theorem with proof and counts the tables it classifies: among the tables of a fixed size, those that need no auxiliary variable number $C_n = 1 + a_n$, with a_n the OEIS sequence A196460. That manuscript is printed in the collection's report `SetTheory/Cardinals/docs/reports/generating-functions-and-asymptotics/oeis-sequence-asymptotics/a196460-clipping-tables`, written together with this Part; its normalization of the table size is stated there. The Hilbert's-tenth programme's review notes that the classification is "for fixed finite clipping tables with unrestricted degree, not arbitrary c.e. languages" [111].]

351.1 The fixed-dimensional extension

Fix an integer $d \geq 1$, $K \geq 2$ and $S \subseteq \{1, \dots, K\}^d$. Inputs $X_1, \dots, X_d$ and witness w are all positive integers. For a cell a , put $J(a) = \{\ell : a_\ell = K\}$, $t(a) = |J(a)|$, and define

$$f_a(X, w) = \sum_{\ell \notin J(a)} (X_\ell - a_\ell)^2 + \left(w - \prod_{\ell \in J(a)} p_K(X_\ell) \right)^2, \quad (351.2)$$

$$Q_S(X, w) = \prod_{a \in S} f_a(X, w).$$

The inner empty product is one. Positivity of w forces every tail factor $p_K(X_\ell)$ to be positive; the fixed-coordinate squares force every other input coordinate. Thus each factor defines exactly its clipped cell, and disjointness proves the same equivalence and uniqueness. Its unique witness is the product over the actual tail coordinates, or one if there are none.

A no-tail factor has degree two; a factor with $t(a) \geq 1$ tails has degree $4Tt(a)$. Degree additivity gives the exact sum of these cell degrees (zero for empty S). In the full table, there are T^d no-tail cells and total tail-coordinate incidence dK^{d-1} , whence

$$\deg Q_S \leq 2T^d + 4dT(T+1)^{d-1}. \quad (351.3)$$

For fixed d this is $O(T^d)$ and the support has at most $\binom{\deg Q_S + d + 1}{d+1}$ monomials. The constants and exponents here may depend on d ; no dimension-uniform polynomial bound is asserted. The fully distributed SOS presentation may still be exponential in $|S|$. At $K = 1$, again use $(w - 1)^2$ or 1.

The infinite-grid necessity proof and the zero-witness construction (351.1) hold verbatim in d coordinates. Consequently Theorem 351.1's exact zero-versus-one classification holds for every fixed finite-dimensional clipping table, with the same single-equation, positive-integer and unrestricted-degree qualifications. [write] (Added 4 October 2026, batch 91: the closed tables of Theorem 351.1 are counted in the research report `SetTheory/Cardinals/docs/reports/generating-functions-and-asymptotics/oeis-sequence-asymptotics/a196460-clipping-tables` (batch-91 manuscript 21, its Part I). For $K = n + 1$ the tables whose exact auxiliary minimum is zero number $C_n = 1 + a_n$, where $a_n = \sum_j \binom{n}{j} (1 + 2^j)^n$ is OEIS A196460, so the zero-witness fraction is asymptotic to $\frac{1}{2} 4^{-n}$; that report restates this theorem and the one-witness construction with proof, as a restatement, and derives fixed-order and uniform asymptotics of a_n . Its "closure dependencies" `ONE_WITNESS.md` and `AUDIT.md` are the files shipped here as `38-low-arity-la-ONE_WITNESS.md` and `38-low-arity-audit-la-AUDIT.md`.)

352 Coefficient, support, storage and circuit costs

All bounds in this section concern the displayed formulas, uniformly in the table at a fixed $K \geq 2$. They are upper bounds, not instancewise comparisons or optimality statements. For an integer

polynomial f , let $\|f\|_1$ denote the sum of absolute values of its collected coefficients. This norm is subadditive and submultiplicative. Integer arithmetic operations are counted separately from bit complexity, and the finite-table preprocessing from Section 348 must be added.

352.1 Tensor coefficients after every affine substitution

Put

$$B_K = (K+1)!, \quad L_K = 2^{K-1}B_K, \quad H_K = L_K^4 + 2B_K^2 + 8(K+1)^2. \quad (352.1)$$

The affine factor $u - i = A - r - (i - 1)$ has norm $i + 1$. Product and triangle bounds, with the binomial sum 2^{K-1} , therefore give

$$\begin{aligned} \|R_K(u)\|_1, \|R_K(v)\|_1 &\leq B_K, \\ \sum_i \|\ell_i(u)\|_1, \sum_i \|\ell_i(v)\|_1 &\leq L_K, \\ \|F_S(u, v)\|_1 &\leq L_K^2. \end{aligned}$$

Each classification residual has norm at most $2(K+1)$, so

$$\|P_S\|_1 \leq H_K. \quad (352.2)$$

Every coefficient magnitude thus needs at most $\lceil \log_2(H_K + 1) \rceil = O(K \log(K+1))$ bits, with an additional sign bit if desired. No substitution-dependent coefficient cost is omitted.

The tensor square has degree at most $2K - 2$ in the pair (A, r) and separately at most $2K - 2$ in (B, s) . Its support is at most $\binom{2K}{2}^2$. The other squares each use only one of those pairs, and have degree at most $2K$. Only the two additional degree layers can lie outside that tensor support envelope; they contribute at most $2[\binom{2K+2}{2} - \binom{2K}{2}] = 8K + 2$. Hence a complete support ceiling is

$$N_K = K^2(2K - 1)^2 + 8K + 2 = O(K^4). \quad (352.3)$$

For example, fixed-field records for coefficients and four exponents need at most

$$N_K [\lceil \log_2(H_K + 1) \rceil + 1 + 4 \lceil \log_2(2K + 1) \rceil] \quad (352.4)$$

bits, apart from a format header. Thus the expanded storage ceiling is $O(K^5 \log(K+1))$. Cancellations may substantially reduce support.

352.2 Tensor factored gates and coefficient generation

Let $m = K^2 - |S|$ be the rejected-cell count. Reuse the $2K$ factors $u - i, v - i$. Computing every one-dimensional basis by multiplying its $K - 1$ factors and its signed binomial coefficient uses at most $2K(K - 1)$ multiplications. The two ranges cost $2(K - 1)$ more, the two classification products two more, the rejected tensor products m , and the five final squares five. Thus

$$\begin{aligned} \text{multiplications} &\leq 2K^2 + m + 5, \\ \text{additions/subtractions} &\leq 2K + 10 + \max(m - 1, 0), \end{aligned} \quad (352.5)$$

including u, v , factors, shifted slacks and the final sum. A uniform combined ceiling is $4K^2 + 2K + 14$. There is no division. Signed binomial constants have $O(K)$ bits each, nodes have $O(\log K)$ bits,

the table has K^2 bits, and a straightforward gate-index encoding uses $O(K^2 \log(K+1))$ bits in all. These counts do not treat large integer operands as fixed-bit objects.

For one explicit expanded-construction procedure, build the K univariate bases by successive linear multiplication in $O(K^3)$ integer coefficient operations. Sum their outer products across the table in $O(K^4)$, and square the bivariate polynomial by naive convolution in $O(K^4)$. Precompute the powers of $A - r + 1$ and $B - s + 1$ through $2K - 2$. Substitute one coordinate by a separable coefficient transform in $O(K^4)$ operations, leaving $O(K^3)$ entries, and then the other in $O(K^5)$. The remaining four squares are smaller. This gives $O(K^5)$ integer coefficient operations after the table is known. Triangle/product bounds, even with the coarse affine-expansion factor 3^{4K} , keep coefficient intermediates at $O(K \log(K+1))$ bits. Index handling is separate; no unit-cost bit-complexity or polynomial-in- $\log T$ assertion follows.

352.3 One-witness heights and expanded storage

Set

$$\begin{aligned} P_K &= (K!)^2, & M_I &= 2K^2 + 4, \\ M_E &= K^2 + (1 + P_K)^2, & M_C &= (1 + P_K^2)^2, \\ C_S &= M_I^{n_I} M_E^{n_E} M_C^{n_C}. \end{aligned} \quad (352.6)$$

All roots of p_K are positive with even multiplicity, so its coefficients alternate in sign and $\|p_K\|_1 = \prod_{h=1}^T (1+h)^2 = P_K$. Equation (350.2) gives norms at most M_I, M_E, M_C for the three cell types: for instance $\|(A-i)^2\|_1 = (i+1)^2 \leq K^2$ and $\|(w - p_K(B))^2\|_1 \leq (1 + P_K)^2$. Therefore

$$\|Q_S\|_1 \leq C_S, \quad \lceil \log_2(C_S + 1) \rceil = O(K^2 \log(K+1)). \quad (352.7)$$

The last bound uses the actual cell counts: $\log C_S \leq T^2 \log M_I + 2T \log M_E + \log M_C$. Applying the corner estimate to all cells would lose this sharper bound. Nonnegative polynomial *values* do not imply nonnegative coefficients.

There are only three actual variables, and total degree D_S . Thus the support is at most $\binom{D_S+3}{3} = O(K^6)$, and a fully expanded sparse record list uses at most

$$\binom{D_S+3}{3} [\lceil \log_2(C_S + 1) \rceil + 1 + 3 \lceil \log_2(D_S + 1) \rceil] \quad (352.8)$$

bits apart from its header, with a constant-size convention for $D_S = 0$. This is $O(K^8 \log(K+1))$. For Q_S^2 , replace D_S, C_S by $2D_S, C_S^2$; the same asymptotic support and storage bounds hold. Horizon zero is constant size.

352.4 One-witness gates and expansion work

Form $A - h, B - h$ and their squares for $1 \leq h \leq T$, then multiply the T squares in each coordinate. This costs $2T$ subtractions and $4T - 2$ multiplications. Preparing the four witness squares in (350.2) costs four subtractions and five multiplications. Interior factors add twice, edge factors once, and the corner is already prepared. For $N = |S| \geq 1$,

$$\begin{aligned} \text{multiplications} &\leq 4T + 3 + \max(N - 1, 0), \\ \text{additions/subtractions} &\leq 2T + 4 + 2n_I + n_E. \end{aligned} \quad (352.9)$$

The combined ceiling is $3T^2 + 10T + 7$. If S is empty, return the constant one with no arithmetic. Squaring Q_S adds one multiplication. The table, constants and gate indices admit an $O(K^2 \log(K + 1))$ -bit factored description. Input-dependent values such as $p_K(A)$ require full precision, so these gates are not a bit-time claim.

For an explicit conservative expansion bound, interior factors have at most seven monomials; edge factors at most $6T + 5$; the corner at most $1 + (2T + 1)^2 + (4T + 1)^2$. This follows by expanding $w^2 - 2wp + p^2$ and the corner analogue. Their support bounds sum over S to $O(K^2)$. Build p_K in $O(K^2)$ coefficient operations. Each intermediate partial product has at most $\binom{D_S+3}{3} = O(K^6)$ monomials, so successive naive monomial convolutions and collection use $O(K^8)$ coefficient operations, with $O(K^2 \log(K + 1))$ -bit coefficient intermediates. A further naive self-convolution for Q_S^2 uses $O(K^{12})$ operations. These are deliberately conservative bounds, with indexing/bookkeeping separate.

Resource, $T \geq 1$	Two-witness P_S	One-witness Q_S	Squared Q_S^2
Native witnesses	2	1	1
Exact degree	$\max(2K, 4, 2 \deg F_S)$	D_S	$2D_S$
Presentation	5 squares	product of SOS factors	1 square
Distributed SOS slots	5	$3^{n_I} 2^{n_E}$	1
Coefficient bits	$O(K \log(K + 1))$	$O(K^2 \log(K + 1))$	same order
Support ceiling	$O(K^4)$	$O(K^6)$	$O(K^6)$
Expanded bits	$O(K^5 \log(K + 1))$	$O(K^8 \log(K + 1))$	same order
Factored gates	$\leq 4K^2 + 2K + 14$	$\leq 3T^2 + 10T + 7$	one extra multiply
Factored bits	$O(K^2 \log(K + 1))$	$O(K^2 \log(K + 1))$	same order
Expansion operations	$O(K^5)$	$O(K^8)$	$O(K^{12})$
Table preprocessing	$\leq K^2 T$ transitions	$\leq K^2 T$ transitions	same table

Table 50: All-variable native resource upper bounds, except where marked exact. $D_S = 2n_I + 4Tn_E + 8Tn_C$ and $K = T + 1$. Gate and expansion counts use ordinary integer arithmetic, not bit complexity. Both one-witness forms have unique positive witnesses; a shorter SOS presentation may cost higher degree.

352.5 Comparison with the accepted three-witness construction

The earlier class-code compiler uses three positive witnesses and six square slots. Its retained upper bounds are degree $4K^2 - 4$, coefficient magnitude bits $O(K^2 \log(K + 1))$, support at most $16K^2 - 5$, and expanded storage $O(K^4 \log(K + 1))$, for $T \geq 1$. They remain correct upper bounds; Report 66 [source 36] did not claim equality in every instance. The separately accepted parity refinement in Report 67 [source 37, Theorem 341.1] gives exact degree $4K^2 - 4$ for even K and $4K^2 - 8$ for odd $K \geq 3$, independently of the acceptance table. Its special horizon-zero degree is two. The full exact-degree proof and accepted audit are retained as comparison sources.

The tensor construction reduces arity and the degree ceiling, and improves the coefficient-bit ceiling, but loses the predecessor's narrow univariate class-code support. Its expanded support/storage upper bounds are worse. The one-witness product saves another variable while increasing the degree and expanded-storage ceilings further. None of these upper-bound comparisons proves that every instance of one construction is larger than every instance of another. Reports 66–68 [sources 36 and 37, and the undelivered Report 68] and their original counts are unchanged.

353 Paid composition with the accepted POWER module

Only this section imports the accepted constructive number-theoretic dependency. The native finite-table results above do not use it. We repeat all base-two residuals and the domain recovery to specify the composed polynomial without an unexpanded POWER predicate.

353.1 Twelve positive leaves, including the output

For positive index C and output o , the module has six directly positive leaves

$$o, g, q_b, q_v, J, q_\alpha,$$

and six distinct positive adapter leaves for the natural aliases

$$d_{wb}, d_{wC}, d_{yC}, q_\sigma, q_\tau, q_r.$$

Each alias z means $z^+ - 1$, where $z^+ \in \mathbb{Z}_{>0}$ is its own actual leaf. Thus the count is twelve including o , or eleven auxiliary leaves beyond a supplied output. The index C is counted separately. In particular q_α is a directly positive integer, not a natural alias. Define only expressions

$$\begin{aligned} \omega &= 2 + d_{wb}, & y &= C + d_{yC}, & \beta &= 1 + 4yq_b, \\ v_p &= y^2 q_v, & t &= C + 4yq_\tau, & M_p &= 2o + J, \\ Z &= 2o + J + 5, & X &= y(Z - 8) + 8o + 4M_p q_r, \\ U &= 4\beta - Z, & V &= q_\alpha X + Uq_\sigma. \end{aligned} \tag{353.1}$$

The complete residuals are

$$\begin{aligned} H_1 &= X^2 - 16 - (Z^2 - 16)y^2, \\ H_2 &= U^2 - 16q_\alpha^2 - (Z^2 - 16)q_\alpha^2 v_p^2, \\ H_3 &= V^2 - 16q_\alpha^2 - 16q_\alpha^2(\beta^2 - 1)t^2, \\ H_4 &= \omega - C - d_{wC}, \\ H_5 &= Z^2 - 16 - 16((\omega + 1)^2 - 1)(\omega g)^2. \end{aligned} \tag{353.2}$$

Every adapter and expression is substituted before treating the sum as a polynomial. No division appears in this formula.

Remark 353.1 ([write] restatement of source 37's Theorem 332.1 for twelve leaves at base two: imported POWER12 theorem). Source 38 states, as its Proposition 6.1: "For positive integers C, o , the five equations (353.2) have eleven positive auxiliary leaves beyond o if and only if $o = 2^{C-1}$. The sum of their squares has exact joint degree twenty. Every accepted (C, o) has infinitely many full module tuples, including when $C = 1$." This is source 37's Theorem 332.1 for $\ell = 12$ at $b = 2$, with its exact degree (339.7) and its infinite fibres (Section 339); the displays (353.1) and (353.2) are source 37's base-two display after (339.2), with source 37's w, v, M, A_*, S, u written $\omega, v_p, M_p, Z, V, u_p$ and $d = 4$.

The accepted proof and its independent audit are retained in the source packet. Their all-exponent dependency is the constructive theorem pair `Pell.matiyasevic` and `Pell.eq_pow_of_pell` in `mathlib4`, at the exact source revision identified in Section 354.3. We import that theorem pair, not a claim of a new Lean execution. The following argument records the local recovery and explains precisely how this theorem is used.

353.2 Recovered integrality, signs and the constructive boundary

[write] [Source 38's argument here is source 37's proof (Sections 334, 335 and 339) at $b = 2$ in the letters just listed, and is not printed twice. Its steps: $\alpha = Z/4$ and $u_p = U/(4q_\alpha)$ have integer squares by $H_5 = 0$ and $H_2 = 0$, so they are integers (source 37's Lemma 337.1); $\alpha > \omega \geq 2$; the sign $u_p > 0$ follows from $\beta = \alpha + q_\alpha u_p \geq 5$ and the *integer* bound $q_\alpha \geq 1$, a step that “would not be valid for arbitrary positive rational leaves”; with $x = y(\alpha - 2) + 2o + M_p q_r$ and $s_p = x + u_p q_\sigma$ one has $X = 4x$ and $V = 4q_\alpha s_p$, and all fifteen equations of the older module are recovered,

$$\begin{aligned}
x^2 &= 1 + (\alpha^2 - 1)y^2, & u_p^2 &= 1 + (\alpha^2 - 1)v_p^2, \\
s_p^2 &= 1 + (\beta^2 - 1)t^2, & \beta &= 1 + 4yq_b, \\
\beta &= \alpha + u_p q_\alpha, & v_p &= y^2 q_v, \\
s_p &= x + u_p q_\sigma, & t &= C + 4yq_r, \\
y &= C + d_{yC}, & \omega &= 2 + d_{wb}, \\
\omega &= C + d_{wC}, & M_p &= 2o + J, \\
\alpha^2 &= 1 + ((\omega + 1)^2 - 1)(\omega g)^2, \\
4\alpha &= M_p + 5, & x &= y(\alpha - 2) + 2o + M_p q_r,
\end{aligned} \tag{353.3}$$

so that the positive-index branch of `Pell.matiyasevic` and `Pell.eq_pow_of_pell` give $2^C = 2o$. For completeness the constructive theorems supply such a tuple, the residue lemma (source 37's Lemma 333.1; for the last congruence $x^2 - [y(\alpha - 2)]^2 = (4\alpha - 5)y^2 + 1 > 0$) makes its quotients nonnegative, and the Pell recurrence

$$\begin{aligned}
X_0(z) &= 1, & Y_0(z) &= 0, \\
X_{n+1}(z) &= zX_n(z) + (z^2 - 1)Y_n(z), & Y_{n+1}(z) &= X_n(z) + zY_n(z)
\end{aligned} \tag{353.4}$$

(source 37's (334.1)) gives, with the old $\alpha, \omega, M_p, g, x, y, u_p, v_p$ held fixed and every integer $k \geq 1$, the progression

$$\begin{aligned}
\beta_k &= \beta_0 + 4yu_p k, & q_{b,k} &= q_{b,0} + u_p k, & q_{\alpha,k} &= q_{\alpha,0} + 4yk, \\
s_k &= X_C(\beta_k), & t_k &= Y_C(\beta_k)
\end{aligned} \tag{353.5}$$

(source 37's (335.1)), which preserves every equation, makes q_α strictly positive and, through the strictly increasing $q_{b,k}$, gives infinitely many full module tuples; this is existential equivalence, not a bijection with the old tuples having $q_\alpha = 0$. Finally the residual degrees are 4, 10, 10, 1, 6, and the term $-J^2 q_\alpha^2 (d_{yC}^+)^4 q_v^2$ of H_2 squares to the coefficient-one monomial

$$J^4 q_\alpha^4 (d_{yC}^+)^8 q_v^4 \tag{353.6}$$

of degree twenty (source 37's (339.7)), so the module has exact degree twenty.]

353.3 Three external gaps and both compiler substitutions

Let g_1, g_2, g_3 be arbitrary positive integer inputs and let $D = g_1 + g_2 + g_3$ be an expression. Introduce positive witnesses A, B and two independent twelve-leaf modules with $(C, o) = (A, p)$ and (B, q) . All their leaves are mutually distinct and distinct from the compiler witnesses. Set

$$G_1 = (20g_1 - D)p - 2D, \quad G_3 = (20g_3 - D)q - 2D, \tag{353.7}$$

and define the common twelve-square decoding block

$$E = G_1^2 + G_3^2 + \sum_{i=1}^5 (H_i^A)^2 + \sum_{i=1}^5 (H_i^B)^2. \quad (353.8)$$

Its paid witness count is $2 + 2 \cdot 12 = 26$, including p, q once each. The three final choices are

$$\mathcal{F}_2 = E + P_S(A, B; r, s), \quad \mathcal{F}_1 = E + Q_S(A, B, w), \quad \mathcal{F}_{1,\square} = E + Q_S(A, B, w)^2. \quad (353.9)$$

Theorem 353.2 (Complete composed equivalence and ledger). *For each choice in (353.9), a positive witness exists exactly when the input is encoded by natural counters a, b satisfying*

$$\frac{g_1}{D} = \frac{1}{20} + \frac{1}{10} 2^{-a}, \quad \frac{g_3}{D} = \frac{1}{20} + \frac{1}{10} 2^{-b}, \quad (353.10)$$

and $(a + 1, b + 1) \in U_S$. When S is a program's table, this is the selected by- T or first-exactly- T predicate. The literal ledgers and exact degrees are

	$\mathcal{F}_2$	$\mathcal{F}_1$	$\mathcal{F}_{1,\square}$
External positive inputs	3	3	3
Positive witnesses	28	27	27
All variables	31	30	30
Final equations	1	1	1
Residual-square slots	17	12	13
Additional block	none	Q_S	none
Exact degree	$\max(20, \deg P_S)$	$\max(20, D_S)$	$\max(20, 2D_S)$

The degree entries involving D_S are for $T \geq 1$; every composition has exact degree twenty at $T = 0$. For $T \geq 1$, the fully distributed $\mathcal{F}_1$ has $12 + 3^{n_1} 2^{n_E}$ square slots. At $T = 0$, its native block is itself a square, so it has thirteen slots there too.

Proof. Every summand in (353.9) is globally nonnegative. A zero therefore forces every decoding residual and the native block to vanish. Remark 353.1 gives $p = 2^{A-1}$, $q = 2^{B-1}$. Equations (353.7), positive outputs and positive D force $20g_1 - D > 0$, $20g_3 - D > 0$ and, by division by positive quantities, exactly (353.10) with $a = A - 1$, $b = B - 1$. The native theorem gives acceptance. Conversely any encoded accepted triple supplies A, B, p, q , the constructive module witnesses and the unique native witnesses, so every required term vanishes. No promise that an arbitrary triple is encoded has been assumed; unencoded triples have no witnesses.

The counts add the native leaves and slots to the paid 26-witness, twelve-square block. Outputs are already in the module counts. All slots and declared variables remain counted in degenerate cases.

The block E has exact degree twenty, as the independent module monomial (353.6) is absent from the other module, the gap residuals and the native variables. The gap squares have degree four. Finally, a nonzero globally nonnegative polynomial has a nonnegative highest homogeneous part: scale the argument by a positive parameter and pass to the leading limit. Such highest parts cannot cancel in a sum of nonnegative polynomials. Apply this to E and each native block to obtain all stated degree equalities, including the product-block case. $\square$

In particular the uniform degree ceilings at $T \geq 1$ are

$$\max(20, 4T), \quad \max(20, 10T^2 + 8T), \quad \max(20, 20T^2 + 16T),$$

respectively. Calling $\mathcal{F}_1$ “thirteen squares” would incorrectly count a whole product-of-SOS block as one square. The thirteen-square presentation is $\mathcal{F}_{1,\square}$ and pays the doubled native degree contribution.

[write] [The decoding block E is source 37’s composition with $\ell = 12$ (26 positive witnesses and twelve squares, Table 49); source 37 adds the three witnesses and six squares of source 36’s compiler (29 witnesses, 18 squares), source 38 the two witnesses and five squares of its tensor compiler (28 and 17) or the one witness of its product (27). The successive ledgers 57, 29, 28 and 27 all stay valid for their own formulas.]

353.4 Unique projections, infinite full fibers, and fixed decoding costs

If a zero exists, the gaps determine

$$p = \frac{2D}{20g_1 - D}, \quad q = \frac{2D}{20g_3 - D}.$$

Injectivity of positive powers of two fixes A, B , and the native theorem fixes (r, s) or w . Thus (A, B, p, q, r, s) for $\mathcal{F}_2$, or (A, B, p, q, w) for either one-witness form, is a unique projection. Nevertheless the full 28- or 27-witness fiber is infinite whenever nonempty: vary (353.5) in one module while holding its index/output, the other module, the inputs and native witnesses fixed. These composed formulas are neither single-fold nor finite-fold.

Each decoding block has a fixed finite coefficient/support list independent of T , with a fixed factored circuit. Adding E increases support by only a T -independent constant and coefficient norm by a fixed additive amount. Therefore the native asymptotic expanded-storage, support and factored-description bounds of Section 352 persist under these compositions. The 30 or 31 total variables do not require a generic dense bound in that many variables. This is a size accounting statement, not a claim of bounded POWER witness height or constant bit-time evaluation.

354 Context, evidence, and the exact boundary

354.1 Elementary ingredients and literature context

The scaled basis in (349.1) is ordinary Lagrange interpolation with a common factorial denominator cleared. NIST’s *Digital Library of Mathematical Functions*, Section 3.3(i), records the standard nodal formula and basis identities [108]. Tensoring those bases prescribes a Boolean-valued table on a rectangular grid. The underlying finite-grid vanishing principle is classical; Alon’s primary paper [110, Lemma 2.1] gives its familiar degree-bounded finite-set form and an induction proof. The infinite-grid implication used here follows from the same one-variable root argument, proved explicitly in Section 351.

Conjunction by a sum of real squares and disjunction by a product of nonnegative factors are elementary algebraic encodings. In the one-witness construction, uniqueness depends additionally on the disjointness of the clipped cells and the positive-integer domain. These contextual observations are not an attribution of the specific formulas or ledgers to the cited works. No exhaustive literature survey, novelty or priority conclusion is offered.

354.2 What is and is not represented

The native theorems quantify over all positive inputs and positive integer witnesses, not a finite sample box. Their proofs establish that domain-wide result. For bounded counter programs,

clipping transports both by-horizon and exact-first-halt semantics, including loops and initially halted programs. The table itself remains fixed data obtained by finite preprocessing.

In particular T is an index of the family. Writing “there exists a horizon T ” outside the family does not encode its varying acceptance table, factorials or products inside one fixed polynomial. This report supplies no unbounded-halting fixed polynomial or single-fold MRDP result. It gives no global witness minimum, minimum degree, minimum coefficient height, universal-polynomial record, efficient POWER witness-height bound, or new physical simulation theorem. The exact native zero-versus-one classification is the narrower theorem proved in Section 351; it does not classify the minimum arity of the gap compositions.

Earlier Reports 66–68 [sources 36 and 37, and the undelivered Report 68] retain their own constructions, resource bounds and scopes. The new formulas are replacements available for comparison, not retroactive recounts or corrections to those reports. No claim about arbitrary unencoded physical trajectories, collision horizons or physical resource costs is made here.

354.3 Pinned arithmetic sources and retained evidence

The full frozen source packet, including PROOF.md, ONE_WITNESS.md, dependencies and recorded exact-algebra evidence, is retained inertly with this article. Its source manifest SHA-256 is

```
b243e39e0fd9698f9d610f0f6d2b17485a98
df8e20a9412c1c38c65877a42c82.
```

The original static review accepts the two-witness proof and composition but explicitly excludes the later one-witness appendix and its executed evidence. A separate full independent audit is included as the controlling review of the complete low-arity packet; its exact pins and completed evidence are recorded in the release README and acceptance record. These are distinct review scopes. [write] [The source packet is shipped under the prefix 38-low-arity-la- (proof 38-low-arity-la-PROOF.md, one-witness note 38-low-arity-la-ONE_WITNESS.md, static review 38-low-arity-la-REVIEW_TWO_WITNESS.md, program code/38-low-arity-la-static_algebra.py) and the full audit under 38-low-arity-audit-la-. The source manifest whose SHA-256 is printed above, science/low-arity/MANIFEST.sha256, is a pure path-to-hash list and is not shipped (it was verified at placement); the audit’s checker hash-pins it, so a rerun of the audit restores the packet from the arrival archive (README). The archive’s exact-degree comparison packet and audit are byte copies of source 37’s exact-degree packet and audit, shipped once under 37-pos-power-ed- and 37-pos-power-audit-ed-. Both reviews were written by the same pipeline as the article; this report does not treat them as external.]

For the imported POWER theorem, the retained source is mathlib4 revision

```
ac77769fabe23cb237559e7f56578dbead91499f,
```

file Mathlib/NumberTheory/PellMatiyasevic.lean. The declarations are Pell.matiyasevic and Pell.eq_pow_of_pell [86]. Its exact inert-file SHA-256 is

```
993760c797ad0ff66fa77064a616fce7795
50bc745cbf6c44e04f392fd0bed0a.
```

The source, original copyright notice and Apache 2.0 license remain included. Its constructive theorem boundary is imported and textually checked; no new proof-assistant run is claimed.

The frozen author’s recorded results include 285 interpolation-node identities, 42 two-witness expansions, 16,200 positive witness-box assignments and 18,432 canonical assignments over all $K = 3$ tables. The separate one-witness evidence includes 24 expansions, 4,348 assignments, all 530 tables through $K = 3$ tested against the closure criterion, and 30 fixed-dimensional degree ledgers. Recorded complete POWER/gap expansions and fixtures corroborate the composed ledgers. These are prior stored results, not report-time executions, and do not replace the symbolic proofs, arbitrary-program table correctness or the imported all-exponent theorem.

354.4 Preservation and reproducible presentation

The source’s preservation record documents a genuine historical interruption: during its original baseline interval, a separate authorized Report68 [undelivered here] release was still being sealed. Seventeen entries changed there, including README bytes, read-only modes and directory times. The original baseline, interrupted log and exact difference are retained; this interval is not described as static. Reports 66/67 and frozen arithmetic dependencies matched that baseline. A separate post-seal Report68 inventory then verified the final 299-file, 31-directory manifest and preserved all 332 later inventory entries. These are different time intervals and different claims.

For Report69 [source 38], the source packet and accepted audit are copied with bytes, modes and nanosecond modification times preserved. All scientific programs, stored expansions, theorem-prover source and prior evidence are inert. Report-owned presentation tools only authenticate, flatten LaTeX, typeset, render and package. They never execute a source scientific checker, author/upstream code, counter interpreter, physical simulator, saved schedule or Lean.

A fresh isolated build uses a new local TeX format and font map, an allowlisted environment, disabled shell escape, three passes, and the union of the format and every pass’s recorder inputs. An externally pinned dependency lock and exact PDF comparison distinguish a locked replay from an authoring bootstrap. Every rendered page receives visual inspection in addition to structural PNG checks. Synthetic release-tool tests and accepted independent mathematical, manuscript and release-tool reviews accompany the release. A fresh post-seal terminal verification is recorded separately after the archive is sealed. All of these checks are distinct from mathematical proof. Reproducible typesetting authenticates the delivered presentation, not the truth of a theorem.

[write] [The delivery README, article source and PDF and the manuscript modules are not shipped; this Part prints the article and this report’s README replaces the delivery README. The release tools are shipped as `code/38-low-arity-tools-*.py`; they expect the delivered layout and refuse Windows paths. The predecessor tools and README of the undelivered Report 68 that the archive carries are shipped as `code/38-low-arity-qa-pred-tools-*.py` and `38-low-arity-qa-pred-tools-README68.md`, as records only. At placement the science checker and the independent audit reproduced their frozen expansions on copies (identical decompressed content): the checker once its historical `/workspace` preservation step was stubbed in a scratch copy, the audit with a one-line path-separator patch (README).]

[write] **Review by the Hilbert’s-tenth programme.** The programme’s scoped intake [111] read source 38’s delivery README completely and, of its article, Sections 348–351, Section 352 up to its comparison subsection, Section 353 from its gap compositions on, and the first three subsections of Section 354; it did not read the module’s recovery argument, and it ran none of the archived programs. Its findings: the tensor construction has exactly two positive witnesses and five residual slots, and the range and slack arguments force the unique clipped cell with no encoded-input promise; the one-witness product proves a unique positive witness in each accepted cell, but “is not one residual square: replacing it by its square doubles degree”; the zero-versus-one

classification is for fixed finite tables with unrestricted degree; the paid arithmetic bounds do not make the growing coefficient table free; and the gap compositions with 28 positive witnesses and 17 squares, or 27 and twelve squares plus the product block, “are finite-horizon predicates”. It reports no defect and names source 38’s finite-table formulas, for deliberately bounded predicates, among the most useful next uses of the batch.

355 Further questions and research (Part XIII)

[write] This section collects the claims bearing on Part XIII that are *not* proved in this report, so that none is dropped silently. Each is stated as an open question with its source credited, the argument its source sketches, and what is missing. The questions the three sources ask themselves are printed in their own sections (Section 331, the last paragraphs of Section 347 and Section 354); the notes there record which of them later sources answer. No claim of the three sources or of the programme’s review was found to be wrong. The write rechecked in exact arithmetic the printed counting formulas of source 36 (Theorem 325.1 for all $X < 3000$ and at $X = 10^6 + 7$, the shell formula (326.1) for $X < 3000$, the primitive scales (324.2) for $a, b < 40$, the slope (327.1) and the four-class constants (328.2) for $M < 40$), the degree and leading coefficient of source 37’s Theorem 341.1 for $2 \leq K \leq 7$, and the ledgers of Table 49 and of Theorem 353.2; all agree. These finite checks support, and do not replace, the printed proofs.

1. Gaussian fluctuations of the encoded-gap deficit (the undelivered Report 68).

Claim. For X drawn uniformly from $\{1, \dots, N\}$, the deficit $E(X) = \rho X - \mathcal{A}(X)$ of source 36’s Theorem 318.3 has mean $\frac{1}{2}(\log_2 N)^2$ and variance $\frac{1637}{5400}(\log_2 N)^3$ to leading order and, normalized, a Gaussian limit law; the inverse deviation $D_n - n/\rho$ of Theorem 329.1, for n uniform up to H , has leading mean $\frac{1125}{1486}(\log_2 H)^2$, variance $\frac{3069375}{4416392}(\log_2 H)^3$ and a Gaussian law. *Credit.* Report 68, *Gaussian fluctuations of encoded gap counts and inverse ranks*, 4 October 2026, by the same pipeline; known here only from its delivery README, carried by source 38 and shipped as 38-low-arity-qa-pred-tools-README68.md. *Sketch, as that README describes it.* The digit form (325.2)–(325.3) reduces E to digit sums in bases 2 and 16; the factor 5 in $X/10$ versus $X/16$ is handled by a five-state multiplication-by-five carry chain, which gives an exact block mean, a rational conditional variance and a martingale decomposition with an $O(m^2)$ variance remainder on complete dyadic blocks; a martingale central limit theorem (cited, with a direct characteristic-function proof for bounded increments) gives the Gaussian law, and prefix mixtures and residue-class moment estimates pass to every cutoff, with variance remainder $O((\log X)^{5/2})$. The inverse statement follows by a common-uniform coupling with uniform short-window bounds, with an L^2 comparison error $O((\log \log(H + 3))^2)$. *A consistency check by the write.* Since $D_n - n/\rho$ is $E(D_n)/\rho$ up to the bounded terms of (329.4), and $\log_2 D_n = \log_2 n + O(1)$, the inverse constants are the forward ones scaled by $1/\rho = 1125/743$ and $1/\rho^2$: indeed $\frac{1}{2} \cdot \frac{1125}{743} = \frac{1125}{1486}$ and $\frac{1637}{5400} \cdot (\frac{1125}{743})^2 = \frac{3069375}{4416392}$ exactly; and the mean $\frac{1}{2}(\log_2 N)^2$ lies, as it must, between the envelopes $\liminf E/(\log_2 X)^2 = 0$ and $\limsup = 1$ of (327.7). *What is missing.* Report 68’s article, proofs and audits are not in this batch or in the repository; the constant 1637/5400, the carry-chain variance and the limit laws are therefore unproved here. Source 36’s question on finer error statistics (Section 331, item 5) stays open until they are supplied.

2. Physical transport. *Claim.* The certificates of the Part also certify bounded instruction-section reachability of a five-live-signal machine on encoded inputs, with elapsed time strictly between kD and $10kD$ and at most $32k$ binary contacts for first halt at $k > 0$ (Sections 323

and 345). *Credit and sketch.* Sources 36 and 37, by composition with source 34’s compiler theorem (Theorem 294.1, Part XII). *What is missing.* Nothing within this report once Part XII’s theorem is granted; but none of the three sources re-audits it, and the Hilbert’s-tenth programme’s review of the signal archives read only selected spans of source 34’s proof [99]. The physical reading stays conditional, as the sources say.

3. **The all-exponent POWER equivalences.** *Claim.* Every module of the Part represents $o = b^{C-1}$ exactly. *Credit and sketch.* Sources 35–38, through the constructive theorems `Pell.matiyasevic` and `Pell.eq_pow_of_pell` of `mathlib4` at `ac77769f`. *What is missing.* The sources read those statements as text and run no Lean; the specialization to the displayed positive-integer residual systems (adapters, natural subtraction, index C versus exponent $C-1$) is proved on paper only. A machine-checked specialization, for instance in the repository’s Lean development, would close this.

A Part I: a minimal worked use of the compiler

The following code can be run from the archive root. ([write] “The archive root” is the delivered layout, with `code/signal_certificates.py`; in this report the module is shipped as `code/07-collision-geometry-signal_certificates.py`, so the snippet runs in a copy that restores the delivered name (README of this report).) It builds the first-collision tournament and constructs a complete certificate without using an external algebra system.

```
import sys
sys.path.insert(0, "code")
from signal_certificates import Machine, compile_skeleton

machine = Machine(
    {"A": 2, "B": 0, "C": -1, "H": 0, "R": 0},
    {
        frozenset(("A", "B")): ("H",),
        frozenset(("B", "C")): ("R",),
        frozenset(("A", "B", "C")): ("R",),
    },
)
skeleton = (((0, 1),),)
cert = compile_skeleton(machine, ("A", "B", "C"), skeleton)
gaps = (1, 1)
slacks = cert.witnesses(gaps)
assert cert.evaluate(gaps, slacks) == 0
assert not cert.accepts((3, 1)) # An earlier BC collision.
assert not cert.accepts((2, 1)) # A simultaneous ABC collision.
print(cert.to_dict())
```

The order of rows in the exported matrices is lexicographic and need not match the order used in the displayed mathematical formula. Row order changes only the order of slack coordinates, not the represented relation or the uniqueness statement.

B Part I: dependency map and claim status

Claim	Principal ingredients	Status in this package
Exact chronology chamber	Intercept induction; affine endpoint lemma; static identity and order validation	Full manuscript proof; exact tests
Event-sparse row bound	Local adjacency replacement count	Full manuscript proof; every exported ledger checked
Unique-slack quadratic	Integer strict inequalities; sum-of-squares zeros	Full manuscript proof; zero and mutation tests
Canonical first-hit quartic	Disjoint first-hit chambers; natural one-hot selector equation	Full manuscript proof; finite union tests
Causal denominator and coefficient bounds	Integer speed differences; induction on causal depth	Full manuscript proof; finite denominator checks
Small integral realization	Homogeneity; rational vertex; Cramer’s rule; Hadamard bound	Full manuscript proof
Robustness and codimension	Linear margin estimate; rank of equalities	Full manuscript proof
Finite-horizon semilinearity	Exact chamber theorem; classical Presburger and semilinear-set equivalence	Derived theorem using cited literature
Exact quasi-polynomial counts	Chamber homogeneity; face inclusion–exclusion; classical Ehrhart theorem	Derived theorem; worked counts checked
Universal signal-machine existence	Durand-Lose’s published construction	External result; rule table not reimplemented
Fixed-arity unbounded Diophantine existence	Computable finite-prefix simulation; classical MRDP	External existence theorem, not new compiler
Global convex-quadratic barrier	Positive semidefinite quadratic completion; Presburger decidability	Full manuscript proof

[write] The row “Global convex-quadratic barrier” is a special case of Part XI of [11] (Remark 13.3); source 07’s proof is kept as a second route. The rows “Exact chronology chamber”, “Event-sparse row bound”, “Unique-slack quadratic”, “Causal denominator and coefficient bounds” and “Finite-horizon semilinearity” have second proofs in Part II (Section 1.2).

C Part II: replay and build guide

The accompanying archive contains this article in editable LaTeX, the compiled PDF, exact Python scripts, expected receipts, example coefficient dictionaries, and a manifest with file hashes. It contains no third-party source PDFs. The scripts require Python 3.10 or later and only the standard library. Assertions must remain enabled; do not run Python with `-O`.

From the extracted package root run:

```
python3 scripts/run_replay.py
```

The runner copies the executable scripts into a temporary directory, executes the suites there, compares fresh JSON receipts to the stored expectations, and writes a summary under `replay-output`. The stored expected receipts and packaged examples are not overwritten. Use `-output DIR` to select a different result directory. The worked-example receipt also contains its concrete witness and explicit rejection reason for the false final-cut schema.

[write] These are the delivered names. In this report the five scripts are shipped as `code/11-signal-certificates-*.py`, `build.sh` as `code/11-signal-certificates-build.sh`, the two example files and the manifest as `data/11-signal-certificates-*.json`, and the four expected receipts as `data/11-signal-certificates-expected-*_receipt.json`. The runner looks for `scripts/`, `examples/` and `expected_receipts/` in the parent of its own directory and runs the scripts by their delivered names, so it runs only in a copy that restores the delivered layout (see the README); it writes nothing into the shipped files. Of the archive contents listed above, the editable source is printed as this Part rather than shipped separately, and the compiled PDF and the checksum file `CHECKSUMS.sha256` are not shipped (the checksum file was verified, 17 of 17 files, at placement); the manifest is shipped, and its payload list still names the delivered paths and the delivered PDF and article source.

The main executable interfaces are:

- `signal_geometry.py`: exact simulator, full-layer compiler, polynomial evaluation and rational Gaussian rank calculation;
- `signal_sparse.py`: event-coordinate and adjacency-lifetime compiler;
- `verify_additional.py`: separately seeded edge cases and sharp height checks;
- `verify_examples.py`: the article’s explicit examples and convention checks.

The low-level compilers use assertions to reject malformed schema partitions or structurally impossible collision directions. They are reference mathematical implementations, not hardened parsers for untrusted inputs. JSON residual dictionaries use `@` for the constant term, `x0`, `x1`, and so on for input positions, and the listed variable names for natural witnesses. A certificate evaluates to the sum of the squares of those rows. No expanded polynomial is required to replay it.

To rebuild the article run:

```
sh build.sh
```

A working TeX Live installation with `pdflatex`, Latin Modern fonts, AMS packages, `geometry`, `microtype`, `booktabs`, `enumitem` and `hyperref` suffices. The build script uses fixed document metadata and a fixed source-date epoch; exact PDF byte equality across different TeX distributions is not promised. The mathematical replay receipts are exact and version-independent within supported Python semantics.

D Part II: claim and source guide

The source-supported context is limited to the following claims. All certificate theorems and proofs in this article are supplied here rather than attributed to the literature.

- Durand-Lose 2009: finite configurations in Definition 1; integer inputs in Section 2; speeds in Figure 4; cyclic-tag simulation and halting in Section 5 and Theorem 5. Common-denominator rescaling is our inference, proved in (19.1).

- Durand-Lose 2007: linear-register event simulation in Section 4 of the author version, especially its ninth PDF page; rational models in Section 5. No Diophantine theorem is attributed to it.
- Durand-Lose 2006: energy and live-population bounds in Definition 3 and the following paragraph; accumulation qualifications in Sections 2.2 and 6.
- Ginsburg and Spanier 1966: equivalence of Presburger-definable and semilinear subsets of $\mathbb{N}^n$, Theorem 1.3, pp. 288–289.

Primary sources were checked on 2 October 2026. The bibliography links to the papers or their official bibliographic records; copyright-bearing PDFs are not redistributed in this package.

[write] In this report the Durand-Lose 2009 and Ginsburg–Spanier entries are merged with Part I’s entries for the same papers [3, 7]; the merged entries add source 11’s editors and links to Part I’s.

E Provenance of this report

[write] The report was built from two manuscripts that arrived together in the archives of commit 808b53ed8 (2 October 2026) and were placed by commit 798b0c5d4, which staged their programs, data and provenance files byte-identical to the delivery. (Parts III and IV were added later from two manuscripts of batch 79, and Parts V and VI from four manuscripts of batch 80; their provenance is at the end of this appendix. [write] Added 4 October 2026: so is that of the later additions, up to Parts IX–XIII of batch 91.)

Source	Archive (batch 78)	Pin	Contributed	Printed as
07 (base)	<code>Collision_Geometry_Diophantine_Signal_Machines.zip</code> , manuscript 07; main file <code>collision_geometry/article.tex</code> , 29-page PDF	<code>f1edb38f9</code>	the chamber theorem, the event-sparse bound, the unique-slack quadratic, denominators and heights, small realizations, the NP bound, robustness, semilinearity, the first-hit quartic, Ehrhart counts, the tournament, the clock, the barrier, nine research directions	Part I with Appendices A and B
11	<code>Signal_Machine_Diophantine_Certificates.zip</code> , manuscript 11; main file <code>signal-diophantine-release/article.tex</code> , 15-page PDF	none named	the same theorem with event-coordinate witnesses, halting semantics, full column rank, the size ledgers, the full-layer construction, the four-speed height bound, the bounded shuttle, positive witnesses, five questions	Part II with Appendices C and D

Source 07’s pin is the full commit `f1edb38f93aa0a362a34b531a600909f6e74608a`, named in its Section 14 and in its delivered `PROVENANCE.md` (shipped as `07-collision-geometry-PROVENANCE.md`); it is an ancestor of the placement commit, and the audit blob it cites is unchanged at the write. Source 11 names no repository snapshot and cites no repository file; its only repository-related statement, that the 75- and 87-operation constructions are not improved, is true at the write.

Where the merge had to choose.

- *Base and order.* Source 07 is the base (Part I): it allows signed rational speeds without a normalization interface, proves more about the chamber, and is the longer manuscript. Source 11 follows as Part II. Every result, proof, example, remark, research question, appendix and limitation of both sources is printed, in the sources' order and with their numbering inside each Part.
- *The common theorem.* Its two statements have different inputs and witnesses, so both are printed, each with its proof. The lemmas of Part II that coincide with lemmas of Part I (Lemmas 19.1, 20.1 and 21.1, Proposition 26.1) open with a [write] note naming their Part I counterpart, and source 11's proofs are kept as second routes. Section 1.2 compares the two routes, including the two shuttles and the two missed-simultaneity examples, which are kept in both Parts because they are different machines.
- *A result already in the collection.* Part I's Theorem 13.1 and Corollary 13.2 are a special case of Part XI of *canonical-diophantine-certificates*. They are printed with source 07's statements and proofs, as a second route, and credited in Remark 13.3; credit notes were added to source 07's abstract and conclusion. Remark 10.3 relates the first-hit quartic to the same Part.
- *Research questions.* Both agendas are printed in their Parts; Table 3 pairs the coinciding ones, and one [write] note re-scopes the part of Part II's Question 3 that both sources already prove.
- *Front matter and bibliography.* The two title pages are replaced by one; each source's title, author line, abstract and keywords open its Part. Section 1 is new. The two bibliographies are merged: source 11's keys DL09 and GS66 are cited as source 07's `durandlose` and `ginsburgspanier`, whose entries now carry source 11's editors and links; entries [11, 12, 13, 14, 15] were added for the cross-references.
- *Typesetting.* Equations are numbered by section in both Parts (source 07 numbered them consecutively). Source 11's macro `\N` is written `\MN` because source 07's `\N` prints $\mathbb{N}_0$; both print the natural numbers $\{0, 1, 2, \dots\}$ as their sources did. No symbol was renamed in the printed text and no normalization changed.
- *Added text.* Every sentence written in the merge is marked [write]: Section 1; the openings of the two Parts; notes in Part I in Sections 2, 5, 6, 7, 10 (and Remark 10.3), 11, 12, 13 (Remark 13.3), 14 (shipped file names, the pin), 15 and 16, and in its abstract; notes in Part II in Sections 17, 18, 19, 20, 21, 24, 25, 26 and 27; notes in the four source appendices; this appendix; and the bibliography entries marked [write]. Labels were added to the first lemma of Part I's Section 5, to one example subsection, the research directions, the conclusion and the second appendix of Part I, and to several sections, one corollary, the questions, the conclusion and the second appendix of Part II.

Files. The report directory holds this text (`article.tex`, `article.pdf`), a README, source 07's `PROVENANCE.md` and source 11's `REPRODUCIBILITY.md` (as `07-collision-geometry-PROVENANCE.md` and `11-signal-certificates-REPRODUCIBILITY.md`), and the two sources' programs in `code/` and data in `data/`, under the prefixes `07-collision-geometry-` and `11-signal-certificates-`. The README maps each delivered name to its shipped name and gives rerun instructions that do not overwrite the shipped files. Source 11's manuscript, both PDFs, source 11's delivered README and both checksum ledgers (source 07's `SHA256SUMS`, 14 of 14 files, and source 11's `CHECKSUMS.sha256`, 17 of 17, both verified at placement) are not shipped; they survive in the

archives of commit 808b53ed8. Source 07’s delivered README was the placed README.md and has been replaced by the report README.

Parts III and IV (batch 79). [write] Two manuscripts of batch 79 of ProveIt’s incoming reports were added on 2 October 2026. Both arrived in the archives of commit aebfa386e and were placed by commit a7ae02511, which staged their programs, data and provenance files byte-identical to the delivery (36 files of source 12, 35 of source 13) and retired the archives from the drop zone. A corrected code edition of source 13, `Sparse_Lattice_Diophantine_Certificates_corrected.zip`, arrived in batch 80 (commit 4e270aa46) and was placed by 8a4e64732 (batch 80, cluster K1), which replaced five of source 13’s files by their corrected bytes and added three (38 files of source 13 now); nothing printed changes (Section 47).

Source	Archive (batch 79)	Pin	Contributed	Printed as
12	<code>Conservative_Signal_Frontend_Corrected.zip</code> , manuscript 16; main file <code>paper/conservative-signal-diophantine.tex</code> with <code>paper/morita-table.tex</code> , 21-page PDF	none named	the literal universal conservative signal machine, the stack and loader arithmetic, the complete event normal form, the pivot-scaled integer lift, the one-coordinate mode control, the canonical quadratic step and horizon packet with its ledger, the finite mode closure	Part III with Appendices F–H
13	<code>Sparse_Lattice_Diophantine_Certificates.zip</code> , manuscript 19; main file <code>paper/sparse-lattice.tex</code> reading five further files, 27-page PDF; corrected code edition <code>Sparse_Lattice_Diophantine_Certificates_corrected.zip</code> (batch 80, manuscript 07)	none named	the sparse finite-horizon quartic certificate and its ledgers, the orthant-exact option, the corrected Morita–Imai interface, the false observer, the paid loader and first pulse, the doubling example, the diameter barriers, the mass-two decision and semilinearity theorems, the post-elimination quartic, five questions	Part IV

Neither manuscript names a ProveIt commit. Source 12’s delivered `CORRECTION.md` cites the programme’s review commit 9f033fa6e, an ancestor of the placement, as the origin of its patch. Source 12’s archive is the corrected edition of batch-79 manuscript 07 (`Conservative_Signal_Diophantine_Frontend.zip`, arrival 2a8a39599): 36 of its 38 files are byte-identical to manuscript 07’s, its article is the same, and its evaluator `code/quadratic_packet.py` is manuscript 07’s with the programme’s patch applied. Manuscript 07 is therefore superseded; nothing of it is shipped, and it survives in its arrival commit. The delivered checksum ledgers verified at placement (manuscript 07, 37 of 37 files; source 12, 44 of 44; source 13, 48 of 48) are not shipped.

Where the batch-79 addition had to choose.

- *Placement and order.* Both manuscripts answer questions of this report (Section 1.11), so they are additions to it rather than a new report; source 12, the signal machine, comes first as Part III, and source 13, its lattice counterpart, follows as Part IV. Every result, proof, example, remark, question, appendix and limitation of both is printed, in the sources’ order.

- *The two editions of source 12.* The corrected edition is printed; its article is identical to the earlier edition's. The correction concerns only the generic evaluator program and is described in Section 37.
- *Author lines.* Source 12's title page describes it as a "Research note prepared with AI assistance for private review"; as the report is public, Part III's opening describes it neutrally as an AI-assisted research note, and the delivered wording is recorded here. Source 13's author line, "Research report prepared with AI assistance for Vladimir", is quoted in Part IV's opening.
- *Re-derived results.* Part III's event lemma and uniform lift, Part IV's comparison and sorting gadgets, its post-elimination quartic and the finite-search step of its cutoff proposition re-derive results of Parts I–II or of *canonical-diophantine-certificates*. They are printed as delivered, with [write] notes naming the earlier result, as second routes; no novelty is claimed for them. The selector/copy construction of Part III is credited to Balas and to Jeroslow and Lowe.
- *Front matter.* The title, subtitle and abstract were widened; Section 1 gained Sections 1.7–1.11, and dated [write] notes were added where batch 79 made earlier sentences stale (Sections 1.1, 1.3, 1.4, 1.5 and 1.6, Table 3, Part I's direction 15.2 and Part II's Questions 1 and 2). Nothing was numbered anew: the notation of Parts III and IV is an uncaptioned table, so that the earlier tables keep their numbers, and Part III's three appendices follow this appendix, so that Appendices A–E keep their letters.
- *Labels.* Source 12's 49 labels are `smc:cs:` plus their delivered names, and source 13's 53 are `smc:s1:` plus theirs; none was dropped or renamed apart from the prefix. The labels added in batch 79 (the two Parts and the five subsections of Section 1) also use `smc:cs:`, although the subsections cover both Parts.
- *Bibliography.* Source 12's keys `d12007` and `d12006` are cited as the existing entries [4, 5] for the same papers, and source 13's `ginsburgspanier` is the existing entry [7]. The sources' other entries are appended as delivered, marked with their source; the entries for Balas, Jeroslow–Lowe and the programme's reviews are added and marked [write].
- *Typesetting.* Both sources' macro `\N` is written `\MN`; it prints the natural numbers $\{0, 1, 2, \dots\}$ as in the sources. Equations are numbered by section. Source 12's table file is printed inline, two forced page breaks of the sources and source 13's own table of contents (with its formatting of contents lines) are omitted, and the titles of Part III's appendices begin with "Part III:". No symbol was renamed and no normalization changed.
- *Added text.* Every sentence written in this addition is marked [write]: the title-page and abstract text on Parts III and IV; Sections 1.7–1.11 and the dated notes listed above; the openings of the two Parts; in Part III, notes after Lemma 34.1, in Section 35 (the uniform lift, the selector/copy credit), after Theorem 36.1, at the end of Section 37 (the shipped release, the correction and the review), and in Appendices F and H; in Part IV, notes in Sections 38 and 39 (Part V of *canonical-diophantine-certificates*), 41 (the review's check of the sign correction), 43, 46 (the second route), 47 (the shipped release and the review) and 48; this paragraph and the bibliography entries marked [write].
- *Excluded data.* Four regenerable files of the deliveries are not shipped: source 12's compact closure records `numeric/MODES.jsonl.gz` and `numeric/EVENTS.jsonl.gz` and source 13's two coefficient-explicit Morita fixtures `replay/core/fixtures/morita_doubling_n0_pulse_T7.json.gz` and `..._n2_pulse_T41.json.gz` (6.84 MB together). The shipped programs

regenerate them with identical decompressed content; the README gives the commands and the retrieval from the arrival commit.

Files added in batch 79. The report directory gained source 12's `CORRECTION.md`, `SOURCE-PROVENANCE.md` and `numeric/PROOF_AND_LEDGER.md` and source 13's `SOURCE-PROVENANCE.md` at its root, with the prefixes `12-conservative-signal-` and `13-sparse-lattice-`, and the two sources' programs in `code/` and records in `data/` under the same prefixes. The README maps each delivered name to its shipped name. Not shipped: both manuscripts and PDFs, both delivered READMEs, the checksum ledgers, source 12's copy of the patch (it is the programme's file [29]), an in-archive duplicate of source 12's machine file, source 13's manifest checker, packager and duplicate Morita audit, the four regenerable files above, and all of manuscript 07.

Parts V and VI (batch 80). [write] Four manuscripts of batch 80 of ProveIt's incoming reports were added on 2 October 2026. All five archives of the cluster arrived in commit `4e270aa46` and were placed by commit `345a9e44e` (batch 80, cluster K2), which staged 116 files (programs, data, provenance and audit notes, and one figure) byte-identical to the delivery (51 of source 14, 28 of source 15, 9 of source 16, 28 of source 17) and retired the five archives from the drop zone. Every delivered checksum ledger verified at placement (manuscript 10: 65 of 65 files; 02: 37 of 37; 05: 12 of 12; 06: 11 of 11; 03: 32 of 32); the ledgers are not shipped.

Source	Archive (batch 80)	Pin	Contributed	Printed as
14	<code>Three_Mass_Reversible_Computation.zip</code> , manuscript 10; main file <code>three-mass-release/three-mass-report.tex</code> (also as four modular files), 24-page PDF	none named	the three-unit reversible compiler, the residue query, the rational gap primitive, the cleanup, the exact counts and microtimes, both radius-one realizations, the fixed-horizon quadratic certificate with its loader, literal examples, six questions	Part V, Sections 49–63
15	<code>Exact_Targets_Three_Mass_Units.zip</code> , manuscript 02; main file <code>clean-target-release/clean-target-report.tex</code> , 15-page PDF	source 14 by SHA-256	the clean return wrapper, the fresh-entry condition, exact-pair undecidability, the first-target clock, the wrapper counts, the compact certificate and its affine lift, the worked example, six questions	Part V, Sections 64–73
16	<code>Single_Unit_Three_Mass_Decidability (1).zip</code> , manuscript 05 (attribution revision of manuscript 06); main file <code>single-unit-three-mass/single-unit-mass-three.tex</code> , 10-page PDF	none named	the stationary-unit frame, the no-split lemma, the mass-three core, uniform timed Presburger definability, the numerical three-particle bound, the quartic corollary, the source comparison	Part VI, Sections 74–82
17	<code>Four_Mass_Decidability_Package.zip</code> , manuscript 03; main file <code>four-mass-bound/four-mass-decidability.tex</code> , 12-page PDF	none named	the seed library and thresholds, the additive gap counter, the dispatch, the expanding-shuttle normal form, anchored observations, the binary radius-six shuttle, the one-witness timing quartic, seven questions	Part VI, Sections 83–92

No manuscript names a ProveIt commit. Source 15 pins source 14 by SHA-256 (its “frozen companion release 9”: L^AT_EX source 8bdfbf66..., PDF 1fc9a5..., checksum ledger c9efd37d...), and all three match source 14’s archive. Source 16 cites source 13 by title and date (“*Sparse Diophantine certificates for finite mass lattice dynamics*, 2 October 2026, Section 9”); source 17 cites source 16 as its companion. Manuscript 06 (`Single_Unit_Three_Mass_Decidability.zip`) is the original of source 16: seven of its twelve files are byte-identical to source 16’s, and its article differs only in the abstract’s attribution, the source comparison and two bibliography entries. It is superseded; nothing of it is shipped, and it survives in its arrival commit.

Where the batch-80 addition had to choose.

- *Placement and order.* All four manuscripts answer questions or non-claims of Part IV (Section 1.16), so they are additions to this report rather than a new report; a separate report on conserved-mass thresholds was the alternative. Source 15 extends source 14 and source 17 builds on source 16, so each pair shares a Part: the universality side (Part V) first, next to Part IV’s Morita–Imai interface, and the decidability side (Part VI) second. Every result,

proof, example, remark, question and limitation of the four sources is printed, in the sources' order, except for the re-proofs below, which are pointers.

- *Re-proofs printed as pointers.* Source 14's proof of the mass-two theorem (its Section 59) and of its bounded-diameter proposition; source 15's two paragraphs recalling source 14's completion and clock, its summary of the mass-two argument and its proof of the forward certificate lemma; and source 17's proofs of its two small-packet lemmas. Each statement is printed, with a bracketed note naming the theorem it repeats; the omitted text survives in the arrival archives. Source 16 is printed in full, with "Same argument as" notes at its lemmas and sections that follow Part IV.
- *The two editions of source 16.* The revision (manuscript 05) is printed; its mathematics is that of manuscript 06, and its attribution is the fuller one.
- *Author lines.* The four author lines ("Mathematical construction and reproducibility report", "A clean return extension of the reversible weighted CA construction", "Decidability theorem and executable checks", "Decidability proof and a binary timing certificate") name no assistant; the Part openings describe the manuscripts as AI-assisted research manuscripts of ProveIt's incoming reports.
- *Companions and bibliography.* Source 15's and source 17's "companion" entries are replaced by references to sources 14 and 16, and source 16's by a reference to Part IV's Section 46. Source 14's `durandlose2006` is the existing entry [5]; the four sources' `moritaimai`, `morita1996`, `alhazovimai`, `cooper` and `ginsburgspanier` are the existing entries of those keys; source 15's `brandt` is source 14's `brandt2020`; source 17's `kong` and `kakenreport` are source 16's `kongthesis` and `kakenfinal`. The other entries are appended as delivered, marked with their source, and the programme's reviews are added and marked [write].
- *Typesetting.* The four sources' macro `\N` is written `\NN`; it prints $N = \{0, 1, 2, \dots\}$ as in the sources. Equations are numbered by section; sources 16 and 17 tagged theirs consecutively, and source 16's untagged reference "(2)" is now a cross-reference. Source 14's table of contents and forced page break are omitted. Source 17's figure is the delivered vector PDF (embedded TrueType fonts, no Type 3 fonts), unchanged. No symbol was renamed and no normalization changed.
- *Front matter.* The title, subtitle and abstract were widened; Section 1 gained Sections 1.12–1.16, and dated [write] notes were added where batch 80 made earlier sentences stale (Sections 1, 1.1, 1.3, 1.4, 1.5, 1.6, 1.10 and 1.11, Table 3, the remark after Theorem 44.1, and Part IV's fourth and fifth questions). Nothing was numbered anew: the new front tables are uncaptioned, the two new Parts follow Part IV, and all appendices keep their letters.
- *Labels.* Source 14's labels are `smc:tm:` plus their delivered names, source 15's `smc:ct:`, source 16's `smc:su:` and source 17's `smc:fm:`; labels were added to unlabelled sections, one lemma, two theorems, two corollaries, two propositions, one figure, the questions and one equation. The two Parts and the five front subsections use `smc:tm:` (for Part VI, `smc:su:part`). The labels of the omitted re-proofs (source 14's `eq:affine`, `eq:nearcount`, `eq:tail`; source 15's `eq:affinefar`) are not used.
- *Added text.* Every sentence written in this addition is marked [write]: the title-page and abstract text on Parts V and VI, Sections 1.12–1.16 and the dated notes listed above, the openings of the two Parts and of the four sources, the bracketed notes inside the four sources, this paragraph and the bibliography entries marked [write].

- *Excluded data.* Five literal exports of source 14 (`examples/cycle_rule.json`, `merge_rule.json`, `trap_rule.json`, `mixed_certificate.json` and `mixed_radius_one_certificate.json`; 20,752,637 bytes together) are not shipped. The shipped programs regenerate the three rules byte for byte and the two certificates identically apart from Windows line ends; the README gives the commands and the retrieval from the arrival commit.

Files added in batch 80. The report directory gained, at its root, source 14's `SOURCE_PROVENANCE.md`, source 15's `CLEAN-TARGET-THEOREM.md`, `SOURCE_PROVENANCE.md` and `audit/INDEPENDENT-AUDIT.md`, source 16's `REVISION.md`, `SOURCE-PROVENANCE.md` and `independent/AUDIT.md`, and source 17's seven proof-detail, provenance and audit notes, with the prefixes `14-three-mass-`, `15-clean-targets-`, `16-single-unit-` and `17-four-mass-`; the four sources' programs in `code/` and records in `data/` under the same prefixes; and a new directory `figures/` with source 17's article figure. The README maps each delivered name to its shipped name. Not shipped: the four manuscripts (and source 14's four modular $\text{\LaTeX}$ files), the four PDFs and delivery READMEs, the checksum ledgers and the manifest checkers of sources 14 and 15, source 15's five vendored programs (byte copies of source 14's), source 14's byte copy of a source-13 receipt, source 17's raster figure, the five excluded exports, and all of manuscript 06.

Source 18 and Part VII (batch 82). [write] Four manuscripts of batch 82 of ProveIt's incoming reports were added on 3 October 2026. All four archives arrived in commit `db37d18c8` and were placed by commit `49dfa8fd6` (batch 82, cluster M4), which staged 119 files (programs, data, provenance and audit notes, and three figures) byte-identical to the delivery (18 of source 18, 17 of source 19, 26 of source 20, 58 of source 21) and retired the four archives from the drop zone. Every delivered checksum ledger verified at placement (manuscript 19: 25 of 25 files; 06: 20 of 20; 17: 31 of 31; 01: 61 of 61); the ledgers are not shipped.

Source	Archive (batch 82)	Pin	Contributed	Printed as
18	Timed_Four_Mass_Quartic_Certificates.zip, manuscript 19 (the pipeline's addendum 13); main file phase-quartic-release/fixed-input-timed-quartics.tex, 12-page PDF	source 17 by title; its source -freeze.json pins its own code	the chart interface and half-open partition, the constant-term lifting compiler with ledgers and selector elimination, signed external pairs, the binary charts and the eight- and four-witness certificates	Part VI, Sections 93–100
19	Dimension_Independent_Particle_Thresholds_Package.zip, manuscript 06 (Report 29); main file dimension-independent-threshold-release-20261003/report29.tex, 22-page PDF	source 17's L ^A T _E X source (803bf0c4...), source 18's (fecc0ae2...), Reports 16 (7e644e82...) and 26 (a156d7d9...)	the $\mathbb{Z}^d$ decision theorem with contact rays, the nonparallel switch set, thresholds, dispatch, termination and drift cutoff; the conditional boundary; the $\mathbb{Z}^d$ timed quartics; six questions	Part VII, Sections 101–117, Appendices I–J
20	Sparse_Orbit_Geometry_and_Exact_Counting_Package.zip, manuscript 17 (Report 30); main file sparse-orbit-geometry-release-20261003/report30.tex, 24-page PDF	source 19's L ^A T _E X source (1e4b9cb1...)	semilinear geometry and sharp counts, quadratic clocks and inverses, first arrivals, drift, the weighted planar automata, first-visit quartics; five questions	Part VII, Sections 118–128, Appendix K
21	Binary_Planar_Four_Particle_Shuttle_Package.zip, manuscript 01 (Report 31); main file binary-planar-shuttle-release-20261003/report31.tex, 14-page PDF	sources 17, 18 and 20 (source 20's L ^A T _E X source 0723338c...)	the binary planar rule and its full-shift proof, the orbit, wedge, counts and inverses, the exact dependency set and degree, the one-step quartic; three questions	Part VII, Sections 129–140

No manuscript names a ProveIt commit. The pins of sources 19–21 are recorded in their `source-lineage.json` files (shipped as `data/19-mass-four-zd-source-lineage.json`, `data/20-orbit-geometry-source-lineage.json` and `data/21-planar-shuttle-source-lineage.json`) and in source 19's lineage appendix; source 17's pin equals the hash of `four-mass-bound/four-mass-decidability.tex` in its arrival `4e270aa46`, and all pins were verified at placement. The two batch-82 manuscripts that source 19 pins besides, Reports 16 and 26, were placed in the collection's report `five-particle-binary-automata` by `bca6383e9` and `7d2b1b245`; Report 16 is printed there as Part II (commit `e6a410588`).

Where the batch-82 addition had to choose.

- *Placement and order.* Source 18 answers source 17's last question and sources 19–21 extend source 17's theorem and build only on it and on one another, so all four are additions to this report. Source 18 is a one-dimensional addendum to source 17, written before source 19, so it is the third source of Part VI; sources 19, 20 and 21 form Part VII in their dependency order, after Part VI and before the appendices. A new report for the batch-82 particle series was the alternative. Every result, proof, example, remark, question and limitation of the four sources

is printed, except for the one pointer below.

- *Re-proof printed as a pointer.* Source 18’s subsection on the radius-six rule, a restatement of source 17’s rule, its conservation proof and its potential certificate; the place states what it repeats, and the omitted text survives in the arrival archive. Source 19’s Appendix I, which restates source 18’s chart partition and compiler for $\mathbb{Z}^d$, is printed in full with notes. Source 19’s re-proof of source 17’s theorem at $d = 1$ is printed in full, because it is the $\mathbb{Z}^d$ proof, with bracketed notes at each shared step; source 18’s statement of the imported normal form and its remark on the return quartic are printed with notes.
- *Byte copies.* Source 18’s three byte copies of source 17’s files are not shipped twice. The normal, optimized and expected copies of several receipts of sources 20 and 21 are shipped as delivered, because their replay harnesses compare them by name.
- *Pipeline numbers and author lines.* The author lines (“Research addendum 13 / Complete fixed input orbits and exact witness accounting”, “Research report 29”, “Research report 30”, “Research report 31”) name no assistant; the openings describe the manuscripts as AI-assisted research manuscripts of ProveIt’s incoming reports. The pipeline’s report numbers are kept in the text with bracketed translations.
- *Companions and bibliography.* Source 18’s “companion report 12” entry and the “Report” entries of sources 20 and 21 are replaced by references to sources 17–20. Source 18’s `kong`, `imai` and `ginsburg` are the existing `kongthesis`, `kakenfinal` and `ginsburgspanier`; source 19’s `kong` and `alhazov` are `kongthesis` and `alhazovimai`; the pebble-automaton entries of sources 19 and 20 (`delorme`, `lacalle`, `blum`) are merged into one entry each, and the other new entries are appended, marked with their sources. The six programme reviews cited are added and marked [write].
- *Typesetting.* The macro `\N` of the four sources is written `\NN` and prints `N`; sources 19–21 printed $\mathbb{Z}_{\geq 0}$, the same set. Their `\norm` (the maximum norm) is renamed, because Part IV’s `\norm` differs; `\file` and source 18’s `\Lift` keep their definitions. Tables of contents, forced page breaks and source 18’s `\Needspace` are omitted. Source 19’s figure is its inline TikZ drawing; the figures of sources 20 and 21 are the delivered vector PDFs (embedded TrueType fonts, no Type 3 fonts), unchanged. Equations are numbered by section. No other symbol was renamed and no normalization changed.
- *Front matter.* The title, subtitle, author line, abstract and status line were widened, Part VI’s title names source 18, and Section 1 gained Sections 1.17–1.21; dated [write] notes were added where batch 82 made earlier sentences stale or answered questions (Sections 1, 1.1, 1.5, 1.12 with a new row of the threshold table, 1.16, Table 3, the opening of Part VI, and source 17’s seven questions). Nothing was numbered anew: the new front tables are uncaptioned, source 18 and Part VII follow every earlier section, their captioned tables and figures follow every earlier one, and the appendices of sources 19 and 20 follow Part III’s, so that all earlier appendices keep their letters.
- *Labels.* Source 18’s labels are `smc:tq:` plus their delivered names, source 19’s `smc:zd:`, source 20’s `smc:og:` and source 21’s `smc:bp:`; labels were added to unlabelled sections, two corollaries and one theorem, and to the questions. The new Part and the five front subsections use `smc:zd:`.
- *Wording.* Source 21’s “answers the binary-realization question”, “This proves induction from

$n = 0$ ” and the asymptotic of its stationary inverse carry bracketed precisions; source 18’s hypothesis $d \geq 7$ carries a note that it is not sharp.

- *Excluded data.* None: the largest shipped file has 267,097 bytes.
- *Added text.* Every sentence written in this addition is marked [write]: the title-page and abstract text on source 18 and Part VII, Sections 1.17–1.21 and the dated notes listed above, the openings of Part VII and of the four sources, the bracketed notes inside the four sources, this paragraph and the bibliography entries marked [write].

Files added in batch 82. The report directory gained, at its root, source 18’s `verification/independent-audit.md`, source 19’s six proof, audit, review and literature notes, source 20’s nine proof, review and supplement notes and source 21’s two proof notes and four audit notes, with the prefixes `18-timed-quartics-`, `19-mass-four-zd-`, `20-orbit-geometry-` and `21-planar-shuttle-`; the four sources’ programs in `code/` and records in `data/` under the same prefixes; and three figures in `figures/`. The README maps each delivered name to its shipped name. Not shipped: the four manuscripts, PDFs, delivery READMEs and checksum ledgers, source 18’s manifest checker and its three byte copies of source-17 files, and source 20’s two supplement checksum manifests.

Part VIII (batch 82, cluster M2). [write] Two further manuscripts of batch 82 were added on 3 October 2026. Both archives arrived in commit `db37d18c8` and were placed by commit `7d2b1b245` (batch 82, cluster M2, which placed four more manuscripts of the same batch in `five-particle-binary-automata`); it staged 61 files byte-identical to the delivery (37 of source 22, 24 of source 23) and retired the archives from the drop zone. At this write both release checksum ledgers verify against the arrival archives (source 22: 103 of 103 files; source 23: 87 of 87), and every shipped file of the two sources equals an archive member byte for byte; the ledgers are not shipped.

Source	Archive (batch 82)	Pin	Contributed	Printed as
22	<code>Unbounded_Compact_Clean_Clocks_Package.zip</code> , manuscript 21 (Report 21); main file <code>unbounded-clean-clock-release-20261003/report21.tex</code> , 14-page PDF	<code>ProveIt</code> <code>ad634b2d1</code> (the research tree’s interface, receipt, review and residue-history proof); source 15’s theorem by SHA-256	the clean-time composition and both directions, the retained chronology and no-wrap clock, infinite fibres, complete ledgers and exact degrees, folded schedules, zero-step circuits, raw-input limits	Part VIII, Sections 141–149
23	<code>Canonical_Histories_and_Infinite_Fibers_Package.zip</code> , manuscript 02 (Report 22); main file <code>canonical-fiber-release-20261003/report22.tex</code> , 14-page PDF	<code>ProveIt</code> <code>ad634b2d1</code> ; source 22’s base and folded manifests by SHA-256	the canonical height and its completeness, outer uniqueness, the entire-fibre projection, fixed-scale native infinitude, canonical ledgers	Part VIII, Sections 150–158

Where the batch-82 M2 addition had to choose.

- *Placement and order.* Both sources answer questions of this report (source 14’s Question 5 and source 15’s Question 5) and rest on source 15’s theorem, so they are an addition to it.

They form a new Part VIII after Part VII and before the appendices, rather than further sources of Part V, because their machinery is the research tree’s residue-history, native AND and Pell arithmetic, not automaton construction. Placing them with Report 25 in `fixed-universal-polynomials` was the alternative; that report prints Report 25 and points here. Source 22 comes first; source 23 is its sequel. Every result, proof, example, remark and limitation of both is printed, except for the two repeated tables below.

- *Repetitions.* Source 23’s tables of the four fixtures (Section 156.1) and of the sanity values (Section 157.2) equal source 22’s and are printed once, with their contents in the bracketed pointers; the omitted tables survive in the arrival archive. Source 22’s restatements of source 15 and of the research tree’s arguments, and source 23’s restatements of source 22, are printed in full with notes. Source 23’s Theorem 155.1, implied by Report 25’s classification in `fixed-universal-polynomials`, is printed in full as a second route.
- *Byte copies.* Source 23 carried 23 files of source 22 byte for byte, and both carried byte copies of the research tree’s pinned files and of source 15’s theorem; none is shipped twice. Two native-fibre source files of source 23 are shipped once, by `fixed-universal-polynomials`.
- *Excluded data.* The 70 emitted circuit files of the two sources (4,189,675 bytes: 34 emitted files of source 22 and eight reference copies, 18 emitted files of source 23 and ten reference copies) are not shipped. The shipped emitters regenerate all of them byte for byte from the repository’s raw receipt in a few seconds, and the shipped emission receipts list the SHA-256 of each; the README gives the commands.
- *Pipeline numbers and author lines.* The author lines name no assistant; the openings describe the manuscripts as AI-assisted research manuscripts of ProveIt’s incoming reports. The pipeline’s report numbers are kept in the text with bracketed translations.
- *Bibliography.* The two sources’ entries for the research tree’s interface and residue-history proof are merged into one entry each (`threemassraw`, `residuehistory`); source 22’s entry for the clean-target theorem is source 15 (Part V); the other entries are appended, marked with their sources, and the three programme notes cited are added and marked [write].
- *Typesetting.* The sources’ macro `\N` is written `\NN`; their `\file` is this report’s; their `\sha` sets a hash as a breakable path instead of loading `seqsplit`, and one file name in source 22’s Section 142 is set as a breakable path for the line width; tables of contents are omitted. Equations are numbered by section. No symbol was renamed and no normalization changed.
- *Front matter.* The title, subtitle, author line, abstract and status line were widened, and Section 1 gained Sections 1.22–1.26; dated [write] notes were added where Part VIII made earlier sentences stale or answered questions (Sections 1, 1.1, 1.4, 1.5, 1.6, 1.11 and 1.15, Table 3, Part II’s Question 2, source 14’s Question 5 and source 15’s Question 5). Nothing was numbered anew: the new front tables are uncaptioned, Part VIII follows every earlier section, and it has no figure, captioned table or appendix.
- *A correction to the batch-82 M4 text.* Six places written for source 19 called Report 26 “placed but not yet written” in `five-particle-binary-automata`; it had been printed there as Part VI (commit d2043bbd0) before that write. They now say so (the threshold table of Section 1.12, Sections 1.16 and 1.17, the note after Corollary 101.2 and Appendix J); the conditional status of source 19’s sharp boundary is unchanged.
- *Labels.* Source 22’s labels are `smc:uc:` plus their delivered names and source 23’s `smc:ch:`;

labels were added to unlabelled sections and subsections and to source 22’s fibre theorem. The new Part and the five front subsections use `smc:uc:`.

- *Added text.* Every sentence written in this addition is marked [write]: the title-page and abstract text on Part VIII, Sections 1.22–1.26 and the dated notes listed above, the opening of Part VIII and of the two sources, the bracketed notes inside them, this paragraph and the bibliography entries marked [write].

Files added in batch 82 (cluster M2). The report directory gained, at its root, source 22’s nine proof, audit, review and README notes and source 23’s eight, with the prefixes `22-clean-clocks-` and `23-canonical-fibers-`; the two sources’ programs in `code/` and records in `data/` under the same prefixes. The README maps each delivered name to its shipped name. Not shipped: the two manuscripts, PDFs and delivery READMEs, the checksum manifests with their checkers, the release tools, the byte copies named above and the 70 regenerable circuit files.

Parts IX–XIII (batch 91, cluster S). [write] Fifteen manuscripts of batch 91 of ProveIt’s incoming reports were added on 4 October 2026, as five Parts. Their archives arrived in commit `0d7f51c44` and were placed by commit `d750d98dd` (batch 91S), which staged 1,568 files byte-identical to the delivery (35,131,787 bytes; 199 at the report root, 212 programs in `code/`, 1,156 records in `data/` and one figure) and retired the fifteen archives from the drop zone. The fragments of the five Parts were drafted separately and assembled in one write; the paragraphs below give each Part’s provenance.

Source	Batch 91	Report	Prefix	Files	Printed as
24	manuscript 01	49	24-reversible-	24	Part IX
25	manuscript 03	51	25-uniform-	41	Part IX
26	manuscript 08	56	26-periodic-	41	Part X
27	manuscript 09	57	27-fivesignal-	49	Part X
28	manuscript 10	58	28-diophantine-	63	Part X
29	manuscript 11	59	29-rotations-	140	Part X
30	manuscript 12	60	30-planar-	202	Part XI
31	manuscript 13	61	31-invertible-	59	Part XI
32	manuscript 14	62	32-projective-	83	Part XI
33	manuscript 15	63	33-homogeneous-	74	Part XI
34	manuscript 16	64	34-branching-	68	Part XII
35	manuscript 17	65	35-clocked-	123	Part XII
36	manuscript 18	66	36-bounded-cert-	78	Part XIII
37	manuscript 19	67	37-pos-power-	257	Part XIII
38	manuscript 20	69	38-low-arity-	266	Part XIII

Where the batch-91 addition had to choose.

- *Placement and order.* All fifteen manuscripts continue Parts of this report or answer its questions, so they are additions to it. Sources 24 and 25 continue Part VI and form Part IX; the signal-machine sources follow in the pipeline’s order as Parts X (Reports 56–59), XI (60–63), XII (64–65) and XIII (66, 67, 69), after Part VIII and before the appendices. Every

result, proof, example, remark, question and limitation of the fifteen sources is printed, except where a restated statement is printed once.

- *One restatement device.* A statement that a source restates from another is printed once, in the Part that owns its theme; the restatement becomes a numbered remark, marked [write], that quotes it, keeps the duplicate’s own label and points to the proof printed elsewhere, and references to it say “Remark”. Genuinely different proofs are kept as marked second routes.
- *Labels.* Every delivered label of a source carries the sub-prefix of its source (`smc:rt:`, `smc:ut:`, `smc:pm:`, `smc:fv:`, `smc:fd:`, `smc:rf:`, `smc:pl:`, `smc:iv:`, `smc:pj:`, `smc:hl:`, `smc:br:`, `smc:ck:`, `smc:bc:`, `smc:pp:`, `smc:la:`); unlabelled numbered statements are `smc:<sub>n<number>`; labels written in the merge carry `smc:ix:` to `smc:xiii:` (the front subsections of Parts X–XIII use `smc:x:`); the Parts are `smc:part:reversible`, `periodic`, `planar`, `clocked` and `nativegap`. No earlier label was renamed or removed.
- *Bibliography.* One key per work. The programme’s intake of the twelve signal and particle archives (commit 49b100cfd) is [99], distinct from the 2 October review [14], which source 26 cites. The mathlib Pell file is one entry [86] for sources 28–30 and 35–38. Becker et al.’s tenth paper is [92] for sources 31–34, Durand-Lose’s sixth is the existing [16], and the papers of Dai and Xia, Ouaknine and Worrell on positivity, and Fijalkow et al. have one entry each [76, 75, 87]; Dudenhefner’s paper [101] serves sources 34, 35 and 37.
- *Macros.* Every new macro is a `\providecommand`, and no earlier macro or colour is redefined. The sources’ `\N` is written `\NN`, their positive integers `\Np`; clashing source macros were renamed (`\norm` as `\enorm` and `\lonenorm`, `\code` as `\Ccode`, `\Power` as `\POWER`, `\ZZ` and `\NN` of source 34 as `\brZ` and `\brN`), and the sources’ colours, among them source 30’s redefinitions of `blue` and `red`, have names of their own.
- *Front matter.* The title, subtitle, author line, abstract and status line were widened, and Section 1 gained Sections 1.27–1.31 and 1.32–1.36; dated [write] notes were added where batch 91 made earlier sentences stale or answered or touched questions (Sections 1, 1.1, 1.4, 1.5, 1.6, 1.12 with the threshold table, 1.15, 1.19, 1.20 and 1.21, Table 3, Part I’s directions 15.4, 15.6 and 15.9, Part II’s Questions 2 and 4, Part III’s Section 35, source 16’s checks, source 17’s rule, quartic and Questions 2, 4 and 6, source 19’s boundary corollary and source 20’s Question 1). Nothing was numbered anew: the new front tables are uncaptioned, the five Parts follow every earlier section, their captioned tables and figures follow every earlier one, and the appendices keep their letters.
- *Unproved and wrong claims.* Under Vladimir’s standing rule of 4 October 2026, no unproved claim of a source, a review or the write was dropped: each Part has a section of further questions (Sections 181, 238, 293.4, 317 and 355) stating such claims as open questions, with their sources, argument sketches and what is missing. Parts IX and XII were drafted before the rule and were scanned against their manuscripts at this assembly.
- *A correction to the placement record.* The message of the placement commit d750d98dd says that Report 67’s and 69’s heavy regenerable evidence was excluded. Only Report 67 (source 37) had excluded heavy files: nine files over 1 MB, six distinct, 34,659,924 bytes; Report 69 (source 38) had none. The placement dossier’s “5 distinct blobs” for the batch is wrong as well: the cluster’s ten heavy files are seven distinct blobs (source 37’s six and one of Report 55, which is placed in *group-theoretic-substrates*).

- *A correction to the batch-82 text.* Two sentences written for source 18 (Section 1.19 and the review-status paragraph of Part VII) say that source 18 has not been reviewed. That was true at that write; the programme took source 18 in later the same day [112], and dated notes now say so. The draft of Part IX had repeated the stale sentence; it now cites the intake.
- *Reciprocal notes applied here.* Notes proposed by the batch-91 writes of `five-particle-binary-automata` (Reports 70 and 71) and `a196460-clipping-tables` are applied in this write: after source 19’s boundary corollary, in source 16’s checks, and in Part XIII (Report 68 and the classification theorem).

Part IX (batch 91, sources 24 and 25). [write] Two manuscripts of batch 91 were added on 4 October 2026. Both archives arrived in commit `0d7f51c44` and were placed in `d750d98dd` (cluster S, with Parts X–XIII).

Source	Archive and main file	Pins
24	<code>A_Sharp_Reversible_Four_Particle_Timing_Threshold_Package.zip</code> , manuscript 01 (the pipeline’s Report 49); main file <code>Research_Report49/manuscript/report49.tex</code> , 14-page PDF	this article, blob <code>ec10c7a5</code> (SHA-256 <code>17d3c0d9...</code> , = commit <code>0f95145cb</code>); <code>review_parallel_particle_reports.md</code> blob <code>30c7366b</code> ; <code>17-four-mass-BINARY-INDEPENDENT-AUDIT.md</code> blob <code>0aa7b970</code>
25	<code>Uniform_Timed_Charts_for_Four_Mass_Units_Package.zip</code> , manuscript 03 (Report 51); main file <code>Research_Report51/manuscript/report51.tex</code> , 21-page PDF	this article, blob <code>ec10c7a5</code> ; source 24’s TeX, SHA-256 <code>c7813317...</code>

All pins equal the repository and delivery bytes (verified at placement and at this write).

- *Placement and order.* Both continue Part VI: source 24 supplies the reversibility that source 17’s shuttle lacks, source 25 answers source 17’s uniform questions conditionally; they form Part IX, before the signal-machine Parts X–XIII of the same batch.
- *Duplicates.* Source 24’s quartic is source 17’s (90.1); it is printed once, and a numbered remark (Remark 163.1) quotes source 24’s statement and proof. Source 24’s prospective-isolation lemma (Report 26’s, in `five-particle-binary-automata`) and its three-particle bound (source 16’s, binary case) and source 25’s restatement of source 16’s and 17’s ingredients are printed in full as marked second routes or restatements. Source 25’s repetition of source 24’s orbit formula is kept inside its proof with a note.
- *Omission in a source.* Source 25 does not cite source 20’s planar drift examples; a note after Proposition 176.1 states the relation.
- *Labels.* Source 24’s 27 labels are `smc:rt:` plus their delivered names and source 25’s 53 are `smc:ut:` plus theirs; write labels carry `smc:ix:`, and the Part label is `smc:part:reversible`. The two appendices are printed as the last sections of their sources.
- *Notation.* Their $\mathbb{N}$ is written $\mathbb{N}$; no symbol was renamed (Section 1.28).
- *Review.* The programme’s routing intake [99] (commit `49b100cfd`) covers both; no other programme commit mentions them.

- *Front matter.* Section 1 gained Sections 1.27–1.31; dated notes were added at source 17’s non-claim, rule and quartic and at Questions 2, 4 and 6, at source 18’s non-claim, at source 20’s Question 1, and in the threshold table.

Files added for Part IX. The report directory gained, at its root, source 24’s three audit and specification notes and source 25’s eleven proof, review and audit notes, with the prefixes 24-reversible- and 25-uniform-; their programs in `code/` (8 and 10) and records in `data/` (12 and 20) under the same prefixes; and source 24’s figure in `figures/`. The README maps each delivered name to its shipped name. Not shipped: both manuscripts, PDFs and delivery READMEs, the checksum manifests, page renders and text dumps, a TeX log, and the byte copies of this article (blob `ec10c7a5`) and of source 24’s manuscript that the archives carried.

Part X (batch 91, sources 26–29). [write] Four manuscripts of batch 91 were added on 4 October 2026. All four archives arrived in commit `0d7f51c44` and were placed in `d750d98dd`.

Source	Archive and main file	Pins
26	<code>Periodic_Signal_Machine_Collision_Certificates_Package.zip</code> , manuscript 08 (the pipeline’s Report 56); main file <code>article/Report56.tex</code> , 15-page PDF	ProveIt <code>e3e58a882</code> (this report’s README and the programme’s finite-schema signal review)
27	<code>Five_Signal_Nonsemialgebraic_Periodic_Validity_Package.zip</code> , manuscript 09 (Report 57); main file <code>Report57.tex</code> with two table files, 20-page PDF	repository searches at <code>7ba8f64e6</code> ; source 26’s proof and audit by SHA-256
28	<code>Five_Signal_Diophantine_Certificate_Source.zip</code> , manuscript 10 (Report 58); main file <code>Report58.tex</code> , 19-page PDF	source 27’s proof and audit by SHA-256; mathlib4 <code>ac77769f</code>
29	<code>Five_Signal_Rational_Rotation_Family_Source.zip</code> , manuscript 11 (Report 59); main file <code>manuscript/report59.tex</code> (the final reviewed edition), 22-page PDF	ProveIt <code>b64f24e59</code> ; source 28’s proof as an inert dependency; mathlib4 <code>ac77769f</code>

- *Placement and order.* Source 26 answers in part Part I’s accumulation direction; sources 27–29 continue it in order. Every result, proof, example, remark, question, appendix and limitation of the four is printed; their appendices are the last sections of each source.
- *Restatement.* Source 29’s display of the fifteen POWER equations is source 28’s system and is printed once, in source 28; source 29’s display became a numbered remark that keeps its labels, and its specialization of the Pell theorems is kept as a marked second account.
- *Renamed symbols.* J (three meanings), the Gaussian quotient, source 27’s normalized coordinates, valid set and rotation, the Pell modulus and source 28’s radix; the opening of Part X lists them.
- *Ledgers.* The witness and equation counts of sources 28 and 29 and of source 30 are compared in Table 16.
- *Questions.* The sources’ questions are printed in place; Section 238 collects them with the claims moved there under the standing rule.

Files added for Part X. The report directory gained the four sources’ proof, audit, review and replay notes at its root (49), their programs in `code/` (54) and records in `data/` (190, among them

source 29’s 45 fixtures) under the prefixes `26-periodic-`, `27-fivesignal-`, `28-diophantine-` and `29-rotations-`. Not shipped: the manuscripts, their table and figure files and earlier editions, the PDFs and delivery READMEs, page renders, TeX logs and checksum manifests, the archives’ copies of earlier packets and of the files of Report 55 (shipped with *group-theoretic-substrates*), and the mathlib Pell source with its licence.

Part XI (batch 91, sources 30–33). [write] Four manuscripts of batch 91 were added on 4 October 2026, placed in `d750d98dd`; their provenance, the choices of the merge and the files shipped are given in Section 293.3, inside the Part. The shipped files carry the prefixes `30-planar-`, `31-invertible-`, `32-projective-` and `33-homogeneous-` (418 files).

Part XII (batch 91, sources 34 and 35). [write] Two manuscripts of batch 91 were added on 4 October 2026, placed in `d750d98dd`.

Source	Archive and main file	Pins
34	<code>Five_signal_branching_source_and_evidence.zip</code> , manuscript 16 (the pipeline’s Report 64); main file <code>Report64/Report64.tex</code> (a flattening of eleven modules), 21-page PDF	source 33’s proof by SHA-256; its own physical and arithmetic packets
35	<code>Clocked_signals_source_and_evidence.zip</code> , manuscript 17 (Report 65); main file <code>Report65/Report65.tex</code> , 20-page PDF	source 34’s physical and trace proofs, the POWER proofs of sources 28 and 30, and mathlib4 <code>ac77769f</code> , all by SHA-256

- *Restatement.* Source 35’s trace lemma restates source 34’s Theorem 301.1 and is a numbered remark (Remark 310.1) with source 35’s short proof; its base-two POWER lemma is kept as a second route for source 28’s lemma.
- *Renamed symbols.* Source 34’s branch tags `\ZZ` and `\NN` are $\mathbb{Z}$ and $\mathbb{N}$; the colours have names of their own. The pipeline’s report numbers carry bracketed source numbers.
- *Questions.* Both sources’ questions are printed in place, with notes where a later source answers them; Section 317, added at this assembly, records the claims moved under the standing rule.

Files added for Part XII. The two sources’ proof, audit and review notes at the report root (28), their programs in `code/` (29) and records in `data/` (134, among them 47 logs), under the prefixes `34-branching-` and `35-clocked-`. Not shipped: the manuscripts and modules, PDFs and delivery READMEs, release inventories, page renders, text dumps and TeX transcripts, and the byte copies of earlier packets (shipped once, at the source that first delivered them) and of the mathlib Pell source.

Part XIII (batch 91, cluster W4). [write] Three manuscripts of batch 91 of ProveIt’s incoming reports were added on 4 October 2026: manuscripts 18, 19 and 20, the research pipeline’s Reports 66, 67 and 69, called sources 36, 37 and 38 here. Their archives arrived in commit `0d7f51c44` and were placed by commit `d750d98dd` (batch 91S, which placed fifteen manuscripts of the batch as Parts IX–XIII), which staged 601 files byte-identical to the delivery (78 of source 36, 257 of source 37, 266 of source 38; 14,733,494 bytes) and retired the archives from the drop zone. No manuscript names a ProveIt commit; each pins its frozen input packets by SHA-256 in its `INPUT_PINS.json`

(shipped as `data/36-bounded-cert-INPUT_PINS.json`, `data/37-pos-power-INPUT_PINS.json` and `data/38-low-arity-INPUT_PINS.json`), and all three pin mathlib4 commit `ac77769f` [86]. Every release inventory verified completely at placement (source 36: 105 of 105 files; 37: 531 of 531; 38: 637 of 637), as did every input pin list and packet manifest; the pure checksum lists are not shipped.

Where the batch-91 W4 addition had to choose.

- *Placement and order.* The three sources continue source 35’s native-gap certificate (Part XII), and by the collection’s rule a continuation goes to the report it continues; they form Part XIII after Part XII, in manuscript order. Source 36 is the base: sources 37 and 38 restate its clipping lemma and compiler and improve its ledgers. Every result, proof, example, remark, question and limitation of the three is printed, except the restatements listed next.
- *Restatements printed once.* Source 36’s Proposition 7.1 (Remark 324.1) and its POWER module’s equivalence, proof and fixture (Remark 323.1) repeat source 35’s Theorem 312.1 and Lemma 309.1; source 37’s Lemma 9.1 and Proposition 9.2 (Remarks 340.1 and 340.2) and source 38’s Lemma 1.2 (Remark 348.2) repeat source 36’s Lemma 319.1 and Theorem 318.1; source 38’s Proposition 6.1 and its recovery argument (Remark 353.1 and the note after it) repeat source 37’s twelve-leaf theorem at base two. Each statement is quoted in a numbered remark with its own label, references to it say “Remark”, and the omitted proofs survive in the arrival archives; labelled displays inside omitted text are kept. Restating displays that later sections compute with (source 37’s Section 340, source 37’s cost bounds) are printed with notes.
- *Byte copies.* The archives carried byte copies of one another (source 36’s bounded-certificate packet in sources 37 and 38; source 37’s exact-degree packet and audit in source 38) and of the proofs of sources 28, 30, 34 and 35; each is shipped once, under the lowest source number. The mathlib Pell source and its licence (nine copies) are third-party files, cited and not shipped.
- *Excluded data.* Nine evidence files of source 37 over 1 MB (six distinct expansions, 34,659,924 bytes in all) are not shipped; the packets’ own programs regenerate them in 3–25 seconds, identical up to line ends, and the README gives the commands. Page renders, empty logs and checksum lists are not shipped either.
- *Report 68.* Source 38’s predecessor, Report 68, was never delivered. Its delivery README, carried by source 38, is shipped as a record; nothing from it is printed (opening of Part XIII).
- *Pipeline numbers and author lines.* The author lines name no assistant; the openings describe the manuscripts as AI-assisted research manuscripts of ProveIt’s incoming reports. The pipeline’s report numbers are kept in the text with bracketed translations.
- *Bibliography.* The three sources’ entries for the mathlib Pell source are merged into one (`mathlibpell`), and those for the DLMF into one (`dmlflagrange`); the other entries are appended with their sources’ keys noted, and the two programme reviews cited are added and marked [write].
- *Typesetting.* The sources’ macros `\N`, `\Pos`, `\pos` and `\norm` are written `\NN`, `\Np`, `\Np` and `\lonenorm` (this report’s `\norm` is Part IV’s); their colours are renamed; tables of contents and page breaks are omitted. Equations are numbered by section. No symbol was renamed

and no normalization changed; a table at the opening of the Part fixes the readings of letters that differ between the sources.

- *Labels.* Source 36's labels are `smc:bc:` plus their delivered names, source 37's `smc:pp:` and source 38's `smc:la:`; none was dropped. The Part is `smc:part:nativegap`; one remark added in the write is `smc:xiii:rem:bcpower`.
- *Added text.* Every sentence written in this addition is marked [write]: the opening of Part XIII and of the three sources, the bracketed notes and restatement remarks inside them, the review notes at the end of each source, this paragraph and the bibliography entries marked [write].

Files added in batch 91 (cluster W4). The report directory gained, at its root, the three sources' proof, review, audit, handoff, literature and README notes (12 of source 36, 24 of source 37, 12 of source 38), with the prefixes `36-bounded-cert-`, `37-pos-power-` and `38-low-arity-`, and their programs in `code/` (11, 23 and 22) and records in `data/` (55, 210 and 232) under the same prefixes. The README maps each delivered name to its shipped name. Not shipped: the three manuscripts and their modules, the PDFs and delivery READMEs, the checksum lists, page renders and empty logs, the byte copies named above, the third-party Pell source and the nine regenerable evidence files.

F Part III: the complete Morita transition data

In the table, bRq_s means write b , move right, and enter q_s ; bLq_s is analogous. A dash is unspecified. The two words halt and null are designated terminal cells, not transitions. Symbol order is the release's digit order `b, Y, N, *, $, 1`, assigned digits 1 through 6.

[write] The delivered article reads this table from a separate file, `paper/morita-table.tex`; it is printed here inline and unchanged. The same table is shipped as data in `data/12-conservative-signal-MORITA_15_6_TABLE.json`.

State	b	Y	N	$*$	$\$$	1
q_0	$\$Lq_2$	$\$Lq_1$	bLq_{11}	—	—	—
q_1	halt	YLq_1	NLq_1	$*Rq_0$	bLq_1	—
q_2	$*Lq_3$	YLq_2	NLq_2	$*Lq_2$	null	—
q_3	bRq_{10}	$1Rq_4$	bRq_5	bRq_8	—	—
q_4	YRq_6	YRq_4	NRq_4	$*Rq_4$	$\$Rq_4$	—
q_5	NRq_6	YRq_5	NRq_5	$*Rq_5$	$\$Rq_5$	—
q_6	bLq_7	—	—	—	—	—
q_7	NLq_3	YLq_7	NLq_7	$*Lq_7$	$\$Lq_7$	YLq_3
q_8	—	YRq_8	NRq_8	$*Rq_8$	$\$Rq_9$	—
q_9	—	YRq_9	NRq_9	$*Rq_9$	YRq_0	—
q_{10}	—	YRq_{10}	NRq_{10}	$*Rq_{10}$	$\$Lq_3$	—
q_{11}	$*Lq_{12}$	YLq_{11}	NLq_{11}	$*Lq_{11}$	$\$Lq_{11}$	—
q_{12}	bRq_{14}	YLq_{12}	NLq_{12}	bRq_{13}	—	—
q_{13}	NRq_0	YRq_{13}	NRq_{13}	$*Rq_{13}$	$\$Rq_{13}$	—
q_{14}	—	YRq_{14}	NRq_{14}	$*Rq_{14}$	$\$Lq_{12}$	—

G Part III: a small complete quadratic packet

Consider the one-dimensional partial-branch presentation of the total map $F(x) = 2x$ on $\mathbb{N}$: branch 0 has guard $x = 0$ and update 0; branch 1 has guard $x > 0$ and update $2x$. The packet has selectors b_0, b_1 , copies z_0, z_1 , and one slack s :

$$P(x, y; b_0, b_1, z_0, z_1, s) = (b_0 + b_1 - 1)^2 + (x - z_0 - z_1)^2 + (y - 2z_1)^2 \\ + z_0^2 + (z_1 - b_1 - s)^2 + b_1 z_0 + b_0 z_1.$$

Here $B = 2$, $d = 1$, $E = 1$, $T = 1$, $W = 0$, so the ledger gives five variables, five squared residuals, and two products. For $x = 0, y = 0$ the unique witness is $(1, 0, 0, 0, 0)$; for $x > 0, y = 2x$ it is $(0, 1, 0, x, x - 1)$. Every other input/output pair has no natural witness. The core replay exhaustively checks 100,000 bounded assignments, including every auxiliary tuple in $\{0, \dots, 4\}^5$ for $0 \leq x < 4$, $0 \leq y < 8$.

The generic event-branch compiler also fully enumerates a three-signal transparent test machine with speeds $2, 0, -2$. Its 19 branches have $E = 20$, $T = 84$, $W = 0$, yielding 160 auxiliary variables, 111 squared affine residuals, and 19 complementarity products. Both pivot and uniform scalings pass 144 valid-germ tests, including 88 legal steps and two simultaneous steps, with every canonical residual exactly zero.

This example illustrates the exact polynomial mechanics without suggesting that a universal branch family has only two branches. For the signal frontend, the same constructor takes the actual finite label-mode and first-collision branch descriptions.

H Part III: the complete finite mode closure and numeric packet

H.1 A finite family covering every encoded run

The naive family of all 114^{18} ordered label words is unnecessary. Starting from the literal initial mode, consider each nonempty subset J of eligible adjacent closing pairs. A pair is eligible only when its exact incoming labels have a defined rule and their speeds are in incoming order. Exclude subsets with consecutive edges, since those would constitute an undefined collision of at least three signals. Replace every selected pair atomically by its speed-sorted output pair and add the resulting mode. Repeating to exhaustion gives the least mode-only closure for this syntactic successor operation.

Every legal event of an encoded run uses one of these nonadjacent eligible subsets, by the complete-collision lemma. Induction therefore places every event mode and branch in the closure. Conversely, every retained branch has exactly the original mode, germ, earliest-collision and rule guards, so it introduces no spurious geometric step. Restricting the integer map to this closed mode family preserves every encoded computation.

The closure is an overapproximation of modes numerically reachable from the specified input family: successive graph edges may be realizable at different gap vectors which cannot be spliced into one orbit. No assertion of reachability of every graph node is made. Nevertheless every branch is individually feasible. If c is its speed-difference vector, choose

$$h_i = \begin{cases} c_i & i \in J, \\ \max(c_i, 0) + 1 & i \notin J, \end{cases} \quad u = q \sum_i h_i.$$

All gaps are strictly positive, its next delay is one, exactly J collides, and the output is an admissible germ. The exporter checks this witness for every branch.

The completed enumeration has 49,700 modes and 80,501 branches: 65,557 single-collision branches and 14,944 branches with two simultaneous disjoint collisions. The maximum graph discovery depth is 321. This depth is a graph statistic, not a computation-time bound or a numeric reachability certificate.

Among the modes, 1,728 have no eligible legal pair. Only 36 of those have no closing edge at all and are collision-free; the other 1,692 are blocked by undefined collisions. Of the 36 collision-free modes, six carry an accepting halt label and 30 carry the null label. The fixed final accepting mode in Section 35 is the target for encoded runs. Neither all graph sinks nor the union of those six label-bearing modes replaces that target.

H.2 Exact guard and witness counts

For each branch the exported guard convention retains one mode equality, $|J| - 1$ nonpivot tie equalities, one strict germ guard for each $c_i \geq 0$, the strict span guard, the strict pivot-gap guard, and all $17 - |J|$ non- J endpoint guards. Some strict guards are redundant; retaining them makes the convention simple and fully reproducible. Nonnegative coordinates are the ambient natural domain, so no weak-gap slacks are introduced.

The resulting exact ledger is

Quantity	Count
Branches B	80,501
Equality guards E	95,445
Strict guards T	2,667,479
Weak guards W	0
Auxiliary variables $19B + T$	4,196,998
All variables including X, Y	4,197,034
Squared affine residuals $37 + E + T$	2,762,961
Complementarity products	80,501

Thus the literal one-step polynomial is fully specified, quadratic, nonnegative on natural arguments, and canonical by Theorem 36.1. For externally fixed K and supplied endpoints, its unrolling has exactly $4,197,016K - 18$ witness variables. This explicit constant is large; the small physical population does not imply a comparably small Diophantine arity.

The untimed affine rows contain 15,323,489 nonzero variable-coefficient entries and one nonzero constant. The maximum absolute affine coefficient has 124 bits. Gap coefficients remain bounded by 24; the larger values arise from finite mode coding. These are sparse affine-circuit statistics, not claims about minimal gate complexity. Adding the optional timestamp coordinate retains the same branch guards and adds one local copy coordinate per branch and two global input/output residuals. Its derived ledger is 4,277,499 auxiliaries and 2,762,963 squared affine residuals per step.

H.3 What the numeric export contains

The mode file gives each mode's 18 label indices, speed differences, exact code and graph index. The compact event file records every branch index, source/target index, canonical pivot and bitmask of the complete set J . Together with the explicit formulas above, these fixed numbers determine every sparse matrix and guard coefficient. The supplied exporter generates the full numeric branch records, including selector, copy and slack indices. A separate circuit file fixes the global variable

order and the polynomial (36.7). All large codes and generated coefficients are serialized as decimal strings.

The shared selector sum is an expression, not another witness variable. Writing its difference with b_r is exactly $\sum_{h \neq r} b_h$, which is nonnegative on all natural assignments. Keeping this expression factored avoids a quadratic-in- B expansion of the complementarity part. The exporter can regenerate every squared affine residual row or answer individual expanded coefficient queries. The verbose branch and full residual-row files were generated and checked during preparation. They are omitted from the compact archive because they are deterministically regenerable from its complete mode/event data and code.

The supplementary incidence calculation gives 1,059,563,096,023 nonzero terms in the full monomial expansion and a maximum absolute coefficient of 248 bits. Those expansion statistics are derived exactly; no trillion-term monomial file was emitted. The actually generated full affine-residual export was checked independently, including all 15,323,489 nonzero variable-coefficient entries.

This is an explicit coefficient description of one fixed step polynomial and each fixed-horizon family. It does not convert the unknown horizon into a fixed number of natural variables, and it does not assert that the mode graph's finite size bounds the length of a universal run.

[write] In this report the machine file, the ledger and the circuit file are shipped as `data/12-conservative-signal-MORITA_18_SIGNAL_MACHINE.json`, `data/12-conservative-signal-LEDGER.json` and `data/12-conservative-signal-POLYNOMIAL_CIRCUIT.json`, and the exporter as `code/12-conservative-signal-compile_packet.py`. The mode file and the compact event file (`numeric/MODES.jsonl.gz` and `numeric/EVENTS.jsonl.gz` in the delivered layout) are not shipped; the exporter's `build` command regenerates them with identical decompressed content (README of this report). The programme's review [29] enumerated the same 49,700 modes and 80,501 branches independently and derived every entry of the ledger above from the closure.

I Part VII: canonical timed quartic certificates (source 19)

The preceding proof supplies more than reachability decidability: it produces exact phase data with a rigid algebraic form. We now apply the constant term lifting compiler already established in Report 13 [source 18, Section 96]. Its one dimensional conclusion is not new here. The finite dimensional extension comes from the proved normal form and a disjoint lexicographic sorting refinement. The complete companion proof and its independent conditional review are included in `scientific/timed/`. [write] [Shipped as `19-mass-four-zd-timed-PROOF.md` and `19-mass-four-zd-timed-REVIEW.md`; with this appendix the fixed-input answer to source 17's Question 7 holds in every finite dimension.]

Theorem I.1 (Fixed input timed charts and quartics). *Fix a CA and input satisfying Theorem 101.1. There is an effective finite family of disjoint charts, each with at most two natural parameters and an integer affine domain, whose time and original frame occupied coordinates are rational polynomials of degree at most two, integer valued on their domains. Occupied sites are strictly lexicographically ordered. The disjoint chart domains map bijectively to the complete timed orbit.*

For every fixed tuple $\mathbf{a} = (a_1, \dots, a_m)$ of nonzero labels, $0 \leq m \leq 4$, there is an effectively obtained integer polynomial

$$C_{\mathbf{a}}(t, X_1, \dots, X_m; w)$$

of ordinary total degree at most four, with natural witnesses w , whose witness fiber is a singleton

exactly when $F^t(c)$ has precisely those lexicographically ordered sites and labels, and is empty otherwise. Here $t \in \mathbb{N}$ and $X_i \in \mathbb{Z}^d$ are external arguments. The witness count depends on the fixed input and rule, not on time or the queried coordinates.

I.1 Half open charts own every time exactly once

[write] [This subsection is source 18's Section 95 in the notation of Section 110.1 (here q_ℓ is source 18's prefix length b_ℓ , and C is its B); new here are the original-frame coordinates in $\mathbb{Z}^d$ and the raw chart count J . It is printed because the counts below use it.] Materialize the finite prefix as one parameter free chart for each $0 \leq t < T_0$. For a translated periodic or independent tail, extend the prefix through all individual transients and choose one common period p . The phase r has the one parameter formula

$$t = T_0 + r + pn, \quad 0 \leq r < p, \quad n \in \mathbb{N},$$

with affine coordinates. Euclidean division determines a unique phase and parameter. The vacuum uses a one clock tail with no occupied coordinates. In the no unit case the same construction already works in the original frame.

For an expanding shuttle, retain the same cycle indexing as in Section 110.1. Let transition ℓ have flight length $h_\ell(n) = a_\ell n + b_\ell$ and fixed encounter prefix length $q_\ell \geq 0$. The flight length is integral for all natural n , so $b_\ell = h_\ell(0)$ and $a_\ell = h_\ell(1) - h_\ell(0)$ are integers. Both $h_\ell(n) > 0$ and $a_\ell > 0$. The cycle start is

$$T(n) = T_0 + A \frac{n(n-1)}{2} + Cn, \quad A, C \in \mathbb{Z}_{>0},$$

and transition ℓ starts at the quadratic integer valued polynomial

$$S_\ell(n) = T(n) + \sum_{h < \ell} (h_h(n) + q_h).$$

The half open cycle intervals $[T(n), T(n+1))$ partition all times from T_0 onward. Each transition owns $[S_\ell(n), S_\ell(n) + h_\ell(n) + q_\ell)$.

If its packet period is p_ℓ , then for each residue $0 \leq r < p_\ell$ the flight chart has parameters $n, j \in \mathbb{N}$, domain and time

$$h_\ell(n) - r - p_\ell j - 1 \geq 0, \quad t = S_\ell(n) + r + p_\ell j. \quad (\text{I.1})$$

It owns only the times strictly before first contact. For $0 \leq s < q_\ell$, a local prefix chart has parameter n and time $S_\ell(n) + h_\ell(n) + s$. If $q_\ell = 0$, contact is the next checkpoint and belongs to the next transition. Thus no boundary is counted twice.

The cycle interval determines n , the transition interval determines ℓ , and division determines the unique flight phase and j or the unique local prefix phase. This proves one chart point per exact time, even if the spatial configuration repeats. Stationary frame coordinates g_v are affine in the chart parameters, and original frame coordinates

$$P_v = g_v + \delta t$$

have degree at most two. If J denotes the raw chart count, the literal construction gives $J = T_0 + p$ for an affine tail and $J = T_0 + \sum_\ell (p_\ell + q_\ell)$ for a shuttle. These input dependent quantities may be enormous.

I.2 Disjoint lexicographic sorting and its cost

For a raw chart with m occupied sites, refine by every permutation π and every list of first differing coordinates $k_a \in \{1, \dots, d\}$ for adjacent sites $v = \pi(a)$, $w = \pi(a + 1)$. Require

$$g_{w,h} - g_{v,h} = 0 \quad (h < k_a), \quad g_{w,k_a} - g_{v,k_a} - 1 \geq 0.$$

The common drift cancels in pair differences, so these conditions are affine. Occupied sites are distinct; hence exactly one order and one first difference list apply at each chart point. Equal labels do not cause multiplicity. Use positive denominator clearing wherever needed.

The sorting factor is 1 when $m \leq 1$, and $m!d^{m-1}$ otherwise. Let B_{ch} be the retained chart count after any fixed label tuple filtering, K its total private parameter slots, M its domain inequalities, and H its domain equalities. Then

$$B_{\text{ch}} \leq 24d^3 J, \quad K \leq 2B_{\text{ch}}, \quad M \leq 4B_{\text{ch}}, \quad H \leq 3(d-1)B_{\text{ch}}. \quad (\text{I.2})$$

The notation B_{ch} is separate from the geometric prefix bound B in the main proof. Incompatible label tuples simply retain no charts.

I.3 The inherited constant term lifting compiler

A slightly more general interface also covers the pattern consequences below. Chart i has private natural parameters z_i , integer polynomial inequalities $A_{ir}(z_i) \geq 0$, integer polynomial equalities $H_{is}(z_i) = 0$, and rational output vector $P_i(z_i) \in \mathbb{Q}^Q$. All degrees are at most two; outputs are integer valued on the domain. Assume the disjoint domains map injectively to the intended external outputs.

Introduce a natural selector e_i , the private parameters, and one natural slack u_{ir} for every inequality. Write

$$E = \sum_i e_i, \quad Z_i = \sum_j z_{ij} + \sum_r u_{ir}, \quad \mathcal{L}_i(f) = f(z_i) - f(0) + f(0)e_i.$$

Only the literal constant is multiplied by the selector, so no quadratic monomial becomes cubic. For each output coordinate choose a positive common denominator L_q across *all* chart polynomials. Define

$$\begin{aligned} R_0 &= E - 1, & R_{ir} &= \mathcal{L}_i(A_{ir}) - u_{ir}, \\ U_{is} &= \mathcal{L}_i(H_{is}), & G_i &= (1 - e_i)Z_i, \\ V_q &= L_q \left(y_q - \sum_i \mathcal{L}_i(P_{iq}) \right). \end{aligned}$$

All residuals have integer coefficients and ordinary degree at most two in the external and witness variables together. Therefore

$$C(y; w) = R_0^2 + \sum_{ir} R_{ir}^2 + \sum_{is} U_{is}^2 + \sum_q V_q^2 + \sum_i G_i^2 \quad (\text{I.3})$$

is an ordinary quartic sum of squares.

At a natural zero, $E = 1$ forces exactly one selector to be 1. Every inactive gate forces $Z_i = 0$, and nonnegativity forces all its parameters and slacks individually to be zero. Its lifted forms vanish.

In the selected branch, the remaining equations enforce precisely its domain, its uniquely determined integer slacks, and its output. Conversely an admissible chart point gives that unique zero, with every inactive private variable zero. Thus the natural zeros are in bijection with admissible chart points over the specified output. Half open ownership and disjoint sorting prove Theorem I.1.

[write] [This is the argument of source 18's Proposition 96.1, whose residuals (96.2)–(96.3) and squared gates (96.5) appear above as U_{is} , V_q and G_i ; source 19 says itself that the one-dimensional conclusion is not new. The programme has read the shipped companion proof 19-mass-four-z d-timed-PROOF.md in full, in a follow-up (original_frame_first_hit30_fixture.md with its independent review review_original_frame_first_hit30_fixture.md, commit 5b4c091e6) [60], and found its half-open chart ownership, common clock, first-difference sorting, constant-term lifts, pooled denominators, uniqueness of the inactive natural witnesses and original-frame padding sound as conditional statements; the general normal form stays an inherited hypothesis.]

There are exactly

$$N_w = B_{\text{ch}} + K + M \leq 7B_{\text{ch}} \leq 168d^3J \quad (\text{I.4})$$

natural witnesses, and $1 + M + H + Q + B_{\text{ch}}$ squared residual slots before deleting identically zero terms. For a fixed m site label tuple, $Q = 1 + dm$. An empty chart family uses $C = 1$ and no witnesses. These counts are deliberately coarse and input dependent.

The polynomial is nonnegative on all real arguments, but its exactness and witness uniqueness are assertions over *natural* witnesses. Replacing them by unrestricted signed difference pairs breaks the inactive branch argument. Pooled denominators are essential: separately scaled branch outputs cannot simply be added without a common scale.

I.4 Whole configurations and signed external coordinates

One polynomial can cover varying site counts and labels by encoding

$$(t, m, a_1, \dots, a_4, X_1, \dots, X_4)$$

with positive integer nonzero label codes, occupied sites sorted, and all unused label and coordinate slots zero. Append those constants to each chart output; then $Q = 6 + 4d$ and the same witness bound applies. Padded zero coordinates remain literally zero and do not acquire the drift δt .

For equality up to translation, subtract the first occupied site. The relative coordinates $P_v - P_1 = g_v - g_1$ are affine, and time remains quadratic. There is no translation witness. The vacuum is handled separately rather than granting it infinitely many arbitrary translation witnesses.

Signed external integer coordinates are valid polynomial inputs. If all external slots must be natural, replace each scalar X by $X^+ - X^-$ and add the squared residual $(X^+ X^-)^2$. The accepted pair is uniquely $(\max(X, 0), \max(-X, 0))$. These are external slots, not extra witnesses. Total degree remains at most four.

I.5 Fixed patterns including prescribed zeros

For integer vectors v, w , there are $2d + 1$ disjoint comparison outcomes: equality, or either sign at their first differing coordinate. Equality uses d coordinate equations; each strict outcome uses earlier coordinate equalities and one inequality with an integer margin of 1. Comparisons of quadratic outputs therefore introduce only degree two domain conditions, within the compiler interface.

For a fixed anchored pattern on s sites, refine every sorted orbit chart by all comparisons of each occupied site with each window site. The full comparison matrix and fixed labels decide its

truth, including zeros. Retain successful matrices and output only t . The refinement is disjoint, and the orbit determines a unique chart point at each time, so the quartic has an empty or singleton natural witness fiber. One may take

$$B_{\text{anch}} \leq B_{\text{ch}}(2d+1)^{4s}, \quad N_{\text{anch}} \leq (7+4s)B_{\text{anch}}.$$

Here B_{ch} refers to all sorted orbit charts before observation filtering.

For a translated pattern with a nonzero prescribed site w_* , every successful placement sends w_* to one of at most four occupied sites carrying the correct label. The candidate shift is $b_i = P_i - w_*$. Refine by the *complete* matrix of comparisons

$$P_v - P_i \quad \text{against} \quad w - w_*$$

for all occupied i, v and all pattern sites w . The drift cancels and these conditions are affine. The matrix determines the success of every candidate. Retain the matrix once if any candidate succeeds, rather than introducing one branch per success. If a placement is also output, choose the least successful occupied site index. This prevents same time placement multiplicity. Bounds are

$$B_{\text{trans}} \leq B_{\text{ch}}(2d+1)^{16s}, \quad N_{\text{trans}} \leq (7+16s)B_{\text{trans}}.$$

A wholly zero translated pattern, including the empty pattern, occurs in every finite support configuration at every time. For occurrence in external natural t alone, the zero polynomial with no witnesses has the single empty witness tuple. There need not be a lexicographically least placement. If a placement is wanted, choose a specified rule that moves its entire window beyond the maximal first support coordinate, with the vacuum handled separately.

These are fixed pattern statements. The chart family and counts may depend on the whole pattern. No uniform arity claim is made for an arbitrary pattern provided as one encoded external input.

I.6 What time uniqueness does not imply

Time is external in every single witness claim. If it is existentially quantified instead, the same construction has one zero for *each occurrence time*. A stationary nonempty configuration already gives an infinite fiber. Thus this appendix does not establish untimed finite foldness, a generic single fold MRDP theorem, a uniform bound over input configurations, or exactness for rational or real witnesses. The explicit chart interface, not bare decidability, is what permits the quartic compiler.

The companion finite checks cover 208 drift restored two dimensional shuttle states, 325 lexicographic refinement cases, 1,215 finite witness cube evaluations, and additional selectors, inactive gates, domains, signed pairs, and pattern placements. Their normal and optimized receipts agree. They test the displayed algebra on fixtures; they do not implement a general CA to chart compiler or prove the dynamical normal form by experiment.

J Part VII: source lineage and audit status (source 19)

The starting proof source is the 3 October 2026 higher dimensional research packet. Its original SHA 256 is

3e8eb41fe43b5b4b61d0e0a5b4bece3b01cf628f834a820ecc369f4da427b30a.

The delivery copy in `scientific/PROOF.md` replaces environment specific provenance paths with report identifiers and updates its introductory delivery description. The literature copy removes the local reading cache note in favor of primary links. `verification/source-lineage.json` records original and delivered hashes. The mathematical exposition in this report follows the corrected proof and makes its determinant formula, dispatch convention, and observation semantics explicit.

The earlier audit corrected two important wording points: compact exits need the actual gap for their later hit test, and a decreasing cycle may exit to a terminal profile as well as to ordinary. Both are incorporated above. The fresh adversarial review, included as `scientific/independent-review.md`, found no missing dynamical case, circular enumeration, or fatal gap in the lower theorem. Its requested zero drift dispatch, common seed anchor convention, and explicit small launch bound are incorporated above. That review does not reprove the inherited upper construction. Source releases were read without modification. The relevant earlier TeX source hashes are:

Report 12 four mass decision precursor

803bf0c4194bb9d0f1942e6ad3eb762c79d7a811bd785742ee63f0c46a775595

[source 17, Part VI: this is the hash of `four-mass-bound/four-mass-decidability.tex` in arrival 4e270aa46]

Report 16 literal reversible source

7e644e82b986903d473a6f99b379f8141e9a71e18fb01ecf3d4e554009563d81

[batch 82, archive `Literal_Universal_Reversible_Source_Package.zip`; printed as Part II of `five-particle-binary-automata`]

Report 26 parallel involution replacement

a156d7d973721f18c896cf71f0e9eedbaa214a613b8c7a77cf4704be90850dab

[batch 82, archive `Two_Parallel_Conservative_Involutions_Package.zip`; printed as Part VI of `five-particle-binary-automata`]

These are proof dependencies for the inherited statements, not redistributed literature. This report does not revise their one dimensional conclusions.

[write] [The delivered files named in this appendix are shipped as `19-mass-four-zd-PROOF.md`, `19-mass-four-zd-literature.md`, `19-mass-four-zd-independent-review.md` and `data/19-mass-four-zd-source-lineage.json`; the README maps every delivered name. The pins of Reports 12, 16 and 26 were verified at placement against the arrival archives. The Hilbert's-tenth programme has since reviewed source 19's proof (note at the end of Section 116) [58].]

K Part VII: canonical first visit quartic certificates (source 20)

The first arrival theorem yields a direct Diophantine consequence that avoids counting repeated visits as different witnesses. We give the construction rather than invoking a generic finite fold or single fold representation theorem. All witnesses below are natural numbers, including zero; the external site coordinates may be signed integers.

Theorem K.1 (One canonical witness per visited site). *For a fixed promised CA and fixed mass at most four input, an ordinary integer polynomial $P(y; w)$ of total degree at most four can be effectively constructed such that*

$$\#\{w \in \mathbb{N}^W : P(y; w) = 0\} = \begin{cases} 1, & y \in S, \\ 0, & y \notin S, \end{cases} \quad y \in \mathbb{Z}^d.$$

The stationary frame first visit time $\tau(y)$ is canonically reconstructible from the unique witness. Alternatively, one can enforce it as an external argument t or move that time argument into one additional canonical natural witness, still at total degree at most four. The polynomial, coefficients, and witness arity depend on this rule and input.

The stronger form with a time argument has a unique natural witness exactly when $y \in S$ and $t = \tau(y)$, and none otherwise. Time is actual elapsed discrete time, not a cycle number. A certificate for all visits followed by existentially forgetting time would not prove this: recurrent sites could retain many or infinitely many time witnesses.

K.1 Disjoint clauses for the spatial domain

More generally, take any effective Presburger set $S \subseteq \mathbb{Z}^d$ with a specified nonnegative integer function f represented by rational quadratic polynomials on finitely many semilinear cells. The first arrival theorem supplies this data for an expanding tail and its prefix. For independent or translated periodic tails the timed occurrence relation is already Presburger; least time selection and Lemma 122.1 give piecewise rational affine $f = \tau$. Finite and empty sets are included.

Apply effective Presburger elimination to all cell formulas. Collect their integer affine comparison forms $\ell_1, \dots, \ell_a$ and congruence expressions g_j modulo fixed positive m_j , $1 \leq j \leq q$. Enumerate the complete outcomes: each ℓ is negative, zero, or positive, and each g_j has exactly one of its m_j residues. Retain an outcome exactly when it belongs to at least one original cell and assign the polynomial of its least indexed containing cell. All such polynomials give f on that outcome. Unsatisfiable outcomes may be deleted.

Converting strict signs by integer unit offsets yields pairwise disjoint branches i of the form

$$A_{ir}(y) \geq 0, \quad H_{is}(y) = 0, \quad L_{iv}(y) \equiv r_{iv} \pmod{m_{iv}}, \quad (\text{K.1})$$

with integer affine forms, $m_{iv} > 0$, and normalized residues. This partitions spatial points, not coefficients of a semilinear generating family. No uniqueness of those generator coefficients is needed.

Let B be the number of branches and let I, H, C be the total occurrences of inequality, equality, and congruence atoms in them. Before redundant atoms are removed, the full outcome construction gives

$$B \leq 3^a \prod_{j=1}^q m_j, \quad I + H = aB, \quad C = qB.$$

The potentially enormous size is not hidden by the constant degree assertion.

K.2 Affine gating with canonical signed quotients

For each branch introduce a natural selector e_i . For each inequality introduce one natural slack u_{ir} , and for each congruence introduce two natural quotient variables q_{iv}^+, q_{iv}^- . Form the residuals

$$\begin{aligned} R_0 &= \sum_i e_i - 1, \\ R_{ir} &= e_i A_{ir}(y) - u_{ir}, \\ S_{is} &= e_i H_{is}(y), \\ Q_{iv} &= e_i (L_{iv}(y) - r_{iv}) - m_{iv} (q_{iv}^+ - q_{iv}^-), \\ K_{iv} &= q_{iv}^+ q_{iv}^-. \end{aligned} \quad (\text{K.2})$$

Let P be the sum of all their squares. Each residual has integer coefficients and total degree at most two, including the external y coordinates, so P has ordinary total degree at most four.

At a natural zero, $R_0 = 0$ selects exactly one branch. On an inactive branch, $R_{ir} = 0$ forces its slack to zero. Its congruence balance forces $q^+ = q^-$, and their zero product forces both to zero. Thus no inactive auxiliary remains free. On the active branch the slack is uniquely $A_{ir}(y)$, the equalities hold, and each congruence fixes an integer quotient

$$q = (L_{iv}(y) - r_{iv})/m_{iv}.$$

The zero product forces its unique natural encoding $(q^+, q^-) = (\max(q, 0), \max(-q, 0))$. The disjoint clauses determine the selector uniquely. Conversely these canonical assignments make every residual vanish for each $y \in S$. A sum of real squares is zero exactly when every residual is zero, proving membership and witness uniqueness.

For $B > 0$ the literal witness and squared residual counts are

$$W_{\text{mem}} = B + I + 2C, \quad R_{\text{mem}} = 1 + I + H + 2C. \quad (\text{K.3})$$

For an empty branch list use $P = 1$ and no witnesses.

[write] [Second route. This canonical semilinear membership quartic is Part IV's compiler of Section 46 (a unique slack for each inequality and the sign-separated Euclidean quotient (46.1)) combined with Part I's selector device (Theorem 10.2); Part IV gates its clauses through a circuit of truth bits, whereas source 20 multiplies each atom by its selector. Part XI of *canonical-diophantine-certificates* gives every semilinear set a single-fold representation of degree two that is nonnegative on the real orthant (Theorem `cdc:of:thm:singlefolds1`; compare Remark 10.3), so degree four is not needed for membership alone. No novelty is claimed for this subsection. New in this appendix are the one shared cycle variable for quadratic first arrivals, the square lifts for general piecewise quadratic time, the untimed first-hit certificates and the original-frame complete-target certificate through whole-orbit injectivity.]

K.3 One shared cycle variable for actual first arrivals

The expanding shuttle application admits a sharper clock implementation than a general quadratic function. Retain on each final tail clause its rational affine functions $n_i(y) = n_*(y)$ and $k_i(y) = T_0 + h_*(y)$. On every prefix first visit singleton, set $n_i = 0$ and k_i equal to its first time. All clauses then share

$$\tau(y) = cn_i(y)^2 + bn_i(y) + k_i(y),$$

with the same c, b , and both affine functions are nonnegative integers on their assigned cells.

Introduce one shared natural variable N . Choose positive integers A, D so that $a_i = An_i$ and $K_i = Dk_i$ are integer affine, and Dc, Db are integers. Add exactly the two squared residuals

$$AN - \sum_i e_i a_i(y), \quad Dt - DcN^2 - DbN - \sum_i e_i K_i(y). \quad (\text{K.4})$$

The unique selector fixes $N = n_i(y)$ and the second residual fixes $t = \tau(y)$. Both residuals have ordinary total degree at most two, so their squares retain degree at most four. Prefix branches work with $N = 0$, without negative offsets or an extra selector. All inactive membership auxiliaries were already forced to zero.

Thus for actual expanding first arrivals, time external requires

$$W = B + I + 2C + 1, \quad R = 3 + I + H + 2C. \quad (\text{K.5})$$

Making time itself a natural witness changes the first count to $B + I + 2C + 2$, with the same residual count. These are counts of variables in the unique witness tuple, not a claim that the tuple has just one variable. Only the extra clock overhead is dimension independent; the clause and atom counts still depend on the full fixed rule and input. Reconstruction of first time from the membership witness alone needs no added variable.

K.4 General piecewise quadratic time outputs

At the unique membership witness, the selected branch already gives the canonical expression $f(y) = \sum_i e_i f_i(y)$. Its total degree may be three, which is harmless for reconstruction but would produce degree six if squared as a residual. To preserve degree four when enforcing time, use shared natural square lifts

$$U_j = y_j^2 \quad (1 \leq j \leq d), \quad U_{jk} = (y_j + y_k)^2 \quad (1 \leq j < k \leq d).$$

Impose these with their quadratic difference residuals. The universal sufficient number is $M = d(d+1)/2$, and each lift is uniquely nonnegative even for signed coordinates. Polarization gives

$$y_j y_k = \frac{1}{2}(U_{jk} - U_j - U_k).$$

Thus every f_i becomes a rational affine expression in (y, U) . Choose one common positive denominator D , including the polarization halves, and write it as $F_i(y, U)/D$ with integer affine F_i . Add the single pooled residual

$$Dt - \sum_i e_i F_i(y, U). \quad (\text{K.6})$$

It has total degree at most two. Adding its square and the square lift residuals to P retains degree at most four and forces exactly $t = f(y)$. All shared lift values are canonical independently of branch selection; none are left as inactive degrees of freedom.

For external time the literal counts are

$$W_{\text{time}} = B + I + 2C + M, \quad R_{\text{time}} = 2 + I + H + 2C + M. \quad (\text{K.7})$$

Moving t to a natural witness adds one to W and leaves R unchanged. For affine cell functions one may take $M = 0$. Unused square lifts can be omitted after specifying a sufficient list.

If all external slots must also be natural, replace each y_j by $Y_j^+ - Y_j^-$ and add the squared product residual $(Y_j^+ Y_j^-)^2$. This accepts exactly the canonical positive and negative part pair. It changes d signed external slots to $2d$ natural slots, adds d residuals, and adds no witnesses. Affine substitution and quadratic products preserve the degree bound.

K.5 First hits of configurations and fixed patterns

The same construction supplies a useful untimed reachability refinement of Report 29 [source 19]. Fix a finite tuple of external observation parameters z , and a time independent observation predicate that is Presburger in z and the finite labeled support of the stationary frame configuration. The schema and tuple length are fixed when compiling the polynomial.

Substituting the finite phase support lists produces a Presburger occurrence relation $\text{Occ}(z, n, h)$ in half open expanding cycles. It may have many witnesses; only its truth is used. A prefix time can satisfy infinitely many z , so the site-only singleton treatment must be replaced. If $P_t(z)$ is the observation predicate evaluated on the actual prefix configuration at time t , assign that time to the Presburger domain

$$P_t(z) \wedge \bigwedge_{0 \leq s < t} \neg P_s(z).$$

These finitely many domains are disjoint and cover all prefix hits. Exclude their union before minimizing (n, h) on the tail. Least cycle and offset are again Presburger functions and become rational affine on effective semilinear cells. Prefix cells use $n_i = 0, k_i = t$; tail cells use $n_i = n_*(z), k_i = T_0 + h_*(z)$. The shared counter construction (K.4) therefore applies unchanged.

Corollary K.2 (Canonical untimed first hit certificates). *For every such fixed schema, its stationary frame first hit domain has an ordinary integer polynomial of degree at most four with exactly one natural witness tuple per accepted external observation and none otherwise. Its first absolute hit time is canonically reconstructible, external, or one additional natural witness. The counts in (K.3) and (K.5) apply to its final disjoint clauses. This includes the following observations.*

1. *Complete finite configuration targets, canonically encoded by a site count $0 \leq m \leq 4$, positive codes for active nonzero labels from the fixed alphabet, and at most four signed coordinate slots, strictly lexicographically sorted, with zero label and coordinate padding beyond m .*
2. *anchors of a fixed finite pattern, including prescribed vacuum symbols. The anchor is external; the pattern's finite offsets and symbols are fixed at compilation.*

For complete targets, cardinality, labels, coordinate equality with all phase support sites, ordering, and padding are finite Presburger conditions. Thus actual untimed stationary frame configuration reachability has a unique natural witness tuple for every reachable canonical target. This does not follow merely by dropping the time argument from an all occurrence certificate.

For a fixed pattern, each nonzero entry requires a matching occupied site at the anchor plus its offset; each zero entry requires absence of all occupied sites there. This is again a finite Presburger predicate. The argument does not cover an arbitrary unbounded external pattern encoding.

There is no quadratic distance growth assertion for general observation parameters. A wholly zero pattern can already hold at infinitely many distant anchors at time zero. The corollary concerns semilinear first hit domains, piecewise quadratic first times, and canonical quartic certificates. As before, it is a fixed rule, fixed input result, and does not by itself assert general expanding original frame reachability under nonzero drift. Complete targets admit the separate corollary below.

K.6 Complete configuration targets in the original frame

For full configurations there is a further argument that removes the frame restriction. It uses exact trajectory injectivity, not a first visit spatial substitution.

Lemma K.3 (An expanding whole configuration orbit is injective). *If the classified tail is expanding, the complete configurations $F^t(c_0)$ are pairwise distinct for all $t \geq 0$, including the initial prefix.*

Proof. At section n the actual two residual unit sites have separation $(x_0 + n\Delta)\nu + z$, with a fixed transverse offset z . Hence the configuration diameter is at least $\|(x_0 + n\Delta)\nu + z\|_\infty$ and tends to

infinity. Translating by $T_n\delta$ leaves this diameter unchanged, so this statement holds in the original frame.

If $F^a(c_0) = F^b(c_0)$ for $a < b$, determinism implies $F^{a+s}(c_0) = F^{b+s}(c_0)$ for every $s \geq 0$. The future then consists of finitely many repeated configurations and has bounded diameter. This contradicts the unbounded section diameters. The argument excludes every repeated pair, even if one or both times precede the eventual tail. $\square$

Corollary K.4 (Original frame untimed configuration certificate). *For every fixed promised CA and input, original frame untimed complete configuration reachability admits an effective ordinary integer polynomial of degree at most four with exactly one natural witness tuple for each reachable canonical target and none otherwise. The first reachability time can be part of that tuple.*

For an expanding tail, the timed chart construction of Report 29 [source 19, Appendix I, whose compiler is source 18's] gives the explicit bound

$$W \leq 7B_{\text{ch}} + 1,$$

where B_{ch} is its retained sorted timed chart count, not the Presburger clause count B in this appendix. No uniform over input arity bound is asserted.

Proof. Report 29's Appendix A [source 19's Appendix I, bound (I.4)] supplies a quartic $C(t, z; w)$ for canonical complete original frame targets z , with exactly one natural witness tuple for each actual time and target pair, and at most $7B_{\text{ch}}$ witness variables. Its half open charts, lexicographic sorting, and inactive branch gates establish that uniqueness. Reclassify the external elapsed time t as one additional natural witness. Lemma K.3 says that a reachable complete target has exactly one occurrence time. Therefore its combined tuple (t, w) is unique, and total degree remains at most four. The time is automatically its first occurrence.

For independent or whole translated periodic tails, the full original frame timed occurrence relation is Presburger: restoring drift preserves the affine phase rays. Select the least occurrence time and apply the disjoint clause construction with a piecewise affine time output. Prefix overlaps and repeated periodic configurations are handled by this first time selection. Finite and empty cases are included. $\square$

This argument is specific to complete configurations. Sites, labels, and pattern observations can recur even when the entire configuration never repeats. The original frame extensions for their first visits therefore do not follow from this corollary. In particular it does not repair a substitution of $y - \delta t$ into a stationary site first visit certificate.

K.7 Application and limits

For expanding shuttles, first remove the finitely many sites first visited in the prefix and then select the lexicographically least cycle and within cycle offset using the Presburger occurrence relation. Equation (122.1) supplies the quadratic cell function. This proves Theorem K.1 for the entire stationary frame set, not just its eventual tail. Affine, periodic, independent, and finite alternatives use their affine first time functions. Vacuum uses the empty polynomial certificate $P = 1$.

If $\delta = 0$, the result directly concerns F . For affine tails, restoring δt preserves affine timed phase rays, so the same construction also works in the original frame. For expanding site and pattern observations, no general original frame first hit corollary is asserted. Complete configurations have the separate argument of Corollary K.4. Substituting $y - \delta t$ into a stationary first visit certificate is

insufficient: a later return to a stationary frame site can create a new original frame site that the first visit certificate omits.

This is a direct elementary construction using disjoint Presburger clauses and explicit square lifting. It is not a generic single fold MRDP theorem, an inference from decidability alone, an assertion for rational or real witnesses, or a uniform over input arity bound. The full supporting note and its independent review are included in the scientific sources. [write] [Shipped as 20-orbit-geometry-first-visit-PR00F.md, 20-orbit-geometry-first-visit-independent-review.md, 20-orbit-geometry-original-frame-PR00F.md and 20-orbit-geometry-original-frame-REVIEW.md.] Finite algebra fixtures check the formula bookkeeping, inactive branch zeros, signed quotients, time lift, and degree counts; they are not a general Presburger elimination or CA classification implementation.

[write] **The programme's intake.** The Hilbert's-tenth programme read the shipped original-frame proof and first-visit companion in full, and the central parts of the geometry proof, in review_original_frame_first_hit30_intake.md (Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/, commit c875bad40, 3 October 2026) [59]. It found no defect in the step of Corollary K.4 that turns the timed certificate into an untimed one through whole-orbit injectivity (Lemma K.3), nor in the separate treatment of affine and periodic tails. It is an intake, not a review of the compiler: it did not review the inherited timed certificate of source 19's Appendix I (whose companion proof it names as the next audit), and it stresses that the witness and residual counts here are not gate counts, that the bound $W \leq 7B_{\text{ch}} + 1$, with $B_{\text{ch}} \leq 24d^3J$ (source 19's (I.2)), depends on the whole automaton and input, and that the shared cycle variable is a lead for a paid comparison, not an established saving. It executed no archived code. In a follow-up (original_frame_first_hit30_fixture.md with its independent review review_original_frame_first_hit30_fixture.md, commit 5b4c091e6) [60], the programme then read source 19's timed proof in full and found no gap in it, conditional on the normal form, and, for one drifted orbit of the weighted planar automaton of Section 123 composed with a unit vertical shift, emitted complete first-hit quartics for complete original-frame targets, the smallest with 35 paid operations (11 multiplications, 24 additions), three natural witnesses, eight squared rows and exact degree four. That is one fixed-input certificate in a decidable class, not a compiler, an arity bound over inputs or a change of the universal bounds.

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- [72] [write] ProveIt research programme, *Input-free height destroys the four direct clean-clock representations*, `Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/three_mass_input_free_height_obstruction.md`, with its program, receipt and independent review in the same directory; commit 735f62a6a, 3 October 2026.
- [73] K. Morita, *Universality of one-dimensional reversible and number-conserving cellular automata*, *Electronic Proceedings in Theoretical Computer Science* **90** (2012), 142–150. <https://doi.org/10.4204/EPTCS.90.12>; <https://arxiv.org/abs/1208.2760>. Definitions in Section 2; the 96-state four-neighbor construction in Section 3. (Sources 24 and 25.)
- [74] Vladimir Reshetnikov, *ProveIt repository*, `SetTheory/Cardinals/docs/reports/hilbert-tenth-problem/signal-machine-collision-certificates/README.md` (the README of this report), at commit e3e58a8820593857650bb7a0193a11ec32db67d6, inspected 4 October 2026. [Pinned source link](#). The release retains the read inert text and its byte hash. (Source 26’s repo.)
- [75] Joël Ouaknine and James Worrell. *Positivity Problems for Low-Order Linear Recurrence Sequences*. arXiv:1307.2779, 2013. In particular, Section 6 on strict positivity. <https://arxiv.org/abs/1307.2779>; author manuscript: <https://www.cs.ox.ac.uk/james.worrell/pos12.pdf>. SODA 2014. Source 30 cites Theorem 4.1, the rational scaling after Theorem 4.2, and Section 6. (Source 26’s positivity and source 30’s entry, merged.)
- [76] L. Dai and B. Xia. *Non-Termination Sets of Simple Linear Loops*. ICTAC 2012, LNCS 7521, 61–73. [doi:10.1007/978-3-642-32943-2_5](https://doi.org/10.1007/978-3-642-32943-2_5); <https://arxiv.org/html/1206.0232v1>. Section 4, Theorem 4.1 and Proposition 3. (Sources 27, 29 and 30, merged.)

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- [78] J. Durand-Lose. *Abstract Geometrical Computation introductory overview*. https://www.univ-orleans.fr/lifo/Members/Jerome.Durand-Lose/Recherche/AGC/intro_AGC.html. (Source 27’s `durandintro`.)
- [79] J. Durand-Lose. *Abstract Geometrical Computation 5: Embedding Computable Analysis*. Natural Computing 10(4), 1261–1273, 2011. [Author manuscript](#). (Sources 27 and 29, merged.)
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- [81] *A five-signal periodic return with rational irrational-angle rotation*, 4 October 2026: source 27’s frozen construction proof, delivered as `science/construction/PROOF.md` (and as `science/sources/five-signal-PROOF.md` in source 28’s archive), SHA-256 `1bea81f8f2e6693d4b7958fb44763eb644663d208fe70c577e0137184e17a725`. [write] Shipped as `27-fivesignal-science-construction-PROOF.md`. (Source 28’s `geometry`.)
- [82] *Independent audit of the five-signal obstruction*, 4 October 2026: delivered as `independent_audit/INDEPENDENT_AUDIT.md` of source 27 (and `science/sources/INDEPENDENT_AUDIT.md` of source 28), SHA-256 `b119df1c0be8685078b011c14e3c71decabce343f9ba9041ee6f97b4c58a5226`. [write] Shipped as `27-fivesignal-independent_audit-INDEPENDENT_AUDIT.md`. (Source 28’s `physicalaudit`.)
- [83] *An explicit degree-12 Diophantine certificate for five-signal infinite validity*, 4 October 2026. Frozen arithmetic proof `science/PROOF.md`, with the four inert DAGs and source receipts. [write] Shipped as `28-diophantine-science-PROOF.md` and `data/28-diophantine-science-evidence-*`. (Source 28’s `science58`.)
- [84] *Independent adversarial audit of five-signal Diophantine certificate 58*, 4 October 2026. `independent_audit/INDEPENDENT_AUDIT.md`; exact-source and arithmetic receipts accompany it. [write] Shipped as `28-diophantine-independent_audit-INDEPENDENT_AUDIT.md` and `data/28-diophantine-independent_audit-*`. (Source 28’s `audit58`.)
- [85] *Independent audit of the expanded POWER specialization*, 4 October 2026. `independent_audit/POWER_AUDIT.md`, with local theorem line references and both-direction domain specialization. [write] Shipped as `28-diophantine-independent_audit-POWER_AUDIT.md`. (Source 28’s `poweraudit`.)
- [86] Mario Carneiro and mathlib contributors, *Pell equation and Matiyasevic’s theorem*, Mathlib/NumberTheory/PellMatiyasevic.lean, mathlib4 commit `ac77769fab23cb237559e7f56578dbead91499f`; theorems `Pell.matiyasevic` (lines 760–766) and `Pell.eq_pow_of_pell` (lines 860–864). Apache License 2.0. File SHA-256 `993760c797ad0ff66fa77064a616fce779550bc745cbf6c44e04f392fd0bed0a`. [Pinned source](#). Read by the sources as inert text; no Lean execution is claimed. (Source 28’s `pell`, source 29’s `mathlib`, and the entries of sources 30, 35, 36, 37 and 38, merged. [write] The archives’ copies of the file and of its licence are not shipped with this report.)

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- [90] W. Calvert, K. Kramer and R. Miller. *The Cardinality of an Oracle in Blum–Shub–Smale Computation*. EPTCS **24**, 56–66, 2010. [doi:10.4204/EPTCS.24.10](https://doi.org/10.4204/EPTCS.24.10). Section 1 and Lemma 1.1. (Source 29’s `calvert`.)
- [91] S. Basu. *Algorithms in Real Algebraic Geometry: A Survey*. [Author-hosted text](#), Section 2.1, Theorem 2.1. [arXiv:1409.1534](https://arxiv.org/abs/1409.1534). (Source 29’s `basu`.)
- [92] F. Becker, T. Besson, J. Durand-Lose, A. Emmanuel, M.-H. Foroughmand-Araabi, S. Goliaei and S. Heydarshahi. *Abstract Geometrical Computation 10: An Intrinsically Universal Family of Signal Machines*. [arXiv:1804.09018](https://arxiv.org/abs/1804.09018), revised 21 March 2019. Section 2, Definitions 1–2 (PDF pp. 4–5), and Example 5 (PDF p. 6). ([write] Merged entry: source 31’s `becker`, source 32’s `becker10`, source 33’s `becker` and source 34’s `Becker18`.)
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- [99] [write] ProveIt research programme, *Bounded intake of twelve signal/particle archives at 0d7f51c44*, Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/review_new_signals_0d7f51c44.md, with its JSON manifest review_new_signals_0d7f51c44.json in the same directory; commit 49b100cfd, 4 October 2026. A routing and scope review of the pipeline’s Reports 49, 51 and 56–65 (sources 24–35, batch-91 manuscripts 01, 03 and 08–17), not a proof certification. (Cited by the drafts of all five batch-91 Parts under three keys, merged.)
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- [101] Andrej Dudenhefner. *Certified Decision Procedures for Two-Counter Machines*. FSCD 2022, LIPIcs 228, 16:1–16:18. Section 2, Definition 2 and Theorem 6 (the instruction model and the undecidability context). [doi:10.4230/LIPIcs.FSCD.2022.16](#). ([write] Merged entry: source 34’s Duden22, source 35’s Dud22 and source 37’s counter.)
- [102] Jérôme Durand-Lose and Aurélien Emmanuel. *Abstract Geometrical Computation 11: Slanted Firing Squad Synchronisation on Signal Machines*. 2021. Section 4, especially the width-proportional bounce delays (manuscript p. 8). [arXiv:2106.11176](#). (Source 35.)
- [103] *Three positive witnesses for bounded two counter halting*. Frozen arithmetic continuation, 4 October 2026, retained unchanged in source 36’s bounded-certificate science packet; main proof SHA-256 233b0f67ce13e81a41132017214db5b34d10c965cd9a3ef56d1cbc9d280ac33c. [write] Shipped as 36-bounded-cert-bc-PROOF.md, with the packet’s other files under the prefix 36-bounded-cert-bc-. (Source 36’s compression.)
- [104] *Counting all encoded positive integer gap initializations*. Frozen arithmetic continuation, 4 October 2026, retained unchanged in source 36’s counting science packet; main proof SHA-256 dbc5c8e1cc692f734eaf795a57d081af242469ad41fdcbcdbed0ab5324d55246. [write] Shipped as 36-bounded-cert-cnt-PROOF.md, with the packet’s other files under the prefix 36-bounded-cert-cnt-. (Source 36’s counting.)
- [105] *Native-gap bounded-halting certificates and exact primitive initialization*. Retained dependency, 4 October 2026; Sections 3 and 6 give the POWER specialization and primitive integer classification; main proof SHA-256 8cc5911e528d0555ee89fe63f0e30b8a3eac4a32e3e4237ee9b00a38107a7d1b. [write] This is source 35’s native-gap proof (Part XII); the byte-identical copy is shipped once, as 35-clocked-science-native-gaps-PROOF.md. (Source 36’s native.)
- [106] *Independent review of the encoded-gap counting continuation*. 4 October 2026. Accepted within its stated arithmetic scope without correction; complete review and independently authored exact-arithmetic evidence retained in source 36’s counting audit packet. [write] Shipped as 36-bounded-cert-audit-cnt-REVIEW.md, code/36-bounded-cert-audit-cnt-check_independent.py and data/36-bounded-cert-audit-cnt-*. Written by the same pipeline as source 36. (Source 36’s countaudit.)
- [107] *Independent audit of the three-witness bounded-halting packet*. 4 October 2026. Accepted without mathematical correction, with the stated dependency boundaries; complete audit,

- independently authored exact-arithmetic evidence and data-only coefficient examples retained in source 36’s bounded-certificate audit packet. [write] Shipped as `36-bounded-cert-audit-bc-AUDIT.md`, `code/36-bounded-cert-audit-bc-check_external.py` and `data/36-bounded-cert-audit-bc-*`. Written by the same pipeline as source 36. (Source 36’s `boundaudit`.)
- [108] NIST Digital Library of Mathematical Functions. Section 3.3(i), Lagrange interpolation, equations 3.3.1–3.3.3_2. <https://dlmf.nist.gov/3.3#i>. Consulted 4 October 2026. (Sources 37 and 38’s `dlmf`; source 38’s entry names the equations and the date.)
- [109] Karol Pāk and Cezary Kaliszyk. *Formalizing a Diophantine Representation of the Set of Prime Numbers*. ITP 2022, LIPIcs 237, Article 26, pp. 26:1–26:8. [doi:10.4230/LIPIcs.ITP.2022.26](https://doi.org/10.4230/LIPIcs.ITP.2022.26). (Source 37’s `pak`.)
- [110] N. Alon. *Combinatorial Nullstellensatz*. *Combinatorics, Probability and Computing* **8** (1999), 7–29, especially Lemma 2.1. Author-hosted primary text: <https://web.math.princeton.edu/~nalon/PDFS/null2.pdf>. (Source 38’s `alon`.)
- [111] [write] ProveIt research programme, *Scoped arithmetic/compiler intake at 0d7f51c44*, `Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/review_new_arithmetic_0d7f51c44.md`, with its manifest (SHA-256 `d19ed0ab...`), commit `a9ab9a698`, 4 October 2026.
- [112] [write] ProveIt research programme, *Scoped intake: source 18 timed four-mass quartics*, `Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/review_timed_four_mass_source18_intake.md`, commit `39f398985`, 3 October 2026 (after the batch-82 write of source 18). It read source 18’s proof in this article and its shipped programs as inert text: “PASS on the conditional chart/compiler argument and the explicit binary certificate”.